# Electrical Engineering Reference

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# Foreword

## The Electric Charge

At the foundation of every topic in this book — every circuit, every signal, every electromagnetic wave — lies a single physical phenomenon: the electric charge. It is one of the most fundamental properties of matter, as intrinsic to certain subatomic particles as mass itself, yet its origins remain one of the deepest mysteries in physics. We do not fully understand *why* charge exists or *why* it comes in precisely two types, positive and negative, that attract and repel with a force that is $10^{36}$ times stronger than gravity. We simply observe that it does, and from this one property, the entire discipline of electrical engineering unfolds.

Consider what electric charge actually gives us. A stationary charge creates an electric field that reaches out through empty space, exerting force on other charges without any physical contact — action at a distance that troubled Newton and fascinated Faraday. Set that charge in motion, and it becomes a current, generating a magnetic field that wraps around it in closed loops, perpendicular to its direction of travel. Accelerate the charge, and it radiates electromagnetic waves that propagate at the speed of light, carrying energy and information across the universe. These three behaviors — the electrostatic field, the magnetic field, and electromagnetic radiation — are not three separate phenomena but three manifestations of the same underlying reality, unified by Maxwell's equations into a single elegant framework.

What makes this especially remarkable is the precision and reliability of electrical phenomena. The charge of an electron, $1.602 \times 10^{-19}$ coulombs, is identical for every electron in the universe — not approximately identical, but exactly so, to the limits of our measurement capability. This uniformity means that electrical systems are inherently predictable and reproducible. A resistor obeys Ohm's Law whether it is in a basement lab in Middleton, Wisconsin or on a spacecraft approaching Mars. A photon emitted by a semiconductor junction carries exactly the energy corresponding to the bandgap of the material. This predictability is what makes engineering possible: we can design, analyze, and build systems with confidence that the underlying physics will behave exactly as the mathematics describes.

The flow of charge through a conductor — something as simple as electrons drifting through copper wire — enables the entire modern world. In a power system, vast numbers of electrons oscillate back and forth sixty times per second, transferring gigawatts of energy from generators to cities hundreds of miles away, with the electromagnetic field doing the actual work of energy transport while the electrons themselves barely move. In a semiconductor, carefully arranged regions of excess and deficit charge create the PN junctions and transistor channels that form the basis of all digital computation — billions of tiny switches on a chip the size of a fingernail, each one exploiting the quantum mechanical behavior of charge carriers in crystalline silicon. In an antenna, oscillating charges launch electromagnetic waves into free space, carrying voice, data, and video to receivers on the other side of the planet or on satellites in orbit.

Every chapter in this book explores a different aspect of how we harness, control, measure, and understand the behavior of electric charge. Power engineering moves charge in bulk to deliver energy. Communications engineering encodes information onto oscillating charges and fields. Semiconductor physics explains how charge behaves in crystalline materials. Control systems regulate the flow of charge to achieve desired outcomes. Circuit analysis provides the mathematical tools to predict what charge will do in any network of components. Signal processing extracts meaning from the time-varying patterns of charge. Electromagnetics describes the fields that charge creates and responds to. Each discipline is a different lens focused on the same underlying phenomenon.

The fact that so much of our technological civilization rests on a single elementary property of matter — a property we can measure with extraordinary precision but cannot fully explain — should give any engineer a sense of wonder. Electric charge is not just the starting point of electrical engineering; it is the thread that connects every topic in this book, from the 800 kV transmission lines that span continents to the nanometer-scale transistors that process billions of operations per second. Understanding this connection transforms electrical engineering from a collection of specialized disciplines into a coherent story about one of nature's most remarkable gifts.

# Introduction

Electrical engineering spans a broad range of disciplines — from power systems and communications to semiconductors, control theory, and signal processing. While the foundational principles remain grounded in circuit theory and electromagnetics, the field has expanded to encompass digital logic, embedded systems, photonics, and beyond.

Power Engineering

Communications Engineering

Semiconductors

Control Systems

Embedded Systems

Digital Logic

Circuit Analysis

Signal Processing

Electromagnetics

Power Electronics

Instrumentation and Measurement

Electric Motors

Operational Amplifiers

National Electrical Code (NEC)

Networking

Antenna Design and Principles

Radar Systems

Optics

Engineering Economics

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# Chapter 1

# Power Engineering

Power engineering is the subfield of electrical engineering concerned with the generation, transmission, distribution, and utilization of electric power. It encompasses the design and operation of the entire electrical grid, from large-scale generating stations that convert primary energy sources into electricity, through high-voltage transmission lines that transport power over long distances, to the distribution networks that deliver electricity to homes, businesses, and industrial facilities. Power engineers work with AC circuit analysis, three-phase systems, transformers, protective relaying, and power electronics to ensure reliable and efficient delivery of electrical energy.

## 1.1 Power Generation

**Y2022 U.S. Electricity Net Generation** (source)

| Source | Million kWh |
|---|---|
| **Fossil Fuels** | |
| Coal | 897,999 |
| Petroleum | 19,173 |
| Natural Gas | 1,579,190 |
| Other Gases | 11,397 |
| **Nuclear Electric Power** | 779,645 |
| **Hydroelectric Pumped Storage** | −5,112 |
| **Renewable Energy** | |
| Conventional Hydroelectric | 251,585 |
| Biomass — Wood | 36,463 |
| Biomass — Waste | 17,790 |
| Geothermal | 15,975 |
| Solar | 115,258 |
| Wind | 378,197 |

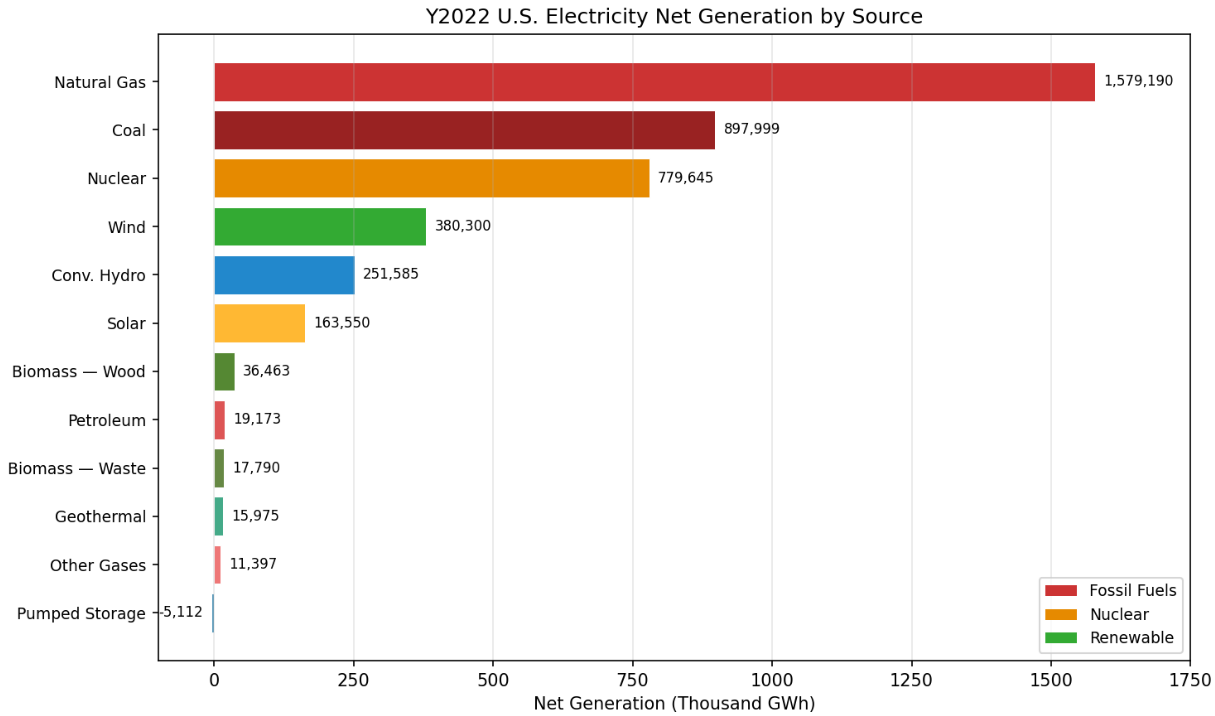



Figure 1.1: U.S. Electricity Generation by Source (2022)

### 1.1.1 Fossil Fuel Plants

Fossil fuel plants generate electricity by burning coal, natural gas, or petroleum to produce steam that drives a turbine connected to a generator. In a typical steam cycle (Rankine cycle), fuel is combusted in a boiler to heat water into high-pressure steam, which expands through a turbine to produce mechanical work, and is then condensed back into water for reuse. Natural gas plants often use a combined-cycle configuration, where exhaust gases from a gas turbine (Brayton cycle) are used to generate steam for a secondary steam turbine, achieving thermal efficiencies above 60%. Coal plants, while less efficient at roughly 33-40%, remain significant baseload generators due to abundant fuel reserves. The primary drawbacks of fossil fuel generation are carbon dioxide emissions, particulate pollution, and dependence on finite fuel supplies.

**Example 1.1.1:** A combined-cycle natural gas plant has a gas turbine (Brayton cycle) with an efficiency of 38% and a steam turbine (Rankine cycle) that recovers 45% of the waste heat from the gas turbine exhaust. If the plant burns natural gas at a thermal input rate of 500 MW, determine the total electrical power output and the overall plant efficiency.
**Solution:**
Gas turbine output: $P_{GT} = 0.38 \times 500 \text{ MW} = 190 \text{ MW}$
Waste heat from gas turbine: $Q_{waste} = 500 - 190 = 310 \text{ MW}$
Steam turbine output: $P_{ST} = 0.45 \times 310 \text{ MW} = 139.5 \text{ MW}$
Total electrical output: $P_{total} = 190 + 139.5 = 329.5 \text{ MW}$
Overall efficiency: $\eta = P_{total} / Q_{in} = 329.5 / 500 = 0.659 = \mathbf{65.9\%}$

### 1.1.2 Hydroelectric Plants

Hydroelectric plants convert the potential energy of water stored at elevation into electricity by directing flow through hydraulic turbines coupled to generators. The available power is proportional to both the volumetric flow rate and the hydraulic head (the vertical distance between the reservoir surface and the turbine), following $P = \rho g Q h$. Conventional impoundment plants use a dam to create a reservoir, while run-of-river installations divert a portion of a river's natural flow with little or no storage. Pumped-storage hydroelectric facilities act as large-scale energy storage: during periods of low demand, water is pumped from a lower reservoir to an upper reservoir, then released through turbines during peak demand, with round-trip efficiencies of 70–85%. Turbine selection depends on head and flow conditions — Francis turbines are used for medium heads (30–600 m), Pelton turbines for high heads (>300 m), and Kaplan turbines for low heads (<60 m) with variable blade pitch for efficiency across a range of flow rates. Hydroelectric plants achieve turbine-generator efficiencies of 85–95%, offer rapid ramping capability for grid frequency regulation, and have operational lifetimes exceeding 50 years with relatively low maintenance costs.

**Example 1.1.2:** A hydroelectric dam has a hydraulic head of 80 meters and a water flow rate of 25 $m^3/s$. The turbine-generator set has an overall efficiency of 88%. Determine the electrical power output. (Use $\rho = 1000\ kg/m^3$ for water and $g = 9.81\ m/s^2$.)
**Solution:**
Available hydraulic power: $P_{hydraulic} = \rho \times g \times Q \times h = 1000 \times 9.81 \times 25 \times 80 = 19{,}620{,}000$ W = 19.62 MW
Electrical output: $P_{elec} = \eta \times P_{hydraulic} = 0.88 \times 19.62$ MW = **17.27 MW**

### 1.1.3 Nuclear Plants

Nuclear power plants generate electricity through nuclear fission, in which heavy atomic nuclei (typically uranium-235 or plutonium-239) are split by neutron bombardment, releasing enormous amounts of thermal energy. This heat is used to produce steam that drives a turbine-generator set, similar to a conventional thermal plant. Nuclear plants operate as baseload generators with capacity factors often exceeding 90%, and they produce no direct carbon dioxide emissions during operation. Reactor designs include pressurized water reactors (PWR), boiling water reactors (BWR), and newer Generation III+ and small modular reactor (SMR) designs that incorporate passive safety systems. The primary challenges of nuclear power are the management of radioactive waste, high capital construction costs, and the long regulatory approval timelines.

**Example 1.1.3:** A nuclear power plant has a rated thermal output of 3400 MW(th) and an electrical output of 1150 MW(e). The plant operates at a capacity factor of 92% over a year (8760 hours). Determine the plant's thermal efficiency and the annual electrical energy generated.
**Solution:**
Thermal efficiency: $\eta = P_{elec} / P_{thermal} = 1150 / 3400 = 0.338$ = **33.8%**
Annual energy at full capacity: $E_{full} = 1150$ MW $\times$ 8760 h = 10,074,000 MWh
Annual energy at 92% capacity factor: $E_{actual} = 0.92 \times 10{,}074{,}000 = 9{,}268{,}080$ MWh = **$9.27 \times 10^6$ MWh**

### 1.1.4 Solar

Solar power generation converts sunlight into electricity using one of two primary technologies: photovoltaic (PV) cells and concentrated solar power (CSP). PV cells use semiconductor materials (typically crystalline silicon) to produce direct current through the photovoltaic effect, which is then converted to alternating current by an inverter for grid connection. CSP systems use mirrors or lenses to concentrate sunlight onto a receiver, heating a working fluid to drive a conventional steam turbine. Solar generation is intermittent and dependent on irradiance, time of day, and weather conditions, making energy storage or supplemental generation necessary for reliable supply. Utility-scale solar installations typically achieve capacity factors of 20-30%, with ongoing cost reductions making solar one of the lowest-cost sources of new electricity generation.

**Example 1.1.4:** A utility-scale solar PV farm has a rated DC capacity of 50 MW. The inverter efficiency is 96%, and the site receives an average peak sun hours (PSH) equivalent of 5.2 hours/day. Determine the average daily AC energy output and the capacity factor.
**Solution:**
Daily DC energy: $E_{DC} = 50\ \text{MW} \times 5.2\ \text{h} = 260\ \text{MWh}$
Daily AC energy: $E_{AC} = 0.96 \times 260 =$ **249.6 MWh**
Hours in a day at full rated output: 24 h
Capacity factor: $CF = E_{AC} / (50\ \text{MW} \times 24\ \text{h}) = 249.6 / 1200 = 0.208 =$ **20.8%**

### 1.1.5 Wind

Wind turbines convert the kinetic energy of moving air into electrical energy using aerodynamic blades mounted on a rotor that drives a generator, either directly or through a gearbox. Modern utility-scale horizontal-axis wind turbines have rated capacities ranging from 2 MW to over 15 MW for offshore installations, with rotor diameters exceeding 200 meters. The power available from wind is proportional to the cube of wind speed ($P = 0.5 \times \rho \times A \times v^3$), but practical extraction is limited by the Betz limit of approximately 59.3% of the total kinetic energy. Wind generation is variable and non-dispatchable, requiring grid balancing through forecasting, energy storage, or complementary generation sources. Onshore wind farms typically achieve capacity factors of 25-45%, while offshore installations benefit from stronger and more consistent winds.

**Example 1.1.5:** A wind turbine has a rotor diameter of 120 m and operates in air with density $\rho = 1.225\ \text{kg/m}^3$. The wind speed is 12 m/s and the turbine's power coefficient is $C_p = 0.42$. Determine the electrical power output.
**Solution:**
Rotor swept area: $A = \pi \times (D/2)^2 = \pi \times (60)^2 = 11{,}309.7\ \text{m}^2$
Available wind power: $P_{wind} = 0.5 \times \rho \times A \times v^3 = 0.5 \times 1.225 \times 11{,}309.7 \times (12)^3$
$P_{wind} = 0.5 \times 1.225 \times 11{,}309.7 \times 1728 = 11{,}970{,}186.48\ \text{W} = 11.97\ \text{MW}$
Electrical output: $P_{elec} = C_p \times P_{wind} = 0.42 \times 11.97 =$ **5.03 MW**

### 1.1.6 Micro Grids

A microgrid is a localized energy system that can operate independently from the main utility grid (islanded mode) or connected to it (grid-tied mode). Microgrids typically integrate distributed energy resources such as solar panels, wind turbines, diesel or natural gas generators, and battery energy storage systems, all managed by a central controller. The controller optimizes power flow, manages load balancing, and coordinates the transition between grid-connected and islanded operation during utility outages. Microgrids improve resilience for critical facilities such as hospitals, military installations, and remote communities that cannot tolerate extended power interruptions. They also enable participation in demand response programs and can reduce energy costs by optimizing the use of local generation and storage resources.

**Example 1.1.6:** A microgrid serves a hospital with a peak load of 800 kW. The microgrid has a 500 kW solar array (currently producing 350 kW), a 400 kWh battery storage system (currently at 90% state of charge), and a 600 kW diesel generator. Determine (a) how much power the battery or diesel generator must supply to meet the peak load, and (b) how long the battery alone could sustain the deficit if the diesel generator is unavailable (assume 90% inverter efficiency and 10% minimum state of charge).
**Solution:**
(a) Deficit = Peak load − Solar output = 800 − 350 = 450 kW
The battery or diesel generator must supply **450 kW**.
(b) Usable battery energy: $E_{usable} = (SOC_{current} - SOC_{min}) \times \text{Capacity} = (0.90 - 0.10) \times 400 = 320$ kWh
AC energy delivered (accounting for inverter efficiency): $E_{AC} = 0.90 \times 320 = 288$ kWh
Duration at 450 kW deficit: t = 288 / 450 = **0.64 hours (approximately 38.4 minutes)**

### 1.1.7 Generator Step-Up (GSU) Transformers

The generator step-up transformer is the critical link between a power plant's generator and the high-voltage transmission system. Generators typically produce electricity at relatively low voltages (11 kV to 25 kV) to keep insulation requirements manageable within the machine, and the GSU steps this up to transmission voltage (115 kV to 765 kV) for efficient long-distance power transfer. GSU transformers are among the largest and most expensive single pieces of equipment in a power plant, with ratings from tens of MVA at small facilities to over 1000 MVA at large nuclear stations. They are typically three-phase, oil-filled units with delta-connected generator-side (low-voltage) windings and wye-connected transmission-side (high-voltage) windings — the delta connection traps third-harmonic currents from the generator, while the wye connection provides a neutral point for system grounding. GSU transformers are designed for continuous operation at full load and must withstand through-fault currents from the transmission system, with impedances typically ranging from 8% to 15% on their own base. Because of their size and long lead times for replacement (often 12 to 18 months), utilities maintain spare transformer programs and monitor GSU health through dissolved gas analysis (DGA), thermal monitoring, and periodic oil testing.

**Example 1.1.7:** A 600 MW natural gas combined-cycle plant has a generator rated at 700 MVA, 22 kV, 0.85 power factor. The GSU transformer is rated 700 MVA, 22 kV delta / 345 kV wye-grounded, with an impedance of 10.5%. Determine (a) the rated current on each side, (b) the maximum symmetrical fault current on the 345 kV side due to a fault at the GSU secondary terminals (assuming an infinite bus on the generator side), and (c) the fault MVA.
**Solution:**
(a) Rated currents:
$I_{LV} = S / (\sqrt{3} \times V_{LV}) = 700 \times 10^6 / (\sqrt{3} \times 22{,}000) =$ **18,370.2 A**

$I_{HV} = S / (\sqrt{3} \times V_{HV}) = 700 \times 10^6 / (\sqrt{3} \times 345{,}000) = \mathbf{1{,}171\ A}$
(b) Maximum symmetrical fault current at 345 kV terminals:
$I_{fault} = I_{HV,rated} / Z_{pu} = 1{,}171 / 0.105 = \mathbf{11{,}152\ A \approx 11.2\ kA}$
(c) Fault MVA:
$S_{fault} = S_{rated} / Z_{pu} = 700 / 0.105 = \mathbf{6{,}667\ MVA \approx 6.7\ GVA}$

### 1.1.8 Auxiliary (Station Service) Transformers

Auxiliary transformers, also called station service transformers or unit auxiliary transformers, supply the internal electrical loads needed to operate the power plant itself. These loads include boiler feed pumps, circulating water pumps, forced-draft and induced-draft fans, coal pulverizers, condensate pumps, control systems, lighting, and HVAC — collectively known as station service or house load, typically consuming 5% to 10% of the plant's gross generation. A generating station usually has two types of auxiliary supply: the unit auxiliary transformer (UAT), which taps off the generator bus between the generator and the GSU and powers loads during normal operation, and the startup/standby auxiliary transformer (SAT), which is fed from the transmission system or a separate substation bus to provide power during startup, shutdown, and emergency conditions when the generator is offline. The UAT is typically rated at 5% to 12% of the generator MVA and steps down from generator voltage (e.g., 22 kV) to medium-voltage auxiliary buses (typically 4.16 kV or 6.9 kV), with further step-downs to 480 V for smaller loads. Automatic transfer schemes (fast bus transfer or residual voltage transfer) switch loads from the UAT to the SAT upon a generator trip to maintain power to critical plant equipment and prevent a cascading shutdown.

**Example 1.1.8:** A 500 MW coal-fired generating unit has a generator rated at 588 MVA, 20 kV. Station service loads total 35 MW at 0.90 power factor lagging. The unit auxiliary transformer (UAT) steps down from 20 kV to 4.16 kV. Determine (a) the station service as a percentage of gross generation, (b) the required UAT MVA rating, and (c) the UAT rated current on the 4.16 kV side.
**Solution:**
(a) Station service percentage:
Station service = $P_{aux} / P_{gross} \times 100 = 35 / 500 \times 100 = \mathbf{7.0\%}$
Net plant output = $500 - 35 = \mathbf{465\ MW}$
(b) Required UAT apparent power:
$S_{UAT} = P_{aux} / PF = 35 / 0.90 = 38.9$ MVA
With 15% margin for future loads and motor starting: $S_{rated} = 38.9 \times 1.15 = \mathbf{44.7\ MVA}$ (select a standard 45 MVA rating)
(c) UAT rated current at 4.16 kV:
$I_{4.16kV} = S_{rated} / (\sqrt{3} \times V) = 45 \times 10^6 / (\sqrt{3} \times 4{,}160) = \mathbf{6{,}245\ A}$

## 1.2 Power Transmission

### 1.2.1 Transmission Lines

Transmission lines carry bulk electrical power from generating stations to substations near load centers, operating at high voltages (69 kV to 765 kV and above) to minimize resistive losses over long distances. They are characterized by four distributed parameters per unit length: series resistance R (conductor ohmic loss), series inductance L (magnetic flux linkage between conductors), shunt capacitance C (electric field between conductors and ground), and shunt conductance G (leakage current through insulators, usually negligible). Whether these parameters can be treated as lumped elements or must be modeled as continuously distributed depends on the electrical length of the line — the ratio of the physical length to the wavelength at the operating frequency (approximately 5,000 km at 60 Hz). Lines shorter than about 80 km are electrically short enough that shunt effects are negligible and a simple series impedance model suffices, while medium lines (80-250 km) require a lumped pi or T equivalent circuit that accounts for charging current, and long lines (over 250 km) demand the full distributed-parameter treatment using hyperbolic functions derived from the telegrapher's equations. Two key parameters emerge from the distributed model: the characteristic impedance $Z_c$ (also called surge impedance), which is the ratio $\sqrt{(L/C)}$ for a lossless line and typically ranges from 250 to 400 Ω for overhead lines, and the propagation constant $\gamma = \sqrt{(zy)}$, which determines attenuation and phase shift per unit length. The surge impedance loading (SIL) — defined as $V_{LL}^2/Z_c$ — represents the power at which the line's distributed capacitive reactive generation exactly balances its inductive reactive absorption, producing a flat voltage profile with unity power factor at both ends. At loads below SIL the line generates net reactive power and the receiving-end voltage rises (the Ferranti effect), while at loads above SIL the line absorbs reactive power and voltage drops, requiring shunt compensation or series capacitors to maintain acceptable voltage profiles.

#### 1.2.1.1 Short Transmission Lines

Short transmission lines are conductors that are less than 80 km in length.

**Example 1.2.1.1:** A 60 km, 69 kV, single-phase short transmission line has a series impedance of $Z = (0.15 + j0.40)\ \Omega/\text{km}$. The line delivers 10 MVA at 69 kV with a 0.85 lagging power factor at the receiving end. Determine the sending-end voltage and the voltage regulation.
**Solution:**
Total line impedance: $Z_{total} = 60 \times (0.15 + j0.40) = 9.0 + j24.0\ \Omega$
Receiving-end voltage: $V_R = 69{,}000$ V
Receiving-end current: $I_R = S / V_R = 10{,}000{,}000 / 69{,}000 = 144.93$ A
Power factor angle: $\theta = \cos^{-1}(0.85) = 31.79°$
$I_R$ in phasor form: $I_R = 144.93\angle{-31.79°}$ A
Sending-end voltage: $V_S = V_R + I_R \times Z_{total}$
$I_R \times Z_{total} = 144.93\angle{-31.79°} \times 25.63\angle 69.44° = 3{,}714.6\angle 37.65°$
$I_R \times Z_{total} = 3{,}714.6 \times (\cos 37.65° + j \sin 37.65°) = 2{,}941.2 + j2{,}268.4$ V
$V_S = 69{,}000 + 2{,}940.4 + j2{,}268.4 = 71{,}940.4 + j2{,}268.4$
$|V_S| = \sqrt{(71{,}940.4^2 + 2{,}268.4^2)} = 71{,}976.2$ V
Voltage regulation: $VR\% = (|V_S| - |V_R|) / |V_R| \times 100 = (71{,}976.2 - 69{,}000) / 69{,}000 \times 100 =$ **4.31%**

#### 1.2.1.2 Medium Transmission Lines

Medium transmission lines range from 80 km to 250 km in length and require consideration of the line's shunt capacitance in addition to series resistance and inductance. The standard model for medium lines is the nominal pi (or nominal T) circuit, where the total shunt capacitance is split equally and placed at each end of the line (pi model) or concentrated at the midpoint (T model). This lumped-parameter approach provides reasonable accuracy for steady-state analysis at power frequencies without the complexity of distributed-parameter models. The ABCD (transmission) parameters for the nominal pi model include the effect of charging current, which becomes significant at these lengths and can cause the receiving-end voltage to rise above the sending-end voltage under light load conditions (the Ferranti effect).

**Example 1.2.1.2:** A 200 km, 230 kV, three-phase medium transmission line has the following per-phase parameters: $z = 0.05 + j0.45\ \Omega/\text{km}$ and $y = j3.4 \times 10^{-6}$ S/km. Using the nominal $\pi$ model, determine the ABCD parameters.
**Solution:**
Total series impedance: $Z = z \times l = (0.05 + j0.45) \times 200 = 10 + j90\ \Omega = 90.55\angle 83.66°\ \Omega$
Total shunt admittance: $Y = y \times l = j3.4 \times 10^{-6} \times 200 = j6.8 \times 10^{-4}$ S
Nominal $\pi$ ABCD parameters:
$A = D = 1 + (Y \times Z) / 2$
$Y \times Z / 2 = (j6.8 \times 10^{-4})(10 + j90) / 2 = (j0.0068 + j^2 0.0612) / 2 = (-0.0612 + j0.0068) / 2$
$Y \times Z / 2 = -0.0306 + j0.0034$
$A = D = 1 - 0.0306 + j0.0034 = \mathbf{0.9694 + j0.0034 = 0.9694\angle 0.20°}$
$B = Z = \mathbf{10 + j90 = 90.55\angle 83.66°\ \Omega}$
$C = Y \times (1 + Y \times Z / 4)$
$Y \times Z / 4 = -0.0153 + j0.0017$
$C = j6.8 \times 10^{-4} \times (1 - 0.0153 + j0.0017) = j6.8 \times 10^{-4} \times (0.9847 + j0.0017)$
$C = j6.696 \times 10^{-4} - 1.156 \times 10^{-6} = \mathbf{-1.156 \times 10^{-6} + j6.696 \times 10^{-4}} \approx \mathbf{6.696 \times 10^{-4}\angle 90.1°\ S}$

### 1.2.1.3 Long Transmission Lines

Long transmission lines exceed 250 km in length and require a distributed-parameter model because lumped approximations introduce unacceptable errors at these distances. The voltage and current at any point along the line are governed by the telegrapher's equations, which yield hyperbolic functions (sinh, cosh) in the exact ABCD parameter expressions. Key characteristics of long lines include the propagation constant ($\gamma$), which determines attenuation and phase shift per unit length, and the characteristic (surge) impedance, which defines the natural loading of the line. At surge impedance loading (SIL), the reactive power generated by the line capacitance exactly equals the reactive power absorbed by the line inductance, resulting in a flat voltage profile. Long lines also exhibit pronounced wave propagation effects, making them susceptible to traveling-wave surges from lightning and switching events.

**Example 1.2.1.3:** A 400 km, 345 kV, three-phase long transmission line has per-phase distributed parameters: $z = 0.03 + j0.35\ \Omega/\text{km}$ and $y = j4.4 \times 10^{-6}$ S/km. Determine the characteristic impedance ($Z_c$) and propagation constant ($\gamma$), and find the surge impedance loading (SIL).
**Solution:**
Characteristic impedance: $Z_c = \sqrt{}(z / y) = \sqrt{}((0.03 + j0.35) / (j4.4 \times 10^{-6}))$
$z / y = (0.03 + j0.35) / (j4.4 \times 10^{-6})$
Multiply numerator and denominator by $-j$: $= (0.03 + j0.35)(-j) / (4.4 \times 10^{-6})$
$= (-j0.03 + 0.35) / (4.4 \times 10^{-6}) = (0.35 - j0.03) / (4.4 \times 10^{-6}) = 79{,}545.5 - j6{,}818.2$
$|z/y| = \sqrt{}(79{,}545.5^2 + 6{,}818.2^2) = 79{,}837.7$
angle $= \tan^{-1}(-6818.2 / 79{,}545.5) = -4.90°$
$Z_c = \sqrt{}(79{,}837.7)\angle(-4.90°/2) = \mathbf{282.6\angle{-2.45}°\ \Omega} \approx \mathbf{282.6\ \Omega}$ (nearly resistive)

Propagation constant: $\gamma = \sqrt{(z \times y)} = \sqrt{((0.03 + j0.35)(j4.4 \times 10^{-6}))}$
$z \times y = (0.03 + j0.35)(j4.4 \times 10^{-6}) = j1.32 \times 10^{-7} - 1.54 \times 10^{-6}$
$= -1.54 \times 10^{-6} + j1.32 \times 10^{-7} = 1.546 \times 10^{-6}\angle 175.1°$
$\gamma = \sqrt{(1.546 \times 10^{-6})}\angle(175.1°/2) = \mathbf{1.243 \times 10^{-3}\angle 87.55°}$ **per km**
$\gamma = \alpha + j\beta \approx 5.31 \times 10^{-5} + j1.242 \times 10^{-3}$ per km
Surge impedance loading (line-to-line voltage):
$SIL = V_{LL}^2 / Z_c = (345{,}000)^2 / 282.6 =$ **421.1 MW**

### 1.2.2 Underground Transmission Lines

Underground transmission lines use high-voltage insulated cables installed in duct banks, tunnels, or direct-buried configurations to transmit bulk power in areas where overhead lines are impractical or prohibited — primarily dense urban corridors, water crossings, airport approach zones, and environmentally sensitive areas. Underground transmission differs fundamentally from underground distribution (§1.3.3) in voltage class (69 kV to 500 kV versus 5-35 kV), cable construction, installation complexity, and the magnitude of special engineering considerations required.

**Cable Construction and Types.** Modern underground transmission cables are predominantly cross-linked polyethylene (XLPE) insulated, which has largely replaced older fluid-filled cable technologies for new installations. XLPE cables are solid-dielectric, maintenance-free, and available at voltages up to 500 kV, with conductor sizes ranging from 500 kcmil to 5,000 kcmil in copper or aluminum. High-pressure fluid-filled (HPFF) pipe-type cables — three insulated conductors pulled through a steel pipe pressurized with dielectric fluid — remain in service at 69-345 kV in many legacy installations, particularly in the northeastern United States. Self-contained fluid-filled (SCFF) cables use a hollow-core conductor with dielectric oil fed through the center, maintaining positive oil pressure to prevent void formation in the insulation. For the highest power transfers, gas-insulated lines (GIL) use $SF_6$ or $SF_6/N_2$ gas mixtures within an aluminum enclosure to achieve ratings up to 5,000 MVA at 500 kV, though at significantly higher cost than cable systems.

**Installation Methods.** Duct bank installations consist of concrete-encased PVC or fiberglass conduits, typically with manholes spaced every 500-800 meters for cable pulling, splicing, and maintenance access. Directional drilling (horizontal directional drilling, HDD) allows cable installation beneath waterways, highways, and environmentally sensitive areas without open trenching, with bore lengths up to 2 km. Cable tunnels provide a walk-in environment for installation and maintenance of multiple cable circuits, often used in dense urban areas where duct bank capacity is exhausted. Direct burial is less common for transmission cables but is used in some rural or suburban settings where soil conditions permit.

**Special Considerations.** Underground transmission requires careful attention to several factors that are either absent or less critical for overhead lines:

- **Thermal management**: Cable ampacity is limited by the maximum allowable conductor temperature (typically 90°C for XLPE) and the ability of the surrounding soil to dissipate heat. Soil thermal resistivity ($\rho_{soil}$, typically 60-120 °C·cm/W) is the dominant factor, and it increases dramatically as soil dries out from cable heating — a phenomenon called thermal runaway. Mutual heating between cables in the same duct bank or trench reduces ampacity by 10-30% compared to a single isolated cable. Thermal analysis follows the Neher-McGrath method (IEEE Std 835) or IEC 60287, and may require special thermal backfill (typically Fluidized Thermal Backfill with $\rho \leq 50$ °C·cm/W) to improve heat dissipation.

- **Charging current**: Underground cables have a much higher capacitance per unit length than overhead lines (typically 20-60 times greater) because the conductor-to-ground spacing in a cable is measured in millimeters rather than meters. This produces a large capacitive charging current that flows even at no load: $I_{charging} = 2\pi fCV$. At 230 kV on an XLPE cable with C = 0.2 μF/km, the charging current is approximately 10 A/km per phase. This limits the practical AC cable length to roughly 30-80 km (depending on voltage) before the entire cable ampacity is consumed by charging current. Shunt reactors are installed to compensate the capacitive reactive power on longer cable circuits.

- **Reactive compensation**: Because of the high charging current, underground cable circuits require shunt reactor compensation at intervals along the route — typically at each end and at intermediate points for cables longer than 20-30 km. The reactive power generated by the cable capacitance can be 10–20 MVAR/km at the highest transmission voltages (345 kV and above), and 4–8 MVAR/km at 230 kV, requiring large shunt reactors (50–200 MVAR each) that add significant cost and land requirements at reactor stations.

- **Fault location and repair**: Locating faults on underground transmission cables is more difficult than on overhead lines, requiring specialized techniques such as time-domain reflectometry (TDR), thumping (applying high-voltage impulses to create an audible discharge at the fault), and acoustic/electromagnetic fault pinpointing. Repair times are measured in weeks rather than hours — a single cable fault typically requires 2-4 weeks to excavate, cut out the damaged section, install a splice, test, and re-energize. This extended outage time is a major consideration in system planning and usually requires redundant cable circuits or backup overhead paths.

- **Joints and terminations**: Cable joints (splices) are required at intervals of 500-1,000 meters (limited by cable manufacturing lengths, drum sizes, and pull tension limits) and are statistically the weakest point in a cable system, accounting for the majority of cable failures. Factory-molded prefabricated joints and cold-shrink technology have improved reliability, but each joint is a potential failure point. Cable terminations at the transition from underground to overhead (called transition stations or riser poles) require stress cones to manage the electric field concentration at the cable end.

- **Cost**: Underground transmission typically costs 5-15 times more than equivalent overhead construction, with 345 kV urban duct bank installations running $15-50 million per mile compared to $1-5 million per mile for overhead. The cost premium includes cable materials, civil works (trenching, duct banks, manholes), longer installation time, and the reactive compensation equipment. This cost difference means underground transmission is reserved for situations where overhead lines are truly not feasible.

**Example 1.2.2:** A 230 kV underground XLPE cable circuit is 25 km long with three single-core cables (one per phase). Each cable has a capacitance of 0.23 μF/km and an ampacity of 800 A (accounting for soil conditions and mutual heating in the duct bank). Determine (a) the charging current per phase, (b) the total three-phase reactive power generated by the cable capacitance, (c) the remaining ampacity available for load current, and (d) the maximum power transfer.
**Solution:**
(a) Charging current per phase:
$I_{charging} = 2\pi fCV_{LN} = 2\pi \times 60 \times (0.23 \times 10^{-6} \times 25) \times (230{,}000/\sqrt{3})$
$C_{total} = 0.23 \times 25 = 5.75\ \mu F$

$I_{charging} = 2\pi \times 60 \times 5.75 \times 10^{-6} \times 132{,}791 =$ **288 A per phase**
(b) Three-phase reactive power:
$Q_{charging} = 3 \times V_{LN} \times I_{charging} = 3 \times 132{,}791 \times 288 = 114.7 \times 10^6$ VAR = **114.7 MVAR**
A shunt reactor of approximately 115 MVAR would be required to compensate this charging current.
(c) Remaining ampacity for load current:
The cable current rating is 800 A. The charging and load currents are roughly 90° apart (charging is capacitive), so:
$I_{load,max} = \sqrt{(I_{rated}{}^2 - I_{charging}{}^2)} = \sqrt{(800^2 - 288^2)} = \sqrt{(640{,}000 - 82{,}944)} = \sqrt{557{,}056} =$ **746 A**
(d) Maximum power transfer:
$S_{max} = \sqrt{3} \times V_{LL} \times I_{load,max} = \sqrt{3} \times 230{,}000 \times 746 = 297 \times 10^6$ VA = **297 MVA**
The charging current reduces the usable capacity from 319 MVA (at 800 A) to 297 MVA — a 7% reduction for this 25 km cable.
At 50 km, the charging current would double to 576 A, leaving only $\sqrt{(800^2 - 576^2)} = 555$ A for load, reducing capacity by 31%.

## 1.3 Power Distribution

### 1.3.1 Substations

Substations are critical nodes in the electric power system where voltage is transformed between transmission and distribution levels, circuits are switched and protected, and power flow is monitored and controlled. The primary function of most substations is voltage transformation — stepping down from transmission voltages (69-765 kV) to distribution voltages (4-35 kV) — but substations also serve as switching points that allow operators to reconfigure the network for maintenance, load balancing, or fault isolation. Substations are classified by their role in the system: transmission substations interconnect high-voltage lines and may include autotransformers for voltage conversion between transmission levels, distribution substations step voltage down to feeder levels for delivery to customers, and switching substations provide interconnection and sectionalizing without voltage transformation. Major equipment categories within a substation include power transformers, circuit breakers, disconnect switches, voltage regulators, instrument transformers (CTs and VTs), surge arresters, capacitor banks, and the control house containing protective relays, metering, SCADA equipment, and battery backup systems. The physical arrangement of buses and breakers — single bus, double bus, main-and-transfer, ring bus, or breaker-and-a-half — determines the substation's reliability and operational flexibility, with higher-reliability configurations used at critical transmission nodes where loss of a single component must not interrupt service. Substation grounding is essential for personnel safety and proper relay operation, requiring a ground grid of buried copper conductors designed to limit step and touch potentials below IEEE 80 thresholds during fault conditions.

#### 1.3.1.1 Transformers

Transformers are static electromagnetic devices that transfer electrical energy between two or more circuits through mutual induction, enabling voltage step-up or step-down while maintaining approximately constant power ($V_1I_1$ approximately equals $V_2I_2$). Power transformers in substations are typically oil-filled for insulation and cooling, with ratings ranging from a few MVA for distribution transformers to over 1000 MVA for large transmission-class units. The turns ratio ($N_1/N_2$) determines the voltage transformation ratio, while the core is constructed from laminated silicon steel to minimize eddy current and hysteresis losses. Three-phase transformers can be configured in delta-wye, wye-delta, wye-wye, or delta-delta connections, each providing different phase shift and grounding characteristics. Transformer nameplate data includes rated power (kVA or MVA), voltage ratings, impedance percentage, cooling class (ONAN, ONAF, etc.), and tap changer range.

**Example 1.3.1.1:** A three-phase, 50 MVA, 138 kV / 13.8 kV, delta-wye transformer has a nameplate impedance of 8.5%. Determine (a) the turns ratio, (b) the rated current on each side, and (c) the maximum symmetrical fault current on the secondary side assuming an infinite bus on the primary.
**Solution:**
(a) Turns ratio: $N_1/N_2 = V_1/V_2$ = 138,000 / 13,800 = **10:1**
(b) Rated currents (three-phase):
Primary: $I_{primary} = S / (\sqrt{3} \times V_{primary})$ = 50,000,000 / ($\sqrt{3} \times$ 138,000) = **209.2 A**
Secondary: $I_{secondary} = S / (\sqrt{3} \times V_{secondary})$ = 50,000,000 / ($\sqrt{3} \times$ 13,800) = **2,091.8 A**
(c) Maximum symmetrical fault current on secondary (infinite bus assumption):
$I_{fault} = I_{rated} / Z_{pu}$ = 2,091.8 / 0.085 = **24,609 A ≈ 24.6 kA**

### 1.3.1.2 Autotransformers

An autotransformer uses a single winding with a tap point to provide voltage transformation, rather than the two electrically isolated windings of a conventional transformer. The portion of the winding common to both the primary and secondary circuits carries only the difference between the primary and secondary currents, which means the winding can be physically smaller and lighter than an equivalent two-winding transformer. The apparent power rating ($S_{auto}$) of the autotransformer is the full load it delivers, but the actual power transferred through electromagnetic coupling ($S_{transformed}$) is only a fraction of the total: $S_{transformed} = S_{auto} \times (1 - N_2/N_1)$ for a step-down configuration. This makes autotransformers significantly more efficient and economical when the voltage ratio is close to 1:1, such as the 138/115 kV or 345/230 kV connections common in transmission systems. The main disadvantage is the lack of galvanic isolation between primary and secondary — a fault on one side is directly coupled to the other — so autotransformers are not used where electrical isolation is required. Common applications include transmission system voltage conversion, voltage regulators (step type), motor starting (reduced voltage), and laboratory variable-voltage supplies (Variacs).

**Example 1.3.1.2:** A single-phase, 100 MVA, 230/115 kV autotransformer supplies a 115 kV load. Determine (a) the turns ratio, (b) the rated load current on the 115 kV side, (c) the current in the common (series) winding, and (d) the power advantage (ratio of autotransformer rating to the equivalent two-winding transformer rating that would handle the same transformed power).
**Solution:**
(a) Turns ratio: $a = N_1/N_2 = V_1/V_2$ = 230/115 = **2:1**
(b) Rated load current at 115 kV:
$I_{load} = S_{auto} / V_2 = 100 \times 10^6$ / 115,000 = **869.6 A**
(c) Current in the series (common) winding:
The series winding carries the difference between load current and source current.
$I_{source} = S_{auto} / V_1 = 100 \times 10^6$ / 230,000 = 434.8 A
$I_{series} = I_{load} - I_{source}$ = 869.6 − 434.8 = **434.8 A**

(d) Transformed power (power actually coupled magnetically):
$S_{transformed} = S_{auto} \times (1 - 1/a) = 100 \times (1 - 1/2) =$ **50 MVA**
Power advantage = $S_{auto}$ / $S_{transformed}$ = 100 / 50 = **2.0**
A two-winding transformer rated at only 50 MVA can deliver 100 MVA as an autotransformer at this 2:1 ratio. The closer the ratio is to 1:1, the greater the advantage — for example, at 230/200 kV the advantage would be $230/(230 - 200) = 7.67$.

#### 1.3.1.3 Circuit Breakers

Circuit breakers are switching devices designed to interrupt fault currents and isolate faulted sections of the power system to prevent equipment damage and maintain system stability. They must be capable of interrupting currents ranging from normal load current to maximum fault current (which can exceed 50 kA in transmission substations) and then withstanding the recovery voltage across their open contacts. Modern high-voltage circuit breakers use $SF_6$ (sulfur hexafluoride) gas as the arc-extinguishing medium due to its superior dielectric strength and thermal properties, while medium-voltage breakers may use vacuum interrupters. Circuit breakers are rated by their nominal voltage, continuous current capacity, short-circuit interrupting capacity, and interrupting time (typically 3-5 cycles at 60 Hz). Protective relays monitor system conditions and send trip signals to circuit breakers when abnormal conditions such as overcurrent, differential current, or distance (impedance) faults are detected.

**Example 1.3.1.3:** A 145 kV substation bus has a rated short-circuit current of 40 kA symmetrical. The circuit breaker must interrupt the fault within 5 cycles at 60 Hz. Determine (a) the interrupting time in milliseconds, and (b) the three-phase short-circuit MVA that the breaker must be rated to handle.
**Solution:**
(a) Interrupting time: $t = 5 \text{ cycles} \times (1/60 \text{ s/cycle}) = 5/60 = 0.0833 \text{ s} =$ **83.3 ms**
(b) Short-circuit MVA:
$S_{fault} = \sqrt{3} \times V_{LL} \times I_{fault} = \sqrt{3} \times 145{,}000 \times 40{,}000 = 10{,}045{,}900{,}000$ VA
$S_{fault}$ = **10,046 MVA ≈ 10.0 GVA**

#### 1.3.1.4 Voltage Regulators

Voltage regulators maintain the distribution voltage within acceptable limits (typically plus or minus 5% of nominal per ANSI C84.1) as load conditions vary throughout the day. The most common type in distribution systems is the step voltage regulator, which is an autotransformer with an automatic tap-changing mechanism that can adjust the output voltage in steps of approximately 0.625% over a range of plus or minus 10%. The regulator control circuit uses a voltage-sensing element, a line drop compensator (to account for voltage drop between the regulator and the load center), and a time-delay mechanism to prevent excessive tap changes during transient fluctuations. Single-phase regulators are used on distribution feeders, while three-phase regulators or load tap changers (LTCs) on substation power transformers provide voltage regulation at the substation level.

**Example 1.3.1.4:** A step voltage regulator on a 7.2 kV (line-to-neutral) distribution feeder is set to maintain 7,200 V at the load center. The line drop compensator settings are R = 3.0 V and X =

9.0 V (on a 120 V base). The current transformer ratio is 200:5. At peak load, the feeder current is 150 A at a power factor of 0.90 lagging. Determine the required regulator output voltage.
**Solution:**
CT ratio: 200:5 = 40:1
Secondary current: $I_{sec}$ = 150 / 40 = 3.75 A
Power factor angle: $\theta = \cos^{-1}(0.90) = 25.84°$
$I_{sec}$ in rectangular form: $I_{sec} = 3.75\angle{-25.84°} = 3.375 - j1.635$ A
Compensator voltage drop (on 120 V base):
$V_{drop} = I_{sec} \times (R + jX) = (3.375 - j1.635)(3.0 + j9.0)$
$= 10.125 + j30.375 - j4.905 + 14.715$
$= 24.84 + j25.47$ V
$|V_{drop}| = \sqrt{(24.84^2 + 25.47^2)} = 35.6$ V
Required regulator output on 120 V base: $V_{reg} = 120 + 35.6 = 155.6$ V
Convert to primary: $V_{out} = (155.6 / 120) \times 7{,}200 =$ **9,336 V**
Since the regulator range is ±10% of 7,200 V (6,480 to 7,920 V), the compensator raises the set point so the regulator output compensates for the line drop.
The regulator output voltage is set to approximately **7,200 + (35.6/120 × 7,200) = 7,200 + 2,136 = 9,336 V**.
Note: In practice, each tap step is 0.625% of 7,200 V = 45 V, so the regulator would need approximately 2,136/45 ≈ 47.5 boost steps, far exceeding the ±16 step maximum (maximum boost = 16 steps × 45 V = 720 V; full raise-to-lower range = 1,440 V).
This indicates the line drop is too large for a single regulator and would require a second regulator or conductor upgrade.
For a more typical 50 A load: $V_{drop}$ ≈ 11.9 V (on 120 V base), and the regulator output would be 7,200 + 714 = **7,914 V**, which is within the ±10% range at tap position +16.

#### 1.3.1.5 Current Transformer (CT)

Current transformers are instrument transformers that produce a secondary current proportional to the primary current flowing through the power conductor, typically standardized at 5 A or 1 A secondary output. CTs enable protective relays and metering equipment to measure high currents (hundreds or thousands of amperes) safely at low, standardized levels. They are specified by their ratio (e.g., 600:5), accuracy class (e.g., 0.3 for metering, C400 for relaying), and burden rating, which defines the maximum secondary impedance the CT can drive while maintaining accuracy. The secondary circuit of a CT must never be left open while primary current is flowing, as this would cause dangerously high voltages on the secondary winding due to the loss of the demagnetizing effect of secondary current. CTs are available in bushing, window (toroidal), and bar-type configurations depending on the installation requirements.

**Example 1.3.1.5:** A 600:5 CT with accuracy class C200 is connected to a relay with an impedance of 2.0 Ω and lead wire resistance of 1.5 Ω (total loop). The primary fault current is 9,000 A. Determine (a) the secondary current, (b) the voltage the CT must develop across the burden, and (c) whether the CT can maintain accuracy.
**Solution:**
(a) Secondary current: $I_{sec} = I_{primary}$ / CT ratio = 9,000 / (600/5) = 9,000 / 120 = **75 A**

(b) Total burden impedance: $Z_{burden} = 2.0 + 1.5 = 3.5\ \Omega$
Voltage across burden: $V_{sec} = I_{sec} \times Z_{burden} = 75 \times 3.5 = \mathbf{262.5\ V}$
(c) The CT accuracy class is C200, meaning it can maintain accuracy up to 200 V across the burden at 20 times rated secondary current ($20 \times 5 = 100$ A).
At 75 A (15 times rated), the required voltage is 262.5 V, which **exceeds the 200 V rating**.
The CT will saturate, producing distorted output. A higher accuracy class (e.g., C400) or lower burden is needed.

#### 1.3.1.6 Voltage Transformer (VT)

Voltage transformers (also called potential transformers, or PTs) step down high primary voltages to standardized low secondary voltages, typically 120 V or 69.3 V, for use by metering and protective relaying equipment. They operate on the same electromagnetic induction principle as power transformers but are designed for high accuracy at very low burden (power output), with accuracy classes such as 0.3 or 0.6 for metering applications. VTs can be electromagnetic (wound-type) for voltages up to about 138 kV, or capacitor voltage transformers (CVTs) at higher voltages, where a capacitive voltage divider reduces the voltage before a small electromagnetic transformer provides the final step-down. The secondary winding of a VT is typically grounded on one terminal for safety, and fuses are installed on both primary and secondary sides to protect against faults. VT ratings include the voltage ratio, thermal burden (maximum VA output), and accuracy class at specified burdens.

**Example 1.3.1.6:** A 14,400:120 V voltage transformer is connected to three relays (each with a burden of 25 VA) and one meter (burden of 50 VA). The VT has a thermal burden rating of 200 VA. Determine (a) the total connected burden, (b) the secondary current drawn, and (c) whether the VT is within its rating.
**Solution:**
(a) Total burden: $S_{total} = 3 \times 25 + 50 = \mathbf{125\ VA}$
(b) Secondary current: $I_{sec} = S_{total} / V_{sec} = 125 / 120 = \mathbf{1.042\ A}$
(c) The total burden of 125 VA is less than the thermal rating of 200 VA, so the VT is **within its rating** with a margin of $200 - 125 = 75$ VA (37.5% margin).

#### 1.3.1.7 Switching

Substation switching equipment includes disconnect switches, load break switches, and bus sectionalizers that provide isolation, operational flexibility, and maintenance access for substation components. Disconnect switches (also called isolators) are used to provide a visible air gap for safe maintenance but can only be operated under no-load conditions since they have no arc-interrupting capability. Load break switches can interrupt normal load currents but are not rated for fault current interruption, making them suitable for routine switching operations such as transferring load between buses or feeders. Bus configurations such as single bus, main-and-transfer, breaker-and-a-half, and ring bus define the arrangement of switches and circuit breakers and determine the level of reliability and operational flexibility of the substation. Proper switching procedures and interlocking schemes are critical to prevent equipment damage and ensure personnel safety during switching operations.

**Example 1.3.1.7:** A breaker-and-a-half substation bus arrangement has two main buses and three bays, each bay containing two breakers and one line connection between the breakers. If one breaker fails and must be isolated for maintenance, determine how many circuits (lines) lose their redundant protection and how many circuits are completely de-energized.
**Solution:**
In a breaker-and-a-half scheme, each line is connected between two circuit breakers in a bay. Each breaker is shared between two circuits (one on each side).
When one breaker is taken out of service:
- The circuit connected to that bay still has its other breaker for protection, so **zero circuits are de-energized**.
- The circuit in that bay operates with only one breaker instead of two, meaning **one circuit loses its redundant breaker protection** (it can still operate but cannot tolerate a second breaker failure).
- All other circuits remain fully protected with their two dedicated breakers.
This demonstrates the high reliability of the breaker-and-a-half scheme: **no circuits are lost** during a single breaker outage.

### 1.3.2 Poles

Distribution poles are the most visible component of the overhead power system, supporting conductors, transformers, and protective equipment along the path from the substation to the end customer. Pole materials include wood (the most common in North America, typically southern yellow pine or Douglas fir), steel (used for higher loads, longer spans, or harsh environments), concrete (spun-cast prestressed, common in coastal and high-wind regions), and fiberglass composite (lightweight and rot-resistant, gaining adoption in environmentally sensitive areas). Standard wood poles in the United States are classified by the ANSI O5.1 standard according to height (typically 35 to 70 feet for distribution) and strength class (Class 1 through Class 10, with Class 1 being the strongest at approximately 4,500 lbs of horizontal load capacity at 2 feet from the top), and are pressure-treated with preservatives such as pentachlorophenol, chromated copper arsenate (CCA), or copper naphthenate to resist decay and insect damage. Poles must be engineered to withstand three categories of mechanical loading: vertical loads from conductor and equipment weight, transverse loads from wind pressure on conductors and the pole itself (with ice accumulation in cold climates), and longitudinal loads from unbalanced conductor tension at dead-end, angle, and junction structures. The NESC defines three loading districts — Heavy, Medium, and Light — each specifying different combinations of ice thickness (0 to 0.5 inches radial), wind pressure (0 to 4 psf), and temperature, which determine the minimum pole class required for a given span and conductor configuration. Span lengths typically range from 100 to 400 feet depending on pole class, conductor size, and terrain, with longer spans requiring stronger poles and potentially guying — the use of down guys anchored to the ground or sidewalk guys to counteract unbalanced loads at corners, dead-ends, and heavy equipment locations. Pole-mounted equipment includes distribution transformers, fuse cutouts, lightning arresters, reclosers, sectionalizers, and capacitor banks, all of which must be installed at proper clearance heights above ground and with adequate working space to comply with the NESC and local utility standards.

**Example 1.3.2:** A 45-foot Class 4 wood distribution pole is to be set in soil with a setting depth of 10% of length plus 2 feet. The pole supports a 167 kVA single-phase transformer (weight 1,200 lbs) mounted at 25 feet above ground, and three 4/0 ACSR conductors. The horizontal wind span is 300 feet, and the NESC Rule 250B wind loading is 4 psf on projected conductor area (diameter 0.563 inches). Determine (a) the setting depth and (b) the total horizontal wind force on the conductors.

**Solution:**

(a) Setting depth: $d = 0.10 \times 45 + 2 = 4.5 + 2 =$ **6.5 feet**

Height above ground: $45 - 6.5 = 38.5$ feet

(b) Projected area of one conductor per foot: $A_{per\_ft} = 0.563$ in / 12 = 0.0469 ft$^2$/ft

Total conductor length (wind span): 300 ft

Projected area per conductor: $A = 0.0469 \times 300 = 14.08$ ft$^2$

Wind force per conductor: $F = 4$ psf $\times$ 14.08 = 56.3 lbs

Total horizontal wind force (3 conductors): $F_{total} = 3 \times 56.3 =$ **168.9 lbs**

### 1.3.3 Underground

Underground distribution systems use insulated cables installed below grade to deliver power in areas where overhead construction is impractical, prohibited, or undesirable — including dense urban centers, new residential subdivisions with underground requirements, airport approach zones, and areas with high aesthetic or environmental sensitivity. The primary cable types used at distribution voltages (5-35 kV) are cross-linked polyethylene (XLPE) and ethylene propylene rubber (EPR) insulated cables, which have largely replaced older paper-insulated lead-covered (PILC) cables due to their superior dielectric performance, moisture resistance, lower weight, and easier splicing and termination. Installation methods include duct bank systems (multiple PVC or HDPE conduits encased in concrete, common in urban areas where future cable pulling and replacement is expected), direct burial (cables with a rugged outer jacket laid in a trench with sand bedding, lower initial cost but difficult to replace), and cable troughs or tunnels in industrial and utility campus environments. Underground systems offer significantly improved reliability against weather-related outages — eliminating exposure to wind, ice storms, falling trees, lightning, and vehicle strikes — but come with substantially higher installation costs (typically 5 to 10 times that of equivalent overhead construction) and present greater challenges when faults do occur. Fault location in underground systems requires specialized techniques such as time-domain reflectometry (TDR), thumper/surge generators, and acoustic or electromagnetic fault pinpointing, and repair times are measured in hours to days compared to minutes for many overhead faults. Cable ampacity (current-carrying capacity) is heavily influenced by the thermal environment — soil thermal resistivity ($\rho_{soil}$, typically 60-120 °C·cm/W), burial depth, spacing between cables or ducts, grouping derating factors for multiple circuits, and ambient soil temperature — requiring thermal analysis per the Neher-McGrath method (NEC 310.60) or IEC 60287. Pad-mounted transformers, switching cabinets, and junction pedestals provide the above-ground interface between the underground system and customer service, designed with tamper-resistant enclosures and visible disconnect points for safe operation in publicly accessible locations.

**Example 1.3.3:** A single-conductor 15 kV, 500 kcmil XLPE underground cable is direct-buried at a depth of 36 inches in soil with thermal resistivity $\rho_{soil} = 90$ °C·cm/W. The cable has a conductor resistance of 0.0277 Ω/1000 ft at 90°C, an insulation thermal resistance of 45.2 °C·cm/W, and the maximum conductor temperature is 90°C with an ambient soil temperature of 25°C. Using a simplified Neher-McGrath approach, estimate the cable ampacity. Assume the total effective thermal resistance from conductor to ambient is $R_{thermal} = 5.8$ °C/W per foot.

**Solution:**
Allowable temperature rise: $\Delta T = T_{max} - T_{ambient} = 90 - 25 = 65°C$
Heat dissipation per foot: $W = \Delta T / R_{thermal} = 65 / 5.8 = 11.21$ W/ft
Since heat is primarily $I^2R$ loss:
$W = I^2 \times R_{conductor} = I^2 \times (0.0277 / 1000)\ \Omega/\text{ft} = I^2 \times 2.77 \times 10^{-5}\ \Omega/\text{ft}$
$I^2 = 11.21 / 2.77 \times 10^{-5} = 404{,}693$
$I = \sqrt{404{,}693} = \mathbf{636\ A}$

### 1.3.4 Three-Phase Connections

Three-phase power systems use two fundamental winding configurations — wye (Y, also called star) and delta (Δ) — to connect generators, transformers, motors, and loads. The choice between wye and delta affects the relationship between line and phase voltages, line and phase currents, neutral availability, and fault behavior, making it one of the most important decisions in power system design.

#### 1.3.4.1 Wye (Star) Connection

In a wye connection, one terminal of each of the three phase windings is connected to a common neutral point, while the other terminal of each winding connects to the respective line conductor. The line-to-line voltage is $\sqrt{3}$ times the phase (line-to-neutral) voltage: $V_{LL} = \sqrt{3} \times V_{LN}$, and the line current equals the phase current: $I_L = I_{phase}$. The neutral point can be grounded to establish a voltage reference, provide a return path for unbalanced currents, and limit overvoltages during ground faults — this is why wye connections are standard on the secondary side of distribution transformers (e.g., 208Y/120 V or 480Y/277 V), where the neutral provides both the safety ground reference and a fourth conductor for single-phase loads. In a balanced wye system, the neutral carries zero current; under unbalanced loading, the neutral carries the vector sum of the three phase currents. Wye connections are also preferred for high-voltage transmission because each winding is insulated only to the line-to-neutral voltage, reducing insulation cost.

**Example 1.3.4.1:** A three-phase, 480Y/277 V wye-connected system supplies a balanced load of 150 kW at 0.90 power factor lagging. Determine (a) the line-to-neutral voltage, (b) the line current, and (c) the neutral current.
**Solution:**
(a) Line-to-neutral voltage:
$V_{LN} = V_{LL} / \sqrt{3} = 480 / 1.732 = \mathbf{277\ V}$
(b) Line current (balanced three-phase):
$S = P / PF = 150 / 0.90 = 166.7$ kVA
$I_L = S / (\sqrt{3} \times V_{LL}) = 166{,}700 / (1.732 \times 480) = \mathbf{200.6\ A}$
(c) Neutral current:
For a balanced load, the three phase currents are equal in magnitude and displaced by 120°. Their vector sum is zero:
$I_N = I_A + I_B + I_C = \mathbf{0\ A}$

### 1.3.4.2 Delta (Δ) Connection

In a delta connection, the three phase windings are connected end-to-end in a closed loop, forming a triangle. Each line conductor connects to the junction of two windings. The line-to-line voltage equals the phase voltage: $V_{LL} = V_{phase}$, and the line current is $\sqrt{3}$ times the phase current: $I_L = \sqrt{3} \times I_{phase}$. Delta connections have no neutral point, so they cannot supply line-to-neutral loads directly and cannot be solidly grounded (though corner grounding or resistance grounding through a grounding transformer is possible). Delta-connected transformer windings trap triplen harmonic currents (3rd, 9th, 15th, etc.) that circulate within the closed delta loop rather than propagating into the power system, which is why the primary winding of distribution transformers is often delta-connected. Delta connections also provide a path for zero-sequence currents during ground faults on the wye side of a delta-wye transformer, enabling ground fault detection. Motors are commonly delta-connected at 480 V to avoid the need for a neutral conductor.

**Example 1.3.4.2:** A three-phase, 480 V delta-connected motor draws 80 A line current at 0.85 power factor lagging. Determine (a) the phase voltage across each motor winding, (b) the current through each winding, and (c) the three-phase power consumed.
**Solution:**
(a) Phase voltage (delta: line voltage equals phase voltage):
$V_{phase} = V_{LL} = \mathbf{480\ V}$
(b) Phase current:
$I_{phase} = I_L / \sqrt{3} = 80 / 1.732 = \mathbf{46.2\ A}$
(c) Three-phase power:
$P = \sqrt{3} \times V_{LL} \times I_L \times PF = 1.732 \times 480 \times 80 \times 0.85 = \mathbf{56{,}533\ W} \approx \mathbf{56.5\ kW}$

### 1.3.4.3 Delta-Wye Transformations

Many power system configurations require converting between delta and wye equivalent circuits for analysis, particularly when combining delta-connected sources with wye-connected loads or vice versa. The transformation formulas relate impedances in a delta configuration ($Z_{AB}$, $Z_{BC}$, $Z_{CA}$) to equivalent wye impedances ($Z_A$, $Z_B$, $Z_C$) and vice versa. For balanced systems where all impedances are equal, the conversion simplifies to $Z_Y = Z_\Delta / 3$ (delta to wye) and $Z_\Delta = 3 \times Z_Y$ (wye to delta). For unbalanced systems, the delta-to-wye conversion is $Z_A = (Z_{AB} \times Z_{CA}) / (Z_{AB} + Z_{BC} + Z_{CA})$, with similar expressions for $Z_B$ and $Z_C$ by cyclic rotation. Transformer connections use specific delta-wye combinations (Dy1, Dy11, Yd1, etc.) that introduce 30° phase shifts between primary and secondary voltages, which must be matched when paralleling transformers.

**Example 1.3.4.3:** A balanced delta-connected load has $Z_\Delta = 30 + j15\ \Omega$ per phase and is connected to a 240 V three-phase source. Determine (a) the equivalent wye impedance per phase, (b) the line current using the wye equivalent, and (c) verify the result using delta phase currents.
**Solution:**
(a) Equivalent wye impedance (balanced system):
$Z_Y = Z_\Delta / 3 = (30 + j15) / 3 = \mathbf{10 + j5\ \Omega}$
$|Z_Y| = \sqrt{(10^2 + 5^2)} = 11.18\ \Omega$
(b) Line current using wye equivalent:
$V_{LN} = V_{LL} / \sqrt{3} = 240 / 1.732 = 138.6\ V$

$I_L = V_{LN} / |Z_Y| = 138.6 / 11.18 =$ **12.40 A**
(c) Verification using delta phase currents:
$|Z_\Delta| = \sqrt{(30^2 + 15^2)} = 33.54\ \Omega$
$I_{phase,\Delta} = V_{LL} / |Z_\Delta| = 240 / 33.54 = 7.16$ A
$I_L = \sqrt{3} \times I_{phase,\Delta} = 1.732 \times 7.16 =$ **12.40 A ✓**

### 1.3.5 AC Analysis

The analysis of AC distribution circuits requires mathematical tools that can represent both the magnitude and phase angle of sinusoidal voltages and currents. Since power system quantities are sinusoidal at a fixed frequency (60 Hz in North America, 50 Hz in most other regions), steady-state analysis is greatly simplified by representing these time-varying quantities as complex numbers known as phasors, enabling the use of algebraic rather than differential equation methods.

#### 1.3.5.1 Complex Numbers and Phasors

Complex numbers provide a compact mathematical representation of AC quantities by encoding both magnitude and phase angle in a single expression of the form $Z = a + jb$, where a is the real (resistive) component and b is the imaginary (reactive) component, with j representing the square root of negative one. The rectangular form (a + jb) is convenient for addition and subtraction of impedances in series, while the polar form (|Z| at angle θ) is preferred for multiplication and division, which arise in series-parallel circuit analysis and power calculations. A phasor is a complex number that represents the amplitude and phase of a sinusoidal waveform relative to a reference, obtained by applying the Euler relation: $V = V_m \times e^{j\theta} = V_m$ at angle θ, where $V_m$ is the peak value and θ is the phase angle. In power engineering, phasors are conventionally expressed as RMS (root-mean-square) values rather than peak values, since RMS values directly relate to average power delivery ($P = V_{rms} \times I_{rms} \times \cos(\theta)$). Impedance, expressed as $Z = R + jX$ (where X is reactance), relates phasor voltage and current through Ohm's law ($V = I \times Z$), enabling systematic analysis of AC circuits using techniques such as Kirchhoff's laws, mesh analysis, and nodal analysis in the phasor domain.

**Example 1.3.5.1:** A series RLC circuit has R = 10 Ω, L = 50 mH, and C = 100 μF, connected to a 120 $V_{rms}$, 60 Hz source. Determine the impedance, current, and power factor.
**Solution:**
Inductive reactance: $X_L = 2\pi fL = 2\pi \times 60 \times 0.050 = 18.85\ \Omega$
Capacitive reactance: $X_C = 1 / (2\pi fC) = 1 / (2\pi \times 60 \times 100 \times 10^{-6}) = 26.53\ \Omega$
Net reactance: $X = X_L - X_C = 18.85 - 26.53 = -7.68\ \Omega$ (capacitive)
Impedance: $Z = R + jX = 10 - j7.68\ \Omega$
$|Z| = \sqrt{(10^2 + 7.68^2)} = \sqrt{(100 + 58.98)} = \sqrt{158.98} = 12.61\ \Omega$
Phase angle: $\theta = \tan^{-1}(-7.68 / 10) = -37.52°$
Z = **12.61∠−37.52° Ω**
Current: I = V / Z = 120∠0° / 12.61∠−37.52° = **9.52∠37.52° A** (current leads voltage)
Power factor: pf = cos(37.52°) = **0.793 leading**

### 1.3.6 Power Factor Correction

Power factor correction (PFC) techniques are employed to improve the power factor of electrical systems, particularly in transmission lines. Power factor is a measure of how effectively electrical power is being utilized, and it is defined as the cosine of the phase angle between voltage and current in an AC circuit. A low power factor can lead to inefficient energy usage, increased losses, and reduced system capacity.

Common power factor correction techniques include capacitor banks, synchronous condensers, Static VAR Compensators (SVCs), Static Synchronous Compensators (STATCOMs), and phase advancers. Capacitor banks — the most widely deployed method — connect capacitors in parallel with the load to supply leading reactive power and offset the lagging reactive component. Synchronous condensers are rotating machines (synchronous generators without a prime mover) whose field excitation can be adjusted to either absorb or supply reactive power. SVCs use thyristors, reactors, and capacitors to provide rapid, continuous reactive compensation with millisecond response to load changes. STATCOMs, based on voltage-source converters, offer faster and more precise reactive power injection or absorption than SVCs and support voltage stability during disturbances. Phase advancers are applied to large induction motors, injecting leading reactive current into the rotor circuit to reduce the lagging reactive demand seen at the stator terminals. Together, these techniques minimize reactive power flow, reduce $I^2R$ losses, improve voltage profiles, and increase the effective capacity of generation and transmission infrastructure.

**Example 1.3.6:** An industrial facility draws 500 kW of real power at a power factor of 0.72 lagging from a 480 V, 60 Hz, three-phase supply. The utility requires the power factor to be corrected to 0.95 lagging. Determine the required capacitor bank size in kVAR and the capacitance per phase for a delta-connected capacitor bank.

**Solution:**

Original reactive power:

$\theta_1 = \cos^{-1}(0.72) = 43.95°$

$Q_1 = P \times \tan(\theta_1) = 500 \times \tan(43.95°) = 500 \times 0.9639 = 481.9$ kVAR

Target reactive power:

$\theta_2 = \cos^{-1}(0.95) = 18.19°$

$Q_2 = P \times \tan(\theta_2) = 500 \times \tan(18.19°) = 500 \times 0.3287 = 164.4$ kVAR

Required capacitor bank: $Q_{cap} = Q_1 - Q_2 = 481.9 - 164.4 =$ **317.5 kVAR**

For a delta-connected bank, each capacitor sees line-to-line voltage:

Q per phase: $Q_{phase} = 317.5 / 3 = 105.8$ kVAR

Capacitive reactance per phase: $X_C = V_{LL}{}^2 / Q_{phase} = (480)^2 / 105{,}800 = 2.177\ \Omega$

Capacitance per phase: $C = 1 / (2\pi f X_C) = 1 / (2\pi \times 60 \times 2.177) =$ **1,219 μF per phase**

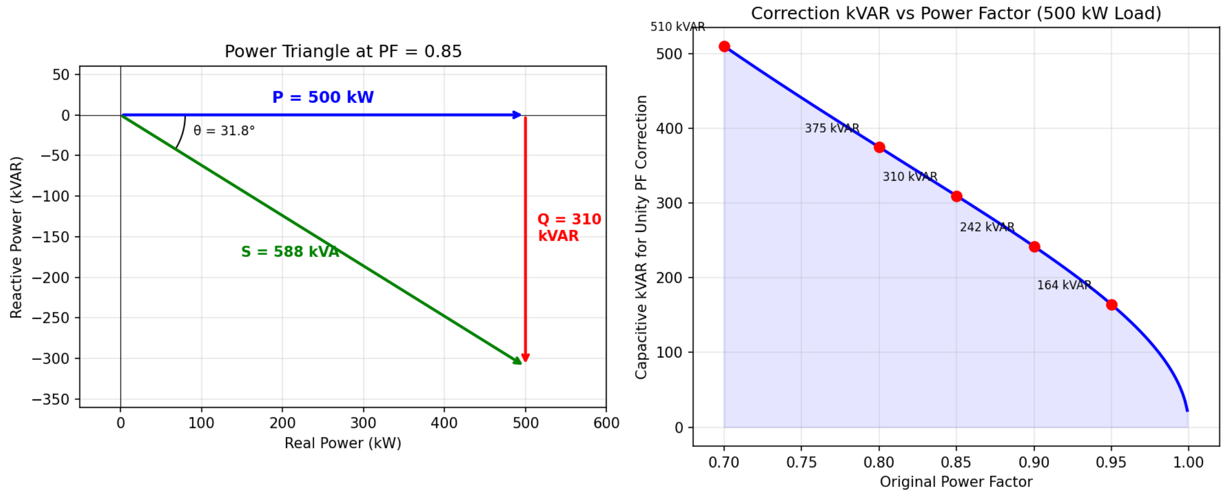



Figure 1.3.6: Power Factor Correction Analysis

## 1.4 Power System Protection

Power system protection is the branch of power engineering concerned with detecting abnormal conditions (faults, overloads, equipment failures) and isolating the affected portion of the system as quickly as possible to prevent equipment damage, maintain system stability, and ensure personnel safety. A protection system consists of instrument transformers (CTs and VTs) that sense system quantities, protective relays that make trip decisions based on measured currents, voltages, impedances, or differential quantities, and circuit breakers that physically interrupt the fault current. The three fundamental requirements of a protection system are dependability (it must operate when required), security (it must not operate when not required), and speed (it must clear faults before they cause cascading damage).

### 1.4.1 Protective Relays

Protective relays are the decision-making elements of a protection system. They continuously monitor electrical quantities provided by instrument transformers and issue trip signals to circuit breakers when abnormal conditions are detected. Modern digital relays (also called numerical or microprocessor-based relays) use sampled current and voltage waveforms processed by algorithms that implement multiple protection functions in a single device, along with oscillography, event recording, and communication capabilities.

The major relay types and their applications are:

- **Overcurrent relays (50/51)**: Operate when current exceeds a pickup threshold. Time-overcurrent relays (51) have an inverse time-current characteristic where trip time decreases as fault current increases. Instantaneous overcurrent relays (50) trip without intentional delay for high-magnitude faults.
- **Distance relays (21)**: Measure the apparent impedance (V/I) seen from the relay location. Since impedance is proportional to line length, distance relays can determine whether a fault is within a protected zone. They are the primary protection for transmission lines, typically with three zones of increasing reach and time delay.

- **Differential relays (87)**: Compare currents entering and leaving a protected element (transformer, bus, generator). Under normal conditions, the currents balance (Kirchhoff's current law); during an internal fault, the differential current exceeds a threshold and the relay operates. Differential protection is the primary protection for transformers, generators, and busbars because it provides fast, selective protection with a clearly defined protection zone.
- **Directional relays (67)**: Determine the direction of power or fault current flow using the phase angle between voltage and current. They are used to provide selectivity in looped or networked systems where fault current can flow in either direction.

**Example 1.4.1:** A time-overcurrent relay (51) protecting a distribution feeder has a pickup current of 600 A and uses the IEEE Very Inverse characteristic: $t = (19.61 / (M^2 - 1) + 0.491) \times TDS$, where $M = I_{fault}/I_{pickup}$ and TDS is the time dial setting. With TDS = 2.0, determine the relay operating time for fault currents of (a) 3,000 A and (b) 6,000 A.
**Solution:**
(a) $M = 3{,}000/600 = 5.0$.
$t = (19.61/(25 - 1) + 0.491) \times 2.0 = (19.61/24 + 0.491) \times 2.0 = (0.817 + 0.491) \times 2.0 =$ **2.62 s**
(b) $M = 6{,}000/600 = 10.0$.
$t = (19.61/(100 - 1) + 0.491) \times 2.0 = (19.61/99 + 0.491) \times 2.0 = (0.198 + 0.491) \times 2.0 =$ **1.38 s**
The inverse characteristic ensures faster tripping for higher fault currents — doubling the fault current from 3,000 A to 6,000 A nearly halved the operating time.

### 1.4.2 Fault Analysis

Fault analysis determines the magnitude and distribution of currents during short-circuit conditions, which is essential for sizing protective equipment and setting relay parameters. The three types of symmetrical component analysis — positive, negative, and zero sequence — are used to analyze unbalanced faults. The four common fault types, in order of severity, are:

| Fault Type | Frequency | Severity | Analysis Method |
|---|---|---|---|
| Single line-to-ground (SLG) | 70-80% | Least | All three sequences |
| Line-to-line (LL) | 15-20% | Moderate | Positive + negative |
| Double line-to-ground (DLG) | 5-10% | High | All three sequences |
| Three-phase (3Φ) | < 5% | Highest | Positive sequence only |

The per-unit system simplifies fault calculations by normalizing all quantities to chosen base values. In the per-unit system, the fault current is $I_{fault} = V_{pre\text{-}fault} / Z_{total}$, where $Z_{total}$ includes the source impedance, transformer impedance, and line impedance in the fault path. The symmetrical fault current (AC component) determines the interrupting rating of circuit breakers, while the asymmetrical fault current (including DC offset) determines the momentary or close-and-latch rating.

**Example 1.4.2:** A 13.8 kV distribution bus is supplied through a transformer with Z = 8% on a 25 MVA base. The source impedance on the same base is 2%. Determine (a) the three-phase fault current at the bus in per-unit and amperes, (b) the fault MVA, and (c) the required circuit breaker

interrupting rating.
**Solution:**
Base current: $I_{base} = S_{base} / (\sqrt{3} \times V_{base}) = 25 \times 10^6 / (\sqrt{3} \times 13{,}800) = 1{,}046$ A.
(a) Total impedance: $Z_{total} = Z_{source} + Z_{transformer} = 0.02 + 0.08 = 0.10$ pu.
$I_{fault} = V/Z = 1.0/0.10 = 10.0$ pu $= 10.0 \times 1{,}046 =$ **10,460 A**
(b) Fault MVA $= S_{base}/Z_{total} = 25/0.10 =$ **250 MVA**
(c) The asymmetrical fault current includes a DC offset factor.
For typical X/R = 15, the asymmetry factor is approximately 1.6: $I_{asym} = 1.6 \times 10{,}460 = 16{,}736$ A.
The breaker must be rated for at least **250 MVA** interrupting capacity and a close-and-latch rating of at least **16.7 kA**.

### 1.4.3 Protection Coordination

Protection coordination (also called selectivity) ensures that the protective device closest to a fault operates first, while upstream devices provide backup protection with sufficient time delay. Proper coordination minimizes the extent of the system affected by a fault — only the faulted section is de-energized while the rest of the system continues to operate. The coordination time interval (CTI) between successive devices is typically 0.2-0.4 seconds for relay-to-relay coordination and 0.1-0.2 seconds for fuse-to-fuse coordination.

Coordination studies plot the time-current characteristics of all protective devices on a single log-log graph (the coordination curve) and verify that they do not overlap. For overcurrent devices, the upstream device must have a longer operating time than the downstream device at all fault current levels within the coordination range. Fuse-fuse coordination requires that the upstream fuse's minimum melting time exceeds the downstream fuse's total clearing time by a factor of at least 1.5. Modern coordination studies use software tools that model relay characteristics, fuse curves, and breaker trip curves to automate the verification process.

**Example 1.4.3:** Two time-overcurrent relays protect series sections of a distribution feeder. Relay B (downstream) has TDS = 1.5 and pickup = 400 A. Relay A (upstream) must coordinate with Relay B with a CTI of 0.3 s. Both use IEEE Moderately Inverse characteristics: $t = (0.0515 / (M^{0.02} - 1) + 0.114) \times TDS$. For a maximum fault current of 4,000 A seen by both relays, determine the required TDS for Relay A (pickup = 600 A).
**Solution:**
Relay B operating time at 4,000 A: $M_B = 4{,}000/400 = 10$.
$t_B = (0.0515/(10^{0.02} - 1) + 0.114) \times 1.5 = (0.0515/(1.0471 - 1) + 0.114) \times 1.5$
$t_B = (0.0515/0.0471 + 0.114) \times 1.5 = (1.093 + 0.114) \times 1.5 =$ **1.81 s**
Required Relay A time: $t_A = t_B + CTI = 1.81 + 0.3 = 2.11$ s.
Relay A at 4,000 A: $M_A = 4{,}000/600 = 6.667$.
$2.11 = (0.0515/(6.667^{0.02} - 1) + 0.114) \times TDS_A$
$6.667^{0.02} = e^{0.02 \times \ln(6.667)} = e^{0.02 \times 1.897} = e^{0.03794} = 1.0387$.
$2.11 = (0.0515/0.0387 + 0.114) \times TDS_A = (1.331 + 0.114) \times TDS_A = 1.445 \times TDS_A$.
$TDS_A = 2.11/1.445 =$ **1.46** (round up to TDS = **1.5** for standard settings).

### 1.4.4 Symmetrical Components

Symmetrical components, developed by Charles Fortescue in 1918, decompose any set of unbalanced three-phase phasors into three balanced sets: positive sequence ($V_1$, normal rotation a-b-c), negative sequence ($V_2$, reverse rotation a-c-b), and zero sequence ($V_0$, all three phasors in phase). The transformation is $[V_0; V_1; V_2] = (1/3) \times [1, 1, 1; 1, a, a^2; 1, a^2, a] \times [V_a; V_b; V_c]$, where $a = 1\angle 120°$ and $a^2 = 1\angle 240°$. Each sequence component sees a different impedance: the positive-sequence impedance $Z_1$ equals the normal balanced impedance, the negative-sequence impedance $Z_2 \approx Z_1$ for transformers and transmission lines (but differs for rotating machines), and the zero-sequence impedance $Z_0$ depends heavily on the grounding configuration and transformer winding connections (delta windings block zero-sequence current).

Symmetrical components simplify unbalanced fault calculations by converting a coupled three-phase problem into three independent single-phase sequence networks. A balanced three-phase fault involves only the positive-sequence network. A single-line-to-ground (SLG) fault connects all three sequence networks in series: $I_{fault} = 3V_1/(Z_1 + Z_2 + Z_0)$. A line-to-line fault connects the positive and negative networks in parallel: $I_{fault} = \sqrt{3} \times V_1/(Z_1 + Z_2)$. These formulations make it possible to analyze unbalanced conditions using sequence impedance data from equipment nameplates and test reports.

**Example 1.4.4:** A 100 MVA, 13.8 kV generator has $Z_1 = j0.15$ pu, $Z_2 = j0.15$ pu, and $Z_0 = j0.05$ pu on the machine base. The generator neutral is grounded through an impedance $Z_g = j0.10$ pu. The pre-fault voltage is 1.0 pu. Calculate the fault current for a single-line-to-ground fault on phase A.
**Solution:**
For an SLG fault, the sequence currents are equal: $I_0 = I_1 = I_2 = V_1/(Z_1 + Z_2 + Z_0 + 3Z_g)$.
Total impedance: $Z_1 + Z_2 + Z_0 + 3Z_g = j0.15 + j0.15 + j0.05 + j0.30 = j0.65$ pu.
Sequence currents: $I_0 = I_1 = I_2 = 1.0/j0.65 = -j1.538$ pu.
Phase A fault current: $I_a = I_0 + I_1 + I_2 = 3 \times (-j1.538) = -j4.615$ pu.
Base current: $I_{base} = S_{base}/(\sqrt{3} \times V_{base})$.
For $S_{base} = 100$ MVA: $I_{base} = 100 \times 10^6/(\sqrt{3} \times 13{,}800) = 4{,}184$ A.
Fault current magnitude: $I_{fault} = 4.615 \times 4{,}184 =$ **19,309 A**.
Phases B and C carry zero current ($I_b = I_c = 0$), confirming the single-line-to-ground fault condition.

## 1.5 Power Quality

Power quality refers to the characteristics of the voltage and current waveforms delivered to end users and their conformance to established standards. Poor power quality can cause equipment malfunctions, overheating, reduced efficiency, and premature failure. The major power quality phenomena include harmonics (waveform distortion), voltage sags and swells (short-duration magnitude variations), transients (fast voltage spikes), flicker (systematic voltage variations), and frequency variations. IEEE Standard 519-2022 establishes limits for harmonic distortion, while IEEE Standard 1159 defines and categorizes power quality events.

### 1.5.1 Harmonics

Harmonics are sinusoidal voltages or currents at integer multiples of the fundamental power system frequency (60 Hz in North America, 50 Hz in most other regions). Nonlinear loads such as variable frequency drives (VFDs), rectifiers, switching power supplies, and LED drivers draw non-sinusoidal currents that contain harmonics. The most significant harmonics in three-phase systems are the 5th (300 Hz), 7th (420 Hz), 11th (660 Hz), and 13th (780 Hz), because triplen harmonics (3rd, 9th, 15th) are trapped by delta transformer windings and do not propagate into the utility system in balanced three-phase circuits.

Total Harmonic Distortion (THD) quantifies the overall distortion level: $THD = \sqrt{(\Sigma V_h^2)} / V_1 \times 100\%$, where $V_h$ is the magnitude of the h-th harmonic and $V_1$ is the fundamental. IEEE 519 limits voltage THD to 5% at the point of common coupling (PCC) for systems below 69 kV, with no individual harmonic exceeding 3%. Current distortion limits depend on the ratio of short-circuit current to load current ($I_{SC}/I_L$), with larger ratios permitting higher distortion because the system is stiffer and less affected.

**Example 1.5.1:** A 6-pulse VFD produces the following harmonic current spectrum (as a percentage of fundamental): 5th = 20%, 7th = 14%, 11th = 9%, 13th = 7.7%, 17th = 5.9%, 19th = 5.3%. The VFD draws 150 A fundamental on a system where $I_{SC}$ = 10,000 A. Determine (a) the current THD, (b) the total demand distortion (TDD), and (c) whether the installation meets IEEE 519 limits (for $I_{SC}/I_L > 50$, the TDD limit is 12%).
**Solution:**
(a) $THD = \sqrt{(20^2 + 14^2 + 9^2 + 7.7^2 + 5.9^2 + 5.3^2)} = \sqrt{(400 + 196 + 81 + 59.3 + 34.8 + 28.1)} = \sqrt{799.2} = \mathbf{28.3\%}$
(b) $TDD = THD \times (I_{fundamental} / I_L)$.
For this example, $I_L$ = maximum demand load current = 150 A.
$I_{SC}/I_L = 10{,}000/150 = 66.7$ (> 50).
$TDD = \sqrt{(\Sigma I_h^2 / I_L^2)} \times 100\%$.
Since fundamental = load current here, TDD = THD = **28.3%**
(c) IEEE 519 limit for $I_{SC}/I_L > 50$: TDD ≤ 12%, individual harmonics: 5th-7th ≤ 15%, 11th-13th ≤ 7%.
The 5th harmonic (20%) exceeds the 15% individual limit, and TDD (28.3%) far exceeds 12%.
The VFD **does not comply** — harmonic mitigation such as a 12-pulse drive, active filter, or passive LC filters would be required.

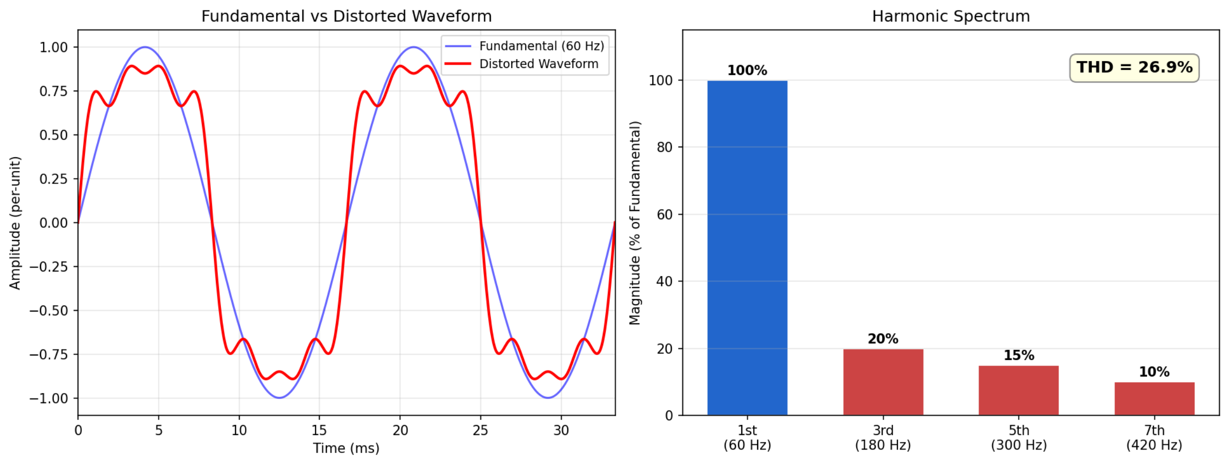



Figure 1.5.1: Harmonic Spectrum and THD

### 1.5.2 Voltage Sags and Swells

A voltage sag (or dip) is a temporary reduction in RMS voltage to between 10% and 90% of nominal for a duration of 0.5 cycles to 1 minute. Voltage sags are the most common power quality event affecting industrial facilities and are primarily caused by faults on adjacent feeders, motor starting, and transformer energization. A voltage swell is a temporary increase in RMS voltage to between 110% and 180% of nominal for the same duration range, typically caused by single line-to-ground faults (which raise the voltage on the unfaulted phases) or load rejection events.

The CBEMA/ITIC curve (now ITI curve from the Information Technology Industry Council) defines the voltage magnitude and duration envelope within which electronic equipment should operate without malfunction. Equipment is expected to ride through sags to 80% of nominal for up to 10 seconds, to 70% for up to 0.5 seconds, and to nearly 0% for up to 1 cycle (about 16.7 ms at 60 Hz). Mitigation strategies include uninterruptible power supplies (UPS), dynamic voltage restorers (DVR), and automatic transfer switches (ATS) that transfer critical loads to an alternate source within a few cycles.

**Example 1.5.2:** A 480 V bus experiences a voltage sag to 65% of nominal lasting 200 ms caused by a fault on an adjacent feeder. A sensitive CNC machine requires a minimum of 80% voltage. Determine (a) the sag magnitude in volts, (b) whether the machine will trip, (c) the energy a DVR must inject to maintain 100% voltage if the load draws 100 kVA, and (d) whether the event falls within the ITIC "prohibited" or "no damage" region.
**Solution:**
(a) Sag voltage: $0.65 \times 480 = \mathbf{312\ V}$ (the voltage drops from 480 V to 312 V).
(b) The machine requires $\geq 80\%$ (384 V). Since 312 V $<$ 384 V, the machine **will trip**.
(c) The DVR must supply the difference: $V_{DVR} = 480 - 312 = 168$ V (35% of nominal).
DVR apparent power: $S_{DVR} = (V_{DVR}/V_{nominal}) \times S_{load} = 0.35 \times 100 = 35$ kVA.
Energy: $E = S_{DVR} \times t = 35 \times 0.2 = \mathbf{7.0\ kJ}$ (assuming unity power factor).
(d) On the ITIC curve, 65% voltage for 200 ms falls **below the lower curve** (outside the acceptable region), confirming equipment is expected to malfunction. A UPS or DVR is needed to protect the CNC machine.

### 1.5.3 Power Quality Monitoring

Power quality monitoring uses specialized instruments to record and analyze voltage, current, and frequency disturbances over time. A power quality analyzer (PQA) captures RMS variations, transients, harmonics, flicker, and power factor with time-stamping, enabling engineers to correlate events with their causes. Monitoring locations are typically selected at the point of common coupling (PCC) with the utility, at the main distribution panel, and at critical or sensitive loads.

IEEE 1159.3 defines the Power Quality Data Interchange Format (PQDIF), a standard file format for exchanging power quality data between different instruments and analysis software. Key measurements include RMS voltage trending (cycle-by-cycle or half-cycle), harmonic spectrum (magnitude and phase of each harmonic up to the 50th), transient capture (high-speed sampling at 1-10 MHz for recording fast voltage spikes), and statistical analysis (SARFI indices that count the number of sags and swells exceeding various thresholds over a period).

**Example 1.5.3:** A manufacturing plant installs a power quality monitor at the 13.8 kV service entrance. Over a 30-day monitoring period, the following events are recorded: 42 voltage sags (15 to 70-80%, 20 to 80-90%, 7 to 50-70%), 3 voltage swells to 110-120%, and average voltage THD of 3.2%. Determine (a) the SARFI-80 index (sags below 80%), (b) the average sag frequency per week, (c) whether the voltage THD meets IEEE 519 limits, and (d) the expected annual cost if each sag below 70% causes a production disruption costing $15,000.
**Solution:**
(a) SARFI-80 counts events where the retained voltage drops below 80% of nominal. The 70-80% category (15 events) falls below the 80% threshold by definition, as does the 50-70% category (7 events). The 80-90% sags (20 events) do not qualify.
SARFI-80 = 15 + 7 = **22 events** over 30 days.
(b) Total sags: 42 in 30 days = 42/4.286 = **9.8 sags per week**.
(c) Voltage THD = 3.2% < 5% limit for systems < 69 kV: **compliant** with IEEE 519.
(d) Sags below 70%: 7 in 30 days → annualized: $7 \times (365/30) = 85.2$/year.
Annual cost: $85.2 \times \$15,000$ = **$1,278,000/year** — easily justifying investment in a UPS or DVR system.

### 1.5.4 Arc Flash Analysis

An arc flash occurs when an electric current passes through ionized air between conductors or from a conductor to ground, creating a plasma arc with temperatures exceeding 20,000°C. The intense thermal energy, pressure blast, and molten metal ejection pose severe hazards to workers performing energized electrical work. NFPA 70E requires arc flash hazard analysis for all electrical equipment likely to be serviced while energized, and IEEE Standard 1584-2018 provides the empirical equations for calculating incident energy based on system parameters.

The IEEE 1584 simplified method estimates incident energy as $E = 4.184 \times C_f \times E_n \times (t/0.2) \times (610^x/D^x)$, where $C_f$ is a calculation factor (1.0 for voltages above 1 kV, 1.5 for voltages at or below 1 kV), $E_n$ is the normalized incident energy, t is the arc duration in seconds, D is the working distance in mm, and x is the distance exponent. The arc duration depends primarily on the clearing time of the upstream protective device — reducing clearing time is the single most effective arc flash mitigation strategy. Other mitigation methods include zone-selective interlocking (which accelerates upstream relay tripping), arc flash detection relays (which sense the arc light and trip in <

5 ms), current-limiting fuses, remote racking of breakers, and increasing the working distance. PPE categories range from Category 1 (4 cal/cm²) through Category 4 (40 cal/cm²), with incident energies above 40 cal/cm² considered too dangerous for energized work.

**Example 1.5.4:** A 480 V switchgear has a bolted fault current of 35 kA. The protective relay clears the fault in 0.25 seconds (15 cycles). The working distance is 610 mm (24 inches). Using the IEEE 1584 simplified method with $E_n = 3.5$ J/cm² (for the given system configuration), $C_f = 1.5$, and distance exponent x = 1.641, determine (a) the incident energy and (b) the required PPE category.
**Solution:**
(a) $E = 4.184 \times C_f \times E_n \times (t/0.2) \times (610^x/D^x)$
$E = 4.184 \times 1.5 \times 3.5 \times (0.25/0.2) \times (610^{1.641}/610^{1.641})$
$E = 4.184 \times 1.5 \times 3.5 \times 1.25 \times 1.0 =$ **27.5 J/cm² = 6.57 cal/cm²**
(b) PPE categories: Cat 1 ≤ 4 cal/cm², Cat 2 ≤ 8 cal/cm², Cat 3 ≤ 25 cal/cm², Cat 4 ≤ 40 cal/cm².
At 6.57 cal/cm², the required PPE is **Category 2** (arc-rated clothing with minimum rating of 8 cal/cm²).
If the relay clearing time were reduced to 0.1 s (5 cycles) using an arc flash relay, the incident energy would drop to 6.57 × (0.1/0.25) = 2.63 cal/cm² — reducing the requirement to Category 1.

## 1.6 HVDC Transmission

High-Voltage Direct Current (HVDC) transmission converts AC power to DC for long-distance transmission and then converts it back to AC at the receiving end. HVDC is preferred over AC for submarine cables (where AC capacitive charging current limits cable lengths to approximately 50-80 km), very long overhead lines (where HVDC becomes economical above approximately 600-800 km due to lower line losses and no reactive power requirements), and asynchronous interconnections between power systems operating at different frequencies or with incompatible AC characteristics.

### 1.6.1 HVDC Fundamentals

An HVDC link consists of a rectifier station (converting AC to DC), a DC transmission line or cable, and an inverter station (converting DC back to AC). The two basic configurations are monopolar (one conductor with ground or sea return) and bipolar (two conductors at $\pm V_{dc}$ with a metallic or ground return, which provides redundancy since each pole can operate independently). Typical HVDC voltage levels range from ±250 kV to ±800 kV, with the highest voltage systems (±1,100 kV in China) transmitting up to 12 GW over distances exceeding 3,000 km.

The advantages of HVDC over AC for long-distance transmission include: no reactive power flow (the DC line has no capacitance or inductance effects at DC), lower losses (only resistive losses in two conductors versus three for AC), no skin effect at DC (the full conductor cross-section carries current), independent control of power flow (magnitude and direction), and the ability to connect asynchronous systems. The disadvantages include the high cost of converter stations (typically $100-200 million each), the generation of harmonics by the converters (requiring large AC and DC filters), and the difficulty of creating multi-terminal DC networks (tapping power off a DC line is more complex than AC).

**Example 1.6.1:** A bipolar HVDC link operates at ±500 kV and transmits 3,000 MW over a distance of 1,500 km. Each conductor has a resistance of 0.012 Ω/km. Determine (a) the DC current per pole, (b) the total $I^2R$ losses, (c) the loss percentage, and (d) the equivalent AC loss for comparison (assuming 3 conductors at 500 kV AC with the same resistance per conductor).
**Solution:**
(a) Each pole carries half the power: $P_{pole} = 3{,}000/2 = 1{,}500$ MW.
$I_{dc} = P_{pole} / V_{dc} = 1{,}500 \times 10^6 / (500 \times 10^3) =$ **3,000 A per pole**
(b) Resistance per conductor: $R = 0.012 \times 1{,}500 = 18\ \Omega$.
Losses per pole: $P_{loss} = I^2R = 3{,}000^2 \times 18 = 162$ MW.
Total losses (two poles): $P_{total\ loss} = 2 \times 162 =$ **324 MW**
(c) Loss percentage: $324/3{,}000 \times 100 =$ **10.8%** (high due to the long distance, but DC has no reactive losses)
(d) For equivalent AC at 500 kV (three-phase), line current: $I_{AC} = 3{,}000 \times 10^6 / (\sqrt{3} \times 500 \times 10^3) = 3{,}464$ A.
AC losses: $3 \times I^2 \times R = 3 \times 3{,}464^2 \times 18 = 3 \times 216.0 \times 10^6 = 648$ MW = 21.6%.
The HVDC link losses (10.8%) are approximately **half** the equivalent AC losses (21.6%), demonstrating the advantage of HVDC for long-distance transmission.

### 1.6.2 Converter Technologies

The two main converter technologies for HVDC are Line-Commutated Converters (LCC) and Voltage Source Converters (VSC). LCC-HVDC uses thyristor valves in a 6-pulse or 12-pulse bridge configuration, where the thyristors are turned on by gate signals but rely on the AC system voltage to turn off (commutate). LCC is a mature technology capable of very high power ratings (up to 12 GW) and voltages (±1,100 kV), but it requires a strong AC system for commutation, consumes reactive power (approximately 50-60% of active power), and can only control the DC current magnitude and direction, not independently control active and reactive power.

VSC-HVDC uses insulated-gate bipolar transistors (IGBTs) in a modular multilevel converter (MMC) topology, where each IGBT can be turned on and off by gate signals, providing full four-quadrant control of active and reactive power. VSC can operate with weak AC systems or even create an AC voltage grid (black start capability), making it ideal for offshore wind farm connections and island power supply. VSC-HVDC ratings have grown rapidly, with current projects reaching ±525 kV and 2,000 MW. The MMC topology uses hundreds of submodules per arm to synthesize a nearly perfect sinusoidal voltage, virtually eliminating the need for harmonic filters.

**Example 1.6.2:** A 12-pulse LCC-HVDC rectifier station operates at a firing angle $\alpha = 15°$ with a commutation overlap angle $\mu = 20°$. The no-load DC voltage (for a 6-pulse bridge) is $V_{d0} = 3\sqrt{2} \times V_{LL} / \pi = 350$ kV. Determine (a) the actual DC voltage per 6-pulse bridge accounting for firing angle and commutation, (b) the total DC voltage for the 12-pulse configuration, and (c) the reactive power consumed by the converter.
**Solution:**
(a) DC voltage per bridge with firing angle and overlap:
$V_d = V_{d0} \times (\cos(\alpha) + \cos(\alpha + \mu))/2 = 350 \times (\cos(15°) + \cos(35°))/2$
$V_d = 350 \times (0.9659 + 0.8192)/2 = 350 \times 0.8926 =$ **312.4 kV per bridge**
(b) A 12-pulse configuration has two 6-pulse bridges in series:

$V_{dc} = 2 \times 312.4 =$ **624.8 kV**
(c) The power factor of an LCC rectifier is approximately:
$\cos(\varphi) \approx (\cos(\alpha) + \cos(\alpha + \mu))/2 = 0.8926$.
$\varphi = \cos^{-1}(0.8926) = 26.8°$.
If $P_{dc}$ = 2,000 MW: $Q = P \times \tan(\varphi) = 2{,}000 \times \tan(26.8°) = 2{,}000 \times 0.505 =$ **1,010 MVAR**
The converter consumes over 1,000 MVAR of reactive power, which must be supplied by capacitor banks and harmonic filters at the converter station.

## 1.7 Load Flow Analysis

Load flow (or power flow) analysis determines the steady-state operating condition of a power system — the voltage magnitude and angle at every bus, and the real and reactive power flowing through every transmission line and transformer. This is the most fundamental power system calculation, performed routinely for system planning, operational studies, and contingency analysis. The results identify overloaded lines, buses with voltage violations, and the reactive power support needed to maintain acceptable voltage profiles.

Each bus in the network is characterized by four quantities: voltage magnitude $|V|$, voltage angle $\delta$, real power injection P, and reactive power injection Q. Of these four, two are specified and two are unknowns. A slack bus (one per system) has fixed $|V|$ and $\delta$ (typically $\delta = 0°$ as the reference), a PV bus (generator) has specified P and $|V|$, and a PQ bus (load) has specified P and Q. The power balance equations at each bus are $P_i = \Sigma|V_i||V_k|(G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k))$ and $Q_i = \Sigma|V_i||V_k|(G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k))$, where $G_{ik} + jB_{ik}$ are elements of the bus admittance matrix $Y_{bus}$. These nonlinear equations are solved iteratively using the Newton-Raphson method (quadratic convergence, typically 3–5 iterations) or the Gauss-Seidel method (simpler but slower convergence).

**Example 1.7:** A 2-bus system has Bus 1 (slack, $V_1 = 1.0\angle 0°$ pu) connected to Bus 2 (PQ load, $P_2 = -1.0$ pu, $Q_2 = -0.5$ pu) through a transmission line with admittance $y_{12} = 5 - j15$ pu. Using the Gauss-Seidel method with an initial guess of $V_2 = 1.0\angle 0°$ pu, perform two iterations to find $V_2$.
**Solution:**
$Y_{bus}$ for a 2-bus system: $Y_{11} = Y_{22} = y_{12} = 5 - j15$, $Y_{12} = Y_{21} = -y_{12} = -5 + j15$.
Gauss-Seidel update: $V_2^{(k+1)} = (1/Y_{22}) \times ((P_2 - jQ_2)/V_2^{(k)*} - Y_{21}V_1)$.
**Iteration 1** ($V_2^{(0)} = 1.0\angle 0°$):
$(P_2 - jQ_2)/V_2^{(0)*} = (-1.0 + j0.5)/1.0 = -1.0 + j0.5$.
$-Y_{21}V_1 = (5 - j15)(1.0) = 5 - j15$.
Sum $= (-1.0 + j0.5) + (5 - j15) = 4.0 - j14.5$.
$V_2^{(1)} = (4.0 - j14.5)/(5 - j15) = (4.0 - j14.5)(5 + j15)/((5)^2 + (15)^2) = (20 + j60 - j72.5 + 217.5)/250 = (237.5 - j12.5)/250 =$ $\mathbf{0.950 - j0.050 = 0.9513\angle{-3.01°}}$ **pu**.
**Iteration 2** ($V_2^{(1)} = 0.950 - j0.050$):
$V_2^{(1)*} = 0.950 + j0.050$.
$(P_2 - jQ_2)/V_2^{(1)*} = (-1.0 + j0.5)/(0.950 + j0.050) = (-1.0 + j0.5)(0.950 - j0.050)/(0.9025 + 0.0025) = (-0.950 + j0.050 + j0.475 + 0.025)/0.9050 = (-0.925 + j0.525)/0.9050 = -1.022 + j0.580$.
Sum $= (-1.022 + j0.580) + (5 - j15) = 3.978 - j14.42$.
$V_2^{(2)} = (3.978 - j14.42)/(5 - j15) = (3.978 - j14.42)(5 + j15)/250 = (19.89 + j59.67 - j72.1 + 216.3)/250 = (236.19 - j12.43)/250 =$ $\mathbf{0.9448 - j0.0497 = 0.9461\angle{-3.01°}}$ **pu**.
The voltage at Bus 2 converges toward approximately $0.946\angle{-3.0°}$ pu, showing a 5.4% voltage

drop due to the load.
Continued iterations (or Newton-Raphson) would refine this result.
The line power flow from Bus 1 to Bus 2 can then be calculated as $S_{12} = V_1(V_1 - V_2)^* \times y_{12}^*$.

## 1.8 SCADA Systems

Supervisory Control and Data Acquisition (SCADA) systems provide centralized monitoring and control of geographically distributed power system assets — generators, substations, transmission lines, and distribution feeders. A SCADA system continuously collects real-time measurements (bus voltages, line currents, power flows, breaker states, transformer tap positions, and equipment temperatures) from Remote Terminal Units (RTUs) and Intelligent Electronic Devices (IEDs) located at field sites, transmits them to a central master station, and presents them to operators through a Human-Machine Interface (HMI). Operators can issue supervisory commands — opening or closing circuit breakers, adjusting transformer taps, changing generator setpoints — that are relayed back to field devices. Modern SCADA systems integrate with Energy Management Systems (EMS) to perform state estimation, automatic generation control (AGC), economic dispatch, and contingency analysis in real time.

### 1.8.1 SCADA Architecture

A typical SCADA architecture consists of four layers: field instrumentation (CTs, VTs, transducers, and sensors), RTUs or substation automation controllers that digitize and aggregate field measurements, a communications network linking remote sites to the control center, and the master station with SCADA servers, databases, and HMI displays. Communication protocols include DNP3 (Distributed Network Protocol), widely used in North American utilities for serial and TCP/IP-based polling of RTUs; IEC 60870-5-101/104, the international equivalent used extensively outside North America; and IEC 61850, a substation automation standard that uses GOOSE (Generic Object Oriented Substation Event) messaging for high-speed peer-to-peer communication between IEDs within a substation. The scan rate — the interval at which the master polls all RTUs — is typically 2–10 seconds for SCADA systems, while IEC 61850 GOOSE messages achieve latencies under 4 ms for protection-critical signals. Redundancy is achieved through dual master stations in hot-standby configuration, redundant communication paths (fiber, microwave, cellular backup), and dual RTU processors.

**Example 1.8.1:** A utility SCADA system monitors 120 substations, each with an RTU reporting 80 analog measurements (voltage, current, power) and 60 digital status points (breaker positions, alarms). The master station polls all RTUs using DNP3 over TCP/IP with a scan cycle of 4 seconds. Each analog measurement requires 4 bytes and each digital point requires 1 bit. Determine the total number of data points, the minimum data throughput required, and the average polling rate per RTU.
**Solution:**
Analog points: $120 \times 80 = 9{,}600$ analog measurements
Digital points: $120 \times 60 = 7{,}200$ status points
Total data points: $9{,}600 + 7{,}200 =$ **16,800 points**
Data per scan cycle:
Analog data: $9{,}600 \times 4$ bytes $= 38{,}400$ bytes

Digital data: 7,200 × 1 bit = 7,200 bits = 900 bytes
Total payload: 38,400 + 900 = 39,300 bytes per scan
With DNP3 framing overhead (~20%): 39,300 × 1.2 = 47,160 bytes per scan
Minimum throughput: 47,160 bytes / 4 s = 11,790 bytes/s = **94.3 kbps**
Average polling rate: 120 RTUs / 4 s = **30 RTUs/s** (one RTU polled every 33.3 ms)

### 1.8.2 SCADA Cybersecurity

Power system SCADA networks are critical infrastructure targets, and cybersecurity has become a fundamental design requirement. The NERC CIP (North American Electric Reliability Corporation Critical Infrastructure Protection) standards mandate security controls for bulk electric system cyber assets, including electronic security perimeters (ESP), access management, security patching, incident response, and configuration change management. Defense-in-depth strategies include network segmentation using firewalls and demilitarized zones (DMZ) between the corporate IT network and the operational technology (OT) SCADA network, encrypted communications using TLS or VPN tunnels for DNP3 Secure Authentication, role-based access control (RBAC) for HMI operator consoles, and intrusion detection systems (IDS) tuned to recognize anomalous SCADA traffic patterns such as unauthorized control commands or abnormal polling frequencies. Air-gapping — physically isolating the SCADA network from the internet — was once standard practice but is increasingly impractical as utilities adopt cloud-based analytics, remote access for maintenance, and wide-area monitoring; instead, managed security interfaces with strict firewall rules, application whitelisting, and continuous monitoring have become the norm.

**Example 1.8.2:** A utility's SCADA cybersecurity audit reveals that the control center firewall processes an average of 2,500 authorized DNP3 sessions per hour. The intrusion detection system (IDS) has a false positive rate of 0.2% and a detection rate (true positive rate) of 98.5% for malicious traffic. During a 24-hour monitoring period, 12 actual intrusion attempts occur among the total traffic. Determine the expected number of detected intrusions, missed intrusions, and false alarms over 24 hours.
**Solution:**
Total authorized sessions in 24 hours: 2,500 × 24 = 60,000 sessions
Actual intrusion attempts: 12
Detected intrusions (true positives): 0.985 × 12 = **11.82** ≈ **12 detected**
Missed intrusions (false negatives): 12 − 11.82 = **0.18** ≈ **0 missed**
Legitimate sessions: 60,000 − 12 = 59,988
False alarms (false positives): 0.002 × 59,988 = **119.98** ≈ **120 false alarms**
Precision: 12 / (12 + 120) = 12 / 132 = **9.1%**
The low precision illustrates the base-rate problem in cybersecurity: even with a low false positive rate, the rarity of actual attacks means most alerts are false alarms, requiring analyst triage to distinguish real threats.

# Chapter 2

# Communications Engineering

Communications engineering is the branch of electrical engineering focused on the transmission, reception, and processing of information-bearing signals across a variety of media including wired, wireless, and optical channels. The discipline spans analog techniques such as amplitude and frequency modulation to modern digital methods including sampling, quantization, error correction, and data compression. At its core, communications engineering applies signal processing theory to reliably move information from a source to a destination while managing constraints such as bandwidth, noise, interference, and power.

## 2.1 Analog Signal Processing

Analog signals are continuous time signals. They represent physical quantities such as voltage, current, or pressure as smoothly varying waveforms without discrete steps. Analog signal processing operates on these continuous waveforms using circuits composed of resistors, capacitors, inductors, and operational amplifiers to perform operations such as amplification, filtering, modulation, and demodulation. Because analog systems work directly with continuous signals, they can be simpler to implement for certain tasks, but they are susceptible to noise accumulation at each processing stage, which limits the distance and fidelity of analog communication links.

### 2.1.1 AM Radio

Amplitude Modulation (AM) is one of the earliest and most fundamental modulation techniques in communications engineering. In AM, the amplitude of a high-frequency carrier signal is varied in proportion to the instantaneous value of the baseband message signal, while the carrier frequency remains constant. Mathematically, the AM signal can be expressed as $s(t) = [A_c + m(t)] \times \cos(2\pi f_c t)$, where $A_c$ is the carrier amplitude, $m(t)$ is the message signal, and $f_c$ is the carrier frequency. The modulation index m, defined as the ratio of the peak message amplitude to the carrier amplitude, must be kept at or below 1.0 to avoid envelope distortion and overmodulation — when $m > 1$, the carrier envelope crosses zero, introducing severe distortion that a simple envelope detector cannot recover. The modulation process is fundamentally a multiplication of the carrier by $(1 + m(t)/A_c)$, which in the frequency domain shifts the baseband message spectrum to be centered on the carrier frequency, creating an upper sideband (USB) and a lower sideband (LSB) symmetric about $f_c$. Because both sidebands carry identical information, the total occupied bandwidth equals twice the highest frequency component of the message signal.

Commercial AM broadcast radio operates in the medium frequency (MF) band from 540 kHz to 1700 kHz, with each station allocated a 10 kHz channel bandwidth, limiting the audio baseband to 5 kHz. Demodulation can be accomplished with a simple envelope detector consisting of a diode and an RC low-pass filter, which is one reason AM radio receivers were historically inexpensive and widely accessible — the crystal radio, requiring no external power source, is the simplest possible AM receiver. However, standard AM (also called Double Sideband Full Carrier, DSB-FC) is not power-efficient because a significant portion of the transmitted power resides in the carrier, which carries no information; at 100% modulation, only one-third of the total power is in the sidebands. Double Sideband Suppressed Carrier (DSB-SC) eliminates the carrier entirely, putting all transmitted power into the information-bearing sidebands, but requires coherent (synchronous) demodulation with a locally generated carrier reference. Single Sideband (SSB) goes further by transmitting only one sideband, halving the bandwidth and further improving power efficiency — SSB is the standard for HF (shortwave) radio, amateur radio, and military communications where spectrum and power are at a premium. Despite its inefficiency, conventional DSB-FC AM remains in active use for commercial broadcast (AM radio), aviation voice communications (where the simple envelope detector ensures reliability), and as a building block for understanding more advanced modulation schemes.

**Example 2.1.1:** An AM broadcast transmitter has a carrier power of 50 kW and is modulated by a single-tone audio signal with a modulation index of m = 0.75. Determine (a) the total transmitted power, (b) the power in each sideband, and (c) the efficiency (percentage of total power carrying information).
**Solution:**
(a) Total transmitted power for a single-tone AM signal:
$P_{total} = P_c \times (1 + m^2/2) = 50 \times (1 + 0.75^2/2) = 50 \times (1 + 0.2813) = 50 \times 1.2813 =$ **64.06 kW**
(b) Power in both sidebands combined: $P_{sb} = P_c \times m^2/2 = 50 \times 0.5625/2 = 50 \times 0.2813 = 14.06$ kW
Power in each sideband: $P_{each} = 14.06 / 2 =$ **7.03 kW**
(c) Efficiency (only sideband power carries information):
$\eta = P_{sb} / P_{total} = 14.06 / 64.06 = 0.2195 =$ **21.95%**
This demonstrates why standard AM is inefficient: at m = 0.75, nearly 78% of the transmitted power is in the carrier, which carries no information.

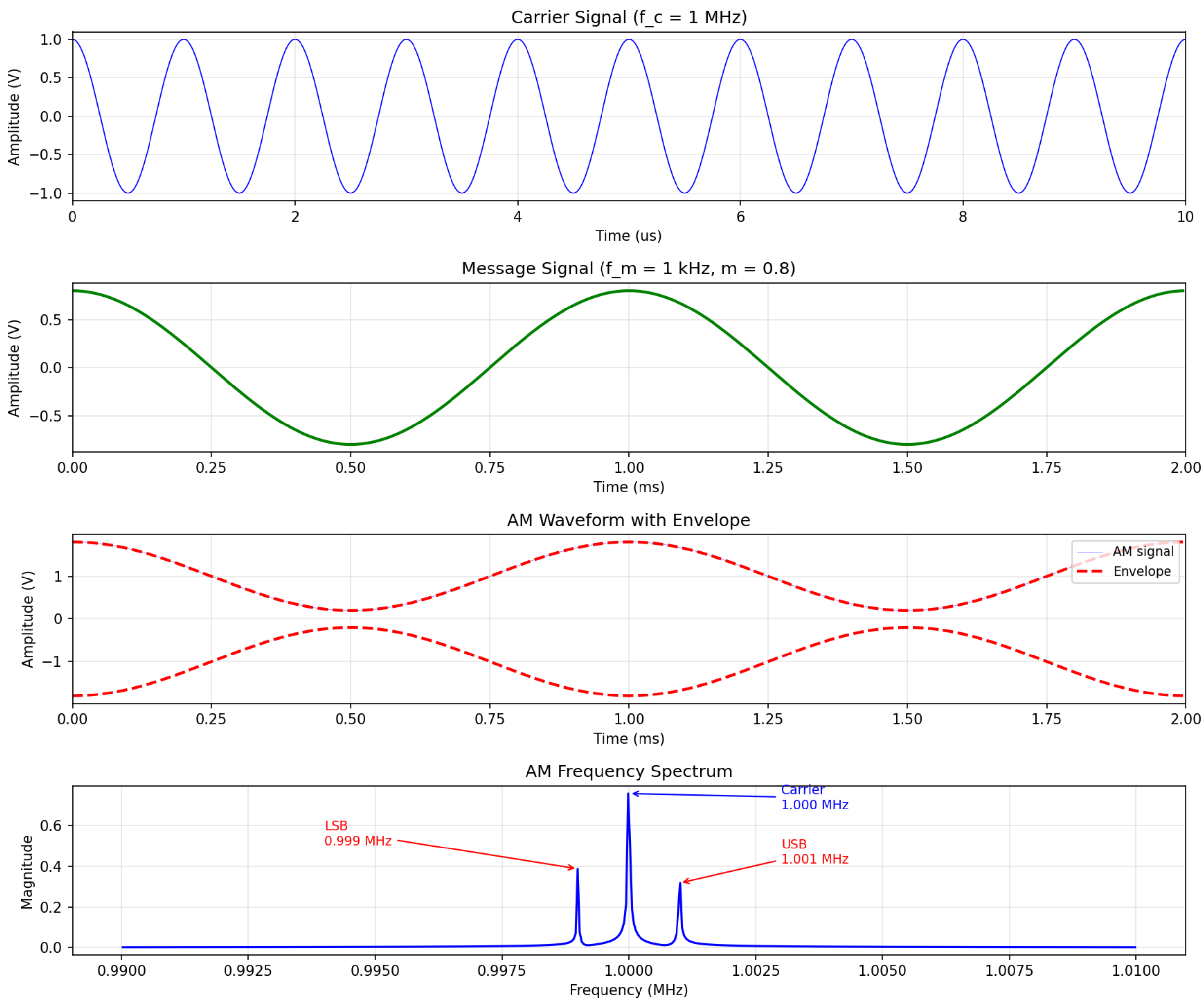



Figure 2.1.1: AM Modulation Waveform and Spectrum

### 2.1.2 FM Radio

Frequency Modulation (FM) encodes information by varying the instantaneous frequency of a carrier signal in proportion to the message signal, while keeping the amplitude constant. The FM signal is expressed as $s(t) = A_c \times \cos(2\pi f_c t + 2\pi k_f \int m(\tau)d\tau)$, where $k_f$ is the frequency sensitivity in Hz/V and $m(t)$ is the message signal. The maximum frequency deviation $\Delta f = k_f \times \max|m(t)|$ determines how far the instantaneous frequency swings from the carrier. The modulation index $\beta = \Delta f / f_m$, where $f_m$ is the highest message frequency, governs the spectral characteristics of the FM signal.

Carson's rule approximates the bandwidth of an FM signal as $B \approx 2(\Delta f + f_m) = 2f_m(\beta + 1)$. Commercial FM broadcast radio operates in the VHF band from 88 to 108 MHz with a maximum frequency deviation of ±75 kHz and audio bandwidth of 15 kHz, giving $\beta = 5$ and a Carson bandwidth of approximately 180 kHz — each station is allocated a 200 kHz channel. FM provides superior noise performance compared to AM because amplitude variations caused by noise can be removed by a limiter before demodulation. The improvement is quantified by the FM noise advantage: for wideband FM with $\beta \gg 1$, the output SNR exceeds the input carrier-to-noise ratio by a factor of $3\beta^2(\beta + 1)$, which is the basis for trading bandwidth for noise performance. Pre-emphasis and de-emphasis filters further improve high-frequency SNR by boosting high frequencies before transmission and attenuating them at the receiver.

**Example 2.1.2:** An FM broadcast station transmits with a maximum frequency deviation of $\Delta f = 75$ kHz and audio bandwidth of $f_m = 15$ kHz. Determine (a) the modulation index, (b) the Carson's rule bandwidth, (c) the number of significant sidebands from the Bessel function table (for $\beta = 5$, there are 8 significant sideband pairs), and (d) the exact bandwidth based on significant sidebands.
**Solution:**
(a) Modulation index: $\beta = \Delta f / f_m = 75{,}000 / 15{,}000 = \mathbf{5}$
(b) Carson's rule bandwidth: $B = 2(\Delta f + f_m) = 2(75{,}000 + 15{,}000) =$ **180 kHz**
(c) For $\beta = 5$, the Bessel function table shows 8 significant sideband pairs ($J_0$ through $J_8$ have magnitudes $> 0.01$).
(d) Exact bandwidth from significant sidebands: $B_{exact} = 2 \times 8 \times f_m = 2 \times 8 \times 15 =$ **240 kHz**
Carson's rule underestimates the bandwidth (180 kHz vs. 240 kHz) but captures approximately 98% of the signal power, which is why FM broadcast channels are spaced at 200 kHz — a practical compromise between Carson's rule and the exact sideband extent.

## 2.2 Digital Signal Processing

A digital signal is created by sampling a continuous signal and representing the samples as discrete values which can be used computationally. The process involves two key steps: sampling, which converts the continuous-time signal into a discrete-time sequence, and quantization, which maps each sample to a finite set of discrete amplitude levels. Digital signal processing (DSP) then manipulates these discrete numerical sequences using algorithms implemented in software or dedicated hardware such as DSP microprocessors and FPGAs. The primary advantages of digital processing over analog include repeatable precision, immunity to noise accumulation during processing stages, the ability to implement adaptive and non-linear algorithms, and straightforward storage and retrieval of signal data.

### 2.2.1 Fourier Analysis

The Fourier series coefficients represent the frequencies present in a signal. Any periodic signal can be decomposed into a sum of sinusoidal components at harmonically related frequencies using the Fourier series, where each coefficient specifies the amplitude and phase of the corresponding harmonic. For aperiodic signals, the Fourier Transform extends this concept by representing the signal as a continuous spectrum of frequencies rather than discrete harmonics. In the digital domain, the Discrete Fourier Transform (DFT) operates on finite-length sampled sequences, and the Fast Fourier Transform (FFT) algorithm computes the DFT efficiently in $O(N \log N)$ operations rather than the $O(N^2)$ required by direct computation. Fourier analysis is foundational in communications engineering for tasks such as spectral analysis, filter design, modulation and demodulation, and characterizing the frequency response of channels and systems.

**Example 2.2.1:** A periodic square wave signal has a fundamental frequency of 1 kHz, amplitude of 5 V (peak-to-peak), and 50% duty cycle. Determine the Fourier series coefficients for the first three non-zero harmonic components, and calculate the RMS voltage of the signal truncated to

these three harmonics.
**Solution:**
A square wave with amplitude A ($\pm A/2$) and 50% duty cycle has the Fourier series:
$x(t) = (2A/\pi) \times [\sin(2\pi f_1 t) + (1/3)\sin(2\pi \cdot 3f_1 t) + (1/5)\sin(2\pi \cdot 5f_1 t) + \ldots]$
With peak-to-peak = 5 V, the amplitude $A/2 = 2.5$ V, so $A = 5$ V, and $2A/\pi = 10/\pi = 3.183$ V.
First three non-zero harmonics (only odd harmonics are present):
1st harmonic (1 kHz): $a_1 = 2A/\pi = 10/\pi =$ **3.183 V peak**
3rd harmonic (3 kHz): $a_3 = 2A/(3\pi) = 10/(3\pi) =$ **1.061 V peak**
5th harmonic (5 kHz): $a_5 = 2A/(5\pi) = 10/(5\pi) =$ **0.637 V peak**
RMS voltage of the truncated signal (sum of sinusoidal RMS values):
$V_{rms} = \sqrt{(a_1^2/2 + a_3^2/2 + a_5^2/2)} = \sqrt{(3.183^2/2 + 1.061^2/2 + 0.637^2/2)}$
$V_{rms} = \sqrt{(5.066 + 0.563 + 0.203)} = \sqrt{5.832} =$ **2.415 V**
For comparison, the true RMS of the square wave is 2.5 V, so the three-harmonic approximation captures $(2.415/2.5)^2 = 93.3\%$ of the total power.

### 2.2.2 Nyquist Sampling

The Nyquist sampling rate is greater than 2 times the maximum signal frequency. More precisely, the Nyquist-Shannon sampling theorem states that a band-limited continuous signal with no frequency content above $f_{max}$ can be perfectly reconstructed from its samples if the sampling rate $f_s$ is greater than $2 \times f_{max}$. This minimum rate of $2 \times f_{max}$ is called the Nyquist rate. If the signal is sampled below this rate, aliasing occurs, where higher-frequency components are misrepresented as lower frequencies in the sampled data, causing irreversible distortion. In practice, anti-aliasing low-pass filters are applied before the analog-to-digital converter to attenuate frequency components above $f_s/2$, and systems typically sample at rates somewhat above the Nyquist rate to provide a guard band that relaxes the filter design requirements.

**Example 2.2.2:** An audio signal contains frequency components from 20 Hz to 18 kHz. It is to be digitized with a sampling rate of 44.1 kHz (CD-quality audio). Determine (a) the Nyquist rate, (b) whether aliasing will occur, and (c) the frequency at which a 20 kHz component would appear in the sampled data.
**Solution:**
(a) Nyquist rate: $f_{Nyquist} = 2 \times f_{max} = 2 \times 18{,}000 =$ **36 kHz**
(b) The sampling rate of 44.1 kHz exceeds the Nyquist rate of 36 kHz, so **no aliasing will occur** for the 18 kHz signal content. The guard band is $44{,}100/2 - 18{,}000 = 4{,}050$ Hz.
(c) If an out-of-band component at 20 kHz is present (not filtered by the anti-aliasing filter):
Check against the Nyquist limit: 20,000 Hz $< f_s/2 = 22{,}050$ Hz, so the 20 kHz component is **below the Nyquist frequency and does not alias** — it appears at 20 kHz in the sampled spectrum.
However, if a 25 kHz component were present:
$f_{alias} = f_s - 25{,}000 = 44{,}100 - 25{,}000 =$ **19,100 Hz** — this would alias and corrupt the audio band, which is why anti-aliasing filters are essential.

### 2.2.3 Quantization and Encoding

Quantization is the process of mapping continuous-amplitude samples to a finite set of discrete levels. After sampling converts a continuous-time signal to a discrete-time sequence, quantization assigns each sample to the nearest available level from a set of $2^N$ levels, where N is the number of bits per sample. The spacing between adjacent levels is the step size $\Delta = (V_{max} - V_{min}) / 2^N$ for uniform quantization, and the resulting quantization error for each sample is bounded by $\pm\Delta/2$. This error behaves as additive white noise with a signal-to-quantization-noise ratio (SQNR) of approximately $6.02N + 1.76$ dB for a full-scale sinusoidal input, meaning each additional bit improves the SQNR by about 6 dB.

Non-uniform quantization allocates finer step sizes to small signal amplitudes and coarser steps to large amplitudes, improving the effective dynamic range for signals like speech that spend most of their time at low levels. The two standard companding laws are μ-law (used in North America and Japan, with $\mu = 255$) and A-law (used in Europe, with $A = 87.6$), both of which are implemented in PCM telephone systems using 8-bit samples at 8 kHz for a bit rate of 64 kbps per voice channel. After quantization, each sample is encoded into a binary codeword for transmission or storage, a process called pulse code modulation (PCM).

**Example 2.2.3:** A 16-bit audio ADC has an input range of ±1 V (2 V full-scale). Determine (a) the quantization step size, (b) the maximum quantization error, (c) the SQNR for a full-scale sinusoidal input, and (d) the dynamic range in dB.
**Solution:**
(a) Step size: $\Delta = 2\text{ V} / 2^{16} = 2 / 65{,}536 =$ **30.52 μV**
(b) Maximum quantization error: $\pm\Delta/2 = \pm 15.26$ μV = **±15.26 μV**
(c) SQNR = 6.02N + 1.76 = 6.02(16) + 1.76 = 96.32 + 1.76 = **98.08 dB**
(d) Dynamic range = $20 \log_{10}(2^N) = 20 \log_{10}(65{,}536) = 20 \times 4.816 =$ **96.33 dB**
The 98 dB SQNR explains why 16-bit audio (CD quality) provides excellent fidelity — the quantization noise floor is nearly 100 dB below a full-scale signal.

## 2.3 Digital Modulation

Digital modulation maps discrete data (bits or symbols) onto analog carrier waveforms for transmission over bandpass channels. Unlike analog modulation where the message is a continuous waveform, digital modulation transmits information as a sequence of symbols, each representing one or more bits. The key performance metrics for digital modulation schemes are bandwidth efficiency (bits/s/Hz), power efficiency (the $E_b/N_0$ required to achieve a target bit error rate), and implementation complexity. The choice of modulation scheme involves a trade-off between these factors, with higher-order schemes achieving greater bandwidth efficiency at the cost of increased power requirements and sensitivity to noise.

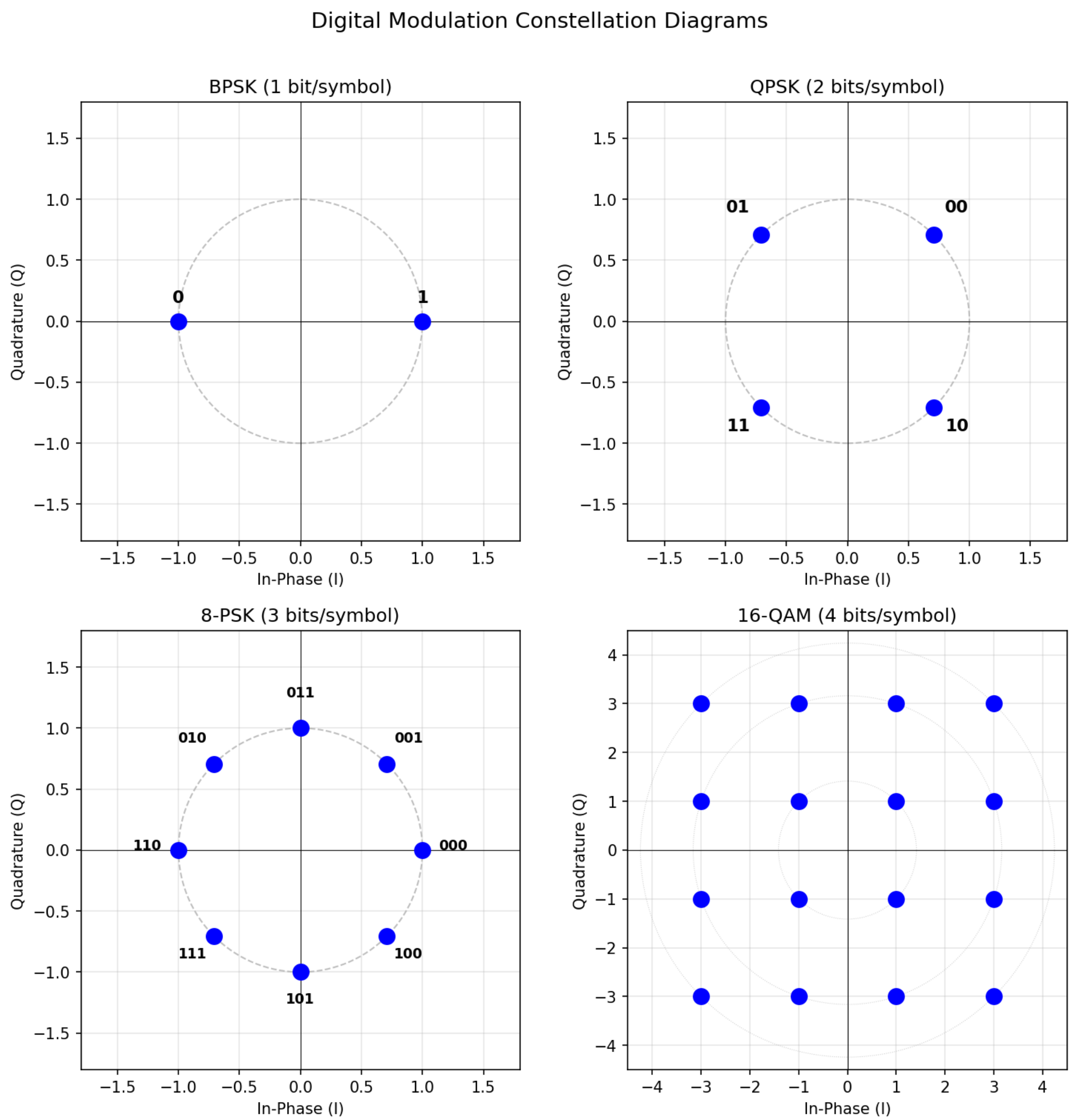



Figure 2.3: Digital Modulation Constellation Diagrams

### 2.3.1 Amplitude Shift Keying (ASK)

Amplitude Shift Keying (ASK) varies the amplitude of a carrier to represent digital data. In binary ASK (also called on-off keying, OOK), a binary 1 is transmitted as the carrier at full amplitude and a binary 0 as zero amplitude (or a reduced amplitude). The transmitted signal is $s(t) = A_k \times \cos(2\pi f_c t)$ during the k-th symbol interval, where $A_k$ takes on one of M discrete amplitude levels for M-ASK. Binary ASK has a bandwidth efficiency of 1 bit/s/Hz and requires the simplest transmitter and receiver, but it is the least power-efficient of the basic digital modulation schemes because the amplitude variations make it highly susceptible to noise and fading. ASK finds application in low-cost, low-data-rate systems such as RFID tags and infrared remote controls.

**Example 2.3.1:** A binary ASK system transmits at 9,600 bps using a carrier at 100 kHz with raised-cosine pulse shaping having a roll-off factor of $\alpha = 0.35$. Determine (a) the required bandwidth, (b) the bandwidth efficiency, and (c) the minimum $E_b/N_0$ for a BER of $10^{-5}$ using coherent detection.
**Solution:**
(a) Bandwidth with raised-cosine shaping: $B = R_b(1 + \alpha) = 9{,}600 \times (1 + 0.35) = 9{,}600 \times 1.35 =$ **12,960 Hz**
(b) Bandwidth efficiency: $\eta = R_b / B = 9{,}600 / 12{,}960 =$ **0.74 bits/s/Hz**
(c) For coherent binary ASK, BER $= Q(\sqrt{(E_b/N_0)})$.
For BER $= 10^{-5}$, the Q-function argument is 4.265, so $E_b/N_0 = 4.265^2 = 18.19 =$ **12.60 dB**.
Binary ASK requires about 3 dB more $E_b/N_0$ than BPSK for the same BER.

### 2.3.2 Frequency Shift Keying (FSK)

Frequency Shift Keying (FSK) encodes digital data by switching the carrier frequency between two or more discrete values. In binary FSK (BFSK), a binary 1 is transmitted at frequency $f_1$ and a binary 0 at frequency $f_2$, with the tone spacing $\Delta f = |f_1 - f_2|$ determining the modulation characteristics. When $\Delta f = R_b/2$ (where $R_b$ is the bit rate), the two tones are orthogonal over each bit period, yielding minimum shift keying (MSK), which achieves the minimum bandwidth for orthogonal FSK. The constant-envelope property of FSK makes it robust against nonlinear amplifier distortion and amplitude fading, which is why it is widely used in wireless and industrial applications.

Gaussian MSK (GMSK) passes the data through a Gaussian low-pass filter before frequency modulation, producing smoother phase transitions and significantly reduced spectral sidelobes compared to standard MSK. GMSK with a bandwidth-time product BT = 0.3 is used in the GSM cellular standard, achieving 99% power containment within a bandwidth of approximately $0.57 \times R_b$. Non-coherent detection of FSK is straightforward using envelope detectors or discriminators, making FSK receivers simpler than PSK receivers at the cost of a 1-3 dB power penalty.

**Example 2.3.2:** A binary FSK system uses frequencies $f_1 = 1{,}200$ Hz (mark) and $f_2 = 2{,}200$ Hz (space) at a data rate of 300 bps (similar to the Bell 103 modem). Determine (a) the frequency deviation, (b) the modulation index $h = \Delta f / R_b$, (c) the Carson's rule bandwidth, and (d) whether the tones are orthogonal.
**Solution:**
(a) Frequency deviation: $\Delta f = |f_1 - f_2| = |1{,}200 - 2{,}200| =$ **1,000 Hz**
(b) Modulation index: $h = \Delta f / R_b = 1{,}000 / 300 =$ **3.33**
(c) Carson's rule bandwidth: $B = 2(\Delta f + R_b) = 2(1{,}000 + 300) =$ **2,600 Hz**
(d) For orthogonal FSK, the minimum tone spacing is $R_b/2 = 150$ Hz for coherent detection or $R_b = 300$ Hz for non-coherent detection.
Since $\Delta f = 1{,}000$ Hz exceeds both thresholds, the tones are **orthogonal** for both coherent and non-coherent detection.
The wide spacing ($h = 3.33$) is spectrally inefficient but provides robust performance at low SNR.

### 2.3.3 Phase Shift Keying (PSK)

Phase Shift Keying (PSK) maps digital data to discrete phase values of the carrier signal. In Binary PSK (BPSK), the carrier phase is 0° for a binary 1 and 180° for a binary 0, providing a phase separation of 180° between the two symbols. BPSK achieves the best power efficiency among binary modulation schemes, requiring an $E_b/N_0$ of only 9.6 dB for a BER of $10^{-5}$ with coherent detection. Quadrature PSK (QPSK) doubles the bandwidth efficiency to 2 bits/s/Hz by using four phase states (0°, 90°, 180°, 270°), where each symbol carries 2 bits. Remarkably, QPSK achieves the same BER performance as BPSK per bit because the 3 dB increase in symbol energy exactly compensates for the reduced phase spacing.

Higher-order PSK (8-PSK, 16-PSK) further increases bandwidth efficiency but with diminishing returns and rapidly increasing power requirements. As the number of phase states M increases, the angular spacing between adjacent symbols shrinks to $2\pi/M$, making the constellation increasingly sensitive to phase noise and requiring higher SNR to maintain the same BER. For $M \geq 8$, hybrid amplitude-phase constellations (QAM) become more power-efficient than pure PSK. Differential PSK (DPSK) encodes data in the phase change between successive symbols rather than absolute phase, simplifying demodulation by eliminating the need for a coherent carrier reference at a cost of approximately 1-2 dB in power efficiency.

**Example 2.3.3:** A QPSK satellite link transmits at a symbol rate of 10 Msymbols/s with raised-cosine pulse shaping ($\alpha = 0.25$). Determine (a) the bit rate, (b) the occupied bandwidth, (c) the bandwidth efficiency, and (d) the required $E_b/N_0$ for BER = $10^{-6}$.
**Solution:**
(a) QPSK carries 2 bits/symbol: $R_b = 2 \times R_s = 2 \times 10 \times 10^6 =$ **20 Mbps**
(b) Occupied bandwidth: $B = R_s(1 + \alpha) = 10 \times 10^6 \times (1 + 0.25) =$ **12.5 MHz**
(c) Bandwidth efficiency: $\eta = R_b / B = 20 / 12.5 =$ **1.6 bits/s/Hz**
(d) For QPSK, BER = $Q(\sqrt{(2E_b/N_0)})$.
For BER = $10^{-6}$, $Q^{-1}(10^{-6}) = 4.753$, so $2E_b/N_0 = 4.753^2 = 22.59$, giving $E_b/N_0 = 11.30 =$ **10.53 dB**.
This is the same $E_b/N_0$ required by BPSK, confirming that QPSK doubles the data rate without a power penalty.

### 2.3.4 Quadrature Amplitude Modulation (QAM)

Quadrature Amplitude Modulation (QAM) combines amplitude and phase modulation by independently modulating two orthogonal carriers (in-phase I and quadrature Q) to create a two-dimensional signal constellation. Each symbol in an M-QAM constellation carries $\log_2(M)$ bits by occupying a unique point in the I-Q plane. Common orders include 16-QAM (4 bits/symbol), 64-QAM (6 bits/symbol), 256-QAM (8 bits/symbol), and 1024-QAM (10 bits/symbol). For square QAM constellations (M = 4, 16, 64, 256, ...), the constellation points form a regular grid, and the minimum Euclidean distance between adjacent symbols determines the error rate performance.

QAM achieves higher bandwidth efficiency than PSK for the same number of constellation points because the rectangular grid arrangement provides better minimum distance between symbols than equally-spaced phase-only constellations. The approximate BER for square M-QAM with Gray coding is BER $\approx (4/\log_2(M)) \times (1 - 1/\sqrt{M}) \times Q(\sqrt{(3 \log_2(M) \times E_b/(N_0(M-1)))})$. QAM is the dominant modulation scheme in modern broadband communications: cable modems (DOCSIS) use up to 4096-QAM, Wi-Fi 6 (802.11ax) uses up to 1024-QAM, and 5G NR uses up to 256-QAM. The trade-off is that higher-order QAM requires significantly better SNR — moving from 64-QAM to 256-QAM gains 33% more bits per symbol but requires approximately 5 dB more $E_b/N_0$.

**Example 2.3.4:** A 64-QAM digital cable TV channel has a symbol rate of 5.057 Msymbols/s (the DOCSIS standard 6 MHz channel). Determine (a) the number of bits per symbol, (b) the raw bit rate, (c) the bandwidth efficiency, and (d) the minimum $E_b/N_0$ for a BER of $10^{-6}$.
**Solution:**
(a) Bits per symbol: $\log_2(64)$ = **6 bits/symbol**
(b) Raw bit rate: $R_b = 6 \times 5.057 \times 10^6$ = **30.34 Mbps**
(c) Bandwidth efficiency: $\eta = R_b / B = 30.34 / 6$ = **5.06 bits/s/Hz**
(d) For 64-QAM, the required $E_b/N_0$ for BER = $10^{-6}$:
Using the approximation: BER $\approx (4/6)(1 - 1/8) \times Q(\sqrt{(3\times6\times E_b/(N_0\times63))})$
Setting BER = $10^{-6}$: $Q(\sqrt{(18E_b/(63N_0))}) \approx 10^{-6} / 0.583 = 1.71 \times 10^{-6}$
$Q^{-1}(1.71 \times 10^{-6}) \approx 4.64$, so $18E_b/(63N_0) = 21.53$, giving $E_b/N_0 = 75.35$ = **18.77 dB**

### 2.3.5 Constellation Diagrams and BER Performance

A constellation diagram is a graphical representation of a digital modulation scheme in the in-phase (I) and quadrature (Q) signal space, where each point represents a unique symbol. Decision regions divide the I-Q plane into zones — the receiver maps each received signal point to the nearest constellation point, and errors occur when noise displaces a point across a decision boundary. Gray coding assigns bit patterns to adjacent constellation points so that they differ by only one bit, ensuring that the most likely symbol errors (to nearest neighbors) cause only a single bit error rather than multiple bit errors.

The bit error rate performance of different modulation schemes reflects the fundamental trade-off between bandwidth efficiency and power efficiency. BPSK and QPSK achieve the same BER per bit (BER = $Q(\sqrt{(2E_b/N_0)})$), requiring $E_b/N_0 \approx 9.6$ dB for BER = $10^{-5}$, but QPSK carries twice the data rate in the same bandwidth. Higher-order schemes require progressively more power: 16-QAM needs approximately 13.4 dB, 64-QAM needs approximately 17.8 dB, and 256-QAM needs approximately 22.5 dB for the same BER = $10^{-5}$. This means each step up in constellation size gains spectral efficiency at the cost of requiring a cleaner channel with higher SNR.

| Modulation | Bits/Symbol | Bandwidth Efficiency | $E_b/N_0$ for BER = $10^{-5}$ |
|---|---|---|---|
| BPSK | 1 | 1 bit/s/Hz | 9.6 dB |
| QPSK | 2 | 2 bits/s/Hz | 9.6 dB |
| 16-QAM | 4 | 4 bits/s/Hz | 13.4 dB |
| 64-QAM | 6 | 6 bits/s/Hz | 17.8 dB |
| 256-QAM | 8 | 8 bits/s/Hz | 22.5 dB |

**Example 2.3.5:** A wireless communication system operates with $E_b/N_0$ = 12 dB and requires a BER no worse than $10^{-5}$. Determine (a) the BER achieved by BPSK, (b) the BER achieved by QPSK, (c) the BER achieved by 16-QAM, and (d) the recommended modulation scheme.
**Solution:**

$E_b/N_0 = 12$ dB $= 10^{12/10} = 15.85$ (linear).
(a) BPSK: BER $= Q(\sqrt{(2 \times 15.85)}) = Q(5.63) =$ **$9.0 \times 10^{-9}$** — well below the $10^{-5}$ requirement.
(b) QPSK: Same BER as BPSK per bit: BER $= Q(\sqrt{(2 \times 15.85)}) =$ **$9.0 \times 10^{-9}$** — also meets the requirement with double the data rate.
(c) 16-QAM: BER $\approx (3/4) \times Q(\sqrt{(4/5 \times 15.85)}) = (3/4) \times Q(\sqrt{(12.68)}) = (3/4) \times Q(3.56) = 0.75 \times 1.85 \times 10^{-4} =$ **$1.39 \times 10^{-4}$** — exceeds the $10^{-5}$ requirement.
(d) **QPSK is recommended.**
It provides 2 bits/s/Hz bandwidth efficiency while meeting the BER target with 2.4 dB of margin.
BPSK also meets the target but wastes half the available spectral efficiency.
16-QAM fails the BER requirement by nearly an order of magnitude at this $E_b/N_0$.

### 2.3.6 MIMO and Spatial Multiplexing

Multiple-Input Multiple-Output (MIMO) systems use multiple antennas at both the transmitter and receiver to exploit the spatial dimension of the wireless channel, dramatically increasing capacity without additional bandwidth or power. The fundamental insight is that in a rich scattering environment (typical of indoor and urban settings), different transmit-receive antenna pairs experience statistically independent fading paths, creating parallel spatial channels through which independent data streams can be simultaneously transmitted. The theoretical capacity of an $N_t \times N_r$ MIMO system in a rich scattering environment is $C = \Sigma \log_2(1 + SNR_i/N_t)$ over $\min(N_t, N_r)$ spatial channels, scaling approximately linearly with $\min(N_t, N_r)$ — a revolutionary result compared to the logarithmic scaling of SISO (single antenna) systems.

MIMO operates in three primary modes. **Spatial multiplexing** transmits independent data streams on each antenna, maximizing throughput — this is the mode used in Wi-Fi 802.11n (up to 4 streams), 802.11ac/ax (up to 8 streams), and LTE (up to 8 layers). **Transmit diversity** sends the same data from multiple antennas with different space-time coding (e.g., Alamouti code for 2 antennas), improving reliability without increasing data rate — used in cell-edge scenarios and safety-critical links. **Beamforming** uses channel state information (CSI) to coherently combine signals at the receiver, increasing the effective SNR by the array gain factor — used in 5G NR with massive MIMO arrays (§16.5.2). The number of independent spatial streams is limited by the rank of the channel matrix H (an $N_r \times N_t$ matrix), which depends on the scattering environment: line-of-sight channels have rank 1 (no multiplexing gain), while rich multipath environments approach full rank. Singular value decomposition (SVD) of H reveals the available parallel channels and their quality.

**Example 2.3.6:** A 4×4 MIMO Wi-Fi system (802.11ac) operates at 5 GHz with 80 MHz bandwidth and 256-QAM modulation (8 bits/symbol) with 5/6 coding rate. The channel supports 4 independent spatial streams. Determine (a) the data rate per spatial stream, (b) the total aggregate data rate, (c) the capacity gain compared to a SISO (1×1) system, and (d) the minimum required SNR per stream for BER $= 10^{-5}$ with 256-QAM.
**Solution:**
(a) For 802.11ac with 80 MHz: 234 data subcarriers, OFDM symbol duration = 3.6 μs (3.2 μs useful + 0.4 μs GI).
Rate per stream $= 234 \times 8 \times (5/6) / 3.6 \times 10^{-6} = 1{,}560 / 3.6 \times 10^{-6} = 433.3 \times 10^6 =$ **433.3 Mbps** per stream (the standard specifies 433.3 Mbps for MCS 9, short GI).
(b) Total data rate $= 4 \times 433.3 =$ **1,733 Mbps** (1.73 Gbps).

This matches the 802.11ac Wave 2 maximum for 80 MHz, 4 spatial streams.
(c) A SISO system with the same bandwidth and modulation achieves 433.3 Mbps.
The MIMO gain is **4×** — a linear scaling with the number of spatial streams, consistent with the theoretical prediction for full-rank channels.
(d) For 256-QAM at BER = $10^{-5}$, the required $E_b/N_0 \approx 22.5$ dB.
Per-stream SNR = $E_b/N_0 + 10\log_{10}(\text{bits/symbol}) - 10\log_{10}(\text{coding gain}) \approx 22.5 + 9.0 - 0.8 =$ **30.7 dB**.
This high SNR requirement limits 256-QAM operation to short-range, high-SNR scenarios — at greater distances, the system adaptively falls back to lower-order modulation (64-QAM, 16-QAM).

## 2.4 Channel Coding and Error Correction

Channel coding adds structured redundancy to transmitted data so that the receiver can detect and correct errors introduced by noise, interference, and fading in the communication channel. Shannon's channel coding theorem proves that error-free communication is theoretically possible at any rate below the channel capacity, provided sufficiently long and sophisticated codes are used. In practice, channel codes are classified into block codes (which process data in fixed-length blocks) and convolutional codes (which process data as a continuous stream). The key metric for any code is its coding gain — the reduction in required $E_b/N_0$ to achieve a given BER compared to uncoded transmission.

### 2.4.1 Error Detection

Error detection codes add check bits that allow the receiver to determine whether errors have occurred during transmission, without necessarily correcting them. The simplest scheme is single parity, where one bit is appended to each data word such that the total number of 1s is even (even parity) or odd (odd parity). Single parity detects all odd numbers of bit errors but misses even numbers of errors, giving an undetected error probability that drops rapidly as the number of errors increases for random error channels.

The Cyclic Redundancy Check (CRC) is a far more powerful error detection code based on polynomial division in GF(2). The transmitter divides the data polynomial by a generator polynomial and appends the remainder as the CRC check value. Standard CRC polynomials include CRC-8 (8-bit), CRC-16 (16-bit), and CRC-32 (32-bit, used in Ethernet and ZIP). A CRC of degree r detects all burst errors of length $\leq r$, all odd numbers of errors (if the generator has an even number of terms), and all double-bit errors (if the generator is a primitive polynomial). The CRC-32 used in Ethernet provides a Hamming distance of 4 for frames up to 12,144 bits, guaranteeing detection of all 1, 2, and 3-bit errors within a frame.

**Example 2.4.1:** An Ethernet frame carries 1,500 bytes of payload data protected by a CRC-32 checksum. If the raw bit error rate of the channel is $10^{-8}$, determine (a) the probability that a frame contains at least one bit error, (b) the probability of an undetected error (assuming the CRC-32 misses errors with probability $2^{-32}$), and (c) the expected time between undetected errors at 1 Gbps.
**Solution:**
(a) Total bits per frame (including headers): approximately 1,538 bytes × 8 = 12,304 bits.
P(one or more errors) = $1 - (1 - 10^{-8})^{12'304} \approx 12{,}304 \times 10^{-8} =$ $\mathbf{1.23 \times 10^{-4}}$ (about 1 in 8,127

frames)
(b) The CRC-32 fails to detect an error pattern with probability approximately $2^{-32} = 2.33 \times 10^{-10}$.
P(undetected error per frame) = P(error) × P(CRC miss) = $1.23 \times 10^{-4} \times 2.33 \times 10^{-10}$ = $\mathbf{2.87 \times 10^{-14}}$
(c) At 1 Gbps with 12,304-bit frames: frame rate = $10^9/12{,}304 \approx 81{,}274$ frames/s.
Mean time between undetected errors = $1 / (81{,}274 \times 2.87 \times 10^{-14}) = 4.29 \times 10^8$ s ≈ **13.6 years**

### 2.4.2 Forward Error Correction (FEC)

Forward Error Correction (FEC) codes enable the receiver to both detect and correct errors without requesting retransmission, which is essential for one-way links (broadcast, deep space) and delay-sensitive applications. A block code (n, k) encodes k data bits into an n-bit codeword by adding n − k parity bits, with code rate $R_c = k/n$. The error-correcting capability depends on the minimum Hamming distance $d_{min}$ of the code: it can correct up to $t = \lfloor (d_{min} - 1)/2 \rfloor$ errors. Hamming codes are the simplest FEC with $d_{min} = 3$ and $t = 1$; the (7,4) Hamming code encodes 4 data bits into 7 bits with a code rate of 0.571.

Reed-Solomon (RS) codes operate on multi-bit symbols rather than individual bits, making them especially effective against burst errors. An RS(n, k) code over $GF(2^m)$ uses m-bit symbols and can correct up to $t = (n - k)/2$ symbol errors regardless of how many bits within each symbol are corrupted. RS(255, 223) with 8-bit symbols is widely used in deep-space communications (CCSDS/NASA standard), satellite data links, and some optical storage systems. (CDs use CIRC with RS(32,28)/RS(28,24) inner/outer codes; DVDs use RS-PC with different parameters.) Convolutional codes process data as a continuous stream through a shift register of constraint length K, producing output bits based on the current input and K − 1 previous inputs. Turbo codes (parallel concatenated convolutional codes) and Low-Density Parity-Check (LDPC) codes approach within fractions of a dB of the Shannon limit and are used in 4G/5G cellular, DVB-S2 satellite TV, and Wi-Fi 6.

**Example 2.4.2:** A deep-space communication link uses a rate-1/2 convolutional code with constraint length K = 7 (the NASA standard) concatenated with an RS(255, 223) outer code. For a channel BER of $10^{-2}$, determine (a) the inner code BER after Viterbi decoding (approximately $10^{-4}$ at this input BER), (b) the symbol error rate into the RS decoder, (c) the number of correctable symbol errors per RS block, and (d) the overall output BER.
**Solution:**
(a) The rate-1/2, K=7 convolutional code with Viterbi decoding achieves approximately $BER_{inner} \approx \mathbf{10^{-4}}$ at an input BER of $10^{-2}$ (a coding gain of about 5.5 dB).
(b) RS(255, 223) uses 8-bit symbols. The probability of a symbol error given a bit error rate of $10^{-4}$:
$P_{symbol\ error} = 1 - (1 - 10^{-4})^8 \approx 8 \times 10^{-4} = \mathbf{8 \times 10^{-4}}$
(c) RS(255, 223) has t = (255 − 223)/2 = **16 correctable symbol errors** per 255-symbol block.
(d) The probability of block failure (more than 16 errors in 255 symbols) is negligibly small at a symbol error rate of $8 \times 10^{-4}$.
Using the binomial distribution, the expected number of symbol errors per block is $255 \times 8 \times 10^{-4} = 0.204$, far below the correction capacity of 16.
The output BER is effectively $\mathbf{< 10^{-12}}$, demonstrating the power of concatenated coding.

### 2.4.3 Interleaving and Burst Error Protection

Interleaving is a technique that rearranges the order of transmitted symbols so that burst errors (which corrupt consecutive bits) are spread across multiple codewords after de-interleaving at the receiver. Most FEC codes are designed to correct random errors, but real channels often produce bursty errors due to fading, impulse noise, or scratches on optical discs. A block interleaver writes data into a matrix row-by-row and reads it out column-by-column, so that a burst error affecting L consecutive bits is distributed across L different codewords, with at most one error per codeword if the interleaving depth $D \geq L$.

Convolutional interleavers are more memory-efficient than block interleavers and introduce less latency, making them preferred in streaming applications. The combination of interleaving with FEC is fundamental to virtually every modern digital communication system. For example, the CD audio system uses Cross-Interleaved Reed-Solomon Coding (CIRC), which combines two levels of RS coding with convolutional interleaving to correct burst errors up to 4,000 consecutive bits (approximately 2.5 mm of track) and conceal bursts up to 13,300 bits.

**Example 2.4.3:** A communication system uses a (15, 11) Hamming code that can correct single-bit errors per codeword. The channel produces burst errors of length up to 5 bits. Design a block interleaver to protect against these bursts and determine (a) the interleaving depth, (b) the interleaver matrix dimensions, and (c) the total latency in bits.
**Solution:**
(a) To spread a burst of $L = 5$ bits across separate codewords, the interleaving depth must be $D \geq L =$ **5 rows**.
(b) The interleaver matrix has D rows $\times$ n columns, where $n = 15$ (codeword length): **5 × 15 matrix** (75 bits total).
(c) The interleaver must fill the entire matrix before reading it out, so the latency $= D \times n = 5 \times 15 =$ **75 bits**.
At both transmitter and receiver, the total end-to-end interleaving latency is $2 \times 75 = 150$ bits.
A burst of 5 consecutive bits in the interleaved stream maps to 5 separate codewords after de-interleaving, placing at most 1 error per codeword — within the correction capability of the Hamming code.

## 2.5 Multiplexing

Multiplexing combines multiple independent signals or data streams onto a single shared transmission medium, allowing efficient use of expensive infrastructure such as fiber optic cables, radio spectrum, and satellite transponders. The fundamental resource being shared differs by technique: frequency, time, or code. Each multiplexing method is characterized by how it partitions the available channel capacity among users and how it handles contention and interference between users sharing the medium.

### 2.5.1 Frequency Division Multiplexing (FDM)

Frequency Division Multiplexing (FDM) assigns each signal a unique frequency band within the total available bandwidth, with guard bands between adjacent channels to prevent spectral overlap. Each user transmits continuously within its allocated band, and the receiver uses bandpass filters to separate the individual channels. Traditional analog FDM was the backbone of the telephone network, where 12 voice channels (each 4 kHz wide with 300 Hz guard bands) were combined into a 48 kHz group, 5 groups into a supergroup (240 kHz), and so on up to master groups carrying 3,600 channels.

Orthogonal Frequency Division Multiplexing (OFDM) is a modern digital variant that uses closely-spaced orthogonal subcarriers, eliminating the need for guard bands between channels. In OFDM, data is distributed across N narrowband subcarriers using the Inverse FFT, and the symbol duration is N times longer than the subcarrier spacing, which provides inherent robustness against multipath fading. OFDM is used in Wi-Fi (802.11a/g/n/ac/ax), 4G LTE, 5G NR, DVB-T digital television, and DOCSIS 3.1 cable systems.

**Example 2.5.1:** An OFDM system (similar to 802.11a Wi-Fi) uses a 20 MHz channel bandwidth with 64 subcarriers, of which 48 carry data, 4 are pilot tones, and 12 are null (guard and DC). The subcarrier spacing is 312.5 kHz and the guard interval (cyclic prefix) is 0.8 μs. Determine (a) the useful symbol duration, (b) the total symbol duration, (c) the raw data rate using 64-QAM (6 bits/subcarrier), and (d) the spectral efficiency.
**Solution:**
(a) Useful symbol duration: $T_{symbol} = 1 / \Delta f = 1 / 312{,}500 = \mathbf{3.2\ \mu s}$
(b) Total symbol duration: $T_{total} = T_{symbol} + T_{guard} = 3.2 + 0.8 = \mathbf{4.0\ \mu s}$
(c) Raw data rate: $R_b$ = (48 data subcarriers × 6 bits/subcarrier) / 4.0 μs = 288 / 4.0 × $10^{-6}$ = **72 Mbps**
(d) Spectral efficiency: $\eta = R_b / B = 72 / 20 =$ **3.6 bits/s/Hz**

### 2.5.2 Time Division Multiplexing (TDM)

Time Division Multiplexing (TDM) assigns each signal a recurring time slot within a repeating frame, so that multiple users take turns transmitting in rapid succession over a single shared channel. Unlike FDM, where each user occupies a narrow frequency band continuously, TDM gives each user the full channel bandwidth but only for a brief, periodically recurring interval — the switching between time slots happens so rapidly that each channel appears to have a continuous, dedicated connection. Synchronous TDM assigns fixed time slots to each channel in every frame regardless of whether that channel has data to send, which simplifies framing and synchronization but wastes capacity when channels are idle. Statistical TDM (also called asynchronous TDM or statistical multiplexing) addresses this inefficiency by dynamically allocating time slots only to channels that have data ready, requiring address headers on each slot to identify the source but achieving significantly higher utilization — typically 2 to 4 times the throughput of synchronous TDM for bursty traffic such as data networking. Frame synchronization is critical in synchronous TDM: the receiver must identify the start of each frame to correctly demultiplex the interleaved channels, which is accomplished through dedicated framing bits or synchronization patterns embedded in the frame structure.

The T-carrier hierarchy, developed by Bell Labs in the 1960s, is the foundational TDM system in North American telecommunications. A DS-0 channel carries one digitized voice call at 64 kbps (8-bit samples at 8 kHz). A DS-1 (T1) frame multiplexes 24 DS-0 channels plus one framing bit into 193 bits per frame, transmitted at 8,000 frames/s for a line rate of 1.544 Mbps. Higher levels aggregate T1s: DS-2 combines 4 DS-1s (6.312 Mbps), and DS-3 (T3) combines 28 DS-1s for 44.736 Mbps — the standard rate for interconnecting telephone switches and early internet backbone links. The European E-carrier hierarchy uses a similar structure but with 32 time slots per frame (30 voice channels plus signaling and synchronization), giving the E1 a rate of 2.048 Mbps. The SONET (Synchronous Optical Network) and SDH (Synchronous Digital Hierarchy) standards extended TDM into the optical domain with synchronous framing, built-in overhead for operations and management, and standardized rates from OC-1/STM-0 (51.84 Mbps) through OC-3 (155.52 Mbps), OC-48 (2.488 Gbps), OC-192 (9.953 Gbps), and OC-768 (39.813 Gbps). While Ethernet and IP-based packet switching have largely replaced TDM for data transport, TDM principles persist in cellular systems (GSM uses TDM within each FDMA channel), passive optical networks (GPON uses upstream TDM), and synchronization-critical applications.

**Example 2.5.2:** A SONET OC-48 link carries 48 STS-1 signals. Each STS-1 frame is 810 bytes (90 columns $\times$ 9 rows) transmitted at 8,000 frames/s. Determine (a) the STS-1 line rate, (b) the OC-48 line rate, (c) the payload capacity of one STS-1 (overhead is 27 bytes per frame), and (d) the maximum number of DS-1 signals that one OC-48 can carry.
**Solution:**
(a) STS-1 line rate: 810 bytes $\times$ 8 bits $\times$ 8,000 frames/s = **51.84 Mbps**
(b) OC-48 line rate: 48 $\times$ 51.84 = **2,488.32 Mbps** (approximately 2.5 Gbps)
(c) STS-1 payload: (810 − 27) bytes $\times$ 8 bits $\times$ 8,000 = 783 $\times$ 8 $\times$ 8,000 = **50.112 Mbps**
(d) Each DS-1 occupies one VT1.5 (1.728 Mbps container capacity, carrying a 1.544 Mbps DS-1 signal).
An STS-1 SPE carries 28 VT1.5s.
An OC-48 carries 48 STS-1s: 48 $\times$ 28 = **1,344 DS-1 signals**.

### 2.5.3 Code Division Multiple Access (CDMA)

Code Division Multiple Access (CDMA) allows multiple users to share the same frequency band and time slots simultaneously by assigning each user a unique pseudo-random spreading code. Each user's data is multiplied by its assigned code (a wideband pseudo-noise sequence), spreading the narrowband data signal across a much wider bandwidth. At the receiver, correlation with the same code despreads the desired signal while all other users' signals (spread with different codes) remain wideband and appear as noise. The processing gain $G_p = W/R$, where W is the spread bandwidth and R is the data rate, quantifies the interference rejection capability of the system.

The two main types of spreading codes are Walsh codes (perfectly orthogonal, used for downlink synchronous channels) and PN sequences (pseudo-orthogonal, used for uplink asynchronous channels). CDMA offers soft capacity limits, graceful degradation under overload, inherent frequency diversity, and resistance to narrowband interference — making it the basis for IS-95 and CDMA2000 (2G/3G cellular) and the underlying access method for GPS satellite navigation. The near-far problem, where a strong nearby transmitter overwhelms a weak distant one, requires precise power control (typically 800 adjustments per second in IS-95) to ensure all users arrive at the base station at approximately equal power levels.

**Example 2.5.3:** A CDMA2000 system operates in a 1.25 MHz band with a data rate of 9.6 kbps per voice user. Determine (a) the processing gain, (b) the processing gain in dB, (c) the maximum number of users per sector if the required $E_b/N_0$ is 7 dB and voice activity factor is 0.4, and (d) the total sector capacity if 3-sector cells are used.
**Solution:**
(a) Processing gain: $G_p$ = W / R = 1,250,000 / 9,600 = **130.2**
(b) Processing gain in dB: 10 $\log_{10}$(130.2) = **21.15 dB**

(c) For a single cell/sector, the approximate capacity is:
$N \approx 1 + (G_p / (E_b/N_0)) \times (1/v)$, where v = voice activity factor = 0.4.
$E_b/N_0 = 10^{7/10} = 5.01$.
$N \approx 1 + (130.2 / 5.01) \times (1/0.4) = 1 + 26.0 \times 2.5 = 1 + 65.0 =$ **66 users per sector**
(d) With 3 sectors per cell and a 3-sector reuse factor of about 2.55× (due to directional antennas):
Total $\approx 66 \times 2.55 \approx$ **168 users per cell**.

### 2.5.4 OFDMA and SC-FDMA

Orthogonal Frequency Division Multiple Access (OFDMA) extends OFDM from a single-user modulation technique to a multi-user access scheme by dynamically allocating subsets of subcarriers to different users based on their channel conditions and traffic demands. In each OFDM symbol interval, the base station assigns groups of contiguous or distributed subcarriers (called resource blocks in LTE or resource units in Wi-Fi 6) to individual users, enabling simultaneous transmission to multiple users within a single OFDM symbol. This fine-grained frequency-domain scheduling exploits frequency-selective fading: by assigning each user the subcarriers where their channel is strongest, OFDMA achieves multi-user diversity gain and significantly higher aggregate throughput than systems that allocate the entire bandwidth to one user at a time.

LTE downlink uses OFDMA with 15 kHz subcarrier spacing and resource blocks of 12 subcarriers × 7 OFDM symbols (0.5 ms slot). A 20 MHz LTE channel has 100 resource blocks (1,200 subcarriers), dynamically distributed among users each millisecond. 5G NR extends this with flexible numerology: subcarrier spacings of 15, 30, 60, 120, or 240 kHz scale the OFDM parameters for different frequency bands and latency requirements. Wi-Fi 6 (802.11ax) introduced OFDMA to Wi-Fi for the first time, using resource units as small as 26 subcarriers (2 MHz) in a 20 MHz channel, enabling simultaneous uplink and downlink transmissions to up to 9 users per 20 MHz channel.

Single Carrier Frequency Division Multiple Access (SC-FDMA) — used for the LTE uplink — applies a DFT precoding stage before the OFDM modulation, effectively converting the multicarrier OFDM signal into a single-carrier waveform in the frequency domain. The key advantage is a lower peak-to-average power ratio (PAPR) compared to OFDMA — typically 2–4 dB lower — which allows the mobile device's power amplifier to operate more efficiently, extending battery life and reducing heat. The trade-off is reduced scheduling flexibility (each user must occupy contiguous subcarriers) and slightly more complex equalization at the receiver.

**Example 2.5.4:** An LTE base station has a 10 MHz downlink channel using OFDMA with 15 kHz subcarrier spacing. The usable bandwidth contains 600 subcarriers grouped into 50 resource blocks (RBs) of 12 subcarriers each. Three users are scheduled: User A (high-throughput, good channel) gets 30 RBs with 64-QAM, 3/4 coding; User B (moderate) gets 15 RBs with 16-QAM, 1/2 coding; User C (cell-edge) gets 5 RBs with QPSK, 1/3 coding. Determine (a) the subcarrier count per user, (b) the data rate per user (14 OFDM symbols per 1 ms subframe, using 11 for data), and (c) the total cell throughput.
**Solution:**
(a) User A: $30 \times 12 = 360$ subcarriers.
User B: $15 \times 12 = 180$ subcarriers.
User C: $5 \times 12 = 60$ subcarriers.
Total: 600 subcarriers (all allocated).

(b) Bits per subcarrier per OFDM symbol:
User A = $6 \times 3/4 = 4.5$ bits; User B = $4 \times 1/2 = 2.0$ bits; User C = $2 \times 1/3 = 0.667$ bits.
Data rate = subcarriers × bits × data symbols / subframe time:
User A: $360 \times 4.5 \times 11 / 1 \times 10^{-3} = 17{,}820 / 10^{-3} =$ **17.82 Mbps**
User B: $180 \times 2.0 \times 11 / 10^{-3} = 3{,}960 / 10^{-3} =$ **3.96 Mbps**
User C: $60 \times 0.667 \times 11 / 10^{-3} = 440 / 10^{-3} =$ **0.44 Mbps**
(c) Total cell throughput: $17.82 + 3.96 + 0.44 =$ **22.22 Mbps**.
This illustrates how OFDMA flexibly partitions resources: User A with a strong channel gets high-order modulation and most resources, while the cell-edge User C receives robust low-rate service using a small allocation, all sharing the same 10 MHz channel simultaneously.

## 2.6 Information Theory

Information theory, founded by Claude Shannon in 1948, establishes the fundamental limits on reliable communication and data compression. It provides the theoretical framework for understanding how much data can be transmitted through a noisy channel and how efficiently a source can be represented. Information theory underpins the design of every modern communication system by establishing bounds that real codes and systems approach but can never exceed.

### 2.6.1 Shannon's Channel Capacity

Shannon's channel capacity theorem states that every communication channel has a maximum rate C (in bits per second) at which information can be transmitted with arbitrarily low error probability. For an additive white Gaussian noise (AWGN) channel with bandwidth B and signal-to-noise ratio SNR, the capacity is $C = B \times \log_2(1 + \text{SNR})$. This deceptively simple formula reveals the fundamental trade-off: capacity increases linearly with bandwidth but only logarithmically with power (SNR). Doubling the bandwidth approximately doubles the capacity, while doubling the transmit power provides only about 1 bit/s/Hz improvement.

The channel capacity represents an existence theorem — it proves that codes exist that can achieve rates up to C with vanishingly small error rates, but it does not prescribe how to construct them. For decades, practical codes operated 3-10 dB from the Shannon limit. Modern turbo codes (1993) and LDPC codes approach within 0.1 dB of the Shannon limit, nearly closing the gap between theory and practice. The spectral efficiency limit $C/B = \log_2(1 + \text{SNR})$ establishes an upper bound on bandwidth efficiency for any modulation and coding scheme.

**Example 2.6.1:** A communication channel has a bandwidth of 4 kHz (standard telephone line) and an SNR of 30 dB. Determine (a) the Shannon capacity, (b) the maximum spectral efficiency, (c) the number of signaling levels needed per symbol to approach capacity (assuming Nyquist signaling at 2B symbols/s), and (d) compare to the actual V.34 modem rate of 33.6 kbps.
**Solution:**
(a) SNR in linear: $10^{30/10} = 1{,}000$.
Capacity: $C = 4{,}000 \times \log_2(1 + 1{,}000) = 4{,}000 \times \log_2(1{,}001) = 4{,}000 \times 9.968 =$ **39.87 kbps**
(b) Maximum spectral efficiency: $C/B = \log_2(1{,}001) =$ **9.97 bits/s/Hz**
(c) At Nyquist symbol rate of $2B = 8{,}000$ symbols/s: bits per symbol needed = $39{,}870 / 8{,}000 = 4.98$, requiring $M = 2^5 = 32$ total constellation points, equivalent to roughly **32-QAM** (between

16-QAM and 64-QAM).
(d) The V.34 modem achieves 33.6 kbps, which is 33.6/39.87 = **84.3%** of the Shannon limit — a testament to the sophisticated trellis-coded modulation used in V.34.

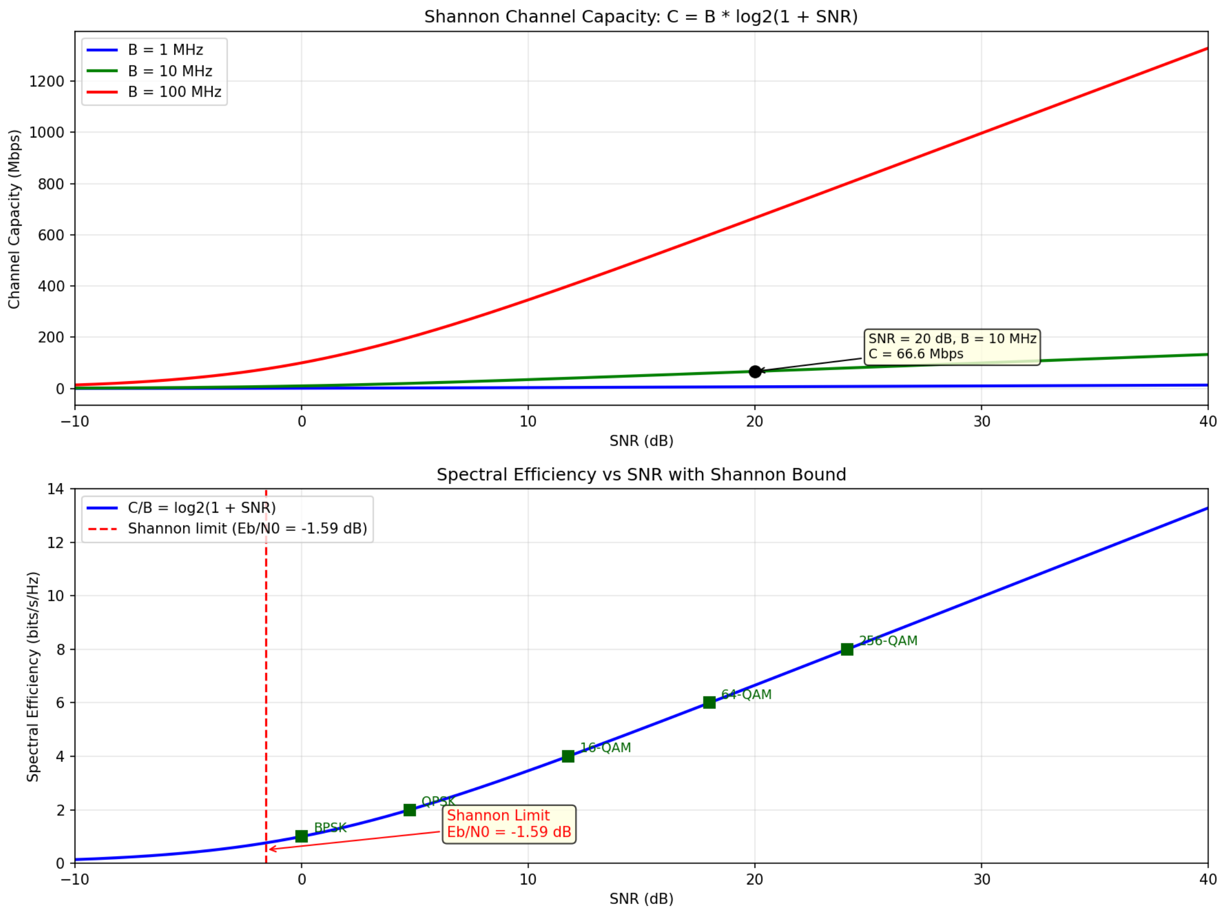



Figure 2.6.1: Shannon Channel Capacity vs SNR

### 2.6.2 Entropy and Source Coding

Entropy H(X) measures the average information content (in bits) per symbol emitted by a discrete source. For a source with M symbols having probabilities $p_1, p_2, \ldots, p_M$, the entropy is $H(X) = -\Sigma\, p_i \times \log_2(p_i)$. Entropy is maximized at $\log_2(M)$ bits/symbol when all symbols are equally likely, and it decreases as the source becomes more predictable (some symbols more probable than others). Shannon's source coding theorem establishes that the entropy H(X) is the minimum average number of bits needed to represent each source symbol without loss — no lossless compression scheme can achieve an average code length below H(X).

Huffman coding constructs an optimal prefix-free code by assigning shorter codewords to more frequent symbols. It achieves an average code length within 1 bit of the entropy. Arithmetic coding can approach the entropy even more closely by encoding entire sequences rather than individual symbols. In practice, modern lossless compression algorithms (LZ77/LZ78, DEFLATE used in ZIP/gzip) exploit both statistical redundancy and structural patterns, routinely compressing English text to about

1-2 bits per character compared to the 8 bits of ASCII. Lossy source coding (e.g., JPEG, MP3, AAC) permits controlled distortion to achieve much higher compression ratios, governed by rate-distortion theory.

**Example 2.6.2:** A binary source emits 0s and 1s with probabilities $P(0) = 0.9$ and $P(1) = 0.1$. Determine (a) the entropy of the source, (b) the maximum lossless compression ratio compared to fixed-length coding, (c) the average Huffman code length (using the trivial assignment: 0→"0", 1→"1"), and (d) the compression achievable by run-length encoding with an average run of 10 zeros between each 1.
**Solution:**
(a) Entropy: $H(X) = -0.9 \times \log_2(0.9) - 0.1 \times \log_2(0.1)$
$H(X) = -0.9 \times (-0.152) - 0.1 \times (-3.322) = 0.137 + 0.332 =$ **0.469 bits/symbol**
(b) Fixed-length coding uses 1 bit/symbol. Compression ratio: 1 / 0.469 = **2.13:1** (maximum theoretical)
(c) For a binary source with unequal probabilities, the trivial Huffman code assigns 1 bit to each symbol, so the average length = 1 bit/symbol — no compression.
To exploit the redundancy, we need to code blocks of symbols: for blocks of $n = 4$, there are 16 possible 4-bit blocks with varying probabilities, and Huffman coding of these blocks achieves closer to $H(X) = 0.469$ bits/source-symbol.
(d) With an average run length of 10 zeros between 1s, we can encode each run using $\lceil \log_2(\text{max run}) \rceil$ bits. For runs averaging 10 symbols, we encode approximately 11 source symbols (10 zeros + one 1) using about 4 bits (for the run length), giving **0.36 bits/symbol** — approaching the entropy limit.

## 2.7 Noise in Communication Systems

Noise is the fundamental limiting factor in all communication systems, establishing a floor below which signals cannot be reliably detected. Understanding noise sources and their propagation through receiver chains is essential for designing systems that achieve the required signal quality. The two key concepts are noise power (how much noise a component or channel introduces) and noise figure (how much a device degrades the signal-to-noise ratio).

### 2.7.1 Thermal Noise and Noise Power

Thermal noise (Johnson-Nyquist noise) is generated by the random motion of charge carriers in any resistive component at a temperature above absolute zero. The available noise power from a resistor at temperature T over a bandwidth B is $N = kTB$, where $k = 1.381 \times 10^{-23}$ J/K is Boltzmann's constant. At the standard reference temperature $T_0 = 290$ K (approximately 17°C), the noise power spectral density is $N_0 = kT_0 = 4.00 \times 10^{-21}$ W/Hz $= -174$ dBm/Hz. This value is a fundamental constant of communications engineering — every receiver operating at room temperature faces at least −174 dBm/Hz of thermal noise.

The open-circuit RMS noise voltage across a resistor R is $V_n = \sqrt{(4kTRB)}$, which shows that noise voltage increases with both resistance and bandwidth. When this resistor drives a matched load (maximum power transfer), the delivered noise power is kTB regardless of the resistance value — a remarkable result that means thermal noise power depends only on temperature and bandwidth. In practice, the total noise in a receiver includes contributions from antenna noise (sky temperature), component thermal noise, and active device noise, all of which are combined using equivalent noise temperature concepts.

**Example 2.7.1:** A 50 Ω receiver front-end has a bandwidth of 10 MHz and operates at T = 290 K. Determine (a) the thermal noise power in watts and dBm, (b) the RMS noise voltage across the 50 Ω input, and (c) the minimum detectable signal if the receiver requires SNR = 10 dB.
**Solution:**
(a) Noise power: $N = kTB = 1.381 \times 10^{-23} \times 290 \times 10 \times 10^6 = \mathbf{4.00 \times 10^{-14}\ W}$
In dBm: $N = 10 \log_{10}(4.00 \times 10^{-14} / 10^{-3}) = 10 \log_{10}(4.00 \times 10^{-11}) = 10 \times (-10.40) = \mathbf{-104.0\ dBm}$
This matches the quick calculation: $-174$ dBm/Hz + $10 \log_{10}(10^7) = -174 + 70 = -104$ dBm.
(b) RMS noise voltage: $V_n = \sqrt{(4kTRB)} = \sqrt{(4 \times 1.381 \times 10^{-23} \times 290 \times 50 \times 10^7)} = \sqrt{(8.01 \times 10^{-12})} = \mathbf{2.83\ \mu V}$
(c) Minimum detectable signal = $N + SNR_{required} = -104.0 + 10 = \mathbf{-94.0\ dBm}$ (or 0.40 pW).
Any signal weaker than this will not achieve the required 10 dB SNR.

### 2.7.2 Noise Figure and Cascaded Systems

The noise figure (NF) of a device quantifies how much it degrades the signal-to-noise ratio, defined as $F = SNR_{in} / SNR_{out}$ (linear ratio) or $NF = 10 \log_{10}(F)$ in dB. An ideal noiseless device has F = 1 (NF = 0 dB), while real amplifiers typically have NF between 0.5 dB (cryogenic LNAs) and 10 dB (mixers). The noise figure can also be expressed through the equivalent noise temperature: $T_e = T_0(F - 1)$, which represents the temperature of a matched resistor at the input that would produce the same amount of added noise as the device itself.

For a cascade of stages, the Friis formula gives the overall noise figure: $F_{total} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2) + \ldots$, where $F_n$ and $G_n$ are the noise figure and available gain of the n-th stage. This formula reveals a critical design principle: the first stage dominates the system noise performance because subsequent stages' noise contributions are divided by the cumulative gain of all preceding stages. This is why a low-noise amplifier (LNA) with high gain is placed at the very front of a receiver chain — its noise figure becomes essentially the system noise figure, and its gain suppresses the noise contributions of the noisier stages that follow.

**Example 2.7.2:** A satellite receiver front-end consists of an LNA (NF = 1.5 dB, gain = 20 dB) followed by a cable (loss = 3 dB) and a mixer (NF = 8 dB). Determine (a) the linear noise figures and gains, (b) the overall system noise figure, (c) the system noise temperature, and (d) the system noise figure if the cable and LNA are swapped (cable first).
**Solution:**
(a) Converting from dB to linear:
LNA: $F_1 = 10^{1.5/10} = 1.413$, $G_1 = 10^{20/10} = 100$
Cable: $F_2 = 10^{3/10} = 2.0$, $G_2 = 10^{-3/10} = 0.5$ (a 3 dB loss has NF = 3 dB and gain = $-3$ dB)
Mixer: $F_3 = 10^{8/10} = 6.31$
(b) Friis formula: $F_{total} = 1.413 + (2.0 - 1)/100 + (6.31 - 1)/(100 \times 0.5)$
$F_{total} = 1.413 + 0.010 + 0.106 = 1.529$
$NF_{total} = 10 \log_{10}(1.529) = \mathbf{1.84\ dB}$
(c) System noise temperature: $T_e = 290 \times (1.529 - 1) = 290 \times 0.529 = \mathbf{153.4\ K}$
(d) If the cable is placed before the LNA: $F_{total} = 2.0 + (1.413 - 1)/0.5 + (6.31 - 1)/(0.5 \times 100) =$

$2.0 + 0.826 + 0.106 = 2.932$, giving $NF = 10\log_{10}(2.932) = \mathbf{4.67\ dB}$.
Moving the lossy cable ahead of the LNA nearly triples the system noise figure (1.84 → 4.67 dB), demonstrating why the LNA must be the first element in the chain.

### 2.7.3 Satellite Link Budget and G/T

Satellite communication links require specialized link budget analysis because of the extreme path distances (36,000 km for geostationary orbit), limited satellite power, and the need to share transponder capacity among many users. The satellite link has two segments: the uplink (earth station to satellite) and the downlink (satellite to earth station), each with its own power budget, path loss, and noise characteristics. The key figure of merit for a satellite receive system is the G/T ratio — the receive antenna gain divided by the system noise temperature, expressed in dB/K — which quantifies the receiver's sensitivity independent of bandwidth.

The carrier-to-noise ratio for a satellite link is: $C/N = EIRP + G/T - L_{path} - k - 10\log_{10}(B)$, where EIRP is the effective isotropic radiated power of the transmitter (dBW), G/T is the receiver figure of merit (dB/K), $L_{path}$ is the free-space path loss (dB), $k = -228.6$ dBW/(K·Hz) is Boltzmann's constant, and B is the noise bandwidth (Hz). For a complete link through a bent-pipe transponder (which amplifies and retransmits without demodulation), the overall C/N is: $1/(C/N)_{total} = 1/(C/N)_{up} + 1/(C/N)_{down} + 1/(C/N)_{intermod}$. Rain attenuation is a critical factor at Ku-band and Ka-band, with typical rain margins of 3–6 dB at Ku-band and 6–12 dB at Ka-band for 99.5–99.9% link availability. The link availability target determines the rain attenuation statistics used, derived from ITU-R rain zone data for the earth station location.

**Example 2.7.3:** A Ku-band satellite TV downlink operates at 12.5 GHz from a geostationary satellite at 36,000 km range. The satellite transponder has EIRP = 52 dBW. The home receiver uses a 60 cm offset dish with 65% aperture efficiency and a system noise temperature of 150 K (including LNB noise). Determine (a) the receive antenna gain, (b) the G/T of the receiver, (c) the free-space path loss, (d) the C/N for a 27 MHz transponder bandwidth, and (e) the rain margin available if the minimum C/N for DVB-S2 QPSK 3/4 is 5.5 dB.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 12.5 \times 10^9 = 0.024$ m.
$G_{rx} = \eta(\pi D/\lambda)^2 = 0.65 \times (\pi \times 0.6/0.024)^2 = 0.65 \times (78.54)^2 = 0.65 \times 6{,}168 = 4{,}009$
$G_{rx}(\text{dBi}) = 10\log_{10}(4{,}009) = \mathbf{36.0\ dBi}$
(b) $G/T = G_{rx} - 10\log_{10}(T_{sys}) = 36.0 - 10\log_{10}(150) = 36.0 - 21.8 = \mathbf{14.2\ dB/K}$
(c) $FSPL = 20\log_{10}(4\pi d/\lambda) = 20\log_{10}(4\pi \times 3.6 \times 10^7/0.024) = 20\log_{10}(1.885 \times 10^{10}) = 20 \times 10.275 = \mathbf{205.5\ dB}$
(d) $C/N = EIRP + G/T - FSPL - k - 10\log_{10}(B)$
$C/N = 52 + 14.2 - 205.5 - (-228.6) - 10\log_{10}(27 \times 10^6)$
$C/N = 52 + 14.2 - 205.5 + 228.6 - 74.3 = \mathbf{15.0\ dB}$
(e) Rain margin $= C/N - C/N_{required} = 15.0 - 5.5 = \mathbf{9.5\ dB}$.
This generous margin accommodates heavy rain events at Ku-band (typical rain attenuation of 3–6 dB for 99.7% availability in temperate climates), ensuring reliable satellite TV reception in most weather conditions.
During extreme downpours exceeding 9.5 dB of rain attenuation, the receiver would lose lock temporarily — the well-known "rain fade" effect.

## 2.8 Link Budget Analysis

A link budget is a systematic accounting of all gains and losses in a communication link from transmitter to receiver, used to determine whether the received signal meets the minimum required level for reliable communication. Link budgets are essential for designing any communication system — wireless, satellite, fiber optic, or wired — and they directly connect the noise and modulation concepts from earlier sections into a complete system-level analysis.

### 2.8.1 Link Budget Fundamentals

The fundamental link budget equation expresses the received power as: $P_{rx} = P_{tx} + G_{tx} - L_{path} + G_{rx} - L_{misc}$ (all in dB), where $P_{tx}$ is the transmit power, $G_{tx}$ and $G_{rx}$ are the transmit and receive antenna gains, $L_{path}$ is the path loss, and $L_{misc}$ accounts for additional losses such as cable losses, atmospheric absorption, and fading margins. For free-space propagation (no obstructions or reflections), the path loss is given by the Friis transmission equation: $\text{FSPL} = 20 \log_{10}(4\pi d/\lambda)$ dB, where d is the distance and $\lambda = c/f$ is the wavelength. At 1 GHz and 1 km, the free-space path loss is approximately 92.4 dB, and it increases by 6 dB for every doubling of distance or frequency.

The system margin is the difference between the actual received power and the receiver sensitivity (the minimum power for acceptable performance): margin $= P_{rx} - P_{sensitivity}$. Receiver sensitivity depends on the noise floor (kTB), the receiver noise figure, and the required SNR (or $E_b/N_0$) for the chosen modulation and coding scheme. A positive margin indicates a working link; a negative margin means the link will fail. Practical designs include a fade margin (typically 10–30 dB depending on reliability requirements) to account for statistical variations in path loss due to multipath fading, rain attenuation, or other time-varying impairments.

**Example 2.8.1:** A 2.4 GHz Wi-Fi access point transmits at $P_{tx} = 20$ dBm with an antenna gain of $G_{tx} = 3$ dBi. The client device has an antenna gain of $G_{rx} = 0$ dBi and a receiver sensitivity of −70 dBm for the 54 Mbps data rate (64-QAM, 3/4 coding). Determine (a) the free-space path loss at 100 m, (b) the received power, (c) the link margin, and (d) the maximum range with a 10 dB fade margin.
**Solution:**
(a) Free-space path loss at 2.4 GHz, 100 m:
$\lambda = c/f = 3 \times 10^8 / 2.4 \times 10^9 = 0.125$ m
$\text{FSPL} = 20 \log_{10}(4\pi \times 100 / 0.125) = 20 \log_{10}(10{,}053) = 20 \times 4.002 =$ **80.0 dB**
(b) Received power: $P_{rx} = 20 + 3 - 80.0 + 0 - 0 =$ **−57.0 dBm**
(c) Link margin: margin $= P_{rx} - P_{sensitivity} = -57.0 - (-70) =$ **13.0 dB**
(d) With a 10 dB fade margin, the allowable path loss $= P_{tx} + G_{tx} + G_{rx} - P_{sensitivity} -$ fade margin $= 20 + 3 + 0 - (-70) - 10 = 83.0$ dB.
Since FSPL = 80.0 dB at 100 m, the additional 3 dB allows $10^{3/20} = 1.41\times$ more distance: $d_{max} = 100 \times 1.41 =$ **141 m**.
In practice, indoor environments add 10–20 dB of wall and obstruction losses, reducing the effective range to 30–50 m at the 54 Mbps rate.

# Chapter 3

# Semiconductors

Semiconductors are materials with electrical conductivity between that of a conductor and an insulator, and they form the foundation of nearly all modern electronic devices. By introducing controlled impurities (doping) into semiconductor crystals, engineers create regions of excess electrons (N-type) and excess holes (P-type) that enable the construction of diodes, transistors, and integrated circuits. The ability to precisely control current flow through semiconductor junctions has made possible everything from simple rectifiers and voltage regulators to microprocessors containing billions of transistors.

## 3.1 Semiconductor Fundamentals

### 3.1.1 Energy Bands and Bandgap

In a crystalline solid, the discrete energy levels of individual atoms merge into continuous energy bands due to the close proximity of atoms in the lattice. The two most important bands are the valence band (the highest energy band that is fully occupied by electrons at absolute zero) and the conduction band (the next higher band, which is empty at absolute zero). The energy gap between them — the bandgap ($E_g$) — determines whether a material is a conductor (no gap or overlapping bands), a semiconductor (small gap, 0.5-3.5 eV), or an insulator (large gap, > 4 eV). At temperatures above absolute zero, thermal energy promotes some electrons from the valence band into the conduction band, creating mobile electrons in the conduction band and mobile holes (vacant electron states) in the valence band. The intrinsic carrier concentration $n_i$ depends exponentially on bandgap and temperature: $n_i = \sqrt{(N_C \times N_V)} \times e^{-Eg/(2kT)}$, where $N_C$ and $N_V$ are the effective density of states in the conduction and valence bands, k is Boltzmann's constant ($8.617 \times 10^{-5}$ eV/K), and T is absolute temperature.

| Material | Bandgap $E_g$ (eV) | $n_i$ at 300 K ($cm^{-3}$) | Type |
|---|---|---|---|
| Germanium (Ge) | 0.66 | $2.4 \times 10^{13}$ | Indirect |
| Silicon (Si) | 1.12 | $1.5 \times 10^{10}$ | Indirect |
| Gallium Arsenide (GaAs) | 1.42 | $1.8 \times 10^{6}$ | Direct |
| Silicon Carbide (SiC) | 3.26 | $\sim 10^{-9}$ | Indirect |
| Gallium Nitride (GaN) | 3.40 | $\sim 10^{-10}$ | Direct |

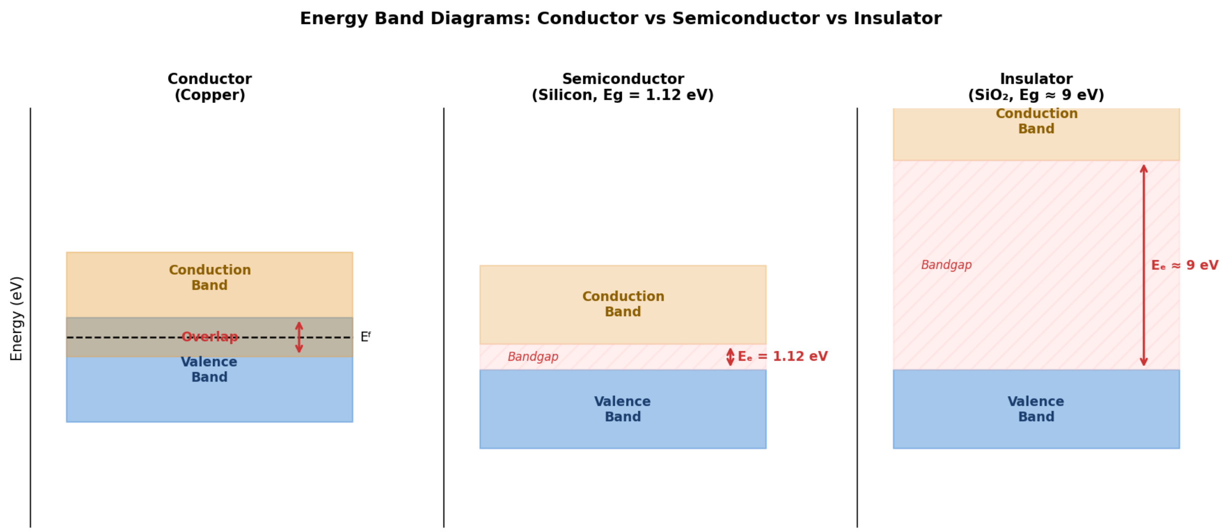



Figure 3.1.1: Energy Band Diagrams: Conductor vs Semiconductor vs Insulator

**Example 3.1.1:** The intrinsic carrier concentration of a semiconductor can be approximated as $n_i \approx B \times T^{3/2} \times e^{-Eg/(2kT)}$, where B is a material constant. For silicon ($E_g = 1.12$ eV) at 300 K, $n_i = 1.5 \times 10^{10}$ cm$^{-3}$. Estimate $n_i$ for silicon at 400 K (elevated operating temperature).
**Solution:**
Taking the ratio at two temperatures (B cancels):
$n_i(T_2)/n_i(T_1) = (T_2/T_1)^{3/2} \times e^{-Eg/(2k) \times (1/T_2 - 1/T_1)}$
$(T_2/T_1)^{3/2} = (400/300)^{3/2} = (1.333)^{3/2} = 1.540$
Exponent: $-E_g/(2k) \times (1/T_2 - 1/T_1) = -1.12/(2 \times 8.617 \times 10^{-5}) \times (1/400 - 1/300)$
$= -6{,}499 \times (0.00250 - 0.00333) = -6{,}499 \times (-8.33 \times 10^{-4}) = 5.416$
$e^{5.416} = 224.9$
$n_i(400\text{ K}) = 1.5 \times 10^{10} \times 1.540 \times 224.9 = \mathbf{5.20 \times 10^{12}\ cm^{-3}}$
The intrinsic carrier concentration increases by a factor of ~346 for a 100 K temperature rise, illustrating why semiconductor device characteristics are strongly temperature dependent.

### 3.1.2 Doping (N-Type and P-Type)

Doping introduces controlled impurity atoms into the semiconductor crystal to dramatically increase the concentration of one carrier type. N-type doping uses donor atoms from Group V of the periodic table (phosphorus, arsenic, antimony for silicon) — each donor has five valence electrons, four of which bond with neighboring silicon atoms while the fifth is loosely bound and easily ionized into the conduction band. P-type doping uses acceptor atoms from Group III (boron, gallium, indium for silicon) — each acceptor has only three valence electrons, creating a hole in the bonding structure that acts as a mobile positive charge carrier. At room temperature, virtually all dopant atoms are ionized, so the majority carrier concentration equals the doping concentration: $n \approx N_D$ for N-type and $p \approx N_A$ for P-type. The minority carrier concentration is determined by the mass action law: $n \times p = n_i^2$, which holds in thermal equilibrium regardless of doping.

**Example 3.1.2:** A silicon wafer is doped with phosphorus at $N_D = 2 \times 10^{16}$ cm$^{-3}$. The electron mobility in this material is $\mu_n = 1{,}250$ cm$^2$/(V·s). Determine (a) the majority and minority carrier concentrations, (b) the resistivity, and (c) the conductivity. Use $n_i = 1.5 \times 10^{10}$ cm$^{-3}$ and $q = 1.602 \times 10^{-19}$ C.
**Solution:**
(a) Majority carriers (electrons): $n \approx N_D =$ **$2 \times 10^{16}$ cm$^{-3}$**
Minority carriers (holes): $p = n_i^2 / n = (1.5 \times 10^{10})^2 / (2 \times 10^{16}) = 2.25 \times 10^{20} / 2 \times 10^{16} =$ **$1.125 \times 10^4$ cm$^{-3}$**
(b) Conductivity (dominated by majority carriers):
$\sigma = q \times n \times \mu_n = 1.602 \times 10^{-19} \times 2 \times 10^{16} \times 1{,}250 =$ **4.006 (Ω·cm)$^{-1}$**
Resistivity: $\rho = 1/\sigma = 1/4.006 =$ **0.250 Ω·cm**

### 3.1.3 Drift and Diffusion

Current flow in semiconductors occurs by two mechanisms: drift and diffusion. Drift current results from the motion of charge carriers under the influence of an electric field — carriers accelerate in the field direction, collide with lattice atoms, and establish an average drift velocity $v_d = \mu \times E$, where $\mu$ is the carrier mobility. The drift current density is $J_{drift} = q \times (n \times \mu_n + p \times \mu_p) \times E$, where the electron and hole components add because electrons move opposite to the field but carry negative charge. Diffusion current results from carrier concentration gradients — carriers naturally move from regions of high concentration to regions of low concentration, similar to gas diffusion. The diffusion current density is $J_{diff} = q \times D_n \times (dn/dx) - q \times D_p \times (dp/dx)$, where D is the diffusion coefficient related to mobility by the Einstein relation: $D = (kT/q) \times \mu$. At room temperature, $kT/q = 0.02585$ V (the thermal voltage $V_T$).

**Example 3.1.3:** A silicon bar doped at $N_D = 10^{16}$ cm$^{-3}$ is 2 mm long with 5 V applied across it. The electron mobility is $\mu_n = 1{,}300$ cm$^2$/(V·s). Determine (a) the electric field, (b) the electron drift velocity, (c) the drift current density, and (d) the electron diffusion coefficient.
**Solution:**
(a) Electric field: $E = V/L = 5/0.2 =$ **25 V/cm**
(b) Drift velocity: $v_d = \mu_n \times E = 1{,}300 \times 25 =$ **32,500 cm/s = 325 m/s**
(c) Drift current density (N-type, electrons dominate):
$J_{drift} = q \times n \times \mu_n \times E = 1.602 \times 10^{-19} \times 10^{16} \times 1{,}300 \times 25$
$= 1.602 \times 10^{-19} \times 3.25 \times 10^{20} =$ **52.1 A/cm$^2$**
(d) Diffusion coefficient (Einstein relation):
$D_n = V_T \times \mu_n = 0.02585 \times 1{,}300 =$ **33.6 cm$^2$/s**

## 3.2 Semiconductor Materials

The choice of semiconductor material determines the fundamental performance limits of a device — its operating frequency, breakdown voltage, thermal tolerance, and optical properties. Silicon dominates electronics due to its abundance, mature processing technology, and excellent native oxide, while III-V and wide-bandgap compounds such as GaAs, GaN, and SiC are chosen where silicon's electron mobility or bandgap is insufficient for the application.

### 3.2.1 Silicon (Si)

Silicon (Si) is a Group IV element with atomic number 14, four valence electrons, and a diamond cubic crystal structure in which each atom forms covalent bonds with four nearest neighbors in a tetrahedral arrangement. With an indirect bandgap of 1.12 eV at 300 K, silicon occupies an ideal middle ground — wide enough to keep intrinsic leakage low ($n_i \approx 1.5 \times 10^{10}$ cm$^{-3}$ at room temperature) yet narrow enough for practical doping and device operation at standard voltages. Silicon is the second most abundant element in the Earth's crust after oxygen, and its native oxide $SiO_2$ is a high-quality electrical insulator that can be thermally grown with an atomically sharp Si/$SiO_2$ interface — this fortuitous property enabled the MOSFET and the entire CMOS revolution, as no other semiconductor forms such an excellent natural gate dielectric. Doping silicon with Group V donors (phosphorus, arsenic, antimony) creates N-type material, while Group III acceptors (boron, gallium, indium) create P-type material, with doping concentrations spanning from $10^{14}$ to $10^{21}$ cm$^{-3}$ for different device regions. Silicon's electron mobility of approximately 1,350 cm$^2$/(V·s) and hole mobility of 480 cm$^2$/(V·s) are modest compared to III-V compounds, but its unmatched manufacturability, mechanical strength, and the mature ecosystem of fabrication processes make it the foundation of CMOS logic, power devices (MOSFETs, IGBTs, thyristors), photovoltaic solar cells, and MEMS sensors — accounting for over 95% of all semiconductor devices produced worldwide.

**Example 3.2.1:** Intrinsic silicon at T = 300 K has an intrinsic carrier concentration of $n_i = 1.5 \times 10^{10}$ cm$^{-3}$. If the silicon is doped with phosphorus (donor) at a concentration of $N_D = 5 \times 10^{16}$ cm$^{-3}$, determine the electron and hole concentrations, and the position of the Fermi level relative to the intrinsic Fermi level (use kT = 0.02585 eV at 300 K).
**Solution:**
Since $N_D \gg n_i$, the electron concentration is approximately:
$n \approx N_D = \mathbf{5 \times 10^{16}\ cm^{-3}}$
Hole concentration (from mass action law $n \times p = n_i^2$):
$p = n_i^2 / n = (1.5 \times 10^{10})^2 / (5 \times 10^{16}) = 2.25 \times 10^{20} / 5 \times 10^{16} = \mathbf{4.5 \times 10^3\ cm^{-3}}$
Fermi level position relative to intrinsic level:
$E_F - E_i = kT \times \ln(n / n_i) = 0.02585 \times \ln(5 \times 10^{16} / 1.5 \times 10^{10})$
$= 0.02585 \times \ln(3.33 \times 10^6) = 0.02585 \times 15.02 = \mathbf{0.388\ eV\ above\ E_i}$

### 3.2.2 Gallium Arsenide (GaAs)

Gallium Arsenide (GaAs) is a III-V compound semiconductor with a direct bandgap of 1.42 eV at 300 K — the direct gap means electrons can transition between the conduction and valence bands by emitting or absorbing a photon without requiring a phonon (lattice vibration) for momentum conservation, making GaAs far superior to silicon for light emission and optical detection. GaAs has an electron mobility of approximately 8,500 cm$^2$/(V·s) at low doping levels, roughly 6 times higher than silicon, which translates directly into faster transit times and higher operating frequencies for transistors. The combination of high mobility and semi-insulating substrate capability (resistivity > $10^7$ Ω·cm, achieved through chromium doping or the native EL2 deep-level defect) makes GaAs the material of choice for monolithic microwave integrated circuits (MMICs) operating from 1 GHz to over 100 GHz, including low-noise amplifiers, power amplifiers, and mixers for radar, satellite, and cellular base station applications. In optoelectronics, the 1.42 eV bandgap corresponds to emission at 870 nm (near-infrared), and the AlGaAs/GaAs material system enables laser diodes, LEDs, and high-efficiency multi-junction solar cells that achieve conversion efficiencies above 45% under concentrated sunlight — making GaAs the standard for space solar panels despite its higher cost. The principal limitations of GaAs are its higher material cost (gallium is far less abundant than silicon, and crystal growth from the melt requires high-pressure arsenic overpressure), its mechanical fragility compared to silicon, the absence of a high-quality native oxide (preventing straightforward MOS device fabrication), and lower thermal conductivity (0.46 W/(cm·K) vs 1.5 W/(cm·K) for silicon), which constrains power dissipation. These trade-offs confine GaAs to applications where its speed and optical properties justify the cost premium, while silicon dominates mainstream digital and power electronics.

**Example 3.2.2:** A GaAs MESFET operates with an electron mobility of $\mu_n$ = 8,500 cm$^2$/(V·s), compared to silicon's $\mu_n$ = 1,350 cm$^2$/(V·s). Both devices have a channel length of L = 0.5 μm. Estimate the transit time for electrons through the channel in each material at an electric field of E = 5 kV/cm, and determine the maximum operating frequency estimate ($f_T \approx 1 / (2\pi \times \tau_{transit})$).
**Solution:**
Electron drift velocity: $v_d = \mu_n \times E$
GaAs: $v_d = 8{,}500 \times 5{,}000 = 4.25 \times 10^7$ cm/s
Transit time: $\tau = L / v_d = 0.5 \times 10^{-4} / 4.25 \times 10^7 = 1.176 \times 10^{-12}$ s = 1.18 ps
$f_T = 1 / (2\pi \times 1.176 \times 10^{-12})$ = **135.2 GHz**
Silicon: $v_d = 1{,}350 \times 5{,}000 = 6.75 \times 10^6$ cm/s
Transit time: $\tau = 0.5 \times 10^{-4} / 6.75 \times 10^6 = 7.407 \times 10^{-12}$ s = 7.41 ps
$f_T = 1 / (2\pi \times 7.407 \times 10^{-12})$ = **21.5 GHz**
The GaAs device has a **6.3× higher** estimated operating frequency than silicon, demonstrating why GaAs is preferred for high-frequency RF applications.

### 3.2.3 Gallium Nitride (GaN)

Gallium Nitride is a wide bandgap semiconductor with a bandgap of approximately 3.4 eV, compared to 1.12 eV for silicon. This wide bandgap allows GaN devices to operate at higher voltages, temperatures, and switching frequencies with lower losses. GaN is widely used in power electronics (high-efficiency power converters and chargers), RF amplifiers (5G base stations, radar, and satellite communications), and optoelectronics (blue and white LEDs, laser diodes). Its high electron mobility, achieved through AlGaN/GaN heterostructures forming a two-dimensional electron gas (2DEG), makes it particularly well-suited for high-frequency High Electron Mobility Transistors (HEMTs).

**Example 3.2.3:** A GaN power transistor switching converter operates at 500 kHz with a drain-source voltage of 400 V and a load current of 10 A. The GaN device has switching losses characterized by $t_{rise}$ = 5 ns and $t_{fall}$ = 8 ns. An equivalent silicon MOSFET has $t_{rise}$ = 30 ns and $t_{fall}$ = 45 ns. Compare the switching power losses for both devices.
**Solution:**
Switching energy per cycle: $E_{sw} = 0.5 \times V_{DS} \times I_D \times (t_{rise} + t_{fall})$
GaN: $E_{sw} = 0.5 \times 400 \times 10 \times (5 \times 10^{-9} + 8 \times 10^{-9}) = 2{,}000 \times 13 \times 10^{-9} = 26$ μJ
Switching power loss: $P_{sw} = E_{sw} \times f_{sw} = 26 \times 10^{-6} \times 500{,}000$ = **13.0 W**
Silicon: $E_{sw} = 0.5 \times 400 \times 10 \times (30 \times 10^{-9} + 45 \times 10^{-9}) = 2{,}000 \times 75 \times 10^{-9} = 150$ μJ
Switching power loss: $P_{sw} = 150 \times 10^{-6} \times 500{,}000$ = **75.0 W**
The GaN device reduces switching losses by a factor of 75.0 / 13.0 = **5.8×**, demonstrating why GaN enables efficient high-frequency power conversion.

## 3.3 PN Junction

When P-type and N-type semiconductor regions are brought into contact, a PN junction is formed — the fundamental building block of diodes, BJTs, MOSFETs, solar cells, and LEDs. The diffusion of majority carriers across the junction creates a charge-depleted region with an internal electric field that opposes further diffusion, establishing an equilibrium condition.

### 3.3.1 Depletion Region and Built-In Potential

At the PN junction, electrons from the N-side diffuse into the P-side and holes from the P-side diffuse into the N-side, leaving behind fixed ionized donor atoms (positive) on the N-side and fixed ionized acceptor atoms (negative) on the P-side. This region of immobile charges, called the depletion region (or space charge region), creates an internal electric field directed from N to P that opposes further diffusion. The resulting built-in potential (also called contact potential or junction potential) is $V_{bi} = (kT/q) \times \ln(N_A \times N_D / n_i^2)$. The depletion width depends on doping concentrations and any externally applied voltage: $W = \sqrt{(2\varepsilon_s/q \times (1/N_A + 1/N_D) \times (V_{bi} - V_A))}$, where $\varepsilon_s$ is the semiconductor permittivity and $V_A$ is the applied voltage (positive for forward bias, negative for reverse bias).

**Example 3.3.1:** A silicon PN junction has $N_A = 5 \times 10^{17}$ cm$^{-3}$ and $N_D = 2 \times 10^{16}$ cm$^{-3}$. At T = 300 K, determine (a) the built-in potential and (b) the depletion width at zero bias. Use $n_i = 1.5 \times 10^{10}$ cm$^{-3}$, $\varepsilon_{Si} = 11.7 \times 8.854 \times 10^{-14}$ F/cm $= 1.036 \times 10^{-12}$ F/cm.
**Solution:**
(a) Built-in potential:
$V_{bi} = V_T \times \ln(N_A \times N_D / n_i^2) = 0.02585 \times \ln(5 \times 10^{17} \times 2 \times 10^{16} / (1.5 \times 10^{10})^2)$
$= 0.02585 \times \ln(10^{34} / 2.25 \times 10^{20}) = 0.02585 \times \ln(4.44 \times 10^{13}) = 0.02585 \times 31.42$
$=$ **0.812 V**
(b) Depletion width at $V_A = 0$:
$W = \sqrt{(2 \times 1.036 \times 10^{-12} / 1.602 \times 10^{-19} \times (1/5 \times 10^{17} + 1/2 \times 10^{16}) \times 0.812)}$
$1/N_A + 1/N_D = 2 \times 10^{-18} + 5 \times 10^{-17} = 5.2 \times 10^{-17}$ cm$^3$
$W = \sqrt{(2 \times 1.036 \times 10^{-12} \times 5.2 \times 10^{-17} \times 0.812 / 1.602 \times 10^{-19})}$
$= \sqrt{(8.748 \times 10^{-29} / 1.602 \times 10^{-19})} = \sqrt{(5.461 \times 10^{-10})} =$ **$2.34 \times 10^{-5}$ cm = 0.234 µm**
The depletion region extends mostly into the lightly doped N-side: $x_N \approx W \times N_A/(N_A + N_D) = 0.234 \times 0.962 = 0.225$ µm on the N-side versus only 0.009 µm on the P-side.

### 3.3.2 Forward and Reverse Bias

When a positive voltage is applied to the P-side relative to the N-side (forward bias), the external field opposes the internal field, reducing the depletion width and the potential barrier. This allows majority carriers to cross the junction, producing an exponentially increasing current described by the Shockley diode equation: $I = I_S \times (e^{V/(nVT)} - 1)$, where $I_S$ is the reverse saturation current (typically $10^{-12}$ to $10^{-15}$ A for silicon), n is the ideality factor (1 for an ideal junction, 1-2 in practice), and $V_T = kT/q \approx 25.85$ mV at 300 K. When a negative voltage is applied (reverse bias), the depletion region widens and only a very small leakage current (approximately $I_S$) flows. If the reverse voltage exceeds the breakdown voltage, the junction enters avalanche or Zener breakdown and current increases rapidly.

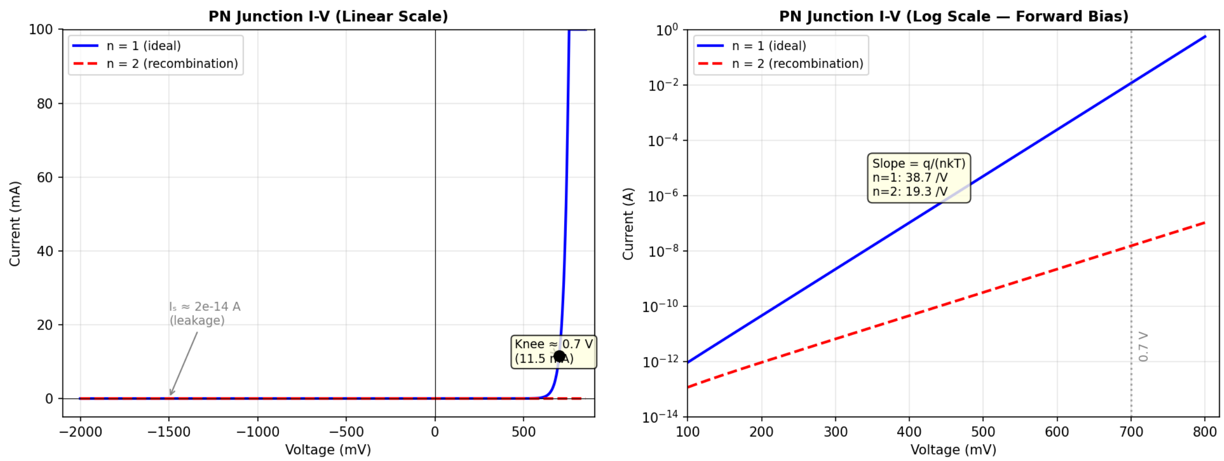



Figure 3.3.1: PN Junction I-V Characteristic

**Example 3.3.2:** A silicon diode has a reverse saturation current $I_S = 2 \times 10^{-14}$ A and an ideality factor n = 1.0 at T = 300 K. Determine the forward current at (a) V = 0.60 V and (b) V = 0.70 V, and (c) calculate the ratio of currents to illustrate the exponential nature of the I-V characteristic.
**Solution:**
$V_T = 0.02585$ V at 300 K.
(a) At V = 0.60 V:
$I = I_S \times (e^{V/VT} - 1) = 2 \times 10^{-14} \times (e^{0.60/0.02585} - 1)$
$= 2 \times 10^{-14} \times (e^{23.21} - 1) = 2 \times 10^{-14} \times 1.207 \times 10^{10} =$ **0.241 mA**
(b) At V = 0.70 V:
$I = 2 \times 10^{-14} \times (e^{0.70/0.02585} - 1) = 2 \times 10^{-14} \times (e^{27.08} - 1)$
$= 2 \times 10^{-14} \times 5.760 \times 10^{11} =$ **11.52 mA**
(c) Current ratio: 11.52/0.241 = **47.8×** — a mere 100 mV increase in forward voltage produces a ~48× increase in current, demonstrating the steep exponential characteristic.

### 3.3.3 Junction Capacitance

A PN junction exhibits two types of capacitance. The depletion (junction) capacitance $C_j$ arises from the charge stored in the depletion region and dominates under reverse bias: $C_j = C_{j0} / (1 - V_A/V_{bi})^m$, where $C_{j0}$ is the zero-bias capacitance, $V_A$ is the applied voltage (negative for reverse bias), and m is the grading coefficient (0.5 for an abrupt junction, 0.33 for a linearly graded junction). The diffusion capacitance $C_d$ arises from minority carrier charge stored near the junction during forward bias and is proportional to the forward current: $C_d = \tau_T \times I / V_T$, where $\tau_T$ is the transit time. Varactor (variable capacitance) diodes are specifically designed to exploit the voltage-dependent depletion capacitance for use in voltage-controlled oscillators (VCOs), tuning circuits, and frequency multipliers.

**Example 3.3.3:** A varactor diode has $C_{j0} = 20$ pF, $V_{bi} = 0.75$ V, and m = 0.5 (abrupt junction). It is used in an LC tank circuit with L = 100 nH. Determine (a) the junction capacitance at reverse bias voltages of 1 V, 5 V, and 15 V, and (b) the corresponding resonant frequencies.
**Solution:**
(a) $C_j = C_{j0} / (1 + V_R/V_{bi})^{0.5}$ (using $V_R = |V_A|$ for reverse bias):

At $V_R$ = 1 V: $C_j = 20 / (1 + 1/0.75)^{0.5} = 20 / (2.333)^{0.5} = 20 / 1.528 =$ **13.09 pF**
At $V_R$ = 5 V: $C_j = 20 / (1 + 5/0.75)^{0.5} = 20 / (7.667)^{0.5} = 20 / 2.769 =$ **7.22 pF**
At $V_R$ = 15 V: $C_j = 20 / (1 + 15/0.75)^{0.5} = 20 / (21.0)^{0.5} = 20 / 4.583 =$ **4.36 pF**
(b) Resonant frequency: $f_0 = 1 / (2\pi\sqrt{(LC)})$
At 1 V: $f_0 = 1 / (2\pi\sqrt{(100 \times 10^{-9} \times 13.09 \times 10^{-12})}) = 1 / (2\pi \times 1.144 \times 10^{-9}) =$ **139.2 MHz**
At 5 V: $f_0 = 1 / (2\pi\sqrt{(100 \times 10^{-9} \times 7.22 \times 10^{-12})}) = 1 / (2\pi \times 8.50 \times 10^{-10}) =$ **187.3 MHz**
At 15 V: $f_0 = 1 / (2\pi\sqrt{(100 \times 10^{-9} \times 4.36 \times 10^{-12})}) = 1 / (2\pi \times 6.604 \times 10^{-10}) =$ **241.1 MHz**
The varactor provides a **1.73:1 frequency tuning range** (241.1/139.2) with a 1-15 V control voltage sweep.

## 3.4 Diodes

A diode is a two-terminal device based on a PN junction that allows current to flow easily in one direction (forward bias) and blocks current in the other direction (reverse bias). A silicon diode will "turn on" when a forward voltage of approximately 0.6 V to 0.7 V is applied and become conductive. Under normal conditions, only a very small leakage current flows when reverse voltage is applied. However, with a sufficiently large reverse voltage, the diode reaches its breakdown voltage and begins conducting in reverse. Zener diodes are specifically designed to operate in this breakdown region and are used as voltage references and regulators.

### 3.4.1 Rectifier Diodes

Standard rectifier diodes are the most common type and are used to convert AC to DC in power supplies. Schottky diodes use a metal-semiconductor junction instead of a P-N junction, providing a lower forward voltage drop (~0.2 V to 0.3 V) and faster switching, making them ideal for high-frequency rectification and clamping circuits. Signal diodes (such as the 1N4148) are optimized for small-signal applications including switching, clipping, and logic circuits. Light Emitting Diodes (LEDs) emit photons when forward biased and are used in indicators, displays, and illumination. Photodiodes operate in reverse bias and generate current proportional to incident light, serving as optical detectors in communications and sensing applications.

**Example 3.4.1:** A full-wave bridge rectifier uses four silicon diodes ($V_f$ = 0.7 V each) and is supplied by a 12.6 $V_{rms}$ AC transformer secondary. The rectifier drives a 100 Ω load through a 1,000 μF filter capacitor at 60 Hz. Determine (a) the peak output voltage, (b) the DC load current, and (c) the peak-to-peak ripple voltage.
**Solution:**
(a) Peak transformer voltage: $V_{peak} = 12.6 \times \sqrt{2} = 17.82$ V
Two diodes conduct at any time in a bridge rectifier:
Peak output voltage: $V_{out\_peak} = 17.82 - 2 \times 0.7 =$ **16.42 V**
(b) DC load current (approximately): $I_{DC} = V_{out\_peak} / R_L = 16.42 / 100 =$ **164.2 mA**
(c) Ripple voltage (full-wave, $f_{ripple} = 2 \times 60 = 120$ Hz):
$V_{ripple} = I_{DC} / (f_{ripple} \times C) = 0.1642 / (120 \times 0.001) =$ **1.37 V peak-to-peak**

### 3.4.2 Zener Diodes

Zener diodes are designed to operate in the reverse-biased breakdown region, providing a stable voltage reference. Below about 5 V, the breakdown mechanism is the Zener effect (quantum tunneling through a thin depletion region in heavily doped junctions); above about 7 V, avalanche breakdown (impact ionization) dominates. Diodes between 5-7 V use a combination of both mechanisms. Zener diodes are characterized by their Zener voltage ($V_Z$), dynamic resistance ($r_z$), power dissipation rating, and temperature coefficient — notably, Zener breakdown has a negative temperature coefficient while avalanche breakdown has a positive one, and 5.1 V Zener diodes are popular because the two effects nearly cancel, providing excellent temperature stability.

**Example 3.4.2:** A 5.1 V Zener diode (with dynamic resistance $r_z = 10\ \Omega$ and maximum power dissipation $P_{max} = 1$ W) is used as a voltage regulator. The unregulated input voltage varies from 9 V to 15 V, and the load resistance is $R_L = 510\ \Omega$. A series resistor $R_S$ is needed. Determine (a) the value of $R_S$ to keep the Zener in regulation over the full input range, and (b) the maximum Zener current and power dissipation.
**Solution:**
(a) Load current: $I_L = V_Z / R_L = 5.1 / 510 = 10$ mA
Minimum Zener current for regulation (practical minimum): $I_{Z\_min} \approx 5$ mA
At minimum input voltage (9 V), $R_S$ must allow $I_{Z\_min} + I_L$ to flow:
$R_S = (V_{in\_min} - V_Z) / (I_{Z\_min} + I_L) = (9 - 5.1) / (0.005 + 0.010) = 3.9 / 0.015 = \mathbf{260\ \Omega}$ (use 270 Ω standard value)
(b) At maximum input voltage (15 V) with $R_S = 270\ \Omega$:
Total current: $I_{total} = (V_{in\_max} - V_Z) / R_S = (15 - 5.1) / 270 = 9.9 / 270 = 36.67$ mA
Zener current: $I_{Z\_max} = I_{total} - I_L = 36.67 - 10 = \mathbf{26.67\ mA}$
Power dissipation: $P_Z = V_Z \times I_{Z\_max} = 5.1 \times 0.02667 = \mathbf{0.136\ W}$ (well within the 1 W rating)

### 3.4.3 Schottky Diodes

Schottky diodes use a metal-semiconductor junction (typically a metal such as platinum, chromium, or molybdenum deposited on N-type silicon) rather than a PN junction. Because there is no minority carrier injection across the junction, Schottky diodes have no reverse recovery time — they switch from conducting to blocking almost instantaneously. The forward voltage drop is significantly lower than silicon PN diodes (0.15-0.45 V versus 0.6-0.7 V), reducing conduction losses in power applications. These characteristics make Schottky diodes the preferred choice for high-frequency switching power supplies, output rectification in DC-DC converters, freewheeling diodes for inductive loads, and reverse polarity protection. The main limitations are lower reverse voltage ratings (typically ≤ 200 V, though SiC Schottky diodes reach 1,200 V) and higher reverse leakage current compared to PN junction diodes, which increases with temperature.

**Example 3.4.3:** A 5 V, 20 A output DC-DC converter uses output rectifier diodes that each carry an average current of 10 A. Compare the power loss using (a) a silicon PN rectifier diode with $V_f = 0.85$ V and $t_{rr} = 35$ ns, versus (b) a Schottky rectifier with $V_f = 0.35$ V and negligible recovery time. The converter switching frequency is 300 kHz with a reverse voltage of 20 V and a di/dt during recovery of 200 A/μs.
**Solution:**

(a) Silicon PN rectifier:
Conduction loss: $P_{cond} = V_f \times I_{avg} = 0.85 \times 10 = 8.5$ W
Reverse recovery loss: $P_{rr} \approx 0.5 \times V_R \times I_{rr} \times t_{rr} \times f_{sw}$
$I_{rr} \approx di/dt \times t_{rr} = 200 \times 10^6 \times 35 \times 10^{-9} = 7.0$ A
$P_{rr} = 0.5 \times 20 \times 7.0 \times 35 \times 10^{-9} \times 300{,}000 = 0.735$ W
Total: $P_{Si} = 8.5 + 0.735 =$ **9.24 W per diode**
(b) Schottky rectifier:
Conduction loss: $P_{cond} = 0.35 \times 10 = 3.5$ W
Recovery loss: $\approx 0$ W (no minority carrier storage)
Total: $P_{Schottky} =$ **3.5 W per diode**
The Schottky diode reduces rectifier losses by $9.24 - 3.5 =$ **5.74 W per diode (62% reduction)**, saving 11.5 W total for the two rectifiers.
This improves converter efficiency by $11.5/(5 \times 20) \times 100 =$ **11.5 percentage points**.

### 3.4.4 Light-Emitting Diodes (LEDs)

Light-emitting diodes emit photons when electrons recombine with holes across a forward-biased PN junction in a direct-bandgap semiconductor. The wavelength (color) of the emitted light is determined by the bandgap energy of the semiconductor material: $\lambda = hc/E_g$, where h is Planck's constant (6.626 $\times 10^{-34}$ J·s), c is the speed of light ($3 \times 10^8$ m/s), and $E_g$ is the bandgap in joules. Common LED materials include GaAs (infrared, ~870 nm), AlGaInP (red-yellow, 570-650 nm), InGaN (green-blue, 450-530 nm), and GaN (violet-UV, 365-405 nm). White LEDs are typically produced by coating a blue InGaN LED with a yellow phosphor (YAG:Ce), with the combination of blue LED emission and yellow phosphor fluorescence perceived as white light. LED efficiency is characterized by wall-plug efficiency (optical power out / electrical power in, typically 30-50% for modern high-power LEDs) and luminous efficacy (lumens per watt, up to 200 lm/W for white LEDs).

**Example 3.4.4:** A white LED luminaire uses 24 high-power LEDs, each rated at 1 W electrical input with a luminous efficacy of 160 lm/W. Each LED operates at $V_f = 3.0$ V and $I_f = 333$ mA. Determine (a) the total luminous output in lumens, (b) the total electrical power, and (c) the equivalent incandescent wattage (assuming 15 lm/W for incandescent bulbs).
**Solution:**
(a) Luminous output per LED: $\Phi = 160$ lm/W $\times$ 1 W = 160 lumens
Total: $\Phi_{total} = 24 \times 160 =$ **3,840 lumens**
(b) Electrical power per LED: $P = V_f \times I_f = 3.0 \times 0.333 = 1.0$ W
Total: $P_{total} = 24 \times 1.0 =$ **24 W**
(c) Equivalent incandescent: $P_{incand} = 3{,}840 / 15 =$ **256 W**
The LED luminaire produces the same light as a 256 W incandescent bulb while consuming only 24 W — a **10.7× reduction** in electrical power.

## 3.5 Transistors

Transistors are the active building blocks of all modern electronic circuits, performing amplification and switching functions by using a small input signal or current to control a much larger output current. The two dominant transistor families are the Bipolar Junction Transistor (BJT), a current-controlled device, and the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET), a voltage-controlled device. Power transistors such as IGBTs and power MOSFETs extend these principles to high-voltage, high-current switching applications.

### 3.5.1 Bipolar Junction Transistor (BJT)

The Bipolar Junction Transistor is a current-controlled three-terminal device (Base, Collector, Emitter) that comes in two polarities: NPN and PNP. In an NPN transistor, a small current flowing into the Base terminal controls a larger current flowing from Collector to Emitter; in a PNP transistor, the current directions are reversed, with current flowing from Emitter to Collector controlled by current flowing out of the Base. The ratio of collector current to base current is the current gain ($\beta$ or $h_{FE}$), typically ranging from 20 to several hundred. BJTs operate in three regions: cutoff (off), active (linear amplification), and saturation (fully on), making them useful for both amplification and switching applications.

**Example 3.5.1:** An NPN BJT common-emitter amplifier has $\beta = 150$, $V_{CC} = 12$ V, $R_C = 2.2$ k$\Omega$, and $R_E = 470\ \Omega$. The base bias network sets $I_B = 40\ \mu$A. Assuming $V_{BE} = 0.7$ V, determine (a) $I_C$, (b) $V_{CE}$, and (c) the voltage gain $A_v$ if the AC input signal is applied to the base and the output is taken at the collector (bypass capacitor across $R_E$ for AC).
**Solution:**
(a) Collector current: $I_C = \beta \times I_B = 150 \times 40 \times 10^{-6} = $ **6.0 mA**
(b) Applying KVL around the collector-emitter loop:
$V_{CE} = V_{CC} - I_C \times (R_C + R_E) = 12 - 0.006 \times (2{,}200 + 470)$
$V_{CE} = 12 - 0.006 \times 2{,}670 = 12 - 16.02 = -4.02$ V
Since $V_{CE}$ is negative, the transistor is in **saturation** (not active mode).
In saturation, $V_{CE\_sat} \approx 0.2$ V.
Recalculating: $I_{C_sat} = (V_{CC} - V_{CE_sat}) / (R_C + R_E) = (12 - 0.2) / 2{,}670 = $ **4.42 mA**
(c) Since the transistor is saturated under the given bias, gain is not defined at this operating point. For the purpose of calculating gain, re-bias the circuit to the edge of saturation using $I_C = 4.42$ mA from part (b):
Transconductance: $g_m = I_C / V_T = 0.00442 / 0.02585 = 171.0$ mA/V
With $R_E$ bypassed: $A_v = -g_m \times R_C = -171.0 \times 10^{-3} \times 2{,}200 = \mathbf{-376.2}$ (inverting)
$|A_v|$ in dB $= 20 \times \log_{10}(376.2) = $ **51.5 dB**

### 3.5.2 MOSFET

The Metal-Oxide-Semiconductor Field-Effect Transistor is a voltage-controlled three-terminal device (Gate, Drain, Source) where a voltage applied to the Gate controls the current flowing between Drain and Source through an electric field. MOSFETs are the most common type of FET and come in enhancement-mode (normally off, requiring a gate voltage to conduct) and depletion-mode (normally on) variants, in both N-channel and P-channel configurations. JFETs (Junction FETs) use a reverse-biased P-N junction as the gate and are depletion-mode devices, often used in low-noise analog front-end circuits. FETs offer extremely high input impedance at the gate, draw virtually no gate current in steady state, and are the fundamental building block of CMOS logic, which dominates modern digital integrated circuits.

**Example 3.5.2:** An N-channel enhancement-mode MOSFET has a threshold voltage $V_{th} = 2.0$ V and a process transconductance parameter $k_n = 0.5$ mA/V². The device operates with $V_{GS} = 5$ V and $V_{DS} = 8$ V. Determine (a) the operating region, (b) the drain current $I_D$, and (c) the small-signal transconductance $g_m$.

**Solution:**

(a) Check operating region:

$V_{GS} - V_{th} = 5 - 2 = 3$ V

$V_{DS} = 8$ V $> V_{GS} - V_{th} = 3$ V, so the MOSFET is in **saturation** (pinch-off region).

(b) Drain current in saturation (ignoring channel-length modulation):

$I_D = (k_n / 2) \times (V_{GS} - V_{th})^2 = (0.5 \times 10^{-3} / 2) \times (3)^2 = 0.25 \times 10^{-3} \times 9 = \mathbf{2.25\ mA}$

(c) Small-signal transconductance:

$g_m = k_n \times (V_{GS} - V_{th}) = 0.5 \times 10^{-3} \times 3 = \mathbf{1.5\ mA/V}$

Alternatively: $g_m = 2 \times I_D / (V_{GS} - V_{th}) = 2 \times 2.25 / 3 = \mathbf{1.5\ mA/V}$

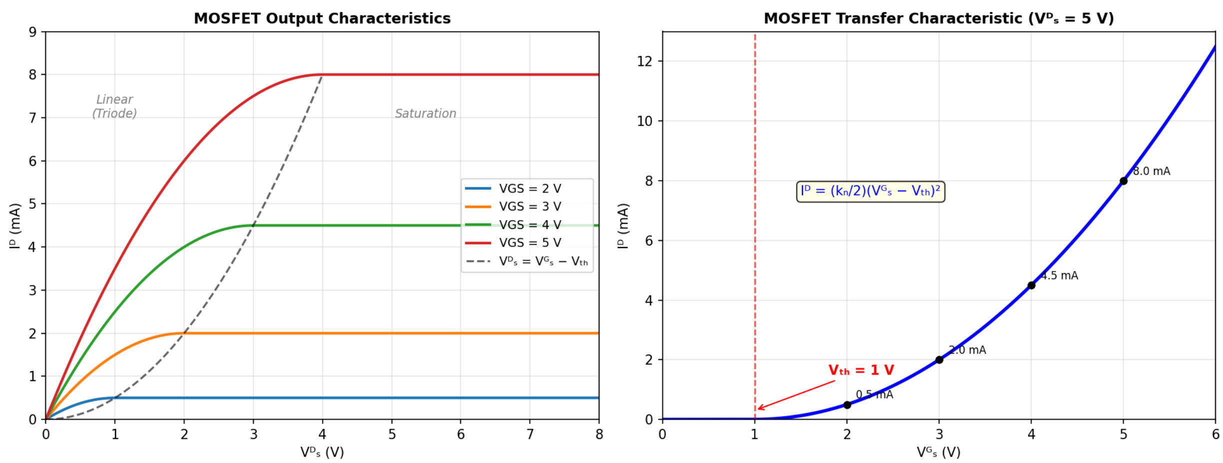



Figure 3.5.2: MOSFET I-V Characteristics

### 3.5.3 Power MOSFETs

Power MOSFETs are designed for high-current, high-voltage switching applications and differ significantly from signal-level MOSFETs in both structure and specifications. They use a vertical device structure (VDMOS or trench MOSFET) where current flows vertically through the silicon die, allowing higher current ratings in a compact package. The key parameter is $R_{DS(on)}$ — the on-state drain-source resistance — which determines conduction losses: $P_{cond} = I_D{}^2 \times R_{DS(on)}$. $R_{DS(on)}$ has a positive temperature coefficient, which provides natural current sharing when MOSFETs are paralleled but means losses increase at elevated temperatures. Switching losses depend on gate charge ($Q_g$), which determines how quickly the gate driver can turn the device on and off: lower $Q_g$ enables faster switching and lower losses at high frequencies. Power MOSFETs also have an intrinsic body diode (a parasitic PN junction between the body and drain) that conducts when $V_{DS}$ goes negative, which is both useful (as a freewheeling diode in half-bridge circuits) and potentially problematic (slow body diode recovery can cause shoot-through losses). The Safe Operating Area (SOA) defines the maximum simultaneous voltage and current the device can withstand without damage.

**Example 3.5.3:** A power MOSFET in a 48 V to 12 V synchronous buck converter operates at $f_{sw}$ = 250 kHz with a load current of 20 A. The high-side MOSFET has $R_{DS(on)} = 8\ m\Omega$, $Q_g = 45$ nC, and switching times $t_{rise} = 15$ ns, $t_{fall} = 20$ ns. The duty cycle is $D = V_{out}/V_{in} = 12/48 = 0.25$. Determine (a) the conduction loss, (b) the switching loss, (c) the gate drive loss ($V_{GS} = 10$ V), and (d) the total power dissipation.
**Solution:**
(a) Conduction loss (high-side conducts for $D \times T_{sw}$):
$I_{RMS} = I_{load} \times \sqrt{D} = 20 \times \sqrt{0.25} = 20 \times 0.5 = 10$ A
$P_{cond} = I_{RMS}{}^2 \times R_{DS(on)} = 100 \times 0.008 = \mathbf{0.80\ W}$
(b) Switching loss (transitions occur at $V_{in}$ and $I_{load}$):
$P_{sw} = 0.5 \times V_{in} \times I_{load} \times (t_{rise} + t_{fall}) \times f_{sw}$
$= 0.5 \times 48 \times 20 \times (15 + 20) \times 10^{-9} \times 250{,}000$
$= 480 \times 35 \times 10^{-9} \times 250{,}000 = \mathbf{4.20\ W}$
(c) Gate drive loss:
$P_{gate} = Q_g \times V_{GS} \times f_{sw} = 45 \times 10^{-9} \times 10 \times 250{,}000 = \mathbf{0.11\ W}$
(d) Total power dissipation:
$P_{total} = 0.80 + 4.20 + 0.11 = \mathbf{5.11\ W}$
Switching losses dominate (82% of total), which is typical for high-voltage converters.
Reducing switching losses requires either lower frequency (increases inductor size) or a MOSFET with lower $Q_g$ (which typically has higher $R_{DS(on)}$ — the classic $R_{DS(on)} \times Q_g$ figure of merit trade-off).

## 3.6 Voltage Regulators

Voltage regulators maintain a stable DC output voltage despite variations in input voltage or load current. Linear regulators (such as the LM7805 and LDO types) operate by dissipating excess voltage as heat across a series pass element, offering low noise and simplicity but with efficiency limited to the ratio of output to input voltage. Switching regulators (buck, boost, and buck-boost topologies) use an inductor, switch, and diode to convert voltage levels with efficiencies typically exceeding 85-95%, but introduce switching noise that must be filtered. The choice between linear and switching regulators involves trade-offs between efficiency, noise, cost, board space, and thermal management requirements.

### 3.6.1 Linear Regulators

Fixed three-terminal linear regulators such as the LM78xx series (positive output) and LM79xx series (negative output) are the simplest voltage regulation solution — the LM7805 provides a fixed 5 V output, the LM7812 provides 12 V, and so on. These regulators require a minimum input-to-output voltage difference (dropout voltage) of approximately 2 V to maintain regulation. Low-Dropout regulators (LDOs) reduce this requirement to as little as 100-300 mV by using a PMOS or PNP pass transistor instead of the NPN Darlington configuration used in the 78xx series. LDOs are essential in battery-powered devices where every millivolt of headroom matters, and in noise-sensitive analog circuits where switching regulator ripple is unacceptable. Key LDO specifications include dropout voltage, output noise (typically 10-100 $\mu V_{rms}$), power supply rejection ratio (PSRR, 40-80 dB), quiescent current (10 $\mu$A to 5 mA), and load/line transient response.

**Example 3.6.1:** A linear regulator powers a 3.3 V, 500 mA microcontroller circuit from a 5.0 V USB supply. Compare (a) an LM1117-3.3 LDO (dropout voltage = 1.2 V at 500 mA) and (b) a standard LM317 configured for 3.3 V (dropout voltage = 2.0 V). For each, determine the power dissipation, efficiency, and whether a heat sink is required (assume a maximum junction temperature of 125°C, ambient temperature of 40°C, and thermal resistance $\theta_{JA}$ = 50°C/W for SOT-223 package).
**Solution:**
(a) LM1117-3.3 LDO:
Input-output differential: $\Delta V = 5.0 - 3.3 = 1.7$ V (> 1.2 V dropout, so it regulates)
Power dissipation: $P_D = \Delta V \times I_{out} = 1.7 \times 0.5 = \mathbf{0.85\ W}$
Efficiency: $\eta = V_{out}/V_{in} = 3.3/5.0 = \mathbf{66.0\%}$
Junction temperature: $T_J = T_A + P_D \times \theta_{JA} = 40 + 0.85 \times 50 = \mathbf{82.5°C}$ (no heat sink needed)
(b) LM317 (adjustable, set to 3.3 V):
Required minimum input: $V_{in\_min} = 3.3 + 2.0 = 5.3$ V
At $V_{in}$ = 5.0 V, the differential is only 1.7 V < 2.0 V dropout — the LM317 **cannot regulate** at this input voltage.
It would require $V_{in} \geq 5.3$ V.
If supplied from a 7.0 V source instead: $P_D = (7.0 - 3.3) \times 0.5 = \mathbf{1.85\ W}$
Efficiency: $\eta = 3.3/7.0 = \mathbf{47.1\%}$
$T_J = 40 + 1.85 \times 50 = 132.5$°C — **exceeds 125°C limit**, a heat sink is required.
This demonstrates why LDOs are preferred when the input-output differential is small.

### 3.6.2 TI TL431 Programmable Reference

The Texas Instruments TL431 is a programmable precision shunt voltage regulator capable of operating over a range of approximately 2.5 V to 36 V. It functions as a voltage-controlled current sink: two external resistors form a voltage divider that sets the output voltage by feeding a fraction of it back to the TL431's reference input, and the device adjusts its sink current to maintain that set voltage at its cathode. The TL431 is widely used in the feedback network of isolated switch-mode power supplies (SMPS), where it works with an optocoupler to regulate the output voltage across the isolation boundary. It also serves as a precision voltage reference, overvoltage clamp, and constant-current source in a variety of power management and signal conditioning circuits.

**Example 3.6.2:** A TL431 is used to regulate the output of a power supply to 12.0 V. The TL431 internal reference voltage is $V_{ref}$ = 2.495 V. The feedback network consists of two resistors forming a voltage divider from the output to the TL431 reference pin. If $R_2$ (lower resistor, from reference pin to ground) is chosen as 10 kΩ, determine the required value of $R_1$ (upper resistor, from output to reference pin), and calculate the cathode current if a 100 Ω series resistor connects a 15 V supply to the cathode (anode grounded).
**Solution:**
At regulation, the voltage at the reference pin equals $V_{ref}$:
$V_{ref} = V_{out} \times R_2 / (R_1 + R_2)$
Solving for $R_1$:
$R_1 = R_2 \times (V_{out} / V_{ref} - 1) = 10{,}000 \times (12.0 / 2.495 - 1) = 10{,}000 \times (4.81 - 1) = 10{,}000 \times 3.81$
$R_1 = \mathbf{38.1\ k\Omega}$ (use 38.3 kΩ or a 36 kΩ + 2.2 kΩ series combination for 38.2 kΩ)
Verify: $V_{out} = V_{ref} \times (R_1 + R_2) / R_2 = 2.495 \times (38{,}200 + 10{,}000) / 10{,}000 = 2.495 \times 4.82 = 12.03$ V (close to 12.0 V)

Cathode current (TL431 acts as a shunt regulator sinking current to maintain 12.0 V):
The divider current: $I_{divider} = V_{out} / (R_1 + R_2) = 12.0 / (38{,}200 + 10{,}000) = 12.0 / 48{,}200 = 0.249$ mA
This is the reference pin bias current.
The cathode current depends on the external circuit driving the cathode.
If a series resistor $R_{series} = 100\ \Omega$ is placed between a 15 V supply and the cathode:
$I_K = (V_{supply} - V_{out}) / R_{series} = (15 - 12) / 100 =$ **30 mA**

## 3.7 Semiconductor Fabrication

Semiconductor fabrication transforms raw silicon into integrated circuits through hundreds of precisely controlled process steps performed in cleanroom environments. A modern fab (fabrication facility) costs \$10–20 billion to build and operates at Class 1 cleanliness (fewer than 1 particle per cubic foot larger than 0.5 μm). The fabrication process involves repeated cycles of deposition, patterning, etching, and doping to build up the transistor structures and interconnect layers that form a complete IC. Understanding these processes is essential for device engineers who design within manufacturing constraints and for process engineers who optimize yield and performance at each technology node.

### 3.7.1 Crystal Growth and Wafer Preparation

The starting material for most semiconductor devices is a single-crystal silicon ingot grown by the Czochralski (CZ) method: a seed crystal is dipped into a crucible of molten silicon (at 1,414°C) and slowly pulled upward while rotating, forming a cylindrical ingot of single-crystal silicon up to 300 mm in diameter and 2 meters long. The pull rate and rotation speed control the crystal diameter and defect density. For ultra-high-purity applications (power devices, radiation detectors), the float-zone (FZ) method passes a molten zone along a polycrystalline rod, achieving resistivities above 10,000 Ω·cm by eliminating crucible contamination. The ingot is sliced into wafers 0.775 mm thick (for 300 mm diameter), polished to sub-nanometer surface roughness, and cleaned. Wafer crystal orientation — (100) for CMOS (lower surface state density) or (111) for bipolar — affects device performance through surface carrier mobility and oxidation rates.

**Example 3.7.1:** A 300 mm diameter silicon wafer is used to fabricate ICs with die dimensions of 15 mm × 15 mm. Estimate the maximum number of dies per wafer, accounting for edge exclusion of 3 mm.
**Solution:**
Usable wafer area: $\pi \times (D/2 - \text{edge})^2 = \pi \times (150 - 3)^2 = \pi \times 147^2 =$ **67,890 mm²**.
Die area: $15 \times 15 = 225\ \text{mm}^2$.
Approximate dies: $67{,}890/225 = 301.7 \approx 302$.
Edge loss correction (dies crossing wafer boundary): subtract approximately $\pi \times D/(\text{die diagonal})$.
Die diagonal $= \sqrt{}(15^2 + 15^2) = 21.2$ mm.
Edge dies lost $\approx \pi \times 294/21.2 \approx 43.6$.
Estimated good dies: $302 - 44 =$ **~258 dies per wafer**.
At a yield of 90%, approximately 232 functional dies would be produced.

### 3.7.2 Lithography and Patterning

Lithography transfers circuit patterns from a photomask onto the wafer surface using light-sensitive photoresist. The process involves coating the wafer with photoresist, exposing it through a photomask (or reticle) using a stepper or scanner, and developing the exposed resist to create the pattern. The minimum feature size is governed by the Rayleigh criterion: $CD_{min} = k_1 \times \lambda/NA$, where $\lambda$ is the exposure wavelength, NA is the numerical aperture of the projection lens, and $k_1$ is a process-dependent factor (theoretical minimum 0.25, practical range 0.3–0.4).

Deep ultraviolet (DUV) lithography at 193 nm with immersion (water between lens and wafer, NA $\approx$ 1.35) has been the workhorse from 65 nm down to 7 nm nodes, using multi-patterning techniques (double and quadruple patterning) to print features smaller than the single-exposure resolution limit. Extreme ultraviolet (EUV) lithography at 13.5 nm uses reflective optics (no lens materials are transparent at this wavelength) and tin-droplet plasma light sources generating over 250 watts of EUV power. EUV enables single-exposure patterning at 7 nm and below, eliminating the cost and complexity of multi-patterning, and is essential for 5 nm, 3 nm, and 2 nm nodes. High-NA EUV (NA = 0.55 vs standard 0.33) extends EUV to sub-2 nm features.

**Example 3.7.2:** Compare the minimum feature size achievable with DUV immersion lithography ($\lambda$ = 193 nm, NA = 1.35, $k_1$ = 0.30) versus standard EUV lithography ($\lambda$ = 13.5 nm, NA = 0.33, $k_1$ = 0.30).
**Solution:**
DUV: $CD_{min} = 0.30 \times 193/1.35 =$ **42.9 nm** (single exposure).
To reach 7 nm node features (~30 nm metal pitch), multi-patterning with 2–4 exposures per layer is required.
EUV: $CD_{min} = 0.30 \times 13.5/0.33 =$ **12.3 nm** (single exposure).
This resolves 7 nm and 5 nm features directly, and high-NA EUV (NA = 0.55) would achieve $0.30 \times 13.5/0.55 =$ **7.4 nm**, enabling 2 nm node patterning.

### 3.7.3 Doping and Ion Implantation

Ion implantation is the primary method for introducing dopant atoms into silicon with precise control of dose (atoms/cm²) and depth profile. Dopant ions (boron for P-type, phosphorus or arsenic for N-type) are accelerated to energies of 1 keV to 3 MeV and directed into the wafer surface. The implanted ions follow an approximately Gaussian depth distribution with a projected range $R_p$ and standard deviation $\Delta R_p$, both determined by the ion species and acceleration energy. Higher energy yields deeper implants; higher dose increases the peak concentration. After implantation, the crystal lattice is damaged by the ion collisions, so a thermal anneal (typically rapid thermal annealing at 900–1,100°C for 1–30 seconds) is required to repair the lattice and electrically activate the dopant atoms.

Thermal diffusion, the older doping method, uses a high-temperature furnace (900–1,200°C) to drive dopant atoms from a surface source into the bulk. The diffusion profile follows Fick's second law, producing a complementary error function (erfc) profile for constant-surface-concentration diffusion or a Gaussian profile for limited-source diffusion. The junction depth $x_j$ where the diffused dopant concentration equals the background doping is approximately $x_j \approx 2\sqrt{(Dt)}$, where D is the diffusion coefficient (cm²/s) and t is the diffusion time.

**Example 3.7.3:** Phosphorus is diffused into a P-type silicon substrate ($N_A = 10^{16}$ cm$^{-3}$) at 1,000°C for 30 minutes. The diffusion coefficient of phosphorus in silicon at 1,000°C is $D = 2.5 \times 10^{-14}$ cm$^2$/s. The surface concentration is $N_s = 10^{20}$ cm$^{-3}$. Estimate the junction depth.
**Solution:**
Diffusion length: $\sqrt{(Dt)} = \sqrt{(2.5 \times 10^{-14} \times 1{,}800)} = \sqrt{(4.5 \times 10^{-11})} = 6.71 \times 10^{-6}$ cm $= 0.0671$ μm.
For a complementary error function profile, the junction occurs where $N(x_j) = N_A$:
$N_s \times \mathrm{erfc}(x_j/(2\sqrt{(Dt)})) = N_A$.
$\mathrm{erfc}(x_j/(2 \times 0.0671\ \mu m)) = 10^{16}/10^{20} = 10^{-4}$.
From erfc tables: $\mathrm{erfc}(2.75) \approx 10^{-4}$, so $x_j/(2 \times 0.0671) = 2.75$.
$x_j = 2.75 \times 0.1342 = \mathbf{0.369\ \mu m}$.
This shallow junction is typical of source/drain implants in submicron CMOS processes.

### 3.7.4 Etching and Deposition

Etching selectively removes material to transfer lithographic patterns into underlying films. Wet etching uses chemical solutions (e.g., buffered HF for $SiO_2$, KOH for silicon) and is isotropic (etches equally in all directions), limiting its use to features larger than ~1 μm. Dry etching — primarily reactive ion etching (RIE) — uses plasma-generated reactive species in a vacuum chamber to achieve anisotropic (directional) etching with nearly vertical sidewalls, essential for sub-100 nm features. Etch selectivity (the ratio of etch rates between the target material and the underlying layer or photoresist) must be high enough to stop precisely at the intended depth.

Thin-film deposition builds up the layers of an IC. Chemical vapor deposition (CVD) uses gas-phase chemical reactions to deposit films: LPCVD (low-pressure) for polysilicon and silicon nitride, PECVD (plasma-enhanced) for $SiO_2$ and $SiN_x$ at lower temperatures compatible with metal layers. Physical vapor deposition (PVD/sputtering) deposits metal films (aluminum, titanium, tantalum) by bombarding a target with argon ions. Atomic layer deposition (ALD) deposits films one atomic layer at a time with self-limiting surface reactions, achieving angstrom-level thickness control essential for high-k gate dielectrics ($HfO_2$) and ultrathin barrier layers at advanced nodes.

**Example 3.7.4:** An RIE process etches $SiO_2$ at a rate of 200 nm/min and the underlying silicon at 10 nm/min. A 500 nm $SiO_2$ film must be completely removed. Calculate (a) the etch time, (b) the selectivity ratio, and (c) the silicon loss if a 20% overetch is applied.
**Solution:**
(a) Etch time: 500/200 = **2.5 minutes**.
(b) Selectivity: $SiO_2$ rate / Si rate = 200/10 = **20:1**.
(c) With 20% overetch: additional time = 0.20 × 2.5 = 0.5 min.
Silicon loss = 10 × 0.5 = **5 nm**.
This is acceptable for most processes but would be significant for ultra-thin body SOI devices where the silicon layer may be only 5–10 nm thick.

### 3.7.5 CMOS Process Integration

A CMOS (Complementary Metal-Oxide-Semiconductor) process fabricates both NMOS and PMOS transistors on the same substrate, enabling logic gates that consume near-zero static power. The twin-well process begins with a lightly doped substrate and forms separate N-wells (for PMOS) and P-wells (for NMOS) through ion implantation and diffusion. Shallow trench isolation (STI) fills etched trenches with $SiO_2$ to electrically isolate adjacent transistors. The gate stack — traditionally polysilicon on $SiO_2$ — has been replaced at 45 nm and below by high-k/metal gate (HKMG), using $HfO_2$ ($k \approx 25$ vs 3.9 for $SiO_2$) and metal gates (TiN, TaN) to reduce gate leakage while maintaining high capacitance.

The front-end-of-line (FEOL) builds the transistors: gate patterning, source/drain implantation with halo and extension implants, silicide formation (NiSi) for low-resistance contacts. The back-end-of-line (BEOL) connects transistors with multiple metal interconnect layers (10–15 layers at advanced nodes), using copper dual-damascene processing with low-k dielectrics ($k \approx 2.5$–$3.0$) to minimize RC delay. Design rules specify minimum metal width, spacing, via sizes, and density requirements that ensure manufacturability and yield.

**Example 3.7.5:** A 7 nm CMOS process has a gate pitch of 54 nm, a minimum metal pitch of 36 nm, and 13 BEOL metal layers. The gate capacitance is 1.2 fF/μm and the total interconnect capacitance for a typical logic gate output is 3.5 fF. With a supply voltage of 0.75 V and an NMOS drive current of 1.1 mA/μm, estimate the gate switching delay.
**Solution:**
Gate delay $\approx C_{total} \times V_{DD} / I_{drive}$.
For a minimum-size inverter with $W_n = 0.054$ μm (one gate pitch):
$C_{gate} = 1.2 \times 0.054 = 0.065$ fF. $C_{total} = C_{gate} + C_{interconnect} = 0.065 + 3.5 = 3.565$ fF.
$I_{drive} = 1.1 \times 0.054 = 0.0594$ mA $= 59.4$ μA.
Gate delay $= 3.565 \times 10^{-15} \times 0.75 / (59.4 \times 10^{-6}) =$ **45 ps**.
At advanced nodes, interconnect capacitance dominates (98% of total), making BEOL optimization critical for performance.

### 3.7.6 Advanced Nodes and Future Directions

As planar MOSFETs scaled below 20 nm, short-channel effects (drain-induced barrier lowering, subthreshold leakage) became unmanageable with conventional gate structures. The industry transitioned to three-dimensional transistor architectures that improve electrostatic control of the channel:

| Architecture | Nodes | Gate Contact | Key Advantage | Limitation |
|---|---|---|---|---|
| Planar MOSFET | >22 nm | Top only | Simple, low cost | High leakage below 22 nm |
| FinFET (tri-gate) | 22–5 nm | Three sides of fin | 3× leakage reduction vs planar | Fin width quantization limits flexibility |
| GAA Nanosheet | 3–2 nm | All four sides of sheet | Adjustable channel width per sheet | Complex process, stacked sheet stress |
| CFET (complementary FET) | <2 nm | Stacked NMOS over PMOS | ~30% area reduction vs GAA | Extreme process complexity |

FinFETs, introduced by Intel at 22 nm in 2012, wrap the gate around three sides of a tall, narrow silicon fin, dramatically reducing leakage current and improving drive current at low supply voltages. Gate-all-around (GAA) nanosheet transistors, adopted at 3 nm by Samsung and at 2 nm by TSMC, stack multiple horizontal silicon nanosheets with the gate surrounding all four sides, providing better electrostatic control and the flexibility to adjust drive strength by varying nanosheet width.

Beyond transistor architecture, advanced packaging is reshaping IC design. Chiplets decompose a large monolithic die into smaller functional tiles (compute, I/O, memory) fabricated at different process nodes and assembled using 2.5D (silicon interposer) or 3D (direct die stacking with hybrid bonding) integration. Through-silicon vias (TSVs) and micro-bumps provide vertical interconnects, while the Universal Chiplet Interconnect Express (UCIe) standard enables multi-vendor chiplet interoperability. High Bandwidth Memory (HBM) stacks DRAM dies using TSVs to achieve bandwidths exceeding 1 TB/s. Backside power delivery networks (BSPDN) route power through the back of the wafer, freeing the front-side BEOL for signal routing and reducing IR drop by 30–50%.

**Example 3.7.6:** A 7 nm FinFET process achieves a transistor density of 91 million transistors per mm². A 3 nm GAA nanosheet process achieves 292 million transistors per mm². A processor design requires 12 billion transistors. Calculate the die area at each node and the percentage area reduction from 7 nm to 3 nm.
**Solution:**
7 nm die area: $12 \times 10^9 / (91 \times 10^6) =$ **131.9 mm²**.
3 nm die area: $12 \times 10^9 / (292 \times 10^6) =$ **41.1 mm²**.
Area reduction: $(131.9 - 41.1)/131.9 \times 100 =$ **68.8%**.
Alternatively, the same 131.9 mm² die at 3 nm could hold $131.9 \times 292 \times 10^6 = 38.5$ billion transistors — enabling significantly more compute capability or allowing a smaller, cheaper die for equivalent functionality.

# Chapter 4

# Control Systems

Control systems engineering deals with the analysis and design of systems that regulate the behavior of dynamic processes to achieve desired performance objectives. A control system takes a reference input (the desired output), compares it to the actual output, and applies corrective action to minimize the difference. Control theory provides the mathematical tools — including transfer functions, frequency response, and state-space methods — to model system dynamics, assess stability, and design controllers that meet specifications for accuracy, speed of response, and robustness to disturbances.

## 4.1 Open Loop

An open-loop control system operates without feedback from the output to the input, meaning the controller has no knowledge of the actual system response. The input command is applied directly to the actuator, and the output is entirely dependent on the accuracy of the system model and the absence of disturbances. Because there is no mechanism to detect or correct errors, open-loop systems are inherently sensitive to parameter variations, external disturbances, and modeling inaccuracies. These systems are simpler and less expensive to implement, making them suitable for applications where the relationship between input and output is well characterized and disturbances are minimal, such as a toaster or a washing machine timer. The transfer function of an open-loop system is simply the forward path transfer function G(s), and the output C(s) = G(s) × R(s), where R(s) is the reference input.

**Example 4.1:** An open-loop motor speed controller has a forward path transfer function G(s) = 50 / (s + 10). A unit step input R(s) = 1/s is applied. Find the steady-state output speed.
**Solution:**
The output is C(s) = G(s) × R(s) = 50 / [s(s + 10)].
Using partial fractions: C(s) = 5/s - 5/(s + 10).
Taking the inverse Laplace transform: c(t) = 5 - 5e^(-10t).
As t approaches infinity, the exponential term decays to zero, so the steady-state output = 5.
If unity gain (output = reference) were desired, the error would be $1 - 5 = -4$ — the system overdrives by a factor of 5 with no mechanism to detect or correct the discrepancy, illustrating the fundamental limitation of open-loop control.

## 4.2 Closed Loop

A closed-loop control system, also known as a feedback control system, continuously measures the output and compares it to the desired reference input to generate an error signal. This error signal is then processed by the controller to adjust the actuator input, driving the output toward the desired value and reducing the effect of disturbances and parameter variations. The canonical form of a closed-loop system with unity feedback has a transfer function of G(s) / (1 + G(s)H(s)), where G(s) is the forward path transfer function and H(s) is the feedback path transfer function. Closed-loop systems offer superior performance in terms of accuracy, stability, and disturbance rejection compared to open-loop systems, at the cost of increased complexity and the potential for instability if not properly designed. Stability analysis techniques such as Routh-Hurwitz criteria, root locus plots, and Bode plots are essential tools for designing closed-loop systems that meet transient and steady-state performance specifications.

**Example 4.2:** A unity feedback closed-loop system (H(s) = 1) has a forward path transfer function G(s) = 100 / [s(s + 5)]. Determine the closed-loop transfer function and the steady-state error to a unit step input.
**Solution:**
The closed-loop transfer function is T(s) = G(s) / [1 + G(s)H(s)] = [100 / s(s + 5)] / [1 + 100 / s(s + 5)] = 100 / ($s^2$ + 5s + 100).
For a unit step input R(s) = 1/s, the steady-state error is $e_{ss}$ = 1 / (1 + $K_p$), where the position error constant $K_p$ = lim(s->0) G(s) = lim(s->0) 100 / [s(s + 5)] = infinity.
Therefore $e_{ss}$ = 1 / (1 + infinity) = 0.
The closed-loop system with a Type 1 open-loop transfer function has zero steady-state error to a step input, demonstrating the superior accuracy of feedback control.

## 4.3 Control Signals

Control signals are standardized test inputs used to characterize and analyze the dynamic behavior of control systems. By applying known input signals and observing the system response, engineers can evaluate performance metrics such as rise time, settling time, overshoot, and steady-state error. The following are the most common control signals used in control system analysis.

### 4.3.1 Step Input

The unit step function u(t) transitions instantaneously from zero to a constant value at time t = 0. Its Laplace transform is 1/s. The step input is the most widely used test signal because it simultaneously excites the transient and steady-state behavior of a system. The step response reveals key performance characteristics including rise time, peak overshoot, settling time, and steady-state error. In practice, many real-world disturbances resemble sudden changes, making the step response a practical measure of how a system handles abrupt input changes.

**Example 4.3.1:** A first-order system has the transfer function G(s) = 20 / (s + 4). Determine the steady-state value and the time constant from the unit step response.
**Solution:**
For a unit step input R(s) = 1/s, the output is C(s) = 20 / [s(s + 4)].
Using partial fractions: C(s) = 5/s - 5/(s + 4).
The time-domain response is c(t) = 5(1 - e^(-4t)).
The time constant $\tau$ = 1/4 = 0.25 s (the system reaches 63.2% of its final value in 0.25 s).

The steady-state value is c(infinity) = 5.
The settling time (to within 2%) is approximately $4\tau = 1.0$ s.

### 4.3.2 Ramp Input

The ramp function $r(t) = t \times u(t)$ increases linearly with time, and its Laplace transform is $1/s^2$. A ramp input tests the ability of a system to track a continuously changing reference, which is important in applications such as antenna tracking systems or CNC machine tool positioning. The steady-state error to a ramp input is characterized by the velocity error constant $K_v$, where a higher $K_v$ indicates better tracking performance. A Type 0 system (no free integrators in the open-loop transfer function) will have an unbounded steady-state error to a ramp input, while a Type 1 system will exhibit a finite constant error.

**Example 4.3.2:** A unity feedback system has an open-loop transfer function G(s) = 200 / [s(s + 10)(s + 20)]. Determine the steady-state error for a unit ramp input r(t) = t.
**Solution:**
This is a Type 1 system (one free integrator in G(s)).
The velocity error constant is $K_v$ = lim(s->0) s * G(s) = lim(s->0) s * 200 / [s(s + 10)(s + 20)] = 200 / (10 * 20) = 1.
The steady-state error to a ramp input is $e_{ss} = 1 / K_v = 1 / 1 = 1$.
The system tracks the ramp with a constant position lag of 1 unit behind the reference.

### 4.3.3 Impulse Input

The unit impulse function $\delta(t)$ is an idealized signal of infinite amplitude and infinitesimal duration with unit area, and its Laplace transform is simply 1. The impulse response is mathematically significant because it is the inverse Laplace transform of the system transfer function itself, meaning the impulse response completely characterizes a linear time-invariant (LTI) system. Any output of an LTI system can be obtained by convolving the impulse response with the input signal. While a true impulse cannot be generated physically, it can be approximated by a short-duration, high-amplitude pulse.

**Example 4.3.3:** A system has the transfer function G(s) = 10 / ($s^2$ + 3s + 2). Find the impulse response g(t).
**Solution:**
Factor the denominator: G(s) = 10 / [(s + 1)(s + 2)].
Using partial fractions: G(s) = 10/(s + 1) - 10/(s + 2).
Since the impulse response is the inverse Laplace transform of G(s) itself (because R(s) = 1 for an impulse), g(t) = 10e^(-t) - 10e^(-2t) for t >= 0.
At t = 0, g(0) = 10 - 10 = 0.
The peak occurs when dg/dt = -10e^(-t) + 20e^(-2t) = 0, giving e^(t) = 2, so t = ln(2) = 0.693 s.
The peak value is g(0.693) = 10(0.5) - 10(0.25) = 2.5.

### 4.3.4 Sinusoidal Input

A sinusoidal input of the form A × sin(ω × t) is used for frequency response analysis, which is fundamental to control system design using Bode plots, Nyquist plots, and Nichols charts. When a sinusoidal input is applied to a stable LTI system, the steady-state output is also sinusoidal at the same frequency but with a different amplitude and phase shift determined by the system transfer function evaluated at s = j*ω. The frequency response reveals the system bandwidth, resonant peaks, gain margin, and phase margin, all of which are critical parameters for assessing relative stability and designing compensators. Frequency domain methods are particularly valuable because they allow engineers to design controllers using experimentally measured data without requiring an explicit mathematical model of the plant.

**Example 4.3.4:** A stable system has the transfer function G(s) = 5 / (s + 2). A sinusoidal input r(t) = 3 sin(4t) is applied. Find the steady-state output.
**Solution:**
Evaluate G(s) at s = jω where ω = 4 rad/s: G(j4) = 5 / (j4 + 2) = 5 / (2 + j4).
The magnitude is |G(j4)| = 5 / sqrt($2^2 + 4^2$) = 5 / sqrt(20) = 5 / (2sqrt(5)) = sqrt(5)/2 = 1.118.
The phase angle is φ = -arctan(4/2) = -arctan(2) = -63.43 degrees.
The steady-state output is $y_{ss}(t)$ = 3 * 1.118 * sin(4t - 63.43 degrees) = 3.354 sin(4t - 63.43 degrees).
The output has the same frequency as the input but with an amplitude scaled by |G(j4)| and a phase lag of 63.43 degrees.

## 4.4 Transfer Functions and Block Diagrams

Transfer functions and block diagrams are the primary tools for modeling and analyzing linear time-invariant (LTI) control systems. A transfer function expresses the input-output relationship of a system in the Laplace domain as a ratio of polynomials, while block diagrams provide a graphical representation of signal flow through interconnected subsystems. Together, they enable systematic analysis and design of complex control systems by breaking them into manageable components.

### 4.4.1 Transfer Function Representation

The transfer function H(s) of an LTI system is defined as the ratio of the Laplace transform of the output Y(s) to the Laplace transform of the input X(s), assuming zero initial conditions: H(s) = Y(s)/X(s) = ($b_m s^m + b_{m-1} s^{m-1} + ... + b_0$) / ($a_n s^n + a_{n-1} s^{n-1} + ... + a_0$). The roots of the numerator polynomial are called zeros and the roots of the denominator polynomial are called poles. The poles determine the stability and natural response modes of the system: poles in the left half of the s-plane produce decaying responses (stable), poles on the imaginary axis produce sustained oscillations (marginally stable), and poles in the right half produce growing responses (unstable).

The order of a system is the highest power of s in the denominator, which equals the number of energy storage elements in the physical system. First-order systems have one pole and are characterized by their time constant τ and DC gain K. Second-order systems have two poles and are characterized by natural frequency $\omega_n$ and damping ratio ζ, which together determine the transient behavior — overdamped (ζ > 1), critically damped (ζ = 1), underdamped (0 < ζ < 1), or undamped (ζ = 0).

**Example 4.4.1:** A system has the transfer function $H(s) = 50(s + 3) / (s^2 + 8s + 25)$. Determine (a) the poles and zeros, (b) the natural frequency and damping ratio, (c) the DC gain, and (d) whether the system is stable.
**Solution:**
(a) Zero: $s + 3 = 0 \rightarrow s = -3$ (one zero at $s = -3$).
Poles: $s^2 + 8s + 25 = 0 \rightarrow s = (-8 \pm \sqrt{(64 - 100)})/2 = (-8 \pm j6)/2 = \mathbf{-4 \pm j3}$ (complex conjugate poles).
(b) Standard form: $s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 8s + 25$.
$\omega_n = \sqrt{25} = \mathbf{5\ rad/s}$, $2\zeta\omega_n = 8$, so $\zeta = 8/(2\times5) = \mathbf{0.8}$ (underdamped).
(c) DC gain: $H(0) = 50(3)/25 = \mathbf{6}$
(d) Both poles are at $s = -4 \pm j3$, which have negative real parts (left half-plane). The system is **stable**.

### 4.4.2 Block Diagram Algebra

Block diagrams represent control systems as interconnected blocks, where each block contains a transfer function and arrows indicate signal flow direction. The three fundamental interconnections are series (cascade), parallel, and feedback. In a series connection, the overall transfer function is the product of individual transfer functions: $H(s) = H_1(s) \times H_2(s)$. In a parallel connection, the outputs are summed: $H(s) = H_1(s) + H_2(s)$. In a negative feedback configuration, the closed-loop transfer function is $H(s) = G(s) / (1 + G(s)H(s))$, where $G(s)$ is the forward path and $H(s)$ is the feedback path.

Block diagram reduction simplifies complex diagrams to a single equivalent transfer function using systematic rules: moving summing junctions and pickoff points, eliminating feedback loops, and combining series/parallel blocks. The reduction process preserves the overall input-output relationship while progressively simplifying the diagram. For systems with multiple inputs and outputs, superposition applies — each input is considered independently while setting all other inputs to zero, and the total output is the sum of individual contributions.

**Example 4.4.2:** A control system has a forward path with two series blocks $G_1(s) = 10/(s + 2)$ and $G_2(s) = 1/(s + 5)$, a unity feedback path $H(s) = 1$, and a disturbance $D(s)$ entering between $G_1$ and $G_2$. Determine (a) the closed-loop transfer function $Y(s)/R(s)$, (b) the disturbance transfer function $Y(s)/D(s)$, and (c) the steady-state response to a unit step disturbance.
**Solution:**
(a) Forward path: $G(s) = G_1(s) \times G_2(s) = 10 / [(s + 2)(s + 5)]$.
Closed-loop: $Y(s)/R(s) = G(s) / (1 + G(s)) = 10 / [(s + 2)(s + 5) + 10] = \mathbf{10 / (s^2 + 7s + 20)}$
(b) For disturbance D entering between $G_1$ and $G_2$:
$Y(s)/D(s) = G_2(s) / (1 + G_1(s)G_2(s)) = (s + 2) / [(s + 2)(s + 5) + 10] = \mathbf{(s + 2) / (s^2 + 7s + 20)}$
(c) Steady-state response to unit step disturbance: $y_{ss} = \lim(s\rightarrow0)\ s \times [Y(s)/D(s)] \times (1/s) = (0 + 2)/(0 + 0 + 20) = \mathbf{0.1}$. The feedback reduces the disturbance effect from what would be $G_2(0) = 0.2$ in open-loop to 0.1 — a factor of 2 attenuation.

### 4.4.3 Signal Flow Graphs

Signal flow graphs (SFGs) provide an alternative to block diagrams for representing system interconnections, using nodes for signals and directed branches for transfer functions. Mason's gain formula computes the overall transfer function directly from the graph without sequential reduction: $T(s) =$

$(1/\Delta) \times \Sigma P_k\Delta_k$, where $P_k$ is the gain of the k-th forward path, $\Delta = 1 -$ (sum of individual loop gains) $+$ (sum of products of non-touching loop gains taken two at a time) $- \ldots$, and $\Delta_k$ is the cofactor for the k-th forward path ($\Delta$ evaluated with all loops touching that path removed).

SFGs are particularly useful when there are multiple forward paths and multiple feedback loops, where block diagram reduction would require many steps. The key terminology includes: a forward path is any path from input to output that does not pass through any node more than once, and a loop is a closed path that starts and ends at the same node.

**Example 4.4.3:** A system has the signal flow graph with: input R to node $X_1$ with gain 1, $X_1$ to $X_2$ with gain $G_1 = 5$, $X_2$ to output Y with gain $G_2 = 2$, $X_2$ to $X_1$ with gain $H_1 = -0.5$ (feedback loop 1), and Y to $X_1$ with gain $H_2 = -0.1$ (feedback loop 2). Apply Mason's gain formula to find Y/R.
**Solution:**
Forward path: $P_1 = 1 \times G_1 \times G_2 = 1 \times 5 \times 2 = 10$.
Loop gains:
$L_1 = G_1 \times H_1 = 5 \times (-0.5) = -2.5$ (loop: $X_1 \to X_2 \to X_1$)
$L_2 = G_1 \times G_2 \times H_2 = 5 \times 2 \times (-0.1) = -1.0$ (loop: $X_1 \to X_2 \to Y \to X_1$)
Non-touching loops: $L_1$ and $L_2$ share nodes $X_1$ and $X_2$, so they are touching. No non-touching pairs.
Graph determinant: $\Delta = 1 - (L_1 + L_2) = 1 - (-2.5 - 1.0) = 1 + 3.5 = 4.5$.
Cofactor: $\Delta_1 = 1$ (all loops touch the forward path).
**$Y/R = P_1\Delta_1/\Delta = 10 \times 1 / 4.5 = 2.22$**

## 4.5 Time-Domain Performance

Time-domain performance specifications define how a control system responds to standard test inputs, particularly the unit step. These specifications provide quantitative criteria for evaluating and comparing controller designs and are directly related to the system's pole locations in the s-plane.

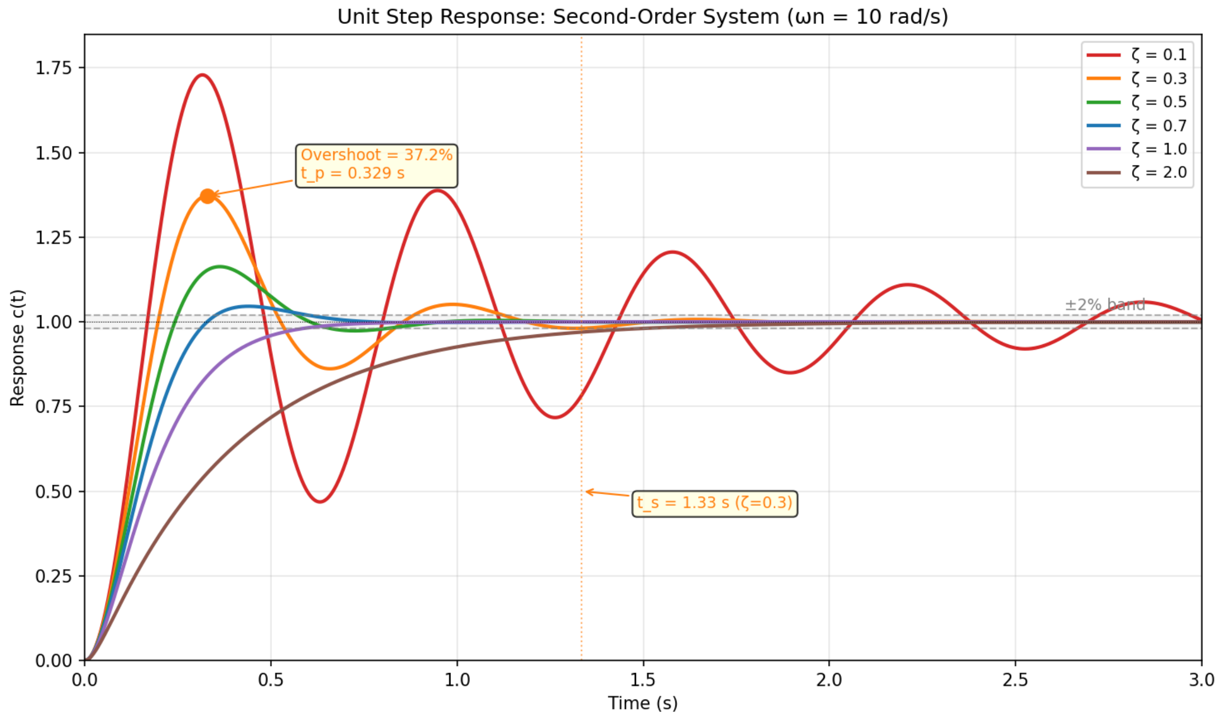



Figure 4.5: Step Response: Second-Order System

### 4.5.1 First-Order System Response

A first-order system has the standard form transfer function $G(s) = K / (\tau s + 1)$, where K is the DC gain and $\tau$ is the time constant. The unit step response is $c(t) = K(1 - e^{-t/\tau})$, which rises monotonically toward the final value K without overshoot. The time constant $\tau$ is the time to reach 63.2% of the final value, and the system reaches 95% at $3\tau$, 98% at $4\tau$, and 99.3% at $5\tau$. The rise time (10% to 90%) is $t_r = 2.2\tau$, and the 2% settling time is $t_s = 4\tau$.

First-order systems appear in many practical applications: thermal systems (heating/cooling), RC circuits, first-order chemical reactions, and liquid level systems. The frequency response of a first-order system has a single break frequency at $\omega = 1/\tau$, beyond which the magnitude rolls off at $-20$ dB/decade.

**Example 4.5.1:** A temperature control system has a transfer function $G(s) = 80 / (25s + 1)$, where the input is heater power (0-100%) and the output is temperature in °C. Determine (a) the steady-state temperature for 50% heater input, (b) the time constant, (c) the time to reach 95% of final temperature, and (d) the 10-90% rise time.
**Solution:**
(a) DC gain $K = 80$, so for 50% input: $T_{ss} = 80 \times 0.5 =$ **40°C**
(b) Time constant: $\tau =$ **25 seconds**
(c) Time to 95%: $t = 3\tau = 3 \times 25 =$ **75 seconds**
(d) Rise time: $t_r = 2.2\tau = 2.2 \times 25 =$ **55 seconds**

### 4.5.2 Second-Order System Response

The standard second-order system has transfer function $G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$, where $\omega_n$ is the undamped natural frequency and $\zeta$ is the damping ratio. The damping ratio determines the character of the step response: for $\zeta < 1$ (underdamped), the system oscillates with decaying amplitude, with the damped frequency $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$. The key performance specifications are:

- **Rise time**: $t_r \approx (1.8 - 0.2\zeta) / \omega_n$ for underdamped systems (approximate)
- **Peak time**: $t_p = \pi / \omega_d = \pi / (\omega_n\sqrt{(1 - \zeta^2)})$
- **Percent overshoot**: $\%OS = 100 \times e^{-\zeta\pi/\sqrt{(1-\zeta^2)}}$
- **Settling time** (2%): $t_s \approx 4 / (\zeta\omega_n)$

These specifications create competing demands: reducing overshoot requires increasing $\zeta$, but this also increases rise time. The poles of the second-order system are at $s = -\zeta\omega_n \pm j\omega_n\sqrt{(1 - \zeta^2)}$, and lines of constant $\zeta$ are radial lines from the origin, while lines of constant $\omega_n$ are circles centered at the origin.

**Example 4.5.2:** A servo motor position control system has a closed-loop transfer function $G(s) = 900 / (s^2 + 24s + 900)$. Determine (a) $\omega_n$ and $\zeta$, (b) the percent overshoot, (c) the peak time, (d) the 2% settling time, and (e) the pole locations.
**Solution:**
(a) Comparing with $\omega_n^2 = 900$ and $2\zeta\omega_n = 24$:
$\omega_n = 30$ rad/s, $\zeta = 24/(2\times30) = \mathbf{0.4}$ (underdamped).
(b) $\%OS = 100 \times e^{-0.4\pi/\sqrt{(1-0.16)}} = 100 \times e^{-0.4\pi/0.9165} = 100 \times e^{-1.371} = 100 \times 0.254 = \mathbf{25.4\%}$
(c) $\omega_d = 30\sqrt{(1 - 0.16)} = 30 \times 0.9165 = 27.49$ rad/s.
$t_p = \pi/\omega_d = \pi/27.49 = \mathbf{0.114\ s}$
(d) $t_s = 4/(\zeta\omega_n) = 4/(0.4 \times 30) = 4/12 = \mathbf{0.333\ s}$
(e) Poles: $s = -12 \pm j27.49 = \mathbf{-12 \pm j27.5}$

### 4.5.3 Steady-State Error

Steady-state error is the difference between the desired reference input and the actual output as time approaches infinity. For unity feedback systems, the steady-state error depends on the system type number (the number of free integrators in the open-loop transfer function) and the type of input signal. The error constants — position constant $K_p$, velocity constant $K_v$, and acceleration constant $K_a$ — quantify the steady-state tracking capability:

| System Type | Step Error ($1/1+K_p$) | Ramp Error ($1/K_v$) | Parabola Error ($1/K_a$) |
|---|---|---|---|
| Type 0 | $1/(1+K_p)$ | $\infty$ | $\infty$ |
| Type 1 | 0 | $1/K_v$ | $\infty$ |
| Type 2 | 0 | 0 | $1/K_a$ |

where $K_p = \lim(s\to0)\ G(s)$, $K_v = \lim(s\to0)\ sG(s)$, and $K_a = \lim(s\to0)\ s^2G(s)$.

**Example 4.5.3:** A unity feedback system has $G(s) = 500(s + 2) / [s^2(s + 10)(s + 25)]$. Determine (a) the system type, (b) the appropriate error constant, and (c) the steady-state error for an input $r(t) = 3t^2$ (a parabolic input with magnitude 3).
**Solution:**
(a) $G(s)$ has two free integrators ($s^2$ in the denominator), so this is a **Type 2** system.
(b) For a Type 2 system, the relevant error constant for a parabolic input is:
$K_a = \lim(s\to0)\, s^2G(s) = \lim(s\to0)\, s^2 \times 500(s+2)/[s^2(s+10)(s+25)] = 500(2)/(10\times25) = \mathbf{4}$
(c) For a parabolic input $r(t) = (A/2)t^2$, where $A = 6$ (since $r(t) = 3t^2 = (6/2)t^2$):
$e_{ss} = A/K_a = 6/4 = \mathbf{1.5}$

## 4.6 PID Control

The Proportional-Integral-Derivative (PID) controller is the most widely used controller in industrial process control, found in over 90% of control loops. The PID controller computes a control signal $u(t)$ based on the error $e(t) = r(t) - y(t)$ as: $u(t) = K_pe(t) + K_i\int e(\tau)d\tau + K_d(de/dt)$, where $K_p$ is the proportional gain, $K_i$ is the integral gain, and $K_d$ is the derivative gain. In the Laplace domain, the PID transfer function is $G_c(s) = K_p + K_i/s + K_ds$.

### 4.6.1 Proportional Control

Proportional control generates a corrective action directly proportional to the error signal: $u(t) = K_p \times e(t)$. Increasing $K_p$ reduces steady-state error and speeds up the response, but excessively high gain leads to oscillations and eventual instability. For a Type 0 system with proportional control, the steady-state error to a step input is $e_{ss} = 1/(1 + K_pG(0))$, which can be made small but never zero. Proportional control alone cannot eliminate steady-state error to a step input in a Type 0 system, which is a fundamental limitation that motivates the addition of integral action.

**Example 4.6.1:** A proportional controller with gain $K_p$ controls a plant $G(s) = 1/(s + 4)$ in a unity feedback configuration. Determine (a) the closed-loop transfer function, (b) the steady-state error to a unit step for $K_p = 20$, and (c) the value of $K_p$ that yields a 2% settling time of 0.5 s.
**Solution:**
(a) Closed-loop: $T(s) = K_pG(s) / (1 + K_pG(s)) = K_p / (s + 4 + K_p) = \mathbf{K_p / (s + 4 + K_p)}$
(b) Position error constant: $K_{pos} = \lim(s\to0)\, K_pG(s) = \lim(s\to0)\, K_p/(s+4) = 20/4 = 5$.
$e_{ss} = 1/(1 + K_{pos}) = 1/(1 + 5) = \mathbf{0.167}$ (16.7% error)
(c) This is first-order with pole at $s = -(4 + K_p)$. Time constant $\tau = 1/(4 + K_p)$.
Settling time: $t_s = 4\tau = 4/(4 + K_p) = 0.5\text{ s} \to 4 + K_p = 8 \to K_p = \mathbf{4}$

### 4.6.2 Integral Control

Integral control adds a term proportional to the accumulated error over time: $u_i(t) = K_i\int e(\tau)d\tau$. The integral action introduces a pole at $s = 0$ (a free integrator) in the controller, which increases the system type by one. This guarantees zero steady-state error for step inputs in systems that would otherwise have finite error with proportional control alone. However, the additional phase lag (−90° at all frequencies) from the integrator reduces the phase margin and can destabilize the system if $K_i$ is too large.

Integral windup is a practical problem that occurs when the actuator saturates (reaches its physical limits) while the integral term continues to accumulate error. When the error eventually changes sign, the large accumulated integral must unwind before the controller can respond appropriately, causing excessive overshoot. Anti-windup strategies include clamping the integrator, conditionally disabling integration during saturation, and back-calculation methods.

**Example 4.6.2:** A PI controller ($K_p = 5$, $K_i = 10$) controls a plant $G(s) = 2/(s + 1)$ with unity feedback. Determine (a) the open-loop transfer function, (b) the system type, (c) the steady-state error to a unit ramp, and (d) the closed-loop poles.
**Solution:**
(a) Controller: $G_c(s) = K_p + K_i/s = (5s + 10)/s$.
Open-loop: $G_{OL}(s) = G_c(s) \times G(s) = 2(5s + 10) / [s(s + 1)] = \mathbf{(10s + 20) / [s(s + 1)]}$
(b) One free integrator in the open-loop → **Type 1** system (proportional-only was Type 0).
(c) $K_v = \lim(s\to0)\ s \times (10s + 20)/[s(s + 1)] = 20/1 = 20$.
$e_{ss} = 1/K_v = 1/20 = \mathbf{0.05}$ (5% position lag for a ramp).
(d) Closed-loop characteristic equation: $s(s + 1) + 10s + 20 = s^2 + 11s + 20 = 0$.
$s = (-11 \pm \sqrt{(121 - 80)})/2 = (-11 \pm \sqrt{41})/2 = (-11 \pm 6.40)/2 = \mathbf{-2.30\ and\ -8.70}$ (both stable, overdamped).

### 4.6.3 Derivative Control

Derivative control adds a term proportional to the rate of change of the error: $u_d(t) = K_d \times de/dt$. Derivative action anticipates future error by responding to its rate of change, providing a damping effect that reduces overshoot and improves transient response. In the frequency domain, the derivative term introduces a zero at $s = 0$, adding positive phase shift that can improve the phase margin. However, derivative control amplifies high-frequency noise, so a first-order low-pass filter is typically applied: the practical derivative term becomes $K_d s / (\tau_f s + 1)$, where $\tau_f$ is a small filter time constant (typically $\tau_f = K_d/(8 \text{ to } 20 \times K_p)$).

Derivative control is never used alone because it provides no corrective action for constant errors. It is always combined with proportional control (PD) or proportional-integral control (PID). The effect of derivative action is often described as "applying the brakes" — it opposes rapid changes in the error, slowing the approach to the setpoint and reducing overshoot.

**Example 4.6.3:** A PD controller ($K_p = 50$, $K_d = 5$) controls a plant $G(s) = 1/(s^2 + 2s)$. Determine (a) the closed-loop transfer function, (b) the natural frequency and damping ratio, (c) the percent overshoot, and (d) compare to proportional-only control ($K_d = 0$).
**Solution:**
(a) PD controller: $G_c(s) = 50 + 5s$.
Closed-loop: $T(s) = (5s + 50)/(s^2 + 2s + 5s + 50) = \mathbf{(5s + 50)/(s^2 + 7s + 50)}$
(b) $\omega_n = \sqrt{50} = 7.07$ rad/s, $2\zeta\omega_n = 7$, $\zeta = 7/(2 \times 7.07) = \mathbf{0.495}$
(c) $\%OS = 100 \times e^{-0.495\pi/\sqrt{(1-0.245)}} = 100 \times e^{-0.495\pi/0.869} = 100 \times e^{-1.789} = \mathbf{16.7\%}$
(d) With $K_d = 0$ (P-only): $T(s) = 50/(s^2 + 2s + 50)$, $\zeta = 2/(2 \times 7.07) = 0.141$.
$\%OS = 100 \times e^{-0.141\pi/0.990} = 100 \times e^{-0.447} = 64.0\%$.
The derivative action reduced overshoot from **64.0% to 16.7%** by increasing the damping ratio from 0.141 to 0.495.

### 4.6.4 PID Tuning

PID tuning determines the values of $K_p$, $K_i$, and $K_d$ that achieve satisfactory closed-loop performance. The Ziegler-Nichols method is the most well-known empirical tuning approach, available in two forms. The first method (step response) applies a step input to the open-loop plant and measures the reaction curve parameters: the dead time L and the time constant T from the maximum slope tangent line. The second method (ultimate gain) increases $K_p$ with $K_i = K_d = 0$ until sustained oscillations occur at the ultimate gain $K_u$ with period $P_u$.

Ziegler-Nichols ultimate gain tuning rules:

| Controller | $K_p$ | $K_i$ | $K_d$ |
|---|---|---|---|
| P | $0.5\ K_u$ | — | — |
| PI | $0.45\ K_u$ | $0.54\ K_u/P_u$ | — |
| PID | $0.6\ K_u$ | $1.2\ K_u/P_u$ | $0.075\ K_u P_u$ |

These rules typically produce a response with 25% overshoot and are used as a starting point for further manual refinement. Other tuning methods include Cohen-Coon, internal model control (IMC), and software-based optimization using performance criteria such as ITAE (Integral of Time-weighted Absolute Error).

**Example 4.6.4:** A temperature control loop has an ultimate gain of $K_u = 8.5$ and an ultimate period of $P_u = 12$ s. Using the Ziegler-Nichols ultimate gain method, determine the PID parameters and the resulting controller transfer function.
**Solution:**
$K_p = 0.6 \times K_u = 0.6 \times 8.5 = \mathbf{5.1}$
$K_i = 1.2 \times K_u\ /\ P_u = 1.2 \times 8.5\ /\ 12 = \mathbf{0.85\ s^{-1}}$
$K_d = 0.075 \times K_u \times P_u = 0.075 \times 8.5 \times 12 = \mathbf{7.65\ s}$
Controller transfer function:
$G_c(s) = 5.1 + 0.85/s + 7.65s = \mathbf{(7.65s^2 + 5.1s + 0.85)\ /\ s}$
The integral time $T_i = K_p/K_i = 5.1/0.85 = 6.0$ s and derivative time $T_d = K_d/K_p = 7.65/5.1 = 1.5$ s, consistent with the alternative Ziegler-Nichols form: $T_i = P_u/2 = 6.0$ s and $T_d = P_u/8 = 1.5$ s.

## 4.7 Stability Analysis

Stability is the most fundamental requirement of any control system — an unstable system produces unbounded outputs and is useless or dangerous. A linear time-invariant system is stable if all poles of its closed-loop transfer function lie in the open left half of the s-plane (negative real parts). Stability analysis methods determine whether a system is stable and how much margin exists before instability, without necessarily computing the exact pole locations.

### 4.7.1 Routh-Hurwitz Criterion

The Routh-Hurwitz criterion determines the number of closed-loop poles in the right half-plane (RHP) by examining the coefficients of the characteristic polynomial without solving for the roots. For a characteristic equation $a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0 = 0$, the Routh array is constructed as a triangular table. A necessary condition for stability is that all coefficients have the same sign. The sufficient condition is that all entries in the first column of the Routh array are positive — the number of sign changes in the first column equals the number of RHP poles.

Special cases arise when a first-column element is zero (replace with a small positive $\varepsilon$ and proceed) or when an entire row is zero (indicating symmetric roots about the origin, such as a pair of purely imaginary poles). The Routh criterion is especially useful for determining the range of a parameter (such as controller gain K) for which the system remains stable.

**Example 4.7.1:** A unity feedback system has the open-loop transfer function $G(s) = K / [s(s + 2)(s + 5)]$. Determine the range of K for stability using the Routh-Hurwitz criterion.
**Solution:**
Closed-loop characteristic equation: $s^3 + 7s^2 + 10s + K = 0$.
Routh array:
| $s^3$ | 1 | 10 |
| $s^2$ | 7 | K |
| $s^1$ | (70 − K)/7 | 0 |
| $s^0$ | K | |
For stability, all first-column entries must be positive:
- $s^3$: $1 > 0$ ✓
- $s^2$: $7 > 0$ ✓
- $s^1$: $(70 - K)/7 > 0 \rightarrow K < 70$
- $s^0$: $K > 0$
Range of K for stability: $\mathbf{0 < K < 70}$. At $K = 70$, the $s^1$ row becomes zero, indicating purely imaginary poles (sustained oscillation at the boundary of stability).

### 4.7.2 Root Locus

The root locus is a graphical method that shows how the closed-loop poles of a system move in the s-plane as a parameter (typically the loop gain K) varies from 0 to infinity. For a system with open-loop transfer function $KG(s)H(s)$, the closed-loop poles satisfy $1 + KG(s)H(s) = 0$, and the root locus is the set of all points s where the angle of $G(s)H(s)$ equals $\pm 180°(2k+1)$. Key construction rules include:

1. The root locus starts at the open-loop poles ($K = 0$) and ends at the open-loop zeros ($K \rightarrow \infty$)
2. The number of branches equals the number of open-loop poles
3. Branches on the real axis lie to the left of an odd number of real poles and zeros
4. Asymptotes approach angles of $(2k+1) \times 180°/(n-m)$ at the centroid $\sigma_a = (\Sigma\text{poles} - \Sigma\text{zeros})/(n-m)$
5. Breakaway/break-in points occur where $dK/ds = 0$

The root locus reveals how gain affects stability (does the locus cross the imaginary axis?), transient response (do the poles move toward higher or lower damping?), and the presence of conditionally stable regions.

**Example 4.7.2:** Plot the root locus for $G(s)H(s) = 1/[s(s + 3)(s + 5)]$ and determine (a) the asymptote angles and centroid, (b) the breakaway point, and (c) the imaginary axis crossing (gain and frequency at which the system becomes unstable).

**Solution:**
Open-loop poles: 0, −3, −5 (n = 3 poles, m = 0 zeros).
(a) Asymptote angles: (2k+1) × 180°/3 = **60°, 180°, 300°**
Centroid: $\sigma_a = (0 - 3 - 5)/3 =$ **−2.67**
(b) Characteristic equation: $K = -s(s+3)(s+5) = -(s^3 + 8s^2 + 15s)$.
$dK/ds = -(3s^2 + 16s + 15) = 0 \rightarrow s = (-16 \pm \sqrt{(256-180)})/6 = (-16 \pm 8.72)/6$.
Breakaway point (on locus, between 0 and −3): s = (−16 + 8.72)/6 = **−1.21**
(c) Applying the Routh–Hurwitz criterion to the characteristic equation $s^3 + 8s^2 + 15s + K = 0$:
Routh $s^1$ row: (8×15 − K)/8 = (120 − K)/8 = 0 → K = **120**.
Auxiliary equation at K = 120: $8s^2 + 120 = 0 \rightarrow s = \pm j\sqrt{15} =$ **±j3.87 rad/s**.

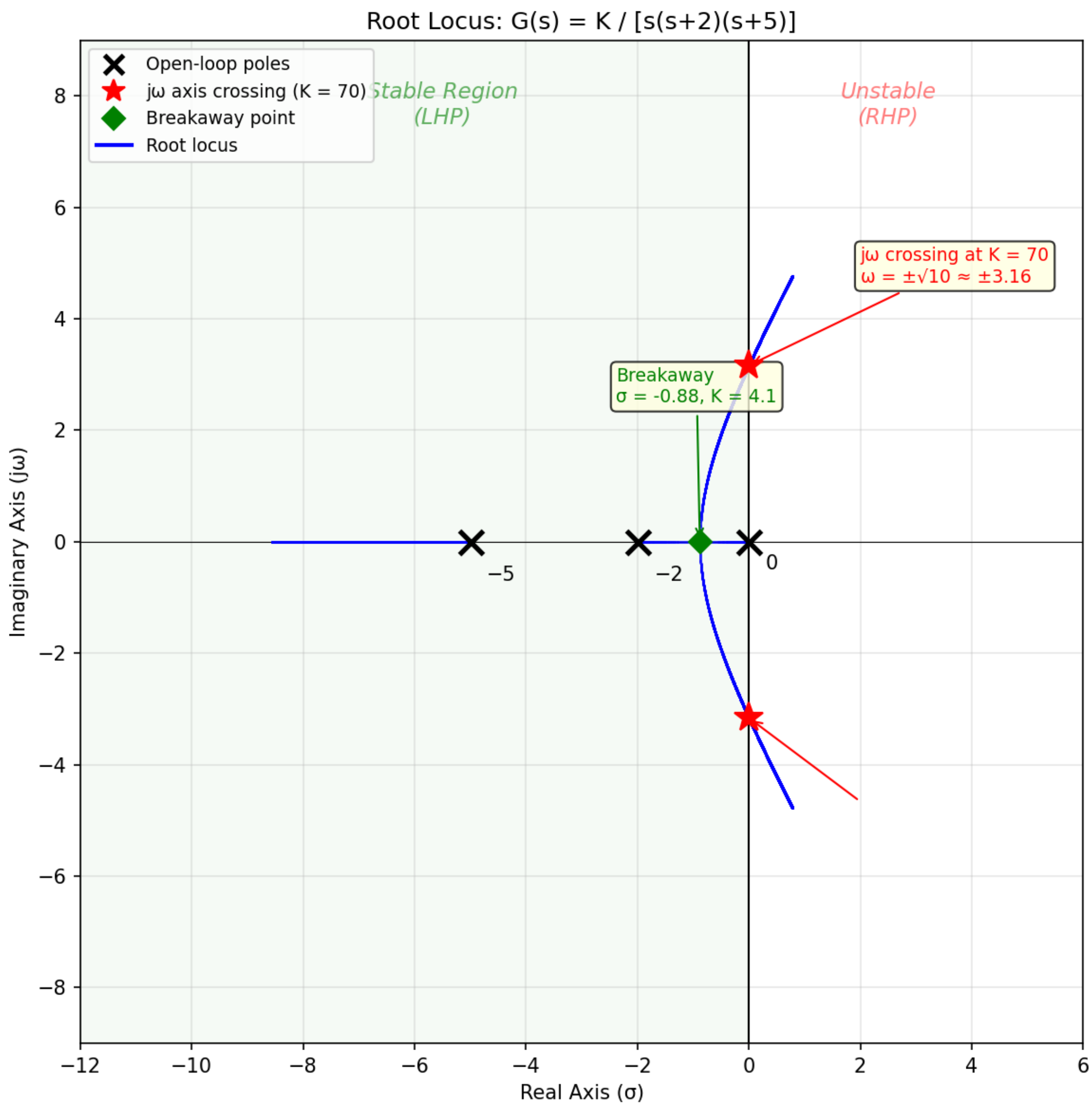



Figure 4.7.2: Root Locus Plot

## 4.8 Frequency Response Design

Frequency response methods analyze and design control systems using the system's response to sinusoidal inputs across a range of frequencies. These methods are powerful because they can be applied using experimentally measured data (no mathematical model required), they provide direct insight into bandwidth, stability margins, and noise rejection, and they form the basis for designing lead, lag, and lead-lag compensators.

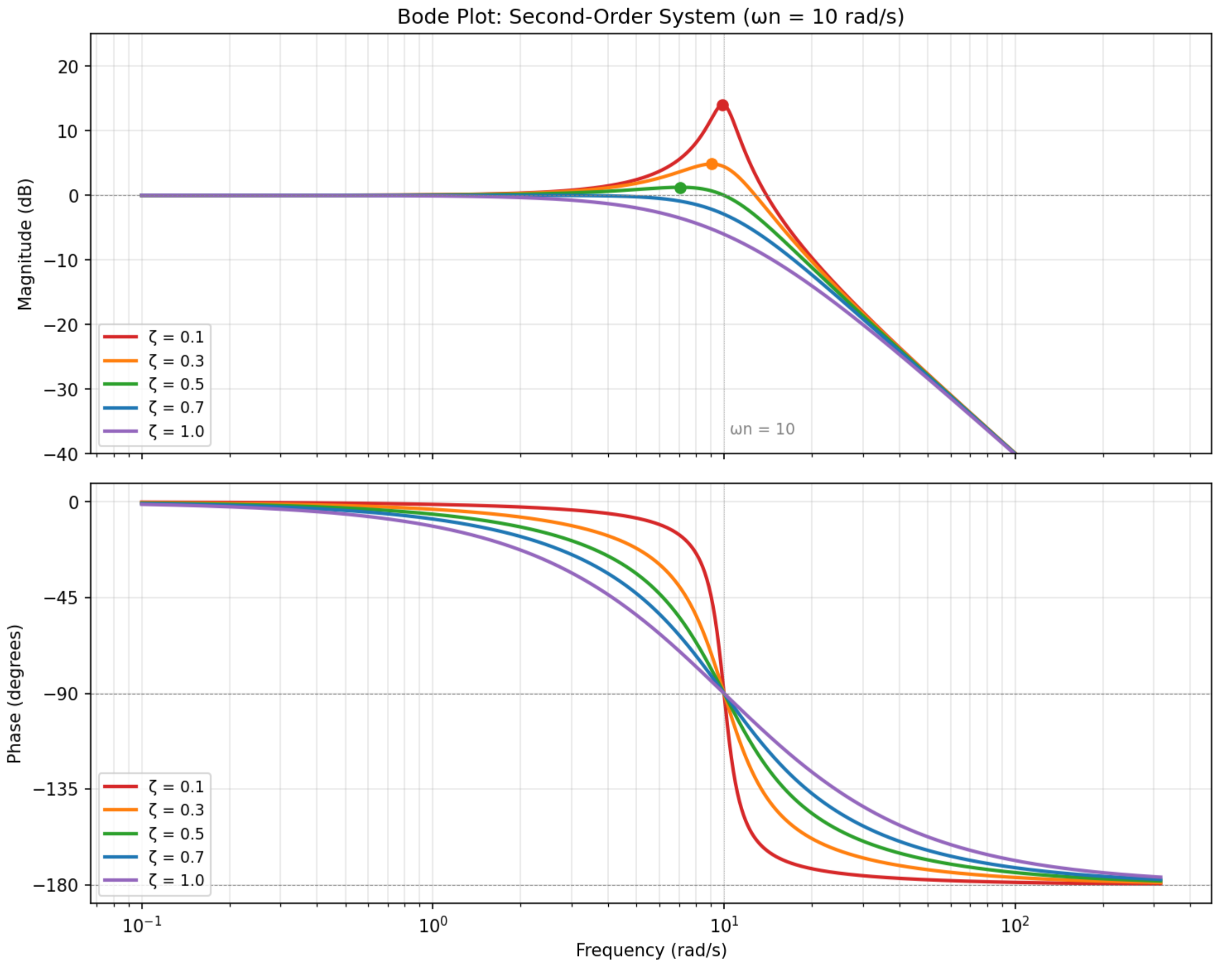



Figure 4.8: Bode Plot: Second-Order System

### 4.8.1 Bode Plots

Bode plots consist of two graphs: magnitude (in dB) versus frequency (log scale) and phase (in degrees) versus frequency (log scale). They are constructed by factoring the transfer function into standard forms and using straight-line approximations. The basic building blocks are: a constant gain K (flat line at $20 \log_{10}|K|$ dB), a pole at the origin (−20 dB/decade slope, −90° phase), a zero at the origin (+20 dB/decade, +90°), a real pole at $s = -a$ (−20 dB/decade above $\omega = a$, −45°/decade phase centered at $\omega = a$), and a real zero (mirror image of a pole). Complex conjugate poles produce a resonant peak near $\omega_n$ with height dependent on $\zeta$.

The bandwidth $\omega_{BW}$ (the frequency where the magnitude drops to −3 dB from its DC value) is directly related to the speed of the closed-loop response: roughly $\omega_{BW} \approx 4/(t_s\zeta)$ for second-order systems. Larger bandwidth means faster response but also greater susceptibility to high-frequency noise.

**Example 4.8.1:** Construct the Bode magnitude and phase asymptotic plots for $G(s) = 100(s + 10) / [s(s + 100)]$ and determine (a) the gain crossover frequency, (b) the DC slope, and (c) the high-frequency slope.

**Solution:**
Rewrite in standard form: $G(s) = 100 \times 10 \times (s/10 + 1) / [s \times 100 \times (s/100 + 1)] = 10(s/10 + 1) / [s(s/100 + 1)]$.
DC gain factor: 10 → 20 dB. Corner frequencies: $\omega_1 = 10$ rad/s (zero), $\omega_2 = 100$ rad/s (pole).
(a) Magnitude at frequency ω: $|G(j\omega)| = 10/\omega$ for $\omega < 10$ (slope −20 dB/dec). At $\omega = 10$: gain = $10/10 = 1 = 0$ dB. The zero at $\omega = 10$ flattens the slope to 0 dB/dec. At $\omega = 100$, the pole resumes −20 dB/dec. Gain crossover (0 dB): occurs at **ω = 10 rad/s** where the magnitude first crosses 0 dB.
(b) DC slope: −20 dB/decade (due to the 1/s pole at origin).
(c) High-frequency slope ($\omega \gg 100$): **−20 dB/decade** (one pole at origin + one zero + one pole = net −20 dB/dec).

### 4.8.2 Gain and Phase Margins

Gain margin (GM) and phase margin (PM) are quantitative measures of how close a stable system is to instability. The gain margin is the amount of additional gain (in dB) needed at the phase crossover frequency (where the phase equals −180°) to make the system unstable. The phase margin is the amount of additional phase lag (in degrees) needed at the gain crossover frequency (where the magnitude equals 0 dB) to reach −180°. Adequate stability margins typically require GM ≥ 6 dB and PM ≥ 30-60°.

For second-order systems, the phase margin is directly related to the damping ratio: $PM \approx 100\zeta$ for $\zeta < 0.7$, or more precisely $\zeta \approx PM/100$ for $PM < 70°$. This provides a direct connection between frequency-domain specifications and time-domain performance. A system with a phase margin of 45° will have a damping ratio of approximately 0.45 and about 20% overshoot.

**Example 4.8.2:** A unity feedback system has $G(s) = 50 / [s(s + 2)(s + 10)]$. Determine (a) the gain crossover frequency, (b) the phase margin, (c) the phase crossover frequency, and (d) the gain margin.
**Solution:**
(a) At gain crossover, $|G(j\omega)| = 1$:
$50 / [\omega \times \sqrt{(\omega^2 + 4)} \times \sqrt{(\omega^2 + 100)}] = 1$.
Simplify: $2{,}500 = \omega^2(\omega^2 + 4)(\omega^2 + 100)$. Trial: $\omega = 2$: $4 \times 8 \times 104 = 3{,}328$ (too high). $\omega = 2.2$: $4.84 \times 8.84 \times 104.84 = 4{,}487$ (too high). $\omega = 1.8$: $3.24 \times 7.24 \times 103.24 = 2{,}422$ (close). By interpolation, $\omega_{gc} \approx$ **1.82 rad/s**.
(b) Phase at $\omega_{gc}$: $\angle G(j1.82) = -90° - \arctan(1.82/2) - \arctan(1.82/10) = -90° - 42.3° - 10.3° = -142.6°$.
PM = 180° − 142.6° = **37.4°** (adequate but modest).
(c) Phase crossover ($\angle G = -180°$): $-90° - \arctan(\omega/2) - \arctan(\omega/10) = -180°$.
$\arctan(\omega/2) + \arctan(\omega/10) = 90°$. This occurs when $(\omega/2)(\omega/10) = 1 \rightarrow \omega^2 = 20 \rightarrow \omega_{pc} =$ **4.47 rad/s**.
(d) $|G(j4.47)| = 50/(4.47 \times \sqrt{(20+4)} \times \sqrt{(20+100)}) = 50/(4.47 \times 4.90 \times 10.95) = 50/239.8 = 0.2085 = -13.6$ dB.
GM = **13.6 dB** (comfortable margin).

### 4.8.3 Nyquist Criterion

The Nyquist stability criterion uses the open-loop frequency response $G(j\omega)H(j\omega)$ plotted as a polar curve (the Nyquist plot) to determine closed-loop stability. The criterion states: the number of closed-loop RHP poles Z equals the number of open-loop RHP poles P plus the number of clockwise encirclements N of the critical point $(-1, 0)$: $Z = P + N$. For a stable open-loop system ($P = 0$), the closed-loop system is stable if and only if the Nyquist plot does not encircle the $(-1, 0)$ point.

The Nyquist criterion is more general than the Routh-Hurwitz criterion because it handles time delays (which introduce infinite-order characteristic equations), open-loop unstable systems, and systems defined only by measured frequency response data. The distance from the Nyquist curve to the $(-1, 0)$ point relates to the gain and phase margins, and the inverse of this minimum distance is the sensitivity peak $M_s$, which should typically be kept below 2 (6 dB).

**Example 4.8.3:** An open-loop stable system has $G(j\omega)H(j\omega)$ that passes through the real axis at $-0.4$ when $\omega = 5$ rad/s. The Nyquist plot makes zero clockwise encirclements of $(-1, 0)$. Determine (a) whether the closed-loop system is stable, (b) the gain margin, and (c) the maximum additional gain before instability.
**Solution:**
(a) $P = 0$ (open-loop stable), $N = 0$ (no encirclements).
$Z = P + N = 0$. The closed-loop system is **stable** (no RHP poles).
(b) The Nyquist plot crosses the negative real axis at $-0.4$. The gain margin is the factor by which the gain could be increased before the plot passes through $(-1, 0)$:
$GM = 1/0.4 = 2.5 = 20\log_{10}(2.5) =$ **7.96 dB**
(c) The system becomes unstable when the gain is multiplied by 2.5, so the maximum additional gain is a factor of **2.5** (or 7.96 dB above the current gain).

### 4.8.4 Lead-Lag Compensation

A lead compensator has the transfer function $G_c(s) = K_c(s + z)/(s + p)$ where $p > z$, placing the zero closer to the origin than the pole. The lead network contributes positive phase shift near the geometric mean frequency $\omega_m = \sqrt{(zp)}$, which increases the phase margin and improves transient response. The maximum phase lead is $\varphi_{max} = \arcsin((p - z)/(p + z))$, and the attenuation factor $\alpha = z/p < 1$ determines the amount of phase boost available (smaller $\alpha$ gives more phase lead but requires more gain compensation).

A lag compensator uses the same form but with $z > p$, placing the pole closer to the origin. The lag network boosts low-frequency gain to reduce steady-state error without significantly affecting the phase margin at the crossover frequency, provided the lag corner frequencies are placed well below the gain crossover. A lead-lag compensator combines both networks to simultaneously improve transient response (via lead) and steady-state accuracy (via lag). The design procedure typically involves: (1) set $K_c$ to meet the steady-state error requirement, (2) determine the additional phase lead needed at the desired crossover frequency, (3) compute $\alpha = (1 - \sin\varphi_{max})/(1 + \sin\varphi_{max})$, and (4) place the compensator zero and pole symmetrically around the crossover frequency.

**Example 4.8.4:** A unity feedback system has $G(s) = 20 / [s(s + 2)]$. The current phase margin is insufficient. Design a lead compensator to achieve a phase margin of 45° at a gain crossover

frequency of $\omega_{gc} = 5$ rad/s.
**Solution:**
Current phase at $\omega = 5$: $\angle G(j5) = -90° - \arctan(5/2) = -90° - 68.2° = -158.2°$. Current PM = $180° - 158.2° = 21.8°$.
Additional phase needed: $45° - 21.8° + 5°$ (safety margin) = $28.2°$.
Compute $\alpha$: $\alpha = (1 - \sin 28.2°)/(1 + \sin 28.2°) = (1 - 0.4726)/(1 + 0.4726) = 0.527/1.473 = \mathbf{0.358}$.
Compensator zero and pole: $z = \omega_{gc}\sqrt{\alpha} = 5 \times 0.598 = 2.99$ rad/s, $p = \omega_{gc}/\sqrt{\alpha} = 5/0.598 = 8.36$ rad/s.
Gain adjustment: $|G(j5)| = 20/(5 \times \sqrt{(29)}) = 20/26.93 = 0.743$. The lead compensator attenuation at $\omega_{gc}$ is $\sqrt{\alpha} = 0.598$, so $K_c = 1/(0.743 \times 0.598) = \mathbf{2.25}$.
Lead compensator: $G_c(s) = 2.25(s + 2.99)/(s + 8.36)$. Verification: new PM $\approx 21.8° + 28.2° = \mathbf{50°}$ (exceeds 45° target due to the 5° safety margin).

## 4.9 State-Space Representation

State-space representation describes a system using a set of first-order differential equations rather than a single higher-order transfer function. This formulation handles multiple-input multiple-output (MIMO) systems naturally, accommodates initial conditions, and provides a framework for modern control design methods including optimal control, adaptive control, and robust control.

### 4.9.1 State-Space Model

The state-space model consists of two matrix equations: the state equation $\dot{x}(t) = Ax(t) + Bu(t)$ and the output equation $y(t) = Cx(t) + Du(t)$, where x is the n×1 state vector, u is the m×1 input vector, y is the p×1 output vector, A is the n×n system matrix, B is the n×m input matrix, C is the p×n output matrix, and D is the p×m feedthrough matrix. The state variables represent the minimum set of variables that completely describe the system's internal condition — typically quantities associated with energy storage elements (inductor currents, capacitor voltages, position, velocity).

The transfer function can be recovered from the state-space model as $H(s) = C(sI - A)^{-1}B + D$. The eigenvalues of the A matrix are the system poles, and the system is stable if all eigenvalues have negative real parts. A key advantage of the state-space approach is that internal system behavior (not visible at the output) can be analyzed through the state variables, enabling detection and control of internal instabilities that transfer-function methods might miss.

**Example 4.9.1:** An RLC circuit has the state-space model with states $x_1$ = capacitor voltage and $x_2$ = inductor current, where L = 0.5 H, C = 0.1 F, R = 3 Ω, input u = source voltage, and output y = capacitor voltage. Write the state-space matrices and determine the eigenvalues.
**Solution:**
From the circuit equations: $L(dx_2/dt) = u - Rx_2 - x_1$ and $C(dx_1/dt) = x_2$.
State equation:
$\dot{x}_1 = (1/C)x_2 = 10x_2$
$\dot{x}_2 = (-1/L)x_1 - (R/L)x_2 + (1/L)u = -2x_1 - 6x_2 + 2u$
$\mathbf{A} = [0, 10; -2, -6]$, $\mathbf{B} = [0; 2]$, $\mathbf{C} = [1, 0]$, $\mathbf{D} = [0]$
Eigenvalues: $\det(sI - A) = s(s + 6) + 20 = s^2 + 6s + 20 = 0$.

$s = (-6 \pm \sqrt{(36 - 80)})/2 = (-6 \pm j6.63)/2 = \mathbf{-3 \pm j3.32}$
Both eigenvalues have negative real parts → system is **stable**. The natural frequency is $\omega_n = \sqrt{20} = 4.47$ rad/s and damping ratio $\zeta = 3/4.47 = 0.671$.

### 4.9.2 Controllability and Observability

Controllability determines whether it is possible to drive the system from any initial state to any desired final state in finite time using the available inputs. A system is controllable if and only if the controllability matrix $C_M = [B, AB, A^2B, ..., A^{n-1}B]$ has full rank (rank n). If a system is not fully controllable, some state variables cannot be influenced by the input, and the system's behavior in those modes is dictated entirely by initial conditions.

Observability determines whether the system's internal states can be uniquely determined from measurements of the output over a finite time interval. A system is observable if and only if the observability matrix $O_M = [C; CA; CA^2; ...; CA^{n-1}]$ has full rank. A system that is both controllable and observable can be fully controlled and monitored using feedback, and it has a minimal realization (the transfer function representation captures all system dynamics).

**Example 4.9.2:** For the RLC circuit from §4.9.1 with $A = [0, 10; -2, -6]$, $B = [0; 2]$, $C = [1, 0]$, determine whether the system is (a) controllable and (b) observable.
**Solution:**
(a) Controllability matrix: $C_M = [B, AB] = [[0; 2], [10\times2; -2\times0 + (-6)\times2]] = [[0, 20]; [2, -12]]$.
$\det(C_M) = 0\times(-12) - 20\times2 = -40 \neq 0$. Rank = 2 = n → **fully controllable**.
(b) Observability matrix: $O_M = [C; CA] = [[1, 0]; [1\times0 + 0\times(-2), 1\times10 + 0\times(-6)]] = [[1, 0]; [0, 10]]$.
$\det(O_M) = 1\times10 - 0\times0 = 10 \neq 0$. Rank = 2 = n → **fully observable**.
Since the system is both controllable and observable, state feedback and observer-based control designs can be applied.

### 4.9.3 State Feedback Control

State feedback control uses the full state vector to compute the control input: $u(t) = -Kx(t) + r(t)$, where K is the 1×n feedback gain vector and r(t) is the reference input. The closed-loop system matrix becomes $A_{CL} = A - BK$, and the eigenvalues of $A_{CL}$ are the closed-loop poles. If the system is controllable, the gain vector K can be chosen to place the closed-loop poles at any desired locations — this is the pole placement theorem. The Ackermann formula provides a direct computation: $K = [0\ 0\ ...\ 1] \times C_M^{-1} \times \varphi(A)$, where $\varphi(A)$ is the desired characteristic polynomial evaluated at the A matrix.

In practice, not all states are directly measurable, so a state observer (or estimator) reconstructs the unmeasured states from the output measurements. The Luenberger observer has the form: $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$, where L is the observer gain vector chosen so that the estimation error decays quickly. The separation principle guarantees that the controller and observer can be designed independently — the combined system's poles are the union of the controller poles and observer poles.

**Example 4.9.3:** For the RLC circuit with $A = [0, 10; -2, -6]$ and $B = [0; 2]$, design a state feedback controller to place the closed-loop poles at $s = -5 \pm j5$ ($\zeta = 0.707$, $\omega_n = 7.07$ rad/s).

**Solution:**
Desired characteristic polynomial: $(s + 5 - j5)(s + 5 + j5) = s^2 + 10s + 50$.
Current characteristic polynomial: $s^2 + 6s + 20$.
$A_{CL} = A - BK$ where $K = [k_1, k_2]$:
$A - BK = [0, 10; -2-2k_1, -6-2k_2]$.
Characteristic equation: $s^2 + (6 + 2k_2)s + (20 + 20k_1) = s^2 + 10s + 50$.
Matching coefficients:
$6 + 2k_2 = 10 \rightarrow k_2 = \mathbf{2}$
$20 + 20k_1 = 50 \rightarrow k_1 = \mathbf{1.5}$
$K = [1.5, 2]$. The control law $u(t) = -1.5x_1(t) - 2x_2(t) + r(t)$ places the closed-loop poles at $s = -5 \pm j5$, achieving $\zeta = 0.707$ (approximately 4.3% overshoot) and $\omega_n = 7.07$ rad/s.

### 4.9.4 Observer Design

When not all state variables are directly measurable, a Luenberger observer reconstructs the full state vector from the available output measurements. The observer has the form $\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - C\hat{x}(t))$, where $\hat{x}$ is the estimated state vector and L is the n×1 observer gain vector. The estimation error $e(t) = x(t) - \hat{x}(t)$ evolves as $\dot{e}(t) = (A - LC)e(t)$, so the observer poles are the eigenvalues of $(A - LC)$. For the estimation error to decay quickly, the observer poles are typically placed 3–5 times faster than the controller poles. The system must be observable (§4.9.2) for arbitrary observer pole placement to be possible. The separation principle guarantees that the controller gain K and observer gain L can be designed independently — the closed-loop system with observer-based feedback has poles at the union of the controller poles (eigenvalues of $A - BK$) and the observer poles (eigenvalues of $A - LC$).

**Example 4.9.4:** For the RLC circuit from §4.9.1 with $A = [0, 10; -2, -6]$, $B = [0; 2]$, and $C = [1, 0]$ (only $x_1$ is measured), design an observer to place the observer poles at $s = -20 \pm j20$, which are approximately 4× faster than the controller poles at $s = -5 \pm j5$ from §4.9.3.
**Solution:**
Desired observer characteristic polynomial: $(s + 20 - j20)(s + 20 + j20) = s^2 + 40s + 800$.
Observer error dynamics matrix: $A - LC$ where $L = [l_1; l_2]$:
$A - LC = [0 - l_1, 10; -2 - l_2, -6]$.
Characteristic equation: $s^2 + (6 + l_1)s + (6l_1 + 10l_2 + 20) = s^2 + 40s + 800$.
Matching coefficients:
$6 + l_1 = 40 \rightarrow l_1 = \mathbf{34}$
$6(34) + 10l_2 + 20 = 800 \rightarrow 224 + 10l_2 = 800 \rightarrow l_2 = \mathbf{57.6}$
$L = [34; 57.6]$. The observer uses the measured output $x_1$ to reconstruct $x_2$, with estimation errors decaying with a time constant of approximately $1/20 = 0.05$ s (compared to the controller's $1/5 = 0.2$ s), ensuring the observer converges well before the controller response settles.

## 4.10 Digital Control Systems

Modern control systems are predominantly implemented using digital processors (microcontrollers, DSPs, FPGAs) that sample the continuous plant output at discrete time intervals, compute the control law using numerical algorithms, and output a control signal through a digital-to-analog converter with a zero-order hold. Digital control offers advantages including flexibility (control laws changed in software), repeatability, noise immunity, and the ability to implement complex algorithms such as adaptive and optimal control that would be impractical with analog circuits.

### 4.10.1 Discrete-Time Systems

A continuous-time plant $G(s)$ preceded by a zero-order hold (ZOH) and sampler with period T produces the discrete-time transfer function $G(z) = (1 - z^{-1})\, Z\{G(s)/s\}$, where $Z\{\cdot\}$ denotes the z-transform. The z-transform variable relates to the Laplace variable by $z = e^{sT}$, which maps the left half of the s-plane (stable region) to the interior of the unit circle in the z-plane. A discrete-time system is stable if and only if all poles of $G(z)$ lie strictly inside the unit circle $|z| < 1$. Poles on the unit circle correspond to sustained oscillations, and poles outside indicate instability.

Continuous-time controllers can be converted to discrete-time using the Tustin (bilinear) approximation $s = (2/T)(z - 1)/(z + 1)$, which preserves the frequency response mapping and maps the $j\omega$ axis in the s-plane to the unit circle in the z-plane without aliasing distortion. The sample period T must be chosen small enough to adequately capture the system dynamics — a common rule of thumb is $T \leq 1/(10 \times f_{BW})$, where $f_{BW}$ is the closed-loop bandwidth.

**Example 4.10.1:** Discretize the continuous-time transfer function $G(s) = 10/(s + 5)$ using the zero-order hold method with a sample period $T = 0.1$ s.
**Solution:**
$G(s)/s = 10/[s(s + 5)]$. Partial fractions: $10/[s(s + 5)] = 2/s - 2/(s + 5)$.
Z-transform: $Z\{2/s - 2/(s + 5)\} = 2z/(z - 1) - 2z/(z - e^{-5T}) = 2z/(z - 1) - 2z/(z - e^{-0.5})$.
$e^{-0.5} = 0.6065$.
$G(z) = (1 - z^{-1})[2z/(z - 1) - 2z/(z - 0.6065)] = 2 - 2(z - 1)/(z - 0.6065)$.
Simplify: $G(z) = 2(z - 0.6065) - 2(z - 1)$ all over $(z - 0.6065) = 2(1 - 0.6065)/(z - 0.6065) =$ $\mathbf{0.787/(z - 0.6065)}$.
The discrete pole at $z = 0.6065$ corresponds to the continuous pole at $s = -5$ (since $e^{-5 \times 0.1} = 0.6065$), and it lies inside the unit circle, confirming stability. The DC gain is $G(1) = 0.787/(1 - 0.6065) = 2.0$, matching the continuous-time DC gain $G(0) = 10/5 = 2.0$.

### 4.10.2 Digital PID Implementation

The continuous PID controller $u(t) = K_p e(t) + K_i \int e(t)dt + K_d de(t)/dt$ is discretized using rectangular integration for the integral term and backward difference for the derivative term, yielding the position form: $u[k] = K_p e[k] + K_i T\, \Sigma e[k] + K_d(e[k] - e[k{-}1])/T$. The velocity (incremental) form $\Delta u[k] = u[k] - u[k{-}1]$ is often preferred because it avoids accumulating the integral sum explicitly and provides natural anti-windup behavior. Anti-windup is essential when the actuator saturates — without it, the integral term continues accumulating error during saturation, causing large overshoot when the error reverses. Common anti-windup strategies include clamping (freezing the integrator when the output is saturated) and back-calculation (reducing the integral term proportionally to the saturation amount).

The sample period T directly affects controller performance. If T is too large, the discrete controller cannot respond to fast disturbances and may introduce excessive phase lag; if T is unnecessarily small, computational burden increases and quantization noise is amplified in the derivative term. The rule of thumb $T \leq 1/(10 \times f_{BW})$ ensures the discrete controller closely approximates the continuous design. For a system with a closed-loop bandwidth of 10 Hz, this requires $T \leq 10$ ms.

**Example 4.10.2:** A motor speed control system has a continuous PID controller with $K_p = 2.0$, $K_i = 5.0$, and $K_d = 0.1$. The sample period is T = 0.05 s. The initial error sequence is e[0] = 10, e[1] = 7, e[2] = 4 (motor approaching setpoint). Compute the control output u[k] for k = 0, 1, and 2 using the position-form digital PID (assume e[−1] = 0 and prior integral sum = 0).

**Solution:**

**k = 0:** Integral sum = e[0] = 10. u[0] = 2.0(10) + 5.0(0.05)(10) + 0.1(10 − 0)/0.05 = 20 + 2.5 + 20 = **42.5**.

**k = 1:** Integral sum = 10 + 7 = 17. u[1] = 2.0(7) + 5.0(0.05)(17) + 0.1(7 − 10)/0.05 = 14 + 4.25 + (−6) = **12.25**.

**k = 2:** Integral sum = 10 + 7 + 4 = 21. u[2] = 2.0(4) + 5.0(0.05)(21) + 0.1(4 − 7)/0.05 = 8 + 5.25 + (−6) = **7.25**.

The control output decreases as the error decreases. The derivative term contributes negative values (−6 at k = 1 and k = 2) because the error is decreasing, providing a braking effect that reduces overshoot. The integral term steadily increases (2.5 → 4.25 → 5.25) to eliminate steady-state error.

The concepts in this chapter are demonstrated on real hardware in Appendix G (TCLab Control Lab), which documents a complete plant identification, IMC tuning, and measured closed-loop response on a dual-heater thermal plant.

# Chapter 5

# Embedded Systems

Embedded systems are special-purpose computing systems designed to perform dedicated functions within a larger mechanical or electrical product. Unlike general-purpose computers, embedded systems are optimized for specific tasks with constraints on size, power consumption, cost, and real-time responsiveness. They combine a microcontroller or microprocessor with firmware, peripherals, and communication interfaces to interact directly with the physical world through sensors and actuators. Embedded systems are ubiquitous in modern life, found in automotive electronics, consumer appliances, medical devices, industrial automation, telecommunications equipment, and aerospace systems.

## 5.1 Microcontrollers

A microcontroller (MCU) is a compact integrated circuit that combines a processor core, memory, and programmable input/output peripherals on a single chip. Unlike general-purpose microprocessors, microcontrollers are designed for dedicated control applications where they interact directly with sensors, actuators, and communication buses in real time. They are found in virtually every electronic product, from household appliances and automotive systems to industrial controllers and medical devices. Microcontrollers typically run firmware stored in on-chip flash memory, often without a full operating system, executing a main loop or responding to hardware interrupts.

### 5.1.1 Architecture

Microcontroller architectures are broadly classified as Harvard or Von Neumann, depending on whether program memory and data memory use separate or shared buses. Harvard architecture, used by AVR and most ARM Cortex-M implementations, provides simultaneous access to instruction and data memory on independent buses, enabling the processor to fetch the next instruction while reading or writing data in the same clock cycle. This parallelism is critical for real-time performance, where deterministic instruction timing allows precise control of interrupt latency and peripheral servicing. Von Neumann architecture uses a single shared bus for both instruction fetches and data access, simplifying the memory map and reducing pin count but creating a fetch/data bottleneck that limits throughput — a tradeoff acceptable in cost-sensitive applications where raw performance is less important than silicon area. Bus width determines how much data the processor can move in a single operation: 8-bit MCUs (PIC16, ATtiny) handle one byte per transfer and are suited for simple control tasks, 16-bit devices (MSP430, dsPIC) balance capability with low power, and 32-bit architectures (ARM Cortex-M, RISC-V) provide the address space and arithmetic width needed for complex algorithms and large memory maps. The instruction set architecture also shapes embedded performance: RISC (Reduced Instruction Set Computer) designs such as ARM and RISC-V execute most instructions in a single clock cycle with a load/store memory model, yielding predictable timing and efficient pipelining, while CISC (Complex Instruction Set Computer) designs like x86 offer powerful multi-step instructions at the cost of variable execution time and more complex decode logic. Most embedded MCUs use memory-mapped I/O, where peripheral registers occupy specific addresses in the same address space as SRAM and flash, allowing the same load/store instructions to configure a timer, read an ADC result, or transmit a UART byte without dedicated I/O instructions. Architecture choice directly impacts power consumption as well — a RISC core completing a task in fewer clock cycles at a lower frequency dissipates less dynamic power (proportional to $V^2 \times f$) than a slower architecture that must run longer or at a higher clock rate to accomplish the same work. Key architectural features include the arithmetic logic unit (ALU), register file, program counter, stack pointer, and interrupt controller, all integrated on the same die with memory and peripherals.

**Example 5.1.1:** A Harvard architecture MCU runs at a clock frequency of 48 MHz and executes most instructions in a single cycle. If a particular routine consists of 120 single-cycle instructions and 30 two-cycle instructions, how long does the routine take to execute?
**Solution:**
Total clock cycles = 120 * 1 + 30 * 2 = 120 + 60 = 180 cycles.
Clock period = 1 / 48 MHz = 20.83 ns.
Execution time = 180 * 20.83 ns = 3.75 μs.
The Harvard architecture enables this single-cycle execution for most instructions because program memory and data memory are accessed simultaneously on separate buses, eliminating the fetch/data contention that would occur in a Von Neumann architecture.

### 5.1.2 Common Families

**ARM Cortex-M** is the dominant 32-bit microcontroller architecture, licensed by ARM and manufactured by vendors including STMicroelectronics (STM32), NXP (LPC, Kinetis), and Texas Instruments (MSP432). The Cortex-M family ranges from the ultra-low-power M0/M0+ for simple control tasks to the high-performance M7 with hardware floating-point and DSP instructions for signal processing applications. **AVR** is an 8-bit RISC architecture developed by Atmel (now Microchip), widely known through the Arduino platform and the ATmega328P. AVR devices are valued for their simple instruction set, single-cycle execution, and extensive community support. **PIC** (Peripheral Interface Controller) from Microchip spans 8-bit (PIC16, PIC18), 16-bit (PIC24, dsPIC), and 32-bit (PIC32) families, with a broad range of integrated peripherals and low power consumption options suited to industrial and automotive applications.

**Example 5.1.2:** An engineer must select an MCU to sample 8 analog sensors at 1 kHz each, perform a 256-point FFT on one channel, and communicate results over SPI at 2 MHz. Which MCU family is most appropriate?
**Solution:**
Total ADC throughput required = 8 channels * 1000 samples/s = 8000 samples/s, which is easily within the capability of all families.
The 256-point FFT requires approximately 256 * $\log_2(256)$ = 256 * 8 = 2048 butterfly operations, each involving multiply-accumulate steps.
An 8-bit AVR lacks hardware multiply efficiency and floating-point support, making it a poor fit.
A Cortex-M4 (e.g., STM32F4 at 168 MHz) provides a hardware FPU and DSP instructions (single-

cycle MAC), completing the FFT in under 0.1 ms.
It also has multiple SPI peripherals supporting well above 2 MHz.
The ARM Cortex-M4 family is the best choice, offering the DSP capability for FFT, sufficient ADC channels, and SPI speed with headroom to spare.

### 5.1.3 Clock System and PLL

The clock system is the heartbeat of a microcontroller, providing the timing reference for the processor core, bus interfaces, and all peripherals. Most MCUs offer multiple clock sources: a high-speed internal RC oscillator (HSI, typically 8-16 MHz, ±1-2% accuracy), a high-speed external crystal oscillator (HSE, typically 4-25 MHz, ±20 ppm accuracy), a low-speed internal RC oscillator (LSI, ~32 kHz for watchdog), and a low-speed external crystal (LSE, 32.768 kHz for the real-time clock). A Phase-Locked Loop (PLL) multiplies the input clock to generate the high-frequency system clock — for example, an 8 MHz HSE can be multiplied to 168 MHz through the PLL. The clock distribution tree routes the system clock through prescalers to create separate clock domains for the AHB (Advanced High-performance Bus), APB1 (low-speed peripheral bus), and APB2 (high-speed peripheral bus), allowing peripherals to run at appropriate speeds while the core runs at maximum frequency. Proper clock configuration is critical for peripheral baud rate accuracy, timer precision, and power consumption — running the core at a lower frequency or switching to the internal RC oscillator during idle periods can significantly reduce power draw.

**Example 5.1.3:** An STM32F4 MCU uses an 8 MHz external crystal (HSE) and a PLL configured with multiplication factor M = 8, N = 336, and P = 2. The AHB prescaler is 1 and the APB1 prescaler is 4. Determine (a) the PLL output (system clock), (b) the AHB bus clock, (c) the APB1 peripheral clock, and (d) the APB1 timer clock.
**Solution:**
(a) PLL output: $f_{PLL} = (HSE / M) \times N / P = (8\text{ MHz} / 8) \times 336 / 2 = 1\text{ MHz} \times 168 =$ **168 MHz**
(b) AHB clock: $f_{AHB} = f_{PLL}$ / AHB prescaler = 168 / 1 = **168 MHz**
(c) APB1 clock: $f_{APB1} = f_{AHB}$ / APB1 prescaler = 168 / 4 = **42 MHz**
(d) APB1 timer clock: When the APB prescaler is not 1, the timer clock is automatically doubled: $f_{timer} = 2 \times f_{APB1} = 2 \times 42 =$ **84 MHz**
This means timers on APB1 have finer resolution (11.9 ns per tick) than the APB1 bus clock would suggest, which must be accounted for when configuring timer prescalers and periods.

## 5.2 Memory

Embedded systems rely on several distinct types of memory, each optimized for a specific role. Non-volatile flash retains the firmware image when power is removed, volatile SRAM provides fast read-write workspace during program execution, and EEPROM bridges the two by offering byte-level write access for configuration data that must survive power cycles without the sector-erase overhead of flash. Efficient use of all three is critical in resource-constrained embedded designs.

### 5.2.1 Flash Memory

Flash memory is non-volatile storage used to hold the firmware (program code) of a microcontroller. It retains data when power is removed and can be electrically erased and reprogrammed, typically with endurance ratings of 10,000 to 100,000 write-erase cycles. Flash is organized in pages or sectors, and erase operations must be performed on entire sectors before new data can be written. In-system programming (ISP) and in-application programming (IAP) allow firmware updates without removing the chip from the circuit. Typical flash sizes in microcontrollers range from a few kilobytes in small 8-bit devices to several megabytes in high-end 32-bit MCUs.

**Example 5.2.1:** A firmware image is 96 KB and must be stored in an MCU's flash memory organized as 128 sectors of 2 KB each. The flash has an endurance of 10,000 write-erase cycles and the firmware is updated once per week via OTA. How many sectors are needed and how long before the flash endurance limit is reached?
**Solution:**
Sectors required = 96 KB / 2 KB = 48 sectors.
Each update erases and rewrites all 48 sectors once.
At one update per week, the flash undergoes 52 erase cycles per year.
Time to endurance limit = 10,000 cycles / 52 cycles per year = 192.3 years.
The flash endurance is far more than adequate for this update frequency.
If updates were instead performed 10 times per day, the endurance would last 10,000 / (10 * 365) = 2.74 years, which may require a wear-leveling strategy or dual-bank flash for reliability.

### 5.2.2 SRAM

Static Random Access Memory (SRAM) is volatile memory used for the stack, heap, and global variables during program execution. SRAM provides fast read and write access (single clock cycle on most MCUs) without the need for refresh cycles, unlike DRAM. It retains data only while power is applied, and its contents are lost on reset or power-down. SRAM sizes in microcontrollers typically range from a few hundred bytes to several hundred kilobytes, and efficient use of SRAM is critical in embedded systems where memory resources are constrained.

**Example 5.2.2:** An MCU has 20 KB of SRAM. The firmware uses 2 KB for the stack, 1.5 KB for global/static variables, and allocates a 512-sample circular buffer of 16-bit ADC readings. Determine the total SRAM usage and remaining memory.
**Solution:**
Circular buffer size = 512 samples * 2 bytes/sample = 1024 bytes = 1 KB.
Total SRAM used = 2 KB (stack) + 1.5 KB (globals) + 1 KB (buffer) = 4.5 KB.
Remaining SRAM = 20 KB - 4.5 KB = 15.5 KB.
This leaves sufficient room for additional buffers or a small RTOS.
As a best practice, the stack should be monitored for overflow by painting it with a known pattern at startup and periodically checking how much has been overwritten during execution.

### 5.2.3 EEPROM

Electrically Erasable Programmable Read-Only Memory (EEPROM) is non-volatile memory designed for storing small amounts of configuration data, calibration values, or operational parameters that must persist across power cycles. Unlike flash, EEPROM can be erased and written at the individual byte level, making it more convenient for frequent small updates. EEPROM endurance is typically rated at 100,000 to 1,000,000 write cycles per byte, with data retention of 20 years or more. Many microcontrollers include a small amount of on-chip EEPROM (typically 256 bytes to a few kilobytes), and external EEPROM ICs (such as the 24Cxx series) are available via I2C or SPI for additional storage.

**Example 5.2.3:** A device stores a 4-byte calibration value and a 2-byte operating hours counter in EEPROM. The counter is incremented and written every hour. The EEPROM is rated for 1,000,000 write cycles per byte. How long before the counter bytes reach their endurance limit?
**Solution:**
The counter bytes are written once per hour, so 24 writes per day and 8,760 writes per year.
Time to endurance limit = 1,000,000 / 8,760 = 114.2 years.
This is well within the product's expected lifetime.
If the write frequency were much higher (e.g., once per second), the limit would be reached in 1,000,000 / (86,400 * 365) = approximately 0.032 years (11.6 days), which would necessitate wear-leveling across multiple EEPROM locations.

## 5.3 Peripherals

Peripherals are the hardware blocks within a microcontroller that extend its capabilities beyond pure computation, enabling it to interact with the physical world and communicate with external devices. They range from simple GPIO pins for digital control to sophisticated conversion and transfer engines — ADCs, DACs, timers, and DMA controllers — that can operate independently of the CPU to maximize throughput and minimize power consumption. Selecting the right MCU for an application requires matching its peripheral set and specifications to the system requirements.

### 5.3.1 GPIO

General Purpose Input/Output (GPIO) pins are the most fundamental peripheral on a microcontroller, providing configurable digital interfaces to external circuitry. Each GPIO pin can typically be configured as an input (reading logic high or low) or an output (driving logic high or low) by writing to direction registers, and the output stage can usually be set to push-pull mode (actively driving both high and low) or open-drain mode (pulling low only, requiring an external pull-up resistor for the high state). Open-drain outputs are essential for wired-OR bus topologies and level-shifting between different voltage domains — for example, I2C requires open-drain lines with pull-ups. Internal pull-up and pull-down resistors (typically 20–50 kΩ) can be enabled on input pins to define a default logic level when the pin is floating, eliminating the need for external resistors in many designs. Electrical characteristics vary by MCU family: drive strength is often configurable (e.g., 2 mA, 4 mA, or 8 mA on STM32), output slew rate can be set to low, medium, or high to control EMI emissions on fast-switching signals, and maximum source/sink current per pin is typically 5-25 mA with an aggregate limit per port or per chip. Most GPIO pins support alternate function multiplexing, where each physical pin can be remapped through configuration registers to serve as a UART TX/RX, SPI clock, I2C data line, PWM output, or ADC input channel, allowing a single MCU package to support diverse peripheral combinations without dedicated pins for every function. Input pins can be configured to trigger hardware interrupts on rising edges, falling edges, or both, enabling event-driven firmware that responds immediately to button presses, sensor alerts, or external timing signals without polling. For mechanical switches and buttons, software or hardware debouncing is required — contact bounce produces multiple transitions over 1-10 ms that would otherwise trigger false interrupts. When interfacing GPIO pins with circuits operating at different voltage levels (e.g., a 3.3 V MCU communicating with 5 V legacy logic), level-shifting circuits or voltage translators are needed to prevent overvoltage damage, and ESD protection diodes (either internal to the MCU or external TVS devices) safeguard pins against electrostatic discharge events that can exceed several kilovolts.

**Example 5.3.1:** A GPIO output pin on a 3.3 V MCU (maximum source current 8 mA) must drive an LED with a forward voltage of 2.0 V and a desired forward current of 10 mA. Design the interface.
**Solution:**
Since the required LED current (10 mA) exceeds the GPIO source limit (8 mA), a transistor driver is needed.
Use an NPN transistor (e.g., 2N2222) with the LED and a current-limiting resistor in the collector circuit.
Collector resistor: $R_C$ = (3.3 V - $V_{LED}$ - $V_{CE}$(sat)) / $I_{LED}$ = (3.3 - 2.0 - 0.2) / 0.010 = 1.1 V / 10 mA = 110 Ω (use 100 Ω standard value, giving $I_{LED}$ = 11 mA).
Base resistor: assuming minimum $h_{FE}$ = 100, $I_B = I_C / h_{FE}$ = 11 mA / 100 = 0.11 mA.
To ensure saturation, use 10x overdrive: $I_B$ = 1.1 mA.
$R_B$ = (3.3 V - $V_{BE}$) / $I_B$ = (3.3 - 0.7) / 0.00110 = 2.36 kΩ (use 2.2 kΩ standard).
The GPIO pin now only sources about 1.2 mA, well within its rating.

### 5.3.2 Timers and PWM

Hardware timers are counter-based peripherals that provide precise timing, event counting, and waveform generation without continuous processor involvement. A timer increments (or decrements) a counter register at each tick of a clock source, which can be the system clock, a prescaled division of it, or an external signal. When the counter reaches a compare value or overflows, the timer can generate an interrupt, toggle an output pin, or trigger other peripherals via hardware. Pulse Width Modulation (PWM) is a timer output mode that generates a periodic digital signal with a controllable duty cycle, widely used for motor speed control, LED dimming, and digital-to-analog conversion. The PWM frequency is set by the timer period, and the duty cycle is set by the compare register value, allowing precise analog-like control from a digital output.

**Example 5.3.2:** An MCU running at 72 MHz uses a 16-bit timer with a prescaler of 72 to generate a 1 kHz PWM signal for motor speed control. Determine the timer period value and the compare register value for a 60% duty cycle.
**Solution:**
Timer clock = 72 MHz / 72 (prescaler) = 1 MHz (1 µs per tick).
For a 1 kHz PWM frequency, the period = 1 / 1 kHz = 1 ms = 1000 timer ticks.
Set the auto-reload register (ARR) = 1000 - 1 = 999.
For 60% duty cycle, compare register (CCR) = 0.60 * 1000 = 600.
The timer counts from 0 to 999 (1000 ticks = 1 ms), and the output is high for counts 0 to 599 (600 µs) and low for counts 600 to 999 (400 µs).
Duty cycle resolution = 1 / 1000 = 0.1%, which is well within 16-bit capacity (max count 65535).

### 5.3.3 ADC

An Analog-to-Digital Converter (ADC) converts a continuous analog voltage into a discrete digital value that the processor can read and process. Most microcontrollers include a successive approximation register (SAR) ADC with 10-bit or 12-bit resolution, providing 1024 or 4096 quantization levels over the input voltage range. The ADC samples the input voltage at a rate determined by the ADC clock and conversion time, and the result is stored in a data register. A multiplexer allows a single ADC to read from multiple analog input pins sequentially. Key specifications include resolution (bits), sampling rate (samples per second), input voltage range (typically 0V to the reference voltage), and integral/differential nonlinearity (INL/DNL) errors.

**Example 5.3.3:** A 12-bit ADC with a reference voltage of 3.3 V reads a digital value of 2482. What is the corresponding analog input voltage, and what is the resolution (smallest detectable voltage change)?
**Solution:**
Resolution (LSB size) = $V_{ref}$ / $2^n$ = 3.3 V / 4096 = 0.8057 mV per count.
Input voltage = digital value * resolution = 2482 * 0.8057 mV = 1.9997 V, approximately 2.000 V.
The ADC can distinguish voltage changes as small as 0.806 mV.
If the signal of interest spans only 0 to 1 V, the effective number of usable codes is 1.0 / 0.0008057 = 1241 counts, equivalent to approximately $\log_2(1241)$ = 10.3 bits of effective resolution over that range.

### 5.3.4 DAC

A Digital-to-Analog Converter (DAC) converts a digital value from the processor into a corresponding analog output voltage. On-chip DACs are commonly 8-bit or 12-bit and are used for generating reference voltages, audio waveforms, control signals for analog circuits, and bias voltages. The output voltage is determined by the formula $V_{out} = (D / 2^n) * V_{ref}$, where D is the digital input code, n is the resolution in bits, and $V_{ref}$ is the reference voltage. DAC outputs typically require an external operational amplifier buffer to drive low-impedance loads. Some MCUs include multiple DAC channels with DMA (Direct Memory Access) support, enabling continuous waveform generation without processor overhead.

**Example 5.3.4:** A 12-bit DAC with $V_{ref}$ = 3.3 V must produce an output of 1.65 V. Determine the required digital input code. If the DAC is used to generate a sine wave at 1 kHz using a lookup table with 100 points per cycle, what is the required DAC update rate?
**Solution:**
Digital code D = ($V_{out}$ / $V_{ref}$) * $2^n$ = (1.65 / 3.3) * 4096 = 0.5 * 4096 = 2048.
Load the value 2048 into the DAC data register to produce 1.65 V at the output.
For the sine wave: update rate = 1 kHz * 100 samples/cycle = 100,000 samples/s = 100 kSPS.
The update interval = 1 / 100,000 = 10 μs.
Using DMA with a timer trigger at 10 μs intervals, the DAC automatically cycles through the 100-entry lookup table without CPU intervention, freeing the processor for other tasks.

### 5.3.5 DMA (Direct Memory Access)

Direct Memory Access (DMA) is a hardware peripheral that transfers data between memory and peripherals (or between memory locations) without processor involvement, freeing the CPU to execute other code during the transfer. A DMA controller contains multiple channels, each configurable with a source address, destination address, transfer size, data width (byte, half-word, or word), and direction (memory-to-peripheral, peripheral-to-memory, or memory-to-memory). Transfers can be triggered by peripheral events (e.g., ADC conversion complete, UART byte received, SPI buffer empty), enabling continuous data streaming. DMA is essential for high-throughput applications such as audio processing, continuous ADC sampling, display frame buffer updates, and communication protocols where the data rate would otherwise overwhelm the CPU. Most Cortex-M MCUs have two DMA controllers with 8-16 channels each, with configurable priority levels and burst transfer modes.

**Example 5.3.5:** An MCU samples a 12-bit ADC at 100 kHz and uses DMA to transfer each sample (16-bit half-word) to a 1024-sample circular buffer in SRAM. Determine (a) the DMA transfer rate in bytes per second, (b) the time to fill the buffer, and (c) the CPU cycles saved per second compared to interrupt-driven transfers (assume 30 CPU cycles per interrupt-driven sample at 168 MHz).
**Solution:**
(a) DMA transfer rate: 100,000 samples/s $\times$ 2 bytes/sample = **200,000 bytes/s = 200 KB/s**
(b) Buffer fill time: 1024 samples / 100,000 samples/s = **10.24 ms**
Using half-transfer and transfer-complete interrupts, the CPU is interrupted only twice per buffer fill (every 5.12 ms) instead of 1024 times.
(c) CPU cycles saved: Interrupt-driven transfers would require 100,000 interrupts/s $\times$ 30 cycles/interrupt = 3,000,000 cycles/s. At 168 MHz, this is 3,000,000 / 168,000,000 = 1.79% of CPU time. With DMA, the CPU is interrupted only 2 $\times$ (100,000/1024) $\approx$ 195 times/s, consuming 195 $\times$ 30 = 5,850 cycles/s — a **99.8% reduction** in CPU overhead for the ADC data transfer.

### 5.3.6 Watchdog Timer

A watchdog timer (WDT) is a safety peripheral that resets the microcontroller if the firmware fails to periodically "kick" or "feed" the watchdog within a specified timeout period. This provides automatic recovery from software hangs, infinite loops, deadlocks, or other fault conditions that would otherwise leave the system unresponsive. The watchdog counter counts down from a programmable reload value, and if it reaches zero before the firmware writes to the refresh register, a system reset is triggered. Most MCUs offer two types: the Independent Watchdog (IWDG), clocked by a separate low-speed internal oscillator so it remains active even if the main clock fails, and the Window Watchdog (WWDG), which not only requires a refresh before timeout but also rejects refreshes that come too early (within a programmable window), detecting both stuck and runaway firmware. Watchdog timers are a fundamental requirement for safety-critical and unattended embedded systems — automotive (ISO 26262), medical (IEC 62304), and industrial (IEC 61508) standards all require watchdog mechanisms.

**Example 5.3.6:** An independent watchdog timer is clocked at 32 kHz (LSI) with a prescaler of 64 and a 12-bit reload value of 4095. Determine (a) the watchdog timeout period, (b) the maximum time the firmware can go between watchdog refreshes, and (c) a suitable refresh interval.
**Solution:**
(a) Watchdog clock after prescaler: $f_{WDT}$ = 32,000 / 64 = 500 Hz (2 ms per tick)
Timeout period: T = (reload + 1) / $f_{WDT}$ = 4096 / 500 = **8.192 seconds**
(b) The firmware must refresh the watchdog before the counter reaches zero, so the maximum time between refreshes is **8.192 seconds**.

(c) In practice, the watchdog should be refreshed well before the timeout to account for worst-case execution paths. A common approach is to refresh at half the timeout period: approximately **4 seconds** (half of 8.192 s). Alternatively, refresh at the end of each pass through the main loop or RTOS health check task. If the main loop period is 100 ms, refreshing every iteration provides a 82× safety margin.

## 5.4 Communication Interfaces

Communication interfaces allow an embedded system to exchange data with sensors, actuators, displays, storage devices, and other processors. Each protocol involves trade-offs among speed, pin count, wiring distance, noise immunity, and implementation complexity, and the choice of interface is frequently dictated by the target peripheral rather than by the MCU. The interfaces covered here span the full spectrum of embedded communication, from the asynchronous simplicity of UART to the high-speed, host-managed USB link.

### 5.4.1 UART

Universal Asynchronous Receiver/Transmitter (UART) is a serial communication interface that transmits and receives data one bit at a time over two signal lines: TX (transmit) and RX (receive). Communication is asynchronous, meaning no shared clock signal is used; instead, both devices must agree on a common baud rate (e.g., 9600, 115200 bits per second), data frame format (typically 8 data bits, no parity, 1 stop bit, abbreviated 8N1), and voltage levels. Each data frame begins with a start bit (logic low), followed by the data bits (LSB first), an optional parity bit for error detection, and one or more stop bits (logic high). UART is simple to implement and is commonly used for debug consoles, GPS modules, Bluetooth modules, and communication between microcontrollers. RS-232 and RS-485 are physical layer standards that define voltage levels and electrical characteristics for UART-based communication over longer distances.

**Example 5.4.1:** An MCU UART is configured at 115200 baud with 8N1 framing. How long does it take to transmit a 64-byte data packet, and what is the effective data throughput?
**Solution:**
Each byte frame consists of 1 start bit + 8 data bits + 1 stop bit = 10 bits per byte.
Total bits to transmit = 64 bytes * 10 bits/byte = 640 bits.
Transmission time = 640 bits / 115200 bits/s = 5.556 ms.
Effective data throughput = (64 * 8 data bits) / 5.556 ms = 512 / 0.005556 = 92,160 bits/s = 92.16 kbps.
The overhead from start and stop bits reduces the effective rate to 80% of the baud rate (92,160 / 115,200 = 0.80).
If parity were added (8E1: 11 bits per byte), the overhead would increase and throughput would drop to 72.7% of the baud rate.

### 5.4.2 SPI

Serial Peripheral Interface (SPI) is a synchronous, full-duplex serial communication bus that uses four signal lines: SCLK (clock), MOSI (Master Out Slave In), MISO (Master In Slave Out), and SS/CS (Slave Select/Chip Select). The master device generates the clock and controls the slave select line, enabling one slave at a time for communication. Data is shifted out on one clock edge and sampled on the opposite edge, with the clock polarity (CPOL) and phase (CPHA) configurable into four modes (Mode 0-3). SPI achieves high data rates (often 10 MHz or more) because of its dedicated clock line and full-duplex operation, making it well-suited for high-speed peripherals such as flash memory, display controllers, SD cards, and ADC/DAC chips. The main drawback is the requirement for a separate chip select line for each slave device, which increases pin usage as the number of peripherals grows.

**Example 5.4.2:** An MCU communicates with an external SPI flash memory at an SPI clock rate of 8 MHz. The flash read command requires sending a 1-byte opcode and a 3-byte address before data is returned. How long does it take to read 256 bytes of data from the flash?
**Solution:**
Total bytes transferred = 1 (opcode) + 3 (address) + 256 (data) = 260 bytes.
Since SPI is full-duplex, each byte requires 8 clock cycles.
Total clock cycles = 260 * 8 = 2080 cycles.
Transfer time = 2080 / 8 MHz = 260 μs.
Effective data throughput for the read portion = 256 bytes / 260 μs = 0.985 bytes/μs = 7.876 Mbps.
The 4-byte command/address overhead is only 1.5% of the total transfer, so for bulk reads the throughput approaches the theoretical maximum of 8 Mbps (1 byte per 8 clock cycles at 8 MHz).

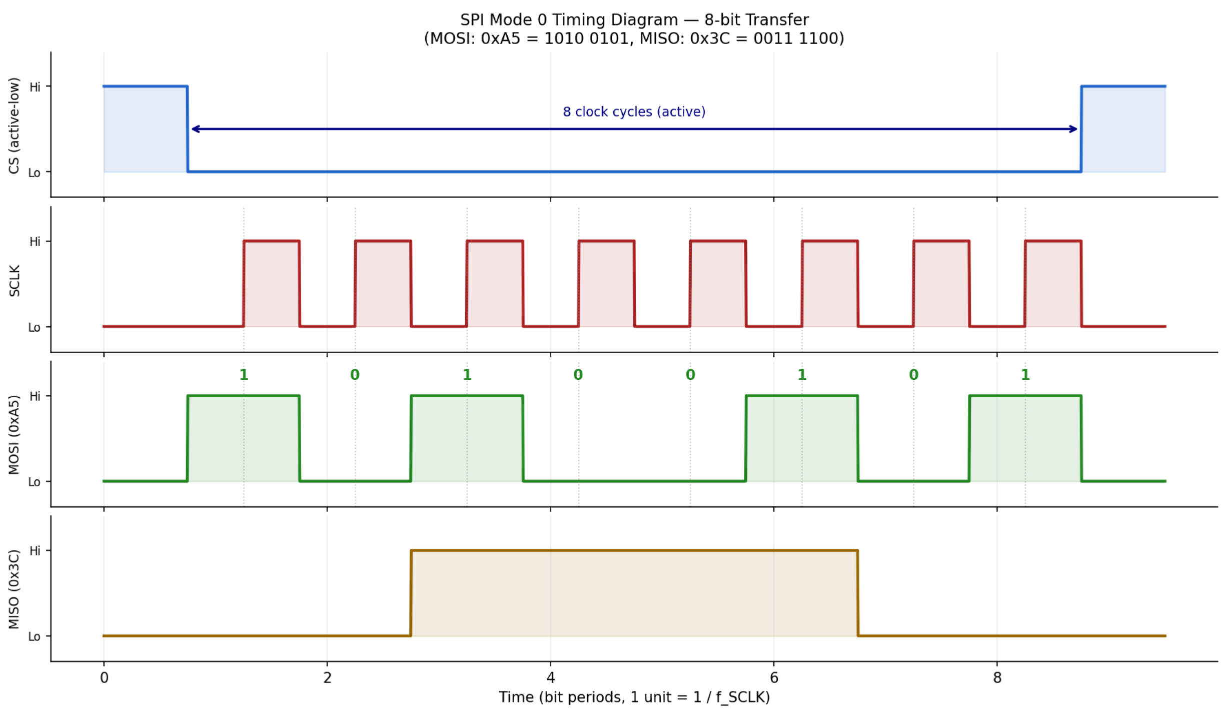



Figure 5.4.2: SPI Mode 0 timing diagram showing an 8-bit transfer (MOSI: 0xA5, MISO: 0x3C). Data is sampled on rising clock edges; the CS line is held low for the duration of the 8-cycle transfer.

### 5.4.3 I2C

Inter-Integrated Circuit (I2C) is a synchronous, half-duplex serial communication bus that uses only two signal lines: SDA (Serial Data) and SCL (Serial Clock). Multiple master and slave devices can share the same two-wire bus, with each slave identified by a unique 7-bit (or 10-bit) address. Communication begins when a master issues a START condition, followed by the slave address and a read/write bit; the addressed slave responds with an ACK (acknowledge) bit, and data bytes are then transferred with an ACK after each byte. Standard I2C operates at 100 kHz, Fast mode at 400 kHz, and Fast-mode Plus at 1 MHz. I2C is widely used for low-speed peripherals such as temperature sensors, EEPROMs, real-time clocks, and OLED displays, where its minimal pin count and multi-device bus topology are advantageous.

**Example 5.4.3:** An MCU reads a 2-byte temperature value from an I2C sensor (7-bit address 0x48) operating in Fast mode (400 kHz). Calculate the minimum time for a complete read transaction.
**Solution:**
An I2C read transaction consists of: START (1 bit) + address byte with R/W=1 (8 bits) + ACK (1 bit) + data byte 1 (8 bits) + ACK (1 bit) + data byte 2 (8 bits) + NACK (1 bit) + STOP (1 bit) = 29 bits.
However, a typical sensor requires a write-then-read sequence: first write the register pointer (START + address W + ACK + register byte + ACK = 19 bits), then repeated START + address R + ACK + 2 data bytes with ACK/NACK + STOP = 29 bits.
Total = 19 + 29 = 48 bit periods.
At 400 kHz: time = 48 / 400,000 = 120 μs.
Including I2C protocol overhead (clock stretching, bus turnaround), a practical measurement is typically 130-150 μs.

### 5.4.4 CAN Bus

Controller Area Network (CAN) is a robust, message-based serial communication protocol originally developed by Bosch for automotive applications. CAN uses a differential two-wire bus (CAN_H and CAN_L) that provides high noise immunity, making it suitable for electrically harsh environments in vehicles, industrial automation, and medical equipment. Messages are broadcast on the bus with an identifier (11-bit standard or 29-bit extended) that determines both the message type and its priority during bus arbitration; lower identifier values have higher priority. CAN implements built-in error detection mechanisms including CRC checks, bit stuffing, frame checks, and acknowledgment verification, with automatic retransmission of corrupted messages. Classical CAN supports data rates up to 1 Mbps, while CAN FD (Flexible Data-rate) extends the payload from 8 to 64 bytes per frame and allows higher bit rates during the data phase.

**Example 5.4.4:** A CAN bus operates at 500 kbps. A standard frame with an 11-bit identifier carries 8 data bytes. Estimate the maximum message throughput on the bus.
**Solution:**
A standard CAN frame consists of: 1 (SOF) + 11 (ID) + 1 (RTR) + 6 (control) + 64 (8 data bytes) + 15 (CRC) + 10 (CRC_delim + ACK + ACK_delim + EOF) + 3 (IFS) = 111 bits minimum.
With worst-case bit stuffing (a stuff bit inserted after every 5 identical bits in the data/ID/CRC fields), approximately 19 additional stuff bits may be added, giving roughly 130 bits per frame.
Maximum throughput = 500,000 bps / 130 bits per frame = 3846 frames/s.
Data throughput = 3846 * 8 bytes = 30,769 bytes/s $\times$ 8 bits/byte $\approx$ 246 kbps effective data rate, which is about 49% of the raw bit rate due to protocol overhead.

### 5.4.5 USB

Universal Serial Bus (USB) is a widely used communication standard that provides both data transfer and power delivery through a single cable. USB operates on a host-device (master-slave) topology, where the host (typically a PC) initiates all transactions and the device (the embedded system) responds. USB device classes define standardized behaviors: CDC (Communications Device Class) emulates a virtual serial port, HID (Human Interface Device) handles keyboards and mice, MSC (Mass Storage Class) presents a filesystem, and vendor-specific classes allow custom protocols. USB 2.0 supports three speeds — Low Speed (1.5 Mbps), Full Speed (12 Mbps), and High Speed (480 Mbps) — while USB 3.x adds SuperSpeed (5-20 Gbps). Most Cortex-M microcontrollers include a Full Speed USB peripheral with an integrated PHY, requiring minimal external components (typically just a USB connector and two 22 Ω series resistors on the data lines). USB device firmware is typically implemented using a middleware stack (such as TinyUSB or the vendor's USB library) that handles the complex protocol state machine, enumeration process, and endpoint management.

**Example 5.4.5:** An embedded data logger uses USB 2.0 Full Speed (12 Mbps) CDC class to stream sensor data to a PC. Each data packet contains 64 bytes (the maximum bulk endpoint packet size for Full Speed). USB bulk transfers have a protocol overhead of approximately 25% (token, handshake, CRC, inter-packet gaps). Determine (a) the maximum number of bulk packets per second, (b) the effective data throughput, and (c) whether this is sufficient to stream 16-bit ADC samples at 100 kHz.

**Solution:**

(a) Effective bit rate: 12 Mbps $\times$ 0.75 (after 25% overhead) = 9.0 Mbps

Each packet: 64 bytes $\times$ 8 bits = 512 bits

Maximum packets/s: 9,000,000 / 512 = **17,578 packets/s**

(b) Effective data throughput: 17,578 $\times$ 64 = 1,125,000 bytes/s = **1.125 MB/s = 9.0 Mbps**

(c) Required throughput for 100 kHz, 16-bit ADC: 100,000 $\times$ 2 = 200,000 bytes/s = 1.6 Mbps

Since 9.0 Mbps » 1.6 Mbps, USB Full Speed is **more than sufficient**, with capacity for 5.6× the required data rate. The data logger could stream up to 562,500 samples/s at 16-bit resolution before saturating the USB link.

## 5.5 Interrupts

An interrupt is a hardware or software signal that causes the processor to suspend its current execution, save its context (program counter and status registers), and jump to a designated interrupt service routine (ISR) to handle the event. Hardware interrupts are generated by peripherals such as GPIO pins, timers, ADCs, and communication interfaces when specific events occur (e.g., a timer overflow, a byte received on UART, or a pin state change). Software interrupts (also called traps or exceptions) are triggered by processor instructions or fault conditions such as divide-by-zero or invalid memory access. Interrupts are assigned priority levels, and a Nested Vectored Interrupt Controller (NVIC) on ARM Cortex-M processors manages prioritization and nesting, allowing higher-priority interrupts to preempt lower-priority ones. ISRs should be kept as short as possible, typically setting a flag or copying data to a buffer, with the main processing deferred to the main loop or a task scheduler to minimize interrupt latency for other events.

**Example 5.5:** An ARM Cortex-M4 MCU running at 168 MHz has three interrupts: Timer (priority 1, ISR execution time 2 μs), UART RX (priority 2, ISR execution time 1.5 μs), and ADC complete (priority 3, ISR execution time 3 μs). If all three trigger simultaneously, determine the total time until all ISRs complete, assuming a context switch overhead of 12 clock cycles per switch.
**Solution:**
Context switch time = 12 cycles / 168 MHz = 71.4 ns per switch.
Execution order (highest priority first): Timer (priority 1) runs first, then UART RX (priority 2), then ADC (priority 3).
Timeline: initial context switch (71.4 ns) + Timer ISR (2 μs) + context switch (71.4 ns) + UART ISR (1.5 μs) + context switch (71.4 ns) + ADC ISR (3 μs) + context switch to main (71.4 ns).
Total = 4 * 0.0714 μs + 2 + 1.5 + 3 = 0.286 + 6.5 = 6.786 μs.
If the UART interrupt arrived during the Timer ISR, it would pend until the Timer ISR completes (since Timer has higher priority), then preempt any lower-priority pending work, demonstrating the NVIC's priority-based nesting behavior.

## 5.6 Real-Time Operating Systems (RTOS)

A Real-Time Operating System (RTOS) is a lightweight operating system designed to guarantee that tasks execute within deterministic time constraints, making it essential for embedded applications with strict timing requirements. Unlike general-purpose operating systems, an RTOS provides a preemptive priority-based scheduler that ensures the highest-priority ready task always runs, with context switch times typically measured in microseconds. Key RTOS primitives include tasks (threads), semaphores, mutexes, message queues, and event flags, which provide structured mechanisms for multitasking, synchronization, and inter-task communication. Popular RTOS platforms for embedded systems include FreeRTOS (open-source, widely used on ARM Cortex-M and ESP32), Zephyr (Linux Foundation, supporting multiple architectures), and ThreadX (now Azure RTOS, certified for safety-critical applications). An RTOS is beneficial when an application requires concurrent handling of multiple time-sensitive activities (e.g., sensor sampling, communication, display updates, and control loops) that would be difficult to manage with a bare-metal super-loop architecture.

**Example 5.6:** A FreeRTOS application on a Cortex-M4 at 168 MHz has three tasks: a control loop task (priority 3, runs every 1 ms for 200 μs), a sensor reading task (priority 2, runs every 10 ms for 500 μs), and a display update task (priority 1, runs every 100 ms for 5 ms). Determine the CPU utilization.
**Solution:**
CPU utilization per task = execution time per period / period.
Control loop: 200 μs / 1 ms = 20.0%.
Sensor reading: 500 μs / 10 ms = 5.0%.
Display update: 5 ms / 100 ms = 5.0%.
Total CPU utilization = 20.0% + 5.0% + 5.0% = 30.0%.
The remaining 70% is spent in the idle task.
Using Rate Monotonic Analysis (RMA), the theoretical schedulability bound for 3 tasks is 3 * (2^(1/3) - 1) = 3 * 0.2599 = 77.98%.
Since 30% is well below 77.98%, the task set is guaranteed schedulable with fixed-priority preemptive scheduling.

The control loop has the highest priority due to its shortest period and strictest deadline.

## 5.7 Power Management

Power management is a critical discipline in battery-powered and energy-harvesting embedded systems, where runtime is directly determined by how efficiently the hardware consumes energy. Microcontrollers provide hardware mechanisms — sleep modes, clock gating, and voltage scaling — that can reduce current consumption by orders of magnitude compared to full-speed operation, while firmware architecture governs how effectively those mechanisms are used. The combination of hardware capability and thoughtful software design determines whether a device lasts hours or years on a single cell.

### 5.7.1 Sleep Modes and Wake-Up Sources

Modern microcontrollers provide multiple low-power sleep modes that progressively disable on-chip subsystems to reduce power consumption when the full processing capability is not needed. A typical Cortex-M MCU offers three to five modes: Sleep mode (CPU halted, peripherals and clocks active, wake on any interrupt, ~1-10 mA), Stop mode (CPU and most clocks halted, SRAM and registers retained, selected peripherals active, wake on EXTI, RTC, or specific peripheral events, ~10-100 μA), and Standby mode (nearly everything powered down except the backup domain and wakeup logic, SRAM contents lost, wake only on specific pins, RTC alarm, or reset, ~1-5 μA). Wake-up sources include external interrupt pins (EXTI), real-time clock (RTC) alarms, watchdog timer, UART activity detection, USB VBUS detection, and comparator threshold crossings. The firmware must configure the wake-up source and any peripheral retention before entering a low-power mode, and typically re-initializes clocks and peripherals upon waking since the PLL and HSE may have been shut down.

**Example 5.7.1:** A battery-powered sensor node uses a 3.3 V, 1000 mAh lithium battery. The MCU wakes every 60 seconds, samples a sensor and transmits data over SPI radio for 50 ms at 15 mA, then returns to Stop mode at 20 μA. Determine (a) the average current consumption, (b) the battery life, and (c) the battery life if the wake interval were reduced to 5 seconds.
**Solution:**
(a) Average current: $I_{avg} = (I_{active} \times t_{active} + I_{sleep} \times t_{sleep}) / T_{total}$
$= (15 \text{ mA} \times 50 \text{ ms} + 0.020 \text{ mA} \times 59{,}950 \text{ ms}) / 60{,}000 \text{ ms}$
$= (0.75 + 1.199) / 60{,}000 = 1.949 \text{ mA·ms} / 60{,}000 \text{ ms} =$ **0.03248 mA = 32.5 μA**
(b) Battery life: $t = C / I_{avg} = 1000 \text{ mAh} / 0.03248 \text{ mA} = 30{,}787$ hours = **3.52 years**
(c) At 5-second interval: $I_{avg} = (15 \times 50 + 0.020 \times 4{,}950) / 5{,}000 = (750 + 99) / 5{,}000 =$ **0.170 mA = 170 μA**
Battery life: 1000 / 0.170 = 5,882 hours = **0.67 years (8 months)**
Reducing the wake interval by 12× increases average current by 5.2× because the active-mode current dominates even at short duty cycles.

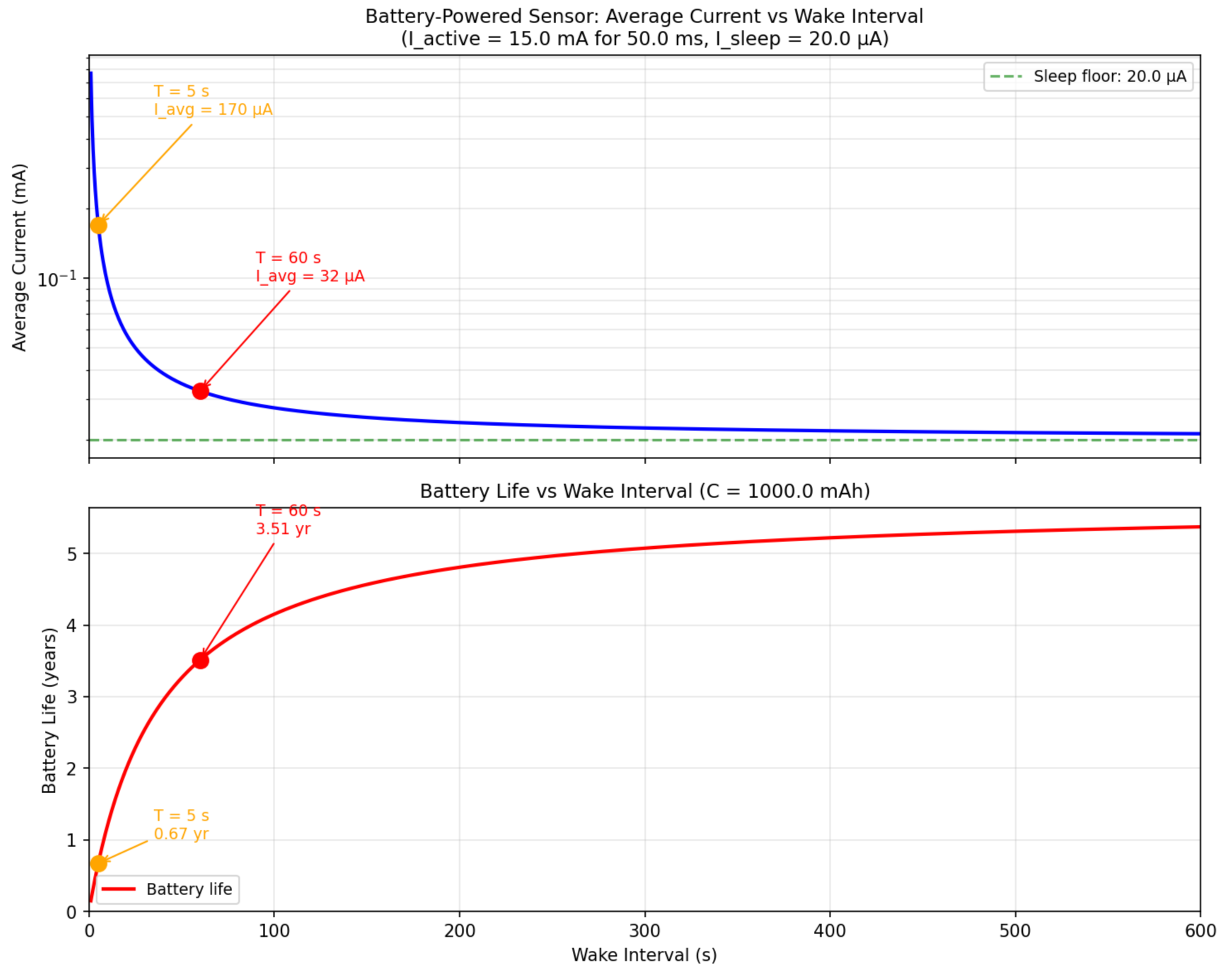



Figure 5.7.1: Average current (log scale) and battery life vs wake interval for a sensor node (Iactive = 15 mA for 50 ms, Isleep = 20 µA, C = 1000 mAh). Markers show the 60 s and 5 s operating points from §5.7.1.

### 5.7.2 Low-Power Design Techniques

Reducing power consumption in embedded systems requires attention at every level: hardware design, clock management, peripheral usage, and firmware architecture. Key techniques include:

- **Clock gating**: Disable clocks to unused peripherals through the RCC (Reset and Clock Control) peripheral. Most MCUs ship with all peripheral clocks enabled by default — disabling unused ones can reduce active current by 10-30%.
- **Voltage scaling**: Many MCUs allow the core voltage to be reduced when running at lower frequencies, providing quadratic power savings since dynamic power is proportional to $V^2 \times f$.
- **Peripheral duty cycling**: Enable peripherals only when needed. For example, power up the ADC, take a measurement, and power it down rather than leaving it continuously active.
- **DMA over interrupts**: Use DMA for data transfers to allow the CPU to enter Sleep mode between transfers instead of waking for each byte.
- **Efficient firmware**: Avoid busy-wait loops (use WFI — Wait For Interrupt — instruction instead), minimize unnecessary memory accesses, and use compiler optimization flags.
- **External hardware**: Use hardware power switches (load switches or LDOs with enable pins) to completely de-power external sensors and radios when not in use, as standby leakage in external

ICs can exceed the MCU's own sleep current.

**Example 5.7.2:** An IoT device runs at 3.3 V with the following current profile: MCU active at 168 MHz (30 mA), MCU active at 48 MHz with voltage scaling (8 mA), MCU Sleep mode (5 mA), MCU Stop mode (20 μA). The firmware spends 10% of its time in computation (requiring 168 MHz), 20% in communication (48 MHz sufficient), and 70% idle. Calculate the average current for (a) no optimization (always at 168 MHz, busy-wait during idle), and (b) optimized (frequency scaling and Stop mode during idle).

**Solution:**

(a) No optimization: $I_{avg} = 30\text{ mA} \times 100\% = \mathbf{30\text{ mA}}$

(b) Optimized:

$I_{avg} = 30 \times 0.10 + 8 \times 0.20 + 0.020 \times 0.70$

$= 3.0 + 1.6 + 0.014 = \mathbf{4.61\text{ mA}}$

Power reduction: $(30 - 4.61) / 30 \times 100 =$ **84.6% reduction**. Battery life improves from 1000/30 = 33 hours to 1000/4.61 = 217 hours — a **6.5× improvement**.

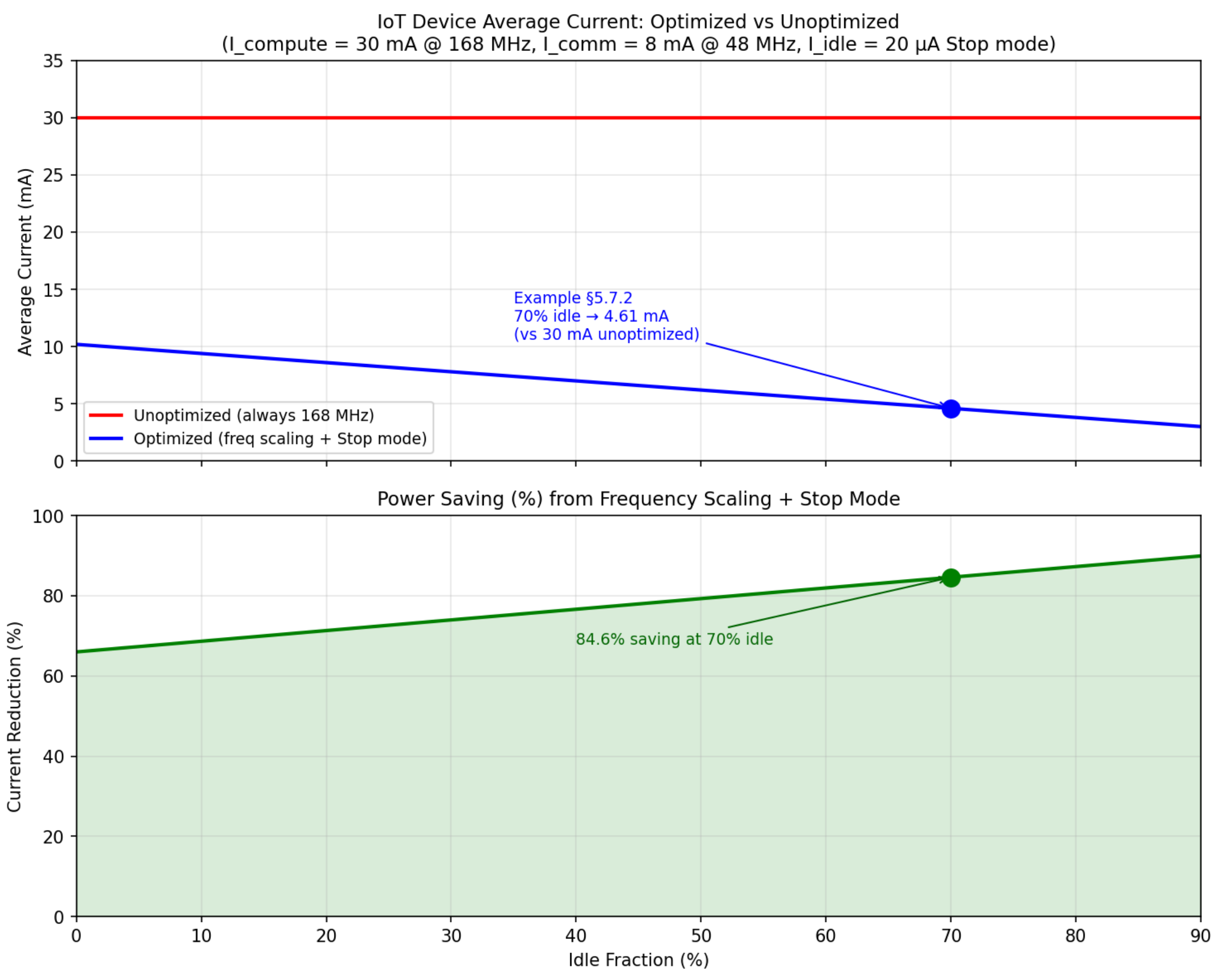



Figure 5.7.2: Average current and percentage power saving vs idle fraction for an IoT device using frequency scaling (168 MHz → 48 MHz) and Stop mode. At 70% idle, the optimized design draws 4.61 mA vs 30 mA unoptimized — an 84.6% reduction.

## 5.8 Development and Debugging

Developing and debugging embedded firmware requires specialized tools that provide visibility into a running system without significantly disrupting its behavior. Unlike application software on a general-purpose OS, embedded code executes directly on hardware with no runtime to catch exceptions or provide debug output, making hardware-assisted debug interfaces and structured firmware update mechanisms essential. The following sections cover the debug and programming infrastructure that underpins modern embedded development.

### 5.8.1 Debug Interfaces (JTAG and SWD)

Debug interfaces provide real-time access to the processor's internal state, enabling developers to set breakpoints, step through code, inspect registers and memory, and flash firmware during development. JTAG (Joint Test Action Group, IEEE 1149.1) is the traditional debug interface using four signal lines: TDI (Test Data In), TDO (Test Data Out), TCK (Test Clock), and TMS (Test Mode Select), plus an optional TRST (Test Reset). JTAG supports daisy-chaining multiple devices on the same bus for boundary scan testing of PCB solder joints. SWD (Serial Wire Debug) is an ARM-specific two-pin alternative (SWDIO and SWCLK) that provides the same debug capabilities as JTAG with fewer pins, making it the standard debug interface for Cortex-M microcontrollers. Debug probes (such as ST-LINK, J-Link, and CMSIS-DAP) bridge between the USB port of the development PC and the target's SWD/JTAG port. ARM CoreSight debug architecture provides features including hardware breakpoints (typically 4-8), data watchpoints, an Instrumentation Trace Macrocell (ITM) for printf-style debug output, and an Embedded Trace Macrocell (ETM) for non-intrusive instruction-level tracing.

**Example 5.8.1:** A developer uses SWD with a J-Link debug probe to flash and debug a Cortex-M4 target. The SWD clock is set to 4 MHz and the firmware image is 256 KB. The flash programming speed through SWD is approximately 50 KB/s for this MCU (including sector erase time). Determine (a) the firmware download time, (b) the SWD raw data rate, and (c) the effective utilization of the SWD link during programming.
**Solution:**
(a) Download time: t = 256 KB / 50 KB/s = **5.12 seconds**
(b) SWD raw data rate: SWD uses a 1-bit data line at 4 MHz. With protocol overhead (packet headers, turnarounds, ACK bits), the effective raw throughput is approximately 50% of the clock rate:
Raw data rate = 4 MHz $\times$ 0.50 / 8 = **250 KB/s**
(c) Effective utilization: 50 / 250 = **20%**. The remaining 80% is consumed by flash erase operations (sector erase takes 16-500 ms per sector depending on the MCU), flash write latency, and read-back verification. This is why flash programming speed is limited by the target's flash controller rather than the SWD link bandwidth.

### 5.8.2 Bootloaders

A bootloader is a small program stored in a protected region of flash memory that executes before the main application firmware and provides a mechanism for updating the application without a dedicated debug probe. On reset, the bootloader runs first and checks for an update trigger — typically a GPIO pin state (e.g., a button held during power-up), a flag set in EEPROM or backup registers, or a command received over a communication interface. If an update is requested, the bootloader receives the new firmware image over UART, USB, CAN, SPI, I2C, or even wireless (OTA — Over-The-Air), verifies its integrity (using CRC-32 or SHA-256 hash), erases the application flash sectors, programs the new image, and then jumps to the application entry point. Many MCUs include a factory-programmed system bootloader in ROM (e.g., STM32 devices support firmware upload via UART and USB DFU from the built-in bootloader). Custom bootloaders add features such as dual-bank (A/B) firmware slots for rollback protection — if the new firmware fails to boot, the bootloader reverts to the previous known-good image.

**Example 5.8.2:** A custom bootloader for a Cortex-M4 MCU occupies the first 32 KB of flash (0x0800_0000 to 0x0800_7FFF). The application starts at 0x0800_8000. The total flash is 512 KB, and a dual-bank scheme reserves 240 KB for each firmware slot with the first 32 KB reserved for the bootloader. The bootloader receives firmware over UART at 115200 baud (8N1). Determine (a) the maximum application size, (b) the time to receive a 200 KB firmware image, and (c) the total update time including flash erase (2 seconds for 240 KB) and programming (50 KB/s).
**Solution:**
(a) Maximum application size per slot: **240 KB**
Memory map: Bootloader (32 KB) + Slot A (240 KB) + Slot B (240 KB) = 512 KB
(b) UART receive time: Effective throughput at 115200 baud, 8N1 = 115,200 / 10 = 11,520 bytes/s
$t_{receive} = 200 \times 1024 / 11{,}520 = 204{,}800 / 11{,}520 =$ **17.78 seconds**
(c) Total update time:
Receive: 17.78 s + Flash erase: 2.0 s + Flash program: 200/50 = 4.0 s + CRC verify: ~0.1 s
Total = **23.9 seconds**
The UART transfer dominates — using USB CDC (1.125 MB/s) would reduce the receive time to 0.18 s and the total update to approximately 6.3 seconds.

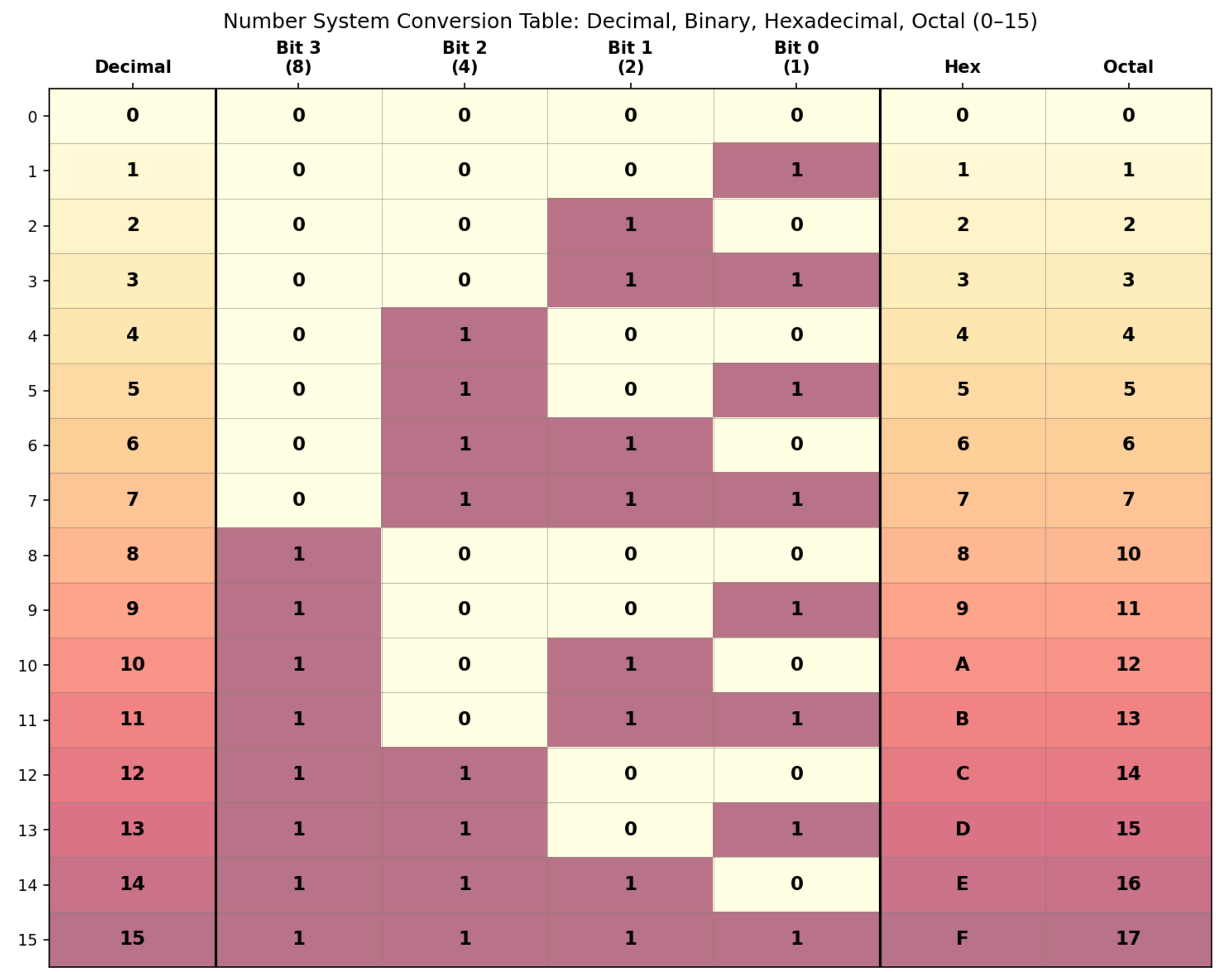
# Chapter 6

# Digital Logic

Digital logic is the foundation of all modern digital systems, from simple combinational circuits to complex microprocessors and FPGAs. It operates on discrete binary values (0 and 1) rather than continuous analog signals, providing noise immunity, reproducibility, and the ability to implement arbitrary computational functions through combinations of basic logic gates. This chapter covers the number systems used to represent digital data, the Boolean algebra that governs logic operations, the fundamental gate types, and the combinational and sequential building blocks from which all digital hardware is constructed.

## 6.1 Number Systems

Digital systems operate on discrete quantities represented as patterns of binary digits, but engineers routinely work with multiple number systems to bridge the gap between hardware-level bit patterns and human-readable values. Binary (base-2) is the native language of digital circuits because each digit maps directly to a voltage level — high or low — making it the only representation that hardware actually implements. Hexadecimal (base-16) and octal (base-8) serve as compact shorthand notations for binary data, with each hex digit representing exactly four bits and each octal digit representing three bits, so engineers can read and write register values, memory addresses, and configuration masks without counting long strings of ones and zeros. Decimal (base-10) remains necessary for human-facing quantities such as clock frequencies, voltage levels, and performance metrics, requiring conversion between bases during design and debugging. Understanding how these number systems relate to one another — and to the physical voltage thresholds in real hardware — is essential for writing firmware, interpreting datasheets, configuring peripherals, and diagnosing digital logic behavior.



Number System Conversion Table: Decimal, Binary, Hexadecimal, Octal (0–15)

| | Decimal | Bit 3 (8) | Bit 2 (4) | Bit 1 (2) | Bit 0 (1) | Hex | Octal |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 2 | 2 | 0 | 0 | 1 | 0 | 2 | 2 |
| 3 | 3 | 0 | 0 | 1 | 1 | 3 | 3 |
| 4 | 4 | 0 | 1 | 0 | 0 | 4 | 4 |
| 5 | 5 | 0 | 1 | 0 | 1 | 5 | 5 |
| 6 | 6 | 0 | 1 | 1 | 0 | 6 | 6 |
| 7 | 7 | 0 | 1 | 1 | 1 | 7 | 7 |
| 8 | 8 | 1 | 0 | 0 | 0 | 8 | 10 |
| 9 | 9 | 1 | 0 | 0 | 1 | 9 | 11 |
| 10 | 10 | 1 | 0 | 1 | 0 | A | 12 |
| 11 | 11 | 1 | 0 | 1 | 1 | B | 13 |
| 12 | 12 | 1 | 1 | 0 | 0 | C | 14 |
| 13 | 13 | 1 | 1 | 0 | 1 | D | 15 |
| 14 | 14 | 1 | 1 | 1 | 0 | E | 16 |
| 15 | 15 | 1 | 1 | 1 | 1 | F | 17 |

Figure 6.1: Number System Conversion Table (Decimal, Binary, Hex, Octal)

### 6.1.1 Binary

The binary number system is a base-2 positional system that uses only two digits, 0 and 1, to represent all values. Each digit position represents a power of 2, with the least significant bit (LSB) at position 0 representing $2^0 = 1$ and each successive position doubling in value. Binary is the native number system of digital electronics because it maps directly to the two voltage states of transistor-based logic circuits — a logic 1 corresponds to a voltage above the high threshold $V_{IH}$ and a logic 0 to a voltage below the low threshold $V_{IL}$, with a forbidden region in between that provides noise immunity. The bit width of a binary number determines its range: an n-bit unsigned integer spans 0 to $2^n - 1$ (for example, an 8-bit value ranges from 0 to 255), while signed integers use two's complement representation, where the most significant bit (MSB) serves as the sign bit, giving a range of $-2^{n-1}$ to $2^{n-1} - 1$ (an 8-bit signed value ranges from −128 to +127). Two's complement is universally used in processors because addition and subtraction circuits work identically for signed and unsigned operands, requiring no special hardware for negative numbers. By convention, the MSB is the leftmost bit (highest power of 2) and the LSB is the rightmost ($2^0$), though data sheets may number bits from 0 at the LSB or from 0 at the MSB depending on the architecture. Conversion from decimal to binary is performed by successive division by 2, recording the remainders from LSB to MSB, and binary arithmetic follows the same positional rules as decimal with carries propagating from right to left. For example, the decimal number 13 is represented as 1101 in binary (8 + 4 + 0 + 1).

**Example 6.1.1:** Convert the decimal number 217 to binary.
**Solution:**
Perform successive division by 2, recording remainders:
217 / 2 = 108 remainder 1 (LSB)
108 / 2 = 54 remainder 0
54 / 2 = 27 remainder 0
27 / 2 = 13 remainder 1
13 / 2 = 6 remainder 1
6 / 2 = 3 remainder 0
3 / 2 = 1 remainder 1
1 / 2 = 0 remainder 1 (MSB)
Reading remainders from MSB to LSB: 217 decimal = 1101 1001 binary.
Verification: 128 + 64 + 0 + 16 + 8 + 0 + 0 + 1 = 217.

### 6.1.2 Hexadecimal

Hexadecimal is a base-16 number system that uses digits 0-9 and letters A-F (representing values 10-15) to provide a compact representation of binary data. Each hexadecimal digit maps directly to a 4-bit binary group (nibble), making it straightforward to convert between binary and hex by simple grouping — no division or multiplication is required, just a one-to-one substitution of each 4-bit nibble for its hex digit. For example, the binary value 1010 1111 splits into 1010 (A) and 1111 (F), giving 0xAF. This direct correspondence makes hex the dominant notation in embedded systems and computer engineering: memory addresses are written in hex (0x0800_0000 for the start of STM32 flash), peripheral register values are documented as hex masks (0x3F to enable bits 0-5), color codes use two hex digits per RGB channel (#FF8000 for orange), and debugger memory dumps display data in hex columns for readability. Two common prefix conventions identify hex values — the 0x prefix used in C, Python, and most programming languages, and the h suffix (such as 3Fh) found in assembly language and some datasheets. Converting from hex to decimal requires expanding each digit by its positional weight (powers of 16), while converting from decimal to hex uses successive division by 16. An 8-bit byte requires exactly two hex digits, a 16-bit half-word requires four, and a 32-bit word requires eight, so hex notation scales cleanly with standard data widths and allows engineers to instantly identify which bits are set in a register or memory value.

**Example 6.1.2:** A 32-bit register contains the binary value 1100 1010 0011 1111 0000 1001 1110 0101. Express this in hexadecimal and determine the decimal value of the upper 8 bits.
**Solution:**
Group into 4-bit nibbles and convert each to hex:
1100 = C, 1010 = A, 0011 = 3, 1111 = F, 0000 = 0, 1001 = 9, 1110 = E, 0101 = 5.
The hexadecimal value is 0xCA3F09E5.
The upper 8 bits are 1100 1010 = 0xCA. Converting to decimal: C = 12, A = 10, so 12 * 16 + 10 = 192 + 10 = 202 decimal.

### 6.1.3 Octal

Octal is a base-8 number system that uses digits 0-7, with each digit representing a 3-bit binary group. While less common than hexadecimal in modern digital systems, octal is still used in Unix/Linux file permissions and some legacy computing contexts. Conversion between binary and octal is performed by grouping binary digits into sets of three from the LSB. For example, the binary value 101 110 is represented as 56 in octal.

**Example 6.1.3:** Convert the octal number 375 to binary and then to decimal.
**Solution:**
Convert each octal digit to its 3-bit binary equivalent:
3 = 011, 7 = 111, 5 = 101.
Binary: 011 111 101 = 11111101 (dropping the leading zero).
Decimal: $3 * 8^2 + 7 * 8^1 + 5 * 8^0 = 3 * 64 + 7 * 8 + 5 * 1 = 192 + 56 + 5 = 253$.
Verification from binary: 128 + 64 + 32 + 16 + 8 + 4 + 0 + 1 = 253.

## 6.2 Boolean Algebra

Boolean algebra is the mathematical foundation of digital logic, operating on variables that can assume only two values: true (1) or false (0). The three fundamental Boolean operations are AND (logical conjunction, denoted A * B or A AND B), OR (logical disjunction, denoted A + B or A OR B), and NOT (logical complement, denoted A' or NOT A). Boolean expressions can be simplified using a set of identities and theorems including the commutative, associative, distributive, identity, complement, idempotent, absorption, and De Morgan's laws. De Morgan's theorems are particularly important in digital design: (A * B)' = A' + B' and (A + B)' = A' * B', allowing conversion between AND and OR forms. Simplification of Boolean expressions directly reduces the number of logic gates required in a circuit, lowering cost, power consumption, and propagation delay.

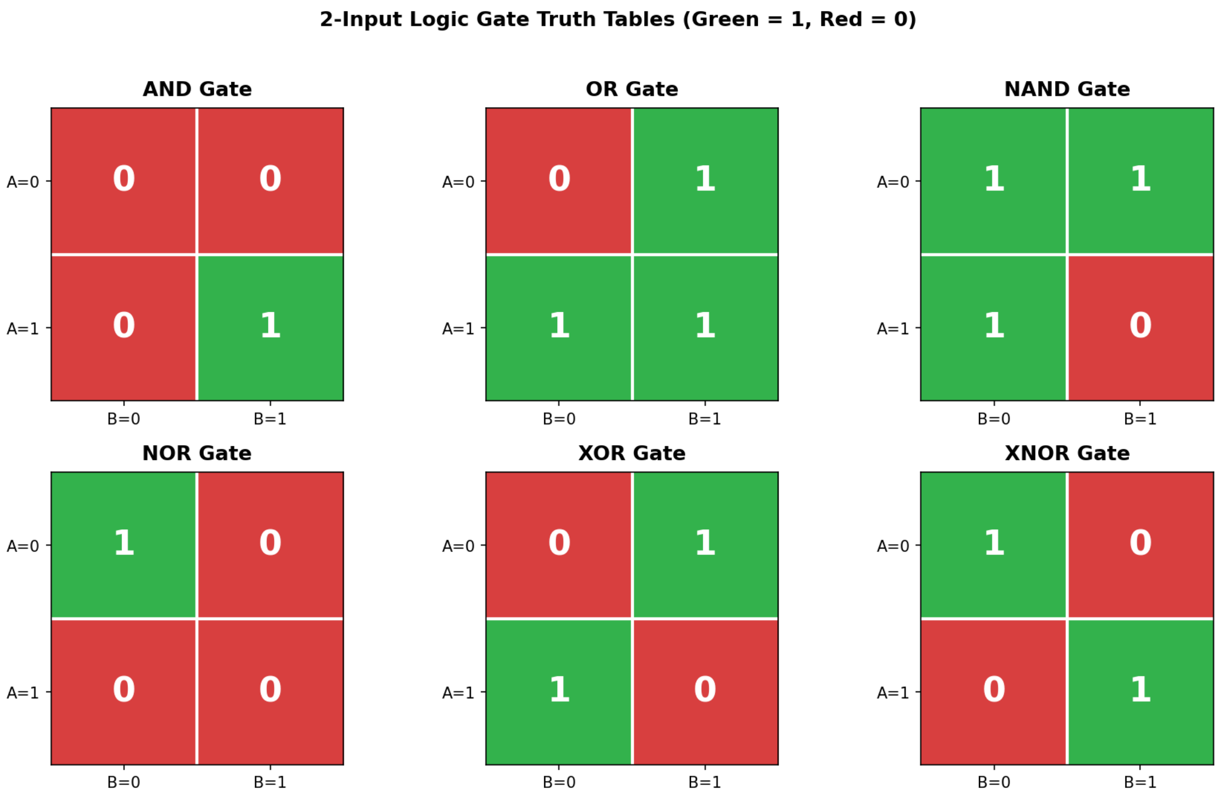



Figure 6.2: Logic Gate Truth Tables

### 6.2.1 Truth Tables

A truth table is a tabular representation that lists all possible combinations of input values and their corresponding output values for a Boolean function. For a function with n input variables, the truth table has $2^n$ rows covering every possible input combination. Truth tables provide a complete and unambiguous specification of a logic function and serve as the starting point for deriving Boolean expressions, designing logic circuits, and verifying equivalence between different implementations. They are also used to define the behavior of standard logic gates and to validate the correctness of combinational circuits.

**Example 6.2.1:** Construct the truth table for the Boolean function F(A, B, C) = A'B + BC'.
**Solution:**
List all $2^3 = 8$ input combinations and evaluate F:

| A | B | C | A' | A'B | C' | BC' | F = A'B + BC' |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |

The function F is 1 for input combinations (A,B,C) = (0,1,0), (0,1,1), and (1,1,0), corresponding to minterms m2, m3, and m6.

### 6.2.2 Karnaugh Maps

A Karnaugh map (K-map) is a graphical method for simplifying Boolean expressions by visually identifying groups of adjacent cells that share common variables. The map arranges truth table outputs in a grid where adjacent cells differ by exactly one variable (Gray code ordering), and groups of 1s in powers of two (1, 2, 4, 8) are circled to form simplified product terms. K-maps are practical for functions of up to four or five variables; beyond that, algorithmic methods such as the Quine-McCluskey algorithm are used. K-maps can also handle "don't care" conditions (inputs that will never occur or whose output value is irrelevant), allowing further simplification by treating them as either 0 or 1 as convenient.

**Example 6.2.2:** Simplify the Boolean function F(A, B, C, D) = sum of minterms (0, 1, 2, 5, 8, 9, 10) using a Karnaugh map.
**Solution:**
Arrange the 4-variable K-map with AB on rows and CD on columns (Gray code order):

```
        CD=00  CD=01  CD=11  CD=10
AB=00 |  1   |  1   |  0   |  1   |
AB=01 |  0   |  1   |  0   |  0   |
AB=11 |  0   |  0   |  0   |  0   |
AB=10 |  1   |  1   |  0   |  1   |
```

Two quads and one pair can be identified:
Group 1 (quad): minterms 0, 1, 8, 9 (AB=00/10, CD=00/01). Rows AB=00 and AB=10 share B=0 (B'=1); columns CD=00 and CD=01 share C=0 (C'=1). Result: B'C'.
Group 2 (quad): minterms 0, 2, 8, 10 (AB=00/10, CD=00/10). Common: B' and D'. Result: B'D'.
Group 3 (pair): minterms 1 and 5 (AB=00/01, CD=01). These cells are adjacent (differ only in the B bit), with common variables A'=1 (both rows have A=0), C'=1, D=1. Result: A'C'D.
Check: B'C' covers {0,1,8,9}. B'D' covers {0,2,8,10}. A'C'D covers {1,5}. All minterms {0,1,2,5,8,9,10} covered.
Simplified expression: **F = B'C' + B'D' + A'C'D**.

## 6.3 Logic Gates

Logic gates are the fundamental building blocks of all digital circuits, each implementing a basic Boolean operation on one or more binary inputs to produce a single binary output. At the physical level, a logic gate is a small circuit of transistors — CMOS technology implements gates using complementary pairs of PMOS and NMOS transistors, where the gate topology (series or parallel connections) determines the Boolean function realized. The three primitive operations — AND, OR, and NOT — can be combined to implement any Boolean function, but in practice the NAND and NOR gates are preferred because they are functionally complete (any function can be built from one gate type alone) and map more efficiently to CMOS transistor structures. Every gate introduces a propagation delay, typically ranging from sub-nanosecond in modern CMOS processes to a few nanoseconds in discrete logic ICs, and the cumulative delay through cascaded gates determines the maximum operating speed of a digital circuit. Fan-in refers to the number of inputs a gate accepts — higher fan-in increases the gate's propagation delay and reduces output drive strength because more transistors are stacked in series. Fan-out is the number of gate inputs that one output can reliably drive; exceeding the fan-out limit degrades voltage levels and increases switching delay, potentially causing logic errors. Complex digital systems — processors, memory controllers, communication interfaces — are built hierarchically from these simple gate primitives: gates form adders and multiplexers, which form ALUs and register files, which form complete datapaths and control units.

### 6.3.1 AND Gate

The AND gate produces a high output (1) only when all of its inputs are high. For a two-input AND gate, the Boolean expression is Y = A * B. The AND function is fundamental to enable logic, where one signal gates (enables or disables) the passage of another signal. AND gates are implemented in CMOS technology using series-connected NMOS transistors and parallel PMOS transistors.

**Example 6.3.1:** A 3-input AND gate has inputs A = 1, B = 1, C = 0. Determine the output. Then write the Boolean expression for a 3-input AND gate and list the number of input combinations that produce a high output.
**Solution:**
Y = A * B * C = 1 * 1 * 0 = 0.
The output is low because not all inputs are high.
The Boolean expression for a 3-input AND is Y = A * B * C.
With 3 inputs there are $2^3 = 8$ possible input combinations.
Only one combination (A=1, B=1, C=1) produces a high output.
The remaining 7 combinations produce a low output.
In general, an n-input AND gate produces a high output for only 1 out of $2^n$ input combinations.

### 6.3.2 OR Gate

The OR gate produces a high output (1) when any one or more of its inputs are high. For a two-input OR gate, the Boolean expression is Y = A + B. OR gates are used in logic circuits where any one of several conditions should trigger an output. In CMOS implementation, OR gates are typically constructed as a NOR gate followed by an inverter.

**Example 6.3.2:** A 4-input OR gate has inputs A = 0, B = 0, C = 0, D = 0. Determine the output. How many input combinations produce a low output for an n-input OR gate?
**Solution:**
Y = A + B + C + D = 0 + 0 + 0 + 0 = 0.
The output is low because none of the inputs are high.
For an n-input OR gate, the output is low only when all inputs are 0, which is exactly 1 combination out of $2^n$.
Therefore, $2^n$ - 1 combinations produce a high output.

For this 4-input OR gate: $2^4$ - 1 = 15 out of 16 combinations produce a high output, and only 1 combination (all zeros) produces a low output.

### 6.3.3 NOT Gate (Inverter)

The NOT gate, or inverter, produces an output that is the logical complement of its input: Y = A'. It is the simplest logic gate, implemented in CMOS with a single complementary pair of PMOS and NMOS transistors. Inverters are fundamental building blocks in digital circuits, used for signal inversion, level shifting, and as components within more complex gates. A chain of an even number of inverters can serve as a buffer to restore signal integrity.

**Example 6.3.3:** A CMOS inverter is powered from $V_{DD}$ = 3.3 V with a switching threshold at $V_{DD}/2$. The inverter has a propagation delay of 5 ns. If 4 inverters are chained together as a buffer, what is the total propagation delay and the output logic level for an input of logic 1?
**Solution:**
Each inverter inverts the signal and adds 5 ns of delay.
Through 4 inverters: input 1 -> 0 -> 1 -> 0 -> 1.
Output = 1 (same as input, since 4 is even).
Total propagation delay = 4 * 5 ns = 20 ns.
The buffer restores the signal to a clean logic level (close to $V_{DD}$ = 3.3 V for logic 1) at the cost of 20 ns latency.
If only 2 inverters were used, the delay would be 10 ns with the same buffering function but less drive capability.

### 6.3.4 NAND Gate

The NAND gate produces a low output (0) only when all inputs are high; otherwise, the output is high. Its Boolean expression is Y = (A * B)'. The NAND gate is functionally complete, meaning any Boolean function can be implemented using only NAND gates. This property makes it the preferred universal gate in CMOS design because it requires fewer transistors than an AND gate (which needs a NAND plus an inverter) and is naturally implemented by the series-NMOS, parallel-PMOS CMOS structure.

**Example 6.3.4:** Implement the function Y = A + B (OR gate) using only 2-input NAND gates.
**Solution:**
Apply De Morgan's theorem: A + B = (A' * B')'. First, generate A' = (A * A)' using a NAND gate with both inputs tied to A.
Similarly, B' = (B * B)'. Then Y = (A' * B')' = NAND(A', B').
Total NAND gates required: 3.
Gate 1: inputs A, A -> output A'.
Gate 2: inputs B, B -> output B'.
Gate 3: inputs A', B' -> output (A' * B')' = A + B.
This demonstrates the functional completeness of the NAND gate.

### 6.3.5 NOR Gate

The NOR gate produces a high output (1) only when all inputs are low; otherwise, the output is low. Its Boolean expression is Y = (A + B)'. Like the NAND gate, the NOR gate is functionally complete and can implement any Boolean function. NOR gates are naturally implemented in CMOS with series-PMOS and parallel-NMOS transistors. Historically, NOR gates were used exclusively in the Apollo Guidance Computer.

**Example 6.3.5:** Implement the AND function Y = A * B using only 2-input NOR gates.
**Solution:**
Apply De Morgan's theorem: A * B = (A' + B')'. First, generate A' = (A + A)' using a NOR gate with both inputs tied to A.
Similarly, B' = (B + B)'. Then compute A' + B' by applying De Morgan's: (A' + B') needs a NOR followed by an inverter, but we can combine steps.
Y = A * B = (A' + B')' = NOR(A', B').
Gate 1: NOR(A, A) = A'.
Gate 2: NOR(B, B) = B'.
Gate 3: NOR(A', B') = (A' + B')' = A * B.
Total NOR gates required: 3.
This confirms the NOR gate's functional completeness, mirroring the NAND implementation of an OR gate.

### 6.3.6 XOR Gate

The Exclusive OR (XOR) gate produces a high output (1) when its inputs differ (one high, one low) and a low output when inputs are the same. Its Boolean expression is Y = A XOR B = A'B + AB'. XOR gates are essential in arithmetic circuits (half adders and full adders), parity generators and checkers, and error detection codes such as CRC. They also function as a controlled inverter: when one input is held high, the XOR gate inverts the other input; when held low, it passes the signal unchanged.

**Example 6.3.6:** A 4-bit even parity generator uses XOR gates to produce a parity bit P such that the total number of 1s in the 5-bit word ($D_3$ $D_2$ $D_1$ $D_0$ P) is even. For the data word $D_3D_2D_1D_0$ = 1011, determine P.
**Solution:**
P = $D_3$ XOR $D_2$ XOR $D_1$ XOR $D_0$.
Step 1: $D_3$ XOR $D_2$ = 1 XOR 0 = 1.
Step 2: result XOR $D_1$ = 1 XOR 1 = 0.
Step 3: result XOR $D_0$ = 0 XOR 1 = 1.
Therefore P = 1. The transmitted word is 1011_1. Verification: the number of 1s in 10111 is 4, which is even. This requires 3 two-input XOR gates connected in a cascade structure, with a propagation delay of 3 gate delays.

### 6.3.7 Logic Families and Voltage Levels

Digital logic gates are manufactured in different technology families, each with distinct voltage thresholds, speed, and power characteristics. The two dominant families are CMOS (Complementary Metal-Oxide-Semiconductor) and TTL (Transistor-Transistor Logic). A gate's input and output voltage levels are specified by four parameters: $V_{IH}$ (minimum input voltage recognized as high), $V_{IL}$ (maximum input voltage recognized as low), $V_{OH}$ (minimum output voltage when driving high), and $V_{OL}$ (maximum output voltage when driving low). The noise margin is the voltage difference between guaranteed output levels and required input levels: $NM_H = V_{OH} - V_{IH}$ and $NM_L = V_{IL} - V_{OL}$. For 5 V CMOS (74HC series), typical values are $V_{OH}$ = 4.9 V, $V_{OL}$ = 0.1 V, $V_{IH}$ = 3.5 V, $V_{IL}$ = 1.5 V, giving noise margins of 1.4 V. For 5 V TTL (74LS series), typical values are $V_{OH}$ = 2.7 V, $V_{OL}$ = 0.5 V, $V_{IH}$ = 2.0 V, $V_{IL}$ = 0.8 V, giving noise margins of 0.7 V and 0.3 V. CMOS gates have near-zero static power dissipation (current flows only during switching) but dynamic power increases with frequency as $P_{dyn} = C_L \times V_{DD}^2 \times f$, where $C_L$ is the load capacitance. Fan-out, the number of gate inputs one output can drive, is effectively unlimited for CMOS at low frequencies (since inputs draw negligible DC current) but is limited at high frequencies by output drive current and capacitive loading. When interfacing between different voltage levels (e.g., 3.3 V and 5 V logic), level-shifting circuits or bidirectional voltage translators are required to prevent damage and ensure valid logic levels.

**Example 6.3.7:** A 74HC00 CMOS NAND gate operates at $V_{DD}$ = 5 V with $V_{OH}$ = 4.9 V, $V_{OL}$ = 0.1 V, $V_{IH}$ = 3.5 V, and $V_{IL}$ = 1.5 V. Calculate the high and low noise margins. If this gate drives a 74LS00 TTL NAND gate with $V_{IH}$ = 2.0 V and $V_{IL}$ = 0.8 V, are the logic levels compatible?
**Solution:**
CMOS noise margins: $NM_H = V_{OH} - V_{IH} = 4.9 - 3.5 = 1.4$ V.
$NM_L = V_{IL} - V_{OL} = 1.5 - 0.1 = 1.4$ V.
For the CMOS-to-TTL interface: The CMOS high output (4.9 V) exceeds the TTL $V_{IH}$ requirement (2.0 V) by 2.9 V — compatible.
The CMOS low output (0.1 V) is well below the TTL $V_{IL}$ requirement (0.8 V) by 0.7 V — compatible.
The CMOS gate can directly drive the TTL gate with generous noise margins.
The reverse (TTL driving CMOS) is problematic: $V_{OH}(\text{TTL}) = 2.7\text{ V} < V_{IH}(\text{CMOS}) = 3.5$ V, so a pull-up resistor or level shifter is needed.

## 6.4 Combinational Circuits

Combinational circuits are logic circuits whose outputs depend solely on the current combination of input values, with no memory of previous states — in contrast to sequential circuits, which incorporate storage elements and whose outputs depend on both current inputs and past history. This memoryless property means that a combinational circuit's behavior is completely specified by a truth table or a set of Boolean expressions: given the same inputs at any point in time, the circuit always produces the same outputs. Design of combinational circuits follows a systematic process: define the desired input-output relationship in a truth table, derive Boolean expressions from the minterms (or maxterms), simplify using Boolean algebra or Karnaugh maps to minimize gate count, and implement the result with physical logic gates. Common combinational building blocks include multiplexers (data selectors), demultiplexers (data distributors), encoders, decoders, and arithmetic circuits such as adders, subtractors, and comparators. These modules serve as the datapath components in processors and digital systems — the ALU, the instruction decoder, the address calculator, and the bus routing logic are all fundamentally combinational. Because combinational outputs can change whenever any input changes (after propagation delay), glitches and hazards must be considered when outputs feed clocked elements, and careful timing analysis ensures that all combinational paths settle before the next clock edge captures the result.

### 6.4.1 Adders

A half adder adds two single-bit inputs (A and B) and produces a sum (S = A XOR B) and a carry output (C = A AND B). A full adder extends this to three inputs (A, B, and carry-in) to produce a sum and carry-out, enabling multi-bit addition by cascading full adders in a ripple-carry configuration. The critical path through a ripple-carry adder grows linearly with the number of bits, limiting speed. Faster alternatives include carry-lookahead adders, which compute carry signals in parallel using generate and propagate logic, reducing the delay to O(log n) for an n-bit addition.

**Example 6.4.1:** Add the 4-bit binary numbers A = 1011 and B = 0110 using a ripple-carry adder. Show the carry chain.
**Solution:**
Add bit by bit from LSB to MSB, propagating carries:
Bit 0: $A_0 + B_0 + C_{in} = 1 + 0 + 0 = 1$, $Sum_0 = 1$, $C_1 = 0$.
Bit 1: $A_1 + B_1 + C_1 = 1 + 1 + 0 = 10$, $Sum_1 = 0$, $C_2 = 1$.
Bit 2: $A_2 + B_2 + C_2 = 0 + 1 + 1 = 10$, $Sum_2 = 0$, $C_3 = 1$.
Bit 3: $A_3 + B_3 + C_3 = 1 + 0 + 1 = 10$, $Sum_3 = 0$, $C_{out} = 1$.
Result: $C_{out}$ $Sum_3$ $Sum_2$ $Sum_1$ $Sum_0$ = 1_0001 = 10001 binary = 17 decimal.
Verification: 1011 (11) + 0110 (6) = 17. The ripple-carry delay is 4 full-adder delays since the carry must propagate through all 4 stages sequentially.

### 6.4.2 Multiplexers

A multiplexer (MUX) is a combinational circuit that selects one of $2^n$ input data lines and routes it to a single output, controlled by n select lines. A 2:1 MUX has the Boolean expression Y = S'A + SB, where S is the select line and A, B are the data inputs. Multiplexers can implement any Boolean function of n variables by using the function's truth table values as data inputs and the variables as select lines. They are fundamental building blocks in data routing, bus architectures, and FPGA lookup tables (LUTs).

**Example 6.4.2:** An 8:1 multiplexer is used to implement the Boolean function F(A, B, C) = sum of minterms (1, 2, 4, 6, 7). Connect variables A, B, C to the select lines $S_2$, $S_1$, $S_0$ respectively. Determine the data input connections.
**Solution:**
The select inputs address the minterms directly. Connect each data input $D_i$ to 1 if minterm i is in the function, and to 0 otherwise:
$D_0$ (ABC = 000, minterm 0) = 0.
$D_1$ (ABC = 001, minterm 1) = 1.
$D_2$ (ABC = 010, minterm 2) = 1.
$D_3$ (ABC = 011, minterm 3) = 0.
$D_4$ (ABC = 100, minterm 4) = 1.
$D_5$ (ABC = 101, minterm 5) = 0.
$D_6$ (ABC = 110, minterm 6) = 1.
$D_7$ (ABC = 111, minterm 7) = 1.
Connect $D_1$, $D_2$, $D_4$, $D_6$, $D_7$ to $V_{CC}$ (logic 1) and $D_0$, $D_3$, $D_5$ to GND (logic 0). The single 8:1 MUX implements the entire 3-variable function with no additional gates.

### 6.4.3 Decoders

A decoder is a combinational circuit that converts an n-bit binary input into one of $2^n$ unique output lines, with exactly one output active for each input combination. A 2-to-4 decoder, for example, takes a 2-bit input and activates one of four output lines. Decoders are used in memory address decoding (selecting a specific memory chip or register), instruction decoding in processors, and seven-segment display drivers. A decoder with an enable input can also function as a demultiplexer, routing a single input signal to one of several outputs.

**Example 6.4.3:** A 3-to-8 decoder is used for memory address decoding. The system has 8 memory chips, each 8 KB in size, mapped to a 64 KB address space starting at address 0x0000. Determine the address range for each chip select output and which decoder output selects the chip at address 0x6000.
**Solution:**
Each chip occupies 8 KB = 0x2000 addresses.
The 3 decoder inputs are connected to address lines $A_{15}$, $A_{14}$, $A_{13}$ (the 3 MSBs of the 16-bit address).
Decoder output mapping:
$Y_0$: $A_{15}$-$A_{13}$ = 000 -> 0x0000 to 0x1FFF (chip 0).
$Y_1$: $A_{15}$-$A_{13}$ = 001 -> 0x2000 to 0x3FFF (chip 1).
$Y_2$: $A_{15}$-$A_{13}$ = 010 -> 0x4000 to 0x5FFF (chip 2).
$Y_3$: $A_{15}$-$A_{13}$ = 011 -> 0x6000 to 0x7FFF (chip 3).
$Y_4$: $A_{15}$-$A_{13}$ = 100 -> 0x8000 to 0x9FFF (chip 4).
$Y_5$ through $Y_7$: 0xA000 to 0xFFFF (chips 5-7).
Address 0x6000 falls in the range 0x6000-0x7FFF, so decoder output $Y_3$ selects the corresponding chip.

### 6.4.4 Encoders

An encoder is the functional complement of a decoder: it converts $2^n$ input lines into an n-bit binary output code. A simple 4-to-2 encoder produces a 2-bit output representing which of its 4 inputs is active, with the assumption that only one input is active at a time. A priority encoder removes this restriction by assigning priority levels to the inputs and encoding the highest-priority active input, outputting both the binary code and a valid flag indicating that at least one input is asserted. Priority encoders are used in interrupt controllers (selecting the highest-priority pending interrupt), keyboard scanners, and leading-zero detectors in floating-point units.

**Example 6.4.4:** An 8-to-3 priority encoder has inputs $I_7$ (highest priority) through $I_0$ (lowest priority) and outputs a 3-bit code $Y_2Y_1Y_0$ plus a valid bit V. If inputs $I_5$, $I_3$, and $I_1$ are simultaneously active (all others inactive), determine the output code and identify which input is encoded.
**Solution:**
The priority encoder selects the highest-priority active input.
Among $I_5$, $I_3$, and $I_1$, the highest priority is $I_5$.
The output code encodes input 5: $Y_2Y_1Y_0$ = 101 (binary for 5).
The valid bit V = 1, indicating at least one input is active.
Inputs $I_3$ and $I_1$ are ignored because they have lower priority than $I_5$.
If $I_5$ were deasserted while $I_3$ and $I_1$ remained active, the output would change to $Y_2Y_1Y_0$ = 011

(encoding $I_3$).
The priority encoder resolves simultaneous assertions without the ambiguity that would cause errors in a simple encoder.

## 6.5 Sequential Circuits

Sequential circuits are logic circuits whose outputs depend on both the current inputs and the history of past inputs, giving them memory capability. They contain storage elements (latches or flip-flops) that maintain state between clock cycles, in addition to combinational logic. Sequential circuits are classified as synchronous (state changes occur only on clock edges) or asynchronous (state changes occur whenever inputs change). All practical digital systems, from simple counters to complex processors, are built from sequential circuits.

### 6.5.1 Latches

A latch is a level-sensitive bistable storage element that can hold one bit of data. The SR (Set-Reset) latch, constructed from cross-coupled NOR or NAND gates, stores a bit based on which input (Set or Reset) was last activated, with the constraint that both inputs must not be asserted simultaneously. The D (Data) latch captures the value of the D input whenever the enable (or gate) signal is active and holds that value when the enable goes inactive. While simple, latches are level-sensitive, meaning they are transparent (input changes pass through to the output) for the entire duration the enable signal is active, which can cause timing problems in synchronous designs.

**Example 6.5.1:** A D latch has Enable = 1. The D input changes in the following sequence: D = 0 at t = 0 ns, D = 1 at t = 15 ns, D = 0 at t = 30 ns. Enable goes low at t = 25 ns. Determine the output Q at t = 10 ns, t = 20 ns, and t = 35 ns.
**Solution:**
While Enable = 1, the latch is transparent and Q follows D.
At t = 10 ns: Enable = 1 and D = 0 (set at t = 0), so Q = 0.
At t = 20 ns: Enable = 1 and D = 1 (changed at t = 15), so Q = 1.
At t = 25 ns: Enable goes low. The latch captures and holds the current value of D, which is 1 at that instant.
At t = 30 ns: D changes to 0, but Enable = 0, so the latch is opaque and Q remains at its latched value.
At t = 35 ns: Q = 1 (held from the value captured when Enable went low). This illustrates the transparency problem: the D change at t = 15 ns passed through to Q because Enable was still high.

### 6.5.2 Flip-Flops

A flip-flop is an edge-triggered bistable storage element that captures its input only on the rising or falling edge of a clock signal, providing well-defined timing for synchronous circuit operation. The D flip-flop is the most common type, capturing the D input value at the clock edge and holding it stable at Q until the next active clock edge. JK flip-flops add the ability to toggle (complement) the stored value when both J and K inputs are high, and T (Toggle) flip-flops toggle the output on each active clock edge when the T input is high. Key timing parameters include setup time (minimum time the input must be stable before the clock edge), hold time (minimum time the input must remain stable after the clock edge), and clock-to-Q delay (propagation delay from the clock edge to the output change). Violating setup or hold times can cause metastability, a condition where the flip-flop enters an indeterminate state that may take an unpredictable amount of time to resolve.

**Example 6.5.2:** A positive-edge-triggered D flip-flop has a setup time $t_{su}$ = 2 ns, hold time $t_h$ = 1 ns, and clock-to-Q delay $t_{cq}$ = 3 ns. The clock period is 10 ns. What is the maximum combinational logic delay allowed between two cascaded flip-flops?
**Solution:**
In a synchronous pipeline, the output of flip-flop 1 passes through combinational logic and must arrive at the input of flip-flop 2 before its setup time.
The timing constraint is: $t_{cq} + t_{logic} + t_{su} <= T_{clk}$.
Solving for maximum logic delay: $t_{logic}(max) = T_{clk} - t_{cq} - t_{su} = 10 - 3 - 2 = 5$ ns.
Any combinational path longer than 5 ns will violate the setup time of the receiving flip-flop, causing potential metastability or incorrect data capture.
The maximum operating frequency is $f_{max} = 1 / (t_{cq} + t_{logic} + t_{su})$.
If the logic delay is 4 ns: $f_{max} = 1 / (3 + 4 + 2) = 1 / 9$ ns = 111.1 MHz.

### 6.5.3 Counters

A counter is a sequential circuit that cycles through a defined sequence of states, typically incrementing or decrementing a binary value on each clock pulse. Binary counters count in natural binary sequence (0, 1, 2, ..., $2^n$ - 1) and are built by cascading flip-flops with appropriate feedback. Synchronous counters clock all flip-flops simultaneously, ensuring clean transitions and predictable timing, while asynchronous (ripple) counters clock each stage from the output of the previous stage, introducing cumulative propagation delay. Counters are used in frequency division, event counting, timing generation, and as the basis for address generators and program counters in processors. Modulo-N counters reset to zero after reaching N-1, enabling division of a clock frequency by any integer value.

**Example 6.5.3:** Design a modulo-6 counter using a 3-bit binary counter. The input clock is 12 MHz. Determine the output frequency and the reset logic.
**Solution:**
A 3-bit binary counter naturally counts from 000 to 111 (modulo-8).
To create a modulo-6 counter, detect the count of 6 (binary 110) and force an asynchronous reset to 000.
Reset logic: RESET = $Q_2 * Q_1 * Q_0$', but since we want to detect 110, we need RESET = $Q_2$ AND $Q_1$ (since $Q_0 = 0$ at count 110, and this is the first time $Q_2$ and $Q_1$ are both high after reset).
More precisely, detect count 110: RESET = $Q_2$ AND $Q_1$ AND (NOT $Q_0$). However, $Q_0$ is 0 at count 6, so RESET = $Q_2$ AND $Q_1$ suffices (count 6 = 110 is the first occurrence of $Q_2$=$Q_1$=1).
The counter cycles: 000, 001, 010, 011, 100, 101, 110 (reset) -> 000.
Count sequence is 0-5 (6 states).
Output frequency = 12 MHz / 6 = 2 MHz, taken from the most significant bit ($Q_2$) or the reset signal.

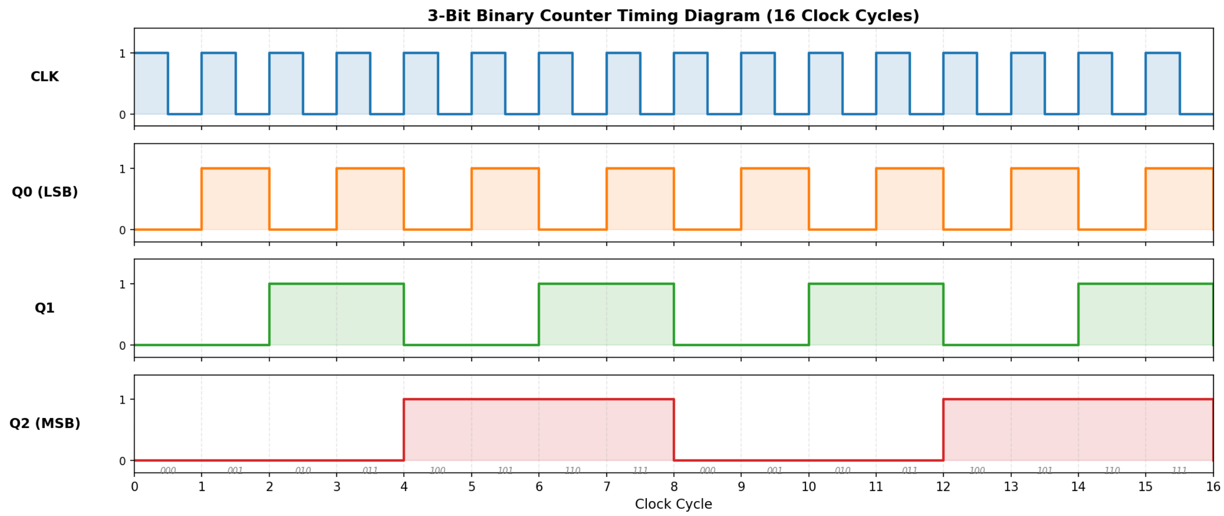



Figure 6.5.3: 3-Bit Binary Counter Timing Diagram

### 6.5.4 Shift Registers

A shift register is a chain of flip-flops where the output of each stage feeds the input of the next, shifting data one position per clock cycle. They support serial-in/serial-out (SISO), serial-in/parallel-out (SIPO), parallel-in/serial-out (PISO), and parallel-in/parallel-out (PIPO) configurations. Shift registers are used for serial-to-parallel and parallel-to-serial data conversion in communication interfaces, temporary data storage, and time delays. Linear Feedback Shift Registers (LFSRs) use XOR feedback from selected tap positions to generate pseudo-random binary sequences, widely used in CRC calculation, scrambling, and built-in self-test (BIST) circuits.

**Example 6.5.4:** An 8-bit serial-in/parallel-out (SIPO) shift register receives the serial bit stream 1 0 1 1 0 1 0 0 (MSB first) at a clock rate of 2 MHz. After 8 clock pulses, what is the parallel output and how long did the conversion take?
**Solution:**
The bits are shifted in one per clock cycle, MSB first. After 8 clock pulses, the register contains the complete byte:
Clock 1: shift in 1 -> register = xxxxxxx1
Clock 2: shift in 0 -> register = xxxxxx10
Clock 3: shift in 1 -> register = xxxxx101
Clock 4: shift in 1 -> register = xxxx1011
Clock 5: shift in 0 -> register = xxx10110
Clock 6: shift in 1 -> register = xx101101
Clock 7: shift in 0 -> register = x1011010
Clock 8: shift in 0 -> register = 10110100
Parallel output after 8 clocks: 1011 0100 = 0xB4 = 180 decimal.
Conversion time = 8 clock cycles / 2 MHz = 4 μs. The serial-to-parallel conversion trades time for pin count, using 1 data pin instead of 8.

### 6.5.5 State Machines

A finite state machine (FSM) is a sequential circuit that transitions between a finite number of states based on its current state and input values, producing outputs according to a defined mapping. Moore machines produce outputs that depend only on the current state, making their outputs synchronous and glitch-free since they change only on clock edges. Mealy machines produce outputs that depend on both the current state and the current inputs, allowing faster response (output can change within a clock cycle when inputs change) but with the risk of glitches on the outputs. FSMs are designed by defining a state diagram (circles for states, arrows for transitions labeled with input conditions), deriving a state table, assigning binary codes to states, and implementing the next-state logic and output logic with flip-flops and combinational gates. State machines are the backbone of digital controllers, protocol handlers, traffic light controllers, vending machines, and the control units within processors.

**Example 6.5.5:** Design a Moore state machine that detects the input sequence "101" on a serial input X, asserting output Z = 1 for one clock cycle when the sequence is detected. The detector should be overlapping, meaning the final "1" of one detection can serve as the first "1" of the next.
**Solution:**
Define four states based on progress toward detecting "101":
S0 = idle (no relevant bits received), Z = 0.
S1 = received "1", Z = 0.
S2 = received "10", Z = 0.
S3 = received "101" (sequence detected), Z = 1.
State transitions (current state, input X → next state):
S0, X=0 → S0. S0, X=1 → S1.
S1, X=0 → S2. S1, X=1 → S1 (restart with new "1").
S2, X=0 → S0 (reset, "100" is not a prefix of "101"). S2, X=1 → S3 (sequence complete).
S3, X=0 → S2 (overlapping: the "1" from S3 serves as start, then "0" → S2). S3, X=1 → S1 (overlapping: restart with new "1").
Using 2-bit state encoding (S0=00, S1=01, S2=10, S3=11) with two D flip-flops $Q_1Q_0$, the next-state equations are:
$Q_1^+ = Q_0 \cdot X'$ (S1→S2 and S3→S2) + $Q_1 \cdot Q_0' \cdot X$ (S2→S3).
After K-map simplification: $Q_1^+ = Q_0 \cdot X' + Q_1 \cdot Q_0' \cdot X$, $Q_0^+ = X$.
Output: $Z = Q_1 \cdot Q_0$ (high only in state S3 = 11). This requires 2 flip-flops, 2 AND gates, 1 OR gate, and 1 inverter.

## 6.6 Programmable Logic

Programmable logic devices allow engineers to implement custom digital logic functions in silicon without designing an application-specific integrated circuit (ASIC). Rather than committing to a fixed gate-level design at manufacturing time, programmable devices can be configured — and often reconfigured — in the field by loading a bitstream that defines connections between logic elements. This flexibility compresses development time and cost: a design can be tested in real hardware, debugged, and updated without a new silicon tape-out. The two dominant families are FPGAs (Field-Programmable Gate Arrays), which use volatile SRAM-based configuration cells and offer the largest capacity and highest performance, and CPLDs (Complex Programmable Logic Devices), which use non-volatile flash and offer instant-on operation with predictable timing. Together they span a wide range of applications from simple glue logic to complete system-on-chip implementations.

### 6.6.1 FPGA

A Field-Programmable Gate Array (FPGA) is an integrated circuit that can be configured by the user after manufacturing to implement virtually any digital logic function. FPGAs consist of an array of configurable logic blocks (CLBs) containing lookup tables (LUTs), flip-flops, and multiplexers, surrounded by programmable interconnects and I/O blocks. The logic function of each CLB is defined by the contents of its LUTs (typically 4-input or 6-input), which can implement any Boolean function of their input variables. FPGAs are programmed using hardware description languages (HDL) such as Verilog or VHDL, which are synthesized into a configuration bitstream that defines the LUT contents and routing connections. They are used for prototyping ASICs, implementing custom digital signal processing, high-speed communication interfaces, and applications requiring hardware-level parallelism and deterministic timing that software cannot achieve.

**Example 6.6.1:** An FPGA has 5000 6-input LUTs and 5000 flip-flops. A design requires 200 instances of an 8-bit register (8 flip-flops each) and 150 instances of a 4-input Boolean function (1 LUT each). Determine the resource utilization.
**Solution:**
Flip-flop usage: 200 registers * 8 flip-flops each = 1600 flip-flops.
Utilization = 1600 / 5000 = 32.0%.
LUT usage: each 4-input Boolean function fits in one 6-input LUT (since a 6-input LUT can implement any function of up to 6 variables).
LUTs used = 150.
Additionally, the 8-bit registers may require LUTs for enable/load logic; assume 1 LUT per register for load control = 200 LUTs.
Total LUTs = 150 + 200 = 350.
LUT utilization = 350 / 5000 = 7.0%.
Both resources are well within capacity.
The design uses 32% of flip-flops and 7% of LUTs, leaving substantial room for additional logic such as state machines, communication interfaces, and debug circuitry.

### 6.6.2 CPLD

A Complex Programmable Logic Device (CPLD) is a programmable logic device that combines multiple programmable logic array blocks with a fixed interconnect structure on a single chip. Unlike FPGAs, CPLDs use non-volatile configuration memory (typically flash), so they are operational immediately at power-up without requiring an external configuration memory. CPLDs offer predictable pin-to-pin timing because of their fixed interconnect architecture, making them well-suited for glue logic, bus interface control, and simple state machines. They are typically smaller than FPGAs (hundreds to a few thousand logic elements) and are used where a small to moderate amount of programmable logic is needed with instant-on capability and deterministic timing.

**Example 6.6.2:** A CPLD with a guaranteed pin-to-pin propagation delay of 7.5 ns is used as bus interface glue logic. It decodes a 16-bit address bus to generate 4 chip select signals and provides a bidirectional 8-bit data bus buffer with read/write control. Estimate the total macrocell usage if each chip select requires 1 macrocell and each data bus buffer bit requires 2 macrocells (one for direction control logic, one for the output). The CPLD has 128 macrocells.

**Solution:**
Chip select logic: 4 chip selects * 1 macrocell each = 4 macrocells.
Data bus buffering: 8 bits * 2 macrocells per bit = 16 macrocells.
Additional control logic (read/write timing, bus enable): estimate 4 macrocells.
Total macrocells used = 4 + 16 + 4 = 24 macrocells.
Utilization = 24 / 128 = 18.75%.
The maximum bus interface speed is limited by the pin-to-pin delay: minimum cycle time = 2 * 7.5 ns = 15 ns (for a read requiring address decode and data return), supporting bus speeds up to 1 / 15 ns = 66.7 MHz.
The instant-on capability ensures the chip selects are valid immediately at power-up, preventing bus contention during system initialization.

### 6.6.3 Timing Analysis

Timing analysis ensures that all signals in a synchronous digital circuit arrive at their destinations within the required time windows. Every logic gate introduces a propagation delay — the time from an input change to the corresponding output change — characterized by $t_{pHL}$ (high-to-low transition) and $t_{pLH}$ (low-to-high transition), which may differ due to asymmetric PMOS/NMOS drive strengths. The critical path is the longest combinational delay between any two flip-flops in the design and determines the maximum clock frequency: $f_{max} = 1 / (t_{cq} + t_{crit} + t_{su})$, where $t_{cq}$ is the clock-to-Q delay of the source flip-flop, $t_{crit}$ is the critical path delay, and $t_{su}$ is the setup time of the destination flip-flop. The hold time constraint ensures that data remains stable long enough after the clock edge: $t_{cq} + t_{min} \geq t_h$, where $t_{min}$ is the minimum combinational delay and $t_h$ is the hold time. Hazards are unwanted momentary glitches in combinational logic outputs caused by unequal propagation delays through different signal paths. A static-1 hazard produces a brief 0 glitch when the output should remain at 1, and a static-0 hazard produces a brief 1 glitch when the output should remain at 0. Hazards can be eliminated by adding redundant product terms identified from the K-map or by registering outputs with flip-flops.

**Example 6.6.3:** A synchronous circuit uses flip-flops with $t_{cq}$ = 1.2 ns, $t_{su}$ = 0.8 ns, and $t_h$ = 0.3 ns. The combinational logic between two flip-flops consists of a 3-level gate network with gate delays of 2.5 ns, 1.8 ns, and 3.0 ns. Determine the critical path delay, maximum clock frequency, and whether the hold time constraint is satisfied assuming the minimum path delay through the logic is 1.5 ns.
**Solution:**
Critical path delay: $t_{crit}$ = 2.5 + 1.8 + 3.0 = 7.3 ns.
Maximum clock frequency: $f_{max} = 1 / (t_{cq} + t_{crit} + t_{su})$ = 1 / (1.2 + 7.3 + 0.8) = 1 / 9.3 ns = 107.5 MHz.
Hold time check: $t_{cq} + t_{min}$ = 1.2 + 1.5 = 2.7 ns $\geq t_h$ = 0.3 ns — satisfied with 2.4 ns of margin.
The design can safely operate at up to 107.5 MHz. If a higher frequency is needed, the critical path must be shortened by using faster gates, reducing logic levels (pipelining), or restructuring the combinational logic.

# Chapter 7

# Circuit Analysis

Circuit analysis is the systematic study of electrical circuits to determine voltages, currents, and power at every point in a network. It provides the essential analytical techniques that underpin all branches of electrical engineering, from power systems and electronics to communications and control. By applying fundamental laws and theorems to circuit models composed of ideal elements such as resistors, capacitors, inductors, and sources, engineers can predict circuit behavior, verify designs, and troubleshoot faults.

## 7.1 Electric Charge and Current

Electric charge and current are the most fundamental quantities in electrical engineering. Every circuit phenomenon — voltage drops, power dissipation, magnetic fields, signal propagation — ultimately traces back to the movement and accumulation of electric charge. Understanding charge, current, and voltage at a conceptual level provides the foundation for all circuit analysis techniques that follow.

### 7.1.1 Electric Charge

Electric charge is an intrinsic property of subatomic particles that gives rise to electromagnetic forces. The two types of charge are positive (carried by protons) and negative (carried by electrons). The SI unit of charge is the coulomb (C), and the elementary charge is $e \approx 1.602 \times 10^{-19}$ C — the magnitude of charge on a single electron or proton. One coulomb equals approximately $6.242 \times 10^{18}$ elementary charges. Charge is conserved (it cannot be created or destroyed, only transferred) and quantized (it always appears in integer multiples of e). In conductors such as copper, the outer valence electrons are loosely bound and free to move through the crystal lattice, forming a "sea" of mobile charge carriers. In insulators, electrons are tightly bound and cannot move freely.

**Example 7.1.1:** A copper wire has a cross-sectional area of 2.08 mm$^2$ (14 AWG). Copper has a free electron density of $n = 8.5 \times 10^{28}$ electrons/m$^3$. If the wire carries a current of 15 A, determine (a) the total charge that flows through the wire in 1 minute, (b) the number of electrons that pass a cross-section in that time, and (c) the drift velocity of the electrons.
**Solution:**
(a) Charge: $Q = I \times t = 15 \times 60 =$ **900 C**
(b) Number of electrons: $N = Q/e = 900 / (1.602 \times 10^{-19}) =$ **$5.62 \times 10^{21}$ electrons**
(c) Drift velocity: $v_d = I / (n \times A \times e) = 15 / (8.5 \times 10^{28} \times 2.08 \times 10^{-6} \times 1.602 \times 10^{-19})$
$v_d = 15 / (2.831 \times 10^4) = 5.30 \times 10^{-4}$ m/s = **0.53 mm/s**

Despite the large current, the electrons drift remarkably slowly — the electrical signal propagates near the speed of light because the electric field pushes all electrons in the wire simultaneously.

### 7.1.2 Current

Electric current is the rate of flow of electric charge past a point in a circuit, defined mathematically as $I = dQ/dt$ and measured in amperes (A), where one ampere equals one coulomb per second. Conventional current direction is defined as the direction positive charges would move — from higher potential (+) to lower potential (−) — which is opposite to the actual electron flow in metallic conductors. This convention, established before the discovery of the electron, is universally used in circuit analysis and causes no practical difficulty as long as it is applied consistently. Current can be direct (DC), where the magnitude and direction are constant, or alternating (AC), where the current periodically reverses direction — most power systems operate with 50 or 60 Hz AC. Current density $J = I/A$ ($A/m^2$) describes how current is distributed across a conductor cross-section and is an important design parameter for preventing overheating. The magnitudes encountered in practice span an enormous range: picoamperes in photodiode dark current, nanoamperes in CMOS leakage, milliamperes in sensor circuits, amperes in household wiring, and kiloamperes in industrial power systems and arc furnaces. Current is always measured by placing an ammeter (or current-sensing resistor) in series with the circuit element of interest, since the instrument must carry the same current as the element being measured. Kirchhoff's Current Law (KCL) — which states that the algebraic sum of all currents entering and leaving any node is zero — is the fundamental conservation principle governing current distribution in every circuit and is developed fully in §7.3.3. For safety and conductor sizing, the National Electrical Code specifies maximum allowable current (ampacity) for each conductor gauge and insulation type.

**Example 7.1.2:** A rechargeable battery with a capacity of 2,500 mAh is charged at a constant current of 500 mA. Determine (a) the charging time, (b) the total charge stored in coulombs, and (c) the number of electrons transferred during charging.
**Solution:**
(a) Charging time: $t$ = capacity / $I$ = 2,500 mAh / 500 mA = **5 hours**
(b) Total charge: $Q = I \times t = 0.5\ A \times (5 \times 3{,}600\ s) = 0.5 \times 18{,}000 =$ **9,000 C**
(c) Number of electrons: $N = Q/e = 9{,}000 / (1.602 \times 10^{-19}) =$ **$5.62 \times 10^{22}$ electrons**

### 7.1.3 Voltage and Energy

Voltage (also called electric potential difference or electromotive force) is the work done per unit charge in moving charge between two points: $V = W/Q$, measured in volts (V), where one volt equals one joule per coulomb. Voltage is always measured between two points — it is a relative quantity, not an absolute one. A voltage source (such as a battery or generator) provides the energy that drives current through a circuit by maintaining a potential difference across its terminals.

The energy delivered to or consumed by a circuit element is $W = Q \times V = V \times I \times t$, measured in joules (J). Power is the rate of energy transfer: $P = dW/dt = V \times I$, measured in watts (W). The kilowatt-hour (kWh) is the practical unit of electrical energy used for billing: 1 kWh = $3.6 \times 10^6$ J. The concepts of charge, current, voltage, and power are related by the fundamental equations: $I = Q/t$, $V = W/Q$, $P = V \times I$, and $W = P \times t$.

**Example 7.1.3:** An electric vehicle battery pack is rated at 75 kWh and operates at a nominal voltage of 400 V. Determine (a) the total stored energy in joules and megajoules, (b) the battery capacity in ampere-hours, (c) the current draw when the motor consumes 150 kW, and (d) the driving time at that power level.
**Solution:**
(a) Energy: W = 75 kWh × 3.6 × $10^6$ J/kWh = **270 × $10^6$ J = 270 MJ**
(b) Capacity: Q = W/V → Ah = kWh/V × 1,000 = 75,000 Wh / 400 V = **187.5 Ah**
(c) Current: I = P/V = 150,000 / 400 = **375 A**
(d) Driving time: t = W/P = 75 kWh / 150 kW = **0.5 hours (30 minutes)**

## 7.2 Passive Components

Passive components are circuit elements that cannot generate energy — they can only store or dissipate it. The three fundamental passive components are resistors (which dissipate energy as heat), capacitors (which store energy in an electric field), and inductors (which store energy in a magnetic field). Understanding their physical construction, ratings, and non-ideal behaviors is essential for selecting appropriate components in practical circuit design.

### 7.2.1 Resistors

A resistor opposes the flow of electric current, converting electrical energy irreversibly into heat. Resistance is measured in ohms (Ω) and is determined by the material's resistivity ρ, length L, and cross-sectional area A according to $R = \rho L/A$. Ohm's Law — $V = IR$ — is the fundamental relationship governing resistor behavior: the voltage across a resistor is directly proportional to the current through it, and the power dissipated is $P = I^2R = V^2/R$. In practical design, selecting the right resistor involves balancing several criteria: the resistance value and tolerance (how closely the actual value matches the nominal), the temperature coefficient (how much resistance drifts with temperature, measured in ppm/°C), the power rating (maximum continuous dissipation without overheating), and the physical package size. Common resistor technologies include carbon film for general-purpose use, metal film for precision and low noise, wirewound for high-power applications, and surface-mount (SMD) thick-film and thin-film for compact PCB designs. Resistors combine simply: in series, resistances add directly ($R_{total} = R_1 + R_2 + \ldots + R_n$), while in parallel they combine as reciprocals ($1/R_{total} = 1/R_1 + 1/R_2 + \ldots + 1/R_n$). Standard resistor values follow the E-series (E12, E24, E96, E192), which are logarithmically spaced to provide approximately uniform percentage coverage across each decade. The most common resistor types are:

| Type | Construction | Tolerance | Temp. Coefficient | Applications |
|---|---|---|---|---|
| Carbon film | Carbon deposited on ceramic | ±5% | −200 to −500 ppm/°C | General purpose |
| Metal film | Metal alloy on ceramic | ±1% or ±0.1% | ±50 to ±100 ppm/°C | Precision circuits |
| Wirewound | Resistance wire on core | ±0.01% to ±5% | ±20 to ±50 ppm/°C | Power, precision |

| Type | Construction | Tolerance | Temp. Coefficient | Applications |
|---|---|---|---|---|
| SMD thick film | Ruthenium oxide on alumina | ±1% to ±5% | ±100 to ±200 ppm/°C | Surface-mount PCBs |
| SMD thin film | Nichrome sputtered on alumina | ±0.1% to ±1% | ±25 ppm/°C | Precision SMD |

The four-band resistor color code encodes resistance value: the first two bands are significant digits, the third is the multiplier (number of zeros), and the fourth indicates tolerance (gold = ±5%, brown = ±1%). For example, red-violet-orange-gold = $27 \times 10^3$ = 27 kΩ ±5%. Five-band and six-band codes provide additional precision and a temperature coefficient band.

Power rating specifies the maximum continuous power a resistor can safely dissipate without exceeding its temperature limit. Standard through-hole ratings are 1/8 W, 1/4 W, 1/2 W, 1 W, and 2 W, while SMD sizes range from 0201 (1/20 W) to 2512 (1 W). Exceeding the power rating causes overheating, resistance drift, and eventual failure. Derating curves reduce the allowable power at elevated ambient temperatures — a typical resistor is derated to zero at 150°C.

**Example 7.2.1:** A circuit requires a 4.7 kΩ resistor that will drop 12 V. Determine (a) the current through the resistor, (b) the power dissipated, (c) the minimum standard power rating, and (d) the color code for a 4-band resistor with ±5% tolerance.
**Solution:**
(a) Current: I = V/R = 12 / 4,700 = **2.55 mA**
(b) Power: P = $V^2$/R = 144 / 4,700 = 0.0306 W = **30.6 mW**
(c) The next standard power rating above 30.6 mW is **1/8 W (125 mW)**, which provides a comfortable 4:1 derating margin.
(d) 4.7 kΩ = $47 \times 10^2$ Ω. First band: 4 = yellow. Second band: 7 = violet. Multiplier: $10^2$ = red. Tolerance: ±5% = gold. Color code: **yellow-violet-red-gold**.

### 7.2.2 Capacitors

A capacitor stores energy in an electric field established between two conductive plates separated by a dielectric material. The capacitance $C = \varepsilon A/d$ depends on the plate area A, the plate separation d, and the permittivity ε of the dielectric, and is measured in farads (F), though practical values range from picofarads (pF) in RF tuning circuits to millifarads (mF) in power supply filtering and up to thousands of farads in supercapacitors used for energy storage. The voltage across a capacitor cannot change instantaneously — the current-voltage relationship $i = C(dv/dt)$ means that a capacitor resists sudden voltage changes by drawing or supplying current proportional to the rate of voltage change, which is the basis for capacitor behavior in AC and transient circuits. The energy stored is $W = \frac{1}{2}CV^2$. In the frequency domain, a capacitor presents an impedance $Z = 1/(j\omega C)$ that decreases with increasing frequency, making capacitors behave as short circuits at high frequencies and open circuits at DC. This frequency-dependent behavior is what makes capacitors essential in filtering (bypassing high-frequency noise to ground), coupling (passing AC signals while blocking DC bias), decoupling (providing local charge reservoirs near ICs to suppress supply noise), and timing circuits (setting RC time constants). Capacitor types are distinguished by their dielectric material, which determines the capacitance range, voltage rating, stability, equivalent series resistance (ESR), and frequency performance — selecting the right type for each application is one of the most important practical skills in circuit design.

| Type | Capacitance Range | Voltage Range | Key Characteristics |
|---|---|---|---|
| Ceramic (C0G/NP0) | 1 pF – 10 nF | 16–500 V | Ultra-stable, low loss, precision |
| Ceramic (X7R) | 100 pF – 100 μF | 6.3–100 V | Moderate stability, general purpose |
| Ceramic (Y5V) | 10 nF – 100 μF | 6.3–50 V | High capacitance density, poor stability |
| Aluminum electrolytic | 0.1 μF – 1 F | 6.3–500 V | Polarized, high capacitance, high ESR |
| Tantalum | 0.1 μF – 1 mF | 2.5–50 V | Polarized, stable, low ESR, failure-prone |
| Film (polyester/PP) | 100 pF – 100 μF | 50–2,000 V | Non-polarized, low loss, excellent stability |
| Supercapacitor (EDLC) | 0.1 F – 3,000 F | 2.5–5.5 V | Ultra-high capacitance, energy storage |

Equivalent Series Resistance (ESR) is the parasitic resistance of a real capacitor, representing losses in the dielectric, plates, and leads. ESR is critical in power supply filtering — a capacitor with high ESR cannot effectively suppress ripple even if its capacitance value is adequate. Aluminum electrolytics have the highest ESR (0.1–5 Ω), while ceramic and film capacitors have very low ESR (< 0.01 Ω). Voltage derating is the practice of selecting a capacitor rated well above the circuit operating voltage to improve reliability and account for the voltage coefficient — ceramic capacitors (especially X7R and Y5V) can lose 50% or more of their rated capacitance near their maximum voltage.

**Example 7.2.2:** A 5 V power supply uses a 100 μF aluminum electrolytic capacitor (ESR = 0.15 Ω) in parallel with a 10 μF ceramic capacitor (ESR = 0.005 Ω) for output filtering. A load step of 2 A occurs. Determine (a) the initial voltage drop from ESR for each capacitor acting alone, (b) the voltage droop from capacitor discharge if the supply takes 50 μs to respond, and (c) why both capacitors are needed.

**Solution:**

(a) ESR voltage drop (instantaneous):

Electrolytic alone: $\Delta V_{ESR} = I \times ESR = 2 \times 0.15 =$ **300 mV**

Ceramic alone: $\Delta V_{ESR} = I \times ESR = 2 \times 0.005 =$ **10 mV**

(b) Voltage droop from discharge (using combined capacitance):

$C_{total} = 100 + 10 = 110$ μF.

$\Delta V = I \times \Delta t / C = 2 \times 50 \times 10^{-6} / (110 \times 10^{-6}) =$ **0.91 V**

(c) The **ceramic capacitor handles the fast ESR transient** (10 mV vs. 300 mV), while the **electrolytic provides bulk charge storage** to limit the voltage droop (100 μF » 10 μF). Together they provide both fast transient response and sustained current delivery — this is why power supply designs use both types in parallel.

### 7.2.3 Inductors

An inductor stores energy in a magnetic field generated by current flowing through a coil of wire. Inductance L is measured in henries (H), though practical values range from nanohenries (nH) for PCB traces and chip inductors to henries for power-line reactors and filter chokes. The voltage across an inductor is proportional to the rate of change of current: $v = L(di/dt)$, which means the current through an inductor cannot change instantaneously — any attempt to abruptly interrupt inductor current produces a large voltage spike, which is why flyback diodes and snubber circuits are essential when switching inductive loads. The energy stored in the magnetic field is $W = ½LI^2$. In the frequency domain, an inductor presents an impedance $Z = j\omega L$ that increases with frequency, making inductors behave as open circuits at high frequencies and short circuits at DC — the dual behavior to capacitors. Inductor construction varies widely by application: air-core inductors avoid saturation effects and are used in RF circuits, ferrite-core inductors provide high inductance in compact packages for switching power supplies and EMI filters, iron-powder cores offer higher saturation current for power applications, and toroidal shapes minimize radiated EMI by containing the magnetic flux within the core. Key specifications for inductor selection include the inductance value, saturation current $I_{sat}$ (above which the core saturates and inductance drops sharply), DC resistance (DCR, which causes $I^2R$ losses and reduces efficiency), and the quality factor $Q = \omega L/R_{DC}$, which measures how much energy is stored versus dissipated per cycle. Inductors are indispensable in switching power converters (as energy-transfer elements), LC filters (for frequency selection), RF matching networks, and current-sensing applications.

| Type | Inductance Range | Current Range | Key Characteristics |
|---|---|---|---|
| Air core | 1 nH – 10 μH | up to 100+ A | No saturation, low inductance, RF applications |
| Ferrite core | 1 μH – 10 mH | 0.1–50 A | High frequency (switching supplies, filters) |
| Iron powder core | 1 μH – 10 mH | 0.5–100 A | Higher saturation than ferrite, moderate frequency |
| Laminated iron core | 1 mH – 100 H | 0.1–1,000 A | Power-line frequency, transformers, reactors |
| Toroidal | 10 μH – 100 mH | 0.1–50 A | Low EMI (contained field), compact, efficient |

The quality factor $Q = \omega\ L\ /\ R_{DC}$ measures how much energy an inductor stores versus how much it dissipates per cycle. Higher Q indicates a more ideal inductor. At higher frequencies, skin effect and proximity effect increase the effective resistance, reducing Q. Saturation current is the DC current at which inductance drops by a specified amount (typically 10–30%) due to magnetic core saturation —

exceeding this current causes the inductor to lose its energy storage ability and behave more like a short circuit, which can be destructive in switching converter applications.

Self-resonant frequency (SRF) is the frequency at which the inductor's parasitic capacitance (between turns) resonates with its inductance, causing the component to behave as a capacitor above this frequency. An inductor should only be used at frequencies well below its SRF — typically below SRF/3.

**Example 7.2.3:** A buck converter switching at 500 kHz requires a 10 μH inductor carrying a DC current of 3 A with 30% peak-to-peak ripple. Determine (a) the peak inductor current, (b) the energy stored at peak current, (c) the minimum required saturation current rating, (d) the inductive reactance at the switching frequency, and (e) the minimum acceptable SRF.
**Solution:**
(a) Ripple current: $\Delta I = 0.30 \times 3 = 0.9$ A. Peak current: $I_{peak} = I_{DC} + \Delta I/2 = 3 + 0.45 =$ **3.45 A**
(b) Energy at peak: $W = ½LI^2 = 0.5 \times 10 \times 10^{-6} \times 3.45^2 =$ **59.5 μJ**
(c) Saturation rating must exceed peak current with margin: $I_{sat} \geq 1.2 \times I_{peak} = 1.2 \times 3.45 =$ **4.14 A minimum** (select a 4.7 A or higher rated inductor)
(d) Inductive reactance: $X_L = 2\pi fL = 2\pi \times 500{,}000 \times 10 \times 10^{-6} =$ **31.4 Ω**
(e) Minimum SRF $\geq 3 \times f_{sw} = 3 \times 500$ kHz = **1.5 MHz**

### 7.2.4 Wire and Cable

Wire is a single metallic conductor, while cable is an assembly of two or more insulated conductors within a common outer jacket or sheath. Copper is the dominant conductor material due to its low resistivity ($\rho = 1.72 \times 10^{-8}$ Ω·m at 20°C), excellent ductility, and good corrosion resistance. Aluminum ($\rho = 2.82 \times 10^{-8}$ Ω·m) is used for larger sizes in power distribution because of its lower cost and weight per unit conductance, but requires larger cross-sections and special connectors to compensate for its higher resistivity and tendency to cold-flow under pressure.

Wire gauge in North America follows the American Wire Gauge (AWG) system, where smaller numbers indicate larger conductors. Each decrease of 3 gauge numbers approximately doubles the cross-sectional area and halves the resistance per unit length. Each decrease of 10 gauge numbers multiplies the area by approximately 10. Common reference points are:

| AWG | Diameter (mm) | Area (mm²) | Resistance (Ω/km at 20°C) | Typical Application |
|---|---|---|---|---|
| 4/0 (0000) | 11.68 | 107.2 | 0.161 | Service entrance, large feeders |
| 2/0 (00) | 9.27 | 67.4 | 0.256 | Subpanels, large branch circuits |
| 1/0 (0) | 8.25 | 53.5 | 0.323 | Service entrance, grounding electrode |
| 4 | 5.19 | 21.2 | 0.815 | Range, dryer circuits (240 V) |
| 6 | 4.11 | 13.3 | 1.296 | Large appliances, subpanels |

| AWG | Diameter (mm) | Area (mm²) | Resistance (Ω/km at 20°C) | Typical Application |
|---|---|---|---|---|
| 10 | 2.59 | 5.26 | 3.277 | Dryers, water heaters, 30 A circuits |
| 12 | 2.05 | 3.31 | 5.211 | General-purpose 20 A branch circuits |
| 14 | 1.63 | 2.08 | 8.286 | General-purpose 15 A branch circuits |
| 18 | 1.02 | 0.823 | 20.95 | Thermostat, low-voltage control |
| 22 | 0.644 | 0.326 | 52.96 | Data, signal, electronics |

Conductors are either solid (a single strand, used for smaller gauges up to about 10 AWG) or stranded (multiple smaller wires twisted together). Stranded wire is more flexible and resistant to fatigue from vibration or repeated bending, making it preferred for portable cords, equipment wiring, and larger gauges. The stranding is specified by the number of strands and their individual gauge — for example, 7/0.0254" means 7 strands each 0.0254 inches in diameter.

**Insulation types** determine the conductor's temperature rating, voltage rating, and suitability for various environments:

| Designation | Material | Temp. Rating | Common Use |
|---|---|---|---|
| THHN | Thermoplastic (nylon jacket) | 90°C dry | Building wire in conduit |
| THWN-2 | Thermoplastic (nylon jacket) | 90°C dry/wet | Indoor and outdoor in conduit |
| XHHW-2 | Cross-linked polyethylene | 90°C dry/wet | Industrial, direct burial |
| NM-B (Romex) | Thermoplastic jacket | 90°C dry | Residential branch circuits |
| UF-B | Thermoplastic jacket | 90°C dry, 75°C wet | Underground feeder, direct burial |
| USE-2 | Cross-linked polyethylene | 90°C wet | Underground service entrance |
| SOOW | Rubber (oil-resistant) | 90°C | Portable power cords, temporary wiring |

| Designation | Material | Temp. Rating | Common Use |
|---|---|---|---|
| MC | Metal clad (aluminum armor) | Per inner conductor | Commercial, industrial feeders |

**Cable assemblies** combine multiple conductors for specific applications. NM-B (non-metallic sheathed cable, commonly called Romex) contains two or three insulated conductors plus a bare ground within a PVC jacket and is the standard residential wiring method. MC (metal-clad) cable uses an interlocking aluminum armor for physical protection in commercial installations. AC (armored cable, BX) uses a flexible steel armor. Tray cable (TC) is rated for installation in cable trays in industrial facilities. For power distribution, medium-voltage cables (5–35 kV) include semiconducting shields and insulation rated for the system voltage class.

**Conduit** provides a raceway for pulling individual conductors, offering physical protection and a grounding path. Common types include EMT (electrical metallic tubing — thin-wall steel, most common commercial), RMC (rigid metal conduit — thick-wall threaded steel, maximum protection), PVC (rigid polyvinyl chloride — non-metallic, corrosion resistant, underground), and ENT (electrical non-metallic tubing — flexible corrugated plastic, concealed in walls).

**Example 7.2.4:** A 120 V, 20 A branch circuit runs 150 feet (45.7 m) from the panel to an outlet using 12 AWG copper THHN in EMT conduit. Determine (a) the one-way conductor resistance, (b) the voltage drop at full load, (c) the voltage drop as a percentage of source voltage, and (d) whether it meets the NEC 3% recommendation for branch circuits.
**Solution:**
(a) 12 AWG resistance: 5.211 Ω/km. One-way length: 45.7 m = 0.0457 km.
$R_{\text{one-way}} = 5.211 \times 0.0457 = \mathbf{0.238\ \Omega}$
(b) Voltage drop (single-phase, round-trip through hot and neutral):
$V_{drop} = 2 \times I \times R = 2 \times 20 \times 0.238 = \mathbf{9.52\ V}$
(c) Percentage: $V_{drop}\% = 9.52/120 \times 100 = \mathbf{7.93\%}$
(d) The NEC recommends $\leq$ 3% for branch circuits (3.6 V at 120 V). At 7.93%, this circuit **does not meet** the recommendation. Solutions include upsizing to 10 AWG (R = 3.277 Ω/km, $V_{drop}$ = 5.99 V = 4.99% — still over 3%) or shortening the run. Using 8 AWG (R = 2.061 Ω/km): $V_{drop} = 2 \times 20 \times 2.061 \times 0.0457 = 3.77$ V = 3.14% — marginal. For a 150-foot 20 A run, **6 AWG** (R = 1.296 Ω/km, $V_{drop}$ = 2.37 V = 1.97%) would comfortably meet the 3% recommendation.

## 7.3 Fundamental Laws

Three fundamental laws govern the behavior of all electrical circuits. Ohm's Law relates voltage, current, and resistance in a single element, while Kirchhoff's two laws — the voltage law (KVL) and the current law (KCL) — express conservation of energy and conservation of charge at the circuit level. Together, these laws form the basis for all circuit analysis methods.

### 7.3.1 Ohm's Law

Ohm's Law states that the voltage across a resistor is directly proportional to the current flowing through it: $V = I \times R$, where V is voltage in volts, I is current in amperes, and R is resistance in ohms

($\Omega$). Equivalently, the current through a resistor is $I = V / R$, and the resistance is $R = V / I$. Ohm's Law applies to resistive elements and is the most frequently used relationship in circuit analysis. The power dissipated by a resistor can be expressed in three equivalent forms: $P = V \times I = I^2 \times R = V^2 / R$.

**Example 7.3.1:** A 470 $\Omega$ resistor carries a current of 25 mA. Find the voltage across it and the power it dissipates.
**Solution:**
Applying Ohm's Law: $V = I \times R = 0.025\ A \times 470\ \Omega = 11.75\ V$.
The power dissipated is $P = I^2 \times R = (0.025)^2 \times 470 = 0.29375\ W \approx 294\ mW$.
Alternatively, $P = V^2 / R = (11.75)^2 / 470 = 138.0625 / 470 \approx 294\ mW$.

### 7.3.2 Kirchhoff's Voltage Law (KVL)

Kirchhoff's Voltage Law states that the algebraic sum of all voltages around any closed loop in a circuit equals zero. This law is a consequence of the conservation of energy: a charge traversing a closed path returns to its starting potential. When applying KVL, voltage rises (such as across a source) are assigned one polarity and voltage drops (such as across a resistor in the direction of current) are assigned the opposite polarity. KVL is the foundation of mesh analysis and loop analysis, enabling the formulation of simultaneous equations that describe the behavior of multi-loop circuits.

**Example 7.3.2:** A series loop contains a 24 V source, a 100 $\Omega$ resistor, a 150 $\Omega$ resistor, and a 50 $\Omega$ resistor. Find the voltage across each resistor.
**Solution:**
The loop current is $I = V_{source} / R_{total} = 24\ V / (100 + 150 + 50)\ \Omega = 24 / 300 = 0.08\ A = 80\ mA$.
By KVL, the sum of voltage drops must equal the source voltage: $V_1 = 0.08 \times 100 = 8\ V$, $V_2 = 0.08 \times 150 = 12\ V$, $V_3 = 0.08 \times 50 = 4\ V$.
Check: $8 + 12 + 4 = 24\ V = V_{source}$.

### 7.3.3 Kirchhoff's Current Law (KCL)

Kirchhoff's Current Law states that the algebraic sum of all currents entering and leaving any node in a circuit equals zero. This law is a consequence of the conservation of charge: charge cannot accumulate at a node. By convention, currents entering a node are positive and currents leaving are negative (or vice versa, as long as the convention is applied consistently). KCL is the foundation of nodal analysis, which formulates equations based on node voltages to solve for all circuit unknowns.

**Example 7.3.3:** At a node in a circuit, three branches meet. Branch 1 carries 3 A into the node, branch 2 carries 1.2 A into the node, and branch 3 carries current out of the node. Find the current in branch 3.
**Solution:** By KCL, the sum of currents entering a node equals the sum of currents leaving: $I_1 + I_2 = I_3$. Therefore $I_3 = 3 + 1.2 = 4.2$ A leaving the node.

## 7.4 DC Circuit Analysis

DC circuit analysis determines the voltages and currents in circuits powered by constant (time-invariant) sources. The fundamental approach is to reduce complex networks to simpler equivalents using series and parallel combinations, then apply Ohm's Law to find the desired quantities.

### 7.4.1 Series Circuits

In a series circuit, components are connected end-to-end so that the same current flows through each element. The total resistance is the sum of the individual resistances: $R_{total} = R_1 + R_2 + \ldots + R_n$. The voltage divides across the series elements in proportion to their resistances, with the voltage across any resistor given by $V_k = V_{source} \times (R_k / R_{total})$. Series circuits have the property that if any element opens (breaks), current flow through the entire circuit ceases.

**Example 7.4.1:** A 12 V battery powers three resistors in series: $R_1 = 220\ \Omega$, $R_2 = 330\ \Omega$, $R_3 = 150\ \Omega$. Find the circuit current and the voltage across $R_2$.
**Solution:**
$R_{total} = 220 + 330 + 150 = 700\ \Omega$.
The series current is $I = 12\text{ V} / 700\ \Omega = 17.14$ mA.
Using the voltage divider, $V_2 = 12 \times (330 / 700) = 12 \times 0.4714 = 5.66$ V.

### 7.4.2 Parallel Circuits

In a parallel circuit, components are connected across the same pair of nodes so that the same voltage appears across each element. The total resistance is found from the reciprocal relation: $1/R_{total} = 1/R_1 + 1/R_2 + \ldots + 1/R_n$. For two resistors in parallel, this simplifies to $R_{total} = (R_1 \times R_2) / (R_1 + R_2)$. The current divides among the parallel branches in inverse proportion to their resistances, with the current through any branch given by $I_k = I_{total} \times (R_{total} / R_k)$. Parallel circuits have the property that if one branch opens, current continues to flow through the remaining branches.

**Example 7.4.2:** Two resistors, $R_1 = 1\ \text{k}\Omega$ and $R_2 = 2.2\ \text{k}\Omega$, are connected in parallel across a 10 V source. Find the total resistance, total current, and the current through each branch.
**Solution:**
$R_{total} = (R_1 \times R_2) / (R_1 + R_2) = (1000 \times 2200) / (1000 + 2200) = 2{,}200{,}000 / 3200 = 687.5\ \Omega$.
Total current: $I_{total} = 10\text{ V} / 687.5\ \Omega = 14.55$ mA.
Branch currents: $I_1 = 10 / 1000 = 10$ mA, $I_2 = 10 / 2200 = 4.55$ mA.
Check: $10 + 4.55 = 14.55$ mA.

### 7.4.3 Series-Parallel Circuits

Series-parallel circuits contain combinations of both series and parallel connections that cannot be reduced to a single series or single parallel network. Analysis proceeds by identifying series and parallel groups, simplifying them step by step from the innermost combinations outward until the entire circuit is reduced to a single equivalent resistance. Once the total current or voltage is found, the process is reversed (working back outward) to determine the voltage and current at each element. Careful identification of which elements share the same current (series) versus the same voltage (parallel) is essential for correct simplification.

**Example 7.4.3:** A circuit has $R_1 = 100\ \Omega$ in series with a parallel combination of $R_2 = 200\ \Omega$ and $R_3 = 300\ \Omega$, all powered by a 20 V source. Find the total current from the source and the voltage across $R_1$.
**Solution:**
First, combine the parallel pair: $R_{23} = (200 \times 300) / (200 + 300) = 60{,}000 / 500 = 120\ \Omega$.
Total resistance: $R_{total} = R_1 + R_{23} = 100 + 120 = 220\ \Omega$.
Total current: $I = 20 / 220 = 90.9$ mA.
Voltage across $R_1$: $V_1 = I \times R_1 = 0.0909 \times 100 = 9.09$ V.
Voltage across the parallel combination: $V_{23} = 20 - 9.09 = 10.91$ V.

## 7.5 Analysis Methods

When circuits are too complex for simple series-parallel reduction, systematic methods based on Kirchhoff's laws provide a structured approach to solving for all unknowns. Nodal analysis applies KCL at nodes, mesh analysis applies KVL around loops, and superposition decomposes multi-source problems into single-source sub-problems.

### 7.5.1 Nodal Analysis

Nodal analysis is a systematic method for solving circuits by writing KCL equations at each non-reference node in terms of node voltages. One node is designated as the reference (ground) node with a voltage of zero, and the remaining node voltages are the unknowns. For a circuit with n non-reference nodes, nodal analysis produces n simultaneous equations. When voltage sources are present, they can be handled by using the supernode technique, which encloses the voltage source and its two nodes in a single region and writes a combined KCL equation for the supernode along with the constraint equation imposed by the source voltage.

**Example 7.5.1:** A circuit has two non-reference nodes ($V_1$ and $V_2$). A 10 mA current source feeds into node $V_1$. A 2 kΩ resistor connects $V_1$ to ground, a 4 kΩ resistor connects $V_1$ to $V_2$, and a 3 kΩ resistor connects $V_2$ to ground. Find $V_1$ and $V_2$ using nodal analysis.
**Solution:**
KCL at node $V_1$: $0.010 = V_1/2000 + (V_1 - V_2)/4000$.
KCL at node $V_2$: $(V_1 - V_2)/4000 = V_2/3000$.
From node $V_2$: $3000(V_1 - V_2) = 4000V_2$, so $3V_1 = 7V_2$, giving $V_1 = (7/3)V_2$.
Substituting into node $V_1$: $0.010 = (7/3)V_2/2000 + ((7/3)V_2 - V_2)/4000 = 7V_2/6000 + (4/3)V_2/4000$ $= 7V_2/6000 + V_2/3000 = 7V_2/6000 + 2V_2/6000 = 9V_2/6000 = 3V_2/2000$.
Therefore $V_2 = 0.010 \times 2000/3 = 6.67$ V and $V_1 = (7/3) \times 6.67 = 15.56$ V.

### 7.5.2 Mesh Analysis

Mesh analysis is a systematic method for solving planar circuits by assigning a circulating current variable (mesh current) to each independent loop and writing KVL equations around each mesh. For a circuit with m independent meshes, mesh analysis produces m simultaneous equations in terms of the mesh currents. Shared elements between adjacent meshes carry the algebraic sum or difference of the two mesh currents. When current sources are present, they can be handled using the supermesh technique, which excludes the current source from the mesh equations and instead writes a combined KVL equation around the supermesh along with the constraint equation imposed by the source current.

**Example 7.5.2:** A single-loop circuit has a 30 V source, a 5 kΩ resistor, and a 10 kΩ resistor in series. A second mesh shares the 10 kΩ resistor and contains a 15 V source and a 20 kΩ resistor. Find the mesh currents $I_1$ (left loop, clockwise) and $I_2$ (right loop, clockwise).
**Solution:**
KVL for mesh 1: $30 = 5000I_1 + 10000(I_1 - I_2)$, giving $30 = 15000I_1 - 10000I_2$ ... (1).
KVL for mesh 2: $15 = 20000I_2 + 10000(I_2 - I_1)$, giving $15 = -10000I_1 + 30000I_2$ ... (2).
From (2): $I_1 = (30000I_2 - 15)/10000 = 3I_2 - 0.0015$.
Substituting into (1): $30 = 15000(3I_2 - 0.0015) - 10000I_2 = 45000I_2 - 22.5 - 10000I_2 = 35000I_2 - 22.5$.
So $35000I_2 = 52.5$, $I_2 = 1.5$ mA.
Then $I_1 = 3(0.0015) - 0.0015 = 0.003$ A = 3 mA.

### 7.5.3 Superposition

The superposition theorem states that in a linear circuit with multiple independent sources, the voltage or current at any element equals the algebraic sum of the contributions from each independent source acting alone, with all other independent sources deactivated. Voltage sources are deactivated by replacing them with short circuits (zero voltage), and current sources are deactivated by replacing them with open circuits (zero current). Dependent sources are never deactivated and remain active in every sub-circuit. Superposition is useful for understanding the individual contribution of each source but requires solving the circuit once for each independent source, making it less efficient than nodal or mesh analysis for circuits with many sources.

**Example 7.5.3:** A 6 kΩ resistor is connected between two sources: a 12 V voltage source on the left and a 6 V voltage source on the right, each with the positive terminal facing the resistor. Find the current through the 6 kΩ resistor using superposition.
**Solution:**
Source 1 alone (replace 6 V source with short circuit): $I_1 = 12$ V / 6 kΩ = 2 mA (flowing left to right).
Source 2 alone (replace 12 V source with short circuit): $I_2 = 6$ V / 6 kΩ = 1 mA (flowing right to left).
By superposition, the net current is $I = I_1 - I_2 = 2 - 1 = 1$ mA flowing left to right.

### 7.5.4 Dependent Sources

A dependent (controlled) source produces a voltage or current that is a function of another voltage or current elsewhere in the circuit. There are four types: voltage-controlled voltage source (VCVS, $V_{out} = \mu V_x$), voltage-controlled current source (VCCS, $I_{out} = g_m V_x$), current-controlled voltage source (CCVS, $V_{out} = rI_x$), and current-controlled current source (CCCS, $I_{out} = \beta I_x$). Dependent sources are essential for modeling real devices: the BJT small-signal model uses a CCCS ($I_c = \beta I_b$), the MOSFET small-signal model uses a VCCS ($I_d = g_m V_{gs}$), and the ideal operational amplifier uses a VCVS ($V_{out} = A_{OL}(V^+ - V^-)$).

Circuits with dependent sources are analyzed using the same nodal and mesh methods as circuits with independent sources, with one critical difference: dependent sources are never deactivated during superposition analysis, since their output depends on circuit variables that change when other sources are turned off. When finding Thévenin or Norton equivalents, the dependent source remains active and the Thévenin resistance must be found by applying a test source (voltage or current) at the output terminals rather than by simply combining resistances.

**Example 7.5.4:** A circuit has a 10 V independent voltage source in series with a 2 kΩ resistor connected to node A. A VCCS with $g_m$ = 5 mA/V, controlled by $V_A$, is connected from ground to node A (injecting current $g_m V_A$ into node A). A 4 kΩ resistor connects node A to ground. Find $V_A$.
**Solution:**
KCL at node A: current from the 10 V source + current from VCCS = current through 4 kΩ to ground.
$(10 - V_A)/2000 + g_m V_A = V_A/4000$.
$(10 - V_A)/2000 + 0.005V_A = V_A/4000$.
Multiply through by 4000: $2(10 - V_A) + 20V_A = V_A$.
$20 - 2V_A + 20V_A = V_A$.
$20 + 18V_A = V_A$, so $17V_A = -20$, $V_A =$ **−1.18 V**. The negative voltage indicates that the VCCS drives enough current to reverse the polarity at node A, demonstrating how dependent sources can produce unexpected results compared to passive circuits.

## 7.6 Circuit Theorems

Circuit theorems provide powerful shortcuts for analyzing complex networks by replacing portions of a circuit with simpler equivalent models. These theorems are particularly useful when analyzing the effect of a single load or source change without resolving the entire circuit.

### 7.6.1 Thévenin's Theorem

Thévenin's theorem states that any linear two-terminal network can be replaced by an equivalent circuit consisting of a single voltage source ($V_{Th}$) in series with a single resistance ($R_{Th}$). The Thévenin voltage $V_{Th}$ is the open-circuit voltage measured across the two terminals with no load connected. The Thévenin resistance $R_{Th}$ is the equivalent resistance seen from the terminals when all independent sources are deactivated (voltage sources shorted, current sources opened). Thévenin's theorem is particularly useful for analyzing the effect of varying a load resistance, since the load sees a simple series circuit regardless of the complexity of the original network.

**Example 7.6.1:** A network consists of a 48 V source in series with a 6 Ω resistor, connected in parallel with a 12 Ω resistor, with output terminals across the 12 Ω resistor. Find the Thévenin equivalent.
**Solution:**
$V_{Th}$ (open-circuit voltage): With no load, the current through the 6 Ω resistor and 12 Ω resistor in series is I = 48 / (6 + 12) = 2.667 A.
The open-circuit voltage across the 12 Ω resistor is $V_{Th}$ = 2.667 × 12 = 32 V.
$R_{Th}$ (deactivate the 48 V source by shorting it): $R_{Th}$ = 6 Ω in parallel with 12 Ω = (6 × 12) / (6 + 12) = 72 / 18 = 4 Ω.

The Thévenin equivalent is a 32 V source in series with a 4 Ω resistor.

### 7.6.2 Norton's Theorem

Norton's theorem states that any linear two-terminal network can be replaced by an equivalent circuit consisting of a single current source ($I_N$) in parallel with a single resistance ($R_N$). The Norton current $I_N$ is the short-circuit current measured through the terminals when they are directly connected. The Norton resistance $R_N$ equals the Thévenin resistance $R_{Th}$, and the two equivalent circuits are related by $I_N = V_{Th} / R_{Th}$. Norton's theorem is the dual of Thévenin's theorem and is preferred when the load is connected in parallel or when current-source representations are more convenient for analysis.

**Example 7.6.2:** Using the Thévenin equivalent from the previous section ($V_{Th} = 32$ V, $R_{Th} = 4$ Ω), find the Norton equivalent circuit.
**Solution:**
The Norton resistance equals the Thévenin resistance: $R_N = R_{Th} = 4$ Ω.
The Norton current is $I_N = V_{Th} / R_{Th} = 32$ V / 4 Ω = 8 A.
The Norton equivalent is an 8 A current source in parallel with a 4 Ω resistor.
Verification: connecting the terminals through a short circuit yields $I_{sc} = 8$ A, and the open-circuit voltage is $V_{oc} = I_N \times R_N = 8 \times 4 = 32\text{ V} = V_{Th}$.

### 7.6.3 Maximum Power Transfer

The maximum power transfer theorem states that a load receives maximum power from a source when the load resistance equals the Thévenin resistance of the source network: $R_{load} = R_{Th}$. Under this condition, the power delivered to the load is $P_{max} = V_{Th}^2 / (4 \times R_{Th})$, and the efficiency is 50% since an equal amount of power is dissipated in the source resistance. While maximum power transfer is critical in communications and signal processing (where signal power is limited), power systems are designed for maximum efficiency rather than maximum power transfer, operating with load resistance much larger than source resistance.

**Example 7.6.3:** An audio amplifier has a Thévenin equivalent output of $V_{Th} = 10$ V and $R_{Th} = 8$ Ω. What load resistance receives maximum power, and how much power is delivered?
**Solution:**
For maximum power transfer, $R_{load} = R_{Th} = 8$ Ω.
The maximum power delivered to the load is $P_{max} = V_{Th}^2 / (4 \times R_{Th}) = (10)^2 / (4 \times 8) = 100 / 32 = 3.125$ W.
The total current is $I = V_{Th} / (R_{Th} + R_{load}) = 10 / 16 = 0.625$ A, and the efficiency is 50% since equal power (3.125 W) is dissipated in the source resistance.

## 7.7 AC Circuit Analysis

AC circuit analysis extends DC methods to circuits driven by sinusoidal sources using the phasor transform and complex impedance. By representing sinusoidal voltages and currents as phasors (see Appendix A), all DC analysis techniques — Ohm's Law, KVL, KCL, nodal and mesh analysis, Thévenin and Norton theorems — apply directly with impedances replacing resistances.

### 7.7.1 Impedance

Impedance (Z) is the AC generalization of resistance, expressed as a complex number Z = R + jX, where R is resistance (real part) and X is reactance (imaginary part), both measured in ohms. Inductive reactance is $X_L = \omega L$ (positive imaginary), and capacitive reactance is $X_C = -1/(\omega C)$ (negative imaginary), where $\omega = 2\pi f$ is the angular frequency. The magnitude $|Z| = \sqrt{(R^2 + X^2)}$ determines the ratio of voltage amplitude to current amplitude, and the angle $\theta = \arctan(X/R)$ determines the phase shift between voltage and current. All DC analysis techniques (Ohm's law, KVL, KCL, nodal analysis, mesh analysis, Thévenin/Norton) apply directly to AC circuits when voltages and currents are expressed as phasors and resistances are replaced by impedances.

**Example 7.7.1:** A series circuit has R = 100 Ω and L = 50 mH driven by a 120 $V_{rms}$, 60 Hz source. Find the impedance, current magnitude, and phase angle.
**Solution:**
$\omega = 2\pi \times 60 = 376.99$ rad/s.
Inductive reactance: $X_L = \omega L = 376.99 \times 0.050 = 18.85\ \Omega$.
Impedance: $Z = 100 + j18.85\ \Omega$.
Magnitude: $|Z| = \sqrt{(100^2 + 18.85^2)} = \sqrt{(10{,}000 + 355.3)} = \sqrt{10{,}355.3} = 101.76\ \Omega$.
Current: $I = V / |Z| = 120 / 101.76 = 1.179\ A_{rms}$.
Phase angle: $\theta = \arctan(18.85 / 100) = \arctan(0.1885) = 10.68°$.
The current lags the voltage by 10.68°.

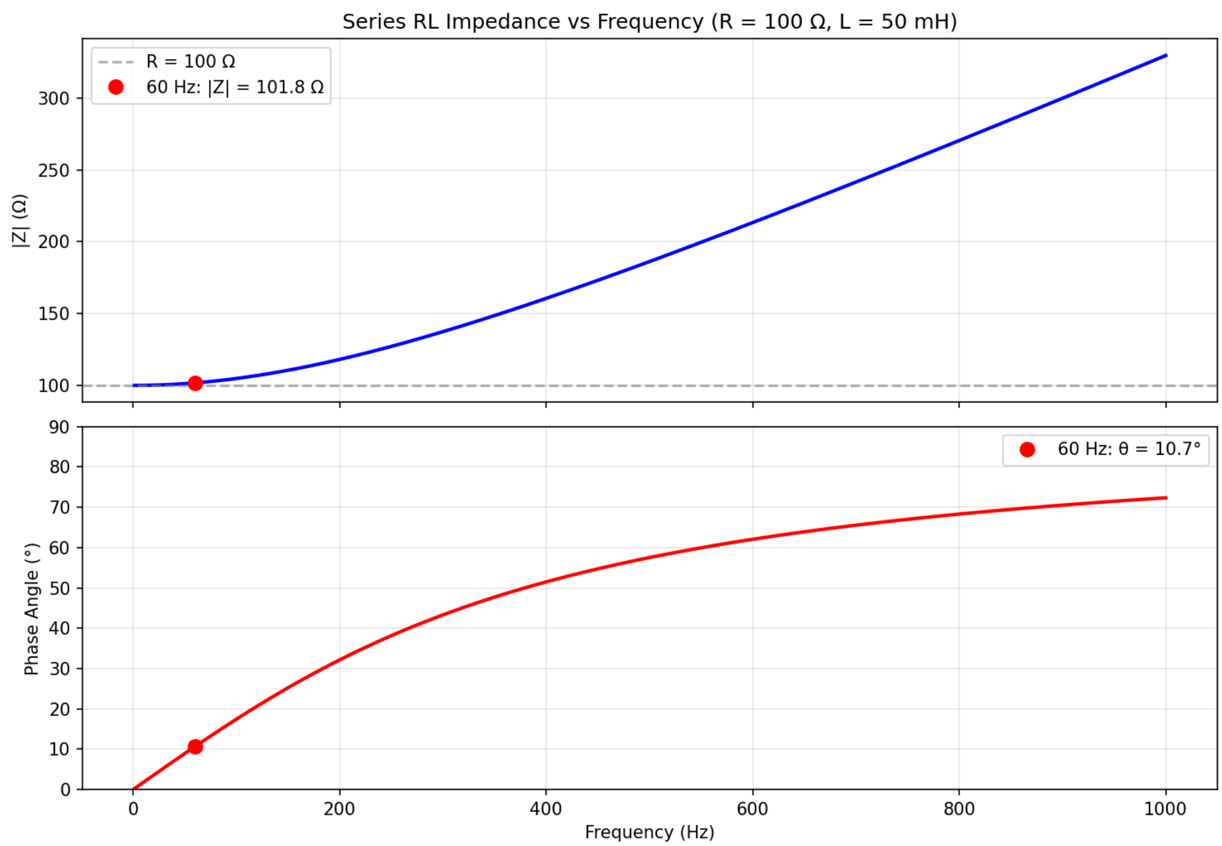



Figure 7.7.1: Series RL Impedance vs Frequency

### 7.7.2 Resonance

Resonance occurs in an AC circuit when the inductive reactance equals the capacitive reactance ($\omega L = 1/\omega C$), causing them to cancel and leaving only the resistive component of impedance. The resonant frequency is $f_0 = 1 / (2\pi\sqrt{(LC)})$. In a series RLC circuit, resonance produces minimum impedance ($Z = R$) and maximum current, making series resonant circuits useful as bandpass filters and in tuning applications. In a parallel RLC circuit, resonance produces maximum impedance and minimum current drawn from the source. The quality factor $Q = f_0$ / bandwidth characterizes the sharpness of the resonance peak, with higher Q indicating a narrower, more selective frequency response.

**Example 7.7.2:** A series RLC circuit has R = 10 Ω, L = 1 mH, and C = 10 nF. Find the resonant frequency, the quality factor, and the bandwidth.
**Solution:**
Resonant frequency: $f_0 = 1 / (2\pi\sqrt{(LC)}) = 1 / (2\pi\sqrt{(0.001 \times 10 \times 10^{-9})}) = 1 / (2\pi\sqrt{(10^{-11})}) = 1 / (2\pi \times 3.162 \times 10^{-6}) = 50{,}329$ Hz ≈ 50.3 kHz.
Quality factor: $Q = (1/R)\sqrt{(L/C)} = (1/10)\sqrt{(0.001 / 10 \times 10^{-9})} = 0.1 \times \sqrt{(100{,}000)} = 0.1 \times 316.2 = 31.62$.
Bandwidth: $BW = f_0 / Q = 50{,}329 / 31.62 = 1{,}591$ Hz ≈ 1.59 kHz.

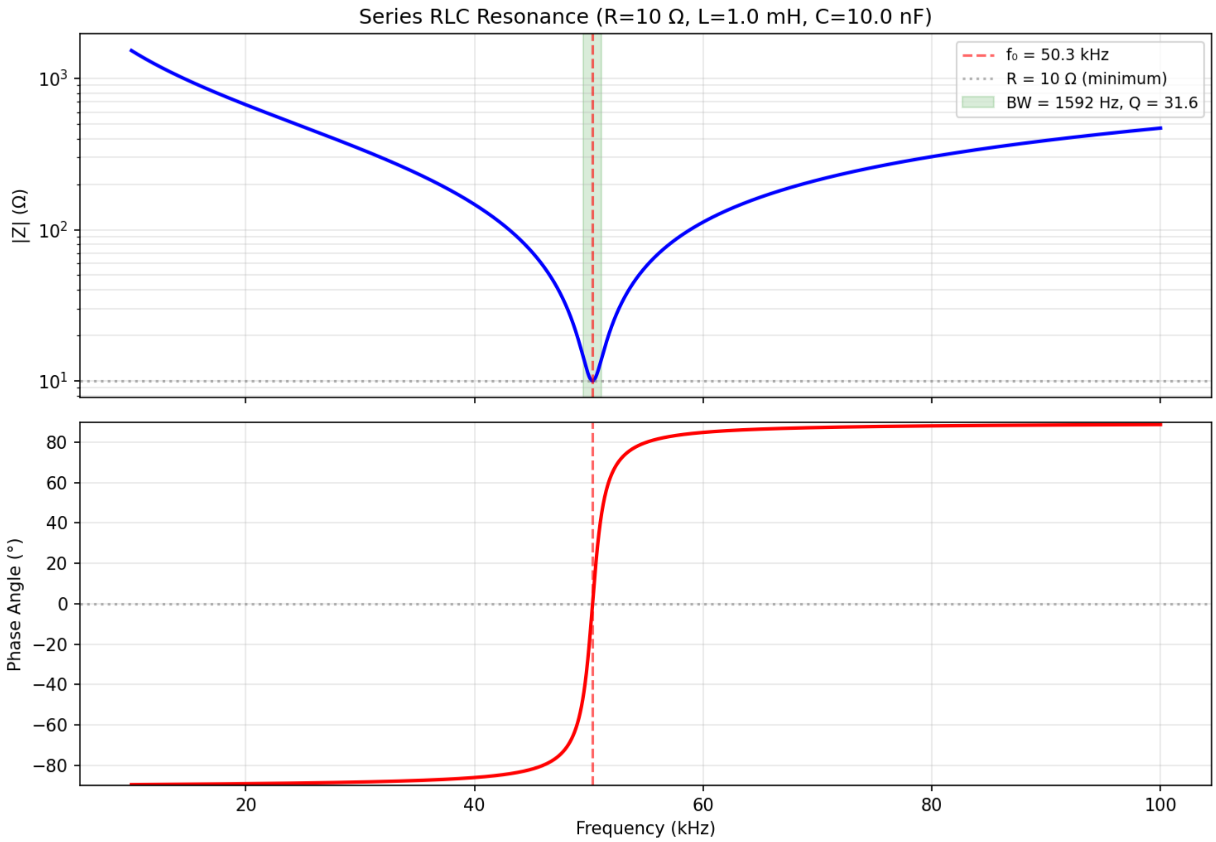



Figure 7.7.2: Series RLC Resonance

### 7.7.3 Power in AC Circuits

In AC circuits, power has three components: real power (P), reactive power (Q), and apparent power (S). Real power (measured in watts, W) is the average power actually consumed by resistive elements: $P = V_{rms} \times I_{rms} \times \cos(\varphi)$, where $\varphi$ is the phase angle between voltage and current. Reactive power (measured in volt-amperes reactive, VAR) represents energy oscillating between the source and reactive elements: $Q = V_{rms} \times I_{rms} \times \sin(\varphi)$. Apparent power (measured in volt-amperes, VA) is the product of RMS voltage and current: $S = V_{rms} \times I_{rms} = \sqrt{(P^2 + Q^2)}$. The power factor, $\cos(\varphi)$, ranges from 0 to 1 and indicates how effectively the circuit converts apparent power into useful real power.

**Example 7.7.3:** A load draws 5 $A_{rms}$ from a 240 $V_{rms}$, 60 Hz supply with a power factor of 0.75 lagging. Find P, Q, and S.
**Solution:**
Apparent power: $S = V_{rms} \times I_{rms} = 240 \times 5 = 1200$ VA.
Real power: $P = S \times \cos(\varphi) = 1200 \times 0.75 = 900$ W.
The phase angle is $\varphi = \arccos(0.75) = 41.41°$.
Reactive power: $Q = S \times \sin(\varphi) = 1200 \times \sin(41.41°) = 1200 \times 0.6614 = 793.7$ VAR (inductive).
Check: $S = \sqrt{(P^2 + Q^2)} = \sqrt{(810{,}000 + 629{,}960)} = \sqrt{1{,}439{,}960} \approx 1200$ VA.

### 7.7.4 Mutual Inductance and Coupled Circuits

When two inductors are placed in close proximity, the changing magnetic flux from one inductor links with the other, creating mutual inductance M measured in henries. The coupling coefficient k $= M/\sqrt{}(L_1L_2)$ ranges from 0 (no coupling) to 1 (perfect coupling), with typical values of 0.95–0.99 for iron-core transformers and 0.01–0.5 for air-core coils. The dot convention indicates the polarity of the mutual coupling: current entering the dotted terminal of one coil induces a voltage with positive polarity at the dotted terminal of the other coil. The voltage equations for coupled coils in the frequency domain are $V_1 = j\omega L_1I_1 + j\omega MI_2$ and $V_2 = j\omega MI_1 + j\omega L_2I_2$ (with + for aiding dots, − for opposing dots).

An ideal transformer (k = 1, no losses) transforms voltages by the turns ratio $V_1/V_2 = N_1/N_2$, transforms currents inversely $I_1/I_2 = N_2/N_1$, and reflects impedances by the square of the turns ratio: $Z_{reflected} = (N_1/N_2)^2 \times Z_{load}$. This impedance transformation is the basis for impedance matching in audio systems, RF circuits, and power distribution.

**Example 7.7.4:** A transformer with $N_1/N_2$ = 5:1, $L_1$ = 100 mH, and k = 0.98 connects a 50 Ω source to an 8 Ω speaker load. Determine (a) the reflected impedance seen by the source, (b) the mutual inductance, and (c) whether this transformer provides a reasonable impedance match.
**Solution:**
(a) Reflected impedance: $Z_{reflected} = (N_1/N_2)^2 \times Z_{load} = (5)^2 \times 8 =$ **200 Ω**.
(b) $L_2 = L_1/(N_1/N_2)^2 = 100$ mH/25 = 4 mH. $M = k\sqrt{}(L_1L_2) = 0.98 \times \sqrt{}(0.100 \times 0.004) = 0.98 \times \sqrt{}(4 \times 10^{-4}) = 0.98 \times 0.02 =$ **19.6 mH**.
(c) The source sees 200 Ω instead of the optimal 50 Ω — this is a **4:1 mismatch**. A turns ratio of $\sqrt{}(50/8)$ = 2.5:1 would provide a perfect match, reflecting 8 Ω to exactly 50 Ω.

## 7.8 Transient Analysis

Transient analysis examines circuit behavior during the transition between steady states, such as when a switch opens or closes. The transient response is governed by the energy storage elements (capacitors and inductors) in the circuit, with the time constant or damping ratio determining how quickly the circuit reaches its new steady state.

### 7.8.1 RC Circuits

When a DC source is suddenly applied to or removed from an RC (resistor-capacitor) circuit, the voltage and current do not change instantaneously but follow an exponential trajectory governed by the time constant $\tau = R \times C$. During charging, the capacitor voltage rises as $V_c(t) = V_s \times (1 - e^{-t/\tau})$, approaching the source voltage asymptotically. During discharging, the capacitor voltage decays as $V_c(t) = V_0 \times e^{-t/\tau}$, where $V_0$ is the initial voltage. After approximately 5 time constants (5τ), the transient is considered complete and the circuit has reached its new steady state. RC circuits are used in timing circuits, filters, coupling and decoupling networks, and power supply smoothing.

**Example 7.8.1:** A 10 μF capacitor, initially uncharged, is connected to a 9 V battery through a 47 kΩ resistor. Find the time constant and the capacitor voltage at t = 0.5 s.
**Solution:**
Time constant: $\tau = R \times C = 47{,}000 \times 10 \times 10^{-6} = 0.47$ s.

At t = 0.5 s: $V_c(0.5) = 9 \times (1 - e^{-0.5/0.47}) = 9 \times (1 - e^{-1.064}) = 9 \times (1 - 0.3453) = 9 \times 0.6547 = 5.89$ V. The capacitor has charged to about 65.5% of the source voltage, which is consistent with being slightly past one time constant.

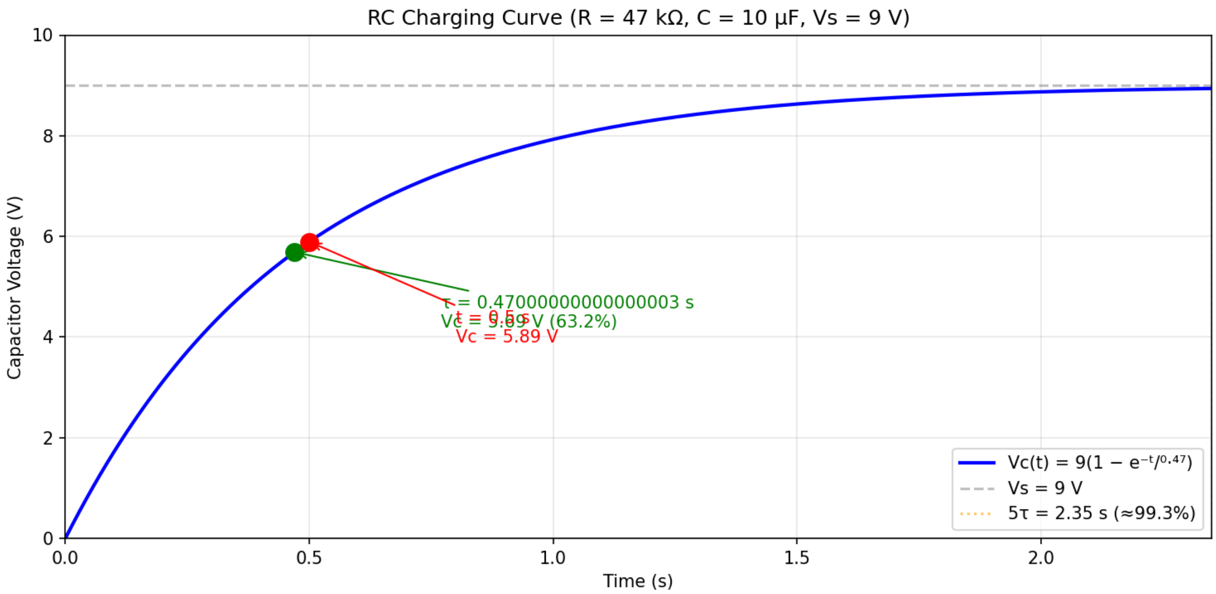



Figure 7.8.1: RC Charging Curve

### 7.8.2 RL Circuits

When a DC source is applied to an RL (resistor-inductor) circuit, the current increases exponentially toward its steady-state value with a time constant $\tau = L / R$. The current during energization follows $I(t) = (V_s / R) \times (1 - e^{-t/\tau})$, while the inductor voltage decays as $V_L(t) = V_s \times e^{-t/\tau}$. The inductor resists changes in current by generating a back-EMF proportional to the rate of current change ($V_L = L \times dI/dt$). After 5τ, the current reaches approximately 99.3% of its final value and the inductor behaves as a short circuit in DC steady state. RL circuits appear in relay coils, motor windings, and inductive loads, where the stored energy ($E = ½ \times L \times I^2$) must be managed during switching to prevent voltage spikes.

**Example 7.8.2:** A 200 mH inductor with 50 Ω of series resistance is connected to a 24 V DC source at t = 0. Find the time constant, the steady-state current, and the current at t = 2 ms.
**Solution:**
Time constant: $\tau = L / R = 0.200 / 50 = 4$ ms.
Steady-state current: $I_{ss} = V_s / R = 24 / 50 = 0.48$ A = 480 mA.
At t = 2 ms: $I(0.002) = 0.48 \times (1 - e^{-0.002/0.004}) = 0.48 \times (1 - e^{-0.5}) = 0.48 \times (1 - 0.6065) = 0.48 \times 0.3935 = 188.9$ mA.
The inductor voltage at t = 2 ms is $V_L = 24 \times e^{-0.5} = 24 \times 0.6065 = 14.56$ V.

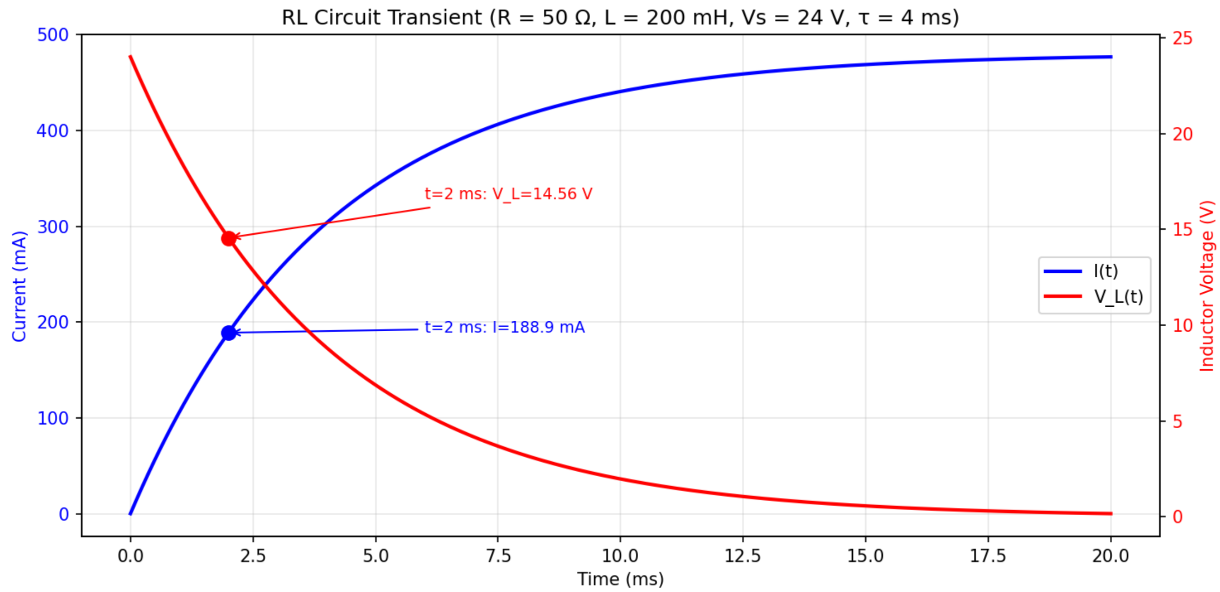



Figure 7.8.2: RL Circuit Transient Response

### 7.8.3 RLC Circuits

RLC circuits containing a resistor, inductor, and capacitor exhibit second-order transient behavior characterized by a second-order differential equation. The nature of the transient response depends on the damping ratio $\zeta$ (zeta), determined by the relative values of R, L, and C. When $\zeta < 1$ (underdamped), the response oscillates with exponentially decaying amplitude at the damped natural frequency. When $\zeta = 1$ (critically damped), the response returns to steady state in the shortest time without oscillation. When $\zeta > 1$ (overdamped), the response decays exponentially without oscillation but more slowly than the critically damped case. The natural resonant frequency $\omega_0 = 1/\sqrt{(LC)}$ and the damping ratio $\zeta = R/(2\sqrt{(L/C)})$ for a series RLC circuit are the key parameters that fully characterize the transient behavior.

**Example 7.8.3:** A series RLC circuit has R = 100 Ω, L = 10 mH, and C = 1 μF. Determine $\omega_0$, $\zeta$, and classify the transient response.
**Solution:**
Natural resonant frequency: $\omega_0 = 1/\sqrt{(LC)} = 1/\sqrt{(0.010 \times 10^{-6})} = 1/\sqrt{(10^{-8})} = 1/(10^{-4}) = 10{,}000$ rad/s.
Damping ratio: $\zeta = R / (2\sqrt{(L/C)}) = 100 / (2 \times \sqrt{(0.010 / 10^{-6})}) = 100 / (2 \times \sqrt{10{,}000}) = 100 / (2 \times 100) = 0.5$.
Since $\zeta = 0.5 < 1$, the circuit is underdamped and will exhibit oscillatory transient behavior with an exponentially decaying envelope.
The damped natural frequency is $\omega_d = \omega_0\sqrt{(1 - \zeta^2)} = 10{,}000 \times \sqrt{(1 - 0.25)} = 10{,}000 \times 0.866 = 8{,}660$ rad/s (about 1,378 Hz).

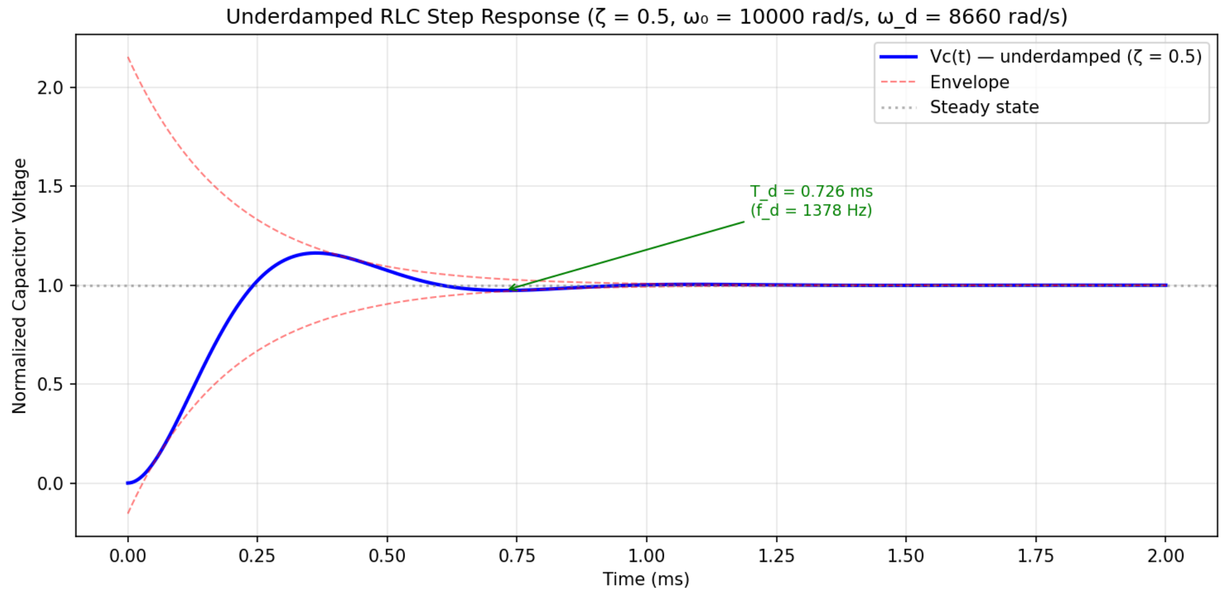



Figure 7.8.3: Underdamped RLC Step Response

# 7.9 Two-Port Networks

A two-port network is a circuit abstraction with four terminals grouped into two ports — an input port and an output port — each carrying a voltage and current. Two-port models are fundamental to characterizing amplifiers, filters, transmission lines, and transformers as "black boxes" described entirely by their terminal relationships, without needing to know the internal circuit topology.

## 7.9.1 Two-Port Parameters

Two-port networks are characterized by parameter sets that relate the port voltages ($V_1$, $V_2$) and currents ($I_1$, $I_2$). Each parameter set is measured under specific terminal conditions:

**Z-parameters (impedance, open-circuit):** $V_1 = Z_{11}I_1 + Z_{12}I_2$ and $V_2 = Z_{21}I_1 + Z_{22}I_2$. Each $Z_{ij}$ is found by open-circuiting one port: $Z_{11} = V_1/I_1|_{I_2=0}$, $Z_{21} = V_2/I_1|_{I_2=0}$, etc. For a reciprocal network (no dependent sources), $Z_{12} = Z_{21}$.

**Y-parameters (admittance, short-circuit):** $I_1 = Y_{11}V_1 + Y_{12}V_2$ and $I_2 = Y_{21}V_1 + Y_{22}V_2$. Each $Y_{ij}$ is found by short-circuiting one port: $Y_{11} = I_1/V_1|_{V_2=0}$, etc. The Y-parameter matrix is the inverse of the Z-parameter matrix.

**h-parameters (hybrid):** $V_1 = h_{11}I_1 + h_{12}V_2$ and $I_2 = h_{21}I_1 + h_{22}V_2$. Hybrid parameters mix open-circuit and short-circuit conditions and are widely used for BJT transistor models, where $h_{11} = h_{ie}$ (input impedance), $h_{21} = h_{fe}$ (current gain), $h_{12} = h_{re}$ (reverse voltage ratio), and $h_{22} = h_{oe}$ (output admittance).

**ABCD-parameters (transmission):** $V_1 = AV_2 - BI_2$ and $I_1 = CV_2 - DI_2$ (with $I_2$ defined as flowing out of port 2). Transmission parameters are uniquely suited for cascaded networks because the overall ABCD matrix is simply the product of the individual ABCD matrices. For a reciprocal network, $AD - BC = 1$.

**Example 7.9.1:** Find the Z-parameters for a T-network consisting of $Z_a = 10\ \Omega$ in series with port 1, $Z_b = 20\ \Omega$ in series with port 2, and $Z_c = 50\ \Omega$ as the shunt element between the two series arms.
**Solution:**
With $I_2 = 0$ (port 2 open): $Z_{11} = V_1/I_1 = Z_a + Z_c = 10 + 50 = \mathbf{60\ \Omega}$.
$Z_{21} = V_2/I_1 = Z_c = \mathbf{50\ \Omega}$ (voltage across the shunt element).
By symmetry of the measurement process: $Z_{22} = Z_b + Z_c = 20 + 50 = \mathbf{70\ \Omega}$.
$Z_{12} = Z_c = \mathbf{50\ \Omega}$.
Since $Z_{12} = Z_{21}$, the network is reciprocal, as expected for a passive circuit with no dependent sources.

### 7.9.2 Two-Port Interconnections

Two-port networks can be interconnected in standard configurations, each governed by a specific parameter set:

**Series connection:** When two two-ports are connected in series at both ports (port currents are shared), the overall Z-parameters add: $Z_{total} = Z_A + Z_B$. This is analogous to series impedances in a one-port circuit.

**Parallel connection:** When two two-ports are connected in parallel at both ports (port voltages are shared), the overall Y-parameters add: $Y_{total} = Y_A + Y_B$. This is analogous to parallel admittances.

**Cascade connection:** When the output port of one network feeds the input port of the next, the overall ABCD matrix is the matrix product of the individual ABCD matrices in order: $[ABCD]_{total} = [ABCD]_A \times [ABCD]_B$. This is the most common interconnection for filters, transmission line segments, and amplifier stages.

Two-port analysis has broad applications: filter sections are cascaded using ABCD matrices to build complex frequency responses, transmission line segments are modeled as ABCD networks for impedance and voltage analysis, and transistor amplifiers use h-parameters for small-signal gain and impedance calculations.

**Example 7.9.2:** Two identical two-port networks, each with ABCD parameters $A = 1$, $B = 10\ \Omega$, $C = 0.02$ S, $D = 1$, are cascaded. Find the overall ABCD parameters and the voltage gain $V_2/V_1$ when the cascade is terminated in a 50 Ω load.
**Solution:**
For a cascade, $[ABCD]_{total} = [ABCD]_1 \times [ABCD]_2$:
$A_T = A_1A_2 + B_1C_2 = (1)(1) + (10)(0.02) = \mathbf{1.2}$
$B_T = A_1B_2 + B_1D_2 = (1)(10) + (10)(1) = \mathbf{20\ \Omega}$
$C_T = C_1A_2 + D_1C_2 = (0.02)(1) + (1)(0.02) = \mathbf{0.04\ S}$
$D_T = C_1B_2 + D_1D_2 = (0.02)(10) + (1)(1) = \mathbf{1.2}$
Verify determinant: $A_TD_T - B_TC_T = 1.2 \times 1.2 - 20 \times 0.04 = 1.44 - 0.80 = \mathbf{0.64}$.
Each individual stage has $AD - BC = 1 - 0.2 = \mathbf{0.8}$, and $(0.8)^2 = 0.64$ ✓ (consistent with cascading two identical stages).
Since $AD - BC = 0.64 \neq 1$, the cascade is **not a reciprocal network** — as expected, because the individual stages are also non-reciprocal ($AD - BC = 0.8 \neq 1$).
With $Z_L = 50\ \Omega$: $V_2/V_1 = Z_L / (A_TZ_L + B_T) = 50 / (1.2 \times 50 + 20) = 50 / 80 = \mathbf{0.625}$ (−4.08 dB).

# Chapter 8

# Signal Processing

Signal processing is the branch of electrical engineering concerned with the analysis, manipulation, and interpretation of signals — quantities that carry information and vary as a function of time, frequency, or space. It provides the mathematical and computational tools to extract useful information from raw data, remove noise and interference, compress data for efficient storage and transmission, and transform signals between different representations. Signal processing techniques are foundational to communications, audio and video systems, radar, medical imaging, control systems, and machine learning.

## 8.1 Signals and Systems

The study of signals and systems provides the mathematical foundation for all of signal processing. A signal is any quantity that varies as a function of one or more independent variables (most commonly time), and a system is any process that transforms an input signal into an output signal. By classifying signals and characterizing systems according to properties such as linearity, time invariance, and causality, engineers can apply powerful general-purpose tools — convolution, correlation, and transform methods — to analyze, design, and optimize a vast range of processing algorithms.

### 8.1.1 Signal Classification

Signals are classified along several dimensions that determine which mathematical tools and processing techniques are appropriate for their analysis. The most fundamental distinction is between continuous-time signals, defined for all values of time t, and discrete-time signals, defined only at specific time instants n (typically uniformly spaced samples produced by an analog-to-digital converter). Orthogonal to this is the amplitude distinction: analog signals take continuous amplitude values, while digital signals are both discrete in time and quantized in amplitude to a finite set of levels. Signals may be deterministic — completely described by a mathematical expression such as $x(t) = A \cos(\omega t + \varphi)$ — or random (stochastic), requiring statistical descriptions such as mean, variance, autocorrelation, and power spectral density. Periodic signals repeat at a fixed interval T such that $x(t) = x(t + T)$, enabling Fourier series decomposition into discrete harmonics, while aperiodic signals require the continuous Fourier transform for frequency analysis. The energy-power classification distinguishes energy signals, which have finite total energy $E = \int|x(t)|^2dt$ and are typically transient (pulses, decaying exponentials), from power signals, which persist indefinitely with finite average power $P = \lim(1/T)\int|x(t)|^2dt$ (sinusoids, periodic waveforms, random noise). Additional classifications include causal versus non-causal (whether the signal is zero for $t < 0$), bounded versus unbounded, and even versus odd symmetry. Correctly classifying a signal is the essential first step in any analysis, because it determines whether to use Fourier series or Fourier transform, continuous or discrete methods, deterministic or statistical tools, and time-domain or frequency-domain representations.

**Example 8.1.1:** Classify the signal $x(t) = 5\cos(2\pi \times 1000t)$ and determine its period, frequency, and average power (assuming a 1 Ω reference).
**Solution:**
The signal is continuous-time, analog, deterministic, and periodic.
The angular frequency is $\omega = 2\pi \times 1000$ rad/s, so the frequency is f = 1000 Hz and the period is T = 1/f = 1 ms.
Since the signal persists indefinitely, it is a power signal.
The average power into 1 Ω is $P = (A^2/2) = (5^2/2) = 12.5$ W, where A = 5 is the peak amplitude (using the RMS value of a sinusoid, $V_{rms} = 5/\sqrt{2} = 3.536$ V, gives $P = V_{rms}^2 / 1 = 12.5$ W).

### 8.1.2 Linear Time-Invariant (LTI) Systems

A system is linear if it obeys the principles of superposition (additivity and homogeneity): the response to a weighted sum of inputs equals the same weighted sum of the individual responses. A system is time-invariant if its behavior does not change with time — a delayed input produces an identically delayed output. LTI systems are the most analytically tractable class of systems because they are completely characterized by their impulse response h(t), and the output for any input x(t) is obtained by convolution: $y(t) = x(t) * h(t)$. In the frequency domain, convolution becomes multiplication: $Y(\omega) = X(\omega) \cdot H(\omega)$, where $H(\omega)$ is the system's frequency response (the Fourier transform of the impulse response). This property makes frequency-domain analysis a powerful tool for understanding and designing LTI systems.

**Example 8.1.2:** An LTI system has impulse response $h(t) = 2e^{-3t}u(t)$, where u(t) is the unit step. Find the frequency response magnitude $|H(\omega)|$ at $\omega = 4$ rad/s.
**Solution:**
The frequency response is the Fourier transform of h(t): $H(\omega) = 2 / (3 + j\omega)$.
At $\omega = 4$: $H(4) = 2 / (3 + j4)$.
The magnitude is $|H(4)| = 2 / \sqrt{(3^2 + 4^2)} = 2 / \sqrt{(9 + 16)} = 2 / \sqrt{25} = 2/5 = 0.4$.
The phase is $\theta = -\arctan(4/3) = -53.13°$.
This means a sinusoidal input at $\omega = 4$ rad/s is attenuated to 40% of its amplitude and shifted by -53.13°.

### 8.1.3 Convolution

Convolution is the mathematical operation that relates the input, output, and impulse response of an LTI system. For continuous-time systems, the convolution integral is $y(t) = \int x(\tau)h(t - \tau)d\tau$, where the integration is over all $\tau$. For discrete-time systems, the convolution sum is $y[n] = \Sigma\, x[k]h[n - k]$, where the summation is over all k. Graphically, convolution involves flipping one signal in time, shifting it, multiplying the two signals point by point, and integrating (or summing) the product. Convolution in the time domain corresponds to multiplication in the frequency domain, which is the basis for efficient frequency-domain filtering using the FFT.

**Example 8.1.3:** Compute the discrete-time convolution $y[n] = x[n] * h[n]$ where $x[n] = \{1, 2, 3\}$ (for $n = 0, 1, 2$) and $h[n] = \{1, 1\}$ (for $n = 0, 1$).
**Solution:**
Apply the convolution sum $y[n] = \Sigma x[k]h[n\text{-}k]$:
$y[0] = x[0]h[0] = 1 \times 1 = 1$.
$y[1] = x[0]h[1] + x[1]h[0] = 1 \times 1 + 2 \times 1 = 3$.
$y[2] = x[1]h[1] + x[2]h[0] = 2 \times 1 + 3 \times 1 = 5$.
$y[3] = x[2]h[1] = 3 \times 1 = 3$.
Therefore $y[n] = \{1, 3, 5, 3\}$ for $n = 0, 1, 2, 3$.
Note the output length is len(x) + len(h) - 1 = 3 + 2 - 1 = 4 samples.

### 8.1.4 Correlation

Correlation measures the similarity between two signals as a function of a time shift applied to one of them. The cross-correlation of signals $x(t)$ and $y(t)$ is defined as $R_{xy}(\tau) = \int x(t)y(t + \tau)dt$, which quantifies how well x aligns with a time-shifted version of y. When a signal is correlated with itself, the result is the autocorrelation $R_{xx}(\tau)$, which reveals periodic structure and is always maximum at zero lag: $R_{xx}(0)$ equals the signal's total energy (or average power for power signals). The autocorrelation and power spectral density form a Fourier transform pair (Wiener-Khinchin theorem), connecting time-domain correlation analysis to frequency-domain spectral analysis. Cross-correlation is widely used in radar and sonar (measuring time delay to determine range), GPS (code acquisition), template matching, and system identification.

**Example 8.1.4:** A radar transmits a pulse $x(t)$ and receives the echo $y(t) = 0.01 \times x(t - 25\ \mu s)$ + noise. The cross-correlation $R_{xy}(\tau)$ peaks at $\tau = 25\ \mu s$. Determine the target range if the signal propagates at the speed of light ($c = 3 \times 10^8$ m/s).
**Solution:**
The cross-correlation peak at $\tau = 25\ \mu s$ indicates the round-trip delay to the target and back.
The round-trip distance is $d_{\text{round-trip}} = c \times \tau = 3 \times 10^8 \times 25 \times 10^{-6} = 7500$ m.
Since the signal travels to the target and returns, the one-way range is $R = d/2 = 7500/2 = 3750$ m = 3.75 km.
The correlation peak height of 0.01 relative to the autocorrelation peak reflects the signal attenuation over the path.

### 8.1.5 Sampling Theorem and Aliasing

The Nyquist-Shannon sampling theorem establishes the fundamental bridge between continuous-time and discrete-time signals: a band-limited signal with no frequency content above $f_{max}$ can be perfectly reconstructed from its samples if the sampling rate satisfies $f_s \geq 2f_{max}$. The critical rate $f_N = 2f_{max}$ is the Nyquist rate, and the corresponding frequency $f_s/2$ is the Nyquist frequency (the highest frequency representable at sampling rate $f_s$). When a signal is sampled, its spectrum is replicated at integer multiples of $f_s$; if $f_s < 2f_{max}$, adjacent spectral copies overlap, causing **aliasing** — high-frequency components fold back into the baseband as spurious low-frequency artifacts that cannot be removed by any post-processing. An **anti-aliasing filter** (analog lowpass with cutoff at $f_s/2$) must precede the sampler to attenuate frequency content above the Nyquist frequency. In practice, the sampling rate is chosen 2–10× above $f_{max}$ to allow a realizable anti-aliasing filter with a gradual roll-off. Reconstruction of the continuous-time signal from its samples uses sinc interpolation: $x(t) = \Sigma\, x[n] \cdot \mathrm{sinc}((t - nT)/T)$, where $T = 1/f_s$. Real DACs approximate this with zero-order hold (staircase output) followed by a reconstruction lowpass filter. **Bandpass sampling** extends the theorem to narrowband signals centered at a carrier frequency: a signal occupying bandwidth B centered at $f_c$ can be sampled at $f_s \geq 2B$ (much less than $2f_c$) provided the sampling rate is chosen so that spectral replicas do not overlap, enabling direct RF digitization in software-defined radio receivers.

**Example 8.1.5:** An audio signal contains frequencies up to 20 kHz. It is sampled at $f_s = 44.1$ kHz (CD standard). A 25 kHz tone from an ultrasonic source leaks into the signal path. Determine whether aliasing occurs and, if so, identify the aliased frequency.
**Solution:**
The Nyquist frequency is $f_s/2 = 44{,}100/2 = 22{,}050$ Hz.
The 25 kHz component exceeds this limit.
If the anti-aliasing filter fails to attenuate it, the aliased frequency is $f_{alias} = f_s - f_{signal} = 44{,}100 - 25{,}000 = 19{,}100$ Hz.
This 19.1 kHz alias falls within the audible band and would appear as a spurious tone that cannot be distinguished from a genuine 19.1 kHz signal.
To prevent this, the anti-aliasing filter must attenuate 25 kHz by at least 90 dB (below the 16-bit noise floor of −96 dB).
CD systems use steep analog filters with a transition band from 20 kHz to 22.05 kHz, which is why the sampling rate was chosen as 44.1 kHz rather than exactly 40 kHz — the extra 10% provides margin for a realizable filter.

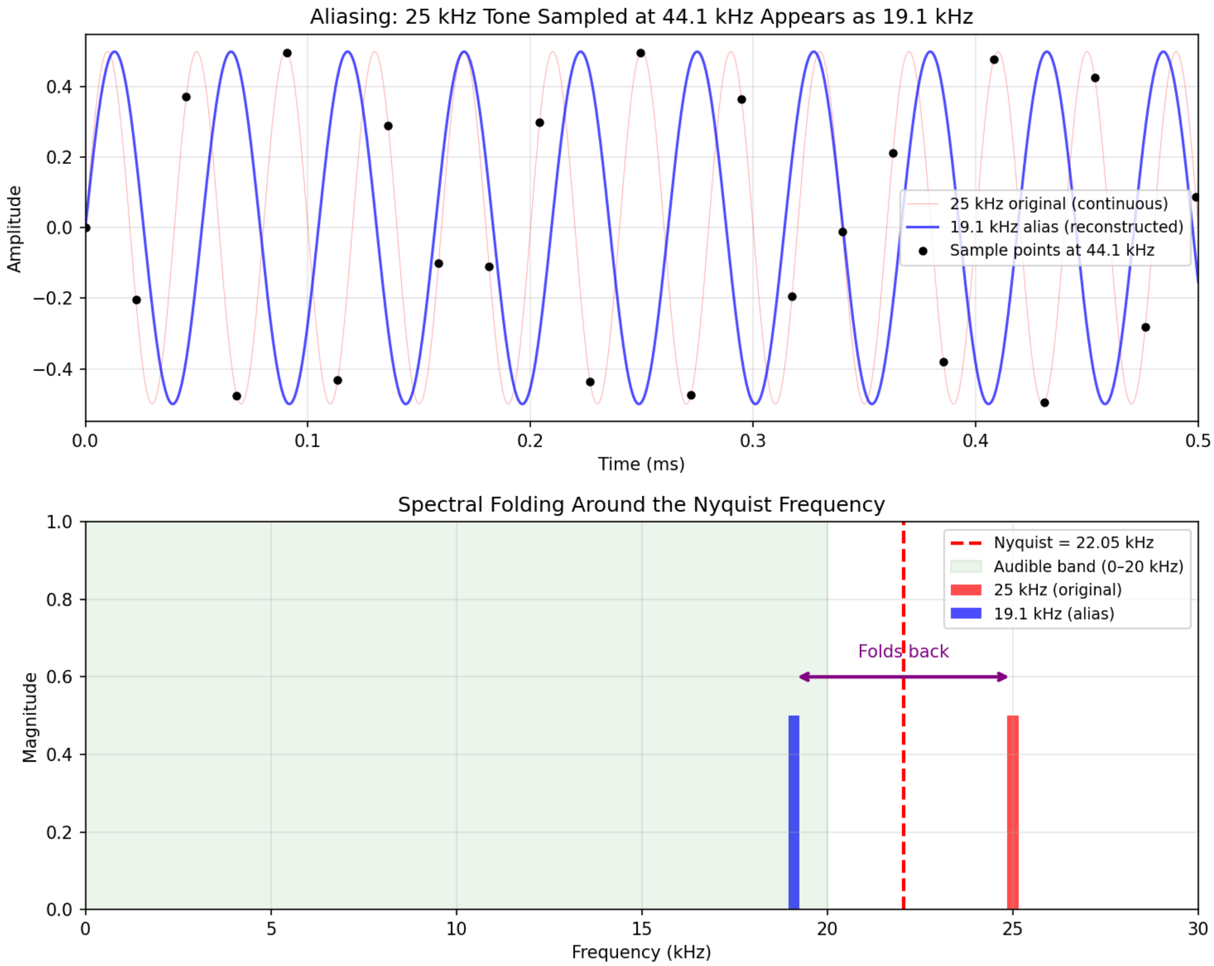



Figure 8.1.5: Sampling and Aliasing

## 8.2 Fourier Analysis

Fourier analysis is the cornerstone of signal processing, providing the framework for decomposing signals into their constituent frequency components. The Fourier series represents periodic signals as sums of harmonically related sinusoids, the Fourier transform extends this to aperiodic signals with continuous spectra, and the Discrete Fourier Transform (DFT) — efficiently computed via the FFT algorithm — makes spectral analysis practical for sampled data. Together, these tools enable engineers to design filters, analyze spectral content, detect signals in noise, and understand the frequency-domain behavior of systems.

### 8.2.1 Fourier Series

The Fourier series is built on a profound insight: any periodic signal, no matter how complex its shape, can be decomposed into a sum of harmonically related sinusoids at integer multiples of the fundamental frequency $f_0 = 1/T$. This means a square wave, a sawtooth, a rectified sine, or any other periodic waveform is fully described by the amplitudes and phases of its harmonic components — the fundamental ($n = 1$), second harmonic ($n = 2$), third harmonic ($n = 3$), and so on. The trigonometric form expresses the series as a sum of cosines and sines with real coefficients $a_n$ and $b_n$, while the complex exponential form uses basis functions $e^{j2\pi nf_0t}$ with complex coefficients $c_n$, which is more compact and better suited for mathematical manipulation. Convergence of the series requires the Dirichlet conditions: the signal must have a finite number of discontinuities, extrema, and finite energy per period. At points of continuity, the series converges exactly to the signal; at discontinuities, it converges to the midpoint of the jump, and the Gibbs phenomenon produces approximately 9% overshoot at the edges of the discontinuity regardless of how many terms are included. Parseval's theorem relates the average power of the periodic signal to the sum of the squared magnitudes of its Fourier coefficients, providing a direct link between time-domain and frequency-domain energy accounting. In practice, the Fourier series is essential for analyzing power system harmonics (quantifying the distortion caused by nonlinear loads like rectifiers and VFDs), designing audio synthesizers (building complex timbres from pure tones), and understanding the spectral content of any repetitive waveform encountered in communications and instrumentation.

**Example 8.2.1:** A periodic square wave has amplitude A = 5 V, period T = 2 ms, and 50% duty cycle. Find the fundamental frequency and the amplitudes of the first three nonzero harmonic components in the trigonometric Fourier series.
**Solution:**
Fundamental frequency: $f_0 = 1/T = 1/0.002 = 500$ Hz.
For a symmetric square wave, the Fourier series contains only odd harmonics with amplitudes $a_n = 4A/(n\pi)$ for odd n.
First harmonic (n = 1): $a_1 = 4 \times 5 \,/\, (1 \times \pi) = 20/\pi = 6.366$ V.
Third harmonic (n = 3): $a_3 = 20 \,/\, (3\pi) = 2.122$ V.
Fifth harmonic (n = 5): $a_5 = 20 \,/\, (5\pi) = 1.273$ V.
The series is $x(t) = 6.366 \sin(2\pi \times 500t) + 2.122 \sin(2\pi \times 1500t) + 1.273 \sin(2\pi \times 2500t) + \ldots$

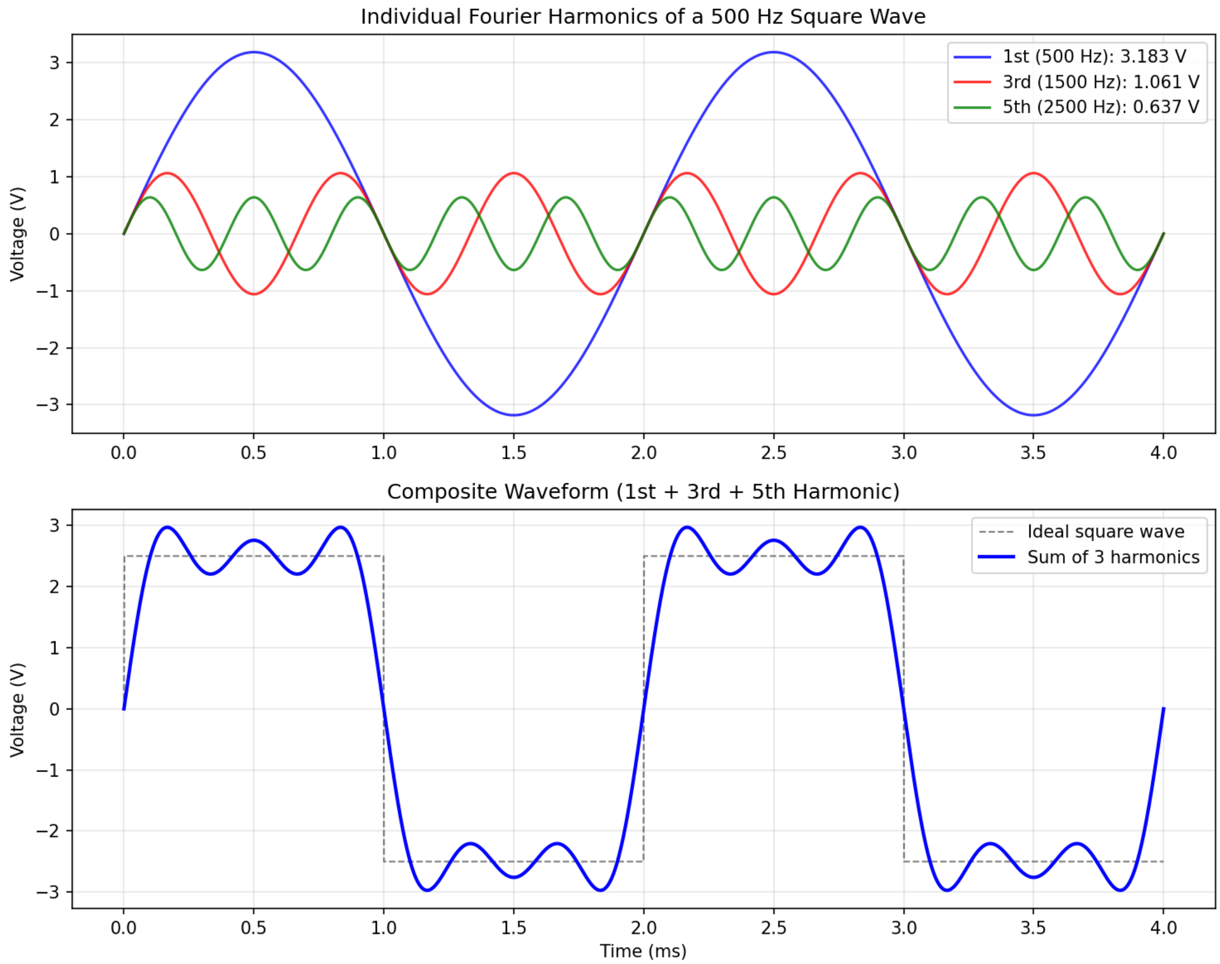



Figure 8.2.1: Fourier Series Harmonics of a Square Wave

### 8.2.2 Fourier Transform

The Fourier Transform extends frequency-domain analysis to aperiodic signals by representing them as a continuous spectrum of frequencies rather than discrete harmonics. The forward transform $X(f) = \int x(t)e^{-j2\pi ft}dt$ decomposes the signal into its frequency components, and the inverse transform $x(t) = \int X(f)e^{j2\pi ft}df$ reconstructs the signal from its spectrum. Important properties include linearity, time shifting (introduces a phase shift), frequency shifting (modulation), time scaling, and the duality between time and frequency domains. The convolution theorem states that convolution in time corresponds to multiplication in frequency and vice versa. Common transform pairs include the rectangular pulse (sinc spectrum), Gaussian (Gaussian spectrum), and the impulse function $\delta(t)$ (flat spectrum equal to 1 for all frequencies).

**Example 8.2.2:** Find the Fourier Transform of a rectangular pulse $x(t) = 1$ for $|t| \leq 0.5$ ms, and 0 otherwise. Determine the first null (zero crossing) of the spectrum.
**Solution:**
The pulse has width $\tau = 1$ ms.
The Fourier Transform is $X(f) = \tau \times \text{sinc}(f\tau) = 0.001 \times \text{sinc}(0.001f)$, where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$.

The magnitude $|X(f)| = 0.001 \times |\text{sinc}(0.001f)|$.
The first null occurs where $\text{sinc}(0.001f) = 0$, i.e., when $\pi \times 0.001f = \pi$, so $f = 1/0.001 = 1000$ Hz = 1 kHz.
The main lobe of the spectrum extends from -1 kHz to +1 kHz, and the spectral width is inversely proportional to the pulse duration.

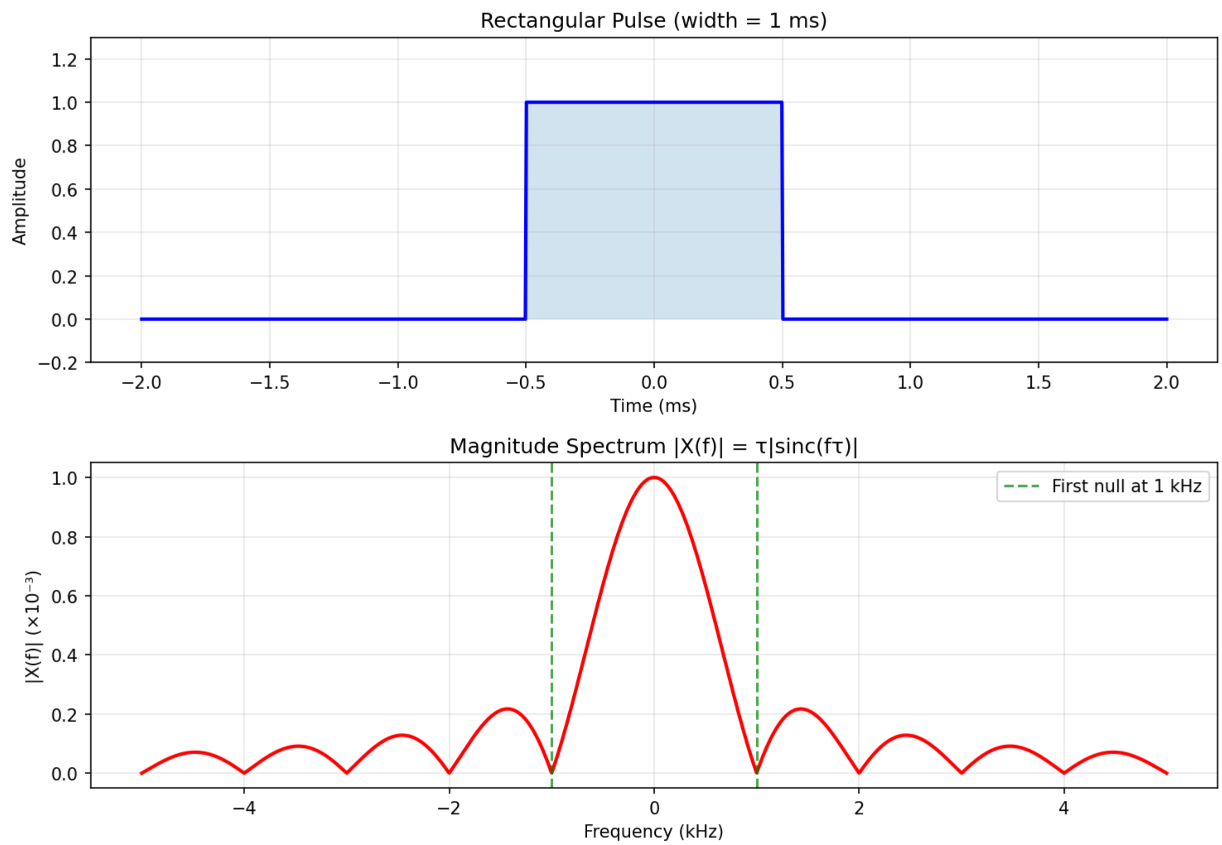



Figure 8.2.2: Rectangular Pulse and Sinc Spectrum

### 8.2.3 Discrete Fourier Transform (DFT)

The Discrete Fourier Transform is the computable form of Fourier analysis for sampled data and is the workhorse of modern digital signal processing. It maps a finite-length sequence of N time-domain samples to N frequency-domain samples, defined as $X[k] = \Sigma\, x[n]e^{-j2\pi kn/N}$ for $k = 0, 1, \ldots, N-1$, and the inverse DFT (IDFT) recovers the original time-domain sequence from the spectral samples. The DFT can be understood as sampling the Discrete-Time Fourier Transform (DTFT) — which is continuous in frequency — at N equally spaced points around the unit circle, yielding a frequency resolution of $\Delta f = f_s/N$, where $f_s$ is the sampling rate. Because the DFT implicitly treats the input sequence as one period of a periodic signal, analyzing a signal that is not exactly periodic within the N-sample window introduces spectral leakage: energy from a single frequency tone spreads into adjacent bins, obscuring nearby weaker components. Windowing functions such as Hanning, Hamming, Blackman, and flat-top are applied to the time-domain data before the DFT to taper the edges of the analysis block and suppress leakage sidelobes, at the cost of slightly widening the main lobe and reducing frequency resolution. Zero-padding — appending zeros to extend the sequence to a longer length — increases the number of frequency-domain samples and provides finer interpolation of the spectral shape, but does not improve the fundamental resolving ability, which is set by the original observation window duration $T = N/f_s$. The DFT's direct computation requires $O(N^2)$ operations, but the Fast Fourier Transform (FFT) algorithm reduces this to $O(N \log N)$, making real-time spectral analysis practical. The DFT is ubiquitous in spectrum analyzers, audio processing, communications receivers (OFDM demodulation), radar signal processing, vibration analysis, and medical imaging.

**Example 8.2.3:** A signal is sampled at $f_s = 8000$ Hz and a 256-point DFT is computed. Find the frequency resolution (bin spacing) and determine which DFT bin index corresponds to a 1500 Hz tone.
**Solution:**
Frequency resolution: $\Delta f = f_s / N = 8000 / 256 = 31.25$ Hz per bin.
The bin index for 1500 Hz: $k = f / \Delta f = 1500 / 31.25 = 48$.
Therefore, the 1500 Hz component appears at bin $k = 48$.
If the signal were zero-padded to $N = 512$, the new resolution would be $\Delta f = 8000/512 = 15.625$ Hz, providing finer spectral interpolation but the same fundamental resolving ability (still limited by the original 256-sample observation window of 32 ms).

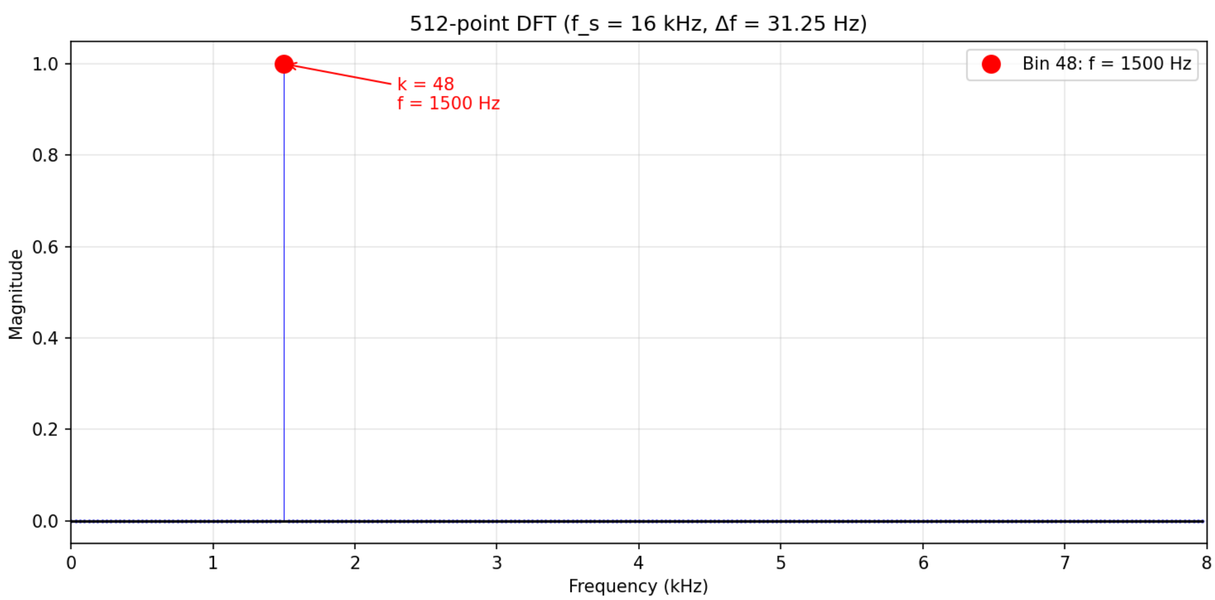



Figure 8.2.3: DFT Frequency Bins

### 8.2.4 Fast Fourier Transform (FFT)

The Fast Fourier Transform is an efficient algorithm for computing the DFT, reducing the computational complexity from $O(N^2)$ to $O(N \log N)$ by exploiting the symmetry and periodicity of the complex exponential basis functions. The most common variant is the radix-2 Cooley-Tukey algorithm, which recursively divides the N-point DFT into two N/2-point DFTs (requiring N to be a power of 2), combines the results using butterfly operations, and achieves dramatic speedup for large N. For example, a 1024-point DFT requires approximately 1,048,576 complex multiplications by direct computation but only about 5,120 using the FFT. The FFT has made real-time spectral analysis practical and is the computational engine behind applications ranging from audio equalizers and speech recognition to radar processing and medical imaging. Split-radix, mixed-radix, and prime-factor algorithms extend FFT efficiency to sequences whose lengths are not powers of 2.

**Example 8.2.4:** Compare the number of complex multiplications required for a 4096-point DFT using direct computation versus the radix-2 FFT.
**Solution:**
Direct DFT: $N^2 = 4096^2 = 16{,}777{,}216$ complex multiplications.
Radix-2 FFT: $(N/2)\log_2 N = (4096/2) \times \log_2(4096) = 2048 \times 12 = 24{,}576$ complex multiplications.
Speedup ratio: 16,777,216 / 24,576 = 682.7×.
The FFT computes the same result nearly 683 times faster, making real-time spectral analysis of a 4096-point block practical even on modest hardware.

### 8.2.5 Discrete Cosine Transform (DCT)

The Discrete Cosine Transform expresses a finite sequence of data points as a sum of cosine functions at different frequencies, producing a purely real-valued transform output (unlike the complex-valued DFT). The most widely used variant, the DCT-II, is defined as $X[k] = \Sigma\, x[n] \cdot \cos[\pi(2n + 1)k / (2N)]$ for $k = 0, 1, \ldots, N-1$, and has the important property that it concentrates the energy of typical real-world signals into a small number of low-frequency coefficients. This energy compaction property makes the DCT the foundation of lossy data compression: JPEG image compression applies the DCT to 8×8 pixel blocks and quantizes the resulting coefficients, discarding high-frequency components that contribute little to visual quality. Similarly, MP3 and AAC audio codecs use the Modified DCT (MDCT) — a variant with overlapping windows that eliminates blocking artifacts — to transform audio samples before perceptual quantization. The DCT is closely related to the DFT of a symmetrically extended sequence, and fast DCT algorithms achieve $O(N \log N)$ complexity by leveraging this relationship. The DCT also appears in spectral analysis as an alternative to the DFT when the signal is real-valued and the phase information is not needed.

**Example 8.2.5:** An 8-point DCT-II is applied to the data block x = {100, 102, 104, 106, 108, 106, 104, 102}. Applying $X[k] = \Sigma\, x[n] \cdot \cos[\pi(2n+1)k/(2N)]$ gives coefficients X = {832.0, −5.13, −12.62, 1.80, 0.00, −1.20, −0.90, 1.02}. If coefficients with magnitude less than 2.0 are set to zero (quantized out), calculate the compression ratio and the reconstruction error.
**Solution:**
Original data requires 8 values to represent.
After thresholding, 3 coefficients are significant: $X[0] = 832.0$, $X[1] = -5.13$, and $X[2] = -12.62$ (the remaining 5 are set to zero).
Compression ratio: $8/3 \approx$ **2.7:1**.
Energy of discarded coefficients: $1.80^2 + 0^2 + 1.20^2 + 0.90^2 + 1.02^2 = 3.24 + 0 + 1.44 + 0.81 + 1.04 = 6.53$.
Total signal energy: $832.0^2 + 5.13^2 + 12.62^2 + 6.53 = 692{,}224 + 26.3 + 159.3 + 6.53 = 692{,}416$.
Energy retained: (692,416 − 6.53) / 692,416 = **99.999%**.
The RMS error per sample is $\sqrt{(6.53/8)} = \sqrt{0.8163} =$ **0.90** (about 0.9% of the mean signal value of 104) — illustrating the DCT's energy compaction property: even discarding 5 of 8 coefficients preserves the signal with negligible error. In practice, smooth correlated data (such as image pixel

blocks in JPEG) yields even stronger compaction, achieving ratios of 10:1 or higher.

### 8.2.6 Hilbert Transform and Analytic Signals

The Hilbert transform is an operation that shifts every frequency component of a real signal by −90° (positive frequencies) or +90° (negative frequencies), producing the quadrature companion of the original signal. For a signal $x(t)$, the Hilbert transform is $\hat{x}(t) = (1/\pi) \int x(\tau)/(t - \tau)\, d\tau$, which in the frequency domain corresponds to multiplication by $-j \cdot \mathrm{sgn}(f)$: the magnitude spectrum is unchanged, but positive frequencies are shifted by −90° and negative frequencies by +90°. The **analytic signal** $z(t) = x(t) + j\hat{x}(t)$ is a complex signal whose spectrum exists only at positive frequencies — the negative-frequency content has been suppressed. From the analytic signal, the **instantaneous envelope** (amplitude) is $A(t) = |z(t)| = \sqrt{(x^2 + \hat{x}^2)}$, and the **instantaneous phase** is $\varphi(t) = \arctan(\hat{x}(t)/x(t))$, from which the **instantaneous frequency** $f_i(t) = (1/2\pi)(d\varphi/dt)$ can be derived. These quantities are essential for analyzing amplitude-modulated and frequency-modulated signals, characterizing non-stationary vibration signals, and demodulating communications waveforms. The Hilbert transform also enables single-sideband (SSB) modulation by cancelling one sideband through phasing, and it is computed efficiently in discrete time by taking the FFT, zeroing the negative-frequency bins, and taking the inverse FFT.

**Example 8.2.6:** A vibration signal $x(t) = (1 + 0.5\cos(2\pi \times 8t)) \times \cos(2\pi \times 200t)$ represents a 200 Hz carrier amplitude-modulated at 8 Hz. Compute the analytic signal and extract the instantaneous envelope to recover the modulation.
**Solution:**
The signal is an AM waveform with carrier $f_c = 200$ Hz and modulation $f_m = 8$ Hz.
The Hilbert transform of $\cos(2\pi f_c t)$ is $\sin(2\pi f_c t)$, so $\hat{x}(t) = (1 + 0.5\cos(2\pi \times 8t)) \times \sin(2\pi \times 200t)$.
The analytic signal is $z(t) = x(t) + j\hat{x}(t) = (1 + 0.5\cos(2\pi \times 8t)) \times e^{j2\pi \times 200t}$.
The instantaneous envelope: $A(t) = |z(t)| = |1 + 0.5\cos(2\pi \times 8t)| \times |e^{j2\pi \times 200t}| = 1 + 0.5\cos(2\pi \times 8t)$.
This cleanly recovers the 8 Hz modulating envelope without rectification or lowpass filtering, ranging from 0.5 (minimum) to 1.5 (maximum).
The modulation index is $m = 0.5/1.0 = 50\%$.
In practice, this is computed by FFT: for $N = 1024$ samples at $f_s = 2048$ Hz, take the FFT, set bins 513–1024 to zero (negative frequencies), double bins 2–512 (positive frequencies), keep bin 1 (DC) unchanged, and take the inverse FFT to obtain $z[n]$.
The envelope is then $A[n] = |z[n]|$.

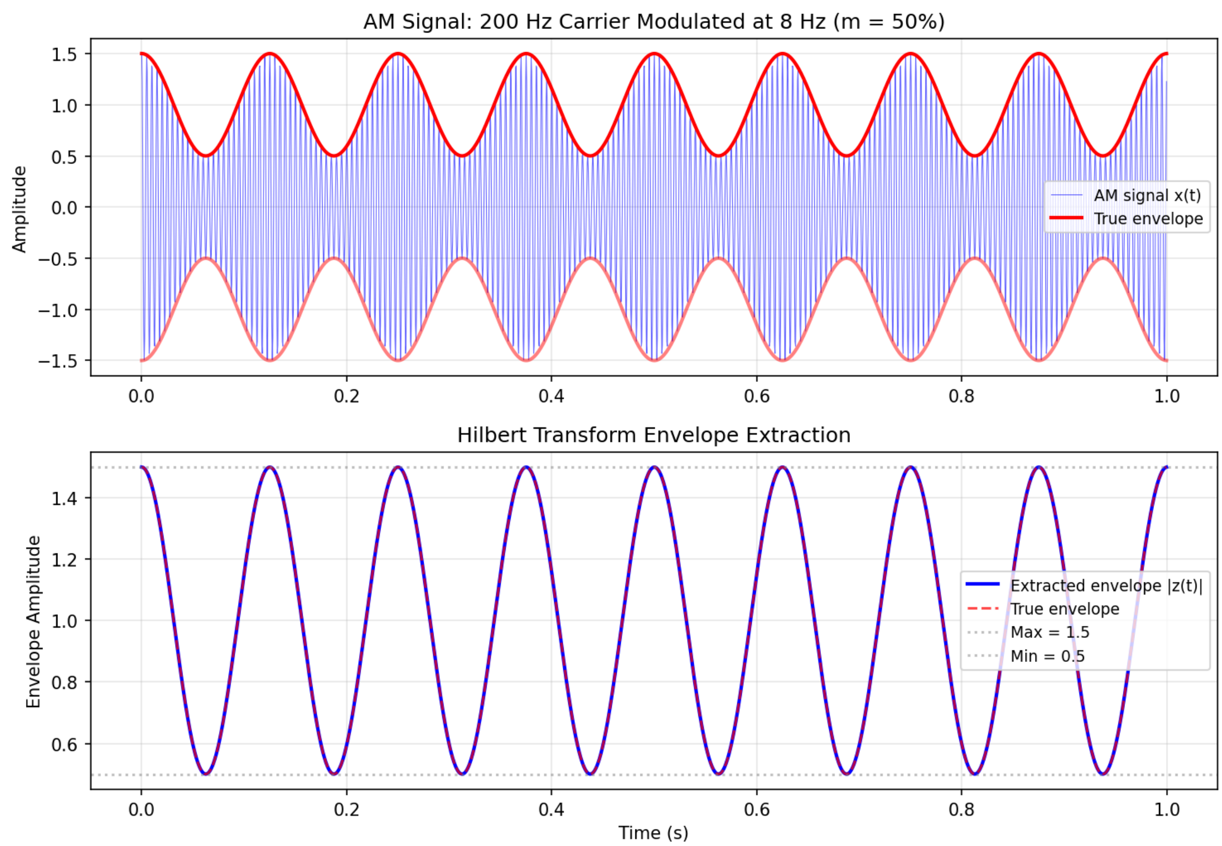



Figure 8.2.6: Hilbert Transform Envelope Extraction

### 8.2.7 Goertzel Algorithm

The Goertzel algorithm is a computationally efficient method for evaluating the DFT at a single frequency bin (or a small number of bins), requiring only O(N) operations per bin compared to the O(N log N) of a full FFT. When only a few specific frequencies need to be detected — rather than the complete spectrum — the Goertzel algorithm is faster than computing the entire FFT and extracting the desired bins. The algorithm reformulates the DFT computation as a second-order IIR filter: for each input sample $x[n]$, the recursion $s[n] = x[n] + 2\cos(2\pi k/N) \times s[n-1] - s[n-2]$ is iterated for $n = 0$ to $N-1$ (with $s[-1] = s[-2] = 0$), and after processing all N samples, the DFT output is $X[k] = s[N-1] - e^{-j2\pi k/N} \times s[N-2]$. The power at bin k can be computed without complex arithmetic: $|X[k]|^2 = s[N-1]^2 + s[N-2]^2 - 2\cos(2\pi k/N) \times s[N-1] \times s[N-2]$. Per frequency bin, the Goertzel algorithm requires N real multiplications and 2N real additions (the recursion), plus a single complex multiply for the final output — compared to $(N/2)\log_2 N$ complex multiplications for the entire FFT. The crossover point where a full FFT becomes more efficient is approximately when the number of bins needed exceeds $2\cdot\log_2(N)$.

The most prominent application is **DTMF (Dual-Tone Multi-Frequency) detection** in telephony, where each key press generates two simultaneous tones from a set of 8 frequencies (697, 770, 852, 941 Hz for rows and 1209, 1336, 1477, 1633 Hz for columns). The Goertzel algorithm evaluates the DFT at these 8 specific frequencies, detecting which two tones are present. Other applications include power line frequency measurement (monitoring 50/60 Hz and harmonics), audio pitch detection, and radar Doppler bin extraction.

**Example 8.2.7:** A DTMF detector samples at $f_s$ = 8000 Hz and uses an N = 205 sample block (25.625 ms). The digit "5" produces tones at 770 Hz and 1336 Hz. Calculate (a) the DFT bin indices for all 8 DTMF frequencies, (b) the number of multiplications for the Goertzel approach versus a full 256-point FFT, and (c) the frequency resolution.
**Solution:**
(a) Bin index for each frequency: $k = \text{round}(f \times N / f_s)$
697 Hz: $k = \text{round}(697 \times 205 / 8000) = \text{round}(17.86) = 18 \rightarrow$ actual $f = 18 \times 8000/205 = 702.4$ Hz
770 Hz: $k = \text{round}(770 \times 205 / 8000) = \text{round}(19.73) = 20 \rightarrow$ actual $f = 20 \times 8000/205 = 780.5$ Hz
852 Hz: $k = \text{round}(852 \times 205 / 8000) = \text{round}(21.83) = 22 \rightarrow$ actual $f = 22 \times 8000/205 = 858.5$ Hz
941 Hz: $k = \text{round}(941 \times 205 / 8000) = \text{round}(24.11) = 24 \rightarrow$ actual $f = 24 \times 8000/205 = 936.6$ Hz
1209 Hz: k = 31, 1336 Hz: k = 34, 1477 Hz: k = 38, 1633 Hz: k = 42
The chosen N = 205 provides bin indices that closely match the DTMF frequencies (< 1.4% error).
(b) Computational cost:
Goertzel (8 bins): $8 \times N = 8 \times 205 =$ **1,640 real multiplications**
256-point FFT (zero-padded): $(N/2) \times \log_2 N = 128 \times 8 =$ **1,024 complex multiplications** = 4,096 real multiplications
Goertzel is **2.5× more efficient** for 8 bins out of 256.
(c) Frequency resolution: $\Delta f = f_s / N = 8000 / 205 =$ **39.0 Hz**, sufficient to distinguish the closest DTMF pair (770 and 852 Hz, separated by 82 Hz).

## 8.3 Laplace Transform

The Laplace transform is the primary tool for analyzing continuous-time linear systems in engineering, extending the concept of frequency-domain analysis to the complex frequency plane $s = \sigma + j\omega$. By converting differential equations into algebraic equations, it simplifies circuit analysis, control system design, and the study of transient and steady-state behavior. The s-domain provides a unified framework for understanding stability, frequency response, and system dynamics through the geometry of poles and zeros.

### 8.3.1 Definition and Properties

The Laplace transform converts a time-domain function x(t) into a complex frequency-domain function X(s), where $s = \sigma + j\omega$ is the complex frequency variable. The one-sided (unilateral) Laplace transform is defined as $X(s) = \int_0^\infty x(t)e^{-st}dt$ and is the standard form used in engineering for causal systems with initial conditions. Key properties include linearity, time differentiation ($sX(s) - x(0^-)$ for the first derivative), time integration ($X(s)/s$), time shifting ($e^{-as}X(s)$), frequency shifting ($X(s-a)$), and the initial and final value theorems. The Laplace transform generalizes the Fourier transform by replacing $j\omega$ with s, enabling analysis of signals and systems that do not have a Fourier transform (such as growing exponentials) and naturally incorporating initial conditions into the solution.

**Example 8.3.1:** Find the Laplace transform of $x(t) = 3e^{-2t}\cos(5t)u(t)$, where u(t) is the unit step function.
**Solution:**
Using the transform pair: $L\{e^{-at}\cos(\omega t)u(t)\} = (s + a) / ((s + a)^2 + \omega^2)$.
Here $a = 2$ and $\omega = 5$.

Applying linearity with the scaling factor of 3: $X(s) = 3(s + 2) / ((s + 2)^2 + 25) = 3(s + 2) / (s^2 + 4s + 4 + 25) = 3(s + 2) / (s^2 + 4s + 29)$.
The region of convergence is $Re\{s\} > -2$.

### 8.3.2 Transfer Functions

A transfer function H(s) is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions: $H(s) = Y(s)/X(s)$. For circuits and systems described by linear constant-coefficient differential equations, the transfer function is a rational function (ratio of polynomials) in s. The roots of the numerator polynomial are the zeros of the system (frequencies where the output is zero), and the roots of the denominator polynomial are the poles (frequencies where the response is theoretically infinite). The locations of poles and zeros in the s-plane completely determine the system's frequency response, stability, and transient behavior. A system is stable if and only if all poles have negative real parts (lie in the left half of the s-plane).

**Example 8.3.2:** A system has transfer function $H(s) = 10(s + 3) / ((s + 1)(s + 5))$. Find the poles, zeros, DC gain, and determine stability.
**Solution:**
Zeros: $s + 3 = 0$, so $z_1 = -3$.
Poles: $s + 1 = 0$ and $s + 5 = 0$, so $p_1 = -1$ and $p_2 = -5$.
Both poles have negative real parts (left half-plane), so the system is stable.
DC gain: $H(0) = 10(0 + 3) / ((0 + 1)(0 + 5)) = 30 / 5 = 6$.
The system amplifies DC signals by a factor of 6 and rolls off at higher frequencies due to the two poles.

### 8.3.3 Inverse Laplace Transform

The inverse Laplace transform recovers the time-domain function from its s-domain representation using partial fraction expansion for rational functions. The process involves factoring the denominator into its roots (poles), decomposing X(s) into a sum of simpler fractions with known inverse transforms, and looking up or computing each term's time-domain equivalent. First-order poles at $s = -a$ correspond to exponential terms $Ae^{-at}$, complex conjugate poles at $s = -\sigma \pm j\omega$ correspond to damped sinusoids, and repeated poles introduce polynomial-times-exponential terms. Tables of common Laplace transform pairs and the linearity property allow most engineering problems to be solved without evaluating the complex contour integral directly.

**Example 8.3.3:** Find the inverse Laplace transform of $X(s) = (5s + 13) / ((s + 1)(s + 3))$.
**Solution:**
Perform partial fraction expansion: $X(s) = A/(s + 1) + B/(s + 3)$.
Multiplying both sides by $(s + 1)(s + 3)$: $5s + 13 = A(s + 3) + B(s + 1)$.
Setting $s = -1$: $5(-1) + 13 = A(2)$, so $8 = 2A$, $A = 4$.
Setting $s = -3$: $5(-3) + 13 = B(-2)$, so $-2 = -2B$, $B = 1$.
Therefore $X(s) = 4/(s + 1) + 1/(s + 3)$.
Using the inverse transform table: $x(t) = 4e^{-t} + e^{-3t}$ for $t \geq 0$.

### 8.3.4 s-Domain Circuit Analysis

The Laplace transform provides a systematic method for analyzing circuits with energy storage elements (capacitors and inductors) by converting integro-differential equations into algebraic equations in s. Each circuit element is replaced by its s-domain impedance: a resistor has impedance R, an inductor has impedance sL (with an initial-condition voltage source $LI_0$ in series), and a capacitor has impedance 1/(sC) (with an initial-condition voltage source $V_0/s$ in series). Once the circuit is transformed, standard techniques — voltage division, current division, nodal analysis, mesh analysis, and Thévenin/Norton equivalents — are applied directly in the s-domain to find the output Y(s). The transfer function H(s) = Y(s)/X(s) characterizes the circuit's frequency response and transient behavior, and the time-domain output is recovered by inverse Laplace transform (partial fraction expansion). This approach unifies DC analysis (s = 0), AC steady-state analysis (s = jω), and transient analysis into a single framework, making it the standard method for analyzing filters, amplifiers, control loops, and power supply feedback networks.

**Example 8.3.4:** A series RLC circuit has R = 100 Ω, L = 10 mH, and C = 1 μF. The input is a step voltage $V_{in}(t) = 10u(t)$ V with zero initial conditions. Find the transfer function $H(s) = V_C(s)/V_{in}(s)$, the natural frequency, damping ratio, and determine whether the response is underdamped, critically damped, or overdamped.
**Solution:**
The s-domain circuit is a voltage divider: $H(s) = (1/sC) / (R + sL + 1/sC) = 1 / (s^2LC + sRC + 1)$.
Substituting values: $H(s) = 1 / (s^2 \times 10^{-2} \times 10^{-6} + s \times 100 \times 10^{-6} + 1) = 1 / (10^{-8}s^2 + 10^{-4}s + 1)$.
Dividing by $10^{-8}$: $H(s) = 10^8 / (s^2 + 10^4s + 10^8)$.
Comparing with the standard form $\omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$: $\omega_n^2 = 10^8$, so $\omega_n = 10^4$ rad/s ($f_n$ = 1,592 Hz).
$2\zeta\omega_n = 10^4$, so $\zeta = 10^4 / (2 \times 10^4) = 0.5$.
Since $\zeta = 0.5 < 1$, the circuit is underdamped.
The damped natural frequency is $\omega_d = \omega_n\sqrt{(1 - \zeta^2)} = 10^4 \times \sqrt{0.75} = 8{,}660$ rad/s.
The step response will oscillate at $f_d$ = 1,378 Hz with an exponential decay envelope $e^{-5000t}$.

## 8.4 Z-Transform

The Z-transform plays the same role for discrete-time systems that the Laplace transform plays for continuous-time systems, providing an algebraic framework for analyzing difference equations, designing digital filters, and studying stability. The z-plane maps directly to the s-plane through the relationship $z = e^{sT}$, with the unit circle in the z-plane corresponding to the jω-axis in the s-plane. Stability, causality, and frequency response of discrete-time systems are all characterized by the locations of poles and zeros relative to the unit circle.

### 8.4.1 Definition and Properties

The Z-transform is the discrete-time counterpart of the Laplace transform, converting a discrete-time sequence x[n] into a function of the complex variable z: $X(z) = \Sigma\, x[n]z^{-n}$. The relationship between the Z-transform and the Laplace transform is $z = e^{sT}$, where T is the sampling period. Key properties include linearity, time shifting ($z^{-k}X(z)$ for a delay of k samples), convolution (multiplication of Z-transforms), and the initial and final value theorems. The region of convergence (ROC) in the z-plane determines which sequences correspond to a given Z-transform and whether the associated system is stable and/or causal. A causal LTI system is stable if and only if all poles of its transfer function H(z) lie inside the unit circle $|z| = 1$.

**Example 8.4.1:** Find the Z-transform of the sequence $x[n] = (0.8)^n u[n]$, where u[n] is the unit step, and determine the region of convergence.
**Solution:**
Using the Z-transform pair for $a^n u[n]$: $X(z) = z / (z - a) = z / (z - 0.8)$, or equivalently $X(z) = 1 / (1 - 0.8z^{-1})$.
The ROC is $|z| > |a| = 0.8$, which is the region outside a circle of radius 0.8 in the z-plane.
Since the ROC includes the unit circle ($|z| = 1 > 0.8$), the sequence has a valid DTFT and represents a stable signal.

### 8.4.2 Discrete-Time Transfer Functions

The transfer function of a discrete-time LTI system is $H(z) = Y(z)/X(z)$, relating the Z-transforms of the output and input sequences. For systems described by linear constant-coefficient difference equations, H(z) is a rational function in z (or $z^{-1}$). The poles and zeros of H(z) in the z-plane determine the system's frequency response (evaluated on the unit circle $z = e^{j\omega T}$), stability, and transient behavior. FIR (Finite Impulse Response) filters have transfer functions with all poles at the origin (H(z) is a polynomial in $z^{-1}$), while IIR (Infinite Impulse Response) filters have non-trivial poles that create feedback and recursion in the time domain.

**Example 8.4.2:** A discrete-time system is described by the difference equation $y[n] = 0.5y[n-1] + x[n]$. Find H(z), the pole location, and determine if the system is stable.
**Solution:**
Taking the Z-transform: $Y(z) = 0.5z^{-1}Y(z) + X(z)$.
Solving for the transfer function: $H(z) = Y(z)/X(z) = 1 / (1 - 0.5z^{-1}) = z / (z - 0.5)$.
The system has one pole at $z = 0.5$ and one zero at $z = 0$.
Since $|0.5| < 1$, the pole lies inside the unit circle, so the system is stable.
This is an IIR system (first-order, single-pole lowpass filter).

### 8.4.3 Inverse Z-Transform

The inverse Z-transform recovers the discrete-time sequence x[n] from its Z-transform X(z), analogous to the inverse Laplace transform for continuous-time signals. For rational $X(z) = B(z)/A(z)$, the most practical method is partial fraction expansion: factor the denominator into its roots, decompose X(z)/z into partial fractions (dividing by z first simplifies the expansion), find the residues, multiply each term by z, and look up the corresponding sequence from a table of Z-transform pairs. A pole at $z = a$ contributes a term of the form $a^n u[n]$ for a causal sequence. Complex conjugate poles produce damped sinusoidal sequences, and repeated poles introduce terms of the form $n \times a^n u[n]$. Long division (power series expansion) provides an alternative computational method: dividing the numerator polynomial by the denominator in ascending powers of $z^{-1}$ directly yields the output samples x[0], x[1], x[2], ..., which is particularly useful for verifying results or when only a few output samples are needed. For systems described by difference equations, long division is equivalent to iterating the difference equation forward from initial conditions.

**Example 8.4.3:** Find the inverse Z-transform of $X(z) = (3z^2 - 4z) / (z^2 - 3z + 2)$, and determine the first three samples x[0], x[1], x[2].
**Solution:**
Factor the denominator: $z^2 - 3z + 2 = (z - 1)(z - 2)$.
Partial fraction expansion of X(z)/z: $X(z)/z = (3z - 4) / [(z - 1)(z - 2)]$.
Setting z = 1: $A = (3 - 4) / (1 - 2) = -1 / -1 = 1$.
Setting z = 2: $B = (6 - 4) / (2 - 1) = 2 / 1 = 2$.
So $X(z)/z = 1/(z - 1) + 2/(z - 2)$, giving $X(z) = z/(z - 1) + 2z/(z - 2)$.
Using the Z-transform pair $z/(z - a) \leftrightarrow a^n u[n]$: $x[n] = 1^n u[n] + 2 \times 2^n u[n] = (1 + 2^{n+1})u[n]$.
Verification: $x[0] = 1 + 2 = 3$, $x[1] = 1 + 4 = 5$, $x[2] = 1 + 8 = 9$.
Check via long division: dividing $(3z^2 - 4z)$ by $(z^2 - 3z + 2)$ gives $3 + 5z^{-1} + 9z^{-2} + \ldots$, confirming $x[0] = 3$, $x[1] = 5$, $x[2] = 9$.

### 8.4.4 Bilinear Transform

The bilinear transform is the standard method for converting a continuous-time (analog) transfer function $H_a(s)$ into an equivalent discrete-time (digital) transfer function H(z), preserving the frequency-domain characteristics while mapping the entire jω-axis of the s-plane onto the unit circle of the z-plane. The substitution $s = (2/T)(z - 1)/(z + 1)$ — where T is the sampling period — maps the left half of the s-plane (stable region) to the interior of the unit circle, ensuring that a stable analog prototype yields a stable digital filter. The inverse mapping $z = (1 + sT/2)/(1 - sT/2)$ confirms this: when $\mathrm{Re}\{s\} < 0$, $|z| < 1$. The bilinear transform introduces **frequency warping**: the relationship between analog frequency Ω and digital frequency ω is nonlinear, $\Omega = (2/T)\tan(\omega/2)$, compressing the infinite analog frequency range into the finite digital range [0, π/T]. To compensate, **pre-warping** adjusts the critical analog prototype frequencies before the transformation so that the digital filter's passband edges, cutoff frequencies, or center frequency land at the desired locations. For a cutoff frequency $\omega_c$ in the digital domain (rad/sample), the pre-warped analog frequency is $\Omega_c = (2/T)\tan(\omega_c/2)$. The bilinear transform is preferred over impulse invariance (which can alias) because it avoids aliasing entirely — the price is the nonlinear frequency mapping, which distorts the shape of wideband filters but is negligible for narrowband designs. All standard IIR filter design procedures (Butterworth, Chebyshev, Elliptic) in DSP software use the bilinear transform internally.

**Example 8.4.4:** Design a first-order digital lowpass filter with a −3 dB cutoff at 2 kHz using the bilinear transform applied to a first-order analog prototype $H_a(s) = \Omega_c/(s + \Omega_c)$. The sampling rate is $f_s = 10$ kHz.
**Solution:**
Digital cutoff frequency: $\omega_c = 2\pi \times 2000/10000 = 0.4\pi$ rad/sample.
Pre-warped analog frequency: $\Omega_c = (2/T)\tan(\omega_c/2) = 2 \times 10000 \times \tan(0.4\pi/2) = 20000 \times \tan(0.2\pi) = 20000 \times 0.7265 = 14531$ rad/s.
Analog prototype: $H_a(s) = 14531/(s + 14531)$.
Apply bilinear transform $s = (2/T)(z - 1)/(z + 1) = 20000(z - 1)/(z + 1)$: $H(z) = 14531 / (20000(z - 1)/(z + 1) + 14531) = 14531(z + 1) / (20000(z - 1) + 14531(z + 1)) = 14531(z + 1) / (34531z - 5469)$.
Dividing numerator and denominator by 34531: $H(z) = 0.4208(z + 1) / (z - 0.1584) = 0.4208(1 + z^{-1}) / (1 - 0.1584z^{-1})$.
Difference equation: $y[n] = 0.1584y[n-1] + 0.4208x[n] + 0.4208x[n-1]$.

Verification at $\omega = 0$ (DC): $H(1) = 0.4208 \times 2 / (1 - 0.1584) = 0.8416/0.8416 = 1.0$ (0 dB, passes DC).
At $\omega = \omega_c = 0.4\pi$: $|H(e^{j0.4\pi})| = 1/\sqrt{2} = 0.707$ ($-3$ dB), confirming the cutoff is correctly placed at 2 kHz.

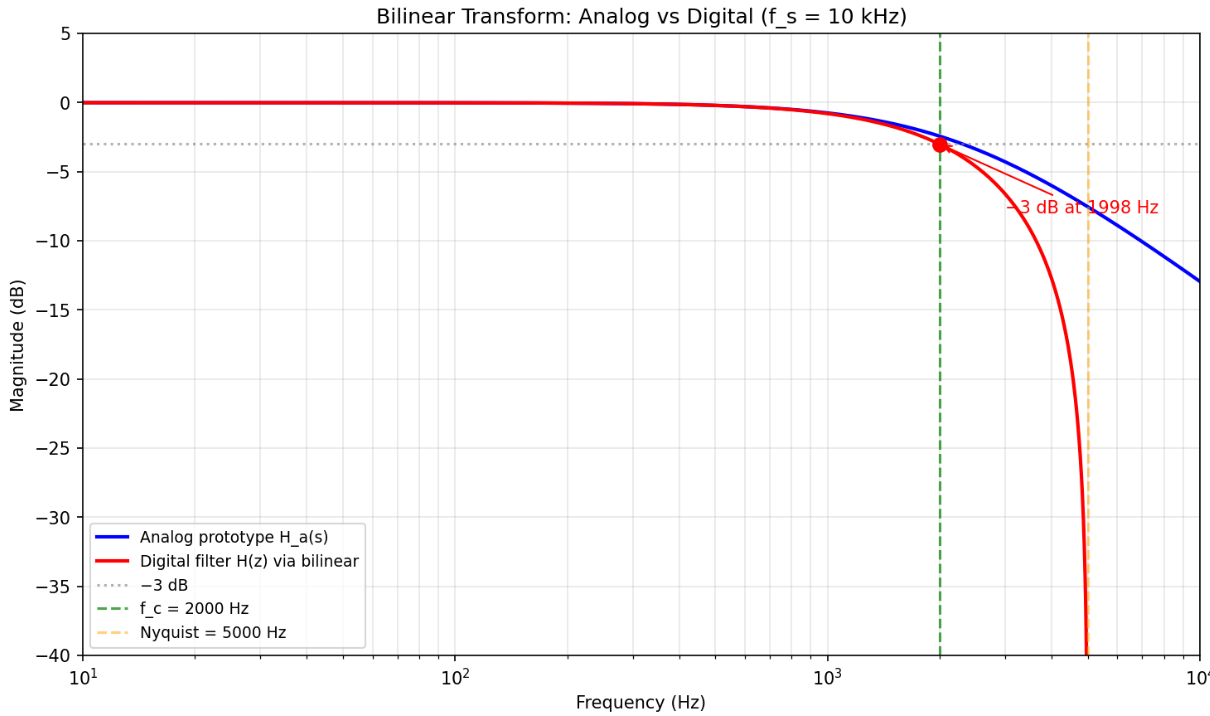



Figure 8.4.4: Bilinear Transform: Analog vs Digital Filter

### 8.4.5 Stability Analysis in the z-Plane

A causal discrete-time LTI system is stable (bounded-input bounded-output, or BIBO stable) if and only if all poles of its transfer function $H(z)$ lie strictly inside the unit circle $|z| < 1$. While direct inspection of pole locations works for low-order systems, higher-order systems require algebraic stability tests analogous to the Routh-Hurwitz criterion in the s-domain. The **Jury stability criterion** tests a polynomial $D(z) = a_0 z^n + a_1 z^{n-1} + \ldots + a_n$ by constructing a triangular array from the coefficients and checking conditions on the array elements. For a polynomial of degree n, the necessary conditions are: $D(1) > 0$, $(-1)^n D(-1) > 0$, and $|a_0| > |a_n|$. The Jury array then provides $n - 2$ additional conditions from the first elements of each row.

For **second-order systems** $H(z) = b_0 / (1 + a_1 z^{-1} + a_2 z^{-2})$, the stability conditions reduce to a simple set of three inequalities known as the **stability triangle**: 1. $1 + a_1 + a_2 > 0$ (ensures no pole at $z = +1$) 2. $1 - a_1 + a_2 > 0$ (ensures no pole at $z = -1$) 3. $|a_2| < 1$ (ensures poles are inside the unit circle)

These three conditions define a triangular region in the $(a_1, a_2)$ coefficient plane within which all second-order systems are stable. This is particularly useful for fixed-point IIR filter design, where coefficient quantization can push $a_1$ and $a_2$ outside the stability triangle.

The **Schur-Cohn stability test** provides an alternative using the reflection coefficients (PARCOR coefficients) of the polynomial: the system is stable if and only if all reflection coefficients have magnitude less than 1. This test is closely related to the lattice filter structure and is computationally efficient for verifying stability after coefficient quantization.

**Example 8.4.5:** A second-order IIR filter has transfer function $H(z) = 1 / (1 - 1.5z^{-1} + 0.72z^{-2})$. Determine (a) whether the filter is stable using the stability triangle conditions, (b) the pole locations, and (c) the maximum value of $a_2$ that maintains stability if $a_1$ is fixed at −1.5.
**Solution:**
(a) Stability triangle check ($a_1 = -1.5$, $a_2 = 0.72$):
Condition 1: $1 + (-1.5) + 0.72 = 0.22 > 0$ ✓
Condition 2: $1 - (-1.5) + 0.72 = 3.22 > 0$ ✓
Condition 3: $|0.72| = 0.72 < 1$ ✓
All three conditions are satisfied → **Stable**
(b) Pole locations from $z^2 - 1.5z + 0.72 = 0$:
$z = (1.5 \pm \sqrt{(2.25 - 2.88)}) / 2 = (1.5 \pm \sqrt{(-0.63)}) / 2 = (1.5 \pm j0.7937) / 2$
$z = 0.75 \pm j0.3969$
$|z| = \sqrt{(0.75^2 + 0.3969^2)} = \sqrt{(0.5625 + 0.1575)} = \sqrt{0.72} = \mathbf{0.8485}$
Both poles lie inside the unit circle at radius 0.8485, confirming stability. The pole angle is $\theta = \arctan(0.3969/0.75) = 27.9°$, corresponding to a resonant frequency of $f_r = \theta/(2\pi) \times f_s$.
(c) From condition 1: $a_2 > -1 - a_1 = -1 + 1.5 = 0.5$. From condition 3: $a_2 < 1$. Combined with condition 2 (always satisfied for $a_1 = -1.5$, $a_2 > 0$): the maximum $a_2$ for stability is $a_2 < \mathbf{1.0}$. In Q15 fixed-point with the pole radius at $\sqrt{a_2} = \sqrt{0.72} = 0.8485$, quantizing $a_2 = 0.72$ to Q15 gives $0.72 \times 32768 = 23593 \rightarrow 23593/32768 = 0.71997$, which remains safely within the stability triangle.

## 8.5 Digital Filters

Digital filters are discrete-time systems designed to selectively modify the spectral content of a signal — passing desired frequency components while attenuating unwanted ones. Unlike analog filters built from resistors, capacitors, and inductors, digital filters are implemented as algorithms in software or dedicated hardware (DSP chips, FPGAs), offering perfect reproducibility, easy reconfigurability, and the ability to realize filter characteristics that are impractical in the analog domain. The two fundamental architectures are FIR (non-recursive, guaranteed stable, linear phase possible) and IIR (recursive, computationally efficient, sharper transitions).

### 8.5.1 FIR Filters

Finite Impulse Response filters compute their output as a weighted sum of a finite number of input samples: $y[n] = \Sigma\, b_k \cdot x[n - k]$ for $k = 0$ to $M$, where the $b_k$ coefficients define the filter's impulse response of length $M + 1$. FIR filters are inherently stable (no feedback, no poles outside the origin) and can be designed to have exactly linear phase (constant group delay), which preserves the waveform shape of the signal — a critical property in applications such as audio processing and data communications. Design methods include the window method (applying a window function to the ideal impulse response), frequency sampling, and the Parks-McClellan (Remez exchange) algorithm for optimal equiripple designs. The primary trade-off is that FIR filters typically require a higher order (more coefficients) than IIR filters to achieve the same transition band sharpness, resulting in greater computational cost and latency.

**Example 8.5.1:** A 4-tap FIR filter has coefficients b = {0.25, 0.25, 0.25, 0.25}. If the input sequence is x[n] = {1, 2, 3, 4, 5, 0, 0, ...}, compute the first four output samples.
**Solution:**
y[n] = 0.25x[n] + 0.25x[n-1] + 0.25x[n-2] + 0.25x[n-3] (a simple moving average).
y[0] = 0.25(1) + 0.25(0) + 0.25(0) + 0.25(0) = 0.25.
y[1] = 0.25(2) + 0.25(1) + 0.25(0) + 0.25(0) = 0.75.
y[2] = 0.25(3) + 0.25(2) + 0.25(1) + 0.25(0) = 1.5.
y[3] = 0.25(4) + 0.25(3) + 0.25(2) + 0.25(1) = 2.5.
This filter computes the 4-point running average, smoothing the input signal.

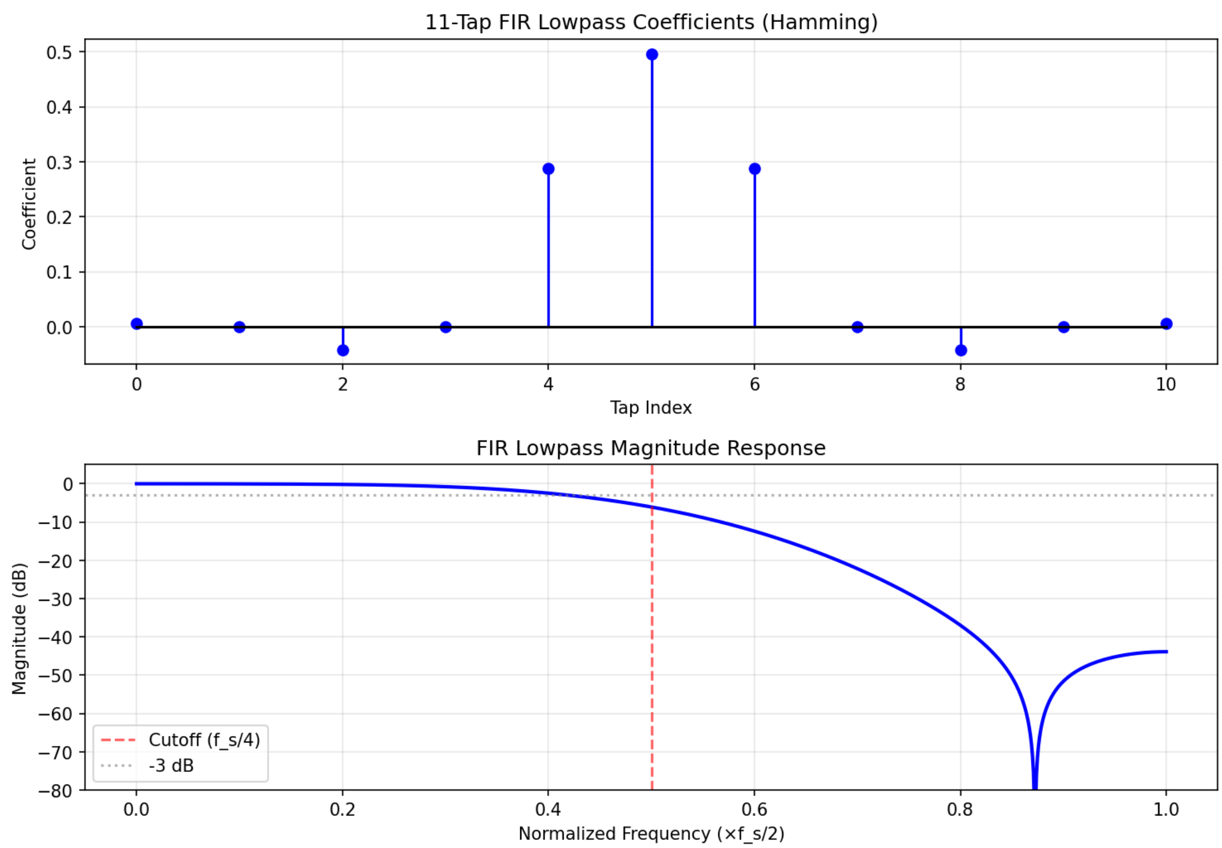



Figure 8.5.1: FIR Lowpass Filter Response

### 8.5.2 IIR Filters

Infinite Impulse Response filters use both feedforward and feedback (recursive) terms: $y[n] = \Sigma\, b_k \cdot x[n - k] - \Sigma\, a_k \cdot y[n - k]$, where the $a_k$ coefficients create the recursive (pole) structure. Because of their poles, IIR filters can achieve sharp frequency transitions with far fewer coefficients than FIR filters, making them computationally efficient for applications with strict selectivity requirements. IIR filters are commonly designed by transforming classical analog filter prototypes (Butterworth, Chebyshev Type I and II, Elliptic) to the digital domain using the bilinear transform, which maps the s-plane to the z-plane while preserving stability. However, IIR filters cannot achieve exactly linear phase, and care must be taken to ensure stability by verifying that all poles remain inside the unit circle.

Quantization of coefficients in fixed-point implementations can shift pole locations and potentially cause instability or limit-cycle oscillations.

**Example 8.5.2:** A first-order IIR lowpass filter is defined by y[n] = 0.1x[n] + 0.9y[n-1]. Find H(z), the pole location, and the magnitude response at DC ($\omega = 0$) and at the Nyquist frequency ($\omega = \pi$).
**Solution:**
$H(z) = 0.1 / (1 - 0.9z^{-1}) = 0.1z / (z - 0.9)$.
The pole is at $z = 0.9$ (inside the unit circle, so stable).
At DC ($z = e^{j0} = 1$): $|H(1)| = 0.1 / |1 - 0.9| = 0.1 / 0.1 = 1.0$ (0 dB, passes DC).
At Nyquist ($z = e^{j\pi} = -1$): $|H(-1)| = 0.1 / |1 + 0.9| = 0.1 / 1.9 = 0.0526$ (-25.6 dB).
This confirms the lowpass behavior with strong attenuation near the Nyquist frequency.

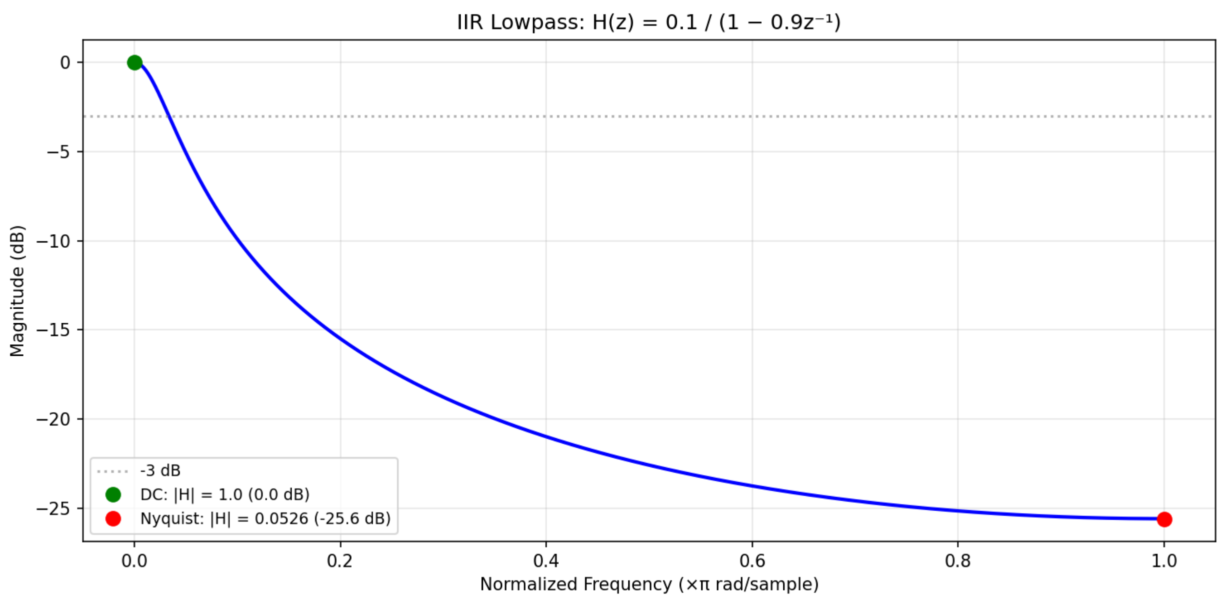



Figure 8.5.2: IIR Lowpass Filter Response

### 8.5.3 Filter Design Specifications

Filter design begins with a set of frequency-domain specifications that define the desired behavior across the frequency range. The passband is the frequency range where signals should pass with minimal attenuation (passband ripple typically specified in dB), and the stopband is the range where signals should be attenuated by at least a minimum amount (stopband attenuation in dB). The transition band is the frequency region between the passband edge and the stopband edge, and a narrower transition band requires a higher-order filter. Common filter types include lowpass (passes frequencies below a cutoff), highpass (passes above a cutoff), bandpass (passes a range of frequencies), and bandstop/notch (rejects a range). The choice between FIR and IIR, the filter order, and the approximation method are determined by the specific requirements for magnitude response, phase linearity, computational budget, and implementation constraints.

**Example 8.5.3:** A digital lowpass filter is needed with passband edge at 2 kHz, stopband edge at 3 kHz, maximum passband ripple of 1 dB, minimum stopband attenuation of 40 dB, and a sampling rate of 16 kHz. Estimate the required order for a Butterworth IIR filter.
**Solution:**
Normalized frequencies: $\Omega_p = 2\pi \times 2000/16000 = 0.25\pi$ rad/sample, $\Omega_s = 2\pi \times 3000/16000 = 0.375\pi$ rad/sample.
Using the bilinear transform pre-warping: $\omega_p = \tan(\Omega_p/2) = \tan(0.125\pi) = 0.4142$, $\omega_s = \tan(\Omega_s/2) = \tan(0.1875\pi) = 0.6682$.
Selectivity ratio: $k = \omega_p/\omega_s = 0.4142/0.6682 = 0.6198$.
The Butterworth order formula: $n \geq \log[(10^{As/10} - 1)/(10^{Ap/10} - 1)] / (2 \times \log(1/k)) = \log[(10^4 - 1)/(10^{0.1} - 1)] / (2 \times \log(1.614)) = \log[9999/0.2589] / (2 \times 0.2080) = \log(38{,}622) / 0.4160 = 4.587 / 0.4160 = 11.03$.
Therefore $n = 12$ (round up to the next integer).
A 12th-order Butterworth filter satisfies the specifications.

### 8.5.4 Multirate Signal Processing

Multirate signal processing involves changing the sampling rate of a discrete-time signal, either by reducing it (decimation) or increasing it (interpolation). Decimation by a factor M reduces the sampling rate by keeping every M-th sample and discarding the rest; an anti-aliasing lowpass filter with cutoff $\pi/M$ must precede the downsampler to prevent aliasing of high-frequency components into the reduced-rate signal. Interpolation by a factor L increases the sampling rate by inserting L - 1 zeros between each original sample (upsampling) followed by a lowpass anti-imaging filter with cutoff $\pi/L$ and gain L to reconstruct the intermediate values. Rational sample rate conversion by L/M is achieved by first interpolating by L, then decimating by M. Multirate techniques are essential in digital audio (CD to DAC conversion at 44.1 kHz to 192 kHz), software-defined radio (channelization of wideband signals), and efficient filter implementations using polyphase decomposition, which reduces computational cost by operating filters at the lower sampling rate.

**Example 8.5.4:** A signal sampled at 48 kHz must be converted to 16 kHz for a telephony codec. Determine the decimation factor, the required anti-aliasing filter cutoff frequency, and the number of output samples for a 1-second input.
**Solution:**
Decimation factor: M = 48,000 / 16,000 = 3.
Before decimating, apply a lowpass filter with cutoff $f_c = f_{s,new} / 2 = 16{,}000 / 2 = 8000$ Hz (or equivalently, the normalized frequency cutoff is $\pi/M = \pi/3$ rad/sample).
This prevents frequencies above 8 kHz from aliasing into the 0–8 kHz band of the decimated signal.
Input samples for 1 second: 48,000.
Output samples: 48,000 / 3 = 16,000 samples.
The anti-aliasing filter must have sufficient stopband attenuation (typically ≥ 60 dB) to suppress aliased components below the noise floor.

### 8.5.5 Fixed-Point Filter Implementation

Many embedded DSP systems and FPGAs implement digital filters using fixed-point arithmetic rather than floating-point, offering lower cost, lower power consumption, and higher throughput at the expense of limited dynamic range and susceptibility to quantization effects. In fixed-point representation, numbers are stored as scaled integers with an implied binary point; a Q15 format, for example, uses 16 bits to represent values from -1.0 to +0.99997 with a resolution of $2^{-15} \approx 30.5 \times 10^{-6}$. **Coefficient quantization** rounds the ideal filter coefficients to the nearest representable fixed-point value, which shifts the actual pole and zero locations from their designed positions — for IIR filters, this can significantly distort the frequency response or even cause instability if a pole moves outside the unit circle. **Roundoff noise** accumulates at each multiply-accumulate operation and appears as additive white noise at the filter output, with variance proportional to $2^{-2B}$ where B is the word length. **Overflow** occurs when intermediate accumulations exceed the fixed-point range, producing severe distortion; saturation arithmetic (clamping to the maximum value) is preferred over wraparound to limit the damage. Cascade (second-order sections) and parallel filter structures are more robust to quantization than high-order direct-form implementations because each second-order section has its own well-separated poles, reducing sensitivity to coefficient rounding. Scaling strategies ($L_1$, $L_2$, or $L_\infty$ norms) set the gain at each stage to prevent overflow while maximizing the signal-to-roundoff-noise ratio.

**Example 8.5.5:** A 6th-order IIR bandpass filter is implemented in Q15 fixed-point (16-bit) on a DSP. The filter is realized as a cascade of three second-order sections (biquads). Each biquad has a 32-bit accumulator. Estimate the output roundoff noise power if each biquad contributes an independent roundoff noise variance of $\sigma^2 = 2^{-2B}/12$, where B = 15 bits, and compare to a single direct-form VI implementation.

**Solution:**

Roundoff noise variance per biquad (one quantization per section output): $\sigma_q{}^2 = 2^{-30} / 12 = 9.31 \times 10^{-10} / 12 = 7.76 \times 10^{-11}$.

For 3 cascaded biquads (assuming unity gain and independent noise), total output noise variance: $\sigma_{total}{}^2 = 3 \times \sigma_q{}^2 = 3 \times 7.76 \times 10^{-11} = 2.33 \times 10^{-10}$.

RMS noise: $\sigma_{total} = \sqrt{(2.33 \times 10^{-10})} = 1.53 \times 10^{-5}$, or about -96.3 dB relative to full scale.

For a direct-form implementation of order 6, there are 6 feedback multiplications each contributing roundoff noise, amplified by the filter's pole locations.

With tightly clustered poles (common in narrowband bandpass filters), the noise gain can be 10-100× higher, yielding output noise 20-40 dB worse than the cascade form.

This is why cascade biquad structures are the standard practice for fixed-point IIR filter implementation.

### 8.5.6 Polyphase Filter Structures

Polyphase decomposition is an efficient implementation technique for multirate filters that reduces computational cost by operating the filter at the lower sampling rate rather than the higher one. In a decimation-by-M system, the standard approach filters at the high input rate and then discards $(M - 1)$ of every M output samples — wasting computation on samples that are immediately thrown away. The polyphase decomposition splits the filter's N coefficients into M subfilters (phases), each containing N/M coefficients, and processes the input directly at the decimated rate. Each phase operates on a different time-offset subset of the input samples, and the M subfilter outputs are summed to produce the final result. This reduces the number of multiply-accumulate operations per output sample from

N to N/M — the same factor as the decimation ratio. For interpolation-by-L, the polyphase structure distributes the filter coefficients across L phases that each produce one of the L output samples per input sample, again computing only the needed outputs at the output rate rather than first upsampling (inserting zeros) and filtering at the high rate. **Polyphase filter banks** generalize this concept to uniformly split a wideband signal into M subbands simultaneously, forming the basis of channelizers in software-defined radio (SDR), audio coding (MP3 uses a 32-band polyphase filter bank), and OFDM modulation/demodulation. The DFT filter bank combines polyphase decomposition with an M-point FFT to efficiently implement M bandpass filters in parallel, enabling real-time channelization of wide RF bandwidths.

**Example 8.5.6:** A 240-tap FIR lowpass filter is used as an anti-aliasing filter before decimation by M = 6 from 48 kS/s to 8 kS/s. Calculate the computational cost (multiplications per output sample) for the direct approach versus the polyphase implementation.
**Solution:**
Direct approach: the filter runs at the input rate of 48 kS/s, requiring 240 multiplications per input sample.
The output rate is 48,000/6 = 8,000 samples/s.
Multiplications per second: $240 \times 48{,}000 = 11{,}520{,}000$.
Per output sample: 11,520,000 / 8,000 = 1,440 multiplications (equivalently, 240 multiplications are computed for each of the 6 input samples, but 5 out of 6 outputs are discarded).
Polyphase approach: the 240 coefficients are distributed into M = 6 subfilters of 240/6 = 40 coefficients each.
Each subfilter runs at the output rate of 8 kS/s.
Multiplications per output sample: $6 \times 40 = 240$.
Multiplications per second: $240 \times 8{,}000 = 1{,}920{,}000$.
Savings: 11,520,000 / 1,920,000 = 6× reduction — exactly equal to the decimation factor M.
The polyphase structure computes only the samples that will actually be used, eliminating all wasted computation.

### 8.5.7 Allpass Filters and Group Delay Equalization

An allpass filter is a system with unity magnitude response at all frequencies — $|H(e^{j\omega})| = 1$ for all $\omega$ — but a frequency-dependent phase response. For a first-order digital allpass filter, the transfer function is $H(z) = (a^* + z^{-1}) / (1 + az^{-1})$, where a is a real coefficient with $|a| < 1$. For a second-order allpass section: $H(z) = (a_2 + a_1z^{-1} + z^{-2}) / (1 + a_1z^{-1} + a_2z^{-2})$ — the numerator coefficients are the denominator coefficients in reversed order, which places the zeros at the reciprocal conjugate locations of the poles ($z_{zero} = 1/z_{pole}^*$), ensuring the unit-circle magnitude is exactly 1. Although allpass filters do not modify the spectrum's magnitude, they reshape its phase, making them essential for **group delay equalization** — the process of flattening the total group delay of a system to make it approximately constant across the passband.

IIR filters (Butterworth, Chebyshev, Elliptic) inherently have nonlinear phase and non-constant group delay, especially near the passband edge where the group delay peaks sharply. This phase distortion spreads pulse edges in time and distorts waveform shapes — unacceptable in applications like data communications (causing intersymbol interference), audio (pre-echo and smearing), and measurement systems. By cascading an allpass equalizer with the IIR filter, the combined group delay can be made nearly constant without affecting the magnitude response. The equalizer design involves: (1) computing the group delay of the original IIR filter, (2) determining the desired flat group delay (equal to the maximum group delay of the IIR filter), (3) designing allpass sections whose group delay fills in the "valleys" to make the total group delay flat, and (4) cascading the allpass sections with the original filter.

Allpass filters also enable **complementary filter pairs**: if H(z) is a lowpass filter, then $G(z) = z^{-N} - H(z)$ is the complementary highpass (power complementary if $|H|^2 + |G|^2 = 1$). Allpass-based complementary pairs use two allpass filters $A_0(z)$ and $A_1(z)$ to create $H_{LP}(z) = (A_0(z^2) + z^{-1}A_1(z^2))/2$ and $H_{HP}(z) = (A_0(z^2) - z^{-1}A_1(z^2))/2$, guaranteeing perfect reconstruction (the sum exactly equals a pure delay). This structure is used in QMF (Quadrature Mirror Filter) banks for audio coding and subband processing.

**Example 8.5.7:** A 4th-order Chebyshev Type I lowpass IIR filter with 0.5 dB passband ripple has a group delay that varies from 8.2 samples at DC to 22.7 samples at the passband edge ($\omega = 0.4\pi$). A cascade of two second-order allpass sections is designed to equalize the group delay to approximately 23 samples (flat). The allpass sections have coefficients $a_{11} = -1.38$, $a_{21} = 0.69$ (section 1) and $a_{12} = -0.85$, $a_{22} = 0.42$ (section 2). Calculate (a) the group delay variation before equalization, (b) the total system order after adding the equalizer, and (c) the latency penalty introduced by equalization.

**Solution:**

(a) Group delay variation before equalization:

$\Delta\tau = \tau_{max} - \tau_{min} = 22.7 - 8.2 =$ **14.5 samples**

At $f_s = 48$ kHz, this corresponds to 14.5 / 48,000 = 0.302 ms of time-domain spreading.

(b) Total system order:

Original IIR filter: 4th order (2 biquad sections)

Allpass equalizer: $2 \times$ 2nd order = 4th order (2 more biquad sections)

Total: **8th order** (4 biquad sections in cascade)

The equalizer doubles the computational cost but does not change the magnitude response.

(c) Latency penalty:

Before equalization: group delay at DC = 8.2 samples

After equalization: group delay $\approx$ 23 samples (flat across passband)

Added latency = 23 − 8.2 = **14.8 samples** = 14.8 / 48,000 = 0.308 ms

This is the fundamental trade-off: achieving flat group delay requires increasing the minimum delay to match the maximum, adding latency equal to the original group delay variation. For audio at 48 kHz, 0.3 ms is imperceptible; for real-time control systems, this added latency may need consideration.

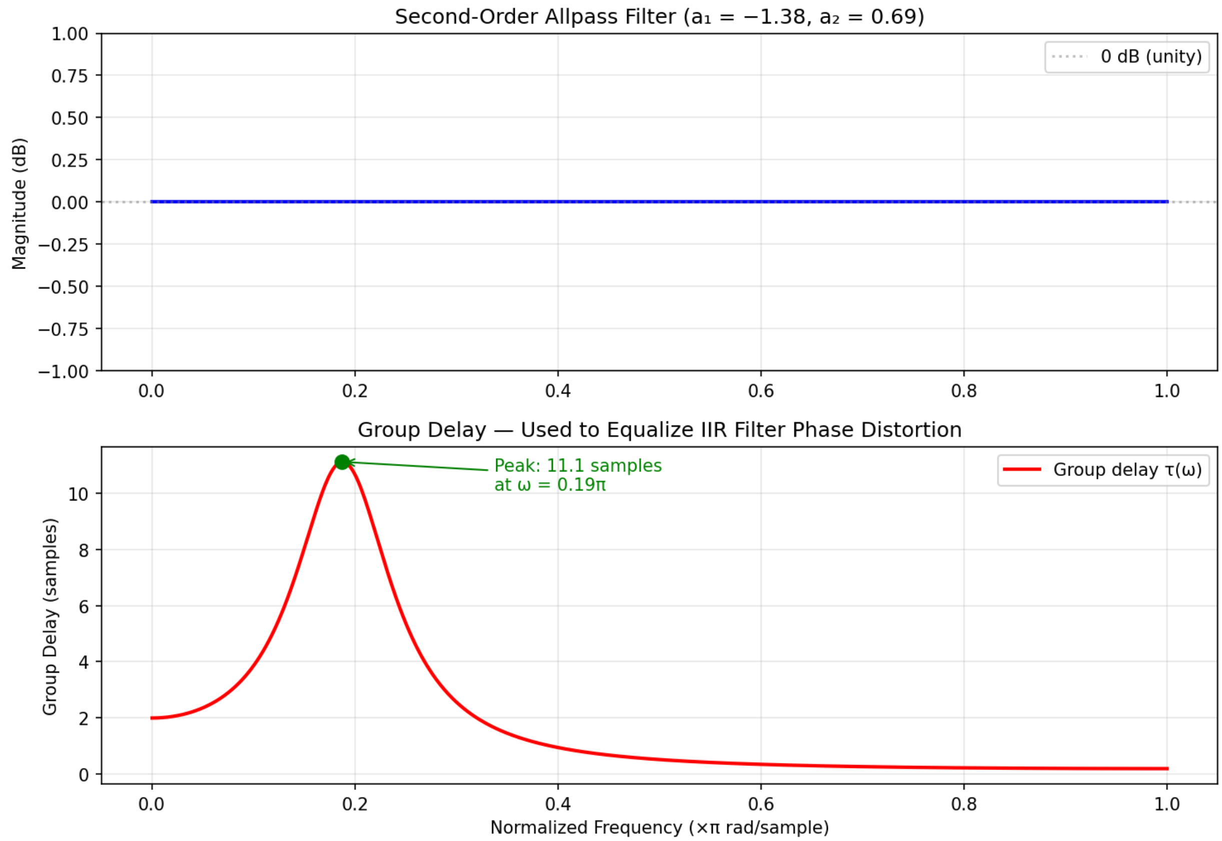



Figure 8.5.7: Allpass Filter: Magnitude and Group Delay

## 8.6 Spectral Analysis

Spectral analysis encompasses the techniques used to estimate and interpret the frequency content of signals, bridging the gap between theoretical transform methods and practical measurement of real-world data. Because real signals are finite in duration, noisy, and often non-stationary, spectral estimation requires careful consideration of resolution, variance, and leakage trade-offs. Methods range from classical periodogram-based approaches to modern parametric and time-frequency techniques.

### 8.6.1 Power Spectral Density

The Power Spectral Density (PSD) describes how the power of a signal is distributed across frequency and is defined as the Fourier transform of the autocorrelation function (Wiener-Khinchin theorem). For deterministic power signals, the PSD is computed as the limit of the squared magnitude of the Fourier transform divided by the observation interval. For random signals, the PSD must be estimated from finite data records using methods such as the periodogram (squared magnitude of the FFT divided by N) or the averaged periodogram (Welch's method), which divides the data into overlapping segments, windows each segment, computes the periodogram of each, and averages the results to reduce variance. The PSD has units of power per unit frequency (e.g., $V^2$/Hz or W/Hz) and integrating the PSD over a frequency band yields the signal power in that band.

**Example 8.6.1:** A white noise signal has a flat PSD of $S(f) = 2 \times 10^{-6}\ V^2/Hz$ over a bandwidth of 0 to 10 kHz. Find the total noise power and the RMS noise voltage.
**Solution:**
Since the PSD is flat, the total power is $P = S(f) \times BW = 2 \times 10^{-6} \times 10{,}000 = 0.02\ V^2$.
The RMS noise voltage is $V_{rms} = \sqrt{P} = \sqrt{0.02} = 0.1414\ V = 141.4\ mV$.
If the bandwidth were reduced to 1 kHz using a lowpass filter, the noise power would drop to $2 \times 10^{-6} \times 1000 = 0.002\ V^2$ ($V_{rms} = 44.7\ mV$), demonstrating why bandwidth limiting is an effective noise-reduction technique.

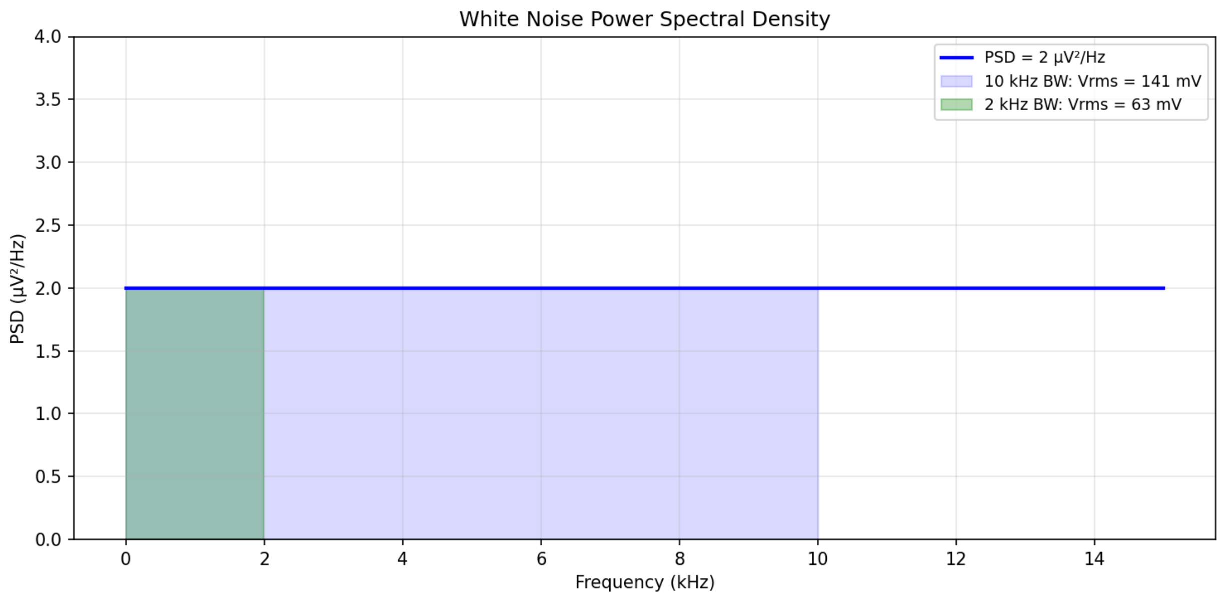



Figure 8.6.1: White Noise Power Spectral Density

### 8.6.2 Windowing

Windowing is the process of multiplying a finite-length signal segment by a window function before computing the DFT, in order to control spectral leakage caused by the abrupt truncation of the signal at the segment boundaries. The rectangular window (no modification) provides the narrowest main lobe and best frequency resolution but has the highest sidelobe levels (-13 dB), allowing energy from strong spectral components to leak into adjacent frequency bins. Tapered windows reduce sidelobe levels at the cost of a wider main lobe: the Hanning window achieves -31 dB sidelobes, the Hamming window -43 dB, and the Blackman window -58 dB. The choice of window involves a trade-off between frequency resolution (main lobe width) and spectral leakage (sidelobe level), determined by the application requirements. The Kaiser window provides a continuously adjustable parameter $\beta$ that allows the designer to smoothly trade off between these competing objectives.

**Example 8.6.2:** A 1024-point FFT is used to analyze a signal sampled at 44.1 kHz. Compare the frequency resolution using a rectangular window versus a Hanning window.
**Solution:**
With a rectangular window, the frequency resolution is approximately $\Delta f = f_s / N = 44{,}100 / 1024$

= 43.07 Hz.
The main lobe width is $2 \times \Delta f = 86.13$ Hz, and the highest sidelobe is at -13 dB.
With a Hanning window, the main lobe widens to approximately $4 \times \Delta f = 172.3$ Hz (twice the rectangular window), but the highest sidelobe drops to -31 dB, an 18 dB improvement in sidelobe suppression.
For resolving two tones spaced 100 Hz apart, the rectangular window succeeds but may show leakage artifacts. With a Hanning window, the Rayleigh resolution criterion is approximately $2 \times \Delta f = 86.1$ Hz, so the 100 Hz separation is marginally above the resolution limit — the two tones may be resolvable depending on their relative amplitudes and SNR, though with more spectral smearing than the rectangular window.

### 8.6.3 Time-Frequency Analysis

Traditional Fourier analysis provides excellent frequency resolution but no time localization — it reveals which frequencies are present in a signal but not when they occur. The Short-Time Fourier Transform (STFT) addresses this by computing the Fourier transform of successive windowed segments of the signal, producing a spectrogram that displays signal energy as a function of both time and frequency. The STFT is subject to the uncertainty principle: a shorter window provides better time resolution but poorer frequency resolution, and vice versa. The Wavelet Transform overcomes this fixed resolution limitation by using analysis functions (wavelets) that are scaled and translated, providing fine time resolution at high frequencies and fine frequency resolution at low frequencies. Wavelets are particularly effective for analyzing transient events, non-stationary signals, and multi-scale phenomena in applications such as speech processing, biomedical signal analysis, image compression (JPEG 2000), and seismic data interpretation.

**Example 8.6.3:** An STFT is performed on a signal sampled at 16 kHz using a 512-sample Hanning window with 50% overlap (256-sample hop). Determine the time resolution, frequency resolution, and the total number of STFT frames for a 2-second recording.
**Solution:**
Window duration: $T_w = 512 / 16{,}000 = 32$ ms.
Frequency resolution: $\Delta f = f_s / N = 16{,}000 / 512 = 31.25$ Hz.
Time resolution (hop size): $\Delta t = 256 / 16{,}000 = 16$ ms.
Total samples in 2 seconds: $2 \times 16{,}000 = 32{,}000$.
Number of frames: $(32{,}000 - 512) / 256 + 1 = 31{,}488 / 256 + 1 = 123 + 1 = 124$ frames.
The resulting spectrogram has 124 time frames and 257 frequency bins ($N/2 + 1$), covering 0 to 8 kHz.

### 8.6.4 Parametric Spectral Estimation

Parametric methods estimate the PSD by fitting a mathematical model to the observed data, rather than directly computing the Fourier transform. The most common approach is the autoregressive (AR) model, which assumes the signal is the output of an all-pole filter driven by white noise: $x[n] = -\Sigma\, a_k x[n-k] + w[n]$, where the $a_k$ are the AR coefficients and $w[n]$ is white noise with variance $\sigma^2$. The AR PSD is then $S(f) = \sigma^2 T / |1 + \Sigma\, a_k e^{-j2\pi fkT}|^2$, which produces smooth spectral estimates with sharp peaks corresponding to the model's poles — ideal for resolving closely spaced sinusoidal components or narrow spectral peaks. The AR model order p determines the number of poles and must be chosen carefully: too low and genuine spectral features are smoothed away, too high and spurious peaks appear from fitting noise. Common AR parameter estimation methods include the Yule-Walker (autocorrelation) method, the Burg method (which guarantees a stable model and is well-suited for short data records), and the covariance method. More advanced subspace methods — MUSIC (Multiple Signal Classification) and ESPRIT (Estimation of Signal Parameters via Rotational Invariance Techniques) — decompose the data correlation matrix into signal and noise subspaces and can resolve closely spaced sinusoids well beyond the Fourier resolution limit, but require knowledge of the number of signal components and assume the signal consists of sinusoids in white noise.

**Example 8.6.4:** Two sinusoids at frequencies $f_1 = 1000$ Hz and $f_2 = 1050$ Hz (50 Hz apart) are sampled at $f_s = 10$ kHz. Only $N = 64$ samples are available. Determine whether classical FFT-based spectral analysis can resolve the two components, and explain why a parametric method may succeed.

**Solution:**

FFT frequency resolution: $\Delta f = f_s / N = 10{,}000 / 64 = 156.25$ Hz.

Since the two sinusoids are separated by only 50 Hz, which is less than one frequency bin (156.25 Hz), the FFT cannot resolve them — they will appear as a single broadened peak.

Even with a rectangular window (narrowest main lobe), the Rayleigh resolution limit is $\Delta f =$ 156.25 Hz, far exceeding the 50 Hz separation.

Zero-padding to 256 or 1024 points would interpolate the spectrum more finely but cannot improve the fundamental resolution beyond 156.25 Hz.

An AR model of order $p = 4$ (two poles per sinusoid) applied via the Burg method estimates the AR coefficients from the 64 samples and places two sharp spectral peaks at the sinusoidal frequencies.

The AR spectral estimate achieves super-resolution because it imposes a model structure (all-pole) that matches the signal, effectively extrapolating the autocorrelation beyond the measured data length.

The MUSIC algorithm with a correlation matrix of size $16 \times 16$ would similarly resolve the two components, provided the SNR is sufficient (typically $> 10$ dB).

### 8.6.5 Cepstral Analysis

The cepstrum is the inverse Fourier transform of the logarithm of the magnitude spectrum, providing a representation in the **quefrency** domain (with units of time) that separates spectral features occurring at different rates. The power cepstrum is defined as $c[n] = \mathrm{IDFT}\{\log|\mathrm{DFT}\{x[n]\}|^2\}$, and the complex cepstrum uses the full complex logarithm (preserving phase) to allow signal reconstruction. Because the logarithm converts the multiplicative combination of spectral envelope and fine structure into an additive sum, the cepstrum cleanly separates slowly varying formant structure (low-quefrency cepstral coefficients) from rapidly varying harmonic or echo structure (high-quefrency peaks called **rahmonics**). In speech processing, the spectral envelope encodes phoneme identity while the harmonic fine structure encodes pitch; cepstral analysis extracts both independently. **Mel-Frequency Cepstral Coefficients (MFCCs)** — computed by applying the DCT to log-energies from a mel-scaled filter bank — are the dominant feature representation in speech recognition, speaker identification, and audio classification. The mel scale warps the linear frequency axis to approximate the nonlinear frequency perception of the human ear, with closer spacing at low frequencies and wider spacing at high frequencies: $f_{mel} = 2595 \times \log_{10}(1 + f/700)$. Typically 12–13 MFCCs (plus delta and delta-delta coefficients) capture the essential spectral shape of each speech frame. Cepstral analysis also detects echoes and periodic structure: an echo delayed by $\tau$ seconds appears as a peak in the cepstrum at quefrency $\tau$, enabling blind echo detection and pitch estimation for musical signals.

**Example 8.6.5:** A speech frame sampled at $f_s = 16$ kHz uses a 25 ms Hamming window ($N = 400$ samples, zero-padded to 512 for FFT). Compute the MFCCs using a 26-channel mel filter bank. If the speaker's fundamental frequency is $f_0 = 125$ Hz, at what quefrency index does the pitch peak appear in the cepstrum?
**Solution:**
The mel filter bank spans 0 to $f_s/2 = 8000$ Hz.
The mel frequency at 8000 Hz is $f_{mel} = 2595 \times \log_{10}(1 + 8000/700) = 2595 \times \log_{10}(12.43) = 2595 \times 1.094 = 2840$ mel.
The 26 triangular filters are uniformly spaced from 0 to 2840 mel, with center frequencies mapped back to Hz.
For each filter, the log-energy is computed from the FFT magnitude spectrum, and the DCT of the 26 log-energies yields the MFCCs: $c_k = \Sigma\,(\log E_m) \times \cos[\pi k(m - 0.5)/26]$ for $k = 0$ to 12, producing 13 coefficients.
The pitch period is $T_0 = 1/f_0 = 1/125 = 8$ ms $= 128$ samples at 16 kHz.
In the 512-point cepstrum, the pitch appears as a peak at quefrency index $n = 128$, corresponding to 8 ms.
This peak is distinct from the vocal tract (formant) information concentrated in the first 3–5 ms of the cepstrum (indices 0–80), demonstrating the cepstrum's ability to separate source (pitch) from filter (formants).

### 8.6.6 Wavelet Transform

The wavelet transform decomposes a signal into scaled and translated versions of a mother wavelet $\psi(t)$, providing multi-resolution analysis that simultaneously captures both time and frequency information with resolution that adapts to the signal content. The continuous wavelet transform (CWT) is defined as $W(a,b) = (1/\sqrt{|a|}) \int x(t)\, \psi^*((t - b)/a)\, dt$, where $a$ is the scale parameter (inversely related to frequency) and $b$ is the translation (time shift). Unlike the STFT, which uses a fixed window size for all frequencies, the wavelet transform uses short windows at high frequencies (fine time resolution, coarse frequency resolution) and long windows at low frequencies (coarse time resolution, fine frequency resolution) — matching the natural structure of many real-world signals such as speech, music, and transients. The **discrete wavelet transform (DWT)** samples the scale and translation on a dyadic grid ($a = 2^j$, $b = k2^j$) and is efficiently implemented by Mallat's filter bank algorithm: the signal is passed through a lowpass filter $h[n]$ (scaling function) and a highpass filter $g[n]$ (wavelet function), each followed by downsampling by 2, producing approximation coefficients (low-frequency content) and detail coefficients (high-frequency content) at each decomposition level. Iterating on the approximation coefficients creates a multi-level decomposition tree. Common wavelet families include Haar (simplest, piecewise constant), Daubechies (compact support, varying smoothness, db1–db20), Symlets (near-symmetric Daubechies), and Coiflets (symmetric with vanishing moments). **Applications** include signal denoising (thresholding small wavelet coefficients to remove noise while preserving edges), data compression (JPEG 2000 uses the CDF 9/7 biorthogonal wavelet), feature extraction for pattern recognition, ECG and EEG analysis in biomedical engineering, transient detection in power systems, and multi-scale edge detection in image processing.

**Example 8.6.6:** A noisy signal $x[n] = s[n] + w[n]$ (clean signal plus white Gaussian noise, $\sigma = 0.5$) is decomposed using a 3-level DWT with Daubechies db4 wavelets. The signal has 1024 samples at $f_s = 8$ kHz. Determine the frequency bands at each decomposition level and describe the soft-thresholding denoising procedure.

**Solution:**

At each DWT level, the signal is split into a low-frequency approximation and a high-frequency detail band, with each level halving the bandwidth and the number of samples.

Level 1: detail $d_1$ covers 2–4 kHz (512 coefficients), approximation $a_1$ covers 0–2 kHz.

Level 2: detail $d_2$ covers 1–2 kHz (256 coefficients), approximation $a_2$ covers 0–1 kHz.

Level 3: detail $d_3$ covers 0.5–1 kHz (128 coefficients), approximation $a_3$ covers 0–0.5 kHz (128 coefficients).

For soft-thresholding denoising, compute the universal threshold $\lambda = \sigma\sqrt{(2 \ln N)} = 0.5 \times \sqrt{(2 \times \ln(1024))} = 0.5 \times \sqrt{(2 \times 6.931)} = 0.5 \times \sqrt{13.863} = 0.5 \times 3.724 = 1.862$.

Apply soft thresholding to each detail coefficient: $\hat{d}[k] = \text{sign}(d[k]) \times \max(|d[k]| - \lambda, 0)$.

Coefficients with magnitude below 1.862 are set to zero (noise), while larger coefficients are shrunk toward zero by $\lambda$ (preserving signal edges).

The approximation coefficients $a_3$ are typically left unmodified.

Reconstruct the denoised signal by applying the inverse DWT using the thresholded detail coefficients and the original approximation coefficients.

This method preserves sharp transients and edges that would be smoothed by a conventional low-pass filter, because the wavelet basis captures localized features at each scale independently.

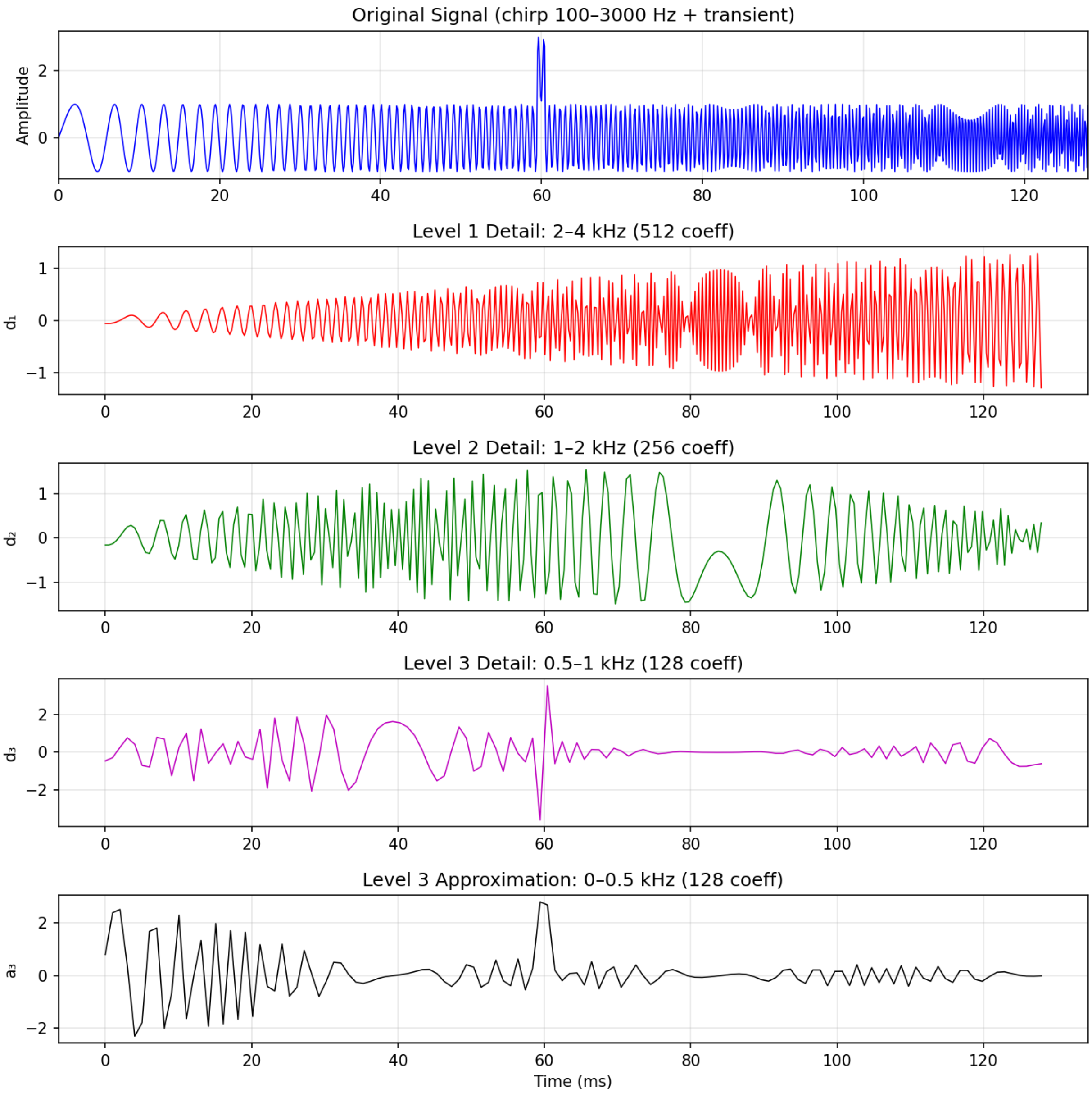



Figure 8.6.6: Wavelet Transform Multi-Level Decomposition

## 8.7 Adaptive Filtering

Adaptive filters are self-adjusting digital filters whose coefficients are updated automatically to minimize an error signal, enabling them to operate in environments where the signal statistics are unknown or time-varying. Unlike fixed filters designed from specifications, adaptive filters learn the optimal filter response from the data itself by iteratively adjusting their weights to minimize a cost function (typically mean squared error). This makes them indispensable for applications where the interference characteristics change over time or cannot be predicted in advance, including noise cancellation, echo cancellation, channel equalization, and system identification.

### 8.7.1 LMS Algorithm

The Least Mean Squares (LMS) algorithm is the most widely used adaptive filtering algorithm due to its simplicity, low computational cost, and robust convergence properties. At each time step n, the LMS algorithm updates the filter weight vector w[n] using the stochastic gradient descent rule: $w[n+1] = w[n] + 2\mu \cdot e[n] \cdot x[n]$, where x[n] is the input vector containing the M most recent input samples, $e[n] = d[n] - w^T[n]x[n]$ is the error between the desired signal d[n] and the filter output, and $\mu$ is the step size (learning rate) that controls the trade-off between convergence speed and steady-state misadjustment. The step size must satisfy $0 < \mu < 1/(M \cdot P_x)$ for convergence, where M is the filter order and $P_x$ is the input signal power. A larger $\mu$ converges faster but produces more residual error; a smaller $\mu$ converges slowly but achieves lower steady-state misadjustment. The Normalized LMS (NLMS) variant divides the step size by the input power estimate, providing more consistent convergence behavior across varying signal levels.

**Example 8.7.1:** An LMS adaptive filter of order M = 4 is used for noise cancellation. The input signal power is $P_x = 0.5$ W. Determine the maximum step size for convergence and suggest a practical step size. If the current weight vector is w = [0.1, -0.2, 0.3, 0.1], the input vector is x = [1.0, 0.5, -0.3, 0.8], and the desired signal is d = 0.7, compute one LMS update.
**Solution:**
Maximum step size: $\mu_{max} = 1 / (M \times P_x) = 1 / (4 \times 0.5) = 0.5$.
A practical step size is typically 10% of the maximum: $\mu = 0.05$.
Filter output: $y = w^Tx = 0.1(1.0) + (-0.2)(0.5) + 0.3(-0.3) + 0.1(0.8) = 0.1 - 0.1 - 0.09 + 0.08 = -0.01$.
Error: $e = d - y = 0.7 - (-0.01) = 0.71$.
Weight update: $w[n+1] = w[n] + 2\mu \cdot e \cdot x = [0.1, -0.2, 0.3, 0.1] + 2(0.05)(0.71)[1.0, 0.5, -0.3, 0.8]$ $= [0.1, -0.2, 0.3, 0.1] + [0.071, 0.0355, -0.0213, 0.0568] = [0.171, -0.1645, 0.2787, 0.1568]$.
The large error indicates the filter has not yet converged; repeated iterations will drive e[n] toward zero.

### 8.7.2 Applications of Adaptive Filters

Adaptive filters are deployed across a wide range of engineering applications where the environment or interference is non-stationary. In **active noise cancellation** (ANC), a reference microphone captures ambient noise, an adaptive filter models the acoustic path to the listener, and the filtered output is played through a speaker in anti-phase to cancel the noise. In **echo cancellation** for telephony and teleconferencing, the adaptive filter models the echo path (acoustic coupling between loudspeaker and microphone) and subtracts the estimated echo from the microphone signal. In **channel equalization** for digital communications, an adaptive equalizer compensates for intersymbol interference (ISI) caused by multipath propagation, using a training sequence to initially converge and then switching to decision-directed mode. In **system identification**, an adaptive filter driven by a known input estimates the impulse response of an unknown system by minimizing the difference between the unknown system's output and the filter's output.

**Example 8.7.2:** A telephone echo canceller uses a 128-tap adaptive filter (representing 16 ms of echo at 8 kHz sampling rate). The echo path delay is approximately 40 ms (320 samples) due to a hybrid coupler reflection. Explain why this filter cannot cancel the echo and determine the minimum filter length required.

**Solution:**
The 128-tap filter can model echo paths up to 128 samples = 128/8000 = 16 ms in duration.
Since the echo arrives at 40 ms (320 samples), this delay exceeds the filter's modeling capability by 320 - 128 = 192 samples.
The filter must be at least 320 taps long to capture the full echo path delay.
In practice, the filter should be longer to account for the echo's decay tail — a typical choice would be 512 taps (64 ms) to model both the delay and the room reverberation.
At 8 kHz with 512 taps, each LMS update requires 512 multiply-accumulate operations, which is easily handled by modern DSP processors.

### 8.7.3 Recursive Least Squares (RLS)

The Recursive Least Squares algorithm is a faster-converging alternative to LMS that minimizes the weighted sum of all past squared errors rather than the instantaneous squared error. At each time step, RLS updates the filter weights using the exact least-squares solution through a matrix inversion lemma (avoiding the expensive direct matrix inversion): the algorithm maintains an inverse correlation matrix P[n] and computes a gain vector k[n] = P[n-1]x[n] / ($\lambda$ + $x^T$[n]P[n-1]x[n]), where $\lambda$ is the forgetting factor (typically 0.95–1.0) that exponentially down-weights older data. The weight update is w[n] = w[n-1] + k[n] × e[n], and the inverse correlation matrix is updated as P[n] = (P[n-1] - k[n]$x^T$[n]P[n-1]) / $\lambda$. RLS converges in approximately M iterations (where M is the filter order) compared to the many hundreds or thousands required by LMS, making it ideal for fast-changing environments such as mobile communication channels. The computational cost is O($M^2$) per sample (versus O(M) for LMS), and numerical stability requires careful implementation — the QR-decomposition-based RLS and square-root RLS variants address potential ill-conditioning of the P matrix. The forgetting factor $\lambda$ controls the effective memory of the algorithm: $\lambda = 1$ gives infinite memory (growing window), while $\lambda < 1$ allows the filter to track time-varying systems by progressively discounting older data.

**Example 8.7.3:** An RLS adaptive equalizer with M = 8 taps and forgetting factor $\lambda = 0.99$ is used to equalize a mobile radio channel. The LMS algorithm with the same filter order requires 500 iterations to converge within 1 dB of the optimal solution. Estimate the RLS convergence time and compare the computational cost per sample of both algorithms.
**Solution:**
RLS convergence: approximately M to 2M iterations = 8 to 16 iterations.
Compared to LMS at 500 iterations, RLS converges roughly 30–60× faster.
Computational cost per sample — LMS: $2M + 1$ multiplications $= 2 \times 8 + 1 = 17$ multiply-accumulate operations (M for output, M for weight update, 1 for error).
RLS: approximately $4M^2 + 4M$ multiplications for the full matrix update $= 4 \times 64 + 32 = 288$ operations.
Cost ratio: RLS/LMS = 288/17 ≈ 17×.
The effective memory length of the RLS filter is approximately $1/(1 - \lambda) = 1/0.01 = 100$ samples.
At a symbol rate of 10 kSymbols/s, this means the equalizer adapts to channel changes within 100/10,000 = 10 ms — adequate for pedestrian mobile channels (coherence time ~50 ms at 900 MHz, 10 km/h) but potentially too slow for vehicular speeds where $\lambda$ should be reduced to 0.95–0.98.

### 8.7.4 Kalman Filter

The Kalman filter is an optimal recursive state estimator that combines a dynamic system model (prediction) with noisy measurements (correction) to produce a minimum-variance estimate of the system state at each time step. Unlike adaptive FIR/IIR filters that estimate filter coefficients, the Kalman filter estimates the internal state vector x[n] of a linear state-space system: **prediction** $\hat{x}[n|n-1] = A\hat{x}[n-1] + Bu[n-1]$ and $P[n|n-1] = AP[n-1]A^T + Q$; **update** $K[n] = P[n|n-1]H^T(HP[n|n-1]H^T + R)^{-1}$, $\hat{x}[n] = \hat{x}[n|n-1] + Kn$, and $P[n] = (I - K[n]H)P[n|n-1]$, where A is the state transition matrix, H is the observation matrix, Q is the process noise covariance, R is the measurement noise covariance, K[n] is the Kalman gain, and P[n] is the error covariance matrix. The Kalman gain optimally balances trust between the model prediction and the measurement: when measurement noise R is large, K is small (trust the model); when process noise Q is large, K is large (trust the measurement). The **Extended Kalman Filter (EKF)** handles nonlinear systems by linearizing around the current estimate using Jacobians, while the **Unscented Kalman Filter (UKF)** uses sigma points to propagate the state distribution through the nonlinearity without linearization, providing better accuracy for highly nonlinear systems. Applications span GPS navigation, inertial measurement unit (IMU) sensor fusion, target tracking, battery state-of-charge estimation, and financial time-series analysis.

**Example 8.7.4:** A GPS receiver tracks position along one axis using a constant-velocity Kalman filter. The state vector is x = [position, velocity]$^T$, the state transition matrix (for $\Delta t = 1$ s) is A = [[1, 1], [0, 1]], the observation matrix is H = [1, 0] (GPS measures position only), the process noise covariance is Q = [[0.25, 0.5], [0.5, 1.0]] $m^2$, and the measurement noise variance is R = 9 $m^2$. At time $n-1$, the estimated state is $\hat{x}$ = [100 m, 2 m/s]$^T$ with error covariance P = [[4, 0], [0, 1]]. A GPS measurement of z = 105 m arrives. Compute the predicted state, Kalman gain, and updated state estimate.
**Solution:**
**Prediction:** $\hat{x}[n|n-1] = A\hat{x}[n-1]$ = [[1,1],[0,1]] × [100, 2]$^T$ = [102, 2]$^T$ (position advances by velocity × $\Delta$t).
$P[n|n-1] = APA^T + Q$ = [[1,1],[0,1]] × [[4,0],[0,1]] × [[1,0],[1,1]] + [[0.25,0.5],[0.5,1.0]] = [[5,1],[1,1]] + [[0.25,0.5],[0.5,1.0]] = [[5.25, 1.5],[1.5, 2.0]].
**Kalman gain:** $S = HP[n|n-1]H^T + R$ = [1,0] × [[5.25,1.5],[1.5,2.0]] × [1,0]$^T$ + 9 = 5.25 + 9 = 14.25.
$K = P[n|n-1]H^T/S$ = [5.25, 1.5]$^T$ / 14.25 = [0.368, 0.105]$^T$.
**Innovation:** $z - H\hat{x}[n|n-1]$ = 105 − 102 = 3 m.
**Update:** $\hat{x}[n]$ = [102, 2]$^T$ + [0.368, 0.105]$^T$ × 3 = [102 + 1.105, 2 + 0.315]$^T$ = [103.1, 2.32]$^T$.
The Kalman filter corrects the predicted position from 102 m toward the measurement of 105 m, but only partially (to 103.1 m) because it accounts for the 9 $m^2$ GPS measurement noise.
The velocity estimate is also updated to 2.32 m/s, reflecting the position innovation's implication that the object may be moving slightly faster than modeled.

### 8.7.5 Wiener Filter

The Wiener filter is the theoretically optimal linear filter for estimating a desired signal from a noisy observation, minimizing the mean squared error (MSE) between the filter output and the desired signal in the statistical (ensemble average) sense. In the frequency domain, the Wiener filter transfer function is $H(f) = S_{xd}(f) / S_{xx}(f)$, where $S_{xd}(f)$ is the cross-power spectral density between the input and the desired signal and $S_{xx}(f)$ is the input auto-power spectral density. For the common case of additive noise — where the observed signal is $x(t) = d(t) + n(t)$ with signal and noise uncorrelated — the Wiener filter simplifies to $H(f) = S_{dd}(f) / (S_{dd}(f) + S_{nn}(f)) = SNR(f) / (1 + SNR(f))$, where $SNR(f) = S_{dd}(f)/S_{nn}(f)$ is the frequency-dependent signal-to-noise ratio. At frequencies where the SNR is high, $H(f) \approx 1$ (pass the signal); where the SNR is low, $H(f) \approx 0$ (suppress the noise). The Wiener filter requires knowledge of the signal and noise power spectra, which must be estimated in practice — this is exactly what adaptive filters (LMS, RLS) accomplish by iteratively converging toward the Wiener solution without explicit spectral estimation. The discrete-time FIR Wiener filter solves the **Wiener-Hopf equation** $Rw = p$, where R is the input autocorrelation matrix, w is the optimal weight vector, and p is the cross-correlation vector between input and desired signal. This matrix equation has the direct solution $w_{opt} = R^{-1}p$, which the LMS algorithm approximates iteratively and the RLS algorithm tracks recursively. Applications include noise reduction in audio and images, channel equalization, linear prediction for speech coding, and optimal interpolation.

**Example 8.7.5:** A signal d(t) with flat PSD $S_{dd}(f) = 1.0 \times 10^{-3}$ V²/Hz is corrupted by additive white noise with PSD $S_{nn}(f) = 2.5 \times 10^{-4}$ V²/Hz over a bandwidth of 0–4 kHz. Design the Wiener filter and calculate the noise reduction (in dB) and the output SNR.
**Solution:**
Since both PSDs are flat (white signal in white noise), the Wiener filter is a simple frequency-independent gain: $H(f) = S_{dd} / (S_{dd} + S_{nn}) = 1.0 \times 10^{-3} / (1.0 \times 10^{-3} + 2.5 \times 10^{-4}) = 1.0 \times 10^{-3} / 1.25 \times 10^{-3} = 0.8$.
The filter passes 80% of the input amplitude at all frequencies.
Input SNR = $S_{dd}/S_{nn} = 1.0 \times 10^{-3} / 2.5 \times 10^{-4} = 4.0$ (6.02 dB).
Signal power at output: $P_{d,out} = H^2 \times S_{dd} \times BW = 0.64 \times 1.0 \times 10^{-3} \times 4000 = 2.56$ W.
Noise power at output: $P_{n,out} = H^2 \times S_{nn} \times BW = 0.64 \times 2.5 \times 10^{-4} \times 4000 = 0.64$ W.
Output SNR = 2.56/0.64 = 4.0 (6.02 dB).
The output SNR equals the input SNR — for white-in-white noise the Wiener filter cannot improve SNR, it only minimizes MSE by scaling.
However, if the signal PSD is concentrated in a narrower band (e.g., 0–1 kHz), the Wiener filter suppresses noise in the 1–4 kHz band where SNR is zero, yielding substantial noise reduction.
With $S_{dd}(f) = 4.0 \times 10^{-3}$ V²/Hz for 0–1 kHz and 0 elsewhere, H(f) = 0.941 in-band and 0 out-of-band, reducing total noise power from 1.0 W to 0.221 W — a 6.5 dB noise reduction.

### 8.7.6 Compressive Sensing

Compressive sensing (CS), also called compressed sampling or sparse recovery, is a signal acquisition framework that enables reconstruction of a signal from far fewer measurements than the Nyquist rate requires, provided the signal is **sparse** in some known basis or transform domain. A signal x of length N is K-sparse if it has at most K nonzero coefficients in a representation basis Ψ (e.g., Fourier, wavelet, or DCT), where $K \ll N$. Instead of sampling at the Nyquist rate, CS acquires $M \ll N$ linear measurements $y = \Phi x$, where Φ is an $M \times N$ measurement matrix, and recovers x by solving the underdetermined system subject to a sparsity constraint. The key theoretical result is that if Φ satisfies the **Restricted Isometry Property (RIP)** — meaning it approximately preserves distances between sparse vectors — then exact recovery is possible from $M \geq C \times K \times \ln(N/K)$ measurements, where C is a small constant. Random Gaussian and random Bernoulli matrices satisfy the RIP with high probability, as do certain structured random matrices (partial Fourier, random convolution). Recovery algorithms solve the optimization problem: minimize $\|x\|_1$ subject to $y = \Phi x$ (basis pursuit), which can be formulated as a linear program, or use greedy iterative methods such as **Orthogonal Matching Pursuit (OMP)** that sequentially identify the support of the sparse signal. For noisy measurements $y = \Phi x + e$, the formulation becomes minimize $\|x\|_1$ subject to $\|y - \Phi x\|_2 \leq \varepsilon$ (basis pursuit denoising, also known as LASSO). Applications include MRI acceleration (reducing scan time by 4–8× by acquiring fewer k-space samples), single-pixel cameras, radar imaging, spectrum sensing in cognitive radio, and seismic data acquisition where sensor deployment is expensive.

**Example 8.7.6:** A signal of length $N = 256$ is known to be $K = 8$ sparse in the DCT domain (only 8 of 256 DCT coefficients are nonzero). A random Gaussian measurement matrix $\Phi$ of size $M \times 256$ is used. Determine the minimum number of measurements M required for reliable recovery, the compression ratio, and compare to the Nyquist requirement.

**Solution:**

Using the CS measurement bound $M \geq C \times K \times \ln(N/K)$ with $C \approx 2$ (practical constant for Gaussian matrices): $M \geq 2 \times 8 \times \ln(256/8) = 16 \times \ln(32) = 16 \times 3.466 = 55.5$, so $M = 56$ measurements.

Compression ratio: $N/M = 256/56 = 4.57:1$.

Compared to Nyquist sampling, which requires all $N = 256$ samples, CS requires only 56 measurements — a 78% reduction.

The measurement matrix $\Phi$ is $56 \times 256$, and each measurement $y_i$ is a random linear combination of all 256 signal values.

Recovery via OMP proceeds iteratively: at each step, the algorithm identifies the column of $\Phi$ most correlated with the current residual $r = y - \Phi\hat{x}$, adds its index to the support set, and solves a least-squares problem over the selected columns.

After $K = 8$ iterations, the algorithm has identified all 8 nonzero DCT coefficients and their values.

In practice, M is chosen 3–5× larger than the theoretical minimum to provide robustness against noise, so $M = 80$–$120$ measurements would be typical for reliable recovery at moderate SNR.

For an MRI application, this means a $256 \times 256$ image with sparsity $K = 2000$ in the wavelet domain could be reconstructed from approximately 20,000 k-space samples instead of 65,536, reducing scan time by roughly 3×.

# Chapter 9

# Electromagnetics

Electromagnetics is the study of electric and magnetic fields, their interactions, and the propagation of electromagnetic waves. It provides the physical foundation for virtually all of electrical engineering, from circuit theory and power transmission to wireless communications, optics, and antenna design. The behavior of electromagnetic fields is governed by Maxwell's equations, a set of four fundamental relationships that unify electricity, magnetism, and light into a single coherent framework.

## 9.1 Electrostatics

Electrostatics deals with electric charges at rest and the fields and potentials they produce. The concepts of electric field, potential, and capacitance developed in electrostatics form the foundation for understanding circuit behavior, dielectric materials, electrostatic discharge (ESD) protection, and energy storage. Coulomb's law and Gauss's law provide the tools to calculate fields from charge distributions, while the concept of electric potential simplifies energy and voltage calculations throughout electrical engineering.

### 9.1.1 Electric Charge and Coulomb's Law

Electric charge is a fundamental property of matter — as intrinsic as mass — and is the ultimate source of all electromagnetic phenomena. It exists in two forms: positive (carried by protons) and negative (carried by electrons), with the elementary charge quantized at $e \approx 1.602 \times 10^{-19}$ C, meaning all observable charges appear in integer multiples of this value. Charge is strictly conserved: it can be transferred between objects but can never be created or destroyed, a principle that underpins Kirchhoff's Current Law in circuit analysis. Coulomb's Law quantifies the electrostatic force between two point charges: $F = q_1q_2/(4\pi\varepsilon_0 r^2)$, where $\varepsilon_0 \approx 8.854 \times 10^{-12}$ F/m is the permittivity of free space and r is the separation distance. The force acts along the line connecting the two charges — like charges repel and unlike charges attract — and follows an inverse-square law identical in form to Newton's law of gravitation, though the electrostatic force is vastly stronger (approximately $10^{36}$ times) at atomic scales. The principle of superposition applies: the total force on any charge due to a collection of other charges is the vector sum of the individual pairwise Coulomb forces, which generalizes naturally from discrete point charges to continuous charge distributions described by charge density $\rho$ ($C/m^3$), surface charge density $\sigma$ ($C/m^2$), or line charge density $\lambda$ (C/m). This transition from discrete to continuous descriptions is essential for analyzing practical structures such as charged conductors, capacitor plates, and semiconductor junctions, and leads directly to the concept of the electric field as a continuous vector field in space.

**Example 9.1.1:** Two point charges, $q_1 = +4\ \mu C$ and $q_2 = -2\ \mu C$, are separated by 30 cm in air. Find the magnitude and direction of the electrostatic force between them.
**Solution:**
$F = k|q_1||q_2|/r^2 = (8.99 \times 10^9)(4 \times 10^{-6})(2 \times 10^{-6})\ /\ (0.30)^2 = (8.99 \times 10^9)(8 \times 10^{-12})\ /\ 0.09 = 0.07192\ /\ 0.09 = 0.799\ N \approx 0.8\ N$.
Since the charges are opposite in sign, the force is attractive, directed along the line connecting the two charges, pulling them toward each other.

### 9.1.2 Electric Field

The electric field E is defined as the force per unit positive test charge at a point in space: $E = F/q$, measured in volts per meter (V/m) or equivalently newtons per coulomb (N/C). For a point charge Q, the electric field at distance r is $E = Q/(4\pi\varepsilon_0 r^2)$ directed radially outward for positive Q. Electric field lines originate on positive charges and terminate on negative charges, with the density of field lines representing the field strength. Gauss's Law states that the total electric flux through any closed surface equals the enclosed charge divided by $\varepsilon_0$: $\oint E \cdot dA = Q_{enclosed}/\varepsilon_0$. Gauss's Law is particularly powerful for calculating electric fields of charge distributions with high symmetry (spherical, cylindrical, or planar).

**Example 9.1.2:** Find the electric field at a distance of 20 cm from a point charge of Q = +5 nC in free space.
**Solution:**
$E = Q\ /\ (4\pi\varepsilon_0 r^2) = kQ/r^2 = (8.99 \times 10^9)(5 \times 10^{-9})\ /\ (0.20)^2 = 44.95\ /\ 0.04 = 1{,}123.75\ V/m \approx 1{,}124\ V/m$.
The field is directed radially outward from the positive charge.
Equivalently using Gauss's Law with a spherical Gaussian surface of radius 0.20 m: $E \times 4\pi r^2 = Q/\varepsilon_0$, so $E = Q\ /\ (4\pi\varepsilon_0 r^2)$, yielding the same result.

### 9.1.3 Electric Potential

Electric potential V is the work done per unit charge in moving a positive test charge from a reference point (typically infinity) to a given location in an electric field: $V = -\int E \cdot dl$. The potential difference (voltage) between two points determines the energy gained or lost by a charge moving between them: $W = q\Delta V$. The electric field is the negative gradient of the potential: $E = -\nabla V$, meaning the field points in the direction of steepest potential decrease. Equipotential surfaces are perpendicular to electric field lines at every point. For a point charge, $V = Q/(4\pi\varepsilon_0 r)$, and for multiple charges, the total potential is the algebraic (scalar) sum of individual potentials – simpler than vector addition of electric fields.

**Example 9.1.3:** Two charges, $q_1 = +3\ \mu C$ at the origin and $q_2 = -1\ \mu C$ at x = 40 cm, lie along the x-axis. Find the electric potential at the midpoint (x = 20 cm).
**Solution:**
The distance from each charge to the midpoint is r = 0.20 m.
$V = V_1 + V_2 = kq_1/r + kq_2/r = (8.99 \times 10^9)(3 \times 10^{-6})\ /\ 0.20 + (8.99 \times 10^9)(-1 \times 10^{-6})\ /\ 0.20 = 134{,}850 + (-44{,}950) = 89{,}900\ V \approx 89.9\ kV$.
The net potential is positive because the larger positive charge dominates at the midpoint.

### 9.1.4 Capacitance

Capacitance is the ability of a structure to store electric charge and energy in an electric field, defined as $C = Q/V$, where Q is the stored charge and V is the voltage across the structure, measured in farads (F). A parallel-plate capacitor has capacitance $C = \varepsilon A/d$, where ε is the permittivity of the dielectric material between the plates, A is the plate area, and d is the plate separation. The energy stored in a capacitor is $W = ½CV^2 = ½Q^2/C = ½QV$. Dielectric materials inserted between capacitor plates increase the capacitance by a factor of the relative permittivity (dielectric constant) $\varepsilon_r$, while also increasing the breakdown voltage. Capacitors in parallel add directly ($C_{total} = C_1 + C_2$), while capacitors in series combine as reciprocals ($1/C_{total} = 1/C_1 + 1/C_2$).

**Example 9.1.4:** A parallel-plate capacitor has plates of area $A = 0.02\ m^2$ separated by $d = 0.5$ mm of a dielectric with $\varepsilon_r = 4.5$. Find the capacitance and the energy stored when charged to 100 V.
**Solution:**
$C = \varepsilon_0\varepsilon_r A/d = (8.854 \times 10^{-12} \times 4.5 \times 0.02) / (0.5 \times 10^{-3}) = (7.969 \times 10^{-13}) / (5 \times 10^{-4}) = 1.594 \times 10^{-9}$ F ≈ 1.59 nF.
Energy stored: $W = ½CV^2 = 0.5 \times 1.594 \times 10^{-9} \times (100)^2 = 0.5 \times 1.594 \times 10^{-9} \times 10{,}000 = 7.97 \times 10^{-6}$ J ≈ 7.97 μJ.

### 9.1.5 Dielectric Materials and Boundary Conditions

A dielectric is an insulating material that becomes polarized when placed in an electric field, aligning its internal molecular dipoles to partially cancel the applied field and thereby increasing the capacitance of any structure containing it. The electric displacement field $D = \varepsilon E = \varepsilon_0\varepsilon_r E$ accounts for both the free-space field and the bound charge effects within the dielectric, where $\varepsilon_r$ (the relative permittivity or dielectric constant) ranges from 1 for vacuum to approximately 2-4 for common PCB substrates (FR-4 ≈ 4.3), 3.9 for $SiO_2$, 7-10 for alumina, and up to several thousand for ferroelectric ceramics ($BaTiO_3$). At the boundary between two different dielectric media, the tangential component of E is continuous ($E_{t1} = E_{t2}$) and the normal component of D is continuous in the absence of free surface charge ($D_{n1} = D_{n2}$, which means $\varepsilon_1 E_{n1} = \varepsilon_2 E_{n2}$). These boundary conditions determine how electric field lines refract at material interfaces: $\tan\theta_1 / \tan\theta_2 = \varepsilon_1/\varepsilon_2$, analogous to Snell's law for optics. The dielectric strength is the maximum electric field a material can withstand before electrical breakdown occurs (e.g., ~3 MV/m for air, ~20 MV/m for polyethylene, ~500 MV/m for $SiO_2$ thin films). Dielectric loss, quantified by the loss tangent $\tan\delta = \varepsilon''/\varepsilon'$ (the ratio of the imaginary to real parts of the complex permittivity), represents energy dissipated as heat in an alternating electric field and is critical in high-frequency applications — low-loss substrates like PTFE ($\tan\delta \approx 0.0002$) are preferred for RF circuits.

**Example 9.1.5:** A parallel-plate capacitor has a plate area of 10 $cm^2$ and a 2 mm gap filled with two dielectric layers: 1 mm of alumina ($\varepsilon_{r1} = 9.0$) and 1 mm of PTFE ($\varepsilon_{r2} = 2.1$). Find the total capacitance and the electric field in each layer when 500 V is applied across the plates.
**Solution:**
The two dielectric layers act as two capacitors in series.
$C_1 = \varepsilon_0\varepsilon_{r1}A/d_1 = 8.854 \times 10^{-12} \times 9.0 \times 10 \times 10^{-4} / 1 \times 10^{-3} = 79.69 \times 10^{-12} / 10^{-3} = 79.69$ pF.
$C_2 = \varepsilon_0\varepsilon_{r2}A/d_2 = 8.854 \times 10^{-12} \times 2.1 \times 10^{-3}\ m^2 / (1 \times 10^{-3}\ m) = 18.59$ pF.
Total: $1/C = 1/C_1 + 1/C_2 = 1/79.69 + 1/18.59 = 0.01255 + 0.05379 = 0.06634\ pF^{-1}$, so C = 15.07

pF.
For the electric fields, the normal component of D is continuous across the boundary: $D = \varepsilon_1 E_1 = \varepsilon_2 E_2$, and $V = E_1 d_1 + E_2 d_2 = 500$ V.
From continuity: $E_1 = (\varepsilon_2/\varepsilon_1)E_2 = (2.1/9.0)E_2 = 0.2333E_2$.
Substituting: $0.2333E_2 \times 10^{-3} + E_2 \times 10^{-3} = 500$, so $E_2(1.2333 \times 10^{-3}) = 500$, $E_2 = 405.4$ kV/m (in PTFE).
$E_1 = 0.2333 \times 405.4 = 94.6$ kV/m (in alumina).
The field is 4.3× stronger in the lower-permittivity PTFE layer — this is why breakdown in multilayer dielectrics typically initiates in the lower-ε material.

### 9.1.6 Laplace's and Poisson's Equations

Laplace's equation $\nabla^2 V = 0$ and Poisson's equation $\nabla^2 V = -\rho/\varepsilon$ govern the spatial distribution of electric potential in charge-free and charge-containing regions, respectively. These second-order partial differential equations, combined with boundary conditions (specified voltages on conductors or specified charge distributions), uniquely determine the potential and hence the electric field everywhere in a region — this is guaranteed by the **uniqueness theorem**, which states that a solution satisfying both the PDE and the boundary conditions is the only solution. Laplace's equation applies in regions free of charge (the dielectric space between capacitor plates, the insulation in a coaxial cable, or the air surrounding a charged conductor), while Poisson's equation applies where space charge exists (inside a semiconductor depletion region, in an electron beam, or in a plasma).

For simple geometries with high symmetry, analytical solutions exist: the potential between coaxial cylinders (radii a and b, inner conductor at $V_0$) is $V(r) = V_0 \ln(b/r)/\ln(b/a)$, and between concentric spheres is $V(r) = V_0\, a(b - r) / (r(b - a))$. For the vast majority of practical geometries — PCB trace cross-sections, high-voltage bushing profiles, IC interconnect structures, electric motor slot geometries — no closed-form solution exists, and numerical methods are used. The **finite difference method (FDM)** discretizes the region onto a rectangular grid and approximates the Laplacian as $V_{i,j} = (V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1})/4$ (the average of the four nearest neighbors in 2D), iterating until convergence. The **finite element method (FEM)** divides the region into triangular or tetrahedral elements, approximates V within each element using polynomial basis functions, and solves a global matrix equation — FEM handles complex geometries, material interfaces, and irregular boundaries far better than FDM and is the standard method in commercial electromagnetic simulation software (ANSYS Maxwell, COMSOL, CST). The **boundary element method (BEM)** discretizes only the surfaces (boundaries) rather than the volume, reducing a 3D problem to a 2D surface mesh, and is efficient for open-region problems like antenna and EMC simulations where the surrounding space extends to infinity.

**Example 9.1.6:** Two parallel conducting plates are separated by 10 mm. The top plate is at V = 100 V and the bottom plate is at V = 0 V. A 5 mm thick dielectric slab ($\varepsilon_r = 4$) rests on the bottom plate, with air ($\varepsilon_r = 1$) filling the upper 5 mm. Using Laplace's equation with the boundary condition that $D_n$ is continuous at the dielectric interface, find the potential and electric field in each region.
**Solution:**
In each region the potential satisfies $\nabla^2 V = 0$ (no free charge), reducing to $d^2V/dz^2 = 0$ in this 1D geometry, so V varies linearly within each layer: $V_1(z) = A_1 z + B_1$ (dielectric, $0 \le z \le 5$ mm) and $V_2(z) = A_2 z + B_2$ (air, 5 mm $\le z \le$ 10 mm).

Boundary conditions: $V_1(0) = 0 \rightarrow B_1 = 0$. $V_2(10\text{ mm}) = 100\text{ V} \rightarrow A_2(0.01) + B_2 = 100$.
At the interface (z = 5 mm): $V_1 = V_2$ (continuity of potential) and $\varepsilon_1 E_1 = \varepsilon_2 E_2$ (continuity of $D_n$).
$E_1 = -dV_1/dz = -A_1$ and $E_2 = -dV_2/dz = -A_2$.
From $D_n$ continuity: $4\varepsilon_0 A_1 = \varepsilon_0 A_2 \rightarrow A_2 = 4A_1$.
From V continuity at z = 5 mm: $A_1(0.005) = A_2(0.005) + B_2$.
From $V_2(0.01) = 100$: $A_2(0.01) + B_2 = 100$.
Solving: $B_2 = 100 - 0.01A_2 = 100 - 0.04A_1$. Then: $0.005A_1 = 0.005(4A_1) + 100 - 0.04A_1 = 0.02A_1 + 100 - 0.04A_1 = 100 - 0.02A_1$.
So $0.005A_1 + 0.02A_1 = 100$, giving $0.025A_1 = 100$, $A_1 = 4{,}000$ V/m.
$A_2 = 4 \times 4{,}000 = 16{,}000$ V/m.
$E_{dielectric}$ = **4,000 V/m** (4 kV/m), $E_{air}$ = **16,000 V/m** (16 kV/m).
V at interface = 4,000 × 0.005 = **20 V**. The electric field is 4× stronger in the air gap (lower ε), concentrating 80% of the voltage drop across the air and only 20% across the dielectric — a critical consideration in high-voltage insulation design where air breakdown (3 MV/m) limits the overall voltage rating.

## 9.2 Magnetostatics

Magnetostatics describes magnetic fields produced by steady (DC) currents and permanent magnets, and the forces these fields exert on moving charges and current-carrying conductors. Magnetic fields are fundamentally different from electric fields — they always form closed loops, exert forces perpendicular to motion, and cannot do work on stationary charges. The concepts of magnetic field intensity, flux, inductance, and magnetic materials are essential for designing transformers, inductors, electric motors, magnetic sensors, and data storage devices.

### 9.2.1 Magnetic Fields

Magnetic fields are produced by moving electric charges — that is, by electric currents — and in turn exert forces on other moving charges, establishing a deep connection between electricity and magnetism that is formalized in Maxwell's equations. The magnetic flux density B (measured in tesla, T) describes the strength and direction of the field, while the magnetic field intensity H (measured in A/m) describes the field produced by free currents independent of the material response; the two are related by $B = \mu H$, where $\mu = \mu_0\mu_r$ is the permeability ($\mu_0 = 4\pi \times 10^{-7}$ H/m for free space). The Biot-Savart law provides the most general method for calculating B from an arbitrary current distribution: $dB = (\mu_0/4\pi)(Idl \times \hat{r})/r^2$, integrating over the current path. For geometries with high symmetry, Ampere's law $\oint H\cdot dl = I_{enclosed}$ offers a far simpler approach, directly yielding the field around long straight conductors ($B = \mu_0 I/(2\pi r)$ with concentric circular field lines), inside solenoids ($B = \mu_0 nI$), and within toroidal coils. A fundamental distinction from electrostatics is that isolated magnetic poles (monopoles) have never been observed — magnetic field lines always form closed loops with no beginning or end, expressed mathematically as $\nabla\cdot B = 0$ (Gauss's Law for magnetism). The force on a charge q moving with velocity v through a magnetic field is the Lorentz force $F = qv \times B$, which is always perpendicular to the velocity and therefore changes the direction of motion without doing work — this is the operating principle behind cyclotrons, mass spectrometers, and Hall-effect sensors. The magnetic flux $\Phi = \int B\cdot dA$ through a surface, measured in webers (Wb), is the quantity that links magnetic fields to electromagnetic induction through Faraday's law, connecting this section to the wave phenomena developed later in the chapter.

**Example 9.2.1:** A long, straight conductor carries a current of 15 A. Find the magnetic field strength at a distance of 5 cm from the conductor.
**Solution:**
$B = \mu_0 I / (2\pi r) = (4\pi \times 10^{-7} \times 15) / (2\pi \times 0.05) = (4\pi \times 10^{-7} \times 15) / (0.1\pi) = (60\pi \times 10^{-7}) / (0.1\pi) = 60 \times 10^{-6}$ T $= 60\ \mu$T.
The field lines form concentric circles around the conductor, with the direction given by the right-hand rule (curling the fingers of the right hand in the direction of field circulation, the thumb points in the direction of current flow).

### 9.2.2 Ampère's Law

Ampère's Law relates the magnetic field circulating around a closed path to the total current passing through any surface bounded by that path: $\oint B \cdot dl = \mu_0 I_{enclosed}$. This law is the magnetic analog of Gauss's Law and is most useful for calculating magnetic fields in configurations with high symmetry, such as long straight conductors, solenoids, and toroidal coils. For a solenoid of n turns per unit length carrying current I, Ampère's Law yields a uniform interior field $B = \mu_0 nI$. Maxwell later added a displacement current term $\mu_0\varepsilon_0(\partial E/\partial t)$ to Ampere's Law to account for time-varying electric fields, completing the set of equations that predict electromagnetic wave propagation.

**Example 9.2.2:** A solenoid has 500 turns, a length of 25 cm, and carries 2 A of current. Find the magnetic field inside the solenoid.
**Solution:**
The number of turns per unit length: $n = N/l = 500 / 0.25 = 2000$ turns/m.
By Ampere's Law, the interior magnetic field is $B = \mu_0 nI = (4\pi \times 10^{-7})(2000)(2) = 4\pi \times 10^{-7} \times 4000 = 16\pi \times 10^{-4} = 5.027 \times 10^{-3}$ T $\approx 5.03$ mT.
The field is uniform and directed along the axis of the solenoid.

### 9.2.3 Magnetic Force

A charge q moving with velocity v in a magnetic field B experiences the Lorentz force: $F = qv \times B$, which is perpendicular to both the velocity and the field, causing the charge to follow a curved path rather than accelerate along the field direction. For a current-carrying conductor of length L in a uniform field, the force is $F = IL \times B$, which is the operating principle behind electric motors, loudspeakers, and galvanometers. Two parallel conductors carrying currents in the same direction attract each other, while currents in opposite directions repel — this force between current-carrying conductors is the basis for the SI definition of the ampere. The torque on a current loop (magnetic dipole) in a uniform field is $\tau = m \times B$, where $m = NIA$ is the magnetic moment (N turns, current I, area A).

**Example 9.2.3:** A straight conductor 0.5 m long carries a current of 10 A and is placed perpendicular to a uniform magnetic field of B = 0.3 T. Find the force on the conductor.
**Solution:**
Since the conductor is perpendicular to the field ($\theta = 90°$), the force is $F = BIL \sin(\theta) = 0.3 \times 10 \times 0.5 \times \sin(90°) = 0.3 \times 10 \times 0.5 \times 1 = 1.5$ N.
The force direction is perpendicular to both the current and the magnetic field, determined by the

right-hand rule (or equivalently, $F = IL \times B$).

### 9.2.4 Inductance

Inductance is the property of a circuit element that opposes changes in current by storing energy in a magnetic field. Self-inductance L relates the voltage induced across a coil to the rate of change of current through it: $V = -L(dI/dt)$, measured in henrys (H). For a solenoid of N turns, length l, cross-sectional area A, and core permeability μ, the inductance is $L = \mu N^2 A/l$. Mutual inductance M quantifies the coupling between two coils, where a changing current in one coil induces a voltage in the other: $V_2 = -M(dI_1/dt)$. The energy stored in an inductor is $W = ½LI^2$, and inductors in series add directly ($L_{total} = L_1 + L_2$ when no mutual coupling exists) while inductors in parallel combine as reciprocals.

**Example 9.2.4:** A solenoid has N = 300 turns, length l = 20 cm, cross-sectional area A = 5 $cm^2$, and an air core ($\mu = \mu_0$). Find the inductance and the energy stored when the current is 2 A.
**Solution:**
$L = \mu_0 N^2 A/l = (4\pi \times 10^{-7})(300^2)(5 \times 10^{-4}) / 0.20 = (4\pi \times 10^{-7})(90{,}000)(5 \times 10^{-4}) / 0.20 = (4\pi \times 10^{-7} \times 45) / 0.20 = (56.55 \times 10^{-6}) / 0.20 = 2.827 \times 10^{-4}$ H ≈ 283 μH.
Energy stored: $W = ½LI^2 = 0.5 \times 2.827 \times 10^{-4} \times (2)^2 = 5.654 \times 10^{-4}$ J ≈ 565 μJ.

### 9.2.5 Magnetic Materials and Hysteresis

Magnetic materials are classified by how they respond to an applied magnetic field H, characterized by the relationship $B = \mu H = \mu_0 \mu_r H$, where $\mu_r$ is the relative permeability. **Diamagnetic** materials (copper, silver, bismuth) have $\mu_r$ slightly less than 1 and weakly oppose the applied field. **Paramagnetic** materials (aluminum, platinum) have $\mu_r$ slightly greater than 1 and weakly align with the field. **Ferromagnetic** materials (iron, nickel, cobalt, and their alloys) have $\mu_r$ ranging from hundreds to hundreds of thousands and exhibit strong magnetization, making them essential for transformer cores, inductors, and permanent magnets. Ferromagnetic materials display a nonlinear B-H relationship and **hysteresis** — when the applied field is cycled, the B-H curve forms a loop because the material retains some magnetization (remanence $B_r$) even after H returns to zero. The coercivity $H_c$ is the reverse field needed to demagnetize the material. Soft magnetic materials (silicon steel, ferrites) have narrow hysteresis loops (low $H_c$), minimizing core losses in transformers and inductors. Hard magnetic materials (NdFeB, SmCo, Alnico) have wide loops (high $H_c$ and $B_r$), making them ideal for permanent magnets in motors, generators, and loudspeakers.

**Example 9.2.5:** A toroidal inductor core made of silicon steel has a mean path length of 20 cm, cross-sectional area of 4 $cm^2$, relative permeability $\mu_r = 3000$, and a winding of 200 turns carrying 0.5 A. Find H, B, and the magnetic flux in the core. If the core saturates at $B_{sat} = 1.8$ T, what is the maximum current before saturation?
**Solution:**
Magnetic field intensity: $H = NI/l = 200 \times 0.5 / 0.20 = 500$ A/m.
Flux density: $B = \mu_0 \mu_r H = 4\pi \times 10^{-7} \times 3000 \times 500 = 1.885$ T.
This exceeds $B_{sat} = 1.8$ T, so in practice the core would already be in saturation, and the actual B would be limited to approximately 1.8 T with $\mu_r$ dropping sharply.

The maximum current before saturation: $B_{sat} = \mu_0\mu_r \times NI_{max}/l$, so $I_{max} = B_{sat} \times l / (\mu_0\mu_r N) = 1.8 \times 0.20 / (4\pi \times 10^{-7} \times 3000 \times 200) = 0.36 / 0.7540 = 0.477$ A.
The core saturates just below 0.5 A; a larger core area or an air gap would increase the saturation current.

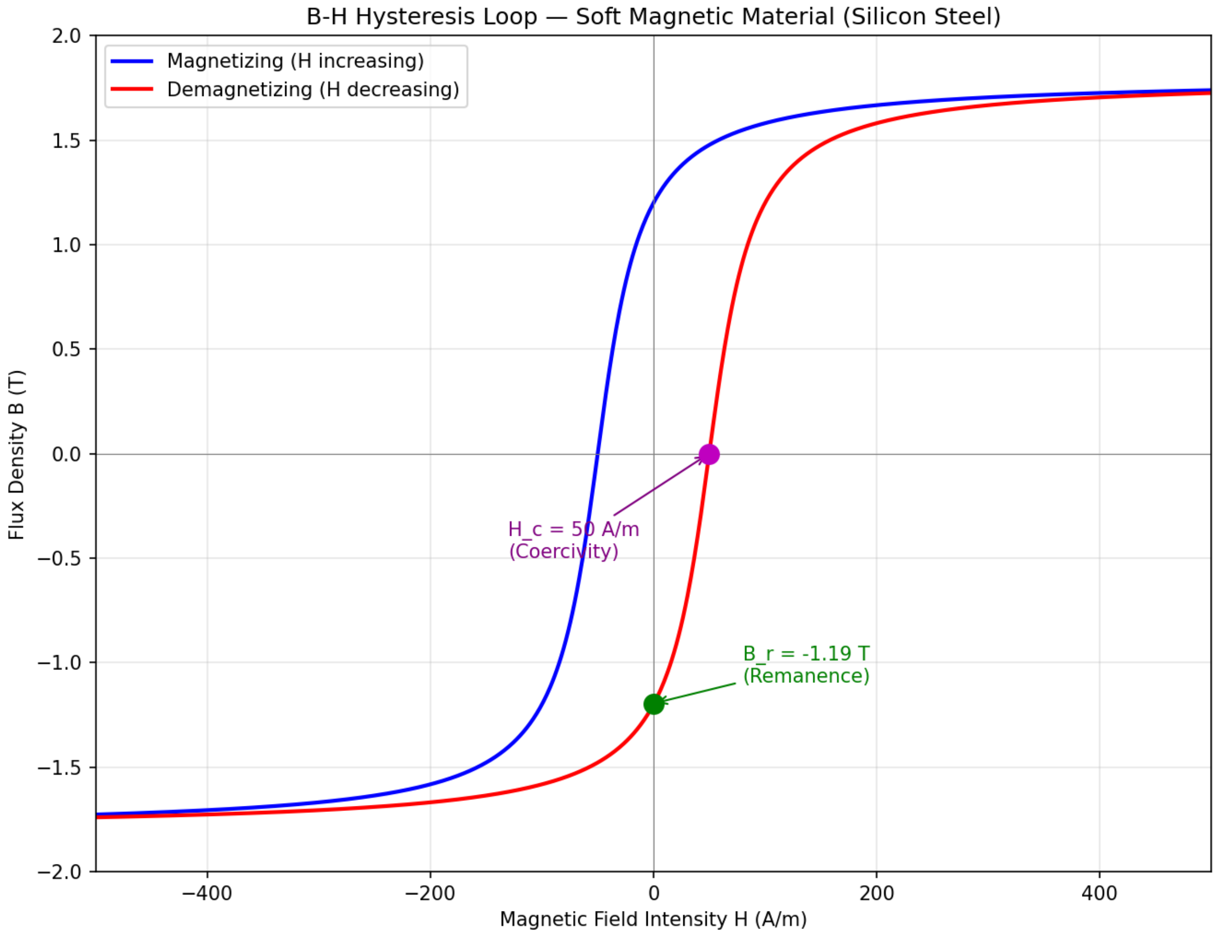



Figure 9.2.5: B-H Hysteresis Loop

### 9.2.6 Magnetic Circuits and Core Design

A magnetic circuit is an analogy to an electric circuit in which magnetic flux $\Phi$ (analogous to current) is driven through a core by magnetomotive force (MMF) $\mathcal{F} = NI$ (analogous to voltage) against reluctance $\mathcal{R} = l/(\mu A)$ (analogous to resistance), where l is the magnetic path length, $\mu$ is the permeability, and A is the cross-sectional area. Ohm's law for magnetic circuits is $\Phi = \mathcal{F}/\mathcal{R}$, and series/parallel reluctance combinations follow the same rules as resistors. **Air gaps** are intentionally introduced in inductor cores to increase the total reluctance, which serves three purposes: storing energy in the gap ($E = ½LI^2$, with most energy in the high-reluctance gap rather than the core), linearizing the inductance (the large gap reluctance dominates over the nonlinear core reluctance, making L nearly independent of current), and increasing the saturation current (since the core operates at lower flux density). The inductance of a gapped core is $L = N^2/(\mathcal{R}_{core} + \mathcal{R}_{gap}) = N^2\mu_0 A/l_g$ when the gap reluctance dominates. **Core materials** for power applications include laminated silicon steel (high $B_{sat} \approx$

1.5–1.8 T, used at 50/60 Hz in power transformers), ferrite (MnZn and NiZn, $B_{sat} \approx 0.3$–$0.5$ T, low core loss at 20 kHz–5 MHz for switching converters), powdered iron (distributed air gap, $B_{sat} \approx 1.0$–$1.4$ T, moderate loss), and amorphous/nanocrystalline alloys ($B_{sat} \approx 1.2$–$1.5$ T, very low loss, used in high-efficiency transformers). Core loss consists of hysteresis loss (proportional to frequency $\times B^n$, where $n \approx 1.6$–$2.5$) and eddy current loss (proportional to $f^2 \times B^2$), and is characterized by the Steinmetz equation $P_{core} = k \times f^\alpha \times B^\beta$ (W/m³), where k, α, and β are material-specific constants.

**Example 9.2.6:** A power inductor for a buck converter must provide $L = 100$ μH at $I_{max} = 5$ A without saturating. The chosen ferrite core (E-core, effective area $A_e = 1.27$ cm², effective path length $l_e = 7.7$ cm, $\mu_r = 2500$, $B_{sat} = 0.35$ T) requires an air gap. Calculate the number of turns, the required air gap length, and verify that the core does not saturate.
**Solution:**
First, determine the minimum inductance factor: $L = N^2\mu_0 A_e/l_g$ when gap dominates, so we need to find N and $l_g$ together.
The peak flux density constraint: $B_{max} = LI_{max}/(NA_e) < B_{sat}$.
So $N > LI_{max}/(B_{sat} \times A_e) = 100 \times 10^{-6} \times 5 / (0.35 \times 1.27 \times 10^{-4}) = 5 \times 10^{-4} / 4.445 \times 10^{-5} = 11.2$ turns — round up to N = 12.
Air gap length: $l_g = N^2\mu_0 A_e/L = 12^2 \times 4\pi \times 10^{-7} \times 1.27 \times 10^{-4} / 100 \times 10^{-6} = 144 \times 1.596 \times 10^{-10} / 10^{-4} = 2.298 \times 10^{-8} / 10^{-4} = 0.230$ mm.
Verify: $\mathcal{R}_{gap} = l_g/(\mu_0 A_e) = 2.3 \times 10^{-4} / (4\pi \times 10^{-7} \times 1.27 \times 10^{-4}) = 1.442 \times 10^6$ A-turns/Wb.
$\mathcal{R}_{core} = l_e/(\mu_0\mu_r A_e) = 0.077 / (4\pi \times 10^{-7} \times 2500 \times 1.27 \times 10^{-4}) = 1.935 \times 10^5$ A-turns/Wb.
Gap reluctance is 7.5× the core reluctance, confirming the gap dominates.
$B_{max} = LI_{max}/(NA_e) = 100 \times 10^{-6} \times 5 / (12 \times 1.27 \times 10^{-4}) = 0.328$ T $< 0.35$ T — the core operates at 94% of saturation, providing adequate margin.

### 9.2.7 Eddy Currents and Induction Heating

When a time-varying magnetic field penetrates an electrically conductive material, it induces circulating currents within the conductor called **eddy currents**, as described by Faraday's law. These currents flow in closed loops perpendicular to the magnetic flux and produce $I^2R$ losses (Joule heating) that dissipate energy within the material. Eddy current losses in transformer and motor cores are proportional to $f^2B^2t^2/\rho$ (where f is frequency, B is peak flux density, t is lamination thickness, and ρ is resistivity), which is why magnetic cores are built from thin laminations (0.35 mm for 60 Hz power transformers, 0.1 mm for 400 Hz aerospace, 25 μm for amorphous metal at 10+ kHz) insulated from each other to break the eddy current paths, or from high-resistivity ferrite materials for frequencies above ~100 kHz. **Induction heating** deliberately exploits eddy current losses: a work coil carrying high-frequency AC (typically 10 kHz–3 MHz) generates a concentrated alternating magnetic field that induces intense eddy currents in a conductive workpiece, heating it from within. The heating is concentrated near the surface to a depth of approximately one skin depth $\delta = \sqrt{(2\rho/(\omega\mu))}$, making induction heating ideal for surface hardening of steel (at 100–500 kHz, $\delta \approx 0.1$–$0.5$ mm), through-heating of billets for forging (at 1–10 kHz, $\delta \approx 2$–$10$ mm), and cooking (induction cooktops at 20–75 kHz). Induction heating is highly efficient (80–90% of input power delivered to the workpiece) because energy is generated directly in the material without thermal contact resistance. **Eddy current non-destructive testing (NDT)** uses a probe coil to induce eddy currents in a conductive part and detects changes in the coil impedance caused by surface cracks, subsurface voids, conductivity variations, or coating thickness differences — the eddy currents are disrupted by defects, altering the reflected impedance measurably.

**Example 9.2.7:** An induction cooktop operates at f = 25 kHz and heats a stainless steel pan ($\rho = 7.2 \times 10^{-7}$ Ω·m, $\mu_r = 600$, density = 7,900 kg/m³, specific heat = 500 J/(kg·°C)). The pan bottom has diameter 200 mm and thickness 3 mm. Calculate the skin depth, the fraction of pan thickness carrying significant current, and the time to raise the pan temperature by 100°C if the cooktop delivers 1,800 W to the pan.
**Solution:**
Skin depth: $\delta = \sqrt{(2\rho/(\omega\mu))} = \sqrt{(2 \times 7.2 \times 10^{-7} / (2\pi \times 25000 \times 600 \times 4\pi \times 10^{-7}))} = \sqrt{(1.44 \times 10^{-6} / (2\pi \times 25000 \times 7.540 \times 10^{-4}))} = \sqrt{(1.44 \times 10^{-6} / 118.4)} = \sqrt{(1.216 \times 10^{-8})} = 0.110$ mm.
The pan thickness is 3 mm = 27.2δ, so the eddy currents are concentrated in the bottom ~0.3 mm (3δ, where approximately 95% of the induced current flows) — the surface heats first, then conducts heat through the pan thickness.
Pan mass: $m = \rho_{density} \times V = 7900 \times \pi(0.1)^2 \times 0.003 = 7900 \times 9.42 \times 10^{-5} = 0.744$ kg.
Energy required: $Q = mc \times \Delta T = 0.744 \times 500 \times 100 = 37{,}200$ J.
Time: $t = Q/P = 37{,}200/1{,}800 = 20.7$ seconds.
The ferromagnetic nature of stainless steel ($\mu_r = 600$) is critical — aluminum ($\mu_r = 1$) would have $\delta \approx 0.52$ mm at 25 kHz (about 5× larger than the stainless steel pan), resulting in less concentrated heating and lower coupling efficiency, which is why standard induction cooktops require ferromagnetic cookware.

## 9.3 Maxwell's Equations

Maxwell's equations are the four fundamental laws governing all classical electromagnetic phenomena, unifying electricity, magnetism, and light into a single theoretical framework. In differential form, they describe how charge densities and current densities create and sustain electric and magnetic fields at every point in space and time. Together with the Lorentz force law and the constitutive relations (relating D to E and B to H in materials), Maxwell's equations provide a complete description of electromagnetic behavior from DC circuits to optical frequencies.

### 9.3.1 Gauss's Law for Electricity

Gauss's Law for electricity states that the electric flux through any closed surface is proportional to the total charge enclosed: $\oint E \cdot dA = Q_{enclosed}/\varepsilon_0$, or in differential form, $\nabla \cdot E = \rho/\varepsilon_0$, where ρ is the volume charge density. This equation implies that electric field lines originate and terminate on electric charges, and that a net outward flux through a closed surface indicates a net positive charge inside. In the absence of free charges, the divergence of E is zero. Gauss's Law is one of the four Maxwell's equations and is the fundamental statement of how electric charges produce electric fields.

**Example 9.3.1:** A hollow conducting sphere of radius 10 cm carries a total charge of Q = +8 nC. Find the electric field at r = 5 cm (inside) and r = 20 cm (outside).
**Solution:**
By Gauss's Law with a spherical Gaussian surface:
Inside (r = 5 cm): The enclosed charge is zero (all charge resides on the outer surface of a conductor), so E = 0.
Outside (r = 20 cm): $E \times 4\pi r^2 = Q/\varepsilon_0$, so $E = Q / (4\pi\varepsilon_0 r^2) = (8.99 \times 10^9)(8 \times 10^{-9}) / (0.20)^2 = 71.92 / 0.04 = 1{,}798$ V/m directed radially outward.

### 9.3.2 Gauss's Law for Magnetism

Gauss's Law for magnetism states that the magnetic flux through any closed surface is always zero: $\oint B \cdot dA = 0$, or in differential form, $\nabla \cdot B = 0$. This equation reflects the experimental observation that magnetic monopoles do not exist — every magnetic field line that enters a closed surface must also exit it. Magnetic field lines always form continuous closed loops, whether produced by permanent magnets or by currents. This is in contrast to electric field lines, which begin on positive charges and end on negative charges.

**Example 9.3.2:** A bar magnet is placed inside a closed cubical box with sides of 20 cm. The magnetic flux entering through three faces is $\Phi_1 = 0.5$ mWb, $\Phi_2 = 0.3$ mWb, and $\Phi_3 = 0.8$ mWb. If the flux through two other faces is $\Phi_4 = -0.4$ mWb and $\Phi_5 = -0.6$ mWb (outgoing), find the flux through the sixth face.
**Solution:**
By Gauss's Law for magnetism, the total flux through any closed surface is zero: $\Phi_1 + \Phi_2 + \Phi_3 + \Phi_4 + \Phi_5 + \Phi_6 = 0$.
Defining incoming as positive: $0.5 + 0.3 + 0.8 + (-0.4) + (-0.6) + \Phi_6 = 0$, so $0.6 + \Phi_6 = 0$, giving $\Phi_6 = -0.6$ mWb (outgoing through face 6).
The net outgoing flux equals the net incoming flux, as required.

### 9.3.3 Faraday's Law of Induction

Faraday's Law states that a time-varying magnetic flux through a circuit induces an electromotive force (EMF) equal to the negative rate of change of the flux: $\text{EMF} = -d\Phi/dt$, or in differential form, $\nabla \times E = -\partial B/\partial t$. The negative sign (Lenz's Law) indicates that the induced EMF opposes the change in flux that produced it, which is a consequence of conservation of energy. Faraday's Law is the operating principle behind transformers, electric generators, and induction motors, where relative motion between conductors and magnetic fields converts between mechanical and electrical energy. It also explains how a time-varying magnetic field produces an electric field, which is essential for electromagnetic wave propagation.

**Example 9.3.3:** A circular loop of wire with radius 8 cm and 50 turns is placed in a magnetic field that decreases uniformly from 0.6 T to 0.2 T in 10 ms. Find the induced EMF.
**Solution:**
The area of the loop: $A = \pi r^2 = \pi(0.08)^2 = 0.02011$ m².
The change in flux per turn: $\Delta\Phi = \Delta B \times A = (0.6 - 0.2) \times 0.02011 = 0.4 \times 0.02011 = 8.044 \times 10^{-3}$ Wb.
The induced EMF: $|\text{EMF}| = N \times |\Delta\Phi/\Delta t| = 50 \times (8.044 \times 10^{-3}) / (10 \times 10^{-3}) = 50 \times 0.8044 = 40.22$ V.
By Lenz's Law, the induced current flows in a direction that creates a magnetic field opposing the decrease (i.e., in the same direction as the original field).

### 9.3.4 Ampère-Maxwell Law

The Ampère-Maxwell Law is the generalization of Ampère's Law that includes Maxwell's displacement current: $\oint B \cdot dl = \mu_0(I_{enclosed} + \varepsilon_0\, d\Phi_E/dt)$, or in differential form, $\nabla \times B = \mu_0 J + \mu_0\varepsilon_0(\partial E/\partial t)$. The displacement current term $\varepsilon_0(\partial E/\partial t)$ accounts for the magnetic field produced by a time-varying electric field, even in the absence of conduction current. This addition was Maxwell's key insight: it completed the symmetry between electric and magnetic fields (a changing B produces E via Faraday's Law, and a changing E produces B via the displacement current) and predicted the existence of electromagnetic waves traveling at the speed of light, $c = 1/\sqrt{(\mu_0\varepsilon_0)}$.

**Example 9.3.4:** A parallel-plate capacitor with plate area $A = 0.01\ m^2$ has an electric field increasing at a rate of $dE/dt = 5 \times 10^{12}$ V/(m·s). Find the displacement current between the plates.
**Solution:**
The displacement current is $I_d = \varepsilon_0 \times A \times (dE/dt) = (8.854 \times 10^{-12})(0.01)(5 \times 10^{12}) = 8.854 \times 10^{-12} \times 5 \times 10^{10} = 0.4427$ A $\approx 443$ mA.
This displacement current produces the same magnetic field around the capacitor as if a conduction current of 443 mA were flowing between the plates, maintaining continuity of current in the circuit.

### 9.3.5 Electromagnetic Boundary Conditions

When electromagnetic fields encounter an interface between two different media, the fields must satisfy boundary conditions derived from Maxwell's equations in integral form. These conditions govern how field components change across the boundary and are essential for solving reflection, refraction, waveguide, and shielding problems. The four boundary conditions are: (1) the tangential component of E is continuous, $\hat{n} \times (E_2 - E_1) = 0$ (from Faraday's law); (2) the tangential component of H is discontinuous by the surface current density, $\hat{n} \times (H_2 - H_1) = J_s$ (from Ampère's law); (3) the normal component of D is discontinuous by the surface charge density, $\hat{n} \cdot (D_2 - D_1) = \rho_s$ (from Gauss's law for electricity); and (4) the normal component of B is continuous, $\hat{n} \cdot (B_2 - B_1) = 0$ (from Gauss's law for magnetism). At a **perfect electric conductor (PEC)** surface, the tangential E-field is zero and the tangential H-field equals the surface current: $E_{tan} = 0$ and $H_{tan} = J_s$. These conditions explain why current flows on the surface of a conductor, why waveguide modes have specific field patterns at the walls, and why an electromagnetic wave reflecting from a PEC undergoes a 180° phase reversal of the electric field. At a **dielectric-dielectric interface** with no free charge or current ($\rho_s = 0$, $J_s = 0$), both $E_{tan}$ and $H_{tan}$ are continuous, while the normal components obey $\varepsilon_1 E_{n1} = \varepsilon_2 E_{n2}$ and $\mu_1 H_{n1} = \mu_2 H_{n2}$. These conditions, combined with the wave equation, yield Snell's law of refraction and the Fresnel reflection coefficients for oblique incidence.

**Example 9.3.5:** A 1 GHz electromagnetic wave propagates in free space ($\mu_1 = \mu_0$, $\varepsilon_1 = \varepsilon_0$) and strikes a ferrite-loaded absorber material ($\mu_2 = 5\mu_0$, $\varepsilon_2 = 12\varepsilon_0$) at normal incidence. The incident magnetic field has a tangential amplitude of $H_i = 10$ mA/m. Find the tangential H and E fields on both sides of the boundary, and the surface current density if the ferrite were replaced by a PEC.
**Solution:**
For the dielectric interface (no surface current), the tangential H is continuous: $H_{tan,1} = H_{tan,2}$.
The total tangential H on side 1 is the sum of incident and reflected fields.
The intrinsic impedances are $\eta_1 = \sqrt{(\mu_0/\varepsilon_0)} = 377\ \Omega$ and $\eta_2 = \sqrt{(5\mu_0/12\varepsilon_0)} = 377\sqrt{(5/12)} = 377 \times 0.6455 = 243.4\ \Omega$.
The reflection coefficient is $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1) = (243.4 - 377)/(243.4 + 377) = -133.6/620.4 = -0.2153$.
The tangential E at the boundary: $E_{tan} = \eta_1 H_i(1 + \Gamma) = 377 \times 0.01 \times (1 - 0.2153) = 377 \times 0.01 \times 0.7847 = 2.958$ V/m (continuous across boundary).

The tangential H at the boundary: $H_{tan} = H_i(1 - \Gamma) = 0.01 \times (1 - (-0.2153)) = 0.01 \times 1.2153 =$ 12.15 mA/m (continuous).
Verification: $E_{tan}/H_{tan} = 2.958/0.01215 = 243.5\ \Omega \approx \eta_2$, confirming the transmitted wave impedance.
If replaced by a PEC: $E_{tan} = 0$ ($\Gamma = -1$), and the surface current density is $J_s = \hat{n} \times H_{tan} = 2H_i =$ 20 mA/m (the incident and reflected H-fields add at the PEC surface).

### 9.3.6 Electromagnetic Potentials

The electromagnetic potentials — the magnetic vector potential **A** and the electric scalar potential φ — provide an alternative formulation of Maxwell's equations that simplifies the calculation of fields from known source distributions and is essential for antenna theory, radiation problems, and quantum electrodynamics. Since $\nabla \cdot B = 0$ always holds, B can be expressed as the curl of a vector potential: $B = \nabla \times A$. Substituting into Faraday's law gives $\nabla \times (E + \partial A/\partial t) = 0$, which means $E + \partial A/\partial t$ is irrotational and can be written as the gradient of a scalar: $E = -\nabla\varphi - \partial A/\partial t$. The potentials are not unique — a **gauge transformation** $A \to A + \nabla\psi$, $\varphi \to \varphi - \partial\psi/\partial t$ leaves the physical fields E and B unchanged for any scalar function ψ. This freedom is exploited by choosing a gauge that simplifies the equations.

The **Lorenz gauge** $\nabla \cdot A + \mu\varepsilon(\partial\varphi/\partial t) = 0$ decouples the equations for A and φ into two independent inhomogeneous wave equations: $\nabla^2 A - \mu\varepsilon(\partial^2 A/\partial t^2) = -\mu J$ and $\nabla^2\varphi - \mu\varepsilon(\partial^2\varphi/\partial t^2) = -\rho/\varepsilon$, where J is the current density and ρ is the charge density. These are solved by the **retarded potentials**, which account for the finite propagation speed of electromagnetic effects: $A(r, t) = (\mu/4\pi) \int J(r', t - |r - r'|/v) / |r - r'|\, dV'$, where the source current is evaluated at the retarded time $t_r = t - |r - r'|/v$ (the time at which the signal left the source to arrive at the observation point at time t). The **Coulomb gauge** $\nabla \cdot A = 0$ makes φ satisfy the instantaneous Poisson equation $\nabla^2\varphi = -\rho/\varepsilon$ (no retardation for the scalar potential), which is useful in electrostatics and circuit-level analysis but complicates the vector potential equation.

For a **magnetic dipole** (small current loop of area A carrying current I), the vector potential at far distances simplifies to $A = (\mu_0\, m \sin\theta)/(4\pi r^2)\, \hat{\varphi}$, where $m = IA$ is the magnetic moment. This leads directly to the familiar dipole radiation fields used in antenna theory. For a **Hertzian dipole** (short current element of length dl carrying current $I_0 e^{j\omega t}$), the retarded vector potential yields the complete near-field and far-field expressions, from which the radiation pattern, radiation resistance, and directivity of all wire antennas are derived.

**Example 9.3.6:** A short current element (Hertzian dipole) of length $dl = 0.05\lambda$ carries a peak current $I_0 = 2$ A at frequency $f = 300$ MHz. Using the vector potential formulation, calculate (a) the radiation resistance, (b) the radiated power, and (c) the maximum electric field magnitude at a distance of 1 km.
**Solution:**
(a) Radiation resistance of a Hertzian dipole:
$R_{rad} = 80\pi^2(dl/\lambda)^2 = 80\pi^2(0.05)^2 = 80 \times 9.8696 \times 0.0025 = \mathbf{1.974\ \Omega}$
(b) Radiated power:
$P_{rad} = ½ I_0^2 R_{rad} = 0.5 \times 4 \times 1.974 = \mathbf{3.95\ W}$
(c) Maximum electric field at r = 1 km:
The far-field electric field of a Hertzian dipole is maximum at θ = 90°:

$|E_{max}| = (\eta_0 I_0 dl) / (2\lambda r) = (377 \times 2 \times 0.05\lambda) / (2\lambda \times 1000)$
$= (377 \times 0.1) / 2000 = 37.7 / 2000 =$ **18.85 mV/m**
Verification via Poynting vector: $S_{max} = E^2/(2\eta_0) = (0.01885)^2/(2 \times 377) = 3.553 \times 10^{-4} / 754 = 4.71 \times 10^{-7}$ W/m². Total radiated power: $P = S_{max} \times 4\pi r^2 / 1.5$ (directivity D = 1.5 for a Hertzian dipole) $= 4.71 \times 10^{-7} \times 4\pi \times 10^6 / 1.5 = 3.96$ W — consistent with part (b).

## 9.4 Electromagnetic Waves

Electromagnetic waves are self-sustaining oscillations of coupled electric and magnetic fields that propagate through space at the speed of light. Predicted by Maxwell's equations and experimentally confirmed by Heinrich Hertz in 1887, electromagnetic waves span an enormous frequency range from ELF radio waves to gamma rays, enabling technologies from broadcast radio and cellular communications to radar, medical imaging, and fiber optics. Understanding wave propagation, the electromagnetic spectrum, polarization, and the interaction of waves with materials is essential for designing antennas, transmission systems, and shielding.

### 9.4.1 Wave Equation

The electromagnetic wave equation is derived from Maxwell's equations in a source-free region, yielding $\nabla^2 E = \mu_0\varepsilon_0(\partial^2 E/\partial t^2)$ for the electric field and an identical equation for the magnetic field. This second-order partial differential equation has solutions in the form of traveling waves propagating at velocity $v = 1/\sqrt{(\mu\varepsilon)}$, which in free space equals the speed of light $c \approx 3 \times 10^8$ m/s. In a plane wave, the electric and magnetic fields are perpendicular to each other and to the direction of propagation, forming a transverse electromagnetic (TEM) wave. The ratio of the electric field magnitude to the magnetic field magnitude is the intrinsic impedance of the medium: $\eta = \sqrt{(\mu/\varepsilon)}$, which equals $\eta_0 \approx 377\ \Omega$ in free space.

**Example 9.4.1:** An electromagnetic wave propagates through a non-magnetic dielectric with $\varepsilon_r = 2.25$ and $\mu_r = 1$. Find the wave velocity, the intrinsic impedance, and the wavelength at 1 GHz.
**Solution:**
Wave velocity: $v = c / \sqrt{(\varepsilon_r\mu_r)} = 3 \times 10^8 / \sqrt{2.25} = 3 \times 10^8 / 1.5 = 2 \times 10^8$ m/s.
Intrinsic impedance: $\eta = \eta_0 / \sqrt{(\varepsilon_r)} = 377 / 1.5 = 251.3\ \Omega$.
Wavelength at 1 GHz: $\lambda = v / f = 2 \times 10^8 / 10^9 = 0.2$ m = 20 cm.
Compared to free space ($\lambda_0 = 30$ cm), the wavelength is shorter by a factor of $\sqrt{\varepsilon_r} = 1.5$.

### 9.4.2 Electromagnetic Spectrum

The electromagnetic spectrum encompasses all electromagnetic waves ordered by frequency (or equivalently, wavelength), from extremely low frequency (ELF) radio waves to gamma rays. Radio waves (3 kHz to 300 GHz) are used for broadcasting, cellular communications, Wi-Fi, radar, and satellite links. Microwaves (300 MHz to 300 GHz) overlap with radio and are used in point-to-point communications, microwave ovens, and radar systems. Infrared radiation (300 GHz to 400 THz) is used in thermal imaging, fiber optic communications, and remote controls. Visible light (400-790 THz) is the narrow band detectable by the human eye. Ultraviolet, X-rays, and gamma rays occupy progressively higher frequencies and are used in medical imaging, sterilization, and scientific instrumentation. All electromagnetic waves travel at the speed of light in vacuum and obey the relationship $c = f\lambda$, where f is frequency and $\lambda$ is wavelength.

**Example 9.4.2:** A 5G cellular system operates at a frequency of 28 GHz (millimeter wave). Find the wavelength in free space and classify the region of the electromagnetic spectrum.
**Solution:**
$\lambda = c / f = (3 \times 10^8) / (28 \times 10^9) = 1.071 \times 10^{-2}$ m = 10.71 mm.
This falls at the boundary of the centimeter-wave and millimeter-wave regions of the microwave spectrum (wavelengths between 1 mm and 10 mm correspond to 30–300 GHz).
At 28 GHz, the wavelength is just above 10 mm, placing it at the boundary between centimeter-wave and millimeter-wave bands, which is why 5G mmWave communications require highly directional antennas and suffer greater atmospheric attenuation than lower-frequency bands.

### 9.4.3 Polarization

Polarization describes the orientation of the electric field vector of an electromagnetic wave as it propagates through space. In linear polarization, the electric field oscillates in a single plane (horizontal or vertical relative to a reference). In circular polarization, the electric field vector rotates at the wave frequency, tracing a circle when viewed along the direction of propagation; it is right-hand circular (RHCP) or left-hand circular (LHCP) depending on the rotation direction. Elliptical polarization is the general case where the tip of the electric field vector traces an ellipse. Polarization is critical in antenna design (an antenna must be matched to the polarization of the incoming wave for maximum reception), optical systems, radar (circular polarization helps reject rain clutter), and display technologies.

**Example 9.4.3:** A vertically polarized antenna receives a signal from a linearly polarized transmitter whose polarization is tilted 30° from vertical. If the incident wave has a power density of 2 $\mu W/m^2$, find the polarization loss factor and the effective received power density.
**Solution:**
The polarization loss factor (PLF) is $\cos^2(\theta)$, where $\theta$ is the angle between the antenna polarization and the wave polarization.
$PLF = \cos^2(30°) = (\sqrt{3}/2)^2 = 3/4 = 0.75$.
The effective received power density is $S_{eff} = PLF \times S_{incident} = 0.75 \times 2\ \mu W/m^2 = 1.5\ \mu W/m^2$.
The polarization mismatch causes a loss of $10 \times \log_{10}(0.75)$ = -1.25 dB, meaning 25% of the available power is lost due to the polarization misalignment.

### 9.4.4 Skin Effect

When an electromagnetic wave impinges on a conductor (or when alternating current flows through a conductor), the fields and current density are confined to a thin layer near the surface, a phenomenon known as the skin effect. The skin depth $\delta = \sqrt{(2/(\omega\mu\sigma))} = 1/\sqrt{(\pi f\mu\sigma)}$ is the distance at which the field amplitude decays to 1/e (36.8%) of its surface value, where f is the frequency, $\mu$ is the permeability, and $\sigma$ is the conductivity of the material. At 60 Hz in copper ($\sigma = 5.8 \times 10^7$ S/m), the skin depth is approximately 8.5 mm, so current distributes relatively uniformly through typical wire cross-sections. At 1 MHz, $\delta \approx 0.066$ mm (66 $\mu$m), and at 1 GHz, $\delta \approx 2.1$ $\mu$m — meaning current flows only in a microscopically thin surface layer. The skin effect increases the effective AC resistance of conductors ($R_{AC}$

$> R_{DC}$) because the current is confined to a smaller cross-sectional area. This is why high-frequency conductors use stranded wire (Litz wire), hollow tubes, or surface plating (silver or gold over copper) to minimize losses. In electromagnetic shielding, a conductor thickness of 5δ provides approximately 99.3% attenuation (about 43 dB) of an incident wave.

**Example 9.4.4:** Calculate the skin depth in copper ($\sigma = 5.8 \times 10^7$ S/m, $\mu_r = 1$) at frequencies of 60 Hz, 1 MHz, and 10 GHz. For a 1 mm diameter copper wire at 1 MHz, estimate the ratio of AC to DC resistance.
**Solution:**
Skin depth formula: $\delta = 1/\sqrt{(\pi f \mu_0 \sigma)} = 1/\sqrt{(\pi \times f \times 4\pi \times 10^{-7} \times 5.8 \times 10^7)}$.
At 60 Hz: $\delta = 1/\sqrt{(\pi \times 60 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7)} = 1/\sqrt{(\pi \times 60 \times 72.88)} = 1/\sqrt{(13{,}738)} = 8.53$ mm.
At 1 MHz: $\delta = 1/\sqrt{(\pi \times 10^6 \times 72.88)} = 1/\sqrt{(2.289 \times 10^8)} = 66.1$ μm.
At 10 GHz: $\delta = 1/\sqrt{(\pi \times 10^{10} \times 72.88)} = 1/\sqrt{(2.289 \times 10^{12})} = 0.661$ μm.
For a 1 mm diameter wire (radius a = 0.5 mm) at 1 MHz where δ = 66.1 μm « a, the current flows in an annular ring of thickness δ.
$R_{AC}/R_{DC} \approx a/(2\delta) = 500/132.2 = 3.78$.
The AC resistance is nearly 4× the DC resistance due to the skin effect.

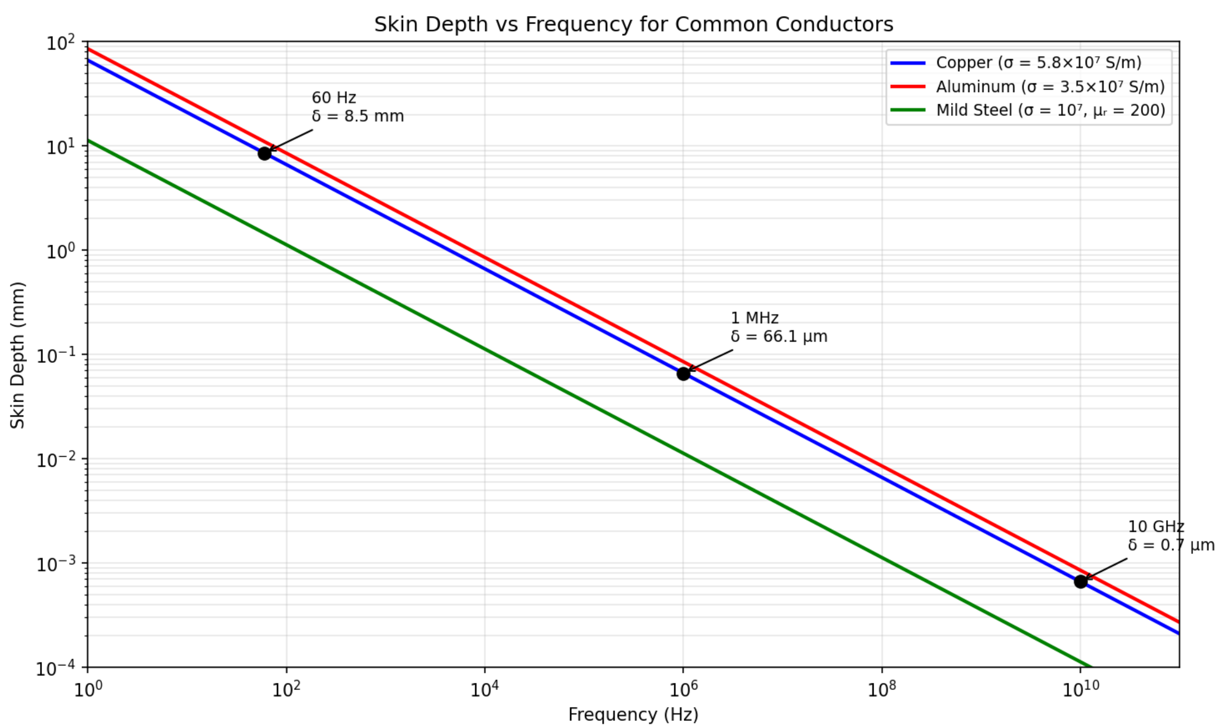



Figure 9.4.4: Skin Depth vs Frequency

### 9.4.5 Wave Reflection and Transmission at Interfaces

When an electromagnetic wave traveling in one medium encounters the boundary with a second medium of different electromagnetic properties, part of the wave energy is reflected and part is transmitted. At **normal incidence** on the boundary between two lossless dielectric media with intrinsic impedances $\eta_1$ and $\eta_2$, the reflection coefficient is $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ and the transmission coefficient is $\tau = 2\eta_2/(\eta_1 + \eta_2)$. The fraction of incident power reflected is $|\Gamma|^2$ and the fraction transmitted is $1 - |\Gamma|^2 = 4\eta_1\eta_2/(\eta_1 + \eta_2)^2$. For a wave in air ($\eta_1$ = 377 Ω) striking glass ($\eta_2 = 377/\sqrt{\varepsilon_r}$), a significant fraction is reflected — this is the physical basis for anti-reflection coatings (§18.2.1). For oblique incidence, the Fresnel equations describe the reflection and transmission coefficients as functions of angle and polarization: the reflection coefficient for parallel (TM) polarization vanishes at Brewster's angle $\theta_B = \arctan(n_2/n_1)$, while perpendicular (TE) polarization has no such zero. At a perfect conductor ($\eta_2 = 0$), the reflection coefficient is $\Gamma = -1$ (total reflection with 180° phase reversal), and the boundary condition requires the tangential electric field to be zero at the surface — the superposition of incident and reflected waves forms a standing wave with the first E-field node at the surface. This standing wave pattern is exploited in cavity resonators, waveguide terminations, and microwave heating (standing wave ovens).

**Example 9.4.5:** A 10 GHz plane wave in free space ($\eta_1$ = 377 Ω) is normally incident on a semi-infinite slab of lossless dielectric with $\varepsilon_r = 9$ and $\mu_r = 1$. Calculate the intrinsic impedance of the dielectric, the reflection and transmission coefficients, and the percentage of power reflected and transmitted.

**Solution:**

Intrinsic impedance of the dielectric: $\eta_2 = \eta_0/\sqrt{\varepsilon_r} = 377/\sqrt{9} = 377/3 = 125.7\ \Omega$.

Reflection coefficient: $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1) = (125.7 - 377)/(125.7 + 377) = -251.3/502.7 = -0.50$.

Transmission coefficient: $\tau = 2\eta_2/(\eta_1 + \eta_2) = 2 \times 125.7/502.7 = 251.4/502.7 = 0.50$.

Power reflected: $|\Gamma|^2 = 0.25 = 25\%$.

Power transmitted: $1 - |\Gamma|^2 = 0.75 = 75\%$.

Verification: the transmitted power can also be calculated as $(\eta_1/\eta_2)|\tau|^2 = (377/125.7) \times 0.25 = 3.0 \times 0.25 = 0.75 = 75\%$.

The negative sign of Γ indicates a 180° phase reversal of the reflected E-field, which occurs because the wave enters a denser medium (higher ε, lower η).

### 9.4.6 Poynting Vector and Electromagnetic Power Flow

The Poynting vector $S = E \times H$ describes the instantaneous power flow per unit area (W/m²) of an electromagnetic wave, with its direction indicating the direction of energy propagation. For a plane wave in a lossless medium, the electric and magnetic fields are perpendicular to each other and to the propagation direction, and the time-averaged Poynting vector is $S_{avg} = ½\mathrm{Re}\{E \times H^*\} = |E_0|^2/(2\eta) = ½\eta|H_0|^2$, where η is the intrinsic impedance of the medium. The total power radiated by an antenna or passing through a surface is obtained by integrating the Poynting vector over that surface: $P = \oint S \cdot dA$. For a point source radiating isotropically with total power $P_t$, the power density at distance r is $S = P_t/(4\pi r^2)$, which defines the inverse-square law for power decay with distance. The **radiation intensity** $U(\theta, \varphi) = r^2 S(r, \theta, \varphi)$ is the power per unit solid angle (W/sr), independent of distance, and provides a convenient measure of the angular distribution of radiated power. The Poynting vector also applies to guided waves: in a coaxial transmission line, the power flows in the dielectric between the conductors (not in the conductors themselves), with S directed axially and concentrated near the inner conductor where E and H are strongest. In a waveguide, the power flow pattern follows the mode structure, and the total transmitted power is obtained by integrating S over the guide cross-section.

**Example 9.4.6:** A 10 GHz radar transmitter radiates 1 kW through an antenna with a gain of 30 dBi. Calculate the peak power density at a distance of 5 km, the electric field strength at that distance, and compare to the ICNIRP occupational exposure limit of 50 W/m² for frequencies above 2 GHz.
**Solution:**
Antenna gain in linear: $G = 10^{30/10} = 1000$.
Peak power density: $S = P_tG/(4\pi r^2) = 1000 \times 1000 / (4\pi \times (5000)^2) = 10^6 / (4\pi \times 2.5 \times 10^7) = 10^6 / 3.1416 \times 10^8 = 3.18 \times 10^{-3}$ W/m² = 3.18 mW/m².
Electric field: $S = E^2/(2\eta_0)$, so $E = \sqrt{(2\eta_0 S)} = \sqrt{(2 \times 377 \times 3.18 \times 10^{-3})} = \sqrt{(2.396)} = 1.55$ V/m.
The power density of 3.18 mW/m² is well below the 50 W/m² ICNIRP limit — by a factor of 15,700 (42 dB).
The minimum safe distance where S = 50 W/m²: $r = \sqrt{(P_tG/(4\pi S_{limit}))} = \sqrt{(10^6/(4\pi \times 50))} = \sqrt{(1592)} = 39.9$ m.
Personnel should not stand within 40 m of the antenna main beam during transmission.

### 9.4.7 Wave Propagation in Lossy Media

When an electromagnetic wave propagates through a lossy medium — one with finite conductivity $\sigma$ or dielectric loss tangent tan $\delta$ — the wave is attenuated exponentially as it travels, described by the complex propagation constant $\gamma = \alpha + j\beta$, where $\alpha$ is the attenuation constant (Np/m) and $\beta$ is the phase constant (rad/m). In a general medium with complex permittivity $\varepsilon_c = \varepsilon' - j\varepsilon'' = \varepsilon'(1 - j \tan \delta)$, where $\varepsilon'' = \sigma/\omega + \varepsilon' \tan \delta$ accounts for both conduction losses and dielectric polarization losses, the propagation constant is $\gamma = j\omega\sqrt{(\mu\varepsilon_c)}$. For a **good conductor** ($\sigma \gg \omega\varepsilon$, i.e., conduction current dominates displacement current), $\alpha \approx \beta \approx \sqrt{(\pi f\mu\sigma)} = 1/\delta$, and the wave attenuates by 1/e (8.686 dB) per skin depth — this is the basis of electromagnetic shielding and the skin effect. For a **low-loss dielectric** ($\sigma \ll \omega\varepsilon$, $\tan \delta \ll 1$), $\alpha \approx (\omega/2)\sqrt{(\mu\varepsilon)} \times \tan \delta = (\pi f \tan \delta \sqrt{(\varepsilon_r)})/c$, and the wave propagates with modest attenuation that increases linearly with frequency and loss tangent — critical for selecting PCB substrate materials, radome designs, and fiber optic cladding. The **loss tangent** $\tan \delta = \varepsilon''/\varepsilon'$ is the figure of merit for dielectric loss: FR-4 PCB material has $\tan \delta \approx 0.02$ (acceptable below ~1 GHz), Rogers RO4003C has $\tan \delta \approx 0.0027$ (suitable to 10+ GHz), and fused silica has $\tan \delta \approx 0.0001$ (used in precision RF windows). In biological tissue, which has both significant conductivity and dielectric loss, electromagnetic wave penetration depth varies dramatically with frequency: at 900 MHz (cellular), the penetration depth in muscle tissue is ~4 cm, dropping to ~2 cm at 2.4 GHz (Wi-Fi) and ~0.5 mm at 60 GHz (5G mmWave), which is why millimeter-wave radiation is absorbed almost entirely by the skin surface. The specific absorption rate (SAR) quantifies the rate of energy absorption in tissue: $SAR = \sigma|E|^2/(2\rho)$ (W/kg), regulated to $\leq 1.6$ W/kg (FCC) or $\leq 2.0$ W/kg (ICNIRP) averaged over 1 g or 10 g of tissue, respectively.

**Example 9.4.7:** A 5 GHz Wi-Fi signal propagates through a concrete wall ($\varepsilon_r = 6.0$, $\tan \delta = 0.04$, $\mu_r = 1$) with thickness 200 mm. Calculate the attenuation constant, the total attenuation through the wall (in dB), and compare to a dry plywood wall ($\varepsilon_r = 2.0$, $\tan \delta = 0.01$, thickness 12 mm).
**Solution:**
For the low-loss dielectric approximation ($\tan \delta \ll 1$): $\alpha = (\pi f \tan \delta \sqrt{\varepsilon_r}) / c = (\pi \times 5 \times 10^9 \times 0.04 \times \sqrt{6.0}) / (3 \times 10^8) = (\pi \times 5 \times 10^9 \times 0.04 \times 2.449) / (3 \times 10^8) = (1.539 \times 10^9) / (3 \times 10^8) = 5.13$ Np/m.

Attenuation through 200 mm: $A = \alpha \times d = 5.13 \times 0.2 = 1.026$ Np $= 1.026 \times 8.686 = 8.91$ dB.
Adding reflection loss at both air-concrete interfaces: $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1) = (377/\sqrt{6} - 377)/(377/\sqrt{6} + 377) = (153.9 - 377)/(153.9 + 377) = -0.420$, so reflection loss per surface $= -10\log_{10}(1 - 0.420^2) = -10\log_{10}(0.824) = 0.84$ dB, total reflection $= 1.68$ dB.
Total wall loss $\approx 8.91 + 1.68 = 10.6$ dB.
For the plywood wall: $\alpha = (\pi \times 5 \times 10^9 \times 0.01 \times \sqrt{2.0}) / (3 \times 10^8) = (2.22 \times 10^8) / (3 \times 10^8) = 0.74$ Np/m.
Attenuation: $0.74 \times 0.012 = 0.0089$ Np $= 0.077$ dB.
With minimal reflection loss (~0.3 dB total), the plywood wall attenuates by only ~0.4 dB — essentially transparent.
This explains why concrete walls severely degrade Wi-Fi coverage while wood-frame construction does not.

## 9.5 Transmission Lines

Transmission lines are structures designed to guide electromagnetic energy from a source to a load with controlled impedance and minimal loss. Unlike ordinary wires at DC or low frequencies, transmission lines at RF and microwave frequencies exhibit wave behavior — signals propagate as voltage and current waves with finite velocity, and impedance mismatches cause reflections that can degrade performance or damage equipment. Understanding characteristic impedance, reflection, standing waves, and impedance matching is essential for RF circuit design, antenna feed systems, high-speed digital interconnects, and power delivery networks.

### 9.5.1 Characteristic Impedance

The characteristic impedance $Z_0$ of a transmission line is the ratio of voltage to current for a wave traveling in one direction along the line, determined by the line's distributed inductance and capacitance per unit length: $Z_0 = \sqrt{(L'/C')}$, where L' and C' are the inductance and capacitance per unit length. For a coaxial cable, $Z_0 = (1/2\pi)\sqrt{(\mu/\varepsilon)} \cdot \ln(b/a)$, where a and b are the inner and outer conductor radii. Common transmission line impedances include 50 Ω (RF and test equipment), 75 Ω (video and cable TV), and 100 Ω (differential data lines). When a transmission line is terminated in a load equal to $Z_0$, all power is delivered to the load with no reflections – the matched condition.

**Example 9.5.1:** A coaxial cable has inner conductor radius a = 0.5 mm, outer conductor radius b = 2.3 mm, and a dielectric with $\varepsilon_r = 2.3$ and $\mu_r = 1$. Find the characteristic impedance.
**Solution:**
$Z_0 = (1/(2\pi)) \times \sqrt{(\mu/\varepsilon)} \times \ln(b/a) = (1/(2\pi)) \times (\eta_0/\sqrt{\varepsilon_r}) \times \ln(b/a) = (1/(2\pi)) \times (377/\sqrt{2.3}) \times \ln(2.3/0.5) = (1/(2\pi)) \times (377/1.517) \times \ln(4.6) = (1/6.2832) \times 248.5 \times 1.526 = (1/6.2832) \times 379.2 = 60.35\ \Omega$.
This is close to the standard 50 Ω coaxial impedance; adjusting the ratio b/a or the dielectric constant would achieve exactly 50 Ω.

### 9.5.2 Reflection and Standing Waves

When a transmission line is terminated in a load impedance $Z_L$ that differs from the characteristic impedance $Z_0$, a portion of the incident wave is reflected back toward the source. The reflection coefficient $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$ quantifies the fraction of the incident voltage wave that is reflected, ranging from -1 (short circuit) through 0 (matched load) to +1 (open circuit). The interaction of the forward and reflected waves creates a standing wave pattern with fixed positions of voltage maxima and minima along the line. The Voltage Standing Wave Ratio (VSWR) = $(1 + |\Gamma|)/(1 - |\Gamma|)$ is a common measure of impedance mismatch, with VSWR = 1 indicating a perfect match. Minimizing reflections is essential in RF systems to maximize power transfer and prevent damage to transmitters.

**Example 9.5.2:** A 50 Ω transmission line is terminated with a 75 Ω load. Find the reflection coefficient, VSWR, and the fraction of incident power reflected.
**Solution:**
Reflection coefficient: $\Gamma = (Z_L - Z_0) / (Z_L + Z_0) = (75 - 50) / (75 + 50) = 25/125 = 0.2$.
VSWR = $(1 + |\Gamma|) / (1 - |\Gamma|) = (1 + 0.2) / (1 - 0.2) = 1.2 / 0.8 = 1.5{:}1$.
Fraction of power reflected: $|\Gamma|^2 = (0.2)^2 = 0.04 = 4\%$.
Therefore 96% of the incident power is delivered to the load.
The return loss is $-20 \times \log_{10}(|\Gamma|) = -20 \times \log_{10}(0.2) = 13.98$ dB.

### 9.5.3 Smith Chart

The Smith Chart is a graphical tool that maps the complex impedance plane onto a polar plot, enabling visual analysis of transmission line impedance matching, reflection coefficients, and VSWR. The chart plots normalized impedance $z = Z/Z_0$ using circles of constant resistance and arcs of constant reactance, with the center representing a perfect match ($z = 1 + j0$). Moving along the transmission line toward the generator corresponds to clockwise rotation on the Smith Chart, with one full revolution representing a half-wavelength of line. The Smith Chart is used to design impedance matching networks (using series/shunt reactive elements or transmission line stubs), determine the input impedance of a terminated line, and convert between impedance and admittance representations. Despite the availability of computational tools, the Smith Chart remains widely taught and used because it provides intuitive visualization of impedance transformation and matching.

**Example 9.5.3:** An antenna with input impedance $Z_L = 25 - j30$ Ω is connected to a 50 Ω transmission line. Find the normalized impedance and locate it on the Smith Chart.
**Solution:**
Normalized impedance: $z = Z_L / Z_0 = (25 - j30) / 50 = 0.5 - j0.6$.
On the Smith Chart, this point lies at the intersection of the r = 0.5 constant-resistance circle and the x = -0.6 constant-reactance arc (in the lower half, indicating capacitive reactance).
The reflection coefficient magnitude is $|\Gamma| = |z - 1| / |z + 1| = |(0.5 - j0.6) - 1| / |(0.5 - j0.6) + 1| = |(-0.5 - j0.6)| / |(1.5 - j0.6)| = \sqrt{(0.25 + 0.36)} / \sqrt{(2.25 + 0.36)} = 0.7810 / 1.6155 = 0.483$.
VSWR = $(1 + 0.483) / (1 - 0.483) = 2.87{:}1$.

### 9.5.4 Waveguides

A waveguide is a hollow metallic structure (typically rectangular or circular cross-section) that guides electromagnetic waves by confining them within the conducting walls, without requiring a center conductor. Waveguides operate above a cutoff frequency $f_c$ determined by their cross-sectional dimensions; below this frequency, waves cannot propagate and are exponentially attenuated (evanescent modes). For a rectangular waveguide with interior dimensions $a \times b$ ($a > b$), the cutoff frequency of the $TE_{mn}$ mode is $f_c = (c/2)\sqrt{((m/a)^2 + (n/b)^2)}$, where m and n are mode indices. The dominant mode (lowest cutoff) is the $TE_{10}$ mode with $f_c = c/(2a)$. Waveguides are preferred over coaxial cables at microwave and millimeter-wave frequencies because they have lower loss (no center conductor, current flows on inner surfaces only), higher power handling capability, and no radiation leakage. Standard waveguide sizes are designated by WR (waveguide rectangular) numbers — for example, WR-90 (22.86 × 10.16 mm) covers the X-band (8.2–12.4 GHz). Waveguides are used in radar transmitters, satellite communication feeds, particle accelerators, and industrial microwave heating.

**Example 9.5.4:** A WR-90 rectangular waveguide has interior dimensions a = 22.86 mm and b = 10.16 mm. Calculate the cutoff frequency of the dominant $TE_{10}$ mode and determine the usable frequency range. Find the guide wavelength at 10 GHz.
**Solution:**
Cutoff frequency ($TE_{10}$): $f_c = c/(2a) = 3 \times 10^8 / (2 \times 0.02286) = 3 \times 10^8 / 0.04572 = 6.562$ GHz.
The usable range is typically $1.25f_c$ to $1.9f_c$ (avoiding near-cutoff dispersion and the next mode $TE_{20}$): 8.2 to 12.5 GHz (the X-band).
The $TE_{20}$ cutoff is $f_{c20} = c/a = 13.12$ GHz, so single-mode operation is guaranteed below 13.12 GHz.
Guide wavelength at 10 GHz: $\lambda_g = \lambda_0/\sqrt{(1 - (f_c/f)^2)} = (c/f)/\sqrt{(1 - (6.562/10)^2)} = 0.03/\sqrt{(1 - 0.4305)} = 0.03/\sqrt{0.5695} = 0.03/0.7547 = 39.75$ mm.
The guide wavelength (39.75 mm) is longer than the free-space wavelength (30 mm), a characteristic of all waveguide modes above cutoff.

### 9.5.5 Microstrip and Stripline

Microstrip and stripline are planar transmission line structures fabricated on printed circuit boards (PCBs) that guide high-frequency signals between components with controlled impedance. A **microstrip** consists of a signal trace on the top surface of a dielectric substrate with a continuous ground plane on the bottom; it is the most common PCB transmission line because the trace is accessible for component mounting. The characteristic impedance depends on the trace width w, substrate height h, and dielectric constant $\varepsilon_r$: an approximate formula for a thin microstrip is $Z_0 \approx (87 / \sqrt{(\varepsilon_r + 1.41)}) \times \ln(5.98h / (0.8w + t))$, where t is the trace thickness. Because the electric field exists partly in the dielectric and partly in the air above the trace, the effective dielectric constant $\varepsilon_{eff}$ is less than $\varepsilon_r$, typically $\varepsilon_{eff} \approx (\varepsilon_r + 1)/2 + (\varepsilon_r - 1)/(2\sqrt{(1 + 12h/w)})$. A **stripline** embeds the signal trace between two ground planes with dielectric filling the entire space, creating a homogeneous environment where $\varepsilon_{eff} = \varepsilon_r$; its impedance is $Z_0 \approx (60/\sqrt{\varepsilon_r}) \times \ln(4b / (0.67\pi(0.8w + t)))$, where b is the ground-to-ground spacing. Stripline provides better shielding and eliminates radiation but requires inner-layer routing and vias for component connections. Typical PCB substrates include FR-4 ($\varepsilon_r \approx 4.3$, $\tan\delta \approx 0.02$ — adequate to ~1 GHz), Rogers RO4003 ($\varepsilon_r \approx 3.55$, $\tan\delta \approx 0.0027$ — suitable to 10+ GHz), and PTFE-based laminates ($\varepsilon_r \approx 2.2$, $\tan\delta \approx 0.0009$ — used for millimeter-wave circuits). Controlled-impedance PCB design is essential for signal integrity in high-speed digital systems (DDR4/5, PCIe, USB 3.x), RF circuits, and any application where trace lengths approach a significant fraction of the signal wavelength.

**Example 9.5.5:** A 50 Ω microstrip line is to be fabricated on an FR-4 substrate ($\varepsilon_r = 4.3$, h = 0.254 mm, t = 0.035 mm copper). Estimate the required trace width and the effective dielectric constant. Find the propagation delay per unit length.

**Solution:**
Using the microstrip impedance formula iteratively (or an online calculator), for $Z_0 = 50\ \Omega$ on FR-4 with h = 0.254 mm: the trace width is approximately w ≈ 0.46 mm (18 mils).
The ratio w/h = 0.46/0.254 = 1.81.
Effective dielectric constant: $\varepsilon_{eff} \approx (4.3 + 1)/2 + (4.3 - 1)/(2\sqrt{(1 + 12/1.81)}) = 2.65 + 3.3/(2\sqrt{7.63}) = 2.65 + 3.3/5.524 = 2.65 + 0.597 = 3.25$.
Propagation velocity: $v = c/\sqrt{\varepsilon_{eff}} = 3 \times 10^8/\sqrt{3.25} = 3 \times 10^8/1.803 = 1.664 \times 10^8$ m/s.
Propagation delay: $t_{pd} = 1/v = 6.01$ ns/m = 6.01 ps/mm ≈ 153 ps/inch.
For a 10 cm trace, the one-way delay is 0.601 ns — significant for signals with rise times below ~2 ns (clock frequencies above ~500 MHz), where impedance-controlled routing becomes essential.

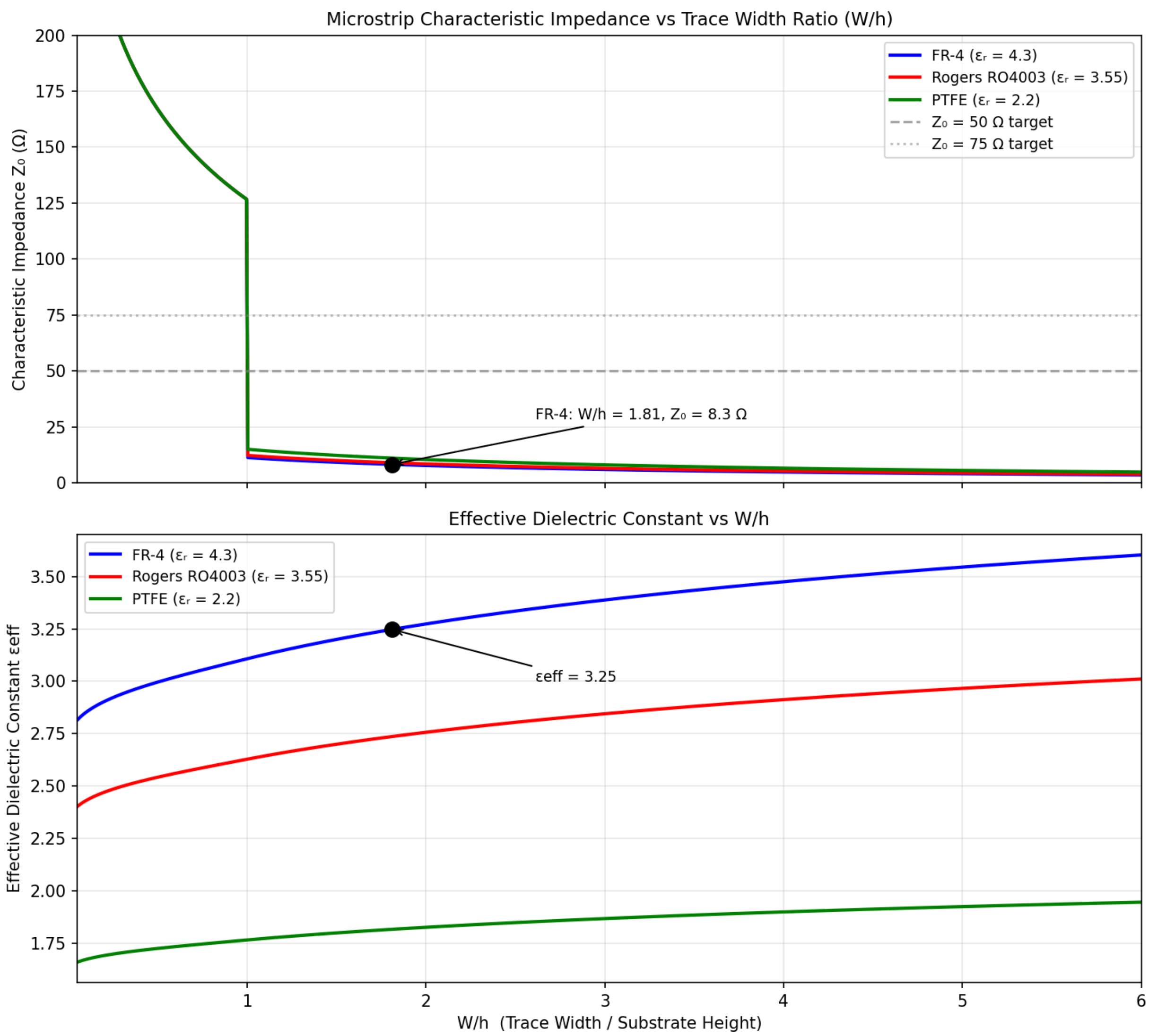



Figure 9.5.5: Microstrip characteristic impedance $Z_0$ and effective dielectric constant εeff vs trace width ratio W/h for FR-4 ($\varepsilon_r = 4.3$), Rogers RO4003 ($\varepsilon_r = 3.55$), and PTFE ($\varepsilon_r = 2.2$). The W/h = 1.81 point (FR-4, 50 Ω) corresponds to the §9.5.5 example.

### 9.5.6 Transmission Line Transients and TDR

When a step voltage is applied to a transmission line, the signal propagates as a traveling wave at velocity $v = 1/\sqrt{(L'C')} \approx c/\sqrt{\varepsilon_{eff}}$, and reflections at impedance discontinuities create a sequence of forward and backward traveling waves described by the lattice (bounce) diagram. A source with impedance $Z_S$ driving a line of impedance $Z_0$ initially launches a voltage step $V^+ = V_S \times Z_0/(Z_S + Z_0)$ onto the line. When this wave arrives at the load after one-way delay $t_d = l/v$, a reflected wave $V^- = \Gamma_L V^+$ travels back, where $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$. The reflected wave then re-reflects at the source with coefficient $\Gamma_S = (Z_S - Z_0)/(Z_S + Z_0)$, and the bouncing continues with diminishing amplitudes until the line reaches steady state. **Time-domain reflectometry** (TDR) exploits this behavior to locate faults and characterize impedance variations: a fast step (rise time < 100 ps for high-resolution instruments) is launched into the line, and the reflected waveform is displayed versus time, with each impedance discontinuity appearing as a step or bump at a time corresponding to twice the electrical distance to the fault. TDR is an essential tool for signal integrity analysis in high-speed digital design, identifying impedance mismatches, stubs, vias, connector transitions, and cable faults with millimeter-level spatial resolution. The characteristic impedance at any point is computed from the reflection coefficient: $Z(t) = Z_0 \times (1 + \Gamma(t))/(1 - \Gamma(t))$.

**Example 9.5.6:** A 1 V step from a 50 Ω source is applied to a 50 Ω coaxial cable (velocity = $2 \times 10^8$ m/s, length = 3 m) terminated in a 150 Ω load. Using a lattice diagram, find the voltage at the load at t = 15 ns (immediately after the incident wave arrives), at t = 45 ns (after the first round trip), and the final steady-state voltage.
**Solution:**
Initial wave: $V^+ = V_S \times Z_0/(Z_S + Z_0) = 1 \times 50/(50 + 50) = 0.5$ V.
One-way delay: $t_d = l/v = 3/(2 \times 10^8) = 15$ ns.
Load reflection coefficient: $\Gamma_L = (150 - 50)/(150 + 50) = 100/200 = 0.5$.
Source reflection coefficient: $\Gamma_S = (50 - 50)/(50 + 50) = 0$ (matched source).
At t = 15 ns (wave arrives at load): $V_L = V^+ + \Gamma_L V^+ = 0.5 + 0.5 \times 0.5 = 0.5 + 0.25 = 0.75$ V.
The reflected wave of 0.25 V travels back and arrives at the source at t = 30 ns. Since $\Gamma_S = 0$, no re-reflection occurs — the source absorbs the reflected wave completely.
At t = 45 ns and beyond, no additional waves arrive at the load. Final steady-state: $V_L = 0.75$ V.
Verification by voltage divider: $V_{L(DC)} = V_S \times Z_L/(Z_S + Z_L) = 1 \times 150/(50 + 150) = 0.75$ V — confirmed.
If the source were mismatched ($\Gamma_S \neq 0$), additional bounces would occur, each diminished by $\Gamma_L \times \Gamma_S$, converging geometrically to the same steady state.

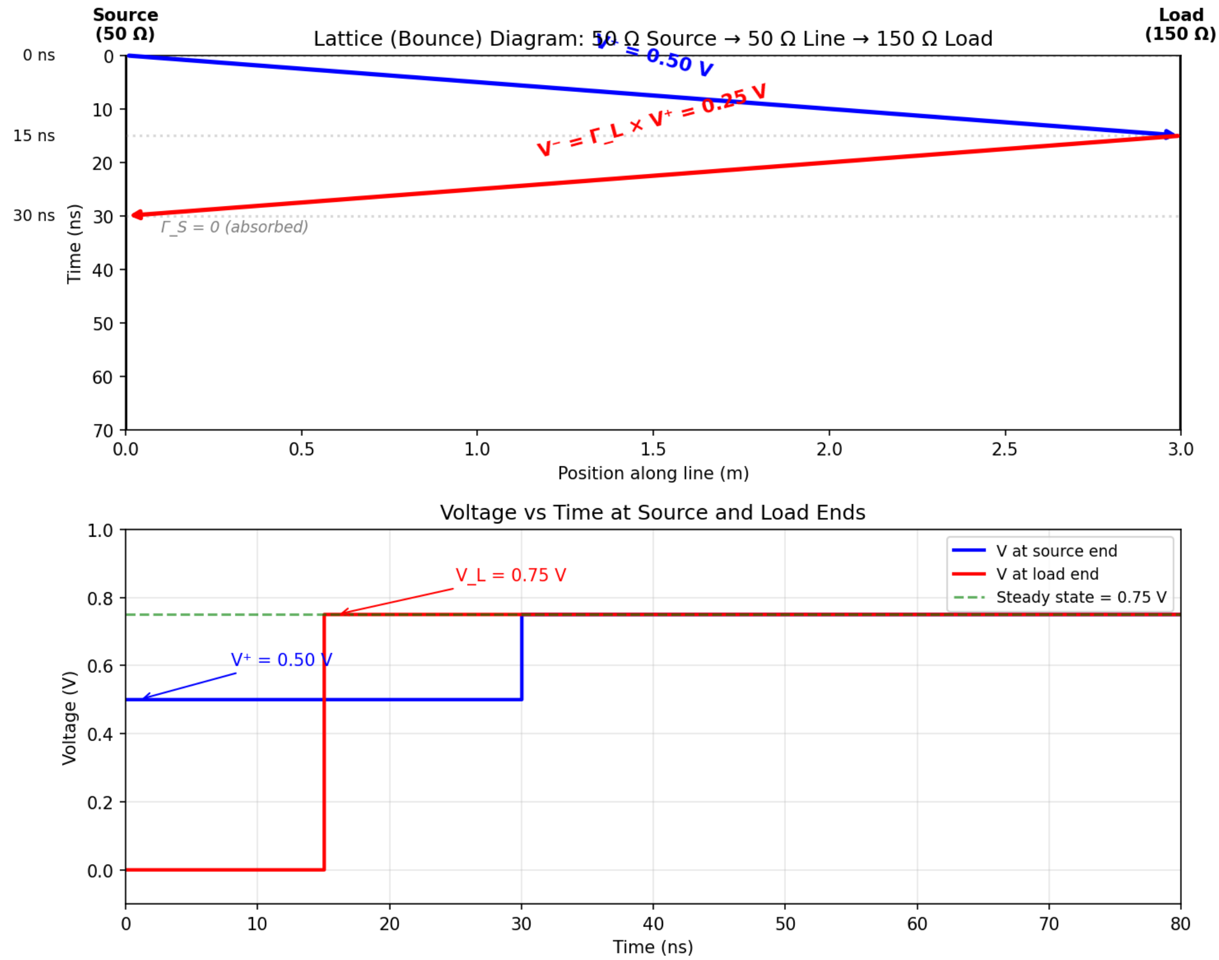



Figure 9.5.6: TDR Lattice Diagram and Voltage Waveforms

### 9.5.7 Impedance Matching Networks

Impedance matching transforms a load impedance to the source impedance (typically 50 Ω) so that maximum power is transferred and reflections are minimized. The simplest narrowband matching technique is the **L-network**, which uses two reactive elements (inductors and/or capacitors) to transform a real load $R_L$ to the desired source resistance $R_S$. For $R_L < R_S$, the shunt element is placed across the load and the series element connects to the source; for $R_L > R_S$, the topology reverses. The design equations use the quality factor $Q = \sqrt{}(R_{high}/R_{low} - 1)$, which also determines the bandwidth: $BW \approx f_0/Q$. The shunt reactance is $X_{shunt} = R_{high}/Q$ and the series reactance is $X_{series} = QR_{low}$, with the choice of inductor or capacitor for each element providing two possible L-network solutions (high-pass or low-pass form). The **quarter-wave transformer** is a transmission line section of length $\lambda/4$ with characteristic impedance $Z_T = \sqrt{}(Z_0 Z_L)$ that transforms a real load $Z_L$ to $Z_0$ at the design frequency. Its bandwidth is inherently narrow (inversely proportional to the impedance ratio), but multi-section quarter-wave transformers with tapered impedances (binomial or Chebyshev) extend the bandwidth. For broadband matching or complex loads, **Pi-networks** and **T-networks** (three-element topologies) provide an additional degree of freedom, allowing independent control of Q and transformation ratio. At microwave frequencies, matching is often performed using **open or shorted stubs** — short transmission line sections that present a purely reactive impedance at their input, positioned along the main line to cancel the load's reactive component. Single-stub matching uses one stub at a calculated distance from the load; double-stub matching uses two stubs at fixed positions, providing more flexibility but with a forbidden region of load impedances.

**Example 9.5.7:** A 915 MHz RFID reader has a 50 Ω output and must drive an antenna with input impedance $Z_L$ = 15 Ω (purely resistive). Design an L-network match and calculate the matching bandwidth.
**Solution:**
Since $R_L = 15\ \Omega < R_S = 50\ \Omega$, the shunt element goes across the load.
$Q = \sqrt{(R_{high}/R_{low} - 1)} = \sqrt{(50/15 - 1)} = \sqrt{(2.333)} = 1.528$.
Shunt reactance: $X_{shunt} = R_{high}/Q = 50/1.528 = 32.72\ \Omega$.
Series reactance: $X_{series} = Q \times R_{low} = 1.528 \times 15 = 22.92\ \Omega$.
**Low-pass solution** (shunt inductor + series capacitor): $L_{shunt} = X_{shunt}/(2\pi f) = 32.72/(2\pi \times 915 \times 10^6) = 5.69$ nH. $C_{series} = 1/(2\pi f X_{series}) = 1/(2\pi \times 915 \times 10^6 \times 22.92) = 7.58$ pF.
**High-pass solution** (shunt capacitor + series inductor): $C_{shunt} = 1/(2\pi f \times 32.72) = 5.32$ pF. $L_{series} = 22.92/(2\pi \times 915 \times 10^6) = 3.99$ nH.
Matching bandwidth: $BW \approx f_0/Q = 915/1.528 = 599$ MHz.
The VSWR stays below 2:1 over approximately ±300 MHz around 915 MHz, which is more than adequate for the ~26 MHz ISM band (902–928 MHz).
The low-pass solution is typically preferred because it provides harmonic rejection.

### 9.5.8 Coupled Transmission Lines and Crosstalk

When two transmission lines run in close proximity — adjacent PCB traces, parallel cable pairs, or neighboring connector pins — their electromagnetic fields interact, coupling energy from the driven (aggressor) line to the quiet (victim) line. This coupling, called **crosstalk**, degrades signal integrity by injecting unwanted noise voltages and currents that can cause false logic transitions, increase bit error rates, or violate EMI emission limits. Crosstalk arises from two coupling mechanisms: **capacitive (electric field) coupling** through the mutual capacitance $C_m$ between the traces, and **inductive (magnetic field) coupling** through the mutual inductance $L_m$ between the current loops. These mechanisms produce two distinct crosstalk components. **Near-end crosstalk (NEXT)** appears at the end of the victim line nearest the aggressor's source, combining forward-traveling capacitive coupling with backward-traveling inductive coupling — both components arrive at the near end simultaneously, reinforcing each other. **Far-end crosstalk (FEXT)** appears at the opposite end of the victim line, where the capacitive and inductive components travel in the same direction but with opposite polarity — in a homogeneous medium (stripline), the two components cancel perfectly, making FEXT zero, but in an inhomogeneous medium (microstrip, where $\varepsilon_{eff} \neq \varepsilon_r$), the velocity difference between even and odd modes prevents perfect cancellation, producing nonzero FEXT.

The crosstalk coefficients for weakly coupled lines are: NEXT coefficient $K_b = (C_m/C_0 + L_m/L_0)/4$ and FEXT coefficient $K_f = (C_m/C_0 - L_m/L_0)/2$, where $C_0$ and $L_0$ are the self-capacitance and self-inductance per unit length, $t_d$ is the coupled-region propagation delay, and $t_r$ is the signal rise time. NEXT has a maximum voltage of $V_{NEXT} = K_b \times V_{step}$ (saturating when the coupled length exceeds the critical length $l_{crit} = t_r v/2$), while FEXT increases linearly with coupled length: $V_{FEXT} = K_f \times V_{step} \times (t_d/t_r)$.

**Differential pairs** are a primary technique for reducing crosstalk susceptibility: two tightly coupled traces carrying equal and opposite signals (differential mode) reject common-mode noise from adjacent aggressors because the noise couples equally to both traces and is cancelled at the differential receiver. The characteristic impedance of a differential pair is $Z_{diff} = 2Z_0(1-k)$, where $Z_0$ is the single-ended impedance and k is the coupling coefficient, typically yielding $Z_{diff} = 85\text{–}100\ \Omega$ for standard PCB designs. Design rules to minimize crosstalk include: maintaining a separation of $\geq 3h$ between adjacent single-ended traces (where h is the height above the ground plane, the "3W" rule), using ground guard traces between sensitive signals, routing differential pairs with consistent tight coupling, and minimizing parallel run lengths of dissimilar signals.

**Example 9.5.8:** Two 50 Ω microstrip traces on a PCB (h = 0.2 mm above the ground plane, $\varepsilon_{eff} = 3.2$) run parallel with edge-to-edge spacing s = 0.3 mm for a coupled length of 80 mm. The aggressor carries a 3.3 V signal with 0.5 ns rise time. The mutual capacitance is $C_m = 25$ pF/m and the mutual inductance is $L_m = 60$ nH/m. The self-capacitance is $C_0 = 120$ pF/m and self-inductance is $L_0 = 300$ nH/m. Calculate (a) the NEXT voltage, (b) the FEXT voltage, and (c) the minimum spacing needed to reduce NEXT below 50 mV.
**Solution:**
(a) NEXT coefficient: $K_b = (C_m/C_0 + L_m/L_0)/4 = (25/120 + 60/300)/4 = (0.2083 + 0.2000)/4 = 0.1021$
Propagation velocity: $v = 1/\sqrt{(L_0C_0)} = 1/\sqrt{(300 \times 10^{-9} \times 120 \times 10^{-12})} = 1/\sqrt{(3.6 \times 10^{-17})} = 1.667 \times 10^8$ m/s
Critical length: $l_{crit} = t_r v/2 = 0.5 \times 10^{-9} \times 1.667 \times 10^8 / 2 = 41.7$ mm
Coupled length (80 mm) > $l_{crit}$ (41.7 mm), so NEXT saturates:
$V_{NEXT} = K_b \times V_{step} = 0.1021 \times 3.3 =$ **337 mV** (10.2% of signal amplitude)
(b) FEXT coefficient: $K_f = (C_m/C_0 - L_m/L_0)/2 = (0.2083 - 0.2000)/2 = 0.00417$
Propagation delay for 80 mm: $t_d = 0.08/1.667 \times 10^8 = 0.480$ ns
$V_{FEXT} = K_f \times V_{step} \times (t_d/t_r) = 0.00417 \times 3.3 \times (0.480/0.5) =$ **13.2 mV**
FEXT is much smaller than NEXT because the capacitive and inductive components nearly cancel ($C_m/C_0 \approx L_m/L_0$ in this case).
(c) Crosstalk decreases approximately as $(h/(h+s))^2$ for adjacent microstrip traces. To reduce NEXT from 337 mV to 50 mV requires $K_{b,new} = 50/3300 = 0.01515$. The ratio $K_{b,new}/K_{b,old} = 0.01515/0.1021 = 0.148$. Since $K_b$ scales roughly as $(h/(h+s))^2$, and the current spacing gives $(0.2/(0.2+0.3))^2 = 0.16$, the new spacing needs $(h/(h+s))^2 = 0.16 \times 0.148 = 0.0237$, so $h/(h+s) = 0.154$, giving $s = h(1/0.154 - 1) = 0.2 \times 5.49 =$ **1.1 mm** (approximately 5.5h spacing, consistent with the "3W" rule where $W \approx 2h$ for these trace dimensions).

## 9.6 Antennas

Antennas are the interface between guided waves on transmission lines and radiated electromagnetic waves in free space. Antenna design involves balancing radiation efficiency, directional gain, bandwidth, impedance matching, and physical size — with the optimal choice depending on the application, from compact chip antennas in smartphones to massive dish antennas for deep-space communication. Chapter 16 provides an in-depth treatment of antenna types; this section introduces the fundamental concepts that connect electromagnetics to antenna behavior.

### 9.6.1 Antenna Fundamentals

An antenna is a transducer that converts guided electromagnetic energy (on a transmission line or waveguide) into radiated electromagnetic waves in free space, and vice versa. The radiation pattern describes the spatial distribution of radiated power as a function of angle, typically characterized by a main lobe (direction of maximum radiation), sidelobes, beamwidth, and front-to-back ratio. Antenna gain G expresses the directivity relative to an isotropic radiator (in dBi) or a half-wave dipole (in dBd), accounting for the antenna's ability to concentrate energy in a preferred direction. The effective aperture $A_e$ relates the antenna's ability to capture incident power to its gain: $A_e = G\lambda^2/(4\pi)$. The Friis transmission equation $P_r = P_t\ G_t\ G_r\ (\lambda/(4\pi d))^2$ relates the received power to the transmitted power, antenna gains, wavelength, and distance, forming the basis of link budget calculations in wireless communications.

**Example 9.6.1:** A Wi-Fi access point transmits 100 mW (20 dBm) at 2.4 GHz with an antenna gain of 5 dBi. A laptop 30 m away has a receive antenna gain of 2 dBi. Find the free-space path loss and the received power.
**Solution:**
Wavelength: $\lambda = c/f = 3 \times 10^8 / 2.4 \times 10^9 = 0.125$ m.
Free-space path loss: $FSPL = (4\pi d/\lambda)^2 = (4\pi \times 30 / 0.125)^2 = (3015.9)^2 = 9.096 \times 10^6$.
In dB: $FSPL = 20\log_{10}(4\pi d/\lambda) = 20 \times \log_{10}(3015.9) = 20 \times 3.479 = 69.6$ dB.
Received power (in dB): $P_r = P_t + G_t + G_r - FSPL = 20 + 5 + 2 - 69.6 = -42.6$ dBm.
Converting: $P_r = 10^{-42.6/10}$ mW $= 5.5 \times 10^{-5}$ mW = **55 nW**, well above typical Wi-Fi receiver sensitivity of about −70 dBm (100 pW).

### 9.6.2 Dipole Antennas

The half-wave dipole is the most fundamental resonant antenna, consisting of two conductive elements each approximately $\lambda/4$ in length, fed at the center. At resonance, a half-wave dipole has an input impedance of approximately $73 + j42.5\ \Omega$, a gain of 2.15 dBi, and a toroidal (doughnut-shaped) radiation pattern with maximum radiation perpendicular to the antenna axis and nulls along the axis. The short dipole (length « λ) has a similar radiation pattern but lower radiation resistance and efficiency. Dipole variants include the folded dipole (higher impedance, wider bandwidth), the sleeve dipole, and dipole arrays. Dipole antennas are widely used as reference antennas, in FM radio reception, and as elements in more complex antenna arrays.

**Example 9.6.2:** Design a half-wave dipole antenna for the FM broadcast band at 100 MHz. Find the total length of the dipole and its radiation resistance.
**Solution:**
Wavelength: $\lambda = c/f = 3 \times 10^8 / 100 \times 10^6 = 3.0$ m.
Half-wave dipole length: $L = \lambda/2 = 3.0 / 2 = 1.5$ m (each arm is $\lambda/4 = 0.75$ m).
In practice, the physical length is shortened by approximately 5% to account for end effects: $L_{practical} \approx 0.95 \times 1.5 = 1.425$ m.
The radiation resistance at resonance is approximately 73 Ω, and the gain is 2.15 dBi (1.64 linear).
The input impedance is approximately $73 + j42.5\ \Omega$; the reactive part can be eliminated by slight length adjustment or a matching network.

### 9.6.3 Antenna Arrays

An antenna array combines multiple individual antenna elements with controlled spacing, amplitude, and phase to achieve radiation characteristics not possible with a single element. By adjusting the relative phase of the signals fed to each element, the main beam can be electronically steered without physically moving the antenna (phased array). The array factor, which depends on the number of elements, spacing, and excitation, multiplies the element pattern to produce the total radiation pattern. Uniform linear arrays with element spacing of $\lambda/2$ and progressive phase shift are the most common configuration, with the beamwidth inversely proportional to the array length and the number of elements. Phased arrays are essential in modern radar (simultaneous tracking of multiple targets), 5G cellular communications (massive MIMO beamforming), satellite communications, and radio astronomy.

**Example 9.6.3:** A uniform linear array of 8 half-wave dipole elements is spaced at $d = \lambda/2$ and operates at 3 GHz. Find the half-power beamwidth (approximate) and the array directivity improvement over a single element.
**Solution:**
Wavelength: $\lambda = c/f = 3 \times 10^8 / 3 \times 10^9 = 0.1$ m = 10 cm.
Element spacing: $d = \lambda/2 = 5$ cm.
Total array length: $L = (N-1) \times d = 7 \times 0.05 = 0.35$ m.
For a broadside uniform linear array, the approximate half-power beamwidth is $\theta_{3dB} \approx 0.886\lambda / (Nd) = 0.886 \times 0.1 / (8 \times 0.05) = 0.0886 / 0.4 = 0.2215$ rad = 12.7°.
The directivity of the array is approximately N times that of a single element: $D_{array} \approx N \times D_{element} = 8 \times 1.64 = 13.12$ (11.2 dBi).
The array focuses the beam into a narrow 12.7° beam, enabling higher gain and spatial selectivity.

### 9.6.4 Near-Field and Far-Field Regions

The electromagnetic field surrounding an antenna is divided into distinct spatial regions with fundamentally different behavior. The **reactive near-field** (Fresnel region) extends from the antenna surface to approximately $r < 0.62\sqrt{(D^3/\lambda)}$, where D is the largest antenna dimension; in this region, the fields store energy reactively (oscillating between electric and magnetic forms each half-cycle), field strength decays as $1/r^2$ or $1/r^3$, and the E/H ratio is not the free-space impedance $\eta_0$ — it can be much higher (for electrically short dipoles, dominated by the electric field) or much lower (for small loop antennas, dominated by the magnetic field). The **radiating near-field** (Fresnel region) spans from the reactive boundary to $r = 2D^2/\lambda$; here, the fields begin to radiate but the angular pattern varies with distance, making this region unsuitable for far-field antenna measurements. The **far-field** (Fraunhofer region) begins at $r > 2D^2/\lambda$ (and $r \gg \lambda$, $r \gg D$); in this region, the wave front is approximately planar, the radiation pattern is independent of distance, E and H are perpendicular and related by $\eta_0 = 377\ \Omega$, and power density decays as $1/r^2$. The far-field boundary distance is critical for antenna measurement: a half-wave dipole at 900 MHz ($D \approx 0.17$ m, $\lambda = 0.33$ m) has a far-field boundary of $2D^2/\lambda = 0.17$ m — close enough for indoor testing. However, a 1 m parabolic dish at 10 GHz ($\lambda = 0.03$ m) has $2D^2/\lambda = 67$ m, requiring either outdoor test ranges, compact antenna test ranges (CATR) using a reflector to create a plane wave in a shorter distance, or near-field scanning with mathematical transformation to far-field patterns. **Near-field communication (NFC)** at 13.56 MHz deliberately operates in the reactive near-field ($\lambda = 22$ m, so all practical distances $\ll \lambda$), using magnetic coupling between loop antennas — the rapid $1/r^3$ field decay provides inherent security by limiting communication to ~10 cm range.

**Example 9.6.4:** A phased array antenna panel measuring 0.5 m × 0.5 m operates at 28 GHz (5G mmWave). Calculate the far-field boundary distance, the reactive near-field extent, and determine the minimum anechoic chamber length for far-field measurements. Also compare to an NFC system at 13.56 MHz with a 50 mm × 50 mm loop antenna.
**Solution:**
At 28 GHz: $\lambda = c/f = 3 \times 10^8 / 28 \times 10^9 = 10.71$ mm.
D = 0.5 m (diagonal of the panel = 0.707 m, but typically the largest linear dimension is used).
Far-field boundary: $r_{ff} = 2D^2/\lambda = 2 \times 0.5^2 / 0.01071 = 0.5/0.01071 = 46.7$ m.
Reactive near-field extent: $r_{reactive} = 0.62\sqrt{(D^3/\lambda)} = 0.62\sqrt{(0.125/0.01071)} = 0.62\sqrt{(11.67)} = 0.62 \times 3.416 = 2.12$ m.
An anechoic chamber for far-field measurement must be at least 47 m long — impractical for indoor facilities. A compact antenna test range (CATR) or planar near-field scanner would be used instead.
For the NFC system: $\lambda = 3 \times 10^8 / 13.56 \times 10^6 = 22.12$ m. D = 0.05 m.
Far-field boundary: $2 \times 0.05^2 / 22.12 = 0.000226$ m = 0.23 mm — essentially at the antenna surface.
Reactive near-field: $0.62\sqrt{(0.05^3/22.12)} = 0.62\sqrt{(5.66 \times 10^{-6})} = 0.62 \times 2.38 \times 10^{-3} = 1.47$ mm.
The entire practical NFC operating range (up to ~100 mm) is deep in the reactive near-field where the magnetic field from the loop antenna decays as $1/r^3$, providing the short-range security characteristic of NFC.
At 100 mm, the field is $(1.47/100)^3 = 3.2 \times 10^{-6}$ times weaker than at the reactive boundary — a 110 dB natural path loss that limits eavesdropping range.

## 9.7 Electromagnetic Compatibility

Electromagnetic compatibility (EMC) is the ability of electronic equipment to function satisfactorily in its electromagnetic environment without introducing intolerable disturbances to other equipment in that environment. Every electronic device is both a potential source of electromagnetic interference (EMI) and a potential victim. EMC engineering applies the principles of electromagnetics — shielding, grounding, filtering, and circuit layout — to ensure that systems meet regulatory emission limits (FCC Part 15, CISPR 32) and immunity requirements (IEC 61000 series). Successful EMC design prevents conducted and radiated interference from degrading system performance.

### 9.7.1 Shielding Effectiveness

Electromagnetic shielding encloses a sensitive circuit or a radiating source in a conductive enclosure to attenuate electric and magnetic fields. The shielding effectiveness (SE) is expressed in dB as the ratio of field strength without the shield to the field strength with the shield: $SE = 20 \log_{10}(E_{incident}/E_{transmitted})$. For a solid conductive sheet of thickness t, the shielding effectiveness has three components: absorption loss $A = 8.686(t/\delta)$ dB (where $\delta$ is the skin depth), reflection loss R (depends on the impedance mismatch between the wave and the shield), and a correction for multiple internal reflections B (significant only when A < 15 dB). For a plane wave incident on a good conductor, the reflection loss is approximately $R = 168 - 10 \log_{10}(f\mu_r/\sigma_r)$ dB, where $\sigma_r$ is the conductivity relative to copper. In practice, shielding effectiveness is limited not by the enclosure walls but by apertures (slots, seams, ventilation holes, cable penetrations) — a slot of length l resonates at $\lambda/2$ and can drastically reduce SE. The rule of thumb is that any aperture longer than $\lambda/20$ at the highest frequency of concern must be treated with EMI gaskets, waveguide-below-cutoff ventilation panels, or filtered connectors.

**Example 9.7.1:** An aluminum enclosure ($\sigma = 3.5 \times 10^7$ S/m, $\mu_r = 1$) with wall thickness t = 1.5 mm must shield against a 100 MHz interference signal. Calculate the skin depth, absorption loss, and determine if the enclosure provides adequate shielding (target SE > 60 dB). Also find the maximum allowable aperture length.
**Solution:**
Skin depth at 100 MHz: $\delta = 1/\sqrt{}(\pi f \mu_0 \sigma) = 1/\sqrt{}(\pi \times 10^8 \times 4\pi \times 10^{-7} \times 3.5 \times 10^7) = 1/\sqrt{}(\pi \times 10^8 \times 43.98) = 1/\sqrt{}(1.381 \times 10^{10}) = 8.51$ µm.
Absorption loss: $A = 8.686 \times (t/\delta) = 8.686 \times (1.5 \times 10^{-3} / 8.51 \times 10^{-6}) = 8.686 \times 176.3 = 1531$ dB.
The absorption loss alone is enormous — even 0.1 mm of aluminum provides over 100 dB at 100 MHz.
Reflection loss for aluminum at 100 MHz: $R \approx 168 - 10 \log_{10}(10^8 \times 1 / 0.604) = 168 - 10 \log_{10}(1.656 \times 10^8) = 168 - 82.2 = 85.8$ dB.
Total SE = A + R = 1531 + 85.8 » 60 dB. The solid walls are not the concern.
Maximum aperture: $\lambda = c/f = 3 \times 10^8/10^8 = 3$ m. $\lambda/20 = 150$ mm.
Any slot, seam, or opening longer than 150 mm will begin to compromise shielding.
At 1 GHz ($\lambda = 30$ cm), the limit drops to 15 mm — highlighting why high-frequency EMC requires tight seam control and extensive use of gaskets.

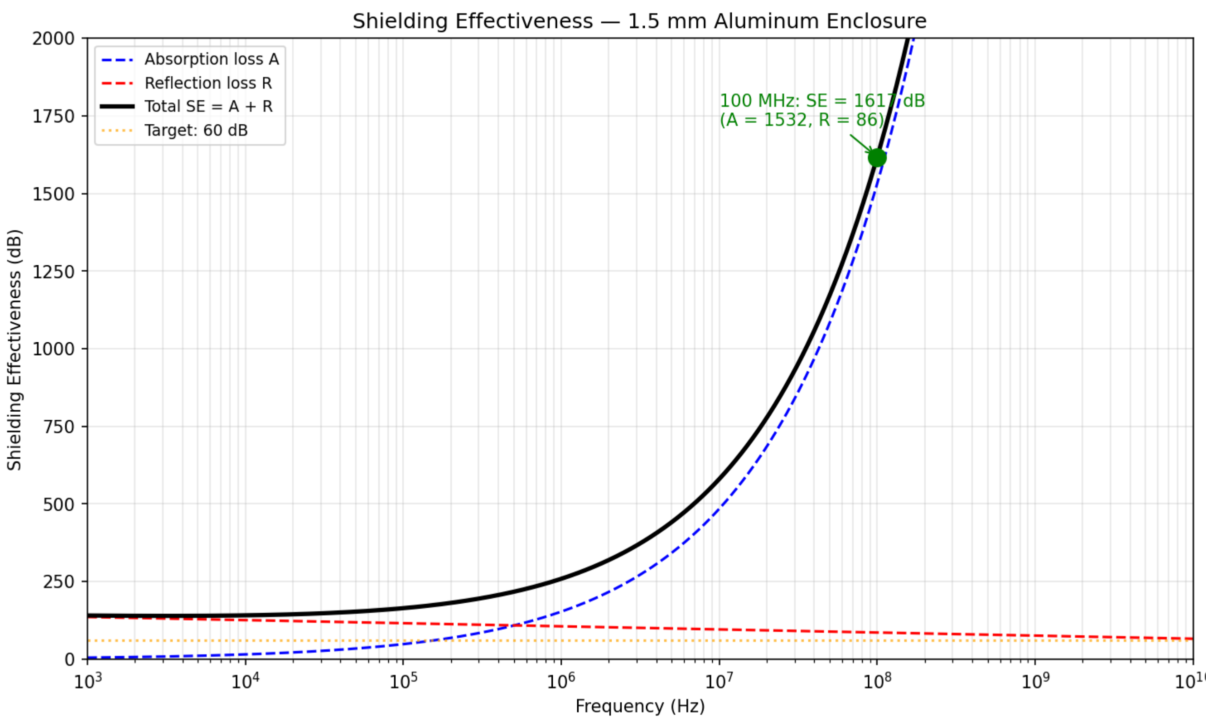



Figure 9.7.1: Shielding Effectiveness vs Frequency

### 9.7.2 EMI Filtering and Conducted Emissions

Conducted emissions are electromagnetic interference currents that travel along power and signal cables, coupling noise from a source (such as a switching power supply) to other equipment sharing the same power distribution network or cable bundle. Conducted emissions are regulated from 150 kHz to 30 MHz (above which radiated emissions dominate) by standards such as CISPR 32 (multimedia equipment) and FCC Part 15 Subpart B. Conducted noise exists in two modes: **differential mode** (noise current flows on the line conductor and returns on the neutral, in the same path as the desired power current) and **common mode** (noise current flows in the same direction on both line and neutral, returning through the ground conductor or parasitic capacitance). Common-mode emissions are typically the dominant source of EMI failures because even small common-mode currents flowing through long cables produce significant radiated emissions. **EMI line filters** suppress conducted emissions using a combination of components: X-capacitors (line-to-neutral) attenuate differential-mode noise, Y-capacitors (line-to-ground) attenuate common-mode noise (limited to a few nF for safety leakage current), and common-mode chokes (coupled inductors with high impedance to common-mode currents but near-zero impedance to differential power current) provide the primary common-mode suppression. A typical single-stage EMI filter provides 20-40 dB of attenuation; multi-stage filters or additional ferrite cores on cables extend this to 50-60 dB. On PCBs, decoupling capacitors (100 nF ceramic close to each IC power pin, plus bulk electrolytics on the board) suppress high-frequency conducted noise at the source, and proper grounding (solid ground planes, star-point or single-point grounding for mixed-signal boards) minimizes ground-coupled interference.

**Example 9.7.2:** A switching power supply produces conducted common-mode noise of 75 dBμV at 1 MHz on its AC input, measured with a LISN (Line Impedance Stabilization Network). The CISPR 32 Class B quasi-peak limit at 1 MHz is 46 dBμV. Design an EMI filter to achieve compliance with 6 dB of margin.
**Solution:**
Required attenuation: 75 - 46 + 6 = 35 dB at 1 MHz.
A common-mode choke is the primary component for common-mode suppression.
The choke impedance at 1 MHz must provide 35 dB attenuation across the 50 Ω LISN impedance.
Attenuation of an L-filter: $A = 20\log_{10}(1 + Z_L/(2 \times R_{LISN}))$. For 35 dB: $10^{\wedge}(35/20) = 1 + Z_L/100$, so $56.23 = 1 + Z_L/100$, $Z_L = 5523\ \Omega$.
At 1 MHz: $L_{CM} = Z_L/(2\pi f) = 5523/(2\pi \times 10^6) = 879\ \mu H$.
A practical common-mode choke of ~1 mH (standard value) provides ≥35 dB at 1 MHz.
Adding Y-capacitors of 2.2 nF (each line to ground) provides additional common-mode attenuation: at 1 MHz, $X_C = 1/(2\pi \times 10^6 \times 2.2 \times 10^{-9}) = 72.3\ \Omega$.
The combined LC filter (1 mH choke + 2.2 nF Y-cap) forms a second-order filter with resonance at $f_0 = 1/(2\pi\sqrt{(LC)}) = 1/(2\pi\sqrt{(10^{-3} \times 2.2 \times 10^{-9})}) = 107$ kHz, providing -40 dB/decade roll-off above resonance and approximately 35 + 20 = 55 dB attenuation at 1 MHz — well beyond the requirement, allowing margin for component tolerances and aging.

### 9.7.3 PCB Layout for EMC

Good PCB layout is the first line of defense against both radiated emissions and susceptibility, and it is far more effective and less costly than adding external shielding or filtering after the design is complete. The fundamental principle is **minimizing loop area** — every current loop acts as both a transmitting and receiving antenna, with radiated emission proportional to the loop area and the square of the frequency. A continuous **ground plane** (an entire copper layer dedicated to the return current) ensures that the return current flows directly beneath the signal trace (the path of least inductance), minimizing the loop area to the trace width times the substrate height. Multi-layer PCB stackups with alternating signal and ground/power planes (e.g., Sig-GND-PWR-Sig for 4-layer, or Sig-

GND-Sig-PWR-GND-Sig for 6-layer) provide low-inductance return paths for every signal. **Decoupling capacitors** (100 nF MLCC at each IC power pin, plus 10 μF bulk capacitors per power rail section) suppress high-frequency supply noise at the source; placement must be as close as possible to the IC pins, with the capacitor's loop area through the vias to the power and ground planes minimized. **Signal routing** rules for EMC include: keeping high-speed clock traces short and away from board edges, avoiding routing traces across splits in reference planes (which forces the return current to detour around the split, creating a large radiating loop), maintaining controlled impedance for signals with rise times faster than $2 \times t_{pd}$ (where $t_{pd}$ is the trace propagation delay), and using guard traces or ground stitching vias to isolate sensitive analog sections from noisy digital sections. **Connector and cable interface** regions require filtering (ferrite beads, feed-through capacitors) and ground stitching at the board edge, since cables are the primary antenna for conducted and radiated emissions.

**Example 9.7.3:** A 4-layer PCB stackup has signal traces on the top layer, a ground plane on layer 2, a power plane on layer 3, and signal traces on the bottom layer. The substrate height between the top signal layer and the ground plane is h = 0.2 mm. A 100 MHz clock trace is 50 mm long on the top layer. Estimate the loop area formed by the trace and its return current, and compare it to a 2-layer board where the return path is 10 mm away from the signal trace.
**Solution:**
4-layer board loop area: $A_1$ = trace length × substrate height = $50 \times 0.2 = 10$ mm$^2$ = $10 \times 10^{-6}$ m$^2$.
The return current flows in the ground plane directly beneath the trace, so the loop height is just the 0.2 mm dielectric thickness.
2-layer board loop area (return path 10 mm away): $A_2 = 50 \times 10 = 500$ mm$^2$ = $500 \times 10^{-6}$ m$^2$.
Ratio: $A_2/A_1 = 500/10 = 50\times$.
Since radiated emission is proportional to loop area, the 2-layer board radiates 50× more (34 dB worse) than the 4-layer board for the same clock signal.
At 100 MHz ($\lambda$ = 3 m), both loops are electrically small (loop dimension « $\lambda$), so the radiated electric field at distance r is approximately $E \propto f^2IA/r$, where I is the current.
For a 20 mA clock signal on the 4-layer board: the estimated radiated field at 3 m is $E \approx 1.316 \times 10^{-14} \times (10^8)^2 \times 0.02 \times 10 \times 10^{-6} / 3 = 8.8$ μV/m — below the FCC Class B limit of ~100 μV/m at 3 m.
The same signal on the 2-layer board would produce ~440 μV/m, exceeding the limit by 13 dB.

### 9.7.4 Electrostatic Discharge (ESD) Protection

Electrostatic discharge is the sudden transfer of charge between two objects at different electrostatic potentials, producing current pulses with peak amplitudes of amperes to tens of amperes and rise times of sub-nanosecond to several nanoseconds. ESD is one of the most common causes of electronic component damage and field failures. The three standard ESD test models are: the **Human Body Model (HBM)**, which simulates a person touching a device (100 pF charged to 2–8 kV, discharged through 1.5 kΩ, producing ~1.3 A peak with ~10 ns rise time); the **Charged Device Model (CDM)**, which simulates a charged IC discharging when it contacts a grounded surface (<1 ns rise, 5–15 A peak, most damaging to gate oxides); and the **Machine Model (MM)**, which simulates discharge from equipment (200 pF, 0 Ω, producing higher peak current). ICs are classified by their ESD withstand voltage (e.g., Class 1A: <250 V HBM, Class 3B: >8 kV HBM per ANSI/ESDA/JEDEC JS-001). **On-chip ESD protection** uses clamp circuits — typically a grounded-gate NMOS (ggNMOS) or silicon-controlled rectifier (SCR) structure — that shunt the ESD current to ground before it reaches the protected circuitry. **Board-level ESD protection** for I/O ports exposed to human contact (USB, HDMI, Ethernet, touchscreens) uses TVS (Transient Voltage Suppressor) diodes that clamp the voltage to a safe level within nanoseconds. Key TVS parameters include breakdown voltage $V_{BR}$, clamping voltage $V_C$ at peak pulse current, junction capacitance (must be low for high-speed data lines — <1 pF for USB 3.x), and peak pulse power rating. **Design practices** to minimize ESD vulnerability include: routing sensitive traces away from board edges and connectors, providing a low-impedance ground path from the connector shell to the ground plane, placing TVS devices as close to the connector as possible, and using series resistors (22–100 Ω) on sensitive inputs to limit peak current.

**Example 9.7.4:** A USB 2.0 data port must survive ±8 kV contact discharge per IEC 61000-4-2. A TVS diode array with $V_{BR} = 6$ V, $V_C = 12$ V at $I_{PP} = 8$ A, and $C_J = 0.5$ pF per line is selected. The USB transceiver IC has an absolute maximum voltage rating of 7 V on data pins. Verify that the TVS protects the IC and assess the signal integrity impact on USB 2.0 (480 Mbps).

**Solution:**

During an 8 kV IEC 61000-4-2 contact discharge, the peak current is approximately 30 A (per the IEC waveform) with a 0.7–1 ns rise time.

The TVS diode must clamp within ~1 ns. At 30 A (well above the specified 8 A), the clamping voltage rises — using the TVS I-V curve, $V_C$ at 30 A is approximately 18–22 V. This exceeds the IC's 7 V limit.

Solution: add a 33 Ω series resistor between the TVS and the IC pin.

With the TVS clamping at 18 V and the 33 Ω resistor, the current flowing into the IC's internal ESD clamp is limited to $I = (18 - 7)/33 = 0.33$ A, which is within the IC's internal ESD handling capability (typically 1+ A for HBM-rated devices).

Signal integrity: the 33 Ω resistor and 0.5 pF TVS capacitance form a lowpass filter with $f_{3dB} = 1/(2\pi RC) = 1/(2\pi \times 33 \times 0.5 \times 10^{-12}) = 9.65$ GHz — well above the USB 2.0 fundamental frequency of 240 MHz, so signal integrity impact is negligible.

For USB 3.x (5 Gbps, fundamental at 2.5 GHz), the 33 Ω resistor would be too large; a 5–10 Ω resistor with a sub-0.3 pF TVS would be required.

### 9.7.5 Grounding and Bonding for EMC

Grounding and bonding are the most fundamental — and most frequently misunderstood — aspects of EMC design. A **ground** is a reference conductor to which signal return currents and fault currents flow; a **bond** is a low-impedance electrical connection between two metallic structures, ensuring they remain at the same potential. At DC and low frequencies, a single-point (star) ground topology minimizes common-impedance coupling by ensuring that return currents from different circuits do not share a conductor segment. At high frequencies (above roughly $f > c/(10 \times \text{longest ground conductor length})$, typically a few MHz), the inductance of long ground wires dominates over their resistance, and a **multi-point ground** (mesh or plane) provides low-impedance return paths by minimizing loop area. Mixed-signal systems often use a **hybrid** approach: a single-point connection between the analog and digital ground domains at low frequencies (preventing digital return currents from flowing through the analog ground), transitioning to multi-point bonding through distributed capacitors at high frequencies. **Bonding** between equipment chassis, cable shields, rack frames, and building structural steel must be low-impedance at the frequencies of concern — bolted joints with clean, bare-metal contact surfaces (no paint or anodize in the contact area) provide bonds below 2.5 mΩ per MIL-STD-188-124B. Ground loops — closed conductive paths formed when two grounded points are connected by both a signal cable shield and the building ground system — allow magnetically induced currents to flow, coupling 50/60 Hz hum and transient noise into sensitive circuits. Ground loop mitigation techniques include differential signaling (rejecting common-mode noise), galvanic isolation (optocouplers, isolated DC-DC converters, fiber optics, isolation transformers), and breaking the loop with a single-point shield ground at the receiver end (for frequencies below ~1 MHz).

**Example 9.7.5:** An industrial measurement system connects a sensor 50 m away to a data acquisition unit. Both are grounded to the building steel, which has a 2 V potential difference at 60 Hz between the two ground points. The sensor signal is 10 mV full scale over a shielded twisted-pair cable with shield resistance of 0.15 Ω. Determine the ground loop current and the resulting noise at the receiver if the shield is grounded at both ends. Propose a solution.
**Solution:**
Ground loop current: $I_{loop} = V_{ground}/R_{shield} = 2/0.15 = 13.3$ A at 60 Hz.
This current flows through the cable shield, inducing a voltage in the signal conductors via mutual inductance between the shield and the twisted pair.
For a typical shielded cable, the transfer impedance is $Z_T \approx 10$ mΩ/m at 60 Hz (dominated by shield resistance), so the induced noise voltage is $V_{noise} = I_{loop} \times Z_T \times \text{length} = 13.3 \times 0.01 \times 50 = 6.65$ V.
This far exceeds the 10 mV signal — the measurement is completely buried in noise.
Even if the shield provides 20 dB of additional common-mode rejection, the noise (0.665 V) still overwhelms the signal by 36 dB.
**Solution:** Use differential signaling with an instrumentation amplifier (CMRR $\geq$ 80 dB at 60 Hz) and ground the shield at the receiver end only, breaking the ground loop.
The 2 V ground difference now appears as common-mode voltage on the signal pair, rejected by the instrumentation amp: $V_{noise} = 2\text{ V} / 10^{80/20} = 2/10{,}000 = 0.2$ mV = 200 μV, which is 2% of the 10 mV full-scale signal (34 dB SNR).
For better performance, add an isolation amplifier (galvanic isolation) at the sensor end to completely eliminate the ground loop, reducing noise to the amplifier's inherent input noise (~1 $\mu V_{rms}$).

# Chapter 10

# Power Electronics

Power electronics is the application of solid-state semiconductor devices to the conversion, control, and conditioning of electric power. It bridges the gap between power engineering and electronics, using switching devices such as diodes, transistors, and thyristors to efficiently convert power between AC and DC forms, regulate voltage and current levels, and control the speed of electric motors. Power electronic converters are found in applications ranging from milliwatt phone chargers to gigawatt HVDC transmission systems, and they are essential enablers of renewable energy integration, electric vehicles, and modern industrial automation.

## 10.1 Switching Devices

Power semiconductor switches are the core building blocks of all power electronic converters. The choice of switching device — diode, MOSFET, IGBT, thyristor, or wide-bandgap transistor — is determined by the application's voltage, current, switching frequency, and efficiency requirements. Each device type occupies a distinct region in the voltage/frequency design space, and understanding their characteristics, loss mechanisms, and gate drive requirements is essential for selecting the optimal device for a given converter topology.

### 10.1.1 Power Diodes

Power diodes are two-terminal devices that conduct current in one direction and block it in the other, serving as the fundamental building block for rectifier circuits. Unlike signal diodes, power diodes are designed to handle high forward currents (tens to thousands of amperes) and high reverse blocking voltages (hundreds to thousands of volts). Key parameters include forward voltage drop (typically 0.7-1.2 V for silicon), reverse recovery time (the time required for the diode to transition from conducting to blocking when reverse biased), and thermal resistance (which determines heat dissipation requirements). Fast-recovery diodes minimize reverse recovery losses in high-frequency switching circuits, while Schottky power diodes offer lower forward voltage drops and negligible reverse recovery time but are limited to lower blocking voltages (typically below 200 V). Silicon carbide (SiC) Schottky diodes combine low forward drop with high voltage ratings and near-zero reverse recovery, making them increasingly popular in high-efficiency power converters.

**Example 10.1.1:** A silicon power diode has a forward voltage drop $V_F = 1.0$ V and carries a continuous forward current of $I_F = 25$ A. The diode has a reverse recovery time $t_{rr} = 50$ ns and is used in a circuit where the reverse voltage is $V_R = 400$ V and the switching frequency is $f_{sw} = 100$ kHz. Estimate the conduction power loss and the reverse recovery power loss.

**Solution:**
Conduction loss: $P_{cond} = V_F \times I_F = 1.0\ V \times 25\ A = 25\ W$.
For reverse recovery loss, the recovered charge is approximately $Q_{rr} \approx ½ \times I_F \times t_{rr} = ½ \times 25 \times 50 \times 10^{-9} = 625\ nC$.
The reverse recovery energy per cycle is $E_{rr} \approx ½ \times Q_{rr} \times V_R = ½ \times 625 \times 10^{-9} \times 400 = 125\ \mu J$.
The reverse recovery power loss is $P_{rr} = E_{rr} \times f_{sw} = 125 \times 10^{-6} \times 100 \times 10^3 = 12.5\ W$.
Total diode power loss $\approx 25 + 12.5 = 37.5\ W$.

### 10.1.2 MOSFETs

Power MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) are voltage-controlled switches widely used in power electronics for their fast switching speeds and ease of gate drive. They conduct current between drain and source when a sufficient gate-to-source voltage exceeds the threshold voltage, and they turn off when the gate voltage is removed. The on-state resistance $R_{DS(on)}$ determines conduction losses and increases with voltage rating, making MOSFETs most efficient at lower voltages (typically below 600 V). Switching losses occur during the transitions between on and off states and are proportional to switching frequency, requiring careful gate drive design to minimize transition times. Enhancement-mode N-channel MOSFETs are the dominant type in power electronics, with silicon superjunction devices, SiC MOSFETs, and GaN HEMTs pushing the boundaries of voltage rating, switching speed, and efficiency.

**Example 10.1.2:** A power MOSFET with $R_{DS(on)} = 45\ m\Omega$ at 25°C is used in a converter switching at $f_{sw} = 200\ kHz$. The drain current is 10 A, the bus voltage is 48 V, and the turn-on and turn-off times are $t_{on} = 20\ ns$ and $t_{off} = 30\ ns$. The $R_{DS(on)}$ temperature coefficient is +0.4%/°C and the junction operates at 100°C. Find the total power loss.
**Solution:**
At 100°C, $R_{DS(on)} = 45\ m\Omega \times [1 + 0.004 \times (100 - 25)] = 45\ m\Omega \times 1.30 = 58.5\ m\Omega$.
Conduction loss: $P_{cond} = I^2 \times R_{DS(on)} = 10^2 \times 0.0585 = 5.85\ W$.
Switching loss: $P_{sw} = ½ \times V \times I \times (t_{on} + t_{off}) \times f_{sw} = ½ \times 48 \times 10 \times (20 + 30) \times 10^{-9} \times 200 \times 10^3 = 2.40\ W$.
Total power loss: $P_{total} = 5.85 + 2.40 = 8.25\ W$.

### 10.1.3 IGBTs

The Insulated Gate Bipolar Transistor (IGBT) combines the voltage-controlled gate of a MOSFET with the high current-carrying capability and low conduction losses of a bipolar transistor. IGBTs are the preferred switching device for medium to high power applications at voltages above 600 V, including motor drives, grid-tied inverters, uninterruptible power supplies, and induction heating. The on-state voltage drop of an IGBT (typically 1.5–3 V) is relatively independent of current rating, giving IGBTs an advantage over MOSFETs at higher voltages where MOSFET $R_{DS(on)}$ becomes prohibitively large. However, IGBTs have a tail current during turn-off (due to stored minority carriers) that limits their maximum practical switching frequency to approximately 20–50 kHz. IGBT modules package multiple devices with antiparallel diodes in standard configurations (half-bridge, full-bridge, six-pack) for convenient use in converter topologies.

**Example 10.1.3:** An IGBT module has $V_{CE(sat)} = 2.0$ V at rated current $I_C = 150$ A, turn-on energy $E_{on} = 30$ mJ, turn-off energy $E_{off} = 20$ mJ per switching event (at $V_{DC} = 600$ V, $I_C = 150$ A), and it operates at $f_{sw} = 10$ kHz with a duty cycle of 0.5. Calculate the conduction and switching losses.
**Solution:**
Conduction loss: $P_{cond} = V_{CE(sat)} \times I_C \times D = 2.0 \times 150 \times 0.5 = 150$ W.
Switching loss: $P_{sw} = (E_{on} + E_{off}) \times f_{sw} = (30 \times 10^{-3} + 20 \times 10^{-3}) \times 10{,}000 = 0.050 \times 10{,}000 = 500$ W.
Total IGBT loss: $P_{total} = 150 + 500 = 650$ W.
The switching losses dominate at this frequency, illustrating why IGBTs are typically limited to moderate switching frequencies.

### 10.1.4 Thyristors

Thyristors (Silicon Controlled Rectifiers, or SCRs) are four-layer PNPN devices that can be triggered into conduction by a gate pulse but remain latched on until the current falls below the holding current. Once conducting, the gate loses control and the thyristor can only be turned off by external circuit conditions that reduce the anode current to zero (natural commutation in AC circuits or forced commutation using auxiliary circuits). Thyristors can handle extremely high voltages (up to 12 kV per device) and currents (up to 6 kA), making them essential in high-power applications such as HVDC transmission, static VAR compensators, and large motor drives. The Gate Turn-Off thyristor (GTO) adds the ability to turn off by applying a negative gate current, while the Integrated Gate-Commutated Thyristor (IGCT) improves on the GTO with faster, more reliable turn-off and lower conduction losses.

**Example 10.1.4:** A thyristor (SCR) is used in a phase-controlled rectifier fed from a 480 $V_{rms}$ single-phase supply. The thyristor has a forward voltage drop $V_T = 1.5$ V, a holding current $I_H = 80$ mA, and a latching current $I_L = 250$ mA. If the load is purely resistive at $R = 10\ \Omega$ and the firing angle is $\alpha = 45°$, what is the average load current and the power dissipated in the thyristor?
**Solution:**
Peak supply voltage: $V_{peak} = 480\sqrt{2} = 678.8$ V.
For a single-phase half-wave controlled rectifier with resistive load, $V_{dc} = (V_{peak} / 2\pi)(1 + \cos\alpha) = (678.8 / 6.283)(1 + \cos 45°) = 108.0 \times 1.707 = 184.4$ V.
Average load current: $I_{dc} = V_{dc} / R = 184.4 / 10 = 18.4$ A, which far exceeds both $I_H$ and $I_L$, confirming the thyristor stays latched.
Thyristor power dissipation: $P_T \approx V_T \times I_{dc} = 1.5 \times 18.4 = 27.6$ W.

### 10.1.5 Wide-Bandgap Semiconductors (SiC and GaN)

Wide-bandgap (WBG) semiconductors — primarily silicon carbide (SiC, bandgap 3.26 eV) and gallium nitride (GaN, bandgap 3.4 eV) — are displacing silicon in power electronics applications where higher efficiency, switching speed, or temperature capability is required. Compared to silicon (bandgap 1.12 eV), WBG materials have approximately 10× higher critical electric field, enabling devices with thinner drift regions and dramatically lower on-resistance at a given voltage rating — or equivalently, much higher voltage ratings in a compact die. **SiC MOSFETs** (available at 650 V to 3.3 kV) offer $R_{DS(on)}$ values 5–10× lower than silicon MOSFETs at the same voltage and can operate at junction temperatures up to 200°C, making them the preferred choice for EV traction inverters, solar string inverters, and fast EV chargers. **SiC Schottky diodes** eliminate reverse recovery losses almost entirely, significantly reducing switching losses in boost PFC stages and bridge rectifiers. **GaN HEMTs** (High Electron Mobility Transistors, typically 100–650 V) leverage the two-dimensional electron gas at the AlGaN/GaN heterojunction to achieve extremely low on-resistance and parasitic capacitance, enabling switching frequencies of 1–10 MHz with minimal losses. GaN devices dominate the compact, high-frequency adapter and charger market (USB-C PD chargers, laptop adapters) and are gaining traction in data center power, lidar drivers, and Class-D audio amplifiers. The primary trade-offs of WBG devices are higher per-device cost (2–5× silicon) and the need for careful PCB layout and gate drive design to exploit the fast switching speeds without excessive ringing or EMI.

**Example 10.1.5:** A 650 V silicon superjunction MOSFET has $R_{DS(on)} = 99$ mΩ and output capacitance $C_{oss} = 65$ pF. A 650 V SiC MOSFET has $R_{DS(on)} = 25$ mΩ and $C_{oss} = 30$ pF. Both are used in a 400 V, 10 A boost PFC stage switching at 100 kHz. Compare the conduction losses and capacitive switching losses.
**Solution:**
Conduction losses (assuming $I_{rms} \approx 7$ A):
Silicon: $P_{cond} = 7^2 \times 0.099 = 4.85$ W.
SiC: $P_{cond} = 7^2 \times 0.025 = 1.23$ W (75% reduction).
Capacitive switching losses: $E_{oss} = ½C_{oss}V^2$ per cycle.
Silicon: $E_{oss} = ½ \times 65 \times 10^{-12} \times 400^2 = 5.2$ μJ.
SiC: $E_{oss} = ½ \times 30 \times 10^{-12} \times 400^2 = 2.4$ μJ.
At 100 kHz: Silicon: $P_{oss} = 5.2 \times 10^{-6} \times 10^5 = 0.52$ W. SiC: $P_{oss} = 2.4 \times 10^{-6} \times 10^5 = 0.24$ W.
Total device losses: Silicon ≈ 5.37 W, SiC ≈ 1.47 W.
The SiC device reduces total MOSFET losses by 73%, enabling higher efficiency or a smaller heat sink.
At 300 kHz switching (feasible with SiC), the inductor and capacitor sizes shrink proportionally, reducing converter volume.

### 10.1.6 Gate Driver Design

Gate drivers are the interface circuits that translate low-power control signals into the high-current, precisely timed pulses needed to switch power MOSFETs and IGBTs on and off. A power MOSFET gate is a capacitive load (total gate charge $Q_g$ typically 10–500 nC) that must be charged and discharged rapidly to minimize switching losses during the Miller plateau region where the drain voltage transitions. Peak gate drive currents of 1–10 A are common, requiring dedicated driver ICs or discrete totem-pole buffer stages. For **low-side** switches (source connected to ground), a simple non-isolated gate driver referenced to the power ground is sufficient. **High-side** switches in half-bridge and full-bridge topologies have their source terminal floating at the switching node voltage, requiring either a **bootstrap** circuit or a fully **isolated** gate driver. A bootstrap driver uses a capacitor ($C_{boot}$) charged from the low-voltage supply through a diode during the low-side on-time; when the high-side switch turns on, the bootstrap capacitor provides gate drive power referenced to the rising source terminal. The bootstrap capacitor must be sized to supply the total gate charge plus quiescent current for the maximum high-side on-time: $C_{boot} \geq Q_{total} / \Delta V_{boot}$, where $\Delta V_{boot}$ is the allowable voltage droop (typically 0.5–1 V). Bootstrap circuits are simple and inexpensive but limited to duty cycles below ~99% because they require periodic low-side on-time to recharge. Fully **isolated gate drivers** use a transformer, optocoupler, or capacitive isolation barrier to transmit the gate signal across a galvanic boundary, powered by an isolated DC-DC converter or integrated isolated supply. They support 100% duty cycle, handle high dv/dt transients (50–200 kV/μs common-mode rejection), and are mandatory for thyristor/IGBT gate drives in medium-voltage applications. **Dead time** — a brief interval (typically 20–500 ns) when both switches in a half-bridge are off — prevents shoot-through (simultaneous conduction creating a short circuit across the DC bus). Dead time is implemented in hardware (resistor-capacitor networks on the gate drive signal) or in the gate driver IC, and must be long enough to ensure complete turn-off of one device before the complementary device turns on, accounting for propagation delay mismatches and temperature variation. Excessive dead time increases body diode conduction losses and output distortion in motor drives.

**Example 10.1.6:** A half-bridge gate driver uses a bootstrap circuit to drive the high-side SiC MOSFET. The MOSFET has $Q_g = 120$ nC, the driver IC quiescent current is $I_Q = 5$ mA, the maximum high-side on-time is 9 μs at $f_{sw} = 100$ kHz, and the allowable bootstrap voltage droop is $\Delta V = 0.5$ V. The bootstrap diode has $V_F = 0.5$ V and the supply is $V_{CC} = 15$ V. Calculate the minimum bootstrap capacitance and the effective gate drive voltage.
**Solution:**
Total charge per cycle: $Q_{total} = Q_g + I_Q \times t_{on(max)} = 120 \times 10^{-9} + 5 \times 10^{-3} \times 9 \times 10^{-6} = 120$ nC + 45 nC = 165 nC.
Minimum bootstrap capacitance: $C_{boot} = Q_{total} / \Delta V = 165 \times 10^{-9} / 0.5 = 330$ nF.
Use a standard 470 nF or 1 μF ceramic capacitor for margin.
The bootstrap voltage when fully charged: $V_{boot} = V_{CC} - V_{F(diode)} = 15 - 0.5 = 14.5$ V.
After the high-side on-time, $V_{boot}$ droops to $14.5 - (165 \times 10^{-9} / 470 \times 10^{-9}) = 14.5 - 0.35 = 14.15$ V, well above the MOSFET's minimum gate drive voltage of 10 V.

## 10.2 Rectifiers

Rectifiers are the most fundamental power electronic converters, transforming AC into DC. They range from simple diode bridges in offline power supplies to high-power thyristor rectifiers in industrial electrochemical plants and HVDC converter stations. Rectifier design involves selecting the topology (half-wave, full-wave, multi-pulse), choosing uncontrolled (diode) or controlled (thyristor) operation, sizing the output filter to meet ripple requirements, and managing harmonic currents drawn from the AC source.

### 10.2.1 Single-Phase Rectifiers

Single-phase rectifiers convert single-phase AC to DC using diodes or thyristors. The half-wave rectifier uses a single diode and conducts only during the positive half-cycle, producing a pulsating DC output with high ripple and a DC voltage of $V_{dc} = V_{peak}/\pi$. The full-wave bridge rectifier uses four diodes and conducts during both half-cycles, doubling the DC output to $V_{dc} = 2V_{peak}/\pi$ with reduced ripple at twice the line frequency. Adding a capacitor filter across the output smooths the ripple, with larger capacitance yielding lower ripple but higher peak diode currents. Phase-controlled rectifiers replace diodes with thyristors, allowing the DC output voltage to be continuously adjusted from maximum to zero (and negative for full converters) by varying the firing angle α: $V_{dc} = (2V_{peak}/\pi)\cos(\alpha)$.

**Example 10.2.1:** A single-phase full-wave bridge rectifier is fed from a 120 $V_{rms}$, 60 Hz source and supplies a load requiring 100 V DC. A capacitor filter is used. Calculate the ideal (no-load) DC output voltage and determine the firing angle α needed if thyristors are used to regulate the

output to exactly 100 V DC.
**Solution:**
$V_{peak} = 120 \times \sqrt{2} = 169.7$ V.
Ideal uncontrolled DC output: $V_{dc} = 2V_{peak}/\pi = 2 \times 169.7 / \pi = 108.0$ V.
For a fully controlled bridge rectifier to produce $V_{dc} = 100$ V: $100 = (2 \times 169.7 / \pi)\cos(\alpha) = 108.0 \cos(\alpha)$.
$\cos(\alpha) = 100/108.0 = 0.926$.
$\alpha = \cos^{-1}(0.926) = 22.2°$.

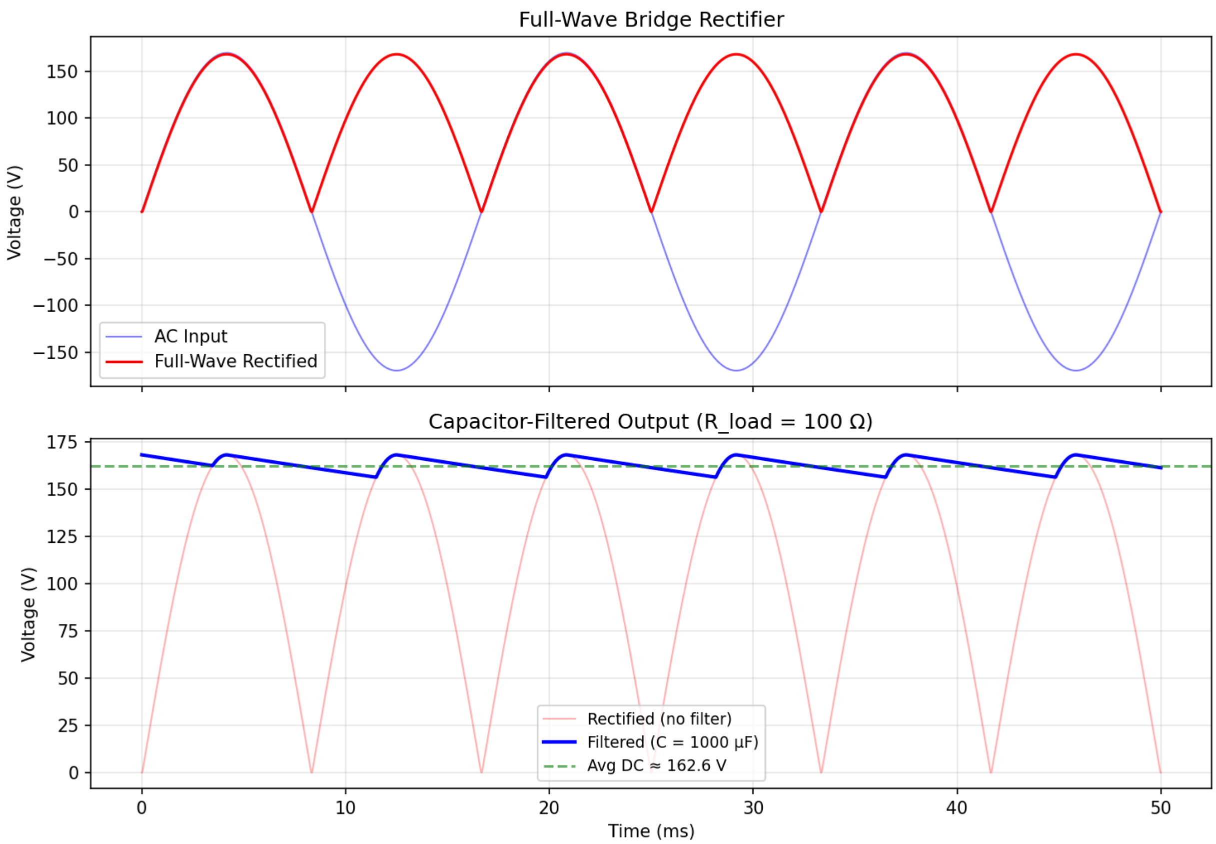



Figure 10.2.1: Full-Wave Bridge Rectifier

### 10.2.2 Three-Phase Rectifiers

Three-phase rectifiers are standard in industrial and high-power applications because they produce lower output ripple and more efficient use of the transformer and source. The three-phase half-wave (three-pulse) rectifier uses three diodes and produces output ripple at three times the line frequency, with $V_{dc} = (3\sqrt{3}/(2\pi))V_{peak}$. The three-phase full-wave bridge (six-pulse) rectifier uses six diodes and produces ripple at six times the line frequency, with $V_{dc} = (3\sqrt{3}/\pi)V_{peak}$, yielding smoother DC with only about 4.2% ripple. Twelve-pulse and higher-pulse rectifiers use transformer phase shifting (typically delta-wye combinations) to further reduce harmonics and ripple. Three-phase controlled rectifiers using thyristors provide adjustable DC output voltage for applications such as DC motor drives, electrochemical processes, and HVDC converter stations.

**Example 10.2.2:** A three-phase full-wave (six-pulse) diode bridge rectifier is fed from a 480 $V_{rms}$ line-to-line, 60 Hz three-phase source. Calculate the average DC output voltage and the ripple frequency.
**Solution:**
The peak line-to-line voltage: $V_{peak} = 480 \times \sqrt{2} = 678.8$ V.
For a six-pulse diode rectifier: $V_{dc} = (3/\pi) \times V_{peak(L\text{-}L)} = (3 / 3.1416) \times 678.8 = 0.9549 \times 678.8 = 648.2$ V (alternatively, $V_{dc} = 3\sqrt{2}/\pi \times V_{rms(L\text{-}L)} = 3 \times 1.4142 / 3.1416 \times 480 = 1.3505 \times 480 = 648.2$ V, confirming the result).
Ripple frequency: $f_{ripple} = 6 \times f_{line} = 6 \times 60 = 360$ Hz.
The peak-to-peak ripple is approximately 4.2% of $V_{dc}$, or about 27.2 V.

### 10.2.3 Rectifier Harmonics and Input Filtering

Non-linear loads such as diode and thyristor rectifiers draw current from the AC source in pulses rather than sinusoidally, injecting harmonic currents back into the power system. A single-phase diode bridge with a capacitor filter draws current only near the peak of the voltage waveform, producing odd harmonics (3rd, 5th, 7th, ...) with a current THD typically exceeding 100% and a displacement power factor near unity but a true power factor of only 0.5-0.65. A six-pulse three-phase diode bridge produces characteristic harmonics of order $h = 6k \pm 1$ (5th, 7th, 11th, 13th, ...) with magnitudes approximately $I_h \approx I_1/h$, yielding a current THD of about 25-30%. IEEE Standard 519 limits the harmonic current distortion that a customer may inject into the utility system at the point of common coupling (PCC), with limits depending on the ratio of short-circuit current to load current ($I_{SC}/I_L$). **Passive harmonic filters** use tuned LC series circuits connected in shunt to provide a low-impedance path for specific harmonics (typically 5th and 7th), diverting them from the supply. **Multi-pulse rectifiers** (12-pulse using delta-wye transformers, 18-pulse, 24-pulse) cancel lower-order harmonics through phase shifting — a 12-pulse rectifier eliminates the 5th and 7th harmonics, reducing THD to approximately 10-12%. **Active harmonic filters** inject equal-and-opposite harmonic currents using a PWM inverter, reducing THD below 5% regardless of load conditions.

**Example 10.2.3:** A 12-pulse rectifier system uses two six-pulse bridges fed from a delta-wye and a delta-delta transformer, each producing 30° of phase shift. The fundamental current drawn by each bridge is 100 A. Calculate the expected 5th, 7th, 11th, and 13th harmonic currents at the input, and compare the THD to a single six-pulse bridge.
**Solution:**
For a single six-pulse bridge, characteristic harmonics: $I_5 = 100/5 = 20$ A, $I_7 = 100/7 = 14.3$ A, $I_{11} = 100/11 = 9.1$ A, $I_{13} = 100/13 = 7.7$ A.
Single-bridge THD $\approx \sqrt{(20^2 + 14.3^2 + 9.1^2 + 7.7^2 + \ldots)} / 100 \approx \sqrt{(400 + 204.5 + 82.8 + 59.3)} / 100 \approx 27.3\%$ (first four harmonics only).
In the 12-pulse system, the 30° phase shift between the two bridges causes the 5th and 7th harmonics to cancel at the AC supply — the negative-sequence 5th harmonic is shifted by $5 \times 30° = 150°$ and the positive-sequence 7th by $7 \times 30° = 210°$ through the delta-wye transformer, and the resulting primary currents sum to zero for both orders.
The 11th and 13th harmonics add in phase: $I_{11} = 2 \times (100/11) = 18.2$ A, $I_{13} = 2 \times (100/13) = 15.4$ A, with fundamental $I_1 = 2 \times 100 = 200$ A.
12-pulse THD $\approx \sqrt{(18.2^2 + 15.4^2)} / 200 = \sqrt{(331.2 + 237.2)} / 200 = 23.8 / 200 = 11.9\%$.

The 12-pulse arrangement reduces THD from ~27% to ~12%, eliminating the dominant 5th and 7th harmonics that are most difficult to filter.

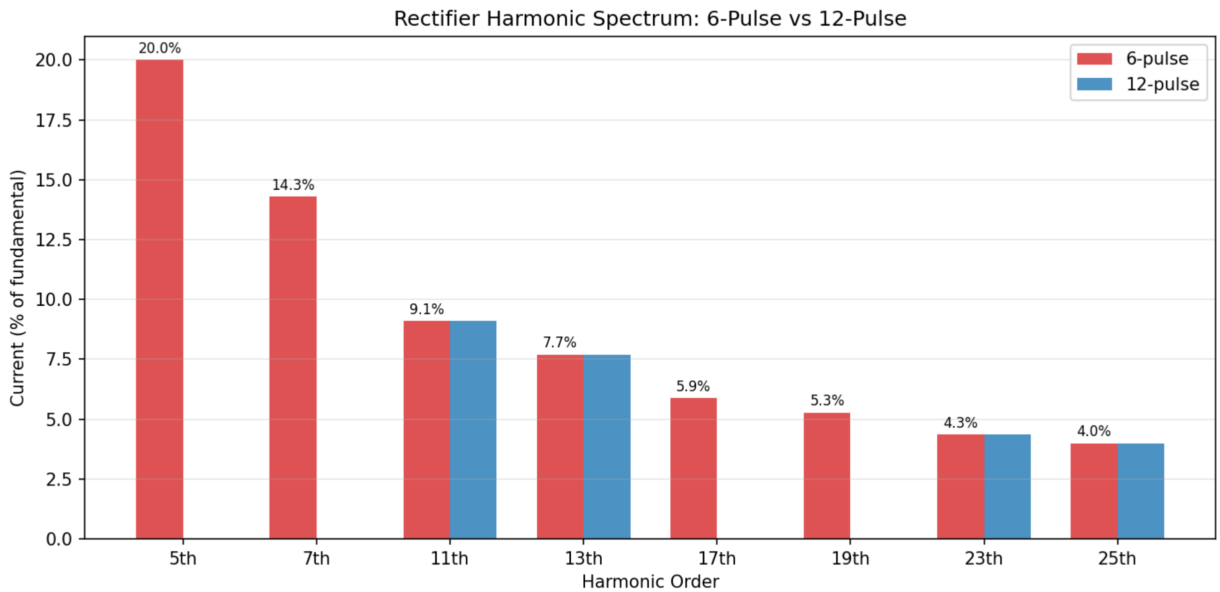



Figure 10.2.3: Rectifier Harmonic Spectrum: 6-Pulse vs 12-Pulse

## 10.3 DC-DC Converters

DC-DC converters change one DC voltage level to another with high efficiency using switching techniques rather than the resistive voltage division of linear regulators. The basic non-isolated topologies — buck, boost, and buck-boost — form the foundation from which dozens of specialized topologies are derived. Isolated converters add a transformer for galvanic isolation and voltage scaling. The choice of topology depends on the input/output voltage ratio, power level, isolation requirements, and performance targets for efficiency, size, and transient response.

### 10.3.1 Buck Converter

The buck (step-down) converter reduces a higher DC input voltage to a lower DC output voltage with high efficiency. It operates by rapidly switching the input voltage across an inductor using a transistor (typically a MOSFET) and a freewheeling diode, with the inductor and output capacitor averaging the switched waveform into a smooth DC output. The output voltage is controlled by the duty cycle D (the fraction of the switching period the transistor is on): $V_{out} = D \times V_{in}$, where $0 < D < 1$. In continuous conduction mode (CCM), the inductor current never reaches zero during a switching cycle, and the output ripple is determined by the inductance, capacitance, and switching frequency. Synchronous buck converters replace the freewheeling diode with a second MOSFET to reduce conduction losses, achieving efficiencies above 95% in modern designs.

**Example 10.3.1:** A buck converter operates from $V_{in}$ = 24 V and must produce $V_{out}$ = 5 V at $I_{out}$ = 3 A. The switching frequency is $f_{sw}$ = 500 kHz. Determine the duty cycle, and calculate the minimum inductance required to maintain continuous conduction mode (CCM) assuming a ripple current of 30% of the output current.
**Solution:**
Duty cycle: $D = V_{out} / V_{in} = 5 / 24 = 0.2083$ (20.83%).
The peak-to-peak inductor ripple current: $\Delta I_L = 0.30 \times I_{out} = 0.30 \times 3 = 0.9$ A.
For a buck converter: $\Delta I_L = (V_{in} - V_{out}) \times D / (L \times f_{sw})$.
Solving for L: $L = (V_{in} - V_{out}) \times D / (\Delta I_L \times f_{sw}) = (24 - 5) \times 0.2083 / (0.9 \times 500{,}000) = 3.958 / 450{,}000 = 8.8\ \mu H$.
Use a standard value of 10 $\mu$H to ensure CCM operation with margin.

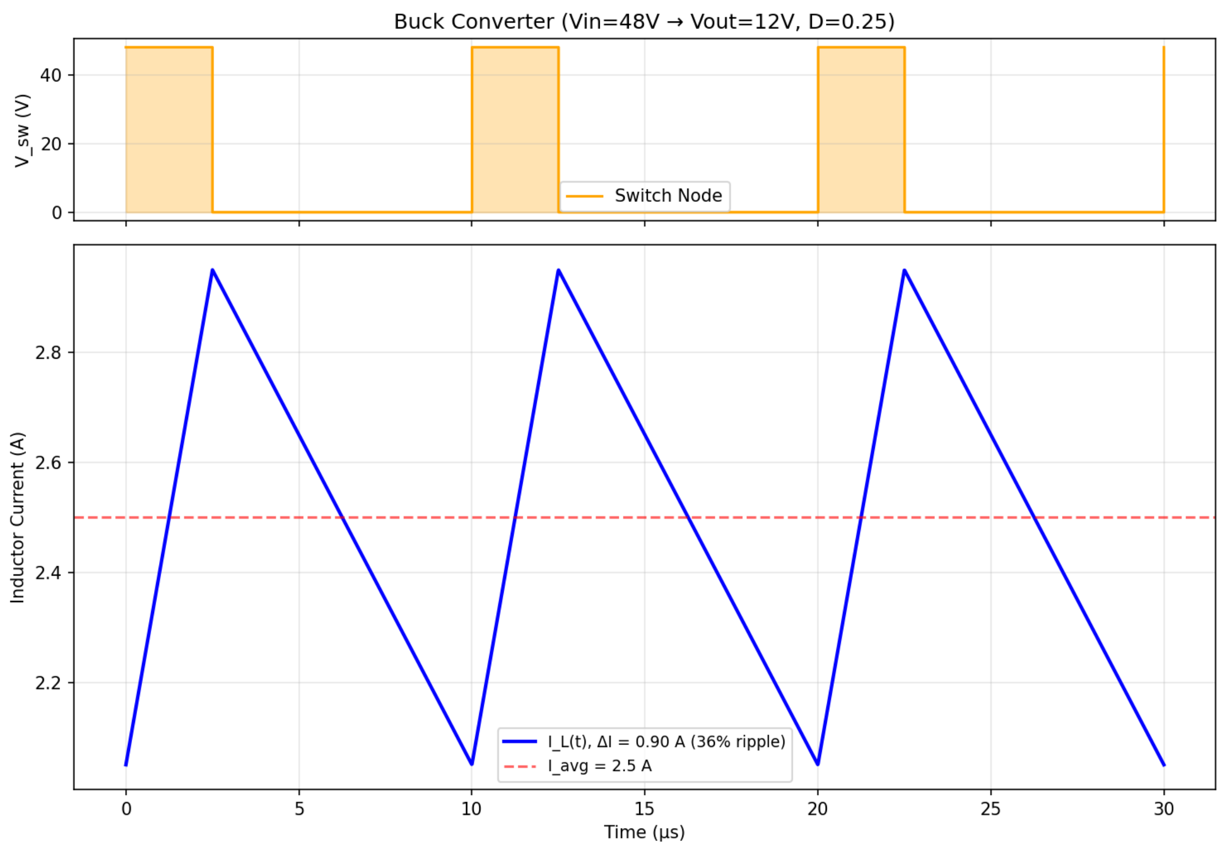



Figure 10.3.1: Buck Converter Inductor Current

### 10.3.2 Boost Converter

The boost (step-up) converter increases a lower DC input voltage to a higher DC output voltage. During the on-time, the switch connects the inductor across the input, storing energy in its magnetic field; during the off-time, the inductor voltage adds to the input voltage and transfers energy through the diode to the output capacitor. The ideal voltage conversion ratio is $V_{out} = V_{in}/(1 - D)$, where D is the duty cycle, allowing theoretically unlimited voltage gain as D approaches 1. In practice, parasitic resistances limit the achievable voltage gain to approximately 4-5× before efficiency degrades significantly.

Boost converters are used in power factor correction front ends, battery-powered systems that require higher voltage rails, solar maximum power point trackers (MPPT), and LED drivers.

**Example 10.3.2:** A boost converter must step up a solar panel voltage of $V_{in}$ = 18 V to $V_{out}$ = 48 V for a battery charging application. The load current is 2 A, and the switching frequency is $f_{sw}$ = 150 kHz. Determine the duty cycle and the input current.
**Solution:**
Duty cycle: $V_{out} = V_{in} / (1 - D)$, so $1 - D = V_{in} / V_{out} = 18 / 48 = 0.375$, giving D = 0.625 (62.5%).
Assuming an ideal (lossless) converter, input power equals output power: $P_{out} = V_{out} \times I_{out} = 48 \times 2 = 96$ W.
$I_{in} = P_{out} / V_{in} = 96 / 18 = 5.33$ A.
Alternatively, $I_{in} = I_{out} / (1 - D) = 2 / 0.375 = 5.33$ A, confirming the result.
The inductor must carry the full 5.33 A input current continuously.

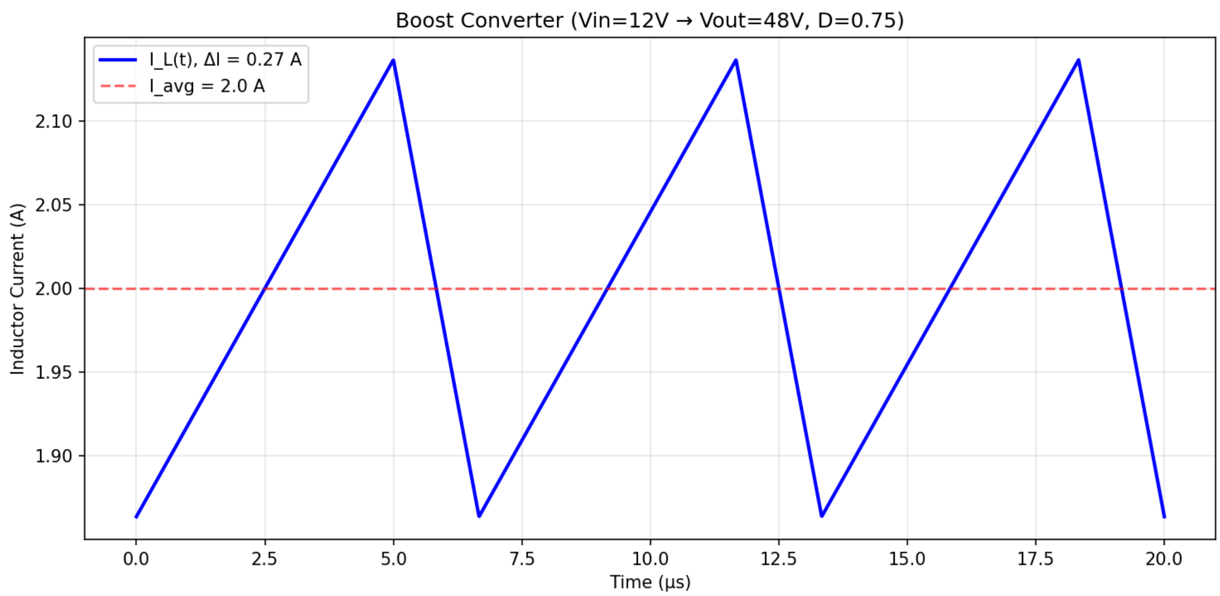



Figure 10.3.2: Boost Converter Inductor Current

### 10.3.3 Buck-Boost Converter

The buck-boost converter can produce an output voltage that is either higher or lower than the input voltage, with the output polarity inverted relative to the input. When the switch is on, the inductor charges from the input; when the switch is off, the inductor discharges into the output through the diode. The ideal voltage conversion ratio is $V_{out} = -V_{in} \times D/(1 - D)$, where the negative sign indicates polarity inversion. The Ćuk converter, SEPIC (Single-Ended Primary-Inductor Converter), and ZETA converter are related topologies that provide non-inverted buck-boost operation using two inductors (or a coupled inductor) and a series capacitor for energy transfer. Four-switch (H-bridge) buck-boost converters avoid polarity inversion and provide seamless transitions between buck and boost modes, commonly used in battery management systems where the battery voltage may be above or below the required output.

**Example 10.3.3:** An inverting buck-boost converter operates from $V_{in}$ = 12 V with a duty cycle D = 0.4. Calculate the output voltage. If the duty cycle is then increased to D = 0.7, what is the new output voltage?
**Solution:**
For an ideal inverting buck-boost converter: $V_{out} = -V_{in} \times D / (1 - D)$.
At D = 0.4: $V_{out} = -12 \times 0.4 / (1 - 0.4) = -12 \times 0.4 / 0.6 = -12 \times 0.667 = -8.0$ V (buck mode, $|V_{out}| < V_{in}$).
At D = 0.7: $V_{out} = -12 \times 0.7 / (1 - 0.7) = -12 \times 0.7 / 0.3 = -12 \times 2.333 = -28.0$ V (boost mode, $|V_{out}| > V_{in}$).
This demonstrates the topology's ability to produce outputs both below and above the input magnitude.

### 10.3.4 Isolated Converters

Isolated DC-DC converters use a transformer to provide galvanic isolation between input and output, enabling voltage scaling through the turns ratio and meeting safety requirements. The flyback converter is the simplest isolated topology, derived from the buck-boost converter with the inductor replaced by a coupled inductor (flyback transformer), widely used in low-power applications (up to ~150 W) such as phone chargers and standby power supplies. The forward converter is derived from the buck converter and uses a true transformer with simultaneous energy transfer, suitable for power levels up to several hundred watts. For higher power levels (500 W to several kilowatts), the half-bridge and full-bridge converters drive the transformer with an AC square wave and use a secondary-side rectifier and filter. Phase-shifted full-bridge converters achieve zero-voltage switching (ZVS) to reduce switching losses, and LLC resonant converters use a resonant tank circuit to achieve both ZVS and zero-current switching (ZCS) for high efficiency across a wide load range.

**Example 10.3.4:** A flyback converter operates from $V_{in}$ = 375 V DC (from a PFC front end) and must produce $V_{out}$ = 12 V at $I_{out}$ = 3 A. The transformer turns ratio is $N_p{:}N_s$ = 20:1, the maximum duty cycle is $D_{max}$ = 0.45, and the switching frequency is $f_{sw}$ = 100 kHz. Verify that the turns ratio is suitable and calculate the peak primary current.
**Solution:**
For a flyback converter in CCM, the output voltage is: $V_{out} = V_{in} \times (N_s/N_p) \times D / (1 - D)$.
At $D = D_{max}$: $V_{out} = 375 \times (1/20) \times 0.45 / (1 - 0.45) = 375 \times 0.05 \times 0.818 = 15.3$ V.
This exceeds 12 V, confirming the turns ratio provides adequate margin.
For $V_{out}$ = 12 V: $12 = 375 \times 0.05 \times D / (1 - D)$, so $D / (1 - D) = 12 / 18.75 = 0.64$, giving D = 0.39.
Output power: $P_{out} = 12 \times 3 = 36$ W. Assuming 90% efficiency, $P_{in} = 36 / 0.90 = 40$ W.
Average primary current: $I_{p(avg)} = P_{in} / V_{in} = 40 / 375 = 0.107$ A.
Peak primary current (CCM, assuming 40% ripple): $I_{p(peak)} \approx I_{p(avg)} / D + \Delta I/2 \approx 0.107 / 0.39 \times (1 + 0.20) \approx 0.329$ A.

### 10.3.5 Resonant Converters

Resonant converters use the interaction between inductors and capacitors (an LC resonant tank) to shape the voltage and current waveforms at the switching devices, enabling zero-voltage switching (ZVS) or zero-current switching (ZCS) to dramatically reduce switching losses. By switching when either the voltage across or the current through the device is zero, the V × I overlap that causes switching losses in hard-switched converters is eliminated, allowing operation at much higher frequencies (500 kHz to several MHz) with high efficiency. The **LLC resonant converter** is the most widely used resonant topology, employing a resonant network of two inductors ($L_r$, $L_m$) and one capacitor ($C_r$) between a half-bridge or full-bridge switching stage and a transformer. The LLC converter achieves ZVS for the primary switches across the entire load range, provides inherent short-circuit protection, and has high efficiency at the resonant frequency where the gain is unity. Frequency modulation (varying $f_{sw}$ relative to the resonant frequency $f_r = 1/(2\pi\sqrt{(L_rC_r)})$) controls the output voltage. Below resonance the converter operates in ZCS mode for the secondary rectifiers; above resonance it maintains primary ZVS but with increasing circulating current. LLC converters dominate in server power supplies, telecom rectifiers, TV power supplies, and EV on-board chargers, typically achieving 95–97% peak efficiency.

**Example 10.3.5:** An LLC resonant converter has $L_r = 50\ \mu H$, $C_r = 10$ nF, and magnetizing inductance $L_m = 250\ \mu H$. The transformer turns ratio is 16:1, $V_{in} = 400$ V (from PFC), and $V_{out} = 12$ V. Calculate the series resonant frequency $f_r$, the ratio $L_m/L_r$, and verify the output voltage at the resonant frequency.
**Solution:**
Series resonant frequency: $f_r = 1/(2\pi\sqrt{(L_rC_r)}) = 1/(2\pi\sqrt{(50 \times 10^{-6} \times 10 \times 10^{-9})}) = 1/(2\pi\sqrt{(500 \times 10^{-15})})$.
Recalculating: $\sqrt{(50 \times 10^{-6} \times 10 \times 10^{-9})} = \sqrt{(5 \times 10^{-13})} = 7.071 \times 10^{-7}$.
$f_r = 1/(2\pi \times 7.071 \times 10^{-7}) = 1/(4.443 \times 10^{-6}) = 225$ kHz.
Inductance ratio: $L_m/L_r = 250/50 = 5$ (a typical value; higher ratios narrow the frequency range but improve efficiency).
At the resonant frequency, the LLC gain for a half-bridge is approximately 1.0, so $V_{out} = (V_{in}/2) \times (N_s/N_p) \times \text{gain} = (400/2) \times (1/16) \times 1.0 = 200 \times 0.0625 = 12.5$ V.
This is close to the 12 V target; the switching frequency will operate slightly above 225 kHz to reduce the gain to exactly $12/12.5 = 0.96$.

### 10.3.6 Converter Control: Voltage Mode and Current Mode

Switching converters require closed-loop feedback control to maintain a regulated output voltage despite variations in input voltage and load current. **Voltage mode control** (VMC) senses the output voltage, compares it to a reference, and adjusts the duty cycle through an error amplifier and PWM comparator. The control loop is a single voltage feedback loop with a Type II or Type III compensator designed to provide adequate phase margin (typically 45–60°) and crossover frequency (typically $f_{sw}/10$ to $f_{sw}/5$). VMC is simple to implement but has slower transient response because load current changes must first appear as output voltage deviations before the loop corrects. **Current mode control** (CMC) adds an inner current loop that senses the inductor current and compares it to the output of the voltage error amplifier, effectively turning the power stage into a voltage-controlled current source. Peak current mode control terminates the on-time when the sensed inductor current reaches the control voltage, providing cycle-by-cycle current limiting, inherent current sharing in parallel converters, and simplified compensator design (the power stage appears as a single-pole system). CMC requires slope compensation to prevent subharmonic oscillation at duty cycles above 50% — a fixed ramp signal is added to the sensed current signal to ensure stability. Average current mode control senses the average inductor current rather than the peak, providing more accurate current regulation and better noise immunity, and is the standard method for boost PFC converters. Modern digital control implementations use ADCs and digital compensators (PID with anti-windup) running on DSPs or dedicated power management ICs, enabling adaptive tuning, nonlinear control, and predictive algorithms that improve transient response beyond what analog loops achieve.

**Example 10.3.6:** A buck converter with $V_{in}$ = 12 V, $V_{out}$ = 3.3 V, L = 4.7 μH, C = 100 μF (ESR = 15 mΩ), and $f_{sw}$ = 500 kHz uses peak current mode control. Calculate the duty cycle, the slope compensation ramp needed to prevent subharmonic oscillation, and the inductor current downslope.
**Solution:**
Duty cycle: $D = V_{out}/V_{in} = 3.3/12 = 0.275$ (27.5%).
Since $D < 0.5$, subharmonic oscillation is not a concern at this operating point, but slope compensation is still good practice for robustness.
Inductor current upslope: $m_1 = (V_{in} - V_{out})/L = (12 - 3.3)/(4.7 \times 10^{-6}) = 1.851$ A/μs.
Inductor current downslope: $m_2 = V_{out}/L = 3.3/(4.7 \times 10^{-6}) = 0.702$ A/μs.
The minimum slope compensation to guarantee stability at all duty cycles is $S_e \geq m_2/2 = 0.702/2 = 0.351$ A/μs.
Converting to a voltage ramp across a 50 mΩ sense resistor: the compensation ramp rate is 0.351 × 0.050 = 17.6 mV/μs, or 17.6 mV/μs × 2 μs (switching period) = 35.1 mV peak ramp added per cycle.

## 10.4 Inverters

Inverters perform the reverse function of rectifiers, converting DC power to AC at a controlled frequency and amplitude. Modern inverters use pulse-width modulation to synthesize high-quality sinusoidal outputs with low harmonic distortion. They are the enabling technology for variable-speed motor drives, grid-tied renewable energy systems, uninterruptible power supplies, and electric vehicle powertrains.

### 10.4.1 Single-Phase Inverters

Single-phase inverters convert DC to single-phase AC using a full-bridge (H-bridge) of four switching devices that alternately connect the load to the positive and negative DC bus. Square-wave inverters produce a simple $\pm V_{dc}$ output but contain significant low-order harmonics that are difficult to filter. Pulse Width Modulation (PWM) inverters switch at high frequency (typically 10-100 kHz) and vary the pulse widths within each half-cycle to synthesize a sinusoidal output with harmonics pushed to high frequencies where they are easily filtered by small inductors and capacitors. Sinusoidal PWM (SPWM) compares a sinusoidal reference to a triangular carrier to generate the switching pattern, with the output fundamental voltage controlled by the modulation index $m_a = V_{reference}/V_{carrier}$. Single-phase inverters are used in residential solar inverters, uninterruptible power supplies (UPS), and small variable-frequency drives.

**Example 10.4.1:** A single-phase full-bridge inverter operates from a DC bus voltage of $V_{dc}$ = 400 V with sinusoidal PWM. The modulation index is $m_a$ = 0.85 and the switching frequency is $f_{sw}$ = 20 kHz. Calculate the fundamental RMS output voltage and the frequency of the lowest-order harmonics in the output.

**Solution:**
For a single-phase SPWM inverter, the peak fundamental output voltage is: $V_{1(peak)} = m_a \times V_{dc} = 0.85 \times 400 = 340$ V.
The fundamental RMS voltage: $V_{1(rms)} = 340 / \sqrt{2} = 240.4$ V.
The switching harmonics in SPWM appear as sidebands around multiples of the switching frequency.
The lowest significant harmonics are at frequencies $f_{sw} \pm f_1$, $2f_{sw} \pm f_1$, etc.
For a 60 Hz fundamental: first harmonic cluster is at 20,000 ± 60 Hz, i.e., 19,940 Hz and 20,060 Hz.
These high-frequency components are easily filtered by a small LC output filter.

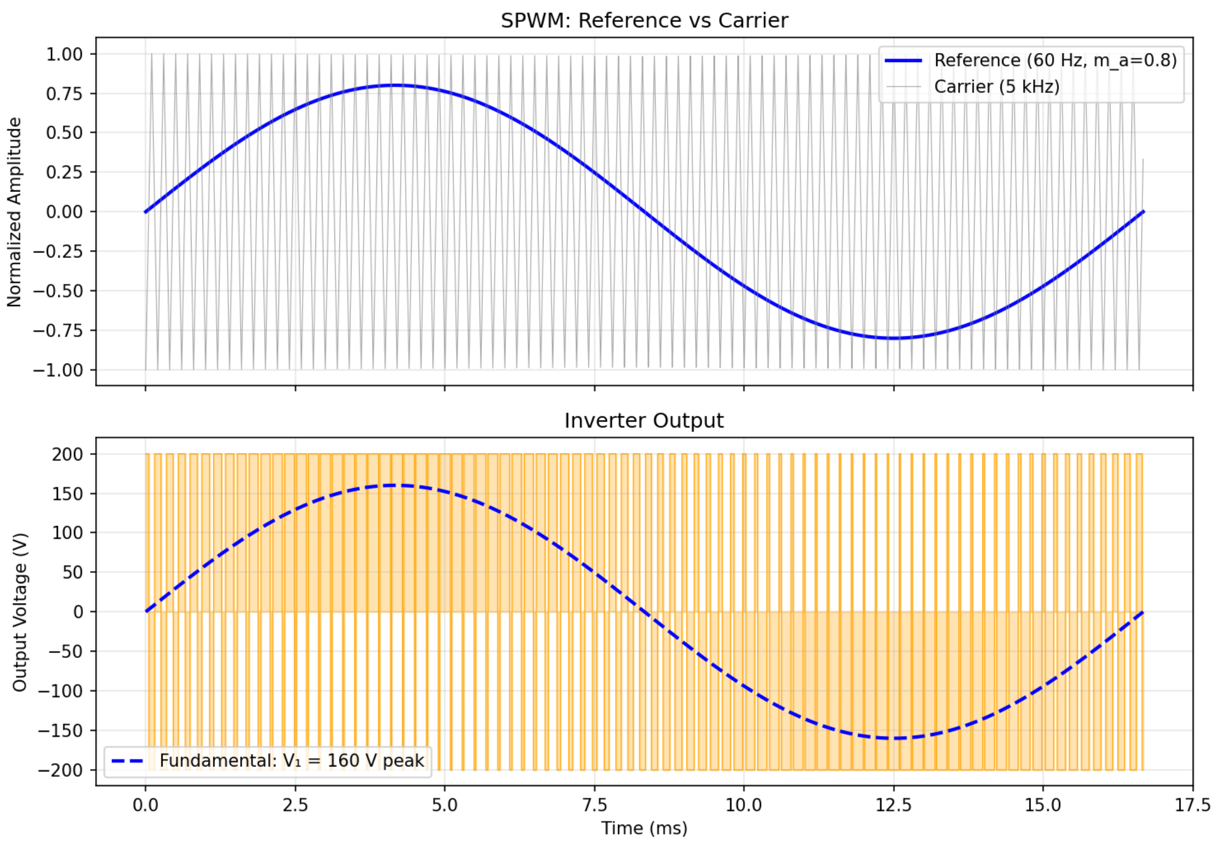



Figure 10.4.1: SPWM Inverter Output

### 10.4.2 Three-Phase Inverters

Three-phase inverters use six switching devices (three half-bridge legs) to produce three-phase AC from a DC source and are the standard topology for industrial motor drives, grid-tied solar and wind inverters, and electric vehicle traction drives. Six-step (120° or 180° conduction) switching produces a quasi-square-wave output with low-order harmonics (5th, 7th, 11th, 13th). Space Vector PWM (SVPWM) is the preferred modulation technique in modern three-phase inverters, providing approximately 15.5% higher DC bus utilization than SPWM and lower harmonic distortion. Multilevel inverters (neutral-point-clamped, flying capacitor, and cascaded H-bridge topologies) synthesize the output waveform using multiple voltage levels, reducing harmonic content and dv/dt stress on the switching devices while enabling operation at higher voltages than individual device ratings would allow. Output LC or LCL filters attenuate the switching-frequency harmonics to meet grid connection standards such as IEEE 1547 and IEC 61000-3-12.

**Example 10.4.2:** A three-phase inverter operates from a $V_{dc}$ = 650 V DC bus using space vector PWM (SVPWM). Calculate the maximum fundamental line-to-line RMS output voltage. Compare this to the output achievable with standard SPWM.
**Solution:**
With SVPWM, the maximum peak fundamental phase voltage is $V_{phase(peak)} = V_{dc} / \sqrt{3} = 650 / 1.732 = 375.3$ V.
The maximum line-to-line RMS voltage: $V_{LL(rms)} = V_{phase(peak)} \times \sqrt{3} / \sqrt{2} = 375.3 \times 1.732 / 1.414 = 375.3 \times 1.225 = 459.7$ V.
With standard SPWM ($m_a = 1.0$): $V_{phase(peak)} = V_{dc} / 2 = 325$ V. $V_{LL(rms)} = 325 \times \sqrt{3} / \sqrt{2} = 325 \times 1.225 = 398.1$ V.
The improvement: 459.7 / 398.1 = 1.155, confirming the 15.5% higher DC bus utilization of SVPWM over SPWM.

### 10.4.3 Multilevel Inverters

Multilevel inverters synthesize the output voltage waveform using three or more discrete voltage levels, reducing the harmonic content and dv/dt stress on the load compared to conventional two-level inverters. By approximating the sinusoidal output with a staircase of smaller voltage steps, each switching transition involves a fraction of the total DC bus voltage, reducing electromagnetic interference, output filter requirements, and insulation stress on motor windings. The three main multilevel topologies are: **Neutral-Point-Clamped (NPC)** inverters, which use clamping diodes to produce three or more voltage levels from a split DC bus with series capacitors; **Flying Capacitor** inverters, which use floating capacitors charged to fractions of the DC bus voltage to create intermediate levels; and **Cascaded H-Bridge (CHB)** inverters, which connect multiple single-phase H-bridge modules in series per phase, each with its own isolated DC source. A three-level NPC inverter (the most common) uses four switches per phase leg and produces output voltage levels of $+V_{dc}/2$, 0, and $-V_{dc}/2$, effectively halving the dv/dt per switching event and eliminating the need for output dv/dt filters in many motor drive applications. Five-level and seven-level topologies further reduce harmonics but increase component count and control complexity. Modular multilevel converters (MMC) — a variant of the CHB approach using dozens to hundreds of submodules per arm — are the dominant topology for HVDC voltage-source converters and STATCOM applications, handling voltages up to ±525 kV with individual submodule voltages of only 1.5-2.5 kV.

**Example 10.4.3:** A three-level NPC inverter operates from a DC bus of $V_{dc}$ = 800 V (±400 V, with a midpoint). Compare the voltage step size and output THD to a two-level inverter at the same bus voltage, both using SPWM at $f_{sw}$ = 5 kHz with a 60 Hz fundamental.
**Solution:**
Two-level inverter: each switching transition steps between +400 V and -400 V, a step of 800 V. The output phase voltage toggles between $+V_{dc}/2$ = +400 V and $-V_{dc}/2$ = -400 V.
dv/dt per transition = 800 V / $t_{rise}$. For a typical 100 ns rise time: dv/dt = $800 / 100 \times 10^{-9}$ = 8 kV/μs.

Three-level NPC: each transition steps between adjacent levels (0 to +400 V, or 0 to -400 V), a step of 400 V — half the two-level value. dv/dt = 400 / 100 × $10^{-9}$ = 4 kV/μs (50% reduction).
The three-level output has an effective switching frequency of 2 × $f_{sw}$ = 10 kHz as seen by the load (each phase produces two switching events per carrier cycle), pushing the lowest harmonic cluster to 10 kHz ± 60 Hz instead of 5 kHz ± 60 Hz.
The weighted THD of the three-level output is typically 40-50% lower than the two-level output at the same switching frequency, or equivalently, the three-level inverter can achieve the same THD at half the switching frequency, reducing switching losses by 50%.

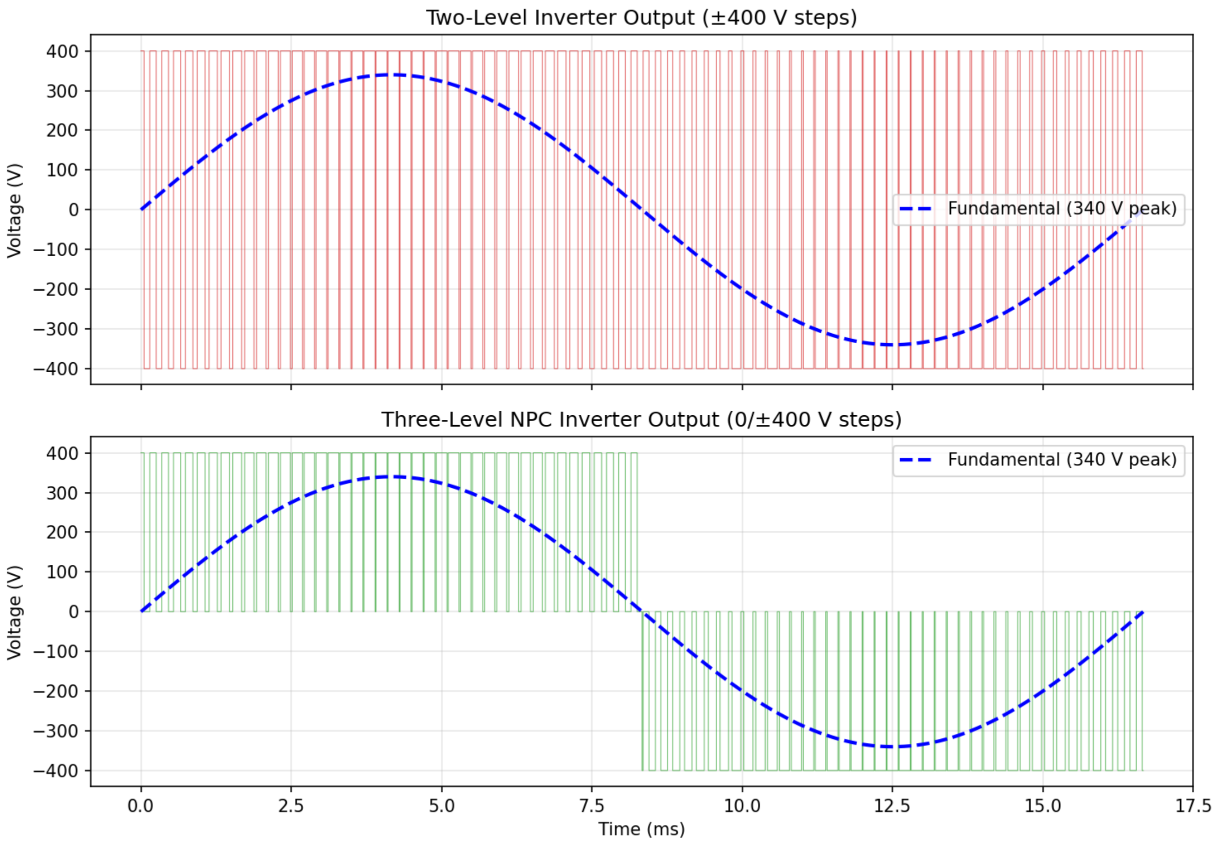



Figure 10.4.3: Multilevel Inverter: 2-Level vs 3-Level NPC

## 10.5 AC-AC Converters

AC-AC converters change the voltage, frequency, or both of an AC power source without an intermediate DC stage (or with a minimal one). Phase-controlled AC voltage controllers adjust the RMS voltage delivered to a load, while cycloconverters and matrix converters provide frequency conversion for specialized motor drive and power conditioning applications.

### 10.5.1 AC Voltage Controllers

AC voltage controllers regulate the RMS value of the AC output voltage by controlling the conduction angle of thyristors or triacs connected in series with the load. Phase-angle control delays the firing of the thyristors by an angle α each half-cycle, reducing the RMS output voltage according to $V_{out} = V_{in} \times \sqrt{((\pi - \alpha + \sin(2\alpha)/2)/\pi)}$ for resistive loads. Common applications include lighting dimmers, heating controls, and soft starters for induction motors that limit the starting current by gradually increasing the voltage. The primary disadvantage of phase-angle control is the generation of low-order harmonics and poor power factor at reduced output voltages. Integral cycle control (burst firing) is an alternative that switches complete half-cycles on and off, producing no additional harmonics but introducing sub-harmonic flicker that limits its use to loads with long thermal time constants such as ovens and heaters.

**Example 10.5.1:** A single-phase AC voltage controller with back-to-back thyristors supplies a 10 Ω resistive heating load from a 240 $V_{rms}$, 50 Hz source. The firing angle is set to α = 90°. Calculate the RMS output voltage and the power delivered to the load.
**Solution:**
For a resistive load with phase-angle control: $V_{out(rms)} = V_{in} \times \sqrt{[(\pi - \alpha + \sin(2\alpha)/2) / \pi]}$.
With $\alpha = 90° = \pi/2$ radians: $\sin(2 \times 90°) = \sin(180°) = 0$.
$V_{out(rms)} = 240 \times \sqrt{[(\pi - \pi/2 + 0) / \pi]} = 240 \times \sqrt{[0.5]} = 240 \times 0.707 = 169.7$ V.
Power delivered: $P = V_{out(rms)}^2 / R = (169.7)^2 / 10 = 28{,}798 / 10 = 2{,}880$ W.
This is exactly 50% of the full power ($240^2/10 = 5{,}760$ W), which makes sense since exactly half the waveform is conducted at α = 90°.

### 10.5.2 Cycloconverters

A cycloconverter converts AC at one frequency directly to AC at a different (typically lower) frequency without an intermediate DC link, using naturally commutated thyristors. It synthesizes the output waveform by selecting segments of the input voltage waveform, with the output frequency limited to approximately one-third of the input frequency for acceptable output quality. Three-phase to single-phase cycloconverters use two back-to-back controlled rectifier bridges (one for positive and one for negative output half-cycles), while three-phase to three-phase configurations use six bridges. Cycloconverters are used in high-power, low-speed motor drives (cement mills, mine hoists, ship propulsion) where their ability to handle megawatt-level power without high-frequency switching is advantageous. Modern matrix converters offer a more compact alternative by providing bidirectional AC-AC conversion with controllable output frequency and amplitude using bidirectional switch arrays without energy storage elements.

**Example 10.5.2:** A three-phase to single-phase cycloconverter is fed from a 60 Hz, 480 $V_{rms}$ supply and must produce a 15 Hz output for a low-speed cement mill drive. Verify this output frequency is achievable and determine the maximum output voltage.
**Solution:**
The maximum recommended output frequency for a cycloconverter is $f_{out(max)} \approx f_{in} / 3 = 60 / 3 = 20$ Hz. Since 15 Hz < 20 Hz, this frequency is achievable with acceptable waveform quality.
The maximum output voltage of a cycloconverter equals the maximum DC output of the constituent controlled rectifier bridges.
For a three-phase fully controlled bridge: $V_{dc(max)} = (3/\pi) \times V_{peak(L\text{-}L)} = 0.9549 \times 678.8 = 648.2$ V (average DC output at firing angle α = 0°).
The RMS output voltage at maximum is approximately $V_{out(rms)} \approx V_{dc(max)} / \sqrt{2} = 648.2 / 1.414 \approx 458.4$ V.

In practice, the output is limited to about 75–80% of the input voltage due to commutation overlap, yielding approximately $V_{out} \approx 0.75 \times 480 = 360\ V_{rms}$.

### 10.5.3 Matrix Converters

A matrix converter is a direct AC-to-AC power converter that connects each output phase to any input phase through a matrix of bidirectional switches, providing frequency and voltage conversion without any intermediate DC energy storage (no DC link capacitor or inductor). For a three-phase to three-phase matrix converter, nine bidirectional switches arranged in a 3×3 matrix connect three input phases to three output phases. Each bidirectional switch is typically realized with two back-to-back IGBTs or MOSFETs with antiparallel diodes (or with reverse-blocking IGBTs that eliminate the need for series diodes). The space vector modulation algorithm selects switch states to synthesize the desired output voltage and frequency while simultaneously controlling the input current to be sinusoidal with a controllable displacement power factor. The maximum voltage transfer ratio is limited to $q_{max} = \sqrt{3}/2 \approx 0.866$ (86.6% of the input voltage) without overmodulation, which is a fundamental limitation compared to back-to-back voltage source converters that can achieve unity or higher voltage ratios. Key advantages of matrix converters include: sinusoidal input and output currents, controllable input power factor (can operate at unity or leading/lagging), compact size due to elimination of bulky DC link capacitors, inherent four-quadrant operation (regeneration capability), and theoretically unlimited output frequency range. Practical challenges include the need for a clamp circuit to protect against overvoltage during commutation, a complex four-step commutation strategy to avoid short-circuiting input phases or open-circuiting inductive loads, and sensitivity to input voltage disturbances. Matrix converters are used in aerospace drives (weight-critical applications), compressor drives, and wind energy systems where bidirectional power flow and compact size are valued.

**Example 10.5.3:** A three-phase matrix converter is fed from a 400 $V_{rms}$ line-to-line, 50 Hz source and must supply a 30 Hz, three-phase output to an induction motor rated at 15 kW. Calculate the maximum output line-to-line voltage, the output line current, and verify the input line current assuming unity input displacement power factor and 97% converter efficiency.
**Solution:**
Maximum output voltage: $V_{out(LL)} = q_{max} \times V_{in(LL)} = 0.866 \times 400 = 346.4\ V_{rms}$.
Output line current at rated power: $I_{out} = P/(\sqrt{3} \times V_{out(LL)}) = 15{,}000/(1.732 \times 346.4) = 15{,}000/599.9 = 25.0$ A.
Input power (accounting for 97% efficiency): $P_{in} = P_{out}/\eta = 15{,}000/0.97 = 15{,}464$ W.
Input line current: $I_{in} = P_{in}/(\sqrt{3} \times V_{in(LL)} \times \cos\varphi) = 15{,}464/(1.732 \times 400 \times 1.0) = 15{,}464/692.8 = 22.3$ A.
The input current (22.3 A) is lower than the output current (25.0 A) because the input voltage is higher than the output voltage — the converter steps down voltage and correspondingly steps up current, conserving power.

## 10.6 Thermal Management

Every watt of power lost in a switching converter is dissipated as heat, and the resulting temperature rise must be managed to keep semiconductor junctions within their safe operating limits. Thermal design uses an electrical analog — thermal resistance (°C/W) models the path from the junction through the package, thermal interface, heat sink, and ultimately to the ambient environment. Proper thermal management is often the factor that limits converter power density and determines physical size.

### 10.6.1 Power Losses

Power losses in switching converters consist of conduction losses and switching losses in the semiconductor devices, core losses and copper losses in magnetic components, and resistive losses in conductors and interconnects. Conduction losses in MOSFETs are $P_{cond} = I_{rms}{}^2 \times R_{DS(on)}$, which increases with temperature due to the positive temperature coefficient of MOSFET on-resistance. Switching losses occur during the transition between on and off states when both voltage across and current through the device are simultaneously non-zero: $P_{sw} = ½ \times V \times I \times (t_{rise} + t_{fall}) \times f_{sw}$. In IGBTs, additional tail current losses occur during turn-off. Gate drive losses, reverse recovery losses in diodes, and dead-time-related losses contribute to overall switching losses. Accurate loss estimation is essential for thermal design, efficiency optimization, and selection of the appropriate switching frequency.

**Example 10.6.1:** A MOSFET in a buck converter has $R_{DS(on)} = 30\ m\Omega$ and carries an RMS current of 8 A. The converter switches at $f_{sw} = 300$ kHz, with $V_{DS} = 48$ V, $I_D = 10$ A (at switching instants), $t_{rise} = 15$ ns, and $t_{fall} = 25$ ns. The gate charge is $Q_g = 30$ nC and the gate drive voltage is $V_{gs} = 10$ V. Calculate total MOSFET losses.
**Solution:**
Conduction loss: $P_{cond} = I_{rms}{}^2 \times R_{DS(on)} = 8^2 \times 0.030 = 1.92$ W.
Switching loss: $P_{sw} = ½ \times V_{DS} \times I_D \times (t_{rise} + t_{fall}) \times f_{sw} = ½ \times 48 \times 10 \times (15 + 25) \times 10^{-9} \times 300 \times 10^3 = ½ \times 48 \times 10 \times 40 \times 10^{-9} \times 3 \times 10^5 = 2.88$ W.
Gate drive loss: $P_{gate} = Q_g \times V_{gs} \times f_{sw} = 30 \times 10^{-9} \times 10 \times 300 \times 10^3 = 0.09$ W.
Total MOSFET loss: $P_{total} = 1.92 + 2.88 + 0.09 = 4.89$ W.

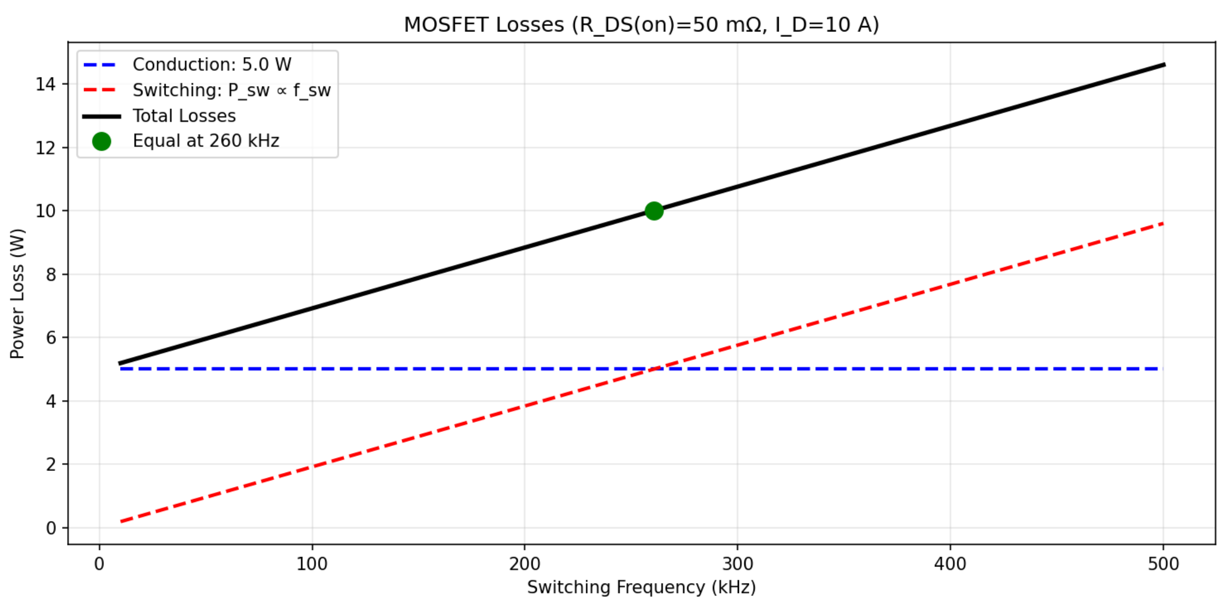



Figure 10.6.1: MOSFET Power Losses vs Switching Frequency

### 10.6.2 Heat Sinks and Cooling

Semiconductor devices must be kept within their rated junction temperature (typically 125-175°C) to ensure reliable operation and adequate lifetime. The thermal path from junction to ambient is modeled as a series of thermal resistances: $R_{\theta JC}$ (junction to case) + $R_{\theta CS}$ (case to heat sink) + $R_{\theta SA}$ (heat sink to ambient), with the total temperature rise $\Delta T = P_{loss} \times (R_{\theta JC} + R_{\theta CS} + R_{\theta SA})$. Heat sinks are passive or active thermal management devices that increase the surface area for heat dissipation to the surrounding air. Natural convection heat sinks rely on buoyancy-driven airflow and are suitable for low to moderate power dissipation, while forced-air cooling using fans can reduce thermal resistance by a factor of 3-10. For high-power-density applications, liquid cooling (cold plates with water or water-glycol mixtures) provides thermal resistances an order of magnitude lower than air cooling, enabling compact designs in electric vehicle inverters, data center power supplies, and high-power industrial drives.

**Example 10.6.2:** An IGBT module dissipates $P_{loss} = 200$ W. The thermal resistances are $R_{\theta JC} = 0.15$ °C/W (junction to case), $R_{\theta CS} = 0.05$ °C/W (case to heat sink, with thermal compound), and $R_{\theta SA} = 0.25$ °C/W (heat sink to ambient). The ambient temperature is $T_A = 40$°C and the maximum junction temperature is $T_{J(max)} = 150$°C. Determine whether the thermal design is adequate.
**Solution:**
Total thermal resistance: $R_{\theta JA} = R_{\theta JC} + R_{\theta CS} + R_{\theta SA} = 0.15 + 0.05 + 0.25 = 0.45$ °C/W.
Temperature rise: $\Delta T = P_{loss} \times R_{\theta JA} = 200 \times 0.45 = 90$°C.
Junction temperature: $T_J = T_A + \Delta T = 40 + 90 = 130$°C.
Since $T_J = 130°C < T_{J(max)} = 150$°C, the design is adequate with a 20°C margin.
The heat sink surface temperature: $T_{HS} = T_A + P_{loss} \times R_{\theta SA} = 40 + 200 \times 0.25 = 90$°C.

### 10.6.3 Thermal Interface Materials and Liquid Cooling

The thermal interface between a semiconductor package and its heat sink is a critical link in the thermal chain, and the choice of thermal interface material (TIM) directly impacts junction temperature and converter reliability. Without a TIM, microscopic surface roughness leaves air gaps (thermal conductivity ~0.025 W/m·K) that create high thermal resistance between the two mating surfaces. **Thermal grease** (silicone-based with metal oxide or ceramic fillers) fills these gaps and provides thermal conductivity of 1–5 W/m·K with thin bond lines (25–75 μm), achieving $R_{\theta CS}$ values of 0.05–0.2 °C/W for typical power module footprints. **Phase-change materials** (PCMs) are solid at room temperature but soften at operating temperatures (typically 50–60°C), flowing into surface irregularities like grease but with the handling convenience of a solid pad; they offer conductivity of 3–5 W/m·K. **Thermal pads** (silicone elastomer with ceramic filler) provide electrical isolation and gap-filling capability with conductivity of 1–6 W/m·K, but their greater thickness (0.25–2 mm) results in higher thermal resistance than grease. **Solder** or **sintered silver** TIMs provide the lowest thermal resistance (conductivity 50–250 W/m·K) and are used for die-attach in power modules; sintered silver interfaces achieve $R_\theta$ values 5–10× lower than thermal grease. For high-power-density converters dissipating hundreds of watts to kilowatts per device — such as EV traction inverters, data center power shelves, and MW-class industrial drives — **liquid cooling** with cold plates replaces air-cooled heat sinks. A cold plate is a thermally conductive metal block (aluminum or copper) with internal channels through which a coolant (water, water-glycol mixture, or dielectric fluid) flows, removing heat by forced convection. Liquid-cooled cold plates achieve thermal resistances of 0.01–0.05 °C/W — an order of magnitude lower than forced-air heat sinks — enabling power densities above 50 W/cm$^2$. The coolant loop includes a pump, reservoir, radiator or chiller, and temperature/flow sensors. Pin-fin and microchannel cold plate designs maximize the wetted surface area within the plate, with microchannel designs achieving heat transfer coefficients above 10,000 $W/m^2{\cdot}K$ but at higher pressure drop. Two-phase cooling (using boiling coolant) and jet impingement are emerging approaches for next-generation power electronics exceeding 200 $W/cm^2$ power density.

**Example 10.6.3:** A SiC power module dissipates 400 W and is mounted on a liquid-cooled cold plate. The thermal interface uses thermal grease with conductivity $k_{TIM} = 3.5$ W/m·K and bond-line thickness t = 50 μm over a 40 mm × 50 mm contact area. The cold plate has $R_{\theta(\text{plate-to-coolant})} = 0.02$ °C/W, and the coolant inlet temperature is 25°C. The module's $R_{\theta JC} = 0.08$ °C/W. Calculate the junction temperature and compare to an air-cooled design with $R_{\theta SA} = 0.15$ °C/W.
**Solution:**
TIM thermal resistance: $R_{\theta CS} = t/(k_{TIM} \times A) = (50 \times 10^{-6})/(3.5 \times 0.040 \times 0.050) = 50 \times 10^{-6} / (7.0 \times 10^{-3}) = 0.00714$ °C/W.
**Liquid-cooled:** Total $R_\theta = R_{\theta JC} + R_{\theta CS} + R_{\theta(\text{plate})} = 0.08 + 0.007 + 0.02 = 0.107$ °C/W. $T_J = 25 + 400 \times 0.107 = 25 + 42.8 = 67.8$°C.
**Air-cooled:** Total $R_\theta = R_{\theta JC} + R_{\theta CS} + R_{\theta SA} = 0.08 + 0.007 + 0.15 = 0.237$ °C/W. $T_J = 40 + 400 \times 0.237 = 40 + 94.8 = 134.8$°C (using 40°C ambient for air-cooled).
The liquid-cooled design runs 67°C cooler, keeping the junction at 67.8°C versus 134.8°C — well within the SiC module's 175°C rating with over 100°C of thermal margin, enabling higher power or a smaller module.

### 10.6.4 EMI and Filtering in Power Converters

Switching power converters are inherent sources of electromagnetic interference (EMI) because their operation depends on rapidly changing voltages and currents. The steep dv/dt and di/dt transitions during switching events generate broadband noise that propagates as **conducted emissions** through the power lines (150 kHz – 30 MHz) and as **radiated emissions** from PCB traces, cables, and component leads (30 MHz – 1 GHz). International standards — CISPR 32 (multimedia equipment), CISPR 11 (industrial, scientific, medical), FCC Part 15 (USA), and EN 55032 (EU) — impose strict limits on both conducted and radiated emissions to prevent interference with other equipment. Conducted emissions are measured with a Line Impedance Stabilization Network (LISN) that presents a defined 50 Ω impedance to the equipment under test, separating the emission measurement from the utility supply. Emissions are categorized as **differential mode** (DM) noise flowing in opposite directions on the line and neutral conductors, and **common mode** (CM) noise flowing in the same direction on both conductors and returning through the ground. DM noise is generated by the pulsating input current of the converter and is filtered by series inductors and parallel capacitors (X-capacitors, typically 0.1–2.2 μF film type) placed across the line and neutral. CM noise arises from parasitic capacitive coupling between switching nodes and the grounded heat sink or chassis, and is filtered by common-mode chokes (coupled inductors that present high impedance to CM current but low impedance to DM current) and Y-capacitors (typically 1–4.7 nF ceramic, limited by safety leakage current requirements) connected from each line to ground. A well-designed EMI filter typically uses a two-stage $\pi$ or T network with both CM and DM filtering elements, achieving 40–80 dB of attenuation in the 150 kHz – 30 MHz band. PCB layout is critical for EMI performance: minimizing the high-di/dt loop area (the "hot loop" between the switching device, freewheeling diode/synchronous rectifier, and input capacitor) reduces radiated emissions, while a solid ground plane provides a low-impedance return path and shielding. Snubber circuits (RC networks across switching devices) dampen voltage ringing caused by parasitic inductance and device capacitance, reducing high-frequency spectral content.

**Example 10.6.4:** A 200 W flyback converter operating at $f_{sw} = 100$ kHz fails conducted emission testing per CISPR 32 Class B, exceeding the quasi-peak limit by 18 dB at the fundamental switching frequency (100 kHz) and by 12 dB at the third harmonic (300 kHz). Design an input EMI filter to achieve compliance with at least 6 dB of margin.
**Solution:**
Required attenuation at 100 kHz: $18 + 6 = 24$ dB. At 300 kHz: $12 + 6 = 18$ dB.
A single-stage LC filter has a roll-off of −40 dB/decade above its corner frequency. For 24 dB of attenuation at 100 kHz, the corner frequency must be below 100 kHz.
Using a second-order LC filter: attenuation $A = (f/f_0)^2$ at frequencies well above $f_0$. 24 dB $= 10^{24/20} = 15.85\times$.
$f_0 = f/\sqrt{A} = 100$ kHz / $\sqrt{15.85} = 100 / 3.98 = 25.1$ kHz.
Choose $L_{DM} = 100$ μH (DM inductor) and calculate C: $f_0 = 1/(2\pi\sqrt{(LC)})$, so $C = 1/(4\pi^2 f_0^2 L) = 1/(4\pi^2 \times (25{,}100)^2 \times 100 \times 10^{-6}) = 1/(4 \times 9.87 \times 6.3 \times 10^8 \times 10^{-4}) = 1/(2.49 \times 10^6) = 0.40$ μF.
Use a 0.47 μF X2 film capacitor.
Verify at 300 kHz: attenuation $= (300/25.1)^2 = (11.95)^2 = 142.8 = 43.1$ dB $\gg 18$ dB required.
Add a common-mode choke (10 mH, rated for line current) and two 2.2 nF Y-capacitors for CM filtering.
Total filter: CM choke → X-cap (0.47 μF) → DM inductor (100 μH) → second X-cap (0.47 μF), providing both CM and DM attenuation with 6+ dB margin.

## 10.7 Power Factor Correction

Power factor correction (PFC) ensures that equipment connected to the AC mains draws current in phase with the voltage and with low harmonic content, minimizing reactive power demand and complying with international harmonic standards (IEC 61000-3-2, EN 61000-3-2). Without PFC, a typical offline power supply with a diode bridge and capacitor filter draws current only at the peaks of the AC voltage, resulting in high peak-to-average current ratios, a power factor as low as 0.5–0.65, and total harmonic distortion (THD) exceeding 100%. Active PFC circuits shape the input current to follow the input voltage waveform, achieving power factors above 0.99 and THD below 5%.

### 10.7.1 Active PFC Topologies

The **boost PFC** converter is the most widely used active PFC topology, operating in continuous conduction mode (CCM) with average current mode control. A boost stage is placed between the diode bridge rectifier and the DC bus capacitor, with a controller that forces the inductor current to track the rectified sinusoidal input voltage shape. The output voltage (typically 380–400 V DC for universal-input designs) must be higher than the peak input voltage (peak of 265 $V_{rms} \times \sqrt{2} = 375$ V for worldwide mains). The duty cycle varies throughout the AC cycle to maintain sinusoidal current draw at each instant. **Critical conduction mode** (CrCM, or transition mode) PFC operates at the boundary between CCM and DCM with variable switching frequency, simplifying the control loop and achieving natural ZVS of the boost diode. CrCM is common in power levels from 75 W to 300 W (LED drivers, small adapters). **Interleaved PFC** uses two or more boost stages operated with phase-shifted switching to reduce input current ripple, output voltage ripple, and component stress, and is used in server power supplies and EV chargers above 1 kW. **Bridgeless PFC** topologies (totem-pole, dual-boost) eliminate the input diode bridge, reducing conduction losses by one diode drop per half-cycle and improving efficiency by 0.5–1%, enabled by GaN or SiC devices that can handle bidirectional current.

**Example 10.7.1:** A boost PFC converter operates from a 230 $V_{rms}$ input (50 Hz) and produces a 400 V DC output at 500 W. The switching frequency is 65 kHz and the boost inductor is 500 µH. Calculate the peak input current, the duty cycle at the peak of the AC input, and the inductor current ripple at that point.

**Solution:**

Input current (assuming PF ≈ 1.0): $I_{in(rms)} = P / V_{in(rms)} = 500 / 230 = 2.174\ A_{rms}$.

Peak input current: $I_{in(peak)} = I_{in(rms)} \times \sqrt{2} = 2.174 \times 1.414 = 3.074$ A.

Peak input voltage: $V_{in(peak)} = 230 \times \sqrt{2} = 325.3$ V.

Duty cycle at the peak of the AC input: $D = 1 - V_{in}/V_{out} = 1 - 325.3/400 = 1 - 0.8133 = 0.1867$ (18.7%).

This is the minimum duty cycle in the AC cycle; at the zero crossings, D approaches 1.0.

Inductor current ripple at the peak: $\Delta I_L = V_{in(peak)} \times D / (L \times f_{sw}) = 325.3 \times 0.1867 / (500 \times 10^{-6} \times 65 \times 10^3) = 60.74 / 32.5 = 1.869\ A_{pp}$.

The ripple ratio is $\Delta I_L / I_{peak} = 1.869 / 3.074 = 60.8\%$, which is higher than desired for CCM operation.

A larger inductor (e.g., 1 mH) would halve the ripple to 30%, or increasing the switching frequency to 130 kHz would achieve the same reduction.

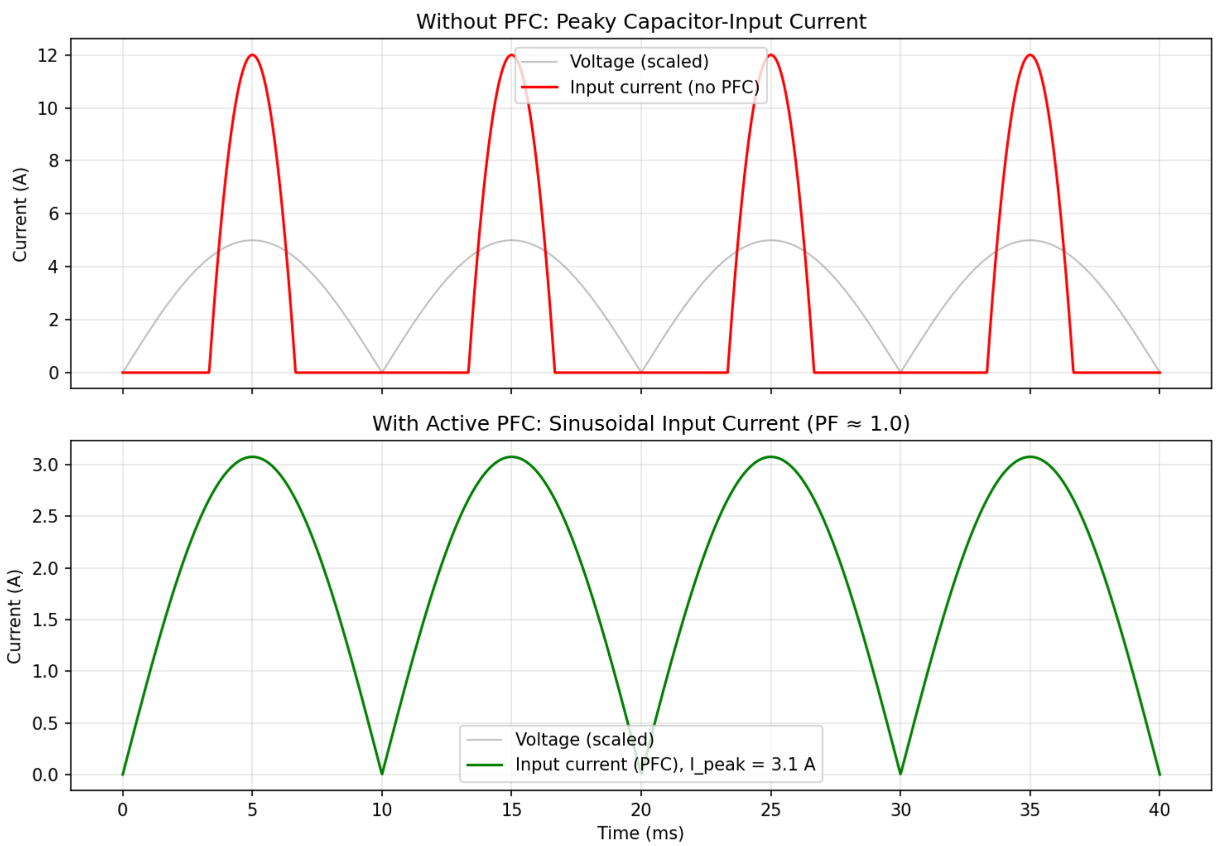



Figure 10.7.1: Active PFC Current Shaping

### 10.7.2 Bridgeless PFC Topologies

Bridgeless PFC topologies eliminate the input diode bridge rectifier, removing two diode forward-voltage drops from the current path and improving overall converter efficiency by 0.5-1.5 percentage points — a significant gain in high-efficiency server, telecom, and EV charger power supplies targeting 80 PLUS Titanium (96%+ efficiency) or equivalent standards. The **totem-pole bridgeless PFC** is the most popular topology, using two active switches (typically GaN HEMTs) in a half-bridge configuration that directly switches the AC line voltage. During the positive half-cycle, one switch operates as the boost switch (high-frequency PWM) while the other acts as the synchronous rectifier; their roles reverse during the negative half-cycle. A second pair of low-frequency switches (silicon MOSFETs or diodes) commutates at the line frequency to steer current during each half-cycle. The totem-pole topology requires devices capable of reverse conduction with low recovery charge — GaN HEMTs are ideal because they have zero reverse recovery loss (the 2DEG channel conducts in reverse without minority carrier storage). SiC MOSFETs can also be used but their body diode has a small reverse recovery charge that must be managed. The **dual-boost bridgeless PFC** uses two boost inductors and two active switches, with each switch handling one half-cycle; it maintains the same ground reference between input and output but requires two inductors. Bridgeless topologies require careful common-mode noise management because the output DC bus is no longer referenced to the AC neutral through the diode bridge, creating high dv/dt common-mode voltages that couple through parasitic capacitances. Dedicated common-mode EMI filtering (Y-capacitors, common-mode chokes) is essential.

**Example 10.7.2:** A 3 kW totem-pole bridgeless PFC converter uses GaN HEMTs with $R_{DS(on)}$ = 50 mΩ (including package) and low-frequency silicon MOSFETs with $R_{DS(on)}$ = 8 mΩ. The input is 230 $V_{rms}$ and the output is 400 V DC. Compare the conduction losses to a conventional boost PFC using a diode bridge with $V_F$ = 0.9 V per diode and a silicon MOSFET with $R_{DS(on)}$ = 50 mΩ.
**Solution:**
Input current: $I_{in(rms)} = P / V_{in} = 3000 / 230 = 13.04$ A.
**Conventional boost PFC:** Diode bridge loss (two diodes conduct at any time): $P_{bridge} = 2 \times V_F \times I_{avg} = 2 \times 0.9 \times (13.04 \times 2\sqrt{2}/\pi) = 2 \times 0.9 \times 11.73 = 21.1$ W (using average rectified current).
Boost MOSFET conduction loss: the MOSFET RMS current $\approx I_{in(rms)} \times \sqrt{D_{avg}}$. With average duty $\approx 0.42$ for 230 V / 400 V: $I_{Q(rms)} \approx 13.04 \times \sqrt{0.42} = 8.45$ A. $P_Q = 8.45^2 \times 0.050 = 3.57$ W.
Total conduction: 21.1 + 3.57 = 24.7 W.
**Totem-pole bridgeless PFC:** High-frequency GaN switch conduction: $P_{GaN} = I_{Q(rms)}^2 \times R_{DS(on)} = 8.45^2 \times 0.050 = 3.57$ W.
Synchronous rectifier GaN (complementary duty): $I_{SR(rms)} \approx 13.04 \times \sqrt{(1 - 0.42)} = 13.04 \times 0.762 = 9.93$ A. $P_{SR} = 9.93^2 \times 0.050 = 4.93$ W.
Low-frequency Si MOSFETs: $P_{LF} = 13.04^2 \times 0.008 = 1.36$ W.
Total conduction: 3.57 + 4.93 + 1.36 = 9.86 W.
The bridgeless topology reduces conduction losses from 24.7 W to 9.86 W — a 60% reduction (14.8 W saved), improving full-load efficiency by 14.8/3000 = 0.49 percentage points.
At lighter loads where diode drops dominate, the improvement is even more pronounced.

### 10.7.3 Power Supply Protection Circuits

Robust protection circuits are essential in power electronic converters to safeguard both the converter and the load from damage during fault conditions, startup transients, and abnormal operating events.

**Overvoltage protection (OVP)** monitors the output voltage and shuts down the converter (or clamps the output with a crowbar circuit) if the voltage exceeds a safe threshold — typically 110–120% of the nominal output. In a crowbar OVP circuit, a thyristor fires when a zener-referenced comparator detects an overvoltage, short-circuiting the output and blowing a fuse or triggering the upstream breaker; this is a last-resort protection used in critical loads like server motherboards where sustained overvoltage is more destructive than a momentary outage. Active OVP in the controller IC reduces the duty cycle or disables switching when the feedback voltage exceeds an internal threshold. **Overcurrent protection (OCP)** limits the output current to prevent damage from short circuits or overloads. In peak current mode control, OCP is inherent — a cycle-by-cycle current limit clamps the peak inductor current within one switching period (microseconds). Hiccup mode protection detects a sustained overcurrent condition, shuts down the converter for a cooldown period (typically 100 ms – 1 s), then attempts a restart; this cycle repeats until the fault clears, limiting average power dissipation during a short circuit to safe levels. **Soft start** gradually ramps the converter's duty cycle or reference voltage over a controlled interval (typically 5–50 ms) after enable, preventing the inrush current surge that would otherwise occur when charging the output capacitor from zero. Without soft start, the instantaneous current into a discharged output capacitor can be many times the rated load current, stressing the switching devices and triggering OCP. Soft start is implemented by slowly ramping an internal voltage reference or by using a capacitor on a dedicated SS pin that charges through an internal current source. **Inrush current limiting** addresses the surge current at initial power-on when bulk input capacitors charge through the AC mains. A negative-temperature-coefficient (NTC) thermistor in series with the input presents high resistance when cold (limiting inrush to 10–20 A), then self-heats to low resistance (0.5–2 Ω) during steady-state operation. For higher-power supplies (>500 W), an active inrush limiter uses a relay or MOSFET to bypass a current-limiting resistor after the capacitors have charged, avoiding the continuous power dissipation of an NTC. **Undervoltage lockout (UVLO)** prevents the converter from operating when the input voltage is too low to sustain proper regulation, avoiding undefined or destructive behavior. UVLO circuits have hysteresis (typically 1–3 V) to prevent oscillation at the turn-on threshold.

**Example 10.7.3:** A 48 V, 20 A power supply has a 2,000 μF output capacitor bank and uses soft start with a 10 ms ramp time. The converter's OCP threshold is set at 25 A. Without soft start, the initial duty cycle would step to the steady-state value immediately. Estimate the peak inrush current without soft start (assuming the converter output impedance during startup is dominated by the MOSFET $R_{DS(on)} = 15\ m\Omega$) and verify that the soft-start ramp keeps the current below the OCP limit.

**Solution:**

**Without soft start:** If the duty cycle steps immediately to the steady-state value, the converter tries to charge the 2,000 μF capacitor from 0 V to 48 V as fast as the inductor and switch allow. The peak current is limited only by the inductor slew rate and OCP; in practice, the current slams into the OCP limit of 25 A on the first switching cycle, triggering hiccup-mode protection and delaying startup.

**With soft start (10 ms ramp):** The output voltage ramps approximately linearly: dV/dt = 48 V / 10 ms = 4,800 V/s.

The capacitor charging current: $I_{cap} = C \times dV/dt = 2{,}000 \times 10^{-6} \times 4{,}800 = 9.6$ A.

Adding the steady-state load current that develops as $V_{out}$ ramps (load current is small initially when $V_{out}$ is low), the peak current occurs near the end of the ramp: $I_{peak} \approx I_{cap} + I_{load} \approx 9.6 + 20 = 29.6$ A, which still exceeds 25 A OCP.

Increasing the soft-start time to 20 ms: $I_{cap} = 2{,}000 \times 10^{-6} \times (48/0.020) = 4.8$ A. $I_{peak} \approx 4.8 + 20 = 24.8$ A, just under the 25 A OCP limit.
Use 25 ms for adequate margin: $I_{cap} = 3.84$ A, $I_{peak} \approx 23.8$ A with 1.2 A margin below OCP.

## 10.8 Battery Management Systems

Battery management systems (BMS) monitor, protect, and optimize rechargeable battery packs used in electric vehicles, grid energy storage, UPS systems, and portable electronics. A BMS measures individual cell voltages, pack current, and temperatures to prevent operation outside safe limits and to maximize the usable capacity and lifetime of the battery. Modern lithium-ion cells can deliver exceptional energy density and cycle life but are intolerant of overcharge, over-discharge, and excessive temperature — conditions that cause irreversible capacity loss, thermal runaway, or fire. The BMS is the critical safety and performance layer between the battery cells and the power electronic converter that charges or discharges them.

### 10.8.1 Battery Cell Characteristics

A rechargeable battery cell converts chemical energy to electrical energy through reversible electrochemical reactions. The key parameters are nominal voltage (determined by cell chemistry), capacity (in amp-hours, the total charge the cell can deliver), energy density (Wh/kg or Wh/L), C-rate (discharge current normalized to capacity — 1C fully discharges the cell in one hour), and cycle life (number of charge-discharge cycles before capacity drops to 80% of initial). Lithium-ion cells dominate modern applications with chemistries including LFP ($LiFePO_4$, 3.2 V nominal, long cycle life, excellent safety), NMC ($LiNiMnCoO_2$, 3.7 V, high energy density), NCA ($LiNiCoAlO_2$, 3.6 V, highest energy density, used in EVs), and LTO ($Li_4Ti_5O_{12}$, 2.4 V, fastest charging, extreme cycle life). The open-circuit voltage (OCV) varies with state of charge following a chemistry-dependent curve, and internal resistance increases with age and low temperature.

**Example 10.8.1:** A battery pack for an electric vehicle uses 96 series-connected NMC cells (3.7 V nominal, 50 Ah each). Determine (a) the nominal pack voltage, (b) the energy capacity in kWh, (c) the maximum continuous discharge current at 3C, and (d) the pack internal resistance if each cell has 2 mΩ internal resistance.
**Solution:**
(a) $V_{pack} = 96 \times 3.7 =$ **355.2 V**
(b) Energy = $V_{pack} \times$ Capacity = $355.2 \times 50 = 17{,}760$ Wh = **17.76 kWh**
(c) Maximum current at 3C: $I_{max} = 3 \times 50 =$ **150 A**
(d) Series cells add resistance: $R_{pack} = 96 \times 0.002 =$ **0.192 Ω**. At 150 A discharge, the voltage drop across internal resistance is $I \times R = 150 \times 0.192 = 28.8$ V, reducing the terminal voltage to $355.2 - 28.8 =$ **326.4 V** under load.

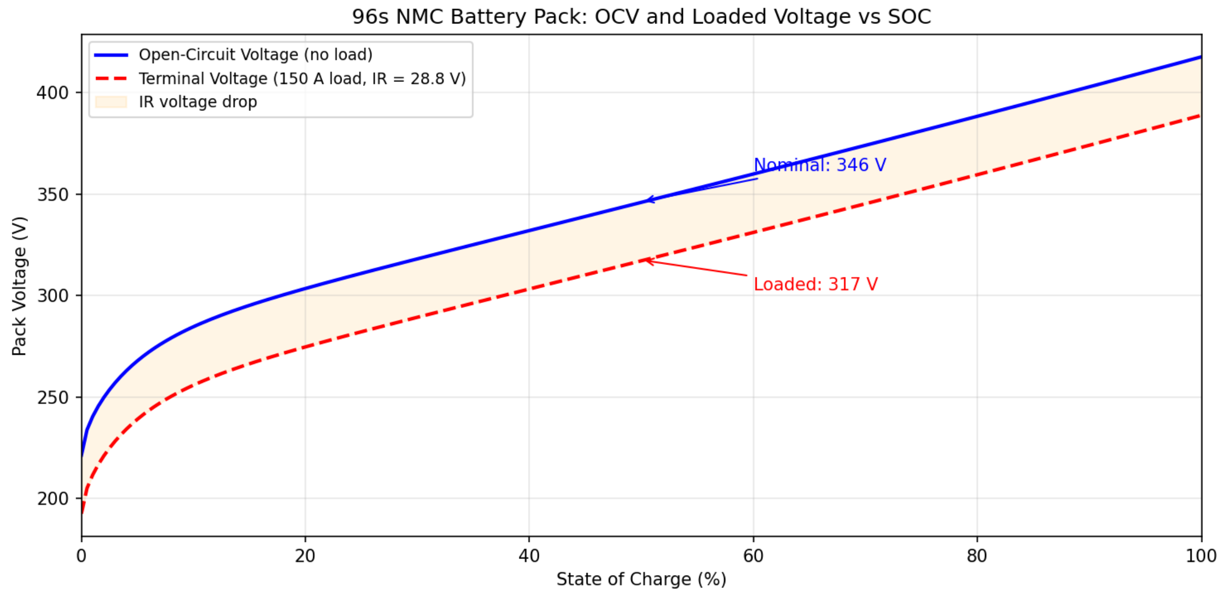



Figure 10.8.1: 96s NMC Pack Voltage vs SOC

### 10.8.2 Cell Balancing

In a series-connected battery pack, manufacturing variations and aging differences cause individual cells to diverge in capacity and state of charge over repeated cycles. Without balancing, the weakest cell limits the entire pack — it reaches the discharge cutoff voltage first (reducing usable capacity) and the charge voltage limit first (preventing full charge of other cells). Cell balancing equalizes the charge across all cells, recovering the capacity lost to cell mismatch. **Passive balancing** dissipates excess energy from higher-charged cells through bleed resistors — it is simple and inexpensive but wastes energy as heat. **Active balancing** transfers charge from higher-charged cells to lower-charged cells using switched-capacitor, transformer-coupled, or inductor-based circuits — it is more efficient but adds cost and complexity.

**Example 10.8.2:** A 12-series lithium-ion pack has cells ranging from 3.95 V to 4.18 V at end of charge. The BMS uses passive balancing with 33 Ω bleed resistors. Determine (a) the balancing current for the highest cell, (b) the power dissipated per balancing resistor, (c) the time to balance a 50 mAh charge difference, and (d) the total energy wasted during balancing.
**Solution:**
(a) Balancing current: $I_{bal} = V_{cell}/R = 4.18/33 =$ **126.7 mA**
(b) Power per resistor: $P = V^2/R = 4.18^2/33 = 17.47/33 =$ **0.529 W**
(c) Time to balance 50 mAh: $t = Q/I = 0.050/0.1267 = 0.3946$ hours $=$ **23.7 minutes**
(d) Energy wasted: $E = P \times t = 0.529 \times 0.3946 = 0.209$ Wh per cell. For all cells requiring balancing (worst case, 11 cells): $E_{total} = 11 \times 0.209 =$ **2.30 Wh**. This is small compared to a typical EV pack capacity (e.g., the 17.76 kWh pack from Example 10.8.1) — about 0.013% of that reference pack.

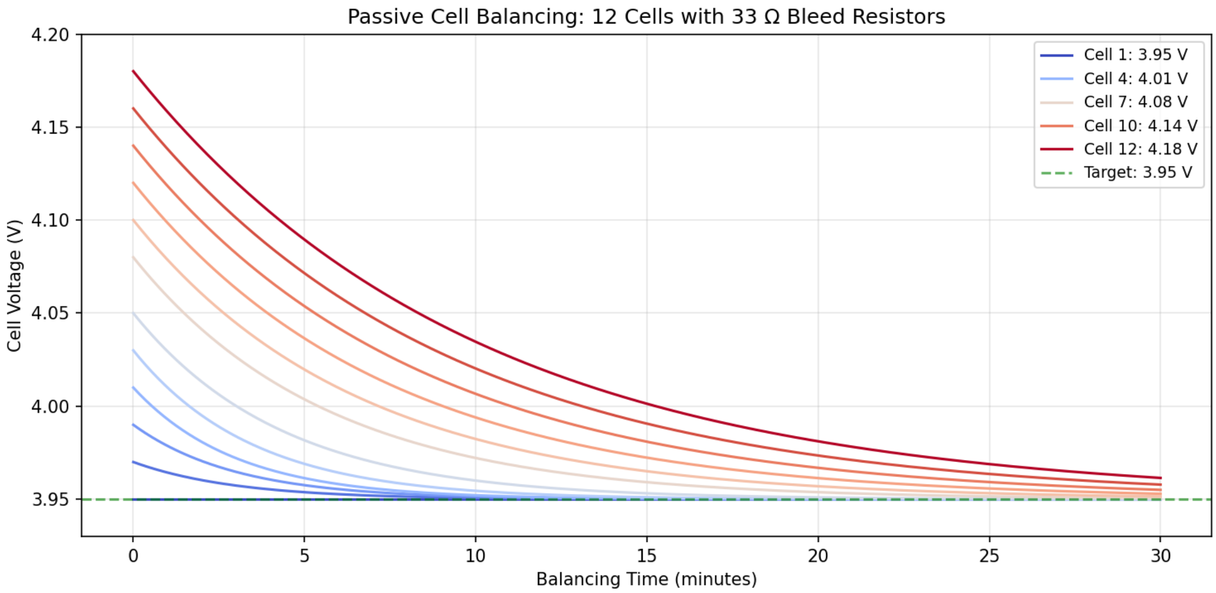



Figure 10.8.2: Passive Cell Balancing Convergence

### 10.8.3 State of Charge Estimation

State of charge (SOC) is the ratio of remaining charge to full capacity, expressed as a percentage: SOC = $Q_{remaining}/Q_{full} \times 100\%$. Accurate SOC estimation is critical for range prediction in EVs, power management in grid storage, and preventing over-discharge. **Coulomb counting** integrates the measured current over time: $SOC(t) = SOC(t_0) - (1/Q_{full}) \int I\,dt$. This method is simple but accumulates drift errors from current sensor offset and requires periodic recalibration. **OCV-based estimation** maps the measured open-circuit voltage to SOC using the cell's OCV-SOC lookup table, but requires the cell to be at rest (no current flow) for an accurate reading. **Kalman filter** methods combine coulomb counting and voltage measurements in a model-based estimator that corrects for drift, providing the most accurate SOC under dynamic load conditions. State of health (SOH) tracks the degradation of maximum capacity over the battery's lifetime: SOH = $Q_{full,current}/Q_{full,new} \times 100\%$.

**Example 10.8.3:** A 100 Ah battery pack starts at SOC = 85%. A current sensor with ±0.5% accuracy measures the discharge. After delivering 60 Ah of charge (by coulomb counting), determine (a) the estimated SOC, (b) the worst-case SOC error from sensor drift after 60 Ah, and (c) the SOC uncertainty range.
**Solution:**
(a) SOC = 85% − (60/100) × 100% = 85% − 60% = **25%**
(b) Current sensor error over 60 Ah: $\Delta Q = 0.005 \times 60 =$ **0.30 Ah**. SOC error = 0.30/100 × 100% = **0.30%**
(c) SOC range: 25% ± 0.30%, or **24.7% to 25.3%**. This is a relatively small error for a single discharge, but over many partial cycles without OCV recalibration, the cumulative drift can reach several percent.

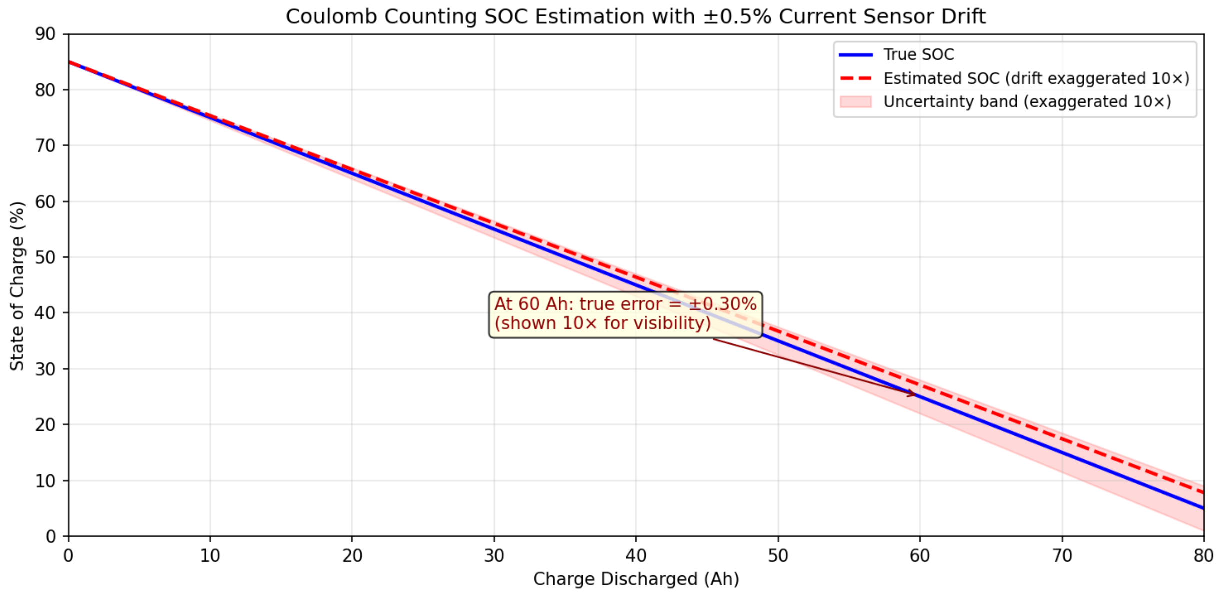



Figure 10.8.3: SOC Estimation with Coulomb Counting Drift

### 10.8.4 BMS Thermal Management

Battery cells must operate within a safe temperature window — typically 0°C to 45°C for charging and −20°C to 60°C for discharging. Temperatures above 60°C accelerate degradation and risk thermal runaway in lithium-ion cells, while temperatures below 0°C dramatically increase internal resistance and can cause lithium plating during charging (permanent damage). The BMS monitors cell temperatures using thermistors or RTDs distributed throughout the pack and controls the thermal management system — which may include liquid cooling (glycol loops), air cooling (fans and ducts), or heating elements (PTC heaters for cold-weather operation). Temperature gradients across the pack cause uneven aging; the BMS may derate charge/discharge current or activate additional cooling to maintain cell-to-cell temperature uniformity within 3–5°C.

**Example 10.8.4:** A battery pack generates 800 W of heat at full discharge rate. The liquid cooling system uses a glycol-water mixture with a flow rate of 6 L/min and a specific heat of 3.4 kJ/(kg·°C). The coolant density is 1.05 kg/L. Determine (a) the coolant temperature rise through the pack, (b) the required inlet temperature to keep cells below 40°C with a 5°C thermal resistance between coolant and cell surface, and (c) the maximum allowable thermal resistance per cell for a 96-cell pack dissipating equal heat.

**Solution:**

(a) Mass flow rate: $\dot{m} = 6 \times 1.05 = 6.3$ kg/min $= 0.105$ kg/s

$\Delta T = \dot{Q}/(\dot{m} \times c_p) = 800/(0.105 \times 3{,}400) = 800/357 = \mathbf{2.24°C}$

(b) Cell surface temperature must be $\leq 40°C$. With 5°C thermal resistance from coolant to cell: $T_{coolant,max} = 40 - 5 = 35°C$. Inlet temperature: $T_{inlet} = 35 - 2.24 = \mathbf{32.76°C}$ (the outlet reaches 35°C).

(c) Heat per cell: $\dot{Q}_{cell} = 800/96 = 8.33$ W

Maximum thermal resistance per cell: $R_\theta = \Delta T/\dot{Q}_{cell} = 5/8.33 = \mathbf{0.60\ °C/W}$

### 10.8.5 Battery Protection and Fault Detection

The BMS protection layer prevents the battery from operating outside its safe operating area (SOA) by monitoring voltage, current, and temperature in real time. **Overvoltage protection** disconnects the charger when any cell exceeds the maximum charge voltage (4.20 V for NMC/NCA, 3.65 V for LFP) to prevent lithium plating, electrolyte decomposition, and thermal runaway. **Undervoltage protection** disconnects the load when any cell drops below the minimum discharge voltage (typically 2.5–3.0 V) to prevent copper dissolution from the anode current collector, which causes permanent internal short circuits on subsequent charging. **Overcurrent protection** limits discharge current to the cell's maximum C-rate and charge current to the manufacturer's specified maximum, using current-sense resistors or Hall-effect sensors feeding the BMS controller. **Temperature protection** restricts charging below 0°C (or reduces charge current to C/10) and shuts down the pack above the maximum operating temperature. The BMS uses contactors (high-voltage relays) or power MOSFETs to disconnect the pack from the load or charger within milliseconds of detecting a fault. Advanced BMS designs also perform **insulation monitoring** on high-voltage packs (>60 V) to detect ground faults, and **short-circuit detection** using di/dt sensing to differentiate a hard short from a normal transient.

**Example 10.8.5:** A 400 V, 100 Ah EV battery pack uses a 0.5 mΩ current-sense resistor and a BMS with a 10-bit ADC referenced to 50 mV full-scale. The pack has overvoltage, undervoltage, and overcurrent protection. Determine (a) the ADC current resolution, (b) the maximum measurable current, (c) the OVP and UVP thresholds for a 96-series NMC pack, and (d) the energy that must be safely dissipated if the pack is disconnected at maximum current.
**Solution:**
(a) ADC resolution: LSB = 50 mV / $2^{10}$ = 50 mV / 1024 = 48.8 μV
Current resolution: $\Delta I = \text{LSB} / R_{sense} = 48.8 \times 10^{-6} / 0.5 \times 10^{-3}$ = **97.7 mA**
(b) Maximum measurable current: $I_{max} = V_{FS} / R_{sense} = 0.050 / 0.0005$ = **100 A** (1C rate for this pack)
(c) OVP threshold: $96 \times 4.20$ V = **403.2 V** (per-cell monitoring triggers at 4.20 V on any cell)
UVP threshold: $96 \times 2.50$ V = **240 V** (per-cell monitoring triggers at 2.50 V on any cell)
(d) Inductive energy in the wiring and contactors at disconnection (assuming 10 μH total circuit inductance at 100 A):
$E = \frac{1}{2}LI^2 = 0.5 \times 10 \times 10^{-6} \times 100^2$ = **0.05 J**. This is small; the main concern is the arc energy across the contactor contacts, which is managed by arc suppression circuits (snubber RC networks or varistors across the contactor).

## 10.9 Battery Energy Storage Systems

Battery energy storage systems (BESS) integrate battery packs, power conversion systems (PCS), thermal management, and control software into grid-connected installations ranging from kilowatt-scale commercial systems to gigawatt-hour utility plants. BESS enables the electrical grid to store surplus energy and dispatch it on demand — replacing peaking gas turbines, firming variable renewable generation, providing ancillary services such as frequency regulation, and deferring transmission and distribution upgrades. While §10.8 addresses cell-level battery management, this section covers the system-level engineering: architecture, power conversion, grid services, sizing methodology, and economic analysis. Lithium-ion technology (particularly LFP for stationary applications) dominates the market due to its declining cost, long cycle life, and high round-trip efficiency, though flow batteries and sodium-ion are emerging for long-duration applications.

### 10.9.1 BESS Architecture and Components

A utility-scale BESS is typically deployed in standardized 20-foot or 40-foot shipping containers, each housing multiple battery racks, a DC bus, and local environmental controls. Each rack contains battery modules connected in series to achieve the target DC bus voltage (typically 600–1,500 V DC), with a rack-level BMS monitoring module voltages and temperatures. Multiple racks connect in parallel to form a DC string feeding the power conversion system (PCS). The PCS is a bidirectional DC-AC inverter that interfaces the battery to the AC grid through a step-up transformer. Supporting systems include an energy management system (EMS) that dispatches charge/discharge commands, HVAC for thermal control, fire detection and suppression (typically clean-agent or aerosol systems), and a site controller that communicates with the grid operator via SCADA or IEEE 2030.5.

**Example 10.9.1:** A utility requires a 50 MW / 200 MWh (4-hour duration) BESS using LFP cells rated at 3.2 V nominal and 280 Ah. Each battery module contains 16 series cells (51.2 V module voltage). Each rack contains 10 series modules (512 V DC bus). Determine (a) the energy per rack, (b) the number of racks required, (c) the total cell count, and (d) the number of 5 MW PCS inverters needed.
**Solution:**
(a) Energy per rack: $E_{rack} = V_{rack} \times C = 512 \times 280 = 143{,}360$ Wh = **143.36 kWh**
(b) Number of racks: $N_{racks} = 200{,}000 / 143.36 = 1{,}395.1 \rightarrow$ **1,396 racks** (round up)
(c) Cells per rack = $16 \times 10 = 160$ cells. Total cells = $1{,}396 \times 160 =$ **223,360 cells**
(d) Number of 5 MW PCS units: $N_{PCS} = 50 / 5 =$ **10 inverters** (each rated 5 MW, paired with ~140 racks)

### 10.9.2 Power Conversion Systems

The power conversion system (PCS) is the bidirectional interface between the DC battery bus and the AC grid. During discharge, the PCS operates as an inverter converting DC to AC; during charge, it operates as a rectifier converting AC to DC. Modern BESS inverters use three-phase, multilevel (typically 3-level NPC or T-type) IGBT or SiC MOSFET topologies with switching frequencies of 2–20 kHz and efficiencies of 95–99% at rated power. The PCS must handle rapid transitions between charge and discharge modes, support four-quadrant operation (active and reactive power in both directions), and comply with grid codes for power quality, fault ride-through, and anti-islanding. AC-coupled BESS connects to the grid through a dedicated inverter and transformer, while DC-coupled solar+storage shares a single inverter with the PV array, reducing hardware cost but limiting independent dispatch.

**Example 10.9.2:** A 10 MW / 40 MWh BESS has a PCS with 96.5% one-way inverter efficiency and 1.2% auxiliary power consumption (cooling, BMS, controls) relative to throughput. The system operates 365 cycles per year at 80% depth of discharge. Determine (a) the round-trip efficiency, (b) the annual energy throughput, (c) the annual energy losses, and (d) the annual auxiliary energy consumption.
**Solution:**
(a) Round-trip efficiency: $\eta_{RT} = \eta_{inv}{}^2 \times (1 - \text{aux})^2 = 0.965^2 \times (1 - 0.012)^2 = 0.9312 \times 0.9761 =$ **90.9%**

(b) Annual energy throughput: $E_{annual} = 40 \times 0.80 \times 365 =$ **11,680 MWh** (discharge side)
(c) Energy losses: $E_{loss} = E_{annual} \times (1 - \eta_{RT}) / \eta_{RT} = 11{,}680 \times (1 - 0.909) / 0.909 = 11{,}680 \times 0.1001$ = **1,169 MWh/year** (energy input exceeds output by this amount)
(d) Auxiliary consumption: $E_{aux} = 11{,}680 \times 2 \times 0.012 =$ **280 MWh/year** (applied to both charge and discharge throughput, hence the factor of 2)

### 10.9.3 BESS Control and Grid Services

A BESS can provide multiple grid services by rapidly adjusting its power output. **Frequency regulation** responds to grid frequency deviations by injecting or absorbing power according to a droop characteristic: $\Delta P/P_{rated} = -\Delta f/(f_{nom} \times R)$, where R is the droop setting (typically 3–5%) and $\Delta f$ is the frequency deviation from nominal. The BESS response time (milliseconds) is far faster than conventional generators (seconds to minutes), making it highly effective for primary frequency response. **Voltage support** provides reactive power (VAR) to maintain local bus voltage within limits. **Peak shaving** discharges during high-demand periods to reduce a facility's peak demand charge. **Renewable smoothing** absorbs rapid PV or wind fluctuations to limit ramp rates seen by the grid. **Arbitrage** charges during low-price hours and discharges during high-price hours, capturing the price spread. Revenue stacking — combining multiple services — is critical for BESS economics.

**Example 10.9.3:** A 20 MW BESS provides primary frequency regulation with a 4% droop setting and ±0.036 Hz deadband around 60 Hz. Determine (a) the power output for a sustained frequency drop to 59.85 Hz, (b) the power output for a frequency rise to 60.10 Hz, and (c) the energy dispatched during a 15-minute event where the average frequency deviation is −0.15 Hz.
**Solution:**
(a) $\Delta f = 60 - 59.85 = 0.15$ Hz. Subtract deadband: $\Delta f_{eff} = 0.15 - 0.036 = 0.114$ Hz.
$\Delta P = P_{rated} \times \Delta f_{eff} / (f_{nom} \times R) = 20 \times 0.114 / (60 \times 0.04) = 20 \times 0.114 / 2.4 =$ **0.95 MW** (discharge, injecting power to support frequency)
(b) $\Delta f = 60.10 - 60 = 0.10$ Hz. $\Delta f_{eff} = 0.10 - 0.036 = 0.064$ Hz.
$\Delta P = 20 \times 0.064 / 2.4 =$ **0.533 MW** (charge, absorbing power to reduce frequency)
(c) For −0.15 Hz average deviation: $\Delta f_{eff} = 0.15 - 0.036 = 0.114$ Hz.
Average power: $P_{avg} = 0.95$ MW. Energy over 15 min: $E = 0.95 \times 15/60 =$ **0.238 MWh**

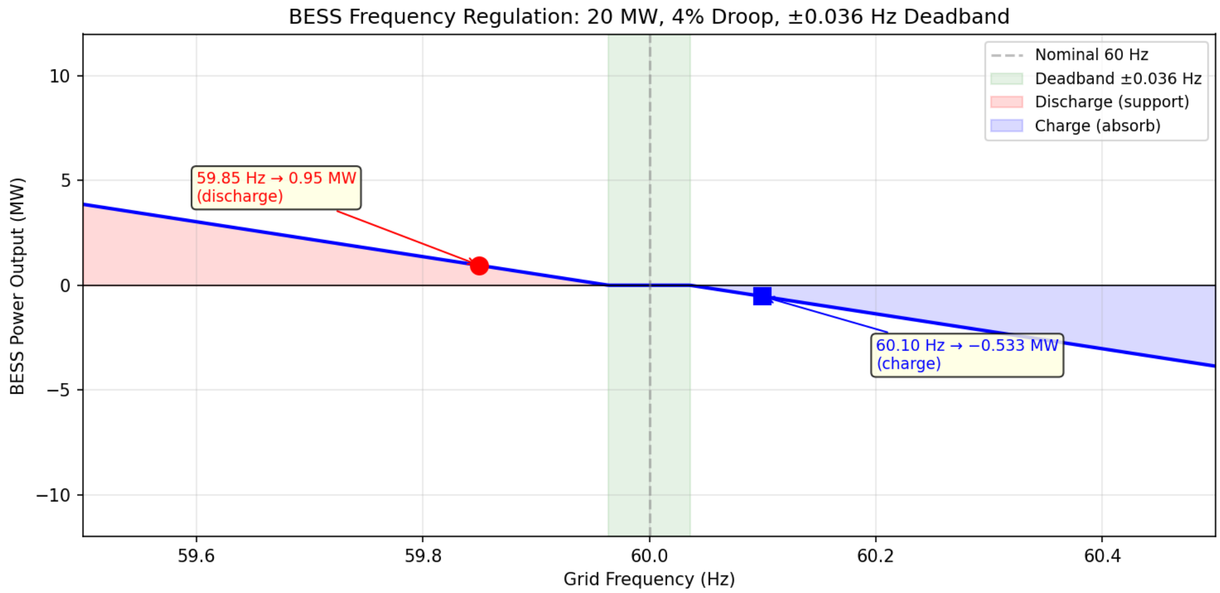



Figure 10.9.3: BESS Frequency Regulation Droop Response

### 10.9.4 BESS Sizing and Duration

BESS sizing requires matching power rating (MW) and energy capacity (MWh) to the target application. The duration (hours) = energy/power determines how long the system can sustain rated output. **Peak shaving** requires analyzing the facility's load profile to identify the magnitude and duration of demand peaks above a target threshold — the BESS must store enough energy to cover the area between the peak and the threshold. **Renewable firming** sizes the battery to absorb surplus generation and fill generation shortfalls over a defined window. **Duration selection** depends on the application: 0.5–1 hour for frequency regulation, 2–4 hours for peak shaving and arbitrage, and 4–12+ hours for renewable firming and capacity replacement. Oversizing by 10–20% accounts for degradation over the project life and ensures the system meets its performance guarantee at end of warranty.

**Example 10.9.4:** A commercial facility has a monthly peak demand of 5.0 MW, but demand exceeds 3.5 MW for only 4 hours per day (peak profile approximated as a trapezoid peaking at 5.0 MW). The utility demand charge is $18/kW-month. Determine (a) the BESS power rating to shave the peak to 3.5 MW, (b) the minimum energy capacity required, (c) the monthly demand charge savings, and (d) the recommended nameplate capacity with a 15% degradation margin.
**Solution:**
(a) Required BESS power: $P_{BESS} = 5.0 - 3.5 = \mathbf{1.5\ MW}$
(b) Energy required: The peak above 3.5 MW over 4 hours forms approximately a triangle with peak excess of 1.5 MW. Energy = $\frac{1}{2} \times 1.5 \times 4 = \mathbf{3.0\ MWh}$
(c) Monthly demand charge savings: $\Delta D = (5.0 - 3.5) \times \$18 = 1.5 \times 18 =$ **$27,000/month** ($324,000/year)
(d) Nameplate capacity with margin: $E_{nameplate} = 3.0 / (1 - 0.15) = 3.0 / 0.85 = \mathbf{3.53\ MWh}$ (specify 3.5 MWh or 4.0 MWh depending on available module sizes)

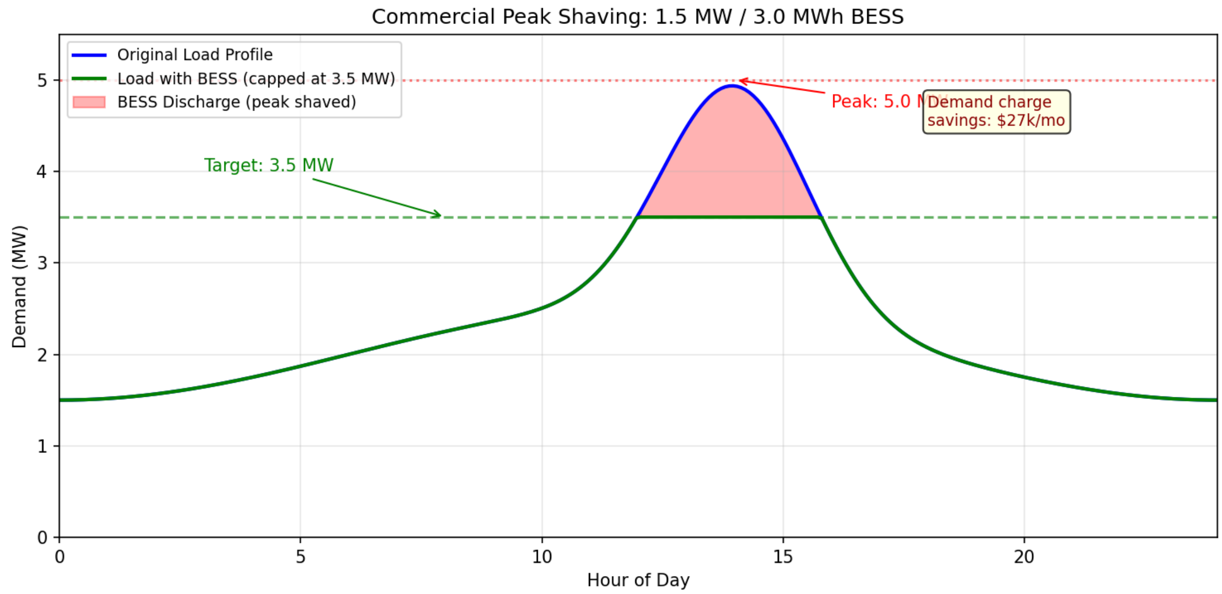



Figure 10.9.4: BESS Peak Shaving Load Profile

### 10.9.5 BESS Economics and Degradation

BESS economics are evaluated using the **levelized cost of storage (LCOS)**, which amortizes all costs over the lifetime energy throughput: LCOS = (Capital + PW of O&M + PW of Augmentation) / PW of Lifetime Energy Discharged. Capital cost is dominated by battery cells ($100–$250/kWh at the pack level for LFP) plus the PCS, transformer, balance of plant, and EPC, bringing total installed cost to $200–$500/kWh depending on scale and duration. **Capacity degradation** from cycling follows an approximately linear model: capacity retention = 1 − (throughput × degradation rate), where degradation rate is typically 0.01–0.03% per full equivalent cycle for LFP. Calendar aging adds 1–2% per year independent of cycling. When cumulative degradation reaches the performance guarantee limit (typically 70–80% of initial capacity after 10–20 years), **augmentation** adds new modules to restore capacity, extending system life. Revenue must exceed LCOS for the project to be viable; revenue stacking across multiple services (arbitrage + regulation + capacity payments) typically yields $80–$200/MWh of throughput.

**Example 10.9.5:** A 100 MWh LFP BESS has an installed cost of $280/kWh, 15-year project life, 1 cycle/day at 80% DOD, 92% round-trip efficiency, 2.5% capacity fade per year (combined cycling and calendar), and O&M of $6/MWh of throughput. Using a discount rate of 8%, determine (a) the total lifetime energy discharged (undiscounted), (b) the LCOS (simplified, without augmentation), and (c) whether the project is viable with stacked revenue of $140/MWh.

**Solution:**

(a) Annual discharge: $E_{yr} = 100 \times 0.80 \times 365 = 29{,}200$ MWh (year 1). With 2.5%/year degradation, average annual capacity over 15 years $\approx 100 \times (1 - 0.025 \times 7.5) = 81.25$ MWh effective. Average annual discharge = $81.25 \times 0.80 \times 365 = 23{,}725$ MWh. Total lifetime: $23{,}725 \times 15 =$ **355,875 MWh**

(b) Capital: $100{,}000 \times \$280 = \$28{,}000{,}000$. Annual O&M: $23{,}725 \times \$6 = \$142{,}350$/year. PW of O&M (P/A, 8%, 15) = $\$142{,}350 \times 8.5595 = \$1{,}218{,}497$. LCOS = ($28,000,000 + $1,218,497) /

355,875 = \$29,218,497 / 355,875 = **\$82.1/MWh**
(c) Revenue of \$140/MWh exceeds LCOS of \$82.1/MWh by \$57.9/MWh, yielding an annual profit margin of 23,725 × \$57.9 = **\$1,373,678/year**. The project is **viable**.

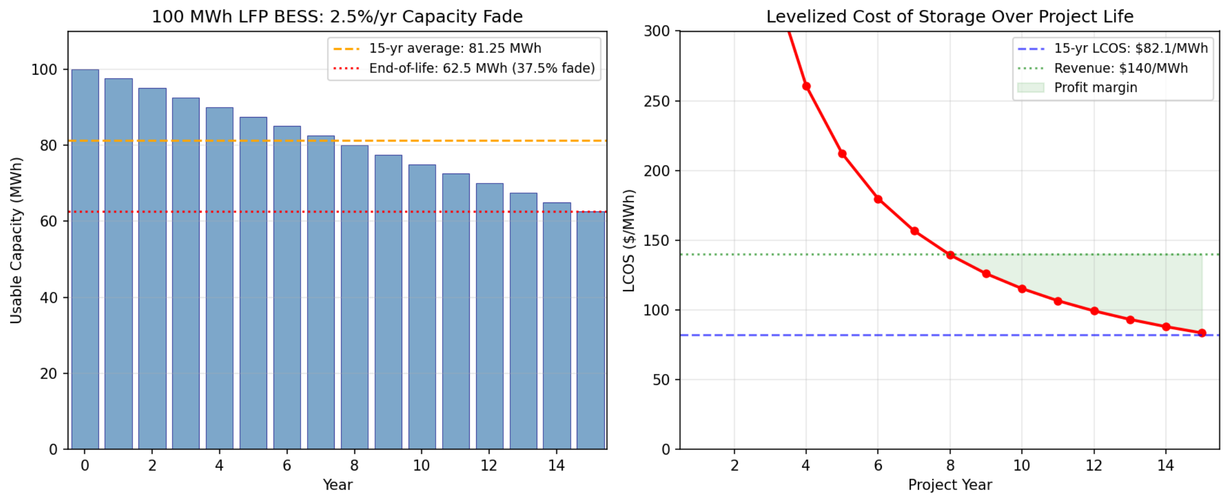



Figure 10.9.5: BESS Capacity Degradation and LCOS

## 10.10 Battery Charging

Battery charging is the controlled process of restoring energy to an electrochemical cell by forcing current through it in the reverse direction of discharge. The charger must regulate voltage, current, and temperature to maximize charge acceptance while preventing damage from overcharging, excessive heat, or lithium plating. Different battery chemistries require different charging algorithms — lithium-ion uses **constant-current/constant-voltage (CC/CV)**, lead-acid uses a three-stage bulk/absorption/float profile, and NiMH relies on **−ΔV detection** to identify full charge. Modern charger design spans from milliwatt USB-powered circuits to megawatt DC fast-charging stations, incorporating power factor correction, galvanic isolation, bidirectional power flow (V2G), and wireless energy transfer. The power electronics topologies, control strategies, and thermal management techniques used in charging systems draw on nearly every topic in this chapter.

### 10.10.1 CC/CV Charging Profile and Charge Termination

The **constant-current/constant-voltage (CC/CV)** algorithm is the standard charging method for lithium-ion batteries. During the **CC phase**, the charger supplies a fixed current (typically 0.5C to 1C, where C is the cell's rated capacity) while the cell voltage rises from its discharged state toward the **charge cutoff voltage** (4.20 V for NMC/NCA, 3.65 V for LFP). Once the cell reaches the cutoff voltage, the charger transitions to the **CV phase**, holding the voltage constant while the current tapers exponentially as the cell approaches full charge. **Charge termination** occurs when the current drops below a threshold (typically C/20 to C/50), indicating that the cell can no longer accept significant charge at the applied voltage. The CC phase typically delivers 60–80% of the cell's capacity in a relatively short time, while the CV phase completes the remaining 20–40% more slowly. **Coulombic efficiency** — the ratio of charge extracted during discharge to charge inserted during charge —

is 99.5–99.9% for healthy lithium-ion cells but decreases with aging and low-temperature operation, serving as a sensitive indicator of side reactions such as SEI growth and lithium plating. For lead-acid batteries, the three-stage charging profile consists of a **bulk** CC phase to ~80% SOC, an **absorption** CV phase at 14.4–14.8 V (for a 12 V battery) to reach 100%, and a **float** stage at 13.2–13.6 V to maintain full charge without overcharging. NiMH cells use a $\mathbf{-\Delta V/\Delta t}$ termination method that detects the slight voltage drop (5–10 mV) that occurs when the cell reaches full charge and exothermic oxygen recombination begins.

**Example 10.10.1:** A 21700 NMC lithium-ion cell has a rated capacity of 5.0 Ah and is charged using a CC/CV algorithm with a CC rate of 0.5C (2.5 A) and a charge cutoff voltage of 4.20 V. The cell starts at 3.0 V (fully discharged) and reaches 4.20 V after 1.6 hours of CC charging. During the CV phase, the current decays exponentially with a time constant $\tau = 0.4$ hours, and charge is terminated when the current drops to C/20 (0.25 A). (a) Calculate the charge delivered during the CC phase. (b) Determine the time to reach charge termination after the CV phase begins. (c) Calculate the charge delivered during the CV phase. (d) If the cell delivers 4.90 Ah on the subsequent discharge, what is the coulombic efficiency?

**Solution:**

(a) Charge in CC phase: $Q_{CC} = I_{CC} \times t_{CC} = 2.5\text{ A} \times 1.6\text{ h} =$ **4.00 Ah** (80% of rated capacity)

(b) CV phase current decays as $I(t) = 2.5 \times e^{-t/0.4}$. Termination when I = 0.25 A:

$0.25 = 2.5 \times e^{-t/0.4} \rightarrow e^{-t/0.4} = 0.1 \rightarrow t = -0.4 \times \ln(0.1) = 0.4 \times 2.303 =$ **0.92 hours** (55 minutes)

(c) Charge in CV phase: $Q_{CV} = \int_0^{0.92} 2.5 \times e^{-t/0.4}\,dt = 2.5 \times 0.4 \times (1 - e^{-0.92/0.4}) = 1.0 \times (1 - 0.1)$ = **0.90 Ah**

(d) Total charge in: $Q_{in} = 4.00 + 0.90 = 4.90$ Ah. Charge out: $Q_{out} = 4.90$ Ah.

Coulombic efficiency: $\eta_C = Q_{out} / Q_{in} = 4.90 / 4.90 =$ **100.0%** — this idealized result indicates negligible side reactions; in practice, $\eta_C \approx 99.5$–99.9% due to minor SEI growth.

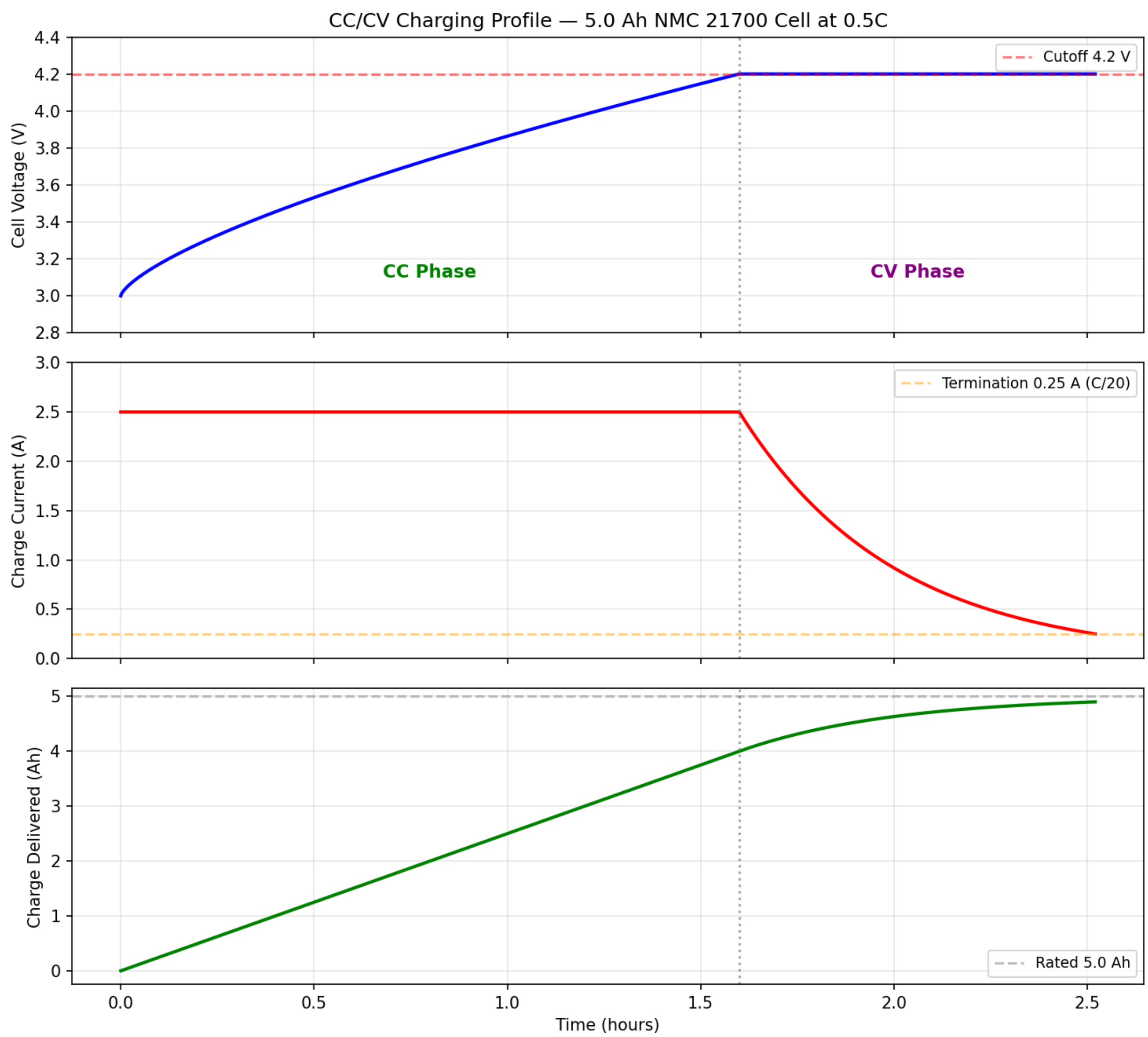



Figure 10.10.1: CC/CV Charging Profile

### 10.10.2 Charger Topologies: On-Board and Off-Board AC/DC

An AC/DC battery charger converts grid-frequency AC power into regulated DC power at the voltage and current required by the battery. The typical architecture is a **two-stage design**: a front-end **power factor correction (PFC)** stage that converts AC to a regulated DC bus (typically 400 V) with near-unity power factor, followed by an isolated **DC-DC converter** stage that steps the voltage down to the battery level and provides galvanic isolation for safety. The PFC stage commonly uses a boost converter topology (see §10.7), while the DC-DC stage favors resonant topologies — the **LLC resonant converter** achieves zero-voltage switching (ZVS) across the full load range with 96–98% efficiency and is the dominant topology for chargers above 1 kW. For bidirectional chargers supporting **vehicle-to-grid (V2G)** operation, the **CLLC resonant converter** provides symmetric bidirectional power flow by adding a second resonant capacitor. **On-board chargers (OBCs)** are integrated into the electric vehicle and are typically rated at 3.3–22 kW for Level 1/Level 2 AC charging, constrained by weight, volume, and cooling capacity. **Off-board chargers** (DC fast-charging stations) bypass the

OBC entirely, connecting directly to the battery through the vehicle's DC charging inlet, and are rated at 50–350 kW or higher. GaN transistors are increasingly replacing silicon MOSFETs in OBC designs at power levels up to 22 kW, enabling switching frequencies above 500 kHz that shrink magnetic components by 50–70%. The **USB Power Delivery (USB-PD)** standard enables up to 240 W charging over a USB-C cable with programmable voltage levels (5/9/15/20/28/36/48 V), governed by digital negotiation between the source and sink.

**Example 10.10.2:** A 3.3 kW on-board charger has a PFC stage with 97% efficiency and an LLC DC-DC stage with 96% efficiency. The charger converts 240 $V_{rms}$ single-phase AC to a 400 V nominal battery pack. (a) Calculate the overall charger efficiency. (b) Determine the AC input current at full load. (c) Calculate the total power dissipated as heat. (d) If the charger enclosure has a thermal resistance of 0.8°C/W to ambient at 35°C, estimate the maximum internal temperature.
**Solution:**
(a) Overall efficiency: $\eta_{total} = \eta_{PFC} \times \eta_{DC\text{-}DC} = 0.97 \times 0.96 = \mathbf{93.1\%}$
(b) AC input power: $P_{in} = P_{out} / \eta_{total} = 3{,}300 / 0.931 = 3{,}544$ W.
AC input current: $I_{AC} = P_{in} / V_{AC} = 3{,}544 / 240 = \mathbf{14.8\ A}$
(c) Total heat dissipated: $P_{loss} = P_{in} - P_{out} = 3{,}544 - 3{,}300 = \mathbf{244\ W}$
(d) Temperature rise: $\Delta T = P_{loss} \times R_\theta = 244 \times 0.8 = 195$°C.
$T_{max} = 35 + 195 = \mathbf{230°C}$ — this is far too high for enclosed operation, indicating that the 0.8°C/W thermal resistance is inadequate. A liquid-cooled or fan-cooled design with $R_\theta \leq 0.15$°C/W would keep the internal temperature below 72°C, which is within the safe operating range for power semiconductors.

### 10.10.3 EV Charging Levels and Standards

Electric vehicle charging infrastructure is categorized into **levels** that define the power delivery capability and connection type. **Level 1** uses a standard 120 V/15 A household outlet (NEMA 5-15) through a portable **EVSE (Electric Vehicle Supply Equipment)** cord set, delivering 1.4–1.9 kW and adding approximately 5–8 km of range per hour — suitable for overnight charging of plug-in hybrids. **Level 2** uses a dedicated 208–240 V circuit with a permanently installed EVSE, delivering 3.3–19.2 kW (typical residential: 7.7 kW on a 40 A circuit) and adding 20–100 km of range per hour. Both Level 1 and Level 2 supply AC power to the vehicle's on-board charger. **DC fast charging (DCFC)** bypasses the OBC and supplies DC power directly to the battery at 50–350 kW (with 800 V architectures enabling up to 500 kW), adding 200–500 km of range in 15–30 minutes. The **SAE J1772** standard defines the Level 1/Level 2 AC connector and **pilot signal** — a 1 kHz PWM square wave on the control pilot (CP) pin whose duty cycle communicates the maximum available current from the EVSE to the vehicle. The **Combined Charging System (CCS)** adds two DC pins below the J1772 connector for DCFC up to 350 kW. **CHAdeMO** is a competing DC fast-charging standard (up to 400 kW) using CAN bus communication, developed in Japan. The **North American Charging Standard (NACS)**, originally Tesla's proprietary connector, was adopted as SAE J3400 and supports both AC and DC charging through a single compact connector rated to 1 MW. The **Megawatt Charging System (MCS)**, standardized as SAE J3068, targets heavy-duty commercial vehicles with charging power up to 3.75 MW at 1,250 V / 3,000 A.

**Example 10.10.3:** A battery electric vehicle has a 75 kWh usable battery capacity and arrives at a charging station with 10% state of charge. Determine the time to charge to 80% SOC using (a)

Level 1 at 1.4 kW, (b) Level 2 at 7.7 kW, (c) DCFC at 150 kW (assuming constant power up to 80% SOC), and (d) the minimum EVSE circuit breaker rating for a Level 2 EVSE on a 240 V circuit delivering 7.7 kW, applying the NEC 125% continuous load rule.
**Solution:**
(a) Energy required: $E = 75 \times (0.80 - 0.10) = 75 \times 0.70 = 52.5$ kWh.
Level 1 time: $t = 52.5 / 1.4 =$ **37.5 hours**
(b) Level 2 time: $t = 52.5 / 7.7 =$ **6.8 hours**
(c) DCFC time: $t = 52.5 / 150 = 0.35$ hours = **21 minutes**
(d) EVSE current: $I = 7{,}700 / 240 = 32.1$ A. NEC continuous load (125%): $I_{CB} = 32.1 \times 1.25 = 40.1$ A → **40 A circuit breaker** (standard size per NEC 240.6(A)), requiring 8 AWG copper conductors (ampacity 50 A at 75°C from NEC Table 310.16)

### 10.10.4 Fast Charging and Thermal Considerations

**Fast charging** at rates above 1C generates significant heat within the cell from **$I^2R$ losses** in the internal resistance, which includes ionic resistance of the electrolyte, interfacial charge-transfer resistance, and electronic resistance of the current collectors and tabs. For a cell with internal resistance $R_{int}$, the instantaneous heat generation rate is approximately $\dot{Q} = I^2R_{int} + I \times T \times (dV_{OCV}/dT)$, where the second term accounts for reversible entropic heating (which can be exothermic or endothermic depending on SOC). At low temperatures (below 10°C), the electrolyte viscosity increases dramatically, raising ionic resistance and reducing lithium-ion diffusivity in the graphite anode. If the charging rate exceeds the anode's ability to intercalate lithium ions, metallic lithium deposits on the anode surface — a process called **lithium plating** — which permanently reduces capacity, increases internal resistance, and in severe cases creates dendrites that can cause internal short circuits. The safe **charge acceptance window** narrows at low temperatures: a cell that safely accepts 2C at 25°C may be limited to 0.3C at 0°C. Battery **pre-conditioning** heats the pack to 15–25°C before fast charging using PTC heaters, waste heat from the motor inverter, or resistive heating of the cells themselves by applying high-frequency AC current. **Pulse charging** protocols alternate short high-current pulses (5–10 seconds) with rest periods (1–3 seconds), allowing lithium-ion concentration gradients to relax and reducing the risk of plating compared to continuous CC charging at the same average rate. Charge rate optimization algorithms use **electrochemical impedance spectroscopy (EIS)** feedback or physics-based models to continuously adjust the charging current based on real-time cell temperature, voltage, and estimated anode potential.

**Example 10.10.4:** A prismatic LFP cell has a capacity of 280 Ah, internal resistance of 0.6 mΩ, and is charged at 1C (280 A). (a) Calculate the $I^2R$ heat generation rate. (b) If the cell has a mass of 5.5 kg and specific heat capacity of 1,100 J/(kg·°C), estimate the adiabatic temperature rise after 20 minutes of CC charging. (c) At 5°C ambient, the internal resistance increases to 1.5 mΩ. Calculate the new heat generation rate. (d) If the maximum safe cell temperature is 45°C and the cell starts at 5°C, determine the maximum continuous charge time at 1C before thermal limits are reached (adiabatic conditions, using the 5°C resistance value).
**Solution:**
(a) Heat generation at 25°C: $\dot{Q} = I^2 \times R_{int} = 280^2 \times 0.0006 = 78{,}400 \times 0.0006 =$ **47.04 W**
(b) Energy dissipated in 20 min: $E = 47.04 \times 20 \times 60 = 56{,}448$ J.
Temperature rise: $\Delta T = E / (m \times c_p) = 56{,}448 / (5.5 \times 1{,}100) = 56{,}448 / 6{,}050 =$ **9.3°C**
(c) At 5°C: $\dot{Q} = 280^2 \times 0.0015 = 78{,}400 \times 0.0015 =$ **117.6 W** (2.5× higher than at 25°C)

(d) Allowable temperature rise: $\Delta T_{max} = 45 - 5 = 40°C$.
Energy budget: $E_{max} = m \times c_p \times \Delta T_{max} = 5.5 \times 1{,}100 \times 40 = 242{,}000$ J.
Maximum charge time: $t = E_{max} / \dot{Q} = 242{,}000 / 117.6 = 2{,}058$ s = **34.3 minutes** — after which the cell must either stop charging or reduce current. In practice, active cooling extends this limit significantly.

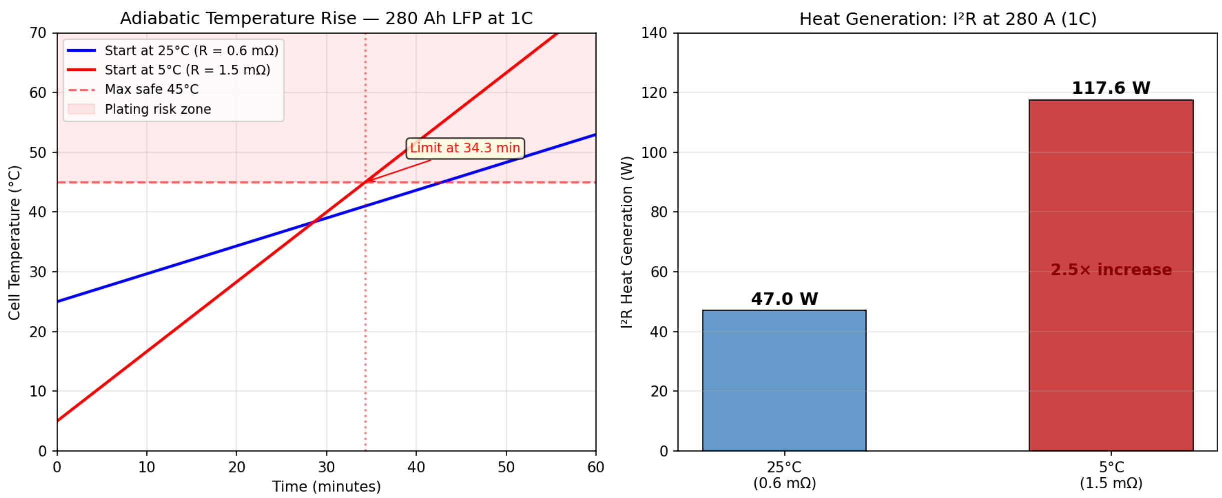



Figure 10.10.4: Fast Charging Thermal Analysis

### 10.10.5 Wireless and Inductive Charging

**Wireless power transfer (WPT)** for battery charging uses **resonant inductive coupling** between a transmitter coil (embedded in a charging pad or road surface) and a receiver coil (mounted on the underside of the device or vehicle). The transmitter drives the primary coil at a resonant frequency (typically 85 kHz for EV charging per SAE J2954, or 100–205 kHz for Qi consumer devices), creating an alternating magnetic field that induces a voltage in the receiver coil through mutual inductance $M = k\sqrt{(L_1 L_2)}$, where k is the **coupling coefficient** (0.1–0.3 for loosely coupled EV systems, 0.5–0.8 for tightly coupled consumer devices). To maximize power transfer efficiency at the resonant frequency, **compensation networks** — series-series (SS), series-parallel (SP), LCC, or double-sided LCC — are added to both coils to cancel reactive impedance and create a resonant condition. The **SS compensation** topology places a capacitor in series with each coil, tuned so that the capacitive reactance equals the inductive reactance at the operating frequency: $C = 1/(\omega^2 L)$. System efficiency depends on the quality factor Q of each coil ($Q = \omega L/R$, typically 200–500 for well-designed Litz-wire coils), the coupling coefficient, and the compensation network losses, with practical WPT systems achieving 85–93% DC-to-DC efficiency. The **Qi standard** (Wireless Power Consortium) defines wireless charging for consumer electronics at power levels from 5 W (baseline) to 15 W (extended power profile), using inductive coupling at 100–205 kHz with in-band communication between receiver and transmitter for power control. **SAE J2954** standardizes wireless EV charging at power levels of WPT1 (3.7 kW), WPT2 (7.7 kW), WPT3 (11 kW), and WPT4 (22 kW) at 85 kHz, specifying ground clearance classes (100–250 mm) and alignment tolerances. **Foreign object detection (FOD)** is critical for safety — metallic objects on the charging pad can absorb energy and overheat; FOD systems use auxiliary sensing coils, quality factor monitoring, or thermal sensors to detect foreign objects and shut down the transmitter.

**Example 10.10.5:** A wireless EV charging system operates at 85 kHz with a transmitter coil inductance $L_1 = 200\ \mu H$, receiver coil inductance $L_2 = 250\ \mu H$, and coupling coefficient $k = 0.20$. The system uses SS compensation. (a) Calculate the mutual inductance. (b) Determine the series compensation capacitors for both coils. (c) If each coil has an AC resistance of 0.15 Ω, calculate the quality factor of each coil. (d) Estimate the maximum achievable efficiency using $\eta_{max} = (k^2Q_1Q_2) / (1 + \sqrt{(1 + k^2Q_1Q_2)})^2$.

**Solution:**

(a) Mutual inductance: $M = k \times \sqrt{(L_1 \times L_2)} = 0.20 \times \sqrt{(200 \times 10^{-6} \times 250 \times 10^{-6})} = 0.20 \times \sqrt{(5.0 \times 10^{-8})} = 0.20 \times 2.236 \times 10^{-4} = \mathbf{44.7\ \mu H}$

(b) Angular frequency: $\omega = 2\pi \times 85{,}000 = 534{,}071$ rad/s.

$C_1 = 1/(\omega^2L_1) = 1/(2.852 \times 10^{11} \times 200 \times 10^{-6}) = 1/(5.704 \times 10^7) = \mathbf{17.5\ nF}$

$C_2 = 1/(\omega^2L_2) = 1/(2.852 \times 10^{11} \times 250 \times 10^{-6}) = 1/(7.130 \times 10^7) = \mathbf{14.0\ nF}$

(c) $Q_1 = \omega L_1/R_1 = 534{,}071 \times 200 \times 10^{-6} / 0.15 = 106.8 / 0.15 = \mathbf{712}$

$Q_2 = \omega L_2/R_2 = 534{,}071 \times 250 \times 10^{-6} / 0.15 = 133.5 / 0.15 = \mathbf{890}$

(d) Product: $k^2Q_1Q_2 = 0.04 \times 712 \times 890 = 0.04 \times 633{,}680 = 25{,}347$.

$\eta_{max} = 25{,}347 / (1 + \sqrt{(1 + 25{,}347)})^2 = 25{,}347 / (1 + 159.2)^2 = 25{,}347 / (160.2)^2 = 25{,}347 / 25{,}664 =$ **98.8%** — this theoretical maximum assumes lossless compensation networks and electronics; practical system DC-to-DC efficiency is 85–93% after accounting for inverter, rectifier, and compensation network losses.

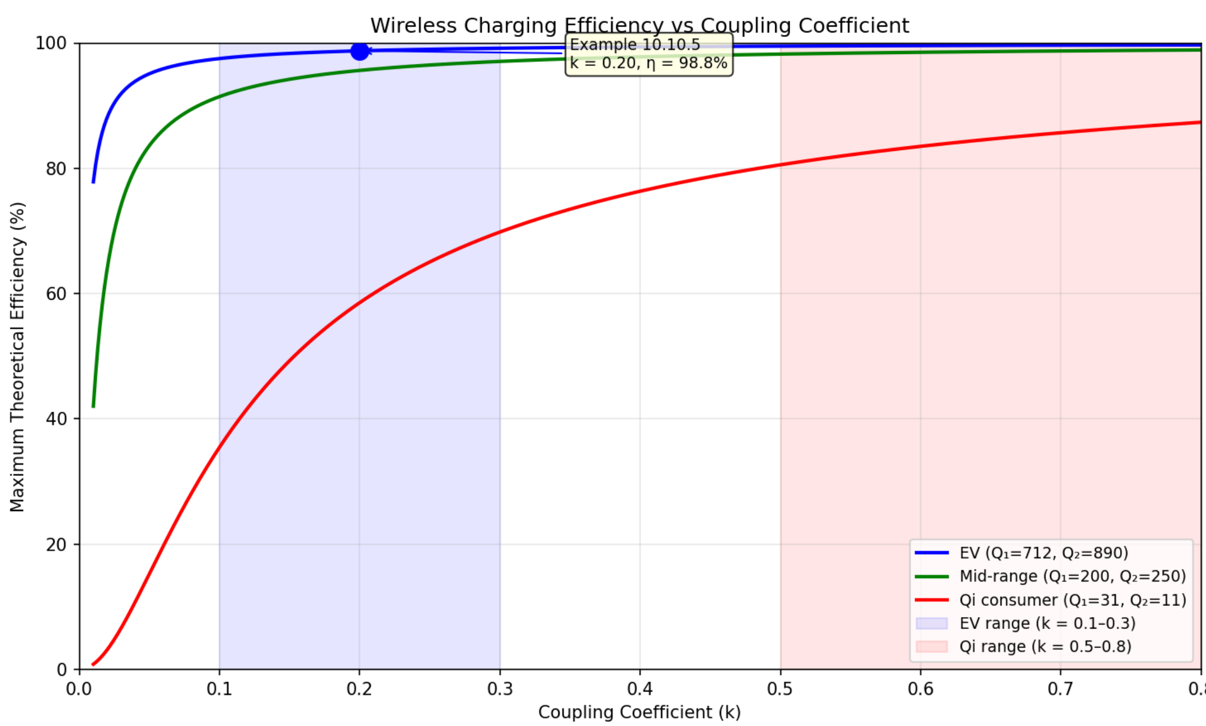



Figure 10.10.5: Wireless Charging Efficiency vs Coupling Coefficient

## 10.11 Supercapacitors

Supercapacitors — also called **ultracapacitors** or **electrochemical double-layer capacitors (EDLC)** — occupy the performance space between conventional electrolytic capacitors and rechargeable batteries, offering capacitances from a few farads to thousands of farads. Their energy density (1–15 Wh/kg) is far below that of lithium-ion batteries (100–250 Wh/kg), but their specific power (500–10,000 W/kg) greatly exceeds battery capability, and their cycle life of 500,000+ cycles dwarfs the 500–2,000 cycles of Li-ion cells. Supercapacitors store energy electrostatically — not chemically — enabling sub-second charge/discharge capability, wide operating temperature ranges (−40 °C to +65 °C), and no risk of thermal runaway. They are widely deployed in regenerative energy recovery for transit vehicles, peak power buffering for industrial drives, uninterruptible power supplies, automotive start-stop systems, and hybrid configurations that pair a battery (high energy) with a supercapacitor (high power).

### 10.11.1 EDLC Fundamentals

An EDLC cell stores charge at the **electrochemical double layer** — the molecular interface between a high-surface-area carbon electrode and an electrolyte. When a voltage is applied, electrolyte ions are adsorbed onto the electrode surface, forming two layers of opposite charge separated by a distance d of roughly 0.3–1 nm. The resulting capacitance follows the standard parallel-plate formula $C = \varepsilon_r \varepsilon_0 A/d$, but because d is so small and the electrode surface area A so large (1,000–3,000 $m^2/g$ for activated carbon), capacitances per unit mass are orders of magnitude higher than in conventional capacitors. Each cell consists of two such electrodes separated by an ion-permeable membrane; the cell capacitance equals half the single-electrode capacitance because the two double layers are in series.

Key operating parameters include: - **Specific capacitance**: 100–300 F/g for activated carbon electrodes - **Cell voltage**: limited to 2.5–2.7 V for organic electrolytes, 1.0–1.2 V for aqueous electrolytes - **ESR** (equivalent series resistance): 1–50 mΩ, which limits peak current and introduces voltage droop - **Self-discharge**: time constant of hours to days (far faster than batteries) - **Cycle life**: > 500,000 full charge/discharge cycles with < 20% capacitance fade

Energy stored in a supercapacitor is $E = ½CV^2$, and a key design consideration is that not all stored energy is recoverable — discharging from $V_{max}$ to $V_{max}/2$ delivers 75% of the total energy, but the terminal voltage halves simultaneously.

**Example 10.11.1:** A symmetric EDLC cell uses two activated carbon electrodes, each with a specific surface area of 1,500 $m^2/g$ and a mass of 5 g. The double-layer capacitance per unit area is 20 $\mu F/cm^2$. (a) Calculate the electrode capacitance. (b) Calculate the cell capacitance. (c) If the cell is charged to 2.7 V, how much energy is stored?
**Solution:**
(a) Total electrode surface area = $1{,}500\ m^2/g \times 5\ g = 7{,}500\ m^2 = 7{,}500 \times (100\ cm)^2/m^2 = 7.5 \times 10^7\ cm^2$.
Electrode capacitance: $C_{electrode} = 20 \times 10^{-6}\ F/cm^2 \times 7.5 \times 10^7\ cm^2 =$ **1,500 F**
(b) Two double layers in series (one per electrode): $C_{cell} = C_{electrode}/2 = 1{,}500/2 =$ **750 F**
(c) $E = ½ \times C_{cell} \times V^2 = ½ \times 750 \times (2.7)^2 = ½ \times 750 \times 7.29 =$ **2,734 J (0.759 Wh)**

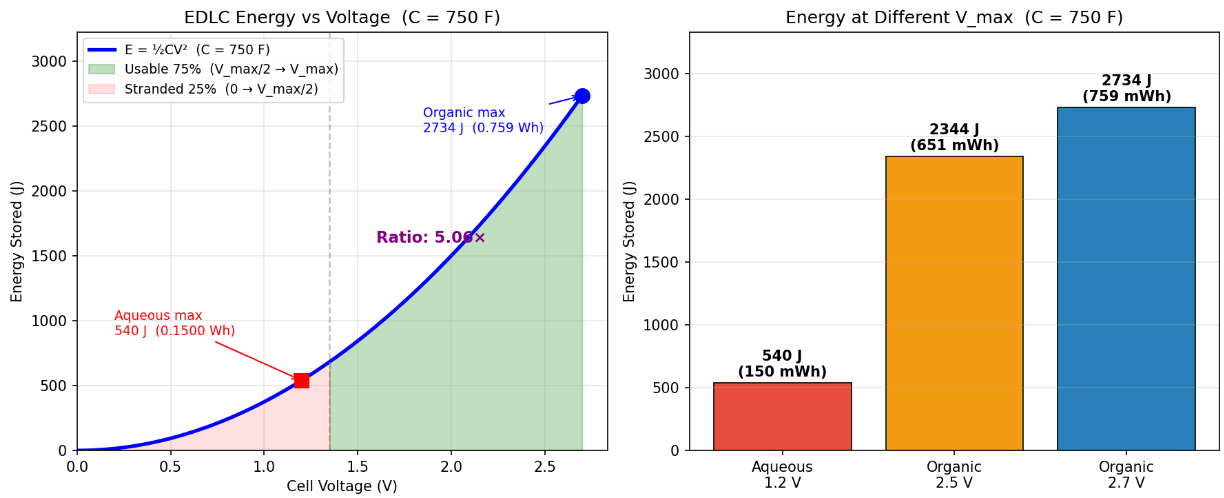



Figure 10.11.1: EDLC Energy Storage vs Voltage — Usable 75% and Electrolyte Comparison

### 10.11.2 Pseudocapacitors and Hybrid Supercapacitors

**Pseudocapacitors** augment double-layer capacitance with **Faradaic charge storage** — reversible electrochemical reactions (surface redox, intercalation, or underpotential deposition) that occur at the electrode surface rather than in the bulk, as in a battery. Common pseudocapacitive materials include $RuO_2$ (ruthenium dioxide, high performance but expensive), $MnO_2$ (manganese dioxide, low cost), and conducting polymers such as polyaniline. Because charge is stored both electrostatically and electrochemically, pseudocapacitors achieve 2–5× higher energy density than EDLC at the cost of slightly reduced cycle life and slower response.

**Lithium-ion capacitors (LICs)** are a prominent hybrid technology combining a pre-lithiated graphite anode (which stores charge like a Li-ion battery) with an activated carbon cathode (which stores charge like an EDLC). The asymmetric design raises the cell voltage to 2.2–3.8 V and triples the energy density versus conventional EDLC, while retaining much of the power advantage. LICs target applications — such as regenerative braking, elevator energy recovery, and industrial UPS — where batteries lack peak-power capability but standard EDLC lack sufficient energy density.

A **Ragone plot** (specific power vs specific energy on a log-log scale) is the standard tool for comparing energy storage technologies and identifying the region each occupies.

**Example 10.11.2:** A 350 F EDLC cell ($V_{max}$ = 2.7 V, mass = 60 g) is compared to a lithium-ion battery cell (specific energy = 150 Wh/kg, specific power = 400 W/kg, mass = 100 g). (a) Calculate the energy stored in each device. (b) Calculate the specific energy of the EDLC. (c) By what factor does the Li-ion cell exceed the EDLC in specific energy?
**Solution:**
(a) EDLC: $E = ½ \times 350 \times (2.7)^2 = ½ \times 350 \times 7.29 =$ **1,275.75 J = 0.354 Wh**
Li-ion battery: $E = 150$ Wh/kg × 0.100 kg = **15.0 Wh**
(b) EDLC specific energy = 0.354 Wh / 0.060 kg = **5.9 Wh/kg**
(c) Ratio = 150 Wh/kg ÷ 5.9 Wh/kg = **25.4×** — the Li-ion cell stores about 25 times more energy per kilogram; however, the EDLC can deliver its energy in a fraction of a second, whereas the battery is rate-limited by ion diffusion.

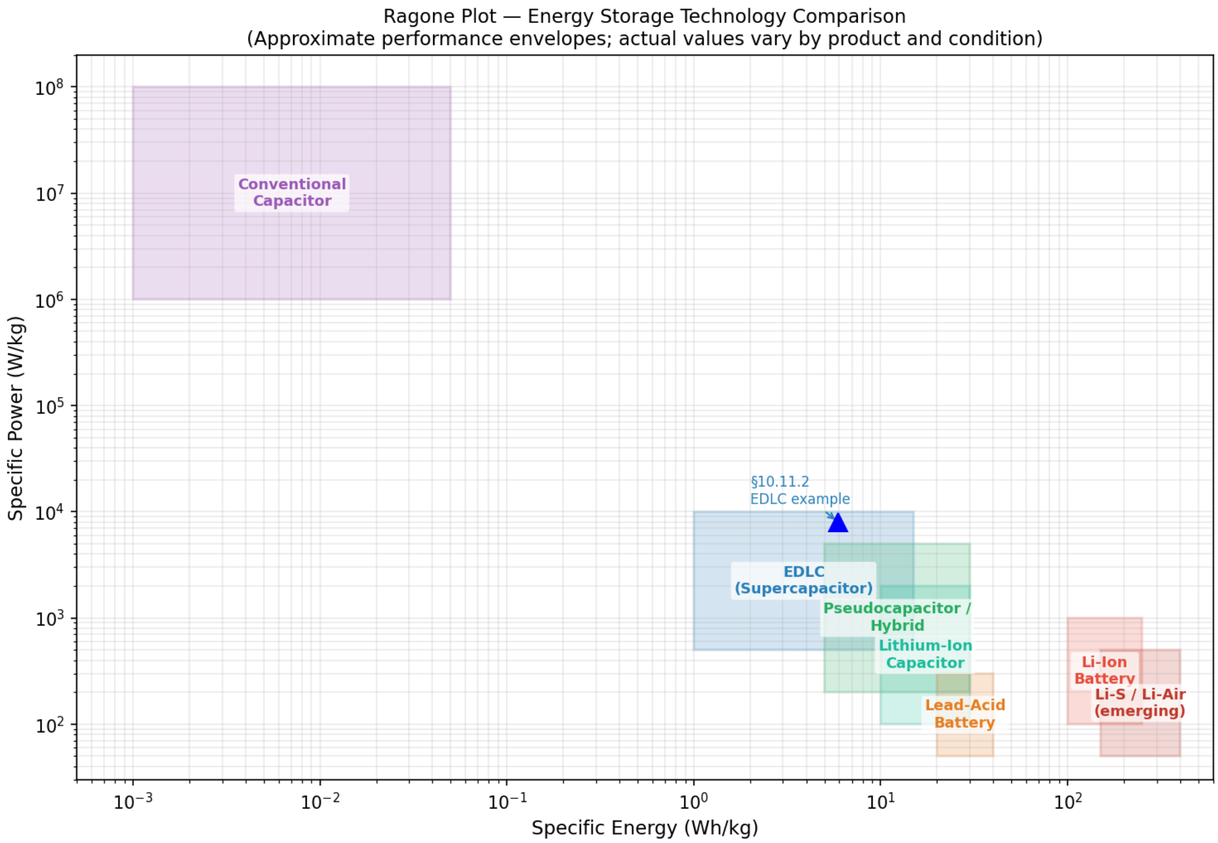



Figure 10.11.2: Ragone Plot — Specific Power vs Specific Energy for Energy Storage Technologies

### 10.11.3 Equivalent Circuit, ESR, and Charge/Discharge

The practical supercapacitor equivalent circuit is a **series RC model**: an ideal capacitor C, a series **equivalent series resistance (ESR)**, and a parallel leakage resistance $R_{leak}$ (typically 100 kΩ–1 MΩ, governing self-discharge rate). The ESR, dominated by electrode contact resistance and electrolyte resistance, limits peak power and causes a voltage step at the start of charge or discharge.

During **constant-current discharge** at current I from initial voltage $V_0$:

$V_C(t) = V_0 - (I/C) \times t$ (capacitor voltage, linear ramp)

$V_{terminal}(t) = V_C(t) - I \times ESR$ (terminal voltage, shifted down by I × ESR)

The maximum power deliverable to a matched load is $P_{max} = V_0^2/(4 \times ESR)$, which highlights that lower ESR directly enables higher peak power. Unlike batteries, which maintain a relatively flat discharge voltage, the supercapacitor terminal voltage drops linearly from $V_0$ to 0 V, requiring a DC-DC converter in most applications to regulate the output voltage as the supercapacitor discharges.

**Example 10.11.3:** A 10 F supercapacitor module has ESR = 50 mΩ and is charged to $V_0 = 16.2$ V (6 cells × 2.7 V). (a) If the module is discharged at a constant 20 A, find the initial terminal voltage. (b) How long until the capacitor voltage falls to 8.1 V (half of initial)? (c) What fraction of total stored energy has been extracted at that point? (d) What is the maximum power available from

this module?
**Solution:**
(a) $V_{terminal}(0) = V_0 - I \times ESR = 16.2 - 20 \times 0.050 = 16.2 - 1.0 = \mathbf{15.2\ V}$
(b) $V_C(t) = 16.2 - (20/10) \times t = 16.2 - 2t$. Setting $V_C = 8.1$:
$8.1 = 16.2 - 2t \rightarrow t = (16.2 - 8.1)/2 = \mathbf{4.05\ s}$
(c) $E_{total} = ½ \times 10 \times 16.2^2 = 1{,}312.2$ J. $E_{remaining} = ½ \times 10 \times 8.1^2 = 328.1$ J.
Fraction extracted = (1,312.2 − 328.1)/1,312.2 = 984.1/1,312.2 = **75%** — discharging to $V_{max}/2$ always extracts exactly 75% of stored energy.
(d) $P_{max} = V_0^2/(4 \times ESR) = 16.2^2/(4 \times 0.050) = 262.44/0.200 = \mathbf{1{,}312\ W}$

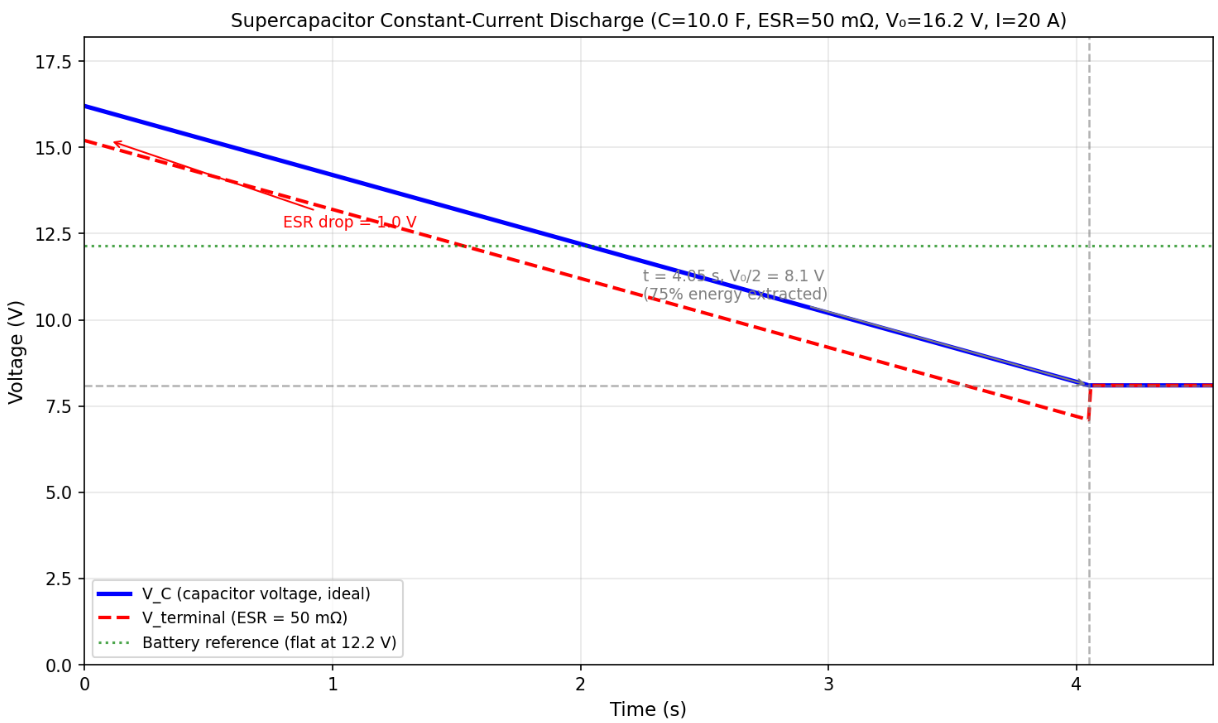



Figure 10.11.3: Supercapacitor Constant-Current Discharge Voltage Profile

### 10.11.4 Series-Parallel Bank Design

Supercapacitors are combined in **series** to increase voltage rating, in **parallel** to increase capacitance and energy, or in series-parallel combinations to meet both requirements. For $n_s$ cells in series with $n_p$ parallel strings:

- $V_{bank} = n_s \times V_{cell}$
- $C_{bank} = n_p \times C_{cell} / n_s$
- $E_{bank} = ½ \times C_{bank} \times V_{bank}^2$

**Cell voltage balancing** is essential for series-connected cells because manufacturing variation in capacitance (±20%) and leakage current causes unequal voltage distribution. **Passive balancing** uses a resistor (typically 100–1,000 Ω) shunted across each cell; while simple, it continuously draws leakage current and reduces efficiency. **Active balancing** uses a switched-capacitor or DC-DC converter topology to shuffle charge between cells, improving efficiency and reducing voltage imbalance to under 10 mV. ESR also adds in series across the string, so total module ESR = $n_s \times ESR_{cell}/n_p$.

**Example 10.11.4:** A supercapacitor energy storage module must deliver at least 100 Wh at a nominal bus voltage of 48 V using cells rated at 2.7 V, 3,000 F each. (a) Determine the minimum number of series cells to meet the voltage requirement. (b) Calculate the string capacitance and energy per string. (c) Determine the number of parallel strings needed. (d) Verify total stored energy.

**Solution:**

(a) Cells per string: $n_s = \lceil 48 / 2.7 \rceil = \lceil 17.78 \rceil =$ **18 cells**; $V_{string} = 18 \times 2.7 = 48.6$ V

(b) $C_{string} = 3{,}000 / 18 =$ **166.7 F**

$E_{string} = ½ \times 166.7 \times (48.6)^2 = ½ \times 166.7 \times 2{,}361.96 =$ **196,869 J = 54.7 Wh**

(c) Parallel strings: $n_p = \lceil 100 / 54.7 \rceil = \lceil 1.83 \rceil =$ **2 strings**

(d) $C_{bank} = 2 \times 166.7 = 333.4$ F; $E_{bank} = ½ \times 333.4 \times (48.6)^2 = 393{,}739$ J = **109.4 Wh ✓** (exceeds 100 Wh requirement)

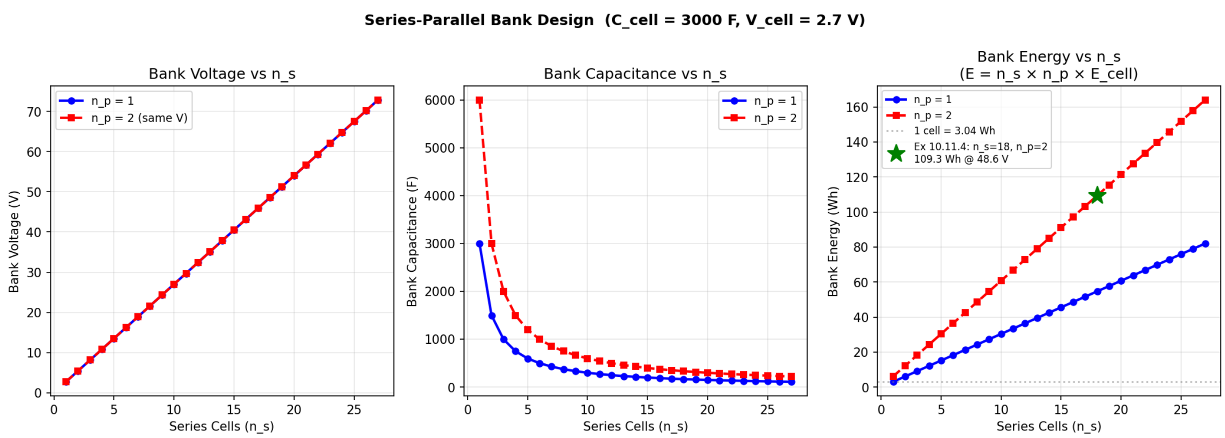



Figure 10.11.4: Series-Parallel Bank Design — Voltage, Capacitance, and Energy vs Series Cell Count

### 10.11.5 Applications: Regenerative Braking and Peak Power Buffering

Supercapacitors excel in **transient energy applications** where energy must be absorbed or delivered in seconds rather than minutes. Key industrial applications include:

- **Regenerative braking** in trains, trams, and hybrid buses: capturing several hundred kilojoules during braking and returning it during acceleration
- **Peak power buffering** in cranes, elevators, and industrial presses: reducing peak current demand from the utility grid and shrinking transformer ratings
- **UPS and ride-through**: bridging the gap during generator startup (typically 10–30 s) for data centers and medical equipment
- **Start-stop automotive systems**: replacing or augmenting lead-acid batteries for engine restart

In a **hybrid battery-supercapacitor system**, the supercapacitor handles fast transients (< 10 s) while the battery supplies the sustained load. This hybrid topology reduces battery peak current, extends battery cycle life by 2–4×, and improves overall system efficiency. The split is governed by the

**energy management strategy** — typically rule-based (frequency splitting) or model-predictive control.

**Example 10.11.5:** A light rail vehicle recovers 40 kJ of kinetic energy during a 5-second braking event. A supercapacitor bank operates between $V_{max}$ = 200 V and $V_{min}$ = 100 V (discharging to half voltage to extract 75% of stored energy). (a) Size the required capacitance. (b) Calculate the average charging current during braking. (c) Verify the energy extracted when the bank is subsequently discharged from 200 V to 100 V.
**Solution:**
(a) Usable energy: $E_{usable} = ½C(V_{max}^2 - V_{min}^2) = ½ \times C \times (200^2 - 100^2) = ½ \times C \times 30{,}000 = 15{,}000C$
Setting $E_{usable}$ = 40,000 J: C = 40,000/15,000 = **2.67 F**
(b) Average power during braking: P = 40,000/5 = 8,000 W.
Average current: $I_{avg} \approx P/V_{avg}$ = 8,000/150 = **53.3 A** ($V_{avg} \approx$ (200+100)/2 = 150 V during charging)
(c) $E_{extracted} = ½ \times 2.67 \times (200^2 - 100^2) = ½ \times 2.67 \times 30{,}000$ = **40,050 J ≈ 40 kJ ✓**

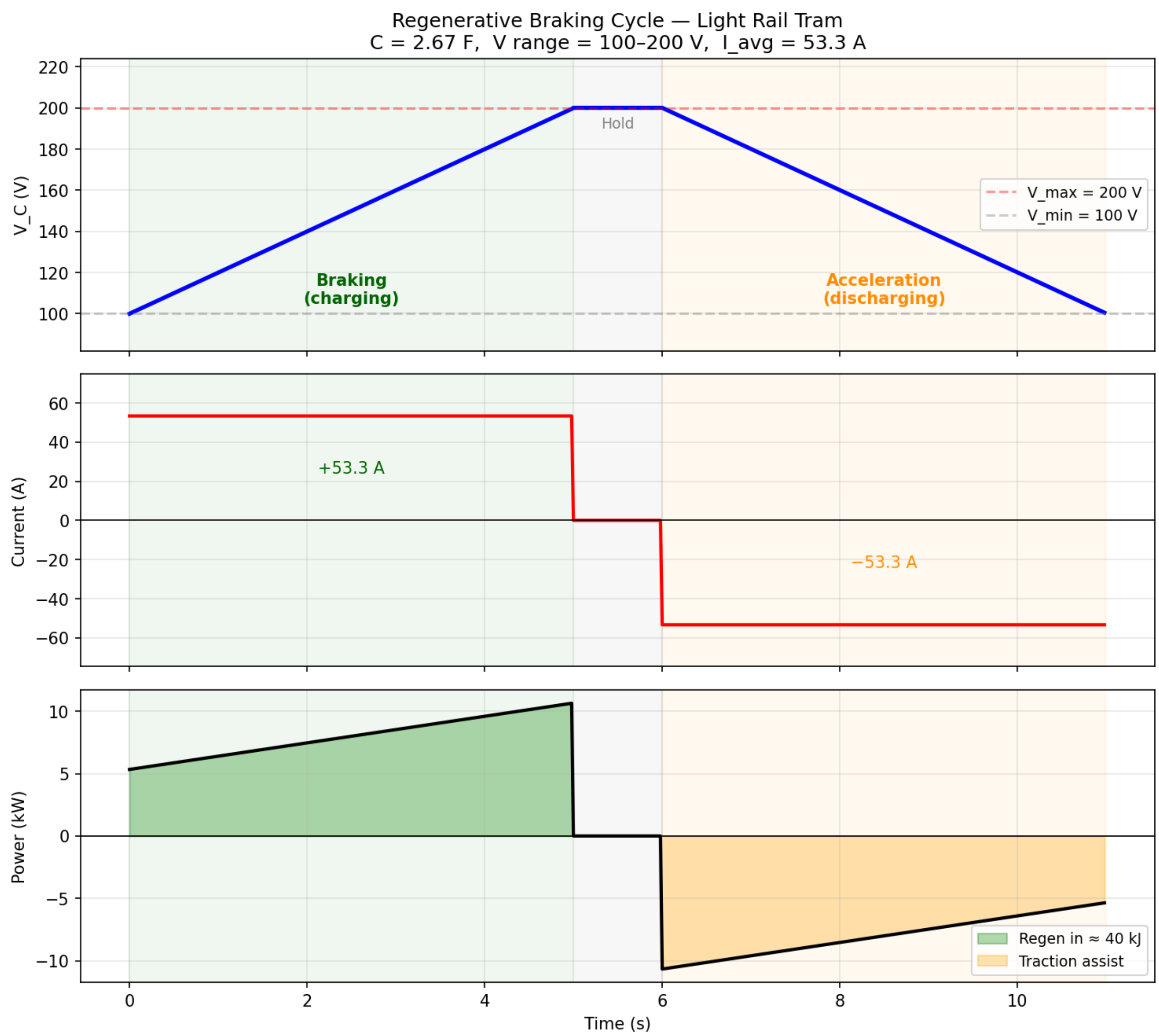



Figure 10.11.5: Regenerative Braking Cycle — Voltage, Current, and Power for a Light Rail Tram

## 10.12 Solar Photovoltaic Systems

Solar photovoltaic (PV) systems convert sunlight directly into electricity using the **photovoltaic effect** — the generation of electron-hole pairs when photons with sufficient energy are absorbed by a semiconductor junction. Power electronics is indispensable at every stage of the energy chain: a **DC-DC converter** with maximum power point tracking (MPPT) extracts the most available power from the array as irradiance and temperature shift continuously, and a **DC-AC inverter** conditions the output for grid injection, AC loads, or battery storage. A PV array behaves as a current source with a nonlinear I-V characteristic whose peak power point moves with every change in operating conditions, making MPPT essential for capturing 97–99% of available energy. Grid-tied inverters must satisfy strict anti-islanding, harmonic distortion, and power quality requirements under IEEE 1547 and UL 1741, while off-grid systems add charge controllers, battery banks, and backup generators for energy autonomy. Solar PV is now among the lowest-cost electricity sources globally, and the power electronics covered throughout this chapter — boost converters (§10.3.2), three-phase inverters (§10.4.2), battery management (§10.8), and energy storage (§10.9) — underlies both utility-scale gigawatt arrays and small rooftop installations.

### 10.12.1 PV Cell Characteristics and the I-V Curve

A photovoltaic cell is a large-area p-n junction operated under illumination. Incident photons with energy exceeding the semiconductor bandgap generate electron-hole pairs; the built-in junction field separates the carriers, producing a photocurrent $I_{ph}$ proportional to irradiance G (W/m²). The cell I-V characteristic is described by the **single-diode model**:

$$I = I_{ph} - I_0(e^{V/(n \cdot VT)} - 1)$$

where $I_0$ is the reverse saturation current, n is the diode ideality factor (1.0–1.5), and $V_T = kT/q \approx$ 25.85 mV at 25 °C. Three parameters define the curve:

- **Short-circuit current** $I_{sc}$: current at V = 0; scales nearly linearly with irradiance G
- **Open-circuit voltage** $V_{oc}$: voltage at I = 0; decreases ~0.3 %/°C for silicon; increases logarithmically with irradiance
- **Maximum power point (MPP)**: ($V_{mp}$, $I_{mp}$) where P = V × I is maximized

The **fill factor** FF = ($V_{mp} \times I_{mp}$) / ($V_{oc} \times I_{sc}$) measures curve "squareness"; commercial silicon cells achieve FF = 0.75–0.85. Cell **efficiency** η = $P_{max}$ / (G × $A_{cell}$) is typically 18–24% for monocrystalline silicon at standard test conditions (STC: G = 1,000 W/m², $T_{cell}$ = 25 °C, AM 1.5 spectrum). A 60-cell series module multiplies $V_{oc}$ and $V_{mp}$ by 60 while leaving $I_{sc}$ unchanged.

**Example 10.12.1:** A 156 mm × 156 mm monocrystalline silicon PV cell (area = 0.0243 m²) has STC parameters: $I_{sc}$ = 9.0 A, $V_{oc}$ = 0.650 V, $I_{mp}$ = 8.5 A, $V_{mp}$ = 0.540 V. (a) Calculate the fill factor. (b) Calculate $P_{max}$. (c) Calculate cell efficiency at STC (G = 1,000 W/m²). (d) At 65 °C, $V_{oc}$ decreases at −2.3 mV/°C from 25 °C. Calculate the new $V_{oc}$ and the percentage drop.
**Solution:**
(a) FF = ($V_{mp} \times I_{mp}$) / ($V_{oc} \times I_{sc}$) = (0.540 × 8.5) / (0.650 × 9.0) = 4.590 / 5.850 = **0.785 (78.5%)**
(b) $P_{max} = V_{mp} \times I_{mp}$ = 0.540 × 8.5 = **4.59 W**
(c) η = $P_{max}$ / (G × A) = 4.59 / (1,000 × 0.0243) = 4.59 / 24.3 = **18.9%**
(d) ΔT = 65 − 25 = 40 °C; $\Delta V_{oc} = -2.3 \times 10^{-3} \times 40 = -0.092$ V
$V_{oc}$(65 °C) = 0.650 − 0.092 = **0.558 V** — a 14.2% reduction, demonstrating why PV output falls significantly on hot days even at constant irradiance.

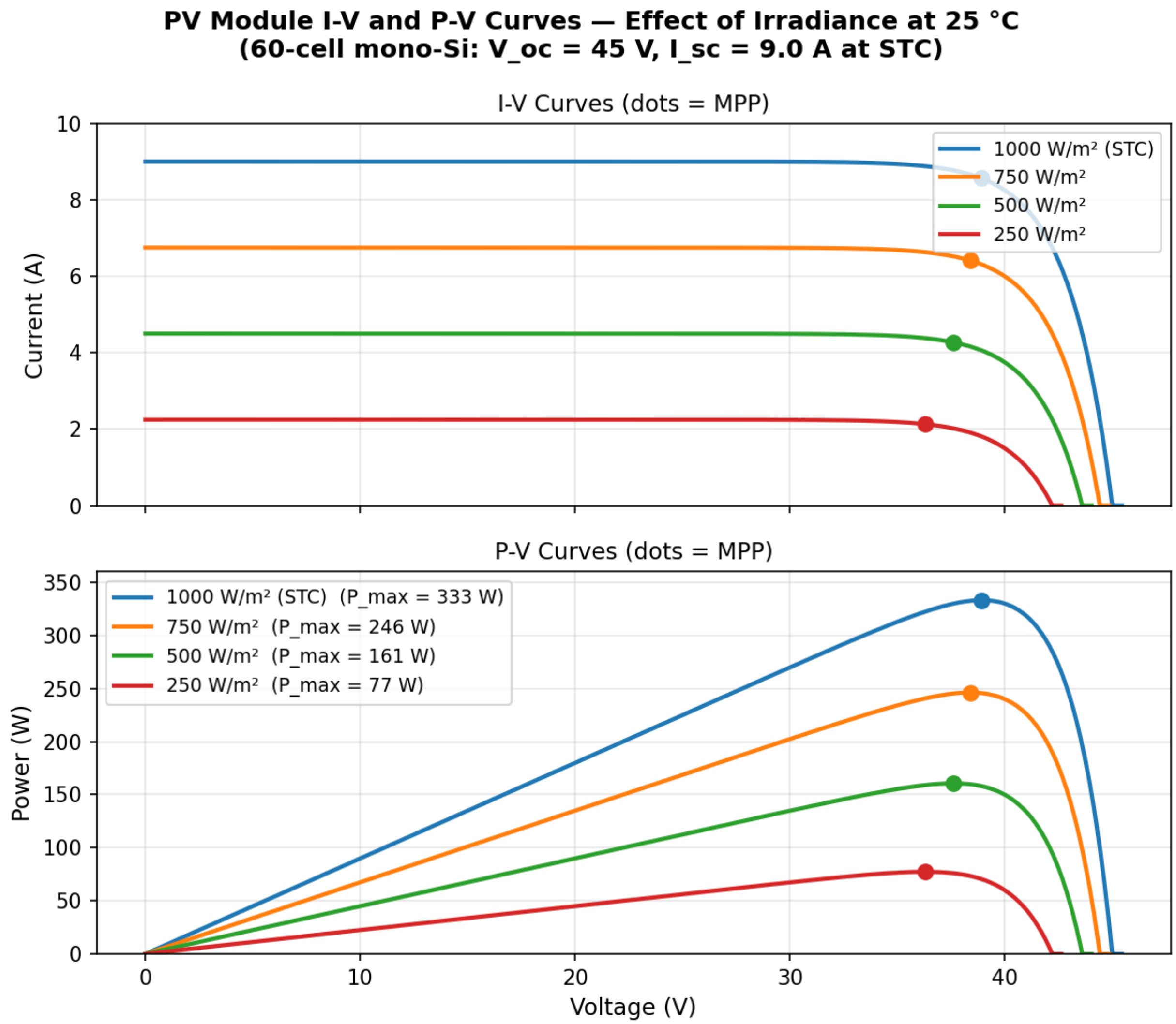



Figure 10.12.1: PV Module I-V and P-V Curves — Effect of Irradiance at 25 °C

### 10.12.2 Maximum Power Point Tracking (MPPT)

A PV module's I-V curve shifts with every change in irradiance and cell temperature, moving the MPP to a new ($V_{mp}$, $I_{mp}$) coordinate. Without active tracking, a converter operating at a fixed duty cycle rarely coincides with the MPP, typically losing 10–30% of available energy. MPPT continuously adjusts the converter operating point to maximize $P = V \times I$.

The most widely deployed algorithm is **Perturb-and-Observe (P&O)**:

1. Measure V and I; compute $P = V \times I$.
2. Perturb voltage by a small step $\Delta V$ (typically 0.5–1 % of $V_{mp}$).
3. Measure new $P'$. If $P' > P$, continue in the same direction; if $P' < P$, reverse direction.
4. Repeat at each control cycle (10–100 ms).

P&O is simple but oscillates with amplitude $\Delta V$ around the MPP in steady state. The **Incremental Conductance (InC)** method eliminates steady-state oscillation by using the analytical MPP condition $dP/dV = 0$, which yields $I + V \times dI/dV = 0$, or equivalently $dI/dV = -I/V$ at the MPP. MPPT efficiency — the ratio of harvested energy to available MPP energy — typically reaches 97–99.5 %. In microinverters and multi-string systems, each module or string has its own MPPT, eliminating losses from module-level mismatch.

**Example 10.12.2:** A PV module has $V_{oc}$ = 45 V, $I_{sc}$ = 9.0 A, $V_{mp}$ = 36 V, $I_{mp}$ = 8.5 A. A boost converter MPPT stage steps up from $V_{pv}$ to a 400 V DC bus with 96% efficiency. (a) Calculate module power at MPP. (b) Calculate the boost converter duty cycle at MPP (ideal). (c) Calculate DC bus power delivered. (d) A P&O algorithm starts at V = 34.0 V with P = 289 W, then steps to V = 34.5 V and measures P = 294 W. State the direction of the next perturbation.
**Solution:**
(a) $P_{mp} = V_{mp} \times I_{mp} = 36 \times 8.5 =$ **306 W**
(b) Ideal boost: $V_{out} = V_{in} / (1 - D) \to 1 - D = 36 / 400 = 0.090 \to D =$ **0.910 (91.0%)**
(c) $P_{dc} = P_{mp} \times \eta = 306 \times 0.96 =$ **293.8 W**
(d) $\Delta P = 294 - 289 = +5$ W $> 0$ with $\Delta V = +0.5$ V → power increased with a voltage increase → **perturb in the same direction (increase V further toward the MPP)**.

### 10.12.3 Grid-Tied PV Inverters

A **grid-tied inverter** synchronizes PV power to the utility grid, injecting current in phase with the grid voltage to achieve unity power factor. Two primary topologies exist:

- **Single-stage**: a DC-AC inverter with built-in MPPT directly converts PV DC to AC; simpler, fewer components, but the DC bus voltage must stay within the inverter's MPPT window at all times
- **Two-stage**: a DC-DC MPPT stage followed by a DC-AC stage; decouples the PV array voltage from the inverter bus, enabling a wider PV operating range and higher overall efficiency

Grid synchronization uses a **phase-locked loop (PLL)** to track grid frequency and phase, ensuring injected current is in phase with the grid voltage. **Anti-islanding protection** is mandatory under IEEE 1547: if the utility goes offline, the inverter must detect the condition and cease energizing the local circuit within 2 seconds, preventing shock hazards for utility workers. Detection methods include passive schemes (over/underfrequency, over/undervoltage) and active schemes (frequency shift, reactive power injection).

Key standards: **IEEE 1547** (interconnection and interoperability), **UL 1741** (inverter safety), **NEC Article 690** (PV system wiring and protection). Modern inverters also support reactive power injection for grid voltage support, low-voltage ride-through (LVRT), and remote curtailment.

**Example 10.12.3:** A single-phase grid-tied PV inverter connects to a 240 $V_{rms}$ / 60 Hz grid with a 400 V DC bus. The inverter outputs 3.0 kW at unity power factor with a 16 kHz switching frequency. (a) Calculate peak grid voltage. (b) Calculate RMS output current. (c) Calculate the SPWM modulation index $m = \hat{V}_{AC}/V_{DC}$. (d) Calculate the number of switching cycles in the 2-second anti-islanding detection window.
**Solution:**
(a) $\hat{V}_{AC} = 240 \times \sqrt{2} =$ **339.4 V**
(b) $I_{rms} = P / V_{rms} = 3{,}000 / 240 =$ **12.5 A**
(c) $m = \hat{V}_{AC} / V_{DC} = 339.4 / 400 =$ **0.849**

(d) $N_{cycles} = f_{sw} \times t = 16{,}000 \times 2 =$ **32,000 switching cycles**

### 10.12.4 Off-Grid and Hybrid PV Systems

An **off-grid (stand-alone) PV system** operates without utility connection, supplying all loads from PV generation and battery storage. The main components are:

- **PV array**: sized to meet daily load energy plus system losses
- **Charge controller**: regulates charging to prevent overcharge and overdischarge; **PWM controllers** connect the array directly to the battery (simple, lower cost, but less efficient when $V_{pv} > V_{battery}$); **MPPT charge controllers** use a DC-DC converter to independently track the array MPP, extracting 10–30% more energy
- **Battery bank**: provides autonomy during nights and low-irradiance periods; typically 1–5 days of storage; lead-acid (50% usable DoD) or lithium iron phosphate (80–90% usable DoD)
- **Inverter/charger**: converts battery DC to AC loads; typically includes a transfer switch for optional AC generator backup

A **hybrid system** adds a diesel or propane generator as backup, operated only when PV generation and battery state of charge cannot meet the load. System sizing begins with daily load energy (Wh/-day), determines battery capacity from autonomy days and allowable DoD, then sizes the PV array to restore the battery within one average peak-sun day.

**Example 10.12.4:** An off-grid cabin has a daily load of 1.2 kWh at a 24 V bus. The system must provide 3 days of autonomy with AGM lead-acid batteries (50% maximum DoD). Peak sun hours = 5 h/day; system efficiency from PV array to load = 80%. (a) Calculate minimum battery capacity in Ah. (b) Calculate minimum PV array peak power. (c) If modules have $V_{mp}$ = 36 V and an MPPT charge controller steps down to 24 V, calculate the required array $I_{mp}$.
**Solution:**
(a) Autonomy energy = 3 × 1,200 = 3,600 Wh. At 50% DoD, total battery energy needed = 3,600 / 0.50 = 7,200 Wh.
Capacity = 7,200 / 24 = **300 Ah**
(b) $E_{pv}$ = 1,200 / 0.80 = 1,500 Wh/day; $P_{pv}$ = 1,500 / 5 = **300 W** (peak array power)
(c) $I_{mp} = P_{pv} / V_{mp}$ = 300 / 36 = **8.33 A** — the MPPT controller draws this current at 36 V and delivers equivalent power at 24 V to the battery (less converter losses).

### 10.12.5 PV Array Wiring, Shading, and Protection

**String configuration** combines modules in series and parallel to meet inverter input requirements. $N_s$ modules in series yield a string voltage of $N_s \times V_{module}$; $N_p$ parallel strings give total array current of $N_p \times I_{string}$. String voltage must fall within the inverter's MPPT window at all temperatures: at low temperature (high $V_{oc}$) the string must not exceed the inverter's maximum DC input voltage; at high temperature (low $V_{mp}$) the string must remain above the MPPT minimum.

**Partial shading** is a critical design issue: one shaded cell in a series string restricts the current of all cells in that string, causing severe power loss and potentially damaging hot spots. **Bypass diodes** — typically one per 18–20 cells, or three per 60-cell module — allow current to flow around a shaded cell group, confining the loss to the affected bypass zone. Without bypass diodes, a single shaded cell could eliminate nearly all of the string's output; with bypass diodes, only the shaded zone (one-third of a module in a three-diode design) is sacrificed.

Additional NEC Article 690 protection requirements include:

- **Arc Fault Circuit Interrupter (AFCI)** — detects series and parallel arcs in DC wiring (NEC 690.11)
- **Rapid Shutdown System (RSS)** — reduces rooftop conductor voltage within 30 seconds to protect first responders (NEC 690.12)
- **Overcurrent protection** — string fuses or breakers required when three or more strings are paralleled

**Example 10.12.5:** A PV string has 20 series-connected modules, each rated: $V_{oc}$ = 45 V, $I_{sc}$ = 9.5 A, $V_{mp}$ = 37 V, $I_{mp}$ = 9.0 A. Each module has 3 bypass diodes protecting 20 cells each. (a) Calculate string $V_{oc}$ and string $P_{mp}$ at full irradiance. (b) One module falls fully into shade, activating its bypass diodes. Calculate the new string power. (c) Without bypass diodes the shaded module limits string current to 1.0 A. Estimate the power loss.
**Solution:**
(a) String $V_{oc} = 20 \times 45 =$ **900 V**; String $P_{mp} = (20 \times 37) \times 9.0 = 740 \times 9.0 =$ **6,660 W (6.66 kW)**
(b) With bypass diodes, the shaded module is bypassed: current remains at $I_{mp}$ = 9.0 A; the shaded module contributes ≈ 0 V and 0 W.
$V_{string} = 19 \times 37 = 703$ V; $P_{shaded} = 703 \times 9.0 =$ **6,327 W** — a loss of 333 W (5.0%), equal to exactly one module's output.
(c) Without bypass diodes, string current is limited to 1.0 A.
$P_{no\text{-}bypass} = 740 \times 1.0 = 740$ W; loss $= 6{,}660 - 740 =$ **5,920 W (88.9%)** — one shaded module eliminates nearly 90% of string output, demonstrating why bypass diodes are mandatory in series strings.

# Chapter 11

# Instrumentation and Measurement

Instrumentation and measurement is the branch of electrical engineering concerned with the accurate sensing, acquisition, and analysis of physical quantities such as voltage, current, resistance, temperature, pressure, and frequency. It encompasses the design of sensors and transducers that convert physical phenomena into electrical signals, the signal conditioning circuits that amplify, filter, and convert those signals, and the measurement instruments and data acquisition systems that display, record, and analyze the results. Precise measurement is fundamental to all engineering disciplines — without the ability to accurately quantify a parameter, it cannot be controlled, optimized, or verified.

## 11.1 Measurement Fundamentals

Every measurement system is characterized by how faithfully it represents the true value of the quantity being measured. Measurement fundamentals encompass the concepts of accuracy, precision, resolution, error analysis, calibration, and signal-to-noise ratio — the tools engineers use to quantify measurement quality, identify error sources, and establish confidence in their results. Understanding these fundamentals is prerequisite to selecting the right sensor, instrument, or data acquisition system for a given application.

### 11.1.1 Accuracy, Precision, and Resolution

Accuracy describes how close a measurement is to the true value of the quantity being measured, while precision describes the repeatability or consistency of measurements under the same conditions. A measurement system can be precise without being accurate (consistent but offset from the true value) or accurate on average without being precise (scattered around the true value). Resolution is the smallest change in the measured quantity that the instrument can detect, determined by the number of bits in a digital system (e.g., a 12-bit ADC with a 10 V range has a resolution of $10/4096 \approx 2.44$ mV). Measurement uncertainty is the quantitative expression of doubt about a measurement result, combining systematic errors (bias) and random errors (noise), and is typically reported as an expanded uncertainty with a stated confidence level (e.g., ±0.5% at 95% confidence).

**Example 11.1.1:** A 16-bit ADC has an input range of 0-5 V. A voltage source is measured 10 times, yielding readings (in volts): 3.3021, 3.3018, 3.3025, 3.3019, 3.3022, 3.3020, 3.3023, 3.3017, 3.3024, 3.3021. The true voltage is 3.3050 V. Determine the resolution, accuracy, and precision.
**Solution:**
Resolution = full-scale range / $2^N = 5 / 2^{16} = 5 / 65{,}536 = 76.3\ \mu$V per count.
Mean of measurements: $\bar{x}$ = (sum of all values) / 10 = 33.021 / 10 = 3.3021 V.

Accuracy (systematic error): |3.3050 - 3.3021| = 2.9 mV, or (2.9 / 3.3050) × 100 = 0.088% of reading.
Precision (standard deviation): $s = \sqrt{[\Sigma(x_i - \bar{x})^2 / (N-1)]}$.
The deviations from the mean are: 0, -0.3, +0.4, -0.2, +0.1, -0.1, +0.2, -0.4, +0.3, 0 mV.
$s = \sqrt{[(0 + 0.09 + 0.16 + 0.04 + 0.01 + 0.01 + 0.04 + 0.16 + 0.09 + 0) \times 10^{-6} / 9]} = \sqrt{[0.0667 \times 10^{-6}]} = 0.258$ mV.
The instrument is precise (small spread) but has a 2.9 mV systematic offset requiring calibration.

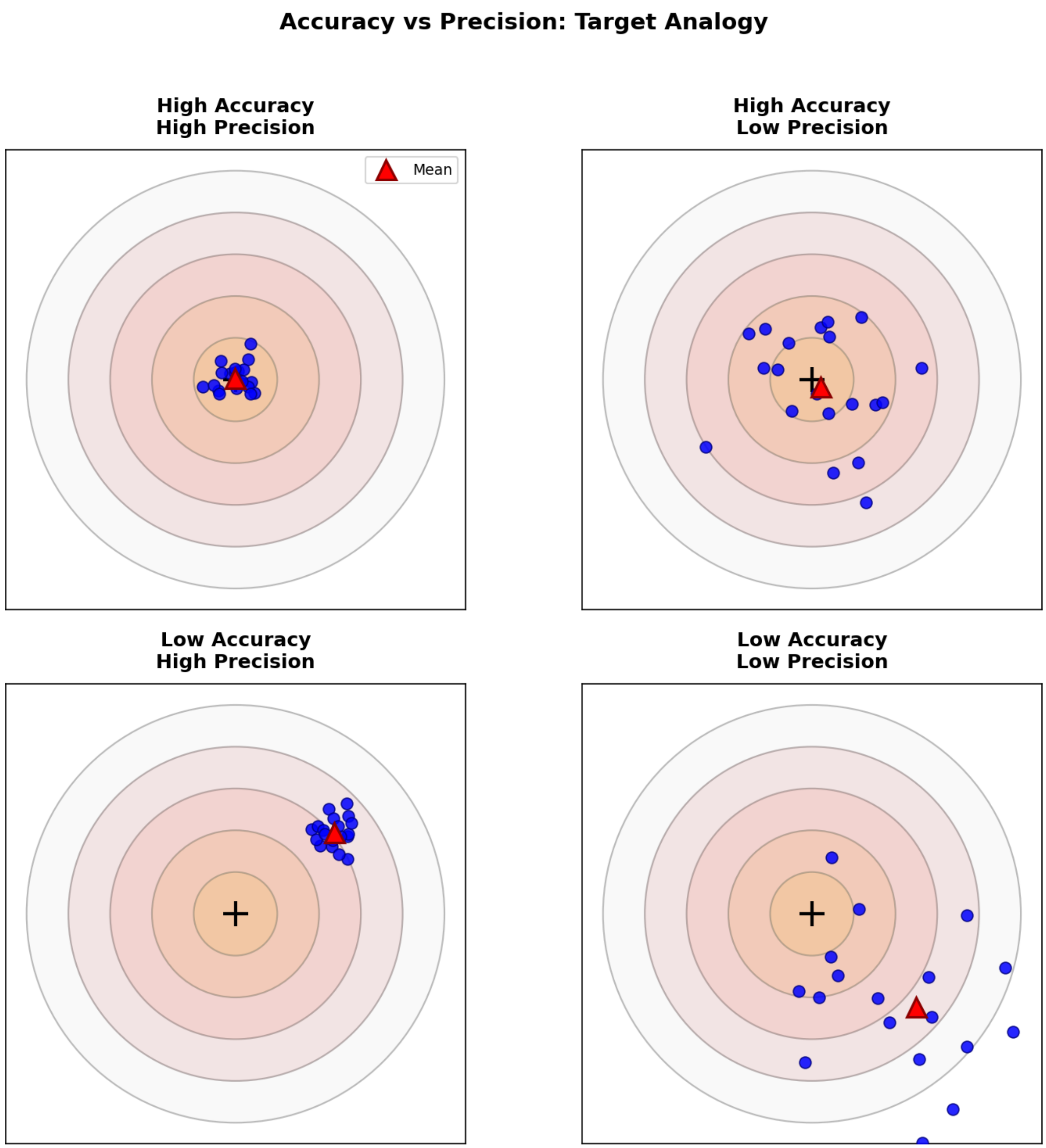



Figure 11.1.1: Accuracy vs Precision: Target Analogy

### 11.1.2 Error and Calibration

Measurement errors are classified as systematic (consistent bias that affects all measurements equally) or random (statistical variations that cause scatter around a mean value). Common sources of systematic error include instrument offset, gain error, nonlinearity, loading effects (where the measurement instrument alters the quantity being measured), and environmental influences such as temperature drift. Calibration is the process of comparing an instrument's readings against a known reference standard and adjusting or documenting the corrections needed to minimize systematic error. Traceability ensures that calibrations are linked through an unbroken chain of comparisons to national or international measurement standards (such as those maintained by NIST or PTB). Regular calibration intervals and proper documentation are essential for maintaining measurement confidence and meeting quality standards such as ISO 17025.

**Example 11.1.2:** A voltmeter is calibrated against a precision reference standard at five points. The reference values and meter readings are: 1.000 V reads 1.003 V, 2.000 V reads 2.005 V, 3.000 V reads 3.008 V, 4.000 V reads 4.010 V, 5.000 V reads 5.013 V. Determine the gain error and offset error assuming a linear error model.
**Solution:**
Using a linear error model: $V_{reading} = G \times V_{true} + V_{offset}$, where G is the gain and $V_{offset}$ is the offset.
Using the first and last calibration points: G = (5.013 - 1.003) / (5.000 - 1.000) = 4.010 / 4.000 = 1.00250.
$V_{offset} = V_{reading} - G \times V_{true} = 1.003 - 1.00250 \times 1.000 = 1.003 - 1.00250 = 0.50$ mV.
Gain error = (G - 1) × 100% = 0.250%.
Verification at 3.000 V: predicted reading = 1.00250 × 3.000 + 0.00050 = 3.0080 V, actual reading = 3.008 V – confirming the linear model.
Corrected measurement: $V_{true} = (V_{reading} - 0.00050) / 1.00250$.

### 11.1.3 Signal-to-Noise Ratio

The Signal-to-Noise Ratio (SNR) quantifies the strength of the desired signal relative to the background noise, expressed in decibels: $SNR(dB) = 20\log_{10}(V_{signal}/V_{noise})$ for voltage measurements, or equivalently $10\log_{10}(P_{signal}/P_{noise})$ for power. A higher SNR indicates a cleaner measurement with less noise corruption. Common noise sources in measurement systems include thermal noise (Johnson-Nyquist noise, proportional to $\sqrt{(4kTR\Delta f)}$), flicker noise (1/f noise, dominant at low frequencies), quantization noise (inherent in analog-to-digital conversion), and electromagnetic interference (EMI) from external sources. Techniques for improving SNR include shielding and grounding, differential measurements to reject common-mode noise, filtering to limit bandwidth to the frequency range of interest, signal averaging to reduce random noise by $\sqrt{N}$ (where N is the number of averages), and lock-in amplification to extract signals buried deep in noise.

**Example 11.1.3:** A thermocouple signal of 5 $mV_{rms}$ is measured through a sensor with 50 kΩ source resistance at 25°C using an amplifier with 100 kHz bandwidth. Calculate the thermal noise voltage and the SNR. How many signal averages are needed to achieve 60 dB SNR?
**Solution:**
Thermal noise voltage: $V_n = \sqrt{(4kTR\Delta f)} = \sqrt{(4 \times 1.381 \times 10^{-23} \times 298 \times 50{,}000 \times 100{,}000)} = \sqrt{(4 \times 1.381 \times 10^{-23} \times 1.49 \times 10^{12})} = \sqrt{(82.3 \times 10^{-12})} = 9.07\ \mu V_{rms}$.

$SNR = 20\log_{10}(V_{signal} / V_{noise}) = 20\log_{10}(5 \times 10^{-3} / 9.07 \times 10^{-6}) = 20\log_{10}(551.3) = 20 \times 2.741 = 54.8$ dB.
To achieve 60 dB: improvement needed = 60 - 54.8 = 5.2 dB, corresponding to a voltage ratio improvement of 10^(5.2/20) = 1.82×.
Since averaging improves SNR by $\sqrt{N}$: $\sqrt{N} = 1.82$, so N = 3.3 – round up to N = 4 averages.

### 11.1.4 Measurement Uncertainty

Measurement uncertainty quantifies the range of values within which the true value of a measurand is expected to lie, following the internationally accepted framework defined by the Guide to the Expression of Uncertainty in Measurement (GUM, ISO/IEC 98-3). **Type A uncertainty** is evaluated by statistical methods — repeated measurements yield a mean and standard deviation, and the standard uncertainty is $u_A = s/\sqrt{N}$, where s is the sample standard deviation and N is the number of observations. **Type B uncertainty** is evaluated by non-statistical methods — manufacturer specifications, calibration certificates, environmental estimates, and engineering judgment — and is converted to a standard uncertainty by dividing the stated tolerance by an appropriate coverage factor (e.g., $\div\sqrt{3}$ for a rectangular distribution, ÷2 for a normal distribution at 95%). The **combined standard uncertainty** $u_c$ is calculated by the root-sum-of-squares (RSS) of all individual uncertainty components: $u_c = \sqrt{(u_1^2 + u_2^2 + \ldots + u_n^2)}$, assuming uncorrelated inputs. The **expanded uncertainty** $U = k \times u_c$ applies a coverage factor k (typically k = 2 for 95% confidence or k = 3 for 99.7%) to define the interval within which the measurand lies with the stated probability. Proper uncertainty analysis is required for ISO 17025 accredited laboratories and is essential for determining measurement capability, pass/fail conformance decisions (guard banding), and calibration hierarchy.

**Example 11.1.4:** A calibrated thermometer measures 85.3°C. The uncertainty budget includes: (1) repeatability from 6 measurements with standard deviation s = 0.12°C, (2) calibration certificate uncertainty of ±0.15°C at k = 2, (3) resolution of 0.1°C (digital display), and (4) drift since last calibration estimated as ±0.08°C (rectangular distribution). Calculate the combined standard uncertainty and the expanded uncertainty at 95% confidence.
**Solution:**
Type A: $u_1 = s/\sqrt{N} = 0.12/\sqrt{6} = 0.049$°C.
Type B components: calibration $u_2 = 0.15/2 = 0.075$°C (convert from k = 2 to standard uncertainty).
Resolution $u_3 = 0.1/(2\sqrt{3}) = 0.029$°C (half the resolution, rectangular distribution).
Drift $u_4 = 0.08/\sqrt{3} = 0.046$°C (rectangular distribution).
Combined standard uncertainty: $u_c = \sqrt{(0.049^2 + 0.075^2 + 0.029^2 + 0.046^2)} = \sqrt{(0.00240 + 0.00563 + 0.00084 + 0.00212)} = \sqrt{(0.01099)} = 0.105$°C.
Expanded uncertainty at 95% (k = 2): $U = 2 \times 0.105 = 0.21$°C.
The measurement result is reported as 85.3 ± 0.2°C (k = 2, 95% confidence).
The calibration certificate is the dominant uncertainty contributor (51% of variance), suggesting that upgrading to a lower-uncertainty reference standard would most effectively improve the overall measurement.

### 11.1.5 Measurement Bridges

A measurement bridge is a circuit that determines an unknown impedance by balancing it against known reference impedances, achieving extraordinary accuracy because the measurement is based on a null (zero) condition rather than on absolute voltage or current readings. The **Wheatstone bridge** is the most fundamental form: four resistors arranged in a diamond with a voltage source across one diagonal and a null detector (galvanometer) across the other. At balance, $R_{unknown} = R_2 \times R_3 / R_1$, and the detector reads zero regardless of source voltage variations — a key advantage over direct measurement. The **Kelvin double bridge** extends the Wheatstone principle to measure very low resistances (micro-ohms to milli-ohms) by adding a second set of ratio arms that eliminates errors from lead and contact resistance, essential for measuring shunt resistors, busbar connections, and superconductor joints. For AC impedance measurement, the **Wien bridge** measures capacitance and dissipation factor by balancing a capacitive arm against resistive references at a specific frequency, while the **Maxwell bridge** (also called the Maxwell-Wien bridge) measures inductance and Q factor by balancing an inductive arm against a capacitive reference — avoiding the need for a standard inductor, which is difficult to construct with known parasitic resistance. The **Schering bridge** is optimized for measuring capacitance and dielectric loss in high-voltage insulation systems, and is the standard method for assessing transformer, cable, and bushing insulation condition. Modern bridge instruments automate the balancing process digitally, but the underlying null-balance principle remains the foundation of precision impedance measurement.

**Example 11.1.5:** A Wheatstone bridge is used to measure an unknown resistance. The bridge has $R_1 = 1{,}000\ \Omega$, $R_2 = 10{,}000\ \Omega$ (decade box), and the standard arm $R_3$ is adjusted to $4{,}728\ \Omega$ for a null reading on the galvanometer. Calculate $R_{unknown}$. If the detector sensitivity is 5 μA/mV of bridge imbalance and the minimum detectable current is 0.1 μA with a 10 V excitation, estimate the measurement resolution.

**Solution:**

At balance: $R_x = R_2 \times R_3 / R_1 = 10{,}000 \times 4{,}728 / 1{,}000 = 47{,}280\ \Omega = 47.28\ k\Omega$.

The ratio $R_2/R_1 = 10{,}000/1{,}000 = 10$ is the bridge multiplier, extending the range of the decade box by 10×.

For resolution, the minimum detectable imbalance voltage: $V_{min} = I_{min}$ / sensitivity = 0.1 μA / 5 μA/mV = 0.02 mV.

Near balance the sensitivity is $dV/dR_x = V_{ex} \times R_3 / (R_3 + R_x)^2$, the exact bridge sensitivity at the operating point.

With $R_3 = 4{,}728\ \Omega$ and $R_x = 47{,}280\ \Omega$: $dV/dR_x = 10 \times 4{,}728 / (52{,}008)^2 = 47{,}280 / (2.705 \times 10^9) = 0.01748$ mV/Ω.

Minimum detectable resistance change: $\Delta R = V_{min} / (dV/dR_x) = 0.02 / 0.01748 = 1.14\ \Omega$.

Resolution as a fraction of $R_x$: 1.14 / 47,280 = 24.1 ppm — demonstrating the exceptional precision achievable with null-balance measurements.

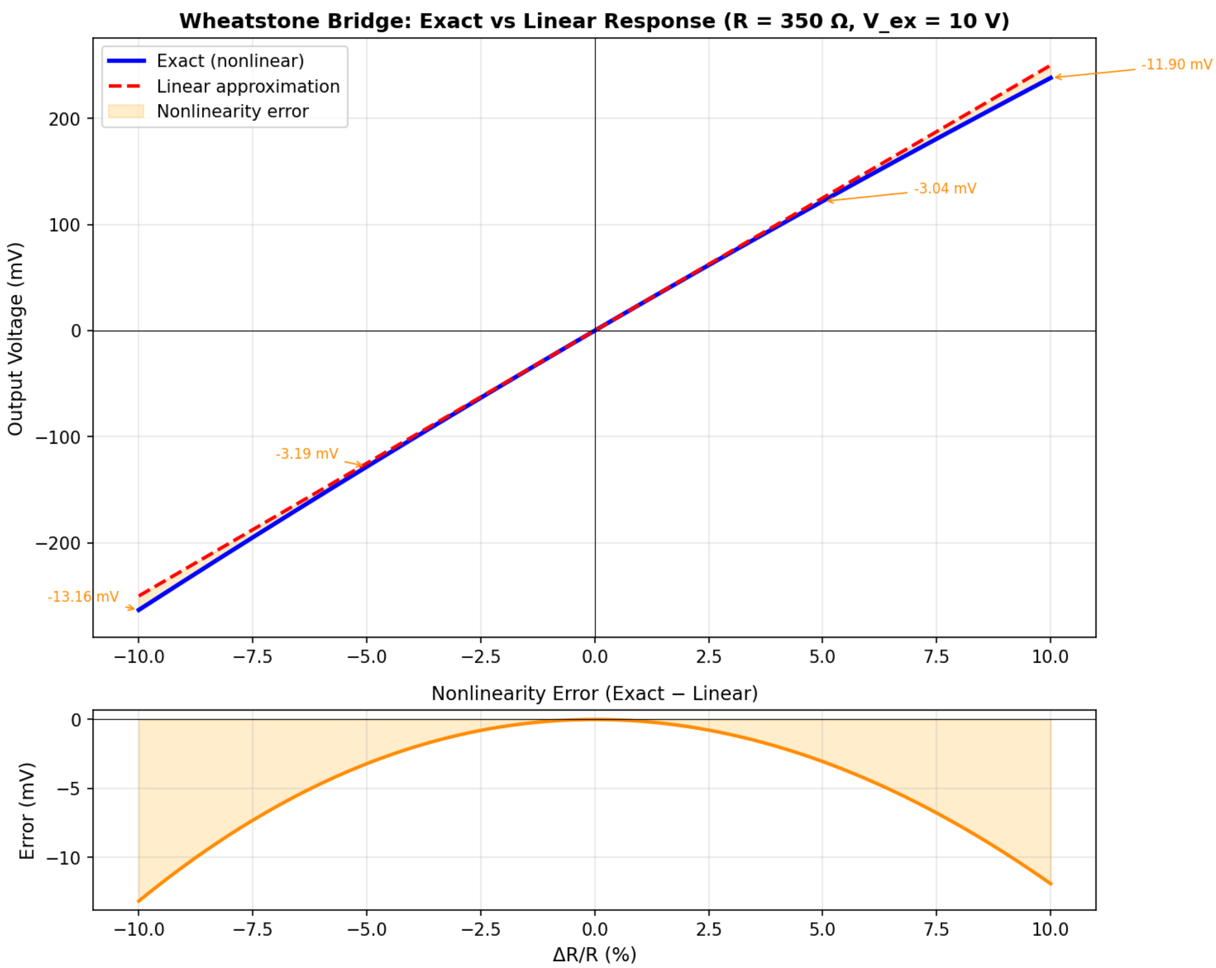



Figure 11.1.5: Wheatstone Bridge Response

## 11.1.6 Grounding and Shielding in Measurement Systems

Grounding and shielding are the primary defenses against noise, interference, and ground-loop errors in precision measurement systems. A ground loop forms whenever two or more points in a measurement circuit are connected to ground at different physical locations, and a potential difference exists between those ground points — even millivolts of ground potential difference can overwhelm microvolt-level sensor signals. Proper grounding topology, cable shielding, and isolation techniques are essential for achieving the full accuracy of any instrumentation system, particularly when measuring low-level signals from thermocouples, strain gauges, and biomedical electrodes in electrically noisy industrial or laboratory environments.

**Single-point grounding** connects all circuit returns to one common ground point (a star topology), eliminating ground loops by ensuring no current flows between ground connections. This approach works well at low frequencies (DC to ~1 MHz) where cable lengths are short relative to the wavelength. **Multi-point grounding** connects each subsystem to the nearest low-impedance ground plane and is necessary at high frequencies (>1 MHz) where lead inductance in long single-point ground wires creates significant impedance — RF and high-speed digital systems universally use multi-point grounding to a solid ground plane. **Hybrid grounding** uses single-point topology at low frequencies and transitions to multi-point at high frequencies via capacitors that bypass the single-point connection above a crossover frequency.

**Cable shielding** intercepts electromagnetic interference before it couples into signal conductors. Braided copper shields provide 70–95% optical coverage and excellent low-frequency magnetic shielding due to their low DC resistance, while aluminum foil shields provide 100% coverage against electric fields but higher resistance that limits low-frequency performance — many high-quality cables combine both (foil + braid). The shield must be grounded, and the grounding method is critical: grounding the shield at one end only (the instrument end) prevents shield current from flowing and is preferred for low-frequency measurements, while grounding at both ends provides better high-frequency shielding but allows ground-loop currents to flow through the shield. A **drain wire** in foil-shielded cables provides a convenient low-resistance termination point for the foil shield. **Driven shields** (guard shields) use an amplifier to actively drive the cable shield to the same potential as the signal conductor, eliminating the cable capacitance effect (since there is no voltage across it) and dramatically improving the common-mode rejection ratio at high source impedances — this technique is standard in electrometer-grade instruments measuring picoampere currents through gigaohm source impedances.

Common-mode rejection (CMR) is the ability of a differential measurement system to reject voltages that appear equally on both input leads relative to ground. An instrumentation amplifier with 100 dB CMRR can reject 100,000:1, but imbalances in source impedance, cable capacitance, or shield connections degrade the effective CMRR. **Isolation** — using transformer-coupled, optically-coupled, or capacitively-coupled amplifiers — provides the ultimate ground-loop solution by completely breaking the galvanic path between the sensor and the measurement system, achieving isolation-mode rejection ratios of 140–160 dB at DC and common-mode voltage ratings of 1,000–5,000 $V_{rms}$.

**Example 11.1.6:** A thermocouple (source impedance $Z_s = 50\ \Omega$) is connected to a data acquisition system 30 m away. The ground potential difference between the sensor location and the DAQ system is $V_{gnd} = 2.5$ V at 60 Hz. The shielded cable has a shield resistance of $R_{shield} = 0.15\ \Omega$ and the shield is grounded at both ends. The DAQ input has CMRR = 90 dB. Calculate the ground-loop current through the shield, the error voltage induced in the signal conductors, and the total measurement error on a 5 mV thermocouple signal. Then determine the error if the shield is grounded at one end only.

**Solution:**

Ground-loop current through shield (grounded at both ends): $I_{loop} = V_{gnd} / R_{shield} = 2.5 / 0.15 = 16.7$ A.

This large current flowing through the shield generates a magnetic field that couples into the signal conductors. The mutual inductance coupling with a typical twisted-pair cable inside the shield induces approximately $V_{induced} \approx M \times dI/dt$. For a 60 Hz sinusoidal current, $dI/dt = 2\pi \times 60 \times 16.7 = 6{,}283$ A/s. With typical mutual coupling of $M \approx 0.1\ \mu$H/m over 30 m: $V_{induced} = 0.1 \times 10^{-6} \times 30 \times 6{,}283 = 18.8$ mV.

Additionally, the CMRR-limited error from the 2.5 V common-mode voltage: $V_{cm_error} = V_{gnd} / 10^{CMRR/20} = 2.5 / 10^{90/20} = 2.5 / 31{,}623 = 79\ \mu$V.

Total error voltage: 18.8 mV + 0.079 mV ≈ 18.9 mV — nearly 4× the 5 mV thermocouple signal, rendering the measurement useless.

With shield grounded at one end only: no shield current flows ($I_{loop} = 0$), so no magnetically induced error. The common-mode voltage still appears on both signal leads, but the differential

error is limited to the CMRR contribution: $V_{error}$ = 79 μV, which is 79/5000 × 100 = 1.6% of the signal — much improved.
Using a fully isolated input (160 dB IMRR) would reduce the error to 2.5 / $10^8$ = 25 nV, achieving 0.0005% error.

## 11.2 Sensors and Transducers

Sensors and transducers are the front end of any measurement system, converting physical quantities — temperature, strain, pressure, position, acceleration, magnetic field, and current — into electrical signals that can be conditioned, digitized, and analyzed. A sensor detects the physical phenomenon, while a transducer converts it into a proportional electrical output (voltage, current, resistance change, or charge). The selection of a sensor involves trade-offs among sensitivity, range, linearity, bandwidth, environmental ruggedness, and cost, and the sensor's characteristics directly determine the achievable accuracy and resolution of the overall measurement system.

### 11.2.1 Temperature Sensors

Temperature is one of the most commonly measured physical quantities in engineering. Thermocouples generate a voltage proportional to the temperature difference between two junctions of dissimilar metals (Seebeck effect), covering wide temperature ranges (-200°C to +2300°C depending on type) with types K, J, T, and E being the most common. Resistance Temperature Detectors (RTDs) use the predictable change in resistance of a pure metal (typically platinum, Pt100 or Pt1000) with temperature, offering high accuracy (±0.1°C) and excellent long-term stability. Thermistors are ceramic semiconductor devices with a large negative (NTC) or positive (PTC) temperature coefficient of resistance, providing high sensitivity over a narrow range. Integrated circuit temperature sensors (such as the LM35 and TMP36) provide a linear voltage output proportional to temperature with built-in signal conditioning, simplifying interfacing with microcontrollers.

**Example 11.2.1:** A Pt100 RTD ($R_0$ = 100 Ω at 0°C, α = 0.00385 Ω/Ω/°C) is used to measure furnace temperature. The measured resistance is 247.1 Ω. Using the linear approximation $R(T) = R_0(1 + \alpha T)$, calculate the temperature. A Type K thermocouple at the same location reads 10.57 mV (reference junction at 0°C). Using the approximate sensitivity of 41 μV/°C, what temperature does the thermocouple indicate?
**Solution:**
For the Pt100: $R(T) = R_0(1 + \alpha T)$, so 247.1 = 100(1 + 0.00385T).
2.471 = 1 + 0.00385T.
T = 1.471 / 0.00385 = 382.1°C.
For the Type K thermocouple: T = V / sensitivity = $10.57 \times 10^{-3} / 41 \times 10^{-6}$ = 257.8°C.
The significant discrepancy (382°C vs. 258°C) suggests the linear thermocouple approximation is too crude at this temperature; using the Type K thermocouple tables, 10.57 mV actually corresponds to approximately 260°C.
This illustrates why RTDs provide superior accuracy and why thermocouple linearization tables or polynomial corrections are essential for precise measurements.

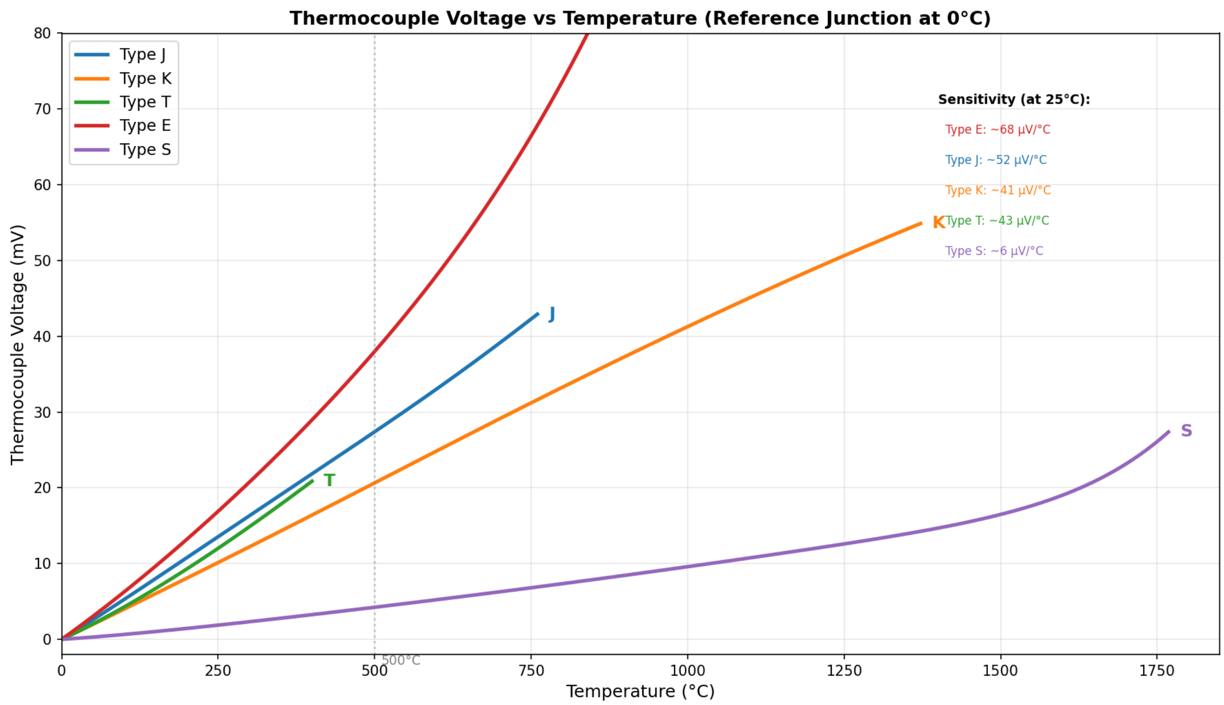



Figure 11.2.1: Thermocouple Voltage vs Temperature

### 11.2.2 Strain and Force Sensors

Strain gauges measure mechanical deformation by detecting the change in electrical resistance of a conductor or semiconductor element bonded to the surface under test. A metallic foil strain gauge has a gauge factor (ratio of fractional resistance change to strain) of approximately 2, while semiconductor strain gauges achieve gauge factors of 100-200 for much greater sensitivity. The Wheatstone bridge is the standard circuit for measuring strain gauge resistance changes, with quarter-bridge, half-bridge, and full-bridge configurations offering increasing sensitivity and temperature compensation. Load cells are force transducers constructed from strain gauges bonded to a precisely machined elastic element (bending beam, shear beam, or compression column), calibrated to produce an output voltage proportional to the applied force or weight. Piezoelectric sensors generate an electric charge proportional to applied mechanical stress and are used for measuring dynamic forces, pressure, and acceleration.

**Example 11.2.2:** A 350 Ω metallic foil strain gauge with a gauge factor GF = 2.1 is bonded to a steel beam (Young's modulus E = 200 GPa). It is connected in a quarter-bridge Wheatstone configuration with an excitation voltage of $V_{ex}$ = 5 V. Under load, the bridge output voltage is measured as $V_{out}$ = 1.8 mV. Calculate the strain and the stress in the beam.
**Solution:**
For a quarter-bridge Wheatstone configuration: $V_{out} = (GF \times \varepsilon / 4) \times V_{ex}$, where ε is the strain.
Solving for strain: $\varepsilon = 4 \times V_{out} / (GF \times V_{ex}) = 4 \times 1.8 \times 10^{-3} / (2.1 \times 5) = 7.2 \times 10^{-3} / 10.5 = 685.7 \times 10^{-6} = 685.7\ \mu\varepsilon$.
Stress: $\sigma = E \times \varepsilon = 200 \times 10^{9} \times 685.7 \times 10^{-6} = 137.1 \times 10^{6}$ Pa = 137.1 MPa.
Resistance change in the gauge: $\Delta R = GF \times \varepsilon \times R = 2.1 \times 685.7 \times 10^{-6} \times 350 = 0.504\ \Omega$ (a change

of only 0.144%).

### 11.2.3 Pressure Sensors

Pressure sensors convert fluid or gas pressure into an electrical signal using various transduction mechanisms. Piezoresistive pressure sensors use a silicon diaphragm with implanted or diffused strain gauges that change resistance under pressure-induced deformation, offering good linearity and compatibility with MEMS fabrication. Capacitive pressure sensors detect the change in capacitance as pressure deflects a diaphragm relative to a fixed electrode, providing high sensitivity and low power consumption. Pressure ranges span from sub-pascal (micromanometers) to hundreds of megapascals (industrial hydraulic systems), with gauge pressure (referenced to atmosphere), absolute pressure (referenced to vacuum), and differential pressure (between two ports) configurations. MEMS pressure sensors have enabled miniaturized, low-cost devices found in automotive tire pressure monitors, altimeters, weather stations, and medical ventilators.

**Example 11.2.3:** A capacitive pressure sensor has a circular diaphragm of diameter 2 mm and a gap of 5 μm at zero pressure. The dielectric is air ($\varepsilon_r = 1$). Calculate the zero-pressure capacitance and the capacitance at a pressure that deflects the diaphragm center by 1 μm (assume parallel-plate approximation with the average gap).
**Solution:**
Plate area: $A = \pi \times (d/2)^2 = \pi \times (1 \times 10^{-3})^2 = 3.1416 \times 10^{-6}$ m².
Zero-pressure capacitance: $C_0 = \varepsilon_0 \times A / g_0 = 8.854 \times 10^{-12} \times 3.1416 \times 10^{-6} / 5 \times 10^{-6} = 27.82 \times 10^{-18} / 5 \times 10^{-6} = 5.56$ pF.
For a uniform diaphragm, the average deflection is approximately 1/3 of the center deflection for clamped circular plates, so the average gap reduction is about $1/3 \times 1$ μm $= 0.333$ μm.
New average gap: $g = 5.0 - 0.333 = 4.667$ μm.
New capacitance: $C = \varepsilon_0 A / g = 8.854 \times 10^{-12} \times 3.1416 \times 10^{-6} / 4.667 \times 10^{-6} = 5.96$ pF.
Capacitance change: $\Delta C = 5.96 - 5.56 = 0.40$ pF (7.2% change), which is readily detectable with a capacitance-to-digital converter IC.

### 11.2.4 Proximity and Position Sensors

Proximity sensors detect the presence or distance of an object without physical contact. Inductive proximity sensors detect metallic objects by sensing changes in an oscillating electromagnetic field, with typical ranges of 1-50 mm. Capacitive proximity sensors detect changes in capacitance caused by nearby objects (metallic or non-metallic), making them suitable for level sensing in tanks and detecting materials through non-metallic barriers. Optical proximity sensors (photoelectric sensors) use emitted and reflected light beams with through-beam, retro-reflective, and diffuse-reflective configurations for detection ranges from millimeters to tens of meters. Linear Variable Differential Transformers (LVDTs) measure linear displacement with high accuracy (±0.1% of full range) using a movable ferromagnetic core within a transformer, providing virtually infinite resolution since the output is a continuous analog signal. Rotary encoders measure angular position using optical or magnetic sensing, available as incremental (producing pulses per revolution) or absolute (providing a unique digital code for each angular position) types.

**Example 11.2.4:** An LVDT has a sensitivity of 25 mV/mm with an excitation of 3 $V_{rms}$ at 5 kHz. The LVDT has a linear range of ±10 mm. If the signal conditioning electronics have a noise floor of 0.5 $mV_{rms}$, what is the maximum measurable displacement, the resolution limited by noise, and the position accuracy if the LVDT nonlinearity is ±0.1% of full range?
**Solution:**
Maximum displacement: ±10 mm (linear range).
Maximum output voltage: $V_{max}$ = 25 mV/mm × 10 mm = 250 mV.
Resolution (limited by noise floor): $\Delta x = V_{noise}$ / sensitivity = 0.5 mV / 25 mV/mm = 0.020 mm = 20 μm.
Nonlinearity error: ±0.1% of full range = ±0.001 × 10 mm = ±0.010 mm = ±10 μm.
Total position accuracy (RSS of noise-limited resolution and nonlinearity): $\pm\sqrt{(20^2 + 10^2)} = \pm\sqrt{(400 + 100)} = \pm\sqrt{500} = \pm 22.4$ μm.
Dynamic range: 20 mm / 0.020 mm = 1000:1, or $20\log_{10}(1000) = 60$ dB.

### 11.2.5 Accelerometers and Gyroscopes

Accelerometers measure acceleration (change in velocity) along one or more axes, while gyroscopes measure angular velocity (rate of rotation). **MEMS accelerometers** use a micromachined proof mass suspended by spring-like flexures; acceleration displaces the proof mass relative to the substrate, and the displacement is detected capacitively or piezoresistively. Typical specifications include measurement range (±2 g to ±200 g), sensitivity (mV/g or LSB/g for digital output), noise density (μg/$\sqrt{}$Hz), and bandwidth (DC to several kHz). **MEMS gyroscopes** use a vibrating structure (tuning fork or ring) and detect the Coriolis force that acts on the vibrating mass when rotated, producing a signal proportional to angular velocity. Inertial Measurement Units (IMUs) combine 3-axis accelerometers and 3-axis gyroscopes (6-DOF) — often with a 3-axis magnetometer (9-DOF) — in a single package for motion tracking, navigation, and stabilization. Applications include smartphone orientation, drone flight control, automotive stability systems (ESC), seismic monitoring, and industrial vibration analysis.

**Example 11.2.5:** A MEMS accelerometer has a sensitivity of 1000 LSB/g, a 16-bit ADC output, a noise density of 100 μg/$\sqrt{}$Hz, and a measurement bandwidth of 500 Hz. Calculate the noise floor in $mg_{rms}$ and the dynamic range.
**Solution:**
Total noise: $a_{noise}$ = noise density × $\sqrt{}$BW = $100 \times 10^{-6} \times \sqrt{500} = 100 \times 10^{-6} \times 22.36 = 2.236 \times 10^{-3}$ g = 2.24 $mg_{rms}$.
Full-scale range for a ±16 g device (using 16-bit output): 32 g.
Dynamic range: DR = $20 \times \log_{10}$(full-scale / noise) = $20 \times \log_{10}(32 / 2.24 \times 10^{-3}) = 20 \times \log_{10}(14{,}286) = 20 \times 4.155 = 83.1$ dB.
The noise floor of 2.24 mg means the sensor can resolve accelerations as small as ~7 mg (3$\sigma$) with 99.7% confidence, sufficient for tilt sensing (where 1° of tilt produces ~17.5 mg) but not for seismology (which requires μg-level resolution with sub-Hz bandwidth).

### 11.2.6 Magnetic Field Sensors

Magnetic field sensors detect the presence, strength, and direction of magnetic fields. **Hall effect sensors** are the most widely used type, producing a voltage proportional to the magnetic flux density perpendicular to the current-carrying semiconductor element: $V_H = K_H \times I \times B$, where $K_H$ is the Hall coefficient. Linear Hall sensors provide an analog output proportional to field strength (typical sensitivity 1–5 mV/mT), while Hall switches provide a digital output that changes state when the field exceeds a threshold — widely used for position sensing, current measurement, and brushless motor commutation. **Magnetoresistive sensors** (AMR, GMR, TMR) exploit the change in electrical resistance caused by an external magnetic field, offering higher sensitivity than Hall sensors (nT range for TMR) and are used in compasses, biomedical detection, and hard disk read heads. **Fluxgate magnetometers** measure DC and low-frequency magnetic fields with very high sensitivity (sub-nT) by detecting the asymmetric saturation of a ferromagnetic core driven by an AC excitation, and are used in geophysical surveys, spacecraft attitude determination, and military applications.

**Example 11.2.6:** A linear Hall effect sensor with sensitivity of 2.5 mV/mT and a quiescent output of 2.5 V (at B = 0) is used to measure the magnetic field from a permanent magnet. The sensor output reads 3.125 V. Calculate the magnetic field and determine the sensor's suitability for measuring Earth's magnetic field (~50 $\mu$T).
**Solution:**
Voltage change from quiescent: $\Delta V = 3.125 - 2.5 = 0.625$ V.
Magnetic field: $B = \Delta V$ / sensitivity $= 0.625 / (2.5 \times 10^{-3}) = 250$ mT $= 0.25$ T.
This is a strong field typical of proximity to a permanent magnet.
For Earth's field (50 $\mu$T = 0.05 mT): the expected output would be $\Delta V = 2.5 \times 10^{-3} \times 0.05 = 0.125$ mV — far below the sensor's typical noise and offset voltage (several mV).
A Hall sensor is not suitable for measuring Earth's field; a magnetoresistive (AMR/TMR) sensor with nT sensitivity or a fluxgate magnetometer would be required.

### 11.2.7 Current Sensors

Current sensors measure electric current without breaking the circuit (non-invasive) or with minimal insertion loss. **Current transformers** (CTs) use electromagnetic coupling to measure AC current: the primary is the conductor carrying the current to be measured, and the secondary winding produces a scaled-down current proportional to the turns ratio (e.g., a 100:5 CT transforms 100 A primary into 5 A secondary). CTs are passive, isolated, and used extensively in power metering and protective relaying. **Hall effect current sensors** place a Hall element in the gap of a magnetic core surrounding the conductor, measuring both AC and DC current with typical accuracy of 1–3%. Closed-loop (null-balance) Hall sensors use a feedback winding to cancel the core flux, achieving accuracy of 0.1–0.5% with bandwidth from DC to 100+ kHz. **Rogowski coils** are flexible, air-core coils wrapped around a conductor that output a voltage proportional to di/dt; the signal is integrated to recover the current waveform. Rogowski coils are lightweight, have no magnetic saturation (linear over enormous current ranges), and are ideal for measuring large transient currents and fault currents. **Shunt resistors** are the simplest current sensors — a precision low-value resistor inserted in the circuit produces a voltage proportional to current (V = IR), but they lack isolation and introduce insertion loss.

**Example 11.2.7:** A closed-loop Hall effect current sensor with a rated primary current of 50 A, an output sensitivity of 40 mV/A, and accuracy of ±0.5% is used to monitor a motor drive. The sensor output reads 1.280 V. Calculate the measured current and the measurement uncertainty. Compare the insertion loss to a 10 m$\Omega$ shunt resistor at the same current.

**Solution:**
Measured current: $I = V_{out}$ / sensitivity = 1.280 / 0.040 = 32.0 A.
Measurement uncertainty: ±0.5% × 32.0 = ±0.16 A, so the current is 32.0 ± 0.16 A.
The Hall sensor has essentially zero insertion loss (the conductor passes through the core aperture).
For a 10 mΩ shunt at 32.0 A: voltage drop = I × R = 32.0 × 0.010 = 0.32 V.
Power dissipation = $I^2R = 32.0^2 \times 0.010 = 10.24$ W.
The shunt dissipates over 10 W of heat and drops 0.32 V from the supply — significant in a power circuit.
The Hall sensor provides galvanic isolation with negligible insertion loss, which is why it is preferred for high-current measurement.

### 11.2.8 Optical and Infrared Sensors

Optical sensors convert light intensity, wavelength, or modulation into electrical signals, enabling non-contact measurement of temperature, position, chemical composition, and physical phenomena. **Pyrometers** (infrared thermometers) measure the thermal radiation emitted by an object to determine its surface temperature without contact, using the Stefan-Boltzmann law (total radiated power $\propto T^4$) or the Planck radiation law (spectral distribution). Single-color pyrometers measure total IR intensity in a band (typically 8–14 μm for low temperatures or 1–5 μm for high temperatures) and require known emissivity, while two-color (ratio) pyrometers compare intensity at two wavelengths to cancel the emissivity dependence — essential for measuring metals, molten glass, and surfaces with uncertain or varying emissivity. Temperature ranges span from −50°C (long-wave IR) to over 3000°C (short-wave for steel mills and furnaces). **Fiber optic sensors** use changes in the light propagating through an optical fiber to measure strain, temperature, pressure, or acceleration. Fiber Bragg Grating (FBG) sensors reflect a narrow wavelength band (Bragg wavelength $\lambda_B = 2n\Lambda$, where n is the effective index and $\Lambda$ is the grating period) that shifts with strain (~1.2 pm/με) and temperature (~10 pm/°C), enabling multiplexing of dozens of sensors along a single fiber. Distributed sensing techniques such as Brillouin Optical Time-Domain Reflectometry (BOTDR) measure strain and temperature continuously along the entire fiber length (up to 50 km) with spatial resolution of 0.5–2 m, making them ideal for structural health monitoring of bridges, pipelines, and dams. **Photoelectric sensors** in industrial automation use emitter-receiver pairs (through-beam, retro-reflective, or diffuse) with modulated LED or laser sources, and **color sensors** detect reflected spectral signatures for sorting and quality inspection.

**Example 11.2.8:** A ratio pyrometer measures thermal radiation from a steel billet at two wavelengths: $\lambda_1 = 0.9$ μm and $\lambda_2 = 1.6$ μm. The detected intensity ratio is $I(\lambda_1)/I(\lambda_2) = 0.180$. Using the Wien approximation $I(\lambda) \propto \varepsilon\, \lambda^{-5} \exp(-c_2/(\lambda T))$ where $c_2 = 14{,}388$ μm·K, calculate the billet temperature. Show why the ratio method eliminates the need to know emissivity.
**Solution:**
Taking the ratio: $I(\lambda_1)/I(\lambda_2) = (\varepsilon_1/\varepsilon_2) \times (\lambda_2/\lambda_1)^5 \times \exp[-c_2/T \times (1/\lambda_1 - 1/\lambda_2)]$.
For a gray body ($\varepsilon_1 = \varepsilon_2$), the emissivity cancels: $0.180 = (1.6/0.9)^5 \times \exp[-14{,}388/T \times (1/0.9 - 1/1.6)]$.
Compute $(1.6/0.9)^5 = (1.778)^5 = 17.77$.
Exponent factor: $1/0.9 - 1/1.6 = 1.111 - 0.625 = 0.4861$ μm$^{-1}$.
So $0.180 = 17.77 \times \exp(-14{,}388 \times 0.4861/T) = 17.77 \times \exp(-6994/T)$.

Solving: $0.180/17.77 = 0.01013 = \exp(-6994/T)$.
$\ln(0.01013) = -4.593 = -6994/T$.
$T = 6994/4.593 = 1523$ K = 1250°C — a typical rolling mill temperature for a steel billet.
The key advantage is that ε cancels in the ratio, so the measurement is independent of surface oxidation, roughness, or viewing angle.

### 11.2.9 Flow Sensors

Flow sensors measure the volumetric or mass flow rate of liquids and gases in pipes, ducts, and open channels — a critical measurement in process control, HVAC, water treatment, oil and gas, and biomedical applications. **Electromagnetic flow meters** (mag meters) apply Faraday's law: a conductive fluid flowing through a magnetic field generates a voltage $V = B \times D \times \bar{v}$, where B is the magnetic flux density, D is the pipe diameter, and $\bar{v}$ is the average fluid velocity. Mag meters have no moving parts, produce no pressure drop, handle slurries and corrosive fluids, and are accurate to ±0.2–0.5%, but require a minimum fluid conductivity of ~5 μS/cm, excluding hydrocarbons and pure water. **Ultrasonic flow meters** use transit-time or Doppler techniques. Transit-time meters measure the difference in propagation time of ultrasonic pulses sent upstream and downstream through the fluid: $\Delta t = 2 \times L \times \bar{v} \times \cos\theta / c^2$, where L is the path length and θ is the beam angle. They are non-invasive (clamp-on versions mount outside the pipe), handle clean liquids, and achieve ±0.5–1% accuracy. **Coriolis flow meters** measure mass flow directly by vibrating a U-shaped or straight tube at its resonant frequency; the Coriolis force induced by mass flow through the vibrating tube causes a phase shift between the inlet and outlet that is proportional to mass flow rate. Coriolis meters are the most accurate flow technology (±0.1% of reading), simultaneously measure density and temperature, and are the standard for custody transfer of hydrocarbons. **Vortex flow meters** detect the frequency of vortices shed alternately from a bluff body (shedder bar) placed in the flow stream; the shedding frequency $f = St \times \bar{v} / d$ is proportional to velocity, where St is the Strouhal number (~0.27) and d is the shedder width. **Differential pressure** flow meters (orifice plates, Venturi tubes, flow nozzles) infer velocity from the pressure drop across a constriction using Bernoulli's principle: $Q = C_d \times A_2 \times \sqrt{(2\Delta P / \rho)}$, where $C_d$ is the discharge coefficient — the oldest and still most widely installed flow technology in industrial plants.

**Example 11.2.9:** An electromagnetic flow meter is installed on a 150 mm (6-inch) diameter pipe carrying water (conductivity 450 μS/cm). The magnetic field is B = 0.05 T, and the measured electrode voltage is 2.35 mV. Calculate the average flow velocity and the volumetric flow rate in $m^3/h$. Then, for a separate higher-flow operating condition on the same pipe, an orifice plate ($\beta = d/D = 0.6$, $C_d = 0.61$) measures ΔP = 18.5 kPa — calculate that flow rate to compare sensor technologies.
**Solution:**
Mag meter (low-flow condition): $\bar{v} = V / (B \times D) = 2.35 \times 10^{-3} / (0.05 \times 0.150) = 2.35 \times 10^{-3} / 7.5 \times 10^{-3} = 0.3133$ m/s.
Pipe area: $A = \pi/4 \times (0.150)^2 = 0.01767\ m^2$.
Volumetric flow: $Q = \bar{v} \times A = 0.3133 \times 0.01767 = 5.536 \times 10^{-3}\ m^3/s = 19.93\ m^3/h$.
Orifice plate (high-flow condition, ΔP = 18.5 kPa): orifice area $A_2 = \pi/4 \times (\beta \times D)^2 = \pi/4 \times (0.090)^2 = 6.362 \times 10^{-3}\ m^2$.
$Q = C_d \times A_2 \times \sqrt{(2\Delta P/\rho)} = 0.61 \times 6.362 \times 10^{-3} \times \sqrt{(2 \times 18{,}500 / 1000)} = 3.881 \times 10^{-3} \times \sqrt{37.0} = 3.881 \times 10^{-3} \times 6.083 = 23.61 \times 10^{-3}\ m^3/s = 85.0\ m^3/h$.

The two sensors are operating at different flow conditions (19.93 vs 85.0 $m^3/h$), illustrating the measurement range of each technology on the same pipe geometry.
The mag meter's advantage is zero pressure drop and no moving parts; the orifice plate permanently reduces pressure by approximately 60% of the measured $\Delta P$ (11.1 kPa), wasting pumping energy.

### 11.2.10 Chemical and Gas Sensors

Chemical and gas sensors detect the presence and concentration of specific chemical species in air, water, or other media, enabling environmental monitoring, industrial safety, process control, and emissions compliance. These sensors face unique challenges compared to physical sensors — they must be selective to the target analyte amid a complex mixture of background gases, they are subject to drift and poisoning over their operational lifetime, and they often require periodic calibration against known reference gases. The diversity of chemical sensing mechanisms reflects the wide range of target species and application requirements, from parts-per-billion detection of toxic gases in semiconductor fabrication to percent-level oxygen monitoring in confined spaces.

**Electrochemical sensors** use a chemical reaction at an electrode-electrolyte interface to generate a current proportional to the target gas concentration. A typical three-electrode cell consists of a sensing (working) electrode, a counter electrode, and a reference electrode separated by an electrolyte (often a liquid acid or polymer gel). The target gas diffuses through a membrane to the sensing electrode, where it is oxidized or reduced, producing a current in the nanoampere to microampere range that is linearly proportional to concentration over 3-4 decades of range. Electrochemical sensors are the industry standard for toxic gas detection (CO, $H_2S$, $NO_2$, $Cl_2$, $SO_2$) in portable safety monitors, offering good selectivity, low power consumption, and moderate cost, with typical lifetimes of 2–3 years before electrolyte depletion requires replacement.

**Metal oxide semiconductor (MOS) gas sensors** use a heated tin dioxide ($SnO_2$) or tungsten trioxide ($WO_3$) sensing element whose electrical conductance changes when target gas molecules adsorb on the surface and react with pre-adsorbed oxygen ions. The sensor requires a heater (typically 200–400°C) to activate the surface reactions, consuming 100–800 mW — a significant disadvantage for battery-powered applications. MOS sensors offer very high sensitivity (sub-ppm for some gases) and long operational life (5–10 years), but suffer from poor selectivity (responding to a broad range of reducing gases) and cross-sensitivity to humidity. Selectivity can be improved by using arrays of sensors with different oxide compositions and operating temperatures, combined with pattern recognition algorithms (electronic nose).

**Nondispersive infrared (NDIR) sensors** exploit the characteristic infrared absorption bands of gas molecules to measure concentration. The sensor passes broadband IR radiation through a gas sample cell and measures the attenuation at the target gas absorption wavelength using a narrowband optical filter and a thermopile or pyroelectric detector. A reference channel at a non-absorbing wavelength compensates for source aging and optical contamination. NDIR sensors are the gold standard for $CO_2$ measurement (absorption at 4.26 μm), and are also used for CO (4.67 μm), $CH_4$ (3.31 μm), and hydrocarbons. They offer excellent selectivity (each gas has a unique absorption fingerprint), no consumable components, and operational lifetimes exceeding 10 years, but are larger and more expensive than electrochemical or MOS sensors.

**Photoionization detectors (PIDs)** use a high-energy UV lamp (typically 10.6 eV) to ionize gas molecules in a sample chamber, generating a current between collection electrodes that is proportional to the total concentration of ionizable volatile organic compounds (VOCs). PIDs respond to a wide range of organic vapors (benzene, toluene, xylene, solvents) with ppb-level sensitivity and fast response (<3 seconds), making them essential for environmental site assessment, industrial hygiene, and hazmat response. However, they provide a total VOC reading rather than identifying individual compounds. **Catalytic bead sensors** (pellistors) detect combustible gases by measuring the heat released when gas oxidizes on a catalyst-coated bead, with the resistance change of an embedded platinum wire measured by a Wheatstone bridge; they are used for LEL (lower explosive limit) monitoring of methane, propane, and hydrogen in oil and gas, mining, and confined-space entry.

**Cross-sensitivity** is the response of a sensor to gases other than the target analyte — for example, an electrochemical CO sensor may also respond to $H_2S$ or $NO_2$, requiring correction factors or scrubber filters. Calibration requires periodic exposure to a certified reference gas (span gas) at a known concentration, typically every 6–12 months, along with a zero-gas (purified air) check to verify baseline. Multi-gas monitors combine several sensing technologies (electrochemical for toxic gases, catalytic bead for combustibles, NDIR for $CO_2$, and galvanic cell for $O_2$) in a single handheld instrument for confined-space entry compliance with OSHA regulations.

**Example 11.2.10:** An electrochemical CO sensor has a sensitivity of 55 nA/ppm with a linear range of 0–500 ppm. The sensor is calibrated at 100 ppm CO (span gas) and produces an output of 5.38 μA. During field use, the sensor output is 2.12 μA. The sensor has a known cross-sensitivity to $H_2$ of 12% (i.e., 100 ppm $H_2$ produces the same response as 12 ppm CO). If the environment contains an estimated 50 ppm $H_2$ background, calculate the indicated CO concentration, the $H_2$ interference, and the corrected CO concentration.
**Solution:**
Calibration check: at 100 ppm CO, expected output = 100 × 55 nA/ppm = 5,500 nA = 5.50 μA.
Actual calibration output: 5.38 μA. Calibration factor: CF = 5.50/5.38 = 1.022 (sensor reads 2.2% low).
Indicated CO concentration from field reading: $CO_{indicated} = I_{field}$ / sensitivity = 2,120 nA / 55 nA/ppm = 38.5 ppm.
Apply calibration correction: $CO_{corrected} = 38.5 \times 1.022 = 39.4$ ppm.
$H_2$ cross-sensitivity interference: 50 ppm $H_2$ × 12% = 6.0 ppm CO-equivalent.
True CO concentration: $CO_{true} = 39.4 - 6.0 = 33.4$ ppm.
The $H_2$ interference accounts for 15% of the calibration-corrected CO reading — a significant error if uncorrected. In practice, multi-gas instruments with both CO and $H_2$ sensors can apply automatic cross-sensitivity compensation. The corrected reading of 33.4 ppm is below the OSHA 8-hour TWA limit of 50 ppm, so the environment is within permissible exposure limits for CO.

## 11.3 Signal Conditioning

Signal conditioning is the intermediate stage between the raw sensor output and the data acquisition system, preparing the signal for accurate digitization. Conditioning circuits amplify low-level sensor signals to match the ADC input range, filter out noise and unwanted frequency components, provide galvanic isolation for safety and ground-loop rejection, and linearize nonlinear sensor outputs. Proper signal conditioning is critical because errors introduced at this stage — offset drift, gain error, insufficient filtering, or inadequate isolation — propagate through the entire measurement chain and cannot be corrected by downstream digital processing.

### 11.3.1 Amplification

Signal conditioning amplifiers scale low-level sensor outputs to voltage ranges suitable for data acquisition systems. Instrumentation amplifiers are precision differential amplifiers with high common-mode rejection ratio (CMRR, typically 80-120 dB), high input impedance (>$10^9$ Ω), and adjustable gain set by a single external resistor, making them ideal for amplifying small differential signals from bridges and thermocouples in the presence of large common-mode voltages. Programmable gain amplifiers (PGAs) allow the gain to be digitally selected, enabling a single data acquisition channel to accommodate sensors with different output ranges. Charge amplifiers convert the charge output of piezoelectric sensors into a proportional voltage, using a high-impedance feedback capacitor around an operational amplifier. Key specifications include gain accuracy, gain drift with temperature, bandwidth, input offset voltage, input bias current, and noise spectral density.

**Example 11.3.1:** An instrumentation amplifier with a gain of G = 500 and CMRR = 100 dB amplifies a 2 mV thermocouple signal. The thermocouple wires pick up a 60 Hz common-mode noise voltage of $V_{cm}$ = 1.5 V. Calculate the desired output signal, the common-mode error at the output, and the signal-to-error ratio.
**Solution:**
Desired output: $V_{out(signal)} = G \times V_{diff} = 500 \times 2$ mV = 1.000 V.
CMRR = 100 dB = 10^(100/20) = 100,000.
Common-mode rejection means the amplifier converts a fraction of $V_{cm}$ to a differential error: $V_{error(input)} = V_{cm}$ / CMRR = 1.5 / 100,000 = 15 μV.
Error at the output: $V_{error(output)} = G \times V_{error(input)} = 500 \times 15$ μV = 7.5 mV.
Signal-to-error ratio: $V_{out(signal)} / V_{error(output)}$ = 1.000 / 0.0075 = 133.3, or $20\log_{10}(133.3)$ = 42.5 dB.
The error is 0.75% of the output signal – acceptable for many industrial measurements but may require additional 60 Hz notch filtering for higher-precision applications.

### 11.3.2 Filtering

Analog filters in signal conditioning remove unwanted frequency components before digitization. Anti-aliasing filters (lowpass) are mandatory before an ADC to attenuate frequencies above the Nyquist frequency ($f_s/2$) and prevent aliasing distortion. Active filters using operational amplifiers implement lowpass, highpass, bandpass, and notch (band-reject) responses with standard topologies including Sallen-Key, Multiple Feedback (MFB), and state-variable configurations. The Butterworth response provides maximally flat passband magnitude, the Chebyshev response allows passband ripple for a steeper transition, and the Bessel response provides maximally flat group delay (linear phase) for waveform preservation. Notch filters at 50/60 Hz are commonly used to reject power line interference in low-frequency measurement systems.

**Example 11.3.2:** A data acquisition system samples a sensor signal with a maximum frequency of interest of 500 Hz. Design a second-order Butterworth anti-aliasing lowpass filter by specifying the cutoff frequency if the ADC sample rate is 5 kS/s, and calculate the attenuation at the Nyquist frequency.
**Solution:**
Nyquist frequency: $f_N = f_s$ / 2 = 5000 / 2 = 2500 Hz.
The anti-aliasing filter must attenuate signals at and above $f_N$ sufficiently (typically > 40 dB).

Set the filter cutoff frequency at $f_c = 1$ kHz (providing a flat passband well above 500 Hz with roll-off beginning before Nyquist).
For a 2nd-order Butterworth filter, the roll-off is -40 dB/decade.
Frequency ratio at Nyquist: $f_N / f_c = 2500 / 1000 = 2.5$.
Butterworth magnitude response: $|H(f)| = 1 / \sqrt{(1 + (f/f_c)^{2n})} = 1 / \sqrt{(1 + 2.5^4)} = 1 / \sqrt{(1 + 39.06)} = 1 / \sqrt{40.06} = 1 / 6.33 = 0.158$.
Attenuation = $20\log_{10}(0.158) = -16.0$ dB.
This is insufficient; a 4th-order Butterworth would give: $1 / \sqrt{(1 + 2.5^8)} = 1 / \sqrt{(1 + 1526)} = 1 / 39.1 = 0.0256$, or -31.8 dB.
For > 40 dB, use a 6th-order filter or increase the sampling rate.

### 11.3.3 Isolation

Signal isolation provides a galvanic barrier between the sensor or field wiring and the measurement system, protecting equipment and personnel from high voltages, breaking ground loops that cause measurement errors, and preventing noise coupling between different parts of the system. Optocouplers use an LED and photodetector across the isolation barrier to transmit analog or digital signals, with isolation ratings typically from 1.5 kV to 5 kV. Capacitive isolation and magnetic (transformer-based) isolation are used in modern digital isolators and isolated amplifiers, offering higher bandwidth and lower power consumption than optocouplers. Isolated power supplies (using small transformers or piezoelectric converters) provide the power needed on the isolated side of the barrier. Isolation is essential in industrial measurement systems, medical instrumentation (patient safety), and high-voltage power system monitoring.

**Example 11.3.3:** A current sensor monitors a 480 V power bus. The sensor output is 0-5 V and must be isolated before connecting to a ground-referenced DAQ system. An isolated amplifier with 2500 $V_{rms}$ isolation rating, 0.05% gain error, ±10 μV/°C offset drift, and 100 kHz bandwidth is selected. If the operating temperature varies by 50°C from calibration, calculate the total measurement error on a 2.5 V signal.
**Solution:**
Gain error contribution: 0.05% × 2.5 V = 1.25 mV.
Offset drift error: ±10 μV/°C × 50°C = ±500 μV = ±0.5 mV.
Total worst-case error: ±(1.25 + 0.5) = ±1.75 mV.
Percent of signal: ±1.75 / 2500 × 100 = ±0.070%.
RSS error (assuming uncorrelated): $\pm\sqrt{(1.25^2 + 0.5^2)} = \pm\sqrt{(1.5625 + 0.25)} = \pm\sqrt{1.8125} = \pm 1.35$ mV, or ±0.054% of the 2.5 V signal.
The isolation barrier provides safety against the 480 V bus while maintaining measurement accuracy well within 0.1%.

### 11.3.4 Linearization

Many sensors have inherently nonlinear transfer characteristics — thermocouples, thermistors, photodetectors, and pressure sensors all produce outputs that are not strictly proportional to the measured quantity over their full range. Linearization corrects these nonlinearities to produce an output that accurately represents the physical variable. **Analog linearization** uses hardware techniques such as logarithmic amplifiers, diode networks, or feedback configurations that shape the transfer function (e.g., a thermistor in one arm of a Wheatstone bridge with appropriate resistor values can partially linearize the response). **Lookup table linearization** stores a table of sensor output vs. true value pairs in firmware and uses interpolation (linear or cubic spline) to convert each reading — this is the most common approach in microcontroller-based systems because it handles arbitrary nonlinearities and is easily updated for different sensor calibrations. **Polynomial linearization** fits a polynomial equation to the sensor characteristic (e.g., the ITS-90 thermocouple reference functions use polynomials of order 8-14) and computes the corrected value algebraically. The Steinhart-Hart equation for thermistors, $1/T = A + B \cdot \ln(R) + C \cdot (\ln(R))^3$, is a widely used three-parameter model that provides ±0.01°C accuracy over a 100°C span with just three calibration points.

**Example 11.3.4:** An NTC thermistor has the following calibration data: $R_1 = 27{,}445\ \Omega$ at $T_1 = 0°C$ (273.15 K), $R_2 = 10{,}000\ \Omega$ at $T_2 = 25°C$ (298.15 K), $R_3 = 4{,}160\ \Omega$ at $T_3 = 50°C$ (323.15 K). Using the Steinhart-Hart equation $1/T = A + B \cdot \ln(R) + C \cdot (\ln(R))^3$, determine the coefficients and calculate the temperature when $R = 6{,}530\ \Omega$.
**Solution:**
Set up three equations using the calibration data.
$\ln(R_1) = \ln(27{,}445) = 10.2199$, $\ln(R_2) = \ln(10{,}000) = 9.2103$, $\ln(R_3) = \ln(4{,}160) = 8.3333$.
The three simultaneous equations:
$1/273.15 = A + 10.2199B + (10.2199)^3C \rightarrow 3.6610 \times 10^{-3} = A + 10.2199B + 1067.4C$.
$1/298.15 = A + 9.2103B + (9.2103)^3C \rightarrow 3.3540 \times 10^{-3} = A + 9.2103B + 781.3C$.
$1/323.15 = A + 8.3333B + (8.3333)^3C \rightarrow 3.0946 \times 10^{-3} = A + 8.3333B + 578.7C$.
Solving the system: $A = 8.404 \times 10^{-4}$, $B = 2.596 \times 10^{-4}$, $C = 1.570 \times 10^{-7}$.
For $R = 6{,}530\ \Omega$: $\ln(6{,}530) = 8.7842$.
$1/T = 8.404 \times 10^{-4} + 2.596 \times 10^{-4} \times 8.7842 + 1.570 \times 10^{-7} \times (8.7842)^3 = 8.404 \times 10^{-4} + 2.280 \times 10^{-3} + 1.064 \times 10^{-4} = 3.227 \times 10^{-3}$.
$T = 1 / 3.227 \times 10^{-3} = 309.9\ K = 36.7°C$.
Without linearization (using a simple linear interpolation between 25°C and 50°C): $T \approx 25 + 25 \times (10{,}000 - 6{,}530) / (10{,}000 - 4{,}160) = 25 + 25 \times 0.594 = 39.9°C$ — a 3.2°C error demonstrating the importance of proper linearization.

### 11.3.5 Analog-to-Digital Converter Selection

Selecting the right ADC architecture is critical for matching the measurement system's requirements for resolution, speed, accuracy, and power consumption. **Successive Approximation Register (SAR) ADCs** use a binary search algorithm with a DAC and comparator, achieving 12–20 bit resolution at sample rates from 10 kS/s to 10+ MS/s with low power consumption — they are the dominant choice for multiplexed data acquisition, sensor interfaces, and battery-powered instruments. **Delta-Sigma (Σ-Δ) ADCs** use oversampling (typically 64× to 256× the Nyquist rate) and noise shaping to push quantization noise out of the band of interest, followed by a digital decimation filter, achieving 16–32 bit resolution but at lower effective sample rates (typically 10 S/s to 100 kS/s) — ideal for precision DC and low-frequency measurements such as weigh scales, temperature, and strain gauge bridges. **Pipeline ADCs** use a cascaded chain of low-resolution stages (1.5–3.5 bits each) with inter-stage residue amplification, achieving 8–16 bit resolution at sample rates from 10 MS/s to 1+ GS/s — suited for oscilloscopes, communications receivers, and waveform capture. Key selection parameters include effective number of bits (ENOB = (SINAD − 1.76) / 6.02), integral nonlinearity (INL), differential nonlinearity (DNL), input bandwidth, latency, and power dissipation per conversion. Modern integrated Σ-Δ ADCs often include a programmable gain amplifier, excitation current sources, and digital filter — providing a nearly complete signal conditioning front end in a single chip.

**Example 11.3.5:** A precision weigh scale requires 0.001% accuracy over a ±20 mV full-scale range from a strain gauge bridge, with a measurement bandwidth of 10 Hz. Compare the minimum resolution requirements and select the appropriate ADC architecture. The system operates from a 3.3 V supply with a 500 μA power budget.
**Solution:**
Required resolution: 0.001% of full scale = 0.001% × 40 mV = 0.4 μV.
Number of bits: $2^N$ = full-scale / resolution = $40 \times 10^{-3}$ / $0.4 \times 10^{-6}$ = 100,000.
N = $\log_2$(100,000) = 16.6 bits minimum — round up to 18 bits practical (with noise margin).
Nyquist rate: 2 × 10 Hz = 20 S/s minimum.
A SAR ADC at 18 bits could work but would require an external low-noise PGA and anti-aliasing filter, consuming significant board space and power.
A 24-bit Σ-Δ ADC (e.g., with 128× oversampling at 2,560 S/s, decimated to 20 S/s) provides: effective resolution at 10 Hz BW ≈ 20+ noise-free bits, integrated PGA (gains of 1–128), built-in digital sinc filter that rejects 50/60 Hz, and typical power consumption of 200–500 μA at 3.3 V.
A pipeline ADC is entirely inappropriate — far too fast (minimum ~10 MS/s) and insufficient resolution (typically 12–14 bits).
**Selection: 24-bit Σ-Δ ADC** — provides the required resolution, built-in signal conditioning, excellent 50/60 Hz rejection, and power consumption within the 500 μA budget.

## 11.4 Measurement Instruments

Measurement instruments are the tools engineers use to observe, quantify, and analyze electrical signals and circuit behavior. The four most common bench instruments — the digital multimeter, oscilloscope, spectrum analyzer, and function generator — each provide a different view of the signal: the DMM measures steady-state magnitude, the oscilloscope shows the time-domain waveform, the spectrum analyzer reveals the frequency-domain content, and the function generator provides controlled stimulus signals. Modern instruments are increasingly digital, offering deep memory, advanced triggering, automated measurements, remote control via GPIB/USB/LAN, and protocol-aware decoding for serial buses.

### 11.4.1 Digital Multimeter (DMM)

The digital multimeter is the most widely used general-purpose measurement instrument, capable of measuring DC and AC voltage, DC and AC current, and resistance, with many models adding capacitance, frequency, temperature, and diode test functions. A DMM operates by converting the quantity being measured into a DC voltage proportional to its value, then digitizing it with an integrating ADC (typically dual-slope or multi-slope) that provides high resolution and excellent rejection of power line frequency interference. Resolution is specified in digits (e.g., a 6½-digit DMM can display values from 0 to 1,999,999, providing approximately 1 μV resolution on the 2 V range). Bench DMMs offer higher accuracy (0.001-0.01%), better resolution (6½ to 8½ digits), and more measurement functions than handheld models (typically 3½ to 4½ digits with 0.1-0.5% accuracy). Input impedance is a critical specification: 10 MΩ is standard for most DMMs, while high-impedance (>10 GΩ) inputs are necessary for measuring high-impedance circuits without loading errors.

**Example 11.4.1:** A 6½-digit DMM on the 10 V range (input impedance 10 MΩ) is used to measure the open-circuit voltage of a sensor with an output impedance of 100 kΩ. The true sensor voltage is 8.500 V. Calculate the displayed reading and the loading error.
**Solution:**
The DMM input impedance forms a voltage divider with the sensor output impedance.
Displayed voltage: $V_{measured} = V_{true} \times R_{DMM} / (R_{source} + R_{DMM}) = 8.500 \times 10{,}000{,}000 / (100{,}000 + 10{,}000{,}000) = 8.500 \times 10^7 / 10.1 \times 10^6 = 8.500 \times 0.99010 = 8.4158$ V.
Loading error: $\Delta V = 8.500 - 8.4158 = 0.0842$ V.
Percent error: $0.0842 / 8.500 \times 100 = 0.99\%$.
This is significant for a precision instrument.
Resolution: 10 V / 1,999,999 counts = 5.0 μV per count, so the DMM itself is far more accurate than the loading error.
Using a high-impedance (>10 GΩ) DMM would reduce the loading error to < 0.001%.

### 11.4.2 Oscilloscope

An oscilloscope displays voltage as a function of time, providing a visual representation of signal waveforms that reveals amplitude, frequency, rise time, noise, distortion, and timing relationships between multiple signals. Modern digital storage oscilloscopes (DSOs) sample the input signal using a high-speed ADC (8-12 bit resolution), store the digitized waveform in memory, and display it on a screen with measurement cursors and automatic measurements. Key specifications include bandwidth (the frequency at which the measured amplitude drops to -3 dB, typically 50 MHz to over 100 GHz), sample rate (typically 1-100+ GS/s, must be at least 2.5-5× the bandwidth for accurate waveform reconstruction), memory depth (determines the time window that can be captured at full sample rate), and number of channels (typically 2-8). Triggering systems (edge, pulse width, logic, protocol) ensure stable display of repetitive waveforms and capture of specific events. Mixed-signal oscilloscopes (MSOs) add digital logic inputs for simultaneous analog and digital signal analysis, and protocol-aware triggering and decoding for serial buses (SPI, I2C, UART, CAN).

**Example 11.4.2:** A digital oscilloscope has a bandwidth of 200 MHz and a sample rate of 1 GS/s. The memory depth is 10 Mpoints. A pulse signal with a rise time of 2 ns must be measured. Determine whether the oscilloscope bandwidth is adequate, the time window at full sample rate, and the maximum time window at reduced sample rate.
**Solution:**
For accurate rise time measurement, the oscilloscope bandwidth should satisfy: $BW \geq 0.35 / t_{rise(actual)}$.
However, the measured rise time is: $t_{rise(measured)} = \sqrt{(t_{rise(signal)}^2 + t_{rise(scope)}^2)}$, where $t_{rise(scope)} = 0.35 / BW = 0.35 / 200 \times 10^6 = 1.75$ ns.
Measured rise time: $t_{rise(measured)} = \sqrt{(2^2 + 1.75^2)} = \sqrt{(4 + 3.0625)} = \sqrt{7.0625} = 2.66$ ns.
The scope significantly affects the measurement (33% error).
A 500 MHz scope would give $t_{rise(scope)} = 0.7$ ns and $t_{rise(measured)} = \sqrt{(4 + 0.49)} = 2.12$ ns (6% error) – much better.
Time window at full sample rate: $T = \text{memory depth} / \text{sample rate} = 10 \times 10^6 / 1 \times 10^9 = 10$ ms.

### 11.4.3 Spectrum Analyzer

A spectrum analyzer displays signal amplitude as a function of frequency, providing a frequency-domain view complementary to the oscilloscope's time-domain display. Swept-tuned spectrum analyzers use a superheterodyne receiver architecture that mixes the input signal with a swept local oscillator and measures the output of a narrowband intermediate-frequency (IF) filter, scanning across the frequency range to build up the spectrum display. FFT-based spectrum analyzers digitize a block of the input signal and compute the frequency spectrum using the Fast Fourier Transform, offering faster measurement speed and the ability to capture transient spectral events. Key specifications include frequency range (DC to 50+ GHz), resolution bandwidth (RBW, the minimum frequency separation between two signals that can be resolved), dynamic range (the range from the noise floor to the maximum input level without distortion, typically 80-120 dB), and phase noise (which determines the ability to measure signals close to a strong carrier). Spectrum analyzers are essential for RF design, EMC testing, wireless communications, and vibration analysis.

**Example 11.4.3:** A spectrum analyzer with a noise floor of -110 dBm (in 1 kHz RBW) is used to measure the second harmonic distortion of an amplifier. The fundamental output at 100 MHz is -5 dBm, and the second harmonic at 200 MHz reads -52 dBm, both measured with RBW = 10 kHz. Calculate the second harmonic distortion and the dynamic range margin above the noise floor.
**Solution:**
Second harmonic distortion: $HD_2 = P_{2nd} - P_{fundamental} = -52 - (-5) = -47$ dBc (47 dB below the carrier).
In percentage: $HD_2 = 10\text{^}(-47/20) \times 100\% = 0.00447 \times 100\% = 0.447\%$.
Noise floor correction for the measurement RBW: noise floor at 10 kHz RBW = $-110 + 10\log_{10}(10{,}000/1000) = -110 + 10 = -100$ dBm.
Signal-to-noise margin for the harmonic measurement: -52 - (-100) = 48 dB above the noise floor, confirming the harmonic measurement is valid and well above the noise.
The maximum measurable dynamic range for this carrier level: -5 - (-100) = 95 dB.

### 11.4.4 Function Generator

A function generator produces standard waveforms (sine, square, triangle, sawtooth, pulse) at selectable frequencies and amplitudes for testing and characterizing electronic circuits and systems. Modern arbitrary waveform generators (AWGs) can produce any user-defined waveform loaded from a digital memory, enabling generation of complex modulated signals, protocol test patterns, and real-world signal replicas. Key specifications include frequency range (typically DC to tens of MHz for general-purpose units, GHz for RF signal generators), amplitude range and resolution, waveform fidelity (harmonic distortion, typically -40 to -70 dBc for sine waves), and output impedance (typically 50 Ω). Frequency accuracy and stability are determined by the internal clock reference (crystal oscillator or optional external reference input), with high-stability OCXO references providing parts-per-billion accuracy. Modulation capabilities (AM, FM, PM, PWM, sweep, burst) allow function generators to serve as stimulus sources for communications, control system, and sensor testing.

**Example 11.4.4:** A function generator with a 50 Ω output impedance is set to produce a 1 MHz sine wave at 2 $V_{pp}$ into a 50 Ω load. The generator's specified harmonic distortion is -55 dBc. Calculate the RMS fundamental voltage across the load, the power delivered to the load, and the

amplitude of the largest harmonic.
**Solution:**
The problem specifies 2 $V_{pp}$ into the 50 Ω load, so $V_{load(pp)} = 2\ V_{pp}$.
$V_{load(rms)} = V_{load(pp)} / (2\sqrt{2}) = 2 / 2.828 = 0.707\ V_{rms}$.
Power into 50 Ω load: $P = V_{rms}^2 / R = 0.707^2 / 50 = 0.500 / 50 = 10.0$ mW = +10.0 dBm.
Harmonic level: -55 dBc means the harmonic is 55 dB below the fundamental.
Harmonic amplitude: $V_{harmonic} = V_{fundamental} \times 10$^(-55/20) $= 0.707 \times 1.778 \times 10^{-3} = 1.26\ mV_{rms}$.

### 11.4.5 Power Analyzer

A power analyzer is a specialized instrument designed to accurately measure electrical power and energy in AC and DC systems, including real power (W), reactive power (var), apparent power (VA), power factor, harmonic content, and efficiency. Unlike a DMM or oscilloscope that measures voltage and current independently, a power analyzer simultaneously samples voltage and current at high speed (typically 1-5 MS/s per channel) and computes power quantities using digital signal processing, correctly handling non-sinusoidal waveforms, phase shifts, and harmonic distortion. Modern power analyzers provide 3-7 power input channels for single-phase, three-phase (3-wire and 4-wire), and multi-stage efficiency measurements (e.g., simultaneously measuring input AC power, DC bus, and output to a motor). Key specifications include basic power accuracy (typically 0.01-0.1%), measurement bandwidth (DC to 1-5 MHz for capturing switching converter harmonics), and harmonic analysis capability (THD and individual harmonics to the 50th or 100th order per IEC 61000-4-7). Power analyzers are essential for motor drive efficiency testing, power supply compliance testing (80 PLUS, Energy Star), IEC 62301 standby power measurement, and power quality audits.

**Example 11.4.5:** A power analyzer measures a single-phase motor drive with the following results: $V_{rms} = 230.5$ V, $I_{rms} = 4.82$ A, real power P = 845 W, and total harmonic distortion of the current $THD_I = 38\%$. Calculate the apparent power, power factor, displacement power factor, and distortion power factor.
**Solution:**
Apparent power: $S = V_{rms} \times I_{rms} = 230.5 \times 4.82 = 1{,}111.0$ VA.
Power factor (total): PF = P / S = 845 / 1,111.0 = 0.761.
For non-sinusoidal currents, the total power factor is the product of displacement power factor ($\cos \varphi_1$, due to phase shift of the fundamental) and distortion power factor (due to harmonics): PF = DPF × DstPF.
The distortion power factor relates to THD: $DstPF = 1 / \sqrt{(1 + THD^2)} = 1 / \sqrt{(1 + 0.38^2)} = 1 / \sqrt{1.1444} = 1 / 1.0698 = 0.935$.
Displacement power factor: DPF = PF / DstPF = 0.761 / 0.935 = 0.814, corresponding to a phase angle of $\cos^{-1}(0.814) = 35.5°$ between fundamental voltage and current.
The motor drive has acceptable displacement power factor but the 38% current THD significantly degrades the overall power factor and may require harmonic filtering to meet IEEE 519 limits.

### 11.4.6 LCR Meter and Impedance Analyzer

An LCR meter measures the inductance (L), capacitance (C), and resistance (R) of passive components, along with derived quantities such as impedance magnitude |Z|, phase angle θ, dissipation factor (D = tan δ), quality factor (Q = 1/D), and equivalent series resistance (ESR). The instrument applies a sinusoidal test signal (typically 0.1–10 $V_{rms}$ at frequencies from 20 Hz to 1+ MHz) to the device under test and simultaneously measures the voltage across and current through it, computing impedance as $Z = V/I = |Z|\angle\theta$. From Z, the meter extracts series or parallel equivalent circuit parameters: for a capacitor, $C_s = -1/(\omega X_s)$ and ESR = $R_s$ in series mode, or $C_p = -B_p/\omega$ and $R_p = 1/G_p$ in parallel mode. **Four-terminal (Kelvin) sensing** eliminates lead resistance errors by using separate force and sense connections, essential for measuring low impedances (<10 Ω). **Impedance analyzers** extend LCR meter capability to broader frequency ranges (1 Hz to 120+ MHz or even GHz with network analyzer techniques), enabling characterization of component behavior across frequency — revealing self-resonant frequencies of capacitors and inductors, parasitic effects, and material properties. Applications include incoming component inspection, PCB parasitic extraction, magnetic core characterization, and electrochemical impedance spectroscopy (EIS) for battery and corrosion analysis.

**Example 11.4.6:** A 100 μF aluminum electrolytic capacitor is measured on an LCR meter at 100 kHz with a 1 $V_{rms}$ test signal. The meter reads $|Z| = 0.185$ Ω and $\theta = -59.2°$. Calculate the ESR, the actual capacitance at 100 kHz, the dissipation factor D, and the quality factor Q. Compare the ESR to the manufacturer's specification of 0.10 Ω maximum.
**Solution:**
Extract series equivalent components from $Z = R_s + jX_s$.
$R_s$ (ESR) $= |Z| \times \cos\theta = 0.185 \times \cos(-59.2°) = 0.185 \times 0.5120 = 0.0947$ Ω.
$X_s = |Z| \times \sin\theta = 0.185 \times \sin(-59.2°) = 0.185 \times (-0.8590) = -0.1589$ Ω.
Capacitance: $C_s = -1/(\omega X_s) = -1/(2\pi \times 100{,}000 \times (-0.1589)) = 1/(99{,}874) = 10.01$ μF.
The capacitance has dropped from 100 μF (at 120 Hz) to ~10 μF at 100 kHz — a 10× reduction typical of aluminum electrolytics, which lose effective capacitance above a few kHz due to internal inductance and dielectric loss.
Dissipation factor: $D = \tan\delta = R_s/|X_s| = 0.0947/0.1589 = 0.596$.
Quality factor: $Q = 1/D = 1.68$.
The ESR of 0.0947 Ω is within the 0.10 Ω specification.
This measurement demonstrates why LCR meters must test at the application frequency — a 120 Hz measurement would show the full 100 μF but would not reveal the high-frequency ESR and capacitance roll-off critical for switching power supply decoupling.

### 11.4.7 Frequency Counter and Time Interval Analyzer

A frequency counter measures the frequency of a periodic signal by counting the number of zero-crossings (or trigger-level crossings) within a precisely timed gate interval. In the **direct count** method, an internal reference oscillator defines the gate time (e.g., 1 s) and the counter tallies input cycles: frequency = count / gate time, with a resolution of ±1 count (±1 Hz for a 1 s gate). The **reciprocal counting** method instead measures the period of N input cycles using a high-speed internal clock (typically 200 MHz–10 GHz with interpolation), then computes frequency as f = N / (measured period), achieving resolution proportional to measurement time regardless of the input frequency — critical for measuring low frequencies where direct counting yields poor resolution. Modern **universal counters** combine both techniques and add time interval measurement (measuring the elapsed time between start and stop trigger events with resolution down to ~20 ps single-shot, improving to sub-picosecond with averaging), totalize counting, phase measurement, and pulse width measurement. The accuracy of all frequency counter measurements ultimately depends on the timebase reference: a standard TCXO provides ±1 ppm stability, an OCXO provides

±0.01 ppm, and a **GPS-disciplined oscillator (GPSDO)** or rubidium reference provides ±0.001 ppb ($10^{-12}$) long-term accuracy traceable to UTC — enabling measurements with 12+ digits of resolution. Frequency counters are essential for calibrating oscillators, characterizing clock jitter, verifying RF transmitter frequencies, and production testing of timing-critical components.

**Example 11.4.7:** A reciprocal frequency counter with a 10 GHz internal timebase measures a 60 Hz power line signal using a gate time of 100 ms. A second measurement is made using the direct count method with a 1 s gate. Compare the frequency resolution of each method. If the timebase is a GPSDO with $1 \times 10^{-11}$ stability, what is the absolute frequency accuracy for a 10.000 000 MHz measurement with a 1 s gate?
**Solution:**
**Reciprocal method** (100 ms gate): During 100 ms, approximately 6 cycles of the 60 Hz signal occur.
The internal clock at 10 GHz counts $10^9$ ticks per 100 ms period.
Period of 6 cycles = 6/60 = 0.1 s, measured with resolution of $1/(10 \times 10^9) = 100$ ps.
Period resolution per cycle: 100 ps / 6 = 16.7 ps.
Frequency resolution: $\Delta f/f = \Delta t/t = 100 \times 10^{-12} / 0.1 = 10^{-9}$, so $\Delta f = 60 \times 10^{-9} = 60$ nHz — 9 digits of resolution.
**Direct count method** (1 s gate): Count = 60. Resolution = ±1 count / 1 s = ±1 Hz. Relative resolution = 1/60 = 1.7%, only 1.8 digits.
The reciprocal method is $10^7\times$ better.
**Absolute accuracy** for 10 MHz with GPSDO: uncertainty = $f \times$ stability = $10^7 \times 10^{-11} = 10^{-4}$ Hz = 0.1 mHz.
The measurement reads 10,000,000.000 0 ± 0.000 1 Hz — 11 significant digits, traceable to the GPS satellite constellation's atomic clocks.

### 11.4.8 Vector Network Analyzer (VNA)

A vector network analyzer measures the complex (magnitude and phase) scattering parameters (S-parameters) of RF and microwave devices, fully characterizing how a device reflects and transmits signals as a function of frequency. The VNA operates by sending a known stimulus signal into the device under test (DUT) through precision test ports and measuring the reflected and transmitted waves using directional couplers and coherent receivers. For a two-port device, the four S-parameters are: $S_{11}$ (input reflection coefficient / return loss), $S_{21}$ (forward transmission / insertion loss or gain), $S_{12}$ (reverse transmission / isolation), and $S_{22}$ (output reflection coefficient). Each parameter is a complex number with magnitude (in dB) and phase (in degrees), enabling the VNA to display results as return loss, insertion loss, VSWR, impedance on a Smith chart, group delay, and time-domain reflectometry (TDR) via inverse FFT. Modern VNAs cover frequencies from 5 Hz to 110+ GHz (with waveguide extensions to 1.5 THz), with dynamic range of 100–140 dB, and typically offer 2 or 4 test ports. **Calibration** is critical to VNA accuracy: the SOLT (Short-Open-Load-Thru) and TRL (Thru-Reflect-Line) calibration methods use known standards to mathematically remove systematic errors from the measurement ports, cables, and adapters — moving the reference plane to the DUT connectors. The VNA is indispensable for characterizing filters, amplifiers, antennas, cables, connectors, PCB transmission lines, and any component where impedance matching and frequency response are critical. Applications extend beyond RF to materials characterization (dielectric constant measurement), biomedical sensing, and semiconductor device modeling.

**Example 11.4.8:** A VNA measures a bandpass filter from 800 MHz to 1200 MHz. At the center frequency of 1000 MHz, $S_{21}$ = -1.2 dB ∠-45° and $S_{11}$ = -18 dB ∠+135°. Calculate the insertion loss, return loss, VSWR at the input port, the fraction of power reflected, and the fraction of power transmitted through the filter.
**Solution:**
Insertion loss = $-S_{21}$ = 1.2 dB.
Return loss = $-S_{11}$ = 18 dB.
Input reflection coefficient magnitude: $|\Gamma| = 10^{S_{11}/20} = 10^{-18/20} = 10^{-0.9} = 0.1259$.
VSWR = $(1 + |\Gamma|) / (1 - |\Gamma|) = (1 + 0.1259) / (1 - 0.1259) = 1.1259 / 0.8741 = 1.288{:}1$.
Power reflected: $|\Gamma|^2 = 0.1259^2 = 0.01585$ = 1.585% of incident power.
Power transmitted: $|S_{21}|^2 = 10^{S_{21}(dB)/10} = 10^{-1.2/10} = 10^{-0.12} = 0.7586$ = 75.86%.
Power absorbed/dissipated in the filter: $1 - |S_{11}|^2 - |S_{21}|^2 = 1 - 0.01585 - 0.7586 = 0.2256$ = 22.56%.
The filter has good input matching (VSWR < 1.3) but dissipates nearly a quarter of the input power — suggesting resistive losses in the filter elements or substrate.
At the band edges (800 and 1200 MHz), $S_{21}$ drops to -25 dB (0.316% power transmission), providing 23.8 dB of out-of-band rejection.

### 11.4.9 Thermal Imaging Cameras

Thermal imaging cameras detect infrared radiation emitted by objects and convert it into a visible image (thermogram) that maps surface temperature across the field of view, enabling non-contact temperature measurement of entire scenes rather than single points. Every object above absolute zero emits infrared radiation according to the Planck radiation law, with the peak wavelength and total radiated power determined by its temperature and surface emissivity. Thermal cameras are indispensable in electrical maintenance — detecting hotspots on switchgear, loose connections, overloaded conductors, and failing transformer bushings before they cause outages or fires — as well as in building diagnostics, mechanical equipment monitoring, process control, and security and surveillance applications.

Two primary detector technologies dominate thermal imaging. **Uncooled microbolometer arrays** use a grid of vanadium oxide (VOx) or amorphous silicon (a-Si) elements suspended on micro-bridges that absorb incident IR radiation, causing a temperature rise that changes the element's electrical resistance. Microbolometers operate at ambient temperature (no cryogenic cooling), are compact and relatively inexpensive, and achieve noise equivalent temperature difference (NETD) values of 30–80 mK at f/1 optics. Standard resolutions range from 160 × 120 to 640 × 480 pixels, with high-end models reaching 1024 × 768. They operate in the long-wave infrared (LWIR) band, 8–14 μm, which coincides with an atmospheric transmission window and the peak emission wavelength of objects near ambient temperature (~10 μm at 300 K per Wien's displacement law). **Cooled detector arrays** use photon detectors — typically indium antimonide (InSb) for the mid-wave infrared (MWIR, 3–5 μm) band or mercury cadmium telluride (MCT/HgCdTe) for MWIR or LWIR — cooled to cryogenic temperatures (typically 77 K using a Stirling-cycle cooler or liquid nitrogen) to suppress thermal noise. Cooled detectors achieve NETD values of 15–25 mK with higher sensitivity, faster response times (<1 ms vs. ~10 ms for microbolometers), and larger format arrays (up to 1280 × 1024 or higher), but are significantly more expensive, bulkier, and require periodic cooler maintenance.

**NETD (Noise Equivalent Temperature Difference)** is the key performance metric for thermal cameras — it represents the smallest temperature difference the camera can resolve, defined as the target temperature change that produces a signal-to-noise ratio of 1. A camera with 50 mK NETD can distinguish temperature differences as small as 0.05°C under ideal conditions. In practice, the minimum resolvable temperature difference depends on spatial averaging, frame integration, and the temperature contrast between the target and background.

**Spectral band selection** affects measurement capability: LWIR (8–14 μm) cameras provide the best performance for objects near ambient temperature and are less affected by solar reflections, making them ideal for outdoor electrical inspections during daylight. MWIR (3–5 μm) cameras offer higher spatial resolution for a given aperture size (shorter wavelength allows smaller diffraction-limited spot size) and better performance at elevated temperatures (>300°C), making them preferred for furnace monitoring, gas detection (many gases have absorption bands in the 3–5 μm region), and military/aerospace applications.

**Emissivity correction** is essential for accurate temperature measurement. Real surfaces emit less radiation than an ideal blackbody at the same temperature, with the ratio defined as emissivity ε (ranging from 0 to 1). Most non-metallic surfaces have $\varepsilon = 0.85$–$0.95$, while polished metals can have ε as low as 0.02–0.10. An uncorrected measurement of a low-emissivity surface will significantly underestimate the actual temperature because the camera receives less radiation than expected. Modern cameras allow the user to set ε, reflected apparent temperature (to compensate for ambient radiation reflected off low-ε surfaces), atmospheric temperature, humidity, and distance to correct for transmission losses.

**Spatial resolution** is defined by the instantaneous field of view (IFOV), which is the angular subtense of a single detector pixel — typically 1–3 mrad for standard lenses. The measurement spot size at a given distance is: spot diameter = IFOV × distance. For accurate temperature measurement, the target must fill at least 3 × 3 pixels (the "spot size ratio" or "distance-to-spot-size ratio" D:S). A camera with 1.4 mrad IFOV viewing a target at 5 m has a pixel footprint of 7 mm, requiring the target to be at least 21 mm across for a valid temperature measurement. Applications in electrical maintenance include scanning distribution panels for overloaded breakers ($\Delta T > 10$°C above adjacent phases indicates a problem), inspecting overhead transmission connections for resistance heating, locating underground cable faults via surface temperature anomalies, and verifying the thermal performance of heat sinks and cooling systems. Infrared thermography programs following NFPA 70B and NETA MTS standards recommend annual thermographic surveys of all electrical distribution equipment rated above 600 V.

**Example 11.4.9:** A thermal imaging camera with NETD = 50 mK and LWIR (8–14 μm) spectral band measures a bolted bus connection in a switchgear panel. The camera settings are $\varepsilon = 0.95$ (oxidized copper), and the reflected apparent temperature is 25°C. The camera reads an apparent temperature of 78°C on the suspect connection, while an adjacent identical connection reads 42°C. The actual emissivity of the suspect connection is $\varepsilon = 0.70$ (partially cleaned/polished surface from recent maintenance). Calculate the corrected temperature of the suspect connection and determine the severity of the hotspot according to NETA guidelines.
**Solution:**
The camera computes temperature assuming $\varepsilon = 0.95$. The total radiation received by the camera from the target is: $W_{total} = \varepsilon_{actual} \times \sigma T^4{}_{actual} + (1 - \varepsilon_{actual}) \times \sigma T^4{}_{reflected}$.
The camera interprets this as: $W_{total} = \varepsilon_{set} \times \sigma T^4{}_{apparent} + (1 - \varepsilon_{set}) \times \sigma T^4{}_{reflected}$.
Setting these equal: $\varepsilon_{set} \times T^4{}_{apparent} + (1 - \varepsilon_{set}) \times T^4{}_{reflected} = \varepsilon_{actual} \times T^4{}_{actual} + (1 - \varepsilon_{actual}) \times T^4{}_{reflected}$.
Converting to Kelvin: $T_{apparent} = 78 + 273.15 = 351.15$ K, $T_{reflected} = 25 + 273.15 = 298.15$ K.
Left side: $0.95 \times (351.15)^4 + 0.05 \times (298.15)^4 = 0.95 \times 1.519 \times 10^{10} + 0.05 \times 7.906 \times 10^9 = 1.443 \times$

$10^{10} + 3.953 \times 10^{8} = 1.483 \times 10^{10}$.
Solving for $T_{actual}$: $\varepsilon_{actual} \times T^4{}_{actual} = 1.483 \times 10^{10} - (1 - 0.70) \times (298.15)^4 = 1.483 \times 10^{10} - 0.30 \times 7.906 \times 10^{9} = 1.483 \times 10^{10} - 2.372 \times 10^{9} = 1.246 \times 10^{10}$.
$T^4{}_{actual} = 1.246 \times 10^{10} / 0.70 = 1.780 \times 10^{10}$.
$T_{actual} = (1.780 \times 10^{10})^{1/4} = 365.3\ K = 92.2°C$.
The corrected temperature is 92.2°C — significantly higher than the 78°C apparent reading due to the lower-than-assumed emissivity.
Temperature rise above the reference connection: $\Delta T = 92.2 - 42 = 50.2°C$.
Per NETA MTS severity criteria: $\Delta T > 40°C$ above a similar component under similar loading is classified as **immediate action required** (the most severe category), indicating a high-resistance connection that should be de-energized and repaired immediately to prevent equipment failure or fire.
This example demonstrates why accurate emissivity settings are critical — the 0.25 difference in emissivity (0.95 assumed vs. 0.70 actual) caused a 14.2°C underestimate that could have led to an incorrect severity classification.

## 11.5 Data Acquisition

Data acquisition (DAQ) bridges the gap between the analog world of sensors and the digital world of computers, converting continuous physical signals into discrete digital data that can be processed, stored, and analyzed by software. A DAQ system encompasses the entire signal chain from the analog front end (multiplexing, sampling, and quantization) through digital communication to the host computer, including the software that configures the hardware, displays real-time data, and logs measurements for later analysis. The performance of a DAQ system is defined by its channel count, sampling rate, resolution, accuracy, and the ability to synchronize measurements across channels and with external events.

### 11.5.1 Sampling and Quantization

Data acquisition systems convert analog sensor signals into digital data for processing, storage, and analysis by a computer. The analog-to-digital converter (ADC) performs two operations: sampling (capturing the instantaneous value of the analog signal at discrete time intervals) and quantization (mapping each sample to the nearest digital code from a finite set of levels). The sampling rate must satisfy the Nyquist criterion ($f_s > 2f_{max}$) to avoid aliasing, and practical systems sample at 5–10× the maximum signal frequency to ease anti-aliasing filter requirements. ADC resolution (typically 12–24 bits for data acquisition) determines the smallest detectable signal change: an N-bit ADC divides the full-scale range into $2^N$ levels. Common ADC architectures for data acquisition include successive approximation (SAR, 12-18 bit, up to several MS/s), delta-sigma (Σ-Δ, 16-24 bit, excellent noise rejection but lower speed), and pipeline (12-16 bit, high speed for waveform capture).

**Example 11.5.1:** A 14-bit SAR ADC with a ±10 V bipolar input range samples a vibration sensor at $f_s = 50$ kS/s. The signal of interest has frequencies up to 2 kHz. Calculate the ADC resolution (LSB), the ideal signal-to-noise-and-distortion ratio (SINAD), and the required anti-aliasing filter cutoff.

**Solution:**
Full-scale range: 20 V (from -10 to +10 V).
Resolution (LSB): 20 V / $2^{14}$ = 20 / 16,384 = 1.221 mV.
Ideal SINAD for an N-bit ADC: SINAD = 6.02N + 1.76 dB = 6.02 × 14 + 1.76 = 84.28 + 1.76 = 86.04 dB.
Nyquist frequency: $f_N = f_s / 2 = 25$ kHz.
The anti-aliasing filter cutoff should be set between the maximum signal frequency and the Nyquist frequency.
A practical choice: $f_c \approx 5$ kHz (2.5× the signal bandwidth), providing a flat passband to 2 kHz and a 5× frequency margin to Nyquist for filter roll-off (just over 2.3 octaves).
A 4th-order Butterworth at 5 kHz gives -80 dB/decade × $\log_{10}(25/5)$ = -80 × 0.699 = -55.9 dB attenuation at Nyquist — a factor of ~625× reduction of any out-of-band signal.

### 11.5.2 Data Acquisition Systems

A complete data acquisition (DAQ) system integrates multiplexing, signal conditioning, ADC conversion, timing, and digital communication into a coordinated measurement platform. Multiplexed systems use an analog multiplexer to sequentially connect multiple sensor channels to a shared ADC, reducing cost at the expense of simultaneous sampling capability. Simultaneous sampling systems provide a dedicated sample-and-hold circuit and ADC for each channel, essential when phase relationships between channels must be preserved (e.g., vibration analysis, power measurements). Timing and triggering control the start of acquisition, sample rate, and synchronization with external events or other instruments. Common DAQ interfaces include USB, PCI/PCIe, and Ethernet, with industrial systems using fieldbus protocols (Modbus, PROFIBUS, EtherCAT) to distribute measurement points across a facility. Software frameworks such as LabVIEW, MATLAB Data Acquisition Toolbox, and Python libraries (nidaqmx, pydaq) provide configuration, real-time display, logging, and analysis of acquired data.

**Example 11.5.2:** A 16-channel data acquisition system uses a multiplexed 16-bit ADC with a maximum throughput of 250 kS/s. Each channel requires a settling time of 2 μs after the multiplexer switches. The signals of interest have a maximum frequency of 1 kHz on all channels. Determine the maximum per-channel sample rate and whether simultaneous sampling is needed.
**Solution:**
Total time per conversion including settling: $t_{ch} = 1/250{,}000 + 2 \times 10^{-6} = 4.0\ \mu s + 2.0\ \mu s = 6.0\ \mu s$ per channel.
Cycle time for all 16 channels: $t_{cycle} = 16 \times 6.0\ \mu s = 96\ \mu s$.
Maximum per-channel sample rate: $f_{ch} = 1 / t_{cycle} = 1 / 96 \times 10^{-6} = 10{,}417$ S/s ≈ 10.4 kS/s per channel.
This is more than 5× the 1 kHz signal bandwidth, satisfying Nyquist with margin.
Phase error between first and last channel: $\Delta t = 15 \times 6.0\ \mu s = 90\ \mu s$.
At 1 kHz: phase skew = $360° \times f \times \Delta t = 360° \times 1000 \times 90 \times 10^{-6} = 32.4°$.
This significant phase skew means **simultaneous sampling (dedicated S/H per channel) is required** if phase relationships between channels must be preserved.

### 11.5.3 Data Logging and Storage

Data logging is the process of recording measurement data over time for later retrieval and analysis, ranging from simple standalone loggers that record a single parameter to enterprise-class systems that continuously capture thousands of channels across an entire facility. **Standalone data loggers** are battery-powered devices with built-in sensors (or sensor inputs), onboard memory (typically 1-64 MB flash), and a real-time clock that timestamp each sample — common for environmental monitoring, cold-chain compliance, and long-term field deployments lasting weeks to months. **PC-based logging** uses DAQ hardware connected to a computer running logging software, offering higher channel counts, faster sample rates, and real-time display but requiring continuous PC operation and power. Storage capacity planning requires calculating the data rate: bytes per second = (channels × sample rate × bytes per sample) + timestamp overhead. File formats include CSV (human-readable, large), TDMS (NI Technical Data Management Streaming, compact with metadata), HDF5 (hierarchical, widely used in science), and proprietary binary formats optimized for specific instruments. For long-duration logging at high sample rates, circular buffer (ring buffer) operation overwrites the oldest data when memory is full, ensuring the most recent data is always captured. Industrial data historians (such as OSIsoft PI or InfluxDB) provide time-series databases optimized for storing and querying billions of timestamped measurements with compression ratios of 10:1 to 50:1.

**Example 11.5.3:** A 32-channel vibration monitoring system logs data continuously from accelerometers sampled at 10 kS/s per channel with 16-bit (2-byte) resolution. Each sample is timestamped with a 4-byte UNIX timestamp per scan (one timestamp for all 32 channels). Calculate the raw data rate in MB/hour, the storage required for 30 days of continuous logging, and the minimum write speed required for the storage medium.
**Solution:**
Data per scan (all 32 channels): 32 channels × 2 bytes = 64 bytes + 4 bytes timestamp = 68 bytes per scan.
Scans per second: 10,000.
Raw data rate: $68 \times 10{,}000 = 680{,}000$ bytes/s = 680 kB/s = 0.680 MB/s.
Per hour: $0.680 \times 3{,}600 = 2{,}448$ MB/hour ≈ 2.45 GB/hour.
Per day: $2.45 \times 24 = 58.8$ GB/day.
For 30 days: $58.8 \times 30 = 1{,}764$ GB ≈ 1.76 TB.
A 2 TB SSD would provide sufficient capacity with margin.
Minimum sustained write speed: 680 kB/s = 0.68 MB/s — easily met by any modern SSD (typical sustained write > 200 MB/s) or even a quality SD card.
With typical 10:1 compression (using lossless algorithms on the redundant sensor data): effective storage ≈ 176 GB for 30 days, fitting on a single 256 GB drive.

### 11.5.4 Automated Test and Remote Instrument Control

Automated test systems use software to configure instruments, execute measurement sequences, collect data, and make pass/fail decisions without operator intervention, enabling high-throughput production testing, regression testing, and complex multi-instrument characterization. The **IEEE 488 (GPIB)** bus was the original standard for instrument communication (1 MB/s, 15 devices per bus, cable lengths to 20 m), and while still widely installed, it has been supplemented by **USB-TMC** (Test and Measurement Class, uses standard USB with GPIB-like semantics), **LXI** (LAN eXtensions for Instrumentation, using Ethernet/TCP-IP for high-speed, long-distance control), and

**PXI** (PCI eXtensions for Instrumentation, a modular rack-based platform with shared high-speed backplane for tightly integrated ATE systems). **SCPI** (Standard Commands for Programmable Instruments) defines a standardized ASCII command syntax across all interfaces — a command like `MEAS:VOLT:DC?` reads a DC voltage measurement on any SCPI-compliant DMM regardless of manufacturer. The **VISA** (Virtual Instrument Software Architecture) library provides a unified API that abstracts the physical interface, allowing the same program code to control instruments over GPIB, USB, LAN, or serial by changing only the resource address string (e.g., `GPIB0::22::INSTR` or `TCPIP::192.168.1.50::INSTR`). Programming environments include LabVIEW (graphical, widely used in test engineering), MATLAB Instrument Control Toolbox, and Python with PyVISA — the latter providing a lightweight, open-source alternative: `import pyvisa; rm = pyvisa.ResourceManager(); dmm = rm.open_resource('TCPIP::192.168.1.50::INSTR'); print(dmm.query('MEAS:VOLT:DC?'))`. **Automated test equipment (ATE)** for production integrates switching matrices (to route signals between DUT pins and instruments), stimulus sources, measurement instruments, and a test executive that sequences test steps, collects results, computes pass/fail limits, and generates test reports with statistical yield analysis.

**Example 11.5.4:** A production test station must measure DC voltage, AC voltage, resistance, and frequency on a circuit board at 8 test points using a DMM and a frequency counter, both controlled via LAN/LXI. Each measurement takes 150 ms (instrument settling + measurement + data transfer). The test sequence requires 8 DC voltage, 4 AC voltage, 4 resistance, and 2 frequency measurements per board. Calculate the total test time per board and the maximum throughput in boards per hour. If parallel testing of 4 boards is implemented using a switching matrix, what is the new throughput?

**Solution:**

Total measurements per board: $8 + 4 + 4 + 2 = 18$ measurements.

Test time per board: $18 \times 150$ ms $= 2.7$ s.

Add 0.5 s for board handler settling and communication overhead: total $= 3.2$ s per board.

Throughput: $3600/3.2 = 1{,}125$ boards/hour.

With parallel testing of 4 boards: the switching matrix routes each test point from one of 4 boards to the instruments in sequence.

Total measurements: $18 \times 4 = 72$.

Test time for 4 boards: $72 \times 150$ ms $+ 0.5$ s $= 10.8 + 0.5 = 11.3$ s.

Per-board time: $11.3/4 = 2.825$ s.

Throughput: $3600/2.825 \times (4/4)$ — wait, the 4 boards are tested sequentially through the mux, not truly in parallel.

Actual throughput: 4 boards per 11.3 s = 1,274 boards/hour — only a 13% improvement because the instruments are the bottleneck.

True parallel testing requires duplicating the instruments: 4 DMMs and 4 counters testing simultaneously would give $4 \times 1{,}125 = 4{,}500$ boards/hour, but at 4× the instrument cost.

A better approach: overlap handler time with test time using a conveyor and dual-site fixturing, achieving **~1,800 boards/hour with a single instrument set**.

# Chapter 12

# Electric Motors

Electric motors convert electrical energy into mechanical energy through the interaction of magnetic fields and current-carrying conductors. They are the most common source of mechanical power in modern industry, transportation, and consumer products, collectively consuming approximately 45% of global electricity generation. Motor selection depends on the application requirements for speed, torque, efficiency, control precision, size, cost, and operating environment, with motor types ranging from simple DC machines to sophisticated synchronous and stepper motors driven by advanced power electronics.

## 12.1 DC Motors

DC motors produce torque by passing current through conductors immersed in a magnetic field, following the Lorentz force law ($F = I \times L \times B$). They offer inherently simple speed control — varying the applied voltage changes the speed, while varying the current changes the torque — making them the earliest motors used for variable-speed applications. The two main categories are brushed DC motors, which use a mechanical commutator to switch current direction in the rotor windings, and brushless DC (BLDC) motors, which use electronic commutation to energize fixed stator windings around a permanent magnet rotor.

### 12.1.1 Brushed DC Motors

Brushed DC motors use a wound rotor (armature) that rotates within a stationary magnetic field produced by permanent magnets or field windings. Current is delivered to the rotating armature through carbon brushes that make sliding contact with a segmented commutator, which mechanically reverses the current direction in each coil as it rotates to maintain continuous torque production. The speed of a brushed DC motor is proportional to the applied voltage minus the back-EMF ($V = E_{back} + I_a \times R_a$), and the torque is proportional to the armature current ($\tau = K_t \times I_a$), making speed and torque control straightforward by varying the supply voltage. Brushed DC motors are simple, inexpensive, and provide excellent low-speed torque, but the brushes and commutator introduce friction, electrical noise (arcing), and require periodic maintenance due to wear. Series-wound, shunt-wound, and compound-wound configurations provide different speed-torque characteristics suited to various applications such as automotive starters (series), machine tools (shunt), and elevators (compound).

**Example 12.1.1:** A shunt-wound DC motor has the following nameplate data: $V_{supply} = 240$ V, $R_a = 0.5\ \Omega$ (armature resistance), and the motor constant $K = 1.2$ V/(rad/s). At no load, the motor draws $I_a = 2$ A. Under full load, the armature current is $I_a = 40$ A. Calculate the no-load speed,

full-load speed, and speed regulation.
**Solution:**
No-load: Back-EMF: $E_{back} = V - I_a \times R_a = 240 - 2 \times 0.5 = 239$ V.
No-load speed: $\omega_{NL} = E_{back} / K = 239 / 1.2 = 199.2$ rad/s $= 199.2 \times 60/(2\pi) = 1{,}902$ RPM.
Full-load: $E_{back} = 240 - 40 \times 0.5 = 220$ V.
Full-load speed: $\omega_{FL} = 220 / 1.2 = 183.3$ rad/s $= 1{,}751$ RPM.
Speed regulation: $SR = (\omega_{NL} - \omega_{FL}) / \omega_{FL} \times 100\% = (199.2 - 183.3) / 183.3 \times 100\% = 8.64\%$.
Full-load torque: $\tau = K \times I_a = 1.2 \times 40 = 48.0$ N·m.

### 12.1.2 Brushless DC Motors (BLDC)

Brushless DC motors eliminate the mechanical commutator by placing permanent magnets on the rotor and the stator windings on the stationary frame, using electronic commutation (via transistor switching) to energize the appropriate stator phases as the rotor turns. Hall effect sensors or back-EMF sensing (sensorless control) detect the rotor position and determine the commutation timing. BLDC motors offer higher efficiency (85-95%), longer life (no brush wear), lower electrical noise, and higher power density compared to brushed DC motors. The trapezoidal back-EMF waveform distinguishes BLDC motors from permanent magnet synchronous motors (which have sinusoidal back-EMF), and they are typically driven with six-step (trapezoidal) commutation from a three-phase inverter. Applications include computer fans, hard disk drives, drone propellers, electric bicycles, and industrial servo systems where high reliability and compact size are required.

**Example 12.1.2:** A BLDC motor has 8 poles, a torque constant $K_t = 0.15$ N·m/A, a back-EMF constant $K_e = 0.15$ V/(rad/s), and a phase resistance of $R_{ph} = 0.8\ \Omega$. The DC bus voltage is 48 V, and the motor drives a load requiring 2 N·m at 1,700 RPM. Calculate the required phase current, the back-EMF, and the motor efficiency (neglecting switching and iron losses).
**Solution:**
Speed in rad/s: $\omega = 1700 \times 2\pi/60 = 178.0$ rad/s.
Required phase current: $I = \tau / K_t = 2.0 / 0.15 = 13.33$ A.
Back-EMF (line-to-line): $E_{back} = K_e \times \omega = 0.15 \times 178.0 = 26.7$ V.
Verify voltage equation (two phases conducting, total resistance $2R_{ph} = 1.6\ \Omega$): $V_{bus} = E_{back} + I \times 2R_{ph} = 26.7 + 13.33 \times 1.6 = 26.7 + 21.3 = 48.0$ V ✓.
Mechanical output power: $P_{mech} = \tau \times \omega = 2.0 \times 178.0 = 356.0$ W.
Copper losses (three phases, two conducting at any time in trapezoidal commutation): $P_{copper} = 2 \times I^2 \times R_{ph} = 2 \times 13.33^2 \times 0.8 = 2 \times 142.1 = 284.2$ W.
Input power: $P_{in} = P_{mech} + P_{copper} = 356.0 + 284.2 = 640.2$ W.
Efficiency: $\eta = 356.0 / 640.2 \times 100\% = 55.6\%$.
The relatively low efficiency reflects the high copper losses at this load current; efficiency improves significantly at lighter loads where $I^2R$ losses are smaller relative to mechanical output.

### 12.1.3 Universal Motors

Universal motors are series-wound DC motors designed to operate on both AC and DC supplies, making them one of the few motor types that can run directly from single-phase household mains without a rectifier or inverter. In a series motor, the field winding and armature winding carry the same current, so when the AC supply reverses polarity, both the field flux and the armature current reverse simultaneously, producing torque in the same direction throughout the cycle. Universal motors achieve very high speeds (up to 20,000-30,000 RPM) and high power-to-weight ratios, making them the dominant motor for handheld power tools (drills, grinders, circular saws, routers), vacuum cleaners, blenders, and hair dryers. Their speed can be controlled simply with a triac-based phase-angle controller, providing variable speed from near-zero to maximum RPM. The main disadvantages are the brush and commutator wear (limiting lifespan to a few hundred hours of continuous operation), high audible noise, radio frequency interference (RFI) from commutator arcing, and reduced efficiency (typically 50-70%) compared to brushless alternatives. In many consumer applications, brushless DC motors with electronic controllers are increasingly replacing universal motors due to longer life, lower noise, and higher efficiency.

**Example 12.1.3:** A universal motor rated at 120 V, 12 A, 20,000 RPM is used in a router. The armature resistance is $R_a = 0.8\ \Omega$ and the series field resistance is $R_f = 0.4\ \Omega$. Calculate the back-EMF at rated speed, the copper losses, the motor output power (assuming total losses are 35% of input), and the torque.
**Solution:**
Total winding resistance: $R_{total} = R_a + R_f = 0.8 + 0.4 = 1.2\ \Omega$.
Back-EMF: $E_{back} = V - I \times R_{total} = 120 - 12 \times 1.2 = 120 - 14.4 = 105.6$ V.
Input power: $P_{in} = V \times I = 120 \times 12 = 1{,}440$ W.
Copper losses: $P_{Cu} = I^2 \times R_{total} = 144 \times 1.2 = 172.8$ W.
Output power (65% of input): $P_{out} = 0.65 \times 1{,}440 = 936$ W = 1.25 HP.
Remaining losses (core, friction, windage): $P_{other} = 1{,}440 - 936 - 172.8 = 331.2$ W.
Speed in rad/s: $\omega = 20{,}000 \times 2\pi/60 = 2{,}094$ rad/s.
Output torque: $\tau = P_{out} / \omega = 936 / 2{,}094 = 0.447$ N·m.
The high speed and relatively low torque are characteristic of universal motors — they develop useful mechanical power through high rotational speed rather than high torque.

## 12.2 AC Motors

AC motors are driven by alternating current and rely on rotating magnetic fields produced by polyphase or single-phase stator windings. The three-phase induction motor dominates industrial applications due to its rugged, maintenance-free construction, while synchronous motors serve applications requiring precise speed control or power factor correction. Single-phase motors fill the residential and light-commercial niche where three-phase power is unavailable, and reluctance motors offer a magnet-free alternative with competitive efficiency for certain variable-speed applications.

### 12.2.1 Induction Motors

The induction motor (also called an asynchronous motor) is the most widely used electric motor in industry due to its rugged construction, low cost, and minimal maintenance requirements. The stator carries three-phase windings that produce a rotating magnetic field at synchronous speed $n_s = 120f/P$, where f is the supply frequency and P is the number of poles. The rotating field induces currents in the rotor conductors (either a squirrel-cage of aluminum or copper bars, or wound rotor with slip rings), and the interaction between the induced rotor currents and the stator field produces torque. The rotor must always turn slower than the synchronous speed for induction to occur; this speed difference is quantified by slip: $s = (n_s - n_r)/n_s$, with typical full-load slip of 2–5% for standard machines. The torque-speed characteristic features a starting torque, a maximum (breakdown) torque, and a nearly linear operating region near synchronous speed where the motor runs during normal loaded operation.

**Example 12.2.1:** A 4-pole, three-phase induction motor is connected to a 60 Hz supply. At full load, the motor speed is measured at 1,740 RPM. The motor delivers 15 HP of mechanical power. Calculate the synchronous speed, slip, rotor electrical frequency, and full-load torque.
**Solution:**
Synchronous speed: $n_s = 120f / P = 120 \times 60 / 4 = 1{,}800$ RPM.
Slip: $s = (n_s - n_r) / n_s = (1800 - 1740) / 1800 = 60 / 1800 = 0.0333$ (3.33%).
Rotor electrical frequency: $f_r = s \times f = 0.0333 \times 60 = 2.0$ Hz.
Mechanical power: $P_{mech} = 15$ HP $\times$ 746 W/HP = 11,190 W.
Rotor speed in rad/s: $\omega_r = 1740 \times 2\pi/60 = 182.2$ rad/s.
Full-load torque: $\tau = P_{mech} / \omega_r = 11{,}190 / 182.2 = 61.4$ N·m.

### 12.2.2 Synchronous Motors

Synchronous motors operate at exactly synchronous speed ($n_s = 120f/P$) regardless of load, maintaining a fixed relationship between rotor position and the rotating stator field. The rotor is magnetized either by DC field windings fed through slip rings (wound-rotor type) or by permanent magnets (PM synchronous motors), and it locks in step with the rotating stator field. Synchronous motors can operate at unity or leading power factor by adjusting the field excitation (overexcited operation), making them useful for power factor correction in industrial plants. Permanent magnet synchronous motors (PMSMs) offer the highest efficiency and power density of any motor type and are the preferred choice for electric vehicle traction drives, industrial servo systems, and high-performance applications. Starting a synchronous motor requires special provisions since it has no starting torque at standstill – methods include variable frequency drives (most common), damper (amortisseur) windings that provide induction motor starting torque, or an auxiliary starting motor.

**Example 12.2.2:** A 6-pole, three-phase synchronous motor operates at 50 Hz. The motor is rated at 500 kW with an efficiency of 95% and is running at unity power factor. The plant also has a 300 kVAR inductive load from induction motors. Calculate the motor speed, the line current at 4160 V, and the required excitation adjustment to correct the entire plant power factor to unity.
**Solution:**
Synchronous speed: $n_s = 120f / P = 120 \times 50 / 6 = 1{,}000$ RPM.
Input power at rated load: $P_{in} = P_{out} / \eta = 500 / 0.95 = 526.3$ kW.
At unity power factor: $S = P = 526.3$ kVA.
Line current: $I_L = S / (\sqrt{3} \times V_{LL}) = 526{,}300 / (1.732 \times 4160) = 526{,}300 / 7{,}205 = 73.0$ A.
To correct the plant power factor to unity, the synchronous motor must supply 300 kVAR of leading reactive power.
Required motor apparent power: $S_{motor} = \sqrt{(P_{in}^2 + Q^2)} = \sqrt{(526.3^2 + 300^2)} = \sqrt{(277{,}012 + 90{,}000)} = \sqrt{367{,}012} = 605.8$ kVA.
New motor power factor: $\cos(\varphi) = 526.3 / 605.8 = 0.869$ leading.
New motor line current: $I_L = 605{,}800 / (1.732 \times 4160) = 84.1$ A.
The field excitation must be increased (overexcited) to generate the required leading reactive power.

### 12.2.3 Single-Phase Motors

Single-phase motors operate from standard single-phase AC power and are used in residential and light commercial applications where three-phase power is unavailable. A single-phase stator winding produces a pulsating rather than rotating magnetic field, so auxiliary means are required to create the phase shift needed for starting torque. Split-phase motors use an auxiliary winding with higher resistance to create a phase displacement, providing moderate starting torque for fans and blowers. Capacitor-start motors add a series capacitor to the auxiliary winding for improved starting torque (belt-driven equipment, compressors), and capacitor-start-capacitor-run motors use two capacitors for both high starting torque and improved running efficiency. Shaded-pole motors use a copper shading band on a portion of each stator pole to create a small phase shift, providing low starting torque at minimal cost for small fans, timers, and light-duty applications. Permanent split-capacitor (PSC) motors use a single run capacitor for both starting and running, offering quiet operation and multiple speed capability for HVAC blowers and fans.

**Example 12.2.3:** A capacitor-start single-phase induction motor operates at 120 V, 60 Hz. The main winding has impedance $Z_m = 6 + j8\ \Omega$, and the auxiliary (start) winding has impedance $Z_{aux} = 12 + j5\ \Omega$. A start capacitor is in series with the auxiliary winding. Calculate the capacitance needed to place the auxiliary winding current exactly 90° ahead of the main winding current for maximum starting torque.
**Solution:**
Main winding impedance angle: $\varphi_m = \tan^{-1}(8/6) = \tan^{-1}(1.333) = 53.1°$.
For the auxiliary current to lead the main current by 90°, the auxiliary impedance angle must be: $\varphi_{aux} = 53.1° - 90° = -36.9°$ (capacitive).
The auxiliary winding alone has: $Z_{aux} = 12 + j5\ \Omega$.
Adding a series capacitor with reactance $X_C$: $Z_{total} = 12 + j(5 - X_C)$.
For $\varphi_{aux} = -36.9°$: $\tan(-36.9°) = (5 - X_C) / 12$.
$-0.750 = (5 - X_C) / 12$.
$5 - X_C = -9.00$.
$X_C = 14.00\ \Omega$.
Capacitance: $C = 1 / (2\pi f X_C) = 1 / (2\pi \times 60 \times 14.00) = 1 / 5{,}278 = 189.5\ \mu F$.
A standard 200 μF start capacitor would be selected.

### 12.2.4 Reluctance Motors

Reluctance motors produce torque by exploiting the tendency of a ferromagnetic rotor to align with the stator magnetic field to minimize the magnetic circuit reluctance. The switched reluctance motor (SRM) has salient poles on both the stator and rotor with concentrated windings on the stator poles and no windings or magnets on the rotor, making it the simplest and most robust motor construction. Each stator phase is energized sequentially by a power electronic converter as the rotor poles approach alignment, producing torque pulses that drive continuous rotation. The synchronous reluctance motor (SynRM) uses a conventional distributed stator winding (identical to an induction motor stator) with a specially designed rotor that has flux barriers punched into the laminations to create a high saliency ratio ($L_d/L_q$), and it operates at synchronous speed when driven by a VFD. SynRM motors achieve IE4 (super-premium) or IE5 efficiency levels without rare-earth permanent magnets, making them attractive for industrial pumps, fans, and compressors where magnet cost and supply-chain concerns are factors.

**Example 12.2.4:** A 4-pole synchronous reluctance motor operates from a 60 Hz VFD supply. The motor has a d-axis inductance $L_d = 120$ mH and a q-axis inductance $L_q = 20$ mH. The rated stator current is 10 A. Calculate the synchronous speed, the saliency ratio, and the maximum reluctance torque per phase using the torque equation $\tau = (3/2) \times (P/2) \times (L_d - L_q) \times I_d \times I_q$, where the maximum torque occurs at $I_d = I_q = I_s/\sqrt{2}$.
**Solution:**
Synchronous speed: $n_s = 120f / P = 120 \times 60 / 4 = 1{,}800$ RPM.
Saliency ratio: $\xi = L_d / L_q = 120 / 20 = 6.0$.
For maximum torque: $I_d = I_q = I_s / \sqrt{2} = 10 / 1.414 = 7.07$ A.
Maximum torque: $\tau = (3/2) \times (P/2) \times (L_d - L_q) \times I_d \times I_q = 1.5 \times 2 \times (0.120 - 0.020) \times 7.07 \times 7.07 = 1.5 \times 2 \times 0.100 \times 50.0 = 15.0$ N·m.
Mechanical power at rated speed: $P_{mech} = \tau \times \omega = 15.0 \times (1800 \times 2\pi/60) = 15.0 \times 188.5 = 2{,}827$ W = 2.83 kW.

### 12.2.5 Linear Motors

A linear motor produces force and motion in a straight line rather than rotational torque, effectively an "unrolled" version of a conventional rotary motor. The **linear induction motor (LIM)** consists of a flat stator (primary) with three-phase windings that produce a traveling magnetic field, and a conductive plate or rail (secondary, typically aluminum backed by steel) in which eddy currents are induced — the interaction between the traveling field and the induced currents produces a thrust force along the direction of travel. The synchronous speed of the traveling field is $v_s = 2\tau_p f$, where $\tau_p$ is the pole pitch and f is the supply frequency. The **linear synchronous motor (LSM)** uses either permanent magnets or electromagnets on the moving part (or the track) to lock in step with the stator's traveling field, providing higher efficiency and force density than LIMs. Linear motors eliminate mechanical transmission elements (gearboxes, ball screws, belts, rack-and-pinion), achieving zero backlash, very high speeds (>500 km/h for maglev trains), and sub-micrometer positioning accuracy in precision applications. **Applications** include magnetic levitation (maglev) trains (Shanghai Transrapid at 431 km/h, SCMaglev at 603 km/h), semiconductor lithography stages (nanometer-level positioning), baggage handling and people movers (airport transit), roller coaster launch systems, and high-speed pick-and-place machines. The main design challenges are the large air gap (compared to rotary motors), end effects that reduce thrust at the edges of the primary, and the need for linear position feedback (linear encoders or laser interferometers) for closed-loop control.

**Example 12.2.5:** A linear synchronous motor for a semiconductor wafer stage has a pole pitch of $\tau_p = 24$ mm, a peak force constant of $K_f = 180$ N/A, a moving mass of 12 kg, and a linear encoder with 1 nm resolution. The stage must accelerate at 50 m/s² to a peak velocity of 2 m/s. Calculate the thrust force required for acceleration, the peak current, and the electrical frequency at peak velocity.
**Solution:**
Thrust force for acceleration: $F = ma = 12 \times 50 = 600$ N.
Adding 20 N for friction and cable drag: $F_{total} = 620$ N.
Peak current: $I = F_{total}/K_f = 620/180 = 3.44$ A.
Electrical frequency at peak velocity: $f = v/(2\tau_p) = 2.0/(2 \times 0.024) = 41.7$ Hz.
Acceleration time to peak velocity: $t = v/a = 2.0/50 = 40$ ms.
Distance during acceleration: $d = ½at^2 = ½ \times 50 \times 0.04^2 = 0.04$ m = 40 mm.

At constant velocity (2 m/s), the continuous thrust needed is only ~20 N (friction), requiring I = 20/180 = 0.11 A — demonstrating that linear motors for positioning are dominated by acceleration forces, not steady-state loads.
The 1 nm encoder resolution provides position feedback far finer than the ~10 nm positioning accuracy achievable by the servo controller.

### 12.2.6 Wound-Rotor Induction Motors

A wound-rotor induction motor replaces the squirrel-cage rotor with a three-phase winding connected to external circuits through slip rings and brushes, providing direct access to the rotor circuit for speed and torque control. By inserting external resistance in series with the rotor windings, the motor's torque-speed characteristic is reshaped: increasing rotor resistance shifts the maximum (breakdown) torque to a lower speed without changing its magnitude, allowing full breakdown torque to be available at standstill for heavy starting loads. This makes wound-rotor motors ideal for **crane and hoist applications** where high starting torque at low speed is essential, **large compressors and mills** where smooth, high-torque starting reduces mechanical stress, and **slip-energy recovery systems** (Scherbius drives) where the rotor slip power is recovered and fed back to the supply through a power electronic converter rather than wasted as heat in resistors. The rotor current at any slip s is $I_2 = sE_2 / \sqrt{(R_{2total}^2 + (sX_2)^2)}$, where $E_2$ is the standstill rotor EMF, $R_{2total}$ is the rotor resistance plus external resistance, and $X_2$ is the rotor leakage reactance. The slip at maximum torque shifts proportionally to the total rotor resistance: $s_{max} = R_{2total} / X_2$, while the breakdown torque remains constant: $T_{max} = 3V_1^2 / (2\omega_s(X_1 + X_2))$, independent of rotor resistance. Modern applications increasingly replace external resistor banks with rotor-side converters that provide stepless speed control and regenerative braking capability, combining the starting advantages of wound-rotor motors with the efficiency of electronic control.

**Example 12.2.6:** A 6-pole, 60 Hz wound-rotor induction motor has a standstill rotor EMF of $E_2$ = 200 V (line-to-line), rotor resistance $R_2 = 0.3\ \Omega$ per phase, and rotor leakage reactance $X_2 = 1.5\ \Omega$ per phase. Calculate the slip at maximum torque with no external resistance, then determine the external resistance per phase needed to achieve maximum torque at standstill (s = 1) for a heavy crane lift.
**Solution:**
Synchronous speed: $n_s = 120 \times 60 / 6 = 1{,}200$ RPM.
Slip at maximum torque (no external resistance): $s_{max} = R_2/X_2 = 0.3/1.5 = 0.2$ (20%).
Speed at max torque: $n = n_s(1 - s_{max}) = 1{,}200 \times 0.8 = 960$ RPM.
For maximum torque at standstill (s = 1): $s_{max} = R_{2total}/X_2 = 1$, so $R_{2total} = X_2 = 1.5\ \Omega$ per phase.
External resistance needed: $R_{ext} = R_{2total} - R_2 = 1.5 - 0.3 = 1.2\ \Omega$ per phase.
Rotor current at starting with external resistance: $I_2 = E_2/(\sqrt{3} \times \sqrt{(R_{2total}^2 + X_2^2)}) = 200/(1.732 \times \sqrt{(1.5^2 + 1.5^2)}) = 200/(1.732 \times 2.121) = 200/3.674 = 54.4$ A per phase.
Without external resistance, starting current would be: $I_2 = 200/(1.732 \times \sqrt{(0.3^2 + 1.5^2)}) = 200/(1.732 \times 1.530) = 75.5$ A — higher current but lower starting torque.
The external resistance reduces starting current by 28% while increasing starting torque to the maximum possible value.
During the lift, the resistors are progressively shorted out in steps as the motor accelerates, until at full speed all external resistance is bypassed and the motor runs as a standard induction motor.

## 12.3 Stepper Motors

Stepper motors are a class of brushless DC motor designed for precise, incremental position control. Unlike continuous-rotation motors that are characterized by speed and torque, stepper motors are defined by their step angle, holding torque, and dynamic torque-speed curve. They operate in open-loop mode without position feedback, relying on the discrete nature of their steps to maintain positional accuracy, which makes them simpler and less expensive than closed-loop servo systems for moderate-precision applications.

### 12.3.1 Types and Operation

Stepper motors are brushless DC motors that divide a full rotation into a large number of discrete, equal steps, enabling precise open-loop position control without feedback sensors. Permanent magnet steppers use a permanent magnet rotor and produce relatively low torque at low cost, suitable for light-duty positioning. Variable reluctance steppers use a soft iron rotor with salient poles that align with the energized stator poles to minimize reluctance, offering fast response but lower torque. Hybrid stepper motors combine both principles — a permanent magnet rotor with fine teeth on its surface — providing the highest step resolution (typically 200 steps/revolution or 1.8° per step) and the best torque characteristics. When a stator phase is energized, the rotor moves to align with the resulting magnetic field; sequentially energizing phases causes the rotor to advance one step at a time.

**Example 12.3.1:** A hybrid stepper motor has 200 steps per revolution and a rated phase current of 2.0 A. The holding torque is specified as 1.8 N·m, and the detent torque is 0.12 N·m. If the motor must rotate exactly 36°, determine the number of full steps required, the step angle, and the maximum load torque that can be applied without losing steps (assume a safety margin of 50%).
**Solution:**
Step angle: $\theta_{step} = 360° / 200 = 1.8°$ per step.
Steps for 36° rotation: $N = 36° / 1.8° = 20$ full steps.
The pull-out torque (dynamic) decreases with speed, but at low speed it approaches the holding torque.
With a 50% safety margin, maximum load torque: $\tau_{load(max)} = 0.50 \times \tau_{holding} = 0.50 \times 1.8 = 0.90$ N·m.
Position accuracy (open-loop): typical non-cumulative error is ±5% of one step = $\pm 0.05 \times 1.8° = \pm 0.09°$.
After 20 steps, the worst-case cumulative position error remains ±0.09° since the error is non-cumulative for a properly loaded stepper motor.

### 12.3.2 Drive Modes

Full-step mode energizes one or two phases at a time, advancing the motor one full step per commutation event (typically 1.8° for a 200-step motor). Half-step mode alternates between one-phase and two-phase energization, doubling the number of steps per revolution to 400 (0.9° per step) with improved smoothness at the cost of slightly unequal torque between half-step positions. Microstepping uses sinusoidal current waveforms to subdivide each full step into smaller increments (typically 4, 8, 16, 32, 64, or 256 microsteps per full step), dramatically improving motion smoothness, reducing vibration and resonance, and enabling step resolutions of 0.007° or finer. Microstepping requires a chopper-type current-regulated driver that precisely controls the current amplitude in each winding according to a sinusoidal profile. Common stepper driver ICs (such as the A4988, DRV8825, and TMC2209) integrate the current regulation, microstepping sequencer, and protection functions into a single package.

**Example 12.3.2:** A stepper motor with 200 full steps/revolution is driven with 16× microstepping. The motor drives a lead screw with a pitch of 2 mm/revolution. Calculate the linear resolution per microstep, the pulse frequency needed for a linear speed of 50 mm/s, and the number of pulses for a 25.4 mm (1 inch) move.
**Solution:**
Microsteps per revolution: $200 \times 16 = 3{,}200$ microsteps/rev.
Linear distance per microstep: d = pitch / microsteps per rev = 2 mm / 3,200 = 0.000625 mm = 0.625 $\mu$m per microstep.
Revolutions per second for 50 mm/s: n = 50 mm/s / 2 mm/rev = 25 rev/s.
Pulse frequency: $f_{pulse} = 25$ rev/s $\times$ 3,200 pulses/rev = 80,000 Hz = 80 kHz.
Pulses for 25.4 mm: N = 25.4 / 0.000625 = 40,640 pulses.
Travel time at 50 mm/s: t = 25.4 / 50 = 0.508 s.
Verification: 40,640 / 80,000 = 0.508 s – confirmed.

### 12.3.3 Applications

Stepper motors are widely used in applications requiring precise, repeatable positioning without the cost and complexity of closed-loop servo systems. CNC machines and 3D printers use stepper motors on each axis to position the tool or print head with sub-millimeter accuracy. Textile machines, pick-and-place equipment, and automated pipetting systems rely on steppers for accurate, programmable motion profiles. Camera focus and zoom mechanisms, robotic joints, and antenna positioners use steppers for their ability to hold position with full torque when stationary (holding torque). The main limitations of stepper motors are reduced torque at high speeds (due to back-EMF and inductance), the potential for missed steps under excessive load (since there is no position feedback in open-loop operation), and audible noise and vibration at certain step rates (resonance), which microstepping and proper driver tuning help mitigate.

**Example 12.3.3:** A 3D printer uses a NEMA 17 stepper motor (1.8°/step, holding torque 0.44 N·m) to drive the X-axis via a GT2 timing belt with a 20-tooth pulley (2 mm belt pitch). The print head mass is 0.3 kg and the desired maximum acceleration is 3,000 mm/s$^2$. Calculate the linear resolution in full-step mode, the force and torque required for acceleration, and whether the motor has adequate torque.
**Solution:**
Pulley circumference: C = 20 teeth $\times$ 2 mm/tooth = 40 mm/rev.
Linear distance per step: d = 40 mm / 200 steps = 0.200 mm per step (200 $\mu$m resolution).
Force for acceleration: $F = m \times a = 0.3 \times 3.0 = 0.9$ N.
Add friction (estimate 0.5 N): $F_{total} \approx 1.4$ N.
Torque at motor: $\tau = F_{total} \times r_{pulley} = 1.4 \times (40 / 2\pi) / 1000 = 1.4 \times 6.366 \times 10^{-3} = 8.9 \times 10^{-3}$ N·m = 8.9 mN·m.
Maximum speed: typical 150 mm/s = 150/40 = 3.75 rev/s = 750 steps/s.
At this speed, available torque is typically 50–70% of holding torque: $\tau_{available} \approx 0.6 \times 0.44 = 0.264$

N·m, far exceeding the required 8.9 mN·m.
The motor is more than adequate.

### 12.3.4 Stepper Motor Resonance and Torque Curves

Stepper motors exhibit speed-dependent torque characteristics and resonance phenomena that are critical to reliable operation. The **pull-in torque curve** defines the maximum load torque at which the motor can start and synchronize without missing steps at a given step rate — it decreases with increasing speed because the windings have limited time to reach full current during each step (due to the L/R electrical time constant). The **pull-out torque curve** defines the maximum load torque the motor can sustain while running at a given speed — it is always higher than the pull-in torque because the rotor has momentum. Between these curves is a "slew range" where the motor can run but cannot start directly; it must be ramped up from a lower speed. **Mid-band resonance** occurs at a characteristic speed (typically 100–400 full steps/s for NEMA 17/23 motors) where the rotor oscillation frequency matches the motor's natural mechanical resonance $f_n = (1/2\pi)\sqrt{(K_h/J)}$, where $K_h$ is the holding torque stiffness and J is the total moment of inertia. At resonance, the rotor oscillates violently, causing missed steps, excessive vibration, and audible noise. Mitigation strategies include: microstepping (distributes the energy impulses more smoothly), adding external damping (viscous dampers), avoiding the resonant speed range in the acceleration profile, increasing the rotor inertia (adding a flywheel, which shifts resonance to a lower frequency), and using advanced driver features such as StealthChop (TMC drivers) that shape the current waveform for quiet operation. The motor's **detent torque** (cogging torque when unpowered) also contributes to vibration in variable-reluctance and hybrid steppers.

**Example 12.3.4:** A NEMA 23 hybrid stepper motor has a rotor inertia of $J_{rotor} = 300$ g·cm² ($3.0 \times 10^{-5}$ kg·m²), a holding torque of 1.9 N·m, and drives a load with a reflected inertia of $J_{load} = 200$ g·cm² ($2.0 \times 10^{-5}$ kg·m²). The step angle is 1.8°. Estimate the natural resonant frequency and the corresponding step rate. If the pull-out torque at 1,000 steps/s is 0.8 N·m, determine the maximum continuous load torque at that speed with a 50% safety margin.
**Solution:**
Total inertia: $J_{total} = J_{rotor} + J_{load} = 3.0 \times 10^{-5} + 2.0 \times 10^{-5} = 5.0 \times 10^{-5}$ kg·m².
Holding torque stiffness: $K_h = T_{hold} / \theta_{step} = 1.9 / (1.8 \times \pi/180) = 1.9 / 0.03142 = 60.5$ N·m/rad.
Natural frequency: $f_n = (1/2\pi)\sqrt{(K_h/J)} = (1/2\pi)\sqrt{(60.5 / 5.0 \times 10^{-5})} = (1/2\pi)\sqrt{(1.21 \times 10^6)} = (1/2\pi) \times 1100 = 175$ Hz.
Resonant step rate: 175 full steps/s (since each step excites the resonance).
At 1.8°/step: resonant speed = 175 × 1.8 / 360 × 60 = 52.5 RPM.
The acceleration profile should ramp quickly through the 150–200 steps/s region.
Maximum continuous load at 1,000 steps/s: $T_{max} = 0.50 \times 0.8 = 0.4$ N·m.
The speed at 1,000 steps/s is 1000 × 1.8 / 360 × 60 = 300 RPM.

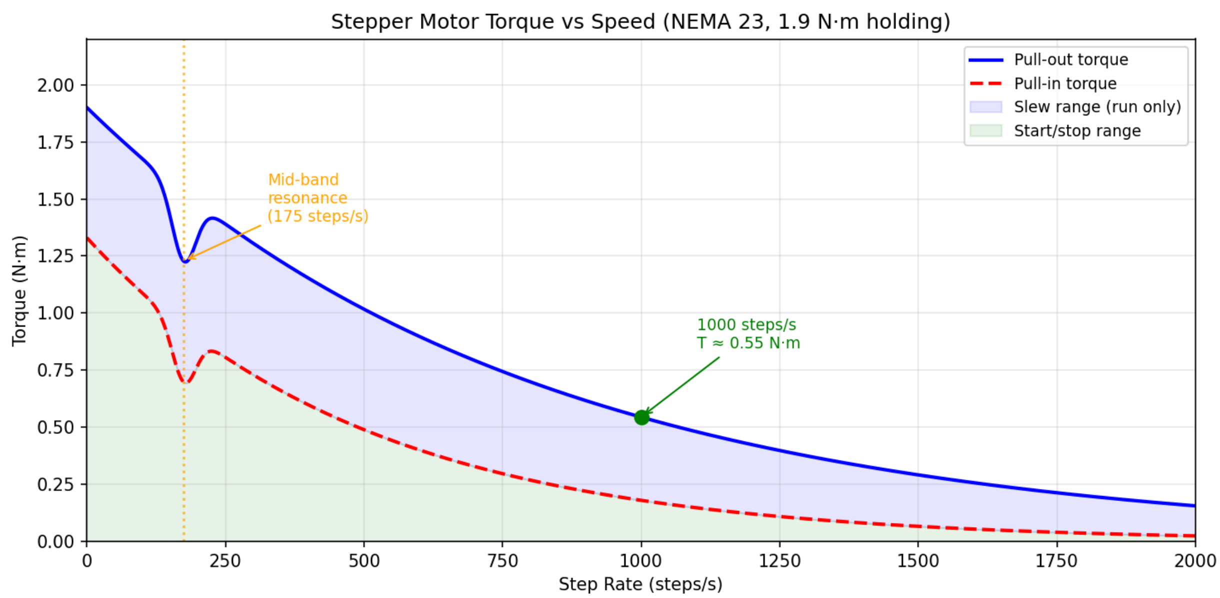



Figure 12.3.4: Stepper Motor Torque vs Speed

## 12.4 Motor Control

Motor control encompasses the power electronic circuits and control algorithms used to start, stop, regulate speed, and control torque of electric motors. The choice of control method depends on the motor type, the application's performance requirements, and cost constraints. Variable frequency drives (VFDs) provide full speed control for AC motors, servo systems deliver precise position and velocity tracking for high-performance motion applications, and soft starters offer a cost-effective solution for reducing inrush current during motor starting.

### 12.4.1 Variable Frequency Drives (VFD)

Variable Frequency Drives control the speed of AC motors by converting fixed-frequency mains power to variable-frequency, variable-voltage output. A typical VFD consists of a rectifier (AC to DC), a DC bus with filter capacitors, and a three-phase PWM inverter that synthesizes the variable-frequency AC output. The motor speed is controlled by varying the output frequency, and the voltage is adjusted proportionally to maintain constant flux (V/f ratio) for consistent torque throughout the speed range. Modern VFDs implement vector control (field-oriented control, or FOC) algorithms that independently control the torque-producing and flux-producing components of the stator current, enabling precise torque and speed control rivaling DC drives. VFDs reduce energy consumption by 20–50% in variable-torque applications (fans, pumps, compressors) compared to throttling or damper control, and they provide soft starting, regenerative braking, and multiple motor protection functions.

**Example 12.4.1:** A 4-pole, 460 V, 60 Hz induction motor rated at 50 HP drives a centrifugal pump. The motor must be operated at 45 Hz to reduce flow. Using constant V/f control, calculate the new operating speed, the voltage applied by the VFD, and the approximate power savings (pump power varies as the cube of speed).

**Solution:**
Base speed at 60 Hz: $n_s = 120 \times 60 / 4 = 1{,}800$ RPM.
New synchronous speed at 45 Hz: $n_{s(new)} = 120 \times 45 / 4 = 1{,}350$ RPM.
Constant V/f ratio: $V/f = 460/60 = 7.667$ V/Hz.
New voltage: $V_{new} = 7.667 \times 45 = 345.0$ V.
Speed ratio: $n_{new}/n_{base} \approx 1350/1800 = 0.75$.
Power ratio for a centrifugal pump (affinity law): $P_{new}/P_{base} = (n_{new}/n_{base})^3 = 0.75^3 = 0.4219$.
New power: $P_{new} = 50 \times 0.4219 = 21.1$ HP.
Power savings: 50 - 21.1 = 28.9 HP (57.8% reduction).
Energy cost savings at \$0.10/kWh running 8,000 hrs/yr: Savings = $28.9 \times 0.746 \times 8000 \times 0.10 =$ \$17,248/yr.

### 12.4.2 Servo Systems

A servo system is a closed-loop motor control system that uses position, velocity, and/or current feedback to precisely track a commanded trajectory. The system typically consists of a servo motor (BLDC or PMSM), a position/velocity feedback device (encoder or resolver), a servo drive (current-regulated inverter with control algorithms), and a motion controller that generates the desired trajectory. The control loop hierarchy includes an inner current loop (fastest, typically 10-20 kHz bandwidth), a velocity loop (intermediate, 100-500 Hz bandwidth), and a position loop (outermost, 10-100 Hz bandwidth), each tuned using PID or advanced control algorithms. Servo systems provide high dynamic response, precise positioning (sub-micrometer in precision applications), and the ability to follow complex multi-axis coordinated motion profiles. Applications include CNC machining, robotics, semiconductor manufacturing, packaging machinery, and any application requiring accurate, responsive motion control.

**Example 12.4.2:** A servo motor with a 2,500-line incremental encoder (quadrature decoded) drives a ball screw with a 5 mm pitch through a 3:1 gear reduction. The required positioning accuracy is $\pm 5$ μm. Determine the encoder resolution in counts per revolution, the linear resolution, and whether the system meets the accuracy requirement.
**Solution:**
Quadrature decoding multiplies the encoder lines by 4: counts/rev = $2{,}500 \times 4 = 10{,}000$ counts/rev at the motor shaft.
With 3:1 gear reduction, counts per revolution at the ball screw: $10{,}000 \times 3 = 30{,}000$ counts/rev (at the output).
Linear resolution per count: d = ball screw pitch / counts per screw rev = 5 mm / 30,000 = 0.000167 mm = 0.167 μm per count.
Since 0.167 μm « $\pm 5$ μm, the encoder resolution is far finer than the required accuracy.
The positioning accuracy will be limited by mechanical factors (ball screw backlash, typically 5-20 μm; thermal expansion; bearing play) rather than encoder resolution.
With a preloaded ball screw (zero backlash) and proper servo tuning, the $\pm 5$ μm target is achievable.

### 12.4.3 Soft Starters

Soft starters limit the inrush current and mechanical stress during the starting of AC induction motors by gradually ramping the applied voltage from a reduced level to full voltage over a programmable time period. They use back-to-back thyristors (SCRs) in each phase to control the voltage applied to the motor by varying the firing angle, similar to a phase-controlled AC voltage controller. During starting, the reduced voltage limits the current to typically 2–4× the full-load current, compared to 6–8× for direct-on-line (DOL) starting. Soft starters also provide controlled deceleration (soft stop) by gradually reducing the voltage, which is beneficial for pump applications where sudden stopping causes water hammer. While less expensive and simpler than VFDs, soft starters do not provide continuous speed control – they are used only during starting and stopping, with the motor running at full speed during normal operation.

**Example 12.4.3:** A 100 HP, 460 V, 3-phase induction motor has a full-load current of $I_{FL} = 124$ A and a DOL starting current of 6× $I_{FL}$. A soft starter is configured to limit starting current to 3× $I_{FL}$ with a ramp time of 10 seconds. Calculate the DOL inrush current, the soft-start limited current, and the initial voltage applied by the soft starter.
**Solution:**
DOL starting current: $I_{start(DOL)} = 6 \times I_{FL} = 6 \times 124 = 744$ A.
Soft-start limited current: $I_{start(soft)} = 3 \times I_{FL} = 3 \times 124 = 372$ A.
Since motor starting current is approximately proportional to applied voltage, the initial voltage ratio: $V_{initial} / V_{rated} = I_{start(soft)} / I_{start(DOL)} = 372 / 744 = 0.50$.
Initial voltage: $V_{initial} = 0.50 \times 460 = 230$ V.
The soft starter ramps the voltage from 230 V to 460 V over 10 seconds.
Starting torque reduction: since torque is proportional to voltage squared, the initial torque is $(0.50)^2 = 0.25$ or 25% of DOL starting torque.
This is adequate for centrifugal pumps and fans (which require low starting torque) but may be insufficient for high-inertia or loaded-start applications.

### 12.4.4 Regenerative Braking

Regenerative braking occurs when a motor operates as a generator, converting the kinetic energy of the load back into electrical energy to decelerate the system. In a VFD-driven application, regeneration happens when the motor speed exceeds the synchronous speed set by the drive (the load is driving the motor), causing power to flow from the motor back to the DC bus. Standard VFDs cannot return energy to the AC mains because their front-end diode rectifiers are unidirectional, so a braking resistor on the DC bus dissipates the regenerated energy as heat. Active front-end (AFE) VFDs replace the diode rectifier with an IGBT-based converter that can return power to the grid, achieving true four-quadrant operation (motoring and braking in both directions). Electric vehicles use regenerative braking extensively, recovering 10–30% of the energy that would otherwise be lost as heat in friction brakes, with the recovered energy stored in the battery. The braking torque available from regeneration depends on the generator back-EMF, the DC bus voltage limits, and the power electronics current rating.

**Example 12.4.4:** An electric vehicle with a mass of 1,800 kg is traveling at 80 km/h (22.2 m/s) and decelerates to 20 km/h (5.56 m/s) using regenerative braking. The motor/generator efficiency during regeneration is 88%, and the battery charging efficiency is 95%. Calculate the kinetic energy recovered, the energy stored in the battery, and the average regenerative braking power if the deceleration takes 8 seconds.

**Solution:**
Initial kinetic energy: $KE_1 = ½mv^2 = 0.5 \times 1800 \times 22.2^2 = 0.5 \times 1800 \times 492.84 = 443{,}556$ J = 443.6 kJ.
Final kinetic energy: $KE_2 = 0.5 \times 1800 \times 5.56^2 = 0.5 \times 1800 \times 30.9 = 27{,}810$ J = 27.8 kJ.
Energy available for recovery: $\Delta KE = 443{,}556 - 27{,}822 = 415{,}734$ J = 415.7 kJ.
Energy stored in battery: $E_{stored} = \Delta KE \times \eta_{motor} \times \eta_{battery} = 415.7 \times 0.88 \times 0.95 = 347.6$ kJ.
Recovery efficiency: $\eta_{total} = 347.6 / 415.7 = 83.6\%$.
Average regenerative braking power: $P_{avg} = \Delta KE / t = 415{,}700 / 8 = 51{,}963$ W ≈ **52.0 kW**.
Average braking force: $F = P_{avg} / v_{avg} = 52{,}000 / ((22.2 + 5.56)/2) = 52{,}000 / 13.89 = 3{,}744$ N.

### 12.4.5 Field-Oriented Control (FOC)

Field-oriented control (FOC), also called vector control, is an advanced motor control technique that decouples the torque-producing and flux-producing components of the stator current, enabling independent control of motor torque and flux similar to a separately excited DC motor. In an AC induction or synchronous motor, the stator current vector is resolved into two orthogonal components in a rotating reference frame aligned with the rotor flux: the d-axis (direct) component $I_d$ controls the rotor flux magnitude, and the q-axis (quadrature) component $I_q$ controls the torque. This decomposition requires knowledge of the rotor flux angle, obtained either from a shaft encoder (direct or sensored FOC) or estimated from the motor voltage and current measurements using an observer algorithm (indirect or sensorless FOC). The Park transform converts three-phase stator quantities (abc) to the two-axis rotating frame (dq) through the intermediate Clarke transform (abc → αβ), and inverse transforms convert the dq-frame voltage commands back to three-phase PWM duty cycles for the inverter. FOC achieves fast dynamic torque response (millisecond-level step changes), smooth low-speed operation down to zero speed, and near-optimal efficiency by maintaining the flux at its rated value during normal operation and reducing it during light loads (flux weakening). It is the standard control method for high-performance drives in electric vehicles, industrial robots, machine tools, and elevator systems.

**Example 12.4.5:** A 4-pole (P = 4), 3-phase PMSM has the following parameters: rated torque $T_{rated}$ = 12 N·m, torque constant $K_t$ = 1.5 N·m/A, rated flux linkage $\lambda_m$ = 0.25 Wb, stator resistance $R_s$ = 0.5 Ω, d-axis inductance $L_d$ = 8 mH, q-axis inductance $L_q$ = 12 mH, and the motor operates at 1,500 RPM. Using FOC with $I_d = 0$ control (simplified — for a true IPM motor, MTPA shifts $I_d$ negative, but $I_d = 0$ is used here for clarity), calculate the required q-axis current, the stator voltage magnitude, and the inverter DC bus voltage needed.
**Solution:**
Required q-axis current for rated torque: $I_q = T_{rated} / K_t = 12 / 1.5 = 8.0$ A.
With $I_d = 0$, the stator current magnitude is: $I_s = \sqrt{(I_d^2 + I_q^2)} = \sqrt{(0 + 64)} = 8.0$ A.
Electrical speed: $\omega_e = (P/2) \times \omega_m$. For a 4-pole motor (P = 4): $\omega_m = 1500 \times 2\pi/60 = 157.1$ rad/s, $\omega_e = 2 \times 157.1 = 314.2$ rad/s.
Voltage equations in the dq frame:
$V_d = R_s \times I_d - \omega_e \times L_q \times I_q = 0.5 \times 0 - 314.2 \times 0.012 \times 8.0 = -30.2$ V.
$V_q = R_s \times I_q + \omega_e \times L_d \times I_d + \omega_e \times \lambda_m = 0.5 \times 8.0 + 0 + 314.2 \times 0.25 = 4.0 + 78.6 = 82.6$ V.
Stator voltage magnitude: $V_s = \sqrt{(V_d^2 + V_q^2)} = \sqrt{(912.0 + 6{,}822.8)} = \sqrt{7{,}734.8} = 87.9$ V (phase peak).

Minimum DC bus voltage (for space vector PWM): $V_{DC} = \sqrt{3} \times V_s = \sqrt{3} \times 87.9 = 152.3$ V. A 160 V or higher DC bus is required.

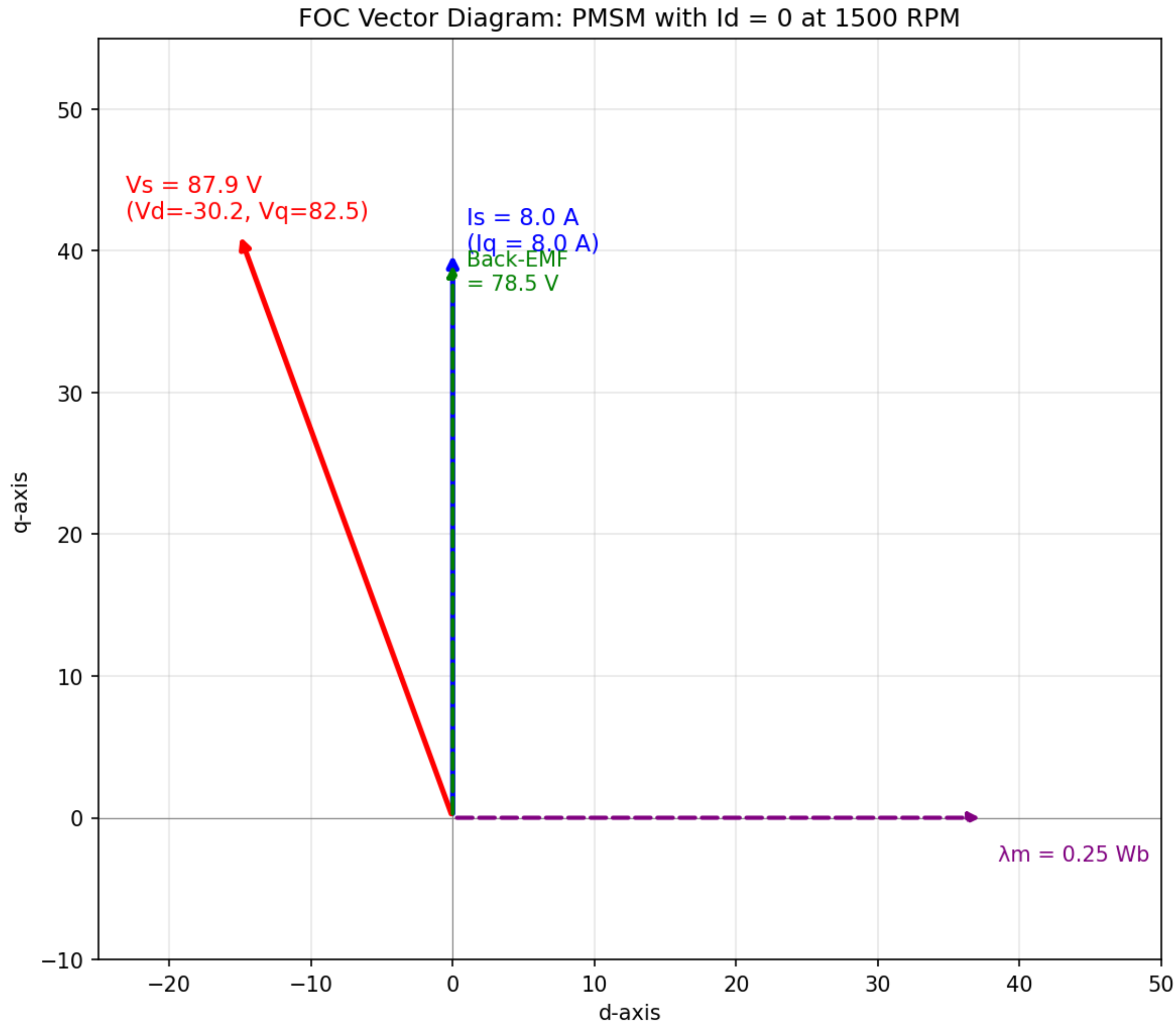



Figure 12.4.5: FOC Vector Diagram

### 12.4.6 Direct Torque Control (DTC)

Direct torque control (DTC) is an alternative to field-oriented control that directly controls the motor's stator flux magnitude and electromagnetic torque without requiring a rotor position sensor or coordinate transformations (Park/Clarke). DTC estimates the stator flux vector and instantaneous torque from measured stator voltages and currents using a voltage model integrator: $\psi_s = \int(V_s - R_s I_s)dt$, and torque is computed as $T_e = (3/2)(P/2)(\psi_{s\alpha} I_{s\beta} - \psi_{s\beta} I_{s\alpha})$. Two hysteresis comparators — one for flux error and one for torque error — compare the estimated values against their references, producing binary outputs that feed an optimal switching table to select one of six active voltage vectors (or two zero vectors) from the inverter. This bang-bang control structure achieves extremely fast torque response (typically 1–2 ms, compared to 5–10 ms for FOC) because the inverter switching state is updated at every control cycle (typically 25–40 kHz sampling) without the delay of PWM modulation. The main drawback of classical DTC is variable switching frequency and higher torque ripple compared to FOC, because the hysteresis bands produce irregular switching patterns. Modern DTC implementations (such as ABB's Direct Torque Control platform) address this with model-predictive extensions that optimize switching sequences over a prediction horizon, reducing torque ripple while maintaining the fast dynamic response. DTC is used in high-performance industrial drives for cranes, winders, rolling mills, and traction applications where rapid torque changes are critical.

**Example 12.4.6:** A DTC-controlled 4-pole induction motor has the following parameters: stator resistance $R_s = 0.4\ \Omega$, rated stator flux $\psi_{s,ref} = 0.85$ Wb, DC bus voltage $V_{DC} = 540$ V, and the flux hysteresis band is ±0.02 Wb and the torque hysteresis band is ±2 N·m. At a given instant, the estimated flux is $\psi_s = 0.83$ Wb and the estimated torque is $T_e = 48$ N·m, while the torque reference is 55 N·m. Determine the hysteresis comparator outputs and the required inverter action.
**Solution:**
Flux error: $\Delta\psi = \psi_{s,ref} - \psi_s = 0.85 - 0.83 = 0.02$ Wb.
Since $\Delta\psi = +0.02$ Wb equals the upper hysteresis limit, the flux comparator output is +1 (increase flux).
Torque error: $\Delta T = T_{ref} - T_e = 55 - 48 = 7$ N·m.
Since $\Delta T = +7$ N·m exceeds the +2 N·m band, the torque comparator output is +1 (increase torque).
With flux = +1, torque = +1, and the current flux sector (determined from the flux angle), the optimal switching table selects a voltage vector that both increases the flux magnitude and advances the flux vector in the direction of rotation — this simultaneously corrects the flux deficit and increases torque.
The magnitude of each active voltage vector is $V_{DC} \times \sqrt{(2/3)} = 540 \times 0.816 = 441$ V.
The torque will increase rapidly (within 1–2 switching cycles at 25 kHz, or ~40–80 μs) until it enters the hysteresis band around the 55 N·m reference, at which point the controller switches to zero vectors or retarding vectors to maintain the torque within ±2 N·m.

### 12.4.7 Sensorless Motor Control

Sensorless motor control eliminates the mechanical position and speed sensors (encoders, resolvers, Hall effect sensors) from the motor drive system by estimating rotor position and velocity from electrical measurements alone. Removing these sensors reduces cost, eliminates a common failure point (encoder cables and connectors are vulnerable to vibration, contamination, and EMI), and enables motor operation in harsh environments such as submersible pumps, sealed compressors, and high-temperature applications where sensors cannot survive. The two fundamental challenges are achieving accurate estimation across the full speed range and maintaining stability during transient load changes.

**Back-EMF sensing** is the simplest sensorless technique, used primarily with BLDC motors under trapezoidal commutation. During six-step commutation, one of the three motor phases is always unenergized (floating), and the back-EMF zero-crossing on that phase indicates the rotor position midpoint between two commutation states. The actual commutation is delayed by 30 electrical degrees from the zero-crossing to align with the optimal switching instant. This method works well above approximately 10–15% of rated speed, where the back-EMF signal is large enough to detect reliably, but fails at standstill and very low speeds because the back-EMF magnitude is proportional to speed:

$E_{back} = K_e \times \omega$.

**Model-based observers** estimate the rotor flux angle and speed from measured stator voltages and currents using mathematical models of the motor. The **Luenberger observer** implements a state-space model of the motor with correction terms proportional to the estimation error: $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$, where L is the observer gain matrix tuned for the desired convergence rate. The **sliding mode observer (SMO)** uses a discontinuous switching function that drives the estimation error to zero along a sliding surface, providing robustness against parameter variations and model inaccuracies — the high-frequency chattering inherent in SMO is filtered to extract the rotor position estimate. The **extended Kalman filter (EKF)** treats the motor model as a nonlinear stochastic system, providing optimal state estimation in the presence of measurement noise, but at higher computational cost. These observers work well at medium to high speeds but degrade below approximately 2–5% of rated speed due to the diminishing back-EMF signal and increased sensitivity to parameter errors (particularly stator resistance $R_s$, which varies with temperature).

**High-frequency injection (HFI)** overcomes the low-speed limitation by exploiting magnetic saliency — the difference between d-axis and q-axis inductances ($L_d \neq L_q$) — to determine rotor position independent of speed. A high-frequency voltage signal (typically 500 Hz to 2 kHz, 5–20 V amplitude) is superimposed on the fundamental excitation, and the resulting high-frequency current response varies with rotor position due to the saliency. Demodulating the high-frequency current component extracts the rotor position angle. HFI provides position estimation down to zero speed and is essential for applications requiring full torque at standstill (e.g., elevator drives, crane hoists). The technique requires measurable saliency ($L_d/L_q \neq 1$), which is inherent in interior permanent magnet (IPM) motors and synchronous reluctance motors, but is minimal in surface-mount PM (SPM) motors where $L_d \approx L_q$. The injected signal produces audible noise and additional losses, which are minimized by keeping the injection amplitude as small as possible while maintaining a reliable signal-to-noise ratio.

**Flux linkage estimation** provides an alternative approach where the stator flux vector is computed by integrating the voltage equation: $\psi_s = \int(V_s - R_s I_s)dt$, and the rotor position is derived from the angle between the estimated stator flux and rotor flux. Pure integration suffers from DC offset drift, so practical implementations use modified integrators (low-pass filters with compensation, or programmable cascaded integrators) that prevent drift while maintaining accuracy at operating frequencies.

Modern sensorless drives typically combine HFI at low speeds (0 to ~10% rated speed) with a model-based observer at medium and high speeds, using a smooth transition algorithm that blends the two estimates across the crossover speed range. This hybrid approach achieves full speed-range sensorless operation with position estimation accuracy of ±5–10 electrical degrees, sufficient for most industrial applications though not matching the sub-degree accuracy of high-resolution encoders.

**Example 12.4.7:** A 6-pole BLDC motor with $K_e = 0.08$ V/(rad/s) is operated with sensorless back-EMF zero-crossing detection. The DC bus voltage is 48 V, the phase resistance is $R_{ph} = 0.6\ \Omega$, and the zero-crossing detection circuit requires a minimum back-EMF of 3.0 V for reliable detection. The motor is driving a fan load at steady state, and the measured back-EMF zero-crossing period (time between consecutive zero-crossings on one phase) is 1.25 ms. Calculate the minimum operating speed for sensorless control and the rotor speed from the zero-crossing measurement.
**Solution:**

Minimum back-EMF for detection: $E_{min} = 3.0$ V.
Minimum speed: $\omega_{min} = E_{min} / K_e = 3.0 / 0.08 = 37.5$ rad/s.
Minimum speed in RPM: $n_{min} = 37.5 \times 60 / (2\pi) = 358$ RPM.
For the zero-crossing measurement: in a 6-pole (3 pole-pair) motor, there are 3 electrical cycles per mechanical revolution.
Each electrical cycle has 6 zero-crossings (one per 60 electrical degrees).
Zero-crossing period: $T_{ZC} = 1.25$ ms = time for 60 electrical degrees.
Electrical period: $T_e = 6 \times 1.25 = 7.50$ ms.
Electrical frequency: $f_e = 1 / 0.00750 = 133.3$ Hz.
Mechanical speed: $n = f_e / (P/2) = 133.3 / 3 = 44.4$ rev/s = 2,667 RPM.
Mechanical speed in rad/s: $\omega = 2{,}667 \times 2\pi / 60 = 279.3$ rad/s.
Verify with back-EMF: $E_{back} = K_e \times \omega = 0.08 \times 279.3 = 22.3$ V — well above the 3.0 V minimum, confirming reliable sensorless operation at this speed.

### 12.4.8 Flux Weakening and Constant-Power Operation

Flux weakening extends the operating speed range of a permanent magnet motor beyond its base speed by injecting negative d-axis current to oppose the rotor magnet flux, effectively reducing the net air-gap flux and the back-EMF so the motor can spin faster within the inverter's voltage limit. Below base speed, the motor operates in the **constant-torque region** where full flux is maintained ($I_d = 0$ for SPM, or MTPA trajectory for IPM) and the available torque is limited by the current rating of the inverter. At base speed, the back-EMF equals the maximum voltage the inverter can produce, and any further speed increase requires reducing the flux. Above base speed, the motor enters the **constant-power region** where torque decreases inversely with speed ($P = \tau \times \omega =$ constant) as the flux is progressively weakened.

The motor's operating limits are described by two constraints in the $I_d$-$I_q$ plane. The **current limit circle** has radius $I_{max}$: $I_d^2 + I_q^2 \le I_{max}^2$, set by the inverter's thermal rating. The **voltage limit ellipse** shrinks with increasing speed: $(L_d I_d + \lambda_m)^2 + (L_q I_q)^2 \le (V_{max}/\omega_e)^2$, where $V_{max} = V_{DC}/\sqrt{3}$ for space vector PWM. As speed increases, the voltage ellipse contracts toward its center at $(-\lambda_m/L_d, 0)$, and the operating point must lie within both the current circle and the voltage ellipse. The **maximum torque per ampere (MTPA)** trajectory optimizes the current angle for maximum torque at each current magnitude — for SPM motors this is simply $I_d = 0$, but for IPM motors (where $L_d \ne L_q$) the MTPA angle shifts into the negative $I_d$ region to exploit reluctance torque: $T = (3/2)(P/2)[\lambda_m I_q + (L_d - L_q) I_d I_q]$.

The **speed range extension ratio** (maximum speed divided by base speed) depends on the motor's characteristic current $I_{ch} = \lambda_m/L_d$. When $I_{ch} = I_{max}$, the voltage ellipse center lies exactly on the current limit circle, enabling theoretically infinite constant-power range — this is the "ideal" flux weakening design point. Practical IPM motors achieve speed range ratios of 2:1 to 4:1, with some traction motors reaching 5:1 or higher through optimized rotor geometry (embedded V-shaped magnets, flux barriers). SPM motors have limited flux weakening capability (typically 1.2:1 to 1.5:1) because their low saliency provides minimal reluctance torque contribution and their characteristic current often exceeds $I_{max}$. In electric vehicle applications, flux weakening enables the motor to deliver high torque for acceleration at low speeds while achieving high highway speeds within the battery voltage constraint, replacing the function of a multi-speed transmission.

**Example 12.4.8:** A 4-pole IPM motor for an electric vehicle has the following parameters: $\lambda_m = 0.18$ Wb, $L_d = 4.0$ mH, $L_q = 10.0$ mH, $I_{max} = 200$ A, $V_{DC} = 350$ V. Calculate the base speed, the characteristic current, and the maximum speed achievable with flux weakening.
**Solution:**
Maximum voltage (phase, for SVPWM): $V_{max} = V_{DC} / \sqrt{3} = 350 / 1.732 = 202.1$ V.
At base speed with $I_d = 0$ (MTPA for simplicity at rated current): the back-EMF determines the voltage limit.
The voltage equation magnitude at $I_d = 0$: $V_s = \omega_e \times \sqrt{(\lambda_m{}^2 + (L_q I_q)^2)}$.
With $I_q = I_{max} = 200$ A: $V_s = \omega_e \times \sqrt{(0.18^2 + (0.010 \times 200)^2)} = \omega_e \times \sqrt{(0.0324 + 4.0)} = \omega_e \times \sqrt{4.0324} = \omega_e \times 2.008$.
At base speed, $V_s = V_{max}$: $\omega_{e,base} = 202.1 / 2.008 = 100.6$ rad/s (electrical).
Mechanical base speed: $\omega_{m,base} = \omega_{e,base} / (P/2) = 100.6 / 2 = 50.3$ rad/s = 480 RPM.
Characteristic current: $I_{ch} = \lambda_m / L_d = 0.18 / 0.004 = 45.0$ A.
Since $I_{ch} = 45\text{ A} < I_{max} = 200$ A, the voltage ellipse center is well inside the current circle, indicating a wide constant-power speed range is achievable.
At maximum speed, the operating point approaches $I_d = -I_{ch} = -45$ A, $I_q \approx 0$ (minimum torque, maximum speed).
Maximum electrical speed: $\omega_{e,max} = V_{max} / (L_d \times |I_d + I_{ch}|)$ — but at $I_d = -I_{max} = -200$ A:
$V_s = \omega_e \times \sqrt{((L_d I_d + \lambda_m)^2 + (L_q I_q)^2)}$ with $I_q = 0$:
$\omega_{e,max} = V_{max} / |L_d \times (-200) + \lambda_m| = 202.1 / |0.004 \times (-200) + 0.18| = 202.1 / |-0.80 + 0.18| = 202.1 / 0.62 = 326.0$ rad/s (electrical).
Mechanical max speed: $\omega_{m,max} = 326.0 / 2 = 163.0$ rad/s = 1,557 RPM.
Speed range ratio: 1,557 / 480 = 3.24:1.
This 3.2:1 speed extension is typical for IPM traction motors, allowing the vehicle to reach highway speeds without a multi-speed gearbox.

## 12.5 Motor Specifications

Motor specifications define the electrical, mechanical, and thermal ratings that determine proper application, installation, and protection of electric motors. Understanding nameplate data is essential for selecting the correct conductor sizing, overcurrent protection, and control equipment. Standardized efficiency classifications and motor protection schemes ensure safe, reliable, and energy-efficient operation throughout the motor's service life.

### 12.5.1 Nameplate Data

The motor nameplate provides the essential specifications needed for proper installation, protection, and application. Key nameplate parameters include rated power (HP or kW), rated voltage and frequency, rated full-load current (FLA), rated speed (RPM), service factor (SF, the permissible continuous overload factor, typically 1.0-1.15), insulation class (B: 130°C, F: 155°C, H: 180°C maximum winding temperature), enclosure type (ODP: Open Drip Proof, TEFC: Totally Enclosed Fan Cooled, TENV: Totally Enclosed Non-Ventilated), and frame size (NEMA or IEC standard dimensions). The NEMA design letter (A, B, C, D) specifies the torque-speed characteristic, with Design B being the most common general-purpose type. The efficiency rating (NEMA Premium, IE3, IE4) indicates the motor's energy conversion efficiency at rated load, with minimum efficiency standards mandated by regulations in most countries.

**Example 12.5.1:** A motor nameplate reads: 25 HP, 460 V, 60 Hz, 3-phase, 1,770 RPM, FLA = 30.8 A, SF = 1.15, Insulation Class F, TEFC, NEMA Design B, Efficiency = 93.0%. Calculate the number of poles, the full-load slip, the input power, the maximum continuous power with service factor, and the maximum winding temperature.

**Solution:**

Number of poles: $n_s$ must be the nearest synchronous speed above 1,770 RPM. $n_s$ = 1,800 RPM, so $P = 120f/n_s = 120 \times 60 / 1800 = 4$ poles.

Full-load slip: s = (1800 - 1770) / 1800 = 30/1800 = 1.67%.

Input power: $P_{in} = P_{out} / \eta = (25 \times 746) / 0.930 = 18{,}650 / 0.930 = 20{,}054$ W = 20.1 kW.

Verify with current: $P_{in} = \sqrt{3} \times V \times I \times PF$. $PF = P_{in} / (\sqrt{3} \times V \times I) = 20{,}054 / (1.732 \times 460 \times 30.8) = 20{,}054 / 24{,}539 = 0.817$ (typical for this motor size).

Maximum continuous power with SF: $P_{max} = 25 \times 1.15 = 28.75$ HP.

Maximum winding temperature (Class F): 155°C (with a 40°C ambient, this allows a 115°C total hot-spot temperature rise, or 105°C average winding rise by resistance measurement plus a 10°C hot-spot allowance).

Power losses: $P_{loss} = P_{in} - P_{out} = 20{,}054 - 18{,}650 = 1{,}404$ W.

### 12.5.2 Motor Protection

Motor protection devices and schemes prevent damage from electrical and thermal faults that can destroy windings, bearings, and insulation. Overcurrent protection (circuit breakers or fuses) interrupts the supply during short circuits and severe overloads that exceed the motor's rated current. Thermal overload relays (bimetallic or electronic) monitor the motor current and trip after a time delay that approximates the motor's thermal capacity, protecting against sustained moderate overloads that cause gradual overheating. Phase loss (single-phasing) protection detects the loss of one supply phase, which causes the motor to draw excessive current on the remaining phases and overheat rapidly. Ground fault protection detects current leaking to ground through damaged insulation, preventing electrical shock and fire hazards. Additional protection functions in electronic motor protection relays include phase reversal, voltage unbalance, undercurrent (indicating loss of load such as a broken belt or pump running dry), stall detection, and winding temperature monitoring via embedded thermistors or RTDs (PTC or Pt100).

**Example 12.5.2:** A 50 HP, 460 V, 3-phase motor has a full-load current of 62 A, a service factor of 1.15, and a locked-rotor current of 6× FLA. Select the thermal overload relay trip class and setting, and determine the minimum circuit breaker size per NEC guidelines.

**Solution:**

Overload relay setting: the trip current is set at 115% of FLA (to allow operation at service factor): $I_{trip} = 1.15 \times 62 = 71.3$ A.

Select a Class 20 thermal overload relay (trips within 20 seconds at 6× setting – standard for most industrial motors).

Verify: at locked-rotor current $I_{LR} = 6 \times 62 = 372$ A, the relay must trip before winding damage occurs.

Circuit breaker sizing (NEC 430.52 for inverse-time breaker, Design B motor): maximum = 250% of FLA = $2.50 \times 62 = 155$ A.

Since 155 A is not a standard ampere rating, NEC 430.52 Exception No. 1 permits the next higher standard size: select 175 A breaker (alternatively, the next lower standard size of 150 A is also

code-compliant but may nuisance-trip during starting).
Short-circuit interrupting capacity must exceed the available fault current at the motor terminals.
The branch circuit conductors must be rated at minimum 125% of FLA = 1.25 × 62 = 77.5 A, requiring 4 AWG copper (rated 85 A at 75°C).

### 12.5.3 Motor Efficiency and Standards

Motor efficiency standards define minimum performance levels to reduce energy waste, since electric motors consume roughly 45% of global electricity. The International Electrotechnical Commission (IEC) standard IEC 60034-30-1 defines four efficiency classes for three-phase induction motors: IE1 (standard), IE2 (high), IE3 (premium), and IE4 (super-premium), with IE5 (ultra-premium) under development. In North America, NEMA MG 1 defines "NEMA Premium" efficiency levels that align closely with IE3. Motor losses consist of stator copper losses ($I^2R$ in the stator windings), rotor copper losses ($I^2R$ in the rotor conductors), core losses (hysteresis and eddy currents in the laminations), friction and windage losses, and stray load losses. Higher-efficiency motors use more copper (larger conductors), thinner and higher-grade electrical steel laminations, tighter air gaps, and optimized rotor slot geometry to reduce these losses. The lifecycle electricity cost of a motor typically exceeds its purchase price within the first few months of operation, making efficiency a critical economic factor.

**Example 12.5.3:** A plant operates a 75 kW, 4-pole, 460 V induction motor continuously (8,760 hours/year) at 75% load. The current IE1 motor has an efficiency of 91.0% at 75% load. A replacement IE4 motor has an efficiency of 95.5% at the same load point. The electricity cost is $0.09/kWh, and the IE4 motor costs $2,800 more than the IE1 replacement. Calculate the annual energy consumption of each motor, the annual savings, and the simple payback period.
**Solution:**
Mechanical output at 75% load: $P_{mech} = 0.75 \times 75 = 56.25$ kW.
IE1 motor input power: $P_{in(IE1)} = P_{mech} / \eta = 56.25 / 0.910 = 61.81$ kW.
IE4 motor input power: $P_{in(IE4)} = 56.25 / 0.955 = 58.90$ kW.
Power savings: $\Delta P = 61.81 - 58.90 = 2.91$ kW.
Annual energy savings: $\Delta E = 2.91 \times 8{,}760 = 25{,}492$ kWh/year.
Annual cost savings: $\Delta C = 25{,}492 \times \$0.09 = \$2{,}294$/year.
Simple payback: $2,800 / $2,294 = 1.22 years.
Over a typical 20-year motor life, the total savings would be approximately 20 × $2,294 = $45,880, a return of over 16× the premium cost.

### 12.5.4 Motor Selection and Sizing

Proper motor selection requires matching the motor's torque-speed characteristics, power rating, duty cycle, and environmental ratings to the driven load. The load torque profile determines the motor type: constant-torque loads (conveyors, hoists, positive-displacement pumps) require motors that deliver rated torque across the full speed range, while variable-torque loads (centrifugal pumps, fans, blowers) follow a torque $\propto$ speed$^2$ relationship and benefit from VFD-driven motors sized at the maximum operating point. The motor must be sized for the worst-case continuous load plus any required overload margin, accounting for the duty cycle — continuous duty (S1), short-time duty (S2), or intermittent duty (S3–S10 per IEC 60034-1) — since intermittent operation allows a smaller motor due to cooling periods between load cycles. The motor's thermal class and enclosure must be compatible with the ambient temperature and environment (dust, moisture, corrosive atmosphere, hazardous locations classified per NEC Article 500). The moment of inertia of the load referred to the motor shaft ($J_{load} = J_{actual} \times (N_{motor}/N_{load})^2$ for geared systems) determines the acceleration torque required: $T_{accel} = (J_{motor} + J_{load}) \times \alpha$, which must not exceed the motor's breakdown torque or the drive's current limit. Oversizing a motor wastes energy (motors are least efficient at light loads) and reduces power factor, while undersizing leads to overheating, shortened insulation life, and nuisance tripping.

**Example 12.5.4:** A conveyor system requires 8 kW of continuous mechanical power at 1,200 RPM. The conveyor has a total reflected inertia of $J_{load} = 0.8$ kg·m² (referred to motor shaft), and the system must accelerate from standstill to full speed in 3 seconds. The motor inertia is $J_{motor} = 0.05$ kg·m². Determine the required motor rating, the acceleration torque, the total torque during acceleration, and verify the motor can deliver it.
**Solution:**
Continuous load torque: $\tau_{load} = P / \omega = 8{,}000 / (1200 \times 2\pi/60) = 8{,}000 / 125.7 = 63.7$ N·m.
Angular acceleration: $\alpha = \omega / t = 125.7 / 3 = 41.9$ rad/s².
Total inertia: $J_{total} = J_{motor} + J_{load} = 0.05 + 0.80 = 0.85$ kg·m².
Acceleration torque: $\tau_{accel} = J_{total} \times \alpha = 0.85 \times 41.9 = 35.6$ N·m.
Total torque during acceleration: $\tau_{total} = \tau_{load} + \tau_{accel} = 63.7 + 35.6 = 99.3$ N·m.
Select a motor rated at least 8 kW continuous with adequate breakdown torque.
A standard 11 kW (15 HP), 4-pole induction motor has a rated torque of approximately 11,000 / 125.7 = 87.5 N·m and a typical breakdown torque of 2.5 × 87.5 = 218.8 N·m.
The acceleration torque of 99.3 N·m is 113% of rated torque, well within the motor's capability.
The 11 kW motor provides a 37.5% power margin for the continuous 8 kW load.

### 12.5.5 Motor Bearings and Vibration Analysis

Bearings support the motor shaft, maintain the air gap between rotor and stator, and are the most common point of failure in electric motors — bearing faults account for approximately 40–50% of all motor failures. **Ball bearings** (deep groove, angular contact) are the standard for small to medium motors, using hardened steel balls rolling between inner and outer races with grease or oil lubrication. **Roller bearings** (cylindrical, spherical, tapered) carry higher radial and axial loads in large motors. **Sleeve (journal) bearings** use an oil film between the shaft and a bronze or babbitt-metal bushing, providing quiet operation and very long life in large, high-speed machines. Bearing life follows the $L_{10}$ rating: the number of revolutions (or hours at a given speed) at which 10% of a population of identical bearings will have failed, calculated as $L_{10} = (C/P)^3 \times 10^6$ revolutions for ball bearings, where C is the dynamic load rating and P is the equivalent bearing load. **Vibration analysis** is the primary predictive maintenance tool for motor bearings: an accelerometer mounted on the bearing housing measures vibration spectra that reveal specific fault signatures. A defect on the outer race produces vibration at the ball pass frequency outer (BPFO), an inner race defect at BPFI, a ball defect at BSF (ball spin frequency), and cage defects at FTF (fundamental train frequency) — each calculated from the bearing geometry, number of balls, and shaft speed. Overall vibration severity is classified by ISO 10816 (velocity in mm/s RMS): Zone A (< 1.8 mm/s, new condition), Zone B (1.8–4.5 mm/s, acceptable), Zone C (4.5–11.2 mm/s, marginal), Zone D (> 11.2 mm/s, danger) for Group 2 machines (15–75 kW). Trending vibration levels over time detects degradation months before catastrophic failure, enabling planned maintenance rather than unplanned downtime.

**Example 12.5.5:** A 30 kW, 4-pole induction motor operates at 1,770 RPM with 6205-2RS deep groove ball bearings (9 balls, ball diameter $d_b = 7.94$ mm, pitch diameter $d_p = 38.5$ mm, contact angle $\alpha = 0°$). The bearing dynamic load rating is C = 14.0 kN, and the radial load on the drive-end bearing is P = 1.2 kN. Calculate the $L_{10}$ bearing life in hours and the ball pass frequency outer race (BPFO).
**Solution:**
$L_{10}$ life in revolutions: $L_{10} = (C/P)^3 \times 10^6 = (14.0/1.2)^3 \times 10^6 = (11.67)^3 \times 10^6 = 1{,}589 \times 10^6$ revolutions.
Convert to hours: $L_{10}(\text{hours}) = L_{10} / (60 \times n) = 1{,}589 \times 10^6 / (60 \times 1{,}770) = 1{,}589 \times 10^6 / 106{,}200 = 14{,}962$ hours $\approx 15{,}000$ hours (about 1.7 years of continuous operation).
$\text{BPFO} = (N_b/2) \times (n/60) \times (1 - d_b \cos \alpha / d_p) = (9/2) \times (1770/60) \times (1 - 7.94/38.5) = 4.5 \times 29.5 \times (1 - 0.2062) = 4.5 \times 29.5 \times 0.7938 = 105.4$ Hz.
A vibration peak at 105.4 Hz (or its harmonics at 210.8, 316.2 Hz) in the bearing housing spectrum indicates an outer race defect developing.
At the current load, the bearing should last ~15,000 hours, but vibration monitoring can detect degradation well before the $L_{10}$ life to prevent unexpected failures.

### 12.5.6 Motor Thermal Modeling and Insulation Life

Motor winding insulation degrades progressively with temperature, and thermal management is the primary factor determining motor service life. The **Arrhenius model** (also known as Montsinger's rule or the "10-degree rule") states that insulation life approximately halves for every 10°C increase above the rated hot-spot temperature: $L = L_{rated} \times 2^{(T_{rated} - T_{actual})/10}$, where $L_{rated}$ is the design life at rated temperature (typically 20,000 hours for Class F insulation at 155°C). Insulation classes define maximum allowable winding temperatures: Class B (130°C), Class F (155°C), and Class H (180°C), each including a 40°C ambient, a temperature rise allowance, and a 10°C hot-spot margin. **Lumped-parameter thermal models** represent the motor as a network of thermal resistances ($R_{th}$ in °C/W) and thermal capacitances ($C_{th}$ in J/°C) analogous to an electrical RC circuit: the winding-to-stator-core resistance, stator-core-to-frame resistance, and frame-to-ambient resistance (convection and radiation) form a series thermal path, with each node having a thermal mass that determines the transient heating and cooling time constants. The dominant winding time constant is typically $\tau_w = R_{th} \times C_{th} =$ 5–30 minutes for small to medium motors, meaning the winding temperature follows $T(t) = T_{ambient} + \Delta T_{ss}(1 - e^{-t/\tau})$ during constant-load operation. For intermittent duty (S3–S10 per IEC 60034-1), the motor can be loaded above its continuous rating during the ON period provided the RMS equivalent heating over the complete cycle does not exceed the continuous thermal limit: the equivalent continuous current is $I_{eq} = \sqrt{(I_1^2 t_1 + I_2^2 t_2 + \dots) / (t_1 + t_2 + \dots)}$. Motor protection relays implement thermal models ($I^2t$ algorithms or first-order thermal replicas) that estimate winding temperature from measured current and time, tripping before the insulation temperature limit is exceeded.

**Example 12.5.6:** A Class F insulation motor (rated hot-spot temperature 155°C) has a design life of 20,000 hours at rated load. The motor operates in a 45°C ambient (5°C above the standard 40°C rating) and carries 110% of rated current continuously. The temperature rise is proportional to $I^2$, so the actual temperature rise is $1.10^2 = 1.21$ times the rated rise. The rated temperature rise is 155 − 10 (hot-spot margin) − 40 = 105°C. Calculate the actual hot-spot temperature and the expected insulation life.

**Solution:**
Actual temperature rise: $\Delta T = 1.21 \times 105 = 127.1$°C.
Actual hot-spot temperature: $T_{actual} = T_{ambient} + \Delta T = 45 + 127.1 = 172.1$°C.
Temperature above rated hot-spot: $\Delta T_{excess} = 172.1 - 155 = 17.1$°C.
Expected insulation life: $L = L_{rated} \times 2^{(Trated - Tactual)/10} = 20{,}000 \times 2^{(155 - 172.1)/10} = 20{,}000 \times 2^{-1.71} = 20{,}000 \times 0.3057 = 6{,}114$ hours.
The combination of elevated ambient (+5°C) and overload (110%) reduces insulation life from 20,000 hours to approximately 6,100 hours — a 69% reduction.
This demonstrates why motors must be derated for high ambient temperatures and why sustained overloads, even modest ones, dramatically shorten motor life.
Operating at the rated conditions would yield: $L = 20{,}000 \times 2^{(155 - (40 + 105 + 10))/10} = 20{,}000 \times 2^0 =$ 20,000 hours as expected.

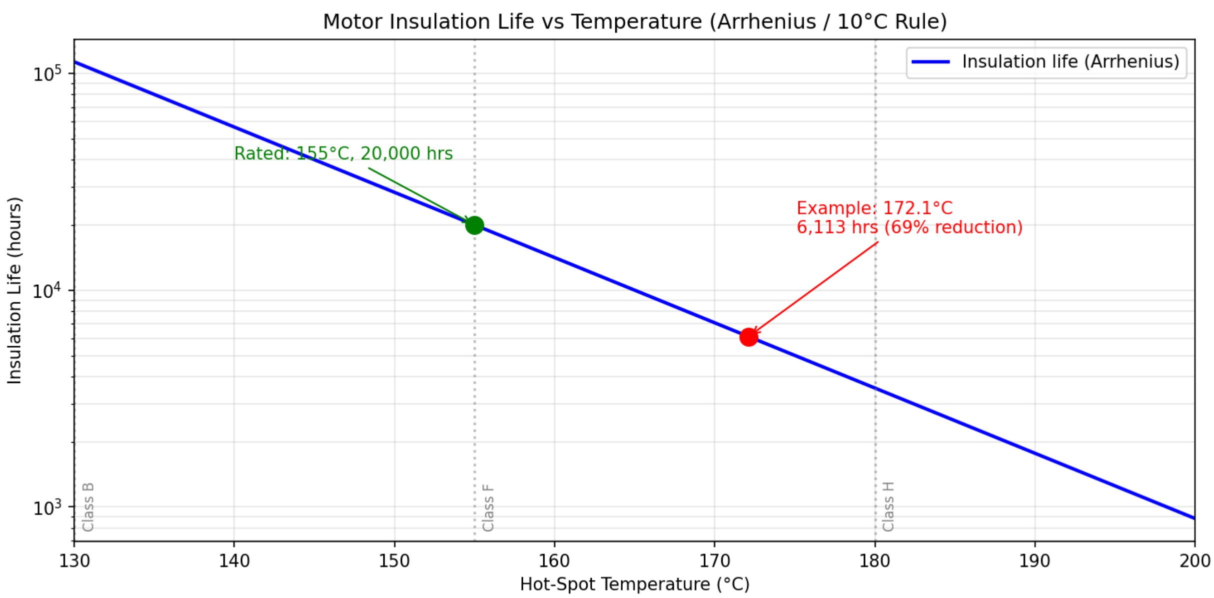



Figure 12.5.6: Motor Insulation Life vs Temperature

### 12.5.7 VFD Effects on Motor Windings

Variable frequency drives produce fast-switching PWM output waveforms that subject motor windings, insulation, and bearings to stresses not present with sinusoidal mains power. Understanding these effects is essential for reliable VFD-motor systems, particularly with long cable runs, high switching frequencies, and motors not rated for inverter duty.

**Voltage reflection and doubling** occurs because the VFD output cable acts as a transmission line at the steep-edged PWM switching frequencies. When a voltage pulse travels down the cable and encounters the motor terminals (a high-impedance mismatch), the pulse reflects and adds to the incoming wave, producing a voltage overshoot at the motor terminals that can reach nearly 2× the DC bus voltage. The reflection magnitude depends on the cable's characteristic impedance $Z_0 = \sqrt{(L_{cable}/C_{cable})}$ (typically 50–100 Ω) relative to the motor's surge impedance (typically 1,000–5,000 Ω). Full voltage doubling occurs when the cable propagation time $t_{prop} = l/v_p$ (where $l$ is the cable length and $v_p$ is the propagation velocity, typically 150–200 m/μs) exceeds half the inverter rise time: $t_{prop} > t_{rise}/2$. For a modern IGBT inverter with a rise time of 100 ns, the critical cable length is $l_{crit} = v_p \times t_{rise}/2 = 170$ m/μs × 50 ns = 8.5 m. Beyond this length, the reflected voltage at the motor terminals approaches 2 × $V_{DC}$, stressing the turn-to-turn insulation in the first few coils of the motor winding.

**dV/dt stress on insulation** is caused by the steep voltage edges of PWM switching. The voltage across the first few turns of a motor winding can be disproportionately large because the high-frequency components of the voltage pulse distribute non-uniformly — up to 80% of the pulse voltage can appear across the first 10–20% of the winding turns due to the inter-turn capacitance forming a capacitive voltage divider at high frequencies. Standard motors with random-wound windings (NEMA MG1 Part 30) are rated for a peak voltage of 1,000 V and a maximum dV/dt of 1,760 V/μs for 460 V class motors. Exceeding these limits causes partial discharge (corona) in the insulation that progressively erodes the enamel coating, leading to turn-to-turn short circuits and eventual winding failure — sometimes within months of installation.

**Bearing currents** are induced by the common-mode voltage generated by the PWM inverter. The three-phase PWM output produces a non-zero common-mode voltage $V_{CM} = (V_a + V_b + V_c)/3$ that oscillates at the switching frequency (typically 4–16 kHz) with amplitude up to $V_{DC}/2$. This common-mode voltage couples through the parasitic capacitance between the stator winding and the rotor ($C_{wr}$, typically 10–100 pF) and between the rotor and the grounded frame through the bearing ($C_{rb}$). The resulting shaft voltage builds up until it exceeds the dielectric breakdown voltage of the bearing lubricant film (typically 5–30 V), causing an electric discharge machining (EDM) event — a tiny spark that pits the bearing race surface. Repeated EDM events produce characteristic "fluting" patterns (washboard-like grooves) on the bearing races, leading to increased vibration, noise, and premature bearing failure within 6–24 months instead of the normal 5–10 year life. Shaft grounding rings (conductive microfiber rings that maintain a low-impedance path from shaft to frame) and insulated bearings (ceramic-coated outer race or hybrid ceramic balls) are the primary mitigation measures.

**Output reactors and dV/dt filters** are installed between the VFD and the motor to protect against voltage reflection and dV/dt stress. A **3% output reactor** (impedance = 3% of motor base impedance) reduces the dV/dt by slowing the voltage rise time and limits the peak voltage overshoot, but does not eliminate reflections on long cables. A **dV/dt filter** (LC low-pass circuit with a cutoff frequency between the PWM switching frequency and the fundamental output frequency) limits the dV/dt to 200–500 V/μs and clamps the peak voltage, providing better protection for cable lengths up to 300 m. A **sinusoidal filter** (also called a sine wave filter) uses a higher-order LC network with a cutoff below the switching frequency to reconstruct a near-sinusoidal waveform, eliminating virtually all high-frequency stress but at higher cost, size, and power loss (1–3% of rated power).

**Inverter-duty motor ratings** per NEMA MG1 Part 31 specify insulation systems designed for VFD operation. Part 31 motors withstand peak voltages of 1,600 V (for 460 V class) with rise times down to 100 ns, compared to 1,000 V for standard Part 30 motors. They use reinforced insulation systems including phase-separation insulation (between coils of different phases), corona-resistant magnet wire with enhanced enamel coatings, and vacuum pressure impregnation (VPI) with high-quality resins that eliminate voids where partial discharge initiates. For motors above 600 V, IEC 60034-18-42 defines Type I insulation (no partial discharge expected during operation) and Type II insulation (designed to withstand partial discharge throughout the motor's life). **Cable length limits** depend on the VFD voltage class, switching frequency, and output filtering: without filters, maximum recommended cable lengths are typically 15–30 m for 460 V drives with fast IGBTs; with a dV/dt filter, this extends to 150–300 m; and with a sinusoidal filter, cable length is essentially unlimited.

**Example 12.5.7:** A 460 V VFD with a DC bus voltage of 650 V and an IGBT rise time of 100 ns drives a standard (Part 30) motor through a 75 m cable. The cable propagation velocity is 170 m/μs. Calculate the cable propagation time, determine whether voltage doubling occurs, estimate the peak voltage at the motor terminals, and specify the required output filter.

**Solution:**

Cable propagation time: $t_{prop} = l / v_p = 75 / (170 \times 10^6) = 441$ ns $= 0.441$ μs.

Critical cable length for voltage doubling: $l_{crit} = v_p \times t_{rise} / 2 = 170 \times 10^6 \times 100 \times 10^{-9} / 2 = 8.5$ m.

Since $l = 75$ m » $l_{crit} = 8.5$ m, full voltage reflection occurs.

Peak voltage at motor terminals: $V_{peak} \approx 2 \times V_{DC} \times (1 - e^{-2tprop/trise})$ for the reflected wave. With $t_{prop}$ » $t_{rise}/2$, the reflection coefficient approaches 1.0.

$V_{peak} \approx 2 \times 650 = 1{,}300$ V.

The standard Part 30 motor is rated for a maximum peak of 1,000 V — the 1,300 V reflected voltage exceeds this by 30%, creating a high risk of insulation failure.

dV/dt at motor terminals (without filter): $dV/dt \approx V_{DC} / t_{rise} = 650 / (100 \times 10^{-9}) = 6{,}500$ V/μs — far exceeding the 1,760 V/μs Part 30 rating.

A **dV/dt filter** must be installed at the VFD output to limit the peak voltage below 1,000 V and reduce the dV/dt below 1,760 V/μs, or the motor must be replaced with an inverter-duty (Part 31) motor rated for 1,600 V peak. For a 75 m cable run, a dV/dt filter is the standard solution and would typically reduce the peak terminal voltage to approximately 750–850 V and the dV/dt to 200–400 V/μs.

Additionally, shaft grounding rings should be installed on the motor to prevent bearing damage from EDM currents.

# Chapter 13

# Operational Amplifiers

The operational amplifier (op-amp) is a high-gain, differential-input, single-ended-output voltage amplifier that serves as the most versatile building block in analog circuit design. Ideally, an op-amp has infinite open-loop gain, infinite input impedance, zero output impedance, and infinite bandwidth, and while real devices deviate from these ideals, modern op-amps come close enough that the ideal model is sufficient for most circuit analysis. By connecting external resistors and capacitors in feedback networks, a single op-amp can be configured to perform amplification, buffering, summation, subtraction, integration, differentiation, filtering, and comparison, making it indispensable in signal conditioning, instrumentation, control systems, and audio electronics.

## 13.1 Ideal Op-Amp Model

### 13.1.1 Characteristics

The ideal op-amp model is based on five assumptions that simplify circuit analysis: infinite open-loop voltage gain ($A \to \infty$), infinite input impedance ($Z_{in} \to \infty$, so no current flows into the input terminals), zero output impedance ($Z_{out} = 0$, so the output can drive any load without voltage drop), infinite bandwidth (the gain is constant at all frequencies), and zero input offset voltage (the output is zero when both inputs are equal). These assumptions lead to two golden rules for analyzing op-amp circuits with negative feedback: (1) the voltage difference between the inverting and non-inverting inputs is zero ($V+ = V-$), and (2) no current flows into either input terminal. These rules, combined with Kirchhoff's laws, allow rapid analysis of virtually any op-amp feedback circuit.

**Example 13.1.1:** An op-amp circuit has the non-inverting input connected to a 2.5 V reference. Using the ideal op-amp golden rules, determine the voltage at the inverting input and the current into the inverting input terminal.
**Solution:**
By the first golden rule (virtual short): $V- = V+ = 2.5$ V.
By the second golden rule (zero input current): $I- = 0$ A.
These two conditions, combined with the external feedback network, are sufficient to solve for all voltages and currents in the circuit.
The virtual short does not mean the inputs are physically connected — it means the feedback forces the differential input voltage to zero.

### 13.1.2 Open-Loop vs. Closed-Loop

In open-loop configuration (no feedback), the op-amp operates at its maximum gain (typically $10^5$ to $10^6$ or 100–120 dB), and even a microvolt difference between the inputs drives the output to one of the supply rails, making it useful only as a comparator. Closed-loop configuration connects a portion of the output back to the inverting input (negative feedback), which reduces the gain to a stable, predictable value determined by the feedback network. The closed-loop gain is $A_{CL} = A_{OL} / (1 + A_{OL} \times \beta)$, where β is the feedback fraction; when $A_{OL} \times \beta \gg 1$, this simplifies to $A_{CL} \approx 1/\beta$, making the gain independent of the op-amp's open-loop characteristics. Negative feedback also increases bandwidth (by the same factor it reduces gain), reduces distortion, stabilizes gain against temperature and supply voltage variations, and modifies the input and output impedances. The gain-bandwidth product (GBW) is approximately constant for a given op-amp: $GBW = A_{CL} \times f_{3dB}$, where $f_{3dB}$ is the closed-loop bandwidth.

**Example 13.1.2:** An op-amp has an open-loop gain of $A_{OL}$ = 200,000 (106 dB) and a gain-bandwidth product of GBW = 5 MHz. It is configured with a feedback network that sets β = 0.01. Calculate the closed-loop gain, the closed-loop bandwidth, and the error between the ideal and actual closed-loop gain.
**Solution:**
Ideal closed-loop gain: $A_{CL(ideal)} = 1/\beta = 1/0.01 = 100$ (40 dB).
Actual closed-loop gain: $A_{CL} = A_{OL} / (1 + A_{OL} \times \beta) = 200{,}000 / (1 + 200{,}000 \times 0.01) = 200{,}000 / 2001 = 99.95$.
Gain error: $(100 - 99.95) / 100 \times 100\% = 0.05\%$.
Closed-loop bandwidth: $f_{3dB} = GBW / A_{CL} = 5{,}000{,}000 / 100 = 50$ kHz.
If the gain were reduced to $A_{CL} = 10$, the bandwidth would increase to 500 kHz, illustrating the gain-bandwidth trade-off.

## 13.2 Inverting Configurations

Inverting configurations connect the input signal to the inverting (−) terminal through an input element, with a feedback element from the output back to the inverting input, while the non-inverting (+) terminal is grounded or connected to a reference. The inverting input acts as a virtual ground (or virtual reference), and all inverting circuits share the property that the output signal is phase-inverted (180°) relative to the input. By choosing different input and feedback elements — resistors, capacitors, or combinations — the same basic topology produces amplifiers, summers, integrators, and differentiators.

### 13.2.1 Inverting Amplifier

The inverting amplifier connects the input signal through a resistor $R_{in}$ to the inverting input, with a feedback resistor $R_f$ from the output to the inverting input, and the non-inverting input grounded. The closed-loop voltage gain is $A_v = -R_f / R_{in}$, where the negative sign indicates a 180° phase inversion between input and output. The input impedance equals $R_{in}$ (since the inverting input is at virtual ground), and the output impedance is approximately zero. The inverting amplifier is one of the most commonly used op-amp configurations, providing precise, stable gain set entirely by the external resistor ratio.

**Example 13.2.1:** Design an inverting amplifier with a gain of -20 using an input resistance of $R_{in}$ = 5 kΩ. Determine $R_f$ and calculate the output voltage for an input of $V_{in}$ = 0.3 V. Also find the current through $R_f$.
**Solution:**
$A_v = -R_f / R_{in}$, so $R_f = |A_v| \times R_{in} = 20 \times 5$ kΩ = 100 kΩ.
Output voltage: $V_{out} = A_v \times V_{in} = -20 \times 0.3 = -6.0$ V.
The inverting input is at virtual ground (0 V), so the current through $R_{in}$ equals: $I = V_{in} / R_{in}$ = 0.3 / 5000 = 60 μA.
By the golden rule (no current into the op-amp), the same 60 μA flows through $R_f$.
Verification: $V_{out} = 0 - I \times R_f = 0 - 60 \times 10^{-6} \times 100{,}000 = -6.0$ V.

### 13.2.2 Summing Amplifier

The summing amplifier extends the inverting configuration by connecting multiple input signals through individual input resistors to the inverting input (the summing junction), with a single feedback resistor $R_f$. The output voltage is $V_{out} = -R_f \times (V_1/R_1 + V_2/R_2 + V_3/R_3 + \ldots)$, where each input contributes independently to the output, weighted by its respective resistor ratio. When all input resistors are equal ($R_1 = R_2 = \ldots = R$), the output is $V_{out} = -(R_f/R) \times (V_1 + V_2 + V_3 + \ldots)$, producing a scaled sum. Summing amplifiers are used in audio mixers, digital-to-analog converters (R-2R ladder DACs), and analog computing circuits where multiple signals must be combined with independent weighting.

**Example 13.2.2:** A summing amplifier has $R_f$ = 47 kΩ, $R_1$ = 10 kΩ, $R_2$ = 22 kΩ, and $R_3$ = 47 kΩ. The inputs are $V_1$ = 1.0 V, $V_2$ = -0.5 V, and $V_3$ = 2.0 V. Calculate the output voltage.
**Solution:**
$V_{out} = -R_f \times (V_1/R_1 + V_2/R_2 + V_3/R_3) = -47{,}000 \times (1.0/10{,}000 + (-0.5)/22{,}000 + 2.0/47{,}000) = -47{,}000 \times (0.0001 + (-0.00002273) + 0.00004255) = -47{,}000 \times 0.00011982 = -5.63$ V.
The contribution from each input: $V_1$ contributes $-47{,}000 \times 0.0001 = -4.70$ V, $V_2$ contributes $-47{,}000 \times (-0.00002273) = +1.07$ V, and $V_3$ contributes $-47{,}000 \times 0.00004255 = -2.00$ V.
Total: -4.70 + 1.07 + (-2.00) = -5.63 V.

### 13.2.3 Integrator

The integrator replaces the feedback resistor of the inverting amplifier with a capacitor C, producing an output that is the time integral of the input signal: $V_{out}(t) = -(1/RC) \int V_{in}(t)dt + V_{initial}$, where R is the input resistance and $V_{initial}$ is the initial capacitor voltage. For a constant DC input, the output ramps linearly; for a sinusoidal input at frequency f, the integrator provides a gain of $1/(2\pi fRC)$ with a 90° phase lag. A practical integrator includes a large resistor ($R_f$, typically 10–100× R) in parallel with the feedback capacitor to provide DC stability and prevent output saturation due to input offset voltage and bias current. Integrators are used in analog computers, PID controllers (the I term), waveform generators (triangle wave from square wave input), and active filters.

**Example 13.2.3:** An op-amp integrator has R = 10 kΩ and C = 100 nF. A constant input of $V_{in}$ = -2.0 V is applied starting at t = 0 with the capacitor initially discharged. Calculate the output voltage at t = 1 ms and t = 5 ms. At what time does the output reach the +12 V supply rail?

**Solution:**
Time constant: RC = 10,000 × 100 × $10^{-9}$ = 1.0 ms.
For a constant input: $V_{out}(t)$ = -(1/RC) × $V_{in}$ × t = -(1/0.001) × (-2.0) × t = 2000t.
At t = 1 ms: $V_{out}$ = 2000 × 0.001 = 2.0 V.
At t = 5 ms: $V_{out}$ = 2000 × 0.005 = 10.0 V.
The output reaches +12 V when: 12 = 2000t, so t = 12/2000 = 6.0 ms.
After 6.0 ms, the output saturates at the positive supply rail.
Ramp rate (slew): $dV_{out}/dt$ = 2000 V/s = 2.0 V/ms.

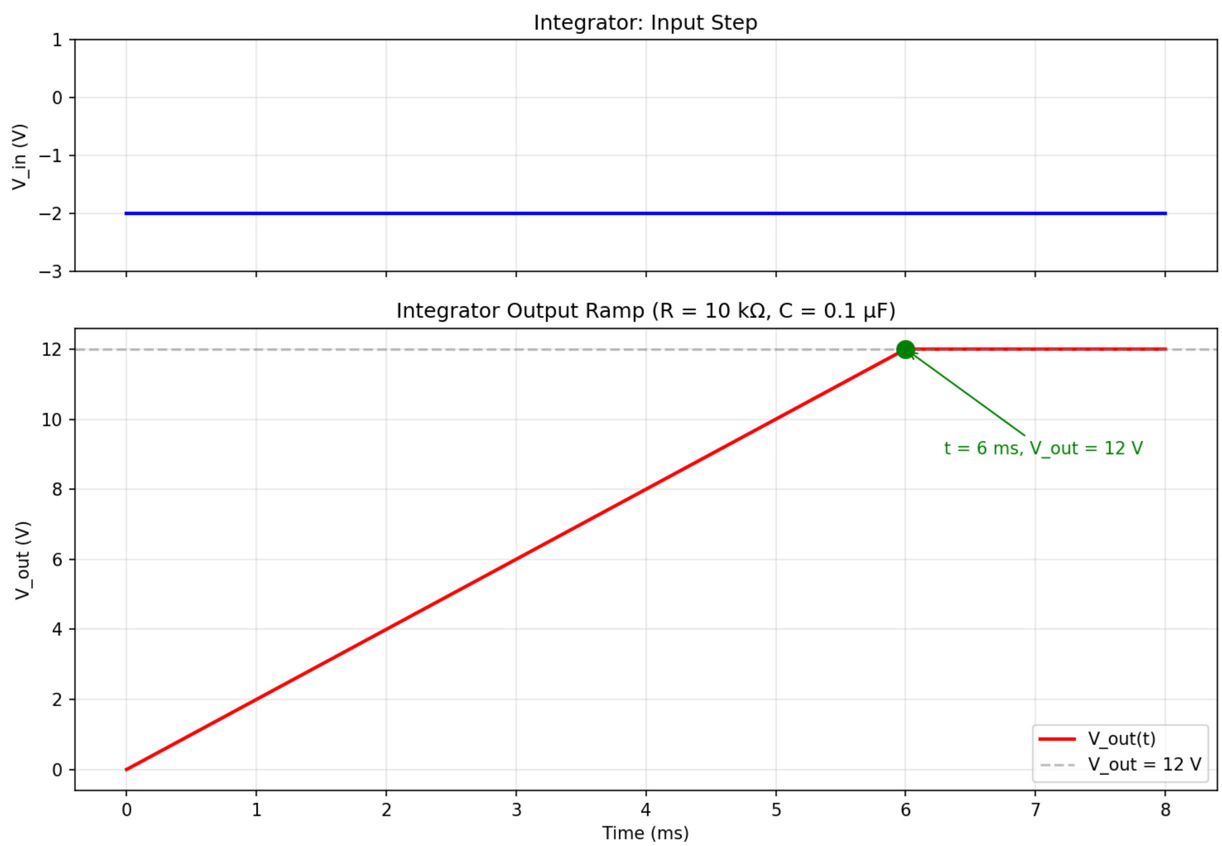



Figure 13.2.3: Op-Amp Integrator Step Response

### 13.2.4 Differentiator

The differentiator swaps the input resistor and feedback capacitor of the integrator, placing a capacitor C at the input and a resistor $R_f$ in the feedback path: $V_{out}(t) = -R_f \times C \times dV_{in}/dt$. The output is proportional to the rate of change of the input signal, making it useful for edge detection, frequency-to-voltage conversion, and the D term in PID controllers. For a sinusoidal input at frequency f, the gain increases linearly with frequency (6 dB/octave), which amplifies high-frequency noise. Practical differentiators include a small series resistor with the input capacitor ($R_s$, typically $R_f/10$ to $R_f/100$) to limit the high-frequency gain and prevent oscillation, creating a bandlimited differentiator that behaves as a true differentiator only below the corner frequency $f_c = 1/(2\pi R_s C)$.

**Example 13.2.4:** A differentiator has C = 10 nF and $R_f$ = 100 kΩ. A triangular wave input with a peak-to-peak amplitude of 4.0 V and a period of 2 ms is applied. Calculate the output voltage during the rising and falling portions of the triangle wave.
**Solution:**
The time constant: $R_f \times C = 100{,}000 \times 10 \times 10^{-9} = 1.0$ ms.
The triangle wave rises from -2 V to +2 V in T/2 = 1 ms, so $dV_{in}/dt$(rising) = 4.0 / 0.001 = 4000 V/s.
During the rising portion: $V_{out} = -R_f \times C \times dV_{in}/dt = -0.001 \times 4000 = -4.0$ V.
During the falling portion: $dV_{in}/dt$ = -4000 V/s, so $V_{out} = -0.001 \times (-4000) = +4.0$ V.
The output is a square wave alternating between -4.0 V and +4.0 V, which is the derivative of the triangular input.
At the triangle wave peaks, the abrupt slope change causes sharp transitions in the output.

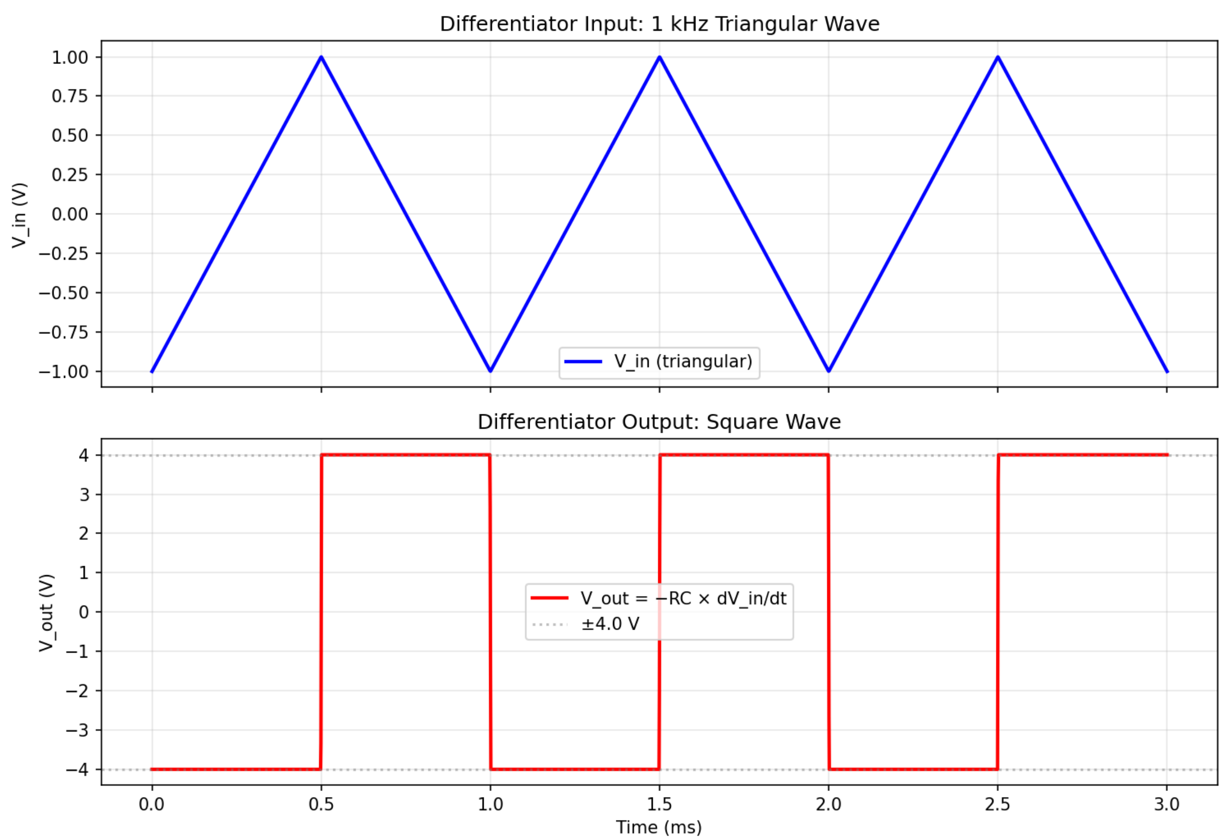



Figure 13.2.4: Op-Amp Differentiator Response

### 13.2.5 Logarithmic and Exponential Amplifiers

The logarithmic amplifier replaces the linear feedback resistor of an inverting amplifier with a diode or BJT collector-base junction, exploiting the exponential I–V relationship of a semiconductor junction to produce an output proportional to the logarithm of the input. With a BJT in the feedback path (collector to inverting input, emitter to output), the output is $V_{out} = -(kT/q) \times \ln(V_{in} / (I_S \times R_{in}))$, where $kT/q \approx 26$ mV at 25°C, $I_S$ is the transistor saturation current, and $R_{in}$ is the input resistor.

The exponential (antilog) amplifier reverses the arrangement, placing the transistor at the input and a resistor in the feedback path, producing $V_{out} = -I_S \times R_f \times e^{-Vinq/(kT)}$. Because $I_S$ is strongly temperature-dependent (approximately doubling every 10°C), practical log amplifiers use a matched transistor pair in a temperature-compensated configuration — one transistor in the log circuit and a second in a reference circuit — to cancel the $I_S$ dependence and produce $V_{out} = -(kT/q) \times \ln(V_{in}/V_{ref})$. Integrated log amplifier ICs (such as the LOG101 and LOG112) include matched transistors, temperature compensation, and scaling amplifiers on a single die. Log and antilog amplifiers enable analog multiplication and division (by adding/subtracting logarithms), decibel-scale measurements, signal compression for wide dynamic range (e.g., audio companders), and RMS-to-DC conversion. The useful input dynamic range spans 3–7 decades (e.g., 1 nA to 10 mA, or 10 $\mu$V to 10 V), limited by the transistor's conformance to the ideal exponential law at low currents and by high-injection effects at high currents.

**Example 13.2.5:** A logarithmic amplifier uses a BJT with $I_S = 10^{-14}$ A in the feedback path and $R_{in} = 10$ kΩ. Calculate the output voltage for input voltages of $V_{in} = 10$ mV, 100 mV, 1 V, and 10 V at 25°C. What is the change in output per decade of input?
**Solution:**
At 25°C, kT/q = 25.69 mV.
Input current: $I_{in} = V_{in}/R_{in}$.
$V_{out} = -(kT/q) \times \ln(I_{in}/I_S) = -0.025693 \times \ln(V_{in}/(R_{in} \times I_S))$.
For $V_{in} = 10$ mV: $I_{in} = 1$ $\mu$A, $V_{out} = -0.025693 \times \ln(10^{-6}/10^{-14}) = -0.025693 \times \ln(10^8) = -0.025693 \times 18.42 = -0.473$ V.
For $V_{in} = 100$ mV: $I_{in} = 10$ $\mu$A, $V_{out} = -0.025693 \times \ln(10^9) = -0.025693 \times 20.72 = -0.532$ V.
For $V_{in} = 1$ V: $I_{in} = 100$ $\mu$A, $V_{out} = -0.025693 \times \ln(10^{10}) = -0.025693 \times 23.03 = -0.592$ V.
For $V_{in} = 10$ V: $I_{in} = 1$ mA, $V_{out} = -0.025693 \times \ln(10^{11}) = -0.025693 \times 25.33 = -0.651$ V.
Change per decade: $\Delta V_{out} = -(kT/q) \times \ln(10) = -0.025693 \times 2.303 = -59.2$ mV/decade.
A scaling amplifier typically multiplies this by a factor of 10–17 to produce a convenient -1 V/decade output.

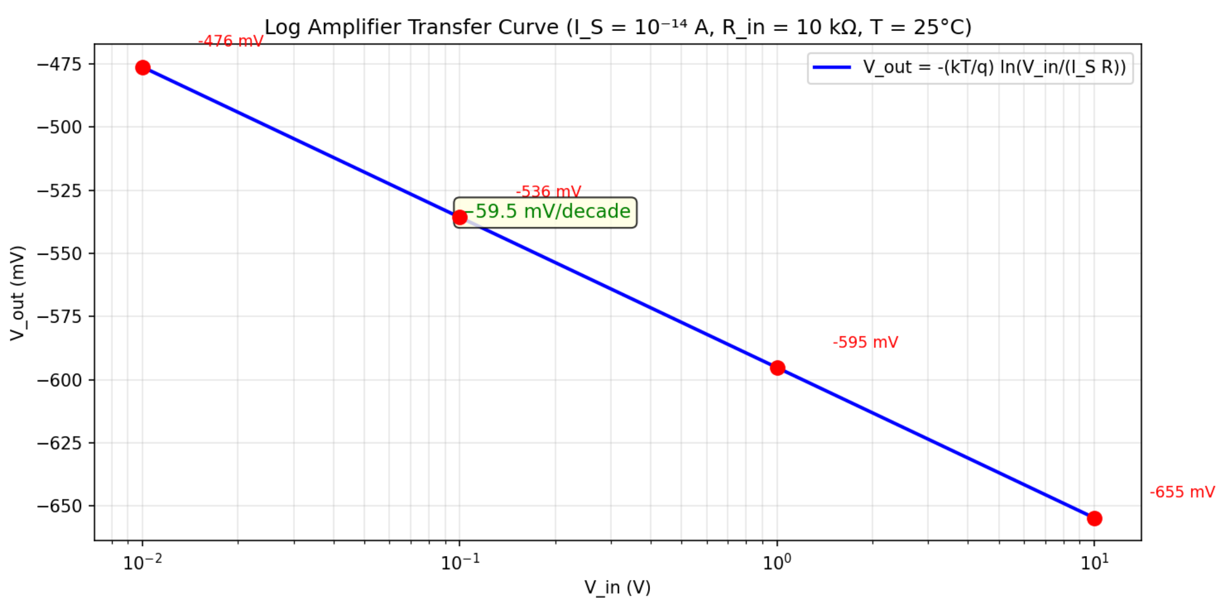



Figure 13.2.5: Log Amplifier Transfer Curve

### 13.2.6 Precision Rectifier

The precision rectifier (also called a superdiode) uses an op-amp to overcome the forward voltage drop of a conventional diode, enabling rectification of millivolt-level signals that would be completely lost in a standard diode's 0.5–0.7 V threshold. In the basic half-wave precision rectifier, a diode is placed in the feedback loop of an inverting amplifier: during the positive input half-cycle, the op-amp drives the diode into conduction and the circuit behaves as an inverting amplifier with gain $-R_f/R_{in}$; during the negative half-cycle, the diode is reverse-biased, the feedback loop opens, and the output is zero (clamped by a pull-down resistor). The effective forward voltage drop seen by the signal is reduced by the op-amp's open-loop gain — a 0.6 V silicon diode divided by $A_{OL}$ = 100,000 yields an effective threshold of only 6 μV. A **full-wave precision rectifier** (absolute value circuit) combines two op-amps: the first produces the inverted half-wave rectified signal, and the second sums the original signal with twice the inverted half-wave to produce $|V_{in}| \times (R_f/R_{in})$. Precision rectifiers are used in AC-to-DC conversion for metering, AM demodulation of low-level signals, and RMS-to-DC converters. The primary limitation is speed — when the diode turns off, the op-amp output must slew from the positive rail to the negative rail (or vice versa) before the diode can conduct again on the next half-cycle, limiting operation to frequencies well below $f_{max} \approx SR / (2\pi V_{sat})$, where $V_{sat}$ is the op-amp saturation voltage.

**Example 13.2.6:** A precision full-wave rectifier (absolute value circuit) uses op-amps with $A_{OL}$ = 100,000 and SR = 10 V/μs. The input is a 200 $mV_{peak}$, 5 kHz sine wave. Calculate the effective diode threshold, the DC output voltage (average of $|V_{in}|$), and the maximum operating frequency if the op-amp saturates at ±12 V.
**Solution:**
Effective diode threshold: $V_{D(eff)} = V_F / A_{OL} = 0.6 / 100{,}000 = 6\ \mu V$ — negligible compared to the 200 mV signal.
For a full-wave rectified sine wave, the average (DC) value is $V_{dc} = 2V_{peak}/\pi = 2 \times 200\text{ mV} / \pi = 127.3\text{ mV}$.
At 5 kHz this is well within the op-amp's capability.
Maximum operating frequency: when the diode turns off, the op-amp must slew from $+V_{sat}$ to $-V_{sat}$ (24 V swing).
Recovery time: $t_{rec} = 2V_{sat} / SR = 24 / (10 \times 10^6) = 2.4\ \mu s$.
For less than 5% distortion, the recovery time should be less than 5% of the half-period: $t_{rec} < 0.05 \times T/2$.
$T/2 > t_{rec} / 0.05 = 2.4\ \mu s / 0.05 = 48\ \mu s$, so $f_{max} < 1/(2 \times 48\ \mu s) = 10.4$ kHz.
A faster op-amp (SR = 50 V/μs) would extend this to ~52 kHz.

## 13.3 Non-Inverting Configurations

Non-inverting configurations apply the input signal to the non-inverting (+) terminal, providing high input impedance and a non-inverted output. The feedback network connects from the output to the inverting terminal, and the gain is always ≥ 1. This topology is preferred when the signal source has high impedance (sensors, voltage dividers, reference circuits) because the op-amp's input draws negligible current.

### 13.3.1 Non-Inverting Amplifier

The non-inverting amplifier applies the input signal to the non-inverting (+) input, with a voltage divider ($R_1$ to ground, $R_f$ from output to inverting input) setting the feedback. The closed-loop gain is $A_v = 1 + R_f/R_1$, which is always greater than or equal to 1 and does not invert the signal. The input impedance is extremely high (approaching the op-amp's open-loop input impedance, typically $>10^9$ Ω for FET-input devices) because the input connects directly to the non-inverting terminal with no input resistor. This high input impedance makes the non-inverting amplifier ideal for buffering high-impedance sources such as sensors and reference circuits.

**Example 13.3.1:** Design a non-inverting amplifier with a gain of 15 using $R_1 = 3.3$ kΩ. Calculate $R_f$, the output voltage for $V_{in} = 0.8$ V, and the input impedance advantage over an equivalent inverting amplifier.
**Solution:**
$A_v = 1 + R_f/R_1$, so $R_f = (A_v - 1) \times R_1 = 14 \times 3{,}300 = 46.2$ kΩ (use 47 kΩ standard value, giving $A_v = 1 + 47{,}000/3{,}300 = 15.24$).
Output voltage: $V_{out} = 15.24 \times 0.8 = 12.19$ V.
Input impedance: for the non-inverting configuration, $Z_{in} \approx Z_{in(op\text{-}amp)} \times (1 + A_{OL} \times \beta)$, which is typically $>10^9$ Ω.
An equivalent inverting amplifier with gain of -15 using $R_{in} = 3.3$ kΩ would have $Z_{in} = R_{in} = 3.3$ kΩ – over 300,000× lower than the non-inverting version.

### 13.3.2 Voltage Follower (Buffer)

The voltage follower (also called a unity-gain buffer) is the simplest non-inverting configuration, with 100% feedback ($R_f = 0$, $R_1 = \infty$, or equivalently the output connected directly to the inverting input). The gain is $A_v = 1$, so $V_{out} = V_{in}$ with no inversion. The voltage follower provides the highest possible input impedance and lowest possible output impedance of any op-amp configuration, making it ideal for impedance matching — isolating a high-impedance source from a low-impedance load without signal attenuation. It is commonly used to buffer voltage references, sensor outputs, and voltage dividers that would otherwise be loaded by downstream circuitry.

**Example 13.3.2:** A voltage divider produces 2.5 V from a 5.0 V supply using two 100 kΩ resistors. A 1 kΩ load is connected. Calculate the output voltage (a) without a buffer and (b) with a voltage follower buffer. What is the loading error without the buffer?
**Solution:**
(a) Without buffer: the 1 kΩ load is in parallel with the lower 100 kΩ resistor.
$R_{parallel} = (100{,}000 \times 1{,}000) / (100{,}000 + 1{,}000) = 990.1$ Ω.
$V_{out} = 5.0 \times 990.1 / (100{,}000 + 990.1) = 5.0 \times 990.1 / 100{,}990 = 0.0490$ V.
Loading error: $(2.5 - 0.049) / 2.5 \times 100\% = 98.0\%$ – the voltage divider is completely collapsed.
(b) With buffer: the buffer input impedance is $>10^9$ Ω, so the divider sees essentially no load.
$V_{out(buffer\ input)} = 2.5$ V.
The buffer drives the 1 kΩ load directly: $V_{out} = 2.5$ V.
Loading error: $\approx 0\%$.
The buffer transforms the 50 kΩ source impedance of the divider to $\approx 0$ Ω.

### 13.3.3 Transimpedance Amplifier

The transimpedance amplifier (TIA) converts an input current to an output voltage, with a transimpedance gain (transfer impedance) of $V_{out}$ = -$I_{in}$ × $R_f$, where $R_f$ is the feedback resistor. The input current source (typically a photodiode, phototransistor, or current-output DAC) connects to the inverting input, which is held at virtual ground, providing a low-impedance termination for the current source. A small feedback capacitor $C_f$ is placed in parallel with $R_f$ to stabilize the circuit against oscillation caused by the input capacitance of the current source ($C_{in}$) interacting with $R_f$. The bandwidth of the TIA is limited by the pole formed by $R_f$ and $C_f$: $f_{3dB} = 1/(2\pi R_f C_f)$, and the optimal $C_f$ for a Butterworth-like response is $C_f = \sqrt{(C_{in} / (2\pi \times R_f \times GBW))}$, where GBW is the op-amp's gain-bandwidth product. Transimpedance amplifiers are essential in fiber optic receivers, optical disc readers, spectroscopy instruments, and any application requiring precise measurement of small currents.

**Example 13.3.3:** A photodiode with a capacitance of $C_{in}$ = 25 pF produces a photocurrent of 10 μA at the operating light level. A TIA with $R_f$ = 100 kΩ is used to convert this current to a voltage. The op-amp has a GBW of 10 MHz. Calculate the output voltage, the optimal feedback capacitance $C_f$ for stability, and the resulting signal bandwidth.
**Solution:**
Output voltage: $V_{out} = -I_{in} \times R_f = -10 \times 10^{-6} \times 100{,}000 = -1.0$ V.
Optimal feedback capacitance: $C_f = \sqrt{(C_{in} / (2\pi \times R_f \times GBW))} = \sqrt{(25 \times 10^{-12} / (2\pi \times 100{,}000 \times 10 \times 10^{6}))} = \sqrt{(25 \times 10^{-12} / 6.283 \times 10^{12})} = \sqrt{(3.979 \times 10^{-24})} = 1.99 \times 10^{-12}$ F ≈ 2.0 pF.
Signal bandwidth: $f_{3dB} = 1/(2\pi R_f C_f) = 1/(2\pi \times 100{,}000 \times 2.0 \times 10^{-12}) = 1/(1.257 \times 10^{-6}) = 796$ kHz.
Noise gain peaking frequency (where instability would occur without $C_f$): $f_p = \sqrt{(GBW / (2\pi R_f C_{in}))} = \sqrt{(10 \times 10^{6} / (2\pi \times 100{,}000 \times 25 \times 10^{-12}))} = \sqrt{(6.37 \times 10^{11})} = 798$ kHz.

### 13.3.4 Programmable Gain Amplifier (PGA)

A programmable gain amplifier allows its gain to be changed dynamically — either digitally via control pins or through software commands — enabling a single signal path to accommodate inputs spanning a wide dynamic range. The simplest PGA uses an analog multiplexer to switch between different feedback resistors in an inverting or non-inverting configuration, providing gain steps set by the resistor ratios (e.g., gains of 1, 2, 4, 8). Integrated PGA ICs (such as the PGA280 and MCP6S26) include laser-trimmed resistor networks with gain accuracy of 0.01–0.1%, an input multiplexer, SPI or $I^2C$ control, and settling times of 1–10 μs after a gain change. In data acquisition systems, the PGA sits between the input multiplexer and the ADC, and the system controller selects the gain for each channel based on the expected signal level — using high gain for small signals (thermocouples, strain gauges) and low gain for large signals (voltage monitors), maximizing the effective dynamic range of the ADC. A critical specification is **gain switching transient** (glitch energy), which is the brief output disturbance that occurs when the gain changes; this transient must settle before the ADC samples, requiring a delay of several time constants after each gain change. The **noise gain** of a PGA increases with the programmed gain, so the input-referred noise contribution of the ADC is divided by the PGA gain — improving the system's noise performance for small signals. Some Σ-Δ ADCs (e.g., ADS1256, AD7190) integrate a PGA on-chip with gains from 1 to 128, eliminating the external PGA and its associated settling and noise issues.

**Example 13.3.4:** A 16-bit SAR ADC with a ±5 V input range and an input-referred noise of 120 $\mu V_{rms}$ is preceded by a PGA with selectable gains of 1, 10, and 100. A thermocouple produces a 5 mV signal. Calculate the effective ADC resolution (ENOB) at each PGA gain setting and determine which gain maximizes measurement quality.
**Solution:**
ADC LSB = 10 V / $2^{16}$ = 152.6 μV.
At PGA gain = 1: signal at ADC = 5 mV (33 LSBs). Input-referred noise = 120 μV, SNR = $20\log_{10}$(5 mV / 120 μV) = 32.4 dB, ENOB = (32.4 − 1.76) / 6.02 = 5.1 bits — very poor, only 33 of 65,536 codes used.
At PGA gain = 10: signal at ADC = 50 mV (328 LSBs). Noise referred to input = 120 μV / 10 = 12 μV. SNR = $20\log_{10}$(5 mV / 12 μV) = 52.4 dB, ENOB = 8.4 bits — significant improvement.
At PGA gain = 100: signal at ADC = 500 mV (3,277 LSBs). Noise referred to input = 120 μV / 100 = 1.2 μV. SNR = $20\log_{10}$(5 mV / 1.2 μV) = 72.4 dB, ENOB = 11.7 bits.
However, the PGA's own noise (e.g., 10 nV/$\sqrt{\text{Hz}}$ over 100 kHz BW = 3.16 $\mu V_{rms}$) now becomes the dominant noise source.
Combined input noise = $\sqrt{(1.2^2 + 3.16^2)}$ = 3.38 $\mu V_{rms}$, SNR = 63.4 dB, ENOB = 10.2 bits.
**Gain = 100** provides the best measurement, improving from 5.1 to 10.2 effective bits — but the PGA's own noise limits further improvement, and a lower-noise PGA or integrated Σ-Δ ADC with on-chip PGA would be needed for higher resolution.

## 13.4 Differential and Instrumentation Amplifiers

Differential amplifiers amplify the difference between two input signals while rejecting common-mode signals present on both inputs. This capability is critical for measuring small signals from sensors (strain gauges, thermocouples, current shunts) that sit on large common-mode voltages or in noisy environments. The basic single-op-amp difference amplifier is limited by resistor matching, while the three-op-amp instrumentation amplifier overcomes this limitation with laser-trimmed internal resistors and a single gain-setting resistor.

### 13.4.1 Difference Amplifier

The difference amplifier uses four resistors arranged so that the op-amp amplifies only the difference between two input signals while rejecting any signal common to both inputs. The basic configuration applies $V_1$ through $R_1$ to the inverting input (with $R_2$ as feedback) and $V_2$ through $R_3$ to the non-inverting input (with $R_4$ to ground). When $R_2/R_1 = R_4/R_3$, the output is $V_{out} = (R_2/R_1)(V_2 - V_1)$. The Common-Mode Rejection Ratio (CMRR) depends critically on resistor matching — a 0.1% mismatch in a gain-of-1 difference amplifier limits CMRR to approximately 66 dB. For this reason, precision difference amplifiers use laser-trimmed thin-film resistor networks integrated with the op-amp.

**Example 13.4.1:** A difference amplifier with $R_1 = R_3 = 10$ kΩ and $R_2 = R_4 = 100$ kΩ measures the voltage across a 0.1 Ω current sense resistor carrying 5 A, with a common-mode voltage of 48 V. If $R_4$ has a 0.1% tolerance error ($R_4$ = 100,100 Ω), calculate the ideal output, the common-mode error, and the effective CMRR.
**Solution:**
Differential signal: $V_{diff} = I \times R_{sense} = 5 \times 0.1 = 0.5$ V.

Gain: $A_{diff} = R_2/R_1 = 100{,}000/10{,}000 = 10$.
Ideal output: $V_{out} = 10 \times 0.5 = 5.0$ V.
Common-mode gain with mismatched $R_4$: $A_{cm} \approx (\Delta R/R) \times (R_2/R_1) / (1 + R_2/R_1) = (0.001) \times 10 / 11 = 0.000909$.
Common-mode error voltage: $V_{error} = A_{cm} \times V_{cm} = 0.000909 \times 48 = 43.6$ mV.
Total output: $5.0 + 0.0436 = 5.044$ V (0.87% error).
CMRR = $A_{diff} / A_{cm} = 10 / 0.000909 = 11{,}000$, or $20\log_{10}(11{,}000) = 80.8$ dB.

### 13.4.2 Instrumentation Amplifier

The instrumentation amplifier (INA) is a precision differential amplifier consisting of three op-amps: two non-inverting input buffers and one difference amplifier output stage. The input buffers provide extremely high input impedance ($>10^9\ \Omega$) on both inputs and allow the differential gain to be set by a single external resistor $R_G$: $A_v = 1 + 2R/R_G$, where R is an internal matched resistor (typically 25 kΩ or 50 kΩ). The output difference stage uses laser-trimmed internal resistors to achieve CMRR of 80-120 dB without requiring external resistor matching. INAs are the standard choice for amplifying small differential signals from Wheatstone bridges, thermocouples, biomedical electrodes, and other sensors in the presence of large common-mode voltages.

**Example 13.4.2:** An INA333 instrumentation amplifier has internal resistors R = 25 kΩ and is configured with $R_G = 1$ kΩ. It amplifies a strain gauge bridge that produces a 2.5 mV differential signal with 2.5 V common mode. The INA has CMRR = 100 dB. Calculate the gain, the desired output, and the common-mode error.
**Solution:**
Gain: $A_v = 1 + 2R/R_G = 1 + 2(25{,}000)/1{,}000 = 1 + 50 = 51$.
Desired output: $V_{out} = 51 \times 2.5$ mV = 127.5 mV.
CMRR = 100 dB = 10^(100/20) = 100,000.
Common-mode gain: $A_{cm} = A_{diff}$ / CMRR = 51 / 100,000 = 0.00051.
Common-mode error at output: $V_{error} = A_{cm} \times V_{cm} = 0.00051 \times 2.5 = 1.275$ mV.
Percentage error: $1.275 / 127.5 \times 100\% = 1.0\%$.
To reduce this error to $< 0.1\%$, a CMRR of at least 120 dB is needed, or the common-mode voltage must be reduced.

### 13.4.3 Current Sense Amplifiers

Current sense amplifiers are specialized differential amplifiers optimized for measuring voltage drops across low-value shunt resistors (typically 1–100 mΩ) in power supply lines, providing an output voltage proportional to the load current. Unlike general-purpose difference amplifiers, current sense amplifiers are designed to operate with high common-mode voltages — **high-side** current sensing measures current between the supply rail and the load, requiring the amplifier to reject common-mode voltages up to the full bus voltage (12 V, 48 V, or even 400 V+), while **low-side** sensing places the shunt between the load and ground, where the common-mode voltage is near zero but the ground path is interrupted. High-side sensing is preferred in most applications because it preserves the load's ground connection and can detect both load current and fault conditions (short to ground). Dedicated current sense amplifier ICs (such as the INA181, INA240, and INA293) integrate gain-setting resistors (gains of 20, 50, 100, or 200 V/V), laser-trimmed for accuracy, and achieve CMRR of 80–140 dB with common-mode input ranges from -4 V to over 100 V. Bidirectional current sense amplifiers can measure current flowing in both directions through the shunt, using a reference voltage at the output midpoint, and are essential in battery management systems (charge and discharge), motor drives (H-bridge current), and regenerative braking systems. Key specifications include gain error (<0.5%), offset voltage referred to input (<25 μV), common-mode rejection ratio, maximum common-mode voltage, bandwidth (50 kHz to 1 MHz), and output voltage range.

**Example 13.4.3:** A high-side current sense amplifier (INA181A3, gain = 100 V/V) monitors load current through a 10 mΩ shunt resistor on a 48 V bus. The amplifier has $V_{OS}$ = 35 μV (referred to input), gain error = 0.3%, and CMRR = 132 dB at DC. The load draws 20 A. Calculate the ideal output voltage, the total output error, and the minimum detectable current.
**Solution:**
Shunt voltage: $V_{shunt} = I \times R_{shunt} = 20 \times 0.010 = 200$ mV.
Ideal output: $V_{out} = V_{shunt} \times \text{Gain} = 0.200 \times 100 = 20.0$ V — this exceeds a typical 5 V supply, so in practice the amplifier would saturate.
Using a 5 mΩ shunt instead: $V_{shunt} = 20 \times 0.005 = 100$ mV, $V_{out} = 100 \text{ mV} \times 100 = 10.0$ V (within a 12 V supply range).
Offset error at output: $V_{OS} \times \text{Gain} = 35 \times 10^{-6} \times 100 = 3.5$ mV.
Gain error: 0.3% × 10.0 V = 30 mV.
CMRR error: CMRR = 132 dB = $3.981 \times 10^6$. Common-mode referred-to-input error = 48/3.981 × $10^6$ = 12.1 μV, at output = 12.1 μV × 100 = 1.21 mV.
Total output error ≈ 3.5 + 30 + 1.2 = 34.7 mV (0.35% of 10.0 V).
Minimum detectable current (limited by offset): $I_{min} = V_{OS}/R_{shunt}$ = 35 μV / 5 mΩ = 7 mA.

## 13.5 Active Filters

Active filters use op-amps with resistors and capacitors to implement frequency-selective circuits that can provide gain, low output impedance, and sharp roll-off characteristics impossible with passive RC networks alone. Unlike passive LC filters, active filters do not require inductors (which are bulky, lossy, and impractical at low frequencies), making them the standard approach for audio, instrumentation, and control system filtering. Common topologies include Sallen-Key (voltage-controlled voltage source), multiple feedback, and state-variable, each offering different trade-offs in component sensitivity, tuning flexibility, and available filter outputs.

### 13.5.1 First-Order Filters

First-order active filters use a single op-amp with one reactive element (capacitor) to provide a frequency response with a -20 dB/decade (6 dB/octave) roll-off. A first-order lowpass filter places a capacitor in parallel with the feedback resistor of an inverting amplifier: the gain is flat at $-R_f/R_{in}$ up to the cutoff frequency $f_c = 1/(2\pi R_f C)$, above which it rolls off at -20 dB/decade. A first-order highpass filter places the capacitor in series with the input resistor: the gain is zero at DC, rises at +20 dB/decade, and levels off at $-R_f/R_{in}$ above $f_c = 1/(2\pi R_{in} C)$. First-order active filters provide gain (unlike passive RC filters which can only attenuate), have a low-impedance output, and avoid loading effects on the source.

**Example 13.5.1:** Design a first-order active lowpass filter with a passband gain of -5 (inverting) and a cutoff frequency of 3.4 kHz (telephone bandwidth). Choose $R_{in}$ and calculate C for a 100 kΩ feedback resistor.
**Solution:**
Passband gain: $A_v = -R_f/R_{in}$, so $R_{in} = R_f / |A_v| = 100{,}000 / 5 = 20$ kΩ.
Cutoff frequency: $f_c = 1/(2\pi R_f C)$, so $C = 1/(2\pi \times R_f \times f_c) = 1/(2\pi \times 100{,}000 \times 3{,}400) = 1 / (2.136 \times 10^9) = 468$ pF (use 470 pF standard value).
Verification: $f_c = 1/(2\pi \times 100{,}000 \times 470 \times 10^{-12}) = 3{,}386$ Hz $\approx 3.4$ kHz.
At f = 10 kHz ($2.95\times f_c$): gain $= -5 / \sqrt{}(1 + (10{,}000/3{,}386)^2) = -5 / \sqrt{}(1 + 8.72) = -5 / 3.12 = -1.60$, or $20\log_{10}(1.60/5) = -9.9$ dB relative to the passband gain.

### 13.5.2 Sallen-Key Filters

The Sallen-Key topology is the most popular second-order active filter structure, using a single op-amp configured as a non-inverting amplifier (typically unity gain) with two resistors and two capacitors forming the frequency-selective network. The standard unity-gain Sallen-Key lowpass filter has the transfer function with cutoff frequency $f_c = 1/(2\pi\sqrt{}(R_1R_2C_1C_2))$ and quality factor $Q = \sqrt{}(R_1R_2C_1C_2) / (C_2(R_1 + R_2))$. By choosing equal-value components ($R_1 = R_2 = R$, $C_1 = C_2 = C$), the design simplifies to $f_c = 1/(2\pi RC)$ with $Q = 0.5$, which gives a critically damped response (no overshoot, fastest settling time). For a Butterworth response ($Q = 0.707$), the capacitor ratio $C_1/C_2 = 2$ is used with equal resistors. The Sallen-Key topology uses minimal components, has low sensitivity to component tolerances at unity gain, and can be cascaded to create higher-order filters.

**Example 13.5.2:** Design a unity-gain Sallen-Key Butterworth lowpass filter with a cutoff frequency of 1 kHz. Use equal resistors and a 10 nF capacitor for $C_2$. Calculate R and $C_1$.
**Solution:**
For a Butterworth response with equal resistors: $C_1 = 2 \times C_2 = 2 \times 10$ nF $= 20$ nF.
Cutoff frequency: $f_c = 1/(2\pi R\sqrt{}(C_1C_2)) = 1/(2\pi R\sqrt{}(20 \times 10^{-9} \times 10 \times 10^{-9}))$.
$\sqrt{}(C_1C_2) = \sqrt{}(200 \times 10^{-18}) = 14.14 \times 10^{-9}$.
So $R = 1/(2\pi \times 1000 \times 14.14 \times 10^{-9}) = 1/(88.86 \times 10^{-6}) = 11.25$ kΩ (use 11 kΩ standard value).
Verification: $f_c = 1/(2\pi \times 11{,}000 \times 14.14 \times 10^{-9}) = 1/(977 \times 10^{-6}) = 1{,}024$ Hz $\approx 1$ kHz.
$Q = \sqrt{}(R^2C_1C_2) / (C_2 \times 2R) = R\sqrt{}(C_1C_2) / (2RC_2) = \sqrt{}(C_1/C_2) / 2 = \sqrt{}2 / 2 = 0.707$, confirming the Butterworth response.

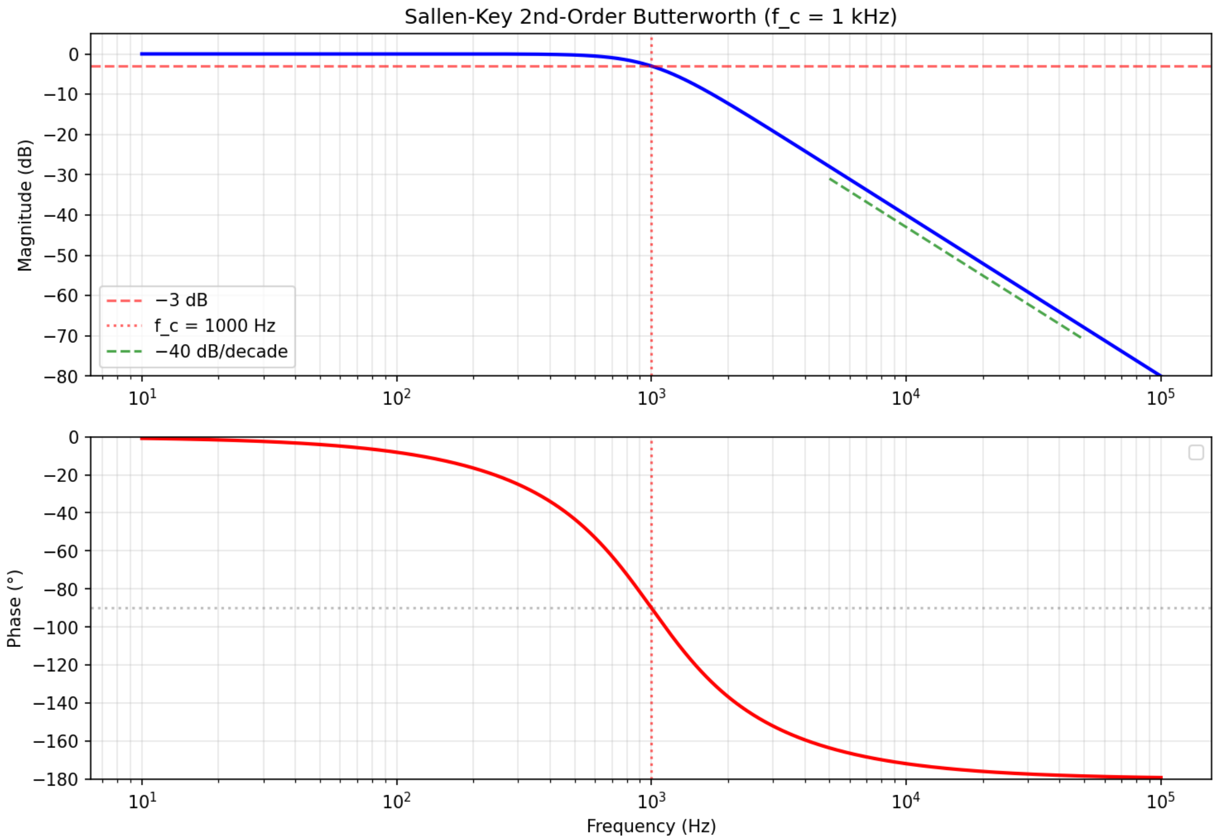



Figure 13.5.2: Sallen-Key Butterworth Bode Plot

### 13.5.3 State-Variable Filters

The state-variable filter uses three op-amps (a summing amplifier and two integrators) connected in a feedback loop to simultaneously provide lowpass, highpass, and bandpass outputs from a single circuit. The center frequency is $f_0 = 1/(2\pi RC)$ (where R and C are the integrator components), the quality factor Q is set by the resistor ratios in the summing amplifier, and the gain of each output is independently controllable. A fourth op-amp can be added to sum the lowpass and highpass outputs with appropriate weighting to produce a bandstop (notch) response. The state-variable topology offers independent control of $f_0$, Q, and gain — changing the frequency does not affect Q, and vice versa — making it ideal for tunable filters, audio equalizers, and measurement instruments where filter parameters must be adjusted without interaction.

**Example 13.5.3:** A state-variable bandpass filter is designed for a center frequency of $f_0 = 5$ kHz with Q = 10. The integrator components are C = 10 nF. Calculate the integrator resistance R, the bandwidth, and the lower and upper -3 dB frequencies.
**Solution:**
Integrator resistance: $R = 1/(2\pi f_0 C) = 1/(2\pi \times 5{,}000 \times 10 \times 10^{-9}) = 1/(314.16 \times 10^{-6}) =$ **3,183 Ω** (use 3.16 kΩ standard value).
Bandwidth: $BW = f_0/Q = 5{,}000/10 =$ **500 Hz**.
Lower -3 dB frequency: $f_L = f_0 - BW/2 = 5{,}000 - 250 =$ **4,750 Hz**.

Upper -3 dB frequency: $f_H = f_0 + BW/2 = 5{,}000 + 250 =$ **5,250 Hz**.
(More precisely for high Q: $f_L = f_0 \times (\sqrt{(1 + 1/(4Q^2))} - 1/(2Q)) = 4{,}756$ Hz and $f_H = f_0 \times (\sqrt{(1 + 1/(4Q^2))} + 1/(2Q)) = 5{,}256$ Hz.)
The bandpass gain at $f_0$ equals $Q \times A_0$, where $A_0$ is the integrator gain.

### 13.5.4 Multiple Feedback Filters

The multiple feedback (MFB) topology uses a single op-amp in an inverting configuration with two feedback paths to implement second-order lowpass, highpass, or bandpass filters. The MFB bandpass filter is particularly popular because it provides a narrow bandpass response with a single op-amp, using two resistors and two capacitors, with the center frequency $f_0 = 1/(2\pi C) \times \sqrt{(1/(R_1R_3))}$, the quality factor $Q = \pi \times f_0 \times C \times R_2$, and a midband gain of $A_0 = -R_2/(2R_1)$. The MFB lowpass filter offers an advantage over Sallen-Key at high Q values because it is less sensitive to op-amp gain-bandwidth limitations and provides an inherently inverting output. Component sensitivity is moderate — the Q and center frequency interact somewhat when components are changed, making independent tuning more difficult than with the state-variable topology. The MFB topology is widely used in audio equalizers, anti-aliasing filters, and communication channel filters.

**Example 13.5.4:** Design a multiple feedback bandpass filter with a center frequency of $f_0 = 2$ kHz, $Q = 5$, and a midband gain of $|A_0| = 10$ using equal capacitors $C = 10$ nF. Calculate $R_1$, $R_2$, $R_3$, and the bandwidth.
**Solution:**
Bandwidth: $BW = f_0 / Q = 2{,}000 / 5 = 400$ Hz.
From the gain equation: $A_0 = -R_2 / (2R_1)$, so $R_2 = 2 \times |A_0| \times R_1$.
From the Q equation: $Q = \pi \times f_0 \times C \times R_2$, so $R_2 = Q / (\pi \times f_0 \times C) = 5 / (\pi \times 2{,}000 \times 10 \times 10^{-9}) = 5 / (62.83 \times 10^{-6}) = 79{,}577\ \Omega \approx 79.6$ kΩ (use 82 kΩ).
From the gain: $R_1 = R_2 / (2 \times |A_0|) = 82{,}000 / 20 = 4{,}100\ \Omega$ (use 3.9 kΩ).
From the center frequency: $f_0 = (1/(2\pi C)) \times \sqrt{(1/(R_1R_3))}$, so $R_3 = 1 / ((2\pi f_0 C)^2 \times R_1) = 1 / ((2\pi \times 2{,}000 \times 10 \times 10^{-9})^2 \times 3{,}900) = 1 / ((125.7 \times 10^{-6})^2 \times 3{,}900) = 1 / (1.580 \times 10^{-8} \times 3{,}900) = 1 / (6.162 \times 10^{-5}) = 16{,}228\ \Omega$ (use 16 kΩ).
Lower and upper -3 dB frequencies: $f_L \approx 2{,}000 - 200 = 1{,}800$ Hz, $f_H \approx 2{,}000 + 200 = 2{,}200$ Hz.

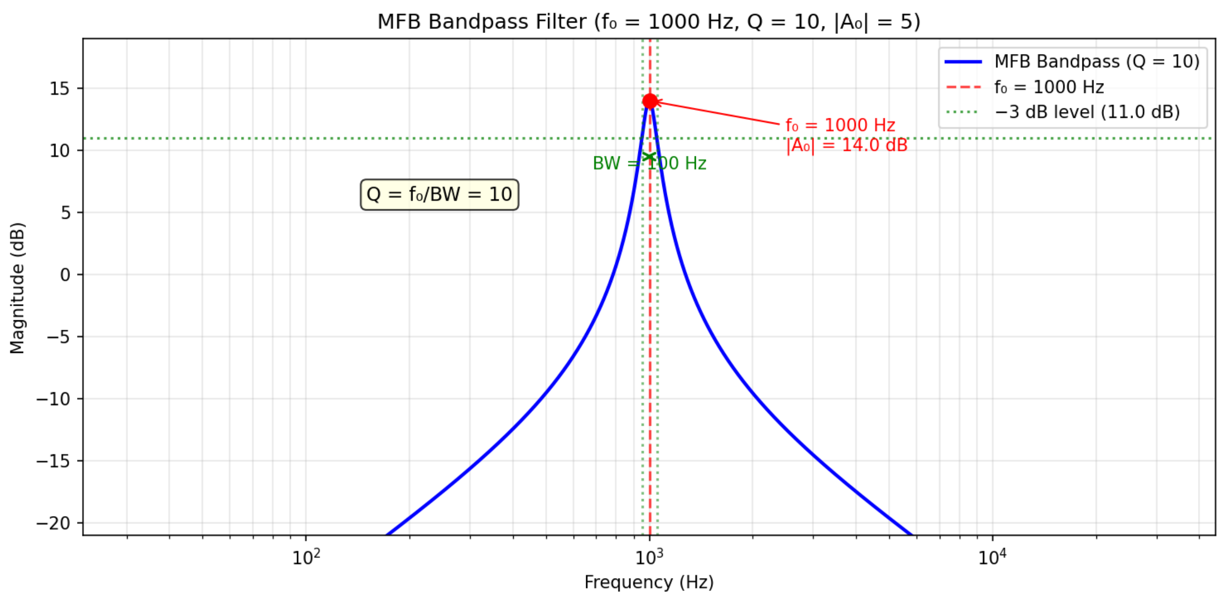



Figure 13.5.4: MFB Bandpass Filter Frequency Response

### 13.5.5 Notch (Band-Reject) Filters

A notch filter (band-reject filter) strongly attenuates a narrow band of frequencies while passing all others, and is widely used to reject power-line interference (50/60 Hz hum), eliminate specific tones in audio processing, and remove carrier frequencies in communication systems. The **Twin-T notch filter** is the most common passive notch network, using two T-shaped RC paths in parallel — one with series resistors and a shunt capacitor (lowpass path), the other with series capacitors and a shunt resistor (highpass path). The two paths produce equal amplitude but opposite phase at the notch frequency $f_0 = 1/(2\pi RC)$, creating a deep null. The passive Twin-T alone has a broad notch (low Q $\approx 0.25$) and limited rejection depth (~20 dB), but placing it in the negative feedback loop of an op-amp with positive feedback dramatically increases the Q and notch depth. With positive feedback factor k applied from the output to the Twin-T center node, the quality factor becomes $Q = 1/(4(1 - k))$, achieving $Q = 5$ at $k = 0.95$ and $Q = 25$ at $k = 0.99$ — narrow enough to reject 60 Hz while preserving adjacent frequencies. An **active Twin-T notch filter** uses a voltage follower buffer to feed a fraction of the output back to the Twin-T junction, with the feedback ratio set by a resistive voltage divider. Alternatively, the state-variable filter (§13.5.3) produces a notch output by summing its lowpass and highpass outputs with equal gain, and offers independently adjustable $f_0$ and Q — a significant advantage when the interfering frequency varies or must be tuned precisely. Notch depth in practical circuits is limited by component matching: the Twin-T requires R and C values matched to better than 1% for depths exceeding 40 dB, and the shunt resistor must be exactly R/2 while the shunt capacitor must be exactly 2C.

**Example 13.5.5:** Design an active Twin-T notch filter to reject 60 Hz power-line hum. Use C = 100 nF and calculate R. The positive feedback fraction is $k = 0.96$. Determine the notch Q, the −3 dB bandwidth, and the attenuation at 50 Hz and 120 Hz.
**Solution:**
Notch frequency: $f_0 = 1/(2\pi RC)$, so $R = 1/(2\pi f_0 C) = 1/(2\pi \times 60 \times 100 \times 10^{-9}) = 1/(37.70 \times 10^{-6})$

$= 26{,}526\ \Omega$.
Use 26.7 kΩ (or 27 kΩ standard, giving $f_0 = 58.9$ Hz — close enough, or use 26.1 kΩ + 470 Ω in series for 26.57 kΩ).
The shunt resistor $= R/2 = 13{,}263\ \Omega$ (use 13.3 kΩ), shunt capacitor $= 2C = 200$ nF.
Quality factor: $Q = 1/(4(1-k)) = 1/(4 \times 0.04) = 6.25$.
Bandwidth: $BW = f_0/Q = 60/6.25 = 9.6$ Hz.
The −3 dB frequencies are approximately $60 \pm 4.8$ Hz = 55.2 Hz and 64.8 Hz.
Attenuation at 50 Hz: the offset from center is $\Delta f = 10$ Hz, normalized $u = 2\Delta f/BW = 20/9.6 = 2.08$. For a second-order notch, the attenuation at offset u is approximately $1/\sqrt{}(1 + 1/u^2)$ of the passband gain $= 1/\sqrt{}(1 + 0.231) = 0.902 = -0.89$ dB (minimal attenuation).
At 120 Hz (2nd harmonic of 60 Hz): $u = 2 \times 60/9.6 = 12.5$, attenuation $\approx 0.997 = -0.026$ dB (essentially no effect on the 2nd harmonic).
The narrow notch removes the 60 Hz fundamental while preserving the rest of the spectrum.

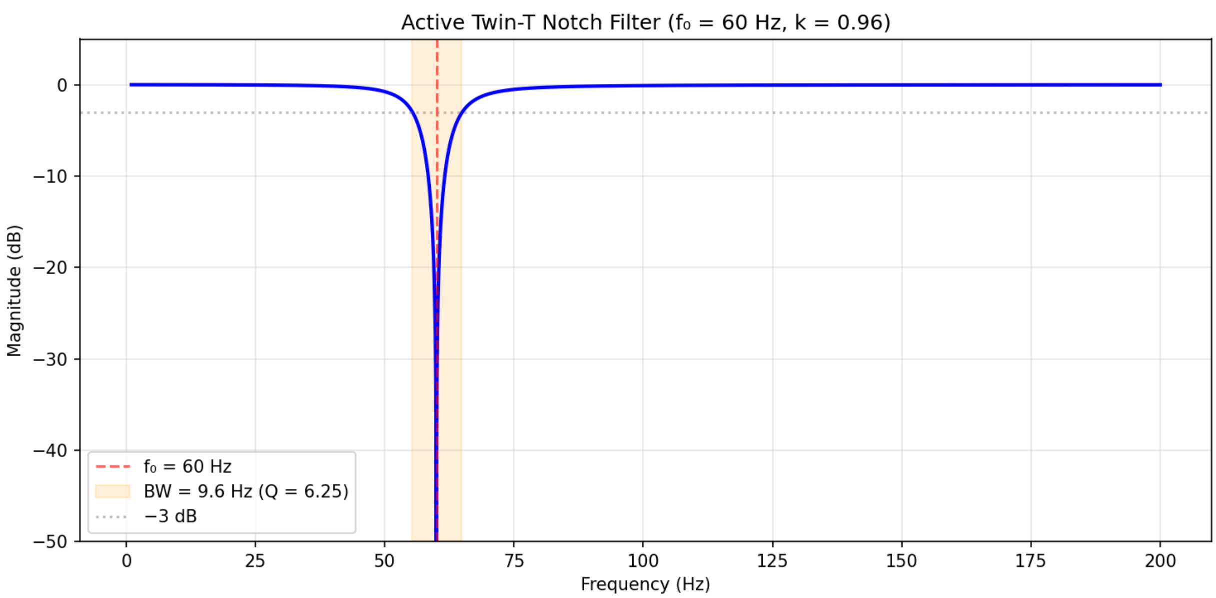



Figure 13.5.5: Active Twin-T Notch Filter

## 13.6 Comparators

Comparators operate the op-amp without negative feedback, exploiting the extremely high open-loop gain to produce a binary output that indicates which of two input voltages is larger. They serve as the interface between the analog and digital domains, converting continuous signals into logic levels for threshold detection, waveform shaping, and analog-to-digital conversion. Adding positive feedback creates a Schmitt trigger with hysteresis, which provides noise immunity essential for reliable switching in real-world noisy environments.

### 13.6.1 Basic Comparator

A comparator operates the op-amp in open-loop (no negative feedback), so the high gain drives the output to one of two saturation levels depending on which input is larger. When $V+ > V-$, the output goes to positive saturation ($V_{OH}$, typically $V_{CC}$ - 1 V for op-amps or $V_{CC}$ for rail-to-rail comparators); when V+ < V−, the output goes to negative saturation ($V_{OL}$, typically $V_{EE}$ + 1 V or ground). Dedicated comparator ICs (such as the LM311, LM339, and TLV3502) are optimized for this function with faster response times (propagation delays of nanoseconds to microseconds), open-collector or open-drain outputs for flexible interfacing, and no compensation capacitor (which would slow the response). Comparators are used in level detection, zero-crossing detection, analog-to-digital conversion, and window comparator circuits.

**Example 13.6.1:** A comparator with $V_{OH}$ = 5.0 V and $V_{OL}$ = 0 V has its inverting input connected to a 2.5 V reference. The non-inverting input receives a 1 kHz sine wave of 3 V peak centered at 2.5 V ($V_{in} = 2.5 + 3\sin(2\pi \times 1000t)$). Determine the output waveform and the fraction of each period the output is high.
**Solution:**
The comparator output is high when $V_{in} > V_{ref}$ = 2.5 V, i.e., when $2.5 + 3\sin(2\pi \times 1000t) > 2.5$, which simplifies to $\sin(2\pi \times 1000t) > 0$.
This is true for $0 < t < T/2$ (the positive half-cycle).
The output is a square wave at 1 kHz with 50% duty cycle, $V_{OH}$ = 5.0 V for the first half-period and $V_{OL}$ = 0 V for the second half-period.
If the reference were shifted to 4.0 V: the condition becomes $3\sin(2\pi \times 1000t) > 1.5$, or $\sin(\omega t) > 0.5$, which is true for $\pi/6 < \omega t < 5\pi/6$.
Duty cycle = $(5\pi/6 - \pi/6) / (2\pi) = (4\pi/6) / (2\pi)$ = 33.3%.

### 13.6.2 Schmitt Trigger

The Schmitt trigger adds positive feedback (from the output to the non-inverting input) to a comparator, creating hysteresis — two different threshold voltages for rising and falling input transitions. The upper threshold $V_{TH}$ (at which the output switches low-to-high) is higher than the lower threshold $V_{TL}$ (at which it switches high-to-low), with the hysteresis voltage $V_H = V_{TH} - V_{TL}$. Hysteresis prevents rapid output oscillation (chatter) when a noisy input signal hovers near the threshold, which is the primary problem with a basic comparator used on noisy signals. An **inverting** Schmitt trigger places the input signal at the inverting terminal; with $R_1$ from the output to the non-inverting input and $R_2$ from the non-inverting input to ground, the thresholds are $V_{TH} = V_{OH} \times R_2/(R_1 + R_2)$ and $V_{TL} = V_{OL} \times R_2/(R_1 + R_2)$. A **non-inverting** Schmitt trigger places the input signal at the non-inverting terminal with a fixed reference at the inverting terminal; positive feedback through $R_1$ and $R_2$ shifts the effective threshold up when the output is high and down when the output is low, producing the same hysteresis behavior with an in-phase output.

**Example 13.6.2:** Design a **non-inverting** Schmitt trigger using a comparator with $V_{OH}$ = 5.0 V and $V_{OL}$ = 0 V that has thresholds of $V_{TH}$ = 3.0 V and $V_{TL}$ = 2.0 V for cleaning up a noisy digital signal. Calculate the required $R_1$ and $R_2$ ratio and the hysteresis voltage. If $R_2$ = 10 kΩ, find $R_1$.
**Solution:**
For this non-inverting Schmitt trigger with a reference at the inverting input: $V_{TH} = V_{ref} + (V_{OH} - V_{ref}) \times R_2/(R_1 + R_2)$ and $V_{TL} = V_{ref} + (V_{OL} - V_{ref}) \times R_2/(R_1 + R_2)$.
With $V_{ref}$ at the midpoint: $V_{ref} = (V_{TH} \times V_{OL} - V_{TL} \times V_{OH}) / (V_{OL} - V_{OH} + V_{TH} - V_{TL}) = (3.0 \times 0 - 2.0 \times 5.0) / (0 - 5.0 + 3.0 - 2.0) = -10 / -4.0 = 2.5$ V.
Feedback ratio: $R_2/(R_1 + R_2) = (V_{TH} - V_{ref}) / (V_{OH} - V_{ref}) = (3.0 - 2.5) / (5.0 - 2.5) = 0.5/2.5 = 0.2$.

With $R_2 = 10\ k\Omega$: $10{,}000/(R_1 + 10{,}000) = 0.2$, so $R_1 + 10{,}000 = 50{,}000$, $R_1 = 40\ k\Omega$.
Hysteresis: $V_H = V_{TH} - V_{TL} = 3.0 - 2.0 = 1.0$ V.
Any noise with amplitude less than 1.0 V peak-to-peak will not cause false triggering.

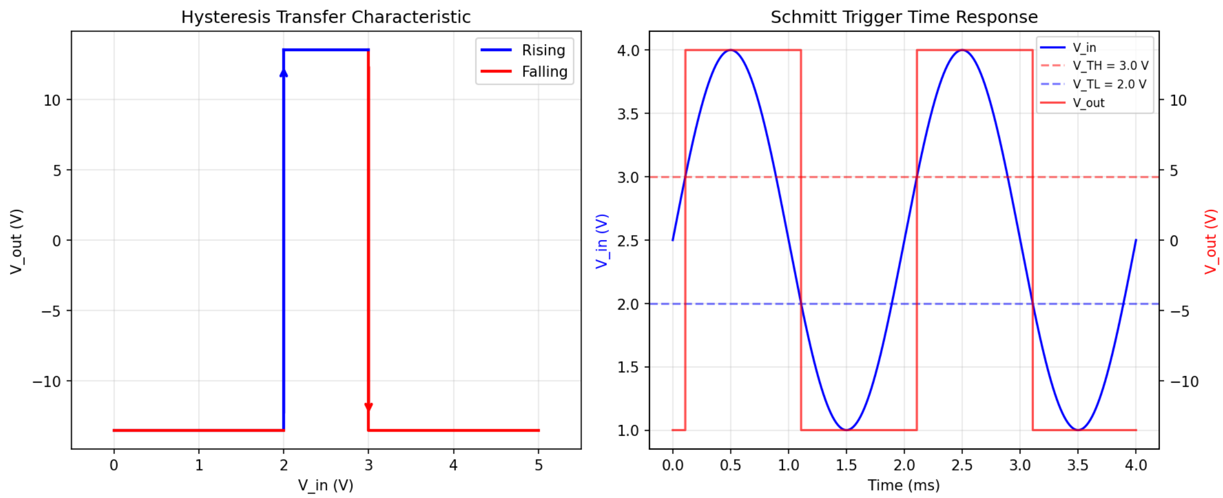



Figure 13.6.2: Schmitt Trigger Hysteresis

### 13.6.3 Wien Bridge and Phase-Shift Oscillators

Op-amp oscillators generate continuous periodic waveforms (sinusoidal, square, or triangular) without an external input signal, using positive feedback to sustain oscillation at a frequency determined by the RC network in the feedback loop. The **Barkhausen criterion** requires that the loop gain equals exactly 1 ($|A\beta| = 1$) and the total phase shift around the loop equals 0° (or 360°) at the oscillation frequency. The **Wien bridge oscillator** is the most popular op-amp sine wave generator, using a series RC and parallel RC network (Wien network) in the positive feedback path that produces zero phase shift at $f_0 = 1/(2\pi RC)$, where the network's attenuation is 1/3. The non-inverting amplifier must therefore provide a gain of exactly 3 ($R_f = 2R_1$) to satisfy the Barkhausen gain condition. In practice, the gain is set slightly above 3 to ensure oscillation startup, and an amplitude stabilization mechanism — either a nonlinear element (small incandescent lamp whose resistance increases with amplitude) or an AGC circuit using a FET or analog multiplier — limits the output to a clean sinusoid. Without stabilization, the gain excess causes clipping distortion. The **phase-shift oscillator** uses three cascaded RC highpass or lowpass sections to provide the required 180° phase shift at the oscillation frequency, combined with an inverting amplifier that contributes the remaining 180°. For three identical RC highpass sections, the oscillation frequency is $f_0 = 1/(2\pi RC\sqrt{6})$, and the required gain is $|A_v| = 29$ ($R_f = 29R_{in}$). Phase-shift oscillators produce lower distortion than Wien bridge oscillators without amplitude stabilization but have a higher minimum gain requirement and less convenient frequency tuning.

**Example 13.6.3:** Design a Wien bridge oscillator for $f_0 = 10$ kHz using $C = 1.5$ nF. Calculate R, and determine $R_f$ if $R_1 = 10\ k\Omega$. If the gain is set 5% above the critical value for reliable startup, what is the actual $R_f$?

**Solution:**
Wien bridge frequency: $f_0 = 1/(2\pi RC)$, so $R = 1/(2\pi f_0 C) = 1/(2\pi \times 10{,}000 \times 1.5 \times 10^{-9}) = 1/(94.25 \times 10^{-6}) = 10{,}610\ \Omega$ (use 10.7 kΩ or 10 kΩ + 620 Ω in series).
Required gain for oscillation: $A_v = 3$ exactly, so $R_f = 2 \times R_1 = 2 \times 10{,}000 = 20$ kΩ.
With 5% gain excess for startup: $A_v = 3 \times 1.05 = 3.15$, $R_f = (3.15 - 1) \times R_1 = 2.15 \times 10{,}000 = 21.5$ kΩ (use 22 kΩ).
Verification: $f_0 = 1/(2\pi \times 10{,}610 \times 1.5 \times 10^{-9}) = 1/(100.00 \times 10^{-6}) = 10{,}000$ Hz = 10 kHz.
The 5% gain excess will cause slight clipping at the supply rails; adding a 1N4148 diode amplitude limiter or a small incandescent lamp (e.g., #327 lamp, $R_{cold} \approx 50\ \Omega$) in the feedback network stabilizes the amplitude for a clean sinusoidal output with THD < 1%.

### 13.6.4 Relaxation Oscillator and Astable Multivibrator

A relaxation oscillator generates square and triangular waves by alternately charging and discharging a capacitor between two threshold voltages set by a Schmitt trigger (positive feedback comparator). The basic op-amp astable multivibrator connects a capacitor from the output to the inverting input through a resistor R (forming an RC integrator), while a resistive voltage divider ($R_1$ and $R_2$) from the output to the non-inverting input establishes the switching thresholds. The capacitor charges toward the current output level through R; when the capacitor voltage reaches the Schmitt trigger threshold, the output snaps to the opposite rail, and the capacitor begins charging in the reverse direction. The resulting output is a square wave, and the voltage across the capacitor is a triangular (or exponential) wave. The oscillation frequency is $f = 1 / (2RC \times \ln((1 + \beta)/(1 - \beta)))$, where $\beta = R_2/(R_1 + R_2)$ is the feedback fraction. For $\beta = 0.5$ ($R_1 = R_2$), $f = 1/(2RC \times \ln(3)) = 1/(2.197RC)$. A separate **triangle wave generator** uses a Schmitt trigger driving an integrator: the Schmitt trigger produces a square wave that feeds an op-amp integrator, whose output ramps linearly up and down between the Schmitt trigger thresholds, producing a high-quality triangle wave with linear slopes (unlike the exponential waveform of the basic astable). The frequency is $f = 1/(4RC) \times (R_2/R_1)$, where R and C are the integrator components and $R_2/R_1$ sets the Schmitt trigger hysteresis. Relaxation oscillators are simpler than Wien bridge oscillators but produce square and triangular waves rather than sinusoids, and are widely used in function generators, pulse-width modulators, and clock sources where sinusoidal purity is not required.

**Example 13.6.4:** An op-amp astable multivibrator uses R = 22 kΩ, C = 10 nF, and the feedback divider has $R_1 = R_2 = 10$ kΩ ($\beta = 0.5$). The op-amp saturates at ±12 V. Calculate the oscillation frequency, the capacitor voltage swing (threshold voltages), and the duty cycle. If $R_1$ is changed to 20 kΩ ($\beta = 1/3$), what is the new frequency?
**Solution:**
Feedback fraction: $\beta = R_2/(R_1 + R_2) = 10{,}000/20{,}000 = 0.5$.
Threshold voltages: $V_{TH} = +\beta \times V_{sat} = 0.5 \times 12 = +6$ V, $V_{TL} = -\beta \times V_{sat} = 0.5 \times (-12) = -6$ V.
The capacitor swings between -6 V and +6 V.
Frequency: $f = 1/(2RC \times \ln((1 + \beta)/(1 - \beta))) = 1/(2 \times 22{,}000 \times 10 \times 10^{-9} \times \ln(1.5/0.5)) = 1/(440 \times 10^{-6} \times \ln(3)) = 1/(440 \times 10^{-6} \times 1.0986) = 1/(483.4 \times 10^{-6}) = 2{,}069$ Hz ≈ 2.07 kHz.
Duty cycle: 50% (symmetric thresholds and equal charging paths).
With $R_1 = 20$ kΩ: $\beta = 10{,}000/30{,}000 = 1/3$. $f = 1/(440\ \mu s \times \ln((4/3)/(2/3))) = 1/(440\ \mu s \times \ln(2)) = 1/(440\ \mu s \times 0.6931) = 1/(304.9\ \mu s) = 3{,}279$ Hz ≈ 3.28 kHz.
The lower β reduces the threshold voltages to ±4 V (smaller capacitor swing) and the ln term

decreases from ln(3) to ln(2), both of which reduce the period. The frequency increases to 3,280 Hz — 59% higher than the $\beta = 0.5$ case.

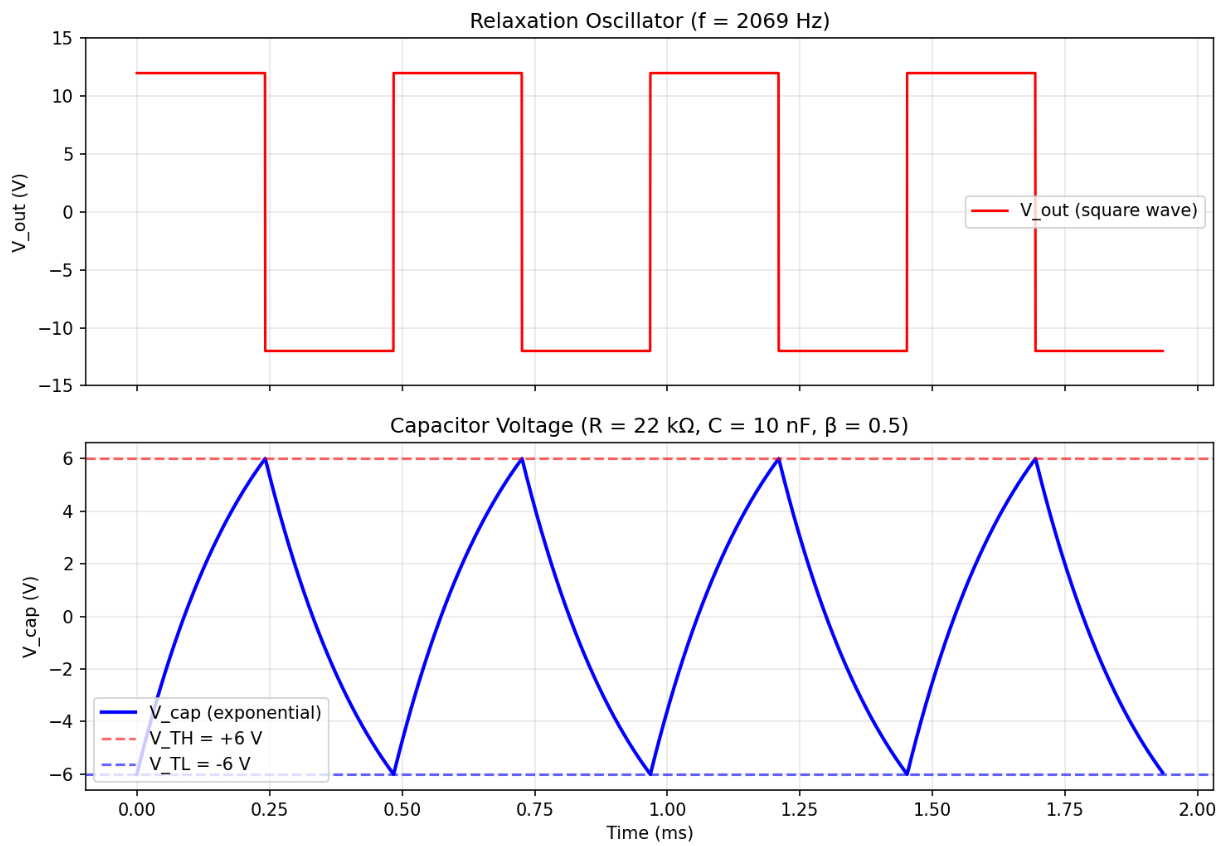



Figure 13.6.4: Relaxation Oscillator Waveforms

## 13.7 Real Op-Amp Limitations

Real op-amps deviate from the ideal model in ways that introduce DC errors, limit bandwidth and output speed, and allow unwanted signals to leak through. Understanding these non-ideal parameters — input offset voltage, bias current, slew rate, CMRR, and PSRR — is essential for predicting circuit accuracy, selecting the right op-amp for an application, and designing compensation techniques that minimize errors. High-precision applications such as instrumentation, data acquisition, and medical electronics require careful attention to these limitations.

### 13.7.1 Input Offset and Bias

Real op-amps have a small but nonzero input offset voltage ($V_{OS}$, typically 0.1–5 mV) that appears as if a small voltage source is connected between the inputs, causing a DC error at the output equal to $V_{OS} \times (1 + R_f/R_{in})$. Input bias current ($I_B$, typically pA for FET-input to μA for bipolar-input op-amps) flows into each input terminal and creates voltage errors when it flows through source resistances. The input offset current ($I_{OS} = |I_{B+} - I_{B-}|$) represents the mismatch between the two input bias currents. The bias current error can be minimized by placing a compensation resistor equal to $R_{in} \parallel R_f$ at the non-inverting input, so that equal bias currents flowing through equal resistances produce a common-mode voltage that is rejected. Auto-zero and chopper-stabilized op-amps achieve offset voltages below 10 μV by continuously measuring and correcting the offset.

**Example 13.7.1:** An inverting amplifier with $R_{in}$ = 10 kΩ and $R_f$ = 1 MΩ (gain = -100) uses a bipolar op-amp with $V_{OS}$ = 2 mV and $I_B$ = 200 nA. Calculate the total output offset voltage with and without a bias compensation resistor at the non-inverting input.
**Solution:**
Without compensation resistor (non-inverting input grounded through 0 Ω):
Output offset due to $V_{OS}$: $V_{out(OS)} = V_{OS} \times (1 + R_f/R_{in}) = 0.002 \times 101 = 202$ mV.
Output offset due to $I_B$ at inverting input: $V_{out(IB)} = I_B \times R_f = 200 \times 10^{-9} \times 1{,}000{,}000 = 200$ mV.
Total output offset: ≈ 202 + 200 = 402 mV (worst case).
With compensation resistor $R_{comp} = R_{in} \parallel R_f = 10{,}000 \parallel 1{,}000{,}000 = 9{,}901$ Ω at the non-inverting input:
The $I_B$ component is now rejected (only $I_{OS}$ matters).
Assuming $I_{OS}$ = 10% of $I_B$ = 20 nA: $V_{out(IOS)} = I_{OS} \times R_f = 20 \times 10^{-9} \times 1{,}000{,}000 = 20$ mV.
Total: ≈ 202 + 20 = **222 mV** — a 45% reduction.

### 13.7.2 Slew Rate

Slew rate (SR) is the maximum rate of change of the output voltage, typically specified in V/μs. It is caused by the limited current available to charge the internal compensation capacitor and is independent of the closed-loop gain. When the required output rate of change exceeds the slew rate, the output cannot follow the input faithfully, causing slew-rate limiting (distortion of fast edges and large-amplitude high-frequency signals). For a sinusoidal output of amplitude $V_p$ at frequency f, the maximum rate of change is $2\pi f V_p$, and the full-power bandwidth is $f_{FP} = SR / (2\pi V_p)$. A general-purpose op-amp like the LM741 has SR = 0.5 V/μs, while high-speed op-amps achieve 1,000-10,000 V/μs.

**Example 13.7.2:** An op-amp with a slew rate of SR = 13 V/μs produces a 10 V peak sine wave. Calculate the full-power bandwidth. If the amplifier is used to amplify a 100 kHz square wave to ±10 V, determine the 10-90% rise time limited by slew rate and whether the output is distorted.
**Solution:**
Full-power bandwidth: $f_{FP} = SR / (2\pi V_p) = 13 \times 10^6 / (2\pi \times 10) = 13 \times 10^6 / 62.83 = 206.9$ kHz.
The op-amp can produce a 10 V peak sine wave without slew-rate distortion up to 207 kHz.
For the square wave: the output must transition from -10 V to +10 V (ΔV = 20 V).
Slew-limited rise time: $t_{rise} = \Delta V / SR = 20 / (13 \times 10^6) = 1.538$ μs.
The 10–90% rise time: $t_{r(10\text{-}90)} = 0.8 \times \Delta V / SR = 16 / (13 \times 10^6)$ s = 1.231 μs.
At 100 kHz, the period is 10 μs and the half-period is 5 μs.
Since the rise time (1.54 μs) is 30.8% of the half-period, the square wave output will have noticeably sloped edges but will reach the full ±10 V before the next transition.

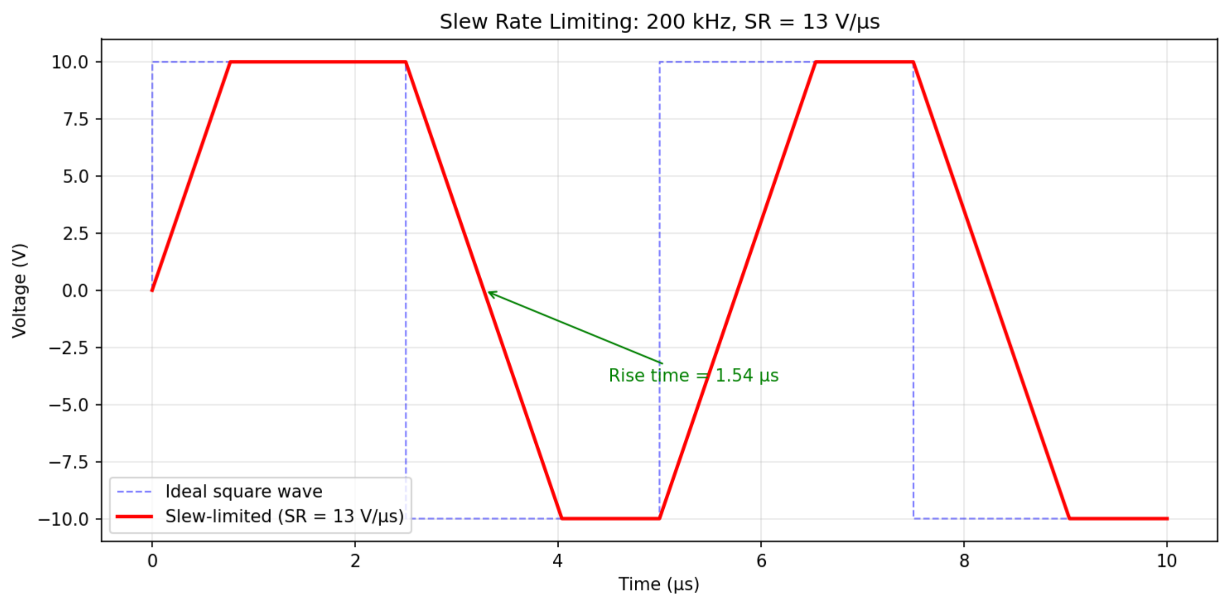



Figure 13.7.2: Slew Rate Limiting

### 13.7.3 Common-Mode Rejection Ratio (CMRR)

The Common-Mode Rejection Ratio quantifies how well an op-amp rejects signals that appear equally on both inputs, defined as CMRR = $A_{diff}$ / $A_{cm}$, typically expressed in decibels: CMRR(dB) = $20\log_{10}(A_{diff} / A_{cm})$. An ideal op-amp has infinite CMRR ($A_{cm} = 0$), meaning common-mode signals produce no output. Real op-amps achieve CMRR of 70–120 dB, which degrades with frequency (typically -20 dB/decade above a few hundred hertz). In practice, the effective CMRR of a circuit depends on both the op-amp's intrinsic CMRR and the matching of external resistors. A common-mode voltage $V_{cm}$ produces an equivalent differential input error of $V_{cm}$ / CMRR, which is then amplified by the closed-loop gain.

**Example 13.7.3:** An op-amp with CMRR = 90 dB is used in a non-inverting amplifier with a gain of 50. The input signal is 10 mV differential riding on a 5 V common-mode voltage (e.g., from a remote sensor). Calculate the desired output, the common-mode error at the output, and the signal-to-error ratio.
**Solution:**
CMRR = 90 dB = 10^(90/20) = 31,623.
Desired output: $V_{out(signal)} = 50 \times 10$ mV = 500 mV.
Equivalent input error from common-mode: $V_{error(input)} = V_{cm}$ / CMRR = 5.0 / 31,623 = 158 μV.
Error at output: $V_{error(output)} = 50 \times 158$ μV = 7.91 mV.
Signal-to-error ratio: 500 / 7.91 = 63.2, or $20\log_{10}(63.2) = 36.0$ dB.
Output accuracy: $7.91/500 \times 100\% = 1.58\%$.
For a 1 mV input signal (50 mV output), the error becomes 15.8% — demonstrating why instrumentation amplifiers with >100 dB CMRR are needed for small signals in the presence of large common-mode voltages.

### 13.7.4 Power Supply Rejection Ratio (PSRR)

The Power Supply Rejection Ratio measures how well an op-amp rejects noise and ripple on its power supply rails, defined as PSRR = $\Delta V_{supply}$ / $\Delta V_{OS}$, where $\Delta V_{OS}$ is the resulting change in input offset voltage caused by a change $\Delta V_{supply}$ in the supply voltage. Expressed in decibels: PSRR(dB) = $20\log_{10}(\Delta V_{supply} / \Delta V_{OS})$. Typical DC PSRR values range from 80 to 120 dB, but PSRR degrades significantly with frequency (typically -20 dB/decade), falling to 40–60 dB at 10 kHz and even lower at higher frequencies. This frequency dependence means that high-frequency switching noise from DC-DC converters can couple through the op-amp's supply pins into the signal path. Proper power supply bypassing (100 nF ceramic capacitor placed as close as possible to each supply pin, supplemented by a 10 μF bulk capacitor) is essential to minimize supply-borne noise. In mixed-signal systems, separate analog and digital supply planes with LC or ferrite-bead filtering are used to prevent digital switching noise from degrading analog performance.

**Example 13.7.4:** An op-amp with DC PSRR = 100 dB and PSRR at 100 kHz = 50 dB is powered by a 5 V supply with 20 $mV_{pp}$ ripple at 100 kHz from a nearby switching regulator. The op-amp is configured as a non-inverting amplifier with a gain of 20. Calculate the equivalent input noise from the supply ripple and the output ripple voltage. Compare with a bypassed supply that reduces the ripple to 0.5 $mV_{pp}$ at 100 kHz.
**Solution:**
PSRR at 100 kHz = 50 dB = 10^(50/20) = 316.2.
Equivalent input offset change: $\Delta V_{OS} = \Delta V_{supply}$ / PSRR = 20 mV / 316.2 = 63.2 $\mu V_{pp}$.
Output ripple: $V_{ripple(out)} = \Delta V_{OS} \times A_{CL}$ = 63.2 μV × 20 = 1.264 $mV_{pp}$.
With proper bypassing (0.5 $mV_{pp}$ ripple): $\Delta V_{OS}$ = 0.5 mV / 316.2 = 1.58 $\mu V_{pp}$.
Output ripple: 1.58 × 20 = 31.6 $\mu V_{pp}$.
The bypass capacitor provides a 40× (32 dB) improvement.
For a 12-bit, 5 V ADC (1 LSB = 1.22 mV), the unbypassed ripple of 1.264 mV exceeds 1 LSB and would corrupt the measurement, while the bypassed ripple of 31.6 μV is well below 1 LSB.

### 13.7.5 Op-Amp Noise Analysis

Every op-amp generates internal noise that sets a fundamental limit on the smallest signal the circuit can resolve. Op-amp noise is modeled as two uncorrelated sources referred to the input: a **voltage noise source** $e_n$ (in nV/$\sqrt{}$Hz) in series with the non-inverting input, and a **current noise source** $i_n$ (in pA/$\sqrt{}$Hz or fA/$\sqrt{}$Hz) flowing into each input terminal. Both have two spectral regions: a flat **white noise** region at higher frequencies and a rising **1/f (flicker) noise** region below a corner frequency $f_c$ (typically 1–100 Hz for bipolar op-amps and 10–1,000 Hz for JFET/CMOS types). In the 1/f region the noise density increases as 1/$\sqrt{}$f, so the power spectral density rises as 1/f — hence the name. Bipolar-input op-amps (e.g., OP27, AD797) have low voltage noise (1–3 nV/$\sqrt{}$Hz) but higher current noise (0.4–2 pA/$\sqrt{}$Hz) due to base current shot noise, making them optimal with low source impedances (<10 kΩ). JFET and CMOS op-amps (e.g., OPA140, ADA4625) have very low current noise (1–10 fA/$\sqrt{}$Hz) but moderately higher voltage noise (4–15 nV/$\sqrt{}$Hz), making them ideal for high-impedance sources (>100 kΩ). The **total input-referred noise** of an op-amp circuit is: $e_{n,total} = \sqrt{}(e_n{}^2 + (i_n \times R_S)^2 + 4kTR_S)$, where $R_S$ is the effective source resistance seen by each input and $4kTR_S$ is the Johnson (thermal) noise of the source resistance (at 25°C, 4kT = $1.66 \times 10^{-20}$ $V^2$/Hz per ohm). To obtain the total RMS noise, the noise spectral density is integrated over the circuit bandwidth: $V_{n,rms} = e_{n,total} \times \sqrt{}(BW)$, where BW is the equivalent noise bandwidth (1.57 × $f_{3dB}$ for a single-pole system). For circuits with significant 1/f noise contribution, the integral includes a $\ln(f_H/f_L)$ term for the 1/f region.

**Example 13.7.5:** A non-inverting amplifier with a gain of 100 uses an op-amp with $e_n = 4$ nV/$\sqrt{}$Hz (white noise) and $i_n = 1.2$ pA/$\sqrt{}$Hz. The source resistance is $R_S = 1$ kΩ and the circuit bandwidth is 10 kHz. The 1/f corner is at 5 Hz (negligible for this bandwidth). Calculate the total input-referred noise, the output noise, and the minimum detectable signal (SNR = 1).
**Solution:**
Source resistance thermal noise density: $e_R = \sqrt{}(4kTR_S) = \sqrt{}(1.66 \times 10^{-20} \times 1{,}000) = \sqrt{}(1.66 \times 10^{-17}) = 4.07$ nV/$\sqrt{}$Hz.
Current noise contribution: $e_{in} = i_n \times R_S = 1.2 \times 10^{-12} \times 1{,}000 = 1.2$ nV/$\sqrt{}$Hz.
Total input-referred noise density: $e_{n,total} = \sqrt{}(4^2 + 1.2^2 + 4.07^2) = \sqrt{}(16 + 1.44 + 16.56) = \sqrt{}34.0 = 5.83$ nV/$\sqrt{}$Hz.
The voltage noise and thermal noise dominate; current noise is negligible at this low source impedance.
Equivalent noise bandwidth (single-pole): $BW_n = 1.57 \times 10{,}000 = 15{,}700$ Hz.
Total input-referred RMS noise: $V_{n,rms} = 5.83 \times 10^{-9} \times \sqrt{}15{,}700 = 5.83 \times 10^{-9} \times 125.3 = 730$ nV $= 0.73\ \mu V_{rms}$.
Output noise: $V_{n,out} = 0.73\ \mu V \times 100 = 73\ \mu V_{rms}$.
Minimum detectable signal (SNR = 0 dB): 0.73 $\mu V_{rms}$ at the input, or about 2.1 $\mu V_{pp}$ (using a 2.9× crest factor).
For a 10 mV signal, the SNR = $20\log_{10}$(10 mV / 0.73 μV) = 82.7 dB.

### 13.7.6 Input and Output Voltage Range Limitations

Real op-amps have finite input and output voltage ranges that constrain circuit operation. The **input common-mode range** (ICMR) specifies the range of voltages that can be applied to both inputs simultaneously without degrading performance — exceeding the ICMR causes a dramatic drop in CMRR, increased offset, or in some op-amps (notably those with PNP input stages), **input phase reversal**, where the output suddenly inverts polarity when an input exceeds the ICMR, potentially latching up positive-feedback circuits. Standard op-amps (e.g., LM741) have an ICMR that extends from about $V_{EE} + 2$ V to $V_{CC} - 2$ V, losing 2 V of headroom at each rail. **Rail-to-rail input** (RRI) op-amps use complementary NMOS/PMOS (or NPN/PNP) input differential pairs that hand off as the input voltage traverses the supply range, extending the ICMR to include both supply rails — but the handoff region (crossover) introduces a step change in $V_{OS}$ and $I_B$, degrading linearity and CMRR in the crossover zone. The **output voltage swing** specifies how close the output can approach the supply rails under load. Standard op-amps with bipolar output stages typically swing to within 1–2 V of each rail (e.g., ±13 V output from ±15 V supplies), while **rail-to-rail output** (RRO) op-amps use CMOS output stages that can swing to within 10–50 mV of each rail with light loads (but the saturation voltage increases with load current: $V_{sat} = I_{load} \times R_{DS(on)}$). Single-supply designs ($V_{CC} = 3.3$ V or 5 V, $V_{EE} = 0$ V) require rail-to-rail input and output (RRIO) op-amps to maximize the usable signal range, because even a 0.5 V headroom loss at each rail reduces the available swing to 2.3 V from a 3.3 V supply — a 30% dynamic range penalty. Key specifications to evaluate include: maximum output current (short-circuit current, typically 20–40 mA), output impedance in the linear region (<1 Ω) vs. near the rails (rising sharply), and capacitive load stability (many RRO op-amps oscillate with loads >100 pF without a series isolation resistor of 10–100 Ω).

**Example 13.7.6:** A single-supply ($V_{CC}$ = 3.3 V, $V_{EE}$ = 0 V) non-inverting amplifier with a gain of 2 uses an op-amp with output swing specifications of $V_{OL}$ = 50 mV and $V_{OH} = V_{CC} - 80$ mV at $I_{load}$ = 5 mA, and $R_{DS(on)} = 40\ \Omega$ for the output FETs. The input is a 0–1.5 V sensor signal biased at 0.825 V. Calculate the maximum undistorted output swing, and determine the output swing and saturation voltage if the load draws 10 mA.

**Solution:**

At $I_{load}$ = 5 mA: $V_{OL}$ = 50 mV, $V_{OH} = 3.3 - 0.08 = 3.22$ V. Output swing $= 3.22 - 0.05 = 3.17\ V_{pp}$.
With gain = 2: output range $= 2 \times 0$ to $2 \times 1.5 = 0$ to 3.0 V. The output can swing from 0.05 V to 3.0 V without clipping (3.0 V < 3.22 V).
At $I_{load}$ = 10 mA: $V_{sat} = I_{load} \times R_{DS(on)} = 0.010 \times 40 = 0.4$ V. $V_{OL} \approx 0.4$ V, $V_{OH} \approx 3.3 - 0.4 = 2.9$ V.
Output swing $= 2.9 - 0.4 = 2.5\ V_{pp}$.
The amplifier output at $2 \times 1.5 = 3.0$ V exceeds $V_{OH}$ = 2.9 V, causing clipping on the positive peaks.
Maximum undistorted input at 10 mA: $V_{in,max} = V_{OH}$ / gain = 2.9 / 2 = 1.45 V.
The 50 mV loss at the low end also clips: $V_{in,min} = V_{OL}$ / gain = 0.4 / 2 = 0.2 V.
The usable input range shrinks from 0–1.5 V to 0.2–1.45 V — a 17% reduction in dynamic range due to the heavier load.
Using a lighter load (>10 kΩ) or a buffer stage would restore the full swing.

# Chapter 14

# National Electrical Code (NEC)

The National Electrical Code (NEC), published as NFPA 70, establishes the minimum requirements for safe electrical installations in the United States and is adopted by reference in most state and local jurisdictions. While the NEC is fundamentally a safety document rather than a design manual, its rules govern the practical implementation of nearly every electrical engineering calculation — from conductor sizing and overcurrent protection to motor circuit design and transformer installations. The code is organized into nine chapters plus annexes, with articles numbered to group related topics (e.g., Article 310 for conductors, Article 430 for motors), and article numbers remain stable across editions even as specific requirements evolve. This chapter covers the engineering calculations and principles behind the most frequently used NEC articles, giving the electrical engineer the quantitative tools to translate a design into a code-compliant installation.

## 14.1 NEC Organization and Structure

### 14.1.1 Code Organization

The NEC's nine-chapter structure follows a logical progression from general definitions and requirements (Chapter 1) through wiring design (Chapter 2), wiring methods (Chapter 3), equipment (Chapter 4), special occupancies (Chapter 5), special equipment (Chapter 6), special conditions (Chapter 7), communications (Chapter 8), and reference tables (Chapter 9). A critical rule is that Chapters 5, 6, and 7 can supplement or modify the requirements of Chapters 1 through 4, meaning that a special-condition article may override a general requirement. Engineers must also understand that the NEC sets minimum standards for safety — it does not guarantee an adequate, efficient, or expandable installation.

**Example 14.1.1:** An engineer is designing a motor circuit in a Class I hazardous (classified) location. Article 430 (Chapter 4) requires motor branch circuit conductors to be rated at 125% of the motor full-load current. Article 501 (Chapter 5) for Class I hazardous locations requires the wiring method to be rigid metal conduit with threaded fittings. If the motor FLC is 28 A, determine the minimum conductor ampacity required by Article 430 and identify which chapter's wiring method rules take precedence.
**Solution:**
Minimum conductor ampacity per Article 430: $I_{min} = 1.25 \times 28 = 35$ A.
Per the NEC hierarchy, Article 501 (Chapter 5) supplements and can modify Article 430 (Chapter 4) requirements.
Therefore, the wiring method must comply with Article 501 (rigid metal conduit with threaded fittings), while the conductor sizing still follows Article 430.

A 10 AWG THWN-2 copper conductor rated at 40 A at 90°C (35 A at 75°C termination limit) satisfies the 35 A minimum ampacity requirement.

### 14.1.2 Key Tables and References

Several NEC tables are used so frequently that engineers should know them by article number. Table 310.16 provides allowable ampacities for insulated conductors rated 0–2000 V in raceway, cable, or direct burial based on conductor size, insulation temperature rating, and conductor material. Table 310.15(C)(1) provides ambient temperature correction factors, and the conduit fill adjustment factors account for mutual heating when multiple conductors share a raceway. Table 250.122 sizes equipment grounding conductors based on the overcurrent protective device rating. Table 430.250 lists full-load currents for three-phase AC motors, and Table 430.52 provides maximum overcurrent device ratings for motor branch circuits. Chapter 9, Table 1 gives conduit fill percentages, and Chapter 9, Tables 4 and 5 provide conduit and conductor dimensions for performing conduit fill calculations.

**Example 14.1.2:** An engineer must size conductors for a 200 A feeder installed in EMT conduit in an ambient temperature of 38°C. The conductors are THWN-2 copper (rated 90°C). Using Table 310.16 and the temperature correction factors, determine the minimum conductor size.
**Solution:**
From Table 310.16, the 75°C column governs for termination purposes per 110.14(C). At 75°C, 3/0 AWG copper is rated at 200 A.
However, the 90°C ampacity can be used for derating purposes. From Table 310.16, 1/0 AWG copper at 90°C is rated at 170 A.
Temperature correction factor for 90°C conductors at 38°C ambient: 0.91.
Corrected ampacity: $170 \times 0.91 = 154.7$ A — insufficient.
For 2/0 AWG at 90°C: $195 \times 0.91 = 177.5$ A — still insufficient.
For 3/0 AWG at 90°C: $225 \times 0.91 = 204.8$ A — exceeds 200 A and does not exceed the 75°C termination rating of 200 A.
Minimum conductor size: 3/0 AWG copper THWN-2.

## 14.2 Conductor Sizing and Ampacity

Proper conductor sizing ensures that wiring can safely carry the required current without excessive heating, while also meeting voltage drop performance goals. The NEC uses a multi-factor approach: base ampacity from Table 310.16 is modified by temperature correction factors for elevated ambient temperatures and conduit fill adjustment factors when multiple conductors share a raceway. The physical conduit size must also accommodate the total cross-sectional area of all conductors per the conduit fill requirements of Chapter 9. Conductor sizing is often the first calculation an electrical engineer performs and the foundation for all downstream protection and equipment selections.

### 14.2.1 Ampacity Tables and Conductor Selection

Table 310.16 is the primary ampacity table for conductors rated 0–2000 V installed in raceway, cable, or direct burial (not more than three current-carrying conductors). The table provides ampacities at three insulation temperature ratings (60, 75, and 90°C) for both copper and aluminum/copper-clad aluminum conductors. Conductor sizing must account for the continuous load rule: conductors supplying continuous loads (loads expected to operate for three hours or more) must have an ampacity of at least 125% of the continuous load plus 100% of the non-continuous load. When selecting a conductor, the engineer must verify that the selected size satisfies all three constraints: ampacity after derating, termination temperature limits, and voltage drop recommendations.

**Example 14.2.1:** A 480 V, three-phase feeder supplies 150 A of continuous load and 40 A of non-continuous load. The conductors are copper THWN-2 (90°C insulation) terminated on equipment rated for 75°C. Determine the minimum required conductor ampacity and select the conductor size from Table 310.16.
**Solution:**
Required ampacity $= 1.25 \times 150 + 1.00 \times 40 = 187.5 + 40 = 227.5$ A.
From Table 310.16 at 75°C (termination limit): 4/0 AWG copper = 230 A. Check: 230 A $\geq$ 227.5 A.
The actual load current is $150 + 40 = 190$ A, and the overcurrent device must be rated at least 227.5 A (next standard size: 250 A per 240.6(A)).
Minimum conductor size: 4/0 AWG copper THWN-2, protected by a 250 A overcurrent device.

### 14.2.2 Temperature Correction Factors

When the ambient temperature exceeds the standard 30°C assumed in Table 310.16, the conductor ampacity must be reduced using correction factors from Table 310.15(C)(1). The correction factor decreases as ambient temperature increases, reflecting the reduced thermal margin available for $I^2R$ heating of the conductor. For conductors with higher insulation temperature ratings, the correction factors are more favorable (less derating) at elevated ambient temperatures, which is one advantage of specifying 90°C insulation even when terminations limit the usable ampacity.

**Example 14.2.2:** A set of 2 AWG THWN-2 copper conductors (90°C rated) is installed in a rooftop conduit where the ambient temperature is 52°C. Determine the corrected ampacity and compare it to the same conductors with THW insulation (75°C rated).
**Solution:**
From Table 310.16 base ampacities: 2 AWG at 90°C = 130 A; 2 AWG at 75°C = 115 A.
Temperature correction factor at 52°C (51–55°C band): for 90°C insulation, factor = 0.76; for 75°C insulation, factor = 0.67.
Corrected ampacity for THWN-2: $130 \times 0.76 = 98.8$ A.
Corrected ampacity for THW: $115 \times 0.67 = 77.1$ A.
The 90°C insulation retains 76% of its base ampacity while the 75°C insulation retains only 67%, demonstrating a 28% advantage in current-carrying capacity for the higher-rated insulation in high-temperature environments.

### 14.2.3 Conduit Fill Adjustment Factors

When more than three current-carrying conductors are installed in a single raceway or cable, the heat generated by each conductor raises the temperature of all adjacent conductors, requiring an ampacity adjustment. For 4–6 conductors the adjustment factor is 0.80, for 7–9 conductors it is 0.70, and the factor continues to decrease as the number of conductors increases. Neutral conductors that carry only unbalanced current in a balanced three-phase system are generally not counted as current-carrying conductors, but a neutral conductor that carries harmonic currents (common with nonlinear loads such as computers and VFDs) must be counted per 310.15(E). The conduit fill adjustment can be combined multiplicatively with the ambient temperature correction factor when both conditions apply simultaneously.

**Example 14.2.3:** An electrical contractor installs 12 current-carrying THWN-2 copper conductors (6 circuits of 2 conductors each) in a single conduit. Each conductor is 8 AWG. The ambient temperature is 35°C. Determine the adjusted ampacity per conductor.
**Solution:**
Base ampacity of 8 AWG THWN-2 copper at 90°C from Table 310.16: 55 A.
Ambient temperature correction factor at 35°C for 90°C insulation: 0.96.
Conduit fill adjustment factor for 12 current-carrying conductors (10–20 range): 0.50.
Adjusted ampacity: $55 \times 0.96 \times 0.50 = 26.4$ A.
If the circuit load exceeds 26.4 A per conductor, the engineer must either upsize the conductors, reduce the number of conductors per conduit, or use separate conduits.
For comparison, with only 3 conductors per conduit (no adjustment): $55 \times 0.96 = 52.8$ A — twice the capacity.

### 14.2.4 Conductor Properties and Resistance

NEC Chapter 9, Table 8 provides DC resistance values for copper and aluminum conductors at 75°C, which are essential for voltage drop calculations. The resistance of a conductor is proportional to its length and inversely proportional to its cross-sectional area, and increases with temperature at approximately 0.393% per degree C for copper. For AC circuits, the effective resistance is slightly higher than the DC resistance due to skin effect (current crowding at the conductor surface at 60 Hz), which becomes significant for conductors larger than 250 kcmil; NEC Chapter 9, Table 9 provides AC impedance values including both resistance and reactance for conductors in various raceway types.

**Example 14.2.4:** A 350 kcmil copper conductor carries 300 A in a steel conduit. From NEC Chapter 9, Table 9, the effective AC impedance is $R = 0.0382$ Ω/1000 ft and $X = 0.0441$ Ω/1000 ft. For a 250-foot one-way run supplying a load at 0.85 power factor lagging, calculate the per-phase voltage drop.
**Solution:**
One-way conductor length: 250 ft.
Effective R per conductor: $0.0382 \times 250/1000 = 0.00955$ Ω.
Effective X per conductor: $0.0441 \times 250/1000 = 0.01103$ Ω.
Per-phase voltage drop: $V_{drop} = I \times (R\cos(\theta) + X\sin(\theta)) = 300 \times (0.00955 \times 0.85 + 0.01103 \times 0.527) = 300 \times (0.00812 + 0.00581) = 300 \times 0.01393 = 4.18$ V per phase.
For a 277 V line-to-neutral system: percent drop $= 4.18/277 \times 100 = 1.51\%$.
For a three-phase 480 V circuit, the line-to-line voltage drop $= \sqrt{3} \times 4.18 = 7.24$ V, or $7.24/480 \times 100 = 1.51\%$.

### 14.2.5 Conduit Fill Calculations

NEC Chapter 9, Table 1 limits the percentage of a conduit's internal cross-sectional area that may be occupied by conductors: 53% for one conductor, 31% for two conductors, and 40% for three or more conductors. These fill limits prevent excessive conductor heating and ensure that conductors can be pulled through the raceway without damage. The calculation compares the total cross-sectional area of all conductors (including insulation, from Chapter 9, Table 5) against the allowable fill area of the conduit (from Chapter 9, Table 4). When conductors of different sizes share a conduit, each conductor's area is summed individually. Nipples (conduit sections $\leq$ 24 inches) are permitted 60% fill per Table 1, Note 4, recognizing the reduced thermal concern for very short runs.

**Example 14.2.5:** An electrician must install the following THWN-2 copper conductors in a single EMT conduit: three 4 AWG (power circuit) and four 12 AWG (control circuit). Determine the minimum EMT conduit size using NEC Chapter 9 tables. From Table 5: 4 AWG THWN-2 area = 0.0824 in$^2$, 12 AWG THWN-2 area = 0.0133 in$^2$.
**Solution:**
Total conductor area: $3 \times 0.0824 + 4 \times 0.0133 = 0.2472 + 0.0532 = 0.3004$ in$^2$.
Total conductors = 7 (three or more), so 40% fill applies.
Required conduit area: 0.3004 / 0.40 = 0.7510 in$^2$.
From Chapter 9, Table 4 for EMT: 3/4" EMT has an internal area of 0.533 in$^2$ (insufficient, $0.533 \times 0.40 = 0.213$ in$^2$). 1" EMT has an internal area of 0.864 in$^2$ (40% fill = 0.346 in$^2$ — sufficient).
Minimum conduit size: 1" EMT.
Verification: fill percentage = $0.3004 / 0.864 \times 100 = 34.8\% < 40\%$ limit.

### 14.2.6 Rooftop Conduit Temperature Adders

Conduits and cables installed on or above rooftops exposed to direct sunlight are subject to significantly higher ambient temperatures than indicated by weather data alone. NEC Table 310.15(C)(1)(a) requires a temperature adder to be applied to the outdoor ambient temperature when conductors or cables are installed in conduit exposed to sunlight on or above rooftops. The adder depends on the distance above the roof surface: conduit on the roof surface (touching) adds 33°C (60°F), ½ inch above the roof adds 22°C (40°F), 3½ inches above adds 17°C (30°F), and 12 inches above adds 14°C (25°F). These adders reflect the heat absorbed and re-radiated by the roofing material, which can raise the air temperature inside the conduit well above the outdoor ambient. Conduit installed more than 36 inches above the roof surface is exempt from the adder per the 2023 NEC. The adjusted ambient temperature (weather ambient + roof adder) is then used with the temperature correction factors from Table 310.15(C)(1) to determine the corrected conductor ampacity. This double penalty — the roof adder pushes the effective ambient far above 30°C, and the correction factor reduces the base ampacity accordingly — frequently forces the engineer to upsize conductors by one to three AWG sizes compared to an indoor installation. Rooftop HVAC equipment, solar panel feeder conduit, and rooftop-mounted antenna cabling are common applications where this derating must be considered.

**Example 14.2.6:** A conduit installed directly on a flat roof surface feeds a rooftop HVAC unit. The outdoor ambient temperature is 40°C. The conductors are 6 AWG THWN-2 copper (90°C rated). Calculate the effective ambient temperature, the temperature correction factor, and the derated ampacity. Compare to the same conductor installed indoors at 30°C.
**Solution:**
Roof surface adder from Table 310.15(C)(1)(a): +33°C for conduit on the roof surface.
Effective ambient: $40 + 33 = 73$°C.
Temperature correction factor for 90°C insulation at 73°C ambient (interpolating from Table

310.15(C)(1)): factor ≈ 0.41.
Base ampacity of 6 AWG THWN-2 at 90°C: 75 A.
Derated ampacity: $75 \times 0.41 =$ **30.8 A**.
Indoor comparison at 30°C ambient: no correction needed, ampacity = 75 A (at 90°C rating), but limited to 65 A by 75°C termination rating.
The rooftop installation loses 59% of its current-carrying capacity.
If the HVAC unit draws 40 A, 6 AWG is insufficient on the rooftop — upgrade to 4 AWG (90°C base = 95 A, derated = $95 \times 0.41 = 39.0$ A — still marginal) or 3 AWG (90°C base = 110 A, derated = $110 \times 0.41 = 45.1$ A — sufficient).
Raising the conduit 3½ inches above the roof on standoffs reduces the adder to 17°C: effective ambient = 57°C, which falls in the 56–60°C band, correction factor ≈ 0.71, and 6 AWG derated ampacity = $75 \times 0.71 = 53.3$ A — sufficient for the 40 A load and saving two wire sizes.

## 14.3 Overcurrent Protection

Overcurrent protection is the NEC's primary mechanism for preventing fires and equipment damage from excessive current flow. Article 240 establishes the rules for sizing and applying fuses and circuit breakers, with standard device ratings listed in 240.6(A). The protection scheme must address both sustained overloads (moderate overcurrent for extended periods) and short-circuit faults (thousands of amperes for fractions of a second). Tap rules, coordination between upstream and downstream devices, and available short-circuit current calculations are critical engineering considerations for any distribution system.

### 14.3.1 Standard Overcurrent Device Ratings

NEC Section 240.6(A) lists the standard ampere ratings for fuses and circuit breakers: 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 110, 125, 150, 175, 200, 225, 250, 300, 350, 400, 450, 500, 600, 700, 800, 1000, 1200, 1600, 2000, 2500, 3000, 4000, 5000, and 6000 A. Per 240.4(B), for circuits rated 800 A or less, the next standard size overcurrent device above the conductor ampacity is permitted, provided the conductors are not supplying multi-outlet receptacle branch circuits. For circuits over 800 A, the overcurrent device rating must not exceed the conductor ampacity.

**Example 14.3.1:** A feeder has a calculated load of 185 A (continuous). The conductors must be sized at 125% of continuous load = 231.25 A. From Table 310.16 (75°C copper): 4/0 AWG = 230 A, 250 kcmil = 255 A. Determine the correct conductor size and overcurrent device rating.
**Solution:**
Required conductor ampacity: $1.25 \times 185 = 231.25$ A.
From Table 310.16 at 75°C: 4/0 AWG = 230 A (insufficient, $230 < 231.25$). 250 kcmil = 255 A (sufficient).
Per 240.4(B), the overcurrent device may be the next standard size above the conductor ampacity: the next standard size above 255 A is 300 A.
However, the overcurrent device must also be rated at least 231.25 A for the continuous load rule.
Select 250 kcmil conductors with a 250 A standard breaker, since $250 \geq 231.25$ and $255 \geq 250$.

### 14.3.2 Conductor Protection and Tap Rules

The general rule is that conductors must be protected at their ampacity by an overcurrent device at the point where they receive their supply (240.4). However, Article 240.21 provides important exceptions known as "tap rules" that allow conductors to be supplied from a larger feeder without overcurrent protection at the tap point, provided specific length and termination conditions are met. The 10-foot tap rule (240.21(B)(1)) allows unprotected tap conductors up to 10 feet if they have an ampacity at least 10% of the upstream overcurrent device and terminate in a single overcurrent device that limits the load to the tap conductor ampacity. The 25-foot tap rule (240.21(B)(2)) allows taps up to 25 feet if the conductors have an ampacity of at least one-third of the upstream overcurrent device rating and are protected from physical damage.

**Example 14.3.2:** A 400 A feeder supplies a tap to a panelboard 20 feet away. The tap terminates in a 100 A main breaker in the panelboard. Determine the minimum tap conductor size per the 25-foot tap rule for copper THWN-2 conductors (75°C terminations).
**Solution:**
The 25-foot tap rule (240.21(B)(2)) requires:
(1) tap conductors not exceeding 25 ft — satisfied (20 ft).
(2) Tap conductor ampacity $\geq 1/3$ of upstream device: $400/3 = 133.3$ A. From Table 310.16 at 75°C: 1 AWG = 130 A (insufficient), 1/0 AWG = 150 A (sufficient).
(3) Tap terminates in a single overcurrent device (100 A breaker) — satisfied.
Minimum tap conductor size: 1/0 AWG copper THWN-2 (rated 150 A $\geq$ 133.3 A).
The panelboard's 100 A main breaker protects the downstream circuits, and the 1/0 AWG tap conductors are rated well above the 100 A load they supply.

### 14.3.3 Overcurrent Device Coordination

Coordination (also called selectivity) ensures that when a fault occurs, only the overcurrent device nearest to the fault operates, while upstream devices remain closed to minimize the extent of the outage. This requires that the time-current characteristics of downstream devices (faster) and upstream devices (slower) do not overlap at any fault current level within the system's range. Engineers use manufacturers' time-current characteristic (TCC) curves to verify coordination by plotting the downstream device curve below (faster than) the upstream device curve with adequate margin.

**Example 14.3.3:** A main breaker is rated at 800 A with an adjustable long-time delay and a downstream feeder breaker is rated at 200 A. At a fault current of 2,000 A, the 200 A breaker clears in 0.05 seconds and the 800 A breaker has a minimum clearing time of 0.10 seconds. At 5,000 A, the 200 A breaker clears in 0.02 seconds and the 800 A breaker clears in 0.04 seconds. Determine whether the two devices coordinate at both fault levels.
**Solution:**
At 2,000 A: coordination ratio = upstream time / downstream time = 0.10 / 0.05 = 2.0:1. The upstream device is 2× slower, providing adequate coordination.
At 5,000 A: coordination ratio = 0.04 / 0.02 = 2.0:1. Again, the upstream device is 2× slower.
Both devices coordinate at these fault levels since the upstream device always clears more slowly, allowing the downstream device to isolate the fault first.
A minimum coordination ratio of 1.5:1 is generally considered acceptable.

If both devices operate in the instantaneous trip region at very high fault currents, coordination may be lost — this must be checked at the maximum available fault current.

### 14.3.4 Short-Circuit Current Calculations

NEC Section 110.9 requires that overcurrent devices have an interrupting rating sufficient for the available fault current at their line terminals, and Section 110.10 requires that equipment withstand the available fault current. Calculating the available short-circuit current at each point in the distribution system is therefore a fundamental NEC compliance task. The point-to-point method estimates fault current by starting with the transformer secondary fault current ($I_{fault} = I_{FLA} / Z_{pu}$) and then reducing it through the impedance of each downstream conductor segment using the formula $f = (1.732 \times L \times I_{fault}) / (C \times V_{LL})$ for three-phase circuits, where L is the conductor length in feet and C is a constant based on conductor material and size. The upstream utility contribution is typically obtained from the utility company as the available fault current at the service entrance, or can be calculated from the transformer kVA and impedance assuming an infinite bus on the primary.

**Example 14.3.4:** A 500 kVA, 480Y/277 V, three-phase transformer has an impedance of 5.75%. The secondary conductors are 500 kcmil copper, 30 feet long, to a main distribution panel (MDP). Calculate the available fault current at the transformer secondary and at the MDP. Assume an infinite primary bus. The “C” constant for 500 kcmil copper is 26,706.
**Solution:**
Transformer secondary full-load current: $I_{FLA} = 500{,}000 / (1.732 \times 480) = 500{,}000 / 831.4 = 601.4$ A.
Available fault current at transformer secondary (infinite bus): $I_{sc(xfmr)} = I_{FLA} / Z_{pu} = 601.4 / 0.0575 = 10{,}459$ A.
Multiplier factor for conductor impedance: $f = (1.732 \times 30 \times 10{,}459) / (26{,}706 \times 480) = 543{,}450 / 12{,}818{,}880 = 0.04239$.
Available fault current at MDP: $I_{sc(MDP)} = I_{sc(xfmr)} / (1 + f) = 10{,}459 / 1.04239 = 10{,}034$ A.
All overcurrent devices at the MDP must have an interrupting rating ≥ 10,034 A (select standard 10 kAIC or 14 kAIC rated devices).
Standard molded-case circuit breakers are typically rated 10 kAIC or 14 kAIC at 480 V.

### 14.3.5 Arc-Fault Circuit Interrupter (AFCI) Protection

Arc-fault circuit interrupters detect dangerous arcing conditions — series arcs (caused by damaged conductors, loose connections, or broken wires) and parallel arcs (line-to-neutral or line-to-ground faults through carbonized insulation) — that may not draw enough current to trip a standard overcurrent device but generate sufficient heat to ignite surrounding combustible materials. NEC Section 210.12 requires AFCI protection for all 120 V, single-phase, 15 A and 20 A branch circuits supplying outlets and devices in dwelling unit bedrooms, living rooms, hallways, closets, kitchens, laundry areas, and most other habitable spaces. The requirement has expanded with each code cycle: the 1999 NEC required AFCI only in bedrooms, while the 2023 NEC requires it in virtually all dwelling unit areas except bathrooms (which require GFCI) and garages. Combination-type AFCIs (the only type currently accepted by the code) detect both series and parallel arcs using electronic algorithms that distinguish dangerous arcing signatures from normal arcing events such as motor brush commutation, switch operation, and plug insertion. AFCIs are available as circuit breakers (the most common form), receptacle-type devices (outlet branch circuit AFCIs), and portable devices. The AFCI must be listed to UL 1699 and installed at the origin of the branch circuit, though 210.12(A) permits alternative means (listed outlet branch circuit AFCI receptacles at the first outlet) when it is impractical to install an AFCI breaker at the origin of the branch circuit. AFCI devices trip at arc signatures with as few as 8 half-cycles of arcing current, providing fire protection that standard breakers cannot offer — a 5 A series arc through a damaged cord in a 15 A circuit would never trip a standard breaker but generates enough heat to ignite wood or fabric.

**Example 14.3.5:** A dwelling unit has a 20 A, 120 V branch circuit supplying six duplex receptacles in a living room. The circuit is wired with 12 AWG NM-B cable (Romex) and is 85 feet from the panelboard to the first outlet. An AFCI breaker is installed per 210.12. If a series arc develops in a damaged section of cord plugged into the farthest outlet, with an arc current of 5 A at 120 V, calculate the arc power, explain why a standard breaker would not trip, and confirm AFCI applicability.
**Solution:**
Arc power: $P_{arc} = V \times I = 120 \times 5 = 600$ W.
A standard 20 A breaker trips thermally at approximately 135% of rating (27 A) after 1 hour, or magnetically at 5–10× rating (100–200 A) for instantaneous trip.
The 5 A arc current is only 25% of the breaker rating — far below any trip threshold — and would dissipate 600 W indefinitely, easily igniting nearby combustible materials (paper ignites at ~230°C, wood at ~300°C).
The AFCI breaker detects the characteristic irregular current waveform of the series arc (random variations in arc impedance as the gap ionizes and deionizes each half-cycle) and trips within 8–60 half-cycles (67–500 ms at 60 Hz).
Per 210.12(A), living room outlets in a dwelling unit require AFCI protection.
The NM-B wiring method is permitted and does not affect the AFCI requirement.

## 14.4 Grounding and Bonding

Grounding and bonding form the safety backbone of any electrical installation, providing a low-impedance fault current return path that enables overcurrent devices to clear ground faults quickly. System grounding connects one conductor of the electrical system to earth, while equipment grounding provides a metallic return path from enclosures and equipment frames back to the source. Bonding ensures electrical continuity between all metallic components so that no dangerous voltage differences can exist between exposed surfaces. Articles 250 and related sections establish one of the most detailed and critical areas of the NEC.

### 14.4.1 System Grounding

System grounding connects one conductor of the electrical system (typically the neutral in a wye-connected system) to earth through a grounding electrode at the service entrance or separately derived system. For solidly grounded systems (the most common configuration for services under 1000 V), the neutral-to-ground connection is made at the service disconnecting means, creating a low-impedance ground fault return path through the grounding electrode conductor and the equipment grounding conductors. The grounding electrode system (Article 250, Part III) requires bonding together all available electrodes — metal water pipe, building steel, concrete-encased electrode (Ufer ground), and ground rods — to create a robust earth connection with the lowest practical impedance.

**Example 14.4.1:** A 208Y/120 V, three-phase, 400 A service has a grounding electrode system consisting of a 10-foot driven ground rod and a concrete-encased electrode. Using Table 250.66, determine the minimum grounding electrode conductor (GEC) size for copper, and calculate the ground fault current if the total ground fault impedance is 0.25 Ω.
**Solution:**
From Table 250.66 for service entrance conductors of 500 kcmil copper (typical for a 400 A service): the minimum grounding electrode conductor is 1/0 AWG copper.
Ground fault current on a 120 V circuit (line to neutral/ground): $I_{fault} = V / Z_{total} = 120 / 0.25 = 480$ A.
This fault current is well above the 20 A branch circuit breaker rating, ensuring the breaker trips quickly.
Note: the majority of the fault current returns through the equipment grounding conductor (metallic path), not through the earth — earth impedance is typically far too high to clear faults reliably, which is why equipment grounding conductors are required.

### 14.4.2 Equipment Grounding Conductors

Equipment grounding conductors (EGCs) provide the low-impedance return path for ground fault current from the point of the fault back to the source (transformer or generator neutral), enabling the overcurrent device to operate and clear the fault. Table 250.122 sizes the EGC based on the rating of the upstream overcurrent device, not on the circuit conductor size, ensuring the grounding conductor can carry the expected fault current for the time required for the overcurrent device to operate. EGCs can be a wire (insulated or bare), a metallic raceway (rigid metal conduit, intermediate metal conduit, or electrical metallic tubing), or the metal armor/sheath of a cable assembly. For larger installations, Section 250.122(B) requires the EGC to be proportionally upsized when ungrounded conductors are increased in size to compensate for voltage drop.

**Example 14.4.2:** A 480 V feeder is protected by a 200 A circuit breaker and uses 3/0 AWG copper conductors in rigid metal conduit. The feeder is 300 feet long, and the ungrounded conductors were increased from the minimum 3/0 AWG to 250 kcmil to limit voltage drop. Determine the minimum EGC size from Table 250.122 and the adjusted EGC size per 250.122(B).
**Solution:**
From Table 250.122, for a 200 A overcurrent device: minimum EGC size = 6 AWG copper.
Per 250.122(B), when conductors are increased in size for voltage drop, the EGC must be proportionally increased: size increase factor = circular mil area of installed conductor / circular mil area of minimum required conductor = 250,000 / 167,800 (3/0 AWG) = 1.49.
Adjusted EGC area = 26,240 (6 AWG) × 1.49 = 39,098 circular mils.
From the wire gauge table: 4 AWG = 41,740 circular mils.
Minimum adjusted EGC: 4 AWG copper.
The rigid metal conduit also serves as an equipment grounding conductor per 250.118, providing a redundant ground fault current path.

### 14.4.3 Ground Fault Protection

Ground fault protection of equipment (GFPE) is required by NEC 230.95 for solidly grounded wye services of more than 150 V to ground but not exceeding 1000 V, where the service disconnect is rated

1000 A or more — this covers the common 480Y/277 V service configuration. GFPE operates on the principle of detecting current imbalance between the phase conductors and the neutral, typically using a zero-sequence current transformer that sums the currents in all conductors passing through it; any net current indicates a ground fault. Unlike ground fault circuit interrupter (GFCI) protection for personnel (which trips at 5 mA), equipment GFPE typically trips at 1200 A or less with a time delay of up to one second, providing protection against arcing ground faults that might not produce enough current to trip standard overcurrent devices.

**Example 14.4.3:** A 480Y/277 V, 2000 A service has ground fault protection set at 800 A pickup with a 0.5-second time delay. An arcing ground fault develops on a feeder with an impedance that limits the fault current to 600 A. Determine whether the GFPE will detect this fault, and calculate the energy ($I^2t$) dissipated if the fault persists for 10 seconds.
**Solution:**
The GFPE pickup is set at 800 A. The fault current is 600 A, which is below the 800 A pickup threshold.
The GFPE will not detect or clear this fault.
Energy at the fault point (assuming the fault persists for 10 seconds): $I^2t = 600^2 \times 10 = 3{,}600{,}000$ $A^2 \cdot s$.
If the GFPE pickup were set at 400 A, the relay would detect the fault and trip after 0.5 seconds: $I^2t = 600^2 \times 0.5 = 180{,}000$ $A^2 \cdot s$ — a 20× reduction in arc energy.
This illustrates the importance of setting GFPE at the lowest practical level.

### 14.4.4 Bonding

Bonding ensures electrical continuity between all metallic components of the electrical system (enclosures, raceways, grounding conductors, and non-current-carrying metal parts) so that fault current can flow freely through a low-impedance path back to the source. Article 250, Part V establishes bonding requirements, and Section 250.104 requires bonding of metal piping systems and structural steel that may become energized. Bonding jumpers must be sized per Table 250.66 (for service equipment) or Table 250.122 (for downstream equipment), and connections must use listed methods such as pressure connectors, clamps, or exothermic welds to ensure reliable, low-resistance joints.

**Example 14.4.4:** A metal water pipe enters a building and runs parallel to the electrical service for 15 feet before connecting to a plastic pipe section. The service is 208Y/120 V, 600 A, with 500 kcmil copper service entrance conductors. Determine the minimum bonding jumper size required to bond the water pipe to the service grounding electrode system per Table 250.66.
**Solution:**
From Table 250.66, for service entrance conductors of 500 kcmil copper: the minimum bonding jumper size is 1/0 AWG copper.
The bonding jumper must be connected to the water pipe on the street side of any plastic pipe section or water meter, ensuring the entire metallic pipe length is bonded to the grounding system.
If the water pipe were the sole grounding electrode, a supplemental electrode (ground rod or concrete-encased electrode) would also be required per 250.53(D)(2), since a metal water pipe electrode must be supplemented.

### 14.4.5 Ground-Fault Circuit Interrupter (GFCI) Protection

A ground-fault circuit interrupter (GFCI) is a personnel protection device that detects leakage current flowing through an unintended path — typically through a person's body to ground — and disconnects the circuit within milliseconds. The GFCI operates on the differential current principle: a toroidal current transformer encircles both the line and neutral conductors, and under normal conditions the currents are equal and opposite, producing zero net flux. When current leaks through a ground fault (e.g., a person touching an energized conductor while grounded), the imbalance is detected and the device trips at a threshold of 4–6 mA (Class A GFCI, the only type permitted by the NEC for personnel protection), interrupting the circuit in approximately 25 ms — fast enough to prevent ventricular fibrillation, which can occur at currents above 30 mA sustained for more than a few hundred milliseconds. NEC Section 210.8 specifies the locations requiring GFCI protection, which has expanded significantly with each code cycle. For **dwelling units** (210.8(A)), the 2023 NEC requires GFCI protection for all 125 V through 250 V, single-phase, 50 A or less receptacles in bathrooms, garages, outdoors, crawl spaces, unfinished basements, kitchens, laundry areas, boathouses, and within 6 feet of sinks, bathtubs, and shower stalls. For **non-dwelling occupancies** (210.8(B)), GFCI protection is required in bathrooms, kitchens, rooftops, outdoors, within 6 feet of sinks, indoor wet locations, locker rooms, and garages/service bays. The 2023 NEC notably expanded GFCI requirements to 250 V circuits (protecting 208 V and 240 V receptacles in addition to 120 V) and to circuits up to 50 A (covering welding outlets, EV chargers, and large appliances). GFCI devices are available as **circuit breakers** (installed in the panelboard, protecting the entire branch circuit), **receptacles** (protecting that outlet and downstream outlets), and **portable devices** (cord-connected for temporary use on construction sites per 590.6). **Nuisance tripping** — false trips caused by accumulated leakage current from long cable runs, VFDs, EMI filters, or multiple appliances — is a common field problem; NEC 210.8 now includes a provision (effective 2023) allowing self-test GFCI devices that automatically verify functionality. The total normal leakage current on a GFCI-protected circuit should be kept below 1 mA to provide adequate margin against the 4–6 mA trip threshold.

**Example 14.4.5:** A kitchen in a dwelling unit has four duplex receptacles on a 20 A, 120 V GFCI-protected branch circuit. The circuit wiring is 12 AWG NM-B, 90 feet from the panel to the first outlet. Each outlet serves an appliance with a typical leakage current of 0.5 mA per appliance. If all four appliances are operating simultaneously, calculate the total leakage current, the remaining margin before nuisance tripping, and the maximum additional leakage from the cable that can be tolerated.

**Solution:**

Total appliance leakage: $4 \times 0.5$ mA = 2.0 mA.

GFCI Class A trip threshold: 5 mA (nominal midpoint of 4–6 mA range).

Remaining margin: $5.0 - 2.0 = 3.0$ mA.

Cable leakage (NM-B in a dry location at 60 Hz): approximately 0.01 mA per foot of cable for typical residential wiring = $90 \times 0.01 = 0.9$ mA.

Total system leakage: $2.0 + 0.9 = 2.9$ mA.

Margin to trip: $5.0 - 2.9 = 2.1$ mA (42% of trip threshold).

This is adequate but not generous — adding a fifth appliance with 0.5 mA leakage would reduce the margin to 1.6 mA, and a damp or degraded cable could push the total above 4 mA, causing nuisance trips.

Best practice: limit total leakage to below 1 mA per circuit by distributing appliances across multiple GFCI circuits.

Per 210.8(A)(7), all kitchen receptacles serving countertop surfaces in a dwelling unit require GFCI protection, and per 210.52(B), at least two 20 A small-appliance branch circuits are required.

## 14.5 Motor Circuits

Motor circuits have unique NEC requirements because motors draw high inrush currents during starting (typically 6–8 times the running current) and may be subject to sustained overloads during operation. Article 430 addresses every aspect of motor circuit design with a layered protection approach: branch circuit conductors sized at 125% of FLC, short-circuit protection sized to pass starting current without tripping, overload protection set to the motor's actual nameplate current, and a disconnecting means within sight of the motor. The use of NEC table FLC values (rather than nameplate current) for conductor and short-circuit protection sizing is a key distinction unique to motor circuits.

### 14.5.1 Motor Full-Load Current Tables

The NEC requires that motor branch circuit conductors and overcurrent devices be sized using the full-load current (FLC) values from NEC Table 430.248 (single-phase) or 430.250 (three-phase), not the actual motor nameplate current. This ensures that the circuit can safely supply any motor of that horsepower and voltage rating, even if the specific motor has a lower nameplate current than the table value. Using the table FLC rather than the nameplate current provides a margin for motor replacement, variations between manufacturers, and the conservative assumption that the motor may operate at its rated load continuously.

**Example 14.5.1:** A 30 HP, 460 V, three-phase motor has a nameplate FLA of 38 A. From NEC Table 430.250, the full-load current for a 30 HP, 460 V motor is 40 A. The motor is Design B with a code letter G (locked-rotor kVA per HP of 5.6–6.29). Calculate the motor branch circuit conductor minimum ampacity and the locked-rotor current.
**Solution:**
Motor branch circuit conductor ampacity per 430.22: $I_{min} = 1.25 \times \text{FLC (from table)} = 1.25 \times 40 = 50$ A.
From Table 310.16 at 75°C: 6 AWG copper = 65 A (sufficient).
Locked-rotor current estimate from code letter G: $kVA_{LR} = 6.0 \text{ (mid-range)} \times 30 \text{ HP} = 180$ kVA.
$I_{LR} = kVA_{LR} / (\sqrt{3} \times V) = 180{,}000 / (1.732 \times 460) = 180{,}000 / 796.9 = 226$ A (approximately 5.65× FLC).
Note that the nameplate FLA (38 A) is lower than the table FLC (40 A), but the NEC requires using the table value of 40 A for all conductor and overcurrent device sizing calculations.

### 14.5.2 Motor Branch Circuit Conductor Sizing

Per NEC 430.22, motor branch circuit conductors must have an ampacity of at least 125% of the motor FLC from the applicable NEC table. For multiple motors on a single feeder, Section 430.24 requires the feeder conductor ampacity to be at least 125% of the largest motor FLC plus 100% of all other motor FLCs plus 100% of any non-motor loads. Temperature correction and conduit fill adjustment factors from Article 310 must be applied on top of the 125% motor sizing requirement, and the final conductor size must also meet voltage drop recommendations for long runs.

**Example 14.5.2:** A motor control center (MCC) feeder supplies three motors: Motor 1 (50 HP, 460 V, three-phase, FLC = 65 A), Motor 2 (25 HP, 460 V, three-phase, FLC = 34 A), and Motor 3 (10 HP, 460 V, three-phase, FLC = 14 A), plus a 30 A continuous lighting load. Determine the minimum feeder conductor ampacity per NEC 430.24.

**Solution:**

Per 430.24: feeder ampacity = 1.25 × largest motor FLC + sum of remaining motor FLCs + non-motor continuous loads × 1.25.

Largest motor: Motor 1 at 65 A.

$I_{feeder} = 1.25 \times 65 + 34 + 14 + 1.25 \times 30 = 81.25 + 34 + 14 + 37.5 = 166.75$ A.

From Table 310.16 at 75°C: 2/0 AWG copper = 175 A (sufficient).

The overcurrent protection for the feeder is determined separately per 430.62: maximum rating = largest motor branch circuit OCPD rating + sum of FLCs of other motors + non-motor loads.

### 14.5.3 Motor Overcurrent Protection

Motor circuits require two levels of overcurrent protection: (1) overload protection (Article 430, Part III), which protects the motor, motor control apparatus, and conductors against excessive heating from sustained overloads, and is typically set at 115–125% of the motor nameplate current; and (2) short-circuit and ground fault protection (Article 430, Part IV), which protects the circuit against high-magnitude fault currents and is sized per Table 430.52 based on the type of overcurrent device and motor design letter. The short-circuit protection is intentionally oversized (up to 250% for inverse-time breakers, 300% for dual-element fuses with Design B motors) to allow the motor starting current (typically 6–8× FLC) to pass without tripping the protective device.

**Example 14.5.3:** A 15 HP, 460 V, three-phase, Design B motor has a full-load current of 21 A (from Table 430.250) and a nameplate FLA of 19.5 A with a service factor of 1.15. Select the overload relay setting and the maximum inverse-time circuit breaker size per Table 430.52.

**Solution:**

Overload protection per 430.32(A)(1): for a motor with service factor ≥ 1.15, the overload relay trips at not more than 125% of nameplate FLA: $I_{OL} = 1.25 \times 19.5 = 24.4$ A (set relay trip to 24.4 A or next available setting).

Short-circuit/ground fault protection per Table 430.52 for Design B motor with inverse-time breaker: maximum = 250% of FLC = $2.50 \times 21 = 52.5$ A.

The next standard breaker size per 240.6(A): 60 A (430.52(C)(1), Exception 1 permits rounding up).

Final motor branch circuit: 10 AWG copper conductors (rated 35 A at 75°C, ≥ 1.25 × 21 = 26.25 A), 60 A inverse-time circuit breaker, and overload relay set at 24.4 A.

### 14.5.4 Motor Disconnecting Means

NEC Article 430, Part IX requires a disconnecting means within sight of the motor location and the driven machinery, capable of disconnecting the motor and its controller from the supply circuit. The disconnect must have an ampere rating of at least 115% of the motor FLC (430.110(A)) and be capable of interrupting the locked-rotor current of the motor if it is a switch type. For motors over 2 HP at 300 V or less, the disconnect must be a motor-circuit switch rated in horsepower, a molded-case circuit breaker, or a molded-case switch.

**Example 14.5.4:** A 75 HP, 460 V, three-phase motor has a full-load current of 96 A from Table 430.250. Determine the minimum disconnect ampere rating and the locked-rotor current the disconnect must be capable of interrupting (motor code letter F, kVA/HP range 5.0–5.59).
**Solution:**
Minimum disconnect rating per 430.110(A): $I_{disc} = 1.15 \times FLC = 1.15 \times 96 = 110.4$ A.
Select the next standard disconnect rating: 200 A (common available size above 110.4 A).
Locked-rotor current: $kVA_{LR} = 5.3$ (mid-range code letter F) $\times$ 75 HP = 397.5 kVA.
$I_{LR} = 397{,}500 / (\sqrt{3} \times 460) = 397{,}500 / 796.9 = 499$ A.
The selected 200 A motor-circuit switch must have an interrupting rating of at least 499 A, which is standard for horsepower-rated motor-circuit switches of this size.
The disconnect must be located within sight (visible and not more than 50 feet) from the motor.

### 14.5.5 Variable Frequency Drive (VFD) Circuit Requirements

Variable frequency drives present unique NEC compliance challenges because of their high-frequency PWM switching output, reflected voltage waves on long motor cables, common-mode currents, and harmonic currents drawn from the supply. NEC Article 430, Part X (Adjustable-Speed Drive Systems) and Section 430.130 address VFD-specific requirements. **Branch circuit conductors** supplying a VFD must be sized at 125% of the VFD's rated input current (not the motor FLC), since the VFD's input current may differ from the motor current due to power factor and efficiency. **Motor leads** between the VFD and motor require careful cable selection: the NEC permits standard THWN/THHN conductors, but in practice VFD-rated cable (with symmetrical ground conductors, heavy insulation rated for the peak voltage stress, and low capacitance) is recommended for runs exceeding 50–100 feet to withstand the reflected voltage waves that can reach 2× the DC bus voltage (~1200 V peak for a 480 V VFD) at the motor terminals. **Overcurrent protection** on the VFD input follows standard motor branch circuit rules (Table 430.52), while the VFD itself provides electronic overload protection for the motor, eliminating the need for a separate thermal overload relay if the VFD is listed for that function per 430.130(A). **Grounding** is critical: 430.130(B) requires the equipment grounding conductor to be a wire-type EGC (conduit alone is insufficient for VFD circuits due to high-frequency common-mode currents that flow through stray capacitance) sized per Table 250.122 or larger. Many VFD manufacturers recommend a symmetrical EGC arrangement (three ground conductors in the motor cable) to provide a low-impedance path for common-mode current and reduce bearing damage from shaft voltage discharge. The VFD input circuit must include appropriate harmonic filtering or must be sized for the harmonic current content per 310.15(E) when the neutral carries triplen harmonics.

**Example 14.5.5:** A 50 HP, 460 V, three-phase induction motor is driven by a VFD with a rated input current of 62 A. The VFD is located 200 feet from the motor. The supply circuit is 250 feet from the MCC to the VFD. Size the supply-side conductors and overcurrent protection per NEC requirements. The motor FLC from Table 430.250 is 65 A.
**Solution:**
**Supply-side conductors** (MCC to VFD): Minimum ampacity = 1.25 × 62 A (VFD input current) = 77.5 A. From Table 310.16 at 75°C: 4 AWG copper = 85 A (sufficient).
**Short-circuit protection** per Table 430.52 for Design B motor with inverse-time breaker: 250% × 65 A (motor FLC from table) = 162.5 A. Next standard size: 175 A.
**Motor-side conductors** (VFD to motor): Sized at 125% of motor FLC = 1.25 × 65 = 81.25 A. From Table 310.16 at 75°C: 4 AWG copper = 85 A.

At 200 feet, reflected voltage waves are a concern — the critical cable length for a 480 V VFD with 100 ns rise time is approximately $L_{crit} = v \times t_{rise} / 2 = (200\ m/\mu s \times 100\ ns) / 2 = 10\ m \approx 33$ feet. Since 200 feet » 33 feet, VFD-rated cable or an output reactor/dv/dt filter is recommended to limit motor terminal voltage to < 1000 V peak.
**Equipment grounding conductor**: Per Table 250.122 for a 175 A OCPD: 6 AWG copper, wire-type per 430.130(B).

### 14.5.6 Hazardous Location Classifications

NEC Articles 500–516 govern electrical installations in locations where flammable gases, vapors, liquids, combustible dusts, or ignitable fibers may be present in sufficient concentrations to create a risk of fire or explosion. The NEC uses two classification systems: the traditional **Class/Division** system (Article 500) and the international **Zone** system (Articles 505/506). In the Class/Division system, **Class I** locations contain flammable gases or vapors (petroleum refineries, paint spray booths, fuel dispensing), **Class II** locations contain combustible dusts (grain elevators, coal handling, metal powder processing), and **Class III** locations contain ignitable fibers or flyings (textile mills, woodworking shops). Each class is subdivided into two Divisions: **Division 1** indicates that hazardous concentrations exist under normal operating conditions or frequently during maintenance, while **Division 2** indicates that hazardous concentrations exist only under abnormal conditions (accidental rupture, equipment failure). The Zone system (more common internationally and increasingly used in the U.S.) divides Class I locations into **Zone 0** (hazardous continuously or for long periods), **Zone 1** (hazardous under normal operations), and **Zone 2** (hazardous only under abnormal conditions), and Class II/III locations into Zones 20, 21, and 22 respectively. Wiring methods in Division 1 are the most restrictive: Class I, Division 1 requires threaded rigid metal conduit or mineral-insulated (MI) cable, with all fittings, boxes, and equipment rated **explosionproof** (designed to contain an internal explosion and cool escaping gases below the ignition temperature) or **intrinsically safe** (energy-limited circuits incapable of igniting the atmosphere). Division 2 permits standard wiring methods (MC cable, EMT in some cases) with sealed fittings at boundaries, and equipment may be **nonincendive** rather than explosionproof. Equipment is further classified by **Group** (A through G based on the specific gas or dust) and by **Temperature Code** (T1 through T6, indicating the maximum surface temperature — e.g., T3A = 180°C, T6 = 85°C — which must be below the auto-ignition temperature of the specific hazardous material).

**Example 14.5.6:** A petroleum refinery has a pump room where flammable vapors (Group D — propane, auto-ignition temperature 450°C) are present under normal operations. The room requires a 20 HP, 460 V motor with a VFD and associated lighting. Classify the location, specify the equipment temperature code, and determine the minimum wiring method requirements for both the motor circuit and lighting.
**Solution:**
Classification: **Class I, Division 1, Group D** — flammable gas (propane) present under normal operating conditions.
Temperature code: propane auto-ignition temperature = 450°C. Equipment surface temperature must be below 450°C. A T3 rating (200°C max surface) provides ample margin and is standard for most Class I motors.
Motor: must be rated for Class I, Division 1, Group D, T3 — an explosionproof motor with an enclosure designed to contain any internal ignition.

VFD: the VFD itself should be located outside the hazardous area (in a general-purpose room) whenever possible; if it must be in the Division 1 area, it requires a purged and pressurized (Type X or Z) enclosure per Article 500.7(D) or an explosionproof enclosure.
Wiring method: **threaded rigid metal conduit (RMC)** with explosionproof fittings at all connection points. All conduit entries to the motor and junction boxes require sealing fittings per 501.15(A) within 18 inches of the enclosure to prevent gas migration through the conduit system.
Lighting: must use explosionproof fixtures rated for Class I, Division 1, Group D. If LED fixtures are used, the fixture must be listed for the classification and the temperature code must account for the LED driver heat.
Conduit seals are required at boundary points where conduit transitions from the Division 1 area to an unclassified area.

## 14.6 Transformer Connections

Transformer overcurrent protection rules in Article 450 differ from standard conductor protection because transformers must tolerate magnetizing inrush current (typically 8–12 times rated current for the first few cycles) without nuisance tripping. The NEC permits either primary-only protection (simpler, less precise) or dual primary-and-secondary protection (more components but tighter coordination). Separately derived systems created by transformers require their own grounding and bonding per Article 250.30, establishing an independent ground fault return path for the secondary system.

### 14.6.1 Transformer Overcurrent Protection (Primary Only)

For transformers rated 1000 V or less, NEC Table 450.3(B) permits primary-only protection at not more than 125% of the primary rated current. If 125% does not correspond to a standard overcurrent device rating, the next standard size up is permitted. Primary-only protection is simpler and less expensive since no secondary overcurrent device is required, but it provides less precise protection of the secondary conductors, which must be sized to handle the full transformer capacity. The primary overcurrent device must have an interrupting rating sufficient for the available fault current at the transformer location.

**Example 14.6.1:** A 75 kVA, 480 V delta primary / 240/120 V secondary, single-phase (center-tapped) transformer is installed with primary-only overcurrent protection. Determine the primary rated current and the maximum overcurrent device size per Table 450.3(B).
**Solution:**
Primary rated current: $I_{primary} = S / V = 75{,}000 / 480 = 156.3$ A.
Maximum primary OCPD per Table 450.3(B): 125% of rated current = $1.25 \times 156.3 = 195.3$ A. The next standard fuse or breaker size per 240.6(A): 200 A.
Secondary rated current: $I_{secondary} = 75{,}000 / 240 = 312.5$ A (line-to-line).
The secondary conductors must be rated for at least 312.5 A since there is no secondary overcurrent protection. From Table 310.16 at 75°C: 400 kcmil copper = 335 A (sufficient for 312.5 A).

### 14.6.2 Transformer Overcurrent Protection (Primary and Secondary)

When overcurrent protection is provided on both the primary and secondary sides, NEC Table 450.3(B) permits the primary device to be rated up to 250% of the primary current, provided the secondary device is rated at no more than 125% of the secondary current. This dual-protection scheme allows the primary device to be sized large enough to withstand magnetizing inrush current without nuisance tripping, while the secondary device provides tighter protection for the transformer and secondary conductors. For transformers with impedance of 6% or less (most distribution transformers), the secondary overcurrent device limits the maximum fault current that can flow through the transformer.

**Example 14.6.2:** A 150 kVA, three-phase, 480 V delta primary / 208Y/120 V secondary transformer has an impedance of 5.5%. Overcurrent protection is provided on both the primary and secondary. Determine the primary and secondary overcurrent device sizes, and calculate the maximum secondary fault current.
**Solution:**
Primary rated current: $I_{primary} = S / (\sqrt{3} \times V) = 150{,}000 / (1.732 \times 480) = 150{,}000 / 831.4 = 180.4$ A.
Maximum primary OCPD (250% with secondary protection): $2.50 \times 180.4 = 451.1$ A. Next standard size above 451.1 A: 500 A.
Secondary rated current: $I_{secondary} = 150{,}000 / (1.732 \times 208) = 150{,}000 / 360.3 = 416.3$ A.
Maximum secondary OCPD (125%): $1.25 \times 416.3 = 520.4$ A. Next standard size: 600 A.
Maximum secondary fault current (infinite bus primary): $I_{fault} = I_{secondary} / Z_{pu} = 416.3 / 0.055 = 7{,}569$ A.
The secondary overcurrent device must have an interrupting rating of at least 7,569 A.

### 14.6.3 Secondary Conductor Protection

Transformer secondary conductors must be protected from overcurrent, either by the transformer primary overcurrent device, the secondary overcurrent device, or the limitations of the transformer itself. Section 240.21(C) provides specific tap rules for transformer secondary conductors, similar to the feeder tap rules in 240.21(B). The most commonly used provision is the 25-foot secondary conductor rule (240.21(C)(6)), which allows secondary conductors up to 25 feet without overcurrent protection at the transformer if the conductor ampacity meets specified minimums and the conductors terminate in a single overcurrent device.

**Example 14.6.3:** A three-phase, 112.5 kVA, 480/208Y/120 V transformer has its primary protected by a 200 A breaker. The secondary conductors run 18 feet to a panelboard with a 400 A main breaker. Determine the minimum secondary conductor size per the 240.21(C)(6) tap rule.
**Solution:**
Secondary rated current: $I_{sec} = 112{,}500 / (1.732 \times 208) = 112{,}500 / 360.3 = 312.3$ A.
Per 240.21(C)(6) (25-foot rule), the NEC minimum secondary conductor ampacity is one-third of the primary OCPD expressed on the secondary voltage basis.
Primary OCPD equivalent on secondary: $200 \times (480/208) = 200 \times 2.308 = 461.5$ A equivalent.
One-third of this: $461.5 / 3 = 153.8$ A — this is the NEC code minimum.
For this installation, the secondary conductors are sized to carry the full transformer secondary current (312.3 A) so the transformer can serve its rated load without restriction.
From Table 310.16 at 75°C: 350 kcmil copper = 310 A (insufficient for 312.3 A). 500 kcmil = 380 A (sufficient).
Minimum secondary conductor: 500 kcmil copper per phase, with 18-foot run to 400 A main breaker.

### 14.6.4 Separately Derived Systems and Grounding

A separately derived system is one that has no direct electrical connection (including a solidly connected grounded conductor) to the supply conductors of another system — the most common example is a transformer with no neutral-to-neutral connection between primary and secondary. Per Article 250.30, separately derived systems must be grounded at the source (transformer secondary) by bonding the neutral to the equipment grounding system and to a grounding electrode. The bonding jumper, system bonding jumper, and grounding electrode conductor are sized per Tables 250.66 and 250.102(C) based on the transformer secondary conductor size. The neutral-to-ground bond must be made at only one point to prevent parallel paths for normal neutral current through the equipment grounding system.

**Example 14.6.4:** A 225 kVA, 480 V delta / 208Y/120 V wye transformer is a separately derived system. The secondary conductors are 350 kcmil copper per phase. Determine the system bonding jumper size and the grounding electrode conductor size per Article 250.30.
**Solution:**
System bonding jumper per 250.30(A)(1) and Table 250.102(C)(1): for 350 kcmil ungrounded conductors, the minimum bonding jumper is 2 AWG copper.
Grounding electrode conductor per 250.30(A)(5) and Table 250.66: for 350 kcmil service-entrance conductors, the minimum GEC is 2 AWG copper.
If a structural metal electrode or concrete-encased electrode is available, 250.30(A)(5) limits the GEC to a maximum of 3/0 AWG copper (no need to be larger regardless of transformer size).
The neutral-to-ground bond is made at the transformer secondary terminal compartment, and the grounded conductor (neutral) is insulated (not bonded to the enclosure) at all downstream points.

### 14.6.5 Delta-Wye Transformer Neutral Sizing

In a delta primary / wye secondary transformer (the most common three-phase distribution configuration), the secondary neutral conductor carries the unbalanced current between the three phases plus any zero-sequence harmonic currents generated by nonlinear loads. For a perfectly balanced linear load, the neutral current is zero and the neutral conductor carries no current. However, in modern buildings with high concentrations of switch-mode power supplies (computers, LED drivers, VFDs), the third harmonic (180 Hz) and its odd multiples (triplen harmonics: 3rd, 9th, 15th, ...) are additive in the neutral rather than cancelling as balanced fundamental currents do. The neutral current from triplen harmonics can be calculated as $I_N = 3 \times I_{3(\text{phase})}$, where $I_3$ is the third harmonic current per phase. In severe cases, the neutral current can exceed the phase current — a building with a 40% third harmonic on each phase will have a neutral current of $3 \times 40\% = 120\%$ of the phase fundamental current. NEC Section 220.61 normally permits reducing the neutral conductor for feeders based on the unbalanced load, but Section 310.15(E) requires counting the neutral as a current-carrying conductor when it carries predominantly harmonic currents, applying the conduit fill derating factors. For this reason, many engineers specify a full-sized or double-sized neutral for transformer secondaries serving nonlinear loads, and some separately derived systems use an oversized neutral bus or a separate neutral for each phase to prevent neutral overheating.

**Example 14.6.5:** A 225 kVA, 480 V delta / 208Y/120 V wye transformer serves a data center with predominantly nonlinear loads. The balanced three-phase load draws 540 A per phase at the secondary, with a measured third harmonic content of 33% per phase. Calculate the neutral current, determine whether the standard neutral conductor is adequately sized, and size the neutral conductor.

**Solution:**

Phase fundamental current: $I_1 = 540$ A.

Third harmonic current per phase: $I_3 = 0.33 \times 540 = 178.2$ A.

Triplen harmonics add in the neutral: $I_{N(3rd)} = 3 \times 178.2 = 534.6$ A.

Total neutral current (dominated by third harmonic): $I_N \approx 534.6$ A.

This is 534.6/540 = 99% of the phase current — the neutral carries nearly as much current as the phase conductors.

A standard neutral sized at 60% of phase ampacity (as permitted by 220.61 for balanced loads) would be rated for only $0.60 \times 540 = 324$ A, resulting in dangerous overheating.

The neutral must be sized for at least 534.6 A.

From Table 310.16 at 75°C: two 350 kcmil copper in parallel ($2 \times 310$ A = 620 A) or one 700 kcmil copper (460 A — insufficient) or one 1000 kcmil copper (545 A — marginal).

**Use two 350 kcmil copper per neutral** (620 A capacity) for adequate capacity and margin.

Per 310.15(E), the neutral counts as a current-carrying conductor for conduit fill derating purposes.

### 14.6.6 K-Factor Transformers and Harmonic Derating

Standard dry-type transformers are designed for sinusoidal (linear) loads, and when they serve nonlinear loads rich in harmonics — computers, VFDs, LED drivers, UPS systems — the harmonic currents cause additional heating in the windings (due to skin effect and proximity effect, which increase winding resistance at higher frequencies) and in the core (due to increased eddy current losses proportional to the square of frequency and flux density). If the extra heating is not accounted for, the transformer's insulation life is shortened and the transformer may overheat. The **K-factor** quantifies the harmonic heating effect as $K = \Sigma(I_h^2 \times h^2) / \Sigma(I_h^2)$, where $I_h$ is the per-unit current of the h-th harmonic and h is the harmonic order. For a purely sinusoidal load, K = 1. Typical K-factors for common load types are: K-1 (linear loads — motors, resistive heating), K-4 (moderate harmonics — mixed commercial loads with some computers), K-13 (high harmonics — data centers, heavy computer loads), and K-20 (severe harmonics — VFDs with SCR front ends, high-density computing). **K-rated transformers** are manufactured with design modifications to withstand the additional harmonic heating: oversized neutrals (200% rated), reduced flux density in the core (lower core losses at harmonic frequencies), transposed and insulated winding conductors (reduced eddy current losses), and additional thermal capacity. A K-13 transformer is typically 15–30% more expensive than a standard K-1 unit of the same kVA rating but avoids the need to derate the standard transformer. For existing standard transformers serving nonlinear loads, the alternative to replacement is **harmonic derating**: the transformer is loaded to a fraction of its nameplate rating to keep the total losses within the thermal design limits. The derated capacity is approximately: $S_{derated} = S_{rated} / \sqrt{K}$, though the exact derating depends on the transformer's eddy current loss factor (typically 5–15% of total load loss for dry-type transformers). ANSI/IEEE C57.110 provides the detailed calculation method for determining the maximum allowable load current in the presence of harmonics. A harmonic survey (using a power quality analyzer to measure the current spectrum at the transformer secondary) is the first step in determining whether a K-rated transformer or harmonic filtering is needed.

**Example 14.6.6:** A standard 150 kVA, 480/208Y/120 V dry-type transformer (K-1 rated, eddy current loss factor $P_{EC\text{-}R}$ = 8% of rated load loss) serves a data center load with the following measured harmonic current spectrum (in per-unit of fundamental): $I_1 = 1.00$, $I_3 = 0.82$, $I_5 = 0.58$, $I_7 = 0.38$, $I_9 = 0.18$, $I_{11} = 0.09$. Calculate the K-factor, the required K-rated transformer, and the derated capacity if a standard transformer must be used.
**Solution:**
K-factor = $\Sigma(I_h^2 \times h^2) / \Sigma(I_h^2)$.
Numerator: $I_1^2{\times}1^2 + I_3^2{\times}3^2 + I_5^2{\times}5^2 + I_7^2{\times}7^2 + I_9^2{\times}9^2 + I_{11}^2{\times}11^2 = 1.00^2{\times}1 + 0.82^2{\times}9 + 0.58^2{\times}25 + 0.38^2{\times}49 + 0.18^2{\times}81 + 0.09^2{\times}121 = 1.000 + 6.052 + 8.410 + 7.076 + 2.624 + 0.980 = 26.14$.
Denominator: $\Sigma(I_h^2) = 1.000 + 0.672 + 0.336 + 0.144 + 0.032 + 0.0081 = 2.192$.
K = 26.14 / 2.192 = **11.9**.
A **K-13** rated transformer is required (next standard K-rating above 11.9).
For derating the existing K-1 transformer per IEEE C57.110: the maximum per-unit load current is $I_{max} = \sqrt{(1 / (1 + P_{EC\text{-}R} \times (K - 1)))} = \sqrt{(1 / (1 + 0.08 \times (11.9 - 1)))} = \sqrt{(1 / (1 + 0.872))} = \sqrt{(1 / 1.872)} = \sqrt{0.534} =$ **0.731 pu**.
Derated capacity: $0.731 \times 150 =$ **109.7 kVA**.
The standard transformer can only safely serve 110 kVA of this harmonic-rich load — a 27% capacity reduction.
If the data center requires the full 150 kVA, either a K-13 replacement transformer or a harmonic filter (active or passive, targeting the 3rd and 5th harmonics) must be installed.

## 14.7 Voltage Drop Calculations

While the NEC does not mandate specific voltage drop limits, Informational Notes in Articles 210 and 215 recommend no more than 3% voltage drop on branch circuits and 5% total (feeder plus branch circuit) for reasonable efficiency of operation. Excessive voltage drop wastes energy as $I^2R$ losses in conductors, reduces motor torque (torque is proportional to voltage squared), dims lighting, and can cause sensitive electronic equipment to malfunction. Voltage drop calculations use conductor resistance and reactance values from NEC Chapter 9, Table 9, with separate formulas for single-phase and three-phase circuits, and account for power factor since reactive current flowing through conductor reactance contributes to the voltage drop.

### 14.7.1 Single-Phase Voltage Drop

For single-phase circuits, the voltage drop is calculated as $V_{drop} = 2 \times I \times L \times (R\cos(\theta) + X\sin(\theta)) / 1000$, where I is the load current in amperes, L is the one-way conductor length in feet, R and X are the conductor resistance and reactance per 1000 feet from NEC Chapter 9, Table 9, and θ is the power factor angle. The factor of 2 accounts for the current flowing through both the line and return conductors. While the NEC does not mandate a maximum voltage drop, Informational Notes in Sections 210.19(A) and 215.2(A)(4) recommend that branch circuit voltage drop not exceed 3% and that the total voltage drop (feeder plus branch circuit) not exceed 5%.

**Example 14.7.1:** A 120 V, single-phase branch circuit supplies a 16 A continuous load at the end of a 150-foot run. The conductors are 12 AWG copper in PVC conduit (R = 1.93 Ω/1000 ft, X = 0.054 Ω/1000 ft from Chapter 9, Table 9). The load power factor is unity. Determine the voltage

drop and whether it meets the NEC 3% recommendation.
**Solution:**
$V_{drop} = 2 \times I \times L \times R / 1000$ (X is negligible at unity PF for small conductors).
$V_{drop} = 2 \times 16 \times 150 \times 1.93 / 1000 = 4800 \times 1.93 / 1000 = 9.26$ V.
Percent voltage drop: $9.26 / 120 \times 100 = 7.72\%$. This exceeds the recommended 3% maximum.
To achieve 3%: $V_{drop(max)} = 0.03 \times 120 = 3.6$ V.
Required resistance: $R_{max} = V_{drop} \times 1000 / (2 \times I \times L) = 3.6 \times 1000 / (2 \times 16 \times 150) = 0.75\ \Omega/1000$ ft.
From Chapter 9, Table 9: 6 AWG copper = 0.491 Ω/1000 ft (sufficient).
Upgrade to 6 AWG copper for a voltage drop of $2 \times 16 \times 150 \times 0.491 / 1000 = 2.36$ V = 1.96%.

### 14.7.2 Three-Phase Voltage Drop

For balanced three-phase circuits, the line-to-line voltage drop is $V_{drop} = \sqrt{3} \times I \times L \times (R\cos(\theta) + X\sin(\theta)) / 1000$, where the $\sqrt{3}$ factor replaces the factor of 2 used in single-phase calculations because the return path is through the other two phases rather than a dedicated neutral conductor. The percent voltage drop is calculated relative to the line-to-line voltage. For three-phase motor circuits, the voltage drop at the motor terminals during starting (when the current is 6–8× the running current) can be significantly larger than the running voltage drop, potentially causing the motor to stall or fail to start if the terminal voltage drops below approximately 80% of rated voltage.

**Example 14.7.2:** A 460 V, three-phase feeder supplies a 100 A load at 0.85 power factor lagging over a 400-foot run. The conductors are 1/0 AWG copper in steel conduit (R = 0.122 Ω/1000 ft, X = 0.0442 Ω/1000 ft from Chapter 9, Table 9). Calculate the voltage drop and the voltage at the load terminals.
**Solution:**
$V_{drop} = \sqrt{3} \times I \times L \times (R\cos(\theta) + X\sin(\theta)) / 1000$. $\cos(\theta) = 0.85$, $\sin(\theta) = 0.527$.
$V_{drop} = 1.732 \times 100 \times 400 \times (0.122 \times 0.85 + 0.0442 \times 0.527) / 1000 = 1.732 \times 100 \times 400 \times (0.1037 + 0.0233) / 1000 = 69{,}280 \times 0.1270 / 1000 = 8.80$ V.
Percent voltage drop: $8.80 / 460 \times 100 = 1.91\%$.
Voltage at load terminals: $V_{load} = 460 - 8.80 = 451.2$ V. This is within the 3% recommendation for feeders.
During motor starting at 600 A inrush: $V_{drop(start)} = 6 \times 8.80 = 52.8$ V, giving a starting voltage of $460 - 52.8 = 407.2$ V = 88.5% of rated voltage — adequate for most motor types.

### 14.7.3 Voltage Drop in Parallel Conductors

For large feeders where a single conductor per phase would be impractical (above 500 kcmil, skin effect losses increase significantly), the NEC permits paralleling conductors per Section 310.10(G), requiring that parallel conductors be the same length, material, size, insulation type, and termination method to ensure equal current distribution. The effective resistance and reactance per phase is divided by the number of parallel conductors, proportionally reducing the voltage drop. Voltage drop calculations for parallel conductors use the per-conductor impedance divided by the number of conductors per phase, then apply the standard voltage drop formula.

**Example 14.7.3:** A 480 V, three-phase, 800 A feeder uses two 500 kcmil copper conductors per phase in parallel, installed in two separate steel conduits. The one-way run is 250 feet, and the load power factor is 0.90 lagging. From Chapter 9, Table 9: R = 0.0293 Ω/1000 ft and X = 0.0439 Ω/1000 ft for 500 kcmil in steel conduit. Calculate the voltage drop.
**Solution:**
Effective impedance per phase (two conductors in parallel): $R_{eff} = 0.0293 / 2 = 0.01465$ Ω/1000 ft.
$X_{eff} = 0.0439 / 2 = 0.02195$ Ω/1000 ft.
$\cos(\theta) = 0.90$, $\sin(\theta) = 0.436$.
$V_{drop} = \sqrt{3} \times 800 \times 250 \times (0.01465 \times 0.90 + 0.02195 \times 0.436) / 1000 = 1.732 \times 800 \times 250 \times (0.01319 + 0.00957) / 1000 = 346{,}400 \times 0.02276 / 1000 = 7.88$ V.
Percent voltage drop: $7.88 / 480 \times 100 = 1.64\%$.
Without parallel conductors, a single 1000 kcmil conductor (R = 0.0186, X = 0.0423) would give: $V_{drop} = 1.732 \times 800 \times 250 \times (0.0186 \times 0.90 + 0.0423 \times 0.436) / 1000 = 346{,}400 \times 0.03518 / 1000 = 12.19$ V = 2.54%, demonstrating the advantage of parallel conductors for both ampacity and voltage drop.

### 14.7.4 Conductor Sizing for Voltage Drop

In many installations, voltage drop — not ampacity — is the controlling factor that determines the required conductor size, particularly for long branch circuit runs to motors, lighting panels, or remote equipment. The design process inverts the voltage drop formula to solve for the maximum allowable conductor resistance, then selects the conductor from NEC Chapter 9, Table 9 that meets this resistance requirement. When the voltage-drop-sized conductor exceeds the ampacity-sized conductor, the larger size governs. A common engineering practice is to size feeders for ≤ 2% drop, leaving 3% margin for branch circuits to stay within the 5% total recommendation.

**Example 14.7.4:** A 208 V, three-phase branch circuit supplies a 42 A continuous load (lighting panel) at the end of a 275-foot run in PVC conduit at unity power factor. The maximum allowable voltage drop is 2%. Determine the minimum conductor size to meet both the ampacity and voltage drop requirements.
**Solution:**
Ampacity requirement: $1.25 \times 42 = 52.5$ A continuous load. From Table 310.16 at 75°C: 6 AWG copper = 65 A (sufficient for ampacity).
Voltage drop requirement: $V_{drop(max)} = 0.02 \times 208 = 4.16$ V.
Rearranging the three-phase voltage drop formula for R (at unity PF, X contribution is negligible):
$R_{max} = V_{drop} \times 1000 / (\sqrt{3} \times I \times L) = 4.16 \times 1000 / (1.732 \times 42 \times 275) = 4{,}160 / 20{,}005 = 0.2080$ Ω/1000 ft.
From Chapter 9, Table 9 (PVC conduit): 6 AWG = 0.510 Ω/1000 ft ($V_{drop} = 1.732 \times 42 \times 275 \times 0.510/1000 = 10.20$ V = 4.90% — too high).
4 AWG = 0.321 Ω/1000 ft ($V_{drop} = 6.42$ V = 3.09% — still exceeds 2%).
2 AWG = 0.206 Ω/1000 ft ($V_{drop} = 4.12$ V = 1.98% — meets the 2% target).
1 AWG = 0.164 Ω/1000 ft ($V_{drop} = 3.28$ V = 1.58%).
Minimum conductor size: 2 AWG copper — governed by voltage drop, not ampacity (2 AWG is rated 115 A, far more than the 52.5 A requirement).

### 14.7.5 Service Load Calculations

NEC Article 220 provides two methods for calculating the total demand load on a building's electrical service: the **standard method** (Article 220, Parts III–IV) and the **optional method** (Article 220, Part V, Section 220.82 for dwelling units). Both methods apply **demand factors** — percentage reductions that account for the statistical improbability that all loads will operate simultaneously at full capacity — to convert the total connected load into the design demand load used for sizing the service entrance conductors and main overcurrent device. The standard method calculates general lighting and receptacle loads from Table 220.12 at 3 VA/ft$^2$ for dwellings (or the applicable VA/ft$^2$ for the occupancy type), applies the demand factors from Table 220.42 (first 3,000 VA at 100%, remainder at 35% for dwellings), adds fixed appliance loads at 75% demand if four or more appliances (220.53), adds the dryer at 5,000 VA minimum per Table 220.54, and adds cooking equipment at reduced demand per Table 220.55 (e.g., one 12 kW range reduces to 8 kW). The largest motor load is added at 125% per 220.50. The **optional method** (220.82) is simpler for existing dwellings or where the total connected load can be determined: the first 10 kVA of general load is taken at 100%, the remainder at 40%, and HVAC is added at 100% of the largest system (heating or cooling, since they do not operate simultaneously per 220.60). For commercial buildings, Article 220 Part IV provides demand factors for specific load types including lighting (Table 220.42), receptacles (Table 220.44: first 10 kVA at 100%, remainder at 50%), and specific equipment loads. The calculated demand load determines the minimum service entrance conductor size (per Table 310.16) and the main overcurrent device rating (per 230.79 — minimum 100 A for a single dwelling).

**Example 14.7.5:** A 2,400 ft$^2$ single-family dwelling has the following loads: general lighting and receptacles (3 VA/ft$^2$), two 20 A small-appliance circuits, one laundry circuit, one 12 kW electric range, one 5.5 kW clothes dryer, one 4.5 kW water heater, one 5 ton (6 kW) air conditioner, and one 10 kW electric furnace. Calculate the service demand load using the standard method and determine the minimum service size.
**Solution:**
**General lighting:** 2,400 ft$^2$ $\times$ 3 VA/ft$^2$ = 7,200 VA.
**Small-appliance circuits:** 2 $\times$ 1,500 VA = 3,000 VA.
**Laundry circuit:** 1,500 VA.
Total general load: 7,200 + 3,000 + 1,500 = 11,700 VA.
Apply Table 220.42 demand factors: first 3,000 VA at 100% = 3,000 VA; remaining 8,700 VA at 35% = 3,045 VA.
Net general load: 3,000 + 3,045 = **6,045 VA**.
**Range:** Table 220.55, Column C for one range $\leq$ 12 kW: demand = **8,000 VA**.
**Dryer:** Table 220.54 for one dryer: 5,500 VA (or 5,000 VA minimum, use nameplate) = **5,500 VA**.
**Water heater:** 4,500 VA (fixed appliance; with fewer than 4 fixed appliances, taken at 100%) = **4,500 VA**.
**Heating/cooling (220.60):** use the larger — furnace 10,000 VA > A/C 6,000 VA, so heating governs: **10,000 VA**.
Total demand: 6,045 + 8,000 + 5,500 + 4,500 + 10,000 = **34,045 VA**.
Service current (240 V single-phase): I = 34,045 / 240 = **141.9 A**.
Minimum service size per 230.79: 150 A or 200 A (the two common residential service sizes above 100 A). Select **200 A** service for capacity margin.
Service entrance conductors: 2/0 AWG copper THWN-2 (rated 175 A at 75°C — insufficient for 200 A OCPD) or 4/0 AWG copper (rated 230 A at 75°C — sufficient). Use **4/0 AWG copper** for

the 200 A service.

## 14.8 Emergency and Standby Power Systems

NEC Articles 700, 701, and 702 establish requirements for power systems that provide electricity when the normal supply fails, with the stringency of requirements proportional to the criticality of the loads served. **Article 700 — Emergency Systems** covers loads whose failure would create a hazard to life safety: egress lighting, exit signs, fire alarm systems, fire pumps, smoke control, and elevator recall. Emergency systems must have an alternate power source that restores power within **10 seconds** of normal supply failure, must have their wiring kept entirely independent from all other wiring (separate raceways, cables, and panelboards, or 2-hour fire-rated enclosures), and must have fully selective (coordinated) overcurrent protection so that a fault on one emergency branch does not trip the entire emergency system. **Article 701 — Legally Required Standby Systems** covers loads whose loss would create hazards or hamper rescue/fire operations but that can tolerate a longer restoration time — typically **60 seconds**. Examples include heating/ventilation for occupied areas, communications systems, and industrial processes whose interruption could create hazards. Wiring for legally required standby may share raceways with other legally required standby circuits but must be independent of normal and emergency circuits. **Article 702 — Optional Standby Systems** covers loads whose loss would cause inconvenience or economic loss but not life-safety hazards — data centers, commercial refrigeration, and building HVAC. Optional standby has no mandated restoration time and no requirement for wiring independence. All three articles require the alternate power source to have sufficient capacity for the connected load, and Article 700.4 requires a written record of periodic testing and maintenance. **Transfer switches** (Article 700.5) are required for emergency and legally required standby systems and must be automatic, listed for emergency use, electrically operated and mechanically held, and approved by the authority having jurisdiction. Generator sizing must account for motor starting inrush (largest motor at locked-rotor current plus running load of all others), the sequence of load transfer (stepped loading to avoid generator overload), and the power factor of the connected load. The generator must be located on the premises and must have sufficient on-site fuel for the required operating duration — typically **2 hours** for emergency systems per 700.12, though the authority having jurisdiction may require more for hospitals (96 hours per NFPA 110 for healthcare facilities).

### 14.8.1 Generator Sizing and Transfer Switch Requirements

Sizing an emergency or standby generator requires calculating the total connected load for each power system category (emergency, legally required standby, optional standby), accounting for motor starting inrush currents, applying appropriate demand and diversity factors, and selecting a generator with adequate kW and kVA capacity at the required power factor. The generator's transient voltage and frequency response during load step changes must also be considered — NFPA 110 requires that the generator maintain voltage within ±10% and frequency within ±3 Hz of rated values during load application and rejection transients. The automatic transfer switch (ATS) monitors the normal source, detects a failure (typically voltage drop below 85% on any phase for more than 1–3 seconds), sends a start signal to the generator, and transfers the load after the generator reaches acceptable voltage and frequency — all within the 10-second window mandated by Article 700. Modern transfer switches include bypass isolation capability (allowing maintenance of the transfer mechanism without interrupting power), programmable time delays (engine start, time to transfer, time to retransfer, engine cool-down), and in-phase transition monitoring (for paralleling applications). For systems with multiple transfer switches, a load priority scheme ensures that the most critical loads (emergency) transfer first, legally required standby loads transfer next, and optional standby loads transfer last — preventing the generator from being overloaded by simultaneous motor starting across all load categories.

**Example 14.8.1:** A hospital requires an emergency generator to serve the following loads: emergency egress lighting (25 kW), fire alarm and communications (10 kW), fire pump — 50 HP, 460 V motor (FLC = 65 A, locked-rotor = 390 A), two elevators — 25 HP each (FLC = 34 A each, locked-rotor = 204 A each), and critical HVAC (80 kW at 0.85 PF). The generator serves a 480Y/277 V, three-phase system. Size the generator in kW and kVA, accounting for motor starting.
**Solution:**
**Running loads:** Lighting: 25 kW. Fire alarm: 10 kW. Fire pump running: $P = \sqrt{3} \times 480 \times 65 \times 0.85 = 45.9$ kW. Elevators running (2): $P = 2 \times \sqrt{3} \times 480 \times 34 \times 0.85 = 2 \times 24.0 = 48.0$ kW. HVAC: 80 kW.
Total running: 25 + 10 + 45.9 + 48.0 + 80 = **208.9 kW**.
**Motor starting analysis** (worst case — fire pump starts with all other loads running): Fire pump starting kVA = $\sqrt{3} \times 480 \times 390 / 1000 = 324.3$ kVA (at ~0.30 PF starting = 97.3 kW).
Running load without fire pump: 208.9 − 45.9 = 163.0 kW.
Total kW during fire pump start: 163.0 + 97.3 = 260.3 kW.
Total kVA during start: 163.0/0.85 + 324.3 = 191.8 + 324.3 = 516.1 kVA (but at mixed PF).
Using a load management strategy that sequences motor starts (fire pump starts first before elevators), the generator must handle: fire pump starting kVA + lighting + fire alarm = 324.3 + 25/0.85 + 10/0.85 = 324.3 + 29.4 + 11.8 = 365.5 kVA peak, then steady-state 208.9 kW / 0.85 = 245.8 kVA.
**Generator selection:** Minimum **continuous rating = 250 kW** (at 0.8 PF = 312.5 kVA) with a **motor starting capability of 375+ kVA** to maintain voltage within the ±10% tolerance during fire pump starting. Select a **300 kW / 375 kVA** standby-rated generator to provide margin.
Per NFPA 110 for Type 10 (healthcare), fuel storage must provide **96 hours** of operation at full load: fuel consumption ≈ 21 gallons/hour for a 300 kW diesel generator, requiring a minimum 2,016-gallon fuel tank.

# Chapter 15

# Networking

Networking is the engineering discipline concerned with interconnecting computing devices and systems to exchange data reliably across physical media and logical protocols. While computer science focuses on software-layer abstractions, the electrical engineering perspective emphasizes the physical transport of signals — the fiber optic links, coaxial cables, and copper pairs that carry bits as light pulses, electromagnetic waves, and voltage transitions. The Open Systems Interconnection (OSI) model provides a layered framework that maps cleanly to EE concerns: from the physical layer's signal integrity, attenuation, and bandwidth through the data link and network layers that frame, address, and route information, to the transport layer protocols that ensure end-to-end delivery. This chapter covers the physical media, optical systems, wireless technologies, and protocol fundamentals that every electrical engineer encounters when designing or supporting networked infrastructure.

## 15.1 The OSI Model

The OSI (Open Systems Interconnection) model is a conceptual framework developed by the International Organization for Standardization (ISO) that divides network communication into seven distinct layers, each with a well-defined role and interface to the layers above and below it. The model provides a common language for describing how data moves from an application on one device to an application on another, regardless of the underlying hardware and software.

### 15.1.1 Layer Overview and Encapsulation

The seven OSI layers, from bottom to top, are: Physical (Layer 1), Data Link (Layer 2), Network (Layer 3), Transport (Layer 4), Session (Layer 5), Presentation (Layer 6), and Application (Layer 7). When data is transmitted, it passes down through the layers at the sender, with each layer adding its own header (and sometimes a trailer) to the data received from the layer above — a process called encapsulation. At the receiver, each layer strips its corresponding header and passes the payload up to the next layer (decapsulation). Layers 5 through 7 are primarily software concerns (session management, data formatting, application interfaces) and are outside the scope of this EE reference. The four lower layers — Physical, Data Link, Network, and Transport — are where electrical engineering intersects with networking, dealing with signals, frames, packets, and segments.

**Example 15.1.1:** An application sends a 1,000-byte payload over an Ethernet network using IPv4 and TCP. The headers added are: TCP (20 bytes), IPv4 (20 bytes), and Ethernet (14-byte header + 4-byte FCS = 18 bytes). Calculate the total frame size and the protocol efficiency (ratio of payload to total frame size).

**Solution:**
Total headers and trailer: 20 (TCP) + 20 (IPv4) + 18 (Ethernet) = 58 bytes
Total frame size: 1,000 + 58 = **1,058 bytes**
Protocol efficiency: $\eta$ = 1,000 / 1,058 = **0.9452 = 94.5%**
Note: Including the 8-byte preamble/SFD and 12-byte inter-frame gap (which are not part of the frame but are required on the wire), the total on-wire overhead is 58 + 20 = 78 bytes, reducing the wire efficiency to 1,000 / 1,078 = 92.8%.

### 15.1.2 Physical Layer (Layer 1)

The physical layer defines the electrical, optical, and mechanical specifications for transmitting raw bits over a communication channel. It specifies signal encoding schemes (NRZ, Manchester, 8b/10b, 64b/66b, PAM-4), voltage or light levels, bit timing, connector types, and cable pinouts. The physical layer does not interpret the meaning of the bits — it only ensures they are placed on the medium and recovered at the far end with acceptable error rates. Key EE parameters at this layer include signal amplitude, characteristic impedance, rise and fall times, jitter, and eye diagram specifications.

**Example 15.1.2:** A PAM-4 physical layer uses 4-level signaling (2 bits per symbol) at a symbol rate of 833 Msymbols/s per wire pair, transmitted across four twisted pairs simultaneously. Calculate (a) the bit rate per pair, (b) the total aggregate bit rate, and (c) verify the bits per symbol.
**Solution:**
(a) Bit rate per pair:
Bits per symbol = $\log_2(4) = 2$
Bit rate per pair = $833 \times 10^6 \times 2$ = **1,666 Mbps = 1.666 Gbps**
(b) Total aggregate bit rate:
Total = 4 pairs × 1.666 Gbps = **6.664 Gbps**
(c) Bits per symbol verification:
PAM-4 has 4 levels: $\{-3, -1, +1, +3\}$
Each symbol encodes $\log_2(4)$ = **2 bits per symbol** ✓

### 15.1.3 Data Link Layer (Layer 2)

The data link layer provides node-to-node data transfer on a shared or point-to-point link, handling framing, physical addressing (MAC addresses), error detection, and media access control. Ethernet is the dominant Layer 2 technology in LANs, using 48-bit MAC addresses and a CRC-32 checksum for frame integrity verification. Switches operate at Layer 2, forwarding frames based on MAC address tables and segmenting collision domains. VLANs (IEEE 802.1Q) extend Layer 2 by inserting a 4-byte tag into the Ethernet frame that includes a 12-bit VLAN ID, enabling logical network segmentation over shared physical infrastructure.

**Example 15.1.3:** An Ethernet switch has a MAC address table capacity of 16,384 entries, each storing a 48-bit MAC address plus an 8-bit port number. Calculate (a) the memory required for the MAC address table and (b) the maximum number of VLANs possible with the 12-bit VLAN ID field in an 802.1Q tag.
**Solution:**

(a) Memory per entry: 48 bits (MAC) + 8 bits (port) = 56 bits = 7 bytes
Total memory: 16,384 × 7 = **114,688 bytes** ≈ **112 KB**
(b) Maximum VLANs:
12-bit VLAN ID: $2^{12} = 4{,}096$ possible values
VLAN 0 and VLAN 4095 are reserved, so usable VLANs = **4,094**

### 15.1.4 Network Layer (Layer 3)

The network layer is responsible for logical addressing, routing, and forwarding packets across multiple interconnected networks (internetworking). IP (Internet Protocol) is the dominant Layer 3 protocol, assigning hierarchical addresses that enable routers to make forwarding decisions based on destination network prefixes. Unlike Layer 2 MAC addresses which are flat and locally significant, Layer 3 IP addresses are structured with network and host portions, enabling scalable routing through aggregation and summarization. The Time-to-Live (TTL) field prevents packets from looping indefinitely by decrementing at each router hop; when TTL reaches zero, the packet is discarded.

**Example 15.1.4:** A router processes packets at a rate of 10 million packets per second (10 Mpps). The average packet size is 512 bytes. Calculate (a) the throughput in Gbps and (b) the maximum number of hops a packet can traverse if the initial TTL is set to 64.
**Solution:**
(a) Throughput:
Bits per packet = 512 × 8 = 4,096 bits
Throughput = $10 \times 10^6 \times 4{,}096 = 40.96 \times 10^9$ bps = **40.96 Gbps**
(b) Maximum hops:
Each router decrements TTL by 1. Starting at TTL = 64:
Maximum hops = **63** successful router hops (each router decrements TTL by 1; the 64th router decrements TTL to 0 and discards the packet without forwarding it)

### 15.1.5 Transport Layer (Layer 4)

The transport layer provides end-to-end communication services between application processes on different hosts, using port numbers (16-bit, range 0–65535) to multiplex multiple connections over a single IP address. TCP provides connection-oriented, reliable delivery with flow control and congestion avoidance, while UDP provides connectionless, best-effort delivery with minimal overhead. The choice between TCP and UDP depends on whether the application requires guaranteed delivery (file transfer, web browsing, email) or prioritizes low latency (voice, video, gaming, DNS queries). Segment size is constrained by the Maximum Transmission Unit (MTU) of the path, typically 1,500 bytes for Ethernet.

**Example 15.1.5:** A TCP connection has a receive window size of 65,535 bytes and the round-trip time (RTT) is 40 ms. Calculate (a) the maximum achievable throughput (window-limited) and (b) the total number of unique port numbers available for simultaneous connections.
**Solution:**
(a) Maximum throughput (bandwidth-delay product limit):
Throughput = Window size / RTT = 65,535 bytes / 0.040 s = 1,638,375 bytes/s

Convert to Mbps: 1,638,375 × 8 = 13,107,000 bps = **13.1 Mbps**
(b) Unique port numbers:
16-bit port field: $2^{16}$ = **65,536 port numbers** (0–65535)
Well-known ports (0–1023) are reserved for standard services, leaving 64,512 for dynamic/ephemeral use.

## 15.2 Physical Media: Copper

Copper cabling has been the foundation of electrical communication since the telegraph era and remains the dominant medium for local-area networking, cable television distribution, and last-mile access. The two principal forms — coaxial cable and twisted-pair cable — each offer distinct impedance characteristics, bandwidth capabilities, and noise immunity suited to different applications.

### 15.2.1 Coaxial Cable

Coaxial cable consists of a central conductor surrounded by a dielectric insulator, a conductive shield (braid, foil, or both), and an outer jacket, forming a transmission line with well-defined characteristic impedance. The two standard impedances are 75 Ω (used in cable television, CATV, and video distribution with RG-6/RG-59) and 50 Ω (used in RF systems, test equipment, and data networks with RG-58/RG-213). The coaxial geometry provides inherent shielding against external electromagnetic interference, as the outer conductor acts as a Faraday cage for the inner conductor. Characteristic impedance is determined by $Z_0 = (138 / \sqrt{\varepsilon_r}) \times \log_{10}(D/d)$, where D is the inner diameter of the shield and d is the outer diameter of the center conductor. The velocity of propagation is $v_p = c / \sqrt{\varepsilon_r}$, where c is the speed of light and $\varepsilon_r$ is the relative permittivity of the dielectric.

**Example 15.2.1:** An RG-6 coaxial cable has a center conductor diameter d = 1.024 mm, a shield inner diameter D = 4.700 mm, and a foam polyethylene dielectric with $\varepsilon_r = 1.45$. Calculate (a) the characteristic impedance and (b) the velocity of propagation as a percentage of the speed of light.
**Solution:**
(a) Characteristic impedance:
$Z_0 = (138 / \sqrt{\varepsilon_r}) \times \log_{10}(D/d) = (138 / \sqrt{1.45}) \times \log_{10}(4.700 / 1.024)$
$\sqrt{1.45} = 1.2042$
138 / 1.2042 = 114.6
$\log_{10}(4.700 / 1.024) = \log_{10}(4.590) = 0.6618$
$Z_0 = 114.6 \times 0.6618 =$ **75.8 Ω ≈ 75 Ω ✓**
(b) Velocity of propagation:
$v_p = c / \sqrt{\varepsilon_r} = c / 1.2042 = 0.8304c$
Velocity of propagation = **83.0% of the speed of light**

### 15.2.2 Twisted-Pair Cable

Twisted-pair cable consists of insulated copper conductors twisted together in pairs to reduce electromagnetic interference and crosstalk between adjacent pairs. Unshielded twisted pair (UTP) relies solely on the twist geometry for noise rejection, while shielded variants (STP, FTP, S/FTP) add metallic foil or braided shields around individual pairs or the entire cable bundle. Category ratings define the maximum supported frequency bandwidth: Cat 5e supports up to 100 MHz, Cat 6 up to 250 MHz, Cat 6A up to 500 MHz, and Cat 8 up to 2 GHz. The twist rate (twists per meter) varies between pairs within the same cable to minimize pair-to-pair crosstalk, and each category specifies tighter performance requirements for attenuation, NEXT, and return loss.

**Example 15.2.2:** A 100 m Cat 6A UTP cable has a maximum attenuation of 32.8 dB at 500 MHz. A signal is transmitted at 0 dBm at 500 MHz. Calculate (a) the received signal power in dBm and (b) the received power in milliwatts.
**Solution:**
(a) Received power in dBm:
$P_{rx} = P_{tx} - \text{Attenuation} = 0 - 32.8 = \mathbf{-32.8\ dBm}$
(b) Convert to milliwatts:
$P_{rx} = 10^{(-32.8/10)} = 10^{-3.28} = 5.25 \times 10^{-4}$ mW $=$ **0.000525 mW = 0.525 μW**

### 15.2.3 Signal Attenuation and Bandwidth

Signal attenuation in copper cables increases with both frequency and distance, following an approximately square-root-of-frequency relationship for coaxial cable (due to skin effect losses dominating) and a more complex function for twisted pair. The bandwidth-distance product characterizes the information-carrying capacity of a medium: doubling the distance approximately halves the usable bandwidth. Insertion loss is specified in dB per unit length (dB/100 m) at specific test frequencies, and the total link loss must remain within the receiver's dynamic range for reliable operation. Skin effect causes current to flow in an increasingly thin layer near the conductor surface at higher frequencies, increasing the effective AC resistance and thus the attenuation.

**Example 15.2.3:** An RG-6 coaxial cable has measured attenuation of 5.6 dB/100 m at 400 MHz and 8.9 dB/100 m at 1 GHz. For a 60 m cable run carrying a 750 MHz CATV signal, estimate the attenuation using the square-root frequency scaling relationship from the 400 MHz specification.
**Solution:**
Square-root frequency scaling: $\alpha(f_2) \approx \alpha(f_1) \times \sqrt{(f_2/f_1)}$
$\alpha(750\text{ MHz}) \approx 5.6 \times \sqrt{(750/400)} = 5.6 \times \sqrt{1.875} = 5.6 \times 1.3693 = 7.67$ dB/100 m
Total attenuation for 60 m:
Loss $= 7.67 \times (60/100) =$ **4.60 dB**
Verification: The estimated 7.67 dB/100 m at 750 MHz falls between the 5.6 (at 400 MHz) and 8.9 (at 1 GHz) specifications, confirming the estimate is reasonable.

### 15.2.4 Crosstalk and Noise

Crosstalk is the unwanted coupling of signals from one conductor pair to another within the same cable or between adjacent cables, and is the primary impairment limiting the performance of twisted-pair Ethernet. Near-End Crosstalk (NEXT) is measured at the same end as the transmitter and is the most critical parameter because it represents interference where the desired signal has not yet been attenuated by cable loss. Far-End Crosstalk (FEXT) is measured at the opposite end and is partially attenuated by cable loss, making it less severe than NEXT. The Attenuation-to-Crosstalk Ratio (ACR) — defined as NEXT minus insertion loss (both in dB) — indicates the signal-to-interference margin available to the receiver, and must be positive for reliable communication.

**Example 15.2.4:** A Cat 6 cable has a NEXT value of 44.3 dB and an insertion loss of 21.7 dB at 250 MHz. Calculate (a) the ACR and (b) determine whether the link meets a minimum ACR requirement of 10 dB.
**Solution:**
(a) Attenuation-to-Crosstalk Ratio:
ACR = NEXT − Insertion Loss = 44.3 − 21.7 = **22.6 dB**
(b) The ACR of 22.6 dB **exceeds the 10 dB minimum** by a margin of 12.6 dB. This means the desired signal arrives at the receiver 22.6 dB stronger than the crosstalk interference, providing comfortable margin for reliable operation at 250 MHz.

### 15.2.5 Cable Testing and Certification

Structured cabling installations must be tested and certified to verify compliance with TIA-568 or ISO 11801 performance standards before being placed into service. Cable certification testers (such as Fluke DSX CableAnalyzer) measure key parameters across the full frequency range of the cable category and issue a pass/fail result against the selected standard. The primary test parameters include wiremap (verifying correct pin-to-pin connectivity and detecting opens, shorts, crossed pairs, and split pairs), insertion loss (signal attenuation from end to end), NEXT (Near-End Crosstalk between pairs), PS-NEXT (Power Sum NEXT from all adjacent pairs), return loss (impedance mismatch reflections), and propagation delay and delay skew between pairs. Two test configurations are defined: the permanent link test (from patch panel IDC to outlet IDC, excluding patch cords) and the channel test (full end-to-end path including patch cords). A permanent link test is preferred for new installations because it certifies only the permanently installed cabling, independent of the patch cords used.

**Example 15.2.5:** A Cat 6A cable installation is tested using a permanent link adapter. The tester reports: insertion loss = 18.2 dB at 500 MHz (limit: 28.1 dB), NEXT = 42.8 dB at 500 MHz (limit: 33.1 dB), and return loss = 14.5 dB at 500 MHz (limit: 12.0 dB). Determine whether each parameter passes and calculate the headroom (margin) for each.
**Solution:**
Insertion loss: 18.2 dB measured vs. 28.1 dB limit. Since lower insertion loss is better, 18.2 < 28.1 — **PASS** with 28.1 − 18.2 = **9.9 dB headroom**.
NEXT: 42.8 dB measured vs. 33.1 dB limit. Since higher NEXT is better (more isolation), 42.8 > 33.1 — **PASS** with 42.8 − 33.1 = **9.7 dB headroom**.
Return loss: 14.5 dB measured vs. 12.0 dB limit. Since higher return loss is better (less reflection), 14.5 > 12.0 — **PASS** with 14.5 − 12.0 = **2.5 dB headroom**.
All three parameters pass with positive margin. The return loss has the least headroom and would be the first parameter to fail if cable quality degrades or connectors are improperly terminated.

## 15.3 Physical Media: Fiber Optics

Fiber optic communication transmits information as pulses of light through thin glass or plastic fibers, offering enormous bandwidth, immunity to electromagnetic interference, low attenuation over long distances, and inherent electrical isolation. Fiber is the dominant medium for telecommunications backbone, metropolitan networks, submarine cables, and increasingly for data center and enterprise interconnects.

### 15.3.1 Optical Fiber Structure and Types

An optical fiber consists of a cylindrical glass or plastic core surrounded by a cladding of slightly lower refractive index, which confines light within the core by total internal reflection. The critical angle for total internal reflection is determined by Snell's law: $\sin \theta_c = n_{clad} / n_{core}$, and the numerical aperture $NA = \sqrt{(n_{core}^2 - n_{clad}^2)}$ defines the cone of light that can enter and propagate through the fiber. Standard telecommunications fiber uses a silica glass core ($n \approx 1.468$) and cladding ($n \approx 1.463$), with a protective acrylate buffer coating and outer jacket. The two fundamental types — single-mode fiber (SMF) and multimode fiber (MMF) — differ in core diameter and the number of propagation modes they support.

**Example 15.3.1:** A step-index optical fiber has a core refractive index $n_{core} = 1.468$ and a cladding refractive index $n_{clad} = 1.463$. Calculate (a) the numerical aperture, (b) the critical angle for total internal reflection, and (c) the maximum acceptance half-angle of the fiber.
**Solution:**
(a) Numerical aperture:
$NA = \sqrt{(n_{core}^2 - n_{clad}^2)} = \sqrt{(1.468^2 - 1.463^2)} = \sqrt{(2.1550 - 2.1404)} = \sqrt{0.01465} = \mathbf{0.121}$
(b) Critical angle:
$\sin \theta_c = n_{clad} / n_{core} = 1.463 / 1.468 = 0.99659$
$\theta_c = \sin^{-1}(0.99659) = \mathbf{85.27°}$
(c) Maximum acceptance half-angle (in air, $n_0 = 1.0$):
$\sin \theta_a = NA / n_0 = 0.121 / 1.0 = 0.121$
$\theta_a = \sin^{-1}(0.121) = \mathbf{6.95°}$

### 15.3.2 Single-Mode Fiber

Single-mode fiber (SMF) has a small core diameter (typically 8–10 µm for an overall 9/125 µm specification) that supports only the fundamental $LP_{01}$ propagation mode, eliminating modal dispersion and enabling transmission over distances exceeding 100 km without regeneration. ITU-T G.652 defines standard SMF with zero-dispersion wavelength near 1310 nm and typical attenuation of 0.35 dB/km at 1310 nm and 0.20 dB/km at 1550 nm. Chromatic dispersion (approximately 17 ps/(nm·km) at 1550 nm) causes different wavelengths within the source's spectral width to travel at slightly different velocities, broadening pulses over distance and limiting the bit-rate-distance product. SMF is the dominant medium for metro, long-haul, and submarine telecommunications, as well as enterprise backbone and data center interconnects exceeding 300 m.

**Example 15.3.2:** A single-mode fiber link operates at 1550 nm over a distance of 80 km. The fiber has an attenuation of 0.22 dB/km and chromatic dispersion of 17 ps/(nm·km). The laser source has a spectral width of 0.1 nm. Calculate (a) the total fiber attenuation and (b) the total chromatic dispersion (pulse broadening) over the link.
**Solution:**
(a) Total fiber attenuation:
$\text{Loss} = 0.22 \text{ dB/km} \times 80 \text{ km} = \mathbf{17.6\ dB}$
(b) Total chromatic dispersion:
$\Delta\tau = D \times L \times \Delta\lambda = 17 \text{ ps/(nm·km)} \times 80 \text{ km} \times 0.1 \text{ nm} = \mathbf{136\ ps}$

At 10 Gbps (bit period = 100 ps), this 136 ps of pulse broadening exceeds the bit period and would cause intersymbol interference, requiring dispersion compensation or a narrower-linewidth laser. At 2.5 Gbps (bit period = 400 ps), the 136 ps broadening is within tolerance.

### 15.3.3 Multimode Fiber

Multimode fiber (MMF) has a larger core diameter (50 μm or 62.5 μm) that supports hundreds of propagation modes, each traveling at a slightly different velocity and creating modal dispersion that limits bandwidth and reach. Graded-index MMF uses a parabolic refractive index profile to equalize mode group velocities, significantly reducing modal dispersion compared to step-index designs. The standard OM categories specify the modal bandwidth as a bandwidth-distance product: OM1 (62.5 μm, 200 MHz·km at 850 nm), OM3 (50 μm, 2,000 MHz·km), OM4 (50 μm, 4,700 MHz·km), and OM5 (50 μm, 4,700 MHz·km with SWDM support at 850–953 nm). MMF is used primarily in enterprise LANs and data centers for distances up to 300–550 m at 10–100 Gbps, where the lower cost of VCSEL transmitters and relaxed alignment tolerances provide economic advantages over SMF.

**Example 15.3.3:** An OM4 multimode fiber has a bandwidth-distance product of 4,700 MHz·km at 850 nm. Determine (a) the maximum modal bandwidth available for a 300 m data center link and (b) the maximum supportable bit rate assuming bandwidth $\approx 0.7 \times$ bit rate for NRZ signaling.
**Solution:**
(a) Modal bandwidth at 300 m:
BW = 4,700 MHz·km / 0.300 km = **15,667 MHz ≈ 15.7 GHz**
(b) Maximum bit rate (NRZ):
Bit rate ≈ BW / 0.7 = 15,667 / 0.7 = **22,381 Mbps ≈ 22.4 Gbps**
This explains why OM4 can support 10GBASE-SR (10 Gbps) up to 400 m and 25GBASE-SR (25 Gbps) up to approximately 100 m, but requires parallel lanes (e.g., $4 \times 25$G) for 100 Gbps.

### 15.3.4 Optical Transmitters and Receivers

Optical transmitters convert electrical signals to modulated light using laser diodes (LD) or light-emitting diodes (LED), with the choice depending on data rate, distance, and cost requirements. Vertical-Cavity Surface-Emitting Lasers (VCSELs) operating at 850 nm are the standard source for multimode fiber links, while Distributed Feedback (DFB) lasers at 1310 nm or 1550 nm serve single-mode applications. Optical receivers use PIN photodiodes or Avalanche Photodiodes (APD) to convert light back to electrical current, with receiver sensitivity specified in dBm as the minimum optical power required to achieve a target bit error rate (typically $10^{-12}$). Transceiver modules (SFP, SFP+, SFP28, QSFP28, QSFP-DD) integrate the transmitter, receiver, and control electronics into a hot-pluggable, standardized form factor with digital diagnostics monitoring (DOM) for real-time power and temperature readings.

**Example 15.3.4:** An SFP+ transceiver for 10GBASE-LR has a minimum transmit power of −1 dBm and a receiver sensitivity of −14.4 dBm at BER = $10^{-12}$. Calculate the total optical power budget available for fiber attenuation, connector losses, and splice losses.
**Solution:**
Power budget = $P_{tx,min} - P_{rx,sensitivity}$

Power budget = (−1) − (−14.4) = **13.4 dB**
This 13.4 dB budget must cover all losses in the link:
- Fiber attenuation (0.35 dB/km at 1310 nm × distance)
- Connector losses (0.3–0.5 dB each)
- Splice losses (0.1 dB each for fusion splices)
- System margin (typically 1–3 dB)

At 0.35 dB/km with two connectors (1 dB) and 2 dB margin: max distance = (13.4 − 1 − 2) / 0.35 = 29.7 km. The 10GBASE-LR standard specifies a maximum reach of 10 km, providing substantial margin.

### 15.3.5 Fiber Link Budgets

A fiber optic link budget accounts for all optical power gains and losses between the transmitter and receiver to verify that the signal arrives with sufficient power for reliable detection. The power budget equals the transmitter output power minus the receiver sensitivity, and must exceed the sum of all losses: fiber attenuation (dB/km × distance), connector losses (typically 0.3–0.5 dB each), splice losses (typically 0.1 dB each for fusion splices), and a system margin (typically 3 dB) for aging, temperature variations, and future repairs. If the link budget is insufficient, options include using a higher-power transmitter, a more sensitive receiver (APD instead of PIN), optical amplifiers, or regenerators. For DWDM systems, the link budget must be calculated per channel, accounting for wavelength-dependent fiber loss and multiplexer/demultiplexer insertion losses.

**Example 15.3.5:** A 40 km single-mode fiber link at 1550 nm uses connectors at each end (0.5 dB each), 4 fusion splices (0.1 dB each), and a fiber attenuation of 0.25 dB/km. The transmitter output is +3 dBm and the receiver sensitivity is −28 dBm. Calculate (a) the total link loss, (b) the power budget, (c) the available margin, and (d) whether the link closes with a 3 dB system margin.
**Solution:**
(a) Total link loss:
Fiber: $0.25 \times 40 = 10.0$ dB
Connectors: $2 \times 0.5 = 1.0$ dB
Splices: $4 \times 0.1 = 0.4$ dB
Total loss = 10.0 + 1.0 + 0.4 = **11.4 dB**
(b) Power budget:
Budget = $P_{tx} - P_{rx,sens} = (+3) - (-28) =$ **31 dB**
(c) Available margin:
Margin = Budget − Total loss = 31 − 11.4 = **19.6 dB**
(d) With 3 dB system margin required: 19.6 > 3, so the link **closes with 16.6 dB of excess margin**. This link is significantly over-engineered; a lower-cost transceiver with less power budget would suffice.

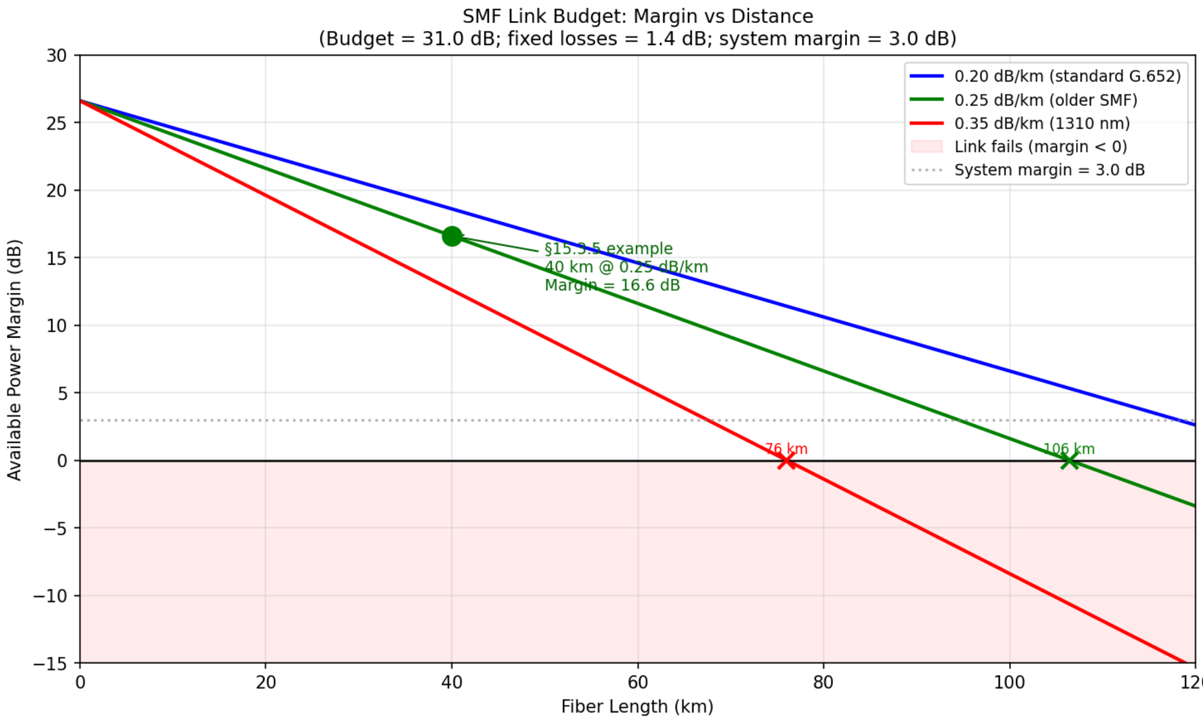



Figure 15.3.5: SMF link power margin vs fiber length for three common attenuation values (0.20, 0.25, 0.35 dB/km). Budget = 31 dB; fixed losses = 1.4 dB (2 connectors + 4 splices); system margin = 3 dB. The §15.3.5 example (40 km, 0.25 dB/km) is marked with a 16.6 dB margin.

### 15.3.6 Fiber Troubleshooting and OTDR

An Optical Time-Domain Reflectometer (OTDR) is the primary instrument for characterizing and troubleshooting fiber optic links, providing a graphical trace of attenuation versus distance along the entire fiber from a single end. The OTDR launches short pulses of light into the fiber and measures the Rayleigh backscatter (continuous low-level reflections from microscopic density variations in the glass) and Fresnel reflections (discrete reflections at interfaces where the refractive index changes abruptly, such as connectors, mechanical splices, and fiber breaks). On the OTDR trace, fiber spans appear as a steadily declining line (the slope equals the fiber attenuation in dB/km), fusion splices appear as small step losses (typically 0.02–0.10 dB with no reflection spike), connectors appear as reflection spikes with an associated insertion loss, and a fiber break or open end appears as a large reflection spike followed by noise floor. The OTDR calculates distance using $d = (c \times t) / (2 \times n)$, where $t$ is the round-trip time of the reflected pulse, $n$ is the group refractive index of the fiber (typically 1.4677 for SMF at 1550 nm), and the factor of 2 accounts for the round trip. Key OTDR specifications include dynamic range (maximum measurable fiber loss, typically 25–45 dB), dead zone (minimum distance after a reflection before the next event can be detected — event dead zone of 0.8–3 m and attenuation dead zone of 5–20 m), and pulse width (shorter pulses give better spatial resolution but less dynamic range).

Common fiber faults and their OTDR signatures:

| Fault | OTDR Signature | Typical Cause |
|---|---|---|
| Fusion splice | Small step loss (0.02–0.10 dB), no reflection | Normal splice point |
| Mechanical splice | Step loss (0.1–0.5 dB) with small reflection | Field-installed splice |
| Connector | Reflection spike + insertion loss (0.3–0.75 dB) | Patch panel, equipment port |
| Dirty connector | Increased reflection and loss vs. baseline | Contamination on endface |
| Macrobend | Localized loss with no reflection | Tight bend, cable pinch, improper routing |
| Fiber break | Large reflection spike, signal drops to noise floor | Cut, crushed, or severed cable |
| Ghosting | False reflection at 2× the distance of a strong reflector | Double reflection artifact |

**Example 15.3.6:** An OTDR with a pulse width of 100 ns is used to test a 25 km single-mode fiber span at 1550 nm (group refractive index n = 1.4677). The trace shows the fiber launch at 0 m, a fusion splice at 8.2 km with 0.05 dB loss, a connector at 15.6 km with 0.40 dB loss, and the fiber end at 25.0 km. The fiber attenuation slope is 0.21 dB/km. Calculate (a) the spatial resolution (pulse length in the fiber), (b) the total end-to-end link loss from the OTDR trace, and (c) the expected optical power at the far end if the launch power is 0 dBm.
**Solution:**
(a) Spatial resolution (one-way pulse length in fiber):
$v_{fiber} = c / n = 3.0 \times 10^8 / 1.4677 = 2.044 \times 10^8$ m/s
Pulse length = $v_{fiber} \times$ pulse width = $2.044 \times 10^8 \times 100 \times 10^{-9} = 20.44$ m
Spatial resolution ≈ **20.4 m** (events closer than this cannot be individually resolved)
(b) Total link loss from OTDR trace:
Fiber attenuation: $0.21 \times 25.0 = 5.25$ dB
Fusion splice: 0.05 dB
Connector: 0.40 dB
Total = 5.25 + 0.05 + 0.40 = **5.70 dB**
(c) Expected power at far end:
$P_{rx} = P_{tx}$ − Total loss = 0 − 5.70 = **−5.70 dBm**
Note: The OTDR measures from one end only. Best practice is to test from both ends and average the splice losses, because OTDR measurements can show apparent “gainers” (negative splice loss) when transitioning from a fiber with a larger mode field diameter to one with a smaller diameter, due to differences in backscatter levels between the two fiber sections.

## 15.4 Dense Wavelength Division Multiplexing (DWDM)

Dense Wavelength Division Multiplexing is an optical transport technology that multiplies the capacity of a single fiber by transmitting multiple independent optical signals simultaneously, each on a different wavelength (color) of light. DWDM is the backbone technology for long-haul telecommunications, submarine cables, and high-capacity metro networks, enabling terabits per second of aggregate capacity over a single fiber pair.

### 15.4.1 WDM Fundamentals and Channel Spacing

Wavelength Division Multiplexing (WDM) transmits multiple independent optical signals simultaneously over a single fiber by assigning each signal a different wavelength, analogous to frequency-division multiplexing in radio systems. Coarse WDM (CWDM) uses wide channel spacing of 20 nm across the 1270–1610 nm range, supporting up to 18 channels without optical amplification. Dense WDM (DWDM) uses tight channel spacing defined by the ITU-T G.694.1 frequency grid, with standard spacings of 100 GHz (≈ 0.8 nm) and 50 GHz (≈ 0.4 nm) in the C-band (1530–1565 nm), supporting 40 to 96 channels per fiber. The C-band is preferred because it coincides with the minimum loss window of silica fiber (≈ 0.20 dB/km) and the gain bandwidth of Erbium-Doped Fiber Amplifiers (EDFAs).

**Example 15.4.1:** A DWDM system uses the ITU-T 100 GHz grid in the C-band from 191.7 THz to 196.1 THz. Calculate (a) the number of channels, (b) the wavelength range in nm (using $c = f \times \lambda$), and (c) the channel spacing in nm at the center of the band.
**Solution:**
(a) Number of channels:
Bandwidth = 196.1 − 191.7 = 4.4 THz = 4,400 GHz
Channels = 4,400 / 100 + 1 = **45 channels**
(b) Wavelength range:
$\lambda_{min} = c / f_{max} = 3 \times 10^8 / (196.1 \times 10^{12}) = 1{,}529.8$ nm
$\lambda_{max} = c / f_{min} = 3 \times 10^8 / (191.7 \times 10^{12}) = 1{,}564.9$ nm
Wavelength range: **1,529.8 nm to 1,564.9 nm** (35.1 nm span)
(c) Channel spacing in nm at band center:
$f_{center} = (191.7 + 196.1) / 2 = 193.9$ THz
$\Delta\lambda = c \times \Delta f / f^2 = (3 \times 10^8 \times 100 \times 10^9) / (193.9 \times 10^{12})^2 = 3 \times 10^{19} / 3.760 \times 10^{28} =$ **0.798 nm ≈ 0.8 nm**

### 15.4.2 Optical Amplifiers (EDFA)

The Erbium-Doped Fiber Amplifier (EDFA) amplifies optical signals directly in the fiber without optical-to-electrical-to-optical (O-E-O) conversion, making it the enabling technology for long-haul DWDM networks. An EDFA consists of a length of erbium-doped fiber pumped by a 980 nm or 1480 nm laser diode; incoming signal photons in the C-band stimulate emission of additional photons at the same wavelength, phase, and direction, providing 15–30 dB of gain per amplifier. EDFAs amplify all DWDM channels simultaneously but add amplified spontaneous emission (ASE) noise, characterized by the noise figure (typically 4–6 dB). The optical signal-to-noise ratio (OSNR) degrades with each amplifier in a chain, ultimately limiting the maximum reach of an amplified system.

**Example 15.4.2:** A DWDM link uses 5 inline EDFAs, each with a gain of 20 dB and a noise figure of 5.5 dB. The input signal power per channel is −2 dBm and the span loss between amplifiers equals the EDFA gain (20 dB). Calculate the OSNR at the receiver in a 0.1 nm reference bandwidth. Use the approximation: $OSNR \approx P_{in} - NF - 10\log_{10}(N) + 58$ dBm (where N is the number of

amplifiers and 58 dBm accounts for the noise bandwidth constant $h\nu \times \Delta\nu$ for 0.1 nm at 1550 nm).
**Solution:**
$\text{OSNR} = P_{in} - \text{NF} - 10\log_{10}(N) + 58$
$P_{in} = -2$ dBm
NF = 5.5 dB
$10\log_{10}(5) = 6.99$ dB
OSNR = $-2 - 5.5 - 6.99 + 58 =$ **43.5 dB**
For 100 Gbps DP-QPSK coherent detection, the required OSNR is approximately 12–15 dB, so this system has more than 28 dB of OSNR margin — indicating either the span count could be increased significantly or the channel power could be reduced to mitigate fiber nonlinearities.

### 15.4.3 DWDM System Design

DWDM system design integrates fiber characteristics, amplifier placement, channel plan, and transceiver specifications to achieve the required capacity over the target distance. Span engineering determines amplifier spacing based on fiber loss (typically 80–120 km spans for terrestrial systems), with total system reach limited by accumulated ASE noise and fiber nonlinear effects such as four-wave mixing and self-phase modulation. Modern coherent transceivers use advanced modulation formats (DP-QPSK, DP-16QAM) and digital signal processing to achieve 100–400 Gbps per channel over thousands of kilometers. The total system capacity equals the per-channel bit rate multiplied by the number of channels.

**Example 15.4.3:** A DWDM system carries 80 channels at 100 Gbps each over 1,200 km of fiber with EDFA amplifier spacing of 80 km. The fiber attenuation is 0.22 dB/km at 1550 nm. Calculate (a) the total system capacity, (b) the number of inline amplifiers required, and (c) the span loss that each EDFA must compensate.
**Solution:**
(a) Total system capacity:
Capacity = 80 channels $\times$ 100 Gbps = **8,000 Gbps = 8 Tbps**
(b) Number of inline amplifiers:
Number of spans = 1,200 / 80 = 15 spans
Inline amplifiers = $15 - 1 =$ **14** (first span has a booster amplifier at the transmitter, last span has a pre-amplifier at the receiver; 14 inline amplifiers in between)
Note: Including booster and pre-amplifier, total amplifiers = $14 + 2 = 16$.
(c) Span loss per amplifier:
Loss per span = $0.22 \times 80 =$ **17.6 dB**
Each EDFA must provide at least 17.6 dB of gain to compensate the span loss.

## 15.5 Ethernet

Ethernet is the dominant networking technology for local-area networks (LANs) and data centers, originally developed at Xerox PARC in the 1970s and standardized as IEEE 802.3. It has evolved from 10 Mbps shared coaxial bus to 400 Gbps point-to-point fiber and copper links while maintaining backward-compatible frame formats.

### 15.5.1 Frame Structure and MAC Addressing

An Ethernet frame consists of a preamble (7 bytes of alternating 1/0 pattern), start-of-frame delimiter (SFD, 1 byte), destination MAC address (6 bytes), source MAC address (6 bytes), EtherType/length field (2 bytes), payload (46–1,500 bytes), and frame check sequence (FCS, 4 bytes of CRC-32). The 48-bit MAC address uniquely identifies each network interface, with the first 24 bits assigned as an Organizationally Unique Identifier (OUI) by IEEE and the remaining 24 bits assigned by the manufacturer. The minimum frame size of 64 bytes (excluding preamble and SFD) was originally required to ensure collision detection in half-duplex CSMA/CD operation, though modern full-duplex Ethernet has eliminated this concern. The 12-byte inter-frame gap (IFG) is required between consecutive frames to allow receiver clock recovery.

**Example 15.5.1:** A network link carries 50,000 Ethernet frames per second, each with the maximum 1,500-byte payload. Calculate (a) the payload throughput in Mbps and (b) the total line rate in Mbps including all overhead (8 bytes preamble/SFD + 14 bytes header + 4 bytes FCS + 12 bytes IFG).

**Solution:**

(a) Payload throughput:

Payload bits per frame = $1{,}500 \times 8 = 12{,}000$ bits

Throughput = $50{,}000 \times 12{,}000 = 600{,}000{,}000$ bps = **600 Mbps**

(b) Total line rate:

Overhead per frame = $8 + 14 + 4 + 12 = 38$ bytes

Total frame on wire = $1{,}500 + 38 = 1{,}538$ bytes = 12,304 bits

Line rate = $50{,}000 \times 12{,}304 = 615{,}200{,}000$ bps = **615.2 Mbps**

Protocol efficiency = 600 / 615.2 = 97.5%. Ethernet overhead is minimal for maximum-size frames but increases significantly for small frames (e.g., 64-byte frames have only 46 bytes of payload, reducing efficiency to 54.8% (46 payload / 84 bytes on-wire)).

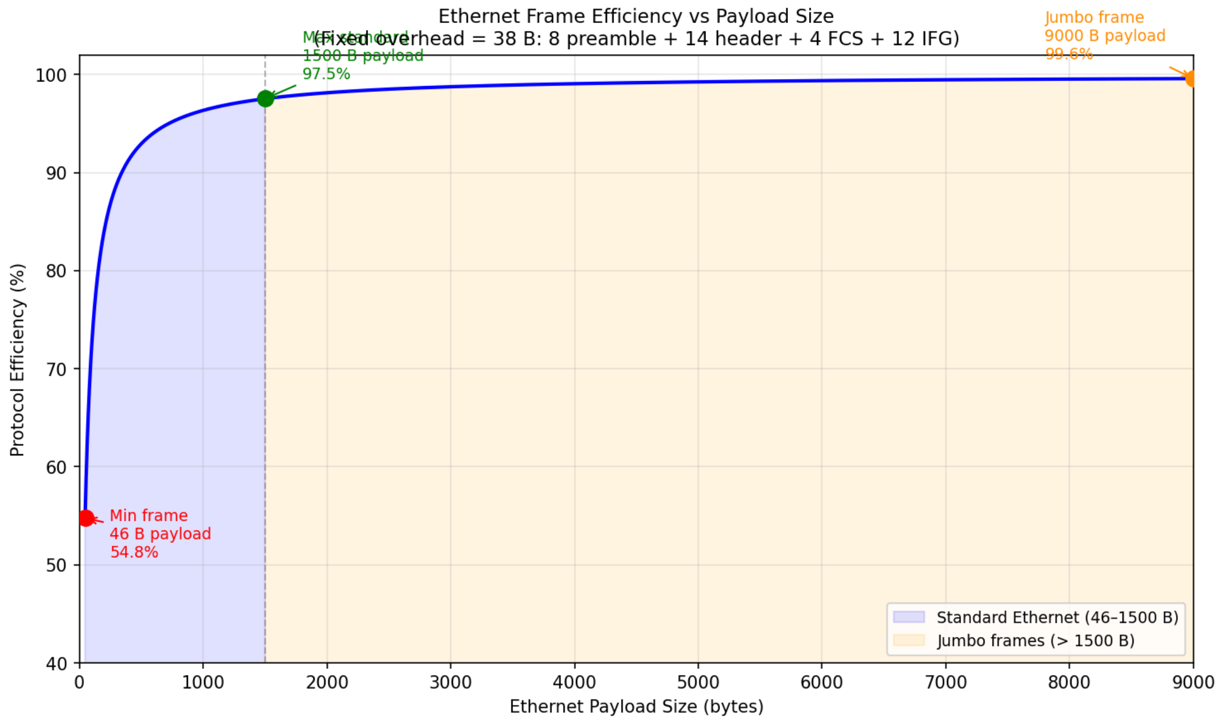



Figure 15.5.1: Ethernet protocol efficiency vs payload size (46–9000 bytes). Fixed on-wire overhead is 38 bytes (8 preamble/SFD + 14 header + 4 FCS + 12 IFG). Efficiency rises from 54.8% at the minimum 46-byte payload to 97.5% at 1500 bytes and exceeds 99% for jumbo frames.

### 15.5.2 Ethernet Physical Standards

Ethernet physical layer standards define the signaling, encoding, and media specifications for each combination of speed and medium, following the naming convention speed-BASE-media (e.g., 1000BASE-T for 1 Gbps over twisted pair, 10GBASE-SR for 10 Gbps short-reach multimode fiber). Key copper standards include 100BASE-TX (100 Mbps, Cat 5, 100 m), 1000BASE-T (1 Gbps, Cat 5e, 100 m), 10GBASE-T (10 Gbps, Cat 6A, 100 m), and 25GBASE-T/40GBASE-T (Cat 8, 30 m). Fiber standards span from 10GBASE-SR (10 Gbps, 300 m on OM3 MMF) and 10GBASE-LR (10 Gbps, 10 km on SMF) to 100GBASE-SR4 (100 Gbps, $4 \times 25$G lanes, 100 m on OM4) and 400GBASE-DR4 (400 Gbps, $4 \times 100$G, 500 m on SMF). Each standard specifies transmit power, receiver sensitivity, and the resulting maximum link distance for a given media type.

**Example 15.5.2:** A data center uses 10GBASE-SR optics (850 nm VCSEL) over OM3 multimode fiber. The minimum transmit power is −7.3 dBm, the receiver sensitivity is −11.1 dBm, and the fiber attenuation at 850 nm is 3.5 dB/km. Calculate (a) the power budget, (b) the maximum fiber-only distance (ignoring connectors), and (c) the maximum distance accounting for 2 connector pairs at 0.5 dB each.
**Solution:**
(a) Power budget:
Budget = $P_{tx,min} - P_{rx,sens} = (-7.3) - (-11.1) =$ **3.8 dB**
(b) Maximum fiber-only distance:
$d_{max}$ = Budget / $\alpha$ = 3.8 / 3.5 = **1.09 km = 1,086 m**

(c) With connector losses:
Available for fiber = 3.8 − (2 × 0.5) = 2.8 dB
$d_{max}$ = 2.8 / 3.5 = **0.80 km = 800 m**
Note: The IEEE 802.3 standard specifies 10GBASE-SR maximum reach as 300 m on OM3, which accounts for modal bandwidth limitations (not just attenuation) and provides margin for worst-case conditions.

### 15.5.3 Power over Ethernet (PoE)

Power over Ethernet (PoE) delivers DC power alongside data over standard twisted-pair Ethernet cabling, eliminating the need for separate power outlets at each device location. IEEE 802.3af (PoE, Type 1) provides up to 15.4 W at the Power Sourcing Equipment (PSE) port with a guaranteed 12.95 W at the Powered Device (PD), sufficient for VoIP phones and basic access points. IEEE 802.3at (PoE+, Type 2) doubles the power to 30 W at the PSE (25.5 W at the PD) for PTZ cameras and advanced APs. IEEE 802.3bt (PoE++, Type 3/4) delivers up to 60 W (Type 3) or 90 W (Type 4) at the PSE using all four pairs, supporting high-power devices such as LED lighting fixtures, thin clients, and outdoor access points. Power is delivered as 48 VDC (nominal, range 44–57 V) over two or four pairs using either Mode A (data pairs, phantom power on pins 1-2 and 3-6) or Mode B (spare pairs, pins 4-5 and 7-8), with the PSE and PD negotiating the power class through a classification handshake. Power loss in the cable follows $P_{loss} = I^2 \times R_{cable}$, and the maximum cable resistance per NEC for PoE installations must account for temperature rise in bundled cables per TIA TSB-184.

**Example 15.5.3:** A PoE+ switch port delivers 30 W (802.3at, Type 2) to a wireless access point over a 75-meter Cat 6A cable run. The DC resistance of Cat 6A is 7.0 Ω/100 m per conductor (two conductors per pair, two pairs used for Mode B power). Calculate the total cable resistance, the current drawn, the power lost in the cable, and the power delivered to the AP.
**Solution:**
Cable resistance per pair (round trip — current flows out on one conductor and returns on the other): $R_{pair} = 2 \times 7.0 \times (75/100) = 2 \times 5.25 = 10.5\ \Omega$.
With two pairs in parallel for power delivery: $R_{cable} = 10.5 / 2 = 5.25\ \Omega$.
PSE output voltage: 50 V (nominal).
Current: I = P / V = 30 / 50 = 0.60 A.
Cable power loss: $P_{loss} = I^2 \times R_{cable} = 0.60^2 \times 5.25 = 0.36 \times 5.25 = 1.89$ W.
Power at the PD: $P_{PD} = 30 - 1.89 =$ **28.11 W**.
Voltage at PD: $V_{PD} = 50 - (0.60 \times 5.25) = 50 - 3.15 =$ **46.85 V**.
Efficiency: $\eta = 28.11 / 30 =$ **93.7%**.
The PD voltage of 46.85 V is above the minimum 42.5 V required by the standard.

### 15.5.4 Ethernet Switching and Spanning Tree

Ethernet switches operate at Layer 2 by learning which MAC addresses are reachable through each port, building a **MAC address table** (also called a CAM table) that maps destination MAC addresses to switch ports. When a frame arrives, the switch records the source MAC and ingress port; if the destination MAC is in the table, the frame is forwarded only to the associated port (unicast forwarding), otherwise it is flooded to all ports except the ingress port (unknown unicast flooding). This learning-and-forwarding process provides microsecond-level latency and full wire-speed performance for known destinations. In networks with redundant links between switches — essential for high availability — Layer 2 loops can form, causing **broadcast storms** where frames circulate endlessly, rapidly saturating bandwidth and crashing the network. The **Spanning Tree Protocol (STP, IEEE 802.1D)** prevents loops by electing a root bridge (the switch with the lowest bridge ID), calculating the shortest path cost from every switch to the root, and placing redundant ports into a blocking state so that the active topology forms a loop-free tree. STP convergence takes 30–50 seconds (listening → learning → forwarding), which is unacceptable for modern networks. **Rapid Spanning Tree Protocol (RSTP, IEEE 802.1w)** reduces convergence to 1–3 seconds by introducing edge ports (directly connected to end devices, transition immediately to forwarding), alternate ports (pre-computed backup root ports), and a proposal/agreement handshake between switches that eliminates the listening and learning timers. **Multiple Spanning Tree Protocol (MSTP, IEEE 802.1s)** maps groups of VLANs to separate spanning tree instances, allowing different VLANs to use different active paths and enabling per-VLAN load balancing across redundant links. Modern data center networks increasingly replace STP with Layer 3 routing protocols (OSPF, BGP) at the access layer or with proprietary multi-chassis link aggregation (MLAG/vPC) that presents two physical switches as a single logical switch, eliminating STP entirely.

**Example 15.5.4:** A network has four switches (S1–S4) connected in a ring: S1↔S2, S2↔S3, S3↔S4, S4↔S1. All links are 1 Gbps with a path cost of 4. Switch bridge priorities are: S1 = 4096, S2 = 8192, S3 = 32768, S4 = 32768. Determine the RSTP root bridge, the root ports on each non-root switch, and which port is placed in the alternate (blocking) state to break the loop.
**Solution:**
Root bridge election: S1 has the lowest bridge priority (4096), so **S1 is the root bridge**.
Root port selection (lowest cost path to root): S2 root port = port facing S1 (cost 4, direct link). S4 root port = port facing S1 (cost 4, direct link).
S3 has two paths to the root: via S2 (cost $4 + 4 = 8$) or via S4 (cost $4 + 4 = 8$). Both paths have equal cost, so S3 selects the path through the neighbor with the lower bridge ID: S2 (8192) < S4 (32768), so **S3's root port faces S2** (cost 8).
Designated ports: on each link segment, the switch closer to the root places its port in the designated (forwarding) role. S1's ports facing S2 and S4 are both designated. S2's port facing S3 is designated (S2 is closer to root than S3).
The remaining port — **S3's port facing S4** (or equivalently S4's port facing S3) — has no role and is placed in the **alternate (blocking) state**. Specifically, S4's port facing S3 becomes designated (S4 has cost 4 vs S3's cost 8 from that segment), so **S3's port facing S4 is alternate/blocking**.
The active tree is: S1↔S2↔S3 and S1↔S4, with the S3↔S4 link blocked. If the S1↔S2 link fails, RSTP promotes S3's alternate port to root port within 1–3 seconds, restoring connectivity through S3↔S4↔S1.

## 15.6 Internet Protocol (IP)

The Internet Protocol provides the logical addressing and routing framework that enables packets to be forwarded across interconnected networks from source to destination. IP operates at Layer 3 and is the universal network-layer protocol of the Internet, with two versions in active use: IPv4 and IPv6.

### 15.6.1 IPv4 Addressing and Subnetting

IPv4 uses 32-bit addresses written in dotted-decimal notation (e.g., 192.168.1.100), divided into a network portion and a host portion by a subnet mask. The subnet mask, expressed as a prefix length (e.g., /24 means 24 network bits and 8 host bits), determines how many subnets and hosts are available: a /24 network provides $2^8 - 2 = 254$ usable host addresses (excluding the network address and broadcast address). Classless Inter-Domain Routing (CIDR) replaced the original classful system (Class A/B/C), allowing arbitrary prefix lengths and enabling efficient allocation through Variable Length Subnet Masks (VLSM). Subnetting is a fundamental skill for configuring routers, switches, firewalls, and any IP-enabled embedded system or IoT device.

**Example 15.6.1:** A company is assigned the network 172.16.0.0/16 and needs to create 60 subnets, each supporting at least 200 hosts. Determine (a) the required subnet mask (prefix length), (b) the number of available subnets, (c) the number of usable hosts per subnet, and (d) the network and broadcast addresses of the first two subnets.
**Solution:**
(a) Determine the minimum host bits: $2^m - 2 \geq 200 \rightarrow m = 8$ ($2^8 - 2 = 254 \geq 200$)
From the /16 base, the remaining bits after host allocation: $32 - 16 - 8 = 8$ subnet bits
Subnets available: $2^8 = 256 \geq 60$ ✓ (meets the requirement with room to spare)
**Prefix length = /24** (256 subnets, 254 usable hosts each)
(b) Available subnets: $2^8 =$ **256 subnets**
(c) Usable hosts per subnet: $2^8 - 2 =$ **254 hosts**
(d) First two subnets:
Subnet 1: Network **172.16.0.0/24**, Broadcast 172.16.0.255, Hosts 172.16.0.1–172.16.0.254
Subnet 2: Network **172.16.1.0/24**, Broadcast 172.16.1.255, Hosts 172.16.1.1–172.16.1.254

### 15.6.2 IPv6 Addressing

IPv6 uses 128-bit addresses written as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:0000:0001:0000:0000:0000:0001), with abbreviation rules: leading zeros within each group may be omitted, and a single sequence of consecutive all-zero groups may be replaced by ::. The vastly larger address space ($2^{128} \approx 3.4 \times 10^{38}$ addresses) eliminates the address exhaustion problem that drove NAT deployment in IPv4. Standard practice assigns a /64 prefix to each subnet, leaving 64 bits for the interface identifier, which can be derived from the device's MAC address using the Modified EUI-64 method or generated randomly for privacy. IPv6 is increasingly relevant as IoT devices, industrial sensors, and embedded systems adopt it for direct end-to-end addressability without NAT.

**Example 15.6.2:** An IoT sensor has the MAC address 00:1A:2B:3C:4D:5E and is on the IPv6 subnet 2001:0db8:abcd:0012::/64. Using the Modified EUI-64 method, derive the full 128-bit IPv6 address and write it in abbreviated notation.
**Solution:**
Step 1 — Split MAC and insert FFFE:
00:1A:2B → 00:1A:2B:**FF:FE**:3C:4D:5E
Step 2 — Flip the 7th bit (Universal/Local bit) of the first byte:
First byte: 00 = 0000 0000 → flip bit 7 (from left, 0-indexed bit 6) → 0000 0010 = 02
Modified: **02**:1A:2B:FF:FE:3C:4D:5E

Step 3 — Form the interface ID (regroup as 16-bit words):
021A:2BFF:FE3C:4D5E
Step 4 — Combine with subnet prefix:
Full address: 2001:0db8:abcd:0012:021A:2BFF:FE3C:4D5E
Abbreviated: **2001:db8:abcd:12:21a:2bff:fe3c:4d5e**

### 15.6.3 IP Routing Protocols

Routing protocols enable routers to automatically discover network topology, exchange reachability information, and compute optimal forwarding paths without manual static route configuration. They are divided into Interior Gateway Protocols (IGPs), which operate within a single autonomous system (AS), and Exterior Gateway Protocols (EGPs), which exchange routes between autonomous systems across the Internet. **OSPF (Open Shortest Path First)**, defined in RFC 2328 (OSPFv2 for IPv4) and RFC 5340 (OSPFv3 for IPv6), is a link-state IGP that builds a complete topological map of the network by flooding Link-State Advertisements (LSAs) to all routers within an area. Each router independently runs Dijkstra's Shortest Path First (SPF) algorithm on this link-state database (LSDB) to compute the shortest-cost path to every destination. OSPF uses a hierarchical area design: Area 0 (the backbone) interconnects all other areas through Area Border Routers (ABRs), reducing the size of the LSDB and limiting the scope of SPF recalculations. Interface costs are typically set inversely proportional to bandwidth — the default cost formula is cost = reference bandwidth / interface bandwidth (e.g., 100 Mbps reference / 1 Gbps = cost 1, / 10 Gbps = cost 0.1, often rounded or using $10^8$ or $10^9$ reference). OSPF converges in seconds after a topology change, making it suitable for enterprise and service provider networks. **BGP (Border Gateway Protocol)**, defined in RFC 4271, is the path-vector EGP that glues the Internet together by exchanging routing information between autonomous systems. BGP selects routes based on a multi-step decision process: highest local preference, shortest AS path length, lowest origin type, lowest MED, eBGP over iBGP, lowest IGP cost to next hop, and lowest router ID as tiebreakers. Unlike OSPF's metric-based shortest path, BGP enables policy-based routing — network operators apply route maps, prefix lists, and community attributes to control which routes are accepted, preferred, and advertised to neighbors. The current Internet BGP routing table contains over 950,000 IPv4 prefixes and 200,000+ IPv6 prefixes, requiring routers with substantial memory and processing capability.

**Example 15.6.3:** An enterprise network has three OSPF areas. A router in Area 1 needs to reach a destination in Area 2. The path options are: (A) Area 1 → ABR1 → Area 0 → ABR2 → Area 2, with link costs $10 + 5 + 5 + 10 = 30$; (B) Area 1 → ABR3 → Area 0 → ABR2 → Area 2, with link costs $5 + 20 + 5 + 10 = 40$. Additionally, BGP receives an external route to the same destination prefix from an eBGP peer with AS path length 3 and local preference 100 (default). The OSPF route has a default administrative distance of 110 and BGP external has 20. Which route is installed in the routing table?
**Solution:**
When multiple routing protocols offer routes to the same destination, the route with the lowest administrative distance (AD) is preferred.
OSPF AD = 110, eBGP AD = 20. Since $20 < 110$, the **eBGP route is installed** in the routing table, regardless of the OSPF cost being lower.
Within OSPF alone, path A (cost 30) would be preferred over path B (cost 40) by the SPF algorithm. The administrative distance hierarchy ensures that directly connected routes (AD 0) are preferred

over static routes (AD 1), which are preferred over eBGP (AD 20), which is preferred over OSPF (AD 110).
If the BGP route is withdrawn, the router falls back to the OSPF path A with cost 30.

### 15.6.4 Network Address Translation (NAT)

Network Address Translation allows hosts with private (RFC 1918) IP addresses — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 — to communicate with the public Internet by translating their source addresses at the network boundary. NAT was originally a stopgap measure to conserve the limited IPv4 address space (approximately 4.3 billion addresses, exhausted by IANA in 2011), but it has become a permanent fixture of network architecture due to its additional benefits of hiding internal topology and providing a basic security boundary. **Static NAT** maps a single private address to a single public address in a permanent one-to-one translation, used for servers that must be reachable from the Internet at a fixed address. **Dynamic NAT** allocates public addresses from a pool on a first-come-first-served basis, with each session receiving a different public address; when the pool is exhausted, new connections are rejected. **Port Address Translation (PAT)**, also called NAT overload, is by far the most common form — it maps thousands of internal hosts to a single public IP address by using unique source port numbers (from the 1024–65535 ephemeral range) to distinguish sessions. A PAT translation table entry contains (private IP, private port, public IP, public port, protocol) and is created when an internal host initiates an outbound connection; reply packets are translated back using the table entry. PAT allows a typical home or small office to share one public IPv4 address among hundreds of devices. **Carrier-Grade NAT (CGNAT, RFC 6598)** applies a second layer of PAT at the ISP level, sharing a smaller pool of public addresses among many subscribers using the reserved 100.64.0.0/10 shared address space. CGNAT conserves public addresses but introduces complications for peer-to-peer applications, inbound connections, and geolocation. NAT limitations include breaking end-to-end connectivity (inbound connections require explicit port forwarding or NAT traversal techniques like STUN/TURN), adding state to the network (the NAT device must maintain per-session translation tables), complicating protocols that embed IP addresses in the payload (FTP, SIP, H.323 — requiring Application Layer Gateways), and increasing latency from translation processing.

**Example 15.6.4:** A small office has 50 workstations using private addresses in the 192.168.1.0/24 range. The ISP provides a single public IPv4 address (203.0.113.5). During peak usage, 35 workstations each have an average of 40 simultaneous TCP/UDP sessions (web browsing, email, SaaS applications). Calculate (a) the total number of NAT translation entries, (b) the percentage of the ephemeral port range consumed, and (c) whether PAT can support this load. If the office grows to 200 workstations with the same session profile, determine the NAT table size and whether a single public IP is still sufficient.
**Solution:**
(a) Total sessions: $35 \times 40 =$ **1,400 simultaneous NAT entries**.
(b) Ephemeral port range: $65535 - 1024 + 1 =$ 64,512 available ports per protocol. For TCP and UDP combined: $2 \times$ 64,512 = 129,024 port mappings. Utilization: 1,400 / 129,024 = **1.09%** — well within capacity.
(c) PAT can easily handle this load; typical consumer NAT devices support 8,000–32,000 simultaneous entries, and enterprise firewalls support 500,000–2,000,000+.
With 200 workstations: assuming the same 70% active ratio and 40 sessions each: $140 \times 40 =$ **5,600**

**entries**, consuming 5,600 / 129,024 = 4.3% of the port space. A single public IP is still sufficient. PAT becomes strained only at tens of thousands of simultaneous sessions (common in large enterprises or CGNAT deployments), at which point multiple public IPs are pooled.

### 15.6.5 CIDR Notation and Route Aggregation

Classless Inter-Domain Routing (CIDR, RFC 4632) is the modern IP addressing scheme that replaced the wasteful classful system (Class A/B/C) by allowing subnet masks at any bit boundary, not just /8, /16, or /24. CIDR notation appends a prefix length to an IP address — **192.168.1.0/24** means the first 24 bits are the network portion and the remaining 8 bits identify hosts. The prefix length directly determines the number of addresses in the block: a /n prefix contains $2^{(32-n)}$ addresses for IPv4. Common CIDR blocks and their sizes:

| Prefix | Addresses | Usable Hosts | Common Use |
|---|---|---|---|
| /32 | 1 | 1 (host route) | Loopback, host-specific route |
| /31 | 2 | 2 (no network/broadcast) | Point-to-point link — RFC 3021 |
| /30 | 4 | 2 | Point-to-point link (legacy) |
| /27 | 32 | 30 | Small department |
| /24 | 256 | 254 | Standard LAN subnet |
| /22 | 1,024 | 1,022 | Large LAN |
| /16 | 65,536 | 65,534 | Campus aggregate |
| /8 | 16,777,216 | 16,777,214 | Major allocation |

To determine the network address from any IP and prefix, a bitwise AND is performed between the IP address and the subnet mask. The subnet mask for /n has n leading 1-bits followed by $(32 - n)$ zero-bits: a /24 mask is 255.255.255.0, a /20 mask is 255.255.240.0, and a /27 mask is 255.255.255.224. The broadcast address is the network address with all host bits set to 1, and usable host addresses range from (network + 1) to (broadcast − 1).

**Route aggregation** (supernetting or summarization) is the critical CIDR application that keeps the global Internet routing table manageable. An ISP that owns the contiguous block 198.51.100.0 through 198.51.103.255 (four /24 networks) can advertise a single /22 route (198.51.100.0/22) to its upstream providers, reducing four routing table entries to one. Aggregation requires contiguous, power-of-two-aligned address blocks: the four /24s must start on a /22 boundary (the first two bits of the third octet must be the same across all four networks). More generally, $2^k$ contiguous /n networks aggregate into one $/(n-k)$ route. Without CIDR aggregation, the global BGP routing table — currently exceeding 950,000 IPv4 prefixes — would be many times larger and would exceed the memory capacity of most routers.

**Example 15.6.5:** An organization is assigned four contiguous /24 networks: 10.4.16.0/24, 10.4.17.0/24, 10.4.18.0/24, and 10.4.19.0/24. (a) Determine the single CIDR summary route that covers all four networks. (b) A host has the address 10.4.18.137/22 — calculate the network address, broadcast address, and usable host range. (c) Can 10.4.16.0/24 and 10.4.18.0/24 (non-

contiguous) be summarized into a single route without including 10.4.17.0/24?
**Solution:**
(a) The four networks span 10.4.16.0 through 10.4.19.255 = $4 \times 256$ = 1,024 addresses. The aggregate prefix: $2^2$ networks of /24 would need /(24 − 2) = /22, since $2^2 = 4$.
Verify alignment: 10.4.16.0 in binary (third octet): 00010000. A /22 boundary requires the last 2 bits of the second-to-last octet to be host bits. 16 = 00010000, so the /22 block starts at 10.4.16.0 (bits 0–1 of the third octet are variable). **Summary route: 10.4.16.0/22**.
(b) Network address: AND the IP with the /22 mask (255.255.252.0): 10.4.18.137 AND 255.255.252.0 = **10.4.16.0** (third octet: 18 AND 252 = 16).
Broadcast: set all host bits to 1: **10.4.19.255** (third octet: 16 OR 3 = 19, fourth octet: 255).
Usable range: **10.4.16.1 – 10.4.19.254** (1,022 hosts).
(c) No — 10.4.16.0 and 10.4.18.0 are not contiguous (10.4.17.0 lies between them). Any CIDR summary that covers both would be 10.4.16.0/22, which includes 10.4.17.0/24 and 10.4.19.0/24 as well. Route aggregation requires contiguous address blocks; skipping 10.4.17.0 would require advertising two separate routes.

### 15.6.6 /31 and /127 Point-to-Point Subnets

Point-to-point links — the inter-router WAN connections and eBGP peering sessions that form the backbone of the Internet — need only two addresses: one for each endpoint. Historically, operators used **/30** subnets (4 addresses: network address, router A, router B, broadcast), wasting 50% of the allocation on reserved addresses that serve no functional purpose on a two-host link. RFC 3021 (2000) eliminated this waste by defining /31 subnets: both of the two addresses in the block are usable as host addresses; there is no designated network address or broadcast address. On a /31, the even address (host bit = 0) and the odd address (host bit = 1) are both assigned to router interfaces, and hosts process packets addressed to either one. For an ISP with 2,000 point-to-point inter-router links, switching from /30 to /31 reduces address consumption from 8,000 addresses to 4,000 — a savings of an entire /19 block.

**eBGP peering links** — sessions between two autonomous systems (ASes) at Internet Exchange Points (IXPs) or over private interconnects — are the most common use case for /31 addressing. Each eBGP peer requires one IP address on the shared link; the two routers form a BGP session using those directly connected addresses as the local and remote neighbor addresses. The absence of network and broadcast addresses in /31 is irrelevant because point-to-point links have no need for broadcast traffic. Many large ISPs and content providers mandate /31 on all inter-AS peering interfaces as a matter of address-conservation policy.

**IPv6 /127 subnets** (RFC 6164, 2011) serve the same role for IPv6 that /31 serves for IPv4. Early IPv6 practice used /64 subnets even for point-to-point links, which wastes $2^{64} - 2$ addresses per link. RFC 6164 defines /127 subnets for inter-router links: both the ::0 and ::1 addresses are usable host addresses, exactly mirroring the /31 model. /127 also closes the "subnet-router anycast" vulnerability present on /126 links (the IPv6 near-equivalent of /30), where the all-zeros address in a /64 is reserved as a subnet-router anycast address. Major network equipment vendors (Cisco, Nokia, Juniper) fully support /31 and /127 on physical and logical interfaces.

| Subnet | Protocol | Addresses | Usable | Wastes |
|---|---|---|---|---|
| /30 | IPv4 | 4 | 2 | 2 (network + broadcast) |
| /31 | IPv4 | 2 | 2 | 0 — RFC 3021 |
| /126 | IPv6 | 4 | 2 | 2 (network + anycast) |
| /127 | IPv6 | 2 | 2 | 0 — RFC 6164 |

**Example 15.6.6:** A regional ISP operates 480 inter-router point-to-point WAN links (backbone links between its own routers) and 120 eBGP peering sessions with external ASes, for a total of 600 two-endpoint links. The ISP has a /20 IPv4 allocation (4,096 addresses) reserved for infrastructure addressing. (a) How many /30 subnets would be needed and what percentage of the /20 would they consume? (b) How many /31 subnets are needed and what percentage is consumed? (c) With /31 addressing, how many additional point-to-point links could the ISP support from the same /20 block? (d) Show the /31 subnet assignment and eBGP neighbor configuration for a peering link between the ISP (AS 65001) and a peer (AS 65002) using the subnet 203.0.113.64/31.

**Solution:**

(a) /30 approach: Each /30 provides 2 usable addresses and consumes 4 total addresses.
600 links $\times$ 4 addresses = **2,400 addresses** required.
A /20 contains $2^{(32-20)}$ = 4,096 addresses.
Utilization: 2,400 / 4,096 = **58.6%** consumed, 1,696 addresses remaining.
(b) /31 approach: Each /31 consumes 2 addresses (both usable).
600 links $\times$ 2 addresses = **1,200 addresses** required.
Utilization: 1,200 / 4,096 = **29.3%** consumed, 2,896 addresses remaining.
(c) Remaining addresses with /31: 4,096 − 1,200 = 2,896 addresses.
Each additional link needs 2 addresses: 2,896 / 2 = **1,448 additional links** supportable.
(With /30, only 1,696 / 4 = 424 additional links would be possible from the same pool.)
(d) The /31 block 203.0.113.64/31 contains exactly two addresses:

- 203.0.113.64 — assigned to ISP router interface (AS 65001 side)
- 203.0.113.65 — assigned to peer router interface (AS 65002 side)

ISP router configuration (Cisco IOS-XR style):

```
interface GigabitEthernet0/0/0
 ipv4 address 203.0.113.64 255.255.255.254   ! /31 mask
!
router bgp 65001
 neighbor 203.0.113.65
  remote-as 65002
  address-family ipv4 unicast
```

Peer router configuration:

```
interface GigabitEthernet0/0/0
 ipv4 address 203.0.113.65 255.255.255.254   ! /31 mask
!
router bgp 65002
 neighbor 203.0.113.64
  remote-as 65001
  address-family ipv4 unicast
```

The /31 mask in dotted-decimal is 255.255.255.254 (31 leading 1-bits). There is no "network address" or "broadcast address" — both 203.0.113.64 and 203.0.113.65 are live interface addresses. The BGP session comes up between the two /31 addresses exactly as it would on a /30, with no behavioral difference visible to the routing protocol.

## 15.7 Transport Protocols

Transport protocols operate at Layer 4 of the OSI model, providing end-to-end data delivery services between application processes. The two dominant transport protocols — TCP and UDP — represent fundamentally different design trade-offs between reliability and simplicity.

### 15.7.1 TCP (Transmission Control Protocol)

TCP provides reliable, ordered, and error-checked delivery of a byte stream between applications, using a three-way handshake (SYN, SYN-ACK, ACK) to establish connections and sequence numbers to detect and retransmit lost segments. The sliding window mechanism allows multiple segments to be in flight simultaneously, with the receiver's advertised window size (up to 65,535 bytes, or up to 1 GB with the window scaling option from RFC 7323) controlling the amount of unacknowledged data. TCP congestion control algorithms (slow start, congestion avoidance, fast retransmit, fast recovery) dynamically adjust the sending rate to avoid overwhelming the network. The effective throughput of a TCP connection is fundamentally limited by the window size and the round-trip time: Throughput $\leq$ Window / RTT, known as the bandwidth-delay product (BDP) constraint.

**Example 15.7.1:** A TCP file transfer operates over a link with 100 Mbps capacity and 20 ms round-trip time (RTT). The TCP receive window is 64 KB (65,535 bytes). Calculate (a) the bandwidth-delay product of the link, (b) the maximum achievable throughput, (c) the link utilization, and (d) the window size needed to fully utilize the 100 Mbps link.
**Solution:**
(a) Bandwidth-delay product:
BDP = Bandwidth $\times$ RTT = $100 \times 10^6 \times 0.020$ = 2,000,000 bits = **250,000 bytes $\approx$ 244 KB**
(b) Maximum throughput (window-limited):
Throughput = Window / RTT = 65,535 / 0.020 = 3,276,750 bytes/s
= 3,276,750 $\times$ 8 = 26,214,000 bps = **26.2 Mbps**
(c) Link utilization:
Utilization = 26.2 / 100 = **26.2%**
(d) Window needed for full utilization:
Window = BDP = **250,000 bytes $\approx$ 244 KB**
This requires the TCP window scaling option (RFC 7323), since the base window field is limited to 65,535 bytes.

### 15.7.2 UDP (User Datagram Protocol)

UDP provides a minimal, connectionless transport service with only 8 bytes of header overhead (source port, destination port, length, and checksum), compared to TCP's 20-byte minimum header. UDP does not provide reliability, ordering, or flow control — datagrams may be lost, duplicated, or arrive out of order, and the application must handle these conditions if required. This simplicity makes UDP ideal for real-time applications where retransmission would add unacceptable latency: VoIP, video streaming, online gaming, DNS queries, SNMP network management, DHCP, and NTP time synchronization. The lack of congestion control means UDP can transmit at whatever rate the application produces, which is both an advantage (predictable timing) and a risk (potential network congestion).

**Example 15.7.2:** A VoIP system uses the G.711 codec (64 kbps, 8,000 samples/s, 8 bits per sample) with 20 ms voice frames (160 bytes of audio per packet). Each packet is encapsulated with 8 bytes UDP, 20 bytes IPv4, and 18 bytes Ethernet (14-byte header + 4-byte FCS). Calculate (a) the total bandwidth per voice call including all headers and (b) the header overhead as a percentage.
**Solution:**
(a) Total bytes per packet:
Payload: 160 bytes (voice)
Headers: 8 (UDP) + 20 (IP) + 18 (Ethernet) = 46 bytes
Total: 160 + 46 = 206 bytes per packet
Packets per second: 1 / 0.020 = 50 packets/s
Total bandwidth = $206 \times 8 \times 50$ = **82,400 bps = 82.4 kbps**
Including preamble/SFD (8 bytes) and IFG (12 bytes): total on-wire = 226 bytes
On-wire bandwidth = $226 \times 8 \times 50$ = **90,400 bps = 90.4 kbps**
(b) Header overhead:
Overhead = (82,400 − 64,000) / 82,400 = 18,400 / 82,400 = **22.3%**
The 64 kbps voice codec requires 82.4 kbps of actual network bandwidth — a 28.8% increase over the payload rate. This is why VoIP system designers sometimes use larger frame sizes (e.g., 30 ms or 40 ms) to improve efficiency, at the cost of increased latency.

### 15.7.3 Sockets

A socket is a software endpoint that provides an application programming interface (API) for sending and receiving data over a network, combining an IP address and a port number into a single addressable entity. The Berkeley sockets API, originally introduced in BSD Unix in 1983, remains the foundation of network programming across virtually all operating systems (Linux, Windows, macOS) and is exposed in languages from C to Python, Java, and Go. A TCP socket connection is uniquely identified by a 4-tuple: (source IP, source port, destination IP, destination port), which allows a single server port (e.g., port 443 for HTTPS) to handle thousands of simultaneous connections from different clients because each connection has a unique combination of source IP and source port. UDP sockets are connectionless — the application sends and receives individual datagrams to and from specific address/port pairs without establishing a persistent connection.

The typical socket lifecycle for TCP follows a client-server pattern:

| Step | Server | Client |
|---|---|---|
| 1 | `socket()` — create socket | `socket()` — create socket |
| 2 | `bind()` — assign local IP:port | (optional bind) |
| 3 | `listen()` — mark as passive, set backlog | |

| Step | Server | Client |
|---|---|---|
| 4 | `accept()` — block until client connects | `connect()` — initiate three-way handshake |
| 5 | `send()` / `recv()` — exchange data | `send()` / `recv()` — exchange data |
| 6 | `close()` — terminate connection | `close()` — terminate connection |

Socket options configure behavior such as `SO_REUSEADDR` (allow immediate rebinding after a server restart), `TCP_NODELAY` (disable Nagle's algorithm for low-latency applications), `SO_RCVBUF` / `SO_SNDBUF` (set receive and send buffer sizes to match the bandwidth-delay product), and `SO_KEEPALIVE` (detect dead connections with periodic probes). For embedded systems and IoT devices, lightweight socket implementations (lwIP, uIP) provide the same API in resource-constrained environments with as little as 10–20 KB of RAM.

**Example 15.7.3:** A server application listens on port 8080 and accepts connections from clients. The server has IP address 10.0.1.50. Three clients connect simultaneously: Client A (192.168.1.10, ephemeral port 49152), Client B (192.168.1.20, ephemeral port 52301), and Client C (192.168.1.10, ephemeral port 49153 — same IP as Client A). Determine (a) the 4-tuple identifying each connection, (b) the total number of sockets the server process holds, and (c) the maximum theoretical number of simultaneous TCP connections to this single server port from a single client IP address.
**Solution:**
(a) Connection 4-tuples:
Connection A: **(192.168.1.10, 49152, 10.0.1.50, 8080)**
Connection B: **(192.168.1.20, 52301, 10.0.1.50, 8080)**
Connection C: **(192.168.1.10, 49153, 10.0.1.50, 8080)**
Each is unique even though all three share the same destination IP and port, because the source IP:port differs.
(b) Server sockets:
1 listening socket (bound to 10.0.1.50:8080, waiting for new connections)

- 3 connected sockets (one per accepted client connection) = **4 sockets total**

(c) Maximum simultaneous connections from one client IP:
The ephemeral port range is 16-bit (0–65535), but typically the OS uses ports 49152–65535 (IANA dynamic range) = 16,384 ports. Each unique source port creates a distinct 4-tuple, so the maximum from a single client IP to a single server port is **16,384 connections** (or up to 65,535 if the full port range is used).

### 15.7.4 WSGI (Web Server Gateway Interface)

The Web Server Gateway Interface (WSGI), defined in PEP 3333, is a standard interface between web servers and Python web applications that decouples the application framework from the HTTP server, allowing any WSGI-compliant application to run on any WSGI-compliant server. Before WSGI, each Python web framework (Zope, Twisted, CherryPy) required its own server integration, making deployment inflexible and framework-locked. A WSGI application is a callable (function or class with a `__call__` method) that accepts two arguments — `environ` (a dictionary containing the HTTP request data, headers, server variables, and CGI-like metadata) and `start_response` (a callback function the application calls to set the HTTP status code and response headers) — and returns an iterable of byte strings as the response body. WSGI servers (Gunicorn, uWSGI, Waitress, mod_wsgi) handle the low-level socket operations, HTTP parsing, process/thread management, and connection pooling, then invoke the WSGI application callable for each request. WSGI middleware is a component that wraps a WSGI application to add cross-cutting functionality such as URL routing, authentication, session management, gzip compression, or logging — the middleware itself is a WSGI application that calls the inner application, forming a composable pipeline.

The WSGI architecture in a typical deployment:

| Layer | Component | Role |
|---|---|---|
| Client | Browser / API client | Sends HTTP request |
| Reverse proxy | Nginx / Apache | TLS termination, static files, load balancing |
| WSGI server | Gunicorn / uWSGI | Process management, socket handling, HTTP parsing |
| WSGI middleware | (optional) | Authentication, compression, routing |
| WSGI application | Flask / Django | Business logic, generates HTTP response |

ASGI (Asynchronous Server Gateway Interface) extends the WSGI model for asynchronous Python applications, supporting WebSockets, HTTP/2, and long-lived connections that WSGI's synchronous request-response model cannot handle. ASGI servers (Uvicorn, Daphne, Hypercorn) serve frameworks like FastAPI, Starlette, and Django Channels.

**Example 15.7.4:** A Gunicorn WSGI server is configured with 4 worker processes, each handling requests synchronously (one request at a time per worker). The average request processing time is 50 ms. Determine (a) the maximum requests per second (RPS) the server can sustain, (b) the number of workers needed to handle 500 RPS, and (c) the recommended worker count using Gunicorn's formula of $(2 \times \text{CPU cores}) + 1$ for a server with 8 CPU cores.
**Solution:**
(a) Maximum RPS with 4 workers:
Each worker handles $1 / 0.050 = 20$ requests/s
Total RPS $= 4 \times 20 =$ **80 requests/s**
(b) Workers for 500 RPS:
Workers $= 500 / 20 =$ **25 workers**
(c) Gunicorn recommended formula:
Workers $= (2 \times 8) + 1 =$ **17 workers**
At 50 ms per request: $17 \times 20 = 340$ RPS. To reach 500 RPS with 17 workers, the average response time would need to decrease to $1 / (500/17) =$ **34 ms**, or async workers (gevent/eventlet) could be used to handle I/O-bound waiting concurrently within each worker.

### 15.7.5 QUIC Protocol

QUIC (originally "Quick UDP Internet Connections") is a transport-layer protocol built on top of UDP that provides reliable, multiplexed, encrypted connections with significantly reduced handshake latency compared to TCP+TLS. Standardized as RFC 9000 (2021) and serving as the transport for HTTP/3 (RFC 9114), QUIC now carries over 30% of all web traffic globally. Traditional HTTPS requires a TCP three-way handshake (1 RTT) followed by a TLS 1.3 handshake (1 RTT), totaling 2 RTTs before the first byte of application data can be sent. QUIC integrates the cryptographic handshake into the transport handshake, achieving a 1-RTT connection setup for new connections and 0-RTT for resumed connections (where the client sends application data in the very first packet using a pre-shared key from a previous session). QUIC eliminates TCP's head-of-line blocking problem by implementing independent, multiplexed streams within a single connection — if a packet carrying data for stream A is lost, only stream A stalls while streams B and C continue receiving data, unlike TCP where a single lost segment blocks all data behind it in the byte stream. Each QUIC connection is identified by a Connection ID (not by the 4-tuple of IP addresses and ports), enabling seamless connection migration when a client changes networks (e.g., switching from Wi-Fi to cellular) without requiring a new handshake. QUIC mandates TLS 1.3 encryption for all connections — there is no unencrypted mode — and encrypts both the payload and most of the transport header fields, making the protocol more resistant to middlebox interference and network ossification than TCP. Flow control operates at both the stream level (limiting bytes per stream) and the connection level (limiting total bytes across all streams), and congestion control uses algorithms similar to TCP (Cubic, BBR) but implemented entirely in user space, allowing rapid iteration without kernel changes.

| Feature | TCP + TLS 1.3 | QUIC |
|---|---|---|
| Handshake latency (new) | 2 RTT (1 TCP + 1 TLS) | 1 RTT |
| Handshake latency (resumed) | 1 RTT (TCP Fast Open + 0-RTT TLS) | 0 RTT |
| Head-of-line blocking | Yes (single byte stream) | No (independent streams) |
| Connection migration | No (tied to 4-tuple) | Yes (Connection ID) |
| Encryption | Optional (TLS layer) | Mandatory (TLS 1.3 built-in) |
| Implementation | Kernel (TCP) + user space (TLS) | User space (entire stack) |
| Middlebox traversal | Subject to NAT/firewall TCP manipulation | UDP-based, fewer middlebox issues |

**Example 15.7.5:** A web browser loads a page that requires 50 resources (HTML, CSS, JS, images) from the same server over a network path with 40 ms RTT and 0.5% packet loss rate. Compare the connection setup time and the impact of packet loss for (a) HTTP/2 over TCP+TLS 1.3 and (b) HTTP/3 over QUIC.
**Solution:**

(a) HTTP/2 over TCP + TLS 1.3:
Connection setup = 1 RTT (TCP SYN/SYN-ACK) + 1 RTT (TLS 1.3) = $2 \times 40$ = **80 ms**
All 50 resources are multiplexed over a single TCP connection. With 0.5% loss probability per packet, the probability that at least one packet is lost over 50 resources (assuming ~10 packets per resource = 500 packets): $P(\geq 1 \text{ loss}) = 1 - (1 - 0.005)^{500} = 1 - 0.995^{500} = 1 - 0.0816 =$ **91.8%**.
When any packet is lost, TCP's head-of-line blocking stalls **all 50 streams** until the retransmission completes (1 additional RTT = 40 ms).
(b) HTTP/3 over QUIC:
Connection setup = 1 RTT = **40 ms** (50% faster)
For resumed connections: 0 RTT = **0 ms** (data sent immediately)
With the same 91.8% probability of at least one lost packet, QUIC's independent stream multiplexing means only the **affected stream(s) stall** — the other 49 resources continue loading unimpeded. Expected streams affected per loss event $\approx$ 1 out of 50, so **98% of resources are unaffected** by any single packet loss, compared to 0% for HTTP/2 over TCP.

## 15.8 Wireless Networking (Wi-Fi)

Wireless local-area networks (WLANs) based on the IEEE 802.11 family of standards provide untethered network connectivity using radio-frequency signals in unlicensed spectrum bands. Wi-Fi technology has evolved through multiple generations, each increasing data rates, spectral efficiency, and multi-user capabilities while operating in the 2.4 GHz, 5 GHz, and 6 GHz bands.

### 15.8.1 Wi-Fi Standards and Comparison

The IEEE 802.11 standards have progressed through several major generations, each identified by both an amendment letter and a Wi-Fi Alliance generation number. Wi-Fi 4 (802.11n, 2009) introduced MIMO (up to 4 spatial streams) and 40 MHz channels in the 2.4 GHz and 5 GHz bands, reaching maximum PHY rates of 600 Mbps. Wi-Fi 5 (802.11ac, 2013) expanded to 80 MHz and 160 MHz channels in 5 GHz only, added MU-MIMO for downlink, and supports up to 8 spatial streams with 256-QAM modulation for a maximum PHY rate of 6,933 Mbps. Wi-Fi 6 (802.11ax, 2020) introduced OFDMA (Orthogonal Frequency Division Multiple Access) for efficient multi-user scheduling, 1024-QAM, and operates in both 2.4 GHz and 5 GHz bands with maximum PHY rates up to 9,608 Mbps. Wi-Fi 7 (802.11be, 2024) added 320 MHz channels in the 6 GHz band, 4096-QAM, and multi-link operation (MLO) for aggregate rates exceeding 46 Gbps.

| Standard | Generation | Bands | Max Channel Width | Modulation | Max Spatial Streams | Max PHY Rate |
|---|---|---|---|---|---|---|
| 802.11n | Wi-Fi 4 | 2.4 / 5 GHz | 40 MHz | 64-QAM | 4 | 600 Mbps |
| 802.11ac | Wi-Fi 5 | 5 GHz | 160 MHz | 256-QAM | 8 | 6,933 Mbps |
| 802.11ax | Wi-Fi 6/6E | 2.4 / 5 / 6 GHz | 160 MHz | 1024-QAM | 8 | 9,608 Mbps |

| Standard | Generation | Bands | Max Channel Width | Modulation | Max Spatial Streams | Max PHY Rate |
|---|---|---|---|---|---|---|
| 802.11be | Wi-Fi 7 | 2.4 / 5 / 6 GHz | 320 MHz | 4096-QAM | 16 | 46,120 Mbps |

**Example 15.8.1:** Compare the spectral efficiency of Wi-Fi 5 (802.11ac) and Wi-Fi 6 (802.11ax) for a single spatial stream on an 80 MHz channel. Wi-Fi 5 uses 256-QAM with 5/6 coding rate and 234 data subcarriers with a 3.2 μs symbol duration (plus 0.8 μs guard interval). Wi-Fi 6 uses 1024-QAM with 5/6 coding rate and 980 data subcarriers with a 12.8 μs symbol duration (plus 0.8 μs guard interval). Calculate the maximum PHY data rate for each.
**Solution:**
**Wi-Fi 5 (802.11ac):**
Bits per subcarrier = $\log_2(256) \times (5/6) = 8 \times 0.8333 = 6.667$ coded bits
Symbol rate = $1 / (3.2 + 0.8)$ μs = $1 / 4.0$ μs = 250,000 symbols/s
Data rate = $234 \times 6.667 \times 250{,}000$ = **390.0 Mbps**
Spectral efficiency = 390 / 80 = 4.88 bps/Hz
**Wi-Fi 6 (802.11ax):**
Bits per subcarrier = $\log_2(1024) \times (5/6) = 10 \times 0.8333 = 8.333$ coded bits
Symbol rate = $1 / (12.8 + 0.8)$ μs = $1 / 13.6$ μs = 73,529 symbols/s
Data rate = $980 \times 8.333 \times 73{,}529$ = **600.5 Mbps**
Spectral efficiency = 600.5 / 80 = 7.51 bps/Hz
Wi-Fi 6 achieves **54.0% higher throughput** and **54.0% better spectral efficiency** than Wi-Fi 5 on the same 80 MHz channel, primarily through denser subcarrier spacing (78.125 kHz vs. 312.5 kHz) and higher-order modulation.

### 15.8.2 RF Propagation and Link Budget

Wireless signal strength decreases with distance due to free-space path loss (FSPL), which increases with both distance and frequency according to FSPL (dB) = $20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$, where d is distance in km and f is frequency in MHz. In indoor environments, additional losses from walls, floors, furniture, and multipath fading further reduce the signal, typically adding 3–15 dB per wall depending on material (drywall ≈ 3 dB, concrete ≈ 10–15 dB). The received signal power must exceed the receiver sensitivity (typically −65 dBm for high-rate operation or −85 dBm for basic connectivity) to maintain a reliable link. A wireless link budget sums the transmitter power, antenna gains, and all path losses to predict the received signal strength and determine the maximum range or required access point density.

**Example 15.8.2:** A Wi-Fi 6 access point transmits at 20 dBm (100 mW) on a 5 GHz channel (5,500 MHz) with an antenna gain of 4 dBi. The client device has an antenna gain of 2 dBi and a receiver sensitivity of −65 dBm for 802.11ax MCS11 (1024-QAM, 5/6). Calculate (a) the free-space path loss at 30 m, (b) the received signal strength (free-space, no obstructions), and (c) the maximum free-space range at which the link can operate at MCS11.
**Solution:**

(a) Free-space path loss at 30 m:
$\text{FSPL} = 20\log_{10}(0.030) + 20\log_{10}(5500) + 32.44$
$= 20 \times (-1.523) + 20 \times 3.740 + 32.44$
$= -30.46 + 74.81 + 32.44 = \mathbf{76.8\ dB}$
(b) Received signal strength at 30 m:
$P_{rx} = P_{tx} + G_{tx} + G_{rx} - \text{FSPL}$
$= 20 + 4 + 2 - 76.8 = \mathbf{-50.8\ dBm}$
This is well above the −65 dBm sensitivity — the link has 14.2 dB of margin at 30 m in free space.
(c) Maximum range for MCS11:
Maximum allowable path loss $= P_{tx} + G_{tx} + G_{rx} - P_{rx,min}$
$= 20 + 4 + 2 - (-65) = 91$ dB
$91 = 20\log_{10}(d) + 20\log_{10}(5500) + 32.44$
$20\log_{10}(d) = 91 - 74.81 - 32.44 = -16.25$
$\log_{10}(d) = -0.8125$
$d = 10^{-0.8125} = 0.1539$ km $=$ **153.9 m** (free space)
In practice, indoor walls and multipath reduce this to approximately 15–30 m for reliable MCS11 operation.

### 15.8.3 Wireless Security

Wireless networks are inherently more vulnerable than wired networks because the radio signal propagates beyond the physical boundaries of the facility, allowing any device within range to intercept frames or attempt to associate with the access point. Wireless security has evolved through several generations, each addressing vulnerabilities of its predecessor. **WEP (Wired Equivalent Privacy)** used a 40-bit or 104-bit static key with RC4 encryption but was fatally flawed — the 24-bit initialization vector (IV) repeated after only $2^{24} \approx 16.7$ million frames, enabling key recovery attacks in minutes. **WPA (Wi-Fi Protected Access)** introduced TKIP (Temporal Key Integrity Protocol) as an interim fix, using per-packet keys and a message integrity check, but TKIP itself was later found vulnerable to certain attacks. **WPA2** (IEEE 802.11i, 2004) replaced TKIP with AES-CCMP (Counter Mode with CBC-MAC Protocol), providing robust 128-bit encryption that remains unbroken when used with strong passphrases. WPA2-Personal uses a Pre-Shared Key (PSK) derived from a passphrase via PBKDF2 with 4,096 iterations, while WPA2-Enterprise uses IEEE 802.1X authentication with a RADIUS server and EAP (Extensible Authentication Protocol) methods such as EAP-TLS (certificate-based, strongest), PEAP (server certificate + inner MSCHAPv2), and EAP-TTLS. **WPA3** (2018) addresses remaining WPA2 weaknesses: WPA3-Personal replaces the PSK four-way handshake with SAE (Simultaneous Authentication of Equals), a zero-knowledge proof protocol based on the Dragonfly key exchange that provides resistance to offline dictionary attacks even with weak passphrases, and forward secrecy (compromising the passphrase does not decrypt previously captured traffic). WPA3-Enterprise mandates 192-bit equivalent cryptographic strength using CNSA (Commercial National Security Algorithm) suite with AES-256-GCM and SHA-384, and requires Protected Management Frames (PMF/802.11w) to prevent deauthentication and disassociation attacks. **802.1X** port-based network access control authenticates devices before granting network access, using a three-party model: the supplicant (client device), the authenticator (access point or switch), and the authentication server (RADIUS). The authenticator blocks all traffic from the supplicant except 802.1X (EAPoL) frames until the RADIUS server validates the supplicant's credentials and returns an Access-Accept with optional VLAN assignment and session parameters.

**Example 15.8.3:** A corporate Wi-Fi network uses WPA3-Enterprise with EAP-TLS (mutual certificate authentication) and a RADIUS server. The AES-256-GCM cipher provides 256-bit key strength. Calculate (a) the number of possible key combinations for a brute-force attack, (b) the time required to exhaust all keys at a rate of $10^{12}$ guesses per second, and (c) compare the security of a WPA2-Personal network with an 8-character lowercase passphrase to WPA3-SAE with the same passphrase.
**Solution:**
(a) AES-256 key combinations: $2^{256} = 1.158 \times 10^{77}$ possible keys.
(b) Time to exhaust: $1.158 \times 10^{77}$ / $10^{12}$ = $1.158 \times 10^{65}$ seconds = $3.67 \times 10^{57}$ years — far beyond the age of the universe ($1.38 \times 10^{10}$ years), confirming that AES-256 is computationally infeasible to brute-force.
(c) WPA2-PSK with 8 lowercase letters: $26^8 = 2.089 \times 10^{11}$ possible passphrases. An offline dictionary attack after capturing the four-way handshake can test $10^6$ passphrases/second (GPU-accelerated), exhausting the keyspace in $2.089 \times 10^{11}$ / $10^6$ = 208,900 seconds ≈ **2.4 days**.
WPA3-SAE with the same passphrase: SAE requires an online interaction with the AP for each guess attempt, limiting the rate to approximately 10 attempts/second before the AP implements rate limiting. Exhaustion time: $2.089 \times 10^{11}$ / 10 = $2.089 \times 10^{10}$ seconds ≈ **662 years**.
WPA3-SAE transforms a weak 2.4-day attack into a 662-year attack by eliminating offline attacks, though a strong passphrase (12+ characters, mixed case/numbers/symbols) is still recommended.

## 15.9 Network Infrastructure

Network infrastructure encompasses the physical equipment that interconnects end devices, provides traffic forwarding, and organizes cabling within facilities. Switches, routers, and structured cabling systems (including patch panels) form the backbone of any wired network, from a single wiring closet to a multi-building campus or data center.

### 15.9.1 Switches

A network switch is a Layer 2 device that forwards Ethernet frames between ports based on destination MAC addresses, maintaining a MAC address table (also called a CAM table) that maps each learned MAC address to the port where it was observed. When a frame arrives, the switch looks up the destination MAC in its table and forwards the frame only to the corresponding port (unicast), or floods the frame to all ports (except the source port) if the destination is unknown or is a broadcast/multicast address. Switches operate in store-and-forward mode (buffering the entire frame and checking the CRC before forwarding) or cut-through mode (forwarding as soon as the destination address is read, reducing latency to as low as 1–2 μs). Managed switches add features such as VLANs (IEEE 802.1Q), Spanning Tree Protocol (STP/RSTP for loop prevention), Link Aggregation (LAG/802.3ad for bonding multiple physical links), Quality of Service (QoS for traffic prioritization), and port security. Layer 3 switches combine switching and routing in a single device, performing inter-VLAN routing at wire speed using hardware-based forwarding (ASICs or FPGAs) rather than software-based routing.

**Example 15.9.1:** A 48-port Gigabit Ethernet switch has a switching fabric rated at 176 Gbps and a forwarding rate of 130.95 Mpps. Determine (a) whether the fabric is non-blocking (can handle full-duplex wire-speed traffic on all ports simultaneously) and (b) the minimum packet size at

which the switch can sustain wire-speed forwarding.
**Solution:**
(a) Non-blocking check:
Full-duplex bandwidth per port = $2 \times 1$ Gbps = 2 Gbps (1 Gbps in each direction)
Total required fabric bandwidth = $48 \times 2 = 96$ Gbps
Switching fabric = 176 Gbps > 96 Gbps → **Non-blocking ✓**
(b) Minimum wire-speed packet size:
Wire-speed packet rate (all 48 ports, full duplex) at minimum 64-byte frames:
Ethernet overhead per frame on wire = 64 bytes + 8 (preamble/SFD) + 12 (IFG) = 84 bytes
Max frames/s per port = $1 \times 10^9 / (84 \times 8) = 1{,}488{,}095$ fps
Total for 48 ports (full duplex, but forwarding counts each frame once): $48 \times 1{,}488{,}095 = 71{,}428{,}571$ fps
Switch forwarding rate = $130.95 \times 10^6$ fps
Wire-speed rate for 48 ports at 64-byte frames = 71.4 Mpps
Since 130.95 Mpps > 71.4 Mpps, the switch can forward at wire speed even at the **minimum 64-byte frame size ✓**

### 15.9.2 Routers

A router is a Layer 3 device that forwards packets between different IP networks (subnets) based on destination IP addresses, using a routing table populated by static configuration or dynamic routing protocols. Unlike switches that operate on MAC addresses within a single broadcast domain, routers make forwarding decisions using the longest-prefix match against the destination IP address, decrement the TTL, recompute the IP header checksum, and rewrite the Layer 2 (MAC) headers for the next hop. Dynamic routing protocols exchange reachability information between routers: Interior Gateway Protocols (IGPs) such as OSPF (Open Shortest Path First) and IS-IS operate within an autonomous system using link-state algorithms, while BGP (Border Gateway Protocol) exchanges routes between autonomous systems across the Internet. Modern routers forward packets at line rate using hardware-based forwarding engines (ASICs with Ternary Content-Addressable Memory, or TCAM), achieving forwarding rates of hundreds of millions to billions of packets per second in core and data center routers.

**Example 15.9.2:** A router has four 100 Gbps interfaces and a forwarding capacity of 600 Mpps. Determine (a) the total bidirectional bandwidth, (b) whether the router can handle wire-speed forwarding of 64-byte packets on all interfaces simultaneously, and (c) the maximum number of full routing table entries (IPv4 prefix + next-hop, each consuming 8 bytes) that can fit in 4 MB of TCAM.
**Solution:**
(a) Total bidirectional bandwidth:
4 interfaces × 100 Gbps × 2 (full duplex) = **800 Gbps**
(b) Wire-speed check at 64-byte packets:
Frames per second per 100G interface = $100 \times 10^9 / (84 \times 8) = 148{,}809{,}524$ fps
Total for 4 interfaces = $4 \times 148{,}809{,}524 = 595{,}238{,}095$ fps ≈ 595.2 Mpps
Router capacity = 600 Mpps > 595.2 Mpps → **Wire-speed at 64 bytes ✓**
(c) TCAM routing table capacity:
Entries = $4 \times 10^6$ bytes / 8 bytes per entry = **500,000 entries**

The current Internet IPv4 routing table contains approximately 950,000 prefixes, so 4 MB of TCAM at 8 bytes/entry would be insufficient for a full Internet routing table. In practice, TCAM entries are wider (40–80 bytes for prefix + mask + action fields), further reducing capacity.

### 15.9.3 Patch Panels and Structured Cabling

A patch panel is a passive (unpowered) device mounted in a rack or cabinet that provides a termination point for horizontal cabling runs from wall outlets, consolidation points, or other equipment. Each port on the patch panel has an insulation displacement contact (IDC) connection on the back (where permanent horizontal cables are punched down) and an RJ-45 jack on the front (where patch cords connect to switch ports). Structured cabling standards (TIA-568, ISO/IEC 11801) define a hierarchical topology: the Equipment Room (ER) or Main Distribution Frame (MDF) houses core switches and serves as the building entry point, Telecommunications Rooms (TR) or Intermediate Distribution Frames (IDF) house access-layer switches on each floor, and horizontal cabling connects the TR to individual work-area outlets within the 90 m permanent link limit (100 m total channel length including patch cords). Patch panels are labeled using a consistent nomenclature: rack identifier, panel position (U-height), and port number (e.g., “MDF-R1-U24-P12” identifies rack 1 in the MDF, patch panel at U-position 24, port 12). Fiber patch panels (fiber distribution frames, or FDFs) serve the same purpose for fiber optic cabling, using LC, SC, or MPO adapters and typically organizing fibers by strand count and cable routing.

| Term | Meaning |
|---|---|
| MDF | Main Distribution Frame — primary equipment room, building entry point |
| IDF | Intermediate Distribution Frame — floor-level telecom room |
| TR | Telecommunications Room (TIA term for IDF) |
| Horizontal cable | Permanent cable from TR patch panel to work-area outlet (max 90 m) |
| Backbone cable | Cable connecting MDF to IDFs (between floors or buildings) |
| Patch cord | Short cable from patch panel to switch or from outlet to device |
| Permanent link | Tested cable path from patch panel IDC to outlet IDC (max 90 m) |
| Channel | Full end-to-end path including patch cords (max 100 m for copper) |
| U-height | Rack unit (1U = 1.75 inches / 44.45 mm); a 24-port panel is typically 1U |
| 110 block | Punch-down termination block for voice/data (alternative to RJ-45 panel) |

**Example 15.9.3:** A three-story office building has an MDF in the basement and one IDF per floor. Each floor has 80 data drops. The horizontal cable runs average 45 m from the IDF to wall outlets. Patch cords are 3 m at the IDF (panel to switch) and 2 m at the work area (outlet to device). Determine (a) the total channel length per drop, (b) whether it meets the TIA-568 100 m channel limit, (c) the number of 24-port patch panels needed per IDF, and (d) the minimum number of 48-port switches per IDF (assuming one switch port per data drop).

**Solution:**

(a) Total channel length:

Channel = IDF patch cord + horizontal cable + work area patch cord

= 3 + 45 + 2 = **50 m**

(b) TIA-568 compliance:

50 m < 100 m → **Compliant ✓** (with 50 m of margin)

Permanent link = 45 m < 90 m → **Compliant ✓**

(c) Patch panels per IDF:

Panels = 80 ports / 24 ports per panel = 3.33 → **4 patch panels** (96 ports, 16 spare)

(d) Switches per IDF:

Switches = 80 ports / 48 ports per switch = 1.67 → **2 switches** (96 ports, 16 spare)

### 15.9.4 VLANs and Network Segmentation

Virtual LANs (VLANs) partition a single physical switch (or a group of interconnected switches) into multiple isolated broadcast domains, each behaving as an independent Layer 2 network. IEEE 802.1Q defines the standard by inserting a 4-byte tag into the Ethernet frame between the source MAC address and the EtherType field; the tag contains a 3-bit Priority Code Point (PCP) for QoS, a 1-bit Drop Eligible Indicator (DEI), and a 12-bit VLAN ID (VID) supporting 4,094 usable VLANs (VIDs 1–4094; VID 0 is priority-tagged only, VID 4095 is reserved). Switch ports are configured as **access ports** (carrying a single untagged VLAN, connected to end devices) or **trunk ports** (carrying multiple tagged VLANs, connected to other switches, routers, or servers). The native VLAN on a trunk is carried untagged for backward compatibility with non-VLAN-aware devices. VLANs provide security through broadcast domain isolation (a host in VLAN 10 cannot directly communicate with VLAN 20 without a Layer 3 router or inter-VLAN routing), reduce unnecessary broadcast traffic, and enable flexible network design independent of physical topology — a user can be in the same VLAN regardless of which floor or building they connect from. **Inter-VLAN routing** requires a Layer 3 device (router or Layer 3 switch) with an interface or Switched Virtual Interface (SVI) in each VLAN. A common design pattern is the “router-on-a-stick” topology, where a single router interface carries all VLANs as 802.1Q sub-interfaces on a trunk link to the switch, though dedicated Layer 3 switch SVIs are preferred for performance. **Private VLANs** (RFC 5765) further subdivide a VLAN into promiscuous, isolated, and community ports for micro-segmentation within a single IP subnet — commonly used in data centers and multi-tenant environments where hosts in the same subnet should not communicate directly.

**Example 15.9.4:** A campus network has three buildings connected by fiber trunk links to a core switch. The network requires four VLANs: VLAN 10 (Engineering, 200 hosts), VLAN 20 (Finance, 50 hosts), VLAN 30 (Guest Wi-Fi, 100 hosts), and VLAN 99 (Management, 20 hosts). Each building has a 48-port access switch trunked to the core. Calculate (a) the number of 802.1Q tag bits used for VLANs vs. total tag bits, (b) the subnet sizes needed for each VLAN using the smallest appropriate CIDR prefix from the 10.0.0.0/8 space, and (c) the total bandwidth overhead of

802.1Q tagging on a trunk carrying 100,000 frames per second.
**Solution:**
(a) VLAN ID field: 12 bits out of the 32-bit (4-byte) 802.1Q tag. VLAN bits / total tag bits = 12/32 = **37.5%**. The remaining bits are: TPID (16 bits, fixed at 0x8100 identifying the frame as tagged), PCP (3 bits), DEI (1 bit).
(b) Subnet sizing:
VLAN 10 (200 hosts): needs $2^n - 2 \geq 200 \rightarrow n = 8$ (254 hosts) → **10.0.10.0/24**.
VLAN 20 (50 hosts): needs $2^n - 2 \geq 50 \rightarrow n = 6$ (62 hosts) → **10.0.20.0/26**.
VLAN 30 (100 hosts): needs $2^n - 2 \geq 100 \rightarrow n = 7$ (126 hosts) → **10.0.30.0/25**.
VLAN 99 (20 hosts): needs $2^n - 2 \geq 20 \rightarrow n = 5$ (30 hosts) → **10.0.99.0/27**.
(c) 802.1Q overhead: each tagged frame adds 4 bytes. At 100,000 fps: overhead = $100{,}000 \times 4 \times 8$ = 3,200,000 bps = **3.2 Mbps**. On a 10 Gbps trunk, this is 3.2/10,000 = 0.032% — negligible.

### 15.9.5 Network Security Fundamentals

Network security protects data in transit, prevents unauthorized access, and detects malicious activity through a layered defense-in-depth strategy combining hardware, software, and policy controls. **Firewalls** are the primary perimeter defense, inspecting traffic and enforcing rules that permit or deny packets based on source/destination IP address, port number, protocol, and connection state. **Stateless packet filters** (ACL-based) evaluate each packet independently against ordered rules — fast but unable to distinguish between legitimate reply traffic and unsolicited inbound connections. **Stateful firewalls** track the state of each TCP/UDP session in a connection table (SYN, ESTABLISHED, RELATED), automatically allowing return traffic for established sessions while blocking unsolicited inbound packets — the standard for modern network firewalls. **Next-generation firewalls (NGFW)** add application-layer inspection (identifying applications regardless of port number), intrusion prevention, URL filtering, and TLS decryption. **Access Control Lists (ACLs)** are ordered rule sets applied to router or switch interfaces that filter traffic by matching packet header fields; rules are processed top-down and the first match determines the action (permit or deny), with an implicit "deny all" at the end. Standard ACLs filter by source IP only, while extended ACLs filter by source IP, destination IP, protocol, and port numbers. **Virtual Private Networks (VPNs)** create encrypted tunnels over untrusted networks (the Internet) to provide confidentiality, integrity, and authentication for remote access and site-to-site connectivity. IPsec VPNs operate at Layer 3 using Authentication Header (AH) for integrity or Encapsulating Security Payload (ESP) for encryption + integrity, with Internet Key Exchange (IKEv2) negotiating the security associations. SSL/TLS VPNs operate at Layers 4–7 and are accessible through standard web browsers, simplifying remote-access deployments. **Intrusion Detection Systems (IDS)** passively monitor network traffic for signatures of known attacks and anomalous behavior, generating alerts for security analysts. **Intrusion Prevention Systems (IPS)** operate inline and can automatically block or drop malicious traffic in real time. Both use signature-based detection (matching known attack patterns) and anomaly-based detection (flagging deviations from a baseline traffic profile). The **802.1X** standard provides port-based network access control, requiring devices to authenticate (via RADIUS) before the switch port grants network access — preventing unauthorized devices from connecting to the LAN.

**Example 15.9.5:** An enterprise firewall has the following stateful ACL rules applied to the outside (Internet-facing) interface, processed in order:
1. Permit TCP from any to 203.0.113.10 port 443 (HTTPS web server)

2. Permit TCP from any to 203.0.113.11 port 25 (SMTP mail server)
3. Permit UDP from any to 203.0.113.12 port 53 (DNS server)
4. Deny IP from any to 203.0.113.0/24 (block all other inbound)
5. Permit IP from 203.0.113.0/24 to any (allow outbound and return traffic)

Determine the firewall's action for each of these packets: (a) TCP SYN to 203.0.113.10:443 from 198.51.100.5:49152, (b) TCP SYN to 203.0.113.10:22 (SSH) from 198.51.100.5:49200, (c) UDP packet to 203.0.113.12:53 from 198.51.100.8:5353, (d) TCP ACK to 203.0.113.50:1025 from 93.184.216.34:80 (reply to outbound web request with an existing session in the state table).

**Solution:**

(a) Matches rule 1 (TCP, destination 203.0.113.10, port 443) → **Permitted**. A new session entry is created in the state table.
(b) Destination is 203.0.113.10 but port 22 (SSH) does not match rule 1 (port 443) or rule 2 (port 25) or rule 3 (UDP port 53). Matches rule 4 (deny all to 203.0.113.0/24) → **Denied (dropped)**. SSH access from the Internet is blocked.
(c) Matches rule 3 (UDP, destination 203.0.113.12, port 53) → **Permitted**.
(d) This is a reply packet to an outbound connection. The stateful firewall checks the session table first — because an established session exists for the outbound HTTP request from 203.0.113.50 to 93.184.216.34:80, the return ACK packet is **permitted as ESTABLISHED return traffic** without needing to match any explicit ACL rule. Without stateful inspection, this packet would match rule 4 and be denied, breaking all outbound web browsing.

### 15.9.6 Software-Defined Networking (SDN)

Software-Defined Networking separates the network's control plane (the intelligence that decides where traffic should go) from the data plane (the hardware that actually forwards packets), centralizing routing decisions in a software-based SDN controller while switches and routers become simple forwarding devices that execute instructions received from the controller. In traditional networking, each device runs its own control-plane protocols (OSPF, BGP, STP) and makes independent forwarding decisions based on distributed state, making network-wide policy changes slow and error-prone — an administrator must configure each device individually. SDN replaces this distributed model with a logically centralized controller that has a global view of the network topology, traffic flows, and device capabilities, enabling programmatic, automated network management through northbound APIs (REST, gRPC) exposed to applications and orchestration systems.

The **OpenFlow** protocol (ONF standard) was the first widely adopted southbound API connecting the SDN controller to forwarding devices. An OpenFlow switch maintains one or more flow tables, each containing flow entries that match packet headers against rules and specify actions (forward to port, drop, modify header, send to controller). When a packet arrives that matches no flow entry, the switch sends it to the controller (a "packet-in" message), which installs a new flow entry to handle subsequent matching packets. Modern SDN has evolved beyond pure OpenFlow to include overlay-based architectures (VXLAN, GENEVE) where the underlay network runs traditional routing and the SDN controller manages virtual network overlays, as well as intent-based networking where administrators specify desired outcomes ("isolate IoT devices from corporate servers") rather than device-level configurations. Data center SDN platforms (VMware NSX, Cisco ACI, OpenStack Neutron) manage thousands of virtual switches across hypervisors, providing microsegmentation, distributed firewalling, and load balancing entirely in software.

| SDN Layer | Function | Examples |
|---|---|---|
| Application layer | Business logic, network apps | Load balancer, firewall policy, traffic engineering |
| Northbound API | Controller ↔ application interface | REST API, gRPC, NETCONF/YANG |
| Control plane | Centralized network intelligence | ONOS, OpenDaylight, VMware NSX, Cisco ACI |
| Southbound API | Controller ↔ switch interface | OpenFlow, OVSDB, P4, gNMI |
| Data plane | Packet forwarding | Open vSwitch (OVS), bare-metal switches, SmartNICs |

**Example 15.9.6:** An SDN controller manages a data center fabric with 96 top-of-rack (ToR) switches, each with $48 \times 25$ Gbps server-facing ports and $6 \times 100$ Gbps uplinks to spine switches. The controller installs flow entries that each consume 512 bytes of TCAM. Each ToR switch has 16 MB of TCAM. Calculate (a) the total server-facing bandwidth in the fabric, (b) the maximum number of flow entries per ToR switch, and (c) the total flow state the controller must manage if each switch averages 60% TCAM utilization.
**Solution:**
(a) Total server-facing bandwidth:
Per ToR: $48 \times 25 = 1{,}200$ Gbps = 1.2 Tbps
Total fabric: $96 \times 1{,}200 = 115{,}200$ Gbps = **115.2 Tbps**
(b) Maximum flow entries per ToR:
Entries = $16 \times 10^6 / 512 =$ **31,250 flow entries**
(c) Total flow state at 60% utilization:
Entries per switch = $31{,}250 \times 0.60 = 18{,}750$
Total across fabric = $96 \times 18{,}750 =$ **1,800,000 flow entries**
Total TCAM state = $1{,}800{,}000 \times 512 = 921.6 \times 10^6$ bytes $\approx$ **922 MB** of flow state managed by the controller.

## 15.10 Network Performance

Network performance metrics quantify how effectively a network transports data between endpoints, bridging the physical and protocol layers into measurable parameters that engineers use for system design, capacity planning, and troubleshooting.

### 15.10.1 Latency and Propagation Delay

Network latency is the total time for a packet to travel from source to destination, composed of propagation delay (signal travel time through the medium), serialization delay (time to place all bits on the wire), processing delay (router/switch forwarding time), and queuing delay (time spent waiting in buffers). Propagation delay depends on the medium: approximately 4.9 μs/km in fiber (speed of light

/ refractive index ≈ c/1.468), 4.5–5.0 μs/km in copper (depending on cable type and velocity factor), and 3.33 μs/km in free space. Serialization delay equals packet size divided by link rate: a 1,500-byte packet takes 12 μs on a 1 Gbps link and 1.2 μs on a 10 Gbps link. For long-distance links, propagation delay dominates; for short, high-speed links, processing and queuing delays are more significant.

**Example 15.10.1:** A 1,500-byte packet traverses a path consisting of 500 km of single-mode fiber (refractive index n = 1.468) and passes through 3 routers, each adding 50 μs of processing delay. The link rate is 10 Gbps. Calculate (a) the propagation delay, (b) the serialization delay, (c) the total one-way latency, and (d) the round-trip time (RTT).
**Solution:**
(a) Propagation delay:
$v_{fiber} = c / n = 3.0 \times 10^8 / 1.468 = 2.044 \times 10^8$ m/s
$t_{prop}$ = distance / velocity = 500,000 / ($2.044 \times 10^8$) = **2,446 μs = 2.446 ms**
(b) Serialization delay:
$t_{serial}$ = packet size / link rate = (1,500 × 8) / ($10 \times 10^9$) = 12,000 / $10^{10}$ = **1.2 μs**
(c) Total one-way latency:
Processing delay = 3 × 50 = 150 μs
$t_{total}$ = 2,446 + 1.2 + 150 = **2,597.2 μs** ≈ **2.60 ms**
(d) Round-trip time:
RTT = $2 \times t_{total}$ = 2 × 2.60 = **5.20 ms**
The propagation delay (2.446 ms) dominates at 94.2% of the total, with processing (5.8%) and serialization (0.05%) being negligible. This demonstrates why the speed of light, not link bandwidth, limits latency on long-distance networks.

### 15.10.2 Throughput and Bandwidth Utilization

Throughput is the actual rate of successful data delivery, which is always less than the raw line rate due to protocol overhead, retransmissions, and congestion. The Shannon-Hartley theorem establishes the theoretical maximum channel capacity: $C = B \times \log_2(1 + SNR)$, where B is the bandwidth in Hz and SNR is the linear signal-to-noise ratio. For practical systems, the spectral efficiency (bps/Hz) depends on the modulation scheme and coding rate: 10GBASE-T achieves approximately 2.5 bps/Hz across its effective bandwidth, while 100G coherent optics using DP-16QAM achieve approximately 4 bps/Hz. Network utilization should typically be kept below 70–80% of capacity to avoid excessive queuing delays and packet loss.

**Example 15.10.2:** A copper Ethernet link has a usable bandwidth of 250 MHz and achieves an SNR of 30 dB. Calculate (a) the Shannon channel capacity and (b) the spectral efficiency of the actual 1000BASE-T (1 Gbps) system operating over this bandwidth.
**Solution:**
(a) Shannon capacity:
SNR (linear) = $10^{(30/10)}$ = 1,000
$C = B \times \log_2(1 + SNR) = 250 \times 10^6 \times \log_2(1{,}001) = 250 \times 10^6 \times 9.967$
C = **2,491.8 Mbps** ≈ **2.49 Gbps**
(b) Spectral efficiency of 1000BASE-T:
$\eta$ = Data rate / Bandwidth = 1,000 / 250 = **4.0 bps/Hz**

Note: 1000BASE-T uses four pairs simultaneously at 250 Mbps each (125 Msymbols/s × PAM-5 encoding), so the per-pair spectral efficiency is 250 / 62.5 ≈ 4.0 bps/Hz (using 62.5 MHz bandwidth per pair). The Shannon limit of 2.49 Gbps confirms that the 1 Gbps rate operates well below the theoretical maximum, providing margin for practical impairments.

### 15.10.3 Bit Error Rate (BER)

Bit Error Rate (BER) is the ratio of erroneously received bits to the total number of bits transmitted, expressed as a probability (e.g., $10^{-12}$ means on average one bit error per trillion bits). For fiber optic links, BER depends on received optical power, receiver noise, and modulation format, with receiver sensitivity defined as the minimum power to achieve a target BER (typically $10^{-12}$ for telecom or $10^{-9}$ pre-FEC for data communications). Forward Error Correction (FEC) adds redundant bits that allow the receiver to detect and correct errors, effectively improving the BER by several orders of magnitude at the cost of small bandwidth overhead (typically 7–25%). The Q-factor relates BER to an equivalent signal-to-noise metric: BER = $0.5 \times \text{erfc}(Q / \sqrt{2})$, with Q = 7.03 corresponding to BER ≈ $10^{-12}$.

**Example 15.10.3:** A 10 Gbps fiber optic link operates continuously for 24 hours. (a) At a BER of $10^{-12}$, calculate the expected number of bit errors and the mean time between errors. (b) If FEC with 7% overhead improves the pre-FEC BER threshold from $10^{-12}$ to $10^{-3}$, how many pre-FEC errors per second would the FEC decoder need to correct at the threshold?
**Solution:**
(a) At BER = $10^{-12}$:
Total bits in 24 hours = $10 \times 10^9 \times 24 \times 3600 = 8.64 \times 10^{14}$ bits
Expected errors = $8.64 \times 10^{14} \times 10^{-12}$ = **864 errors per day**
Mean time between errors = $1 / (10 \times 10^9 \times 10^{-12}) = 1 / 0.01$ = **100 seconds**
(b) At pre-FEC BER = $10^{-3}$:
Effective line rate with 7% FEC overhead = $10 \times 1.07 = 10.7$ Gbps
Pre-FEC errors per second = $10.7 \times 10^9 \times 10^{-3}$ = **$10.7 \times 10^6$ = 10.7 million errors/s**
The FEC decoder must correct 10.7 million errors per second at the threshold BER while reducing the post-FEC BER to below $10^{-12}$, demonstrating the remarkable power of modern error correction codes.

### 15.10.4 Network Timing and Synchronization

Precise time synchronization across networked devices is critical for distributed measurement systems, power grid protection, financial trading, telecommunications, and industrial control. The Network Time Protocol (NTP) synchronizes clocks over packet networks to within 1–10 ms of UTC using a hierarchical stratum system, where stratum 0 devices are reference clocks (GPS receivers, atomic clocks), stratum 1 servers connect directly to reference clocks, and each subsequent stratum adds another hop. The Precision Time Protocol (PTP, IEEE 1588) achieves sub-microsecond synchronization (typically 10–100 ns) by using hardware timestamping in network interface cards and PTP-aware switches (transparent clocks or boundary clocks) to measure and compensate for network path delay asymmetry. Synchronous Ethernet (SyncE, ITU-T G.8261/G.8262) distributes a frequency reference through the Ethernet physical layer by locking the transmit clock to a recovered clock from the upstream link, providing the frequency accuracy required by telecommunications equipment (±4.6 ppm for free-running, ±0.01 ppm when locked). PTP and SyncE are often combined — SyncE provides frequency synchronization while PTP provides phase/time synchronization — enabling applications such as 5G radio access networks that require ±1.5 μs time alignment between base stations.

**Example 15.10.4:** A PTP grandmaster clock synchronizes a boundary clock over a network path with a measured mean path delay of 2.35 μs and a delay asymmetry of 120 ns. The boundary clock's oscillator has a free-running accuracy of ±10 ppm. Calculate (a) the time error introduced by the delay asymmetry if not compensated, (b) the clock drift in microseconds over a 1-hour PTP holdover period (if the grandmaster becomes unreachable), and (c) the maximum time between PTP sync messages to maintain ±1 μs accuracy.
**Solution:**
(a) Delay asymmetry error: PTP assumes symmetric path delays. An asymmetry of 120 ns causes a time offset of asymmetry / 2 = 120 / 2 = **60 ns**. This error is constant and can be corrected if the asymmetry is known and configured.
(b) Clock drift during holdover: drift rate = ±10 ppm = ±10 μs per second. Over 1 hour: drift = $10 \times 10^{-6} \times 3{,}600$ = **0.036 s = 36 ms**. To maintain ±1 μs during holdover: maximum holdover time = 1 μs / (10 μs/s) = **0.1 s = 100 ms**.
(c) Maximum sync interval for ±1 μs: the sync message corrects the accumulated drift. To keep error < 1 μs with ±10 ppm drift: $T_{sync} < 1$ μs / (10 μs/s) = **0.1 s**.
PTP sync messages should be sent at intervals ≤ 100 ms (typically configured at 1 per second or 8 per second for high-accuracy applications). With a better oscillator (±0.1 ppm OCXO), the interval extends to $T_{sync} < 1 / 0.1 = 10$ s.

### 15.10.5 Quality of Service (QoS)

Quality of Service is a set of mechanisms that allow network devices to differentiate traffic classes and provide preferential treatment to delay-sensitive, loss-sensitive, or business-critical applications over best-effort traffic. Without QoS, all packets compete equally for bandwidth and buffer space, and during congestion, latency-sensitive traffic (VoIP, video conferencing, industrial control) suffers the same packet loss and delay as bulk file transfers or software updates. QoS operates through three fundamental mechanisms: **classification and marking** (identifying traffic and tagging it with a priority), **queuing and scheduling** (placing packets into different queues and determining the order of transmission), and **policing and shaping** (enforcing rate limits and smoothing traffic bursts).

**Classification and Marking:** Traffic is classified at the network edge by inspecting packet headers (source/destination IP, port number, DSCP field, 802.1p/PCP bits) or using deep packet inspection (DPI) for application-layer identification. The Differentiated Services Code Point (DSCP) is a 6-bit field in the IP header (part of the 8-bit Type of Service / Traffic Class byte) that encodes 64 possible per-hop behaviors (PHBs). The most common PHBs are: Expedited Forwarding (EF, DSCP 46) for low-latency traffic like VoIP, Assured Forwarding (AF classes AF11–AF43, 12 codepoints) for guaranteed-delivery business traffic with drop probability tiers, and Default/Best Effort (DSCP 0) for standard traffic. At Layer 2, the IEEE 802.1p Priority Code Point (PCP) provides 3 bits (8 priority levels, 0–7) within the 802.1Q VLAN tag for Ethernet frame prioritization.

**Queuing and Scheduling:** Switches and routers place classified packets into multiple output queues (typically 4–8 per port) and use scheduling algorithms to determine which queue is serviced next. **Strict Priority (SP)** always services the highest-priority queue first — guarantees minimum latency for voice/video but can starve lower-priority queues if the high-priority queue is never empty. **Weighted Round Robin (WRR)** services queues in rotation but gives more slots to higher-weight queues, preventing starvation while still providing differentiation. **Weighted Fair Queuing (WFQ)** allocates bandwidth proportionally to each flow based on assigned weights. **Low Latency Queuing (LLQ)** combines a strict priority queue for real-time traffic with a weighted fair queuing system for remaining traffic classes — the standard for modern enterprise QoS. **Weighted Random Early Detection (WRED)** proactively drops packets from lower-priority flows before the queue is full, using the DSCP or IP precedence value to set different drop probability curves per traffic class, avoiding TCP global synchronization (where all flows reduce their window simultaneously after tail-drop).

**Policing and Shaping:** A **policer** enforces a rate limit by measuring traffic against a token bucket and immediately dropping or re-marking packets that exceed the committed information rate (CIR). A **shaper** also enforces a rate limit but buffers excess traffic and transmits it later, smoothing bursts at the cost of added latency. Policing is typically applied at ingress (network edge), while shaping is applied at egress.

| QoS Class | DSCP | PHB | Application | Latency | Loss |
|---|---|---|---|---|---|
| Voice | 46 (EF) | EF | VoIP | $< 150$ ms | $< 1\%$ |
| Video | 34 (AF41) | AF41 | Video conferencing | $< 200$ ms | $< 2\%$ |
| Signaling | 24 (CS3) | CS3 | Call setup, routing | $< 500$ ms | $< 1\%$ |
| Business | 18 (AF21) | AF21 | ERP, database | $< 500$ ms | $< 3\%$ |
| Best Effort | 0 (BE) | BE | Web, email, files | No guarantee | No guarantee |
| Scavenger | 8 (CS1) | CS1 | Backup, updates | No guarantee | Higher drop |

*PHB abbreviations: **EF** = Expedited Forwarding (strict priority, minimum latency); **AF** = Assured Forwarding (guaranteed delivery with drop-probability tiers, classes AF1x–AF4x); **CS** = Class Selector (backward-compatible with IP Precedence); **BE** = Best Effort (default, no guarantees).*

**Example 15.10.5:** An enterprise WAN link has a capacity of 100 Mbps and carries three traffic classes: VoIP (EF, requiring $< 150$ ms latency and $< 1\%$ loss), business applications (AF21), and best-effort traffic (BE). The network carries 50 concurrent G.711 VoIP calls, each requiring 87.2 kbps (64 kbps codec + IP/UDP/RTP overhead with 20 ms packetization). The QoS policy allocates the LLQ strict priority queue at 10% of link capacity for EF, 50% WFQ for AF21, and 40% WFQ for BE. Calculate (a) the bandwidth consumed by VoIP, (b) whether the LLQ allocation is sufficient, (c) the serialization delay for a 1,500-byte best-effort packet, and (d) the maximum queuing delay for a voice packet if 3 best-effort packets are ahead of it in the interface output queue (without QoS) vs. with LLQ.

**Solution:**

(a) VoIP bandwidth:

50 calls $\times$ 87.2 kbps = 4,360 kbps = **4.36 Mbps**

(b) LLQ allocation check:

LLQ bandwidth = $100 \times 0.10 = 10$ Mbps

VoIP requirement = 4.36 Mbps

Utilization = 4.36 / 10 = 43.6% → **Sufficient ✓** (with 56.4% headroom for burst absorption)
(c) Serialization delay for 1,500-byte packet:
$t_{serial} = (1{,}500 \times 8) / (100 \times 10^6) = 12{,}000 / 10^8 = \mathbf{0.12\ ms}$
(d) Queuing delay comparison:
**Without QoS:** Voice packet waits behind 3 × 1,500-byte packets:
Delay = 3 × 0.12 = **0.36 ms** (plus risk of loss if queue is full)
**With LLQ:** Voice packet is in the strict priority queue and is serviced immediately after the current packet finishes transmitting. Maximum wait = serialization of one MTU-sized packet already on the wire = **0.12 ms**. LLQ reduces the worst-case queuing delay by 3× in this scenario. For a heavily loaded link with 50+ packets queued, the improvement is far greater — without QoS, voice delay could reach 6+ ms per hop, while LLQ keeps it under 0.12 ms per hop.

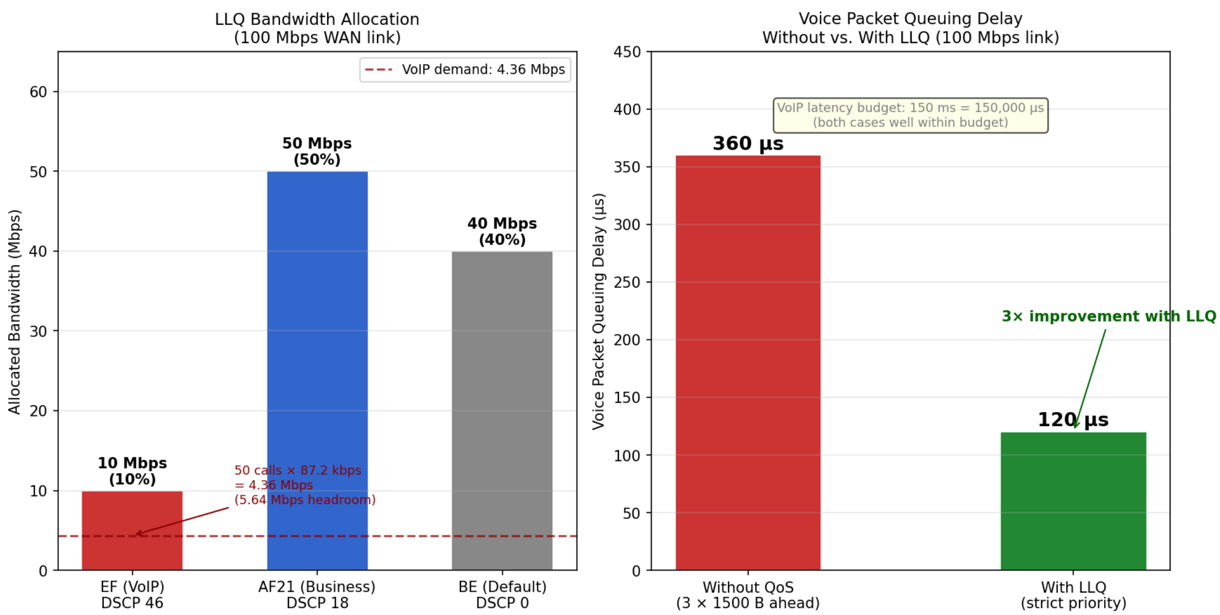



Figure 15.10.5: LLQ bandwidth allocation (left) and voice packet queuing delay comparison (right) for a 100 Mbps WAN link. EF receives 10 Mbps (43.6% utilized by 50 VoIP calls), AF21 receives 50 Mbps, and BE receives 40 Mbps. LLQ reduces worst-case voice queuing delay from 360 μs to 120 μs — a 3× improvement.

# Chapter 16

# Antenna Design and Principles

Antenna engineering bridges the gap between guided electromagnetic waves in transmission lines and radiated waves in free space. An antenna is fundamentally a transducer that converts electrical energy into electromagnetic radiation (transmitting) or captures radiation and converts it back to electrical signals (receiving). The design of an antenna involves balancing competing requirements including gain, bandwidth, impedance matching, polarization, physical size, and cost. This chapter covers the key antenna types — from simple wire antennas to sophisticated phased arrays — along with the fundamental parameters, design equations, and practical considerations that every engineer needs for antenna selection and design.

## 16.1 Antenna Fundamentals

The performance of any antenna is characterized by a set of fundamental parameters that describe how it radiates or receives electromagnetic energy. Understanding these parameters is essential for selecting the right antenna for a given application and for designing communication, radar, and sensing systems.

### 16.1.1 Radiation Mechanism and Radiation Patterns

An antenna radiates electromagnetic waves when time-varying current flows through a conductor, causing accelerating charges that produce changing electric and magnetic fields that propagate outward. The region immediately surrounding the antenna (within a distance of approximately $2D^2/\lambda$, where D is the largest antenna dimension) is the near-field region, where the fields are complex and do not exhibit plane-wave behavior. Beyond this distance is the far-field (Fraunhofer) region, where the radiated fields decay as $1/r$ and the radiation pattern is fully formed and independent of distance.

The radiation pattern is a graphical representation of the antenna's radiated power as a function of direction, typically plotted in polar or rectangular coordinates. Key pattern features include the main lobe (the direction of maximum radiation), sidelobes (minor lobes adjacent to the main lobe), back lobe (radiation in the direction opposite the main lobe), and nulls (directions of zero or minimum radiation). The half-power beamwidth (HPBW) is the angular width of the main lobe between the −3 dB points, and the first null beamwidth (FNBW) extends between the first nulls on either side of the main lobe. The sidelobe level (SLL) is typically expressed in dB below the main lobe peak, with values of −13 dB for uniform illumination and −20 to −40 dB for tapered or optimized distributions.

**Example 16.1.1:** A directional antenna has a measured HPBW of 30° in the E-plane and 35° in the H-plane. The first sidelobe is 15 dB below the main lobe peak. Estimate (a) the directivity

using the approximate formula $D \approx 32{,}400 / (\theta_E \times \theta_H)$, and (b) the sidelobe power relative to the main lobe.
**Solution:**
(a) Directivity: $D \approx 32{,}400 / (30 \times 35) = 32{,}400 / 1{,}050 = 30.86$ (linear)
$D\ (dB) = 10 \log_{10}(30.86) =$ **14.9 dBi**
(b) Sidelobe level $= -15$ dB means the sidelobe power is $10^{-1.5} = 0.0316$ times the main lobe power, or **3.16%** of the peak power density. This is a moderate sidelobe level, typical of practical horn or reflector antennas without special sidelobe suppression techniques.

### 16.1.2 Antenna Parameters

Directivity (D) measures how strongly an antenna focuses radiation in its peak direction compared to an isotropic radiator: $D = 4\pi U_{max} / P_{rad}$, where $U_{max}$ is the maximum radiation intensity and $P_{rad}$ is the total radiated power. Antenna gain (G) accounts for efficiency losses: $G = \eta D$, where the radiation efficiency $\eta$ (typically 0.5–0.99) includes ohmic losses in the conductors and dielectric, but not mismatch losses. The effective isotropic radiated power (EIRP) is $EIRP = P_t G_t$, representing the equivalent power an isotropic antenna would need to produce the same peak radiation intensity.

The effective aperture $A_e = G\lambda^2/(4\pi)$ relates gain to the equivalent area the antenna presents to an incoming wave. Radiation resistance $R_{rad}$ is the equivalent resistance that would dissipate the same power as the antenna radiates, and the input impedance $Z_{in} = R_{rad} + R_{loss} + jX_{ant}$ includes both radiation resistance, loss resistance, and reactance. Antenna bandwidth is typically defined as the frequency range over which VSWR remains below 2:1 (return loss > 9.5 dB), and it ranges from < 1% for resonant patches to > 100% for wideband designs like log-periodic arrays. Polarization describes the orientation of the electric field vector: linear (vertical or horizontal), circular (RHCP or LHCP), or elliptical, with a polarization loss factor $\cos^2(\Delta\varphi)$ for linear misalignment or 3 dB loss for circular-to-linear reception.

**Example 16.1.2:** A parabolic dish antenna with a diameter of 2 m and aperture efficiency of 0.60 operates at 6 GHz. Determine (a) the gain, (b) the effective aperture, (c) the EIRP if fed with 10 W, and (d) the HPBW.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 6 \times 10^9 = 0.05$ m.
$G = \eta(\pi D/\lambda)^2 = 0.60 \times (\pi \times 2 / 0.05)^2 = 0.60 \times (125.66)^2 = 0.60 \times 15{,}791 = 9{,}475$ (linear)
$G\ (dBi) = 10 \log_{10}(9{,}475) =$ **39.8 dBi**
(b) Effective aperture: $A_e = G\lambda^2/(4\pi) = 9{,}475 \times 0.05^2 / (4\pi) = 9{,}475 \times 0.0025 / 12.566 =$ **1.88 m²**
This equals $\eta \times A_{physical} = 0.60 \times \pi(1)^2 = 1.88\ m^2$, confirming the result.
(c) $EIRP = P_t \times G = 10 \times 9{,}475 = 94{,}750\ W = 10 + 39.8 =$ **49.8 dBW** (or 79.8 dBm) EIRP
(d) $HPBW \approx 70\lambda/D = 70 \times 0.05 / 2 =$ **1.75°**

### 16.1.3 Friis Transmission Equation and Link Budgets

The Friis transmission equation relates the received power to the transmitted power for a free-space radio link: $P_r/P_t = G_t G_r(\lambda/(4\pi d))^2$, where $G_t$ and $G_r$ are the transmit and receive antenna gains, $\lambda$ is the wavelength, and d is the distance. The term $(\lambda/(4\pi d))^2$ is the free-space path loss (FSPL) expressed as a power ratio, and in decibels the equation becomes: $P_r(dBm) = P_t(dBm) + G_t(dBi) + G_r(dBi) - FSPL(dB)$, where $FSPL = 20 \log_{10}(4\pi d/\lambda)$. This equation assumes line-of-sight propagation with no reflections, diffraction, or atmospheric absorption, and it forms the foundation of all link budget analyses (see also Ch. 2, §2.8).

For radar systems, the radar range equation extends Friis to account for the round-trip path and target scattering: $P_r = P_t G_t^2 \lambda^2 \sigma / ((4\pi)^3 R^4)$, where $\sigma$ is the radar cross section (RCS) of the target in $m^2$ and R is the range. The $R^4$ dependence (compared to $R^2$ for one-way links) means radar performance degrades rapidly with distance, making high-gain antennas and powerful transmitters essential for long-range detection.

**Example 16.1.3:** A Ku-band satellite downlink operates at 12 GHz with a satellite EIRP of 52 dBW. The ground station uses a 0.6 m dish with 60% aperture efficiency. The orbital altitude is 35,786 km (geostationary). Determine (a) the receive antenna gain, (b) the free-space path loss, and (c) the received power.

**Solution:**

(a) $\lambda = c/f = 3 \times 10^8 / 12 \times 10^9 = 0.025$ m.

$G_r = 0.60 \times (\pi \times 0.6 / 0.025)^2 = 0.60 \times (75.40)^2 = 0.60 \times 5{,}685 = 3{,}411$

$G_r$(dBi) = 10 $\log_{10}$(3,411) = **35.3 dBi**

(b) FSPL = 20 $\log_{10}(4\pi \times 35{,}786{,}000 / 0.025)$ = 20 $\log_{10}(1.80 \times 10^{10})$ = 20 × 10.255 = **205.1 dB**

(c) $P_r$ = EIRP + $G_r$ − FSPL = 52 + 35.3 − 205.1 = **−117.8 dBW** (or −87.8 dBm)

This received power level is typical for satellite TV reception, where the receiver noise floor is around −130 dBW (kTB with B ≈ 30 MHz), providing adequate carrier-to-noise ratio for demodulation.

## 16.2 Wire Antennas

Wire antennas are the simplest and most widely used antenna type, consisting of straight or curved conductors carrying RF current. Their low cost, ease of fabrication, and well-understood behavior make them the default choice for frequencies from HF through UHF.

### 16.2.1 Dipole Antennas

The dipole is the most fundamental antenna type, consisting of two collinear conductors fed at the center. A short dipole (total length L « λ) has a radiation resistance of $R_{rad} = 20\pi^2(L/\lambda)^2$, which is very small (e.g., 2 Ω for L = λ/10), making it inefficient due to the dominance of ohmic losses. The half-wave dipole (L = λ/2) is the practical standard: its input impedance is 73 + j42.5 Ω at exact resonance, and shortening the element to approximately 0.95 × λ/2 eliminates the reactive component, yielding a purely resistive 73 Ω input. The half-wave dipole has a gain of 2.15 dBi (1.64 linear) and a donut-shaped omnidirectional radiation pattern in the plane perpendicular to the antenna axis.

The folded dipole consists of two parallel half-wave elements connected at the ends, with the feed applied to only one element. The folded geometry transforms the input impedance to approximately 4 × 73 = 292 Ω (a convenient match to 300 Ω twin-lead transmission line) while providing significantly wider bandwidth than a simple dipole — typically 10–20% compared to 5–8% for a standard dipole. This impedance transformation occurs because the folded dipole supports both antenna-mode and transmission-line-mode currents, and the four-fold impedance increase applies to any two-conductor folded element with equal conductor diameters.

**Example 16.2.1:** Design a half-wave dipole for the 2 m amateur radio band at 146 MHz. Determine (a) the theoretical half-wavelength, (b) the practical element length (using the 0.95 shortening factor), (c) the radiation resistance, and (d) the gain in dBd (relative to a dipole).
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 146 \times 10^6 = 2.055$ m. Half-wavelength = **1.027 m** (total tip-to-tip).
(b) Practical length = $0.95 \times 1.027$ = **0.976 m** (97.6 cm). Each arm is 48.8 cm from the feed point.
(c) Radiation resistance: $R_{rad}$ = **73 Ω** (the standard value for a resonant half-wave dipole). With the shortened length eliminating the $+j42.5\ \Omega$ reactance, $Z_{in} \approx 73 + j0\ \Omega$.
(d) The gain of a half-wave dipole is 2.15 dBi by definition, so relative to itself: **0 dBd**. The dBd unit (decibels relative to a dipole) is commonly used in amateur radio and commercial antenna specifications, where 0 dBd = 2.15 dBi.

### 16.2.2 Monopole and Ground Plane Antennas

A monopole antenna is one half of a dipole mounted perpendicular to a conducting ground plane, which acts as a mirror to create the equivalent of a full dipole through image theory. The quarter-wave monopole ($L = \lambda/4$) has an input impedance of approximately $36.5 + j21\ \Omega$ — exactly half the dipole impedance because the ground plane eliminates the lower hemisphere of radiation, concentrating all power into the upper half-space. This doubles the directivity compared to a dipole, giving the monopole a gain of 5.15 dBi (3 dBd), while the radiation pattern is omnidirectional in azimuth with a maximum at the horizon.

The ground plane size and quality significantly affect monopole performance. An infinite perfect ground plane produces the ideal 36.5 Ω impedance, but practical ground planes (vehicle roofs, metal enclosures, or radial wire systems) are finite and imperfect. For VHF/UHF applications, a ground plane extending at least $\lambda/4$ in radius is adequate. AM broadcast towers use extensive buried radial wire ground systems (typically 120 radials, each $\lambda/4$ long) to minimize ground losses. Whip antennas on vehicles are practical monopoles using the vehicle body as the ground plane, while ground-plane antennas with drooping radials are popular for base stations because angling the radials downward raises the input impedance closer to 50 Ω.

**Example 16.2.2:** Design a quarter-wave monopole antenna for the 900 MHz GSM cellular band. Determine (a) the element length, (b) the input impedance, (c) the gain, and (d) the minimum ground plane radius.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 900 \times 10^6 = 0.333$ m. Quarter-wave length = 0.333/4 = **83.3 mm** (approximately 8.3 cm).
(b) Input impedance: $Z_{in} \approx$ **36.5 Ω** (resistive at resonance). This is a reasonable match to 50 Ω coax (VSWR ≈ 1.37:1, return loss = 16 dB), which is acceptable for most applications without additional matching.
(c) Gain: **5.15 dBi** (or 3.0 dBd) — 3 dB more than a half-wave dipole due to the ground plane concentrating radiation into the upper hemisphere.
(d) Minimum ground plane radius ≈ $\lambda/4$ = **83.3 mm**. This is compact enough for integration into cellular base station equipment or handheld device enclosures.

### 16.2.3 Loop Antennas

Loop antennas consist of one or more turns of wire forming a closed loop, and their behavior depends dramatically on the loop circumference relative to the wavelength. A small loop (circumference C « λ, typically $C < \lambda/10$) behaves as a magnetic dipole, with a radiation pattern identical in shape to an electric dipole but rotated 90° — maximum radiation is in the plane of the loop, with nulls along the loop axis. The radiation resistance of a small loop is $R_{rad} = 320\pi^4(A/\lambda^2)^2$, where A is the loop area, which is extremely small for electrically small loops (< 1 Ω), making efficiency a major challenge without ferrite loading.

A full-wave loop (circumference $C = \lambda$) is a resonant antenna with gain of approximately 3.4 dBi (1.3 dB above a dipole), input impedance around 100 Ω, and a radiation pattern with maximum broadside to the plane of the loop. Full-wave loops are popular in HF amateur radio for their quiet receive characteristics and modest gain. At low frequencies and for near-field applications, small loops serve as the basis for RFID tag antennas and NFC inductive coupling, where the mutual inductance between reader and tag coils enables wireless power transfer and data communication at ranges of a few centimeters to a meter.

**Example 16.2.3:** An NFC reader antenna is a circular loop with radius 25 mm and 3 turns, operating at 13.56 MHz. Determine (a) the loop circumference in wavelengths, (b) the radiation resistance for a single turn, (c) the total radiation resistance for 3 turns, and (d) whether this qualifies as a small loop.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 13.56 \times 10^6 = 22.12$ m. Circumference $= 2\pi \times 0.025 = 0.157$ m.
$C/\lambda = 0.157 / 22.12 = \mathbf{0.0071}$ — far less than 0.1.
(b) Loop area $A = \pi \times 0.025^2 = 1.963 \times 10^{-3}$ m².
$R_{rad} = 320\pi^4(A/\lambda^2)^2 = 320 \times 97.41 \times (1.963 \times 10^{-3} / 489.3)^2 = 31{,}171 \times (4.01 \times 10^{-6})^2 = 31{,}171 \times 1.61 \times 10^{-11} = \mathbf{5.01 \times 10^{-7}\ \Omega}$
(c) For N turns, $R_{rad}$ scales as $N^2$: $R_{total} = 9 \times 5.01 \times 10^{-7} = \mathbf{4.51 \times 10^{-6}\ \Omega}$ (4.51 μΩ).
(d) **Yes**, with $C/\lambda = 0.0071$, this is an extremely small loop. The negligible radiation resistance confirms that NFC operates by near-field inductive coupling (mutual inductance), not radiation — the antenna functions as a coupling coil, not a radiator.

### 16.2.4 Yagi-Uda Antennas

The Yagi-Uda antenna is a directional wire array consisting of a single driven element (typically a half-wave dipole or folded dipole), one or more parasitic reflector elements behind the driven element, and one or more parasitic director elements in front. The parasitic elements are not electrically connected to the feedline — they receive energy by mutual coupling from the driven element and reradiate it with specific phase relationships that create constructive interference in the forward direction and destructive interference rearward. The reflector element is slightly longer than λ/2 (typically 5% longer, making it inductively loaded), which causes its reradiated field to add constructively in the forward direction. Director elements are slightly shorter than λ/2 (typically 5–10% shorter, capacitively loaded), and each successive director adds approximately 1 dB of gain while narrowing the beamwidth. The spacing between elements is typically 0.15λ to 0.25λ for the reflector-to-driven distance and 0.2λ to 0.35λ between directors. A typical 3-element Yagi (one reflector, one driven, one director) achieves approximately 7.5 dBi gain with a front-to-back ratio of 15–20 dB, while a 10-element design reaches approximately 13 dBi with a front-to-back ratio exceeding 25 dB. The input impedance of a Yagi drops as more elements are added due to mutual coupling — a simple dipole-driven Yagi may have an impedance of 20–30 Ω, while a folded dipole driven element provides the 4:1 impedance transformation that brings the input closer to 50 Ω and increases bandwidth. Yagi-Uda antennas are the standard design for terrestrial television reception (VHF/UHF), amateur radio directional communication, point-to-point wireless links, and radio astronomy at meter wavelengths.

**Example 16.2.4:** Design a 5-element Yagi-Uda antenna for the 2 m amateur radio band at 144 MHz. Determine (a) the wavelength and approximate total boom length, (b) the element lengths (reflector, driven, and 3 directors), (c) the expected gain, and (d) the input impedance improvement using a folded dipole driven element.

**Solution:**

(a) $\lambda = c/f = 3 \times 10^8 / 144 \times 10^6 = 2.083$ m.

Element spacings: reflector-to-driven = $0.20\lambda = 0.417$ m; director spacing = $0.25\lambda = 0.521$ m.

Total boom length = $0.417 + 3 \times 0.521 = 0.417 + 1.563 =$ **1.98 m** (approximately $1\lambda$).

(b) Element lengths: reflector = $1.05 \times \lambda/2 = 0.525\lambda = 1.094$ m (**109.4 cm**); driven element = $0.95 \times \lambda/2 = 0.475\lambda = 0.989$ m (**98.9 cm**); director 1 = $0.45\lambda = 0.937$ m (**93.8 cm**); director 2 = $0.44\lambda = 0.917$ m (**91.7 cm**); director 3 = $0.43\lambda = 0.896$ m (**89.6 cm**).

Directors are progressively shorter to maintain proper phasing.

(c) A 5-element Yagi typically achieves approximately **10.5 dBi** (8.3 dBd) gain with a front-to-back ratio of 20–22 dB and a 3 dB beamwidth of approximately 52° in the E-plane and 62° in the H-plane.

(d) With a simple dipole driven element, $Z_{in} \approx 24$ Ω (due to mutual coupling lowering the impedance).

Using a folded dipole: $Z_{in} \approx 4 \times 24 =$ **96 Ω**, which can be matched to a 50 Ω feedline with a simple beta or gamma match.

The folded dipole also increases the usable bandwidth from approximately 2% to 5% of the center frequency.

## 16.3 Aperture Antennas

Aperture antennas radiate through an open area (aperture) over which the electromagnetic field distribution determines the radiation pattern. These antennas are used at microwave and millimeter-wave frequencies where their physical dimensions are manageable relative to the wavelength.

### 16.3.1 Horn Antennas

A horn antenna is formed by flaring the end of a rectangular or circular waveguide, creating a gradual transition from the guided wave in the waveguide to free-space radiation. The three basic types are the E-plane sectoral horn (flared in the E-field direction), H-plane sectoral horn (flared in the H-field direction), and pyramidal horn (flared in both planes). The pyramidal horn is most common, with gain approximately $G = (4\pi A_e)/\lambda^2 = (4\pi \times \eta_{ap} \times a \times b)/\lambda^2$, where a and b are the aperture dimensions and $\eta_{ap} \approx 0.51$ for an optimum-gain pyramidal horn.

Horn antennas are valued for their predictable performance, moderate gain (10–25 dBi), wide bandwidth (typically > 2:1), and low VSWR. They serve as primary feeds for parabolic reflectors, as standard gain reference antennas for calibrating other antennas, and as direct radiators in EMC testing chambers. The optimum horn maximizes gain for a given horn length by balancing the aperture size against the phase error across the aperture — too large an aperture for a given length introduces destructive phase variations that reduce the effective gain.

**Example 16.3.1:** Design a pyramidal horn antenna operating at 10 GHz (X-band) with a target gain of 20 dBi. Determine (a) the required effective aperture area, (b) the physical aperture dimensions assuming a square aperture with $\eta_{ap} = 0.51$, and (c) the HPBW.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 10 \times 10^9 = 0.03$ m. $G = 10^{20/10} = 100$ (linear).
$A_e = G\lambda^2/(4\pi) = 100 \times 0.03^2 / (4\pi) = 100 \times 9 \times 10^{-4} / 12.566 = \mathbf{7.16 \times 10^{-3}\ m^2}$ (71.6 $cm^2$)
(b) Physical aperture: $A_{phys} = A_e / \eta_{ap} = 7.16 \times 10^{-3} / 0.51 = 1.40 \times 10^{-2}\ m^2$.
For a square aperture: $a = b = \sqrt{(1.40 \times 10^{-2})} = \mathbf{0.119\ m}$ (11.9 cm × 11.9 cm).
(c) HPBW $\approx 70\lambda/a = 70 \times 0.03 / 0.119 = \mathbf{17.6°}$ in both planes (symmetric for a square aperture).

### 16.3.2 Parabolic Reflector Antennas

The parabolic reflector antenna uses a paraboloidal dish to focus incoming plane waves to a single focal point (or equivalently, to collimate radiation from a feed antenna at the focus into a narrow beam). The dish gain is $G = \eta(\pi D/\lambda)^2$, where D is the dish diameter and the aperture efficiency $\eta$ typically ranges from 0.55 to 0.70, accounting for spillover, illumination taper, surface roughness, and blockage losses. The half-power beamwidth is approximately $\theta \approx 70\lambda/D$ degrees, showing that larger dishes produce narrower beams. The f/D ratio (focal length to diameter) is a key design parameter: typical values of 0.3–0.5 provide a good compromise between feed illumination angle, spillover loss, and mechanical depth.

Three common feed configurations are used. Prime focus places the feed at the focal point, which is simple but causes aperture blockage. The Cassegrain design adds a convex hyperbolic subreflector near the focal point that redirects the feed radiation back to the main reflector, reducing feed line losses and allowing the feed electronics to be mounted behind the dish. The offset reflector uses a section of a larger paraboloid so the feed and support structure do not block the aperture, eliminating blockage losses and improving sidelobe performance — this is the standard configuration for consumer satellite TV dishes and modern earth station antennas.

**Example 16.3.2:** A Ku-band satellite TV dish has a diameter of 1.2 m and operates at 12 GHz. Assuming an aperture efficiency of 0.60, determine (a) the gain, (b) the HPBW, (c) the effective aperture, and (d) the pointing accuracy required to keep the gain within 1 dB of peak.
**Solution:**
(a) $\lambda = 0.025$ m. $G = 0.60 \times (\pi \times 1.2 / 0.025)^2 = 0.60 \times (150.80)^2 = 0.60 \times 22{,}740 = 13{,}644$
$G(\text{dBi}) = 10 \log_{10}(13{,}644) = \mathbf{41.3\ dBi}$
(b) HPBW $\approx 70 \times 0.025 / 1.2 = \mathbf{1.46°}$
(c) $A_e = \eta \times \pi(D/2)^2 = 0.60 \times \pi \times 0.36 = \mathbf{0.679\ m^2}$
(d) The 1 dB beamwidth is approximately 0.58 × HPBW ≈ 0.85°. To keep the gain within 1 dB of peak, the pointing accuracy must be within **±0.42°** — this explains why satellite dishes require precise alignment and why wind loading is a design concern.

### 16.3.3 Slot and Cavity-Backed Antennas

A slot antenna is formed by cutting a narrow rectangular opening in a conducting ground plane and exciting it with a transmission line or waveguide. By Babinet's principle, the radiation pattern of a slot is complementary to that of a dipole of the same dimensions — a half-wave slot (length $\approx \lambda/2$) radiates broadside with a pattern identical to a dipole but with the E-plane and H-plane interchanged. The input impedance of a slot is related to its complementary dipole by $Z_{slot} \times Z_{dipole} = \eta^2/4$, where $\eta = 377\ \Omega$ is the impedance of free space. For a half-wave slot, this gives $Z_{slot} = 377^2/(4 \times 73) \approx 486\ \Omega$, which is a high impedance that requires a matching network for 50 Ω systems. Slots are fed by microstrip lines, coplanar waveguide (CPW), or waveguide coupling apertures. A major advantage is that slot antennas can be flush-mounted in metallic surfaces — aircraft fuselages, vehicle bodies, and array ground planes — without protruding elements that create aerodynamic drag or mechanical vulnerability.

A cavity-backed slot places a metallic cavity behind the slot, eliminating backward radiation and increasing gain by 2–3 dB compared to an unbacked slot. The cavity dimensions are chosen so that no cavity modes resonate near the operating frequency, ensuring clean impedance behavior. Typical cavity depth is $\lambda/4$, creating a short-circuit back wall that reflects energy forward. Cavity-backed slots achieve gains of 5–7 dBi with bandwidths of 5–15% depending on cavity volume and slot shape. Variations include the cavity-backed annular slot (for circular polarization), the T-shaped slot (for dual-band operation), and dielectric-loaded cavities that reduce physical depth while maintaining performance.

**Example 16.3.3:** A half-wave slot antenna is cut into an aluminum ground plane for an aircraft L-band transponder at 1.09 GHz. Determine (a) the slot length, (b) the slot impedance from Babinet's principle, (c) the impedance with a $\lambda/4$ deep cavity backing, and (d) a suitable matching approach for 50 Ω coax.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 1.09 \times 10^9 = 0.2752$ m. Slot length $= \lambda/2 =$ **137.6 mm**. Slot width is typically $\lambda/20$ to $\lambda/10$, so $w \approx 14$–28 mm.
(b) From Babinet's principle: $Z_{slot} = \eta^2/(4 \times Z_{dipole}) = 377^2/(4 \times 73) = 142{,}129/292 =$ **486.7 Ω**.
(c) The cavity backing eliminates the back lobe, effectively doubling the forward radiation. The cavity modifies the impedance — for a $\lambda/4$ deep cavity, the impedance drops to approximately $Z_{slot}/2 =$ **243 Ω** because the short-circuit back wall at $\lambda/4$ presents an open circuit at the slot plane, maintaining slot resonance while doubling the radiation conductance — the in-phase image source at $\lambda/2$ distance constructively doubles the forward-radiated power for the same slot voltage, halving the radiation resistance.
(d) A 243 Ω to 50 Ω transformation requires an impedance ratio of 4.86:1. A quarter-wave transformer with $Z_{match} = \sqrt{(50 \times 243)} = \sqrt{12{,}150} =$ **110.2 Ω** provides the match. Alternatively, an offset microstrip feed positioned at a point along the slot where the impedance is lower can achieve a direct 50 Ω match without external components.

## 16.4 Printed and Microstrip Antennas

Microstrip (patch) antennas are fabricated on printed circuit board substrates, making them lightweight, low-profile, conformable to surfaces, and inexpensive to mass-produce. These properties have made them the dominant antenna type for mobile devices, GPS receivers, Wi-Fi equipment, and phased array systems.

### 16.4.1 Rectangular Patch Antennas

A rectangular microstrip patch antenna consists of a conducting patch on one side of a dielectric substrate with a ground plane on the other side. The patch acts as a resonant cavity bounded by the ground plane below, the patch above, and open-circuit radiating edges on the sides. The resonant length is approximately $L \approx c/(2f_r\sqrt{\varepsilon_{r,eff}}) - 2\Delta L$, where $\varepsilon_{r,eff}$ is the effective dielectric constant (accounting for fringing fields extending into air) and $\Delta L$ is the length extension due to fringing at each radiating edge.

The patch width W affects radiation efficiency and impedance, with a typical choice of $W = c/(2f_r) \times \sqrt{(2/(\varepsilon_r + 1))}$ providing good radiation efficiency. The input impedance at the radiating edge is typically 150–300 Ω, which can be matched to 50 Ω through inset feeding (cutting a notch from the edge toward the center) or probe feeding at the appropriate position. Patch antenna bandwidth is inherently narrow (1–5% for standard substrates) because the resonant cavity has a high Q factor, but thicker substrates with lower $\varepsilon_r$ increase bandwidth at the cost of larger size. Typical gain ranges from 6 to 9 dBi for a single patch element.

**Example 16.4.1:** Design a rectangular patch antenna on FR-4 substrate ($\varepsilon_r = 4.4$, thickness h = 1.6 mm) for 2.4 GHz Wi-Fi. Determine (a) the patch width, (b) the effective dielectric constant, (c) the fringing length extension, and (d) the patch length.
**Solution:**
(a) Patch width: $W = c/(2f_r) \times \sqrt{(2/(\varepsilon_r + 1))} = (3 \times 10^8)/(2 \times 2.4 \times 10^9) \times \sqrt{(2/5.4)}$
$W = 0.0625 \times 0.6086 = \mathbf{38.0\ mm}$
(b) Effective dielectric constant:
$\varepsilon_{r,eff} = (\varepsilon_r + 1)/2 + (\varepsilon_r - 1)/2 \times (1 + 12h/W)^{-0.5}$
$\varepsilon_{r,eff} = 5.4/2 + 3.4/2 \times (1 + 12 \times 1.6/38.0)^{-0.5} = 2.7 + 1.7 \times (1.505)^{-0.5} = 2.7 + 1.7 \times 0.815 = \mathbf{4.09}$
(c) Length extension: $\Delta L = 0.412h \times (\varepsilon_{r,eff} + 0.3)(W/h + 0.264) / ((\varepsilon_{r,eff} - 0.258)(W/h + 0.8))$
$\Delta L = 0.412 \times 1.6 \times (4.39)(24.014) / ((3.832)(24.55)) = 0.659 \times 105.42 / 94.07 = \mathbf{0.740\ mm}$
(d) Patch length: $L = c/(2f_r\sqrt{\varepsilon_{r,eff}}) - 2\Delta L = (3 \times 10^8)/(2 \times 2.4 \times 10^9 \times \sqrt{4.09}) - 2 \times 0.740$
$L = 0.0625/2.022 - 1.480 = 30.91 - 1.48 = \mathbf{29.4\ mm}$
The final patch dimensions are approximately 38.0 mm × 29.4 mm on standard 1.6 mm FR-4 — compact enough for integration into Wi-Fi routers and access points.

### 16.4.2 Patch Antenna Variations

The circular patch antenna resonates when its effective radius satisfies the condition $a_{eff} = 1.8412c/(2\pi f_r\sqrt{\varepsilon_r})$, where the factor 1.8412 corresponds to the first zero of the derivative of the Bessel function $J_1$. Circular patches are commonly used for GPS receivers because they can easily generate circular polarization, which matches the RHCP signal transmitted by GPS satellites. The physical radius requires a fringing correction similar to the rectangular patch.

Circular polarization from a patch antenna can be achieved by two primary methods. The first uses a square (or nearly square) patch with truncated opposite corners, which excites two orthogonal modes with equal amplitude and 90° phase difference. The second uses dual orthogonal feeds with a 90° hybrid coupler to excite two orthogonal modes simultaneously. Stacked patches — multiple patch layers separated by foam or air — increase bandwidth to 10–25% by creating multiple coupled resonances. Patch antenna arrays combine multiple elements with a corporate or series feed network to achieve higher gain (18–25 dBi) and beam shaping, forming the basis of 5G panel antennas and Wi-Fi sector antennas.

**Example 16.4.2:** Design a circularly polarized square patch antenna with truncated corners for GPS L1 reception at 1.575 GHz on a substrate with $\varepsilon_r = 2.2$ and h = 3.175 mm (Rogers RT/duroid 5880). Determine (a) the patch side length for a square patch, and (b) the approximate truncation dimension (truncation length is typically $\Delta C \approx L/(2Q)$, where $Q \approx c\sqrt{\varepsilon_r}/(4hf_r)$ for the patch quality factor).
**Solution:**
(a) For a square patch at resonance:
$L = c/(2f_r\sqrt{\varepsilon_{r,eff}})$. With $\varepsilon_r = 2.2$, h = 3.175 mm, and assuming W = L (square):
$\varepsilon_{r,eff} \approx (2.2 + 1)/2 + (2.2 - 1)/2 \times (1 + 12 \times 3.175/L)^{-0.5}$
Starting with an estimate: $L \approx c/(2f_r\sqrt{\varepsilon_r}) = 3 \times 10^8/(2 \times 1.575 \times 10^9 \times 1.483) =$ **64.2 mm**
After fringing correction and iteration: L ≈ **62.0 mm** (62.0 mm × 62.0 mm square patch).
(b) Patch quality factor: $Q \approx c\sqrt{\varepsilon_r}/(4hf_r) = 3 \times 10^8 \times 1.483 / (4 \times 3.175 \times 10^{-3} \times 1.575 \times 10^9) =$ 22.3
Truncation: $\Delta C \approx L/(2Q) = 62.0 / (2 \times 22.3) =$ **1.39 mm**. Each truncated corner removes a right triangle with legs of approximately 1.4 mm, creating the two orthogonal degenerate modes needed for circular polarization.

## 16.5 Antenna Arrays

Antenna arrays combine multiple radiating elements to achieve higher gain, beam steering capability, and adaptive pattern control that single elements cannot provide. Arrays are fundamental to modern radar, satellite communications, 5G cellular, and Wi-Fi MIMO systems.

### 16.5.1 Array Theory and Pattern Multiplication

The total radiation pattern of an antenna array equals the product of the individual element pattern and the array factor (AF), a principle known as pattern multiplication. For a uniform linear array (ULA) of N identical elements spaced d apart along a line, the array factor is $AF = \sin(N\psi/2) / (N \sin(\psi/2))$, where $\psi = kd\cos\theta + \beta$, $k = 2\pi/\lambda$ is the wavenumber, $\theta$ is the angle from the array axis, and $\beta$ is the progressive phase shift between elements. The main beam direction $\theta_0$ is determined by $\psi = 0$, giving $\cos\theta_0 = -\beta/(kd)$.

A broadside array ($\beta = 0$) radiates perpendicular to the array axis, while an end-fire array ($\beta = -kd$) radiates along the array axis. The first-null beamwidth for a broadside array is approximately FNBW $\approx 2\lambda/(Nd)$, and the directivity is $D \approx 2Nd/\lambda$. Grating lobes — full-strength replicas of the main beam at undesired angles — appear when the element spacing d exceeds $\lambda/(1 + |\sin\theta_s|)$, where $\theta_s$ is the scan angle from broadside. For an unsteered broadside array, the condition $d < \lambda$ prevents grating lobes; for arrays that must scan to ±60°, the spacing must be reduced to $d < 0.536\lambda$.

**Example 16.5.1:** A 10-element uniform linear broadside array operates at 3 GHz with half-wavelength spacing ($d = \lambda/2$). Determine (a) the array factor first-null beamwidth, (b) the directivity of the array factor, (c) the maximum scan angle before a grating lobe appears, and (d) the directivity if the elements are half-wave dipoles.

**Solution:**
(a) $\lambda = c/f = 0.10$ m, $d = 0.05$ m.
FNBW $\approx 2\lambda/(Nd) = 2 \times 0.10 / (10 \times 0.05) =$ **0.40 radians = 22.9°**
HPBW $\approx 0.886\lambda/(Nd) = 0.886 \times 0.10 / 0.50 =$ **0.177 radians = 10.2°**
(b) Array factor directivity: $D_{AF} = 2Nd/\lambda = 2 \times 10 \times 0.05 / 0.10 =$ **10.0 (10.0 dB)**
(c) Grating lobe condition: $d < \lambda/(1 + |\sin\theta_s|)$. With $d = \lambda/2$:
$0.5 < 1/(1 + \sin\theta_s)$, so $1 + \sin\theta_s < 2$, giving $\sin\theta_s < 1$, meaning $\theta_s < 90°$.
**The array can scan the full visible range (±90°) without grating lobes at d = λ/2 spacing.**
(d) Total directivity $\approx D_{AF} \times D_{element} = 10.0 \times 1.64$ (dipole) $= 16.4 =$ **12.1 dBi**. In practice, mutual coupling between closely-spaced elements modifies the element patterns, and the actual directivity will differ slightly from this simple product.

### 16.5.2 Beamforming and Smart Antennas

Beamforming electronically steers and shapes an antenna array's beam by controlling the amplitude and phase (or time delay) of each element's signal. Analog beamforming uses RF phase shifters at each element to steer the beam in one direction at a time — simple and cost-effective, used in radar and early 5G systems. Digital beamforming digitizes the signal at each element and applies complex weights in the digital domain, enabling simultaneous formation of multiple independent beams pointing in different directions. Hybrid beamforming combines analog subarrays with digital processing, balancing performance and cost for massive MIMO deployments.

MIMO (Multiple-Input Multiple-Output) antenna systems use multiple antennas at both transmitter and receiver to exploit multipath propagation rather than fighting it. Spatial multiplexing MIMO transmits independent data streams on different spatial channels, multiplying throughput by the number of parallel streams (up to $\min(N_t, N_r)$). Massive MIMO, deployed in 5G NR, uses 64 to 256 antenna elements at the base station to simultaneously serve many users through spatial division, achieving spectral efficiencies 5–10× higher than conventional systems. The array gain of a massive MIMO system is approximately $10\log_{10}(N)$ dB, where N is the number of elements, providing both coverage improvement and interference suppression.

**Example 16.5.2:** A 5G base station uses a 64-element massive MIMO antenna panel operating at 3.5 GHz, arranged as an 8 × 8 planar array with λ/2 spacing. Determine (a) the physical panel dimensions, (b) the broadside array gain assuming 2.15 dBi element gain and 90% array efficiency, (c) the HPBW in azimuth and elevation, and (d) the number of simultaneous user beams possible.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 3.5 \times 10^9 = 0.0857$ m. Spacing $d = \lambda/2 = 42.9$ mm.
Panel size $= 8 \times 42.9$ mm $= 343$ mm per side $\approx$ **34.3 cm × 34.3 cm**.
(b) Array gain: $G = \eta \times N \times G_{element} = 0.90 \times 64 \times 1.64 = 94.5$ (linear)
$G(\text{dBi}) = 10\log_{10}(94.5) =$ **19.8 dBi**. Alternatively: $10\log_{10}(64) + 2.15 - 0.46 = 18.06 + 2.15 - 0.46 = 19.8$ dBi.
(c) For an 8-element linear array at λ/2: HPBW $\approx 0.886\lambda/(8 \times \lambda/2) = 0.886/4 = 0.222$ rad = **12.7°** in both azimuth and elevation (symmetric for a square array).
(d) With digital beamforming, the maximum number of independent beams equals the number of RF chains. With full digital beamforming (one RF chain per element), up to **64 simultaneous beams** are possible, though practical deployments typically serve 8–16 users simultaneously due

to inter-user interference and processing constraints.

### 16.5.3 Phased Array Scanning and Grating Lobes

Phased array antennas steer the main beam electronically by applying a progressive phase shift β between adjacent elements, directing the beam to an angle $\theta_0$ from broadside where $\sin \theta_0 = -\beta\lambda/(2\pi d)$. The scan volume is the range of angles over which the array can steer while maintaining acceptable performance — typically ±60° from broadside for most systems, limited by element pattern roll-off, impedance mismatch, and grating lobe onset. As the beam scans away from broadside, the projected aperture decreases as $\cos \theta_0$, reducing the effective gain by approximately $10 \log_{10}(\cos \theta_0)$ dB. At 60° scan, this represents a 3 dB gain reduction, and the beamwidth broadens by a factor of $1/\cos \theta_0$.

Grating lobes are secondary main beams that appear when the element spacing is large enough for a second solution to the array factor maximum condition. For a ULA scanning to angle $\theta_0$, grating lobes appear at angles $\theta_{GL}$ where $\sin \theta_{GL} = \sin \theta_0 + n\lambda/d$ (for integer $n \neq 0$). To keep the first grating lobe outside the visible region ($|\sin \theta| \leq 1$), the spacing must satisfy $d < \lambda/(1 + |\sin \theta_0|)$. For a maximum scan of ±60°, this requires $d < 0.536\lambda$; for ±90° (full hemisphere), $d < \lambda/2$. In practice, element spacing of 0.5λ to 0.6λ is standard, with 0.5λ being the safe choice for wide-scan applications. Phase shifters at each element introduce quantization errors due to their discrete phase states (typically 3-bit = 8 states to 6-bit = 64 states). The average sidelobe level due to phase quantization is approximately −6B dB below the main beam for B-bit phase shifters — a 4-bit shifter produces quantization sidelobes at approximately −24 dB, which is adequate for most radar and communication systems.

**Example 16.5.3:** A phased array radar operates at 10 GHz with 32 elements spaced at $d = 0.55\lambda$. The beam is steered to $\theta_0 = 45°$ from broadside. Determine (a) the progressive phase shift β required, (b) whether a grating lobe enters the visible region, (c) the grating lobe angle if one exists, and (d) the gain reduction due to scanning.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 10 \times 10^9 = 0.03$ m. $d = 0.55 \times 0.03 = 0.0165$ m.
$\beta = -kd \sin \theta_0 = -(2\pi/0.03) \times 0.0165 \times \sin 45° = -209.44 \times 0.0165 \times 0.7071 =$ **−2.443 rad (−139.9°)**.
(b) Grating lobe condition: $\sin \theta_{GL} = \sin 45° + \lambda/d = 0.7071 + 1/0.55 = 0.7071 + 1.818 = 2.525$. Since $|\sin \theta_{GL}| > 1$ for $n = +1$, the first positive-order grating lobe is outside visible space. For $n = -1$: $\sin \theta_{GL} = 0.7071 - 1.818 = -1.111$. Since $|-1.111| > 1$, this grating lobe is also just outside visible space but **very close to entering** (only 0.111 beyond the boundary). At slightly larger scan angles or frequencies, it would appear.
(c) The $n = -1$ grating lobe would enter visible space when $\sin \theta_0 - \lambda/d = -1$, or $\sin \theta_0 = \lambda/d - 1 = 1.818 - 1 = 0.818$, giving $\theta_0 =$ **54.9°**. Beyond this scan angle, a grating lobe appears at −90° and sweeps inward. The maximum grating-lobe-free scan angle is therefore ±54.9° for this 0.55λ spacing.
(d) Gain reduction at $\theta_0 = 45°$: $\Delta G = 10 \log_{10}(\cos 45°) = 10 \log_{10}(0.7071) =$ **−1.5 dB**. The broadside beamwidth of $0.886\lambda/(Nd) = 0.886 \times 0.03/(32 \times 0.0165) = 0.0503$ rad $= 2.88°$ broadens to $2.88°/\cos 45° =$ **4.07°** at the scanned angle.

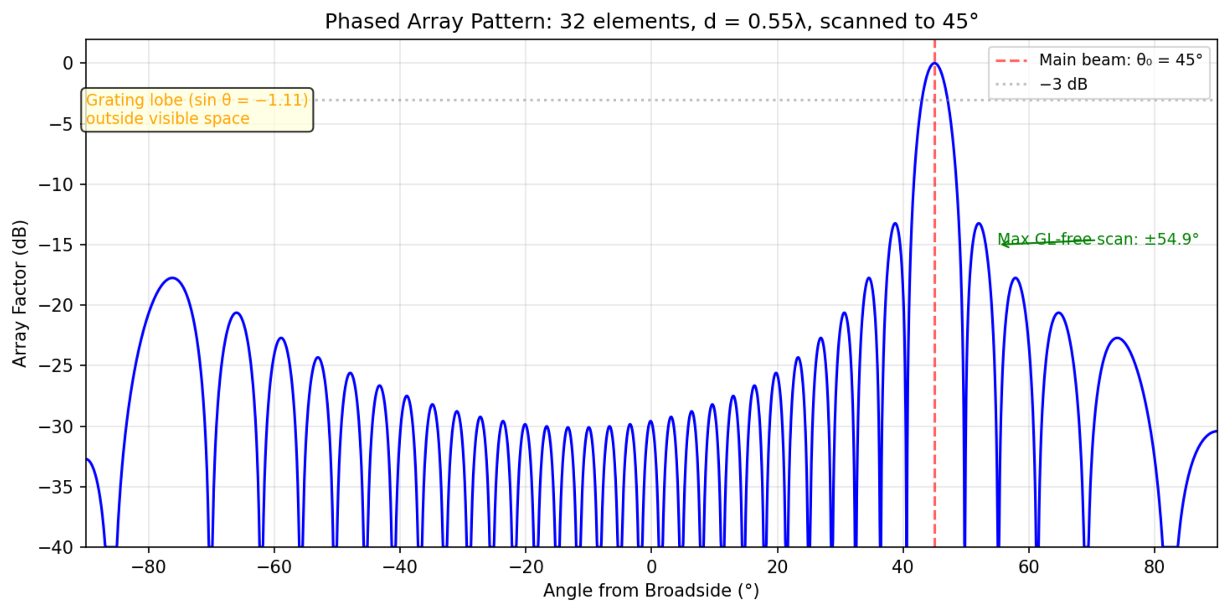



Figure 16.5.3: Phased Array Pattern

## 16.6 Impedance Matching and Practical Design

Practical antenna design requires not only understanding the radiation characteristics but also ensuring efficient power transfer between the antenna and the transmission line, and verifying performance through measurement.

### 16.6.1 Antenna Impedance Matching

The voltage standing wave ratio (VSWR) quantifies the impedance mismatch between an antenna and its feedline: $VSWR = (1 + |\Gamma|)/(1 - |\Gamma|)$, where $\Gamma = (Z_{ant} - Z_0)/(Z_{ant} + Z_0)$ is the reflection coefficient. The return loss $RL = -20\log_{10}|\Gamma|$ represents the fraction of power reflected back toward the transmitter in dB, with $RL > 10$ dB ($VSWR < 2{:}1$) considered acceptable and $RL > 15$ dB ($VSWR < 1.43{:}1$) considered good. The mismatch loss — the fraction of power actually delivered to the antenna — is $ML = -10\log_{10}(1 - |\Gamma|^2)$; at VSWR = 2:1, only 11% of power is reflected and mismatch loss is just 0.51 dB.

Matching networks transform the antenna impedance to the characteristic impedance of the feedline (typically 50 Ω). The L-network (one series and one shunt reactive element) is the simplest, matching any impedance at a single frequency with the component values determined by $Q = \sqrt{(R_{high}/R_{low} - 1)}$. The quarter-wave transformer uses a transmission line section of impedance $Z_{match} = \sqrt{(Z_0 \times Z_{ant})}$ and length $\lambda/4$ to provide a broadband match. A balun (balanced-to-unbalanced transformer) is required when connecting a balanced antenna (dipole, loop) to an unbalanced feedline (coaxial cable), with common types including the sleeve balun, folded balun, and ferrite-core balun.

**Example 16.6.1:** A half-wave dipole ($Z_{ant} = 73$ Ω) is to be matched to a 50 Ω coaxial feedline at 145 MHz using an L-network. Determine (a) the VSWR without matching, (b) the return loss without matching, (c) the L-network component values, and (d) an alternative quarter-wave transformer

impedance.
**Solution:**
(a) $\Gamma = (73 - 50)/(73 + 50) = 23/123 = 0.187$
VSWR $= (1 + 0.187)/(1 - 0.187) = 1.187/0.813 =$ **1.46:1**
(b) RL $= -20 \log_{10}(0.187) = -20 \times (-0.728) =$ **14.6 dB** — already acceptable without matching, but an L-network can improve it.
(c) L-network with series L and shunt C (matching 50 Ω to 73 Ω):
$Q = \sqrt{(73/50 - 1)} = \sqrt{(0.46)} = 0.678$
$X_L = Q \times R_{low} = 0.678 \times 50 = 33.9\ \Omega \rightarrow L = 33.9/(2\pi \times 145 \times 10^6) =$ **37.2 nH**
$X_C = R_{high}/Q = 73/0.678 = 107.7\ \Omega \rightarrow C = 1/(2\pi \times 145 \times 10^6 \times 107.7) =$ **10.2 pF**
(d) Quarter-wave transformer: $Z_{match} = \sqrt{(50 \times 73)} = \sqrt{(3{,}650)} =$ **60.4 Ω**. A 60.4 Ω transmission line section of length $\lambda/4 = 0.517$ m at 145 MHz provides the match, though finding or fabricating this impedance is less convenient than standard 50 or 75 Ω cable.

### 16.6.2 Antenna Measurement and Testing

Antenna radiation pattern measurement requires controlled environments to eliminate reflections and interference. An anechoic chamber lined with RF-absorbing material (typically pyramidal foam absorbers effective from 100 MHz to 40+ GHz) provides a reflection-free test zone, and is the standard for precision measurements of gain, pattern, polarization, and efficiency. Outdoor antenna ranges measure larger antennas using an elevated range (transmit and receive antennas on towers) or a slant range, with the test distance satisfying the far-field criterion $d > 2D^2/\lambda$ to ensure plane-wave illumination. Near-field scanning measures the amplitude and phase of the antenna's field on a surface close to the antenna (planar, cylindrical, or spherical scan) and mathematically transforms the data to compute the far-field pattern — essential for large arrays that would require impractically long far-field distances.

Antenna impedance is measured using a vector network analyzer (VNA), which measures the complex reflection coefficient $S_{11}$ at the antenna input port across frequency. The VNA displays return loss, VSWR, and impedance (on a Smith Chart) versus frequency, revealing the resonant frequency (where the impedance is purely resistive), the impedance bandwidth, and any parasitic resonances. For a two-port system (such as an antenna with separate transmit and receive feeds), $S_{21}$ provides the isolation or coupling between ports. Gain measurement uses either the comparison method (comparing the antenna under test to a reference antenna of known gain) or the three-antenna method (measuring all three pairs of unknown antennas to solve for individual gains without a reference standard).

**Example 16.6.2:** A VNA measurement of a patch antenna shows $S_{11} = -14$ dB at the design frequency of 2.45 GHz. Determine (a) the reflection coefficient magnitude, (b) the VSWR, (c) the percentage of incident power reflected, and (d) the mismatch loss.
**Solution:**
(a) $|\Gamma| = 10^{-14/20} = 10^{-0.7} =$ **0.200**
(b) VSWR $= (1 + 0.200)/(1 - 0.200) = 1.200/0.800 =$ **1.50:1**
(c) Reflected power $= |\Gamma|^2 = 0.200^2 = 0.040 =$ **4.0%** — meaning 96% of incident power is delivered to the antenna.
(d) Mismatch loss $= -10 \log_{10}(1 - |\Gamma|^2) = -10 \log_{10}(0.960) = -10 \times (-0.0177) =$ **0.18 dB**. This is a well-matched antenna — the 0.18 dB mismatch loss is negligible in most link budgets, confirming

that $S_{11} = -14$ dB represents good performance.

# Chapter 17

# Radar Systems

Radar (Radio Detection And Ranging) uses electromagnetic waves to detect, locate, and measure the velocity of objects at distances ranging from centimeters to hundreds of kilometers. A radar system transmits a known waveform toward a target, then analyzes the reflected echo to extract information about the target's range, direction, speed, and physical characteristics. Since its development during World War II, radar has become essential to aviation safety, weather forecasting, military defense, autonomous vehicles, and scientific remote sensing. This chapter covers the fundamental principles of radar operation, the major waveform types and signal processing techniques, and practical radar applications with comparisons of system parameters across different use cases.

## 17.1 Radar Fundamentals

The performance of any radar system is governed by a set of fundamental relationships between transmitted power, antenna characteristics, target properties, and receiver sensitivity. These equations determine the maximum detection range, range and velocity resolution, and the ability to distinguish targets from noise and clutter.

### 17.1.1 Radar Range Equation

The radar range equation relates the received echo power to the system and target parameters: $P_r = P_t G_t G_r \lambda^2 \sigma \,/\, ((4\pi)^3 R^4)$, where $P_t$ is the peak transmit power, $G_t$ and $G_r$ are the transmit and receive antenna gains (often the same antenna, so $G^2$ replaces $G_t G_r$), $\lambda$ is the wavelength, $\sigma$ is the radar cross section (RCS) of the target in $m^2$, and R is the range to the target. The $R^4$ dependence is the defining characteristic of radar — doubling the detection range requires 16 times the transmit power, making radar inherently power-hungry for long-range applications.

The maximum detection range occurs when the received power equals the minimum detectable signal $S_{min}$, giving: $R_{max} = (P_t G^2 \lambda^2 \sigma / ((4\pi)^3 S_{min}))^{1/4}$. The minimum detectable signal depends on the receiver noise floor ($kT_sB_n$, where $T_s$ is the system noise temperature and $B_n$ is the noise bandwidth) and the required signal-to-noise ratio $SNR_{min}$ for a given probability of detection $P_d$ and false alarm probability $P_{fa}$. For a typical surveillance radar with $P_d = 0.9$ and $P_{fa} = 10^{-6}$, the required single-pulse SNR is approximately 13.2 dB.

**Example 17.1.1:** An airport surveillance radar operates at 2.8 GHz (S-band) with a peak transmit power of 1.4 MW, antenna gain of 34 dBi, and system noise temperature of 600 K. The noise bandwidth is 1 MHz and the required SNR is 13 dB. Determine (a) the wavelength, (b) the minimum detectable signal, (c) the maximum detection range for an aircraft with RCS $\sigma = 10$ $m^2$, and (d)

the range for a small drone with $\sigma = 0.01$ m$^2$.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 2.8 \times 10^9 =$ **0.107 m**
(b) $S_{min} = kT_sB_n \times SNR_{min} = 1.381 \times 10^{-23} \times 600 \times 10^6 \times 10^{13/10}$
$S_{min} = 8.286 \times 10^{-15} \times 19.95 =$ **$1.65 \times 10^{-13}$ W** ($-97.8$ dBm)
(c) $G = 10^{34/10} = 2{,}512$ (linear).
$R_{max} = (1.4 \times 10^6 \times 2{,}512^2 \times 0.107^2 \times 10 / ((4\pi)^3 \times 1.65 \times 10^{-13}))^{1/4}$
Numerator $= 1.4 \times 10^6 \times 6.31 \times 10^6 \times 0.01145 \times 10 = 1.011 \times 10^{12}$
Denominator $= 1{,}984 \times 1.65 \times 10^{-13} = 3.274 \times 10^{-10}$
$R_{max} = (1.011 \times 10^{12} / 3.274 \times 10^{-10})^{1/4} = (3.088 \times 10^{21})^{1/4} =$ **236 km**
(d) For $\sigma = 0.01$ m$^2$: the range scales as $\sigma^{1/4}$, so $R_{drone} = 236 / (1000)^{1/4} = 236 / 5.62 =$ **42.0 km**. The 1,000× smaller RCS reduces detection range by a factor of $(1000)^{1/4} = 5.62$ — illustrating why small drones are difficult to detect with conventional surveillance radar.

### 17.1.2 Radar Cross Section

Radar cross section (RCS) quantifies how effectively a target reflects radar energy back toward the transmitter, defined as $\sigma = 4\pi R^2 \times (P_{reflected}/P_{incident})$, where the ratio represents the reflected power density at the target normalized to the incident power density. RCS depends on the target's size, shape, material, orientation relative to the radar, and the radar frequency. Common reference values include: a flat plate of area A has peak RCS of $\sigma = 4\pi A^2/\lambda^2$ (when perpendicular to the beam), a sphere of radius $a \gg \lambda$ has $\sigma = \pi a^2$ (independent of frequency), and a corner reflector with face dimension L has $\sigma = 12\pi L^4/\lambda^2$ (very large for its size due to triple-bounce retroreflection).

Typical RCS values span many orders of magnitude: insects ~$10^{-5}$ m$^2$ ($-50$ dBsm), birds ~$10^{-2}$ m$^2$ ($-20$ dBsm), small drones ~0.01 m$^2$ ($-20$ dBsm), humans ~1 m$^2$ (0 dBsm), automobiles ~10–200 m$^2$ (10–23 dBsm), small aircraft ~1–10 m$^2$ (0–10 dBsm), commercial jetliners ~10–100 m$^2$ (10–20 dBsm), and large ships ~10,000–100,000 m$^2$ (40–50 dBsm). Stealth aircraft use shaping (angled surfaces to deflect energy away from the radar), radar-absorbing materials (RAM), and edge treatments to reduce RCS to 0.001–0.01 m$^2$ ($-30$ to $-20$ dBsm). RCS fluctuation models (Swerling models I–IV) describe how RCS varies over time due to target aspect changes, with Swerling I/II assuming scan-to-scan or pulse-to-pulse independence.

**Example 17.1.2:** A metallic sphere with a radius of 0.5 m is used as a radar calibration target at 10 GHz. Determine (a) whether the sphere is in the optical region ($a \gg \lambda$), (b) the RCS, (c) the RCS in dBsm, and (d) compare to a 0.5 m × 0.5 m flat plate oriented perpendicular to the beam.
**Solution:**
(a) $\lambda = c/f = 0.03$ m. $a/\lambda = 0.5/0.03 = 16.7 \gg 1$, so **yes**, the sphere is in the optical region.
(b) Sphere RCS: $\sigma_{sphere} = \pi a^2 = \pi \times 0.25 =$ **0.785 m²**
(c) $\sigma$ (dBsm) $= 10 \log_{10}(0.785) =$ **−1.05 dBsm**
(d) Flat plate RCS: $A = 0.25$ m$^2$, $\sigma_{plate} = 4\pi A^2/\lambda^2 = 4\pi \times 0.0625 / 9 \times 10^{-4} =$ **873 m² (29.4 dBsm)**. The flat plate has 1,112× larger RCS than the sphere of comparable size because it coherently reflects energy back toward the radar. This illustrates why corner reflectors (which act as retroreflecting flat plates) are used to enhance the radar visibility of small boats, buoys, and calibration targets.

### 17.1.3 Doppler Effect and Velocity Measurement

When a target moves relative to the radar, the reflected signal experiences a frequency shift proportional to the radial velocity. The Doppler frequency shift is $f_d = 2v_r f_0/c = 2v_r/\lambda$, where $v_r$ is the radial component of the target's velocity (positive for approaching targets), $f_0$ is the transmit frequency, and the factor of 2 accounts for the round-trip path. This Doppler shift is the basis for measuring target velocity and for distinguishing moving targets from stationary clutter.

The maximum unambiguous velocity (without aliasing) for a pulsed radar is $v_{max} = \lambda \times PRF / 4$, where PRF is the pulse repetition frequency. Higher PRF allows measurement of faster targets but reduces the maximum unambiguous range ($R_{ua} = c/(2 \times PRF)$), creating the fundamental range-Doppler ambiguity: a radar cannot simultaneously have long unambiguous range and high unambiguous velocity with a single PRF. Medium-PRF radars (8–30 kHz) resolve this by using multiple staggered PRFs and processing the returns jointly. The velocity resolution of a radar is $\Delta v = \lambda/(2T_{dwell})$, where $T_{dwell}$ is the coherent integration time — longer observation times yield finer velocity resolution.

**Example 17.1.3:** An X-band police speed radar operates at 10.525 GHz in CW (continuous wave) mode. Determine (a) the wavelength, (b) the Doppler shift for a vehicle approaching at 120 km/h, (c) the minimum detectable velocity if the Doppler filter has a 3 dB bandwidth of 25 Hz, and (d) the Doppler shift for a vehicle receding at 60 km/h.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 10.525 \times 10^9 =$ **0.02850 m** (28.50 mm)
(b) $v_r = 120$ km/h $= 33.33$ m/s. $f_d = 2 \times 33.33 / 0.02850 =$ **2,339 Hz** (2.34 kHz)
(c) Minimum detectable velocity: $v_{min} = \lambda \times \Delta f / 2 = 0.02850 \times 25 / 2 =$ **0.356 m/s** (1.28 km/h). This allows detection of walking-speed targets.
(d) $v_r = -60$ km/h $= -16.67$ m/s (negative for receding). $f_d = 2 \times (-16.67) / 0.02850 =$ **−1,170 Hz**. The negative frequency shift indicates the target is moving away. The radar distinguishes approaching and receding targets by the sign of the Doppler shift, typically using I/Q (quadrature) demodulation.

### 17.1.4 Radar Clutter and Noise

Clutter is the unwanted radar return from objects other than the target of interest — ground surfaces, sea waves, precipitation, birds, insects, and chaff. Unlike receiver noise, which is independent of the transmitted signal, clutter power scales directly with transmit power and cannot be reduced by increasing the radar's energy budget. Surface clutter (ground and sea returns) enters the radar through the antenna's sidelobes and mainlobe when the beam intersects the earth's surface. The clutter RCS per unit area is characterized by the normalized clutter coefficient $\sigma^0$ (sigma-zero), measured in $m^2/m^2$ or dB, which depends on grazing angle, surface roughness, frequency, and polarization. Typical values range from −40 dB for smooth calm sea at low grazing angles to −10 dB for rough terrain at steep angles.

Sea clutter presents particular challenges because it is non-stationary and has heavy-tailed amplitude statistics. At low sea states and high grazing angles, sea clutter follows a Rayleigh distribution, but at low grazing angles and high sea states the amplitude follows a Weibull or K-distribution with occasional very large returns ("sea spikes") that mimic small boat targets. The signal-to-clutter ratio (SCR) for a target in surface clutter is $SCR = \sigma_{target} / (\sigma^0 \times A_c)$, where $A_c$ is the clutter cell area illuminated by the radar ($A_c = R \times \theta_{az} \times c\tau/(2 \cos \psi)$ for a pulsed radar at grazing angle $\psi$). Volume clutter from precipitation fills the radar resolution cell with a distributed reflectivity that is proportional to the cell volume.

The system noise temperature $T_{sys} = T_{ant} + T_r$ determines the receiver noise floor, where $T_{ant}$ is the antenna noise temperature (including sky, ground, and atmospheric contributions) and $T_r = T_0(F - 1)$ is the receiver noise temperature derived from the noise figure F. A low-noise amplifier (LNA) with a noise figure of 1.5 dB at the antenna port gives $T_r = 290 \times (1.413 - 1) = 120$ K. The system noise figure of a cascaded receiver chain follows the Friis noise formula: $F_{sys} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2) + \ldots$, emphasizing that the first-stage LNA dominates overall noise performance.

**Example 17.1.4:** A coastal surveillance radar operates at X-band (9.4 GHz) with a pulse width of 100 ns and azimuth beamwidth of 1.5°. A small boat with RCS $\sigma = 3$ m² is at a range of 8 km at a grazing angle of 2°. The normalized sea clutter coefficient at sea state 3 is $\sigma^0 = -30$ dB (0.001 m²/m²). Determine (a) the clutter cell area, (b) the total clutter RCS in the cell, (c) the signal-to-clutter ratio, and (d) whether MTI processing with 25 dB improvement factor can detect the boat.
**Solution:**
(a) Clutter cell area: $A_c = R \times \theta_{az} \times c\tau/(2 \cos \psi) = 8{,}000 \times (1.5 \times \pi/180) \times (3 \times 10^8 \times 100 \times 10^{-9})/(2 \times \cos 2°)$
$A_c = 8{,}000 \times 0.02618 \times 30/1.9994 = 8{,}000 \times 0.02618 \times 15.005 =$ **3,143 m²**
(b) Total clutter RCS: $\sigma_c = \sigma^0 \times A_c = 0.001 \times 3{,}143 =$ **3.14 m²** (5.0 dBsm)
(c) SCR $= \sigma_{target}/\sigma_c = 3/3.14 =$ **0.955 (−0.2 dB)**. The boat return is approximately equal to the clutter — it cannot be detected by amplitude alone.
(d) With 25 dB MTI improvement: output SCR = −0.2 + 25 = **24.8 dB**. Since a typical detection threshold requires SCR > 13 dB, the boat is now **clearly detectable** after MTI processing. The boat's motion (even at low speed) produces a Doppler shift that separates it from stationary sea clutter.

## 17.2 Radar Waveforms and Modulation

The choice of transmitted waveform determines a radar's range resolution, velocity resolution, and ability to suppress clutter and interference. Each waveform type represents a different trade-off between these capabilities, hardware complexity, and signal processing requirements.

### 17.2.1 Pulsed Radar

Pulsed radar transmits short bursts of RF energy and measures the time delay of the echo to determine target range: $R = c\tau/2$, where $\tau$ is the round-trip time delay and the factor of 2 accounts for the two-way path. The range resolution — the ability to distinguish two targets at different ranges — is $\Delta R = c\tau_p/2$, where $\tau_p$ is the pulse width. A 1 μs pulse gives $\Delta R = 150$ m, while a 10 ns pulse gives $\Delta R = 1.5$ m. The maximum unambiguous range is $R_{ua} = c/(2 \times PRF)$, limited by the requirement that each echo must return before the next pulse is transmitted.

The peak power of a pulsed radar can be very high (kilowatts to megawatts), but the average power $P_{avg} = P_{peak} \times \tau_p \times PRF = P_{peak} \times$ duty cycle is much lower (typically 0.1–10% duty cycle). The pulse energy $E = P_{peak} \times \tau_p$ determines the detection capability: a longer pulse contains more energy for detection but has poorer range resolution. This conflict between detection range (favoring long pulses with high energy) and range resolution (favoring short pulses) motivates pulse compression techniques

(§17.2.3). Pulse integration — averaging N return pulses — improves the detection SNR by a factor of N for coherent integration or approximately $\sqrt{N}$ for non-coherent integration.

**Example 17.2.1:** A marine navigation radar operates at 9.4 GHz (X-band) with a peak power of 25 kW, pulse width of 0.5 μs, and PRF of 2,000 Hz. Determine (a) the range resolution, (b) the maximum unambiguous range, (c) the duty cycle, (d) the average transmit power, and (e) the detection range improvement from non-coherent integration of 20 pulses.
**Solution:**
(a) Range resolution: $\Delta R = c\tau_p/2 = 3 \times 10^8 \times 0.5 \times 10^{-6} / 2 =$ **75 m**
(b) Maximum unambiguous range: $R_{ua} = c/(2 \times PRF) = 3 \times 10^8 / (2 \times 2{,}000) =$ **75 km**
(c) Duty cycle $= \tau_p \times PRF = 0.5 \times 10^{-6} \times 2{,}000 =$ **0.001 (0.1%)**
(d) Average power: $P_{avg} = 25{,}000 \times 0.001 =$ **25 W**
(e) Non-coherent integration gain $\approx \sqrt{N} = \sqrt{20} = 4.47$ (6.5 dB). Since range scales as $SNR^{1/4}$, the range improvement factor $= 4.47^{1/4} =$ **1.45×** (45% increase in detection range).

### 17.2.2 Continuous Wave (CW) and FMCW Radar

Continuous wave (CW) radar transmits a constant-frequency signal and measures the Doppler shift of the echo to determine target velocity, but it cannot measure range because there is no timing reference in the unmodulated waveform. CW radar is the simplest radar type, requiring only a stable oscillator, a transmit/receive antenna (or separate antennas), and a mixer that produces the Doppler frequency as a beat note. CW radars are used in police speed radars, industrial motion detectors, and missile seekers where velocity is the primary measurement.

Frequency Modulated Continuous Wave (FMCW) radar adds range measurement capability by sweeping the transmit frequency linearly over a bandwidth B during a sweep time $T_{sweep}$. The received echo is mixed with the current transmit signal, producing a beat frequency $f_b = (2RB)/(cT_{sweep})$ that is proportional to range. The range resolution is $\Delta R = c/(2B)$ — determined by the sweep bandwidth, not the pulse width — so a 1 GHz bandwidth FMCW radar achieves 15 cm range resolution with milliwatts of transmit power. FMCW radar can simultaneously measure range and velocity using triangular modulation (alternating up-sweep and down-sweep), where the sum of beat frequencies gives range and the difference gives velocity. FMCW is the dominant technology for automotive radar (77 GHz), drone altitude sensors, level measurement in industrial tanks, and short-range surveillance.

**Example 17.2.2:** An automotive FMCW radar operates at 77 GHz with a sweep bandwidth of 4 GHz and sweep time of 40 μs. Determine (a) the range resolution, (b) the beat frequency for a vehicle at 50 m range, (c) the maximum unambiguous range for a 12-bit ADC sampling at 40 MHz, and (d) the maximum beat frequency.
**Solution:**
(a) Range resolution: $\Delta R = c/(2B) = 3 \times 10^8 / (2 \times 4 \times 10^9) =$ **3.75 cm**. This fine resolution enables the radar to distinguish individual pedestrians, vehicles, and road features.
(b) Beat frequency: $f_b = 2RB/(cT_{sweep}) = 2 \times 50 \times 4 \times 10^9 / (3 \times 10^8 \times 40 \times 10^{-6}) = 4 \times 10^{11} / 1.2 \times 10^4 =$ **33.3 MHz**
(c) Maximum beat frequency $= f_s/2 = 20$ MHz (Nyquist limit).
$R_{max} = f_{b,max} \times cT_{sweep} / (2B) = 20 \times 10^6 \times 3 \times 10^8 \times 40 \times 10^{-6} / (2 \times 4 \times 10^9) =$ **30 m**
(d) The beat frequency for 50 m (33.3 MHz) exceeds the 20 MHz Nyquist limit at this ADC rate,

so the ADC sample rate would need to be increased to at least 67 MHz, or the sweep time lengthened. With $T_{sweep} = 80$ μs: $f_b = 16.7$ MHz, safely within the ADC bandwidth, and $R_{max} = 60$ m — sufficient for automotive forward-collision applications.

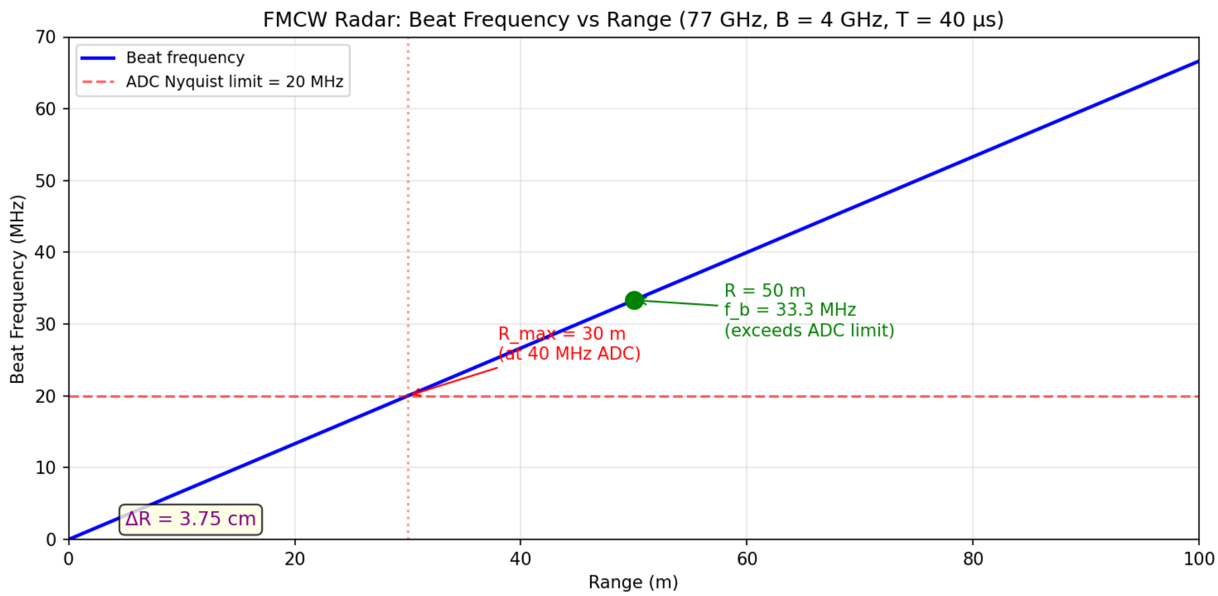



Figure 17.2.2: FMCW Beat Frequency vs Range

### 17.2.3 Pulse Compression

Pulse compression resolves the conflict between detection range (which requires high pulse energy, favoring long pulses) and range resolution (which requires short pulses) by transmitting a long, coded pulse and compressing it in the receiver to achieve the resolution of a much shorter pulse. The compression ratio equals the time-bandwidth product: $CR = B\tau_p$, where B is the bandwidth of the coded waveform and $\tau_p$ is the pulse duration. The compressed pulse width is 1/B, giving a range resolution of $\Delta R = c/(2B)$ regardless of the transmitted pulse length.

The most common pulse compression waveform is the linear frequency modulation (LFM) chirp, where the frequency sweeps linearly across bandwidth B during pulse duration $\tau_p$. A matched filter (dispersive delay line or digital correlator) in the receiver compresses the chirp to a narrow pulse with a peak power gain equal to the compression ratio. For example, a 10 μs chirp with 10 MHz bandwidth has CR = 100 (20 dB), producing the same range resolution as a 0.1 μs pulse but with 100× more energy. Phase-coded pulses (e.g., Barker codes, polyphase codes) achieve similar compression by modulating the phase of the transmitted pulse in a pseudo-random pattern. Barker codes of length N provide compression ratios up to N with sidelobe levels of 1/N, but the longest known Barker code has only N = 13.

**Example 17.2.3:** A military surveillance radar transmits a 50 μs LFM chirp pulse with 5 MHz bandwidth at a peak power of 100 kW. Determine (a) the compression ratio, (b) the range resolution, (c) the equivalent short-pulse peak power (the effective peak power after compression), (d)

the compressed pulse width, and (e) compare to an uncompressed 50 μs pulse.
**Solution:**
(a) Compression ratio: $CR = B\tau_p = 5 \times 10^6 \times 50 \times 10^{-6} =$ **250** (24 dB)
(b) Range resolution: $\Delta R = c/(2B) = 3 \times 10^8 / (2 \times 5 \times 10^6) =$ **30 m**
(c) Effective peak power after compression: $P_{eff} = P_{peak} \times CR = 100 \times 10^3 \times 250 =$ **25 MW equivalent**. The matched filter concentrates the energy of the long pulse into the compressed pulse width.
(d) Compressed pulse width: $\tau_{compressed} = 1/B = 1 / (5 \times 10^6) =$ **0.2 μs**
(e) Without compression, the 50 μs pulse has range resolution $\Delta R = c\tau_p/2 = 3 \times 10^8 \times 50 \times 10^{-6} / 2 = 7{,}500$ m — **250× worse** than with compression. To achieve 30 m resolution without compression would require a 0.2 μs pulse with 25 MW peak power, which is impractical. Pulse compression provides the detection energy of a long pulse with the resolution of a short pulse.

## 17.3 Radar Signal Processing

Signal processing transforms raw radar returns into useful target information by extracting echoes from noise, suppressing clutter from stationary objects, and forming images. Modern radars rely heavily on digital signal processing (DSP) to achieve performance that would be impossible with analog techniques alone.

### 17.3.1 Moving Target Indication (MTI) and Clutter Rejection

Moving Target Indication (MTI) exploits the Doppler shift to distinguish moving targets from stationary clutter (ground returns, buildings, weather). A basic MTI filter subtracts consecutive pulse returns — since stationary clutter produces identical echoes from pulse to pulse, the subtraction cancels clutter while preserving the Doppler-shifted returns from moving targets. The MTI improvement factor quantifies clutter suppression: $I = (S/C)_{out} / (S/C)_{in}$, where S/C is the signal-to-clutter ratio. A single-delay canceller achieves 20–25 dB improvement, while a double canceller reaches 30–40 dB.

MTI has "blind speeds" at velocities where the Doppler shift is a multiple of the PRF ($f_d = n \times PRF$), causing the target return to look identical to clutter. The first blind speed is $v_{blind} = \lambda \times PRF / 2$. Staggered PRF techniques eliminate blind speeds by using two or more alternating PRFs. Moving Target Detection (MTD) extends MTI by using a bank of Doppler filters (typically an FFT across N pulses in a coherent processing interval) to provide both clutter rejection and velocity measurement, forming the basis of modern air traffic control radars like the ASR-9 and ASR-11.

**Example 17.3.1:** An S-band air surveillance radar operates at 3.0 GHz with a PRF of 1,000 Hz and uses a 2-pulse MTI canceller. Determine (a) the first blind speed, (b) the second blind speed, (c) the improvement factor if the clutter-to-noise ratio (CNR) is 40 dB and the canceller provides 25 dB of improvement, and (d) a staggered PRF ratio to eliminate the first blind speed.
**Solution:**
(a) $\lambda = 0.10$ m. First blind speed: $v_{blind1} = \lambda \times PRF / 2 = 0.10 \times 1{,}000 / 2 =$ **50 m/s** (180 km/h)
(b) Second blind speed: $v_{blind2} = 2 \times v_{blind1} =$ **100 m/s** (360 km/h). Aircraft at these speeds would be invisible to the MTI filter.
(c) With 25 dB clutter improvement: output S/C = input S/C + 25 dB. If input CNR = 40 dB (clutter 40 dB above noise) and the target is at noise level (S/C = −40 dB input), then output S/C

= −40 + 25 = **−15 dB**. The target is still below clutter, requiring additional integration or MTD processing to detect.
(d) Using PRFs of 1,000 Hz and 1,250 Hz (ratio 4:5): The blind speeds of $PRF_1$ are 50, 100, 150... m/s. The blind speeds of $PRF_2$ are 62.5, 125, 187.5... m/s. The first common blind speed is **250 m/s** (900 km/h) — well above typical aircraft speeds, effectively eliminating the blind speed problem for normal targets.

### 17.3.2 Constant False Alarm Rate (CFAR) Detection

In a radar receiver, the detection threshold must adapt to varying noise and clutter levels across range and angle to maintain a constant probability of false alarm. A fixed threshold would produce excessive false alarms in regions of high clutter and miss targets in quiet regions. CFAR processors estimate the local noise/clutter level by averaging the power in reference cells surrounding the cell under test (CUT), then set the detection threshold as a multiple of this estimate: threshold = $\alpha \times P_{noise,est}$, where $\alpha$ is chosen to achieve the desired $P_{fa}$.

Cell-averaging CFAR (CA-CFAR) averages N reference cells on both sides of the CUT (typically N = 16–32 total), excluding guard cells immediately adjacent to the CUT to prevent target energy from biasing the estimate. For N reference cells and a desired $P_{fa}$, the threshold multiplier is $\alpha = N \times ({P_{fa}}^{-1/N} - 1)$. CA-CFAR works well in homogeneous noise but degrades at clutter edges. Greatest-of CFAR (GO-CFAR) selects the larger of the leading and trailing window averages, providing better performance at clutter boundaries. Ordered-statistics CFAR (OS-CFAR) ranks the reference cells by power and selects the k-th largest value, offering robustness against multiple interfering targets in the reference window.

**Example 17.3.2:** A CA-CFAR processor uses N = 16 reference cells (8 leading, 8 trailing) with 2 guard cells on each side of the cell under test. The desired false alarm probability is $P_{fa} = 10^{-6}$. Determine (a) the threshold multiplier $\alpha$, (b) the threshold in dB above the estimated noise floor, (c) the CFAR loss compared to an ideal fixed-threshold detector, and (d) the total number of cells in the CFAR window.
**Solution:**
(a) $\alpha = N \times ({P_{fa}}^{-1/N} - 1) = 16 \times ((10^{-6})^{-1/16} - 1) = 16 \times (10^{6/16} - 1) = 16 \times (10^{0.375} - 1)$
$\alpha = 16 \times (2.371 - 1) = 16 \times 1.371 =$ **21.94**
(b) Threshold = $10 \log_{10}(21.94)$ = **13.4 dB** above the estimated noise floor.
(c) The CFAR loss (additional SNR required compared to a known-noise-level detector) is approximately $10 \log_{10}(1 + 2/N)$ dB, which for N = 16 gives CFAR loss $\approx 10 \log_{10}(1 + 2/16) = 10 \log_{10}(1.125)$ = **0.51 dB**. This small loss is the penalty for estimating the noise level rather than knowing it exactly.
(d) Total cells = 1 (CUT) + 4 (guard cells) + 16 (reference cells) = **21 range cells** in the sliding window.

### 17.3.3 Synthetic Aperture Radar (SAR)

Synthetic Aperture Radar (SAR) achieves fine azimuth (cross-range) resolution from a moving platform by coherently processing echoes collected over a distance along the flight path, synthesizing the effect of a very large antenna. A real-aperture radar's azimuth resolution is limited to $\theta_{az} \times R \approx \lambda R/D$, where D is the antenna length, requiring impractically large antennas for fine resolution at long ranges. SAR overcomes this by recording the amplitude and phase of echoes as the platform moves, then using matched filtering to focus the synthetic aperture, achieving a cross-range resolution of $\delta_{cr} = D/2$ — remarkably independent of range and wavelength.

This counterintuitive result means that a smaller antenna actually gives finer SAR resolution, because a smaller antenna has a wider beam that illuminates a target for a longer portion of the flight path, creating a longer synthetic aperture. Stripmap SAR continuously images a swath parallel to the flight path, while spotlight SAR steers the beam to dwell on a specific area, achieving finer resolution at the expense of area coverage. SAR is used for satellite Earth observation (Sentinel-1 at 5 m resolution, TerraSAR-X at 0.25 m), military reconnaissance, terrain mapping, ocean monitoring, and interferometric applications (InSAR) that measure ground displacement to millimeter accuracy for earthquake and subsidence monitoring.

**Example 17.3.3:** An airborne SAR operates at 5.3 GHz (C-band) from an altitude of 8 km with a velocity of 200 m/s. The antenna has a length of 2 m and the transmitted bandwidth is 100 MHz. Determine (a) the wavelength, (b) the cross-range (azimuth) resolution, (c) the range resolution, (d) the synthetic aperture length for a target at 20 km slant range, and (e) the coherent integration time.
**Solution:**
(a) $\lambda = c/f = 3 \times 10^8 / 5.3 \times 10^9 =$ **0.0566 m**
(b) Cross-range resolution: $\delta_{cr} = D/2 = 2/2 =$ **1.0 m** — independent of range.
(c) Range resolution: $\delta_r = c/(2B) = 3 \times 10^8 / (2 \times 100 \times 10^6) =$ **1.5 m**
(d) Synthetic aperture length: $L_{SA} = \lambda R/D = 0.0566 \times 20{,}000 / 2 =$ **566 m**. The radar must coherently process echoes over a 566 m flight path segment to achieve 1.0 m resolution.
(e) Integration time: $T_{int} = L_{SA}/v = 566/200 =$ **2.83 seconds**. During this time, the platform motion must be known to a fraction of a wavelength ($< \lambda/4 \approx 14$ mm) for proper focusing, requiring precise inertial navigation or autofocus algorithms.

### 17.3.4 Target Tracking and Data Association

Radar detection produces a stream of plots (detections) at each scan — position measurements corrupted by noise and mixed with false alarms. A tracker converts these plots into tracks, estimating each target's state (position, velocity, acceleration) and predicting its future location. Track-while-scan (TWS) processing maintains multiple target tracks simultaneously by associating new detections with existing tracks at each scan update. The tracking cycle consists of prediction (projecting existing tracks forward to the time of the next scan), gating (defining a region around each predicted position where an associated detection is expected), association (assigning detections to tracks), and update (refining the track state using the associated measurement).

The Kalman filter is the standard estimator for radar tracking, providing the minimum-variance state estimate for linear systems with Gaussian noise. The filter maintains a state vector $\hat{x}$ (typically $[x, \dot{x}, y, \dot{y}]$ for 2D tracking) and a covariance matrix P that quantifies the estimation uncertainty. At each update: $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K(z_k - H\hat{x}_{k|k-1})$, where K is the Kalman gain, $z_k$ is the measurement, and H is the measurement matrix. The innovation $(z_k - H\hat{x}_{k|k-1})$ represents the difference between the actual measurement and the predicted measurement. For maneuvering targets, the extended Kalman filter (EKF) or interacting multiple model (IMM) filter handles nonlinear dynamics and mode switching between straight-line and turning motion.

Data association — deciding which detection belongs to which track — becomes challenging in dense multi-target environments. The Global Nearest Neighbor (GNN) algorithm assigns each detection to the closest predicted track position (within the gate), minimizing the total assignment cost. The Joint Probabilistic Data Association (JPDA) filter handles ambiguous associations by computing the probability that each detection originated from each track and using a weighted combination of all possible associations. Track initiation uses M-of-N logic: a new track is confirmed if M detections fall within a consistent trajectory over N consecutive scans (typically 3-of-5 or 4-of-7).

**Example 17.3.4:** A surveillance radar tracks an aircraft at a range of 100 km with a range accuracy of $\sigma_R = 50$ m and azimuth accuracy of $\sigma_\theta = 0.3°$. The scan interval is 5 seconds and the aircraft velocity is 250 m/s. Determine (a) the cross-range position accuracy, (b) the predicted position uncertainty after one scan (assuming constant velocity), (c) the gate size for 99% probability of containing the true target position, and (d) the number of false alarms expected in the gate if $P_{fa} = 10^{-6}$ per resolution cell.

**Solution:**

(a) Cross-range accuracy: $\sigma_{cr} = R \times \sigma_\theta = 100{,}000 \times (0.3 \times \pi/180) = 100{,}000 \times 0.00524 = \mathbf{524\ m}$. The cross-range accuracy is much worse than the range accuracy, as is typical for surveillance radars.

(b) During one scan interval, the aircraft moves $\Delta x = v \times T = 250 \times 5 = 1{,}250$ m. The predicted position uncertainty grows due to velocity estimation error. If the velocity accuracy is $\sigma_v \approx \sigma_{cr}/T = 524/5 = 105$ m/s, then the position prediction uncertainty after one scan is $\sigma_{pred} = \sqrt{}(\sigma_{cr}{}^2 + (\sigma_v \times T)^2) = \sqrt{}(524^2 + 525^2) = \mathbf{742\ m}$ in cross-range. In range, $\sigma_{pred} \approx \sqrt{}(50^2 + (\sigma_{vr} \times 5)^2)$, which is much smaller.

(c) For a 2D Gaussian distribution, 99% probability requires a gate radius of approximately $3.035\sigma$ (chi-squared with 2 DOF at $p = 0.99$). Gate dimensions: range gate $= 3.035 \times 50 = 152$ m; azimuth gate $= 3.035 \times 742 = \mathbf{2{,}252\ m}$. The gate area is approximately $\pi \times 152 \times 2{,}252 = \mathbf{1.08 \times 10^6\ m^2}$.

(d) Number of resolution cells in gate: approximately $1.08 \times 10^6 / (150 \times 524) \approx 13.7$ cells. Expected false alarms per gate $= 13.7 \times 10^{-6} = \mathbf{1.37 \times 10^{-5}}$ — essentially zero, confirming that association is unambiguous in this scenario. In dense traffic or with higher false alarm rates, multiple detections within the gate require JPDA or multi-hypothesis tracking.

## 17.4 Radar Applications

Radar systems serve diverse applications spanning civil aviation, meteorology, automotive safety, and subsurface imaging. Each application domain drives unique requirements for frequency, resolution, update rate, power, and cost, resulting in vastly different system architectures.

### 17.4.1 Air Traffic Control and Surveillance Radar

Air Traffic Control (ATC) radar provides the position and identity of aircraft in controlled airspace, forming the backbone of aviation safety. Primary Surveillance Radar (PSR) detects aircraft by their radar echoes without requiring any onboard equipment, operating at S-band (2.7–2.9 GHz) for en-route surveillance with detection ranges of 100–450 km, or at L-band (1.2–1.4 GHz) for long-range air defense. Secondary Surveillance Radar (SSR) interrogates aircraft transponders at 1,030 MHz and receives coded replies at 1,090 MHz, providing aircraft identity (Mode A), altitude (Mode C), and full ADS-B data (Mode S).

Airport surveillance uses short-range S-band PSR (ASR-9, ASR-11) with 60 nmi range and 4.8 s rotation period, combined with SSR for identification. Airport Surface Detection Equipment (ASDE-X) uses X-band radar and multilateration to track aircraft and vehicles on runways and taxiways with 1 m accuracy. Precision Approach Radar (PAR) provides azimuth and elevation guidance for landing in poor visibility, using Ku-band for high angular accuracy. Modern ATC systems are transitioning to ADS-B (Automatic Dependent Surveillance-Broadcast), where aircraft continuously broadcast GPS-derived position, reducing dependence on ground-based radar.

**Example 17.4.1:** An ASR-11 airport surveillance radar has the following parameters: frequency 2.8 GHz, peak power 25 kW (solid-state), antenna rotation rate 12.5 RPM, azimuth beamwidth 1.4°, range 60 nmi (111 km), and PRF 1,000 Hz. Determine (a) the scan time, (b) the number of pulses on target per scan (hits per scan), (c) the integration gain for non-coherent integration, and (d) the range resolution for a 1 $\mu$s pulse width.
**Solution:**
(a) Scan time = 60/RPM = 60/12.5 = **4.8 seconds** per revolution.
(b) Hits per scan = PRF × $\theta_{az}$ / (360° × RPM/60) = 1,000 × 1.4 / (360 × 12.5/60) = 1,400 / 75 = **18.7** ≈ **19 pulses** on target per scan.
(c) Non-coherent integration gain ≈ $\sqrt{N} = \sqrt{19}$ = **4.36 (6.4 dB)**. This improves the effective SNR beyond what a single pulse provides.
(d) Range resolution: $\Delta R = c\tau/2 = 3 \times 10^8 \times 10^{-6} / 2$ = **150 m**. This is adequate for separating aircraft in terminal airspace, where minimum radar separation is 3 nmi (5.6 km) laterally.

### 17.4.2 Weather Radar

Weather radar detects precipitation (rain, snow, hail) by measuring the radar reflectivity of hydrometeors — water droplets and ice particles in the atmosphere. The weather radar equation modifies the standard radar range equation by replacing the discrete target RCS with a distributed reflectivity factor Z (measured in $mm^6/m^3$), giving $P_r = C \times Z / R^2$, where C is a system constant. The reflectivity factor Z is related to rainfall rate $R_r$ by the Marshall-Palmer relation $Z = 200R_r^{1.6}$, where $R_r$ is in mm/h and Z is in $mm^6/m^3$. Reflectivity is commonly expressed in dBZ, where dBZ = 10 $\log_{10}(Z)$.

The NEXRAD (WSR-88D) network of 160 S-band (2.7–3.0 GHz) Doppler weather radars covers the continental United States, operating with 750 kW peak power, a 28-foot (8.5 m) parabolic antenna (gain ≈ 45 dBi), and Doppler processing to measure wind speeds within storms. Dual-polarization capability (transmitting both horizontal and vertical polarizations) allows NEXRAD to distinguish rain from hail, snow, and debris, significantly improving precipitation estimation and tornado detection. X-band (9.3–9.5 GHz) weather radars are used for gap-filling in urban areas and at airports, while airborne weather radars (C-band or X-band) in commercial aircraft detect turbulence and thunderstorms along the flight path.

**Example 17.4.2:** A NEXRAD WSR-88D radar detects a thunderstorm cell at 100 km range with a measured reflectivity of 55 dBZ. Determine (a) the reflectivity in linear units, (b) the estimated rainfall rate using the Marshall-Palmer relation, (c) the classification of the storm (light rain < 20 dBZ, moderate rain 20–40 dBZ, heavy rain 40–50 dBZ, very heavy rain/hail > 50 dBZ), and (d) the Doppler velocity resolution if the dwell time is 50 ms.
**Solution:**

(a) $Z = 10^{55/10} = 10^{5.5} = \mathbf{316{,}228\ mm^6/m^3}$
(b) $Z = 200R_r^{1.6}$, so $R_r = (Z/200)^{1/1.6} = (316{,}228/200)^{0.625} = (1{,}581)^{0.625}$
$\ln(1{,}581) = 7.366$, $\times\, 0.625 = 4.604$, $R_r = e^{4.604} =$ **100 mm/h** (3.9 inches/hour)
(c) At 55 dBZ, this is classified as **very heavy rain, likely containing hail**. NWS severe thunderstorm warnings are typically issued for reflectivities > 50 dBZ.
(d) $\lambda = 0.107$ m (at 2.8 GHz). Velocity resolution: $\Delta v = \lambda/(2T_{dwell}) = 0.107 / (2 \times 0.050) =$ **1.07 m/s** (3.9 km/h). This resolution is sufficient to detect the rotational signatures within mesocyclones (tornado precursors), which typically have velocity differentials of 15+ m/s.

### 17.4.3 Automotive Radar

Automotive radar enables advanced driver assistance systems (ADAS) and autonomous driving by detecting vehicles, pedestrians, cyclists, and road infrastructure in all weather conditions. The 77 GHz band (76–81 GHz) has become the global standard for automotive radar, offering a good balance of range resolution (< 10 cm with 4 GHz bandwidth), compact antenna size ($\lambda \approx 3.9$ mm), and regulatory availability. FMCW is the dominant waveform because it provides simultaneous range and velocity measurement with low transmit power (typically 10–13 dBm EIRP) and low-cost silicon germanium (SiGe) or CMOS integrated circuits.

Modern automotive radar systems use multiple operating modes on a single platform. Long-range radar (LRR) at 77 GHz covers 150–300 m with a narrow beam (3–4° azimuth) for adaptive cruise control and highway collision avoidance. Medium-range radar (MRR) covers 30–100 m with moderate beamwidth (15–30°) for cross-traffic alerts and lane-change assistance. Short-range radar (SRR) covers 0.2–30 m with wide beamwidth (60–150°) for parking assistance, blind-spot detection, and rear collision warning. A typical Level 2+ ADAS vehicle uses 5–6 radar sensors (1 forward LRR, 2 front-corner MRR, 2 rear-corner SRR) combined with cameras and ultrasonic sensors in a sensor fusion architecture.

| Feature | Long-Range (LRR) | Medium-Range (MRR) | Short-Range (SRR) |
|---|---|---|---|
| Range | 150–300 m | 30–100 m | 0.2–30 m |
| Azimuth FOV | ±8° | ±40° | ±75° |
| Range resolution | 0.3–1.0 m | 0.1–0.5 m | 0.05–0.1 m |
| Bandwidth | 500 MHz–1 GHz | 1–2 GHz | 4 GHz |
| Application | ACC, AEB | Cross-traffic, lane change | Parking, blind spot |

**Example 17.4.3:** A 77 GHz FMCW long-range automotive radar has a bandwidth of 1 GHz, sweep time of 50 μs, maximum range of 200 m, and transmit antenna array gain of 20 dBi. Determine (a) the range resolution, (b) the velocity resolution using 128 chirps in a frame (frame time = 128 × 50 μs), (c) the beat frequency for a target at 100 m, and (d) the maximum unambiguous velocity.
**Solution:**
(a) Range resolution: $\Delta R = c/(2B) = 3 \times 10^8 / (2 \times 10^9) =$ **15 cm**
(b) $\lambda = 3.9$ mm. Frame time $T_{frame} = 128 \times 50 \times 10^{-6} = 6.4$ ms.

Velocity resolution: $\Delta v = \lambda/(2T_{frame}) = 3.9 \times 10^{-3} / (2 \times 6.4 \times 10^{-3})$ = **0.30 m/s** (1.1 km/h)
(c) Beat frequency: $f_b = 2RB/(cT_{sweep}) = 2 \times 100 \times 10^9 / (3 \times 10^8 \times 50 \times 10^{-6}) = 2 \times 10^{11} / 1.5 \times 10^4$ = **13.3 MHz**
(d) Maximum unambiguous velocity: $v_{max} = \lambda/(4T_{sweep}) = 3.9 \times 10^{-3} / (4 \times 50 \times 10^{-6})$ = **19.5 m/s** (70.2 km/h). For highway speeds exceeding this, velocity ambiguity is resolved by varying the chirp repetition interval or using multiple chirp slopes.

### 17.4.4 Ground-Penetrating Radar (GPR)

Ground-Penetrating Radar (GPR) transmits electromagnetic pulses into the ground and records reflections from subsurface interfaces where the dielectric properties change — such as boundaries between soil layers, buried pipes, voids, or embedded objects. GPR typically operates at frequencies from 10 MHz to 4 GHz, with the choice of frequency representing a fundamental trade-off between penetration depth (which decreases with frequency due to soil attenuation) and resolution (which improves with frequency). Low-frequency GPR (10–100 MHz) can penetrate 10–50 m in dry sand or rock but achieves only meter-scale resolution, while high-frequency GPR (1–4 GHz) offers centimeter resolution but penetrates only 0.1–1 m in typical soils.

GPR uses ultra-wideband (UWB) impulse waveforms — very short pulses (0.5–5 ns) containing energy across a wide bandwidth — rather than the narrowband waveforms used in conventional radar. The reflection coefficient at a dielectric interface depends on the contrast in relative permittivity: $\Gamma = (\sqrt{\varepsilon_{r2}} - \sqrt{\varepsilon_{r1}}) / (\sqrt{\varepsilon_{r2}} + \sqrt{\varepsilon_{r1}})$. Typical permittivity values are: air $\varepsilon_r = 1$, dry sand 3–5, wet soil 10–30, concrete 6–11, water 81, and metal $\infty$ (perfect reflector). Applications include utility mapping (locating buried pipes and cables before excavation), road and bridge inspection (detecting voids and rebar corrosion), archaeology, forensic investigation, and military landmine detection.

**Example 17.4.4:** A 400 MHz GPR system is used to locate a buried water pipe in clay soil ($\varepsilon_r = 12$, attenuation $\alpha = 5$ dB/m). The pipe is at a depth of 1.5 m. Determine (a) the propagation velocity in the soil, (b) the round-trip time to the pipe, (c) the two-way attenuation, (d) the vertical resolution for a pulse bandwidth of 400 MHz, and (e) whether the pipe is detectable if the system dynamic range is 90 dB.
**Solution:**
(a) Velocity: $v = c/\sqrt{\varepsilon_r} = 3 \times 10^8 / \sqrt{12} = 3 \times 10^8 / 3.464$ = **$8.66 \times 10^7$ m/s** (29% of the speed of light)
(b) Round-trip time: $t = 2d/v = 2 \times 1.5 / 8.66 \times 10^7$ = **34.6 ns**
(c) Two-way attenuation: $L = 2 \times \alpha \times d = 2 \times 5 \times 1.5$ = **15 dB**
(d) Vertical resolution: $\delta_v = v/(2B) = 8.66 \times 10^7 / (2 \times 400 \times 10^6)$ = **10.8 cm**. This is sufficient to resolve the pipe diameter (typically 10–30 cm for water mains).
(e) The 15 dB two-way soil attenuation is well within the 90 dB dynamic range, so **yes**, the pipe is easily detectable. The maximum penetration depth at this attenuation rate is approximately $90/(2 \times 5) = 9$ m, though the practical limit in wet clay is typically 2–4 m due to additional scattering and dispersion losses.

### 17.4.5 Missile Defense and Space Surveillance Radar

Ballistic missile defense requires radar systems with extraordinary range, precision, and discrimination capability to detect, track, and classify warheads and decoys during the midcourse phase of flight in the exoatmospheric environment. The AN/FPS-132 Upgraded Early Warning Radar (UEWR) operates at UHF (420–450 MHz) with a 32 m × 32 m phased array face producing approximately 40 dBi gain and peak power of 30 MW, detecting ballistic missiles at ranges exceeding 5,000 km. The Sea-Based X-Band Radar (SBX) operates at X-band (9.5 GHz) with a 12.8 m diameter phased array containing approximately 45,000 transmit/receive modules, producing a pencil beam of 0.1° that can track a baseball-sized object at 4,000 km range and discriminate warheads from decoys by measuring fine target features.

The AN/TPY-2 radar, the sensor for the THAAD (Terminal High Altitude Area Defense) system, operates at X-band with an active electronically scanned array (AESA) of approximately 25,000 gallium arsenide T/R modules. In forward-based mode it provides early detection and tracking of ballistic missiles during boost phase at ranges exceeding 1,000 km; in terminal mode it provides fire-control quality tracks for THAAD interceptor guidance. The Space Fence (AN/FSY-3) at Kwajalein Atoll is an S-band phased array radar designed to detect and catalog small objects in low Earth orbit (LEO), tracking items as small as 5 cm at ranges up to 40,000 km to maintain the space debris catalog. These systems share common features: AESA architecture with thousands of solid-state T/R modules, digital beamforming for multiple simultaneous beams, wideband waveforms for high range resolution and target feature extraction, and sophisticated discrimination algorithms that analyze RCS, polarization, micro-Doppler, and ballistic trajectory to distinguish warheads from decoys and debris.

**Example 17.4.5:** A ground-based X-band missile defense radar (similar to SBX) has an antenna diameter of 12.8 m, aperture efficiency of 0.65, peak transmit power of 2 MW, and system noise temperature of 400 K. The target is a warhead with RCS $\sigma = 0.01$ m$^2$ at a range of 2,000 km. Determine (a) the antenna gain, (b) the received signal power for a single pulse, (c) the single-pulse SNR for a 1 MHz noise bandwidth, and (d) the number of pulses required for coherent integration to achieve 20 dB SNR.

**Solution:**

(a) $\lambda = c/f = 3 \times 10^8 / 9.5 \times 10^9 = 0.03158$ m.

$G = 0.65 \times (\pi \times 12.8/0.03158)^2 = 0.65 \times 1{,}273.3^2 = 0.65 \times 1{,}621{,}300 = 1{,}053{,}800$

$G(\text{dBi}) = 10\log_{10}(1{,}053{,}800) =$ **60.2 dBi**

(b) $P_r = P_t G^2 \lambda^2 \sigma / ((4\pi)^3 R^4) = 2 \times 10^6 \times (1{,}053{,}800)^2 \times 0.03158^2 \times 0.01 / ((4\pi)^3 \times (2 \times 10^6)^4)$

Numerator $= 2 \times 10^6 \times 1.110 \times 10^{12} \times 9.97 \times 10^{-4} \times 0.01 = 2.213 \times 10^{13}$

Denominator $= 1{,}984 \times 1.6 \times 10^{25} = 3.174 \times 10^{28}$

$P_r = 2.213 \times 10^{13} / 3.174 \times 10^{28} =$ $\mathbf{6.97 \times 10^{-16}}$ **W** ($-151.6$ dBW)

(c) Noise power: $N = kT_sB_n = 1.381 \times 10^{-23} \times 400 \times 10^6 = 5.52 \times 10^{-15}$ W.

Single-pulse SNR $= 6.97 \times 10^{-16} / 5.52 \times 10^{-15} =$ **0.126 (−9.0 dB)**. The target is well below the noise floor on a single pulse.

(d) Coherent integration gain required: $20 - (-9.0) = 29.0$ dB. For coherent integration, gain $= 10\log_{10}(N)$, so $N = 10^{29.0/10} =$ **794 pulses**. At a PRF of 10 kHz, this requires a dwell time of 0.079 seconds. This illustrates why missile defense radars use long coherent integration times and extremely high-gain antennas to detect small targets at extreme ranges.

### 17.4.6 LiDAR (Light Detection and Ranging)

LiDAR uses laser pulses — typically at near-infrared wavelengths (905 nm or 1550 nm) — rather than radio-frequency waves to measure the range, velocity, and reflectivity of targets with centimeter-level accuracy. The operating principle is identical to pulsed radar: a short laser pulse is transmitted, the round-trip time of flight (ToF) to a reflective surface is measured, and range is computed as $R = ct/2$. Because the wavelength is approximately 10,000× shorter than microwave radar (1 μm vs. 1 cm), LiDAR achieves angular resolution orders of magnitude finer for a given aperture size, enabling dense 3D point cloud generation with millions of points per second. The range equation for LiDAR follows the same form as radar: $P_r = P_t \rho A_r / (\pi R^2)$ for a diffuse (Lambertian) target, where $\rho$ is the target reflectivity (0.1–0.9 depending on surface) and $A_r$ is the receiver aperture area.

Three main scanning architectures are used. Mechanical LiDAR rotates a laser/detector assembly (or a spinning mirror) through 360° to create a full surround point cloud, achieving 100–300 m range with 0.1° angular resolution (Velodyne, Ouster). Solid-state LiDAR uses MEMS mirrors, optical phased arrays (OPA), or flash illumination to scan without moving parts, offering greater reliability and lower cost but typically with a narrower field of view (60–120°). Flash LiDAR illuminates the entire scene simultaneously with a single wide-beam laser pulse and uses a 2D detector array (similar to a camera sensor) to capture range at every pixel simultaneously — ideal for very fast 3D imaging at short range but limited in range by the eye-safe power distributed across the full field.

The choice of wavelength involves a safety-performance trade-off. The 905 nm systems use inexpensive silicon avalanche photodiode (APD) detectors and low-cost GaAs/InGaAs laser diodes, but the wavelength falls within the retinal hazard zone, limiting transmit power to Class 1 eye-safe levels (~1 mW average for a narrow beam). The 1550 nm systems are eye-safe at much higher power levels (the cornea absorbs 1550 nm before it reaches the retina), enabling longer range, but require more expensive InGaAs detectors and erbium-doped fiber laser sources. Frequency-modulated continuous wave (FMCW) LiDAR — analogous to FMCW radar — is an emerging approach that achieves simultaneous range and velocity measurement with coherent detection, providing shot-noise-limited sensitivity and immunity to ambient sunlight interference.

**Example 17.4.6:** A 905 nm pulsed LiDAR for autonomous vehicles has a transmit pulse energy of 4 μJ, pulse width of 5 ns, receiver aperture diameter of 25 mm, and detector NEP (noise-equivalent power) of 0.2 nW/$\sqrt{\text{Hz}}$. A vehicle with reflectivity $\rho = 0.3$ is at a range of 150 m. Determine (a) the range resolution, (b) the received power for a Lambertian target, (c) the receiver SNR assuming a detection bandwidth of 200 MHz, and (d) the maximum detection range for SNR = 10.
**Solution:**
(a) Range resolution: $\Delta R = c\tau/2 = 3 \times 10^8 \times 5 \times 10^{-9} / 2 =$ **0.75 m**. In practice, timing electronics with sub-nanosecond precision achieve 1–5 cm range accuracy using threshold or peak detection on the return pulse.
(b) Received power for a Lambertian target: $P_r = P_t \times \rho \times A_r / (\pi \times R^2)$
$P_t = E/\tau = 4 \times 10^{-6} / 5 \times 10^{-9} = 800$ W (peak power).
$A_r = \pi(0.0125)^2 = 4.91 \times 10^{-4}$ m$^2$.
$P_r = 800 \times 0.3 \times 4.91 \times 10^{-4} / (\pi \times 150^2) = 0.1178 / 70{,}686 = \mathbf{1.67 \times 10^{-6}\ W}$ (1.67 μW)
(c) Noise floor: $N = NEP \times \sqrt{B} = 0.2 \times 10^{-9} \times \sqrt{(200 \times 10^6)} = 0.2 \times 10^{-9} \times 14{,}142 = 2.83 \times 10^{-6}$ W.
$SNR = P_r/N = 1.67 \times 10^{-6} / 2.83 \times 10^{-6} =$ **0.59 (−2.3 dB)**. The target is just below the noise floor at 150 m.
(d) For SNR = 10: $P_r$ must equal $10 \times N = 2.83 \times 10^{-5}$ W. Since $P_r \propto 1/R^2$, $R_{max} = R \times \sqrt{(P_r/(10 \times N))} = 150 \times \sqrt{(1.67 \times 10^{-6} / 2.83 \times 10^{-5})} = 150 \times \sqrt{0.059} = 150 \times 0.243 =$ **36.4 m**. For 150 m range at SNR = 10, the system would need approximately $(150/36.4)^2 = 17\times$ more pulse energy (68 μJ) or averaging of ~17 pulses, which at a pulse rate of 500 kHz takes only 34 μs — well within

the scan time per point.

## 17.5 Radar System Comparison

Selecting the appropriate radar technology requires matching the system parameters to the application requirements. Frequency band, waveform type, power level, antenna configuration, and signal processing all interact to determine what a radar system can and cannot do.

### 17.5.1 Frequency Band Selection

Radar frequency bands are designated by letter codes established during WWII and standardized by IEEE. Lower frequencies provide longer range and better penetration through weather and foliage but require larger antennas for a given beamwidth. Higher frequencies offer finer resolution and more compact hardware but suffer greater atmospheric attenuation, particularly from rain and atmospheric gases.

| Band | Frequency | Wavelength | Key Properties | Primary Applications |
|---|---|---|---|---|
| HF | 3–30 MHz | 10–100 m | Over-the-horizon propagation, ionospheric reflection | OTH radar, ocean surveillance |
| VHF | 30–300 MHz | 1–10 m | Foliage penetration, long range, low resolution | Early warning, stealth detection |
| UHF | 300 MHz–1 GHz | 30 cm–1 m | Good range, moderate resolution | Wind profilers, foliage penetration |
| L | 1–2 GHz | 15–30 cm | Good range, weather tolerance | Long-range surveillance, ATC (SSR) |
| S | 2–4 GHz | 7.5–15 cm | Balance of range and resolution | ATC (PSR), weather radar (NEXRAD) |
| C | 4–8 GHz | 3.75–7.5 cm | Good resolution, moderate range | Weather radar, satellite SAR |
| X | 8–12 GHz | 2.5–3.75 cm | High resolution, moderate rain attenuation | Marine radar, airborne radar, fire control |
| Ku | 12–18 GHz | 1.7–2.5 cm | Fine resolution, significant rain attenuation | Satellite altimetry, mapping |
| K | 18–27 GHz | 1.1–1.7 cm | Water vapor absorption at 22.24 GHz | Rarely used (absorption band) |
| Ka | 27–40 GHz | 7.5–11 mm | Very fine resolution, high rain attenuation | Airport surface radar, cloud radar |

| Band | Frequency | Wavelength | Key Properties | Primary Applications |
|---|---|---|---|---|
| W | 75–110 GHz | 2.7–4 mm | Extremely fine resolution, compact | Automotive (77 GHz), security screening |

The atmospheric attenuation window at 77 GHz (between the oxygen absorption line at 60 GHz and oxygen absorption line at 118.75 GHz) makes this frequency ideal for automotive radar: it provides millimeter-wave resolution with acceptable atmospheric loss for ranges up to 300 m. Rain attenuation becomes significant above X-band — at 10 mm/h rainfall, the attenuation (per ITU-R P.838-3) is approximately 0.002 dB/km at S-band, 0.22 dB/km at X-band, and ~1.3 dB/km at Ka-band, which is why weather radars use S-band or C-band to see through heavy precipitation.

**Example 17.5.1:** A radar system must detect aircraft at 200 km range with 100 m range resolution. The antenna aperture is limited to 3 m. Determine (a) the minimum bandwidth required, (b) the beamwidth at S-band (3 GHz), X-band (10 GHz), and Ku-band (15 GHz), (c) which band offers the best compromise, and (d) the rain attenuation at each band for moderate rain (4 mm/h) over 200 km.

**Solution:**
(a) Minimum bandwidth: $B = c/(2\Delta R) = 3 \times 10^8 / (2 \times 100) =$ **1.5 MHz**
(b) Beamwidth ($\theta \approx 70\lambda/D$):
S-band: $\theta = 70 \times 0.10 / 3 =$ **2.33°**
X-band: $\theta = 70 \times 0.03 / 3 =$ **0.70°**
Ku-band: $\theta = 70 \times 0.02 / 3 =$ **0.47°**
(c) **S-band** offers the best compromise: the 2.33° beamwidth provides adequate angular resolution for aircraft tracking, atmospheric attenuation is minimal, and the frequency is allocated for surveillance radar. X-band would give a narrower beam but suffers more rain attenuation, limiting all-weather performance.
(d) Approximate two-way rain attenuation for 4 mm/h over 200 km:
S-band: 0.00075 dB/km × 400 km = **0.30 dB** (negligible)
X-band: 0.071 dB/km × 400 km = **28 dB** (severe — reduces detection range by ~80%, from 200 km to ~40 km)
Ku-band: 0.2 dB/km × 400 km = **80 dB** (completely blocks the signal in heavy rain)

### 17.5.2 Radar Systems Comparison

The following table summarizes key parameters across common radar applications, illustrating how different requirements drive fundamentally different system designs.

| Parameter | ATC (ASR-11) | Weather (NEXRAD) | Marine Nav | Automotive (77 GHz) | Police Speed | GPR |
|---|---|---|---|---|---|---|
| **Frequency** | 2.8 GHz (S) | 2.8 GHz (S) | 9.4 GHz (X) | 77 GHz (W) | 10.5 GHz (X) | 400 MHz–4 GHz |

| Parameter | ATC (ASR-11) | Weather (NEXRAD) | Marine Nav | Automotive (77 GHz) | Police Speed | GPR |
|---|---|---|---|---|---|---|
| **Waveform** | Pulsed | Pulsed | Pulsed | FMCW | CW | Impulse (UWB) |
| **Peak power** | 25 kW | 750 kW | 25 kW | 10 mW | 100 mW | 10–100 mW |
| **Range** | 111 km | 460 km | 75 km | 300 m | 300 m | 0.1–50 m |
| **Range resolution** | 150 m | 250 m | 75 m | 4 cm | N/A (CW) | 5–50 cm |
| **Antenna** | Reflector (5 m) | Dish (8.5 m) | Slotted array | Patch array | Horn/patch | Bowtie/horn |
| **Scan** | Mechanical 360° | Mechanical 360° | Mechanical 360° | Electronic ±8° | Fixed beam | Manual sweep |
| **Doppler** | Yes (MTD) | Yes (dual-pol) | Optional | Yes | Yes (primary) | No |
| **Cost** | \$2–5M | \$5–10M | \$2–20K | \$10–50 | \$200–2K | \$10–50K |
| **Key metric** | $P_d$ at range | Reflectivity (dBZ) | Target detection | Range, velocity | Velocity accuracy | Depth, resolution |

Key observations from the comparison:

**Power spans 8 orders of magnitude** — from 10 mW (automotive FMCW) to 750 kW (NEXRAD) — driven by the $R^4$ range dependence. The NEXRAD radar must detect weak distributed targets (raindrops) at hundreds of kilometers, while automotive radar needs only 300 m range with strong reflectors (vehicles).

**Waveform matches the application** — Pulsed radar dominates long-range systems where high peak power and unambiguous range are critical. FMCW dominates short-range applications where low cost, high resolution, and simultaneous range-velocity are needed. CW is sufficient when only velocity measurement is required.

**Antenna size is proportional to range/frequency** — The NEXRAD 8.5 m dish produces a 1° beam at S-band for precise storm cell location, while the automotive 77 GHz patch array achieves a comparable 4° beam from a package smaller than a smartphone.

**Example 17.5.2:** A system engineer must select a radar for a harbor surveillance application requiring detection of small boats (RCS = 2 m$^2$) at 10 km range in moderate rain (4 mm/h), with a range resolution of 10 m and a budget of \$50K. Evaluate whether (a) X-band pulsed, (b) S-band pulsed, or (c) X-band FMCW is the better choice.

**Solution:**

(a) **X-band pulsed** (9.4 GHz): Well-established marine radar technology. Range resolution = 10 m requires $\tau_p = 67$ ns or pulse compression. Rain attenuation at X-band in 4 mm/h rain over 10 km (two-way, 20 km path) $\approx 0.06 \times 20 =$ **1.2 dB** — acceptable. Commercial marine radars in this class cost \$5–20K. **Best choice for this application.**

(b) **S-band pulsed** (3 GHz): Rain attenuation is negligible (0.1 dB two-way), but achieving 10 m resolution requires a bandwidth of 15 MHz (pulse width 67 ns) and the antenna must be 3× larger for the same beamwidth. S-band harbor radars cost \$100–500K — **over budget**.

(c) **X-band FMCW**: Could achieve 10 m resolution with only 15 MHz bandwidth and milliwatt power levels. However, achieving 10 km range with FMCW requires careful design to manage transmitter-receiver leakage and phase noise. Emerging solid-state X-band FMCW radars in the \$20–50K range are becoming viable for this application. **Possible alternative, but less proven than pulsed.**

**Recommendation:** X-band pulsed marine radar at \$10–15K, with 25 kW magnetron transmitter, 4-foot slotted waveguide array antenna, and standard marine radar display. This is a mature, proven technology that meets all requirements within budget.

# Chapter 18

# Optics

Optics is the branch of physics and engineering concerned with the generation, propagation, manipulation, and detection of light — electromagnetic radiation in and near the visible spectrum (roughly 380–750 nm wavelength). For electrical engineers, optics is foundational to fiber-optic communications, laser systems, imaging sensors, lidar, photovoltaics, and display technologies. The field spans geometric optics (ray-based analysis of lenses and mirrors), wave optics (interference, diffraction, and polarization phenomena), and photonics (the generation and detection of photons using semiconductor devices). This chapter covers the optical principles most relevant to electrical engineering practice, from Snell's law and lens equations to laser operation, photodetector design, fiber-optic communication, and optical system engineering.

## 18.1 Geometric Optics

Geometric optics treats light as rays that travel in straight lines through uniform media, bending at interfaces according to the laws of reflection and refraction. This approximation is valid when the optical elements (lenses, mirrors, apertures) are much larger than the wavelength of light, making it the primary design tool for imaging systems, illumination optics, and optical instruments. Ray tracing through sequences of surfaces allows engineers to predict image location, magnification, and aberrations without solving Maxwell's equations.

### 18.1.1 Reflection and Refraction (Snell's Law)

When a light ray encounters the boundary between two media with different refractive indices, part of the ray is reflected and part is transmitted (refracted). The law of reflection states that the angle of incidence equals the angle of reflection ($\theta_i = \theta_r$), both measured from the surface normal. Snell's law governs the refraction angle: $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$, where $n_1$ and $n_2$ are the refractive indices of the incident and transmitted media. The refractive index $n = c/v$ relates the speed of light in vacuum ($c = 3 \times 10^8$ m/s) to its speed v in the medium. Common refractive indices include air ($n \approx 1.000$), water ($n \approx 1.333$), glass ($n \approx 1.5$), and silicon ($n \approx 3.5$ at 1550 nm). When light passes from a denser to a less dense medium ($n_1 > n_2$), total internal reflection occurs at angles greater than the critical angle $\theta_c = \arcsin(n_2/n_1)$ — the principle underlying optical fiber waveguiding.

**Example 18.1.1:** A light ray in glass ($n_1 = 1.52$) strikes the glass-air interface at an angle of 35° from the normal. Determine the refraction angle in air and the critical angle for total internal reflection.

**Solution:**
Applying Snell's law: $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$, so $1.52 \times \sin(35°) = 1.00 \times \sin(\theta_2)$.
$\sin(\theta_2) = 1.52 \times 0.5736 = 0.8719$.
$\theta_2 = \arcsin(0.8719) = 60.7°$.
The refracted ray bends away from the normal as expected when entering a less dense medium.
Critical angle: $\theta_c = \arcsin(n_2/n_1) = \arcsin(1.00/1.52) = \arcsin(0.6579) = 41.1°$.
Since $35° < 41.1°$, the ray is below the critical angle and refraction occurs. At angles $\geq 41.1°$, total internal reflection would occur.

### 18.1.2 Lenses and Imaging

A thin lens focuses or diverges light rays according to the thin lens equation: $1/f = 1/d_o + 1/d_i$, where f is the focal length, $d_o$ is the object distance, and $d_i$ is the image distance (all measured from the lens). Converging (convex) lenses have positive focal lengths and bring parallel rays to a real focus, while diverging (concave) lenses have negative focal lengths and cause parallel rays to appear to diverge from a virtual focus. The lateral magnification is $m = -d_i/d_o$, where a negative value indicates an inverted image. The lensmaker's equation relates the focal length to the lens geometry: $1/f = (n - 1)[1/R_1 - 1/R_2]$, where n is the refractive index and $R_1$, $R_2$ are the radii of curvature of the two surfaces. For multi-element lens systems, the combined focal length of two thin lenses separated by distance d is found from: $1/f_{total} = 1/f_1 + 1/f_2 - d/(f_1 \times f_2)$. The f-number (f/# = f/D, where D is the aperture diameter) determines the light-gathering ability and depth of field of the lens.

**Example 18.1.2:** A converging lens with focal length f = 50 mm forms an image of an object placed 200 mm from the lens. Find the image distance, magnification, and determine whether the image is real or virtual, upright or inverted.
**Solution:**
Using the thin lens equation: $1/d_i = 1/f - 1/d_o = 1/50 - 1/200 = (4 - 1)/200 = 3/200$.
$d_i = 200/3 = 66.7$ mm.
Since $d_i$ is positive, the image is real (formed on the opposite side of the lens from the object).
Magnification: $m = -d_i/d_o = -66.7/200 = -0.333$.
The negative magnification indicates the image is inverted, and $|m| = 0.333$ means the image is one-third the size of the object.
This is the typical configuration for a camera lens imaging a distant scene.

### 18.1.3 Mirrors and Curved Surfaces

Curved mirrors follow the mirror equation: $1/f = 1/d_o + 1/d_i$, which is identical in form to the thin lens equation but with different sign conventions. For a spherical mirror, the focal length is $f = R/2$, where R is the radius of curvature (positive for concave mirrors, negative for convex). Concave (converging) mirrors focus parallel rays to a real focal point in front of the mirror, making them useful for telescopes, solar concentrators, and satellite dish feeds. Convex (diverging) mirrors produce virtual, upright, reduced images and are used for wide-angle viewing (vehicle side mirrors, security mirrors). Parabolic mirrors eliminate spherical aberration for on-axis rays by ensuring all parallel rays converge to a single focal point, making them the preferred shape for astronomical telescopes and high-power laser focusing systems. The reflectance at a surface depends on the materials and angle of incidence; metallic coatings (aluminum, silver, gold) provide broadband reflectivity, while dielectric multilayer coatings achieve very high reflectivity (> 99.9%) at specific wavelengths.

**Example 18.1.3:** A concave mirror has a radius of curvature R = 40 cm. An object is placed 30 cm from the mirror. Find the image distance, magnification, and describe the image.
**Solution:**
Focal length: $f = R/2 = 40/2 = 20$ cm.
Mirror equation: $1/d_i = 1/f - 1/d_o = 1/20 - 1/30 = (3 - 2)/60 = 1/60$.
$d_i = 60$ cm. The positive image distance means the image is real (in front of the mirror).
Magnification: $m = -d_i/d_o = -60/30 = -2.0$.
The image is inverted (negative m) and twice the size of the object ($|m| = 2$).
The object is between f and 2f from the mirror, producing a magnified real image beyond 2f. This configuration is used in dental examination mirrors and projectors where a real, enlarged image is needed. Shaving and makeup mirrors instead require the object inside the focal length ($d_o < f$), which produces a virtual, upright, magnified image.

### 18.1.4 Prisms and Dispersive Elements

A prism separates white light into its constituent wavelengths by exploiting the wavelength dependence of the refractive index (material dispersion). When a collimated beam enters a prism at angle $\theta_i$, each wavelength refracts by a different amount according to Snell's law, producing an angular spread of colors at the exit face. The angular dispersion of a prism is $d\theta/d\lambda = (t/d)(dn/d\lambda)$, where t is the base length, d is the beam diameter, and $dn/d\lambda$ is the material dispersion of the glass. For a prism at minimum deviation (where the ray passes symmetrically through the prism), the deviation angle is $\delta_{min} = 2 \arcsin(n \sin(\alpha/2)) - \alpha$, where $\alpha$ is the prism apex angle. Common prism materials include BK7 glass ($n \approx 1.517$ at 546 nm, low dispersion), SF11 flint glass ($n \approx 1.785$, high dispersion for greater spectral separation), and fused silica ($n \approx 1.458$, excellent UV transmission for spectrometry). Prisms are used in spectrometers, wavelength tuning of lasers, beam steering, and binoculars (Porro and roof prisms for image inversion without dispersion).

Diffraction gratings provide much higher spectral resolution than prisms by exploiting the interference of light diffracted from thousands of periodic grooves. The grating equation $d \sin(\theta_m) = m\lambda$ (where d is the groove spacing, m is the diffraction order) determines the angles at which constructive interference occurs for each wavelength. The resolving power $R = mN$ (where N is the total number of illuminated grooves) determines the ability to distinguish closely spaced wavelengths: $\Delta\lambda_{min} = \lambda/R$. A typical research-grade grating with 1200 grooves/mm and 50 mm illuminated width has N = 60,000 grooves and resolving power R = 60,000 in first order, separating wavelengths as close as 0.009 nm at 550 nm. Blazed gratings concentrate most of the diffracted energy into a single order by angling the groove facets, achieving efficiencies of 60–90% at the blaze wavelength. Volume holographic gratings and arrayed waveguide gratings (AWGs) are the key dispersive elements in DWDM fiber-optic systems for wavelength multiplexing and demultiplexing.

**Example 18.1.4:** A spectrometer uses a diffraction grating with 600 grooves/mm and a 25 mm illuminated width. Light at $\lambda = 589.0$ nm and $\lambda = 589.6$ nm (the sodium D doublet) is analyzed in first order. Determine (a) the grating spacing, (b) the diffraction angle for 589.0 nm, (c) the resolving power, and (d) whether the grating can resolve the sodium D doublet.
**Solution:**

(a) Grating spacing: $d = 1/600$ mm = 1.667 μm.
(b) Diffraction angle: $\sin(\theta_1) = m\lambda/d = 1 \times 589.0 \times 10^{-9} / 1.667 \times 10^{-6} = 0.3534$. $\theta_1 = \arcsin(0.3534)$ = **20.69°**.
(c) Total grooves: $N = 600 \times 25 = 15{,}000$. Resolving power: $R = mN = 1 \times 15{,}000 =$ **15,000**. Minimum resolvable wavelength difference: $\Delta\lambda_{min} = \lambda/R = 589.0/15{,}000 = 0.0393$ nm.
(d) The sodium D doublet separation is $\Delta\lambda = 589.6 - 589.0 = 0.6$ nm. Since 0.6 nm » 0.039 nm, the grating **easily resolves** the doublet. The required resolving power would be only $R = 589/0.6 = 982$, which needs just 982 grooves — even a small grating section would suffice. This grating could resolve lines as close as 0.039 nm apart, making it suitable for analyzing fine spectral structure in atomic emission and absorption spectra.

## 18.2 Wave Optics

Wave optics treats light as an electromagnetic wave, accounting for phenomena that geometric optics cannot explain: interference (constructive and destructive combination of waves), diffraction (bending of light around obstacles and through apertures), and polarization (the orientation of the electric field oscillation). These effects become significant when feature sizes approach the wavelength of light (hundreds of nanometers for visible light) and are critical to the design of optical coatings, diffraction gratings, interferometric sensors, and photolithography systems used in semiconductor fabrication.

### 18.2.1 Interference

Interference occurs when two or more coherent light waves overlap, producing a pattern of alternating bright (constructive) and dark (destructive) regions determined by the phase difference between the waves. In Young's double-slit experiment, light passing through two narrow slits separated by distance d produces an interference pattern on a screen at distance L, with bright fringes at angles $\theta$ satisfying $d \sin(\theta) = m\lambda$ ($m = 0, \pm 1, \pm 2, \ldots$) and dark fringes at $d \sin(\theta) = (m + ½)\lambda$. The fringe spacing on the screen is $\Delta y = \lambda L/d$. Thin-film interference occurs when light reflects from the top and bottom surfaces of a thin transparent film (e.g., anti-reflection coatings, soap bubbles), with constructive or destructive interference depending on the film thickness, refractive index, and wavelength. Anti-reflection coatings use a quarter-wave thickness ($t = \lambda/4n_{film}$) with $n_{film} = \sqrt{(n_1 \times n_2)}$ to cancel reflections by destructive interference, reducing surface reflections from ~4% to near zero at the design wavelength.

**Example 18.2.1:** A thin anti-reflection coating of $MgF_2$ ($n = 1.38$) is applied to a glass lens ($n = 1.52$). Design the coating for minimum reflection at $\lambda = 550$ nm (green light). Calculate the coating thickness and the residual reflectance.
**Solution:**
For a quarter-wave anti-reflection coating, the thickness is $t = \lambda / (4 \times n_{film}) = 550 / (4 \times 1.38) = 550 / 5.52 = 99.6$ nm.
The ideal refractive index for zero reflectance would be $n_{ideal} = \sqrt{(1.00 \times 1.52)} = \sqrt{1.52} = 1.233$.
Since $n_{MgF_2} = 1.38 \neq 1.233$, the reflectance is not perfectly zero.
The residual reflectance at normal incidence is $R = [(n_{film}{}^2 - n_1 \times n_2) / (n_{film}{}^2 + n_1 \times n_2)]^2 = [(1.38^2 - 1.00 \times 1.52) / (1.38^2 + 1.00 \times 1.52)]^2 = [(1.904 - 1.52) / (1.904 + 1.52)]^2 = [0.384 / 3.424]^2 = 0.1121^2 = 0.0126 = 1.26\%$.

This is a significant improvement over the uncoated reflectance of ~4.3%.

### 18.2.2 Diffraction

Diffraction is the bending and spreading of light waves as they pass through apertures or around obstacles, a consequence of the Huygens-Fresnel principle that every point on a wavefront acts as a source of secondary wavelets. For a single slit of width a, the far-field (Fraunhofer) diffraction pattern has intensity minima at angles $\theta$ where $a \sin(\theta) = m\lambda$ ($m = \pm1, \pm2, \ldots$). The angular width of the central maximum is $2\lambda/a$, which widens as the slit narrows. For a circular aperture of diameter D (relevant to lenses and telescopes), the diffraction pattern is an Airy disk with angular radius to the first minimum of $\theta = 1.22\lambda/D$. This sets the Rayleigh criterion for angular resolution — two point sources can just be resolved when separated by this angle. Diffraction gratings consist of many equally spaced slits (or grooves) and produce sharp spectral peaks at angles given by $d \sin(\theta) = m\lambda$, where d is the grating spacing and m is the diffraction order, enabling high-resolution wavelength measurement in spectrometers.

**Example 18.2.2:** A telescope has an objective lens diameter of D = 200 mm. Determine the angular resolution limit at $\lambda = 550$ nm and the minimum resolvable separation of two stars at a distance that subtends 1 arc-second.
**Solution:**
Rayleigh criterion: $\theta_{min} = 1.22\lambda/D = 1.22 \times 550 \times 10^{-9} / 0.200 = 671 \times 10^{-9} / 0.200 = 3.355 \times 10^{-6}$ rad.
Converting to arc-seconds: $\theta = 3.355 \times 10^{-6} \times (206{,}265 \text{ arcsec/rad}) = 0.692$ arc-seconds.
Since 0.692 arcsec < 1 arcsec, the telescope can resolve two stars separated by 1 arc-second — the optics are not the limiting factor.
In practice, atmospheric turbulence (seeing) typically limits ground-based resolution to 1–2 arc-seconds, which is why adaptive optics systems are used to correct wavefront distortion.

### 18.2.3 Polarization

Polarization describes the orientation of the electric field vector in an electromagnetic wave. In unpolarized light (from the sun or an incandescent bulb), the electric field oscillates in random directions perpendicular to the propagation direction. A linear polarizer transmits only the component of the electric field along its transmission axis, reducing the intensity of unpolarized light by half (Malus's law: $I = I_0 \cos^2(\theta)$, where $\theta$ is the angle between the polarization direction and the polarizer axis). Polarization by reflection occurs at Brewster's angle $\theta_B = \arctan(n_2/n_1)$, where the reflected light is completely polarized perpendicular to the plane of incidence. Circular and elliptical polarization result from two perpendicular linear components with a phase difference (90° for circular). Quarter-wave plates convert between linear and circular polarization and are essential components in optical isolators, CD/DVD readers, and 3D display systems. Polarization-maintaining fibers and polarization beam splitters are critical in fiber-optic communication and sensing systems.

**Example 18.2.3:** Unpolarized light with intensity $I_0 = 100$ mW/cm$^2$ passes through two linear polarizers. The first polarizer has its transmission axis vertical, and the second is tilted 60° from vertical. Find the intensity after each polarizer.

**Solution:**
After the first polarizer, unpolarized light is reduced by half: $I_1 = I_0/2 = 100/2 = 50$ mW/cm$^2$.
The transmitted light is now vertically polarized.
After the second polarizer (Malus's law): $I_2 = I_1 \cos^2(60°) = 50 \times \cos^2(60°) = 50 \times (0.5)^2 = 50 \times 0.25 = 12.5$ mW/cm$^2$.
The overall transmission is $I_2/I_0 = 12.5/100 = 12.5\%$.
If a third polarizer at 30° from vertical were inserted between the two, it would actually increase the final output — a counterintuitive result demonstrating that polarizers can "rotate" polarization in stages.

### 18.2.4 Thin-Film Optical Coatings

Thin-film optical coatings exploit interference in multilayer dielectric stacks to engineer the spectral reflectance and transmittance of optical surfaces with high precision. While a single quarter-wave layer provides a basic anti-reflection (AR) coating (§18.2.1), practical applications require multilayer designs — alternating high-index (H, e.g., $TiO_2$, n ≈ 2.3) and low-index (L, e.g., $SiO_2$, n ≈ 1.46) quarter-wave layers — to achieve broadband AR coatings, high-reflectivity (HR) mirrors, bandpass filters, and dichroic beam splitters. A dielectric HR mirror using a stack of $(HL)^N H$ quarter-wave pairs achieves reflectivity $R = [(n_H/n_L)^{2N} - 1]^2 / [(n_H/n_L)^{2N} + 1]^2$, approaching 99.99% with sufficient layers ($N \geq 15$). Bandpass filters (also called interference filters) combine HR stacks with a spacer layer to create a Fabry-Perot cavity that transmits only a narrow wavelength band — used in DWDM channel multiplexers, fluorescence microscopy, and astronomical narrowband imaging. Dichroic mirrors (also called hot/cold mirrors or color-separating mirrors) use carefully designed multilayer stacks to reflect one wavelength band while transmitting another, enabling beam combining in projectors, separating pump and signal wavelengths in laser systems, and splitting RGB channels in 3-CCD cameras. All thin-film coatings are deposited by physical vapor deposition (PVD) techniques — electron-beam evaporation, ion-beam sputtering, or magnetron sputtering — with in-situ optical monitoring to control layer thicknesses to sub-nanometer precision.

**Example 18.2.4:** A dielectric HR mirror for a Nd:YAG laser (λ = 1064 nm) uses alternating quarter-wave layers of $TiO_2$ ($n_H = 2.30$) and $SiO_2$ ($n_L = 1.46$) on a glass substrate. Calculate the reflectivity for N = 8 and N = 15 layer pairs, and the physical thickness of each quarter-wave layer.
**Solution:**
Reflectivity for N = 8: ratio = $n_H/n_L = 2.30/1.46 = 1.575$.
$R_8 = [(1.575^{16} - 1) / (1.575^{16} + 1)]^2$.
$1.575^{16} = (1.575^4)^4 = (6.159)^4 = (6.159^2)^2 = (37.93)^2 = 1{,}438.7$.
$R_8 = [(1438.7 - 1) / (1438.7 + 1)]^2 = [1437.7 / 1439.7]^2 = [0.99861]^2 = 0.99722 = 99.72\%$.
For N = 15: $1.575^{30} = 1.575^{16} \times 1.575^{14}$.
$1.575^{14} = 1.575^{16} / 1.575^2 = 1438.7 / 2.481 = 579.9$.
$1.575^{30} = 1438.7 \times 579.9 = 834{,}190$.
$R_{15} = [(834{,}190 - 1) / (834{,}190 + 1)]^2 \approx (0.999998)^2 = 99.9996\%$.
Quarter-wave physical thicknesses: $t_H = \lambda/(4n_H) = 1064/(4 \times 2.30) = 115.7$ nm, $t_L = \lambda/(4n_L) = 1064/(4 \times 1.46) = 182.2$ nm.
Total stack thickness for 15 pairs (30 layers + 1): $15 \times (115.7 + 182.2) + 115.7 = 15 \times 297.9 + 115.7 = 4{,}584$ nm ≈ 4.6 μm.

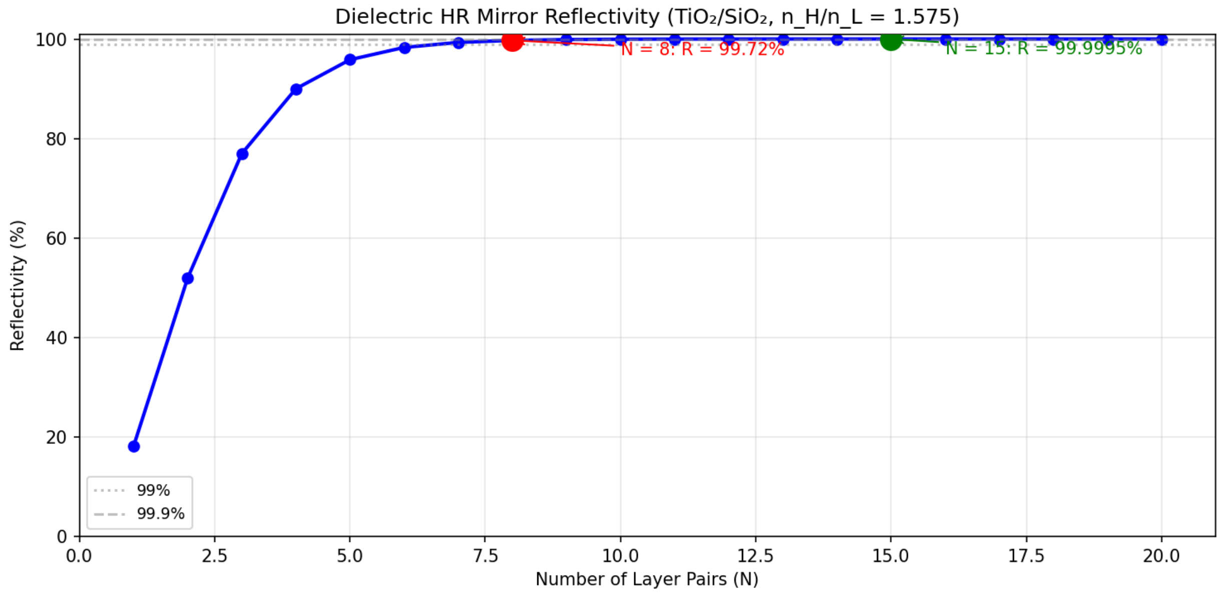



Figure 18.2.4: Dielectric Mirror Reflectivity

## 18.3 Lasers

Lasers (Light Amplification by Stimulated Emission of Radiation) produce coherent, monochromatic, and highly directional light through the process of stimulated emission in a gain medium placed within an optical resonator. Unlike thermal light sources, laser output has extremely narrow spectral linewidth, spatial coherence across the entire beam, and temporal coherence over long path lengths. These properties make lasers indispensable in fiber-optic communications (transmitter sources), manufacturing (cutting, welding, 3D printing), medicine (surgery, diagnostics), metrology (interferometry, lidar), and semiconductor fabrication (photolithography).

### 18.3.1 Stimulated Emission and Laser Operation

Laser operation requires three conditions: a gain medium with a population inversion (more atoms in the excited state than the ground state), an energy source (pump) to maintain the inversion, and an optical resonator (cavity) to provide feedback. In stimulated emission, an incoming photon with energy $E = hf$ (matching the transition energy) triggers an excited atom to emit a second photon with identical frequency, phase, direction, and polarization — this is the amplification mechanism. The gain medium can be a gas (HeNe, $CO_2$, argon), solid crystal (Nd:YAG, Ti:sapphire), semiconductor (GaAs, InGaAsP), liquid dye, or optical fiber doped with rare earth ions (erbium, ytterbium). The optical cavity, typically formed by two mirrors (one highly reflective, one partially transmitting as the output coupler), provides the feedback that allows stimulated emission to build up into a coherent beam. The threshold condition requires that the round-trip gain equals the round-trip losses (mirror transmission, scattering, absorption). Above threshold, the output power increases linearly with pump power, characterized by the slope efficiency $\eta_s$.

**Example 18.3.1:** A Nd:YAG laser ($\lambda = 1064$ nm) has a cavity with mirrors of reflectivity $R_1 = 99.9\%$ and $R_2 = 95\%$ (output coupler), internal round-trip loss of 2%, and a small-signal gain coefficient of $g_0 = 0.5\ cm^{-1}$ in a 10 cm gain medium. Determine whether the laser reaches threshold.
**Solution:**
Round-trip gain: $G = \exp(2 \times g_0 \times L) = \exp(2 \times 0.5 \times 10) = \exp(10) = 22{,}026$ (expressed as single-pass gain, the gain is $\exp(g_0L) = \exp(5) = 148.4$).
Round-trip loss factor: the light must reflect from both mirrors and survive internal losses: $\delta = R_1 \times R_2 \times (1 - \text{internal loss}) = 0.999 \times 0.95 \times 0.98 = 0.930$.
The threshold condition requires that the single-pass gain squared times the loss factor equals 1: $G^2_{threshold} \times \delta = 1$, so $G_{threshold} = 1/\sqrt{\delta} = 1/\sqrt{0.930} = 1.037$.
The actual single-pass gain is $\exp(5) = 148.4$ » 1.037.
The laser is far above threshold, so it will oscillate strongly. In practice, gain saturation will clamp the intracavity gain to the threshold value.

### 18.3.2 Laser Types and Characteristics

Lasers are categorized by their gain medium, which determines the output wavelength, power capability, efficiency, and beam quality. **Gas lasers** include the helium-neon (HeNe, 632.8 nm, milliwatts, excellent beam quality for alignment and interferometry), $CO_2$ (10.6 μm, kilowatts, industrial cutting and welding), and excimer lasers (UV, 193–351 nm, photolithography and eye surgery). **Solid-state lasers** use crystalline or glass hosts doped with ions: Nd:YAG (1064 nm, Q-switched for pulsed applications), Ti:sapphire (tunable 650–1100 nm, ultrafast pulses), and fiber lasers (ytterbium or erbium-doped, highly efficient, scalable to multi-kilowatt CW power). **Semiconductor lasers** (laser diodes) are the most widely used laser type by volume, spanning wavelengths from UV (GaN, 405 nm) to mid-IR (InGaAsSb), with direct electrical pumping, compact size, high efficiency (> 50%), and modulation bandwidth exceeding 10 GHz — making them the standard source for fiber-optic communications, barcode scanners, laser printers, and optical disc drives. **VCSELs** (Vertical-Cavity Surface-Emitting Lasers) emit perpendicular to the chip surface, enabling wafer-level testing and 2D array fabrication for high-speed data links and 3D sensing (face recognition, lidar).

**Example 18.3.2:** A semiconductor laser diode emits 20 mW of optical power at $\lambda = 1550$ nm when driven with 50 mA at 1.5 V. Calculate the wall-plug efficiency, the photon energy, and the number of photons emitted per second.
**Solution:**
Electrical input power: $P_{elec} = I \times V = 0.050 \times 1.5 = 75$ mW.
Wall-plug efficiency: $\eta = P_{opt}/P_{elec} = 20/75 = 26.7\%$.
Photon energy: $E = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (1550 \times 10^{-9}) = 1.988 \times 10^{-25} / 1.55 \times 10^{-6} = 1.282 \times 10^{-19}$ J = 0.80 eV.
Photon rate: $N = P_{opt}/E = 20 \times 10^{-3} / 1.282 \times 10^{-19} = 1.56 \times 10^{17}$ photons/s.
Each photon carries very little energy, so even modest optical power involves enormous photon counts.

### 18.3.3 Laser Safety Classifications

Lasers are classified by their potential to cause eye or skin injury, following the IEC 60825-1 / ANSI Z136.1 standards. **Class 1** lasers are safe under all conditions of normal use (e.g., enclosed CD/DVD players, laser printers). **Class 1M** are safe for the unaided eye but may be hazardous when viewed with optical instruments (magnifiers or telescopes). **Class 2** emit visible light (400–700 nm) up to 1 mW, relying on the blink reflex (0.25 s) for eye protection. **Class 3R** (formerly 3A) emit up to 5 mW visible or higher power at other wavelengths, posing a slight risk of eye injury. **Class 3B** lasers (5–500 mW CW) can cause immediate eye injury from direct or specular reflection and require protective eyewear and controlled access. **Class 4** lasers exceed 500 mW and can cause eye and skin burns, ignite materials, and produce hazardous fumes — requiring full engineering controls, interlocks, eyewear, and training. The Maximum Permissible Exposure (MPE) depends on wavelength, exposure duration, and beam characteristics. The Nominal Ocular Hazard Distance (NOHD) is the distance beyond which the beam irradiance falls below the MPE.

**Example 18.3.3:** A Class 3B laser emits a 100 mW beam at 532 nm with a beam divergence of 1.5 mrad. The MPE for a 0.25 s exposure at 532 nm is 2.5 mW/cm². Calculate the NOHD.
**Solution:**
The beam diameter at distance r from the laser is approximately $d(r) = d_0 + \theta \times r$, where $d_0$ is the initial beam diameter (assume small, ~1 mm) and $\theta$ is the full-angle divergence.
The beam area is $A = \pi(d/2)^2$.
At the NOHD, the irradiance equals the MPE: $P/(\pi r^2\theta^2/4) = \text{MPE}$ (approximating $d \approx \theta r$ for large r).
Solving: $r = \sqrt{(4P/(\pi \times \theta^2 \times \text{MPE}))} = \sqrt{(4 \times 0.1 / (\pi \times (1.5 \times 10^{-3})^2 \times 25\ \text{W/m}^2))} = \sqrt{(0.4 / (\pi \times 2.25 \times 10^{-6} \times 25))} = \sqrt{(0.4 / 1.767 \times 10^{-4})} = \sqrt{(2264)} = 47.6$ m.
(MPE = 2.5 mW/cm² = 25 W/m² in SI units.)
The laser beam remains hazardous to the naked eye for approximately 48 meters.
Protective eyewear rated OD ≥ 2 at 532 nm is typically required for Class 3B green lasers within the NOHD; the exact OD depends on working distance and actual beam irradiance at that point ($\text{OD} = \log_{10}(I_{beam}/\text{MPE})$).

### 18.3.4 Fiber Lasers and Ultrafast Lasers

Fiber lasers use an optical fiber doped with rare-earth ions (ytterbium, erbium, thulium) as the gain medium, pumped by semiconductor laser diodes coupled into the fiber cladding. The fiber geometry provides excellent thermal management (high surface-area-to-volume ratio), long interaction length (meters of gain fiber), and inherent single-mode beam quality ($M^2 \approx 1.0$) because the fiber core acts as a spatial filter. Ytterbium-doped fiber lasers operating at 1060–1080 nm dominate industrial laser cutting, welding, and additive manufacturing, with CW powers now exceeding 100 kW from a single fiber and wall-plug efficiencies of 30–50% — far superior to $CO_2$ lasers (10–15% efficiency) for metal processing. The beam is delivered through a flexible fiber cable (50–200 μm core), eliminating the complex mirror-based beam delivery systems required by $CO_2$ lasers. Erbium-doped fiber lasers and amplifiers (EDFAs) at 1530–1565 nm are the backbone of optical telecommunications (§18.5.3), while thulium-doped fibers (1900–2100 nm) serve medical and materials processing applications in the "eye-safe" mid-IR band.

Ultrafast lasers produce pulses as short as femtoseconds ($10^{-15}$ s) through the technique of mode-locking, in which a large number of longitudinal cavity modes are forced to oscillate in phase. The resulting pulse width is inversely proportional to the laser bandwidth: $\tau_p \approx 0.44/\Delta f$ (for transform-limited Gaussian pulses). A Ti:sapphire laser with a 100 nm bandwidth centered at 800 nm produces pulses as short as ~10 fs, with peak powers exceeding gigawatts despite pulse energies of only nano-joules. Chirped pulse amplification (CPA) — invented by Strickland and Mourou (2018 Nobel Prize) — stretches the ultrashort pulse temporally before amplification (to avoid fiber/crystal damage), amplifies the stretched pulse, then compresses it back to femtosecond duration, achieving peak powers up to petawatts ($10^{15}$ W). Applications of ultrafast lasers include precision micromachining (cold ablation with minimal heat-affected zone), multiphoton microscopy in biology, attosecond science, optical coherence tomography (OCT) in ophthalmology, and frequency comb generation for precision spectroscopy and optical clocks.

**Example 18.3.4:** A mode-locked ytterbium fiber laser produces 200 fs pulses at a repetition rate of 80 MHz with an average output power of 1.6 W at $\lambda = 1040$ nm. Calculate (a) the pulse energy, (b) the peak power, (c) the spectral bandwidth required for transform-limited pulses, and (d) the number of longitudinal modes locked together (cavity length = 1.875 m).
**Solution:**
(a) Pulse energy: $E = P_{avg}/f_{rep} = 1.6 / (80 \times 10^6) =$ **20 nJ** per pulse.
(b) Peak power: $P_{peak} = E/\tau_p = 20 \times 10^{-9} / 200 \times 10^{-15} =$ **100 kW**. From just 20 nJ of energy, the ultrashort pulse duration concentrates the energy into an enormous peak power.
(c) Transform-limited bandwidth: $\Delta f = 0.44/\tau_p = 0.44 / (200 \times 10^{-15}) = 2.2 \times 10^{12}$ Hz = **2.2 THz**. In wavelength: $\Delta\lambda = \lambda^2 \Delta f / c = (1040 \times 10^{-9})^2 \times 2.2 \times 10^{12} / (3 \times 10^8) =$ **7.93 nm**. The laser must support at least this bandwidth for the pulses to be transform-limited.
(d) Cavity mode spacing: $\Delta f_{mode} = c/(2L) = 3 \times 10^8/(2 \times 1.875) = 80$ MHz (matching the repetition rate, as expected). Number of locked modes: $N = \Delta f/\Delta f_{mode} = 2.2 \times 10^{12}/(80 \times 10^6) =$ **27,500 modes** oscillating in phase. The coherent addition of 27,500 modes at a single instant is what creates the enormous peak power relative to the average power.

## 18.4 Photodetectors

Photodetectors convert optical signals into electrical signals, forming the receiver side of any optical system — from fiber-optic communication links to digital cameras and solar cells. The fundamental mechanism is the absorption of photons by a semiconductor material, generating electron-hole pairs that produce a photocurrent proportional to the incident optical power. Key performance parameters include responsivity (A/W), quantum efficiency (percentage of photons that generate carriers), bandwidth (speed of response), dark current (noise current without illumination), and noise-equivalent power (NEP, the minimum detectable optical power).

### 18.4.1 Photodiodes (PIN and Avalanche)

A PIN photodiode consists of a p-type, intrinsic (undoped), and n-type semiconductor layer. Photons absorbed in the wide intrinsic region generate electron-hole pairs that are swept out by the reverse-bias electric field, producing a photocurrent. The responsivity is $R = \eta q\lambda/(hc)$, where $\eta$ is the quantum efficiency, q is the electron charge, $\lambda$ is the wavelength, h is Planck's constant, and c is the speed of light. Silicon PIN photodiodes cover the visible to near-IR range (400–1100 nm), while InGaAs devices cover the telecom bands (900–1700 nm). The bandwidth is limited by the carrier transit time across the intrinsic region and the RC time constant: $f_{3dB} \approx 1/(2\pi \times \max(t_{transit}, RC))$. Avalanche photodiodes (APDs) provide internal gain M (typically 10–100) through impact ionization, where accelerated carriers create additional electron-hole pairs. The APD responsivity is $R_{APD} = M \times R_{PIN}$, but the avalanche process adds excess noise characterized by the excess noise factor F(M). APDs are used in long-haul fiber receivers and lidar where the internal gain improves the signal-to-noise ratio.

**Example 18.4.1:** An InGaAs PIN photodiode has a quantum efficiency of 85% at $\lambda = 1550$ nm. Calculate the responsivity and the photocurrent generated by 10 $\mu$W of incident optical power.
**Solution:**
Responsivity: $R = \eta q\lambda/(hc) = 0.85 \times 1.602 \times 10^{-19} \times 1550 \times 10^{-9} / (6.626 \times 10^{-34} \times 3 \times 10^{8}) = 0.85 \times 1.602 \times 10^{-19} \times 1.55 \times 10^{-6} / (1.988 \times 10^{-25}) = 0.85 \times 1.249 = 1.062$ A/W.
Photocurrent: $I_{ph} = R \times P = 1.062 \times 10 \times 10^{-6} = 10.62\ \mu$A.
For comparison, a silicon detector at 850 nm with $\eta = 0.7$ would have $R = 0.7 \times 1.602 \times 10^{-19} \times 850 \times 10^{-9} / (1.988 \times 10^{-25}) = 0.48$ A/W — lower responsivity because each photon carries more energy at shorter wavelengths.

### 18.4.2 Phototransistors and Photomultipliers

Phototransistors combine a photodiode with a transistor amplifier in a single device, providing current gain ($\beta$, typically 100–1000) at the expense of bandwidth (usually limited to tens of kHz). The base-collector junction acts as the photodiode, and the photocurrent is amplified by the transistor action, making phototransistors suitable for low-speed applications such as optocouplers, light barriers, and ambient light sensing. Photomultiplier tubes (PMTs) are vacuum-tube devices that achieve extremely high gain ($10^6$–$10^8$) through a cascade of secondary-emission dynodes. An incident photon strikes the photocathode, releasing a photoelectron, which is accelerated and multiplied through a chain of 8–14 dynodes, each multiplying the electron count by a factor of 3–6. PMTs offer single-photon sensitivity, very low dark current when cooled, and nanosecond time resolution, making them the detector of choice for scintillation counters (nuclear physics), fluorescence spectroscopy, astronomy, and photon-counting applications.

**Example 18.4.2:** A photomultiplier tube has 10 dynodes, each with a secondary emission ratio of $\delta = 4$. Calculate the total gain and the anode current for an incident photon rate of $10^6$ photons/s, assuming a photocathode quantum efficiency of 25%.
**Solution:**
Total gain: $M = \delta^{10} = 4^{10} = 1{,}048{,}576 \approx 10^6$.
Photoelectron rate: $N_{pe} = \eta \times$ photon rate $= 0.25 \times 10^6 = 250{,}000$ photoelectrons/s.
Anode current: $I_{anode} = N_{pe} \times M \times q = 2.5 \times 10^5 \times 1.049 \times 10^6 \times 1.602 \times 10^{-19} = 42.0 \times 10^{-9}$ A $=$ 42.0 nA.
This measurable current is produced from just $10^6$ photons/s incident on the cathode, demonstrating the extraordinary sensitivity of PMTs.
At this gain, the signal from a single photon produces a pulse of ~$10^6$ electrons (160 fC), easily detected by standard electronics.

### 18.4.3 Image Sensors (CCD and CMOS)

Image sensors are two-dimensional arrays of photodetectors that capture spatial light patterns for digital imaging. **CCD** (Charge-Coupled Device) sensors accumulate photogenerated charge in potential wells during exposure, then shift the charge packet sequentially through the array to a single output amplifier for readout. CCDs achieve excellent noise performance (low read noise), high fill factor, and uniform response, making them the standard for scientific imaging, astronomy, and medical imaging. **CMOS** (Complementary Metal-Oxide-Semiconductor) image sensors integrate a photodiode and amplifier circuit at each pixel, enabling parallel readout, lower power consumption, on-chip signal processing (ADC, noise reduction, auto-exposure), and integration with other digital circuits on the same chip. CMOS sensors dominate consumer electronics (smartphones, webcams, DSLRs) due to lower cost, faster readout, lower power, and the ability to fabricate the sensor and processing on a single die. Key specifications include pixel count (megapixels), pixel size (μm), full-well capacity (electrons), read noise (electrons RMS), dynamic range (dB), and quantum efficiency as a function of wavelength.

**Example 18.4.3:** A CMOS image sensor has 12-megapixel resolution with 1.4 μm pixel pitch, full-well capacity of 8000 electrons, and read noise of 2 electrons RMS. Calculate the sensor active area dimensions (assuming 4:3 aspect ratio) and the dynamic range.
**Solution:**
Total pixels: $12 \times 10^6$. For 4:3 aspect ratio: $N_x \times N_y = 12 \times 10^6$ with $N_x/N_y = 4/3$.
$N_y = \sqrt{(12 \times 10^6 \times 3/4)} = \sqrt{(9 \times 10^6)} = 3000$. $N_x = 4000$.
Sensor dimensions: W = 4000 × 1.4 μm = 5.6 mm, H = 3000 × 1.4 μm = 4.2 mm (a 1/2.3" class sensor).
Dynamic range: $DR = 20 \times \log_{10}(\text{full-well / read noise}) = 20 \times \log_{10}(8000/2) = 20 \times \log_{10}(4000) = 20 \times 3.602 = 72.0$ dB.
This corresponds to approximately 12 stops (72/6.02), meaning the sensor can capture detail from the darkest shadows to the brightest highlights across a 4000:1 intensity range.

### 18.4.4 Noise in Optical Detection

Photodetector noise determines the minimum optical power that can be reliably detected, setting fundamental limits on fiber-optic link reach, lidar range, and imaging sensor sensitivity. The dominant noise sources in optical receivers are **shot noise**, **thermal noise**, and **dark current noise**. Shot noise arises from the discrete nature of photon arrival and carrier generation: $i_{shot} = \sqrt{(2qI_{ph}B)}$, where q is the electron charge, $I_{ph}$ is the photocurrent, and B is the electrical bandwidth. Thermal (Johnson) noise is generated by the receiver's load resistance and transimpedance amplifier: $i_{thermal} = \sqrt{(4k_BTB/R_L)}$, where $k_B$ is Boltzmann's constant, T is the absolute temperature, and $R_L$ is the load resistance. Dark current noise behaves as additional shot noise: $i_{dark} = \sqrt{(2qI_dB)}$. The total RMS noise current is $i_n = \sqrt{(i_{shot}^2 + i_{thermal}^2 + i_{dark}^2)}$. The **noise-equivalent power** (NEP) is the optical power that produces a signal-to-noise ratio of 1 in a 1 Hz bandwidth: $NEP = i_n/(R \times \sqrt{B})$, in units of $W/\sqrt{Hz}$. The **specific detectivity** $D^* = \sqrt{A}/NEP$ (where A is the detector area) allows comparison across detector sizes. For avalanche photodiodes, the noise is multiplied by the excess noise factor: $i_{shot,APD} = M \times \sqrt{(2qI_{ph}F(M)B)}$, where $F(M) = M^x$ (x = 0.2–0.7 depending on material). The optimal APD gain balances the signal amplification against the excess noise to maximize the SNR.

**Example 18.4.4:** An InGaAs PIN photodiode with responsivity R = 1.0 A/W and dark current $I_d$ = 5 nA is used in a 10 Gbps receiver with bandwidth B = 7.5 GHz, load resistance $R_L$ = 50 Ω, and temperature T = 300 K. Calculate the thermal noise, the minimum detectable power for SNR = 1 (NEP), and the receiver sensitivity for BER = $10^{-9}$ (requiring Q = 6, or SNR = $Q^2 \approx 36$).
**Solution:**
Thermal noise: $i_{thermal} = \sqrt{(4k_BTB/R_L)} = \sqrt{(4 \times 1.381 \times 10^{-23} \times 300 \times 7.5 \times 10^9 / 50)} = \sqrt{(4 \times 1.381 \times 10^{-23} \times 300 \times 1.5 \times 10^8)} = \sqrt{(2.486 \times 10^{-12})} = 1.577$ μA RMS.
Dark current noise: $i_{dark} = \sqrt{(2 \times 1.602 \times 10^{-19} \times 5 \times 10^{-9} \times 7.5 \times 10^9)} = \sqrt{(1.201 \times 10^{-17})} = 3.47$

nA (negligible versus thermal noise).
In the thermal-noise-limited regime, minimum detectable power (SNR = 1) over bandwidth B: $P_{min} = i_{thermal} / R = 1.577 \times 10^{-6} / 1.0 = 1.577\ \mu W = -28.0$ dBm.
For BER = $10^{-9}$ (Q = 6): $P_{sensitivity} = Q \times i_{thermal} / R = 6 \times 1.577 \times 10^{-6} / 1.0 = 9.46\ \mu W = -20.2$ dBm.
A transimpedance amplifier (TIA) with higher effective $R_L$ (e.g., 500 Ω) would reduce thermal noise by $\sqrt{10}$ and improve sensitivity by 5 dB.

## 18.5 Optical Fiber and Communication

Optical fiber communication transmits information as modulated light through glass or plastic fibers, offering enormous bandwidth (terabits per second), immunity to electromagnetic interference, low attenuation (< 0.2 dB/km at 1550 nm), and physical security against tapping. Optical fibers form the backbone of the global telecommunications infrastructure, carrying over 95% of intercontinental data traffic. While Chapter 15 covers fiber from a networking perspective (cable types, link budgets, OTDR), this section examines the underlying optical physics of fiber propagation, dispersion, and amplification.

### 18.5.1 Fiber Modes and Propagation

An optical fiber guides light by total internal reflection at the core-cladding boundary, where the core (refractive index $n_1$) has a slightly higher index than the surrounding cladding ($n_2$). The numerical aperture NA = $\sqrt{(n_1^2 - n_2^2)}$ determines the maximum acceptance angle for light entering the fiber: $\theta_{max}$ = arcsin(NA). The V number (normalized frequency) $V = (\pi d/\lambda) \times NA$ determines the number of guided modes, where d is the core diameter. Single-mode fiber (SMF) operates with $V < 2.405$, requiring a small core diameter (typically 8–10 μm at 1550 nm), and supports only the fundamental $LP_{01}$ mode — eliminating intermodal dispersion and enabling long-distance, high-bandwidth transmission. Multimode fiber (MMF) has a larger core (50 or 62.5 μm), supports hundreds of modes, and is used for shorter links (< 2 km) due to modal dispersion limiting bandwidth. Graded-index multimode fiber uses a parabolic refractive index profile in the core to equalize the group velocities of different modes, significantly reducing modal dispersion compared to step-index fiber.

**Example 18.5.1:** A step-index single-mode fiber has core diameter d = 9 μm, core index $n_1$ = 1.4677, and cladding index $n_2$ = 1.4624. Calculate the numerical aperture, maximum acceptance angle, and verify single-mode operation at λ = 1550 nm.
**Solution:**
Numerical aperture: NA = $\sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.4677^2 - 1.4624^2)} = \sqrt{(0.01553)} = 0.1246$.
Maximum acceptance angle: $\theta_{max}$ = arcsin(0.1246) = 7.16°.
V number: $V = \pi d \times NA / \lambda = \pi \times 9 \times 10^{-6} \times 0.1246 / (1550 \times 10^{-9}) = 3.523 \times 10^{-6} / 1.55 \times 10^{-6} = 2.273$.
Since V = 2.273 < 2.405, the fiber is single-mode at 1550 nm.
At 1310 nm, V = 2.273 × (1550/1310) = 2.690 > 2.405, so the fiber would support two modes at 1310 nm (though in practice it is designed to be single-mode at both telecom windows).

### 18.5.2 Attenuation and Dispersion

Fiber attenuation is measured in dB/km and limits the maximum transmission distance before signal regeneration or amplification is needed. Silica fiber has three low-loss transmission windows: the O-band centered at 1310 nm (~0.35 dB/km), the C-band at 1550 nm (~0.20 dB/km, the absolute minimum), and the L-band at 1575–1625 nm (~0.22 dB/km). Attenuation is caused by Rayleigh scattering (proportional to $1/\lambda^4$, dominant at shorter wavelengths), absorption by $OH^-$ ions and silica (water peak at 1383 nm, eliminated in modern low-water-peak fibers), and bending losses (macro-bends and micro-bends). **Chromatic dispersion** causes pulse broadening because different wavelength components travel at different velocities; it has two contributions: material dispersion (wavelength dependence of the refractive index) and waveguide dispersion (wavelength dependence of the mode propagation constant). Standard SMF has zero chromatic dispersion near 1310 nm and ~17 ps/(nm·km) at 1550 nm. Dispersion-shifted fiber (DSF) moves the zero-dispersion wavelength to 1550 nm. **Polarization mode dispersion** (PMD) arises from fiber birefringence causing the two polarization modes to travel at slightly different velocities, typically 0.1–1.0 ps/$\sqrt{\text{km}}$.

**Example 18.5.2:** A single-mode fiber link operates at 1550 nm over 80 km with fiber attenuation of 0.22 dB/km and chromatic dispersion of 17 ps/(nm·km). The laser source has a spectral width of 0.1 nm. Calculate the total attenuation and the pulse broadening due to chromatic dispersion.
**Solution:**
Total attenuation: $\alpha_{total} = 0.22 \times 80 = 17.6$ dB.
Chromatic dispersion broadening: $\Delta\tau = D \times L \times \Delta\lambda = 17 \times 80 \times 0.1 = 136$ ps.
For a 10 Gbps NRZ signal (bit period = 100 ps), this 136 ps broadening exceeds the bit period, causing severe intersymbol interference.
Solutions include: (1) using a narrower linewidth source (DFB laser, $\Delta\lambda < 0.01$ nm), reducing broadening to 13.6 ps; (2) dispersion-compensating fiber (DCF) with negative dispersion to cancel the accumulated dispersion; or (3) digital signal processing (DSP) at the receiver for coherent detection systems.

### 18.5.3 Optical Amplifiers and WDM

Optical amplifiers boost the signal directly in the optical domain without converting to electrical and back (O-E-O), enabling long-haul transmission over thousands of kilometers. The **erbium-doped fiber amplifier** (EDFA) is the dominant technology, providing 20–40 dB gain across the C-band (1530–1565 nm) by pumping a section of erbium-doped fiber with 980 nm or 1480 nm laser diodes. EDFAs amplify all wavelength channels simultaneously, making them ideal for wavelength-division multiplexing (WDM) systems. **Raman amplifiers** use stimulated Raman scattering in the transmission fiber itself (pumped at a wavelength ~13 THz below the signal), providing distributed gain that improves the signal-to-noise ratio. **Semiconductor optical amplifiers** (SOAs) are compact and integrable but suffer from crosstalk between WDM channels due to gain dynamics. WDM combines multiple wavelength channels onto a single fiber, with Dense WDM (DWDM) systems carrying 80–160 channels at 50 GHz or 100 GHz spacing, achieving aggregate capacities exceeding 10 Tbps per fiber. The noise figure of an EDFA is typically 4–6 dB, setting a fundamental limit on the achievable optical signal-to-noise ratio (OSNR) after cascaded amplifier spans.

**Example 18.5.3:** A DWDM system uses 80 channels at 100 GHz spacing in the C-band, each modulated at 100 Gbps using coherent DP-QPSK. The link has 10 amplifier spans of 80 km each, with EDFAs having 16 dB gain and 5 dB noise figure. Calculate the total capacity and the OSNR after 10 spans (assuming 0 dBm launch power per channel and 0.2 dB/km fiber loss).
**Solution:**
Total capacity: 80 channels $\times$ 100 Gbps = 8 Tbps aggregate.
Span loss: $0.2 \times 80 = 16$ dB (matched by the 16 dB EDFA gain).
Using the standard EDFA cascade formula: OSNR = 58 + $P_{launch}$(dBm) − $L_{span}$(dB) − NF(dB) − $10\log_{10}(N) = 58 + 0 - 16 - 5 - 10 = 27$ dB.
(The 58 dB constant equals $-10\log_{10}(hfB_{ref}) = -10\log_{10}(6.626 \times 10^{-34} \times 193.1 \times 10^{12} \times 12.5 \times 10^{9} \times 10^{3}) = -10\log_{10}(1.599 \times 10^{-6}$ mW$) = +58.0$, for reference bandwidth $B_{ref} = 12.5$ GHz = 0.1 nm at 1550 nm; the $\times 10^3$ factor converts watts to milliwatts, consistent with $P_{launch}$ in dBm.)
For DP-QPSK at 100 Gbps, the required OSNR is approximately 12–14 dB, so the system has 13–15 dB of OSNR margin — sufficient for reliable operation.

### 18.5.4 Coherent Optical Communication

Coherent optical communication uses a local oscillator (LO) laser at the receiver to mix with the incoming signal, recovering both amplitude and phase information — unlike direct detection receivers that measure only optical power (intensity). By preserving the full optical field (amplitude, phase, and polarization), coherent detection enables advanced modulation formats that encode multiple bits per symbol, dramatically increasing spectral efficiency and capacity. The coherent receiver uses a 90° optical hybrid to split the signal and LO into in-phase (I) and quadrature (Q) components on both X and Y polarization tributaries, producing four electrical outputs that are digitized by high-speed ADCs (typically 64–100 GS/s, 8-bit resolution). Digital signal processing (DSP) in the receiver ASIC then performs chromatic dispersion compensation, polarization demultiplexing, carrier frequency and phase recovery, equalization, and forward error correction (FEC) decoding — all in the digital domain.

The key modulation formats are dual-polarization quadrature phase-shift keying (DP-QPSK, 4 bits/symbol, SE $\approx$ 2 b/s/Hz), DP-16QAM (8 bits/symbol, SE $\approx$ 4 b/s/Hz), and DP-64QAM (12 bits/symbol, SE $\approx$ 6 b/s/Hz). Higher-order formats carry more bits per symbol but require higher OSNR — DP-QPSK needs approximately 12 dB OSNR at BER = $10^{-2}$ (pre-FEC), while DP-16QAM needs approximately 18 dB and DP-64QAM approximately 24 dB. Modern coherent transceivers (400G ZR, 400G ZR+, and 800G) operate at 400–800 Gbps per wavelength on a single fiber pair. The 400G-ZR standard uses DP-16QAM at ~60 GBaud on a single wavelength in a QSFP-DD pluggable form factor, targeting data center interconnects (DCI) up to 120 km. Long-haul submarine systems use probabilistic constellation shaping (PCS) to approach the Shannon limit, adjusting the symbol probability distribution to maximize mutual information for a given OSNR.

**Example 18.5.4:** A coherent optical transceiver operates at 64 GBaud using DP-16QAM modulation at 1550 nm. Each polarization carries 16QAM (4 bits/symbol), and both X and Y polarizations are used. Determine (a) the raw data rate before FEC overhead, (b) the net data rate with 25% FEC overhead, (c) the spectral efficiency for a 75 GHz channel spacing, and (d) the required OSNR for BER = $4 \times 10^{-2}$ (soft-decision FEC threshold) assuming 1.5 dB implementation penalty.
**Solution:**
(a) Raw data rate: 64 GBaud $\times$ 4 bits/symbol $\times$ 2 polarizations = **512 Gbps**.

(b) Net data rate: 512 / 1.25 = **409.6 Gbps** (approximately 400G net). The 25% FEC overhead provides powerful error correction that allows operation at pre-FEC BER up to ~$4 \times 10^{-2}$, correcting to post-FEC BER < $10^{-15}$.
(c) Spectral efficiency: SE = 409.6 / 75 = **5.46 b/s/Hz**. This approaches the theoretical maximum for 16QAM (8 b/s/Hz with both polarizations), reduced by FEC overhead, guard bands, and Nyquist filtering.
(d) Theoretical required OSNR for DP-16QAM at BER = $4 \times 10^{-2}$ is approximately 15.5 dB (in 0.1 nm reference bandwidth). With 1.5 dB implementation penalty: required OSNR = 15.5 + 1.5 = **17.0 dB**. For a link with EDFAs (NF = 5 dB) and 80 km spans at 0 dBm launch power, the maximum number of spans is limited by: OSNR = $58 + 0 - 16 - 5 - 10 \log_{10}(N) \geq 17$, giving $10 \log_{10}(N) \leq 20$, so $N \leq 100$ spans or **8,000 km** — sufficient for transoceanic submarine links.

## 18.6 Optical System Design

Optical system design integrates the principles of geometric optics, wave optics, and photonics to create functional imaging and illumination systems. Engineers must balance competing requirements: resolution (limited by diffraction), field of view, light-gathering ability (f-number), aberration correction, size, weight, and cost. Modern optical design relies heavily on ray-tracing software (Zemax, Code V) that propagates thousands of rays through multi-element systems and optimizes surface curvatures, spacings, and glass types to minimize aberrations.

### 18.6.1 Optical Power and Radiometry

Radiometry quantifies the propagation of electromagnetic radiation through optical systems using a hierarchy of related quantities. **Radiant flux** ($\Phi$, watts) is the total power emitted, transmitted, or received. **Irradiance** (E, W/m$^2$) is the power incident per unit area on a surface. **Radiant intensity** (I, W/sr) is the power emitted per unit solid angle from a point source. **Radiance** (L, W/(m$^2$·sr)) is the power per unit area per unit solid angle, and is conserved through lossless optical systems (a consequence of the brightness theorem). For imaging systems, the image irradiance is $E_{image} = (\pi/4) \times L \times (1/f\#^2)$, showing that lower f-numbers produce brighter images. **Photometry** weights radiometric quantities by the human eye's spectral response (the luminous efficiency function $V(\lambda)$, peaking at 555 nm), producing quantities measured in lumens, lux, and candela. The luminous efficacy of radiation relates lumens to watts: 683 lm/W at 555 nm for monochromatic light, and typically 80–200 lm/W for practical white LED sources.

**Example 18.6.1:** A point source emits 100 W of radiant flux uniformly into a hemisphere ($2\pi$ steradians). Calculate the radiant intensity and the irradiance at a distance of 2 m on axis.
**Solution:**
Radiant intensity: $I = \Phi / \Omega = 100 / (2\pi) = 15.92$ W/sr.
Irradiance at distance r follows the inverse-square law: $E = I / r^2 = 15.92 / (2^2) = 15.92 / 4 = 3.98$ W/m$^2$.
Alternatively, the total flux through a hemisphere at r = 2 m must still be 100 W: $E = \Phi / (2\pi r^2) = 100 / (2\pi \times 4) = 3.98$ W/m$^2$.
At 5 m: $E = 15.92 / 25 = 0.637$ W/m$^2$.
The inverse-square law is fundamental to all optical link budget calculations.

### 18.6.2 Optical Link Budget

An optical link budget accounts for all gains and losses between the transmitter and receiver to verify that a fiber-optic or free-space optical link will operate with adequate margin. The link equation is: $P_{received} = P_{transmit} - \alpha_{fiber} \times L - L_{connectors} - L_{splices} - L_{other} + G_{amplifiers}$, where all quantities are in dB or dBm. The link operates correctly when $P_{received} \geq P_{sensitivity} + M_{system}$, where $P_{sensitivity}$ is the receiver's minimum detectable power at the required BER and $M_{system}$ is the system margin (typically 3–6 dB to account for component aging, temperature variations, repair splices, and future degradation). For fiber links, loss contributions include fiber attenuation (0.2–0.35 dB/km depending on wavelength), connector losses (0.3–0.5 dB per mated pair for PC/APC connectors), fusion splice losses (0.05–0.1 dB each), and passive component insertion losses (WDM mux/demux, optical switches, splitters). Chromatic dispersion and PMD impose bandwidth-distance limits separate from the power budget, requiring a dispersion budget check for high-speed links ($\geq$ 10 Gbps). For amplified links, the OSNR budget replaces the simple power budget, tracking the accumulated amplified spontaneous emission (ASE) noise through cascaded EDFAs. Free-space optical links additionally account for geometric spreading loss (proportional to distance squared), atmospheric absorption and scattering, and pointing/tracking losses.

**Example 18.6.2:** Design a single-mode fiber link operating at 1550 nm over 40 km with the following components: DFB laser transmit power = +3 dBm, fiber attenuation = 0.25 dB/km, 4 connectors at 0.5 dB each, 2 fusion splices at 0.1 dB each, and receiver sensitivity = -28 dBm at 2.5 Gbps (BER = $10^{-12}$). Calculate the total loss, received power, and system margin.
**Solution:**
Fiber loss: $0.25 \times 40 = 10.0$ dB.
Connector losses: $4 \times 0.5 = 2.0$ dB.
Splice losses: $2 \times 0.1 = 0.2$ dB.
Total link loss: $10.0 + 2.0 + 0.2 = 12.2$ dB.
Received power: $P_{rx} = P_{tx} - L_{total} = +3 - 12.2 = -9.2$ dBm.
System margin: $M = P_{rx} - P_{sensitivity} = -9.2 - (-28) = 18.8$ dB.
This exceeds the typical 3–6 dB requirement, providing substantial margin for future splices, connector aging, and cable repairs.
The link is power-budget limited at approximately $(P_{tx} - P_{sensitivity} - M_{required}) / \alpha = (3 + 28 - 6) / 0.25 = 100$ km maximum reach.
Dispersion check: $\Delta\tau = D \times L \times \Delta\lambda = 17 \times 40 \times 0.01 = 6.8$ ps (with DFB laser $\Delta\lambda = 0.01$ nm), well within the 400 ps bit period at 2.5 Gbps.

### 18.6.3 Lens Design and Aberrations

Real lenses deviate from the ideal thin-lens behavior due to aberrations — systematic errors in how rays are focused. The five primary (Seidel) monochromatic aberrations are: **Spherical aberration** (marginal rays focus closer to the lens than paraxial rays, producing a blurred image), **Coma** (off-axis points form comet-shaped blur patterns), **Astigmatism** (tangential and sagittal rays focus at different distances for off-axis points), **Field curvature** (the image surface is curved rather than flat), and **Distortion** (straight lines in the object map to curved lines in the image — barrel or pincushion). **Chromatic aberration** arises because the refractive index varies with wavelength (dispersion), causing different colors to focus at different points; it is corrected using achromatic doublets (a combination of crown and flint glass elements with different dispersions). Aspherical surfaces (departing from a simple sphere) can dramatically reduce spherical aberration with fewer elements, and are commonly produced by precision molding of glass or polymer optics. Modern camera lenses use 6–15 elements to correct these aberrations across the entire field of view and wavelength range.

**Example 18.6.3:** A singlet lens with focal length f = 100 mm and diameter D = 25 mm is used to focus a collimated beam. The lens has a longitudinal spherical aberration of LSA = 0.5 mm (marginal rays focus 0.5 mm closer). Estimate the blur circle diameter at the paraxial focus and the f-number.
**Solution:**
f-number: f/# = f/D = 100/25 = f/4.
The marginal rays converge to a point 0.5 mm in front of the paraxial focus.
At the paraxial focal plane, the marginal rays form a cone with half-angle $\theta \approx (D/2)/f = 12.5/100 = 0.125$ rad.
The blur circle diameter at the paraxial focus is approximately $d_{blur} = LSA \times (D/f) = 0.5 \times (25/100) = 0.5 \times 0.25 = 0.125$ mm = 125 μm.
For comparison, the diffraction-limited Airy disk diameter at λ = 550 nm is $d_{Airy} = 2.44 \times \lambda \times f/\# = 2.44 \times 0.55 \times 10^{-3} \times 4 = 5.37$ μm.
The aberration blur (125 μm) is 23× the diffraction limit, confirming that this simple lens is severely aberration-limited.
Stopping down to f/8 (halving the aperture from D = 25 mm to 12.5 mm) would reduce the spherical aberration by approximately 8× — since transverse blur scales as $D^3$, halving D reduces it by $2^3 = 8$.

# Chapter 19

# Engineering Economics

Engineering economics applies quantitative techniques to evaluate the financial viability of engineering projects and investments. It provides a systematic framework for comparing alternatives that differ in timing, magnitude, and duration of cash flows — enabling engineers to make rational decisions about equipment purchases, infrastructure upgrades, and capital investments. These methods are essential for electrical engineers who must justify projects ranging from substation upgrades and cable replacements to renewable energy installations and power plant construction.

## 19.1 Time Value of Money

The time value of money is the fundamental principle that a dollar available today is worth more than a dollar received in the future, because today's dollar can be invested to earn interest. This concept underlies all engineering economic analysis — it is the reason cash flows occurring at different times cannot simply be added together. Instead, they must be converted to equivalent values at a common point in time using an appropriate interest rate. The interest rate reflects the opportunity cost of capital: the return that could be earned by investing the money elsewhere.

### 19.1.1 Simple Interest

Simple interest is calculated only on the original principal amount, not on any accumulated interest. The total interest earned is $I = P \times i \times n$, where P is the principal, i is the interest rate per period, and n is the number of periods. The total amount owed after n periods is $F = P(1 + i \times n)$. Simple interest is rarely used in engineering economic analysis but provides an important conceptual foundation for understanding compound interest.

**Example 19.1.1:** A utility borrows \$500,000 for a substation capacitor bank upgrade at 6% simple annual interest for 3 years. Determine (a) the total interest paid and (b) the total amount owed at maturity.
**Solution:**
(a) Total interest: $I = P \times i \times n = 500{,}000 \times 0.06 \times 3 =$ **\$90,000**
(b) Total amount owed: $F = P + I = 500{,}000 + 90{,}000 =$ **\$590,000**

### 19.1.2 Compound Interest

Compound interest is calculated on both the original principal and any previously accumulated interest. The future value after n periods is $F = P(1 + i)^n$, where interest earned in each period is added to the principal and earns interest in subsequent periods. Compounding causes exponential growth rather than linear growth, and the difference between simple and compound interest becomes significant over long time horizons. Nearly all engineering economic analyses use compound interest.

**Example 19.1.2:** An electrical contractor invests \$100,000 in a certificate of deposit at 5% annual interest compounded quarterly. Determine (a) the account balance after 4 years and (b) the total interest earned.
**Solution:**
(a) Quarterly interest rate: $i = 0.05/4 = 0.0125$. Number of periods: $n = 4 \times 4 = 16$.
$F = P(1+i)^n = 100{,}000 \times (1.0125)^{16} = 100{,}000 \times 1.2199 = \mathbf{\$121{,}989}$
(b) Total interest earned: $I = F - P = 121{,}989 - 100{,}000 = \mathbf{\$21{,}989}$

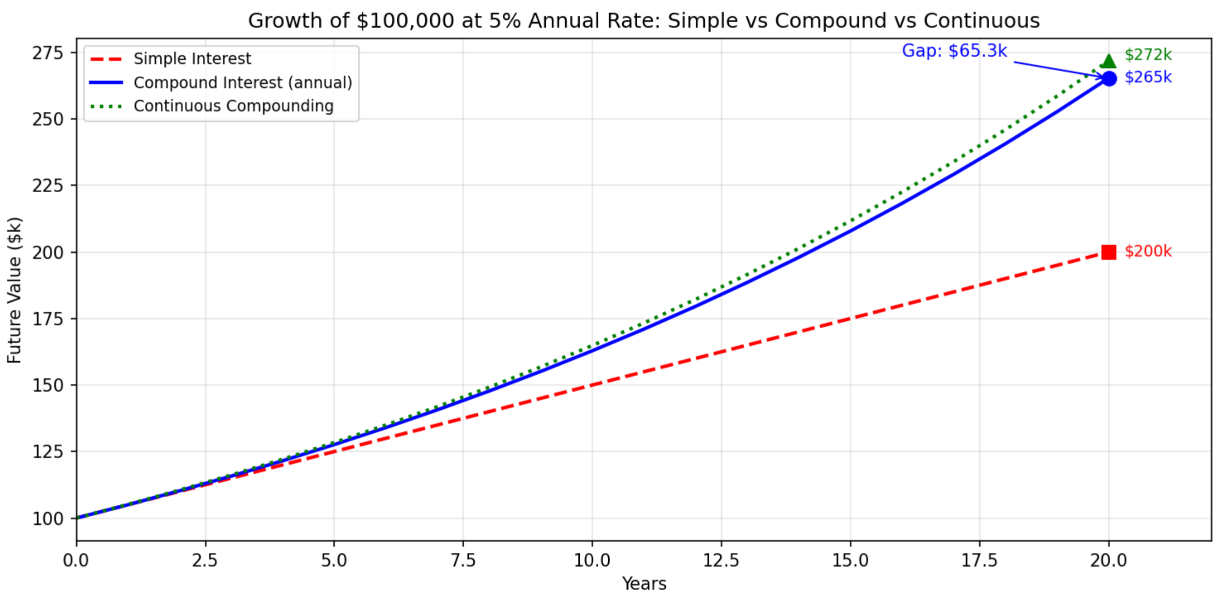



Figure 19.1.2: Simple vs Compound vs Continuous Interest Growth

### 19.1.3 Nominal and Effective Interest Rates

When interest is compounded more frequently than once per year, the stated annual rate is the nominal rate ($r$), and the actual annual yield is the effective rate ($i_{eff}$). The relationship is $i_{eff} = (1 + r/m)^m - 1$, where $m$ is the number of compounding periods per year. The effective rate is always greater than or equal to the nominal rate, with equality only when $m = 1$. Engineers must convert to effective rates when comparing financing options with different compounding frequencies.

**Example 19.1.3:** A utility equipment lease quotes 12% annual interest compounded monthly. Determine (a) the effective annual interest rate and (b) the effective semiannual rate.
**Solution:**
(a) Effective annual rate: $i_{eff} = (1 + 0.12/12)^{12} - 1 = (1.01)^{12} - 1 = 1.12683 - 1 = \mathbf{12.68\%}$
(b) Effective semiannual rate: $i_{semi} = (1 + 0.12/12)^6 - 1 = (1.01)^6 - 1 = 1.06152 - 1 = \mathbf{6.15\%}$

### 19.1.4 Continuous Compounding

Continuous compounding represents the theoretical limit as the number of compounding periods per year approaches infinity. The future value is $F = Pe^{rn}$, where r is the nominal annual rate and n is the number of years. The effective annual rate under continuous compounding is $i_{eff} = e^r - 1$. Continuous compounding is used in some financial models, bond pricing, and as a mathematical convenience in economic analysis.

**Example 19.1.4:** A power company invests \$2,000,000 in an infrastructure fund at 8% continuously compounded interest. Determine (a) the fund value after 5 years, (b) the effective annual interest rate, and (c) the total interest earned.
**Solution:**
(a) $F = Pe^{rn} = 2{,}000{,}000 \times e^{0.08\times 5} = 2{,}000{,}000 \times e^{0.40} = 2{,}000{,}000 \times 1.4918 =$ **\$2,983,649**
(b) $i_{eff} = e^{0.08} - 1 = 1.08329 - 1 =$ **8.33%**
(c) Total interest: $I = 2{,}983{,}649 - 2{,}000{,}000 =$ **\$983,649**

## 19.2 Cash Flow Diagrams and Economic Equivalence

Cash flow diagrams and economic equivalence are the two foundational tools for structuring and solving engineering economic problems. A cash flow diagram provides a visual timeline of all monetary inflows and outflows, while equivalence allows comparison of cash flows occurring at different points in time. Together they transform complex financial scenarios into systematic calculations that support rational decision-making.

### 19.2.1 Cash Flow Diagram Conventions

A cash flow diagram is a horizontal timeline where the x-axis represents time periods (usually years) and vertical arrows represent cash flows. Upward arrows indicate cash inflows (revenues, savings, salvage values), and downward arrows indicate cash outflows (costs, investments, expenses). Time zero is the present, and cash flows are assumed to occur at the end of each period unless stated otherwise. The diagram provides a clear visual summary that prevents errors in setting up economic equations.

**Example 19.2.1:** A distribution utility purchases a \$150,000 pad-mounted transformer that generates \$35,000/year in reduced losses over 6 years, with a \$10,000 salvage value at the end of year 6. Describe the cash flow diagram.
**Solution:**
Time 0: ↓ \$150,000 (initial cost, downward arrow)
Years 1–6: ↑ \$35,000/year (annual savings, upward arrows)
Year 6: ↑ \$10,000 (salvage value, additional upward arrow)
The net cash flow in year 6 is \$35,000 + \$10,000 = **\$45,000** upward. Total undiscounted inflows $= 6 \times 35{,}000 + 10{,}000 =$ **\$220,000**, which exceeds the \$150,000 cost, but a time-value-of-money analysis is needed to determine true profitability.

### 19.2.2 Economic Equivalence

Two cash flow patterns are economically equivalent if they have the same value when converted to a common point in time at a given interest rate. Equivalence means the decision-maker is indifferent between the two patterns. This concept allows engineers to compare lump-sum payments against installment plans, evaluate lease-versus-buy decisions, and convert irregular cash flows into uniform series for easier comparison. The equivalence relationship depends on the interest rate — cash flows that are equivalent at one rate are generally not equivalent at another.

**Example 19.2.2:** A municipality can pay $800,000 now or $250,000 per year for 4 years for a distribution feeder upgrade. At i = 8%, determine (a) the present value of the installment plan and (b) which option is more economical.
**Solution:**
(a) $P = A \times (P/A, 8\%, 4) = 250{,}000 \times [(1.08^4 - 1) / (0.08 \times 1.08^4)]$
$= 250{,}000 \times [(1.3605 - 1) / (0.08 \times 1.3605)]$
$= 250{,}000 \times [0.3605 / 0.10884]$
$= 250{,}000 \times 3.3121 =$ **$828,028**
(b) The lump-sum payment of $800,000 is less than the present value of the installment plan ($828,028), so the **lump-sum payment is more economical** by $28,028.

## 19.3 Single Payment and Uniform Series Factors

Engineering economic analysis uses six standard compound interest factors to convert between present, future, and annual values. These factors — F/P, P/F, P/A, A/P, F/A, and A/F — form the computational backbone of all equivalence calculations. Gradient series factors extend the analysis to cash flows that increase by a constant amount (arithmetic gradient) or a constant percentage (geometric gradient) each period. Mastering these factors enables rapid evaluation of virtually any cash flow pattern encountered in engineering practice.

### 19.3.1 Single Payment Factors (F/P and P/F)

The single payment compound amount factor $(F/P, i, n) = (1 + i)^n$ converts a present value to a future value. Its reciprocal, the single payment present worth factor $(P/F, i, n) = 1/(1 + i)^n$, converts a future value to a present value. These are the simplest and most frequently used factors in engineering economics. The F/P factor answers "what will this investment grow to?" while the P/F factor answers "what is a future payment worth today?"

**Example 19.3.1:** A solar farm developer needs $5,000,000 in 7 years for a panel replacement cycle. If funds earn 6% annually, determine (a) the amount that must be invested today, and (b) the amount of interest earned over the 7 years.
**Solution:**
(a) $P = F \times (P/F, 6\%, 7) = 5{,}000{,}000 \times 1/(1.06)^7 = 5{,}000{,}000 \times 1/1.5036 = 5{,}000{,}000 \times 0.6651 =$ **$3,325,500**
(b) Interest earned: $I = F - P = 5{,}000{,}000 - 3{,}325{,}500 =$ **$1,674,500**

### 19.3.2 Uniform Series Factors (P/A and A/P)

The uniform series present worth factor $(P/A, i, n) = [(1 + i)^n - 1] / [i(1 + i)^n]$ converts a series of equal annual payments into a single present value. Its reciprocal, the capital recovery factor $(A/P, i, n) = [i(1 + i)^n] / [(1 + i)^n - 1]$, converts a present value into a series of equal annual payments. The

A/P factor is widely used for loan amortization and for expressing the annual cost of owning an asset (capital recovery cost).

**Example 19.3.2:** A manufacturing plant installs a $400,000 variable frequency drive system financed over 10 years at 7% annual interest. Determine (a) the annual payment and (b) the total amount paid over the loan term.
**Solution:**
(a) $A = P \times (A/P, 7\%, 10) = 400{,}000 \times [0.07 \times 1.07^{10}] / [1.07^{10} - 1]$
$= 400{,}000 \times [0.07 \times 1.9672] / [1.9672 - 1]$
$= 400{,}000 \times 0.13771 / 0.9672$
$= 400{,}000 \times 0.14238 =$ **$56,952/year**
(b) Total paid: $56{,}952 \times 10 =$ **$569,520**, of which $169,520 is interest.

### 19.3.3 Sinking Fund and Compound Amount (A/F and F/A)

The sinking fund factor $(A/F, i, n) = i / [(1 + i)^n - 1]$ determines the uniform annual deposit needed to accumulate a target future amount. Its reciprocal, the uniform series compound amount factor $(F/A, i, n) = [(1 + i)^n - 1] / i$, finds the future value of a series of equal annual deposits. Sinking funds are commonly used by utilities to accumulate reserves for major equipment replacements, plant decommissioning, or regulatory compliance upgrades.

**Example 19.3.3:** A utility must accumulate $3,000,000 in 12 years for a generator overhaul. At 5% interest, determine (a) the required uniform annual deposit and (b) the total interest earned by the sinking fund.
**Solution:**
(a) $A = F \times (A/F, 5\%, 12) = 3{,}000{,}000 \times 0.05 / [(1.05)^{12} - 1]$
$= 3{,}000{,}000 \times 0.05 / [1.7959 - 1]$
$= 3{,}000{,}000 \times 0.05 / 0.7959$
$= 3{,}000{,}000 \times 0.06283 =$ **$188,476/year**
(b) Total deposits: $188{,}476 \times 12 = \$2{,}261{,}712$. Interest earned: $3{,}000{,}000 - 2{,}261{,}712 =$ **$738,288**

### 19.3.4 Arithmetic Gradient Series

An arithmetic gradient series has cash flows that increase by a constant amount G each period: the cash flow in period t is $A_1 + (t - 1)G$, where $A_1$ is the base amount in period 1. The present worth of the gradient component alone is $P_G = G \times [(1 + i)^n - in - 1] / [i^2(1 + i)^n]$, often written as $G \times (P/G, i, n)$. The total present worth is $P = A_1 \times (P/A, i, n) + G \times (P/G, i, n)$. Arithmetic gradients model maintenance costs that increase by a fixed dollar amount each year due to aging equipment.

**Example 19.3.4:** Maintenance costs for a backup generator start at $5,000 in year 1 and increase by $1,200/year thereafter. At i = 8% over 10 years, determine (a) the maintenance cost in year 10 and (b) the present worth of all maintenance costs.
**Solution:**
(a) Cost in year 10: $A_{10} = 5{,}000 + (10 - 1) \times 1{,}200 = 5{,}000 + 10{,}800 =$ **$15,800**

(b) $P = A_1 \times (P/A, 8\%, 10) + G \times (P/G, 8\%, 10)$
$(P/A, 8\%, 10) = [(1.08)^{10} - 1] / [0.08 \times (1.08)^{10}] = [2.1589 - 1] / [0.08 \times 2.1589] = 1.1589 / 0.17271 = 6.7101$
$(P/G, 8\%, 10) = [(1.08)^{10} - 0.08 \times 10 - 1] / [0.08^2 \times (1.08)^{10}] = [2.1589 - 0.8 - 1] / [0.0064 \times 2.1589] = 0.3589 / 0.013817 = 25.977$
$P = 5{,}000 \times 6.7101 + 1{,}200 \times 25.977 = 33{,}551 + 31{,}172 =$ **\$64,723**

### 19.3.5 Geometric Gradient Series

A geometric gradient series has cash flows that increase by a constant percentage g each period: the cash flow in period t is $A_1(1 + g)^{t-1}$. When $g \neq i$, the present worth is $P = A_1 \times [1 - (1 + g)^n(1 + i)^{-n}] / (i - g)$. When $g = i$, the present worth simplifies to $P = A_1 \times n / (1 + i)$. Geometric gradients model costs that escalate at a fixed percentage rate, such as energy prices, labor costs, and material costs subject to inflation.

**Example 19.3.5:** Electricity costs for a data center are \$200,000 in year 1 and escalate at 4%/year. At i = 9%, determine the present worth of 15 years of electricity costs.
**Solution:**
$P = A_1 \times [1 - (1 + g)^n(1 + i)^{-n}] / (i - g)$
$= 200{,}000 \times [1 - (1.04)^{15}(1.09)^{-15}] / (0.09 - 0.04)$
$(1.04)^{15} = 1.8009$, $(1.09)^{15} = 3.6425$
$= 200{,}000 \times [1 - 1.8009/3.6425] / 0.05$
$= 200{,}000 \times [1 - 0.4944] / 0.05$
$= 200{,}000 \times 0.5056 / 0.05$
$= 200{,}000 \times 10.112 =$ **\$2,022,400**

## 19.4 Present Worth Analysis

Present worth (PW) analysis converts all cash flows to their equivalent values at time zero using a minimum attractive rate of return (MARR). A project is economically justified if its net present value (NPV) is positive — meaning the present value of benefits exceeds the present value of costs. For mutually exclusive alternatives, the option with the highest NPV is preferred. PW analysis is the most widely used evaluation method in engineering economics because it provides a direct measure of the economic value added by a project.

### 19.4.1 Net Present Value (NPV)

The net present value of a project is the sum of all cash flows discounted to time zero: $NPV = \Sigma CF_t / (1 + i)^t$, where $CF_t$ is the net cash flow in period t and i is the MARR. A positive NPV indicates the project earns more than the MARR and should be accepted. A negative NPV means the project does not meet the minimum return requirement. For a single project, the accept/reject criterion is simply $NPV \geq 0$.

**Example 19.4.1:** A utility evaluates a \$2,500,000 battery energy storage system that saves \$450,000/year in peak demand charges over 10 years with a \$200,000 salvage value. At MARR

= 10%, determine (a) the NPV and (b) whether the project is justified.
**Solution:**
(a) NPV = −2,500,000 + 450,000 × (P/A, 10%, 10) + 200,000 × (P/F, 10%, 10)
(P/A, 10%, 10) = $[(1.10)^{10} - 1] / [0.10 \times (1.10)^{10}] = [2.5937 - 1] / [0.10 \times 2.5937] = 1.5937 / 0.25937$ = 6.1446
(P/F, 10%, 10) = $1/(1.10)^{10} = 1/2.5937 = 0.3855$
NPV = −2,500,000 + 450,000 × 6.1446 + 200,000 × 0.3855
= −2,500,000 + 2,765,070 + 77,100 = **$342,170**
(b) Since NPV = $342,170 > 0, the project **is justified** — it earns more than the 10% MARR.

### 19.4.2 Comparing Alternatives with Equal Lives

When two or more mutually exclusive alternatives have the same service life, PW analysis compares them directly by computing the NPV of each alternative at the same MARR. The alternative with the highest (least negative) NPV is selected. For cost-only alternatives where revenues are identical, the analysis minimizes the present worth of costs. All alternatives must be compared over the same time period and at the same MARR.

**Example 19.4.2:** Compare two cable insulation options for a 5-mile distribution line over 20 years at i = 8%. Option A: $1,200,000 first cost, $40,000/year maintenance. Option B: $900,000 first cost, $70,000/year maintenance. Determine which option is more economical.
**Solution:**
$PW_A$ = −1,200,000 − 40,000 × (P/A, 8%, 20)
(P/A, 8%, 20) = $[(1.08)^{20} - 1] / [0.08 \times (1.08)^{20}] = [4.6610 - 1] / [0.08 \times 4.6610] = 3.6610 / 0.37288$ = 9.8181
$PW_A$ = −1,200,000 − 40,000 × 9.8181 = −1,200,000 − 392,724 = **−$1,592,724**
$PW_B$ = −900,000 − 70,000 × 9.8181 = −900,000 − 687,267 = **−$1,587,267**
**Option B is more economical** by $5,457 in present worth, despite higher annual maintenance, because the lower first cost more than compensates.

### 19.4.3 Comparing Alternatives with Unequal Lives

When mutually exclusive alternatives have different service lives, direct PW comparison is invalid because the alternatives provide service for different durations. The least common multiple (LCM) method repeats each alternative over the LCM of their lives so that all alternatives span the same total period. Alternatively, the annual worth method (§19.5) avoids this complication entirely. The LCM approach assumes identical replacement at the same cost, which may not hold if technology or prices change.

**Example 19.4.3:** Compare a 15-year overhead line rebuild ($600,000, $25,000/year maintenance) with a 30-year underground cable ($1,800,000, $8,000/year maintenance) at i = 6%. Use the LCM method to select the more economical option.
**Solution:**
LCM of 15 and 30 is 30 years. The overhead line is repeated twice (years 0–15 and 15–30).

$PW_{overhead} = -600{,}000 - 600{,}000 \times (P/F, 6\%, 15) - 25{,}000 \times (P/A, 6\%, 30)$
$(P/F, 6\%, 15) = 1/(1.06)^{15} = 1/2.3966 = 0.4173$
$(P/A, 6\%, 30) = [(1.06)^{30} - 1] / [0.06 \times (1.06)^{30}] = [5.7435 - 1] / [0.06 \times 5.7435] = 4.7435 / 0.34461 = 13.765$
$PW_{overhead} = -600{,}000 - 600{,}000 \times 0.4173 - 25{,}000 \times 13.765$
$= -600{,}000 - 250{,}380 - 344{,}125 =$ **−$1,194,505**
$PW_{underground} = -1{,}800{,}000 - 8{,}000 \times (P/A, 6\%, 30)$
$= -1{,}800{,}000 - 8{,}000 \times 13.765 = -1{,}800{,}000 - 110{,}120 =$ **−$1,910,120**
The **overhead line rebuild is more economical** by $715,615 in present worth over the 30-year study period.

## 19.5 Annual Worth Analysis

Annual worth (AW) analysis converts all cash flows into an equivalent uniform annual amount. The AW method is particularly convenient for comparing alternatives with different service lives because the AW of one life cycle is the same as the AW for any number of repeating life cycles, provided the costs and revenues remain unchanged. A project is economically justified if its $AW \geq 0$. For mutually exclusive alternatives, the one with the highest AW (or least negative AW for cost-only comparisons) is selected.

### 19.5.1 Capital Recovery and Annual Worth

The annual cost of owning an asset has two components: the capital recovery cost (the equivalent annual cost of the initial investment minus salvage value) and the annual operating cost. Capital recovery cost is $CR = (P - S) \times (A/P, i, n) + S \times i$, where P is the first cost, S is the salvage value, n is the useful life, and i is the interest rate. The total AW is $AW = -CR -$ Annual O&M + Annual Revenue.

**Example 19.5.1:** A 500 kW diesel generator costs $350,000, has a 12-year life, annual O&M of $28,000, and a salvage value of $40,000. At i = 9%, determine (a) the capital recovery cost and (b) the total annual worth.
**Solution:**
(a) $CR = (P - S) \times (A/P, 9\%, 12) + S \times i$
$(A/P, 9\%, 12) = [0.09 \times (1.09)^{12}] / [(1.09)^{12} - 1] = [0.09 \times 2.8127] / [2.8127 - 1] = 0.25314 / 1.8127 = 0.13965$
$CR = (350{,}000 - 40{,}000) \times 0.13965 + 40{,}000 \times 0.09$
$= 310{,}000 \times 0.13965 + 3{,}600 = 43{,}292 + 3{,}600 =$ **$46,892/year**
(b) AW = −CR − Annual O&M = −46,892 − 28,000 = **−$74,892/year**

### 19.5.2 Comparing Alternatives by Annual Worth

The AW method is the simplest approach for comparing alternatives with different service lives because no LCM calculation is needed — each alternative is evaluated over its own life. The alternative with the numerically highest (least negative) AW is preferred. This method implicitly assumes that each alternative can be replaced at the same cost at the end of its life.

**Example 19.5.2:** Compare two motor choices for a pump station at i = 7%. Standard motor: \$8,000, 10-year life, \$3,200/year energy cost. Premium motor: \$12,000, 10-year life, \$2,400/year energy cost. Neither has salvage value. Determine which is more economical.
**Solution:**
$(A/P, 7\%, 10) = [0.07 \times (1.07)^{10}] / [(1.07)^{10} - 1] = [0.07 \times 1.9672] / [1.9672 - 1] = 0.13770 / 0.9672 = 0.14238$
$AW_{standard} = -8{,}000 \times 0.14238 - 3{,}200 = -1{,}139 - 3{,}200 = \mathbf{-\$4{,}339/year}$
$AW_{premium} = -12{,}000 \times 0.14238 - 2{,}400 = -1{,}709 - 2{,}400 = \mathbf{-\$4{,}109/year}$
The **premium efficiency motor is more economical** by \$230/year. The energy savings of \$800/year more than offsets the higher capital recovery cost of \$570/year.

## 19.6 Rate of Return Analysis

Rate of return (ROR) analysis finds the interest rate at which the net present value of a project's cash flows equals zero. This rate, called the internal rate of return (IRR), represents the effective yield of the investment. A project is accepted if its IRR exceeds the MARR. For comparing mutually exclusive alternatives, incremental rate of return analysis must be used — comparing alternatives by their individual IRRs can lead to incorrect decisions. The IRR method is popular because it expresses economic merit as a percentage, which is intuitive and easy to communicate to decision-makers.

### 19.6.1 Internal Rate of Return (IRR) for Single Projects

The IRR is the interest rate i* that satisfies $\Sigma\ CF_t / (1 + i^*)^t = 0$, where $CF_t$ is the net cash flow in period t. For simple investments (one sign change in the cash flow sequence), a unique positive IRR exists and can be found by trial and error, interpolation, or numerical methods. If IRR ≥ MARR, the project is economically justified.

**Example 19.6.1:** A \$180,000 power factor correction system saves \$38,000/year in utility demand charges for 8 years with no salvage value. Determine (a) the IRR and (b) whether the project is acceptable at MARR = 12%.
**Solution:**
(a) Set NPV = 0: $0 = -180{,}000 + 38{,}000 \times (P/A, i, 8)\ (P/A, i, 8) = 180{,}000/38{,}000 = 4.7368$
Try i = 12%: $(P/A, 12\%, 8) = 4.9676$ (NPV > 0)
Try i = 15%: $(P/A, 15\%, 8) = 4.4873$ (NPV < 0)
Interpolate: $i^* = 12\% + 3\% \times (4.9676 - 4.7368)/(4.9676 - 4.4873) = 12\% + 3\% \times 0.2308/0.4803 = 12\% + 1.44\% = \mathbf{13.44\%}$
(b) Since IRR = 13.44% > MARR = 12%, the project **is acceptable**.

### 19.6.2 Incremental Rate of Return for Mutually Exclusive Alternatives

When comparing mutually exclusive alternatives, the alternative with the higher IRR is not necessarily the better choice. Instead, incremental analysis evaluates the rate of return on the additional investment required by the more expensive alternative. The incremental cash flow is $\Delta CF = CF_B - CF_A$, where B has the higher first cost. If the incremental IRR (ΔIRR) exceeds the MARR, the extra investment is justified and the more expensive alternative is selected.

**Example 19.6.2:** Two relay protection upgrade options over 8 years. Option A: \$60,000 cost, \$15,000/year savings. Option B: \$100,000 cost, \$23,000/year savings. Neither has salvage value. Use incremental IRR analysis at MARR = 10% to select the better option.
**Solution:**
Incremental cash flow (B − A): $\Delta$Cost = −\$40,000, $\Delta$Savings = +\$8,000/year.
Set NPV of increment = 0: $0 = -40{,}000 + 8{,}000 \times (P/A, \Delta i, 8)$ $(P/A, \Delta i, 8) = 40{,}000/8{,}000 = 5.0000$
Try i = 10%: $(P/A, 10\%, 8) = 5.3349$ (NPV > 0)
Try i = 12%: $(P/A, 12\%, 8) = 4.9676$ (NPV < 0)
Interpolate: $\Delta i^* = 10\% + 2\% \times (5.3349 - 5.0000)/(5.3349 - 4.9676) = 10\% + 2\% \times 0.3349/0.3673$ $= 10\% + 1.82\% = \mathbf{11.82\%}$
Since $\Delta$IRR = 11.82% > MARR = 10%, the extra \$40,000 investment is justified. **Select Option B.**

### 19.6.3 Multiple Rate of Return and Modified IRR

When a cash flow sequence has more than one sign change, multiple IRR values may exist (Descartes' rule of signs). This occurs in projects with large intermediate expenditures or end-of-life decommissioning costs. The modified internal rate of return (MIRR) resolves this ambiguity by separating the financing rate (for negative cash flows) from the reinvestment rate (for positive cash flows), producing a single unique rate.

**Example 19.6.3:** A cogeneration project has an initial cost of \$5,000,000, net revenues of \$1,200,000/year for years 1–8, then a \$3,000,000 decommissioning cost in year 9. Determine (a) the number of sign changes, and (b) the MIRR using a reinvestment rate of 10% and a finance rate of 8%.
**Solution:**
(a) Cash flow signs: − + + + + + + + + −. There are **2 sign changes** (−→+ and +→−), so up to 2 IRR values may exist.
(b) Future value of positive cash flows at reinvestment rate (10%):
$FV_{positive} = 1{,}200{,}000 \times (F/A, 10\%, 8) \times (F/P, 10\%, 1)$
$(F/A, 10\%, 8) = [(1.10)^8 - 1] / 0.10 = [2.1436 - 1] / 0.10 = 11.436$
$FV_{positive} = 1{,}200{,}000 \times 11.436 \times 1.10 = 1{,}200{,}000 \times 12.580 = \$15{,}095{,}520$
Present value of negative cash flows at finance rate (8%):
$PV_{negative} = 5{,}000{,}000 + 3{,}000{,}000 \times (P/F, 8\%, 9) = 5{,}000{,}000 + 3{,}000{,}000 / (1.08)^9$
$= 5{,}000{,}000 + 3{,}000{,}000 / 1.9990 = 5{,}000{,}000 + 1{,}500{,}750 = \$6{,}500{,}750$
$MIRR = (FV_{positive} / PV_{negative})^{1/9} - 1 = (15{,}095{,}520 / 6{,}500{,}750)^{1/9} - 1$
$= (2.3222)^{1/9} - 1 = 1.0981 - 1 = \mathbf{9.81\%}$

## 19.7 Benefit-Cost Analysis

Benefit-cost (B/C) analysis is the standard evaluation method for public-sector projects such as utility infrastructure, grid modernization, and government-funded energy programs. The B/C ratio compares the present worth (or annual worth) of benefits to the present worth (or annual worth) of costs. A project is justified when B/C $\geq$ 1.0. For comparing multiple alternatives, incremental B/C analysis ensures that each additional increment of investment produces benefits that exceed costs. Unlike private-sector analysis, public projects consider societal benefits such as reduced outage costs, environmental improvements, and public safety enhancements.

### 19.7.1 Conventional Benefit-Cost Ratio

The conventional B/C ratio is defined as B/C = (Benefits − Disbenefits) / Costs, where all terms are expressed in present worth or annual worth. Benefits are advantages to the public, disbenefits are disadvantages, and costs are expenditures by the government or utility. A modified B/C ratio moves operating and maintenance costs to the numerator: $B/C_{modified}$ = (Benefits − Disbenefits − O&M) / Initial Investment. Both formulations give the same accept/reject decision.

**Example 19.7.1:** A city invests $4,000,000 in a smart grid system that reduces outage costs by $700,000/year and saves $150,000/year in operations over 15 years. At i = 5%, determine the conventional B/C ratio.
**Solution:**
Annual benefits: B = $700,000 + $150,000 = $850,000/year
AW of cost: $C = 4{,}000{,}000 \times (A/P, 5\%, 15) = 4{,}000{,}000 \times [0.05 \times (1.05)^{15}] / [(1.05)^{15} - 1]$
$= 4{,}000{,}000 \times [0.05 \times 2.0789] / [2.0789 - 1]$
$= 4{,}000{,}000 \times 0.10395 / 1.0789$
$= 4{,}000{,}000 \times 0.09634 = \$385{,}360$/year
B/C = 850,000 / 385,360 = **2.21**
Since B/C = 2.21 > 1.0, the project **is justified**.

### 19.7.2 Incremental B/C Analysis for Multiple Alternatives

When comparing mutually exclusive public-sector alternatives, incremental B/C analysis is required. Alternatives are ranked by increasing cost, and each increment of investment is evaluated. Starting with the lowest-cost alternative as the base, the incremental B/C ratio $\Delta B/\Delta C$ is computed for each successive alternative. If $\Delta B/\Delta C \geq 1.0$, the higher-cost alternative is selected as the new base; otherwise, it is eliminated.

**Example 19.7.2:** Three substation automation levels are proposed over 20 years at i = 6%. Level 1: $1,000,000 cost, $180,000/year benefits. Level 2: $2,500,000 cost, $400,000/year benefits. Level 3: $4,000,000 cost, $600,000/year benefits. Use incremental B/C analysis to select the best alternative.
**Solution:**
$(A/P, 6\%, 20) = [0.06 \times (1.06)^{20}] / [(1.06)^{20} - 1] = [0.06 \times 3.2071] / [3.2071 - 1] = 0.19243 / 2.2071 = 0.08718$
AW of costs: $C_1 = 1{,}000{,}000 \times 0.08718 = \$87{,}180$; $C_2 = 2{,}500{,}000 \times 0.08718 = \$217{,}950$; $C_3 = 4{,}000{,}000 \times 0.08718 = \$348{,}720$
Level 1 vs. do-nothing: B/C = 180,000/87,180 = 2.06 ≥ 1.0 → **Level 1 justified** (new base).
Level 2 vs. Level 1: $\Delta B/\Delta C$ = (400,000 − 180,000)/(217,950 − 87,180) = 220,000/130,770 = 1.68 ≥ 1.0 → **Level 2 justified** (new base).
Level 3 vs. Level 2: $\Delta B/\Delta C$ = (600,000 − 400,000)/(348,720 − 217,950) = 200,000/130,770 = 1.53 ≥ 1.0 → **Level 3 justified**.
**Select Level 3** — each incremental investment is justified.

## 19.8 Depreciation

Depreciation is the systematic allocation of an asset's cost over its useful life for accounting and tax purposes. It recognizes that physical assets lose value over time due to wear, obsolescence, and deterioration. Depreciation is not a cash flow itself, but it affects cash flow through its impact on taxable income — higher depreciation charges reduce taxes and increase after-tax cash flow. Engineers must understand depreciation methods to perform accurate after-tax economic analyses and to comply with tax regulations governing capital asset recovery.

### 19.8.1 Straight-Line Depreciation

Straight-line (SL) depreciation allocates an equal amount of the asset's depreciable cost to each year of its useful life. The annual depreciation charge is $D = (P - S)/n$, where P is the cost basis, S is the salvage value, and n is the useful life. The book value at the end of year t is $BV_t = P - t \times D$. Straight-line depreciation is the simplest method and is commonly used for financial reporting and for assets that lose value uniformly over time.

**Example 19.8.1:** A 2 MVA pad-mounted transformer costs $85,000 with a salvage value of $5,000 and a useful life of 20 years. Determine (a) the annual depreciation charge and (b) the book value after 8 years.
**Solution:**
(a) $D = (P - S)/n = (85{,}000 - 5{,}000)/20 = 80{,}000/20 =$ **$4,000/year**
(b) $BV_8 = P - 8 \times D = 85{,}000 - 8 \times 4{,}000 = 85{,}000 - 32{,}000 =$ **$53,000**

### 19.8.2 Declining Balance Depreciation

Declining balance (DB) depreciation applies a fixed percentage rate to the current book value each year, producing larger charges in early years and smaller charges later. The depreciation in year t is $D_t = d \times BV_{t-1}$, where d is the depreciation rate (typically 1.5/n for 150% DB or 2/n for 200% double declining balance). The book value is $BV_t = P(1 - d)^t$. Declining balance methods never depreciate below the salvage value; a switch to straight-line may be made when SL produces a larger deduction.

**Example 19.8.2:** A $120,000 CNC wire-bending machine for motor winding production uses 200% declining balance (DDB) depreciation over a 10-year life. Determine the depreciation and book value for years 1 through 4.
**Solution:**
DDB rate: $d = 2/n = 2/10 = 0.20$

| Year | Book Value (start) | Depreciation (20%) | Book Value (end) |
|---|---|---|---|
| 1 | $120,000 | $24,000 | $96,000 |
| 2 | $96,000 | $19,200 | $76,800 |
| 3 | $76,800 | $15,360 | $61,440 |
| 4 | $61,440 | $12,288 | $49,152 |

After 4 years, the book value is **49,152** and total depreciation charged is $120,000 − $49,152 = **$70,848**.

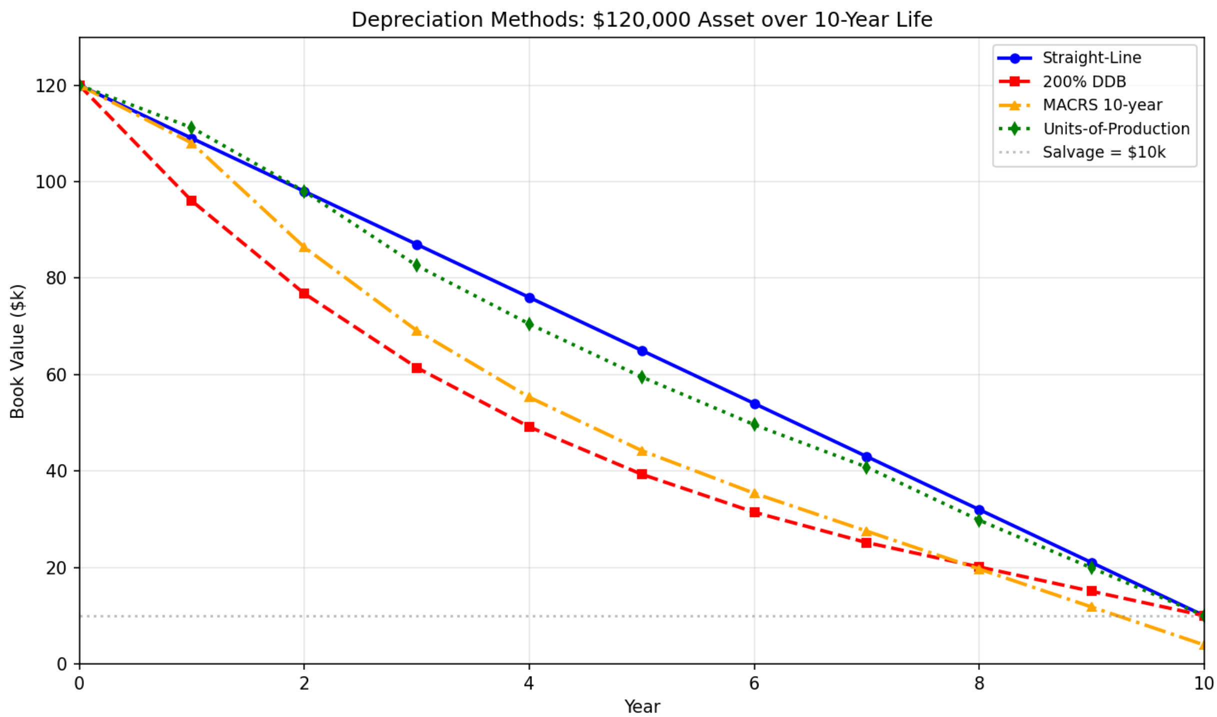



Figure 19.8.2: Depreciation Methods: Book Value Comparison

### 19.8.3 MACRS Depreciation

The Modified Accelerated Cost Recovery System (MACRS) is the depreciation method required by U.S. tax law for most business assets. MACRS uses prescribed recovery periods and percentage tables based on declining balance switching to straight-line. Common recovery periods are 3, 5, 7, 10, 15, and 20 years. MACRS assumes zero salvage value and uses a half-year convention (only half a year's depreciation in the first and last years). The MACRS percentages for common recovery periods are published by the IRS.

**Example 19.8.3:** A utility installs a $1,200,000 natural gas turbine classified as 15-year MACRS property. Using the MACRS percentage table, determine the depreciation charges for years 1 through 5.
**Solution:**
MACRS 15-year percentages: Year 1 = 5.00%, Year 2 = 9.50%, Year 3 = 8.55%, Year 4 = 7.70%, Year 5 = 6.93%

| Year | MACRS % | Depreciation | Cumulative |
|---|---|---|---|
| 1 | 5.00% | $60,000 | $60,000 |

| | | | |
|---|---|---|---|
| 2 | 9.50% | \$114,000 | \$174,000 |
| 3 | 8.55% | \$102,600 | \$276,600 |
| 4 | 7.70% | \$92,400 | \$369,000 |
| 5 | 6.93% | \$83,160 | \$452,160 |

After 5 years, the cumulative depreciation is **\$452,160**, representing 37.68% of the cost basis.

### 19.8.4 Units-of-Production Depreciation

Units-of-production (UOP) depreciation allocates cost based on actual usage rather than time elapsed. The depreciation rate per unit is $d_{unit} = (P - S)$ / Total units, and the depreciation in any period is D = $d_{unit} \times$ Units produced. This method is appropriate for assets whose wear is driven by usage rather than age, such as test equipment, generators measured by operating hours, or production machinery measured by output quantity.

**Example 19.8.4:** A \$60,000 test instrument rated for 50,000 test cycles has a salvage value of \$5,000. It is used for 7,500 cycles in year 1 and 9,200 cycles in year 2. Determine (a) the depreciation rate per cycle and (b) the depreciation for each year.
**Solution:**
(a) $d_{unit} = (P - S)$ / Total units = (60,000 − 5,000) / 50,000 = 55,000 / 50,000 = **\$1.10/cycle**
(b) Year 1: $D_1 = 1.10 \times 7{,}500 =$ **\$8,250**
Year 2: $D_2 = 1.10 \times 9{,}200 =$ **\$10,120**
Book value after year 2: $BV_2 = 60{,}000 - 8{,}250 - 10{,}120 =$ **\$41,630**

## 19.9 Taxes and After-Tax Analysis

Income taxes significantly affect the economic viability of engineering projects because they reduce the net cash flow from revenues and savings. After-tax analysis accounts for the tax impact of operating income, depreciation deductions, and gains or losses on asset disposal. The after-tax cash flow (ATCF) in any year equals the before-tax cash flow (BTCF) minus the income tax: ATCF = BTCF − Tax, where Tax = (Taxable Income) × Tax Rate. Since depreciation is a non-cash deduction that reduces taxable income, the choice of depreciation method directly affects the timing and magnitude of after-tax cash flows.

### 19.9.1 After-Tax Cash Flow Analysis

The after-tax cash flow for each year is computed as ATCF = BTCF − (BTCF − D) × T, where BTCF is the before-tax cash flow, D is the depreciation charge, and T is the marginal tax rate. This simplifies to ATCF = BTCF × (1 − T) + D × T, showing that depreciation provides a tax shield equal to D × T. The after-tax NPV is computed using the after-tax MARR, which is lower than the before-tax MARR.

**Example 19.9.1:** A \$500,000 power monitoring system generates \$120,000/year in energy savings over 7 years with straight-line depreciation and no salvage value. At a 25% tax rate and after-tax

MARR of 8%, determine the after-tax NPV.
**Solution:**
Annual depreciation: D = 500,000/7 = \$71,429/year
Taxable income: TI = 120,000 − 71,429 = \$48,571
Tax: T = 48,571 × 0.25 = \$12,143
ATCF = 120,000 − 12,143 = \$107,857/year
Alternatively: ATCF = 120,000 × (1 − 0.25) + 71,429 × 0.25 = 90,000 + 17,857 = \$107,857 ✓
After-tax NPV = −500,000 + 107,857 × (P/A, 8%, 7)
(P/A, 8%, 7) = $[(1.08)^7 - 1] / [0.08 \times (1.08)^7] = [1.7138 - 1] / [0.08 \times 1.7138] = 0.7138 / 0.13710 = 5.2064$
After-tax NPV = −500,000 + 107,857 × 5.2064 = −500,000 + 561,547 = **\$61,547**
The project is justified on an after-tax basis (NPV > 0).

### 19.9.2 Tax Credits and Incentives

Tax credits directly reduce the tax liability dollar-for-dollar, unlike deductions that only reduce taxable income. The Investment Tax Credit (ITC) for solar energy, for example, allows a percentage of the system cost to be subtracted from taxes owed. Tax credits are treated as a positive cash flow in the year they are applied. When combined with accelerated depreciation (such as MACRS), tax incentives can significantly improve project economics — particularly for renewable energy installations.

**Example 19.9.2:** A commercial building installs a \$300,000 rooftop solar array qualifying for a 30% investment tax credit. Annual energy savings are \$45,000 over 25 years. Using simplified 5-year straight-line depreciation (on 85% of cost after ITC adjustment), a 24% tax rate, and after-tax MARR of 6%, determine the after-tax NPV for the first 5 years of cash flows plus the PW of remaining savings.
**Solution:**
ITC: 300,000 × 0.30 = \$90,000 (received in year 1)
Depreciable basis: 300,000 × 0.85 = \$255,000 (ITC reduces basis by half the credit under some rules; using 85% here)
Annual depreciation (years 1–5): D = 255,000/5 = \$51,000
Years 1–5 ATCF: BTCF = \$45,000; TI = 45,000 − 51,000 = −\$6,000 (tax savings)
Tax = −6,000 × 0.24 = −\$1,440 (tax benefit); ATCF = 45,000 + 1,440 = \$46,440
Year 1 additional: +\$90,000 ITC → Year 1 ATCF = 46,440 + 90,000 = \$136,440
Years 6–25 ATCF: No depreciation; ATCF = 45,000 × (1 − 0.24) = \$34,200/year
NPV = −300,000 + 136,440/(1.06) + 46,440 × (P/A, 6%, 4) × (P/F, 6%, 1) + 34,200 × (P/A, 6%, 20) × (P/F, 6%, 5)
= −300,000 + 128,717 + 46,440 × 3.4651 × 0.9434 + 34,200 × 11.470 × 0.7473
= −300,000 + 128,717 + 151,811 + 293,146 = **\$273,674**
The solar array is highly justified with the tax credit.

### 19.9.3 Gain and Loss on Disposal

When an asset is sold for more than its book value, the difference is a taxable gain; when sold for less, the difference is a deductible loss. The tax on disposal is Tax = (Selling Price − Book Value) × Tax Rate. A gain increases tax liability, and a loss provides a tax benefit. The after-tax salvage value is $S_{AT}$

= Selling Price − Tax on gain (or + Tax benefit from loss). Disposal tax effects must be included in the final year's cash flow for accurate after-tax analysis.

**Example 19.9.3:** A \$200,000 testing system is depreciated using 5-year MACRS and sold after 3 years for \$90,000. Determine (a) the book value at disposal, (b) the gain or loss, and (c) the after-tax salvage value at a 25% tax rate.
**Solution:**
(a) MACRS 5-year percentages: Year 1 = 20%, Year 2 = 32%, Year 3 = 19.2%
Cumulative depreciation: $(0.20 + 0.32 + 0.192) \times 200{,}000 = 0.712 \times 200{,}000 = \$142{,}400$
$BV_3 = 200{,}000 - 142{,}400 = \mathbf{\$57{,}600}$
(b) Selling price (\$90,000) > Book value (\$57,600) → **Gain of \$32,400**
(c) Tax on gain: $32{,}400 \times 0.25 = \$8{,}100$
After-tax salvage: $S_{AT} = 90{,}000 - 8{,}100 = \mathbf{\$81{,}900}$

## 19.10 Replacement Analysis

Replacement analysis determines the optimal time to retire existing equipment (the defender) and replace it with new equipment (the challenger). The key insight is that the defender's current market value — not its original cost — is the relevant investment for keeping it. The economic service life (ESL) is the number of years that minimizes the equivalent uniform annual cost, and it provides the basis for comparing the defender and challenger on an annual-cost basis. Replacement decisions arise when maintenance costs are rising, efficiency is declining, or newer technology offers better performance.

### 19.10.1 Economic Service Life

The economic service life is the number of years n that minimizes the total equivalent uniform annual cost (EUAC), which includes capital recovery and annual operating costs. As an asset ages, its capital recovery cost decreases (the asset has already been largely paid for) but operating costs typically increase. The ESL occurs at the crossover point where the sum is minimized. For the challenger, the ESL determines the AW used in comparison; for the defender, only the remaining economic life is relevant.

**Example 19.10.1:** A 20-year-old distribution transformer has a current market value of \$8,000 and annual maintenance costs that start at \$3,000 in year 1 and increase by \$800/year. At i = 10%, find the economic service life by computing the EUAC for 1 through 5 years of additional service (assume zero salvage in all cases).
**Solution:**
(A/P, 10%, n) values: n=1: 1.1000; n=2: 0.57619; n=3: 0.40211; n=4: 0.31547; n=5: 0.26380

| Years | CR = 8,000 × (A/P) | AW of Maint. | EUAC |
|---|---|---|---|
| 1 | \$8,800 | \$3,000 | \$11,800 |
| 2 | \$4,610 | \$3,381 | \$7,991 |
| 3 | \$3,217 | \$3,749 | \$6,966 |

| | | | |
|---|---|---|---|
| 4 | \$2,524 | \$4,105 | \$6,629 |
| 5 | \$2,110 | \$4,448 | \$6,558 |

AW of maintenance for n years uses the gradient: AW = 3,000 + 800 × (A/G, 10%, n)
(A/G, 10%, n): n=1: 0; n=2: 0.4762; n=3: 0.9366; n=4: 1.3812; n=5: 1.8101
The EUAC continues to decrease through 5 years, so the ESL is ≥ **5 years** (at 5 years, EUAC = **\$6,558/year**). In practice, the ESL would be found by extending the analysis until EUAC begins to increase.

### 19.10.2 Defender-Challenger Comparison

The defender is the existing equipment with its current market value as the "investment" in keeping it, and the challenger is the new replacement. The decision rule is: if the challenger's EUAC at its ESL is less than the defender's EUAC at its ESL, replace now. If the defender has a lower EUAC, keep it — but re-evaluate annually as the defender ages and its costs increase. The sunk cost of the original purchase is irrelevant; only the current market value matters.

**Example 19.10.2:** A utility's existing switchgear (defender) has a market value of \$25,000 and an EUAC of \$18,000/year for its remaining economic life of 3 years. A new solid-state switchgear (challenger) costs \$120,000 with an EUAC of \$15,500/year over a 15-year life. At i = 10%, determine whether to replace now.
**Solution:**
Defender EUAC: **\$18,000/year** (for remaining ESL of 3 years)
Challenger EUAC: **\$15,500/year** (for 15-year ESL)
Since the challenger EUAC (\$15,500) < defender EUAC (\$18,000), the annual savings from replacement are \$2,500/year. **Replace now.** The new switchgear costs less on an annualized basis than keeping the old equipment.

### 19.10.3 Replacement with a Study Period

When management imposes a fixed planning horizon (study period), the analysis must consider what happens at the end of the period. If the defender cannot serve the full study period, a replacement (possibly the challenger) must fill the remaining years. All feasible combinations of defender and challenger service over the study period are evaluated, and the combination with the lowest total EUAC or PW of costs is selected.

**Example 19.10.3:** Management sets a 10-year study period. The defender can serve 4 more years with EUAC = \$22,000/year, and the challenger has a 10-year life with EUAC = \$19,000/year. Compare (a) replacing now versus (b) replacing at year 4. Use i = 8%.
**Solution:**
(a) Replace now: Challenger serves all 10 years.
PW = 19,000 × (P/A, 8%, 10) = 19,000 × 6.7101 = **\$127,492**
(b) Replace at year 4: Defender serves years 1–4, challenger serves years 5–10.
PW = 22,000 × (P/A, 8%, 4) + 19,000 × (P/A, 8%, 6) × (P/F, 8%, 4)
(P/A, 8%, 4) = 3.3121; (P/A, 8%, 6) = 4.6229; (P/F, 8%, 4) = 0.7350

PW = 22,000 × 3.3121 + 19,000 × 4.6229 × 0.7350
= 72,867 + 64,559 = **$137,426**
**Replace now** — it saves $137,426 − $127,492 = $9,934 in present worth over the study period.

## 19.11 Inflation and Price Changes

Inflation is the sustained increase in the general price level, causing purchasing power to decline over time. Engineering economic analysis must account for inflation to avoid overstating the real value of future cash flows. Two approaches exist: the actual-dollar (current-dollar) method uses the market interest rate and cash flows stated in nominal terms, while the constant-dollar (real-dollar) method uses the real interest rate and cash flows stated in today's purchasing power. Both approaches yield the same present worth when applied correctly. The relationship between the market rate $i_f$, real rate $i_r$, and inflation rate f is $(1 + i_f) = (1 + i_r)(1 + f)$.

### 19.11.1 Inflation-Adjusted Analysis

When inflation is present, the market (nominal) interest rate includes an inflation premium. The real interest rate can be found from $i_r = (i_f - f) / (1 + f)$. In the actual-dollar approach, future cash flows are escalated at the inflation rate and discounted at the market rate. In the constant-dollar approach, cash flows remain in today's dollars and are discounted at the real rate. Both methods must produce the same present worth.

**Example 19.11.1:** A utility budgets $50,000/year in today's dollars for cable maintenance. With 3% annual inflation and a market MARR of 10%, compute the present worth of 8 years of maintenance using (a) the constant-dollar approach and (b) the actual-dollar approach.
**Solution:**
(a) Constant-dollar approach: Real rate $i_r = (0.10 - 0.03)/(1.03) = 0.07/1.03 = 6.796\%$
PW = 50,000 × (P/A, 6.796%, 8)
$(P/A, 6.796\%, 8) = [(1.06796)^8 - 1] / [0.06796 \times (1.06796)^8] = [1.6922 - 1] / [0.06796 \times 1.6922] = 0.6922 / 0.11501 = 6.019$
PW = 50,000 × 6.019 = **$300,950**
(b) Actual-dollar approach: Each year's cost is escalated at 3%.
$PW = \Sigma\, [50{,}000 \times (1.03)^t / (1.10)^t]$ for t = 1 to 8
$= 50{,}000 \times \Sigma\, [(1.03/1.10)^t] = 50{,}000 \times \Sigma\, [(0.93636)^t]$
This is a geometric series with ratio 0.93636:
$PW = 50{,}000 \times 0.93636 \times [1 - (0.93636)^8] / [1 - 0.93636]$
$= 50{,}000 \times 0.93636 \times [1 - 0.5910] / 0.06364$
= 50,000 × 0.93636 × 6.428 = **$300,900**
The two approaches agree within rounding (≈ $300,950), confirming consistency.

### 19.11.2 Differential Escalation

Differential escalation occurs when the price of a specific commodity increases faster or slower than the general inflation rate. The differential escalation rate $e_d$ is the rate above inflation: the actual escalation rate is approximately $f + e_d$. When a cost component escalates at a rate different from general inflation, it must be treated as a geometric gradient series using its specific escalation rate rather than the general inflation rate.

**Example 19.11.2:** Copper conductor costs are escalating at 5%/year while general inflation is 3%/year. A project requires $200,000 of copper in year 1. At a market rate of 11%, find the present worth of copper purchases over 6 years.
**Solution:**
This is a geometric gradient with g = 5% (copper escalation) and i = 11% (market rate).
$P = A_1 \times [1 - (1+g)^n(1+i)^{-n}] / (i - g)$
$= 200{,}000 \times [1 - (1.05)^6(1.11)^{-6}] / (0.11 - 0.05)$
$(1.05)^6 = 1.3401; (1.11)^6 = 1.8704$
$= 200{,}000 \times [1 - 1.3401/1.8704] / 0.06$
$= 200{,}000 \times [1 - 0.7165] / 0.06$
$= 200{,}000 \times 0.2835 / 0.06$
$= 200{,}000 \times 4.725 =$ **$945,000**
The differential escalation rate above inflation is $e_d = 5\% - 3\% = 2\%$/year, meaning copper is becoming more expensive in real terms.

## 19.12 Capital Budgeting and Project Selection

Capital budgeting is the process of selecting which engineering projects to fund when the total investment capital is limited. Unlike mutually exclusive alternatives where exactly one must be chosen, capital budgeting involves selecting a portfolio of independent projects that maximizes the total economic value within a budget constraint. The minimum attractive rate of return (MARR) serves as the hurdle rate that separates acceptable from unacceptable investments, and it reflects the opportunity cost of capital, the cost of borrowing, and a risk premium appropriate to the project type.

### 19.12.1 Independent Project Selection Under Budget Constraints

When capital is limited, the optimal portfolio is the combination of independent projects that maximizes total NPV without exceeding the budget. The approach is to compute the NPV of each project, then evaluate all feasible combinations (bundles) that fit within the budget. For small numbers of projects, enumeration is practical; for large portfolios, the projects can be ranked by the profitability index (PI = NPV/Investment) and selected in order until the budget is exhausted.

**Example 19.12.1:** A utility has $3,000,000 in capital and five independent project proposals. Select the portfolio that maximizes total NPV within the budget.

| Project | Investment | NPV |
|---|---|---|
| A — Substation upgrade | $800,000 | $120,000 |
| B — Feeder reconductoring | $1,200,000 | $210,000 |
| C — SCADA upgrade | $600,000 | $95,000 |
| D — Capacitor banks | $400,000 | $55,000 |

| Project | Investment | NPV |
|---|---|---|
| E — Battery storage | $1,500,000 | $190,000 |

**Solution:** Profitability index (PI = NPV/Investment): A = 0.150, B = 0.175, C = 0.158, D = 0.138, E = 0.127

Rank by PI: B (0.175), C (0.158), A (0.150), D (0.138), E (0.127)

Select in order: B ($1,200,000) + C ($600,000) + A ($800,000) + D ($400,000) = $3,000,000 ✓ Total NPV = 210,000 + 95,000 + 120,000 + 55,000 = **$480,000**

Verify no better combination: B + E = $2,700,000 (NPV $400,000) + C or D fits → B + E + C = $3,300,000 (exceeds budget) → B + E + D = $3,100,000 (exceeds budget). So B + C + A + D at **$480,000 NPV** is the optimal portfolio.

### 19.12.2 Minimum Attractive Rate of Return (MARR)

The MARR is the minimum return that a project must earn to be considered acceptable. It is typically set above the cost of capital to account for project risk, opportunity cost, and management expectations. The MARR may be set as the weighted average cost of capital (WACC) plus a risk premium, or based on the rate earned by the company's best rejected project. A project is accepted if its IRR exceeds the MARR or its NPV at the MARR is positive.

**Example 19.12.2:** A company can borrow at 7% and earns 12% on its best current investments. With a risk premium of 3% for a new power electronics product line, determine (a) the appropriate MARR and (b) whether a $500,000 project returning $95,000/year for 8 years is acceptable.
**Solution:**
(a) The opportunity cost of capital is 12% (best current return), plus a 3% risk premium → MARR = 12% + 3% = **15%**
(b) NPV = −500,000 + 95,000 × (P/A, 15%, 8)
(P/A, 15%, 8) = $[(1.15)^8 - 1] / [0.15 \times (1.15)^8] = [3.0590 - 1] / [0.15 \times 3.0590] = 2.0590 / 0.45885 = 4.4873$
NPV = −500,000 + 95,000 × 4.4873 = −500,000 + 426,294 = **−$73,706**
Since NPV < 0, the project **does not meet the 15% MARR** and should be rejected.

## 19.13 Breakeven and Sensitivity Analysis

Breakeven and sensitivity analysis address the inherent uncertainty in engineering economic estimates. Breakeven analysis finds the critical value of a parameter (such as production volume, operating hours, or energy price) at which two alternatives are equally attractive or a single project's NPV equals zero. Sensitivity analysis systematically varies one or more input parameters to determine how much the economic outcome changes, revealing which parameters most influence the decision. These techniques help engineers assess risk and make more robust investment decisions.

### 19.13.1 Breakeven Analysis

Breakeven analysis determines the value of a key variable at which two alternatives have equal annual worth, equal NPV, or equal total cost. This critical value defines the threshold above or below which one alternative becomes preferable. Breakeven analysis is commonly used to evaluate energy-efficient equipment upgrades, make-versus-buy decisions, and minimum production volume requirements.

**Example 19.13.1:** A factory considers replacing a 92%-efficient motor ($6,000) with a 96%-efficient motor ($10,000) for a 50 HP load running at 80% average loading. Electricity costs $0.10/kWh and i = 8% over a 10-year life with no salvage value. Find the breakeven operating hours per year.

**Solution:**

Power input (standard): $P_1 = 50 \times 0.746 \times 0.80 / 0.92 = 32.43$ kW

Power input (premium): $P_2 = 50 \times 0.746 \times 0.80 / 0.96 = 31.08$ kW

Power savings: $\Delta P = 32.43 - 31.08 = 1.35$ kW

Annual energy savings for H hours: $\Delta E = 1.35 \times H \times \$0.10 = \$0.135 \times H$

Capital recovery of extra cost: $\Delta CR = (10{,}000 - 6{,}000) \times (A/P, 8\%, 10) = 4{,}000 \times 0.14903 =$ \$596.12/year

Breakeven: $0.135 \times H = 596.12$

$H = 596.12 / 0.135 =$ **4,416 hours/year**

If the motor operates more than 4,416 hours/year, the premium motor is more economical. At 6,000 hours/year, annual savings exceed the cost difference.

### 19.13.2 Single-Parameter Sensitivity Analysis

Single-parameter sensitivity (or "what-if") analysis varies one input at a time while holding all others at their base values. For each parameter, the NPV (or AW) is recalculated over a range of values (typically ±10% to ±30%). The results are often plotted on a spider diagram, where steeper slopes indicate higher sensitivity. Parameters with the greatest impact on the economic outcome deserve the most careful estimation and risk management.

**Example 19.13.2:** A $1,000,000 solar carport project has expected annual savings of $160,000 over 15 years with no salvage value. At MARR = 8%, determine (a) the base-case NPV, and (b) the NPV if annual savings vary by ±20%.

**Solution:**

(a) $(P/A, 8\%, 15) = [(1.08)^{15} - 1] / [0.08 \times (1.08)^{15}] = [3.1722 - 1] / [0.08 \times 3.1722] = 2.1722 / 0.25378 = 8.5595$

Base NPV $= -1{,}000{,}000 + 160{,}000 \times 8.5595 = -1{,}000{,}000 + 1{,}369{,}520 =$ **\$369,520**

(b) At −20% savings ($128,000/year):

NPV $= -1{,}000{,}000 + 128{,}000 \times 8.5595 = -1{,}000{,}000 + 1{,}095{,}616 =$ **\$95,616** (still positive)

At +20% savings ($192,000/year):

NPV $= -1{,}000{,}000 + 192{,}000 \times 8.5595 = -1{,}000{,}000 + 1{,}643{,}424 =$ **\$643,424**

The NPV swings from $95,616 to $643,424 — a range of $547,808 — for a ±20% change in savings. The project remains viable even at the pessimistic estimate.

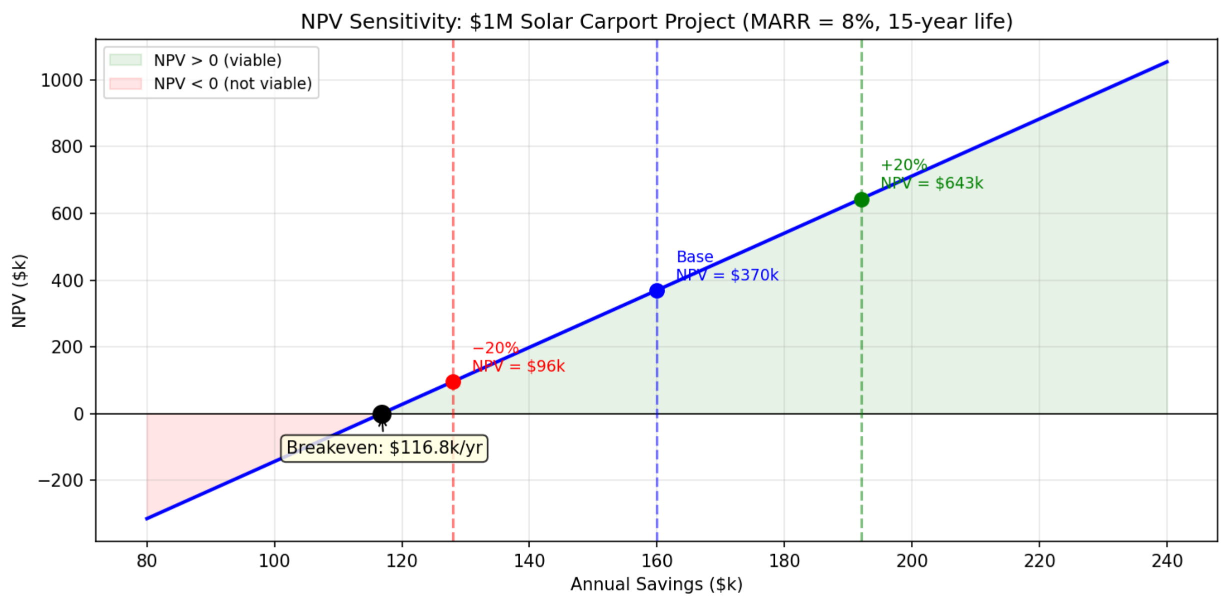



Figure 19.13.2: NPV Sensitivity Analysis

### 19.13.3 Scenario Analysis

Scenario analysis evaluates a project under several coherent sets of assumptions (typically pessimistic, most likely, and optimistic) where multiple parameters vary simultaneously. Unlike single-parameter sensitivity, scenario analysis captures the combined effect of changes in related variables. Each scenario produces a distinct NPV, providing decision-makers with a range of possible outcomes and their associated assumptions.

**Example 19.13.3:** A wind turbine installation ($3,000,000, 20-year life, no salvage) has annual revenues dependent on capacity factor: pessimistic (25%, $280,000/year), most likely (32%, $360,000/year), optimistic (38%, $430,000/year). At $i = 9\%$, compute the NPV for each scenario.
**Solution:**
$(P/A, 9\%, 20) = [(1.09)^{20} - 1] / [0.09 \times (1.09)^{20}] = [5.6044 - 1] / [0.09 \times 5.6044] = 4.6044 / 0.50440 = 9.1285$
Pessimistic: NPV = −3,000,000 + 280,000 × 9.1285 = −3,000,000 + 2,555,980 = **−$444,020** (not viable)
Most likely: NPV = −3,000,000 + 360,000 × 9.1285 = −3,000,000 + 3,286,260 = **$286,260** (viable)
Optimistic: NPV = −3,000,000 + 430,000 × 9.1285 = −3,000,000 + 3,925,255 = **$925,255** (highly viable)
The project is viable in the most-likely and optimistic scenarios but not in the pessimistic case. The decision depends on confidence in the capacity factor estimate and risk tolerance.

## 19.14 Life Cycle Cost Analysis

Life cycle cost (LCC) analysis evaluates the total cost of owning, operating, maintaining, and disposing of an asset over its entire useful life. Unlike simple first-cost comparisons, LCC captures the long-term economic impact of design decisions, material choices, and maintenance strategies. It is the standard method for major infrastructure procurement in utilities, government agencies, and industrial facilities. The levelized cost of energy (LCOE) is a specialized form of LCC used to compare electricity generation technologies on a per-unit-energy basis.

### 19.14.1 Life Cycle Cost Components

The major LCC categories are: (1) acquisition costs (purchase price, installation, commissioning), (2) operating costs (energy, labor, consumables), (3) maintenance costs (preventive and corrective), (4) periodic overhaul or replacement costs, and (5) disposal costs (decommissioning, removal, environmental remediation). All costs are converted to present worth at the discount rate for comparison. The option with the lowest total LCC is preferred.

**Example 19.14.1:** Compare two UPS systems for a hospital data center over 15 years at i = 6%.

| Cost Category | System A | System B |
|---|---|---|
| Purchase | \$80,000 | \$120,000 |
| Installation | \$15,000 | \$10,000 |
| Annual maintenance | \$4,000/year | \$2,000/year |
| Battery replacement | \$12,000 at years 5, 10 | \$8,000 at years 7, 14 |
| Disposal | \$3,000 at year 15 | \$2,000 at year 15 |

**Solution:** (P/A, 6%, 15) = 9.7122; (P/F, 6%, 5) = 0.7473; (P/F, 6%, 7) = 0.6651 (P/F, 6%, 10) = 0.5584; (P/F, 6%, 14) = 0.4423; (P/F, 6%, 15) = 0.4173

$LCC_A = 80{,}000 + 15{,}000 + 4{,}000 \times 9.7122 + 12{,}000 \times (0.7473 + 0.5584) + 3{,}000 \times 0.4173 = 95{,}000 + 38{,}849 + 12{,}000 \times 1.3057 + 1{,}252 = 95{,}000 + 38{,}849 + 15{,}668 + 1{,}252 =$ **\$150,769**

$LCC_B = 120{,}000 + 10{,}000 + 2{,}000 \times 9.7122 + 8{,}000 \times (0.6651 + 0.4423) + 2{,}000 \times 0.4173 = 130{,}000 + 19{,}424 + 8{,}000 \times 1.1074 + 835 = 130{,}000 + 19{,}424 + 8{,}859 + 835 =$ **\$159,118**

**System A has a lower life cycle cost** by \$8,349, despite the higher annual maintenance, because its lower purchase price and installation cost more than compensate.

### 19.14.2 Levelized Cost of Energy (LCOE)

The levelized cost of energy is the constant per-unit cost of electricity that equates the present worth of all lifetime costs with the present worth of all lifetime energy production: LCOE = PW(Costs) / PW(Energy). It allows direct comparison of generation technologies with different capital costs, operating characteristics, and lifetimes. LCOE is expressed in \$/MWh or ¢/kWh and is widely used in utility planning and renewable energy project evaluation.

**Example 19.14.2:** A 10 MW solar PV plant costs \$12,000,000, has annual O&M of \$150,000 escalating at 2%/year, produces 18,000 MWh/year with 0.5%/year degradation, and has a 25-year life. At a discount rate of 7%, compute the LCOE.

**Solution:**
PW of capital: \$12,000,000
PW of O&M (geometric gradient, $g = 2\%$, $i = 7\%$):
$PW_{O\&M} = 150{,}000 \times [1 - (1.02)^{25}(1.07)^{-25}] / (0.07 - 0.02)$
$(1.02)^{25} = 1.6406$; $(1.07)^{25} = 5.4274$
$= 150{,}000 \times [1 - 1.6406/5.4274] / 0.05$
$= 150{,}000 \times [1 - 0.3023] / 0.05$
$= 150{,}000 \times 0.6977 / 0.05 = 150{,}000 \times 13.954 = \$2{,}093{,}100$
PW of total costs: $12{,}000{,}000 + 2{,}093{,}100 = \$14{,}093{,}100$
PW of energy production (geometric gradient with $g = -0.5\%$, $i = 7\%$):
$PW_{energy} = 18{,}000 \times [1 - (0.995)^{25}(1.07)^{-25}] / (0.07 - (-0.005))$
$(0.995)^{25} = 0.8822$
$= 18{,}000 \times [1 - 0.8822/5.4274] / 0.075$
$= 18{,}000 \times [1 - 0.1626] / 0.075$
$= 18{,}000 \times 0.8374 / 0.075 = 18{,}000 \times 11.165 = 200{,}970$ MWh (present-worth equivalent)
LCOE = 14,093,100 / 200,970 = **\$70.12/MWh**

# Appendix A

# Imaginary Numbers and Phasors

Imaginary numbers and phasors are essential mathematical tools that appear throughout electrical engineering. Complex numbers provide a natural way to represent quantities that have both magnitude and direction, such as AC voltages, currents, and impedances. Phasors extend this concept by encoding the amplitude and phase of sinusoidal signals as complex numbers, transforming differential equations into simple algebraic operations. This appendix establishes the mathematical foundations used extensively in circuit analysis, signal processing, power systems, and control theory.

## A.1 Imaginary Numbers

### A.1.1 The Imaginary Unit

The imaginary unit j is defined as the square root of negative one: $j = \sqrt{(-1)}$, or equivalently, $j^2 = -1$. In electrical engineering, the letter j is used instead of i (which is reserved for current) to avoid confusion. The imaginary unit enables the representation of numbers that have no position on the real number line but are indispensable for describing oscillatory and rotational phenomena. Any imaginary number is a real number multiplied by j, such as j3, j7.5, or −j4.2.

**Example A.1.1:** Simplify $\sqrt{(-49)}$ and express the result as an imaginary number.
**Solution:**
$\sqrt{(-49)} = \sqrt{(49 \times (-1))} = \sqrt{49} \times \sqrt{(-1)} = 7 \times j = j7$.
The result is the imaginary number j7, which lies on the imaginary axis at a distance of 7 units from the origin.

### A.1.2 Powers of j

The powers of j follow a repeating cycle of four: $j^1 = j$, $j^2 = -1$, $j^3 = -j$, $j^4 = 1$, and then the cycle repeats with $j^5 = j$, $j^6 = -1$, and so on. This cyclic property means that any power of j can be reduced by dividing the exponent by 4 and using the remainder: $j^n = j^{(n \bmod 4)}$. Negative powers follow the same pattern: $j^{-1} = -j$, $j^{-2} = -1$, $j^{-3} = j$. Understanding this cycle is essential for simplifying expressions that arise in AC circuit analysis and signal processing.

**Example A.1.2:** Simplify $j^7 + j^{10} + j^{-3}$.
**Solution:**
$j^7$: $7 \bmod 4 = 3$, so $j^7 = j^3 = -j$.
$j^{10}$: $10 \bmod 4 = 2$, so $j^{10} = j^2 = -1$.

$j^{-3}$: $-3 \bmod 4 = 1$, so $j^{-3} = j^1 = j$.
Therefore $j^7 + j^{10} + j^{-3} = -j + (-1) + j = -1$.

## A.2 Complex Numbers

### A.2.1 Rectangular Form

A complex number in rectangular form is written as $Z = a + jb$, where a is the real part (Re{Z}) and b is the imaginary part (Im{Z}). Geometrically, complex numbers are represented on the complex plane with the real part plotted on the horizontal axis and the imaginary part on the vertical axis. The real part and imaginary part are extracted as $a = \text{Re}\{Z\}$ and $b = \text{Im}\{Z\}$. In circuit analysis, the real part of an impedance represents resistance and the imaginary part represents reactance, making rectangular form a direct representation of physical circuit properties.

**Example A.2.1:** An impedance is $Z = 47 + j22\ \Omega$. Identify the resistance and reactance, and determine the magnitude and angle.
**Solution:**
The resistance is $R = \text{Re}\{Z\} = 47\ \Omega$ and the reactance is $X = \text{Im}\{Z\} = 22\ \Omega$ (inductive, since positive).
The magnitude is $|Z| = \sqrt{(47^2 + 22^2)} = \sqrt{(2209 + 484)} = \sqrt{2693} = 51.89\ \Omega$.
The angle is $\theta = \arctan(22/47) = \arctan(0.4681) = 25.08°$.
So $Z = 51.89\angle 25.08°\ \Omega$.

### A.2.2 Complex Arithmetic

Complex numbers are added and subtracted by combining their real and imaginary parts separately: $(a + jb) + (c + jd) = (a + c) + j(b + d)$. Multiplication uses the distributive property and the fact that $j^2 = -1$: $(a + jb)(c + jd) = (ac - bd) + j(ad + bc)$. Division is performed by multiplying the numerator and denominator by the complex conjugate of the denominator: $(a + jb)/(c + jd) = (a + jb)(c - jd)/(c^2 + d^2)$. Addition and subtraction are most efficient in rectangular form, while multiplication and division are often easier in polar form.

**Example A.2.2:** Given $Z_1 = 3 + j4$ and $Z_2 = 6 - j2$, compute $Z_1 + Z_2$, $Z_1 \times Z_2$, and $Z_1 / Z_2$.
**Solution:**
Addition: $Z_1 + Z_2 = (3 + 6) + j(4 + (-2)) = 9 + j2$.
Multiplication: $Z_1 \times Z_2 = (3)(6) - (4)(-2) + j((3)(-2) + (4)(6)) = 18 + 8 + j(-6 + 24) = 26 + j18$.
Division: $Z_1 / Z_2 = (3 + j4)(6 + j2) / (6^2 + 2^2) = ((18 - 8) + j(6 + 24)) / 40 = (10 + j30) / 40 = 0.25 + j0.75$.

### A.2.3 Complex Conjugate

The complex conjugate of $Z = a + jb$ is $Z^* = a - jb$, obtained by negating the imaginary part. Geometrically, conjugation reflects a complex number across the real axis. The product of a complex number and its conjugate yields a real number: $Z \times Z^* = a^2 + b^2 = |Z|^2$. This property is fundamental in circuit analysis for rationalizing complex fractions (dividing complex numbers) and in power calculations where complex power is $S = V \times I$, *with I* being the conjugate of the current phasor.

**Example A.2.3:** Given $Z = 5 + j12$, find Z*, the product* $Z \times Z$, and verify that $Z \times Z^* = |Z|^2$.
**Solution:**
$Z^* = 5 - j12$.
$Z \times Z^* = (5 + j12)(5 - j12) = 25 - j60 + j60 - j^2 144 = 25 + 144 = 169$.
Magnitude: $|Z| = \sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$.
Therefore $|Z|^2 = 169$, which confirms $Z \times Z^* = |Z|^2$.

## A.3 Polar and Exponential Forms

### A.3.1 Polar Form

A complex number can be expressed in polar form as $Z = |Z|\angle\theta$, where $|Z| = \sqrt{(a^2 + b^2)}$ is the magnitude and $\theta = \arctan(b/a)$ is the angle (phase) measured from the positive real axis. When computing θ, care must be taken with the quadrant: if $a < 0$, the angle must be adjusted by adding 180° (or $\pi$ radians). Multiplication in polar form is straightforward: $Z_1 \times Z_2 = |Z_1||Z_2|\angle(\theta_1 + \theta_2)$, and division is $Z_1 / Z_2 = (|Z_1|/|Z_2|)\angle(\theta_1 - \theta_2)$. Polar form is the preferred representation for expressing AC voltages, currents, and impedances because it directly displays the magnitude and phase shift.

**Example A.3.1:** Convert $Z = -3 + j4$ to polar form and express the result in degrees.
**Solution:**
Magnitude: $|Z| = \sqrt{((-3)^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.
The reference angle is $\arctan(4/3) = 53.13°$.
Since the real part is negative and the imaginary part is positive, Z lies in the second quadrant, so $\theta = 180° - 53.13° = 126.87°$.
Therefore $Z = 5\angle 126.87°$.

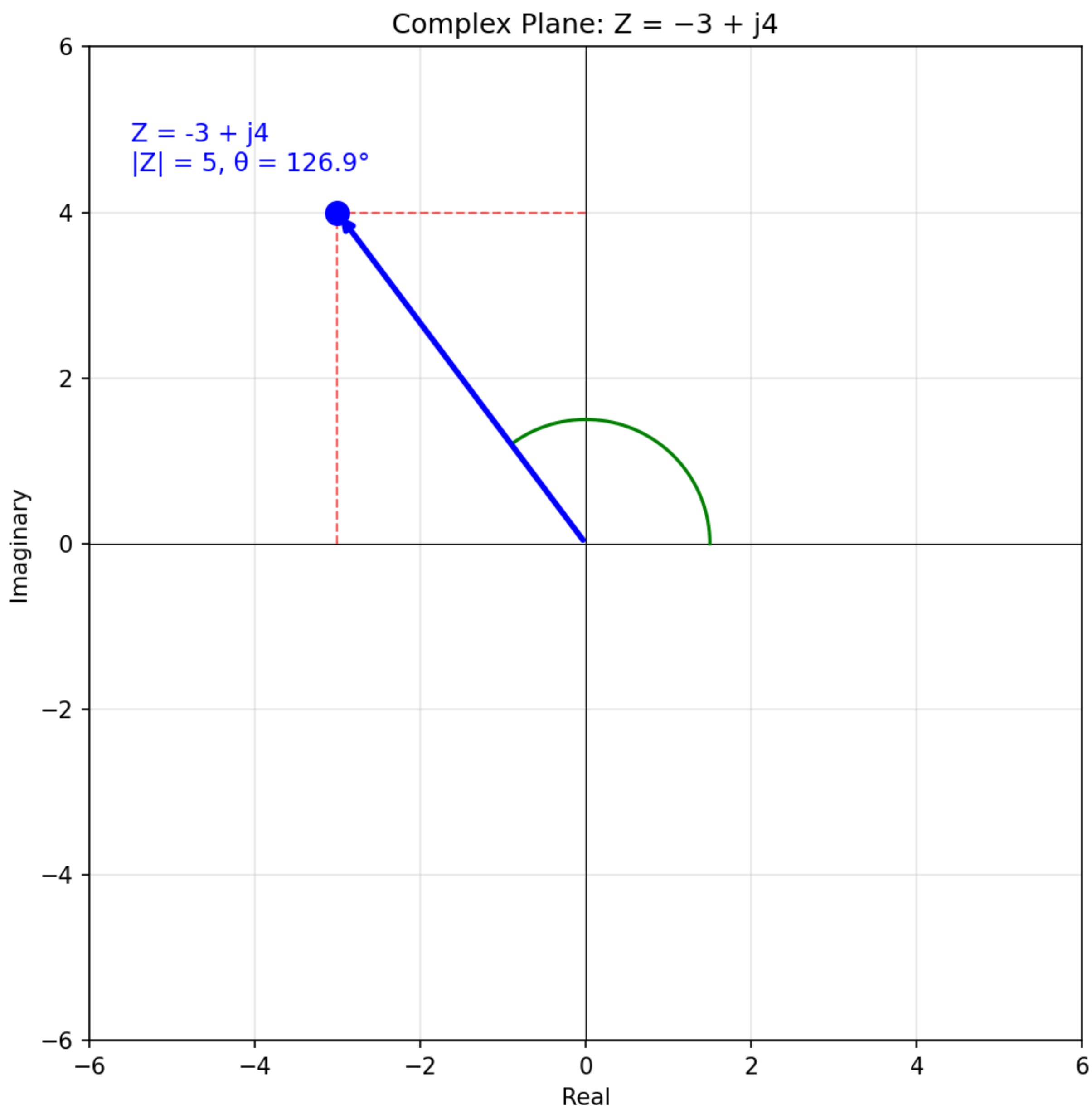



Figure A.3.1: Complex Plane: Polar Form

### A.3.2 Euler's Formula

Euler's formula states that $e^{j\theta} = \cos(\theta) + j\sin(\theta)$, establishing a direct connection between exponential functions and trigonometric functions. This relationship allows a complex number to be written in exponential form as $Z = |Z| \times e^{j\theta}$, which is mathematically equivalent to polar form. Euler's formula is the foundation of phasor analysis: a sinusoidal signal $v(t) = V_m \cos(\omega t + \varphi)$ is represented by the phasor $V = V_m \times e^{j\varphi}$. The special case $e^{j\pi} + 1 = 0$ (Euler's identity) connects the five most fundamental constants in mathematics.

**Example A.3.2:** Express $10\angle 60°$ in exponential form and verify by converting back to rectangular form.
**Solution:**
Exponential form: $Z = 10 \times e^{j60°} = 10 \times e^{j\pi/3}$.
Converting to rectangular: $Z = 10(\cos 60° + j \sin 60°) = 10(0.5 + j0.866) = 5 + j8.66$.
Verification: $|Z| = \sqrt{(5^2 + 8.66^2)} = \sqrt{(25 + 75)} = \sqrt{100} = 10$ and $\theta = \arctan(8.66/5) = \arctan(1.732) = 60°$.
The result matches the original polar form.

### A.3.3 Conversion Between Forms

Converting between rectangular and polar forms is a routine operation in AC analysis. Rectangular to polar: given $Z = a + jb$, the magnitude is $|Z| = \sqrt{(a^2 + b^2)}$ and the angle is $\theta = \text{atan2}(b, a)$, where the atan2 function correctly handles all four quadrants. Polar to rectangular: given $Z = |Z|\angle\theta$, the real part is $a = |Z| \cos(\theta)$ and the imaginary part is $b = |Z| \sin(\theta)$. Engineers frequently switch between forms during a single analysis: using polar form for multiplication/division steps and rectangular form for addition/subtraction steps.

**Example A.3.3:** Two impedances in series are $Z_1 = 20\angle 45°\ \Omega$ and $Z_2 = 15\angle{-30°}\ \Omega$. Find the total impedance in both rectangular and polar form.
**Solution:**
Convert to rectangular for addition.
$Z_1 = 20 \cos 45° + j20 \sin 45° = 14.14 + j14.14\ \Omega$.
$Z_2 = 15 \cos(-30°) + j15 \sin(-30°) = 12.99 - j7.5\ \Omega$.
Total: $Z_{total} = (14.14 + 12.99) + j(14.14 - 7.5) = 27.13 + j6.64\ \Omega$.
Converting back to polar: $|Z_{total}| = \sqrt{(27.13^2 + 6.64^2)} = \sqrt{(736.0 + 44.1)} = \sqrt{780.1} = 27.93\ \Omega$.
$\theta = \arctan(6.64 / 27.13) = 13.76°$.
Therefore $Z_{total} = 27.93\angle 13.76°\ \Omega$.

## A.4 Phasors

### A.4.1 Sinusoidal Representation

A sinusoidal signal in the time domain is described by $v(t) = V_m \cos(\omega t + \varphi)$, where $V_m$ is the peak amplitude, $\omega = 2\pi f$ is the angular frequency in rad/s, and $\varphi$ is the phase angle in degrees or radians. Since all voltages and currents in a linear AC circuit operate at the same frequency, the frequency information ($\omega t$) is redundant once established and can be suppressed. This key insight allows the time-domain sinusoid to be represented compactly as a phasor: a complex number that captures only the amplitude and phase, dramatically simplifying AC circuit analysis.

**Example A.4.1:** A voltage waveform is $v(t) = 170 \cos(377t + 30°)$ V. Identify the peak amplitude, frequency, period, and phase angle.
**Solution:**
Peak amplitude: $V_m = 170$ V.
Angular frequency: $\omega = 377$ rad/s, so the frequency is $f = \omega / (2\pi) = 377 / 6.2832 = 60$ Hz.

Period: $T = 1/f = 1/60 = 16.67$ ms.
Phase angle: $\varphi = 30°$ (leading the reference cosine by 30°).
The RMS value is $V_{rms} = V_m / \sqrt{2} = 170 / 1.414 = 120.2$ V.

### A.4.2 Phasor Notation

A phasor is a complex number that represents a sinusoidal quantity by its amplitude and phase angle, with the time-varying component ($\omega t$) implicit. The time-domain signal $v(t) = V_m \cos(\omega t + \varphi)$ is represented by the phasor $V = V_m\angle\varphi$ (using peak values) or $V = V_{rms}\angle\varphi$ (using RMS values). In power engineering, RMS phasors are standard because they simplify power calculations ($P = V_{rms} \times I_{rms} \times \cos(\varphi)$). Phasor notation transforms calculus operations into algebra: differentiation in time (d/dt) becomes multiplication by $j\omega$, and integration becomes division by $j\omega$, enabling AC circuit analysis using the same techniques as DC analysis.

**Example A.4.2:** Express $v(t) = 311 \cos(314t - 45°)$ V as an RMS phasor, and convert the phasor $V = 24\angle 60°\ V_{rms}$ back to a time-domain expression at 50 Hz.
**Solution:**
For $v(t) = 311 \cos(314t - 45°)$: $V_{rms} = 311 / \sqrt{2} = 219.9$ V, so the phasor is $V = 219.9\angle{-45°}\ V_{rms}$.
For the reverse: $V = 24\angle 60°\ V_{rms}$, so $V_m = 24 \times \sqrt{2} = 33.94$ V.
At 50 Hz, $\omega = 2\pi \times 50 = 314.16$ rad/s.
The time-domain expression is $v(t) = 33.94 \cos(314.16t + 60°)$ V.

### A.4.3 Phasor Arithmetic

Phasor addition and subtraction are performed in rectangular form by adding or subtracting the real and imaginary components separately. Phasor multiplication and division are performed in polar form by multiplying/dividing magnitudes and adding/subtracting angles. When combining voltages in series (KVL) or currents at a node (KCL), phasors must be added as vectors, not as scalar magnitudes. The resultant magnitude of two phasors at different angles is always less than or equal to the sum of their individual magnitudes, with equality occurring only when both phasors are in phase.

**Example A.4.3:** Two voltage sources in series produce $V_1 = 100\angle 0°$ V and $V_2 = 60\angle 90°$ V. Find the total voltage phasor and its magnitude.
**Solution:**
Convert to rectangular: $V_1 = 100 + j0$ V, $V_2 = 0 + j60$ V.
Total: $V_{total} = 100 + j60$ V.
Magnitude: $|V_{total}| = \sqrt{(100^2 + 60^2)} = \sqrt{(10{,}000 + 3{,}600)} = \sqrt{13{,}600} = 116.6$ V.
Angle: $\theta = \arctan(60/100) = 30.96°$.
So $V_{total} = 116.6\angle 30.96°$ V.
Note that the combined voltage (116.6 V) is less than the arithmetic sum ($100 + 60 = 160$ V) because the sources are 90° out of phase.

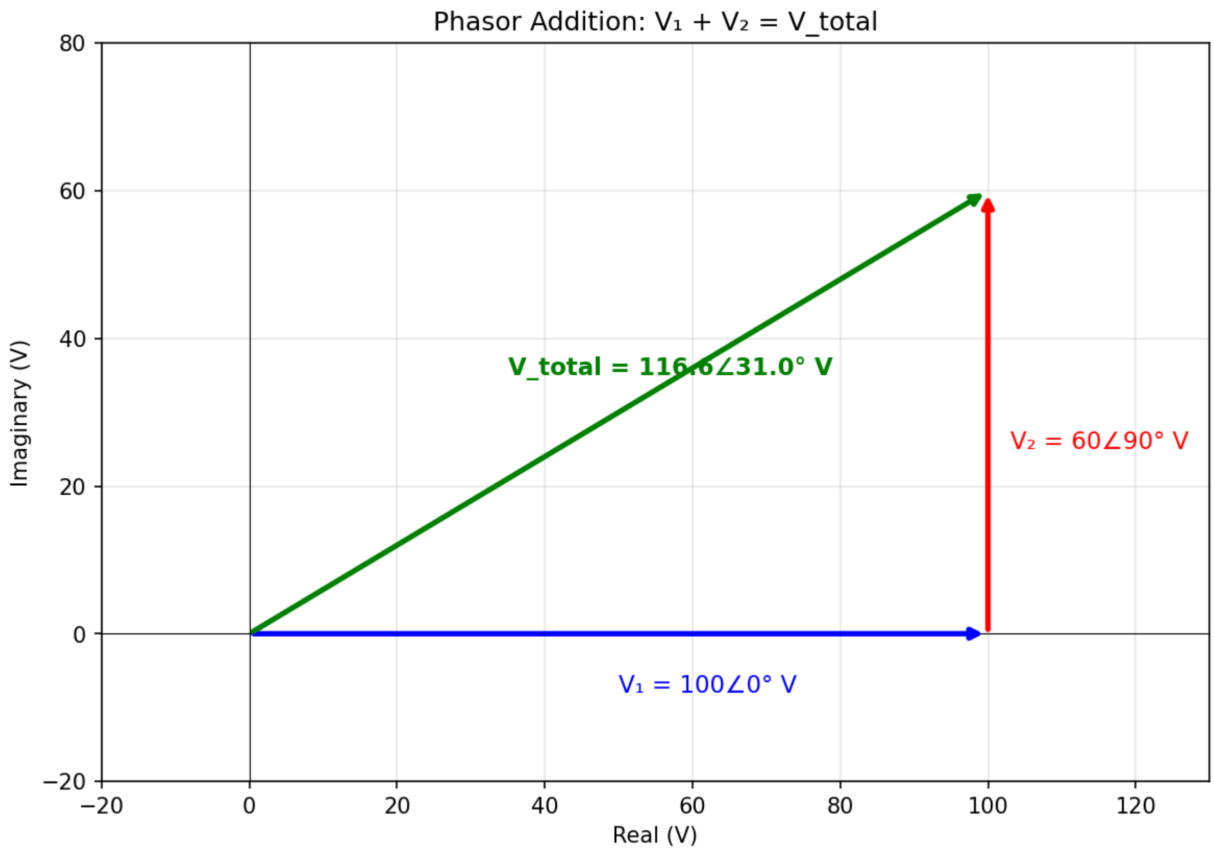



Figure A.4.3: Phasor Addition

## A.4.4 Phasor Diagrams

A phasor diagram is a graphical representation of phasors plotted on the complex plane, with the real axis horizontal and the imaginary axis vertical. Each phasor is drawn as an arrow from the origin, with its length proportional to the magnitude and its angle measured counterclockwise from the positive real axis. Phasor diagrams provide visual insight into phase relationships between voltages and currents: in a purely resistive circuit, voltage and current phasors are aligned; in an inductive circuit, current lags voltage; in a capacitive circuit, current leads voltage. They are particularly useful for understanding power factor, resonance conditions, and the effect of adding reactive compensation.

**Example A.4.4:** A series RL circuit carries $I = 5\angle 0°$ A through $R = 30\ \Omega$ and $X_L = 40\ \Omega$. Find the voltage across each element and the total voltage, then describe the phasor diagram.
**Solution:**
$V_R = I \times R = 5\angle 0° \times 30 = 150\angle 0°$ V (in phase with current).
$V_L = I \times jX_L = 5\angle 0° \times 40\angle 90° = 200\angle 90°$ V (leads current by 90°).
Total: $V_{total} = 150 + j200$ V $= 250\angle 53.13°$ V.
On the phasor diagram, the current phasor (reference) points along the positive real axis. $V_R$ is aligned with the current, $V_L$ points straight up (90° ahead), and $V_{total}$ is the diagonal of the rectangle formed by $V_R$ and $V_L$, at 53.13° above the real axis.

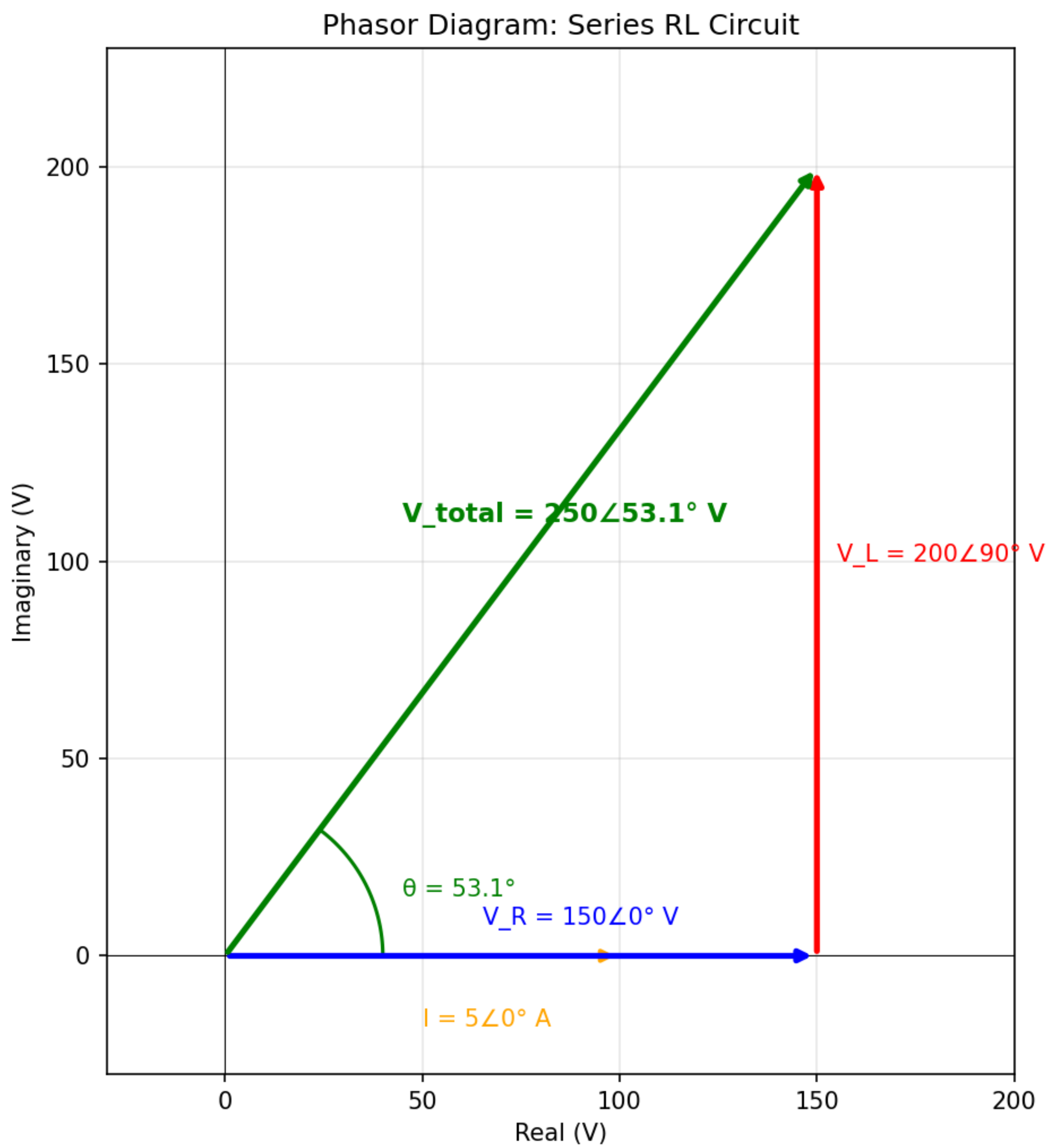



Figure A.4.4: Series RL Phasor Diagram

## A.5 Applications in Circuit Analysis

### A.5.1 Impedance and Admittance

Impedance $Z = R + jX$ generalizes resistance to AC circuits, where R is resistance and X is reactance (positive for inductors, negative for capacitors). The impedance of a resistor is $Z_R = R$, an inductor is $Z_L = j\omega L$, and a capacitor is $Z_C = 1/(j\omega C) = -j/(\omega C)$. Admittance $Y = 1/Z = G + jB$ is the reciprocal of impedance, where G is conductance and B is susceptance. Impedances in series add directly ($Z_{total}$

$= Z_1 + Z_2 + \ldots$), while admittances in parallel add directly ($Y_{total} = Y_1 + Y_2 + \ldots$), exactly analogous to resistances and conductances in DC circuits.

**Example A.5.1:** A 100 Ω resistor, a 10 mH inductor, and a 50 μF capacitor are connected in series at 1 kHz. Find the total impedance and admittance.
**Solution:**
$\omega = 2\pi \times 1000 = 6283.2$ rad/s.
$Z_R = 100\ \Omega$.
$Z_L = j\omega L = j \times 6283.2 \times 0.010 = j62.83\ \Omega$.
$Z_C = -j/(\omega C) = -j/(6283.2 \times 50 \times 10^{-6}) = -j/(0.31416) = -j3.18\ \Omega$.
Total: $Z_{total} = 100 + j62.83 - j3.18 = 100 + j59.65\ \Omega = 116.4\angle 30.82°\ \Omega$.
Admittance: $Y = 1/Z_{total} = 1/116.4\angle 30.82° = 8.59 \times 10^{-3}\angle -30.82°$ S $= (7.38 - j4.40) \times 10^{-3}$ S.

### A.5.2 Voltage and Current Phasors

In phasor-domain circuit analysis, Ohm's law becomes $V = I \times Z$, and all DC analysis methods (KVL, KCL, nodal analysis, mesh analysis, superposition, Thevenin/Norton) apply directly with complex-valued voltages, currents, and impedances. The voltage across an element and the current through it are related by the impedance of that element: the magnitude ratio $|V|/|I| = |Z|$ determines the amplitude relationship, while the angle difference $\angle V - \angle I = \angle Z$ determines the phase relationship. In a resistor the voltage and current are in phase, across an inductor the voltage leads the current by 90°, and across a capacitor the voltage lags the current by 90°.

**Example A.5.2:** A source $V_s = 120\angle 0°\ V_{rms}$ at 60 Hz drives a series circuit of $R = 20\ \Omega$ and $C = 100\ \mu F$. Find the current phasor and the voltage across the capacitor.
**Solution:**
$\omega = 2\pi \times 60 = 376.99$ rad/s.
$Z_C = -j/(\omega C) = -j/(376.99 \times 100 \times 10^{-6}) = -j26.53\ \Omega$.
$Z_{total} = 20 - j26.53\ \Omega$.
$|Z_{total}| = \sqrt{(20^2 + 26.53^2)} = \sqrt{(400 + 703.84)} = \sqrt{1103.8} = 33.22\ \Omega$.
$\theta = \arctan(-26.53/20) = -52.99°$.
Current: $I = V_s / Z_{total} = 120\angle 0° / 33.22\angle -52.99° = 3.61\angle 52.99°\ A_{rms}$ (current leads voltage, confirming capacitive circuit).
Voltage across capacitor: $V_C = I \times Z_C = 3.61\angle 52.99° \times 26.53\angle -90° = 95.77\angle -37.01°\ V_{rms}$.

### A.5.3 Power in Phasor Form

Complex power is defined as $S = V \times I^* = P + jQ$, where $I^*$ is the complex conjugate of the current phasor, P is real power in watts (W), and Q is reactive power in volt-amperes reactive (VAR). The magnitude $|S| = V_{rms} \times I_{rms}$ is the apparent power in volt-amperes (VA). The power factor is $pf = P/|S| = \cos(\theta)$, where $\theta$ is the angle of the complex power (equivalently, the phase difference between voltage and current). A positive Q indicates a lagging (inductive) load that absorbs reactive power, while a negative Q indicates a leading (capacitive) load that supplies reactive power. The complex power formulation unifies all power relationships into a single expression and enables systematic power analysis of multi-element circuits.

**Example A.5.3:** A load draws $I = 10\angle{-25°}$ $A_{rms}$ from a source $V = 240\angle 0°$ $V_{rms}$. Calculate the complex power, real power, reactive power, apparent power, and power factor.
**Solution:**
Complex power: $S = V \times I^* = 240\angle 0° \times 10\angle 25° = 2400\angle 25°$ VA. (Note: $I^* = 10\angle{+25°}$ since the conjugate negates the angle.)
Real power: $P = |S| \cos(25°) = 2400 \times 0.9063 = 2175$ W.
Reactive power: $Q = |S| \sin(25°) = 2400 \times 0.4226 = 1014$ VAR (positive, so inductive/lagging).
Apparent power: $|S| = 2400$ VA.
Power factor: $pf = \cos(25°) = 0.906$ lagging.
The load consumes 2175 W of real power while exchanging 1014 VAR of reactive power with the source.

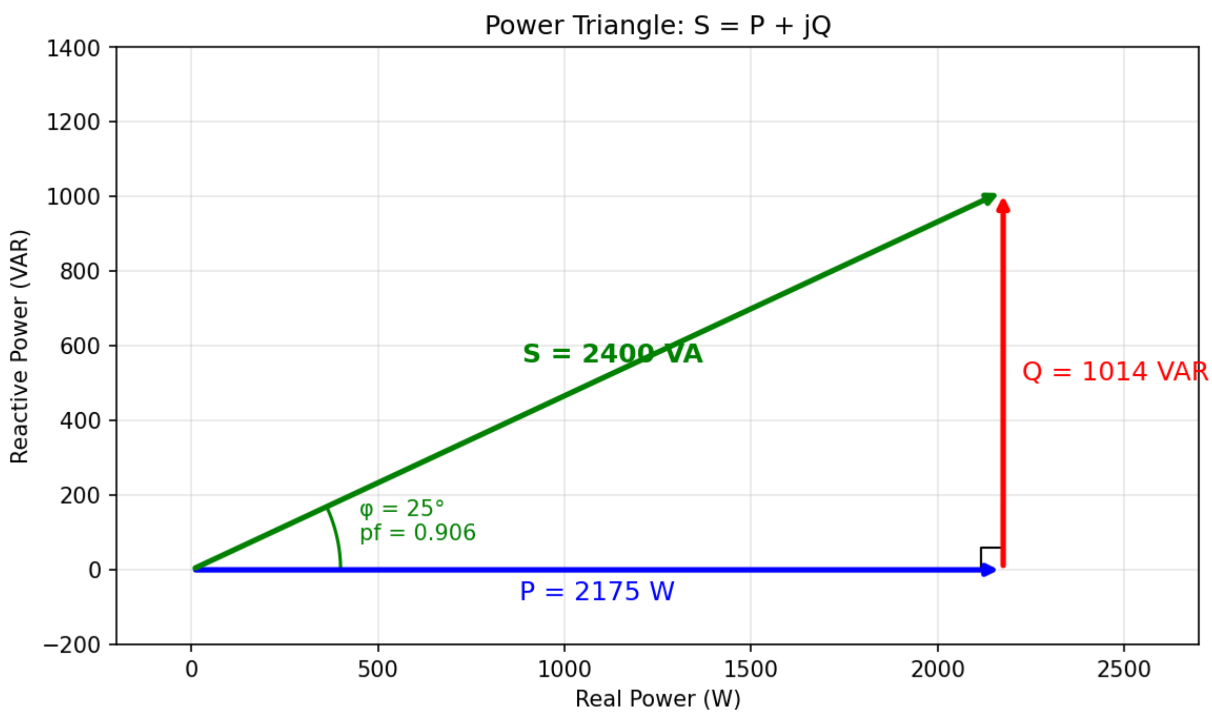



Figure A.5.3: Power Triangle

# Appendix B

# Arctangent and atan2

Computing angles from rectangular coordinates is one of the most common operations in AC circuit analysis, yet it contains a subtle pitfall that can produce incorrect results. The standard arctangent function (atan) returns angles in only two quadrants, making it unable to distinguish between points in opposite quadrants that share the same tangent ratio. The two-argument arctangent function (atan2) resolves this ambiguity by considering the signs of both coordinates independently, returning the correct angle in all four quadrants. This appendix explains the difference between these two functions and demonstrates why atan2 is the reliable choice for electrical engineering calculations.

## B.1 The Arctangent Function

### B.1.1 Definition and Range

The arctangent function, written as $\arctan(x)$ or $\tan^{-1}(x)$, is the inverse of the tangent function. Given a ratio $x = b/a$ (imaginary part divided by real part), $\arctan(x)$ returns the angle $\theta$ whose tangent equals $x$. The range of arctan is limited to $-90° < \theta < +90°$ (or $-\pi/2 < \theta < +\pi/2$ in radians), corresponding to quadrants I and IV of the complex plane. This restricted range ensures that arctan is a proper function (single-valued), but it means the output is always between $-90°$ and $+90°$ regardless of where the original complex number actually lies.

**Example B.1.1:** Calculate $\arctan(1)$ and $\arctan(-1)$, and identify which quadrant each result falls in.
**Solution:**
$\arctan(1) = 45°$, which lies in quadrant I (positive real, positive imaginary).
$\arctan(-1) = -45°$, which lies in quadrant IV (positive real, negative imaginary).
Both results fall within the range $-90°$ to $+90°$, confirming that arctan can only return angles in quadrants I and IV.

### B.1.2 Quadrant Ambiguity

The fundamental problem with arctan is that $\tan(\theta) = b/a$ produces the same ratio for two angles that are 180° apart. For example, a phasor at 135° has components $a = -1$ (real) and $b = +1$ (imaginary), so $b/a = (+1)/(-1) = -1$, and $\arctan(-1) = -45°$, missing the true angle by 180°. This occurs because dividing b by a discards the individual signs: the ratio $(-4)/(-3)$ is indistinguishable from $4/3$, and $(-4)/3$ is indistinguishable from $4/(-3)$. In circuit analysis, this means an impedance in the second quadrant (negative resistance with positive reactance) or third quadrant (both negative) would be assigned an incorrect angle if only $\arctan(X/R)$ is used. Engineers using arctan must manually inspect the signs of the real and imaginary parts and add or subtract 180° to correct the angle when the real part is negative.

**Example B.1.2:** Two impedances have the same tangent ratio: $Z_1 = 3 + j4\ \Omega$ and $Z_2 = -3 - j4\ \Omega$. Show that arctan gives the same angle for both and determine the correct angles.
**Solution:**
For $Z_1$: $\arctan(4/3) = \arctan(1.333) = 53.13°$.
For $Z_2$: $\arctan(-4/-3) = \arctan(1.333) = 53.13°$.
Both return 53.13°, but $Z_1$ is in quadrant I while $Z_2$ is in quadrant III.
The correct angle for $Z_2$ is $53.13° - 180° = -126.87°$ (or equivalently 233.13°).
Using arctan alone without checking the signs of the real and imaginary parts gives a 180° error for $Z_2$.

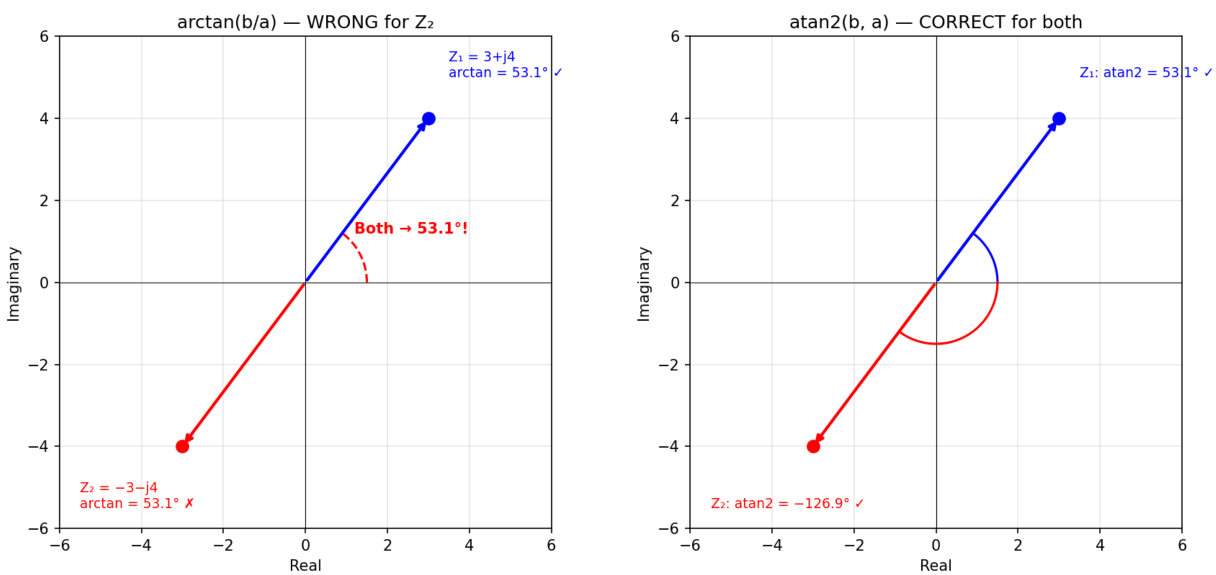



Figure B.1.2: Quadrant Ambiguity: arctan vs atan2

## B.2 The Two-Argument Arctangent

### B.2.1 Definition and Full-Circle Range

The two-argument arctangent function atan2(b, a) takes the imaginary part b and the real part a as separate arguments, preserving their individual signs. It returns the angle θ in the range $-180° < \theta \leq +180°$ (or $-\pi < \theta \leq +\pi$ in radians), covering all four quadrants of the complex plane. The function is defined for all (a, b) pairs except (0, 0) and is equivalent to arctan(b/a) with automatic quadrant correction. In most programming languages and scientific calculators, atan2 is available as a built-in function: atan2(y, x) in C/Python/MATLAB, or ATAN2(y, x) in spreadsheets.

**Example B.2.1:** Compute atan2(4, 3), atan2(−4, −3), atan2(4, −3), and atan2(−4, 3), and verify that each returns the correct quadrant.

**Solution:**
$\text{atan2}(4, 3) = 53.13°$ (quadrant I: $a > 0$, $b > 0$).
$\text{atan2}(-4, -3) = -126.87°$ (quadrant III: $a < 0$, $b < 0$).
$\text{atan2}(4, -3) = 126.87°$ (quadrant II: $a < 0$, $b > 0$).
$\text{atan2}(-4, 3) = -53.13°$ (quadrant IV: $a > 0$, $b < 0$).
All four results are distinct and correctly identify the quadrant, unlike arctan which would return only $53.13°$ or $-53.13°$ for all four cases.

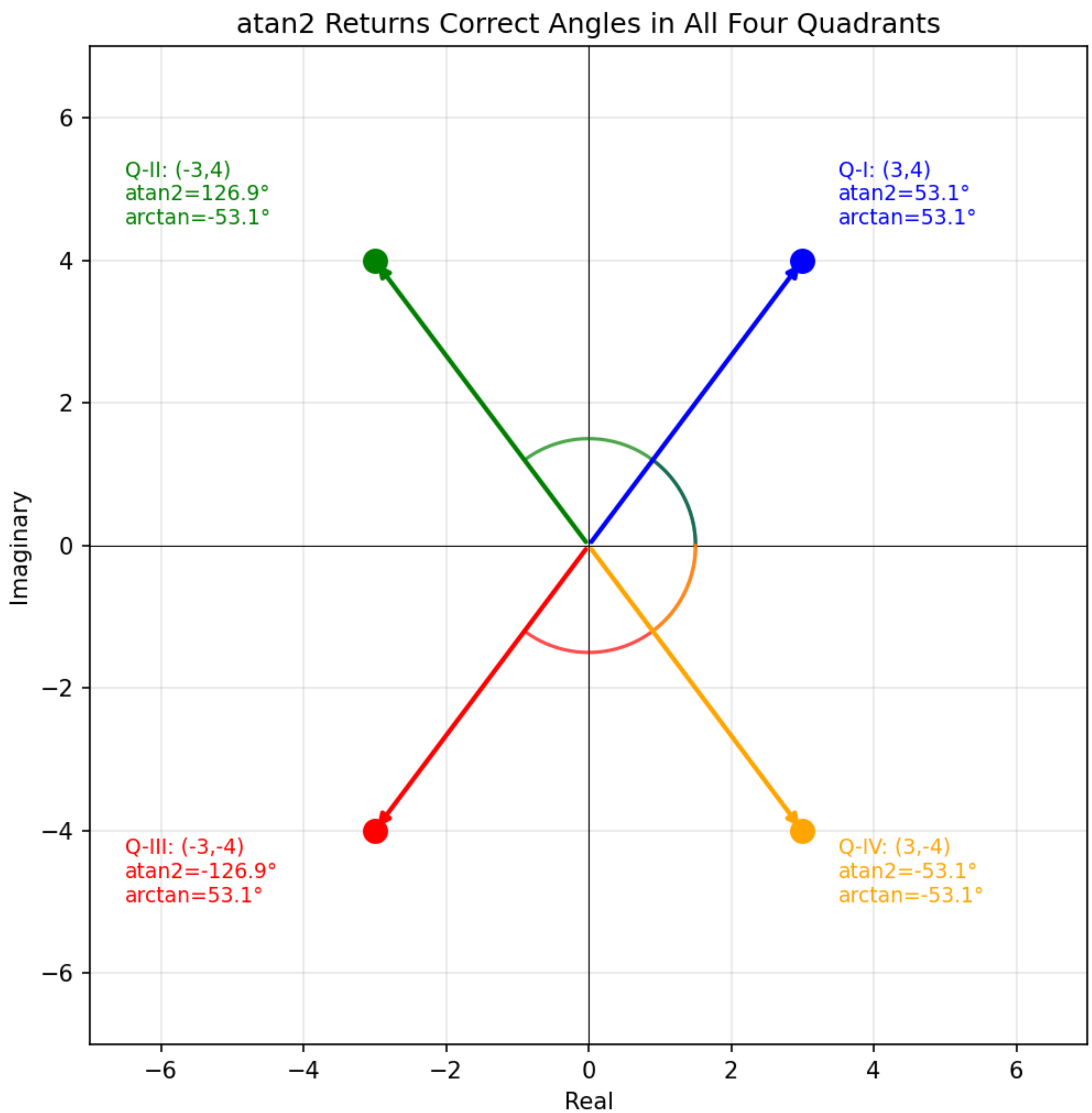



Figure B.2.1: atan2 in All Four Quadrants

### B.2.2 Quadrant Resolution

The atan2 function determines the correct quadrant by examining the signs of a and b independently before computing the angle. When $a > 0$, the angle is simply $\arctan(b/a)$ (quadrants I and IV). When $a < 0$ and $b \geq 0$, the angle is $\arctan(b/a) + 180°$ (quadrant II). When $a < 0$ and $b < 0$, the angle is $\arctan(b/a) - 180°$ (quadrant III). When $a = 0$, the angle is $+90°$ if $b > 0$ or $-90°$ if $b < 0$. This logic is built into the atan2 function, so the engineer does not need to perform manual quadrant checks — the function always returns the geometrically correct angle.

**Example B.2.2:** An impedance $Z = -10 + j5\ \Omega$ lies in quadrant II. Compute the angle using both arctan and atan2, and show the correction needed for arctan.
**Solution:**
Using arctan: $\theta = \arctan(5/(-10)) = \arctan(-0.5) = -26.57°$. This incorrectly places Z in quadrant IV.
Manual correction: since $a < 0$, add 180°: $\theta = -26.57° + 180° = 153.43°$ (quadrant II, correct).
Using atan2: $\theta = \text{atan2}(5, -10) = 153.43°$ directly, with no manual correction needed.
The magnitude is $|Z| = \sqrt{(100 + 25)} = \sqrt{125} = 11.18\ \Omega$, so $Z = 11.18\angle 153.43°\ \Omega$.

### B.2.3 Special Cases and Axis Points

The atan2 function handles points on the real and imaginary axes correctly, where $\arctan(b/a)$ would encounter division by zero or ambiguity. For a purely real positive number ($a > 0$, $b = 0$), atan2 returns 0°. For a purely real negative number ($a < 0$, $b = 0$), atan2 returns 180° (or ±180°). For a purely imaginary positive number ($a = 0$, $b > 0$), atan2 returns 90°. For a purely imaginary negative number ($a = 0$, $b < 0$), atan2 returns −90°. These axis cases arise frequently in circuit analysis: a pure resistance has 0° impedance angle, a pure inductance has +90°, and a pure capacitance has −90°.

**Example B.2.3:** Find the impedance angle for a 50 mH inductor and a 100 μF capacitor at 60 Hz using atan2.
**Solution:**
For the inductor: $Z_L = j\omega L = j \times 2\pi \times 60 \times 0.050 = j18.85\ \Omega$, so $a = 0$ and $b = 18.85$.
$\text{atan2}(18.85, 0) = 90°$, confirming the inductor impedance has a +90° angle (voltage leads current).
For the capacitor: $Z_C = -j/(\omega C) = -j/(2\pi \times 60 \times 100 \times 10^{-6}) = -j26.53\ \Omega$, so $a = 0$ and $b = -26.53$.
$\text{atan2}(-26.53, 0) = -90°$, confirming the capacitor impedance has a −90° angle (voltage lags current).

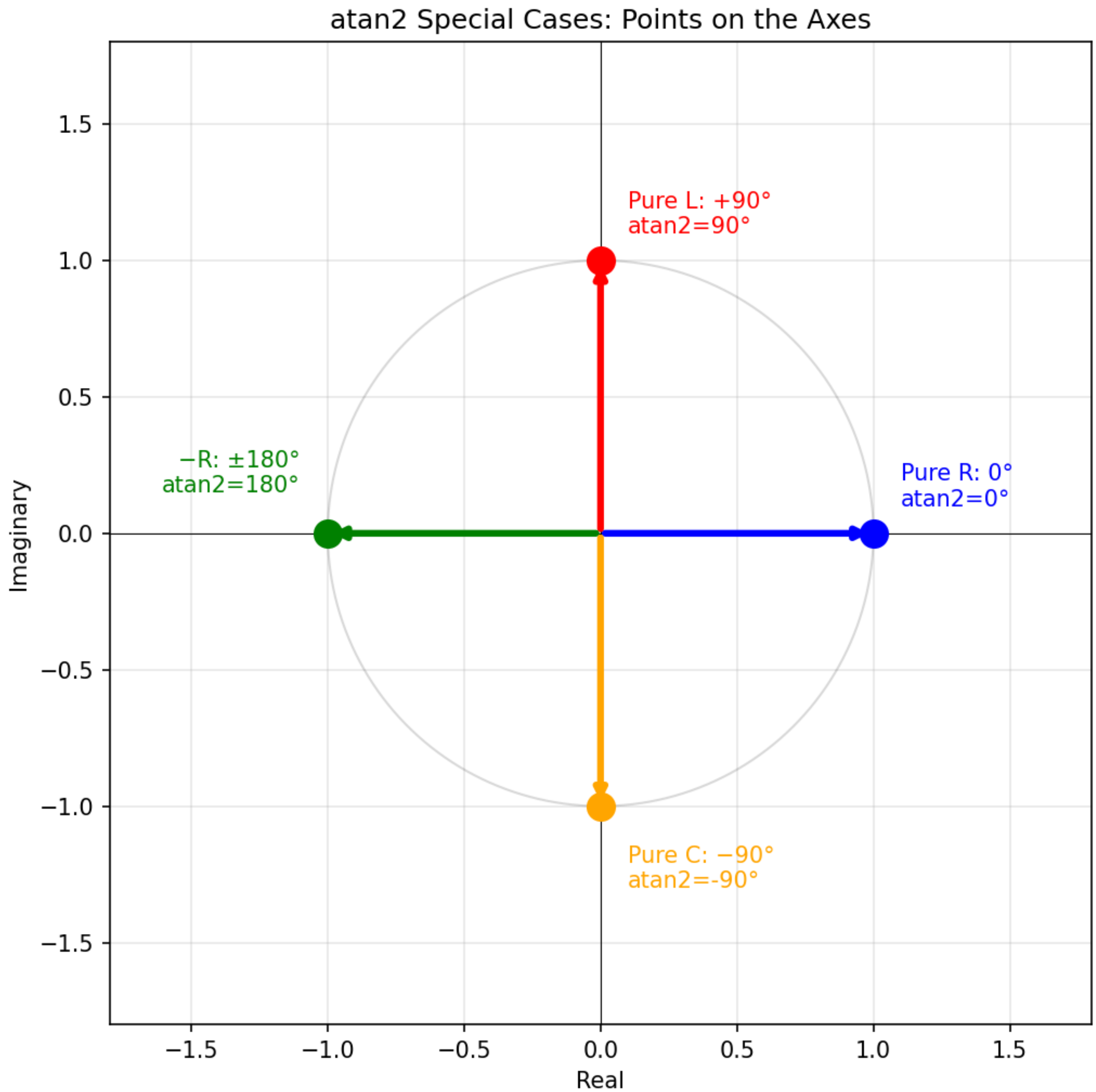



Figure B.2.3: atan2 Special Cases: Axis Points

## B.3 Applications in Electrical Engineering

### B.3.1 Impedance Angle Calculation

When converting an impedance from rectangular form $Z = R + jX$ to polar form $Z = |Z|\angle\theta$, the angle $\theta$ should be computed using atan2(X, R) rather than arctan(X/R). For passive circuits where $R \geq 0$, arctan(X/R) happens to give correct results because the impedance always lies in quadrants I or IV (inductive or capacitive with positive resistance). However, in circuits with negative resistance (such as active circuits with oscillators, tunnel diodes, or negative impedance converters), the impedance can fall in quadrants II or III, and atan2 becomes essential for correctness.

**Example B.3.1:** A negative impedance converter produces an effective impedance of $Z = -50 + j30\ \Omega$. Calculate the polar form using atan2 and show the error that arctan would produce.
**Solution:**
Magnitude: $|Z| = \sqrt{((-50)^2 + 30^2)} = \sqrt{(2500 + 900)} = \sqrt{3400} = 58.31\ \Omega$.
Using atan2: $\theta = \text{atan2}(30, -50) = 180° - \arctan(30/50) = 180° - 30.96° = 149.04°$. So $Z = 58.31\angle 149.04°\ \Omega$ (quadrant II, correct).
Using arctan: $\theta = \arctan(30/(-50)) = \arctan(-0.6) = -30.96°$ (quadrant IV, wrong quadrant).
The arctan result has a 180° error because it cannot distinguish between $(-50, 30)$ in quadrant II and $(50, -30)$ in quadrant IV.

### B.3.2 Phasor Angle from Rectangular Components

When measuring or computing AC voltages and currents, the in-phase (real) and quadrature (imaginary) components are often obtained separately — for example, from an I/Q demodulator, a lock-in amplifier, or a digital signal processing algorithm. Converting these rectangular components to a magnitude-and-angle phasor requires atan2 to correctly determine the phase angle across all four quadrants. This is especially important in communications engineering where phase angles span the full 360° range, and in power systems where current phasors can lag or lead the voltage reference by any angle.

**Example B.3.2:** A lock-in amplifier measures a signal with in-phase component $I = -2.5$ mV and quadrature component $Q = -4.33$ mV. Find the signal magnitude and phase angle.
**Solution:**
Magnitude: $|V| = \sqrt{((-2.5)^2 + (-4.33)^2)} = \sqrt{(6.25 + 18.75)} = \sqrt{25} = 5.0$ mV.
Phase angle: $\theta = \text{atan2}(-4.33, -2.5) = -(180° - \arctan(4.33/2.5)) = -(180° - 60°) = -120°$ (quadrant III).
The phasor is $V = 5.0\angle -120°$ mV.
Using $\arctan(-4.33/-2.5) = \arctan(1.732) = 60°$ would incorrectly place the phasor in quadrant I, a 180° error.

### B.3.3 Power Factor Angle

The power factor angle $\varphi$ between voltage and current phasors determines whether a load is inductive (lagging) or capacitive (leading). When complex power $S = P + jQ$ is computed from real power P and reactive power Q, the power factor angle is $\varphi = \text{atan2}(Q, P)$. For typical loads where $P > 0$, $\arctan(Q/P)$ gives the correct result. However, in systems with regenerative loads (motors returning energy to the grid), P can be negative, and atan2 is needed to correctly identify the power flow direction and the associated angle in quadrants II or III.

**Example B.3.3:** A regenerative drive returns power to the grid during braking, producing $S = -800 + j600$ VA (negative real power, positive reactive power). Find the power factor angle and compare the arctan and atan2 results.
**Solution:**
Using atan2: $\varphi = \text{atan2}(600, -800) = 180° - \arctan(600/800) = 180° - 36.87° = 143.13°$. This correctly indicates the operating point is in quadrant II (power flowing back to the source with

inductive reactive power).
Using arctan: $\varphi = \arctan(600/(-800)) = \arctan(-0.75) = -36.87°$, which incorrectly suggests quadrant IV (consuming power with capacitive reactive power).
The apparent power is $|S| = \sqrt{(800^2 + 600^2)} = \sqrt{1{,}000{,}000} = 1000$ VA, and the power factor is $\cos(143.13°) = -0.8$, where the negative sign indicates reverse power flow.

# Appendix C

# Decibels

The decibel (dB) is a logarithmic unit used throughout electrical engineering to express ratios of power, voltage, or current. Because engineering systems routinely span many orders of magnitude — from microvolt sensor signals to megawatt transmitters, or from nanovolt noise floors to volt-level outputs — linear scales become unwieldy. The logarithmic decibel scale compresses these enormous ranges into manageable numbers and converts multiplication and division of gains and losses into simple addition and subtraction. This appendix covers the definition, common reference levels, useful rules of thumb, and practical applications of decibels in electrical engineering.

## C.1 Definition and Fundamentals

### C.1.1 Power Ratios

The decibel was originally defined as a ratio of two power levels: $\text{dB} = 10 \times \log_{10}(P_2/P_1)$, where $P_1$ is the reference power and $P_2$ is the measured power. A positive dB value indicates a gain ($P_2 > P_1$), while a negative dB value indicates a loss or attenuation ($P_2 < P_1$). A ratio of 0 dB means the two power levels are equal. The factor of 10 in the formula scales the unit so that 1 dB represents a power ratio of approximately 1.259:1, which is a conveniently small increment for practical measurements.

**Example C.1.1:** An amplifier receives 5 mW of input power and delivers 800 mW of output power. Express the power gain in decibels.
**Solution:** $G = 10 \times \log_{10}(P_{out}/P_{in}) = 10 \times \log_{10}(800/5) = 10 \times \log_{10}(160) = 10 \times 2.204 = 22.04$ dB. The amplifier provides 22.04 dB of power gain.

### C.1.2 Voltage and Current Ratios

Since power is proportional to voltage squared ($P = V^2/R$) and current squared ($P = I^2R$), the decibel formula for voltage and current ratios includes a factor of 20 instead of 10: $\text{dB} = 20 \times \log_{10}(V_2/V_1)$ or $\text{dB} = 20 \times \log_{10}(I_2/I_1)$. This ensures consistency with the power definition — doubling the voltage across the same impedance quadruples the power, which is 6.02 dB by either formula. The voltage/current form assumes equal impedances at the measurement points; when impedances differ, the power ratio formula should be used directly.

**Example C.1.2:** A filter attenuates a 2 $V_{rms}$ input signal to 0.35 $V_{rms}$ at the output. Express the voltage gain in decibels.

**Solution:** $A_v = 20 \times \log_{10}(V_{out}/V_{in}) = 20 \times \log_{10}(0.35/2) = 20 \times \log_{10}(0.175) = 20 \times (-0.757) = -15.14$ dB. The filter attenuates the signal by 15.14 dB.

### C.1.3 Common Decibel Values

Several decibel values appear so frequently that engineers memorize them as rules of thumb. A 3 dB increase represents a doubling of power (ratio 2:1), while −3 dB represents halving the power — this is why filter cutoff frequencies are called "3 dB points." A 10 dB increase represents a tenfold increase in power, and 20 dB represents a hundredfold increase. For voltage, 6 dB represents a doubling (ratio 2:1) and 20 dB represents a tenfold increase. These reference points allow engineers to quickly estimate gains and losses without a calculator.

| dB | Power Ratio | Voltage Ratio |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 1.259 | 1.122 |
| 3 | 2 | 1.414 ($\sqrt{2}$) |
| 6 | 4 | 2 |
| 10 | 10 | 3.162 |
| 20 | 100 | 10 |
| 30 | 1,000 | 31.62 |
| 40 | 10,000 | 100 |
| −3 | 0.5 | 0.707 ($1/\sqrt{2}$) |
| −10 | 0.1 | 0.316 |
| −20 | 0.01 | 0.1 |

**Example C.1.3:** An amplifier has a gain of 23 dB. Using the rules of thumb, estimate the power gain without a calculator.
**Solution:**
23 dB = 20 dB + 3 dB.
A 20 dB gain is a power ratio of 100, and a 3 dB gain doubles the power.
Therefore the total power gain is approximately $100 \times 2 = 200$.
The exact value is $10^{23/10} = 10^{2.3} = 199.5$, confirming the estimate.

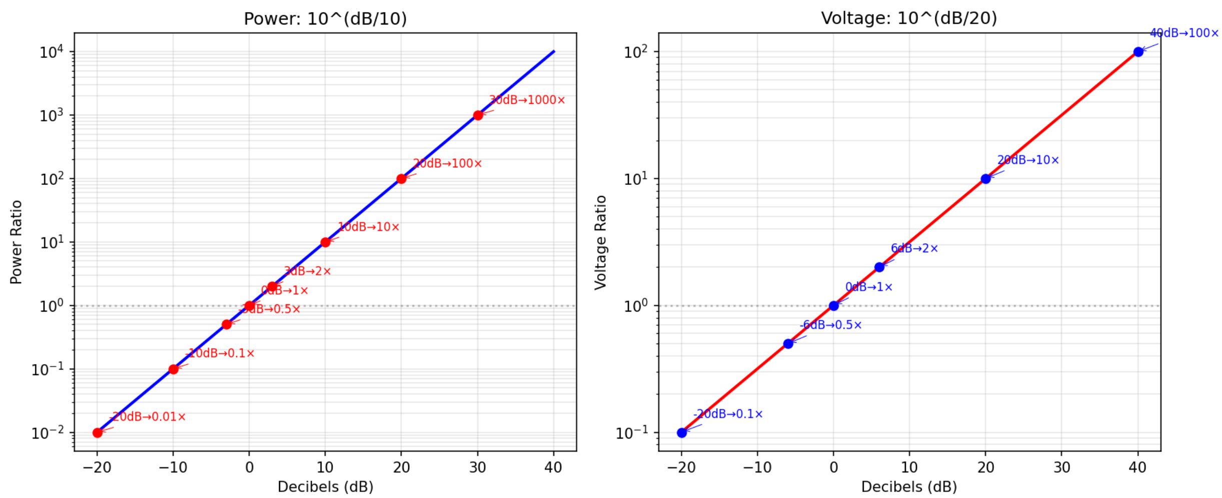



Figure C.1.3: Decibel Scale: Power and Voltage Ratios

## C.2 Absolute Reference Levels

### C.2.1 dBm — Decibels Referenced to 1 mW

dBm expresses absolute power referenced to 1 milliwatt: $P(dBm) = 10 \times \log_{10}(P/1\ mW)$. A value of 0 dBm equals exactly 1 mW, +30 dBm equals 1 W, and −30 dBm equals 1 μW. dBm is the most widely used absolute power unit in RF engineering, telecommunications, and fiber optics. It is independent of impedance — 0 dBm is always 1 mW regardless of whether the system impedance is 50 Ω, 75 Ω, or 600 Ω. When converting to voltage, the impedance must be specified: in a 50 Ω system, 0 dBm corresponds to 224 $mV_{rms}$.

**Example C.2.1:** A wireless transmitter outputs +20 dBm into a 50 Ω antenna. Find the output power in watts and the RMS voltage across the antenna.
**Solution:** Power: $P = 1\ mW \times 10^{20/10} = 1\ mW \times 100 = 100\ mW = 0.1\ W$. Voltage: $V_{rms} = \sqrt{}(P \times R) = \sqrt{}(0.1 \times 50) = \sqrt{5} = 2.236\ V$.

### C.2.2 dBW — Decibels Referenced to 1 W

dBW expresses absolute power referenced to 1 watt: $P(dBW) = 10 \times \log_{10}(P/1\ W)$. Since dBW and dBm share the same logarithmic scale but differ by a factor of 1000 in the reference, the conversion is simply dBW = dBm − 30. dBW is commonly used in high-power applications such as broadcast transmitters, satellite links, and power systems, where milliwatt references would produce inconveniently large numbers. For example, a 50 kW broadcast transmitter is +47 dBW (or equivalently +77 dBm).

**Example C.2.2:** A satellite transponder has an output power of 20 W. Express this in dBW and dBm.
**Solution:**
dBW: $P = 10 \times \log_{10}(20/1) = 10 \times 1.301 = 13.01$ dBW.
dBm: $P = 13.01 + 30 = 43.01$ dBm.

Verification: $10^{43.01/10} \times 1$ mW $= 10^{4.301}$ mW $= 20{,}000$ mW $= 20$ W.

### C.2.3 dBV and dBμV — Voltage References

dBV expresses voltage referenced to 1 volt: $V(\text{dBV}) = 20 \times \log_{10}(V/1\ \text{V})$. dBμV (decibels referenced to 1 microvolt) is used for small signals: $V(\text{dB}\mu\text{V}) = 20 \times \log_{10}(V/1\ \mu\text{V})$. The conversion between them is $\text{dB}\mu\text{V} = \text{dBV} + 120$. dBV is common in audio engineering, where $0\ \text{dBV} = 1\ V_{rms}$ is a standard reference level. dBμV is prevalent in EMC (electromagnetic compatibility) testing and antenna measurements, where signal levels are typically in the microvolt to millivolt range. The related unit dBmV (referenced to 1 millivolt) is standard in cable television systems.

**Example C.2.3:** An EMC test measures a radiated emission of 42 dBμV/m at 100 MHz. Convert this to dBV/m and find the field strength in V/m.
**Solution:**
$\text{dBV} = \text{dB}\mu\text{V} - 120 = 42 - 120 = -78$ dBV/m.
Field strength: $E = 1\ \mu\text{V/m} \times 10^{42/20} = 10^{-6} \times 10^{2.1} = 10^{-6} \times 125.9 = 125.9\ \mu\text{V/m}$.
This is a relatively weak emission typical of unintentional radiators.

## C.3 Decibel Arithmetic

### C.3.1 Cascaded Gains and Losses

The primary advantage of decibels is that gains and losses in a cascaded system are simply added. If a signal passes through stages with gains $G_1$, $G_2$, and $G_3$ in dB, the total system gain is $G_{total} = G_1 + G_2 + G_3$. This replaces multiplication of linear ratios with addition, making system-level calculations straightforward. For example, a transmitter chain with a +20 dB amplifier, a −3 dB cable loss, and a +12 dBi antenna gain has a total gain of $20 + (-3) + 12 = +29$ dB relative to the input.

**Example C.3.1:** A fiber optic link consists of a +3 dBm laser source, a −0.5 dB connector loss, 20 km of fiber with 0.3 dB/km attenuation, a −0.5 dB connector loss, and a receiver with −28 dBm sensitivity. Determine the received power and the link margin.
**Solution:**
Fiber loss: $20 \times 0.3 = 6$ dB.
Total loss: $0.5 + 6 + 0.5 = 7$ dB.
Received power: $+3$ dBm $- 7$ dB $= -4$ dBm.
Link margin: received power − sensitivity $= -4 - (-28) = 24$ dB.
The link has 24 dB of margin above the minimum receiver sensitivity, providing a comfortable buffer for aging and environmental degradation.

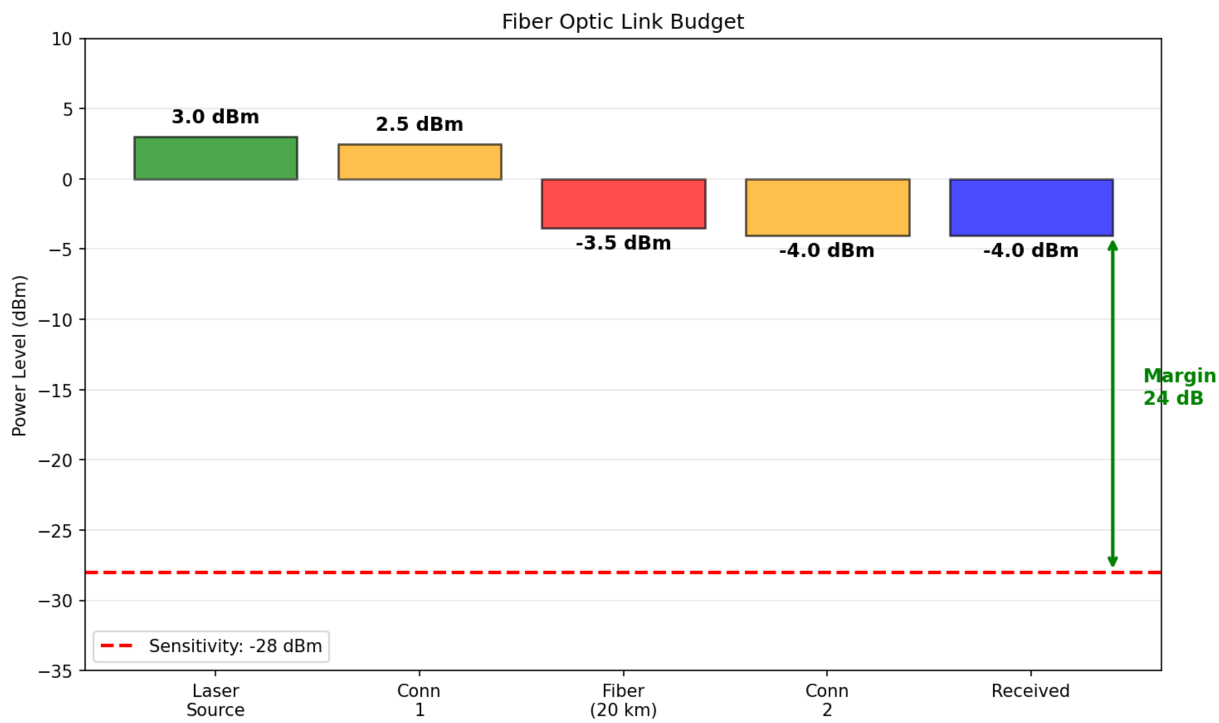



Figure C.3.1: Fiber Optic Link Budget

## C.3.2 Converting Between Linear and Decibel

Converting from decibels to linear ratios reverses the logarithm: for power, the linear ratio is $10^{\mathrm{dB}/10}$; for voltage or current, the linear ratio is $10^{\mathrm{dB}/20}$. Converting from linear to decibels applies the logarithm: for power, $\mathrm{dB} = 10 \times \log_{10}(\text{ratio})$; for voltage, $\mathrm{dB} = 20 \times \log_{10}(\text{ratio})$. A common mistake is using the wrong formula — applying the factor of 20 to a power ratio or 10 to a voltage ratio. The rule is: use 10 for power (and energy, intensity), use 20 for field quantities (voltage, current, electric field, pressure).

**Example C.3.2:** A receiver specification states a noise figure of 4.5 dB. Convert this to a linear noise factor F, and compute the equivalent noise temperature ($T_0 = 290$ K).
**Solution:**
Linear noise factor: $F = 10^{4.5/10} = 10^{0.45} = 2.818$.
Equivalent noise temperature: $T_e = T_0 \times (F - 1) = 290 \times (2.818 - 1) = 290 \times 1.818 = 527.2$ K.
The receiver adds 527.2 K of equivalent noise to the signal.

## C.3.3 Adding Powers in Decibels

When two or more signals combine (such as noise from independent sources or signals arriving at a combiner), their powers add in the linear domain, not in decibels. To add powers expressed in dBm, each must first be converted to milliwatts, summed, and then converted back to dBm: $P_{total}(\mathrm{dBm}) = 10 \times \log_{10}(10^{P_1/10} + 10^{P_2/10})$. Two equal power levels combine to produce a result 3 dB higher than either one (doubling of power). This distinction between adding dB (for cascaded stages) and adding powers (for combining signals) is a frequent source of errors.

**Example C.3.3:** Two uncorrelated noise sources produce −90 dBm and −87 dBm at the input of a receiver. Find the total noise power.
**Solution:**
Convert to linear: $P_1 = 10^{-90/10}$ mW = $10^{-9}$ mW = 1.000 pW.
$P_2 = 10^{-87/10}$ mW = $1.995 \times 10^{-9}$ mW = 1.995 pW.
Total: $P_{total} = 1.000 + 1.995 = 2.995$ pW = $2.995 \times 10^{-9}$ mW.
$P_{total} = 10 \times \log_{10}(2.995 \times 10^{-9}) = 10 \times (-8.524) = -85.24$ dBm.
The combined noise is −85.24 dBm, which is 1.76 dB above the stronger source alone — less than 3 dB because the two sources are not equal in power.

## C.4 Applications in Electrical Engineering

### C.4.1 Amplifier Gain and Frequency Response

Amplifier gain is universally expressed in decibels, whether for voltage gain ($A_v$ in dB), current gain, or power gain. The frequency response of an amplifier is plotted as gain in dB versus frequency on a logarithmic scale (Bode plot), where the bandwidth is defined by the frequencies at which the gain drops 3 dB below the midband value. A gain roll-off of −20 dB/decade indicates a first-order response (single pole), while −40 dB/decade indicates a second-order response (two poles). The gain-bandwidth product (GBW) of an op-amp is constant: if the gain is halved (−6 dB), the bandwidth doubles.

**Example C.4.1:** An op-amp has an open-loop gain of 100 dB and a gain-bandwidth product of 10 MHz. Find the closed-loop bandwidth when configured for a gain of 40 dB.
**Solution:**
Open-loop gain in linear: $A_{OL} = 10^{100/20} = 10^5 = 100{,}000$.
Closed-loop gain in linear: $A_{CL} = 10^{40/20} = 10^2 = 100$.
Closed-loop bandwidth: BW = GBW / $A_{CL}$ = 10 MHz / 100 = 100 kHz.
The unity-gain frequency (0 dB) equals the GBW: $f_{unity} = 10$ MHz.
At 40 dB gain, the amplifier is usable up to 100 kHz before the gain begins to roll off.

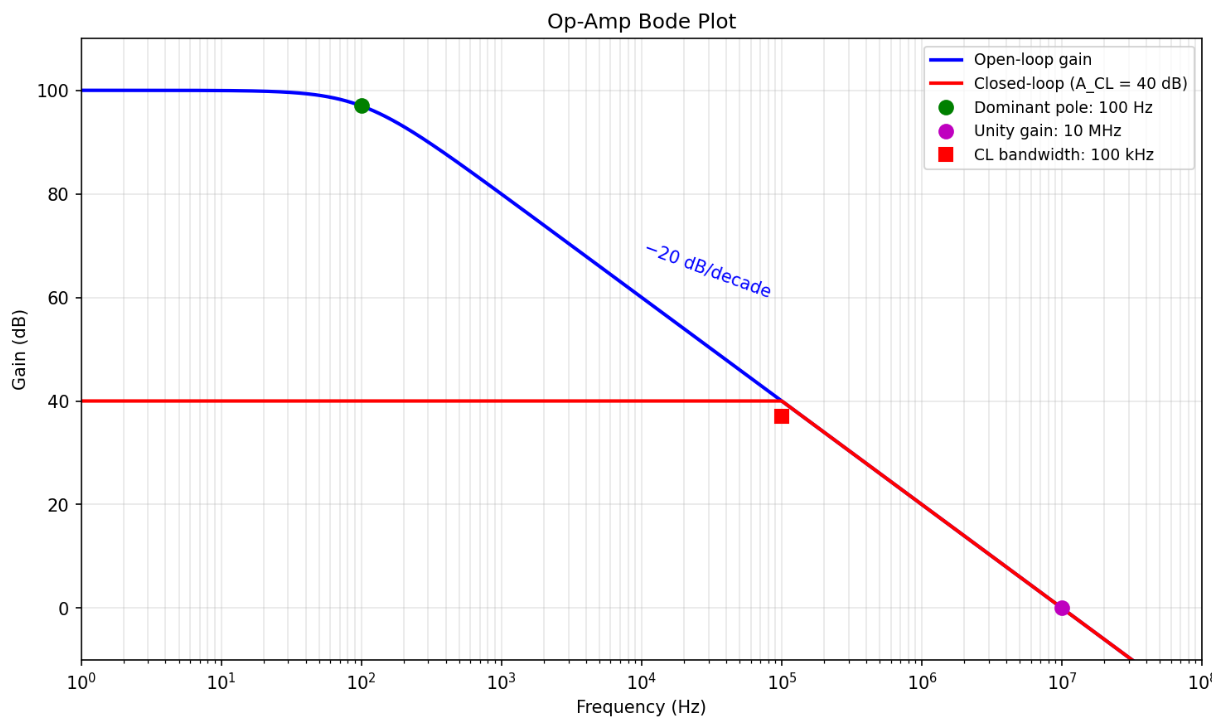



Figure C.4.1: Op-Amp Bode Plot

## C.4.2 Signal-to-Noise Ratio

Signal-to-noise ratio (SNR) is the ratio of signal power to noise power, expressed in decibels: SNR = 10 $\times \log_{10}(P_{signal}/P_{noise})$ or equivalently $20 \times \log_{10}(V_{signal}/V_{noise})$. A higher SNR indicates a cleaner signal. In audio systems, a 90 dB SNR means the signal power is $10^9$ times the noise power. In digital communications, the bit error rate (BER) depends on the SNR per bit ($E_b/N_0$), with typical requirements ranging from 7 dB for basic QPSK to over 20 dB for 256-QAM. Dynamic range, the ratio of the largest to smallest signal a system can handle, is also expressed in dB.

**Example C.4.2:** An ADC has a full-scale voltage of 3.3 V and an RMS noise floor of 0.5 mV. Calculate the SNR in dB and the effective number of bits (ENOB).
**Solution:**
$\text{SNR} = 20 \times \log_{10}(V_{signal}/V_{noise}) = 20 \times \log_{10}(3.3/0.0005) = 20 \times \log_{10}(6600) = 20 \times 3.8195 = 76.39$ dB.
ENOB = (SNR − 1.76) / 6.02 = (76.39 − 1.76) / 6.02 = 74.63 / 6.02 = 12.4 bits.
The ADC performs equivalently to an ideal 12.4-bit converter, meaning roughly 12 bits of useful resolution.

## C.4.3 Link Budgets

A link budget is a systematic accounting of all gains and losses in a communication system from transmitter to receiver, expressed entirely in decibels. The received power is computed as: $P_{rx} = P_{tx} + G_{tx} - L_{path} + G_{rx} - L_{misc}$, where all terms are in dB or dBm. Free-space path loss for a radio link is $L_{path} = 20 \times \log_{10}(4\pi d/\lambda)$, where d is the distance and λ is the wavelength. Link budgets allow engineers to predict system performance, determine required transmit power, and evaluate the feasibility of a communication link before deployment.

**Example C.4.3:** A 5.8 GHz point-to-point wireless link spans 2 km. The transmitter outputs +24 dBm, the transmit antenna has +18 dBi gain, and the receive antenna has +18 dBi gain. Cable losses total 2 dB at each end. Calculate the free-space path loss and received power.
**Solution:**
Wavelength: $\lambda = c/f = 3 \times 10^8 / 5.8 \times 10^9 = 0.05172$ m.
Free-space path loss: $L_{path} = 20 \times \log_{10}(4\pi \times 2000 / 0.05172) = 20 \times \log_{10}(4.856 \times 10^5) = 20 \times 5.686 = 113.7$ dB.
Received power: $P_{rx} = +24 + 18 - 2 - 113.7 + 18 - 2 = -57.7$ dBm.
Converting: $P_{rx} = 10^{-57.7/10} \times 1$ mW $= 1.7 \times 10^{-6}$ mW $= 1.7$ nW.
This is a typical received power level for a short-range microwave link.

# Appendix D

# Unit Prefixes and SI Units

Electrical engineering relies on the International System of Units (SI) to express physical quantities consistently and unambiguously. Because EE spans an extraordinary range of magnitudes — from femtofarad capacitances in integrated circuits to gigawatt power plants, and from nanosecond switching times to megahertz frequencies — the SI prefix system is indispensable for writing compact, readable values. This appendix provides a reference for the SI base and derived units most relevant to electrical engineering, the complete set of SI prefixes, and the conventions for correct usage.

## D.1 SI Base Units

### D.1.1 The Seven Base Units

The SI system is built on seven base units from which all other units are derived. The four most frequently encountered in electrical engineering are the meter (m) for length, the kilogram (kg) for mass, the second (s) for time, and the ampere (A) for electric current. The ampere is defined by fixing the elementary charge to exactly $1.602176634 \times 10^{-19}$ coulombs, making it the fundamental electrical quantity in SI. The kelvin (K) for temperature, the mole (mol) for amount of substance, and the candela (cd) for luminous intensity complete the set but appear less often in core EE work.

| Base Unit | Symbol | Quantity |
|---|---|---|
| meter | m | length |
| kilogram | kg | mass |
| second | s | time |
| ampere | A | electric current |
| kelvin | K | temperature |
| mole | mol | amount of substance |
| candela | cd | luminous intensity |

**Example D.1.1:** A copper wire has a length of 250 m, a cross-sectional area of 2.5 mm$^2$, and carries a current of 15 A. Express the cross-sectional area in base SI units (m$^2$).
**Solution:**
1 mm = $10^{-3}$ m, so 1 mm$^2$ = $(10^{-3})^2 = 10^{-6}$ m$^2$.
Therefore 2.5 mm$^2$ = $2.5 \times 10^{-6}$ m$^2$.

The resistivity of copper is $\rho = 1.68 \times 10^{-8}$ Ω·m, giving $R = \rho L/A = (1.68 \times 10^{-8} \times 250) / (2.5 \times 10^{-6}) = (4.2 \times 10^{-6}) / (2.5 \times 10^{-6}) = 1.68$ Ω.
The voltage drop is $V = IR = 15 \times 1.68 = 25.2$ V.

### D.1.2 The Ampere and Electrical Quantities

The ampere is the SI base unit for electric current, defined since 2019 by fixing the elementary charge $e = 1.602176634 \times 10^{-19}$ C exactly. One ampere equals one coulomb of charge per second (1 A = 1 C/s). All other electrical units derive from the ampere combined with the meter, kilogram, and second: the volt is $kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-1}$, the ohm is $kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-2}$, and the farad is $s^4{\cdot}A^2{\cdot}kg^{-1}{\cdot}m^{-2}$. While engineers rarely need these base-unit decompositions, they are essential for dimensional analysis and verifying that equations are physically consistent.

**Example D.1.2:** Verify that the unit of power (watt) is consistent in the equation $P = V \times I$ by expressing both sides in SI base units.
**Solution:** The watt is defined as $W = kg{\cdot}m^2{\cdot}s^{-3}$. The volt is $V = kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-1}$, and the ampere is A. Therefore $V \times A = (kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-1}) \times A = kg{\cdot}m^2{\cdot}s^{-3} = W$. The units are consistent, confirming that $P = V \times I$ is dimensionally correct.

## D.2 SI Derived Units for Electrical Engineering

### D.2.1 Voltage, Resistance, and Power

The volt (V) is the unit of electric potential difference and electromotive force, defined as one watt per ampere (1 V = 1 W/A). The ohm (Ω) is the unit of electrical resistance, defined as one volt per ampere (1 Ω = 1 V/A). The watt (W) is the unit of power, defined as one joule per second (1 W = 1 J/s). The siemens (S) is the unit of electrical conductance, the reciprocal of the ohm (1 S = 1/Ω = 1 A/V). These four units, together with the ampere, form the core set used in virtually every circuit analysis problem.

| Unit | Symbol | Quantity | Definition |
|---|---|---|---|
| volt | V | voltage, EMF | $W/A = kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-1}$ |
| ohm | Ω | resistance | $V/A = kg{\cdot}m^2{\cdot}s^{-3}{\cdot}A^{-2}$ |
| watt | W | power | $J/s = kg{\cdot}m^2{\cdot}s^{-3}$ |
| siemens | S | conductance | $A/V = \Omega^{-1}$ |

**Example D.2.1:** A 12 V battery is connected to a 4 Ω load. Calculate the current, power, and conductance, verifying the units at each step.
**Solution:**
Current: $I = V/R = 12$ V / 4 Ω = 3 A.
Power: $P = V \times I = 12$ V × 3 A = 36 W, or equivalently $P = I^2 \times R = 9 \times 4 = 36$ W, or $P = V^2/R = 144/4 = 36$ W.
Conductance: $G = 1/R = 1/4$ Ω = 0.25 S.

Verification using conductance: I = G × V = 0.25 S × 12 V = 3 A.

### D.2.2 Capacitance, Inductance, and Charge

The farad (F) is the unit of capacitance, defined as one coulomb per volt (1 F = 1 C/V). The henry (H) is the unit of inductance, defined as one volt-second per ampere (1 H = 1 V·s/A). The coulomb (C) is the unit of electric charge, defined as one ampere-second (1 C = 1 A·s). Practical capacitors range from picofarads (pF) in RF circuits to millifarads (mF) in power supply filters, while practical inductors range from nanohenries (nH) in RF circuits to henries (H) in power transformers. The weber (Wb) for magnetic flux and the tesla (T) for magnetic flux density are also derived from these units.

| Unit | Symbol | Quantity | Definition |
|---|---|---|---|
| farad | F | capacitance | $C/V = s^4 \cdot A^2 \cdot kg^{-1} \cdot m^{-2}$ |
| henry | H | inductance | $V \cdot s/A = kg \cdot m^2 \cdot s^{-2} \cdot A^{-2}$ |
| coulomb | C | charge | $A \cdot s$ |
| weber | Wb | magnetic flux | $V \cdot s = kg \cdot m^2 \cdot s^{-2} \cdot A^{-1}$ |
| tesla | T | magnetic flux density | $Wb/m^2 = kg \cdot s^{-2} \cdot A^{-1}$ |

**Example D.2.2:** A 470 μF capacitor is charged to 25 V. Calculate the stored charge and energy, expressing the results with appropriate SI prefixes.
**Solution:**
Charge: $Q = C \times V = 470 \times 10^{-6}$ F × 25 V = $11.75 \times 10^{-3}$ C = 11.75 mC.
Energy: $E = ½CV^2 = 0.5 \times 470 \times 10^{-6} \times 25^2 = 0.5 \times 470 \times 10^{-6} \times 625 = 146.9 \times 10^{-3}$ J = 146.9 mJ.

### D.2.3 Frequency and Time

The hertz (Hz) is the unit of frequency, defined as one cycle per second (1 Hz = 1 $s^{-1}$). Electrical engineering uses frequencies from millihertz (mHz) in seismology and power system oscillations to terahertz (THz) in optical communications. The radian per second (rad/s) is the unit of angular frequency, related to hertz by $\omega = 2\pi f$. Time-related quantities include the period $T = 1/f$ and the time constant $\tau$ for exponential transients. The second is the SI base unit for time, but engineers commonly work with milliseconds, microseconds, nanoseconds, and picoseconds.

| Unit | Symbol | Quantity | Typical EE Range |
|---|---|---|---|
| hertz | Hz | frequency | mHz to THz |
| rad/s | — | angular frequency | mrad/s to Trad/s |
| second | s | time, period | ps to ks |

**Example D.2.3:** A microcontroller operates at a clock frequency of 168 MHz. Find the clock period in nanoseconds and the angular frequency.
**Solution:**
Period: $T = 1/f = 1/(168 \times 10^6) = 5.952 \times 10^{-9}$ s = 5.952 ns.
Angular frequency: $\omega = 2\pi f = 2\pi \times 168 \times 10^6 = 1.0556 \times 10^9$ rad/s $\approx$ 1.056 Grad/s.
Each clock cycle is approximately 6 ns, meaning instructions that take multiple cycles complete in tens of nanoseconds.

## D.3 SI Prefixes

### D.3.1 Prefix Table

SI prefixes represent powers of 10, allowing any unit to be scaled to a convenient magnitude. The prefixes range from quecto ($10^{-30}$) to quetta ($10^{30}$), though electrical engineering most commonly uses pico through giga. Each prefix is a fixed power of $10^3$ (or $10^{-3}$), with the exception of deci, centi, deca, and hecto, which are rarely used in EE. Prefixes attach directly to the unit symbol without a space or separator: 4.7 kΩ, 100 pF, 50 μH.

| Prefix | Symbol | Factor | Example |
|---|---|---|---|
| tera | T | $10^{12}$ | THz (terahertz) |
| giga | G | $10^9$ | GHz (gigahertz) |
| mega | M | $10^6$ | MΩ (megohm), MW (megawatt) |
| kilo | k | $10^3$ | kΩ (kilohm), kHz (kilohertz) |
| — | — | $10^0$ | V, A, Ω, F, H |
| milli | m | $10^{-3}$ | mA (milliamp), mV (millivolt) |
| micro | μ | $10^{-6}$ | μF (microfarad), μH (microhenry) |
| nano | n | $10^{-9}$ | nF (nanofarad), ns (nanosecond) |
| pico | p | $10^{-12}$ | pF (picofarad), pW (picowatt) |
| femto | f | $10^{-15}$ | fF (femtofarad), fs (femtosecond) |

**Example D.3.1:** Express 0.0000047 F, 2,200,000 Ω, and 0.000000033 H using appropriate SI prefixes.
**Solution:**
0.0000047 F = $4.7 \times 10^{-6}$ F = 4.7 μF.
2,200,000 Ω = $2.2 \times 10^6$ Ω = 2.2 MΩ.
0.000000033 H = $33 \times 10^{-9}$ H = 33 nH.
The convention is to choose the prefix that places the numeric value between 1 and 999, keeping one to three significant digits before the decimal point.

### D.3.2 Prefix Arithmetic

When performing calculations with prefixed units, each prefix must be converted to its power of 10 before arithmetic operations. A common approach is to convert all values to base SI units, perform the calculation, and then apply the appropriate prefix to the result. For frequently combined prefixes, shortcuts are useful: kilo × milli = 1 ($10^3 \times 10^{-3}$), mega × micro = 1, and milli × milli = micro. When dividing, prefixes subtract as exponents: milli / kilo = micro ($10^{-3} / 10^3 = 10^{-6}$). Dimensional analysis with prefixes prevents order-of-magnitude errors, which are among the most common mistakes in engineering calculations.

**Example D.3.2:** Calculate the current through a 4.7 kΩ resistor with 3.3 V across it, and the time constant of that resistor with a 100 nF capacitor.
**Solution:**
Current: $I = V/R = 3.3\text{ V} / 4.7\text{ k}\Omega = 3.3 / (4.7 \times 10^3) = 0.702 \times 10^{-3}\text{ A} = 702\ \mu\text{A}$.
Time constant: $\tau = RC = 4.7\text{ k}\Omega \times 100\text{ nF} = 4.7 \times 10^3 \times 100 \times 10^{-9} = 4.7 \times 10^{-4}\text{ s} = 470\ \mu\text{s}$.
Using the shortcut: $\text{k}\Omega \times \text{nF} = 10^3 \times 10^{-9} = 10^{-6} = \mu\text{s}$, so $\tau = 4.7 \times 100\ \mu\text{s} = 470\ \mu\text{s}$.

### D.3.3 Engineering Notation

Engineering notation is a convention where numbers are expressed with exponents that are multiples of 3, aligning directly with SI prefixes. For example, 0.0022 A is written as $2.2 \times 10^{-3}$ A = 2.2 mA, not $22 \times 10^{-4}$ A or $0.22 \times 10^{-2}$ A. This differs from scientific notation, which allows any exponent. Engineering notation keeps the mantissa between 1 and 999.999 and the exponent as a multiple of 3 (..., $10^{-6}$, $10^{-3}$, $10^0$, $10^3$, $10^6$, ...). Most scientific calculators have an ENG mode that displays results in this format, and it is the standard practice in electrical engineering for expressing results clearly.

**Example D.3.3:** Convert the following scientific notation values to engineering notation with SI prefixes: $5.6 \times 10^{-5}$ V, $3.2 \times 10^7$ Hz, and $8.1 \times 10^{-11}$ F.
**Solution:**
$5.6 \times 10^{-5}\text{ V} = 56 \times 10^{-6}\text{ V} = 56\ \mu\text{V}$.
$3.2 \times 10^7\text{ Hz} = 32 \times 10^6\text{ Hz} = 32\text{ MHz}$.
$8.1 \times 10^{-11}\text{ F} = 81 \times 10^{-12}\text{ F} = 81\text{ pF}$.
In each case, the exponent is shifted to the nearest multiple of 3, and the mantissa is adjusted accordingly.

## D.4 Common Unit Conversions

### D.4.1 Energy and Charge Units

The joule (J) is the SI unit of energy, but electrical engineers also encounter the watt-hour (Wh) and kilowatt-hour (kWh) for energy billing, and the electron-volt (eV) in semiconductor physics. The conversions are: 1 Wh = 3600 J, 1 kWh = $3.6 \times 10^6$ J, and 1 eV = $1.602 \times 10^{-19}$ J. Battery capacity is commonly rated in ampere-hours (Ah) or milliampere-hours (mAh), where 1 Ah = 3600 C. The energy stored in a battery is approximately E = capacity × nominal voltage, so a 3000 mAh battery at 3.7 V stores about 11.1 Wh.

**Example D.4.1:** A lithium-ion battery pack is rated at 5000 mAh and 11.1 V (3S configuration). Calculate the stored energy in watt-hours, joules, and kilowatt-hours.
**Solution:**
Energy: $E = 5000\text{ mAh} \times 11.1\text{ V} = 55{,}500\text{ mWh} = 55.5\text{ Wh}$.
In joules: $55.5 \times 3600 = 199{,}800\text{ J} \approx 200\text{ kJ}$.
In kilowatt-hours: $55.5 / 1000 = 0.0555\text{ kWh}$.
For comparison, this is enough energy to power a 10 W LED light for about 5.5 hours.

### D.4.2 Temperature Scales

Electrical engineering uses three temperature scales: Celsius (°C) for ambient and operating temperatures, Kelvin (K) for absolute temperatures in noise calculations and thermodynamics, and occasionally Fahrenheit (°F) for component datasheets from American manufacturers. The conversions are: $K = °C + 273.15$, $°F = °C \times 9/5 + 32$, and $°C = (°F - 32) \times 5/9$. Thermal resistance in datasheets is given in °C/W (equivalent to K/W since the scale divisions are identical), and junction temperature calculations require consistent units throughout.

**Example D.4.2:** A power MOSFET has a maximum junction temperature of 175°C, the ambient temperature is 45°C, and the thermal resistance from junction to ambient is $R_{\theta JA} = 62°C/W$. Find the maximum allowable power dissipation and express the junction temperature in Kelvin.
**Solution:**
Maximum power: $P_{max} = (T_{j,max} - T_{ambient}) / R_{\theta JA} = (175 - 45) / 62 = 130 / 62 = 2.097\text{ W}$.
Junction temperature in Kelvin: $T_j = 175 + 273.15 = 448.15\text{ K}$.
The MOSFET can dissipate approximately 2.1 W before reaching its maximum rated junction temperature.

# Appendix E

# Getting Started with Python and marimo

The `scripts/` directory in this book contains interactive notebooks built with marimo, a reactive Python notebook framework. These notebooks visualize the example problems from the chapters with graphs and plots, making it easier to build intuition about circuit behavior, signal processing, and other EE concepts. This appendix walks through installing Python and marimo from scratch, running the notebooks, and understanding how they work.

## E.1 Installing Python

### E.1.1 Download and Install

Python is a free, open-source programming language available for Windows, macOS, and Linux. Visit python.org/downloads and download the latest stable release (Python 3.12 or newer is recommended). On Windows, check the box "Add Python to PATH" during installation — this allows you to run Python from any command prompt. On macOS, Python can also be installed via Homebrew with `brew install python`. To verify the installation, open a terminal (Command Prompt on Windows, Terminal on macOS/Linux) and type `python3 --version`, which should display the installed version number.

**Example E.1.1:** Verify that Python is installed and check the version.
**Solution:** Open a terminal and run:
```
python3 --version
```
Expected output: `Python 3.12.x` (or similar). If you see "command not found," Python is not installed or not in your PATH. On Windows, try `python --version` instead (without the `3`).

### E.1.2 Virtual Environments

A virtual environment is an isolated Python installation that keeps project dependencies separate from your system Python. This prevents version conflicts between projects. Create a virtual environment in the `scripts/` directory, activate it, and install dependencies there. The virtual environment lives in a folder (typically `.venv`) that should not be committed to version control.

**Example E.1.2:** Create and activate a virtual environment, then install the required packages.
**Solution:**

```
cd scripts
python3 -m venv .venv
```

Activate it:
- macOS/Linux: `source .venv/bin/activate`
- Windows: `.venv\Scripts\activate`

Your terminal prompt will show `(.venv)` indicating the environment is active. Then install dependencies:

```
pip install -r requirements.txt
```

This installs marimo, numpy, and matplotlib into the virtual environment.

## E.2 Installing and Running marimo

### E.2.1 What is marimo

marimo is a reactive Python notebook where cells automatically re-run when their dependencies change. Unlike traditional Jupyter notebooks (stored as JSON), marimo notebooks are pure Python files (`.py`) that can be version-controlled with Git, run as scripts, and edited with any text editor. Each cell is a Python function decorated with `@app.cell`, and marimo automatically tracks which cells depend on which variables. When you modify a cell, all downstream cells update instantly.

**Example E.2.1:** Install marimo and verify the installation.
**Solution:**

```
pip install marimo
marimo --version
```

Expected output: `marimo 0.19.4` (or newer). marimo is included in the `requirements.txt` file, so if you followed the virtual environment setup in E.1.2, it is already installed.

### E.2.2 Running a Notebook

marimo notebooks can be opened in two modes: edit mode (for modifying code) and run mode (read-only, like a presentation). Edit mode opens the notebook in your browser with an interactive code editor. Run mode displays the outputs without showing the code cells, which is useful for viewing the graphs without the underlying Python code.

**Example E.2.2:** Open the Chapter 7 Circuit Analysis notebook in edit mode.
**Solution:**

```
cd scripts
marimo edit 07_circuit_analysis.py
```

This opens your default web browser with the marimo editor. You will see markdown cells with descriptions and code cells with matplotlib graphs. To run in read-only mode instead:

```
marimo run 07_circuit_analysis.py
```

### E.2.3 Navigating the Interface

The marimo editor displays cells vertically in order. Markdown cells show formatted text, and code cells show Python code with their outputs (graphs, tables, or text) below. The cell dependency graph (accessible from the menu) shows how cells are connected — if cell B uses a variable defined in cell A, then changing cell A will automatically re-run cell B. You can add new cells, reorder cells, or modify existing code. Changes are saved to the `.py` file when you press Ctrl+S (or Cmd+S on macOS).

**Example E.2.3:** Modify a parameter in the Chapter 7 notebook and observe the graph update.
**Solution:** Open `07_circuit_analysis.py` in edit mode. In the RC Circuit cell, change `R_rc = 47e3` to `R_rc = 100e3` (increasing the resistance by about 2.1×). The time constant $\tau$ changes from 0.47 s to 1.0 s, and the charging curve graph automatically updates to show a slower exponential rise. Press Ctrl+Z to undo the change, or Ctrl+S to save.

# E.3 Understanding the Script Structure

## E.3.1 Imports and App Initialization

Every marimo script begins with the same boilerplate: importing the `marimo` module, declaring the version it was generated with, and creating an `App` instance. The first cell imports the common libraries (marimo as `mo`, numpy as `np`, matplotlib.pyplot as `plt`). These imports are available to all subsequent cells through marimo's dependency tracking — any cell that lists `np` or `plt` in its function parameters automatically receives the imported modules.

**Example E.3.1:** Examine the standard header of a marimo script.
**Solution:** The first lines of every script are:

```
import marimo
__generated_with = "0.19.4"
app = marimo.App()
```

Followed by the imports cell:

```
@app.cell
def _():
    import marimo as mo
    import numpy as np
    import matplotlib.pyplot as plt
    return mo, np, plt
```

The `return` statement exports `mo`, `np`, and `plt` for use by other cells.

## E.3.2 Markdown and Code Cells

Markdown cells use `mo.md()` to display formatted text with headings, bold text, and equations. Code cells perform computations and generate graphs. A typical pattern is a markdown cell describing the example, followed by a code cell that computes and plots the result. The `@app.cell` decorator marks each function as a notebook cell, and the function's parameters declare which variables it needs from other cells.

**Example E.3.2:** Identify the cell pattern used for each example problem.
**Solution:** Each example follows this pattern:

```python
# Markdown cell: description
@app.cell
def _(mo):
    mo.md("## Section Title\n\nProblem description...")
    return

# Code cell: computation and graph
@app.cell
def _(np, plt):
    # Set up parameters
    R = 100  # Ω
    # Compute values
    t = np.linspace(0, 1, 1000)
    v = R * np.exp(-t)
    # Create graph
    fig, ax = plt.subplots(figsize=(10, 5))
    ax.plot(t, v)
    ax.set_xlabel("Time (s)")
    fig
    return
```

The last expression in the code cell (`fig`) is what marimo displays as the cell output.

### E.3.3 Running as a Python Script

marimo notebooks can be executed as standard Python scripts from the command line. The `if __name__ == "__main__": app.run()` block at the end of each file enables this. Running as a script executes all cells in dependency order and launches the notebook viewer. This is useful for batch processing, automated report generation, or running on a remote server.

**Example E.3.3:** Run a notebook as a script and export it to HTML.
**Solution:** Run as a script:
`python 07_circuit_analysis.py`
Export to a static HTML file (no Python required to view):
`marimo export html 07_circuit_analysis.py -o circuit_analysis.html`
The HTML file contains all the graphs and text and can be opened in any web browser, shared by email, or hosted on a website.

## E.4 Modifying and Extending Scripts

### E.4.1 Changing Parameters

The simplest way to explore is to modify the numerical parameters in existing cells. Each code cell begins with clearly labeled constants (resistance, capacitance, frequency, etc.) that can be changed to see how the graph responds. Because marimo is reactive, changing a parameter value and pressing Shift+Enter immediately updates the graph. This is a powerful way to build intuition — for example, increasing the damping ratio in an RLC circuit from 0.5 to 0.9 visually shows the transition from oscillatory to nearly critically damped behavior.

**Example E.4.1:** Explore the effect of changing the quality factor in the resonance example.
**Solution:** In `07_circuit_analysis.py`, find the resonance cell and change `R_rlc_res = 10` to `R_rlc_res = 50`. The quality factor Q drops from 31.62 to 6.32 and the resonance peak becomes broader (higher bandwidth). Change it to `R_rlc_res = 1` to see Q = 316.2 with an extremely sharp, narrow peak. Each change updates the graph in real time.

### E.4.2 Adding New Cells

To add a new visualization, insert a new cell in the marimo editor (click the + button between cells). Write a markdown cell for the description and a code cell for the graph. Any variable defined in an existing cell can be reused by listing it as a function parameter. New cells are saved to the `.py` file along with the rest of the notebook.

**Example E.4.2:** Add a cell that plots the current waveform alongside the existing voltage graph in the RC circuit example.
**Solution:** In the marimo editor, add a new cell after the RC charging graph:

```
@app.cell
def _(np, plt):
    R_rc = 47e3
    C_rc = 10e-6
    Vs_rc = 9
    tau_rc = R_rc * C_rc
    t = np.linspace(0, 5 * tau_rc, 500)
    I_rc = (Vs_rc / R_rc) * np.exp(-t / tau_rc)
    fig, ax = plt.subplots(figsize=(10, 5))
    ax.plot(t, I_rc * 1e6, "r-", linewidth=2)
    ax.set_xlabel("Time (s)")
    ax.set_ylabel("Current (μA)")
    ax.set_title("RC Circuit Charging Current")
    ax.grid(True, alpha=0.3)
    fig
    return
```

The graph shows the exponential decay of charging current from 191 μA to near zero.

# Appendix F

# Matrix Operations

Matrix algebra is a fundamental tool in electrical engineering, enabling the compact representation and systematic solution of systems of linear equations that arise throughout circuit analysis, control theory, signal processing, and power systems. Nodal and mesh analysis of circuits with many nodes or loops produce systems of simultaneous equations that are naturally expressed in matrix form. State-space representations of control systems use matrices to describe dynamic behavior, and digital signal processing relies on matrix operations for filter design and spectral analysis. This appendix covers the essential matrix operations used in these applications.

## F.1 Matrix Fundamentals

### F.1.1 Definitions and Notation

A matrix is a rectangular array of numbers arranged in rows and columns, written as A with dimensions $m \times n$ (m rows, n columns). The element in row i, column j is denoted $a_{ij}$. A square matrix has equal rows and columns ($m = n$). A column vector is an $n \times 1$ matrix, and a row vector is a $1 \times n$ matrix. In circuit analysis, the conductance matrix (G) is square and symmetric, the unknown vector (V) contains node voltages, and the source vector (I) contains current source values, forming the system $GV = I$.

**Example F.1.1:** Write the nodal admittance matrix for a circuit with three nodes where $Y_{12} = 0.1$ S, $Y_{13} = 0.2$ S, and $Y_{23} = 0.05$ S connect the node pairs, and a 0.25 S admittance connects node 1 to ground.
**Solution:**
The diagonal entries are the sum of all admittances connected to each node, and the off-diagonal entries are the negative of the admittance between node pairs.
$Y_{11} = 0.25 + 0.1 + 0.2 = 0.55$ S, $Y_{22} = 0.1 + 0.05 = 0.15$ S, $Y_{33} = 0.2 + 0.05 = 0.25$ S.
Off-diagonal: $Y_{12} = Y_{21} = -0.1$, $Y_{13} = Y_{31} = -0.2$, $Y_{23} = Y_{32} = -0.05$.
The matrix is:
$Y = [0.55, -0.1, -0.2; -0.1, 0.15, -0.05; -0.2, -0.05, 0.25]$

### F.1.2 Special Matrices

Several special matrix types appear frequently in EE applications. The identity matrix I has ones on the diagonal and zeros elsewhere; multiplying by I leaves any matrix unchanged ($AI = A$). A diagonal matrix has non-zero entries only on the main diagonal, useful for representing independent scaling of variables. A symmetric matrix equals its own transpose ($A = A^T$), which occurs in admittance and impedance matrices of reciprocal networks. A sparse matrix has mostly zero entries, typical of large power system networks where each bus connects to only a few neighbors.

**Example F.1.2:** Given a $3 \times 3$ identity matrix I and a diagonal matrix D = diag(2, 5, 10), compute $D \times I$ and verify the result.
**Solution:** $D \times I = D$, since multiplying by the identity matrix leaves any matrix unchanged. $D \times I = [2, 0, 0; 0, 5, 0; 0, 0, 10]$. This is the same as D. The diagonal matrix D could represent three independent gain stages of 2, 5, and 10 in a signal processing chain.

## F.2 Matrix Arithmetic

### F.2.1 Addition and Subtraction

Matrices of the same dimensions are added (or subtracted) element by element: if $C = A + B$, then $c_{ij} = a_{ij} + b_{ij}$. Matrix addition is commutative ($A + B = B + A$) and associative ($A + (B + C) = (A + B) + C$). In circuit analysis, superposition of multiple source contributions involves adding the individual voltage or current vectors: $V_{total} = V_1 + V_2 + V_3$, where each $V_k$ is the response to source k acting alone.

**Example F.2.1:** Two current source vectors for a three-node circuit are $I_1 = [0.5, 0, -0.2]^T$ A and $I_2 = [0, 0.3, 0.1]^T$ A. Find the total source vector.
**Solution:** $I_{total} = I_1 + I_2 = [0.5 + 0, 0 + 0.3, -0.2 + 0.1]^T = [0.5, 0.3, -0.1]^T$ A. Node 1 has 0.5 A injected, node 2 has 0.3 A injected, and node 3 has 0.1 A extracted (negative sign indicates current leaving the node).

### F.2.2 Scalar Multiplication

Multiplying a matrix by a scalar k scales every element: if $B = kA$, then $b_{ij} = k \times a_{ij}$. This operation is used when scaling a system of equations, such as converting units or normalizing values. In per-unit systems used in power engineering, all quantities are divided by their base values, which is equivalent to multiplying the system matrix by a scalar reciprocal.

**Example F.2.2:** A resistance matrix (in ohms) is $R = [100, 20; 20, 50]$. Convert to a conductance matrix G by computing $R^{-1}$, and also express R in kilohms using scalar multiplication.
**Solution:** R in kilohms: $R_{k\Omega} = (1/1000) \times R = [0.1, 0.02; 0.02, 0.05]$ kΩ. For the conductance matrix, we need the inverse (covered in F.3.3), but the scalar multiplication simply divides each element by 1000.

### F.2.3 Matrix Multiplication

Matrix multiplication $C = A \times B$ requires that the number of columns in A equals the number of rows in B. If A is $m \times p$ and B is $p \times n$, then C is $m \times n$, with each element $c_{ij} = \Sigma(a_{ik} \times b_{kj})$ for $k = 1$ to p. Matrix multiplication is not commutative ($AB \neq BA$ in general) but is associative ($A(BC) = (AB)C$) and distributive ($A(B + C) = AB + AC$). In circuit analysis, computing $V = Z \times I$ multiplies the impedance matrix by the current vector to obtain node voltages.

**Example F.2.3:** A 2-node circuit has impedance matrix Z = [10, 3; 3, 8] Ω and current vector I = $[2, 1]^T$ A. Find the voltage vector V = Z × I.
**Solution:**
$V_1 = 10 \times 2 + 3 \times 1 = 20 + 3 = 23$ V.
$V_2 = 3 \times 2 + 8 \times 1 = 6 + 8 = 14$ V.
Therefore $V = [23, 14]^T$ V. Node 1 is at 23 V and node 2 is at 14 V with respect to the reference node.

## F.3 Matrix Operations

### F.3.1 Transpose

The transpose of matrix A, written $A^T$, is formed by interchanging rows and columns: the element in row i, column j of A becomes the element in row j, column i of $A^T$. For an m × n matrix, the transpose is n × m. Key properties include $(AB)^T = B^T A^T$ and $(A^T)^T = A$. In EE, the admittance matrix of a reciprocal network (containing only R, L, C, and ideal transformers) is symmetric, meaning $Y = Y^T$. The transpose also appears in least-squares estimation ($A^T A$) used in signal processing and system identification.

**Example F.3.1:** Given A = [1, 4; 2, 5; 3, 6], find $A^T$ and verify that $(A^T)^T = A$.
**Solution:** A is 3 × 2, so $A^T$ is 2 × 3: $A^T$ = [1, 2, 3; 4, 5, 6]. Taking the transpose again: $(A^T)^T$ = [1, 4; 2, 5; 3, 6] = A, confirming the double-transpose property.

### F.3.2 Determinant

The determinant of a square matrix A, written det(A) or |A|, is a scalar value that indicates whether the matrix is invertible (det ≠ 0) or singular (det = 0). For a 2 × 2 matrix [a, b; c, d], the determinant is ad − bc. For larger matrices, the determinant is computed by cofactor expansion along any row or column. In circuit analysis, a zero determinant means the system of equations has no unique solution, indicating a dependent or inconsistent set of equations (for example, a circuit with a floating node or redundant constraint).

**Example F.3.2:** Compute the determinant of the mesh impedance matrix Z = [15, −10; −10, 30] from a two-mesh circuit.
**Solution:**
det(Z) = (15)(30) − (−10)(−10) = 450 − 100 = 350.
Since det(Z) ≠ 0, the system has a unique solution and the mesh currents can be determined.
The determinant appears in Cramer's rule: $I_1 = \det(Z_1)/\det(Z)$ and $I_2 = \det(Z_2)/\det(Z)$, where $Z_1$ and $Z_2$ are matrices with the source vector substituted into the appropriate column.

### F.3.3 Inverse

The inverse of a square matrix A, written $A^{-1}$, satisfies $A \times A^{-1} = A^{-1} \times A = I$ (the identity matrix). A matrix has an inverse if and only if its determinant is non-zero. For a 2 × 2 matrix [a, b; c, d], the inverse is (1/det) × [d, −b; −c, a]. The inverse is the primary tool for solving linear systems: given AX

= B, the solution is $X = A^{-1}B$. In nodal analysis, $YV = I$ is solved as $V = Y^{-1}I$, giving all node voltages simultaneously.

**Example F.3.3:** Solve the system $YV = I$ where $Y = [0.15, -0.05; -0.05, 0.25]$ S and $I = [1, 2]^T$ A.
**Solution:**
$\det(Y) = (0.15)(0.25) - (-0.05)(-0.05) = 0.0375 - 0.0025 = 0.035$.
$Y^{-1} = (1/0.035) \times [0.25, 0.05; 0.05, 0.15] = [7.143, 1.429; 1.429, 4.286]$.
$V = Y^{-1}I$: $V_1 = 7.143 \times 1 + 1.429 \times 2 = 7.143 + 2.857 = 10.0$ V.
$V_2 = 1.429 \times 1 + 4.286 \times 2 = 1.429 + 8.571 = 10.0$ V.
Both nodes are at 10.0 V.

## F.4 Solving Linear Systems

### F.4.1 Gaussian Elimination

Gaussian elimination transforms a system of equations into upper triangular form through a sequence of row operations: swapping rows, multiplying a row by a non-zero scalar, and adding a multiple of one row to another. Once in upper triangular form, the unknowns are found by back-substitution starting from the last equation. This method is computationally efficient ($O(n^3)$ for n unknowns) and is the foundation of numerical linear algebra. Partial pivoting (choosing the largest element as the pivot) improves numerical stability.

**Example F.4.1:** Solve the system $2V_1 + V_2 = 5$ and $V_1 + 3V_2 = 10$ using Gaussian elimination.
**Solution:**
Augmented matrix: [2, 1, 5; 1, 3, 10].
Multiply row 2 by 2: [2, 1, 5; 2, 6, 20].
Subtract row 1 from row 2: [2, 1, 5; 0, 5, 15].
Back-substitute: $5V_2 = 15$, so $V_2 = 3$. Then $2V_1 + 3 = 5$, so $V_1 = 1$.
Verification: $2(1) + 3 = 5$ and $1 + 3(3) = 10$.

### F.4.2 Cramer's Rule

Cramer's rule solves a system of n equations in n unknowns by computing $n + 1$ determinants. For the system $AX = B$, each unknown $x_k = \det(A_k) / \det(A)$, where $A_k$ is the matrix A with its k-th column replaced by B. While computationally expensive for large systems ($O(n \times n!)$ without optimization), Cramer's rule is useful for hand calculations with $2 \times 2$ and $3 \times 3$ systems and provides theoretical insight into when solutions exist and how they depend on the system parameters.

**Example F.4.2:** Use Cramer's rule to solve the two-mesh circuit equations: $15I_1 - 10I_2 = 30$ and $-10I_1 + 30I_2 = 15$.
**Solution:**
$\det(A) = (15)(30) - (-10)(-10) = 450 - 100 = 350$.
For $I_1$: replace column 1 with $[30, 15]^T$. $\det(A_1) = (30)(30) - (-10)(15) = 900 + 150 = 1050$. $I_1 = 1050/350 = 3.0$ A.
For $I_2$: replace column 2 with $[30, 15]^T$. $\det(A_2) = (15)(15) - (30)(-10) = 225 + 300 = 525$. $I_2 =$

$525/350 = 1.5$ A.
Verification: $15(3) - 10(1.5) = 45 - 15 = 30$ and $-10(3) + 30(1.5) = -30 + 45 = 15$.

# F.5 Applications in Electrical Engineering

## F.5.1 Nodal Analysis Matrix Form

The nodal admittance equation YV = I is the standard matrix formulation of nodal analysis. The admittance matrix Y is constructed by inspection: diagonal element $Y_{kk}$ is the sum of all admittances connected to node k, and off-diagonal element $Y_{kj} = Y_{jk}$ is the negative of the admittance between nodes k and j. The voltage vector V contains the unknown node voltages, and the current vector I contains the net current injected at each node from independent current sources. For n non-reference nodes, Y is $n \times n$, V is $n \times 1$, and I is $n \times 1$.

**Example F.5.1:** A circuit has three nodes. A 2 A source feeds into node 1, and a 1 A source feeds into node 3. Conductances are: $G_{12} = 0.5$ S, $G_{13} = 0.2$ S, $G_{23} = 0.1$ S, with $G_{10} = 0.3$ S (node 1 to ground). Set up and solve YV = I.
**Solution:**
$Y = [0.3 + 0.5 + 0.2, -0.5, -0.2; -0.5, 0.5 + 0.1, -0.1; -0.2, -0.1, 0.2 + 0.1] = [1.0, -0.5, -0.2; -0.5, 0.6, -0.1; -0.2, -0.1, 0.3]$.
$I = [2, 0, 1]^T$.
Solving YV = I (by Gaussian elimination or matrix inverse): $V_1 = 10.00$ V, $V_2 = 10.59$ V, $V_3 = 13.53$ V.

## F.5.2 Two-Port Network Parameters

Two-port networks are characterized by $2 \times 2$ parameter matrices that relate the port voltages and currents. The Z-parameters (impedance) relate $[V_1; V_2] = Z \times [I_1; I_2]$, the Y-parameters (admittance) relate $[I_1; I_2] = Y \times [V_1; V_2]$, and the ABCD parameters relate $[V_1; I_1] = [A, B; C, D] \times [V_2; I_2]$. Cascading two-port networks is accomplished by multiplying their ABCD matrices: the overall ABCD matrix of networks in cascade is the product of the individual matrices, making matrix multiplication the natural operation for analyzing transmission line sections, filter stages, and amplifier chains.

**Example F.5.2:** Two transmission line sections have ABCD matrices $M_1 = [1, 50; 0, 1]$ and $M_2 = [1, 0; 0.02, 1]$. Find the overall ABCD matrix of the cascade.
**Solution:**
$M_{total} = M_1 \times M_2$.
$A = 1 \times 1 + 50 \times 0.02 = 1 + 1 = 2$.
$B = 1 \times 0 + 50 \times 1 = 50$.
$C = 0 \times 1 + 1 \times 0.02 = 0.02$.
$D = 0 \times 0 + 1 \times 1 = 1$.
$M_{total} = [2, 50; 0.02, 1]$.
Verification: $\det(M_{total}) = 2 \times 1 - 50 \times 0.02 = 2 - 1 = 1$, confirming a reciprocal network (det = 1 for lossless reciprocal two-ports).

### F.5.3 State-Space Representation

Linear time-invariant systems in control theory are described by the state-space equations: $x'(t) = Ax(t) + Bu(t)$ and $y(t) = Cx(t) + Du(t)$, where x is the state vector, u is the input vector, y is the output vector, and A, B, C, D are constant matrices. The A matrix (state matrix) determines the system dynamics — its eigenvalues are the system poles. The B matrix maps inputs to state derivatives, C maps states to outputs, and D provides direct feedthrough. For an n-th order system with p inputs and q outputs, A is $n \times n$, B is $n \times p$, C is $q \times n$, and D is $q \times p$.

**Example F.5.3:** A series RLC circuit ($R = 2\ \Omega$, $L = 1$ H, $C = 0.5$ F) driven by voltage source u(t) has state variables $x_1$ = capacitor voltage and $x_2$ = inductor current, with output y = capacitor voltage. Write the state-space matrices.
**Solution:**
From circuit equations: $dx_1/dt = (1/C)x_2 = 2x_2$ and $dx_2/dt = (1/L)(u - Rx_2 - x_1) = -x_1 - 2x_2 + u$.
Output: $y = x_1$.
Therefore: $A = [0, 2; -1, -2]$, $B = [0; 1]$, $C = [1, 0]$, $D = [0]$.
The eigenvalues of A (system poles) are found from $\det(A - \lambda I) = 0$: $\lambda^2 + 2\lambda + 2 = 0$, giving $\lambda = -1 \pm j1$, confirming an underdamped second-order system with $\omega_0 = \sqrt{2}$ rad/s and $\zeta = 1/\sqrt{2} \approx 0.707$.

# Appendix G

# TCLab Control Lab

The TCLab (Temperature Control Lab) is an Arduino-based dual heater/temperature shield that makes Chapter 4 control theory physical. Each heater channel drives a small resistive element attached to an aluminium surface, and each sensor returns the surface temperature in degrees Celsius over USB serial. The board's thermal dynamics are slow (time constant $\tau \approx 143$ s) and well-behaved, making it an ideal plant for identifying a first-order plus dead-time (FOPDT) model and verifying a tuned PID controller against analytical predictions. This appendix documents the full identification-to-control workflow: step test, FOPDT fit, Internal Model Control (IMC) tuning, and measured closed-loop results including a multi-setpoint schedule.

## G.1 Hardware and Software

### G.1.1 TCLab Hardware

The TCLab shield mounts on an Arduino Uno and exposes two independently controllable heater channels (Q1, Q2, each 0–100 %) and two temperature sensors (T1, T2, each returning °C). Passive fin cooling is the only heat-removal mechanism — there is no active fan — so the board cools slowly after a heater reduction. The USB serial connection is used for both firmware commands and data streaming; only one process can hold the port at a time. The `tclab` Python package provides a context-manager API (`TCLab()` for the real board, `TCLabModel()` for the built-in simulator) that handles serial framing and unit conversion transparently.

**Example G.1.1:** Verify that the TCLab board is detected and streaming temperatures.
**Solution:** With the board connected via USB, run:
`uv run python3 -c "import tclab; lab = tclab.TCLab(); print(lab.T1, lab.T2); lab.close()"`
Expected output: two temperatures near ambient (e.g., `24.2 23.8`). If the connection fails, check that no other process (Arduino IDE, another Python session) holds the serial port, and that the `tclab` package is installed (`uv pip install tclab`).

### G.1.2 Software Setup

The `tclab_control` package (public repo `sbj-ee/tclab_control`) wraps the board with four command-line tools: `tclab-step` (open-loop step test), `tclab-fit` (FOPDT identification), `tclab-tune` (IMC/Ziegler-Nichols tuning comparison), and `tclab-run` (closed-loop PID on the real board or simulator). The project uses `uv` for dependency management — a virtual environment with no system-level `pip`.

**Example G.1.2:** Install the package and run a 60-second closed-loop test on the simulator.
**Solution:**
```
git clone https://github.com/sbj-ee/tclab_control
cd tclab_control
uv venv
uv pip install -e ".[hardware,dev]"
uv run tclab-run --kp 2.7 --ki 0.018 --kd 25 --setpoint 50 --sim --seconds 60
```
The simulator uses a built-in FOPDT model and requires no hardware. Substitute `--hardware` (or omit `--sim`) to connect to the real board.

## G.2 Plant Identification

### G.2.1 Open-Loop Step Test

A step test drives one heater from 0 % to a fixed power level and records the temperature response at 1-sample-per-second. The test must start from thermal equilibrium (temperature stable at ambient) and run long enough to observe the full exponential rise — typically 5–8 time constants (≥ 600 s for the TCLab). The heater step magnitude should be large enough to give a clear response above sensor noise (50–70 % is typical) while keeping the final temperature safely below the hardware limit (≈ 65 °C for extended operation).

**Example G.2.1:** Run a 600-second step test on the real board at 60 % heater power.
**Solution:**
```
uv run tclab-step --hardware --seconds 600 --power 60 --out data/step_hw.csv
```
The output CSV has columns `t, T1, T2, Q1, Q2`. The temperature starts near ambient (≈ 24 °C), shows a dead-time delay of ≈ 20 s, then rises with a first-order exponential approach to a new steady state.

### G.2.2 FOPDT Model Identification

The FOPDT transfer function is $G(s) = K\,e^{-\theta s} / (\tau s + 1)$, where K (°C/%) is the static gain, τ (s) is the time constant, and θ (s) is the dead time. The identification fits these three parameters to the step-test data using a graphical initialisation followed by a nonlinear least-squares refinement. The static gain is $K = \Delta T_{ss} / \Delta u$; the initial guesses for τ and θ come from the 28.3 % and 63.2 % response times on the tangent-line approximation.

**Example G.2.2:** Fit the FOPDT model to the measured step-test data.
**Solution:**
```
uv run tclab-fit data/step_hw.csv --plot data/fit_hw.png
```
Measured result (TCLab unit, 2026-05-31, 60 % step from 24 °C ambient):
- K = 0.694 °C/% — steady-state gain
- τ = 142.9 s — first-order time constant
- θ = 19.9 s — dead time (sensor and thermal lag)
- RMSE (root mean square error) = 0.43 °C

The steady-state output required to hold any setpoint $T_{sp}$ from ambient $T_0$ is $u_{ss} = (T_{sp} - T_0) / K$.

For $T_{sp}$ = 50 °C and $T_0$ = 23 °C, $u_{ss}$ = 27 / 0.694 = 38.9 %.

## G.3 Controller Design

### G.3.1 IMC Tuning from the FOPDT Model

The Internal Model Control (IMC) λ-tuning rule converts the FOPDT parameters into PID gains with a single closed-loop speed knob λ (the desired closed-loop time constant). Larger λ gives slower, more robust response; smaller λ gives faster response at the cost of sensitivity to model mismatch. The formulas are:

$T_i = \tau + \theta/2$
$T_d = \tau(\theta/2) / T_i$
$K_p = (1/K) \times T_i / (\lambda + \theta/2)$

where λ defaults to max(0.5τ, θ) as a conservative starting point. The discrete controller runs at 1 Hz with derivative-on-measurement (eliminating the setpoint kick) and clamping-based anti-windup (integral frozen when the heater output saturates at 0 or 100 %).

**Example G.3.1:** Compute the IMC PID gains for the measured TCLab plant.
**Solution:** Using K = 0.694, τ = 142.9 s, θ = 19.9 s, and λ = max(0.5 × 142.9, 19.9) = 71.5 s:
$T_i$ = 142.9 + 9.95 = 152.8 s
$T_d$ = 142.9 × 9.95 / 152.8 = 9.3 s
$K_p$ = (1/0.694) × 152.8 / (71.5 + 9.95) = 1.441 × 152.8 / 81.45 = **2.71**
Per-second gains: $k_i = K_p/T_i = 0.0177\ s^{-1}$, $k_d = K_pT_d$ = 25.2 s.

### G.3.2 Ziegler-Nichols Comparison

The Ziegler-Nichols step-response method also derives PID gains from the FOPDT parameters but uses empirical correlations tuned for aggressive response: $K_p = 1.2\tau/(KL)$, $T_i = 2L$, $T_d = 0.5L$, where L is the effective dead time from the tangent-line method (here L ≈ θ = 19.9 s). The result is a much higher proportional gain with a shorter integral time, producing faster rise at the cost of significant overshoot.

**Example G.3.2:** Compare IMC and Ziegler-Nichols gain sets for the TCLab plant.
**Solution:** Computed from the fitted FOPDT:

| Method | $K_p$ | $T_i$ (s) | $T_d$ (s) | Sim overshoot | Sim settle ±2 % |
|---|---|---|---|---|---|
| IMC (λ = 71.5 s) | 2.71 | 152.8 | 9.3 | 0.4 % | 244 s |
| Ziegler-Nichols | 12.46 | 39.7 | 9.9 | 5.1 % | 163 s |

IMC is the preferred starting point for the TCLab because the plant model is accurate and low overshoot is valued over raw speed. Ziegler-Nichols is available via `--tuning zn` for comparison.

## G.4 Measured Results

### G.4.1 Single Setpoint Step Response

The first hardware run drives T1 from ambient to 50 °C using IMC gains with `tclab-run`. The preheat option (`--preheat 80`) runs the heater open-loop at 80 % until the temperature reaches a threshold (default: setpoint − 5 °C), then hands off to the PID with the integral seeded at $u_{ss}$ so the heater output transitions smoothly without a sudden drop.

**Example G.4.1:** Run a single-setpoint closed-loop step to 50 °C on the real board.
**Solution:**

```
uv run tclab-run --from-fit data/step_hw.csv --setpoint 50 --tuning imc \
                 --out data/run_hw.csv --plot data/run_hw.png
```

Measured result (run_hw.csv, 600 samples, 873 s):
- Overshoot: +0.2 °C (0.4 %)
- Steady-state error: −0.03 °C
- Settling ±2 % band: 587 s
- Heater at steady state: ≈ 38 % (matches $u_{ss}$ = 38.9 % from the model)

The real hardware settles approximately 2.4× slower than the FOPDT simulation predicts (587 s measured vs. 244 s simulated), reflecting additional thermal mass and heat losses not captured by the simple first-order model.

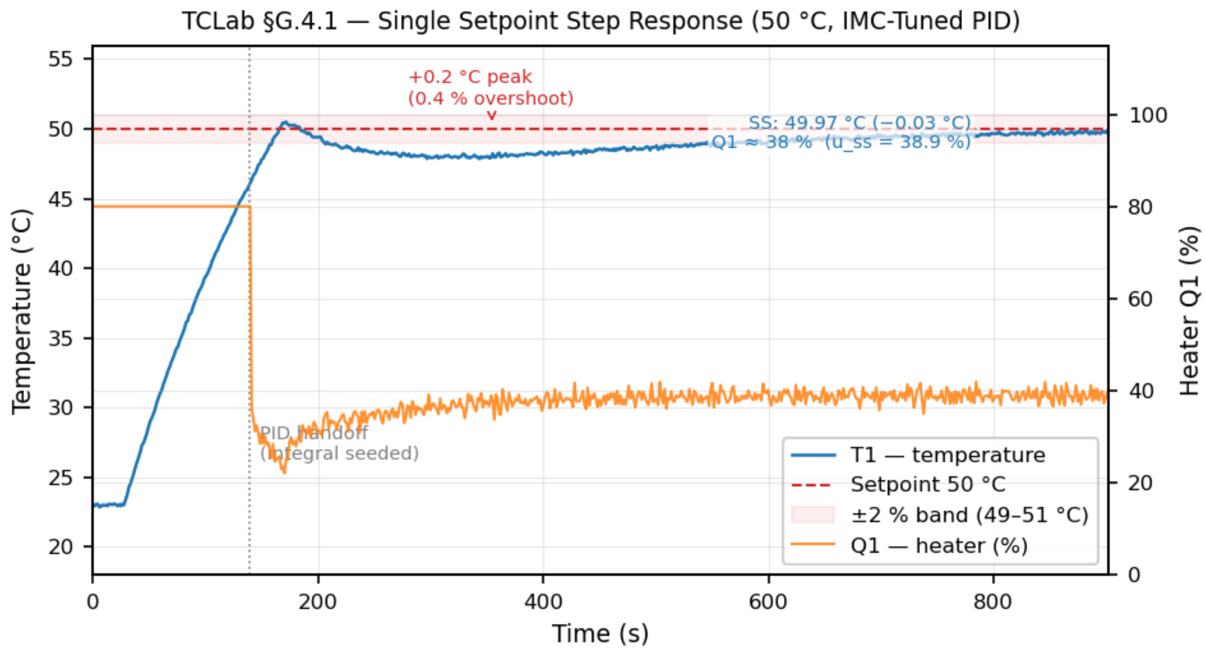



Figure G.4.1: TCLab single setpoint step response — temperature T1 and heater output Q1 vs. time. Preheat at 80 % drives the plant from ambient to 46 °C; the PID (with integral seeded at uss) takes over at the dotted boundary. The ±2 % settling band (49–51 °C) is shaded; steady-state heater output of ≈ 38 % matches the model prediction of uss = 38.9 %.

## G.4.2 Multi-Setpoint Schedule

The multi-setpoint schedule cycles T1 through a sequence of setpoints in a single run. The PID integral is not reset between segments to avoid output bumps at transitions; instead, whenever the new setpoint is above the current temperature, the integral is re-seeded at $u_{ss}$ for the new setpoint (bumpless transfer). The schedule below ramps to 50 °C, cools passively toward 35 °C, then heats back to 40 °C.

**Example G.4.2:** Run the multi-setpoint schedule on the real board.
**Solution:**

```
uv run tclab-run --from-fit data/step_hw.csv \
                 --schedule "50:300" "35:500" "40:500" \
                 --out data/schedule_hw.csv --plot data/schedule_hw.png \
                 --preheat 80:46
```

Measured result (schedule_hw.csv, 2026-06-02, 1365 samples, 1963 s total):

| Segment | Duration | Peak | Overshoot | SS (last 60 s) | SS error |
|---|---|---|---|---|---|
| Preheat | 85 s | — | — | — | — |
| 50 °C | 301 s | 50.5 °C | +0.5 °C | 49.32 °C | −0.68 °C |
| 35 °C | 500 s | 36.0 °C (min) | — | 36.18 °C | +1.18 °C |
| 40 °C | 1077 s | 40.3 °C | +0.3 °C | 40.04 °C | +0.04 °C |

The 35 °C segment did not fully reach its setpoint in 500 iterations because passive cooling from 49.6 °C is limited by natural convection; extending to 650 iterations ($\approx$ 650 s) is sufficient per simulation. The 50 °C and 40 °C segments track their setpoints within ±0.7 °C.

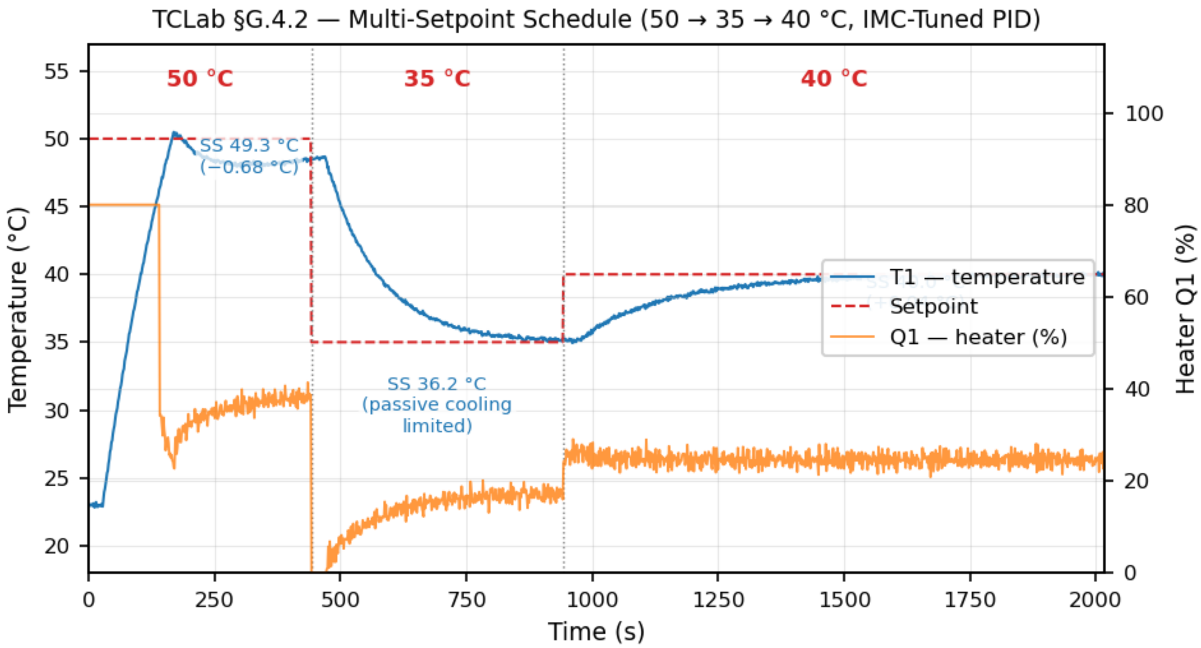



Figure G.4.2: TCLab multi-setpoint schedule — T1 and Q1 across all three segments. The heater cuts to near zero at the 35 °C transition (passive cooling only), then stabilises at $\approx$ 17 % (uss for 35 °C). Bumpless transfer seeds the integral at each upward setpoint transition, eliminating the output dip described in §G.5.1.

## G.5 Implementation Notes

### G.5.1 Bumpless Transfer at Mode Transitions

When control transitions from open-loop (preheat at 80 %) to closed-loop PID, a cold integral (starting at zero) produces an initial output of only $K_p \times e \approx 2.71 \times 5 = 13.6$ % — a sudden drop from 80 %. The delay buffer in the plant still carries 80 % signals for the next $\approx$ 20 s (dead time), but the heater command has already collapsed, causing the temperature to dip $\approx$ 7 °C before the integrator catches up over $T_i = 153$ s.

The fix is to pre-seed the integral at handoff so the first PID output equals the plant steady-state requirement $u_{ss}$, not the preheat power and not zero. Setting $I_0 = u_{ss} - K_p \times e$ gives an initial output exactly equal to $u_{ss}$ regardless of the error at the handoff point, and the integrator then winds naturally as the temperature approaches the setpoint. The same bumpless seed applies at segment boundaries where the setpoint rises above the current temperature — without it, the integral wound down during a cooling segment produces a similarly low initial output when the next heating segment starts.

**Example G.5.1:** Compute the integral seed for a bumpless handoff at T = 46 °C targeting 50 °C.
**Solution:** $u_{ss} = (50 - 23) / 0.694 = 38.9$ %, error $e = 50 - 46 = 4$ °C.
Seed: $I_0 = 38.9 - 2.71 \times 4 = 38.9 - 10.8 =$ **28.1 %**.
First PID output $= K_p \times e + I_0 = 10.8 + 28.1 =$ **38.9 %**.
Without the seed, the first output would be $K_p \times e = 10.8$ %, causing the temperature to continue falling for $\approx T_i = 153$ s before the integrator recovers.

## G.5.2 Model–Hardware Discrepancy

The measured settling time (587 s to ±2 %) is approximately 2.4× the FOPDT simulation prediction (244 s). The FOPDT captures the dominant first-order lag and dead time but omits secondary thermal effects: heat spreading laterally across the aluminium surface, radiation losses, and non-linear convection that depends on temperature difference with ambient. The model is accurate enough for gain selection (the steady-state heater output of ≈ 38 % closely matches $u_{ss} = 38.9$ %) but underestimates the time needed for full settling.

In practice, schedule segment durations should be set from hardware observation rather than simulation. The simulation is useful for comparing tuning rules (IMC vs. Ziegler-Nichols) and for verifying that bumpless transfer initialisation removes the dip before committing to a long hardware run.

**Example G.5.2:** Estimate whether a 300-iteration segment is sufficient to settle at 50 °C on the real board.
**Solution:** The FOPDT simulation predicts settling at ≈ 244 s. Applying the observed 2.4× hardware factor gives a rough estimate of 244 × 2.4 ≈ 586 s required. At 1.44 s per iteration (serial communication overhead adds ≈ 0.44 s to the nominal 1 s sleep), 300 iterations ≈ 432 s real time — short of the ≈ 586 s estimate. The measured SS error at 300 iterations is −0.68 °C, confirming the prediction. Extending to 450 iterations (≈ 648 s) would approach full settling; for a demonstration run, −0.68 °C is acceptable.

# Dictionary of Terms

A reference glossary of technical terms, acronyms, and abbreviations used throughout this book. Terms are listed alphabetically with the chapter or appendix where they are primarily discussed indicated in parentheses.

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**802.1X** — IEEE port-based network access control standard that authenticates devices before granting network access using a three-party model: supplicant (client), authenticator (switch/AP), and authentication server (RADIUS). Supports EAP methods including EAP-TLS, PEAP, and EAP-TTLS. (Ch. 15)

**Aberration (Optical)** — Systematic deviation of light rays from ideal focusing in a lens or mirror system. The five Seidel aberrations are spherical aberration, coma, astigmatism, field curvature, and distortion; chromatic aberration arises from wavelength-dependent refraction. (Ch. 18)

**AC (Alternating Current)** — Electric current that periodically reverses direction, characterized by frequency (Hz) and amplitude. Standard power frequencies are 60 Hz (North America) and 50 Hz (most other regions). (Ch. 1, 7)

**Accelerometer** — A sensor that measures acceleration along one or more axes using a micromachined proof mass and capacitive or piezoresistive detection. MEMS accelerometers are used in smartphones, drones, automotive stability systems, and vibration analysis. (Ch. 11)

**Accuracy** — The closeness of a measured value to the true value of the quantity being measured, often expressed as a percentage of full scale or reading. (Ch. 11)

**ACR (Attenuation-to-Crosstalk Ratio)** — The difference between NEXT and insertion loss in dB on a twisted-pair cable; a positive ACR indicates the desired signal is stronger than crosstalk interference. (Ch. 15)

**Adaptive Filter** — A digital filter whose coefficients are automatically adjusted to minimize an error signal, enabling operation in environments with unknown or time-varying signal statistics. Applications include noise cancellation, echo cancellation, and channel equalization. (Ch. 8)

**ADC (Analog-to-Digital Converter)** — A peripheral or circuit that converts a continuous analog voltage to a discrete digital value, characterized by resolution (bits), sampling rate, and reference voltage. (Ch. 5, 11)

**Admittance (Y)** — The reciprocal of impedance, $Y = G + jB$, measured in siemens (S), where G is conductance and B is susceptance. Used in nodal analysis of AC circuits. (Ch. 7, App. A)

**After-Tax Cash Flow (ATCF)** — The net cash flow remaining after income taxes, computed as ATCF

= BTCF − (BTCF − D) × T, where BTCF is before-tax cash flow, D is depreciation, and T is the tax rate. (Ch. 19)

**Aliasing** — Distortion that occurs when a signal is sampled below the Nyquist rate (less than twice the highest frequency), causing high-frequency components to appear as false low-frequency artifacts. (Ch. 2, 8)

**Allpass Filter** — A filter with unity magnitude response at all frequencies ($|H(e^{j\omega})| = 1$) but frequency-dependent phase. Used for group delay equalization of IIR filters and constructing complementary filter pairs. Zeros are placed at reciprocal conjugate locations of the poles. (Ch. 8)

**AM (Amplitude Modulation)** — A modulation technique that encodes information by varying the amplitude of a carrier wave, used in AM radio broadcasting. (Ch. 2)

**Ampacity** — The maximum current a conductor can carry continuously under specified conditions of use (ambient temperature, insulation type, installation method) without exceeding its rated temperature. (Ch. 1, 14)

**Ampere (A)** — The SI base unit of electric current, defined as the flow of one coulomb of charge per second. (App. D)

**Ampere's Law** — A Maxwell equation relating the magnetic field circulation around a closed path to the enclosed current: $\oint B \cdot dl = \mu_0 I_{enc}$. (Ch. 9)

**Annual Worth (AW)** — The equivalent uniform annual value of all cash flows over a project's life, used to compare alternatives with different service lives without requiring a least common multiple calculation. (Ch. 19)

**Antenna** — A transducer that converts guided electromagnetic energy (in a transmission line or waveguide) to radiated electromagnetic waves in free space, or vice versa. Characterized by gain, directivity, bandwidth, and radiation pattern. (Ch. 9, 16)

**Antenna Gain** — The ratio of radiation intensity in a given direction to the intensity of an isotropic radiator with the same total power, expressed in dBi. Gain = efficiency × directivity. (Ch. 9, 16)

**Anti-Islanding** — A protection requirement for grid-tied inverters mandating that the inverter cease energizing a local circuit within 2 seconds whenever the utility grid goes offline, preventing shock hazards for utility workers and unsynchronized reclosure damage. Detection methods include passive (over/underfrequency, over/undervoltage) and active (frequency shift, reactive power injection) schemes. Required by IEEE 1547 and UL 1741. (Ch. 10)

**Anti-Windup** — A control strategy that prevents the integral term of a PID controller from accumulating error when the actuator is saturated. Common methods include clamping (freezing the integrator) and back-calculation (reducing the integral proportionally to saturation). (Ch. 4)

**Arc Flash** — An electrical explosion caused by current flowing through ionized air between conductors, producing temperatures exceeding 20,000°C. Arc flash hazard analysis per IEEE 1584 calculates incident energy to determine required PPE levels. (Ch. 1)

**Arc-Fault Circuit Interrupter (AFCI)** — A circuit breaker or receptacle that detects dangerous arcing signatures (series and parallel arcs) and trips to prevent fires. Required by NEC 210.12 in most dwelling unit branch circuits. Listed to UL 1699. (Ch. 14)

**Arithmetic Gradient** — A cash flow pattern where amounts increase by a constant dollar value G each period, with the present worth factor (P/G, i, n) converting the gradient to present value. (Ch.

19)

**ARM Cortex-M** — A family of 32-bit RISC processor cores designed by ARM Holdings for embedded microcontroller applications, with variants ranging from the low-power M0 to the high-performance M7. (Ch. 5)

**ASGI (Asynchronous Server Gateway Interface)** — An extension of the WSGI standard for asynchronous Python web applications, supporting WebSockets, HTTP/2, and long-lived connections. Servers include Uvicorn and Hypercorn. (Ch. 15)

**ASK (Amplitude Shift Keying)** — A digital modulation technique that encodes data by varying the amplitude of a carrier signal. Binary ASK (on-off keying) transmits the carrier at full amplitude for a 1 and zero amplitude for a 0. (Ch. 2)

**atan2(y, x)** — The two-argument arctangent function that returns the angle in all four quadrants (−180° to +180°) by using the signs of both arguments, resolving the quadrant ambiguity of the single-argument arctan function. (App. B)

**Augmentation (BESS)** — The strategy of adding new battery modules to an existing BESS installation to compensate for capacity fade over time, maintaining the system's nameplate energy rating throughout its contracted life. (Ch. 10)

**Autocorrelation** — The correlation of a signal with a time-shifted copy of itself, $R_{xx}(\tau)$, revealing periodic structure and signal power. Related to the power spectral density via the Wiener-Khinchin theorem. (Ch. 8)

**Autotransformer** — A transformer with a single winding tapped at an intermediate point, where the primary and secondary circuits share a common winding section. More compact and efficient than a two-winding transformer when the voltage ratio is close to 1:1, but provides no galvanic isolation. (Ch. 1)

**AVR** — An 8-bit RISC microcontroller architecture originally developed by Atmel (now Microchip), widely used in Arduino development boards. (Ch. 5)

**AWG (American Wire Gauge)** — A standardized wire sizing system where smaller gauge numbers indicate larger conductor diameters. Common sizes range from 14 AWG (residential wiring) to 4/0 AWG (large feeders). (Ch. 14)

**Babinet's Principle** — The radiation pattern of a slot antenna is complementary to that of a dipole of the same dimensions, with E-plane and H-plane interchanged. The slot and dipole impedances are related by $Z_{slot} \times Z_{dipole} = \eta^2/4$, where $\eta = 377\ \Omega$. (Ch. 16)

**Back EMF** — The voltage generated by a rotating motor that opposes the applied voltage, proportional to rotor speed. Also called counter-electromotive force (CEMF). (Ch. 12)

**Balun** — A transformer or network that converts between balanced (equal and opposite signals, e.g., dipole antenna) and unbalanced (single-ended, e.g., coaxial cable) transmission systems. Common types include sleeve, folded, and ferrite-core baluns. (Ch. 16)

**Bandgap Energy** — The energy difference between the valence band and conduction band of a semiconductor material, determining its electrical and optical properties. Silicon has a bandgap of 1.12 eV. (Ch. 3)

**Bandwidth** — The range of frequencies over which a system, signal, or medium operates effectively. In networking, often used loosely to mean data transfer rate. (Ch. 2, 8, 15)

**Battery Energy Storage System (BESS)** — A grid-scale or commercial energy storage installation comprising battery modules, a power conversion system (bidirectional inverter), battery management system hierarchy, thermal management, and control systems. Used for peak shaving, frequency regulation, renewable firming, and energy arbitrage. (Ch. 10)

**Battery Management System (BMS)** — An electronic system that monitors and controls a rechargeable battery pack, managing cell voltages, currents, temperatures, state of charge estimation, cell balancing, and fault protection to ensure safe operation and maximize battery life. (Ch. 10)

**Beamforming** — A signal processing technique that uses an antenna array to steer and shape the radiated beam by controlling the amplitude and phase of each element. Analog beamforming uses RF phase shifters; digital beamforming uses per-element DSP for simultaneous multi-beam capability. (Ch. 16)

**Beamwidth** — The angular width of an antenna's main lobe, typically measured at the half-power (−3 dB) points (HPBW). Narrower beamwidth indicates higher directivity. Approximated as HPBW $\approx 70\lambda/D$ for aperture antennas. (Ch. 16)

**Bearing Fault Frequency** — Characteristic vibration frequencies generated by defects in rolling-element bearings. BPFO (ball pass frequency outer race) and BPFI (ball pass frequency inner race) depend on bearing geometry, number of rolling elements, and shaft speed. Monitoring these frequencies with spectrum analysis enables predictive maintenance before catastrophic failure. (Ch. 12)

**Benefit-Cost Ratio (B/C)** — The ratio of a public project's benefits to its costs, both expressed in present worth or annual worth; a project is justified when $B/C \geq 1.0$. (Ch. 19)

**BER (Bit Error Rate)** — The ratio of incorrectly received bits to the total number of bits transmitted, expressed as a probability (e.g., $10^{-12}$). (Ch. 2, 15)

**BGP (Border Gateway Protocol)** — The exterior gateway routing protocol used between autonomous systems on the Internet, making routing decisions based on path attributes and policy rules. (Ch. 15)

**Bilinear Transform** — A mapping $s = (2/T)(z - 1)/(z + 1)$ that converts continuous-time transfer functions to discrete-time equivalents, preserving stability by mapping the left half s-plane to the interior of the unit circle. Introduces frequency warping corrected by pre-warping critical frequencies. (Ch. 8)

**Binary** — Base-2 number system using digits 0 and 1, the fundamental representation of digital data. (Ch. 6)

**BJT (Bipolar Junction Transistor)** — A three-terminal semiconductor device (NPN or PNP) that uses both electron and hole carriers for current amplification, controlled by base current. (Ch. 3)

**BLDC (Brushless DC Motor)** — A DC motor that uses electronic commutation (via Hall sensors or back-EMF sensing) instead of mechanical brushes, providing higher efficiency, longer life, and lower maintenance. (Ch. 12)

**Bode Plot** — A pair of graphs showing the magnitude (in dB) and phase (in degrees) of a system's frequency response as a function of frequency on a logarithmic scale. (Ch. 4, 13)

**Bonding** — The permanent joining of metallic parts to form an electrically conductive path that ensures electrical continuity and the capacity to safely conduct any fault current. (Ch. 14)

**Boolean Algebra** — A mathematical system operating on binary variables (0 and 1) using logical operations AND, OR, and NOT, forming the theoretical foundation of digital logic circuits. (Ch. 6)

**Boost Converter** — A step-up DC-DC switching converter where the output voltage is greater than the input voltage, with $V_{out} = V_{in} / (1 - D)$. (Ch. 10)

**Bootloader** — A small program in protected flash memory that executes before the main application firmware, providing a mechanism for updating firmware over UART, USB, CAN, or wirelessly (OTA) without a debug probe. (Ch. 5)

**Breakeven Analysis** — A technique that determines the critical value of a parameter at which two alternatives are equally attractive or a project's NPV equals zero. (Ch. 19)

**Brewster's Angle** — The angle of incidence $\theta_B = \arctan(n_2/n_1)$ at which reflected light is completely polarized perpendicular to the plane of incidence, with zero p-polarization reflection. (Ch. 18)

**Bridgeless PFC** — A power factor correction topology that eliminates the input diode bridge rectifier, reducing conduction losses by one or two diode drops per half-cycle. The totem-pole bridgeless PFC using GaN HEMTs is the dominant topology for high-efficiency (96%+) server and EV charger power supplies. (Ch. 10)

**Bumpless Transfer** — A technique for transitioning smoothly between control modes (e.g., open-loop to closed-loop PID) by pre-seeding the integrator state so the controller output is continuous at the handoff point. Without it, the integral starts at zero and the output drops suddenly, causing a transient error that takes several integral time constants to recover. (Ch. 4, App. G)

**Buck Converter** — A step-down DC-DC switching converter where the output voltage is less than the input voltage, with $V_{out} = D \times V_{in}$, where D is the duty cycle. (Ch. 10)

**Bypass Diode (PV)** — A diode placed in parallel with a group of series-connected PV cells (typically 18–20 cells per diode) to provide an alternate current path when those cells are shaded or damaged. Without bypass diodes, one shaded cell can eliminate nearly all output from its series string; with bypass diodes, the shaded cell group is bypassed and the power loss is limited to the bypassed section. Modern 60-cell modules include three bypass diodes in the junction box. (Ch. 10)

**C-Rate** — A measure of the rate at which a battery is charged or discharged relative to its capacity. A 1C rate means the battery is fully charged or discharged in one hour; 0.5C takes two hours; 2C takes 30 minutes. Higher C-rates increase stress and reduce cycle life. (Ch. 10)

**Calibration** — The process of comparing an instrument's measurements against a known standard and adjusting or documenting any deviations to ensure accuracy. (Ch. 11)

**CAN Bus (Controller Area Network)** — A robust serial communication protocol developed for automotive applications, using differential signaling and message-based arbitration to allow multiple nodes on a shared bus. (Ch. 5)

**Capacitance (C)** — The ability of a component or system to store electric charge, defined as $C = Q/V$, measured in farads (F). (Ch. 7, 9)

**Capital Recovery Factor (A/P)** — The compound interest factor $[i(1+i)^n]/[(1+i)^n-1]$ that converts a present value into an equivalent series of uniform annual payments. (Ch. 19)

**Carrier Wave** — A high-frequency sinusoidal signal used to carry information through modulation (amplitude, frequency, or phase). (Ch. 2)

**CC/CV (Constant-Current/Constant-Voltage)** — The standard charging algorithm for lithium-ion batteries. The charger first supplies a fixed current (CC phase) until the cell reaches its cutoff voltage, then holds that voltage constant (CV phase) while current tapers exponentially. Charge terminates when the current drops below a threshold (typically C/20 to C/50). (Ch. 10)

**CCD (Charge-Coupled Device)** — An image sensor that accumulates photogenerated charge in potential wells and transfers packets sequentially to a single output amplifier, providing low noise and high uniformity for scientific and medical imaging. (Ch. 18)

**CCS (Combined Charging System)** — An EV DC fast-charging standard that extends the SAE J1772 AC connector with two additional DC pins, enabling charging at up to 350 kW. Uses PLC (Power Line Communication) for high-level digital communication between vehicle and charger. (Ch. 10)

**CDMA (Code Division Multiple Access)** — A multiple access technique where each user's signal is spread across a wide bandwidth using a unique pseudo-random code, allowing simultaneous transmission in the same frequency band. The processing gain provides interference rejection. (Ch. 2)

**Cell Balancing** — The process of equalizing charge levels across series-connected battery cells to prevent capacity loss and safety hazards from cell voltage divergence. Passive balancing dissipates excess energy through resistors; active balancing transfers energy between cells using capacitors, inductors, or transformers for higher efficiency. (Ch. 10)

**Cepstrum** — The inverse Fourier transform of the logarithm of the magnitude spectrum, mapping a signal into the quefrency domain (units of time). Separates slowly varying spectral envelope (low quefrency) from fine harmonic structure (high quefrency). Mel-frequency cepstral coefficients (MFCCs) derived from the cepstrum are the standard feature representation for speech recognition and speaker identification. (Ch. 8)

**CFAR (Constant False Alarm Rate)** — A radar detection technique that adapts the detection threshold to the local noise/clutter environment by estimating noise power from reference cells surrounding the cell under test, maintaining a constant false alarm probability. (Ch. 17)

**CHAdeMO** — A DC fast-charging standard for electric vehicles developed in Japan, using CAN bus communication and supporting bidirectional power flow (V2G). Capable of charging rates up to 400 kW. Being superseded by CCS and NACS in most markets. (Ch. 10)

**Charge Termination** — The criterion used to end a battery charging cycle. For lithium-ion (CC/CV), charging terminates when the taper current drops below C/20 to C/50. For NiMH, a $-\Delta V/\Delta t$ voltage drop of 5–10 mV signals full charge. For lead-acid, termination is based on return current dropping below a threshold during the absorption stage. (Ch. 10)

**Chemical Sensor** — A transducer that converts the concentration of a chemical species into an electrical signal. Types include pH electrodes (glass membrane, −59.16 mV/pH at 25°C), conductivity cells (AC excitation, cell constant $K_{cell}$), dissolved oxygen sensors (electrochemical or optical luminescence quenching), and gas detectors (catalytic bead, electrochemical, MOS, NDIR). (Ch. 11)

**Chiplet** — A small, modular die designed to be combined with other chiplets in an advanced package using 2.5D (interposer) or 3D (direct stacking) integration. Enables mixing process nodes and functions (compute, I/O, memory) for cost and performance optimization. (Ch. 3)

**CIDR (Classless Inter-Domain Routing)** — An IP addressing scheme (RFC 4632) that replaces the original classful system, using variable-length prefix notation (e.g., /24) to specify the network/host boundary at any bit position. A /n prefix contains $2^{(32-n)}$ addresses. CIDR enables route aggregation (supernetting), where contiguous address blocks are advertised as a single shorter prefix to reduce routing table size. (Ch. 15)

**Circuit Breaker** — A switching device designed to automatically interrupt fault currents and isolate faulted sections of the power system, rated by voltage, continuous current, and interrupting capacity. (Ch. 1, 14)

**Clutter (Radar)** — Unwanted radar returns from the environment (ground, sea, precipitation, birds) rather than the targets of interest. Characterized by the normalized clutter coefficient $\sigma^0$ ($m^2/m^2$). Sea clutter exhibits non-Rayleigh statistics with heavy tails, requiring specialized detection algorithms. (Ch. 17)

**CMOS (Complementary Metal-Oxide-Semiconductor)** — A logic family using complementary pairs of PMOS and NMOS transistors, offering near-zero static power dissipation, high noise margins, and rail-to-rail output swings. Dynamic power increases with switching frequency as $P = C_L \times V_{DD}^2 \times f$. (Ch. 3, 6)

**CMRR (Common-Mode Rejection Ratio)** — The ratio of differential gain to common-mode gain in an amplifier, expressed in dB. A higher CMRR indicates better rejection of signals common to both inputs. (Ch. 13)

**Coaxial Cable** — A transmission line consisting of a center conductor, dielectric insulator, conductive shield, and outer jacket. Standard impedances are 50 Ω (RF/data) and 75 Ω (video/CATV). (Ch. 15)

**Coherent Optical Detection** — A fiber-optic receiver technique that mixes the incoming signal with a local oscillator laser to recover both amplitude and phase, enabling advanced modulation formats (DP-QPSK, DP-16QAM) and digital signal processing for dispersion compensation and polarization demultiplexing. (Ch. 18)

**Combinational Circuit** — A digital logic circuit whose output depends only on the current combination of inputs, with no memory or feedback (e.g., adders, multiplexers, decoders). (Ch. 6)

**Commutator** — A mechanical rotary switch in a brushed DC motor that reverses the current direction in the armature windings as the rotor turns, maintaining continuous torque. (Ch. 12)

**Complex Conjugate** — For a complex number $z = a + jb$, its conjugate is $z^* = a - jb$. Multiplying a complex number by its conjugate yields a real number: $z \times z^* = a^2 + b^2$. (App. A)

**Complex Number** — A number of the form $z = a + jb$, where a is the real part, b is the imaginary part, and $j = \sqrt{(-1)}$. Essential for AC circuit analysis. (App. A)

**Complex Power (S)** — The product of phasor voltage and conjugate phasor current, $S = P + jQ$, where P is real power (watts) and Q is reactive power (VAR). Measured in volt-amperes (VA). (App. A)

**Compound Interest** — Interest calculated on both the original principal and accumulated interest, producing exponential growth described by $F = P(1+i)^n$. (Ch. 19)

**Compressive Sensing** — A signal acquisition framework that recovers sparse signals from far fewer measurements than the Nyquist rate requires, using random measurement matrices and $\ell_1$-minimization or greedy algorithms (OMP). Requires $M \geq C \times K \times \log(N/K)$ measurements for a K-sparse signal of length N. (Ch. 8)

**Conducted Emissions** — Electromagnetic interference currents that travel along power and signal cables, regulated from 150 kHz to 30 MHz by CISPR 32 and FCC Part 15. Suppressed using EMI line filters with X-capacitors (differential mode), Y-capacitors (common mode), and common-mode chokes. (Ch. 9)

**Conduit Fill** — The percentage of a conduit's internal cross-sectional area occupied by conductors, limited by NEC Chapter 9, Table 1 to 53% for one conductor, 31% for two, and 40% for three or more, to prevent overheating and facilitate conductor pulling. (Ch. 14)

**Constellation Diagram** — A graphical representation of a digital modulation scheme in the I-Q (in-phase/quadrature) signal space, where each point represents a unique symbol. Used to visualize decision regions, assess signal quality, and compare modulation schemes (BPSK, QPSK, 16-QAM, 64-QAM). (Ch. 2)

**Continuous Compounding** — The theoretical limit of compound interest as compounding periods approach infinity, with future value F = Pe^(rn) and effective rate i = e^r − 1. (Ch. 19)

**Controllability** — A property of a dynamic system indicating whether it is possible to drive the system from any initial state to any desired final state using the available control inputs. Determined by the rank of the controllability matrix $[B, AB, A^2B, \ldots]$. (Ch. 4)

**Convolution** — A mathematical operation that relates the output of an LTI system to its input and impulse response: $y(t) = x(t) * h(t)$. In the frequency domain, convolution becomes multiplication. (Ch. 8)

**Coriolis Flow Meter** — A mass flow sensor that vibrates a tube at its resonant frequency; the Coriolis force from flowing fluid produces a phase shift between inlet and outlet proportional to mass flow rate. Provides ±0.1% accuracy and simultaneous density measurement, making it the standard for hydrocarbon custody transfer. (Ch. 11)

**Coulomb Counting** — A state of charge estimation method that integrates battery current over time ($SOC = SOC_0 - \int I \cdot dt / Q_{rated}$), providing real-time tracking but accumulating drift error from current sensor offset, requiring periodic recalibration via open-circuit voltage or Kalman filtering. (Ch. 10)

**Coulomb's Law** — The fundamental electrostatic law giving the force between two point charges: $F = kq_1q_2/r^2$, where $k \approx 8.99 \times 10^9$ N·m²/C². (Ch. 9)

**Coulombic Efficiency** — The ratio of charge extracted from a battery during discharge to charge inserted during the preceding charge cycle, expressed as a percentage. Healthy lithium-ion cells achieve 99.5–99.9%; values below 99% indicate accelerated side reactions (SEI growth, lithium plating) and degradation. (Ch. 10)

**Counter** — A sequential digital circuit that counts clock pulses and outputs the count in binary, available in up, down, and up/down configurations. (Ch. 6)

**Coupled Transmission Lines** — Two or more transmission lines in close proximity where electromagnetic fields interact, causing crosstalk through capacitive and inductive coupling; characterized by even-mode and odd-mode impedances. (Ch. 9)

**Coupling Coefficient (k)** — A dimensionless parameter ($0 \leq k \leq 1$) indicating how much magnetic flux links two coupled inductors: $k = M/\sqrt{(L_1L_2)}$. A value of $k = 1$ represents perfect coupling (ideal transformer); practical power transformers achieve $k \approx 0.95$–$0.99$. (Ch. 7)

**CPLD (Complex Programmable Logic Device)** — A programmable digital logic device with a simpler, more predictable timing structure than an FPGA, suitable for glue logic and control applications. (Ch. 6)

**CRC (Cyclic Redundancy Check)** — An error detection code based on polynomial division in GF(2), used in Ethernet (CRC-32), USB, and most digital communication protocols. Detects all burst errors of length ≤ r bits for an r-bit CRC. (Ch. 2)

**Critical Path** — The longest combinational logic delay path between any two flip-flops in a synchronous digital circuit, determining the maximum clock frequency: $f_{max} = 1 / (t_{cq} + t_{crit} + t_{su})$. (Ch. 6)

**Cross-Correlation** — A measure of similarity between two signals as a function of a time shift, used in radar (range finding), GPS (code acquisition), and system identification to detect time delays. (Ch. 8)

**Crosstalk** — Unwanted signal coupling between adjacent conductors or circuits caused by mutual capacitance (electric field) and mutual inductance (magnetic field). Measured as NEXT (near-end) and FEXT (far-end). NEXT saturates for coupled lengths exceeding the critical length; FEXT increases linearly with coupled length and vanishes in homogeneous media (stripline). (Ch. 9, 15)

**CT (Current Transformer)** — See *Current Transformer (CT)*.

**Current Mode Control (CMC)** — A converter control method that adds an inner loop sensing inductor current, with peak CMC terminating the switch on-time when current reaches the error amplifier output. Provides cycle-by-cycle current limiting and simplified compensation but requires slope compensation above 50% duty cycle. (Ch. 10)

**Current Sense Amplifier** — A specialized differential amplifier optimized for measuring voltage across low-value shunt resistors (1–100 mΩ) in power lines. High-side types reject common-mode voltages up to hundreds of volts while providing fixed gains (20–200 V/V) with sub-percent accuracy. (Ch. 13)

**Current Sensor** — A device that measures electric current non-invasively (Hall effect sensors, current transformers, Rogowski coils) or with minimal insertion loss (shunt resistors). Hall sensors measure AC and DC; CTs and Rogowski coils measure AC only. (Ch. 11)

**Current Transformer (CT)** — A transformer-based current sensor where the primary is the conductor being measured and the secondary produces a proportional scaled-down current (e.g., 100:5 ratio). Used in power metering and protective relaying. (Ch. 1, 11)

**Cycloconverter** — A power electronics converter that directly converts AC at one frequency to AC at a different (lower) frequency without an intermediate DC link. (Ch. 10)

**Czochralski Method** — The primary technique for growing single-crystal silicon ingots by pulling a seed crystal from a crucible of molten silicon while rotating, producing ingots up to 300 mm in diameter for IC wafer fabrication. (Ch. 3)

**DAC (Digital-to-Analog Converter)** — A circuit that converts a digital value to a corresponding analog voltage or current, used in signal generation and control systems. (Ch. 5, 11)

**Damping Ratio (ζ)** — A dimensionless parameter describing how oscillations in a second-order system decay: underdamped ($\zeta < 1$), critically damped ($\zeta = 1$), or overdamped ($\zeta > 1$). (Ch. 4)

**DAQ (Data Acquisition System)** — A system that converts physical signals (temperature, pressure, voltage) to digital data through sensors, signal conditioning, and ADCs for computer processing. (Ch. 11)

**Data Association** — The process of assigning radar detections to existing tracks in a multi-target tracking system; algorithms include nearest-neighbor, probabilistic data association (PDA), and joint probabilistic data association (JPDA). (Ch. 17)

**Data Logger** — A standalone or PC-based device that records measurement data over time with timestamps for later retrieval and analysis. Standalone loggers use battery power and onboard flash memory for long-term field deployments; PC-based loggers offer higher channel counts and real-time display. (Ch. 11)

**dB (Decibel)** — A logarithmic unit expressing the ratio of two power levels: $dB = 10 \log_{10}(P_2/P_1)$. For voltage ratios: $dB = 20 \log_{10}(V_2/V_1)$. (App. C)

**dBm** — An absolute power level referenced to 1 milliwatt: $dBm = 10 \log_{10}(P/1\ mW)$. 0 dBm = 1 mW, +30 dBm = 1 W. (App. C)

**DC (Direct Current)** — Electric current flowing in one constant direction, produced by batteries, solar cells, and rectified AC power supplies. (Ch. 7)

**DCT (Discrete Cosine Transform)** — A transform that expresses a data sequence as a sum of cosine functions, producing real-valued coefficients with strong energy compaction. The foundation of lossy compression in JPEG (images) and MDCT-based audio codecs (MP3, AAC). (Ch. 8)

**Dead Zone** — In OTDR testing, the minimum distance after a reflective event before the next event can be detected. Event dead zone is typically 0.8–3 m; attenuation dead zone is 5–20 m. (Ch. 15)

**Decimation** — The process of reducing the sampling rate of a discrete-time signal by a factor M, requiring an anti-aliasing lowpass filter before discarding samples to prevent spectral aliasing. (Ch. 8)

**Declining Balance Depreciation** — An accelerated depreciation method that applies a fixed percentage rate to the current book value each year, producing larger charges in early years. (Ch. 19)

**Decoder** — A combinational logic circuit that converts an n-bit binary input to one of $2^n$ output lines, used for memory address decoding and device selection. (Ch. 6)

**Defender** — In replacement analysis, the currently owned asset being evaluated for continued service versus replacement by a challenger. (Ch. 19)

**Delta Connection (Δ)** — A three-phase winding configuration where phase windings are connected end-to-end in a closed triangle. $V_{LL} = V_{phase}$, $I_L = \sqrt{3} \times I_{phase}$. No neutral point available. (Ch. 1)

**Demand Factor** — The ratio of the maximum demand of a system to the total connected load, expressed as a percentage. NEC Article 220 provides demand factors for general lighting (Table 220.42), cooking equipment (Table 220.55), and other loads to reduce service and feeder sizing from the sum of all connected loads. (Ch. 14)

**Dependent Source** — A voltage or current source whose value is controlled by a voltage or current elsewhere in the circuit. Four types exist: VCVS, VCCS, CCVS, and CCCS. Used to model transistor small-signal behavior and amplifier gain. (Ch. 7)

**Depletion Region** — The charge-depleted zone at a PN junction where mobile carriers have diffused away, leaving fixed ionized donor and acceptor atoms that create an internal electric field opposing further diffusion. Its width varies with applied voltage. (Ch. 3)

**Depreciation** — The systematic allocation of an asset's cost over its useful life for accounting and tax purposes, reducing taxable income by a non-cash deduction. (Ch. 19)

**Depth of Discharge (DOD)** — The percentage of a battery's total capacity that has been discharged, where 80% DOD means 80% of the energy has been used and 20% remains. Deeper DOD per cycle accelerates capacity degradation; most BESS installations operate at 80–90% DOD. (Ch. 10)

**Determinant** — A scalar value computed from a square matrix that indicates whether the matrix is invertible (non-zero determinant) and appears in Cramer's rule for solving linear systems. (App. F)

**DFT (Discrete Fourier Transform)** — The Fourier transform applied to a finite-length discrete-time signal, producing a discrete frequency spectrum. Computed efficiently using the FFT algorithm. (Ch. 8)

**DHCP (Dynamic Host Configuration Protocol)** — A network protocol that automatically assigns IP addresses and other configuration parameters to devices joining a network. (Ch. 15)

**Dielectric Constant (Relative Permittivity)** — The ratio $\varepsilon_r = \varepsilon/\varepsilon_0$ that quantifies a material's ability to store electric field energy relative to free space. Determines capacitance, wave velocity, and characteristic impedance in dielectric-filled structures. Common values: air $\approx$ 1, FR-4 $\approx$ 4.3, alumina $\approx$ 9, $BaTiO_3 \approx$ 1000+. (Ch. 9)

**Differential Amplifier** — An amplifier that produces an output proportional to the difference between two input signals while rejecting signals common to both inputs. (Ch. 13)

**Differential Escalation** — The rate at which a specific cost escalates above or below the general inflation rate, requiring separate treatment in economic analysis. (Ch. 19)

**Diffraction** — The bending and spreading of light waves as they pass through apertures or around obstacles. For a circular aperture of diameter D, the angular resolution limit (Rayleigh criterion) is $\theta = 1.22\lambda/D$. (Ch. 18)

**Diffraction Grating** — An optical element with many periodic grooves that disperses light into its constituent wavelengths by diffraction. The grating equation $d \sin(\theta) = m\lambda$ determines diffraction angles. Resolving power $R = mN$, where N is the number of grooves. Used in spectrometers and DWDM demultiplexers. (Ch. 18)

**Diffusion Current** — Current flow in a semiconductor caused by carrier concentration gradients, where carriers move from regions of high concentration to low concentration. Governed by Fick's law: $J_{diff} = qD(dn/dx)$. (Ch. 3)

**Diode** — A two-terminal semiconductor device that allows current to flow primarily in one direction (forward bias), blocking current in the reverse direction until breakdown voltage is reached. (Ch. 3)

**Dipole Antenna** — A fundamental antenna type consisting of two collinear conductors, typically $\lambda/2$ in total length, producing an omnidirectional radiation pattern in the azimuthal plane. Input impedance is $73 + j42.5\ \Omega$ with gain of 2.15 dBi. (Ch. 9, 16)

**Direct Torque Control (DTC)** — A motor control method that directly controls stator flux and electromagnetic torque using hysteresis comparators and an optimal switching table, without coordinate transformations or a rotor position sensor. Achieves 1–2 ms torque response, faster than FOC, but with variable switching frequency. (Ch. 12)

**Directivity** — The ratio of an antenna's maximum radiation intensity to the average radiation intensity over all directions: $D = 4\pi U_{max}/P_{rad}$. Related to gain by $G = \eta D$, where $\eta$ is radiation efficiency. (Ch. 16)

**DMA (Direct Memory Access)** — A hardware peripheral that transfers data between memory and peripherals without CPU involvement, freeing the processor for other tasks and enabling high-throughput data streaming with minimal overhead. (Ch. 5)

**DMM (Digital Multimeter)** — A measuring instrument capable of reading voltage (AC/DC), current (AC/DC), and resistance, with additional functions such as continuity, diode test, and capacitance. (Ch. 11)

**DNP3 (Distributed Network Protocol)** — A SCADA communication protocol widely used in North American electric utilities for polling Remote Terminal Units (RTUs) over serial or TCP/IP links. Supports unsolicited responses, time-stamped events, and secure authentication. (Ch. 1)

**DNS (Domain Name System)** — A hierarchical naming system that translates human-readable domain names (e.g., example.com) to IP addresses, typically using UDP port 53. (Ch. 15)

**Doping** — The intentional introduction of impurity atoms into a semiconductor to modify its electrical conductivity, creating N-type (excess electrons) or P-type (excess holes) material. (Ch. 3)

**Doppler Shift (Radar)** — The frequency change in a radar echo caused by target motion: $f_d = 2v_r/\lambda$, where $v_r$ is the radial velocity. Used for velocity measurement and moving target indication. (Ch. 17)

**Drift Current** — Current flow in a semiconductor caused by the motion of charge carriers under an applied electric field. The drift current density is $J_{drift} = qn\mu E$, where $\mu$ is carrier mobility. (Ch. 3)

**Duty Cycle (D)** — The fraction of a switching period during which the switch is in the ON state, expressed as a ratio (0 to 1) or percentage. Controls the output voltage in DC-DC converters. (Ch. 10)

**DWDM (Dense Wavelength Division Multiplexing)** — An optical transport technology that transmits multiple data channels simultaneously over a single fiber using closely spaced wavelengths (typically 100 GHz or 50 GHz spacing) in the C-band (1530–1565 nm). (Ch. 15)

**Economic Service Life (ESL)** — The number of years of asset retention that minimizes the equivalent uniform annual cost, balancing decreasing capital recovery against increasing operating costs. (Ch. 19)

**Eddy Currents** — Circulating currents induced in a conductor by a time-varying magnetic field (Faraday's law). Cause $I^2R$ losses proportional to $f^2B^2t^2/\rho$ in transformer and motor cores, mitigated by laminations or high-resistivity ferrite. Deliberately exploited in induction heating and eddy current NDT. (Ch. 9)

**EDFA (Erbium-Doped Fiber Amplifier)** — An optical amplifier that uses erbium-doped fiber pumped by a 980 nm or 1480 nm laser to amplify signals in the C-band without optical-to-electrical conversion. Provides 15–30 dB gain with 4–6 dB noise figure. (Ch. 15, 18)

**EDLC (Electrochemical Double-Layer Capacitor)** — A supercapacitor that stores energy electrostatically at the interface between a high-surface-area carbon electrode (activated carbon, 1,000–3,000 $m^2/g$) and an electrolyte. The electric double layer — two charge sheets separated by ~0.3–1 nm —

yields capacitances from a few farads to thousands of farads. Cell voltage is limited to 2.5–2.7 V (organic electrolyte) or 1.0–1.2 V (aqueous). Also called an ultracapacitor. (Ch. 10)

**EEPROM (Electrically Erasable Programmable Read-Only Memory)** — Non-volatile memory that can be electrically erased and reprogrammed byte-by-byte, used for storing configuration data and calibration values. (Ch. 5)

**Effective Interest Rate** — The actual annual yield accounting for compounding frequency, computed as $i_{eff} = (1 + r/m)^m - 1$ where r is the nominal rate and m is the compounding periods per year. (Ch. 19)

**EGC (Equipment Grounding Conductor)** — A conductor that provides a low-impedance ground-fault return path from equipment enclosures back to the electrical source, sized per NEC Table 250.122. (Ch. 14)

**EIRP (Effective Isotropic Radiated Power)** — The product of transmit power and antenna gain: $EIRP = P_t G_t$. Represents the equivalent power an isotropic antenna would need to produce the same peak radiation intensity. (Ch. 16)

**Electric Field (E)** — A vector field representing the force per unit positive charge at each point in space, measured in volts per meter (V/m). (Ch. 9)

**Electric Potential (V)** — The work per unit charge required to move a test charge from infinity to a point in an electric field, measured in volts. (Ch. 9)

**Electromagnetic Compatibility (EMC)** — The ability of electronic equipment to function satisfactorily in its electromagnetic environment without introducing intolerable interference to other devices. Addressed through shielding, filtering, grounding, and circuit layout per FCC/CISPR/IEC standards. (Ch. 9)

**Electromagnetic Wave** — A self-propagating wave of oscillating electric and magnetic fields perpendicular to each other and to the direction of propagation, traveling at the speed of light in vacuum. (Ch. 9)

**Electrostatic Discharge (ESD)** — The sudden transfer of charge between objects at different electrostatic potentials, producing nanosecond-scale current pulses that can damage semiconductor devices. Standard test models include the Human Body Model (HBM, 100 pF/1.5 kΩ), Charged Device Model (CDM), and Machine Model (MM). Protection uses TVS diodes and on-chip clamp circuits. (Ch. 9)

**Encapsulation** — The process of adding protocol headers and trailers as data passes down through the OSI layers, with each layer wrapping the data from the layer above. (Ch. 15)

**Encoder** — A combinational logic circuit that converts $2^n$ input lines into an n-bit binary output code. A priority encoder resolves multiple simultaneous inputs by encoding the highest-priority active input. (Ch. 6)

**Engineering Notation** — A form of scientific notation where the exponent is restricted to multiples of 3, aligning with SI prefixes (kilo, mega, giga, milli, micro, nano). (App. D)

**Entropy** — In information theory, the average information content per symbol emitted by a source: $H(X) = -\Sigma\, p_i \log_2(p_i)$. Represents the theoretical minimum number of bits per symbol for lossless compression. (Ch. 2)

**Equivalent Series Resistance (ESR)** — The lumped resistive component of a capacitor or supercapacitor equivalent circuit, combining electrode contact resistance, electrolyte resistance, and current-collector resistance. ESR limits peak power ($P_{max} = V^2/(4 \cdot ESR)$), causes a voltage step at the start of charge or discharge, and is measured in milliohms for supercapacitors (1–50 mΩ) and microfarad capacitors. (Ch. 10)

**Equivalent Uniform Annual Cost (EUAC)** — The annualized total cost of owning and operating an asset, including capital recovery and operating expenses, used in replacement and life cycle analysis. (Ch. 19)

**Ethernet** — The dominant LAN technology standardized as IEEE 802.3, using CSMA/CD (historically) or full-duplex switching, with frame sizes of 64–1,518 bytes and speeds from 10 Mbps to 400 Gbps. (Ch. 15)

**Euler's Formula** — The mathematical identity $e^{j\theta} = \cos\theta + j\sin\theta$, which links exponential, trigonometric, and complex number representations. Fundamental to phasor analysis. (App. A)

**EUV Lithography (Extreme Ultraviolet)** — A semiconductor patterning technology using 13.5 nm wavelength light with reflective optics, enabling single-exposure patterning of features below 40 nm. Essential for 7 nm and smaller process nodes. (Ch. 3)

**EVSE (Electric Vehicle Supply Equipment)** — The infrastructure that delivers electrical energy to an electric vehicle, including the connector, cable, control electronics, and pilot signal circuitry. Level 1 EVSE uses 120 V household outlets; Level 2 uses 208–240 V dedicated circuits; DC fast chargers bypass the on-board charger entirely. (Ch. 10)

**Fade Margin** — The additional signal level (in dB) included in a link budget beyond the minimum required for reliable operation, accounting for time-varying impairments such as multipath fading, rain attenuation, and atmospheric variations. Typical values range from 10–30 dB depending on reliability requirements. (Ch. 2)

**Fan-Out** — The maximum number of gate inputs that a single logic gate output can drive while maintaining valid voltage levels. Effectively unlimited for CMOS at low frequencies but limited by capacitive loading at high frequencies. (Ch. 6)

**Farad (F)** — The SI derived unit of capacitance. One farad is the capacitance that stores one coulomb of charge at one volt. Practical values range from picofarads (pF) to millifarads (mF). (App. D)

**Faradaic Charge Storage** — Charge storage resulting from reversible electrochemical reactions at or near an electrode surface, including surface redox reactions, intercalation, and underpotential deposition. Pseudocapacitors exploit Faradaic storage to achieve 2–5× higher energy density than purely electrostatic EDLC, at the cost of slightly reduced cycle life and slower response. (Ch. 10)

**Fast Charging** — Charging a battery at rates above 1C, generating significant $I^2R$ heat and risking lithium plating at low temperatures. Requires active thermal management and charge acceptance algorithms that adjust current based on cell temperature and anode potential. (Ch. 10)

**FEC (Forward Error Correction)** — A coding technique that adds redundant data to transmitted messages, allowing the receiver to detect and correct errors without retransmission. Common FEC overheads are 7–25%. (Ch. 15)

**Feedback** — The process of returning a portion of a system's output to its input for comparison with a reference signal, forming the basis of closed-loop control. (Ch. 4)

**Ferromagnetic** — A class of materials (iron, nickel, cobalt, and their alloys) with very high relative permeability ($\mu_r \gg 1$) that exhibit strong magnetization, hysteresis, and magnetic saturation. Used in transformer cores, inductors, and permanent magnets. (Ch. 9)

**FET (Field-Effect Transistor)** — A voltage-controlled semiconductor device where current flow between drain and source is controlled by the electric field from a gate terminal. Includes JFET and MOSFET types. (Ch. 3)

**FFT (Fast Fourier Transform)** — An efficient algorithm for computing the DFT with $O(N \log_2 N)$ complexity instead of $O(N^2)$, making real-time spectral analysis practical. (Ch. 8)

**Fiber Laser** — A laser using rare-earth-doped optical fiber (ytterbium, erbium, thulium) as the gain medium, pumped by semiconductor diodes. Offers excellent beam quality ($M^2 \approx 1$), high efficiency (30–50%), and scalability to >100 kW CW power for industrial cutting and welding. (Ch. 18)

**Fiber Optic Cable** — A cable containing one or more glass or plastic fibers that transmit data as pulses of light, offering high bandwidth, low attenuation, EMI immunity, and electrical isolation. (Ch. 15)

**Field-Oriented Control (FOC)** — An advanced motor control technique that decouples the torque-producing (q-axis) and flux-producing (d-axis) components of stator current in a rotating reference frame aligned with the rotor flux, enabling independent torque and flux control comparable to a separately excited DC motor. Also called vector control. (Ch. 12)

**Fill Factor (FF)** — A figure of merit for a PV cell or module defined as $FF = (V_{mp} \times I_{mp}) / (V_{oc} \times I_{sc})$, where $(V_{mp}, I_{mp})$ is the maximum power point and $V_{oc}$, $I_{sc}$ are open-circuit voltage and short-circuit current. FF measures the "squareness" of the I-V curve; commercial monocrystalline silicon cells achieve FF = 0.75–0.85. Higher series resistance and lower shunt resistance reduce FF. (Ch. 10)

**FinFET** — A three-dimensional transistor structure where the gate wraps around three sides of a tall, narrow silicon fin, providing superior electrostatic control compared to planar MOSFETs. Used from 22 nm through 5 nm process nodes. (Ch. 3)

**FIR Filter (Finite Impulse Response)** — A digital filter whose impulse response has finite duration, implemented as a weighted sum of delayed input samples. Always stable and can provide linear phase. (Ch. 8)

**Fixed-Point Arithmetic** — A number representation using scaled integers with an implied binary point (e.g., Q15 uses 16 bits for values from -1.0 to +0.99997). Used in embedded DSP and FPGAs for digital filter implementation, offering lower cost and power than floating-point at the expense of limited dynamic range and susceptibility to quantization effects. (Ch. 8)

**Flash Memory** — Non-volatile semiconductor memory that can be electrically erased and reprogrammed in blocks, used for program storage in microcontrollers. (Ch. 5)

**FLC (Full-Load Current)** — The current drawn by a motor operating at its rated load, voltage, and frequency. NEC motor circuit sizing is based on FLC table values, not nameplate current. (Ch. 12, 14)

**Flip-Flop** — An edge-triggered bistable memory element that captures and stores one bit of data on the active clock edge. Types include D (data), JK, and T (toggle). (Ch. 6)

**Flux Weakening** — A motor control technique that reduces rotor flux above base speed to prevent voltage saturation of the inverter, enabling constant-power operation at speeds beyond the motor's rated speed. Implemented in FOC by injecting negative d-axis current. IPM motors with high saliency ratios achieve the widest constant-power speed ranges (3:1 to 5:1 typical for EV traction). (Ch. 12)

**FM (Frequency Modulation)** — A modulation technique that encodes information by varying the instantaneous frequency of a carrier signal. FM provides superior noise immunity compared to AM by trading bandwidth for SNR improvement. Used in FM radio broadcasting (88-108 MHz). (Ch. 2)

**FMCW (Frequency Modulated Continuous Wave)** — A radar waveform that sweeps the transmit frequency linearly over a bandwidth B, measuring range from the beat frequency between transmitted and received signals. Range resolution = c/(2B). Dominant technology for automotive radar at 77 GHz. (Ch. 17)

**FOPDT (First-Order Plus Dead Time)** — A transfer function model $G(s) = K e^{-\theta s} / (\tau s + 1)$ characterising a process by its static gain K, time constant $\tau$, and dead time $\theta$. Widely used for industrial process identification because the three parameters can be determined from a single open-loop step test, and they directly feed IMC and Ziegler-Nichols tuning rules. (Ch. 4, App. G)

**FOD (Foreign Object Detection)** — A safety system in wireless charging that detects metallic or living objects on the charging pad that could absorb energy and overheat. Methods include auxiliary sensing coils, quality factor monitoring, and thermal sensors. Required by SAE J2954 and Qi standards. (Ch. 10)

**Fourier Series** — The representation of a periodic signal as an infinite sum of harmonically related sinusoidal components (fundamental frequency plus integer multiples). (Ch. 8)

**Fourier Transform** — A mathematical operation that decomposes a continuous-time signal into its constituent frequencies, transforming from the time domain to the frequency domain. (Ch. 8)

**FPGA (Field-Programmable Gate Array)** — A reconfigurable integrated circuit containing programmable logic blocks, interconnects, and I/O blocks that can implement arbitrary digital logic designs after manufacture. (Ch. 6)

**Free-Space Path Loss (FSPL)** — The attenuation of radio waves due to spreading over distance in free space: $\text{FSPL (dB)} = 20 \log_{10}(4\pi d/\lambda) = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$ (d in km, f in MHz). (Ch. 2, 15)

**Frequency Counter** — An instrument that measures signal frequency by counting zero-crossings within a timed gate interval. Reciprocal counters measure the period using a high-speed internal clock for superior resolution at low frequencies. Accuracy depends on the timebase reference: TCXO (~1 ppm), OCXO (~0.01 ppm), or GPS-disciplined oscillator (~$10^{-12}$ long-term). (Ch. 11)

**Frequency Regulation** — A grid ancillary service in which a BESS or generator adjusts its power output in response to system frequency deviations to maintain the grid at its nominal frequency (50 or 60 Hz). Primary regulation uses droop control; secondary regulation (AGC) restores frequency to the setpoint. (Ch. 10)

**Frequency Response** — The characterization of a system's output magnitude and phase as a function of input frequency, typically displayed as a Bode plot. (Ch. 4, 8, 13)

**Fresnel Equations** — The equations describing the reflection and transmission coefficients of electromagnetic waves at the boundary between two media as a function of incidence angle and polarization. At normal incidence: $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$. The reflection vanishes for TM polarization at Brewster's angle. (Ch. 9)

**Friis Formula** — The equation for calculating the overall noise figure of cascaded stages: $F_{total} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2) + \ldots$, showing that the first stage dominates system noise performance when it has sufficient gain. (Ch. 2)

**FSK (Frequency Shift Keying)** — A digital modulation technique that encodes data by switching the carrier frequency between discrete values. Provides constant-envelope signals robust against nonlinear distortion. GMSK (Gaussian MSK) is used in GSM cellular. (Ch. 2)

**FSPL** — See Free-Space Path Loss. (Ch. 2, 15)

**Fusion Splice** — A permanent, low-loss joint between two optical fibers made by melting and fusing the fiber ends together with an electric arc. Typical loss: 0.02–0.10 dB. (Ch. 15)

**G/T (Gain-to-Noise Temperature Ratio)** — The figure of merit for a satellite or radio receiver system, equal to the antenna gain divided by the system noise temperature (dB/K). Higher G/T indicates better sensitivity. Determines the carrier-to-noise ratio independent of bandwidth. (Ch. 2)

**GAA (Gate-All-Around)** — A transistor architecture where the gate completely surrounds the channel on all four sides using stacked horizontal nanosheets, providing maximum electrostatic control. Adopted at 3 nm and below, succeeding FinFET. (Ch. 3)

**GaAs (Gallium Arsenide)** — A III-V compound semiconductor with a bandgap of 1.42 eV, used for high-frequency amplifiers, RF circuits, and optoelectronics. (Ch. 3)

**Gain Margin** — The additional gain (in dB) that can be applied to a control system before it becomes unstable, measured at the phase crossover frequency where the phase angle equals −180°. Adequate stability typically requires GM $\geq$ 6 dB. (Ch. 4)

**GaN (Gallium Nitride)** — A wide-bandgap semiconductor (3.4 eV) used in high-power, high-frequency power electronics and RF amplifiers due to its high breakdown voltage and electron mobility. GaN HEMTs enable MHz-range switching in compact adapters and chargers. (Ch. 3, 10)

**Gate Driver** — An interface circuit that translates low-power control signals into high-current pulses to rapidly charge and discharge the gate capacitance of power MOSFETs and IGBTs. Bootstrap gate drivers use a capacitor recharged during the low-side on-time to power the high-side driver, while isolated gate drivers use transformer or capacitive barriers to provide galvanic isolation and high dv/dt common-mode rejection. (Ch. 10)

**Gaussian Elimination** — An algorithm for solving systems of linear equations by systematically reducing the augmented matrix to row-echelon form through elementary row operations. (App. F)

**Geometric Gradient** — A cash flow pattern where amounts increase by a constant percentage g each period, modeling inflation-driven cost escalation. (Ch. 19)

**GFCI (Ground Fault Circuit Interrupter)** — A protective device that detects leakage current (typically 5 mA) between hot and neutral conductors and rapidly disconnects the circuit to prevent electric shock. (Ch. 14)

**Goertzel Algorithm** — An efficient O(N)-per-bin method for computing the DFT at a single frequency, implemented as a second-order IIR recursion. More efficient than a full FFT when fewer than $\log_2(2N)$ frequency bins are needed. Primary application is DTMF tone detection in telephony. (Ch. 8)

**GPIO (General Purpose Input/Output)** — Configurable digital pins on a microcontroller that can be individually set as inputs (reading external signals) or outputs (driving external devices). (Ch. 5)

**GPR (Ground-Penetrating Radar)** — A radar technique that transmits UWB impulse waveforms into the ground to detect subsurface interfaces and buried objects. Operates at 10 MHz–4 GHz, trading penetration depth for resolution. Used for utility mapping, road inspection, and archaeology. (Ch. 17)

**Grating Lobe** — An unwanted secondary main beam in an antenna array that appears when element spacing exceeds $\lambda/(1 + |\sin\theta_0|)$, where $\theta_0$ is the scan angle from broadside. At half-wavelength spacing, grating lobes are suppressed for all scan angles. (Ch. 16)

**Ground Loop** — A closed conductive path formed when two points in a system are connected to ground through different paths, allowing magnetically induced or ground-potential-difference currents to flow and couple noise (typically 50/60 Hz hum) into signal circuits. Mitigated by differential signaling, galvanic isolation, or single-point shield grounding. (Ch. 9, 11)

**Grounding** — The intentional connection of an electrical system or equipment to the earth through a grounding electrode, providing a reference potential and a fault current return path for safety. (Ch. 14)

**GSU (Generator Step-Up Transformer)** — A large power transformer at a generating station that steps up generator voltage (typically 11–25 kV) to transmission voltage (115–765 kV). (Ch. 1)

**Gyroscope (MEMS)** — A sensor that measures angular velocity using a vibrating structure and detecting Coriolis forces. Combined with accelerometers in Inertial Measurement Units (IMUs) for motion tracking, navigation, and stabilization. (Ch. 11)

**Hall Effect Sensor** — A magnetic field sensor that produces a voltage proportional to the perpendicular magnetic flux density through a current-carrying semiconductor. Used for position sensing, current measurement, and brushless motor commutation. (Ch. 11)

**Hazardous Location** — An area where flammable gases, vapors, dusts, or fibers may be present in sufficient quantity to create an explosion risk. NEC Article 500 classifies by material type (Class I gases, Class II dusts, Class III fibers), probability of presence (Division 1 normal, Division 2 abnormal), and gas/dust group (A–G). The IEC Zone system (Zones 0/1/2 for gases, Zones 20/21/22 for dusts) provides an alternative classification accepted in NEC Article 505. (Ch. 14)

**Henry (H)** — The SI derived unit of inductance. One henry is the inductance that produces one volt of EMF when the current changes at one ampere per second. (App. D)

**Hertz (Hz)** — The SI derived unit of frequency, equal to one cycle per second. (App. D)

**Hexadecimal** — Base-16 number system using digits 0–9 and letters A–F, providing a compact representation of binary data (each hex digit = 4 bits). (Ch. 6)

**Hilbert Transform** — An operation that shifts every frequency component of a real signal by ±90°, producing the quadrature companion signal. Combined with the original to form the analytic signal $z(t) = x(t) + j\hat{x}(t)$, from which instantaneous envelope $A(t) = |z(t)|$ and instantaneous frequency can be extracted. Used in AM/FM demodulation, vibration analysis, and single-sideband modulation. (Ch. 8)

**HMI (Human-Machine Interface)** — The operator console in a SCADA system that displays real-time power system data (one-line diagrams, alarm lists, trend charts) and provides controls for supervisory commands such as opening breakers or adjusting transformer taps. (Ch. 1)

**Hold Time ($t_h$)** — The minimum time a data input must remain stable after the active clock edge of a flip-flop to ensure reliable capture. Violating hold time can cause metastability. (Ch. 6)

**Horn Antenna** — An aperture antenna formed by flaring the end of a waveguide, providing moderate gain (10–25 dBi), wide bandwidth (> 2:1), and predictable performance. Used as feeds for parabolic reflectors, calibration standards, and EMC testing. (Ch. 16)

**HVDC (High-Voltage Direct Current)** — A power transmission technology that converts AC to DC for long-distance transmission and back to AC at the receiving end. Preferred for submarine cables, very long overhead lines (>600 km), and asynchronous system interconnections. Uses LCC (thyristor) or VSC (IGBT) converter technology. (Ch. 1)

**Hysteresis (Magnetic)** — The phenomenon where the magnetization of a ferromagnetic material depends on its history, forming a B-H loop when the applied field is cycled. Characterized by remanence $B_r$ (residual flux density) and coercivity $H_c$ (field to demagnetize). (Ch. 9)

**I2C (Inter-Integrated Circuit)** — A two-wire synchronous serial communication bus (SDA for data, SCL for clock) supporting multiple masters and slaves on the same bus with 7-bit or 10-bit addressing. (Ch. 5)

**IDF (Intermediate Distribution Frame)** — A floor-level telecommunications room in a building's structured cabling system, housing access-layer switches and patch panels that serve the work areas on that floor. (Ch. 15)

**IE Efficiency Classes** — International efficiency classification for electric motors defined by IEC 60034-30-1: IE1 (standard), IE2 (high), IE3 (premium), IE4 (super-premium), and IE5 (ultra-premium). NEMA Premium aligns with IE3. Higher classes use more copper and better steel to reduce losses. (Ch. 12)

**IEC 61850** — An international standard for substation automation that defines a common data model and communication services for IEDs, including GOOSE (Generic Object Oriented Substation Event) messaging for high-speed peer-to-peer tripping and interlocking with latencies under 4 ms. (Ch. 1)

**IED (Intelligent Electronic Device)** — A microprocessor-based device in a substation (protective relay, meter, breaker controller) that acquires data, communicates via protocols such as DNP3 or IEC 61850, and can execute local control logic. (Ch. 1)

**IGBT (Insulated Gate Bipolar Transistor)** — A power semiconductor switch that combines the voltage-controlled gate drive of a MOSFET with the high-current, low-saturation-voltage characteristics of a BJT. Used in motor drives and inverters. (Ch. 10)

**IIR Filter (Infinite Impulse Response)** — A digital filter that uses feedback (recursive computation), producing an impulse response of theoretically infinite duration. More efficient than FIR for the same frequency response but can be unstable. (Ch. 8)

**IMC (Internal Model Control)** — A PID tuning method that derives gains from an identified plant model and a single closed-loop speed parameter $\lambda$. For a FOPDT plant the formulas are $T_i = \tau + \theta/2$, $K_p = T_i / [K(\lambda + \theta/2)]$, $T_d = \tau(\theta/2)/T_i$. Increasing $\lambda$ gives a slower, more robust controller; $\lambda = 0.5\tau$ is a common default. (Ch. 4, App. G)

**Imaginary Unit (j)** — The square root of $-1$, denoted j in electrical engineering (i in mathematics) to avoid confusion with current. Forms the basis of complex number representation for AC analysis. (App. A)

**Impedance (Z)** — The total opposition to current flow in an AC circuit, combining resistance and reactance as a complex quantity: $Z = R + jX$, measured in ohms ($\Omega$). (Ch. 7, App. A)

**Impedance Matching Network** — A circuit (L-network, pi-network, T-network, or transformer) inserted between a source and load to maximize power transfer by transforming the load impedance to the complex conjugate of the source impedance; commonly designed using Smith charts. (Ch. 9)

**Impulse Response** — The output of a system when the input is a Dirac delta function (unit impulse). For LTI systems, the impulse response completely characterizes the system's behavior. (Ch. 4, 8)

**Incident Energy** — The thermal energy per unit area ($cal/cm^2$ or $J/cm^2$) at a specific working distance from an arc flash, used to determine the required PPE category per NFPA 70E and IEEE 1584. (Ch. 1)

**Incremental Rate of Return** — The internal rate of return earned on the additional investment required by a more expensive alternative, used to compare mutually exclusive options. (Ch. 19)

**Inductance (L)** — The property of a circuit element that opposes changes in current by storing energy in a magnetic field. $L = N\Phi/I$, measured in henries (H). (Ch. 9)

**Induction Motor** — An AC motor where the rotor current is induced by the rotating magnetic field of the stator, causing the rotor to turn at slightly less than synchronous speed (slip). The most common industrial motor type. (Ch. 12)

**Instrumentation Amplifier** — A precision differential amplifier with very high input impedance, high CMRR, and adjustable gain set by a single resistor. Typically implemented with three op-amps. (Ch. 13)

**Insulation Class** — A thermal rating system for motor winding insulation defining maximum allowable hot-spot temperature: Class B (130°C), Class F (155°C), Class H (180°C). Insulation life approximately halves for every 10°C above the rated temperature (Arrhenius/Montsinger rule). (Ch. 12)

**Integrator** — An op-amp circuit whose output is proportional to the time integral of the input signal, implemented with a capacitor in the feedback path: $V_{out} = -(1/RC) \int V_{in}\, dt$. (Ch. 13)

**Internal Rate of Return (IRR)** — The interest rate at which a project's net present value equals zero, representing the effective yield of the investment; accepted when IRR exceeds the MARR. (Ch. 19)

**Interrupt** — A hardware or software signal that causes a processor to suspend its current execution, save context, and execute a designated interrupt service routine (ISR). (Ch. 5)

**Inverse Z-Transform** — The operation that recovers a discrete-time sequence x[n] from its Z-transform X(z), typically performed by partial fraction expansion of X(z)/z followed by table lookup, or by long division (power series expansion) for direct computation of output samples. (Ch. 8)

**Inverter** — In power electronics, a circuit that converts DC power to AC power. In digital logic, a NOT gate that outputs the complement of its input. (Ch. 6, 10)

**Investment Tax Credit (ITC)** — A dollar-for-dollar reduction in tax liability based on a percentage of a qualifying investment, commonly applied to renewable energy systems. (Ch. 19)

**Ion Implantation** — The primary doping method in modern semiconductor fabrication, accelerating dopant ions (boron, phosphorus, arsenic) into the silicon substrate with precise control of dose ($atoms/cm^2$) and depth profile through acceleration energy. (Ch. 3)

**IP Address** — A numerical label assigned to each device on a network for identification and routing. IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses. (Ch. 15)

**IPv4** — Internet Protocol version 4, using 32-bit addresses written in dotted-decimal notation (e.g., 192.168.1.1), providing approximately 4.3 billion unique addresses. (Ch. 15)

**IPv6** — Internet Protocol version 6, using 128-bit addresses written in colon-hexadecimal notation (e.g., 2001:db8::1), providing $3.4 \times 10^{38}$ addresses. (Ch. 15)

**Irradiance** — The radiant power incident per unit area on a surface, measured in $W/m^2$. Follows the inverse-square law for point sources: $E = I/r^2$. (Ch. 18)

**JTAG (Joint Test Action Group)** — A debug and boundary scan interface (IEEE 1149.1) using four signal lines (TDI, TDO, TCK, TMS) that provides real-time access to processor state, breakpoints, memory inspection, and flash programming. (Ch. 5)

**Jury Stability Criterion** — An algebraic test for discrete-time system stability analogous to Routh-Hurwitz for continuous-time systems. Constructs a triangular array from the characteristic polynomial coefficients to determine whether all roots lie inside the unit circle. For second-order systems, reduces to the stability triangle conditions: $1 + a_1 + a_2 > 0$, $1 - a_1 + a_2 > 0$, $|a_2| < 1$. (Ch. 8)

**K-Factor Transformer** — A transformer designed to withstand the additional heating caused by harmonic currents from nonlinear loads. The K-factor $K = \Sigma(I_h{}^2 \times h^2)/\Sigma(I_h{}^2)$ quantifies harmonic heating; standard ratings are K-1, K-4, K-13, and K-20. Features include oversized neutrals, reduced core flux density, and transposed windings to minimize eddy current losses. (Ch. 14)

**Kalman Filter** — An optimal recursive estimator for linear systems with Gaussian noise that maintains a state estimate and covariance, updating each with new measurements via a predict-correct cycle. Used in radar target tracking, GPS navigation, IMU sensor fusion, and battery state estimation. The extended Kalman filter (EKF) handles nonlinear systems via Jacobian linearization. (Ch. 8, 17)

**Karnaugh Map (K-map)** — A graphical method for simplifying Boolean expressions by arranging truth table entries in a grid that allows visual identification of adjacent minterms for grouping. (Ch. 6)

**KCL (Kirchhoff's Current Law)** — The principle that the algebraic sum of all currents entering and leaving any node in a circuit equals zero. (Ch. 7)

**kcmil** — Thousand circular mils, a unit of area for large electrical conductors (formerly MCM). One kcmil equals the area of a circle with a diameter of one mil (0.001 inch), multiplied by 1,000. (Ch. 14)

**KVL (Kirchhoff's Voltage Law)** — The principle that the algebraic sum of all voltages around any closed loop in a circuit equals zero. (Ch. 7)

**L-Network** — A two-element impedance matching network using one series and one shunt reactive component (inductor or capacitor) to transform a load impedance to a desired source impedance. The matching $Q = \sqrt{(R_{high}/R_{low} - 1)}$ determines both the element values and the matching bandwidth. (Ch. 9)

**LAG (Link Aggregation Group)** — A technique (IEEE 802.3ad/802.1AX) that bundles multiple physical network links into a single logical link to increase bandwidth and provide redundancy. (Ch. 15)

**Lag Compensator** — A frequency-domain controller $G_c(s) = K(s + z)/(s + p)$ with $z > p$ that boosts low-frequency gain to reduce steady-state error without significantly affecting phase margin at the crossover frequency. (Ch. 4)

**Laplace Transform** — An integral transform that converts a time-domain function f(t) into a complex-frequency-domain function F(s), where $s = \sigma + j\omega$. Enables algebraic analysis of differential equations and transfer functions. (Ch. 4, 8)

**Laplace's Equation** — The second-order PDE $\nabla^2 V = 0$ governing electric potential in charge-free regions. Combined with boundary conditions, it uniquely determines the field distribution. Solved analytically for simple geometries and numerically (FEM, FDM, BEM) for complex structures like PCB traces, high-voltage bushings, and IC interconnects. (Ch. 9)

**Laser** — Light Amplification by Stimulated Emission of Radiation. A device that produces coherent, monochromatic light through stimulated emission in a gain medium within an optical resonator. Types include gas (HeNe, $CO_2$), solid-state (Nd:YAG), semiconductor (laser diodes), and fiber lasers. (Ch. 18)

**Latch** — A level-sensitive bistable memory element that is transparent (passes input to output) when the enable signal is active. Simpler than a flip-flop but susceptible to timing hazards. (Ch. 6)

**Latency** — The total time delay for data to travel from source to destination, comprising propagation, serialization, processing, and queuing delays. (Ch. 15)

**LCR Meter** — An instrument that measures inductance (L), capacitance (C), and resistance (R) of passive components by applying a sinusoidal test signal and computing impedance $Z = V/I$. Also reports dissipation factor (D), quality factor (Q), and ESR. Four-terminal Kelvin sensing eliminates lead resistance errors for low-impedance measurements. (Ch. 11)

**LDO (Low-Dropout Regulator)** — A linear voltage regulator that can operate with a very small input-to-output voltage differential (100-300 mV), using a PMOS or PNP pass transistor. Preferred for battery-powered and noise-sensitive applications. (Ch. 3)

**Lead Compensator** — A frequency-domain controller $G_c(s) = K(s + z)/(s + p)$ with $p > z$ that adds positive phase shift near the geometric mean frequency $\sqrt{(zp)}$ to increase phase margin and improve transient response. (Ch. 4)

**LED (Light-Emitting Diode)** — A semiconductor diode that emits photons when forward biased, with the emitted wavelength determined by the bandgap of the semiconductor material. Used in indicators, displays, and general illumination with efficacies up to 200 lm/W. (Ch. 3)

**Levelized Cost of Energy (LCOE)** — The constant per-unit cost of electricity that equates the present worth of all lifetime costs with all lifetime energy production, expressed in $/MWh. (Ch. 19)

**Levelized Cost of Storage (LCOS)** — The constant per-unit cost of energy delivered from a storage system over its lifetime, calculated as LCOS = Total Lifetime Costs / Total Lifetime Energy Delivered, expressed in $/MWh. Accounts for capital cost, O&M, degradation, round-trip efficiency, and replacement costs. (Ch. 10)

**LiDAR (Light Detection and Ranging)** — A remote sensing technology that uses pulsed or FMCW laser light (typically 905 nm or 1550 nm) to measure range with centimeter-level accuracy, generating dense 3D point clouds. Key technology for autonomous vehicles, topographic mapping, and forestry. (Ch. 17)

**Life Cycle Cost (LCC)** — The total present worth of all costs associated with an asset over its entire life, including acquisition, operation, maintenance, and disposal. (Ch. 19)

**Linear Motor** — An electric motor that produces force and motion in a straight line rather than rotation, created by "unrolling" a rotary motor stator into a flat or tubular geometry. Linear induction motors (LIMs) produce a traveling magnetic field at synchronous velocity $v_s = 2\tau_p f$, while linear synchronous motors (LSMs) use permanent magnets or electromagnets on the translator for zero-backlash, sub-micron positioning. Applications include maglev trains, semiconductor lithography stages, and baggage handling systems. (Ch. 12)

**Linear Regulator** — A voltage regulator that maintains a stable output by dissipating excess voltage as heat across a series pass element. Simple and low-noise but limited in efficiency to $V_{out}/V_{in}$. Includes fixed (LM78xx) and adjustable (LM317) types. (Ch. 3)

**Linearization** — The process of correcting a nonlinear sensor transfer characteristic to produce an output proportional to the measured quantity. Methods include lookup tables with interpolation, polynomial curve fitting (e.g., Steinhart-Hart equation for thermistors), and analog hardware shaping circuits. (Ch. 11)

**Link Budget** — An accounting of all gains and losses in a communication link from transmitter to receiver, expressed in dB, used to determine whether the received signal meets the minimum required level. (Ch. 2, 15, App. C)

**LISN (Line Impedance Stabilization Network)** — A measurement device that presents a defined impedance (typically 50 Ω) to equipment under test for conducted EMI testing, isolating the measurement from the utility supply while coupling the device's conducted emissions to a spectrum analyzer or EMI receiver per CISPR standards. (Ch. 10)

**Lithium Plating** — The deposition of metallic lithium on the surface of a battery anode when the charging rate exceeds the anode's ability to intercalate lithium ions, particularly at low temperatures. Permanently reduces capacity, increases internal resistance, and can create dendrites that cause internal short circuits. (Ch. 10)

**Lithium-Ion Capacitor (LIC)** — A hybrid energy storage device combining a pre-lithiated graphite anode (battery-like intercalation) with an activated carbon cathode (EDLC-like double-layer storage). The asymmetric design raises the cell voltage to 2.2–3.8 V, achieving 3× the energy density of a conventional EDLC while retaining much of its peak-power advantage. Targets applications — elevator energy recovery, regenerative braking, industrial UPS — that require more energy than standard EDLC but more power than lithium-ion batteries. (Ch. 10)

**LLC Resonant Converter** — An isolated DC-DC converter using a resonant tank of two inductors and one capacitor to achieve zero-voltage switching across the full load range. Widely used in server power supplies, telecom rectifiers, and EV chargers for its high efficiency (95–97%). (Ch. 10)

**LMS (Least Mean Squares)** — The most widely used adaptive filtering algorithm, updating filter weights using stochastic gradient descent: $w[n+1] = w[n] + 2\mu e[n]x[n]$. Simple, low computational cost, with convergence controlled by the step size $\mu$. (Ch. 8)

**Load Cell** — A transducer that converts mechanical force or weight into an electrical signal, typically using strain gauges arranged in a Wheatstone bridge configuration. (Ch. 11)

**Load Flow (Power Flow)** — The fundamental power system analysis that determines steady-state voltage magnitudes and angles at all buses and power flows through all branches, solved iteratively using Newton-Raphson or Gauss-Seidel methods. (Ch. 1)

**Locked-Rotor Current** — The current drawn by a motor at the instant of starting when the rotor is stationary, typically 4–8 times the full-load current. (Ch. 12, 14)

**Logarithmic Amplifier** — An op-amp circuit that uses a BJT or diode in the feedback path to produce an output proportional to the logarithm of the input current or voltage. Provides a useful dynamic range of 3–7 decades and outputs approximately 59.16 mV/decade (= 2.303 × $V_T$ = 2.303 × 25.69 mV) at 25°C before scaling. (Ch. 13)

**Logic Gate** — A fundamental digital circuit that implements a Boolean function (AND, OR, NOT, NAND, NOR, XOR, XNOR) with one or more inputs and one output. (Ch. 6)

**Loss Tangent (tan δ)** — The ratio of the imaginary to real parts of complex permittivity ($\varepsilon''/\varepsilon'$), quantifying dielectric energy dissipation in an alternating electric field. Low tan δ materials (PTFE ≈ 0.0002, Rogers ≈ 0.003) are preferred for RF circuits; FR-4 (tan δ ≈ 0.02) is adequate below ~1 GHz. (Ch. 9)

**LTI System (Linear Time-Invariant)** — A system that obeys the principles of superposition (linearity) and time-shift invariance, allowing complete characterization by its impulse response or transfer function. (Ch. 8)

**Luenberger Observer** — A state estimator that reconstructs unmeasured state variables from output measurements using the equation $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$, where L is the observer gain vector. Requires the system to be observable. (Ch. 4)

**LVDT (Linear Variable Differential Transformer)** — A position sensor that uses electromagnetic coupling between a movable core and three coils (one primary, two secondary) to produce an output voltage proportional to core displacement. (Ch. 11)

**MAC Address** — A 48-bit hardware address permanently assigned to a network interface, written as six pairs of hexadecimal digits (e.g., 00:1A:2B:3C:4D:5E). The first 24 bits are the OUI identifying the manufacturer. (Ch. 15)

**MACRS** — Modified Accelerated Cost Recovery System, the U.S. tax depreciation method using prescribed recovery periods, declining balance switching to straight-line, and a half-year convention. (Ch. 19)

**Magnetic Circuit** — An analogy to electric circuits where magnetic flux Φ (analogous to current) is driven by magnetomotive force $\mathcal{F} = NI$ (analogous to voltage) through reluctance $\mathcal{R} = l/(\mu A)$ (analogous to resistance). Used to design inductor and transformer cores, with air gaps added to store energy, linearize inductance, and increase saturation current. (Ch. 9)

**Magnetic Field (B)** — A vector field representing the magnetic influence on moving charges and magnetic materials, measured in tesla (T) or gauss (1 T = 10,000 gauss). (Ch. 9)

**Matrix** — A rectangular array of numbers arranged in rows and columns, used to represent and solve systems of linear equations, circuit analysis problems, and state-space models. (App. F)

**Matrix Converter** — A direct AC-to-AC converter using a matrix of bidirectional switches (9 switches for 3φ-to-3φ) to provide frequency and voltage conversion without intermediate DC energy storage. Maximum voltage ratio is $\sqrt{3}/2 \approx 0.866$. Offers sinusoidal I/O, controllable power factor, and four-quadrant operation. (Ch. 10)

**Matrix Inverse ($A^{-1}$)** — The matrix satisfying $A \times A^{-1} = I$ (identity matrix). Exists only for square matrices with non-zero determinant. Used in solving AX = B as $X = A^{-1}B$. (App. F)

**Maximum Power Point (MPP)** — The operating point ($V_{mp}$, $I_{mp}$) on a PV module's I-V curve where the product P = V × I is maximized. The MPP shifts with irradiance and temperature, requiring active maximum power point tracking (MPPT) to harvest available energy efficiently. At STC, a 60-cell module typically has $V_{mp} \approx$ 30–38 V. (Ch. 10)

**Maximum Power Point Tracking (MPPT)** — An algorithm implemented in a DC-DC converter that continuously adjusts its duty cycle to operate a PV array at its maximum power point as irradiance and temperature change. Common algorithms include Perturb-and-Observe (P&O) and Incremental Conductance (InC). MPPT efficiency typically reaches 97–99.5%. (Ch. 10)

**Maximum Power Transfer** — The theorem stating that maximum power is delivered to a load when the load impedance equals the complex conjugate of the source impedance (or equals the source resistance for resistive circuits). (Ch. 7)

**Maxwell's Equations** — The four fundamental equations governing all electromagnetic phenomena: Gauss's law for electricity, Gauss's law for magnetism, Faraday's law of induction, and the Ampere-Maxwell law. (Ch. 9)

**MDF (Main Distribution Frame)** — The primary equipment room in a building's structured cabling system, serving as the building entry point for external services and housing core network switches. (Ch. 15)

**Mealy Machine** — A finite state machine whose outputs depend on both the current state and the current inputs, allowing faster response than a Moore machine but with the potential for output glitches during input transitions. (Ch. 6)

**Measurement Uncertainty** — The quantitative expression of doubt about a measurement result, evaluated using the GUM (Guide to the Expression of Uncertainty in Measurement) framework. Type A uncertainty is evaluated statistically from repeated measurements; Type B is evaluated from calibration certificates, specifications, and engineering judgment. Combined standard uncertainty is the RSS of all components; expanded uncertainty $U = k \times u_c$ (typically $k = 2$ for 95% confidence). (Ch. 11)

**Mesh Analysis** — A systematic circuit analysis method that applies KVL around each independent loop (mesh) to establish equations for the unknown mesh currents. (Ch. 7)

**Microcontroller** — A single-chip computer integrating a CPU, program memory (Flash), data memory (SRAM), and peripherals (GPIO, timers, ADC, communication interfaces) for embedded applications. (Ch. 5)

**Microgrid** — A local energy grid with defined electrical boundaries that can connect to and disconnect from the main utility grid, operating autonomously using local generation and storage. (Ch. 1)

**Microstrip** — A planar transmission line consisting of a signal trace on the top surface of a dielectric substrate with a ground plane on the bottom, used for controlled-impedance interconnects on PCBs at RF and high-speed digital frequencies. Characteristic impedance depends on trace width, substrate height, and dielectric constant. (Ch. 9)

**MIMO (Multiple-Input Multiple-Output)** — A wireless technique using multiple antennas at both transmitter and receiver to increase throughput and reliability through spatial multiplexing and diversity. Capacity scales linearly with $\min(N_t, N_r)$ in rich scattering. Massive MIMO (64–256 elements) is fundamental to 5G NR. (Ch. 2, 15, 16)

**Minimum Attractive Rate of Return (MARR)** — The minimum return a project must earn to be considered acceptable, typically set above the cost of capital to account for risk and opportunity cost. (Ch. 19)

**MMF (Multimode Fiber)** — Optical fiber with a 50 $\mu$m or 62.5 $\mu$m core that supports multiple propagation modes. Bandwidth limited by modal dispersion. OM3/OM4/OM5 grades support 10–100 Gbps over data center distances. (Ch. 15)

**Mode-Locking** — A laser technique that forces many longitudinal cavity modes to oscillate in phase, producing a train of ultrashort pulses (femtoseconds to picoseconds). Pulse width is inversely proportional to laser bandwidth: $\tau \approx 0.44/\Delta f$. Enables peak powers of gigawatts to petawatts (with CPA). (Ch. 18)

**Modified Internal Rate of Return (MIRR)** — A rate of return measure that resolves multiple-IRR ambiguity by separating the finance rate for negative cash flows from the reinvestment rate for positive cash flows. (Ch. 19)

**Modulation Index** — The ratio of the modulating signal amplitude to the carrier amplitude in AM systems; determines the depth of modulation and sideband power. (Ch. 2)

**Monopole Antenna** — A wire antenna consisting of a quarter-wavelength element mounted perpendicular to a ground plane. Has half the impedance of a dipole ($\approx 36.5\ \Omega$) and 3 dB higher gain (5.15 dBi) due to ground-plane image effect. (Ch. 16)

**Moore Machine** — A finite state machine whose outputs depend only on the current state, providing glitch-free synchronous outputs that change only on clock edges. (Ch. 6)

**MOSFET (Metal-Oxide-Semiconductor FET)** — A voltage-controlled transistor widely used as a switch in power electronics, operating at high switching frequencies with very low gate drive power. (Ch. 3, 10)

**MTI (Moving Target Indication)** — A radar signal processing technique that suppresses echoes from stationary clutter by exploiting Doppler shift differences between moving targets and the fixed background. Uses pulse-to-pulse cancellation or Doppler filter banks. (Ch. 17)

**MTU (Maximum Transmission Unit)** — The largest packet or frame size (in bytes) that can be transmitted on a network segment without fragmentation. Ethernet MTU is typically 1,500 bytes. (Ch. 15)

**Multilevel Inverter** — A power converter topology that synthesizes the output voltage using three or more discrete voltage levels, reducing harmonic content and dv/dt stress compared to two-level inverters. Main types: neutral-point-clamped (NPC), flying capacitor, and cascaded H-bridge (CHB). Modular multilevel converters (MMC) are used in HVDC systems. (Ch. 10)

**Multiphase Converter** — A DC-DC converter topology using N parallel phases with switching signals phase-shifted by 360°/N, reducing input and output ripple by cancellation and increasing effective ripple frequency to $N \times f_{sw}$. The standard architecture for CPU/GPU voltage regulator modules (VRMs), typically using 6–16 interleaved buck phases with DCR current sensing and adaptive voltage positioning. (Ch. 10)

**Multiple Feedback (MFB) Filter** — A second-order active filter topology using a single op-amp in an inverting configuration with two feedback paths. Particularly effective for bandpass filters with a single op-amp, and preferred over Sallen-Key at high Q values due to lower sensitivity to op-amp gain-bandwidth limitations. (Ch. 13)

**Multiplexer (MUX)** — A combinational logic circuit that selects one of several input signals and forwards it to a single output, based on select line values. (Ch. 6)

**Multirate Signal Processing** — Techniques for changing the sampling rate of a discrete-time signal through decimation (rate reduction) and interpolation (rate increase), used in digital audio, software-defined radio, and efficient filter implementations. (Ch. 8)

**Mutual Inductance (M)** — The property of two coupled inductors where a changing current in one induces a voltage in the other: $V_2 = M(dI_1/dt)$. Related to self-inductances by $M = k\sqrt{(L_1L_2)}$, where k is the coupling coefficient. (Ch. 7)

**NACS (North American Charging Standard)** — An EV charging connector standard (SAE J3400), originally Tesla's proprietary design, supporting both AC and DC charging through a single compact connector rated to 1 MW. Adopted by most major automakers for the North American market. (Ch. 10)

**NAT (Network Address Translation)** — A technique that maps private (RFC 1918) IP addresses to public addresses at the network boundary. Static NAT provides one-to-one mappings for servers; Port Address Translation (PAT/overload) maps thousands of internal hosts to a single public IP using unique source port numbers. Carrier-Grade NAT (CGNAT) applies a second PAT layer at the ISP. NAT breaks end-to-end connectivity and requires ALGs for protocols that embed addresses in payloads. (Ch. 15)

**Natural Frequency ($\omega_0$)** — The frequency at which an undamped second-order system oscillates freely, $\omega_0 = 1/\sqrt{(LC)}$ for an RLC circuit. (Ch. 4, 7)

**Near-Field Region** — The region surrounding an antenna where the electromagnetic field stores energy reactively and does not behave as a propagating wave. Extends to approximately $r = 2D^2/\lambda$, beyond which the far-field (Fraunhofer) region begins with 1/r power decay and a stable radiation pattern. (Ch. 9, 16)

**NEC (National Electrical Code)** — NFPA 70, the U.S. standard for safe electrical installation practices, covering wiring methods, conductor sizing, overcurrent protection, grounding, and equipment installation. (Ch. 14)

**NERC CIP (Critical Infrastructure Protection)** — A set of mandatory cybersecurity standards issued by the North American Electric Reliability Corporation for bulk electric system cyber assets, covering electronic security perimeters, access control, security patching, incident response, and configuration management. (Ch. 1)

**Net Present Value (NPV)** — The sum of all cash flows discounted to time zero at the MARR; a positive NPV indicates the project earns more than the minimum required return. (Ch. 19)

**NEXT (Near-End Crosstalk)** — Crosstalk measured at the same end of a cable as the signal source, representing the worst-case interference because the transmitted signal is at full strength. (Ch. 15)

**Nodal Analysis** — A systematic circuit analysis method that applies KCL at each node to establish equations for the unknown node voltages, using a conductance (admittance) matrix. (Ch. 7, App. F)

**Noise Figure (NF)** — A measure of how much a device degrades the signal-to-noise ratio, defined as $NF = 10 \log_{10}(SNR_{in}/SNR_{out})$ in dB. An ideal noiseless device has NF = 0 dB. Related to equivalent noise temperature by $T_e = T_0(F - 1)$. (Ch. 2)

**Noise Margin** — The voltage difference between a logic gate's guaranteed output level and the required input level of the receiving gate. $NM_H = V_{OH} - V_{IH}$ and $NM_L = V_{IL} - V_{OL}$. Higher noise margins provide greater immunity to electrical noise. (Ch. 6)

**Noise-Equivalent Power (NEP)** — The optical power incident on a photodetector that produces a signal-to-noise ratio of 1 in a 1 Hz bandwidth, expressed in $W/\sqrt{Hz}$. Lower NEP indicates higher detector sensitivity. Related to specific detectivity by $D^* = \sqrt{A}/NEP$, where A is the detector area. (Ch. 18)

**Nominal Interest Rate** — The stated annual interest rate before accounting for compounding frequency, equal to the periodic rate times the number of compounding periods per year. (Ch. 19)

**Norton's Theorem** — Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source in parallel with a resistance. $I_N = V_{Th}/R_{Th}$. (Ch. 7)

**Notch Filter (Band-Reject Filter)** — A frequency-selective circuit that strongly attenuates a narrow band of frequencies while passing all others. The active Twin-T notch filter uses an RC Twin-T network in an op-amp feedback loop with positive feedback to achieve high Q (narrow rejection bandwidth), commonly used to remove 50/60 Hz power-line hum from instrumentation signals. (Ch. 13)

**NRZ (Non-Return-to-Zero)** — A binary line encoding where the signal level is held constant for the entire bit period: one level for a 1, another for a 0. Simple but lacks inherent clock recovery. (Ch. 15)

**Numerical Aperture (NA)** — A dimensionless number characterizing the light-gathering ability of an optical fiber: $NA = \sqrt{(n_{core}^2 - n_{clad}^2)}$. Determines the maximum acceptance angle. (Ch. 15, 18)

**Nyquist Sampling Theorem** — A bandlimited signal can be perfectly reconstructed from its samples if the sampling rate is greater than twice the highest frequency component: $f_s > 2f_{max}$. (Ch. 2, 8)

**OBC (On-Board Charger)** — An AC/DC converter integrated into an electric vehicle that converts grid AC power to DC for battery charging. Typically rated at 3.3–22 kW for Level 1/Level 2 charging, using a two-stage architecture (PFC + isolated DC-DC). DC fast charging bypasses the OBC entirely. (Ch. 10)

**Observability** — A property of a dynamic system indicating whether the internal states can be uniquely determined from measurements of the output over a finite time interval. Determined by the rank of the observability matrix $[C; CA; CA^2; \ldots]$. (Ch. 4)

**OCPD (Overcurrent Protective Device)** — A circuit breaker or fuse designed to open a circuit when current exceeds a specified value, protecting conductors and equipment from overloads and short circuits. (Ch. 14)

**OFDM (Orthogonal Frequency Division Multiplexing)** — A digital multiplexing technique that distributes data across many closely-spaced orthogonal subcarriers using the Inverse FFT. Provides robustness against multipath fading. Used in Wi-Fi, LTE, 5G NR, and DVB-T. (Ch. 2)

**OFDMA (Orthogonal Frequency Division Multiple Access)** — A multi-user variant of OFDM that dynamically assigns subsets of subcarriers (resource blocks) to different users based on channel conditions. Used in LTE downlink, 5G NR, and Wi-Fi 6 (802.11ax). Enables frequency-domain scheduling and multi-user diversity. (Ch. 2, 15)

**Ohm (Ω)** — The SI derived unit of electrical resistance. One ohm is the resistance that produces one volt of drop per ampere of current: $V = IR$. (App. D)

**Ohm's Law** — The fundamental circuit relationship $V = I \times R$, stating that voltage across a resistor is proportional to the current through it. Extended to AC circuits as $V = I \times Z$ using complex impedance. (Ch. 7)

**Op-Amp (Operational Amplifier)** — A high-gain DC-coupled differential voltage amplifier with very high input impedance and low output impedance, used as a building block for analog circuits (amplifiers, filters, comparators, oscillators). (Ch. 13)

**Optical Link Budget** — An accounting of all optical gains and losses in a fiber-optic or free-space link, verifying that received power exceeds the receiver sensitivity plus a system margin (typically 3–6 dB). Loss contributions include fiber attenuation, connector and splice losses, and passive component insertion losses. (Ch. 15, 18)

**OSI Model (Open Systems Interconnection)** — A seven-layer conceptual framework for network communication: Physical, Data Link, Network, Transport, Session, Presentation, Application. (Ch. 15)

**OSNR (Optical Signal-to-Noise Ratio)** — The ratio of signal power to ASE noise power in a specified optical bandwidth (typically 0.1 nm), used to assess the quality of DWDM optical links. (Ch. 15)

**OSPF (Open Shortest Path First)** — A link-state interior gateway routing protocol that uses Dijkstra's algorithm to compute the shortest path to each destination within an autonomous system. (Ch. 15)

**OTDR (Optical Time-Domain Reflectometer)** — A fiber optic test instrument that characterizes a fiber link by launching optical pulses and analyzing the backscattered and reflected light to map events (splices, connectors, breaks) and measure attenuation as a function of distance. (Ch. 15)

**OUI (Organizationally Unique Identifier)** — The first 24 bits (3 bytes) of a MAC address, assigned by IEEE to identify the manufacturer or organization that produced the network interface. (Ch. 15)

**Overcurrent Protection** — The use of fuses or circuit breakers to automatically disconnect a circuit when current exceeds safe levels, preventing conductor overheating and equipment damage. (Ch. 14)

**Overshoot (Percent)** — The amount by which a system's step response exceeds its final steady-state value, expressed as a percentage. For a second-order system: $\%OS = 100 \times e^{-\zeta\pi/\sqrt{(1-\zeta^2)}}$. (Ch. 4)

**PAM-4 (Pulse Amplitude Modulation, 4-level)** — A signaling method using four distinct amplitude levels to encode 2 bits per symbol, used in 10GBASE-T Ethernet and 400G optical interfaces. (Ch. 15)

**Parabolic Reflector Antenna** — A high-gain antenna using a paraboloidal dish to focus electromagnetic waves to a focal point. Gain $= \eta(\pi D/\lambda)^2$, where D is the dish diameter and $\eta$ is aperture efficiency (0.55–0.70). Used in satellite communications, radar, and radio astronomy. (Ch. 16)

**Parametric Spectral Estimation** — A class of PSD estimation methods that fit a mathematical model (typically autoregressive) to observed data, achieving higher frequency resolution than FFT-based methods on short data records. Includes AR/Burg methods and subspace methods (MUSIC, ESPRIT). (Ch. 8)

**Patch Antenna** — A low-profile antenna fabricated on a printed circuit board substrate, consisting of a conducting patch over a ground plane. Typical gain is 6–9 dBi with narrow bandwidth (1–5%). Widely used in mobile devices, GPS, Wi-Fi, and phased arrays. (Ch. 16)

**Patch Panel** — A passive (unpowered) device mounted in a rack that provides a termination point for horizontal cabling, with IDC punchdown connections on the back and RJ-45 jacks (or fiber adapters) on the front. (Ch. 15)

**Peak Shaving** — The use of energy storage or demand response to reduce peak electrical demand, typically to lower demand charges on commercial utility bills. A BESS discharges during peak periods and recharges during off-peak hours. (Ch. 10)

**Perturb-and-Observe (P&O)** — The most widely deployed MPPT algorithm for PV systems. The controller periodically perturbs the array voltage by a small step $\Delta V$, measures the resulting change in power $\Delta P$, and continues in the same direction if $\Delta P > 0$ or reverses if $\Delta P < 0$. Simple to implement but produces steady-state oscillation of amplitude $\Delta V$ around the MPP. (Ch. 10)

**Phase Margin** — The amount of additional phase lag (in degrees) at the gain crossover frequency that would make a control system unstable. Adequate stability typically requires PM $\geq$ 30-60°. For second-order systems, PM $\approx 100\zeta$ for $\zeta < 0.7$. (Ch. 4)

**Phasor** — A complex number representing the amplitude and phase of a steady-state sinusoidal signal, typically expressed in RMS magnitude and phase angle: $V = V_{rms}\angle\theta$. (App. A)

**Photodiode** — A semiconductor device that converts incident photons into electrical current through absorption in a reverse-biased PN junction. PIN photodiodes have a wide intrinsic region for high-speed response; avalanche photodiodes (APDs) provide internal gain through impact ionization. (Ch. 18)

**Photolithography** — The process of transferring circuit patterns onto a semiconductor wafer by exposing light-sensitive photoresist through a photomask, then developing the resist to create a patterned etch mask. Resolution limited by $CD_{min} = k_1\lambda/NA$. (Ch. 3)

**Photovoltaic (PV) Cell** — A semiconductor p-n junction device that generates electricity from sunlight through the photovoltaic effect. Incident photons generate electron-hole pairs that are separated by the junction's built-in electric field, producing a photocurrent proportional to irradiance. Key parameters are short-circuit current $I_{sc}$, open-circuit voltage $V_{oc}$, fill factor FF, and efficiency $\eta$. Commercial monocrystalline silicon cells achieve 18–24% efficiency at STC. (Ch. 10)

**PID Controller** — A feedback controller that computes a control signal as the sum of proportional, integral, and derivative terms of the error signal, providing fast response with zero steady-state error. (Ch. 4)

**PIN Photodiode** — A semiconductor photodetector with a P-I-N junction structure that converts incident light to electrical current. Used in fiber optic receivers for its linear response and fast speed. (Ch. 15)

**PLL (Phase-Locked Loop)** — A clock generation circuit that multiplies a reference frequency to produce a higher-frequency output clock. In MCUs, the PLL multiplies the crystal oscillator frequency to generate the system clock (e.g., 8 MHz $\times$ 21 = 168 MHz). (Ch. 5)

**PN Junction** — The interface between P-type and N-type semiconductor regions, forming the fundamental building block of diodes, BJTs, and MOSFETs. Characterized by a depletion region, built-in potential, and exponential I-V behavior under forward bias. (Ch. 3)

**Polar Form** — A representation of a complex number as magnitude and angle: $z = r\angle\theta$, where $r = |z|$ and $\theta = \text{atan2}(b, a)$. Convenient for multiplication and division. (App. A)

**Poles and Zeros** — The values of the complex frequency variable s (or z) that make a transfer function infinite (poles) or zero (zeros). Pole locations determine system stability and transient behavior. (Ch. 4, 8)

**Polyphase Filter** — An efficient multirate filter structure that decomposes an N-tap filter into M subfilters of N/M taps each, operating at the lower sampling rate to avoid wasting computation on samples that will be discarded (decimation) or on zero-valued inputs (interpolation). Reduces computational cost by a factor of M. (Ch. 8)

**Power Analyzer** — A precision instrument that simultaneously samples voltage and current to compute real power, reactive power, apparent power, power factor, and harmonic content for AC and DC systems. Provides accurate measurements on non-sinusoidal waveforms from motor drives, power supplies, and switching converters. (Ch. 11)

**Power Factor (PF)** — The ratio of real power (W) to apparent power (VA) in an AC circuit, equal to $\cos\theta$ where $\theta$ is the phase angle between voltage and current. PF = 1.0 is ideal. (Ch. 1)

**Power Factor Correction (PFC)** — The practice of improving a system's power factor toward unity. In power systems, achieved by adding capacitors to offset inductive loads. In power electronics, active PFC circuits (boost PFC) shape the input current to follow the voltage waveform, achieving PF > 0.99 and meeting IEC 61000-3-2 harmonic limits. (Ch. 1, 10)

**Power MOSFET** — A MOSFET designed for high-current, high-voltage switching using a vertical device structure (VDMOS or trench). Key parameters include $R_{DS(on)}$ (conduction loss), gate charge $Q_g$ (switching speed), and body diode characteristics. (Ch. 3, 10)

**Power over Ethernet (PoE)** — A technology that delivers DC power over twisted-pair Ethernet cabling alongside data, defined by IEEE 802.3af (15.4 W), 802.3at (30 W), and 802.3bt (60/90 W). Uses 48 VDC nominal with classification handshake between PSE and PD. (Ch. 15)

**Power Spectral Density (PSD)** — The distribution of signal power across frequency, expressed in watts per hertz (W/Hz), obtained from the Fourier transform of the autocorrelation function. (Ch. 8)

**Power Supply Rejection Ratio (PSRR)** — A measure of how well an op-amp rejects noise and ripple on its power supply rails, defined as $PSRR = \Delta V_{supply}/\Delta V_{OS}$. Typical DC values are 80–120 dB but degrade with frequency at -20 dB/decade. (Ch. 13)

**Power Supply Sequencing** — The practice of enabling multiple voltage rails in a specific order and timing to prevent latch-up, excessive inrush, or damage to sensitive ICs during startup and shutdown. Strategies include sequential (daisy-chained PG signals), ratiometric (all rails ramp proportionally), and simultaneous. PMBus provides digital control and telemetry for multi-rail power management. (Ch. 10)

**Poynting Vector (S)** — The cross product $S = E \times H$, representing the instantaneous electromagnetic power flow per unit area (W/m²). The time-averaged Poynting vector $S_{avg} = |E_0|^2/(2\eta)$ gives the power density of a plane wave. Used to calculate radiated power, radiation intensity, and RF safety exposure limits. (Ch. 9)

**Precision** — The repeatability of a measurement — the degree to which repeated measurements under identical conditions yield the same result. High precision does not imply high accuracy. (Ch. 11)

**Precision Rectifier** — An op-amp circuit (also called a superdiode) that overcomes the forward voltage drop of a conventional diode, enabling rectification of millivolt-level signals. The op-amp's open-loop gain effectively reduces the diode threshold by a factor of $A_{OL}$ (e.g., 0.6 V becomes 6 μV). Used for precision AC-to-DC conversion, AM demodulation, and absolute value circuits. (Ch. 13)

**Precision Time Protocol (PTP)** — IEEE 1588 protocol for sub-microsecond clock synchronization over packet networks, using hardware timestamping and boundary/transparent clocks to measure and compensate for network delay. Often combined with SyncE for phase and frequency synchronization in 5G and telecom. (Ch. 15)

**Present Worth Factor (P/F)** — The compound interest factor $1/(1+i)^n$ that converts a future value to its equivalent present value. (Ch. 19)

**Prism** — A transparent optical element with flat, polished surfaces that refracts light; used to separate white light into its constituent wavelengths (dispersion) or to redirect beams at specific angles based on Snell's law and total internal reflection. (Ch. 18)

**Profitability Index** — The ratio of a project's NPV to its initial investment (PI = NPV/Investment), used to rank independent projects for capital budgeting under budget constraints. (Ch. 19)

**Programmable Gain Amplifier (PGA)** — An amplifier whose gain can be changed dynamically via digital control (SPI/$I^2C$), typically by switching between laser-trimmed feedback resistors using an analog multiplexer. Used between input multiplexers and ADCs in data acquisition systems to maximize effective dynamic range by matching gain to signal level. (Ch. 13)

**Propagation Delay** — The time for a signal to travel through a medium or circuit. In fiber optics: $t = d \times n/c$. In digital logic: the time from an input change to the corresponding output change of a gate ($t_{pHL}$, $t_{pLH}$), determining the critical path and maximum clock frequency. (Ch. 6, 15)

**Protective Relay** — An intelligent device that monitors electrical quantities (current, voltage, impedance, frequency) via instrument transformers and issues trip signals to circuit breakers when abnormal conditions are detected. Types include overcurrent (50/51), distance (21), differential (87), and directional (67). (Ch. 1)

**PSD** — See Power Spectral Density. (Ch. 8)

**Pseudocapacitor** — A supercapacitor electrode that stores charge through reversible Faradaic reactions at the electrode surface rather than purely electrostatic double-layer adsorption. Common pseudocapacitive materials include $RuO_2$ (high performance, expensive), $MnO_2$ (low cost), and conducting polymers (polyaniline, polypyrrole). Achieves 2–5× higher energy density than EDLC at the cost of slightly reduced cycle life. (Ch. 10)

**PSK (Phase Shift Keying)** — A digital modulation technique that encodes data in the phase of a carrier signal. BPSK uses two phases (1 bit/symbol), QPSK uses four phases (2 bits/symbol). QPSK achieves the same BER per bit as BPSK while doubling bandwidth efficiency. (Ch. 2)

**Pulse Compression** — A radar technique that transmits a long, coded waveform (LFM chirp or phase code) and compresses it in the receiver to achieve the range resolution of a much shorter pulse while maintaining the energy of the long pulse. Compression ratio = time-bandwidth product $B\tau$. (Ch. 17)

**PWM (Pulse Width Modulation)** — A technique for encoding an analog signal level as the duty cycle of a digital pulse train, used in motor control, power conversion, and DAC approximation. (Ch. 5, 10)

**Pyrometer** — A non-contact temperature sensor that measures thermal radiation emitted by an object's surface. Single-color pyrometers require known emissivity, while two-color (ratio) pyrometers compare intensity at two wavelengths to cancel emissivity dependence. Ranges from −50°C to over 3000°C. (Ch. 11)

**Q-Factor (Quality Factor)** — A dimensionless parameter expressing the ratio of energy stored to energy dissipated per cycle in a resonant circuit: $Q = \omega_0 L/R$ (series RLC) or $Q = R/(\omega_0 L)$ (parallel

RLC). Higher Q means narrower bandwidth and sharper resonance. (Ch. 7)

**QAM (Quadrature Amplitude Modulation)** — A modulation scheme that encodes data in both the amplitude and phase of a carrier, used in Wi-Fi (up to 4096-QAM in Wi-Fi 7), cable modems, and coherent optical transceivers. (Ch. 15)

**Quality of Service (QoS)** — Network mechanisms that differentiate traffic classes and provide preferential treatment through classification/marking (DSCP, 802.1p), queuing/scheduling (strict priority, WFQ, LLQ), and policing/shaping. Enables latency-sensitive traffic (VoIP, video) to receive priority over best-effort traffic during congestion. (Ch. 15)

**Quantization** — The process of mapping continuous amplitude values to a finite set of discrete levels in an ADC, introducing quantization noise proportional to the step size. (Ch. 2, 11)

**QUIC** — A transport-layer protocol built on UDP that provides reliable, multiplexed, encrypted connections with 1-RTT (or 0-RTT resumed) handshake latency. Eliminates TCP's head-of-line blocking through independent streams, mandates TLS 1.3, supports connection migration via Connection IDs, and serves as the transport for HTTP/3. (Ch. 15)

**Radar** — A system that uses transmitted electromagnetic waves and their echoes to detect, locate, and measure the velocity of objects. The received power follows an $R^4$ range dependence for point targets. (Ch. 17)

**Radar Clutter** — Unwanted radar echoes from the environment (ground, sea, rain, birds) that can mask target returns; characterized by its reflectivity ($\sigma^0$ for surface clutter) and mitigated using MTI filtering, Doppler processing, and CFAR detection. (Ch. 17)

**Radar Cross Section (RCS)** — A measure of a target's reflectivity to radar: $\sigma = 4\pi R^2(P_{reflected}/P_{incident})$, expressed in $m^2$ or dBsm. Depends on target size, shape, material, and frequency. Typical values: person ~1 $m^2$, aircraft ~1–100 $m^2$, stealth aircraft ~0.001–0.01 $m^2$. (Ch. 17)

**Radiation Pattern** — A graphical representation of an antenna's radiated power as a function of direction, showing the main lobe, sidelobes, nulls, and back lobe. Characterized by beamwidth (HPBW), sidelobe level, and front-to-back ratio. (Ch. 16)

**Ragone Plot** — A log-log chart comparing energy storage technologies by plotting specific power (W/kg) on the vertical axis against specific energy (Wh/kg) on the horizontal axis. Each technology occupies a characteristic region: conventional capacitors (extreme power, minimal energy) → EDLC → pseudocapacitors/LIC → batteries → fuel cells (high energy, low power). Used to select the appropriate storage technology for a given application. (Ch. 10)

**Rail-to-Rail Op-Amp** — An op-amp whose input common-mode range and/or output swing extends to within millivolts of the supply rails ($V_{CC}$ and $V_{EE}$), maximizing usable signal range in single-supply and low-voltage designs. RRIO (rail-to-rail input and output) types use complementary input pairs and CMOS output stages. (Ch. 13)

**Rayleigh Backscatter** — Low-level continuous reflections of light in an optical fiber caused by microscopic density variations in the glass, used by OTDRs to measure fiber attenuation along the length. (Ch. 15)

**Reactance (X)** — The imaginary component of impedance, representing opposition to AC current due to energy storage in inductors ($X_L = \omega L$, positive) or capacitors ($X_C = -1/\omega C$, negative). Measured in ohms. (Ch. 7)

**Reactive Power (Q)** — The power that oscillates between source and reactive elements (inductors and capacitors) without performing useful work. Measured in volt-ampere reactive (VAR). (Ch. 1, 7)

**Real Power (P)** — The average power consumed by resistive elements and converted to useful work or heat, measured in watts (W). $P = VI\cos\theta$. (Ch. 1, 7)

**Receiver Sensitivity** — The minimum input signal power at which a receiver can achieve the required performance (typically a specified BER or SNR), determined by the noise floor, noise figure, and required signal-to-noise ratio. (Ch. 2)

**Rectifier** — A circuit that converts AC to DC using diodes. Configurations include half-wave, full-wave center-tap, and bridge rectifiers for single-phase, and six-pulse for three-phase. (Ch. 10)

**Recursive Least Squares (RLS)** — An adaptive filtering algorithm that minimizes the exponentially weighted sum of all past squared errors, converging in approximately M iterations (filter order) compared to hundreds for LMS. Uses a forgetting factor $\lambda$ (0.95–1.0) and costs $O(M^2)$ per sample. (Ch. 8)

**Reflection Coefficient (Γ)** — The ratio of reflected wave amplitude to incident wave amplitude at a discontinuity in a transmission line: $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$. (Ch. 9)

**Refractive Index (n)** — The ratio of the speed of light in vacuum to its speed in a medium: $n = c/v$. Determines how light bends at interfaces (Snell's law) and governs optical fiber waveguiding. Common values: air ≈ 1.000, glass ≈ 1.5, silicon ≈ 3.5. (Ch. 18)

**Regenerative Braking** — A technique in which a motor operates as a generator during deceleration, converting kinetic energy back into electrical energy. The recovered energy is either dissipated in a braking resistor or returned to the supply through an active front-end converter. (Ch. 12)

**Relaxation Oscillator** — An oscillator that generates square and triangular waves by alternately charging and discharging a capacitor between two threshold voltages set by a Schmitt trigger. The frequency is $f = 1/(2RC \times \ln((1+\beta)/(1-\beta)))$, where $\beta$ is the positive feedback fraction. Used in function generators, PWM circuits, and clock sources. (Ch. 13)

**Reluctance Motor** — An AC motor that produces torque by exploiting the tendency of a ferromagnetic rotor to align with the stator magnetic field to minimize reluctance. Types include the switched reluctance motor (SRM) with salient poles and the synchronous reluctance motor (SynRM) with flux barriers in the rotor laminations. (Ch. 12)

**Replacement Analysis** — The decision framework for determining when to retire existing equipment and replace it with new equipment, based on economic service life and annual cost comparison. (Ch. 19)

**Resolution** — The smallest change in a measured quantity that an instrument can detect and display, determined by the number of bits (for digital instruments) or scale divisions (for analog instruments). (Ch. 11)

**Resonance** — The condition in an AC circuit where inductive reactance equals capacitive reactance ($X_L = X_C$), producing maximum current (series) or maximum impedance (parallel) at the resonant frequency $f_0 = 1/(2\pi\sqrt{(LC)})$. (Ch. 7)

**Rise Time** — The time for a system's step response to go from 10% to 90% of its final value. For a first-order system: $t_r = 2.2\tau$. Inversely related to system bandwidth. (Ch. 4)

**RMS (Root-Mean-Square)** — The effective value of an AC waveform, equal to $V_{peak}/\sqrt{2}$ for a sinusoid. The RMS value produces the same heating effect as an equivalent DC value. (App. A)

**Rogowski Coil** — A flexible, air-core coil wrapped around a conductor that outputs a voltage proportional to di/dt; integration recovers the current waveform. Linear over enormous current ranges (no magnetic saturation) and ideal for measuring transient and fault currents. (Ch. 11)

**Root Locus** — A graphical method showing how closed-loop poles move in the s-plane as a parameter (typically loop gain K) varies from 0 to infinity. Branches start at open-loop poles and end at open-loop zeros. (Ch. 4)

**Round-Trip Efficiency (RTE)** — The ratio of energy delivered by a storage system during discharge to the energy consumed during charging, expressed as a percentage. For lithium-ion BESS, typical RTE is 85–92%, with losses occurring in the inverter, battery internal resistance, and auxiliary systems. (Ch. 10)

**Router** — A Layer 3 network device that forwards packets between different IP networks based on destination IP address, using routing tables populated by static configuration or dynamic routing protocols (OSPF, BGP). (Ch. 15)

**Routh-Hurwitz Criterion** — An algebraic stability test that determines the number of right half-plane poles of a polynomial by constructing a Routh array and counting sign changes in the first column. Used to find the range of gain K for stability. (Ch. 4)

**RTD (Resistance Temperature Detector)** — A temperature sensor that measures temperature by correlating the resistance of a metallic element (typically platinum, Pt100 or Pt1000) with temperature. More accurate and stable than thermocouples. (Ch. 11)

**RTOS (Real-Time Operating System)** — An operating system that guarantees task execution within specified time constraints (deadlines), providing deterministic scheduling, inter-task communication, and resource management for embedded systems. (Ch. 5)

**RTU (Remote Terminal Unit)** — A ruggedized microprocessor-based device installed at a remote field site (substation, pumping station) that acquires analog and digital measurements from local instruments, executes supervisory commands from the SCADA master station, and communicates via DNP3 or IEC 60870-5 protocols. (Ch. 1)

**SAE J1772** — The SAE standard defining the Level 1 and Level 2 AC charging connector for electric vehicles in North America, including the control pilot (CP) signal — a 1 kHz PWM square wave whose duty cycle communicates the maximum available current from the EVSE to the vehicle. (Ch. 10)

**SAE J2954** — The SAE standard for wireless (inductive) EV charging at 85 kHz, defining power levels WPT1 (3.7 kW) through WPT4 (22 kW), ground clearance classes (100–250 mm), alignment tolerances, and foreign object detection requirements. (Ch. 10)

**Sallen-Key Filter** — A popular second-order active filter topology using a single op-amp with positive feedback through a voltage follower configuration. Available in low-pass, high-pass, and bandpass forms. (Ch. 13)

**Salvage Value** — The estimated market value of an asset at the end of its useful life, subtracted from the cost basis when calculating depreciation. (Ch. 19)

**Sampling Rate ($f_s$)** — The number of samples taken per second when converting an analog signal to digital form, measured in samples per second (S/s) or hertz (Hz). (Ch. 2, 8, 11)

**SAR (Synthetic Aperture Radar)** — A radar imaging technique that achieves fine cross-range resolution by coherently processing echoes collected as the radar platform moves, synthesizing a large effective antenna aperture. Azimuth resolution = D/2 (antenna length / 2), independent of range. Used for satellite Earth observation and military reconnaissance. (Ch. 17)

**SC-FDMA (Single Carrier Frequency Division Multiple Access)** — A DFT-precoded OFDM uplink access scheme used in LTE, producing a single-carrier waveform with 2–4 dB lower peak-to-average power ratio (PAPR) than OFDMA, improving mobile device power amplifier efficiency and battery life. (Ch. 2)

**SCADA (Supervisory Control and Data Acquisition)** — A centralized system for real-time monitoring and control of geographically distributed infrastructure (power grids, pipelines, water systems). Collects measurements from RTUs/IEDs, displays data on HMI screens, and relays operator commands to field devices. Modern SCADA integrates with EMS for state estimation, AGC, and economic dispatch. (Ch. 1)

**Schmitt Trigger** — A comparator circuit with hysteresis (two different threshold voltages for rising and falling inputs), providing noise immunity and clean output transitions for noisy or slowly changing signals. (Ch. 13)

**Schottky Diode** — A diode using a metal-semiconductor junction instead of a PN junction, providing lower forward voltage drop (0.15-0.45 V), no reverse recovery time, and faster switching. Widely used in high-frequency power supply rectification. (Ch. 3)

**SCPI (Standard Commands for Programmable Instruments)** — A standardized ASCII command syntax for controlling test instruments over GPIB, USB-TMC, LAN/LXI, or serial interfaces. Commands follow a hierarchical tree structure (e.g., `MEAS:VOLT:DC?`), enabling manufacturer-independent instrument programming via VISA libraries and languages such as Python (PyVISA), LabVIEW, and MATLAB. (Ch. 11)

**Self-Discharge** — The gradual loss of stored charge in a capacitor or battery without an external load, caused by leakage currents through the dielectric or electrolyte. In supercapacitors, self-discharge is governed by a parallel leakage resistance (100 kΩ–1 MΩ) and reduces the terminal voltage with a time constant of hours to days — far faster than batteries (weeks to months). (Ch. 10)

**Sensitivity Analysis** — A technique that systematically varies input parameters to determine how much the economic outcome changes, identifying the most influential variables. (Ch. 19)

**Sensorless Motor Control** — Control techniques that estimate rotor position and speed without mechanical sensors (encoders, resolvers). Methods include back-EMF estimation (voltage model observers, sliding-mode observers, extended Kalman filters) for medium/high-speed operation, and high-frequency injection (HFI) exploiting rotor saliency for zero/low-speed operation. (Ch. 12)

**Sequential Circuit** — A digital logic circuit whose output depends on both the current inputs and the previous state (stored in memory elements such as flip-flops or latches). (Ch. 6)

**Settling Time** — The time for a system's step response to reach and remain within a specified percentage (typically 2% or 5%) of its final value. For a second-order system: $t_s \approx 4/(\zeta\omega_n)$ for 2% criterion. (Ch. 4)

**Setup Time ($t_{su}$)** — The minimum time a data input must be stable before the active clock edge of a flip-flop to ensure reliable capture. Part of the critical path timing constraint along with clock-to-Q delay. (Ch. 6)

**Shannon-Hartley Theorem** — The theoretical maximum data rate of a communication channel: $C = B \times \log_2(1 + SNR)$, where B is bandwidth in Hz and SNR is the linear signal-to-noise ratio. (Ch. 15)

**Shift Register** — A sequential circuit consisting of cascaded flip-flops that shifts stored data by one position per clock cycle. Used for serial-to-parallel conversion, data buffering, and sequence generation. (Ch. 6)

**Short-Circuit Current** — The maximum current that can flow at a given point in an electrical distribution system during a bolted fault, determined by the source impedance (transformer, utility, and conductor). NEC 110.9 requires overcurrent devices to have an interrupting rating at least equal to the available short-circuit current. (Ch. 14)

**SiC (Silicon Carbide)** — A wide-bandgap semiconductor (3.26 eV) with 10× the critical electric field of silicon, enabling power MOSFETs and Schottky diodes with dramatically lower on-resistance and switching losses at 650 V–3.3 kV ratings. Used in EV inverters, solar inverters, and fast chargers. (Ch. 3, 10)

**Siemens (S)** — The SI derived unit of electrical conductance (reciprocal of ohm). One siemens equals one ampere per volt. (App. D)

**Silicon (Si)** — The most widely used semiconductor material, with a bandgap of 1.12 eV, forming the basis of most diodes, transistors, integrated circuits, and solar cells. (Ch. 3)

**Simple Interest** — Interest calculated only on the original principal, producing linear growth described by $I = P \times i \times n$. (Ch. 19)

**Sinking Fund Factor (A/F)** — The compound interest factor $i/[(1+i)^n-1]$ that determines the uniform annual deposit needed to accumulate a target future amount. (Ch. 19)

**Skin Effect** — The tendency of AC current to concentrate near the surface of a conductor at higher frequencies, increasing effective resistance and attenuation. The skin depth $\delta = 1/\sqrt{(\pi f \mu \sigma)}$ decreases with frequency. (Ch. 9, 15)

**Sleep Mode** — A low-power operating state in a microcontroller where the CPU is halted and selected peripherals and clocks are disabled to reduce power consumption. Wake-up is triggered by interrupts, RTC alarms, or external pin events. (Ch. 5)

**Slew Rate** — The maximum rate of change of an op-amp's output voltage, expressed in V/μs. Limits the maximum undistorted output frequency for large-signal operation. (Ch. 13)

**Slip** — The difference between synchronous speed and actual rotor speed in an induction motor, expressed as a fraction: $s = (n_s - n_r)/n_s$. Typical full-load slip is 2–5%. (Ch. 12)

**Slot Antenna** — An antenna formed by cutting a narrow rectangular opening in a conducting ground plane. By Babinet's principle, a half-wave slot has a complementary radiation pattern to a dipole. Cavity-backed slots eliminate backward radiation and are used for flush-mounted aircraft and vehicle installations. (Ch. 16)

**SMF (Single-Mode Fiber)** — Optical fiber with an 8–10 μm core that supports only one propagation mode, eliminating modal dispersion. Standard attenuation: 0.35 dB/km at 1310 nm, 0.20 dB/km at 1550 nm. Supports transmission over 100+ km. (Ch. 15)

**Smith Chart** — A graphical tool that maps complex impedance or reflection coefficient onto a circular chart, enabling visual analysis of transmission line matching, VSWR, and impedance transformation. (Ch. 9)

**Snell's Law** — The law of refraction relating the angles of incidence and transmission at an interface between two media: $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$, where $n_1$, $n_2$ are the refractive indices and $\theta_1$, $\theta_2$ are the angles from the surface normal. (Ch. 18)

**SNMP (Simple Network Management Protocol)** — A protocol for monitoring and managing network devices (routers, switches, servers), using UDP to exchange management information between agents and a management station. (Ch. 15)

**SNR (Signal-to-Noise Ratio)** — The ratio of desired signal power to noise power, expressed in dB: $SNR = 10 \log_{10}(P_{signal}/P_{noise})$. Higher SNR indicates a cleaner signal. (Ch. 11, App. C)

**Socket** — A software endpoint combining an IP address and port number that provides the API for network communication. A TCP connection is identified by a 4-tuple: (source IP, source port, destination IP, destination port). (Ch. 15)

**Soft Start** — A controlled startup technique in power converters that gradually ramps the duty cycle or reference voltage over a defined interval (typically 5–50 ms), limiting inrush current into discharged output capacitors and preventing overcurrent trips during power-on. Also used in motor starters to reduce starting current by ramping applied voltage. (Ch. 10, 12)

**Software-Defined Networking (SDN)** — A network architecture that separates the control plane (routing decisions) from the data plane (packet forwarding), centralizing network intelligence in a software controller that programs forwarding devices via southbound APIs (OpenFlow, gNMI). Enables programmatic network management, automation, and policy enforcement through northbound REST/gRPC APIs. (Ch. 15)

**SONET (Synchronous Optical Networking)** — A standardized TDM transport protocol for fiber optic networks. The base rate STS-1 is 51.84 Mbps; higher rates include OC-3 (155 Mbps), OC-48 (2.5 Gbps), and OC-192 (10 Gbps). (Ch. 2)

**Spanning Tree Protocol (STP)** — IEEE 802.1D protocol that prevents Layer 2 loops in networks with redundant switch links by electing a root bridge and placing redundant ports in a blocking state to form a loop-free tree topology. Rapid STP (802.1w) converges in 1–3 seconds vs 30–50 seconds for classic STP. Multiple STP (802.1s) maps VLAN groups to separate spanning tree instances for per-VLAN load balancing. (Ch. 15)

**Specific Energy** — Energy stored per unit mass, expressed in watt-hours per kilogram (Wh/kg) or joules per kilogram (J/kg). The key figure of merit for comparing energy storage technologies: EDLC supercapacitors 1–15 Wh/kg; Li-ion batteries 100–250 Wh/kg; lithium-ion capacitors 10–30 Wh/kg. (Ch. 10)

**Specific Power** — Peak power deliverable per unit mass, expressed in watts per kilogram (W/kg). The key figure of merit for high-rate applications: EDLC supercapacitors 500–10,000 W/kg; Li-ion batteries 100–1,000 W/kg. Plotted against specific energy on a Ragone plot to compare technologies. (Ch. 10)

**Spectral Leakage** — The spreading of energy across frequency bins in a DFT caused by analyzing a signal over a finite observation window. Reduced by applying window functions (Hanning, Hamming, Blackman). (Ch. 8)

**SPI (Serial Peripheral Interface)** — A four-wire synchronous serial communication protocol (MOSI, MISO, SCLK, CS) providing full-duplex, high-speed data transfer between a master and one or more slave devices. (Ch. 5)

**SRAM (Static Random-Access Memory)** — Fast volatile memory that retains data as long as power is supplied, without requiring refresh cycles. Used for data memory (variables, stack) in microcontrollers. (Ch. 5)

**Standard Test Conditions (STC)** — The standardized reference conditions under which PV cells and modules are rated: irradiance G = 1,000 W/m$^2$, cell temperature T = 25 °C, and AM 1.5 solar spectrum. PV nameplate power (watts peak, $W_p$) is always specified at STC. Real-world output is typically 10–25% lower due to elevated cell temperatures and sub-STC irradiance. (Ch. 10)

**State Machine (Finite State Machine)** — A sequential circuit that transitions between a finite number of states based on current state and inputs, producing outputs according to a defined mapping. Moore machines have state-dependent outputs; Mealy machines have state-and-input-dependent outputs. (Ch. 6)

**State of Charge (SOC)** — The remaining capacity of a battery expressed as a percentage of its rated capacity (0–100%). Estimated via coulomb counting, open-circuit voltage lookup, or model-based methods such as extended Kalman filters. (Ch. 10)

**State of Health (SOH)** — A measure of a battery's condition relative to its original specifications, typically expressed as the ratio of current maximum capacity to rated capacity. Decreases over time due to calendar aging and cycle aging from SEI layer growth and lithium plating. (Ch. 10)

**State-Space Representation** — A mathematical model of a dynamic system using first-order matrix differential equations: $\dot{x} = Ax + Bu$, $y = Cx + Du$, where x is the state vector, u is input, and y is output. (Ch. 4, App. F)

**Stepper Motor** — A brushless DC motor that divides a full rotation into a fixed number of equal steps, providing precise open-loop position control without feedback sensors. Types include permanent magnet, variable reluctance, and hybrid. (Ch. 12)

**Straight-Line Depreciation** — The simplest depreciation method, allocating equal annual charges of $(P-S)/n$ over the asset's useful life. (Ch. 19)

**Strain Gauge** — A resistive sensor whose resistance changes proportionally to mechanical deformation (strain), typically bonded to a structure and measured in a Wheatstone bridge circuit. Gauge factor $\approx 2$ for metallic foil gauges. (Ch. 11)

**Stripline** — A transmission line structure consisting of a flat conductor sandwiched between two ground planes with a dielectric fill; provides inherent shielding and supports TEM propagation, commonly used in multilayer PCBs and microwave circuits. (Ch. 9)

**Structured Cabling** — A standardized approach to building telecommunications infrastructure following TIA-568/ISO 11801, organizing cables into horizontal runs (max 90 m), backbone links, and equipment rooms (MDF/IDF). (Ch. 15)

**Subnet Mask** — A 32-bit value that divides an IP address into network and host portions. Written in dotted-decimal (e.g., 255.255.255.0) or prefix notation (e.g., /24). (Ch. 15)

**Supercapacitor** — An electrochemical energy storage device occupying the performance space between conventional capacitors and batteries, offering capacitances from a few farads to thousands of farads, specific power of 500–10,000 W/kg, specific energy of 1–15 Wh/kg, and cycle lives exceeding 500,000 cycles. Charge storage is primarily electrostatic (EDLC) or a combination of electrostatic and

Faradaic (pseudocapacitor). Energy stored: $E = ½CV^2$; discharging to $V_{max}/2$ extracts exactly 75% of stored energy. Also called an ultracapacitor or EDLC. Applications include regenerative braking, peak power buffering, UPS ride-through, and hybrid battery-supercapacitor systems. (Ch. 10)

**Superposition** — The principle that in a linear circuit with multiple independent sources, the total response equals the algebraic sum of responses due to each source acting individually (with others deactivated). (Ch. 7)

**SWD (Serial Wire Debug)** — An ARM-specific two-pin debug interface (SWDIO and SWCLK) providing the same debug capabilities as JTAG with fewer pins, used as the standard debug interface for Cortex-M microcontrollers. (Ch. 5)

**Switch (Network)** — A Layer 2 device that forwards Ethernet frames between ports based on destination MAC addresses, using a MAC address table (CAM table) for lookup. (Ch. 15)

**Symmetrical Components** — A mathematical transformation that decomposes unbalanced three-phase phasors into three balanced sets: positive sequence (normal rotation), negative sequence (reverse rotation), and zero sequence (in phase). Used to analyze unbalanced faults. (Ch. 1)

**Synchronous Motor** — An AC motor that runs at exactly synchronous speed ($n_s = 120f/p$ RPM), with the rotor locked to the rotating magnetic field. Requires excitation (DC field or permanent magnets). (Ch. 12)

**Tap Rule** — An NEC exception (240.21(B)) allowing conductors smaller than the overcurrent device rating for limited lengths (10-foot or 25-foot tap rules), provided specific conditions are met. (Ch. 14)

**Target Tracking** — The process of estimating a radar target's position, velocity, and trajectory over time using sequential measurements; implemented with Kalman filters or alpha-beta trackers and requiring data association in multi-target environments. (Ch. 17)

**TCLab (Temperature Control Lab)** — An Arduino-based dual heater/temperature shield used as an educational control systems plant. Two resistive heaters (Q1, Q2, 0–100 %) and two surface temperature sensors (T1, T2) are accessible over USB serial. The plant behaves as a FOPDT system with time constant $\tau \approx 143$ s and dead time $\theta \approx 20$ s, making it well-suited for step-test identification and PID tuning exercises. (App. G)

**TCAM (Ternary Content-Addressable Memory)** — Specialized memory in routers and switches that performs parallel lookups on all entries simultaneously, enabling wire-speed forwarding table lookups with ternary (0, 1, don't-care) matching. (Ch. 15)

**TCP (Transmission Control Protocol)** — A connection-oriented, reliable Layer 4 protocol that provides ordered delivery, flow control, and congestion avoidance using sequence numbers, acknowledgments, and a sliding window mechanism. (Ch. 15)

**TDM (Time Division Multiplexing)** — A multiplexing technique that assigns each signal a recurring time slot within a repeating frame, allowing multiple signals to share a single channel. The T-carrier (DS-1/T1: 1.544 Mbps) and SONET/SDH hierarchies are TDM systems. (Ch. 2)

**TDR (Time-Domain Reflectometry)** — A measurement technique that launches a fast step into a transmission line and analyzes the reflected waveform to locate impedance discontinuities, faults, and connector transitions. Each discontinuity appears as a step at a time corresponding to twice the electrical distance; the characteristic impedance at any point is $Z = Z_0(1 + \Gamma)/(1 - \Gamma)$. (Ch. 9)

**THD (Total Harmonic Distortion)** — A measure of waveform distortion: THD = $(\sqrt{(\Sigma V_h^2)})/V_1 \times$ 100%. IEEE 519 limits voltage THD to 5% at the point of common coupling for systems below 69 kV. (Ch. 1)

**Thermal Imaging Camera** — An instrument that captures infrared radiation emitted by objects to produce a two-dimensional temperature map (thermogram). Uses cooled (InSb, HgCdTe) or uncooled (microbolometer) detector arrays. Key parameter: NETD (Noise Equivalent Temperature Difference), typically 20–50 mK. Requires surface emissivity compensation for accurate temperature measurement. (Ch. 11)

**Thermal Interface Material (TIM)** — A material placed between a semiconductor package and heat sink to fill microscopic air gaps and reduce thermal resistance. Types include thermal grease (1–5 W/m·K), phase-change materials, thermal pads, and sintered silver (up to 250 W/m·K). (Ch. 10)

**Thermal Noise** — Random electrical noise generated by the thermal agitation of charge carriers in any conductor at a temperature above absolute zero, also called Johnson-Nyquist noise. The noise power is N = kTB, where k is Boltzmann's constant, T is temperature in kelvin, and B is bandwidth in hertz. (Ch. 2)

**Thermal Runaway** — An uncontrolled exothermic chain reaction in a lithium-ion battery cell where internal temperature rises above ~130°C, causing separator melting, internal short circuit, and potentially fire or explosion. BMS protection circuits monitor cell voltages and temperatures to detect early signs and disconnect the pack. (Ch. 10)

**Thermocouple** — A temperature sensor formed by joining two dissimilar metals, producing a voltage proportional to temperature difference (Seebeck effect). Common types: J, K, T, E, N. (Ch. 11)

**Thin-Film Optical Coating** — A multilayer stack of alternating high-index and low-index dielectric materials (each a quarter-wave optical thickness) deposited on optical surfaces to engineer spectral reflectance and transmittance. Used to create anti-reflection coatings, high-reflectivity mirrors (> 99.99%), bandpass filters, and dichroic beam splitters. (Ch. 18)

**Throughput** — The actual rate of successful data delivery over a network link, always less than the raw line rate due to protocol overhead, retransmissions, and congestion. (Ch. 15)

**Thyristor (SCR)** — A four-layer (PNPN) semiconductor device that conducts when triggered by a gate pulse and continues conducting until forward current drops below the holding current. Used in AC power control and rectifier circuits. (Ch. 10)

**Thévenin's Theorem** — Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source ($V_{Th}$) in series with a resistance ($R_{Th}$). (Ch. 7)

**Time Constant (τ)** — The time for an exponential response to reach 63.2% of its final value (charging) or decay to 36.8% (discharging). $\tau = RC$ for RC circuits, $\tau = L/R$ for RL circuits. (Ch. 7)

**Time Value of Money** — The fundamental principle that a dollar available today is worth more than a dollar received in the future due to its earning potential through investment. (Ch. 19)

**TL431** — A programmable precision shunt voltage regulator from Texas Instruments, widely used as a voltage reference and error amplifier in power supply feedback circuits. (Ch. 3)

**Torque** — The rotational force produced by a motor, measured in newton-meters (N·m). Related to power and speed by $T = P / \omega = P \times 60 / (2\pi \times n)$. (Ch. 12)

**Total Internal Reflection** — The complete reflection of light at an interface when traveling from a denser to a less dense medium at an angle exceeding the critical angle $\theta_c = \arcsin(n_2/n_1)$. The fundamental waveguiding mechanism in optical fibers. (Ch. 18)

**Transfer Function** — The ratio of the Laplace transform of a system's output to its input (assuming zero initial conditions): $H(s) = Y(s)/X(s)$. Characterizes the input-output behavior of an LTI system. (Ch. 4, 8)

**Transfer Switch (ATS)** — An automatic transfer switch that monitors the normal power source and transfers the load to an alternate source (generator) when the normal supply fails. Emergency systems (NEC Article 700) require transfer within 10 seconds. Modern ATS units include bypass isolation, programmable time delays, and in-phase transition monitoring. (Ch. 14)

**Transformer** — A static electromagnetic device that transfers electrical energy between circuits through mutual induction, enabling voltage step-up or step-down while maintaining approximately constant power. (Ch. 1)

**Transient Response** — The temporary circuit behavior that occurs during the transition between steady states, governed by the natural response of energy-storage elements (capacitors and inductors). (Ch. 7)

**Transimpedance Amplifier (TIA)** — An op-amp circuit that converts an input current to an output voltage with gain $V_{out} = -I_{in} \times R_f$. Used with photodiodes, phototransistors, and current-output sensors. A feedback capacitor $C_f$ is required for stability. (Ch. 13)

**Transmission Line** — A pair of conductors designed to carry electromagnetic signals from source to load with controlled impedance. In power systems, high-voltage lines carrying bulk power over long distances. (Ch. 1, 9, 15)

**Triplen Harmonics** — Multiples of the third harmonic (3rd, 6th, 9th, 12th, 15th, ...) that are zero-sequence. In a balanced three-phase system, even triplens cancel; odd triplens (3rd, 9th, 15th, ...) are zero-sequence and additive in the neutral conductor of a three-phase wye system. In systems with nonlinear loads, the neutral current from triplens can exceed the phase current, requiring oversized neutral conductors. (Ch. 14)

**Truth Table** — A tabular representation of a Boolean function showing the output value for every possible combination of input values. (Ch. 6)

**TTL (Time to Live)** — A field in the IP packet header that is decremented by each router; when it reaches zero, the packet is discarded. Prevents routing loops from causing packets to circulate indefinitely. (Ch. 15)

**TTL (Transistor-Transistor Logic)** — A logic family built from bipolar junction transistors, historically dominant in digital design. The 74LS series operates at 5 V with $V_{OH}$ = 2.7 V, $V_{OL}$ = 0.5 V, and lower noise margins than CMOS. (Ch. 6)

**Twisted-Pair Cable** — Copper cable consisting of insulated conductor pairs twisted together to reduce electromagnetic interference and crosstalk. Categories: Cat 5e (100 MHz), Cat 6 (250 MHz), Cat 6A (500 MHz), Cat 8 (2 GHz). (Ch. 15)

**Two-Port Network** — A circuit abstraction with four terminals grouped into an input port and an output port, characterized by parameter sets (Z, Y, h, ABCD) that relate port voltages and currents. Used to model amplifiers, filters, transformers, and transmission line segments. (Ch. 7)

**UART (Universal Asynchronous Receiver/Transmitter)** — A hardware peripheral for asynchronous serial communication using start/stop bits for framing, configurable baud rate, and optional parity for error detection. Common in embedded systems and RS-232 interfaces. (Ch. 5)

**UDP (User Datagram Protocol)** — A connectionless, unreliable Layer 4 protocol with minimal overhead (8-byte header), used for real-time applications (VoIP, video, gaming) and lightweight queries (DNS, SNMP, NTP). (Ch. 15)

**Ultrafast Laser** — A laser producing pulses with durations from femtoseconds ($10^{-15}$ s) to picoseconds ($10^{-12}$ s), enabling applications in precision micromachining, nonlinear optics, and time-resolved spectroscopy; mode-locked oscillators with chirped pulse amplification (CPA) achieve peak powers exceeding terawatts. (Ch. 18)

**Units-of-Production Depreciation** — A depreciation method that allocates cost based on actual usage rather than time elapsed, appropriate for assets whose wear is driven by usage. (Ch. 19)

**Universal Motor** — A series-wound DC motor designed to operate on both AC and DC supplies, achieving high speeds (up to 20,000–30,000 RPM) and high power-to-weight ratios. Commonly used in handheld power tools, vacuum cleaners, and blenders, with speed controlled by triac phase-angle controllers. (Ch. 12)

**USB (Universal Serial Bus)** — A communication standard providing data transfer and power delivery through a single cable, using a host-device topology. Speeds range from Full Speed (12 Mbps) to SuperSpeed (5-20 Gbps). Common device classes include CDC, HID, and MSC. (Ch. 5, 15)

**Varactor Diode** — A PN junction diode designed to exploit the voltage-dependent depletion capacitance under reverse bias, used in voltage-controlled oscillators (VCOs), tuning circuits, and frequency multipliers. (Ch. 3)

**VCSEL (Vertical-Cavity Surface-Emitting Laser)** — A semiconductor laser that emits light perpendicular to the chip surface at 850 nm, used as the standard optical source for multimode fiber links due to low cost and high modulation speed. (Ch. 15, 18)

**Vector Network Analyzer (VNA)** — An instrument that measures the complex (magnitude and phase) S-parameters of RF/microwave devices, characterizing reflection and transmission as a function of frequency. Results are displayed as return loss, insertion loss, VSWR, Smith chart impedance, and group delay. Requires SOLT or TRL calibration to remove systematic errors. (Ch. 11)

**Vector Potential (A)** — A vector field defined by $B = \nabla \times A$, providing an alternative formulation of Maxwell's equations that simplifies radiation and antenna calculations. Combined with the scalar potential $\varphi$ (where $E = -\nabla\varphi - \partial A/\partial t$), the retarded potentials give the fields produced by arbitrary time-varying source distributions. (Ch. 9)

**VFD (Variable Frequency Drive)** — A power electronics system that controls AC motor speed by varying the frequency and voltage of the power supplied to the motor. Consists of a rectifier, DC bus, and inverter. PWM output causes reflected wave voltage doubling on long cables and bearing currents from common-mode voltage; mitigation includes dv/dt filters, sine-wave filters, insulated bearings, and shaft grounding brushes. (Ch. 12)

**Virtual Ground** — A node in an op-amp circuit (typically the inverting input) that is held at ground potential by negative feedback, without being directly connected to ground. (Ch. 13)

**VLAN (Virtual LAN)** — A logical subdivision of a physical network at Layer 2, implemented using

IEEE 802.1Q tags (12-bit VLAN ID) to isolate broadcast domains without separate physical infrastructure. (Ch. 15)

**Volt (V)** — The SI derived unit of electric potential difference. One volt is the potential difference that drives one ampere through one ohm of resistance. (App. D)

**Voltage Drop** — The reduction in voltage along a conductor due to its resistance (and reactance for AC), calculated as $V_{drop} = I \times Z \times L$. NEC recommends limiting voltage drop to 3% for branch circuits and 5% total. (Ch. 14)

**Voltage Follower (Buffer)** — A non-inverting op-amp configuration with unity gain ($A = 1$) and very high input impedance, used to isolate a high-impedance source from a low-impedance load. (Ch. 13)

**Voltage Mode Control** — A power converter control strategy where the duty cycle is determined by comparing the output voltage error signal to a fixed-frequency sawtooth ramp; simpler than current mode control but requires slope compensation at duty cycles above 50% and has slower transient response due to the output LC filter pole. (Ch. 10)

**Voltage Regulator** — A circuit that maintains a constant output voltage despite variations in input voltage or load current. Types include linear regulators, switching regulators (buck, boost), and shunt regulators (TL431). (Ch. 3)

**Voltage Sag** — A temporary reduction in RMS voltage to between 10% and 90% of nominal lasting 0.5 cycles to 1 minute, typically caused by faults on adjacent feeders or motor starting. The most common power quality disturbance affecting industrial facilities. (Ch. 1)

**Voltage Transformer (VT)** — An instrument transformer that steps down high primary voltages to standardized secondary voltages (typically 120 V) for metering and protective relaying. Also called a potential transformer (PT). (Ch. 1)

**VSWR (Voltage Standing Wave Ratio)** — A measure of impedance mismatch on a transmission line: $VSWR = (1 + |\Gamma|)/(1 - |\Gamma|)$, where $\Gamma$ is the reflection coefficient. VSWR = 1:1 is a perfect match; VSWR < 2:1 (return loss > 10 dB) is generally acceptable. (Ch. 16)

**VT** — See Voltage Transformer. (Ch. 1)

**Watchdog Timer (WDT)** — A safety peripheral that resets the microcontroller if firmware fails to refresh it within a specified timeout period, providing automatic recovery from software hangs and fault conditions. Required by most safety-critical standards. (Ch. 5)

**Watt (W)** — The SI derived unit of power, equal to one joule per second. In electrical circuits, $P = V \times I$ (DC) or $P = V_{rms} \times I_{rms} \times \cos\theta$ (AC). (App. D)

**Watt-hour (Wh)** — A unit of energy equal to one watt sustained for one hour, or 3,600 joules. Commonly used for electrical energy billing (kWh). (App. D)

**Waveguide** — A hollow metallic structure (rectangular or circular) that guides electromagnetic waves by confining them within conducting walls. Operates above a cutoff frequency $f_c$ determined by cross-sectional dimensions; preferred over coaxial cables at microwave frequencies for lower loss and higher power handling. (Ch. 9)

**Wavelet Transform** — A multi-resolution analysis that decomposes a signal into scaled and translated wavelets, providing adaptive time-frequency resolution (fine time resolution at high frequencies, fine frequency resolution at low frequencies). The DWT uses Mallat's filter bank algorithm for efficient computation. Used in denoising, JPEG 2000 compression, and transient detection. (Ch. 8)

**Wheatstone Bridge** — A four-resistor circuit that measures an unknown resistance by adjusting known reference resistors until a null detector reads zero, achieving sub-ppm accuracy. At balance, $R_x = R_2 \times R_3/R_1$. Extended forms include the Kelvin bridge (low resistance), Wien bridge (capacitance), Maxwell bridge (inductance), and Schering bridge (high-voltage insulation). (Ch. 11)

**Wi-Fi** — Wireless local-area networking technology based on the IEEE 802.11 family of standards, operating in the 2.4 GHz, 5 GHz, and 6 GHz unlicensed bands. Generations include Wi-Fi 4 (802.11n), Wi-Fi 5 (802.11ac), Wi-Fi 6/6E (802.11ax), and Wi-Fi 7 (802.11be). (Ch. 15)

**Wide-Bandgap Semiconductor** — A semiconductor material with a bandgap significantly larger than silicon's 1.12 eV, such as silicon carbide (SiC, 3.26 eV) and gallium nitride (GaN, 3.4 eV); enables higher blocking voltages, switching frequencies, and operating temperatures with lower conduction and switching losses. (Ch. 10)

**Wien Bridge Oscillator** — An op-amp sine wave oscillator using a Wien network (series RC + parallel RC) in the positive feedback path, oscillating at $f_0 = 1/(2\pi RC)$ when the non-inverting amplifier gain equals exactly 3. Amplitude stabilization (lamp, FET AGC) is required for low-distortion output. (Ch. 13)

**Wiener Filter** — The theoretically optimal linear filter that minimizes mean squared error between its output and a desired signal. In the frequency domain, $H(f) = S_{xd}(f)/S_{xx}(f)$. For additive noise with signal and noise uncorrelated, simplifies to $H(f) = SNR(f)/(1 + SNR(f))$, passing frequencies with high SNR and suppressing those with low SNR. The LMS and RLS adaptive algorithms converge toward the Wiener solution iteratively. (Ch. 8)

**Windowing** — The application of a window function (Hanning, Hamming, Blackman, Kaiser) to a finite-length signal before DFT computation, reducing spectral leakage at the expense of frequency resolution. (Ch. 8)

**Wireless Power Transfer (WPT)** — The transmission of electrical energy across an air gap using time-varying magnetic fields between loosely coupled coils (coupling coefficient k = 0.1–0.6). Series-series resonant compensation cancels reactive impedance to maximize efficiency. Standards include Qi (consumer electronics, 5–15 W at 110–205 kHz) and SAE J2954 (EV charging, 3.7–22 kW at 85 kHz). (Ch. 10)

**Wound-Rotor Motor** — An induction motor with a three-phase rotor winding connected to external circuits through slip rings, allowing insertion of external resistance for torque-speed control. Provides maximum torque at standstill for crane and hoist applications, and enables slip-energy recovery (Scherbius drives). (Ch. 12)

**WPA3 (Wi-Fi Protected Access 3)** — The current Wi-Fi security standard, replacing WPA2. WPA3-Personal uses SAE (Simultaneous Authentication of Equals) to resist offline dictionary attacks and provide forward secrecy. WPA3-Enterprise mandates 192-bit cryptographic strength with AES-256-GCM. (Ch. 15)

**WSGI (Web Server Gateway Interface)** — A Python standard (PEP 3333) defining the interface between web servers and Python web applications, enabling any WSGI-compliant application (Flask, Django) to run on any WSGI-compliant server (Gunicorn, uWSGI). (Ch. 15)

**Wye Connection (Y)** — A three-phase winding configuration where one terminal of each phase is connected to a common neutral point. $V_{LL} = \sqrt{3} \times V_{LN}$, $I_L = I_{phase}$. Provides a neutral for grounding and single-phase loads. (Ch. 1)

**Yagi-Uda Antenna** — A directional wire antenna array with a single driven element (dipole or folded dipole), a reflector element (~5% longer than $\lambda/2$), and one or more director elements (~5–10% shorter). Parasitic elements create forward gain through mutual coupling. A 5-element Yagi achieves ~10.5 dBi. (Ch. 16)

**Z-Transform** — The discrete-time equivalent of the Laplace transform, converting a discrete-time sequence x[n] to a function of the complex variable z: $X(z) = \Sigma\, x[n]z^{-n}$. Used for digital filter analysis and design. (Ch. 8)

**Zener Diode** — A specially designed diode intended to operate in reverse breakdown at a precise voltage (the Zener voltage), providing voltage regulation and voltage reference functions. (Ch. 3)

**Zero-Order Hold (ZOH)** — A circuit that holds a sampled value constant for one sample period T until the next sample arrives, producing a staircase approximation of the continuous signal. Used in digital control to interface discrete-time controllers with continuous-time plants. (Ch. 4)

**Ziegler-Nichols Method** — An empirical PID tuning procedure using either the step response (dead time and time constant) or the ultimate gain method (finding the gain $K_u$ and period $P_u$ that produce sustained oscillations) to determine initial $K_p$, $K_i$, and $K_d$ values. (Ch. 4)

**ZVS (Zero-Voltage Switching)** — A soft-switching technique where a power transistor turns on only after its drain-source voltage has been reduced to zero by resonant action, eliminating turn-on switching losses and enabling higher switching frequencies with high efficiency. (Ch. 10)